



The Watchmaker's Guide to Population Genetics

Build every algorithm from first principles

Kevin Korfmann

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Kevin Korfmann

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“Die Tat ist alles, nichts der Ruhm.”

--- Johann Wolfgang von Goethe, *Faust II*

PREFACE

Population genetics is blessed with powerful algorithms --- but cursed with inaccessible ones. Many of the field's most important methods live inside papers and codebases that assume years of specialized training to read, let alone reimplement. They rarely come with manuals, guided tours, or on-ramps for the curious outsider --- or even for the insider who works on a different corner of the field. This book is an attempt to change that: to make the algorithms not only open but genuinely *accessible*, with all the prerequisites laid out and every derivation shown in full.

I assembled these chapters during my transition from the University of Oregon to the University of Pennsylvania, at my own expense and in my own time. The book is freely available because I believe science should be. It was not, however, free to create --- and I mention this only to underscore that the motivation was personal before it was practical. I wanted to understand these algorithms more deeply myself. Writing them out, gear by gear, was the surest way I knew how.

Nothing here is meant to diminish the original work. Each algorithm in this book represents a serious intellectual achievement --- the kind that earns PhD titles and advances entire subfields. The goal is translation, not judgment: to take ideas that were expressed for expert audiences and re-express them for anyone willing to learn. The content may contain errors --- mathematical, conceptual, or otherwise --- and should ideally be read in combination with the original journal articles, which remain the authoritative source for each method.

The book is accompanied by `watchgen`, a Python package that provides minimal, self-contained implementations of every algorithm covered. These mini implementations are not production tools; they are pedagogical companions --- small enough to read in one sitting, complete enough to run on toy examples, and tested enough to give you confidence that the math on the page actually works. Think of them as the movements you build on the workbench: not meant for sale, but meant to teach your hands what your eyes have read.

Finally, this project is an open invitation. I welcome collaborators who want to cross-check derivations, correct mistakes, improve explanations, add chapters, or simply point out where things could be clearer. The ambition is a living resource that grows more accurate and more useful over time, built by the community it is meant to serve.

Looking ahead. The mini implementations in `watchgen` are pedagogical --- deliberately simple, deliberately slow. But the landscape is shifting fast. AI models are growing more capable at an extraordinary pace, and within the next year it may become realistic to go further: to use these same models to produce a unified, production-grade software package that brings the algorithms covered here under a single roof --- correct, tested, interoperable, and maintained. This book, with its explicit derivations and reference implementations, is designed to serve as the foundation for exactly that kind of effort.

On versioning. This is a living draft under chapter-by-chapter technical audit. The online verification table records completed source, implementation, and automated-test reviews. These reviews are not a substitute for independent domain-expert review, and no derivation should be treated as error-free. Substantial contributors will be credited as the project develops.

Kevin Korfmann
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A watchmaker doesn't just use a watch – they understand every gear, every spring, every mechanism. Nothing is a black box.

Welcome to *The Watchmaker's Guide to Population Genetics*.

The Watchmaker's Guide is a hands-on, build-it-yourself guide to the algorithms behind modern population genetics. Every concept is explained from first principles, every method implemented from scratch in Python, every abstraction earned rather than assumed. Think of it as an apprenticeship: you start with raw materials – basic math, simple code – and end up with working mechanisms that you built with your own hands.

By the end you won't just *run* the tools – you'll know how to *build* them. And like a watchmaker, once you've built something by hand, you can fix it, modify it, and trust it completely.

THE WATCHMAKER'S PHILOSOPHY

1.1 Why Build It Yourself?

Population genetics has a problem. The field's most important algorithms are locked inside software packages that few people truly understand. Researchers run tools, get numbers, and publish papers – but ask them to explain what happens inside the black box, and you'll get a shrug or a hand-wave toward a 40-page supplement.

This isn't anyone's fault. The methods *are* complex. The math *is* deep. The implementations *are* thousands of lines of optimized code. It's rational to treat them as black boxes.

But it's also dangerous.

When you don't understand the mechanism, you can't:

- Diagnose when results are wrong
- Adapt methods to new problems
- Know which assumptions are violated by your data
- Improve upon existing approaches
- Teach the next generation how the tools actually work

Imagine a watchmaker who can sell watches but cannot open one. Who can read the time but cannot explain why the hands move. That's the state of much of computational population genetics today: we read the output, but we can't explain the mechanism.

1.2 The Watchmaker's Way

A watchmaker who only buys watches is a watch *dealer*, not a watchmaker. The craft is in the building. The understanding is in the assembly. And the confidence – the ability to trust your results, diagnose problems, and push the boundaries of what's possible – comes only from having built the mechanism yourself, gear by gear.

In this guide, every algorithm is a **Timepiece** – a mechanism you disassemble, understand, and reassemble with your own hands. We don't skip steps. We don't hand-wave. We don't say “it can be shown that...” and move on.

Here's what we promise:

We WILL	We will NOT
Derive every equation	Skip “obvious” steps
Implement every algorithm	Use library functions as black boxes
Explain every assumption	Hide behind “standard results”
Test every implementation	Trust code without verification
Build intuition first	Start with the most general case
Teach the math as we go	Assume you already know everything

1.3 The Gears of Understanding

Every watch, no matter how complex, is built from smaller parts: gears, springs, escapements, jewel bearings. Individually, each part is simple enough to hold in your hand and understand completely. The complexity of a watch arises not from any single part, but from how the parts mesh together.

The same is true for algorithms.

Each Timepiece is built from **gears** – smaller mechanisms that mesh together. We build the smallest gears first, test them, then combine them into larger mechanisms. This mirrors how real algorithms are designed, and how understanding actually develops: from the simple to the complex, with each step resting firmly on the one before it.

We organize these gears into four categories, borrowing from horology:

1. **Escapement** – The core mathematical insight that makes the whole thing tick. In a watch, the escapement regulates the release of energy. In our algorithms, this is the key formula or theorem that everything else depends on.
2. **Gear Train** – The computational machinery that turns insight into algorithm. In a watch, the gear train transmits energy from the mainspring to the hands. Here, it’s the sequence of computational steps that transforms the core insight into a working procedure.
3. **Mainspring** – The data structures and implementation details that store energy and state. In a watch, the mainspring stores the energy that drives everything. In code, these are the arrays, tables, and structures that hold the data the algorithm works on.
4. **Case and Dial** – The interface: inputs, outputs, and interpretation. The case protects the mechanism; the dial presents the result. For us, this is how the algorithm connects to real data and how we interpret what it tells us.

You wouldn’t try to understand a watch by staring at the finished product through the crystal. You’d take it apart, lay out every piece on the bench, understand what each does, and carefully put it back together. That’s exactly what we do here.

1.4 On Mathematical Rigor

We don’t shy away from math – we embrace it. But we also believe that mathematics is a *tool*, not a gatekeeping mechanism. Every formula in this book is here because it helps you understand something, not because it makes us look clever.

Our approach to mathematics:

- **Every equation is motivated before it’s stated.** We always explain *why* we need a formula before writing it down. What question does it answer? What would we be missing without it?
- **Every derivation is accompanied by intuition.** Alongside the formal steps, we provide a plain-English narrative: “this term counts the probability of survival, and this term counts the probability of the event happening right now.”
- **Every formula is implemented in code so you can see it work.** Mathematics on paper can feel abstract. Mathematics in Python, producing numbers you can check, feels concrete and real. Code is our way of testing understanding: if you can implement a formula correctly, you understand it.

- **Notation is consistent and explicitly defined.** We use the same symbols for the same concepts throughout the book, and we define every symbol when it first appears.

When you encounter a formula in this book, you should be able to:

1. Say in plain English what it computes
2. Explain why it has the form it does
3. Implement it in Python
4. Verify it produces correct results on simple examples

If any of these fail, we've failed as authors. Like a watch that doesn't keep time, a formula you can't use is a formula that isn't working.

1.5 On Teaching Probability and Calculus

This book uses probability and calculus extensively – they are essential tools for understanding how genetic algorithms work. But we don't assume you arrive with these tools already sharp and polished in your toolkit.

Here's our approach:

For probability: We introduce each concept as it naturally arises. The first time we need the exponential distribution, we explain what it is, where it comes from, why it appears in our context, and what its parameters mean. The same for Bayes' theorem, conditional probability, Markov chains, and every other concept. We provide "Probability Aside" boxes that give you the background you need, right when you need it.

For calculus: We use derivatives and integrals, but we always explain what they mean in context. A derivative is a rate of change – we'll say what's changing and how fast. An integral is a sum – we'll say what's being summed and why. When we differentiate or integrate, we show the steps and explain the rules being used.

For Python: We explain every non-obvious construct the first time it appears. If you know basic Python (variables, loops, functions), you have enough to start. We'll explain NumPy array operations, list comprehensions, and anything else as we go.

Think of it this way: a master watchmaker teaching an apprentice doesn't assume the apprentice already knows metallurgy, thermodynamics, and materials science. Instead, the master teaches these things *as they become relevant* – explaining the properties of steel when it's time to shape a spring, explaining thermal expansion when it's time to fit a jewel bearing. Context makes learning stick.

That's our philosophy: teach every tool in the context where it's needed, so the knowledge arrives precisely when it's useful.

1.6 On Python Implementations

We use Python because it reads like pseudocode. Our implementations prioritize **clarity over performance**. Real production tools use C++ for speed – but you can't learn from code you can't read.

Every implementation follows these rules:

- **No magic imports.** We use NumPy and SciPy, nothing else. Every helper function is written from scratch in front of you. When we call a NumPy function for the first time, we explain what it does.
- **No premature optimization.** We write the naive version first, understand it, then optimize only where the math demands it. The naive version often reveals the structure of the algorithm more clearly than the optimized one.
- **No hidden state.** Every variable is named descriptively, every parameter is documented. You should be able to read the code like a recipe – each line corresponding to a step in the mathematical derivation.

- **Every code block can be run.** The code in this book is not pseudocode dressed up to look like Python. It runs. It produces the numbers we claim. You can type it into a notebook and verify for yourself.

A note on production code

The code in this book is *educational* code. It is correct, tested, and complete – but it is not optimized for production use. Real tools like SINGER, Relate, and tsinfer use C++ with careful memory management for good reason. Our Python implementations exist to help you understand the algorithms, not to replace the production tools.

Think of the difference like this: a watchmaking student first builds a movement from brass using hand tools. Later, they may work with modern CNC machines and exotic alloys. But the hand-built movement teaches them something that no factory ever could – the deep understanding of *why* each part has the shape it does.

1.7 Your Journey

You're about to build algorithms that took decades to develop. You'll derive equations that fill supplementary materials. You'll implement methods that exist as thousands of lines of C++.

It won't be easy. But neither is watchmaking. The reward, in both cases, is the same: the profound satisfaction of understanding a complex mechanism so completely that you could build it again from memory.

When you're done, you'll own the knowledge completely – no black boxes, no hand-waving, no mystery, just gears you built yourself, ticking reliably, because you understand every one.

Let's open the case and get to work.

THE WORKBENCH (PREREQUISITES)

Before you can build a Timepiece, you need the right tools on your workbench.

A watchmaker's bench holds files, pliers, tweezers, loupes – each one essential for specific tasks, each one mastered individually before being used in combination. Our workbench holds mathematical and biological concepts: the foundational ideas that every algorithm in this book is built upon.

This section covers eight topics. Each one is self-contained and builds from the ground up – we don't assume you've seen these ideas before. If you *have* encountered them, the treatment here will sharpen your understanding and connect the concepts directly to the algorithms we'll build later.

The suggested reading order is:

1. **Likelihood-Based Probabilistic Inference** – The inferential logic that unifies every Timepiece. Before diving into any specific algorithm, you need to understand the framework they all share: the likelihood function, maximum likelihood estimation, and Bayesian inference. This chapter explains how we go from observed genetic data to conclusions about evolutionary history, and situates the likelihood-based approach alongside the emerging neural-network paradigm. (*Start here – it frames everything that follows.*)
2. **Coalescent Theory** – How to think about ancestry backwards in time. This is the biological foundation: the idea that the genealogical history of a sample can be described by a branching tree, and that the shape of this tree is governed by simple probabilistic rules. We'll introduce the exponential distribution, the Poisson process, and the fundamental connection between population size and coalescence time. (*This is our most important tool – it goes into every Timepiece.*)
3. **Ancestral Recombination Graphs** – When a single tree isn't enough. Recombination means that different parts of the genome can have different genealogical histories. An ARG captures this full picture: the complete history of a sample, including all the places where the tree changes. We'll explain what recombination is, why it complicates things, and how the tree sequence data structure makes the complexity manageable.
4. **Hidden Markov Models** – The computational engine behind most methods. An HMM is a mathematical framework for inferring hidden information from noisy observations. We'll build one from scratch, implement the forward algorithm, and show how a clever trick (the Li-Stephens structure) makes it fast enough for genomic data. If you've never seen an HMM before, this chapter will give you everything you need.
5. **The Sequentially Markov Coalescent** – Making the impossible tractable. The full coalescent with recombination is not Markov, which means we can't use HMMs directly. The SMC is an approximation that restores the Markov property at the cost of a tiny amount of accuracy. We'll explain exactly what's lost and why it doesn't matter much.
6. **The Diffusion Approximation** – When N is large, the discrete Wright-Fisher model converges to a continuous diffusion process governed by a partial differential equation (the Fokker-Planck equation). This chapter develops the diffusion limit, stochastic differential equations, boundary conditions, stationary distributions, and finite-difference numerical methods. (*Essential for moments, dadi, and momi2.*)
7. **Ordinary Differential Equations** – Many Timepieces reduce their core computation to solving a system of ODEs. This chapter covers ODE fundamentals, Euler's method, Runge-Kutta solvers, coupled systems, stiffness, and the matrix exponential. (*Essential for moments; useful for SMC++ and momi2.*)

8. **Markov Chain Monte Carlo** – When the posterior distribution over genealogies is too complex to compute directly, MCMC provides a principled way to sample from it. This chapter covers Bayesian inference, the Metropolis-Hastings algorithm, Gibbs sampling, and convergence diagnostics. (*Essential for ARGweaver, SINGER, and PHLASH.*)

The first five tools – probabilistic inference, the coalescent, the ARG, the HMM, and the SMC – form the inferential, biological, and computational backbone of the book. The last three – diffusion theory, ODEs, and MCMC – provide the mathematical machinery that specific Timepieces rely on. Together, these eight instruments equip you to build any Timepiece in the collection.

i Do I need to read all of these?

It depends on which Timepiece you want to build. **Probabilistic Inference** and **Coalescent Theory** are essential for everything – start there. **HMMs** are needed for PSMC, SINGER, and ARGweaver. **ARGs** and the **SMC** are needed for SINGER and tsinfer. **The Diffusion Approximation** and **ODEs** are needed for moments and dadi. **MCMC** is needed for ARGweaver, SINGER, and PHLASH. If you're not sure, start with Probabilistic Inference and Coalescent Theory, then work through the first five in order – they build naturally on each other. The last three can be read independently as needed.

2.1 Likelihood-Based Probabilistic Inference

“The data are fixed; the parameters vary. The likelihood surface ranks their support within the model.”

A watchmaker presented with a broken watch doesn't guess randomly. She examines the symptoms – the watch runs 5 minutes fast per day, the second hand stutters at 12 o'clock – and asks: *which internal configurations would produce these specific symptoms?* The configurations that best explain the symptoms are the most likely diagnoses.

This is the logic of likelihood-based inference, and it is the inferential backbone of every inferential Timepiece in this book. Given observed genetic data (the symptoms), we want to find the demographic history, genealogy, or set of parameters (the internal configuration) that best explains what we see. This chapter introduces the framework that makes this possible: the likelihood function, maximum likelihood estimation, and Bayesian inference. It also situates these approaches within the broader landscape of modern inference, including the neural-network-driven methods that have emerged as an alternative paradigm.

Unlike the other prerequisites in this section, this chapter is *not* about a single mathematical topic. It is about the inferential logic that connects all the other topics together. The exponential distribution, the Poisson process, the HMM forward algorithm, the diffusion equation – each of these is a tool on the workbench. Likelihood-based inference is the *method* by which these tools are put to work: the discipline of asking, for each tool, “what does the data tell us about the parameters?”

2.1.1 Why Likelihood?

Every inference method in this book follows the same inferential logic. The simulation Timepieces instead run the generative model in the forward direction, from parameters to synthetic data.

1. **Propose a model** with parameters θ (a demographic history, a genealogy, a set of population sizes and split times).
2. **Derive the probability** of observing the data D under this model: $P(D | \theta)$. This is the **likelihood function**.
3. **Find the parameters** that make the data most probable (maximum likelihood), or characterize the full range of plausible parameters (Bayesian inference).

The likelihood function is the bridge between models and data. It transforms an abstract mathematical model – “the ancestral population had size 10,000 and split 2,000 generations ago” – into a concrete, quantitative prediction that can be compared to the genome sequences in your FASTA file.

i Biology Aside – Why population genetics is well-suited to likelihood

Population genetics models are **generative**: they describe the mechanistic process that produces genetic data. The coalescent tells us how lineages merge backward in time. The mutation model tells us how differences accumulate on branches. The recombination model tells us how the genealogy changes along the genome. Because these models are fully specified probability models, they define a probability mass or density for possible data under each admissible parameter value. This is what makes likelihood-based inference possible. It does *not* mean that every likelihood can be evaluated exactly: for many realistic models latent variables must be marginalized by an HMM or Monte Carlo method, or the full likelihood is replaced by an approximation or composite likelihood. Summary statistics such as F_{ST} and Tajima's D compress the data and can diagnose departures from simple null models, but a value of either statistic generally does not identify one unique demographic or selective mechanism.

2.1.2 The Likelihood Function

For a parameter vector θ (population sizes, divergence times, migration rates, mutation rates, etc.), the **likelihood function** is:

$$\mathcal{L}(\theta) = P(D \mid \theta)$$

This reads: “the probability of observing data D if the true parameters were θ .”

For discrete data, $P(D \mid \theta)$ is a probability mass. For continuous data it denotes a density at the observed value; the probability of any one exact continuous outcome is zero. In either case, the likelihood need not integrate to one over θ .

Crucially, the likelihood is a function of θ , not of D . The data are fixed (you've already sequenced the genomes); the parameters are what you're trying to learn. Different parameter values give different likelihoods, and the likelihood surface – the landscape of $\mathcal{L}(\theta)$ over all possible θ – tells you which parameters are consistent with your data and which are not.

In practice, we work with the **log-likelihood**:

$$\ell(\theta) = \log P(D \mid \theta)$$

because probabilities are often tiny numbers (the probability of a specific genome sequence is astronomically small), and products of probabilities become sums of log-probabilities, which are numerically stable.

When the data consist of n independent observations $D = (d_1, d_2, \dots, d_n)$, the likelihood factorizes:

$$\mathcal{L}(\theta) = \prod_{i=1}^n P(d_i \mid \theta) \quad \implies \quad \ell(\theta) = \sum_{i=1}^n \log P(d_i \mid \theta)$$

This factorization – turning products into sums via logarithms – is the reason log-likelihoods are so convenient. We will use it repeatedly throughout this chapter and the rest of the book.

i Plain-language summary – What the likelihood surface looks like

Imagine a mountain landscape where the height at each point represents how well a particular demographic history explains your data. The peak of the tallest mountain is the maximum likelihood estimate – the single best-fitting history. A broad, flat-topped mountain means many similar histories fit equally well (the data are not very informative about that parameter). A sharp peak means the data strongly constrain the parameter. A ridge connecting two peaks means two different models are hard to distinguish. The entire shape of this landscape – not just its peak – contains information about what the data can and cannot tell you.

Each inferential Timepiece constructs an objective differently:

Timepiece	Data D	Likelihood $P(D \mid \theta)$
PSMC	Het/hom sequence along genome	HMM forward algorithm (coalescent HMM)
dadi / moments / momi2	Site frequency spectrum (SFS)	Poisson or multinomial over SFS entries
ARGweaver / SINGER	Sequence alignment	Product over sites given a proposed ARG
tsdate	Mutations on tree sequence edges	Poisson edge likelihood times a coalescent prior
PHLASH	Heterozygosity sequences, SFS, and optionally LD summaries	Composite PSMC, SFS, and optional LD terms

The rest of this chapter builds your fluency with the distributions that appear in this table. By the end, you will have derived, coded, and visualized likelihoods for the exponential, Poisson, and gamma distributions – the three workhorses of population genetics inference – and you will have seen how Bayesian inference, conjugate priors, and composite likelihoods extend the framework.

2.1.3 The Toolkit: Key Distributions

Before tackling any Timepiece, you need hands-on experience with the probability distributions that generate genetic data. This section works through each one from first principles, derives the likelihood and its MLE, and implements everything in Python. The examples are drawn directly from population genetics so that every formula you encounter here will reappear, in recognizable form, in the Timepiece chapters.

The Exponential Distribution: Coalescence Waiting Times

The most fundamental random variable in population genetics is the **waiting time to coalescence** – how many generations back in time until two lineages find a common ancestor. Under the standard coalescent model (see *Coalescent Theory*), this waiting time is exponentially distributed.

The exponential distribution with rate parameter $\lambda > 0$ has probability density:

$$f(t \mid \lambda) = \lambda e^{-\lambda t}, \quad t \geq 0$$

Where does this formula come from? The exponential arises whenever we have a process where “something happens at a constant rate per unit time.” The $e^{-\lambda t}$ is the probability of *surviving* (not coalescing) for time t – it decays exponentially because at every instant there is a constant hazard rate λ of the event occurring. The leading λ converts this survival function into a density: it is the probability of the event happening in the infinitesimal interval $[t, t + dt)$.

Its mean and variance can be computed by integration:

$$\mathbb{E}[t] = \int_0^\infty t \cdot \lambda e^{-\lambda t} dt = \frac{1}{\lambda}, \quad \text{Var}(t) = \frac{1}{\lambda^2}$$

The mean $1/\lambda$ is intuitive: a higher rate means shorter waiting times. The variance equals the mean squared, which means the standard deviation equals the mean – exponential waiting times are inherently noisy.

```
import numpy as np

# Visualize the exponential density for different rates
def exponential_pdf(t, lam):
    """Probability density of Exponential(lambda) at time t."""
    return lam * np.exp(-lam * t)

t_grid = np.linspace(0, 5, 200)
```

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```
# Rate = 1 (coalescent units, N_e = reference size)
density_1 = exponential_pdf(t_grid, lam=1.0)
print(f"Rate=1.0: mean={1/1.0:.1f}, density at t=0: {density_1[0]:.2f}")

# Rate = 0.5 (larger population, slower coalescence)
density_05 = exponential_pdf(t_grid, lam=0.5)
print(f"Rate=0.5: mean={1/0.5:.1f}, density at t=0: {density_05[0]:.2f}")

# Rate = 2.0 (smaller population, faster coalescence)
density_2 = exponential_pdf(t_grid, lam=2.0)
print(f"Rate=2.0: mean={1/2.0:.1f}, density at t=0: {density_2[0]:.2f}")

# Verify the mean by simulation
np.random.seed(42)
samples = np.random.exponential(scale=1.0/1.0, size=10000)
print(f"\nSimulated mean (rate=1): {np.mean(samples):.3f} (expected: 1.0)")
print(f"Simulated std: {np.std(samples):.3f} (expected: 1.0)")
```

i Biology Aside – Rate and population size

In the coalescent, the rate of coalescence for a pair of lineages in a population of effective size N_e is $\lambda = 1/(2N_e)$ per generation (in a diploid model). When we measure time in units of $2N_e$ generations (coalescent units), the rate becomes $\lambda = 1$. Larger populations have smaller coalescence rates – it takes longer, on average, for two lineages to find a common ancestor in a large population than in a small one.

The likelihood. Suppose we observe n independent coalescence times $t_1, t_2, \dots, t_n$, each drawn from an exponential distribution with unknown rate λ . Because the observations are independent, the joint probability (the likelihood) is the product of the individual densities:

$$\mathcal{L}(\lambda) = \prod_{i=1}^n f(t_i | \lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i}$$

Taking the logarithm converts the product into a sum – each factor $\lambda e^{-\lambda t_i}$ contributes $\log \lambda + (-\lambda t_i)$ to the log-likelihood:

$$\ell(\lambda) = \sum_{i=1}^n \log(\lambda e^{-\lambda t_i}) = \sum_{i=1}^n [\log \lambda - \lambda t_i] = n \log \lambda - \lambda \sum_{i=1}^n t_i$$

This is a function of the single unknown λ , with the data $t_1, \dots, t_n$ baked in as constants. The two terms pull in opposite directions: $n \log \lambda$ increases with λ (favoring a high rate), while $-\lambda \sum t_i$ decreases with λ (penalizing a rate that predicts shorter times than observed). The MLE is where these two forces balance.

Finding the MLE. To find the maximum, we take the derivative with respect to λ , set it to zero, and solve. The derivative of $n \log \lambda$ is n/λ (using the rule $d/dx[\log x] = 1/x$), and the derivative of $-\lambda \sum t_i$ is $-\sum t_i$:

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n t_i = 0 \quad \implies \quad \hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n t_i} = \frac{1}{\bar{t}}$$

To confirm this is a maximum (not a minimum), check the second derivative: $d^2\ell/d\lambda^2 = -n/\lambda^2 < 0$. The negative sign confirms the log-likelihood is concave, so the critical point is indeed the peak.

The MLE of the rate is the reciprocal of the sample mean. This makes intuitive sense: if lineages coalesce quickly (small $\bar{t}$), the rate must be high (small population); if they coalesce slowly (large $\bar{t}$), the rate must be low (large population).

i Biology Aside – Estimating population size from coalescence times

Since $\lambda = 1/(2N_e)$, the MLE of the effective population size is $\hat{N}_e = \bar{t}/2$ when the observed times are in generations. If times have already been divided by $2N_{\text{ref}}$, the fitted rate estimates N_{ref}/N_e ; converting back to an absolute N_e requires the reference scale. This is the simplest possible demographic estimator: observe how long lineages take to coalesce, and infer the population size from the average. Every inferential Timepiece based on coalescence times elaborates this idea while accounting for unobserved genealogies, recombination, and dependence.

```
import numpy as np

def exponential_log_likelihood(lam, times):
    """Log-likelihood of exponential rate parameter lambda.

    Parameters
    -----
    lam : float
        Rate parameter (must be positive).
    times : array-like
        Observed waiting times.

    Returns
    -----
    ll : float
        Log-likelihood.
    """
    times = np.asarray(times)
    n = len(times)
    return n * np.log(lam) - lam * np.sum(times)

# Simulate coalescence times for a population with N_e = 10,000
# In coalescent units (2*N_e generations), rate = 1
np.random.seed(42)
true_rate = 1.0 # coalescent units
n_pairs = 50 # observe 50 independent pairs
coal_times = np.random.exponential(scale=1.0/true_rate, size=n_pairs)

# MLE
rate_mle = 1.0 / np.mean(coal_times)
print(f"True rate: {true_rate:.4f}")
print(f"MLE rate: {rate_mle:.4f}")
print(f"Mean coal time: {np.mean(coal_times):.4f} (expected: 1.0)")

# Scan the likelihood surface
rates = np.linspace(0.3, 2.5, 200)
ll_values = [exponential_log_likelihood(r, coal_times) for r in rates]
best_idx = np.argmax(ll_values)
print(f"Grid-search MLE: {rates[best_idx]:.4f}")
```

i Plain-language summary – Reading a likelihood curve

The code above computes the log-likelihood at 200 candidate rate values. If you plot `ll_values` against `rates`,

you get a curve with a single peak at the MLE. The width of the peak tells you how precisely the data constrain the rate: a narrow peak (lots of data) means the rate is well-determined; a broad peak (little data) means many rates are roughly equally plausible. Try changing `n_pairs` from 50 to 5 and watch the peak broaden.

The Poisson Distribution: Mutations and the SFS

Mutations accumulate on the branches of a genealogy as a **Poisson process**. If a branch has length ℓ generations and the mutation rate is μ per site per generation, the expected number of mutations on that branch is $\mu\ell$ per site, or $\mu\ell L$ across a region of L sites. If branch length is expressed in units of $2N_{\text{ref}}$ generations, it must first be multiplied by $2N_{\text{ref}}$ (or, equivalently, paired with the scaled mutation rate $\theta = 4N_{\text{ref}}\mu$ for diploids).

The Poisson distribution with mean μ has probability mass function:

$$P(k \mid \mu) = \frac{\mu^k e^{-\mu}}{k!}, \quad k = 0, 1, 2, \dots$$

This formula has three pieces: μ^k counts how likely it is to get exactly k events (each event contributes one factor of μ); $e^{-\mu}$ is the probability of *no* events occurring (the “silence” between mutations); and $k!$ corrects for the fact that the k events are interchangeable (order doesn’t matter).

```
import numpy as np
from scipy.special import factorial

# Compute Poisson probabilities by hand
def poisson_pmf(k, mu):
    """P(k | mu) = mu^k * exp(-mu) / k!"""
    return mu**k * np.exp(-mu) / factorial(k, exact=True)

# Example: expected 3 mutations on a branch
mu = 3.0
for k in range(8):
    p = poisson_pmf(k, mu)
    print(f" P(k={k} | mu={mu}) = {p:.4f}")

# The most probable outcome is k=2 or k=3 (near the mean)
# But k=0 (no mutations) still has probability ~5%
```

Biology Aside – Why Poisson?

The Poisson distribution describes “the number of events in a fixed interval when events occur independently at a constant average rate.” The usual mutation model idealizes mutations this way: conditional on a genealogy and a specified rate, events on disjoint branch segments are independent Poisson increments. Real mutation rates vary with sequence context, genomic position, and time, so this is a model rather than a literal property of every mutation. The infinite-sites model assumes that each mutation hits a new site, so *conditional on a fixed genealogy* the number of segregating sites in a genomic region is Poisson with a mean determined by its total branch length. After integrating over a random genealogy, the count is generally a Poisson mixture rather than exactly Poisson.

From one branch to the whole SFS. The Poisson model for a single branch extends naturally to the entire site frequency spectrum. The SFS counts, for each frequency k from 1 to $n - 1$, how many polymorphic sites have exactly k copies of the derived allele in a sample of n chromosomes.

Under the Poisson random-field approximation, mutations land on branches as Poisson events and frequency-class counts are modeled as independent. A mutation on a branch that subtends k leaves out of n total produces a site at frequency

k/n . The resulting working model is:

$$D_k \sim \text{Poisson}(\xi_k(\theta))$$

where $\xi_k(\theta)$ is the expected count – the total expected branch length subtending exactly k leaves, multiplied by the mutation rate. The parameter θ encodes the demographic history that determines these branch lengths.

Deriving the SFS log-likelihood. Under the working assumption that the $n - 1$ SFS entries are independent Poisson variables, the joint probability of the entire observed SFS $\mathbf{D} = (D_1, \dots, D_{n-1})$ is the product of $n - 1$ Poisson probabilities:

$$P(\mathbf{D} \mid \theta) = \prod_{k=1}^{n-1} \frac{\xi_k(\theta)^{D_k} e^{-\xi_k(\theta)}}{D_k!}$$

Taking the log and expanding:

$$\ell_{\text{SFS}}(\theta) = \sum_{k=1}^{n-1} \left[\underbrace{D_k \log \xi_k(\theta)}_{\text{data match}} - \underbrace{\xi_k(\theta)}_{\text{expected count}} - \underbrace{\log(D_k!)}_{\text{constant}} \right]$$

The first term rewards models that predict high expected counts ξ_k where we observe many sites D_k . The second term penalizes models that predict too many total sites (overprediction). The third term is a constant that depends only on the data and can be dropped during optimization:

$$\ell_{\text{SFS}}(\theta) = \sum_{k=1}^{n-1} [D_k \log \xi_k(\theta) - \xi_k(\theta)] + \text{const}$$

This is the Poisson random-field likelihood available in `dadi` and used in closely related SFS workflows. These packages also commonly use a multinomial likelihood conditional on the total number of segregating sites; in that case only the normalized SFS shape enters the objective [Gutenkunst *et al.*, 2009, Jouganous *et al.*, 2017, Kamm *et al.*, 2019]. Here is the Poisson form in Python:

```
import numpy as np
from scipy.special import gammaln

def sfs_log_likelihood(D_obs, xi_expected):
    """Poisson log-likelihood of observed SFS given expected SFS.

    Computes: sum_k [D_k * log(xi_k) - xi_k - log(D_k!)]
    """
    xi = np.maximum(xi_expected, 1e-300) # avoid log(0)
    return np.sum(D_obs * np.log(xi) - xi - gammaln(D_obs + 1))

# Quick example: neutral SFS with theta=100, n=10
k = np.arange(1, 10) # frequency classes 1..9
xi = 100.0 / k # expected: xi_k = theta/k
D = np.array([105, 48, 30, 25, 21, 17, 12, 14, 10]) # "observed"

ll = sfs_log_likelihood(D, xi)
print(f"SFS log-likelihood: {ll:.2f}")
```

Now let's build up the mathematical machinery behind this function, starting from the simplest case.

Single Poisson MLE. Before tackling the full SFS, let's derive the MLE for the simplest case: n independent observations $k_1, \dots, k_n$, each drawn from a Poisson with the same unknown mean μ . The log-likelihood is:

$$\ell(\mu) = \sum_{i=1}^n [k_i \log \mu - \mu - \log(k_i!)]$$

Grouping terms (and dropping the constant $\sum \log(k_i!)$):

$$\ell(\mu) = \left(\sum_i k_i \right) \log \mu - n\mu + \text{const}$$

This has the same structure as the exponential log-likelihood: one term that grows with μ (the $\log \mu$ term) and one that shrinks (the $-n\mu$ penalty). Taking the derivative and setting to zero:

$$\frac{d\ell}{d\mu} = \frac{\sum_i k_i}{\mu} - n = 0 \quad \Rightarrow \quad \hat{\mu}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n k_i = \bar{k}$$

The MLE of the Poisson mean is the sample mean – again, an intuitive result. The second derivative $d^2\ell/d\mu^2 = -\sum k_i/\mu^2 < 0$ confirms this is a maximum.

```
# Verify the Poisson MLE
np.random.seed(42)
mu_true = 7.0
observations = np.random.poisson(mu_true, size=50)
mu_mle = np.mean(observations)
print(f"True mu: {mu_true}")
print(f"MLE mu: {mu_mle:.2f} (= sample mean of {len(observations)} obs)")
```

Connecting the Poisson MLE to the SFS. For the SFS, the situation is slightly different: each entry D_k is Poisson with its own mean $\xi_k(\theta)$, and these means are linked through the demographic model. We can't just take the sample mean – instead, we search over θ to find the demographic history that makes all the $\xi_k(\theta)$ values simultaneously consistent with the observed D_k .

```
import numpy as np
from scipy.special import gammaln # log(k!) = gammaln(k+1)

def poisson_log_likelihood(mu, counts):
    """Poisson log-likelihood for a vector of observed counts.

    Parameters
    -----
    mu : float or ndarray
        Expected count(s). If scalar, same mean for all observations.
        If array, must match shape of counts.
    counts : ndarray
        Observed counts.

    Returns
    -----
    ll : float
        Log-likelihood.
    """
    counts = np.asarray(counts, dtype=float)
    mu = np.asarray(mu, dtype=float)
    # Full Poisson log-likelihood: k*log(mu) - mu - log(k!)
    return np.sum(counts * np.log(np.maximum(mu, 1e-300)) - mu
                  - gammaln(counts + 1))

# -----
# Example: SFS under the standard neutral model
```

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```

# Under constant population size, the expected SFS is  $\xi_k = \theta/k$ 
# (Watterson's result). Here  $\theta = 4N_e\mu L$ .
# -----
n_samples = 20          # number of haploid chromosomes
theta_true = 200.0      # true population-scaled mutation rate

# Expected SFS:  $\xi_k = \theta / k$  for  $k = 1, \dots, n-1$ 
k_values = np.arange(1, n_samples)
xi_expected = theta_true / k_values

# Simulate observed SFS by Poisson sampling
np.random.seed(42)
D_observed = np.random.poisson(xi_expected)

print("Frequency k: ", k_values[:8])
print("Expected xi: ", np.round(xi_expected[:8], 1))
print("Observed D:   ", D_observed[:8])

# Compute SFS log-likelihood at the true theta
ll_true = poisson_log_likelihood(xi_expected, D_observed)
print(f"\nLog-likelihood at true theta={theta_true}: {ll_true:.2f}")

# Scan over candidate theta values
thetas = np.linspace(100, 350, 300)
ll_scan = []
for th in thetas:
    xi_candidate = th / k_values
    ll_scan.append(poisson_log_likelihood(xi_candidate, D_observed))
ll_scan = np.array(ll_scan)

theta_mle = thetas[np.argmax(ll_scan)]
print(f"MLE theta (grid search): {theta_mle:.1f}")

# Analytical MLE: total observed mutations / sum(1/k)
harmonic = np.sum(1.0 / k_values)
S = np.sum(D_observed) # total segregating sites
theta_mle_exact = S / harmonic
print(f"MLE theta (analytical): {theta_mle_exact:.1f}")
print(f"This is Watterson's estimator!")

```

Biology Aside – Watterson's estimator

Under the independent-Poisson SFS working model, the analytical MLE of θ is $S / \sum_{k=1}^{n-1} 1/k$, where S is the total number of segregating sites. The same expression is **Watterson's estimator**, originally derived as an unbiased estimator under the infinite-sites neutral model [Watterson, 1975]; calling it an MLE without naming the Poisson working model would overstate the result. It is the starting point for neutrality tests like Tajima's D, which compares Watterson's estimator (based on the total number of segregating sites) with a frequency-weighted estimator (based on the average pairwise differences). When the two disagree, it signals a departure from the standard neutral model – for example, a recent population expansion or a selective sweep – but finite-sample variation, linkage, sequencing error, and other violations can also create a difference. Tajima's D standardizes the contrast by its neutral-model variance rather than treating every nonzero difference as evidence.

The Gamma Distribution: Ages and Rates

The **gamma distribution** appears throughout population genetics as a flexible model for positive continuous quantities. In current `tsdate`, gamma distributions approximate marginal posteriors for node ages; they are not, in general, the exact coalescent prior or exact posterior [Pope *et al.*, 2026].

A gamma random variable with shape $\alpha > 0$ and rate $\beta > 0$ has density:

$$f(t \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}, \quad t > 0$$

where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$ is the gamma function. The mean is α/β and the variance is α/β^2 .

i Plain-language summary – What shape and rate control

The **shape** α controls the form of the distribution:

- $\alpha = 1$: the gamma reduces to an exponential (our coalescence time distribution).
- $\alpha < 1$: the density spikes near zero and has a heavy tail – useful for modeling quantities that are often very small but occasionally very large.
- $\alpha > 1$: the density has a peak away from zero, forming a bell-shaped curve skewed to the right.

The **rate** β controls the scale: larger β compresses the distribution toward zero (shorter times), smaller β stretches it out (longer times).

In `tsdate`, the gamma distribution is used to approximate the posterior distribution of each node's age. The shape and rate are updated iteratively by Expectation Propagation as the algorithm absorbs information from mutations and the coalescent prior.

The exponential distribution is a special case: $\text{Gamma}(1, \lambda) = \text{Exponential}(\lambda)$. This means the exponential likelihood we derived above is a special case of the gamma likelihood with known $\alpha = 1$.

Deriving the gamma log-likelihood. For n independent observations $t_1, \dots, t_n$ from a $\text{Gamma}(\alpha, \beta)$, the joint density is:

$$\mathcal{L}(\alpha, \beta) = \prod_{i=1}^n \frac{\beta^\alpha}{\Gamma(\alpha)} t_i^{\alpha-1} e^{-\beta t_i}$$

Taking the logarithm, each factor contributes $\alpha \log \beta - \log \Gamma(\alpha) + (\alpha - 1) \log t_i - \beta t_i$:

$$\ell(\alpha, \beta) = n\alpha \log \beta - n \log \Gamma(\alpha) + (\alpha - 1) \sum_{i=1}^n \log t_i - \beta \sum_{i=1}^n t_i$$

The four terms have clear roles: $n\alpha \log \beta$ and $-n \log \Gamma(\alpha)$ are normalization terms that depend on the parameters but not on individual data points; $(\alpha - 1) \sum \log t_i$ measures how the data agree with the shape parameter (larger α favors larger values of t); and $-\beta \sum t_i$ penalizes large rate parameters when the data contain large values.

Partial MLEs. To find the MLE of β for a given α , take the partial derivative with respect to β :

$$\frac{\partial \ell}{\partial \beta} = \frac{n\alpha}{\beta} - \sum t_i = 0 \quad \implies \quad \hat{\beta}(\alpha) = \frac{n\alpha}{\sum t_i}$$

This is analogous to the exponential MLE (which is the special case $\alpha = 1$). However, there is no closed-form MLE for α itself – the derivative $\partial \ell / \partial \alpha$ involves the **digamma function** $\psi(\alpha) = d \log \Gamma(\alpha) / d\alpha$, giving the equation $\log \hat{\beta} - \psi(\alpha) = \frac{1}{n} \sum \log t_i$ which must be solved numerically. The code below uses profile likelihood: for each candidate α , plug in the optimal $\hat{\beta}(\alpha)$ and maximize over α alone.


```

import numpy as np
from scipy.special import gammaln, digamma
from scipy.optimize import minimize_scalar

def gamma_log_likelihood(alpha, beta, data):
    """Log-likelihood for Gamma(alpha, beta) observations.

    Parameters
    -----
    alpha : float
        Shape parameter.
    beta : float
        Rate parameter.
    data : ndarray
        Observed positive values.

    Returns
    -----
    ll : float
        Log-likelihood.
    """
    data = np.asarray(data)
    n = len(data)
    return (n * alpha * np.log(beta)
            - n * gammaln(alpha)
            + (alpha - 1) * np.sum(np.log(data))
            - beta * np.sum(data))

def gamma_mle(data):
    """Compute the MLE of Gamma(alpha, beta) by profiling.

    For fixed alpha, the MLE of beta is  $n \cdot \alpha / \text{sum}(\text{data})$ .
    We optimize over alpha using this profile likelihood.
    """
    data = np.asarray(data)
    n = len(data)
    sum_data = np.sum(data)
    sum_log_data = np.sum(np.log(data))

    def neg_profile_ll(alpha):
        beta_hat = n * alpha / sum_data
        return -(n * alpha * np.log(beta_hat)
                - n * gammaln(alpha)
                + (alpha - 1) * sum_log_data
                - beta_hat * sum_data)

    result = minimize_scalar(neg_profile_ll, bounds=(0.01, 100),
                             method='bounded')

    alpha_hat = result.x
    beta_hat = n * alpha_hat / sum_data
    return alpha_hat, beta_hat

# Simulate node ages from a gamma prior (as in tsdate)

```

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```

np.random.seed(42)
true_alpha, true_beta = 3.0, 0.5 # mean = 6.0, variance = 12.0
node_ages = np.random.gamma(shape=true_alpha, scale=1.0/true_beta, size=100)

alpha_hat, beta_hat = gamma_mle(node_ages)
print(f"True:   alpha={true_alpha}, beta={true_beta}, mean={true_alpha/true_beta}")
print(f"MLE:    alpha={alpha_hat:.3f}, beta={beta_hat:.3f}, "
      f"mean={alpha_hat/beta_hat:.3f}")
print(f"Sample mean: {np.mean(node_ages) :.3f}")

```

i Biology Aside – Why gamma for node ages?

Under a constant-size Kingman coalescent, waiting times while there are different numbers of ancestral lineages are exponential but have *different* rates. Their sum is therefore generally hypoexponential, not gamma; a gamma law is exact only for sums of independent exponentials with a common rate. `tsdate` uses the gamma family as a tractable approximation because it has positive support, flexible shape, and a convenient exponential-family representation for moment matching [Pope *et al.*, 2026].

The Gaussian Distribution: Smoothness Priors

The **Gaussian** (normal) distribution appears in population genetics both as a data model and as a prior. A common regularizer for a discretized demographic history is a Gaussian random walk on adjacent log population sizes; the example below illustrates that generic construction.

The Gaussian density for a scalar x with mean μ and variance σ^2 is:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

For a vector $\mathbf{x}$ of dimension d , the multivariate Gaussian with mean $\boldsymbol{\mu}$ and covariance matrix Σ has density:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

i Biology Aside – Smoothness in demographic history

Let $\mathbf{h} = (\log N_1, \log N_2, \dots, \log N_M)$ denote log population sizes in adjacent epochs. A first-order Gaussian random-walk prior assigns independent normal increments $h_{k+1} - h_k \sim N(0, \sigma^2)$.

Concretely, the prior penalizes large *differences* between adjacent entries:

$$\log p(\mathbf{h}) \propto -\frac{1}{2\sigma^2} \sum_{k=1}^{M-1} (h_{k+1} - h_k)^2$$

This prior says “I expect the demographic history to change gradually.” The hyperparameter σ^2 controls how much change is allowed per time step. The difference penalty alone is *improper* because adding a constant to every h_k changes no increment; a proper prior also anchors one level or adds a separate level penalty. PHLASH’s published model instead uses random low-dimensional time discretizations and averages posterior histories; its software also exposes optional quadratic penalties, so the generic random-walk example should not be mistaken for the complete PHLASH prior [Terhorst, 2025].

Gaussian MLE. For n observations $x_1, \dots, x_n$ from $N(\mu, \sigma^2)$, the log-likelihood is:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

The first term is a normalization penalty that grows with σ^2 (wider distributions spread probability thinner). The second term measures how well μ fits the data – it is minimized when μ is at the center of the data.

Taking the derivative with respect to μ (with σ^2 held fixed):

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0 \quad \implies \quad \hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

And with respect to σ^2 (with $\mu = \hat{\mu}$):

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \bar{x})^2 = 0 \quad \implies \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

These are the sample mean and (biased) sample variance – the same “differentiate, set to zero, solve” recipe we’ve used throughout.

```
import numpy as np

def gaussian_log_likelihood(mu, sigma2, data):
    """Log-likelihood for N(mu, sigma^2) observations."""
    data = np.asarray(data)
    n = len(data)
    return (-0.5 * n * np.log(2 * np.pi * sigma2)
           - 0.5 * np.sum((data - mu)**2) / sigma2)

def smoothness_prior_log_density(h, sigma=1.0):
    """Log-density of the Gaussian random-walk smoothness prior.

    Penalizes large jumps between adjacent log-population sizes.

    Parameters
    -----
    h : ndarray, shape (M,)
        Log-population sizes at M time points.
    sigma : float
        Smoothness scale (standard deviation of allowed changes).

    Returns
    -----
    lp : float
        Log prior density (up to a constant).
    """
    diffs = np.diff(h)
    return -0.5 * np.sum(diffs**2) / sigma**2

# Demonstrate: smooth vs. rough demographic histories
M = 30 # time points
h_smooth = np.zeros(M)
h_smooth[10:20] = -0.5 # gentle bottleneck
```

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```

h_rough = np.zeros(M)
h_rough[:,2] = 1.0 # wild oscillations
h_rough[1::2] = -1.0

sigma = 0.5
lp_smooth = smoothness_prior_log_density(h_smooth, sigma)
lp_rough = smoothness_prior_log_density(h_rough, sigma)
print(f"Smooth history prior: {lp_smooth:.2f}")
print(f"Rough history prior: {lp_rough:.2f}")
print(f"The smooth history is exp({lp_smooth - lp_rough:.0f}) times "
      f"more probable under the prior")

```

2.1.4 Maximum Likelihood Estimation (MLE)

The previous section showed the MLE recipe for individual distributions. Here we formalize it and show how it works for a more realistic problem.

The **maximum likelihood estimate** (MLE) is the parameter value that maximizes the likelihood:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} P(D \mid \theta)$$

Equivalently, it maximizes the log-likelihood (since log is monotonically increasing). The recipe is:

1. Write down the log-likelihood $\ell(\theta)$.
2. Take the derivative $d\ell/d\theta$ (or gradient, for vector θ).
3. Set the derivative to zero and solve for θ .
4. Verify that the solution is a maximum (the second derivative is negative).

When step 3 has a closed-form solution (as for the exponential, Poisson, and Gaussian), the MLE is called **analytically tractable**. When it doesn't (as for the gamma shape parameter, or for complex demographic models), we use numerical optimization – gradient ascent, Newton's method, or specialized algorithms like `scipy.optimize.minimize`.

Under correct specification, identifiability, and the usual regularity conditions, the MLE has attractive large-sample properties:

- **Consistency:** as the amount of independent information grows, the MLE converges to the true parameter value.
- **Efficiency:** in regular parametric models, the MLE asymptotically attains the inverse Fisher-information bound. This is not a claim that it has the smallest variance among every conceivable consistent estimator.
- **Invariance:** if $\hat{\theta}$ is the MLE of θ , then $f(\hat{\theta})$ is an MLE under a one-to-one reparameterization f ; more general transformations require profiling over parameter values that map to the same transformed value.

In practice, `dadi`, `moments`, and `mom2` support likelihood-based optimization of demographic parameters using Poisson or multinomial SFS objectives [Gutenkunst *et al.*, 2009, Jouganous *et al.*, 2017, Kamm *et al.*, 2019].

Practical caveat – Local optima

The likelihood surface for complex demographic models is rarely convex. It can have multiple peaks (local optima), ridges, and flat regions. A gradient-based optimizer starting from a single initial point may find a local optimum rather than the global one. This is why `mom2` and `dadi` recommend running the optimizer from multiple random starting points and keeping the best result. It is also why simpler models (fewer parameters) are often preferred: they have smoother likelihood surfaces with fewer local optima.

Worked Example: Inferring the SFS Scale

Here is a complete MLE example using an exact result rather than a fabricated demographic approximation. Under the standard neutral, constant-size, infinite-sites model, the expected unfolded SFS is $\xi_k = \theta/k$, where the regional scale $\theta = 4N_e\mu L$ for a diploid population. Given an observed SFS, we estimate θ by maximizing its Poisson likelihood. Real `dadi` and `moments` analyses first compute the expected SFS shape under a chosen demographic model and then optimize its parameters; the toy calculation here isolates the likelihood step [Gutenkunst *et al.*, 2009, Jouganous *et al.*, 2017].

```
import numpy as np
from scipy.special import gammaln

def expected_sfs_constant(n, theta):
    """Exact expected unfolded SFS under constant population size."""
    k = np.arange(1, n)
    return theta / k

def sfs_poisson_loglik(D_obs, xi_expected):
    """Poisson log-likelihood of observed SFS given expected."""
    xi = np.maximum(xi_expected, 1e-300)
    return np.sum(D_obs * np.log(xi) - xi - gammaln(D_obs + 1))

# Generate an observed SFS under the exact constant-size model
np.random.seed(42)
n = 30
theta_true = 500.0

xi_true = expected_sfs_constant(n, theta_true)
D_obs = np.random.poisson(xi_true)

# Scan theta values and compute the log-likelihood for each
thetas = np.linspace(200.0, 800.0, 400)
lls = [sfs_poisson_loglik(D_obs, expected_sfs_constant(n, theta))
        for theta in thetas]
lls = np.array(lls)

theta_mle_grid = thetas[np.argmax(lls)]
harmonic = np.sum(1.0 / np.arange(1, n))
theta_mle_exact = np.sum(D_obs) / harmonic
print(f"True theta:           {theta_true:.2f}")
print(f"Grid-search MLE:        {theta_mle_grid:.2f}")
print(f"Analytical MLE:         {theta_mle_exact:.2f}")
print(f"Log-likelihood at MLE:   {np.max(lls):.2f}")
print(f"Log-likelihood at true: {sfs_poisson_loglik(D_obs, xi_true):.2f}")
```

Fisher Information and Confidence Intervals

The MLE tells you the *best* parameter value, but not how confident you should be in it. The **Fisher information** quantifies the precision of the MLE.

For a single parameter θ , the Fisher information is:

$$I(\theta) = -\mathbb{E} \left[\frac{d^2 \ell}{d\theta^2} \right]$$

What does this mean? The second derivative $d^2 \ell / d\theta^2$ measures the **curvature** of the log-likelihood curve. At the MLE (the peak), the curvature is always negative (the peak bends downward). The more negative it is, the sharper the peak, and

the more precisely the data pin down θ . The expectation $\mathbb{E}[\cdot]$ averages this curvature over all possible datasets – it measures the curvature we'd *expect* to see. The negative sign flips this to a positive number, giving us the **Fisher information**: a measure of how much information the data carry about θ .

For n independent observations, the Fisher information scales linearly: $I_n(\theta) = n \cdot I_1(\theta)$, where I_1 is the information from a single observation. More data means more information.

The key result (proven in statistical theory): for large samples, the MLE is approximately normally distributed with variance $1/I(\theta)$:

$$\hat{\theta}_{\text{MLE}} \sim N\left(\theta, \frac{1}{I(\theta)}\right)$$

This connects the curvature of the likelihood to the uncertainty in the estimate: sharp peak (large I) means small variance (precise MLE); flat peak (small I) means large variance (imprecise MLE).

Replacing the unknown information by its value at the MLE gives the familiar large-sample Wald **95% confidence interval**:

$$\hat{\theta} \pm 1.96 \cdot \frac{1}{\sqrt{I(\hat{\theta})}}$$

where 1.96 is the z -value for 95% coverage under the normal distribution, and $1/\sqrt{I(\hat{\theta})}$ is the **standard error** of the MLE. Wald intervals can perform poorly near parameter boundaries or with little data. For a positive rate, an interval constructed on $\log \lambda$ or an exact/profile-likelihood interval avoids a negative lower endpoint.

i Plain-language summary – Fisher information as a ruler

The Fisher information measures how “informative” the data are about the parameter. If you double the number of observations, you roughly double the Fisher information and halve the variance of the MLE – meaning the confidence interval shrinks by a factor of $\sqrt{2}$. This is why more data gives more precise estimates. It's also why some parameters are harder to estimate than others: if the likelihood surface is flat in some direction (low Fisher information), even a large dataset won't pin down that parameter precisely.

Example: Fisher information for the exponential. Let's work through the full calculation. We showed that the log-likelihood is $\ell(\lambda) = n \log \lambda - \lambda \sum t_i$. We already computed the first derivative: $d\ell/d\lambda = n/\lambda - \sum t_i$. Differentiating again:

$$\frac{d^2\ell}{d\lambda^2} = -\frac{n}{\lambda^2}$$

This is already non-random (it doesn't depend on the data t_i), so the expectation is just itself: $I(\lambda) = n/\lambda^2$. The *asymptotic* variance of the MLE is $1/I(\lambda) = \lambda^2/n$, and its asymptotic standard error is $\lambda/\sqrt{n}$. The exact finite-sample distribution of the reciprocal sample mean is skewed.

The 95% CI is $\hat{\lambda} \pm 1.96 \cdot \hat{\lambda}/\sqrt{n}$. Notice that the CI width is proportional to $1/\sqrt{n}$ – quadrupling the data cuts the CI width in half.

```
import numpy as np

# Confidence interval for the exponential rate parameter
np.random.seed(42)
true_rate = 1.0
n_obs = 50
times = np.random.exponential(scale=1.0/true_rate, size=n_obs)
```

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```

rate_mle = 1.0 / np.mean(times)
fisher_info = n_obs / rate_mle**2
se = 1.0 / np.sqrt(fisher_info) # standard error
ci_low = rate_mle - 1.96 * se
ci_high = rate_mle + 1.96 * se

print(f"True rate:           {true_rate:.4f}")
print(f"MLE rate:            {rate_mle:.4f}")
print(f"Fisher info:         {fisher_info:.2f}")
print(f"Standard error:       {se:.4f}")
print(f"95% CI:               [{ci_low:.4f}, {ci_high:.4f}]")
print(f"True rate in CI: {ci_low <= true_rate <= ci_high}")

# Demonstrate: more data -> tighter CI
for n in [10, 50, 200, 1000]:
    t = np.random.exponential(scale=1.0/true_rate, size=n)
    lam_hat = 1.0 / np.mean(t)
    se_n = lam_hat / np.sqrt(n)
    print(f"  n={n:4d}:  MLE={lam_hat:.3f}, "
          f"  CI width={2*1.96*se_n:.3f}")

```

2.1.5 Bayesian Inference

Maximum likelihood gives a single best answer. **Bayesian inference** goes further: it characterizes the entire range of plausible parameter values by computing the **posterior distribution**:

$$P(\theta | D) = \frac{P(D | \theta) P(\theta)}{P(D)}$$

This equation (Bayes' theorem) has three components:

- **Likelihood** $P(D | \theta)$: how well the parameters explain the data (same as above).
- **Prior** $P(\theta)$: what we believed about the parameters *before* seeing the data. This encodes biological knowledge or constraints – for example, population sizes must be positive, or the demographic history should be smooth.
- **Posterior** $P(\theta | D)$: what we believe about the parameters *after* seeing the data. This is the target of inference.

The denominator $P(D) = \int P(D | \theta) P(\theta) d\theta$ (the **marginal likelihood** or **evidence**) is a normalizing constant that ensures the posterior integrates to 1. It is typically intractable to compute directly, which is why methods like MCMC (see *Markov Chain Monte Carlo*) and SVGD (see *Stein Variational Gradient Descent (SVGD)*) are needed.

Plain-language summary – Prior, likelihood, posterior

Think of it as a conversation between two sources of information:

- The **prior** is your background knowledge: “population sizes are typically between 1,000 and 1,000,000” or “the demographic history should be smooth, not wildly oscillating.”
- The **likelihood** is what the data say: “these particular allele frequencies are 100 times more probable under a bottleneck model than under constant size.”
- The **posterior** is the synthesis: “given both my background knowledge and the data, the population most likely experienced a bottleneck of ~5,000 individuals, with a 95% credible interval of 2,000–15,000.”

The posterior gives you not just a point estimate but a full picture of uncertainty. This is why Bayesian methods (ARGweaver, SINGER, PHLASH) can provide credible intervals and uncertainty bands, while pure MLE methods (dadi, moments) require additional steps like bootstrapping to quantify uncertainty.

Conjugate Priors: When Bayesian Inference Has Closed-Form Solutions

In general, computing the posterior requires numerical methods (MCMC, EP, SVGD). But for certain combinations of likelihood and prior, the posterior has the **same functional form** as the prior. These are called **conjugate priors**, and they give exact, closed-form Bayesian updates.

Conjugate priors are not just a mathematical curiosity. The simple gamma–Poisson update below also explains why gamma factors are convenient in message-passing algorithms. It is important, however, to distinguish that one-variable conjugate calculation from `tsdate`'s actual edge factor, which depends on the *difference* between a parent and child time and is handled by Expectation Propagation and moment matching [Pope *et al.*, 2026].

Likelihood	Conjugate prior	Posterior	Used in
Poisson(λ)	Gamma(α, β)	Gamma(α', β')	Exact for one unknown exposure
Binomial(n, p)	Beta(a, b)	Beta(a', b')	(general)
Normal(μ, σ^2)	Normal(μ_0, σ_0^2)	Normal(μ', σ'^2)	Generic Gaussian models

The Gamma-Poisson Conjugacy

This is a useful exact conjugacy and a stepping stone toward understanding gamma approximations in node-dating algorithms.

Setup. Suppose the number of mutations k on a branch of length t follows a Poisson distribution with rate μt :

$$P(k | t) = \frac{(\mu t)^k e^{-\mu t}}{k!}$$

and we place a gamma prior on t :

$$p(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}$$

Deriving the posterior. By Bayes' theorem:

$$\begin{aligned} p(t | k) &\propto P(k | t) p(t) \\ &\propto (\mu t)^k e^{-\mu t} \cdot t^{\alpha-1} e^{-\beta t} \\ &= \mu^k \cdot t^{k+\alpha-1} \cdot e^{-(\mu+\beta)t} \end{aligned}$$

Dropping constants that don't depend on t , we recognize this as the kernel of a gamma distribution with updated parameters:

$$t | k \sim \text{Gamma}(\alpha + k, \beta + \mu)$$

This is the Bayesian update rule:

- **Shape update:** $\alpha' = \alpha + k$ (each observed mutation increments the shape by 1).
- **Rate update:** $\beta' = \beta + \mu$ (the mutation rate adds to the rate parameter).

i Biology Aside – A simplified branch-length update

Consider a branch in a genealogy with 3 observed mutations and a per-branch mutation rate of $\mu = 0.5$ per coalescent unit. If our prior on the branch length is $\text{Gamma}(2, 1)$ (mean 2, moderate uncertainty), then after observing the 3 mutations, the posterior is $\text{Gamma}(2 + 3, 1 + 0.5) = \text{Gamma}(5, 1.5)$.

The posterior mean shifts from $2/1 = 2.0$ to $5/1.5 = 3.33$ – the data (3 mutations suggesting a longer branch) have pulled the estimate upward from the prior. If we had observed 0 mutations instead, the posterior would be $\text{Gamma}(2, 1.5)$ with mean $2/1.5 = 1.33$ – the data (no mutations suggesting a shorter branch) have pulled the estimate downward.

This calculation treats the branch length t itself as the one unknown. In `tsdate`, an edge length is $t_p - t_c$, so its Poisson factor couples two node ages and is not conjugate to independent gamma marginals for t_p and t_c . The variational-gamma algorithm removes an edge's current approximate factors, combines the true coupled factor with the remaining `cavity` distribution, and moment-matches new gamma marginals. Those are approximate EP updates, not repeated $(\alpha + k, \beta + \mu)$ updates [Pope *et al.*, 2026].

```
import numpy as np
from scipy.stats import gamma as gamma_dist

def gamma_poisson_update(alpha_prior, beta_prior, k_observed, mu):
    """Bayesian update: Gamma prior + Poisson likelihood -> Gamma posterior.

    Parameters
    -----
    alpha_prior : float
        Prior shape parameter.
    beta_prior : float
        Prior rate parameter.
    k_observed : int
        Number of observed mutations.
    mu : float
        Per-branch mutation rate.

    Returns
    -----
    alpha_post : float
        Posterior shape.
    beta_post : float
        Posterior rate.
    """
    return alpha_prior + k_observed, beta_prior + mu

# Demonstrate the Bayesian update for branch length estimation
alpha_prior, beta_prior = 2.0, 1.0
mu = 0.5

print("Prior: Gamma({}, {}) -> mean={:.2f}, var={:.2f}".format(
    alpha_prior, beta_prior,
    alpha_prior/beta_prior, alpha_prior/beta_prior**2))

for k in [0, 1, 3, 10]:
    a_post, b_post = gamma_poisson_update(alpha_prior, beta_prior, k, mu)
```

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```

mean_post = a_post / b_post
var_post = a_post / b_post**2
print(f"  k={k:2d} mutations -> Gamma({a_post:.1f}, {b_post:.1f}), "
      f"mean={mean_post:.2f}, var={var_post:.2f}")

# Show how the posterior concentrates with more data
print("\nMultiple branches, each with mu=0.5:")
alpha, beta = 2.0, 1.0 # start from prior
for i, k in enumerate([2, 1, 3, 0, 4, 2, 1, 3, 2, 2]):
    alpha, beta = gamma_poisson_update(alpha, beta, k, mu)
    print(f"  After branch {i+1} (k={k}): Gamma({alpha:.1f}, {beta:.1f}), "
          f"mean={alpha/beta:.3f}, CI=[ "
          f"  {gamma_dist.ppf(0.025, alpha, scale=1/beta):.3f}, "
          f"  {gamma_dist.ppf(0.975, alpha, scale=1/beta):.3f}]"
    )

```

The Beta-Binomial Conjugacy

This conjugacy is less central to this book but appears frequently in population genetics and is instructive.

Setup. Observe k successes in n trials (e.g., k copies of the derived allele in a sample of n chromosomes):

$$P(k | p) = \binom{n}{k} p^k (1 - p)^{n-k}$$

with a Beta prior on the success probability p :

$$p(p) = \frac{p^{a-1} (1 - p)^{b-1}}{B(a, b)}$$

where $B(a, b)$ is the Beta function (a normalizing constant).

Deriving the posterior. Applying Bayes' theorem, the posterior is proportional to the product of likelihood and prior:

$$\begin{aligned}
 p(p | k) &\propto P(k | p) \cdot p(p) \\
 &\propto p^k (1 - p)^{n-k} \cdot p^{a-1} (1 - p)^{b-1} \\
 &= p^{(a+k)-1} \cdot (1 - p)^{(b+n-k)-1}
 \end{aligned}$$

We collected the powers of p and $(1 - p)$ by adding exponents. The result is the kernel of a Beta distribution:

$$p | k \sim \text{Beta}(a + k, b + n - k)$$

The update rule is: add the observed successes to a , and add the observed failures to b . Calling a and b “pseudo-counts” is useful shorthand, but the exact offset depends on what is being matched: the beta density has exponents $a - 1$ and $b - 1$, whereas its mean has denominator $a + b$.

```

import numpy as np
from scipy.stats import beta as beta_dist

def beta_binomial_update(a_prior, b_prior, k, n):
    """Bayesian update: Beta prior + Binomial likelihood -> Beta posterior."""
    return a_prior + k, b_prior + n - k

# Example: estimate allele frequency from a sample
# Prior: Beta(1, 1) = Uniform(0, 1) -- no prior information

```

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```

a_prior, b_prior = 1.0, 1.0

# Observe 7 derived alleles out of 20 chromosomes
k_obs, n_obs = 7, 20
a_post, b_post = beta_binomial_update(a_prior, b_prior, k_obs, n_obs)

mean_post = a_post / (a_post + b_post)
ci = beta_dist.ppf([0.025, 0.975], a_post, b_post)

print(f"Observed: {k_obs}/{n_obs} derived alleles")
print(f"MLE frequency: {k_obs/n_obs:.3f}")
print(f"Posterior mean: {mean_post:.3f}")
print(f"95% credible interval: [{ci[0]:.3f}, {ci[1]:.3f}]")

# With a stronger prior: "I expect low-frequency variants"
# Beta(1, 10) has mean 0.09
a_prior2, b_prior2 = 1.0, 10.0
a_post2, b_post2 = beta_binomial_update(a_prior2, b_prior2, k_obs, n_obs)
mean_post2 = a_post2 / (a_post2 + b_post2)
ci2 = beta_dist.ppf([0.025, 0.975], a_post2, b_post2)
print(f"\nWith informative prior Beta(1,10):")
print(f"Posterior mean: {mean_post2:.3f}")
print(f"95% credible interval: [{ci2[0]:.3f}, {ci2[1]:.3f}]")
print("(The informative prior pulls the estimate toward lower frequencies)")

```

i Plain-language summary – Why conjugate priors matter

Conjugate priors are special because they keep the math tractable. When the prior and posterior belong to the same family, the Bayesian update reduces to updating two numbers (the parameters of the distribution) rather than computing an integral over all possible parameter values. This is computationally free – no MCMC needed, no approximation required.

`tsdate` gains efficiency from representing approximate node marginals with two gamma natural parameters, but its EP projection also evaluates moments of a coupled parent–child edge factor. The current implementation contains distinct rootward, leafward, and two-node projection routines; it does not apply the scalar update above verbatim [Pope *et al.*, 2026].

Bayesian inference in this book appears in:

- **ARGweaver**: MCMC sampling over ARGs, with a coalescent prior on the genealogy
- **SINGER**: MCMC sampling with data-informed proposals (SGPR)
- **tsdate**: Expectation Propagation with gamma approximations to node-age marginals and coupled Poisson edge factors
- **PHLASH**: SVGD targeting a posterior over randomly discretized population size histories

2.1.6 Composite and Approximate Likelihoods

In many settings, computing the exact likelihood is too expensive. Several Timepieces use **approximations** that retain the essential structure while being computationally feasible.

- **Composite likelihood**: multiplies valid component likelihoods without modeling their full joint dependence. For

example, `mom2` uses the SFS while ignoring linkage among SNPs, and PHLASH multiplies marginal PSMC likelihoods across genomes and adds SFS and optional LD terms [Kamm *et al.*, 2019, Terhorst, 2025]. Composite-likelihood estimators can be consistent under correct component models, identifiability, and suitable dependence conditions, but this is not automatic. Naive inverse-Hessian standard errors generally require a sandwich/Godambe correction or a block bootstrap.

- **Pairwise and conditional likelihoods:** PSMC models the two haplotypes in one diploid genome. SMC++ does not simply repeat a two-sample PSMC: it tracks the coalescence time of a distinguished pair while its emissions use the conditional SFS of the remaining, undistinguished lineages. Multiple choices of the distinguished pair can then be combined as a composite likelihood [Li and Durbin, 2011, Terhorst *et al.*, 2016].
- **Copying-model objective:** `tsinfer` uses Li–Stephens-style matching to choose parsimonious copying paths when building and matching ancestors. It is an algorithmic approximation for topology inference, not a calibrated likelihood for demographic parameters [Kelleher *et al.*, 2019, Li and Stephens, 2003].

These approximations are not defects – they are engineering choices that make inference possible at genomic scale. Understanding *which* approximation each tool makes, and what it costs in terms of statistical efficiency, is essential for interpreting results correctly.

Worked Example: Composite Likelihood from Two Data Sources

PHLASH combines marginal PSMC likelihoods with an SFS penalty and, when requested, an LD-summary term [Terhorst, 2025]. Here we build a deliberately simpler two-summary example to show the arithmetic of a composite objective; it is not a reimplementation of PHLASH.

Suppose we want to estimate the effective population size N_e (or equivalently, the coalescence rate $\lambda = 1/(2N_e)$) using two summaries of genetic data:

1. **The SFS:** from a sample of n chromosomes, we observe counts of segregating sites at each frequency.
2. **Pairwise heterozygosity:** for a single diploid, we observe the fraction of sites that are heterozygous, which depends on the expected pairwise coalescence time.

Each source provides its own likelihood. The composite likelihood simply multiplies them (adds the log-likelihoods):

$$\ell_{\text{comp}}(\lambda) = \ell_{\text{SFS}}(\lambda) + \ell_{\text{het}}(\lambda)$$

```
import numpy as np
from scipy.special import gammaln

def sfs_log_likelihood(theta, D_obs, n):
    """Poisson SFS log-likelihood under constant population size.

    Under the neutral coalescent,  $\xi_k = \theta / k$ .
    """
    k = np.arange(1, n)
    xi = theta / k
    xi = np.maximum(xi, 1e-300)
    return np.sum(D_obs * np.log(xi) - xi - gammaln(D_obs + 1))

def het_log_likelihood(theta, n_het, n_sites):
    """Binomial log-likelihood for heterozygosity.

    Marginalizing  $\exp(-2\mu T)$  over an exponential pairwise
    coalescence time gives  $p_{\text{het}} = \theta_{\text{site}} / (1 + \theta_{\text{site}})$ ,
    where  $\theta_{\text{site}} = \theta_{\text{region}} / n_{\text{sites}}$ .
```

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```

"""
# Expected fraction of heterozygous sites
theta_site = theta / n_sites
p_het = theta_site / (1.0 + theta_site)
p_het = np.clip(p_het, 1e-300, 1 - 1e-300)
n_hom = n_sites - n_het
return n_het * np.log(p_het) + n_hom * np.log(1.0 - p_het)

# Simulate data from a population with theta_true = 200
np.random.seed(42)
theta_true = 200.0
n_samples = 20
n_sites = 100_000

# Source 1: SFS
k = np.arange(1, n_samples)
xi_true = theta_true / k
D_obs = np.random.poisson(xi_true)

# Source 2: heterozygosity
theta_site_true = theta_true / n_sites
p_het_true = theta_site_true / (1.0 + theta_site_true)
n_het = np.random.binomial(n_sites, p_het_true)

# Compute individual and composite log-likelihoods
thetas = np.linspace(50, 400, 300)
ll_sfs = [sfs_log_likelihood(th, D_obs, n_samples) for th in thetas]
ll_het = [het_log_likelihood(th, n_het, n_sites) for th in thetas]
ll_comp = [s + h for s, h in zip(ll_sfs, ll_het)]

ll_sfs = np.array(ll_sfs)
ll_het = np.array(ll_het)
ll_comp = np.array(ll_comp)

theta_mle_sfs = thetas[np.argmax(ll_sfs)]
theta_mle_het = thetas[np.argmax(ll_het)]
theta_mle_comp = thetas[np.argmax(ll_comp)]

print(f"True theta:           {theta_true:.1f}")
print(f"MLE from SFS only:      {theta_mle_sfs:.1f}")
print(f"MLE from het only:       {theta_mle_het:.1f}")
print(f"MLE from composite:     {theta_mle_comp:.1f}")
print(f"\nThe composite MLE balances the two data sources.")
print(f"If they agree, the composite is sharper (more precise).")
print(f"If they disagree, the composite finds a compromise.")

```

Plain-language summary – Why composite likelihoods work

A composite likelihood combines multiple incomplete views of the same underlying process. Each view provides partial information: the SFS captures the frequency distribution but ignores linkage, while the pairwise HMM captures linkage but only for one pair. By adding their log-likelihoods, we get an estimate that benefits from both sources.

The catch is that the two summaries are not truly independent (they arise from overlapping genealogical information), so curvature alone does not determine estimator variance. Under suitable conditions the composite estimator remains consistent, but uncertainty is governed by both score variability and expected curvature. A posterior formed from an unadjusted composite likelihood can likewise be miscalibrated; using SVGD does not by itself restore the missing dependence. PHLASH therefore evaluates the calibration of its composite-likelihood posterior empirically [Terhorst, 2025].

2.1.7 The Other Paradigm: Neural Networks and Amortized Inference

The inferential Timepieces in this book mostly follow a classical paradigm: **define a generative model, derive an exact or approximate objective, then optimize or sample**. Simulation-based inference (SBI) offers another route when a simulator is available but its likelihood is unavailable or too costly to evaluate. Some SBI procedures use neural networks and amortize the cost of inference across many datasets; others, including classical approximate Bayesian computation (ABC), are neither neural nor amortized.

The key idea

Instead of running a new optimizer or sampler from scratch for each dataset, an amortized estimator trains a network to learn a reusable conditional mapping. Depending on the method, its output may be a point estimate, a likelihood ratio, a likelihood surrogate, or an approximate posterior:

Classical (this book):	Data	->	Likelihood function	->	Optimizer/MCMC	->	Parameters
Amortized:	Data	->	Trained network	->	Estimate or distribution		

The training process is:

1. **Simulate** thousands (or millions) of datasets from the generative model under randomly sampled parameters.
2. **Train** a neural network to predict the parameters (or the full posterior) from each simulated dataset.
3. **Apply** the trained network to observed data, then validate calibration and simulation-to-observation fit.

The terms are related but not synonyms. **Simulation-based inference** is the umbrella; **neural posterior estimation**, **neural likelihood estimation**, and **neural ratio estimation** are different SBI objectives. ABC accepts or weights simulations using a discrepancy and need not contain a neural density estimator. “Likelihood-free” means that inference does not evaluate the likelihood, not that the generative model lacks one.

What amortized inference does well

- **Speed at inference time.** Once trained, a network can make each subsequent conditional-density or parameter evaluation inexpensive. The up-front simulation and training cost is worthwhile mainly when it can be reused.
- **Flexibility with summary statistics.** Neural networks can learn which features of the data are informative, without requiring the user to derive a likelihood function. This is powerful for complex models where the likelihood is intractable.
- **Access to simulator-defined models.** Complex migration or selection models can be considered even when their analytical likelihood is unavailable, provided the simulator is accurate and the parameter space can be covered adequately.

What likelihood-based inference does well

- **Transparency.** Every step of the inference is mathematically specified. You can inspect the likelihood surface, diagnose convergence, understand why a particular estimate was returned, and know exactly which assumptions are made. Neural networks are harder to interrogate – their internal representations are opaque.

- **Established diagnostics and asymptotics.** For regular, identifiable, correctly specified models, MLE and exact Bayesian procedures have mature theory. MCMC is only exact in the limit when its transition kernel targets the intended posterior and the chain actually converges; SVGD is itself an approximation.
- **No training-distribution shift.** A likelihood can be evaluated wherever its numerical method and parameterization are valid, whereas an amortized estimator may be unreliable when observed data lie outside its prior- predictive training distribution. Likelihood methods can still fail through model misspecification, non-identifiability, or numerical approximation.
- **No amortization budget.** Analytical and deterministic likelihood methods avoid a large reusable training set, although some likelihood estimators themselves use Monte Carlo simulation.
- **Interpretable uncertainty target.** Exact likelihood and Bayesian methods define the target distribution explicitly. Neither composite likelihoods, finite MCMC runs, SVGD particles, nor neural posteriors are automatically calibrated; each needs diagnostics appropriate to its approximation.

Biology Aside – When to use which paradigm

Prefer a tractable likelihood-based method when:

- You need calibrated uncertainty and the likelihood approximation has been validated for the setting
- The model is well-specified and the likelihood can be computed or well-approximated
- Transparency and reproducibility are paramount
- You are studying a small number of populations with a well-defined model

Consider amortized neural SBI when:

- The model is too complex for any analytical likelihood (e.g., many populations with continuous migration and selection)
- You need to screen many populations quickly and will follow up interesting cases with targeted diagnostics or follow-up inference
- You want to explore which summary statistics are informative
- Speed at inference time is critical (e.g., real-time analysis pipelines)

In practice, the two paradigms are **complementary**, not competing. Amortized methods can provide fast initial estimates that serve as starting points for likelihood-based refinement. Simulation-based calibration, posterior-predictive checks, and coverage tests on held-out simulations can assess neural estimators; agreement with a tractable likelihood in a simplified model is another useful check.

Why this book focuses on the likelihood approach

This book is about understanding mechanisms – opening the watch and seeing every gear. Likelihood-based inference is tied closely to a generative model: the likelihood is that model's probability mass or density evaluated at the observed data and viewed as a function of its parameters. Every equation you derive, every algorithm you implement, directly encodes the biological process that generated the data. When you build a PSMC from scratch, you understand *exactly* how population size history maps to the probability of observing a heterozygous site at a given genomic position. That understanding doesn't come from training a neural network – it comes from building the gear train yourself.

Moreover, every amortized method ultimately relies on a generative model to produce training data. Understanding how that model works – how coalescence, mutation, and recombination generate genetic variation – is essential even if you ultimately use a neural network for inference. The foundations in this book are prerequisites for *both* paradigms.

2.1.8 Summary

This chapter has equipped you with the inferential toolkit used by the inference Timepieces. Here is what you should take away:

Four distributions that generate genetic data:

Distribution	Role in population genetics	MLE	Where it appears
Exponential(λ)	Coalescence waiting times	$\hat{\lambda} = 1/\bar{t}$	PSMC, ARGweaver, SINGER
Poisson(μ)	Mutation counts, SFS entries	$\hat{\mu} = \bar{k}$	dadi, moments, momi2, tsdate
Gamma(α, β)	Node ages, branch lengths	Numerical	tsdate (approximate marginals)
Gaussian(μ, σ^2)	Data models and regularizing priors	$\hat{\mu} = \bar{x}$	Generic demographic regularization

Three inference strategies:

1. **MLE** – find the peak of the likelihood surface. Used by dadi, moments, momi2. Fast but gives only a point estimate.
2. **Bayesian inference** – target a posterior distribution, exactly or by an explicitly stated approximation. Used by ARGweaver, SINGER, tsdate, and PHLASH.
3. **Composite likelihood** – combine valid component likelihoods without specifying their full joint dependence. Used by PHLASH, momi2. A practical compromise when the exact likelihood is intractable.

One key conjugacy: for a single unknown Poisson exposure, a gamma prior gives a gamma posterior. This motivates convenient gamma representations, but tsdate must approximate its coupled parent–child factors by moment matching.

The inferential logic, in four lines:

1. **Model:** Define a mechanistic model of how genetic data arise (coalescent + mutation + recombination + demography).
2. **Likelihood:** Compute (or approximate) $P(\text{data} \mid \theta)$ – how probable the observed data are under each candidate parameter set.
3. **Optimize or sample:** Find the best parameters (MLE) or characterize the full posterior (Bayesian inference via MCMC, EP, or SVGD).
4. **Interpret:** Translate the statistical results back into biological conclusions about population sizes, divergence times, migration rates, and selection pressures.

Every inferential Timepiece implements steps 1–3 in a different way, tailored to different data types and biological questions. The data are fixed when the likelihood is viewed as a function of parameters; relative likelihood then measures support *within the stated model*. It does not establish that the model itself is true.

Next: *Coalescent Theory* – the biological foundation that every Timepiece depends on.

2.2 Coalescent Theory

To understand the present, you must run the clock backwards.

2.2.1 The Big Idea

Most of biology thinks forward in time: mutations arise, organisms reproduce, populations evolve. But in population genetics, some of the most powerful insights come from thinking **backwards**.

Here's the intuition. Imagine you've collected DNA from several individuals in a population. Each of those DNA sequences had a "parent" sequence in the previous generation, and that parent had a parent, and so on. If you trace these

lineages backwards through time, something remarkable happens: they converge. Lineages that were separate in the present share a common ancestor in the past. Eventually, if you go back far enough, *all* the lineages trace back to a single ancestor.

This process of lineages merging as we look backwards is called **coalescence** (from “to coalesce” – to come together, to merge). The mathematical theory that describes it is called **coalescent theory**, and it is the escapement mechanism at the heart of every Timepiece in this book.

i Why “escapement”?

In a mechanical watch, the escapement is the component that regulates the release of energy from the mainspring. It's what makes the watch *tick* – the fundamental oscillation that everything else depends on. Coalescent theory plays the same role in population genetics: it's the mathematical heartbeat that drives every algorithm we'll build.

Why think backwards? Because it's far more efficient. A population might contain millions of individuals, but we've only sampled a handful. Going forward, we'd need to simulate the entire population. Going backwards, we only track the lineages of our actual samples – and those lineages simplify quickly as they merge.

2.2.2 The Wright-Fisher Model (Forward in Time)

Before going backwards, let's establish the forward-time model that the coalescent approximates. This model is deliberately simple – it strips away all the biological complexity (mating, geography, natural selection) to capture just the essence of random inheritance.

Consider a population of N diploid individuals. “Diploid” means each individual carries two copies of each chromosome – one inherited from each parent. Since we're tracking a single genetic locus (a specific position on the genome), there are $2N$ gene copies in the population.

i Why $2N$?

Diploid organisms (including humans) carry two copies of each chromosome. When we track a single locus, there are $2N$ copies in the population – two per individual. In this idealized model N is both the census size and the parameter controlling drift. In real populations, N_e is an **effective** size chosen to match a specified aspect of the ancestry or drift process; $2N_e$ is a coalescent time scale, not generally the literal number of gene copies. For now, $2N$ is the actual number of copies in the Wright–Fisher population.

In the **Wright-Fisher model**, reproduction works like this:

1. **Generations are discrete and non-overlapping.** The entire population is replaced at once – all parents die, all offspring are born simultaneously.
2. **Each gene copy in the next generation chooses its parent uniformly at random** from the $2N$ copies in the current generation. “Uniformly at random” means every gene copy in the parental generation has an equal chance $1/(2N)$ of being chosen as the parent of any given offspring copy.
3. **The population size is constant** – there are always exactly $2N$ gene copies.

This is obviously not how real biology works. But its simplicity is its strength: it captures the essential randomness of inheritance (which gene copy gets passed on is largely a matter of chance), and many of the conclusions we draw from it are remarkably robust to the simplifying assumptions.

Let's simulate this model in Python. If you're new to NumPy, don't worry – we'll explain each function as we use it.

```
import numpy as np
```

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```
def wright_fisher_forward(two_N, n_generations):
    """Simulate the Wright-Fisher model forward in time.

    Parameters
    -----
    two_N : int
        Total number of gene copies (2 * diploid population size).
    n_generations : int
        Number of generations to simulate.

    Returns
    -----
    parent_table : ndarray of shape (n_generations, two_N)
        parent_table[g, i] = parental-copy index of offspring copy i for
        the transition represented by row g.
    """
    # np.zeros creates an array filled with zeros.
    # dtype=int means the entries are integers (parent indices).
    parent_table = np.zeros((n_generations, two_N), dtype=int)

    for g in range(n_generations):
        # np.random.randint(0, two_N, size=two_N) generates 'two_N'
        # random integers, each between 0 and two_N-1 (inclusive).
        # This is the "each gene copy picks a random parent" step.
        parent_table[g] = np.random.randint(0, two_N, size=two_N)

    return parent_table

# Try it: 10 gene copies, 20 generations
# np.random.seed(42) makes the random numbers reproducible --
# you'll get the same result every time you run this code.
np.random.seed(42)
parents = wright_fisher_forward(10, 20)
print("Parent of each gene copy in the first simulated transition:")
print(parents[0])
```

Notice what the code does: in each generation, every gene copy independently picks a random parent from the previous generation. Some parents get picked multiple times (their genes are passed on to multiple offspring), while others get picked zero times (their lineage dies out). This randomness is the engine that drives genetic drift.

2.2.3 Going Backwards: The Coalescent

Now the key insight. Instead of tracking the entire population forward, we **sample** n lineages and trace them **backwards**. At each generation, two lineages that chose the same parent **coalesce** into one.

This is like tracing the hands of a clock backwards: the minute and hour hands start at different positions (our sampled lineages) and, as we rewind, we watch for the moments when they align (coalescence events). Each alignment reduces the number of independent lineages by one, until finally there's just a single hand pointing to the most recent common ancestor.

The probability that two specific lineages coalesce in a given generation

Let's work out the most basic question: if we pick two specific lineages from our sample, what is the probability that they share a parent in the immediately preceding generation?

In a population of $2N$ gene copies, consider two specific lineages. Each one independently chooses a parent uniformly at random from the $2N$ copies in the previous generation.

The first lineage picks some parent – it doesn't matter which. The second lineage picks each of the $2N$ possible parents with equal probability $1/(2N)$. It picks the *same* parent as the first lineage with probability:

$$P(\text{coalesce in one generation}) = \frac{1}{2N}$$

and a *different* parent with probability:

$$P(\text{do not coalesce}) = 1 - \frac{1}{2N} = \frac{2N - 1}{2N}$$

i Why uniform and independent?

This follows directly from the Wright-Fisher model: each gene copy in the next generation picks its parent *independently* and *uniformly* from the $2N$ copies. The “uniformly” gives us $1/(2N)$ and the “independently” lets us treat the two draws separately. Sampling *with* replacement is essential here: sampling parental copies without replacement would forbid the two lineages from choosing the same copy and hence would eliminate coalescence in that generation.

For an idealized population with $N = 10,000$, we get $2N = 20,000$, so the probability of coalescence in any given generation is just $1/20,000 = 0.00005$. That's tiny! This means that for most generations, nothing happens – the two lineages just independently trace back to different parents. But over thousands of generations, coalescence becomes inevitable.

Waiting time to coalescence

How many generations do we wait until two lineages coalesce? Let's think about this carefully, because it introduces an important probability distribution.

For coalescence to happen for the *first time* at generation t , two things must be true:

1. The lineages did **not** coalesce in each of the first $t - 1$ generations
2. They **did** coalesce in generation t

Since each generation is independent (the Wright-Fisher model has no memory), we can multiply the probabilities:

$$P(T_2 = t) = \underbrace{\left(1 - \frac{1}{2N}\right)^{t-1}}_{\text{survived } t-1 \text{ generations}} \cdot \underbrace{\frac{1}{2N}}_{\text{coalesced at } t}$$

i Probability aside: the geometric distribution

This is a **geometric distribution** with success probability $p = 1/(2N)$. In general, if you repeat an experiment independently until the first success, and the probability of success on each trial is p , then the number of trials until the first success follows a geometric distribution:

$$P(T = t) = (1 - p)^{t-1} \cdot p, \quad t = 1, 2, 3, \dots$$

Its mean (expected value) is $E[T] = 1/p$. The intuition: if each trial succeeds with probability p , you expect to need about $1/p$ trials. For our coalescent, $p = 1/(2N)$, so the expected waiting time is $2N$ generations. In a population of 10,000 diploid individuals, two lineages take an average of 20,000 generations to coalesce.

For large N , this geometric distribution is well approximated by something smoother and easier to work with: the **exponential distribution**. Let's derive this carefully, because the exponential distribution will appear in nearly every chapter of this book.

The geometric-to-exponential limit. We want to show that as $N \rightarrow \infty$, the geometric distribution converges to an exponential distribution when time is measured in units of $2N$ generations.

The key is to **rescale time**. Instead of counting individual generations, we measure time in units of $2N$ generations. Define:

$$\tau = \frac{t}{2N}$$

So $\tau = 1$ corresponds to $2N$ generations, $\tau = 0.5$ corresponds to N generations, and so on.

Now, what is the probability that two lineages have *not* coalesced after $t = 2N\tau$ generations?

$$P(T_2 > t) = \left(1 - \frac{1}{2N}\right)^t = \left(1 - \frac{1}{2N}\right)^{2N\tau}$$

Here we use a beautiful fact from calculus:

Calculus aside: the exponential limit

One of the most important limits in mathematics is:

$$\lim_{m \rightarrow \infty} \left(1 - \frac{1}{m}\right)^m = e^{-1} \approx 0.3679$$

More generally, for any constant c :

$$\lim_{m \rightarrow \infty} \left(1 - \frac{c}{m}\right)^m = e^{-c}$$

This limit connects discrete compound processes to the exponential function. It says: if something small ($1/m$) happens repeatedly (m times), the combined effect is exponential.

Setting $m = 2N$ and using this limit:

$$\left(1 - \frac{1}{2N}\right)^{2N\tau} = \left[\left(1 - \frac{1}{2N}\right)^{2N}\right]^\tau \xrightarrow{N \rightarrow \infty} [e^{-1}]^\tau = e^{-\tau}$$

Therefore, in the limit:

$$P(T_2/(2N) > \tau) \rightarrow e^{-\tau}$$

Probability aside: the exponential distribution

A random variable X follows an **exponential distribution** with rate λ (written $X \sim \text{Exp}(\lambda)$) if:

$$P(X > x) = e^{-\lambda x} \quad \text{for } x \geq 0$$

Key properties:

- **Mean:** $E[X] = 1/\lambda$
- **Variance:** $\text{Var}(X) = 1/\lambda^2$
- **Memoryless property:** $P(X > s + t \mid X > s) = P(X > t)$. Knowing you've already waited s units doesn't change the distribution of additional waiting time. This is what makes the exponential distribution the continuous analogue of the geometric.
- **Density:** $f(x) = \lambda e^{-\lambda x}$ (obtained by differentiating $-P(X > x)$)

The “rate” λ controls how quickly events happen. Higher rate = shorter waits. Rate 1 means “on average, one event per unit time.”

Since $P(T_2/(2N) > \tau) \rightarrow e^{-\tau}$, we have:

$$T_2/(2N) \xrightarrow{d} \text{Exp}(1) \quad \text{as } N \rightarrow \infty$$

This tells us: the coalescence time of two lineages, measured in units of $2N$ generations, is approximately exponentially distributed with rate 1.

i Why scale by $2N$?

The probability of coalescence per generation is $1/(2N)$. The expected waiting time is $\mathbb{E}[T_2] = 2N$ generations (the mean of a geometric with parameter p is $1/p$). After rescaling by $2N$, the expected coalescence time becomes $2N/(2N) = 1$, matching the mean of $\text{Exp}(1)$. This rescaling is the natural “clock speed” of the coalescent – it makes the coalescence rate for a pair exactly 1, which simplifies all subsequent mathematics enormously.

From now on in this chapter, unless stated otherwise, time is measured in **coalescent units** of $2N$ generations. Other chapters sometimes use generations directly or state a different conventional scaling.

i How good is the approximation?

The convergence is pointwise in τ , not uniform over all time: the relative error eventually grows in the far tail. For the interval $0 \leq \tau \leq 2$, the relative error in the survival function is about 1% or less when $2N = 100$, and about 0.1% or less when $2N = 1000$. Typical effective sizes make the approximation excellent over ordinary coalescent time horizons.

Let's verify this with simulation. We'll directly simulate pairs of lineages picking random parents until they coalesce, repeat many times, and check that the distribution of waiting times matches the exponential.

```
def coalescence_time_two_lineages(two_N, n_replicates=10000):
    """Simulate coalescence times for pairs of lineages.

    Each replicate: two lineages independently pick random parents
    each generation until they pick the same one (coalescence).

    Parameters
    -----
    two_N : int
        Total number of gene copies (2N).
    n_replicates : int
```

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```

    Number of independent simulations to run.

Returns
-----
times : ndarray
    Coalescence times in generations.
"""
times = np.zeros(n_replicates)
for rep in range(n_replicates):
    t = 0
    while True:
        t += 1
        # Both lineages pick a parent at random
        parent1 = np.random.randint(0, two_N)
        parent2 = np.random.randint(0, two_N)
        if parent1 == parent2:
            break
    times[rep] = t
return times

two_N = 100
times = coalescence_time_two_lineages(two_N, n_replicates=50000)

# Convert to coalescent time units (divide by 2N)
scaled_times = times / two_N

print(f"Mean coalescence time (in 2N generations): {scaled_times.mean():.3f}")
print(f"Expected (Exp(1) mean): 1.000")
print(f"Variance: {scaled_times.var():.3f}")
print(f"Expected (Exp(1) variance): 1.000")

```

You should see the mean and variance very close to 1.000, confirming that the exponential approximation works well even for $2N = 100$.

2.2.4 The Coalescent with n Samples

So far we've considered just two lineages. What happens with n samples?

The key insight in Kingman's large-population limit is that mergers are binary and exchangeable [Kingman, 1982]. When there are k lineages remaining, we count the possible pairs and determine the rate of the next merger. The finite Wright-Fisher ancestry can contain simultaneous or multiple mergers; their probabilities vanish under the usual fixed-sample, $N \rightarrow \infty$ scaling.

Counting pairs. The number of distinct pairs you can form from k items is the **binomial coefficient**:

$$\binom{k}{2} = \frac{k!}{2!(k-2)!} = \frac{k(k-1)}{2}$$

i Why $k(k-1)/2$?

To choose a pair from k items: pick the first item (k choices), then pick the second ($k-1$ choices). But order doesn't matter (the pair $\{A, B\}$ is the same as $\{B, A\}$), so divide by 2. For $k = 5$: $\binom{5}{2} = 5 \times 4/2 = 10$ pairs.

In the limiting continuous-time chain, each pair has instantaneous merger hazard 1 in these coalescent units. Equivalently, one may construct independent rate-1 exponential clocks for the currently available pairs. Therefore the **total coalescence rate** is:

$$\lambda_k = \binom{k}{2} = \frac{k(k-1)}{2}$$

Probability aside: rates of competing exponentials

When multiple independent exponential events are “racing” to happen first, the time until the first event is also exponential, with a rate equal to the **sum** of the individual rates. This is a fundamental property:

If $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$ are independent, then $\min(X_1, X_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$.

With $\binom{k}{2}$ independent pairs, each with rate 1, the first coalescence happens at rate $\binom{k}{2}$.

The waiting time until the next coalescence event (any pair) is:

$$T_k \sim \text{Exp}\left(\binom{k}{2}\right)$$

This means the expected waiting time is $E[T_k] = 2/(k(k-1))$. Notice how this decreases rapidly as k grows: with many lineages, coalescence events happen quickly because there are so many possible pairs. Like the ticking of a clock that gradually slows down – the coalescent starts with rapid events and progressively decelerates.

When a coalescence event does occur, the pair that coalesces is chosen uniformly at random from all $\binom{k}{2}$ pairs (because all pairs are racing at the same rate, each is equally likely to win).

The total tree height. The time to the most recent common ancestor (MRCA) is the sum of all the waiting times:

$$T_{\text{MRCA}} = \sum_{k=2}^n T_k$$

and its expectation is:

$$\mathbb{E}[T_{\text{MRCA}}] = \sum_{k=2}^n \mathbb{E}[T_k] = \sum_{k=2}^n \frac{2}{k(k-1)}$$

Let's evaluate this sum. The technique is called **partial fractions** – a way to split a complicated fraction into simpler pieces that cancel nicely.

We decompose:

$$\frac{1}{k(k-1)} = \frac{A}{k-1} + \frac{B}{k}$$

Multiplying both sides by $k(k-1)$:

$$1 = Ak + B(k-1)$$

Setting $k = 0$: $1 = -B$, so $B = -1$. Setting $k = 1$: $1 = A$, so $A = 1$. Therefore:

$$\frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$$

This is a **telescoping sum** – a sum where most terms cancel in pairs, like interlocking gears where each tooth advances exactly as far as the previous tooth retreats. Substituting:

$$\begin{aligned}\sum_{k=2}^n \frac{2}{k(k-1)} &= 2 \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) \\ &= 2 \left[\left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n} \right) \right] \\ &= 2 \left(\frac{1}{1} - \frac{1}{n} \right) = 2 \left(1 - \frac{1}{n} \right)\end{aligned}$$

Most terms cancel in pairs: the $-1/2$ from the first term cancels the $+1/2$ from the second, the $-1/3$ from the second cancels the $+1/3$ from the third, and so on. Only the very first and very last terms survive.

Key insight: diminishing returns of sampling

As $n \rightarrow \infty$, the expected TMRCA converges to $2(1 - 1/n) \rightarrow 2$ (in coalescent time units). Adding more samples barely changes the tree height! Here are the numbers:

- $n = 2$: $\mathbb{E}[T_{\text{MRCA}}] = 1.000$
- $n = 5$: $\mathbb{E}[T_{\text{MRCA}}] = 1.600$
- $n = 10$: $\mathbb{E}[T_{\text{MRCA}}] = 1.800$
- $n = 100$: $\mathbb{E}[T_{\text{MRCA}}] = 1.980$
- $n = 1000$: $\mathbb{E}[T_{\text{MRCA}}] = 1.998$

The diminishing returns are dramatic. Most coalescent events happen quickly when there are many lineages (the rate $k(k-1)/2$ is huge), so the tree rapidly narrows to just a few lineages. The last few coalescences take most of the time.

This is like the final adjustments in watchmaking: the bulk of the mechanism comes together quickly, but the last few precision alignments take as long as all the earlier work combined.

Let's implement the full coalescent simulation:

```
def simulate_coalescent(n):
    """Simulate a coalescent tree for n samples.

    This simulates Kingman's coalescent: starting with n lineages,
    we repeatedly wait an exponential time and merge a random pair,
    until only one lineage (the MRCA) remains.

    Parameters
    -----
    n : int
        Number of samples (leaf nodes).

    Returns
    -----
    events : list of (time, child1, child2, parent)
        Coalescence events in chronological order (going back in time).
        'time' is in coalescent units (multiples of 2N generations).
    """
```

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```

# Active lineages, labeled 0 to n-1
lineages = list(range(n))
next_label = n # labels for internal (ancestor) nodes
events = []
current_time = 0.0

while len(lineages) > 1:
    k = len(lineages)
    rate = k * (k - 1) / 2 # binom(k, 2)

    # Sample waiting time from Exp(rate).
    # np.random.exponential(scale) draws from Exp with mean = scale,
    # so scale = 1/rate gives rate = rate.
    wait = np.random.exponential(1.0 / rate)
    current_time += wait

    # Choose a random pair to coalesce.
    # np.random.choice(n, size=2, replace=False) picks 2 distinct
    # indices from {0, 1, ..., n-1}.
    i, j = np.random.choice(len(lineages), size=2, replace=False)
    child1 = lineages[i]
    child2 = lineages[j]

    # Create parent node
    parent = next_label
    next_label += 1
    events.append((current_time, child1, child2, parent))

    # Remove children, add parent
    lineages = [l for idx, l in enumerate(lineages)
                 if idx != i and idx != j]
    lineages.append(parent)

return events

# Simulate and print
np.random.seed(42)
events = simulate_coalescent(5)
print("Coalescence events (time, child1, child2, parent):")
for t, c1, c2, p in events:
    print(f"  t={t:.4f}: lineages {c1} and {c2} -> {p}")
print(f"\nTMRCA = {events[-1][0]:.4f}")
print(f"Expected TMRCA = {2*(1 - 1/5):.4f}")

```

Read through the output carefully. Early waiting times are shorter *on average* because there are more possible pairs, while the final waiting time has the largest mean. A single realization need not appear in that order. The rate drops from $\binom{5}{2} = 10$ to $\binom{2}{2} = 1$.

2.2.5 Expected Number of Lineages at Time t

A useful approximation for the SINGER algorithm asks: given n samples, how many lineages remain at time t on a typical coalescent genealogy?

Let $K(t)$ be the random lineage count. Its exact mean obeys $d\mathbb{E}[K]/dt = -\mathbb{E}[K(K-1)]/2$, which is not closed because it depends on the second moment. The deterministic approximation replaces $\mathbb{E}[K(K-1)]$ by $\mathbb{E}[K](\mathbb{E}[K] - 1)$. Writing the resulting smooth approximation as $\lambda(t)$ gives the **ordinary differential equation (ODE)** [Jewett and Rosenberg, 2014]:

$$\frac{d\lambda}{dt} = -\binom{\lambda}{2} = -\frac{\lambda(\lambda-1)}{2}$$

with initial condition $\lambda(0) = n$.

i Calculus aside: what is a differential equation?

A differential equation relates a quantity to its rate of change. Here, $d\lambda/dt$ is the rate at which the number of lineages changes over time. The equation says: the rate of decrease equals the number of mean-field number of possible pairs, $\binom{\lambda}{2}$. For the stochastic process, a coalescence reduces K by 1 and occurs at rate $\binom{K}{2}$; replacing the random quadratic rate by the quadratic of the mean is the approximation. The minus sign indicates that λ decreases.

Solving a differential equation means finding a function $\lambda(t)$ that satisfies this relationship. We do this below using a technique called “separation of variables.”

This is a **separable ODE**, meaning we can move all the λ terms to one side and all the t terms to the other. Separating variables:

$$\frac{d\lambda}{\lambda(\lambda-1)} = -\frac{dt}{2}$$

i Calculus aside: separation of variables

This technique works when a differential equation has the form $f(\lambda) d\lambda = g(t) dt$. We can integrate both sides independently. The left side becomes an integral in λ , the right side an integral in t .

Using partial fractions on the left (the same technique we used for the telescoping sum $-\frac{1}{\lambda(\lambda-1)} = \frac{1}{\lambda-1} - \frac{1}{\lambda}$):

$$\left(\frac{1}{\lambda-1} - \frac{1}{\lambda}\right) d\lambda = -\frac{dt}{2}$$

Integrating both sides:

$$\ln(\lambda-1) - \ln(\lambda) = -\frac{t}{2} + C$$

i Calculus aside: the integral $\int \frac{1}{x} dx = \ln|x|$

The natural logarithm $\ln(x)$ is the function whose derivative is $1/x$. Equivalently, $\int \frac{1}{x} dx = \ln|x| + C$. This is one of the most important integrals in mathematics, and it appears throughout this book.

Using the logarithm rule $\ln(a) - \ln(b) = \ln(a/b)$:

$$\ln\left(\frac{\lambda-1}{\lambda}\right) = -\frac{t}{2} + C$$

At $t = 0$, $\lambda = n$, so $C = \ln\left(\frac{n-1}{n}\right)$. Exponentiating both sides (applying e^x to undo $\ln$):

$$\frac{\lambda - 1}{\lambda} = \frac{n - 1}{n} e^{-t/2}$$

Solving for λ (rewrite the left side as $1 - 1/\lambda$):

$$1 - \frac{1}{\lambda} = \frac{n - 1}{n} e^{-t/2}$$

$$\frac{1}{\lambda} = 1 - \frac{n - 1}{n} e^{-t/2} = \frac{n - (n - 1)e^{-t/2}}{n}$$

Inverting:

$$\lambda(t) = \frac{n}{n - (n - 1)e^{-t/2}} = \frac{n}{n + (1 - n)e^{-t/2}}$$

where the last step uses $-(n - 1) = 1 - n$.

Let's verify this formula by comparing it to simulation:

```
def expected_lineages(t, n):
    """Mean-field approximation to the lineage count at time t.

    Uses the deterministic mean-field approximation analyzed by
    Jewett and Rosenberg (2014).

    Parameters
    -----
    t : float
        Time in coalescent units.
    n : int
        Number of initial samples.

    Returns
    -----
    float
        Approximate expected number of lineages at time t.
    """
    return n / (n + (1 - n) * np.exp(-t / 2))

def simulate_lineage_count(n, t, n_replicates=10000):
    """Estimate E[lineages at time t] by simulation.

    Simulates the coalescent n_replicates times, counts how many
    lineages remain at time t, and returns the average.
    """
    counts = np.zeros(n_replicates)
    for rep in range(n_replicates):
        k = n
        current_time = 0.0
        while k > 1 and current_time < t:
            rate = k * (k - 1) / 2
            wait = np.random.exponential(1.0 / rate)
            if current_time + wait > t:
                break
```

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```

        current_time += wait
        k -= 1
        counts[rep] = k
    return counts.mean()

n = 50
for t in [0.01, 0.05, 0.1, 0.5, 1.0, 2.0]:
    approx = expected_lineages(t, n)
    simulated = simulate_lineage_count(n, t, n_replicates=5000)
    print(f"t={t:.2f}: approx={approx:.2f}, simulated={simulated:.2f}")

```

The agreement should be close for this grid at $n = 50$. It is not an identity: accuracy depends on time and sample size, and the exact expected lineage count is obtained from the pure-death transition probabilities rather than this closed ODE [Jewett and Rosenberg, 2014].

2.2.6 Mutations on the Coalescent Tree

So far, the coalescent tells us about the genealogical tree – its shape and timing. But what we actually *observe* in real data is not the tree itself, but **mutations**: differences between DNA sequences at specific positions.

Mutations are added to the coalescent tree here under the **infinite-sites model**:

- Each mutation occurs at a unique position on the genome (no position mutates twice)
- Mutations arise as a **Poisson process** along each branch of the tree
- The rate is governed by $\theta = 4N_e\mu$, where μ is the per-base-pair, per-generation mutation rate

i Where does $\theta = 4N_e\mu$ come from?

In the Wright-Fisher model, each base pair mutates with probability μ per generation. A branch of length ℓ in coalescent time units corresponds to $2N_e \cdot \ell$ generations (since 1 coalescent unit = $2N_e$ generations). The expected number of mutations on this branch at a per base pair is:

$$\mu \times 2N_e\ell = \frac{4N_e\mu}{2} \cdot \ell = \frac{\theta}{2} \cdot \ell$$

The factor of $4N_e$ appears because θ is defined as $4N_e\mu$ (a convention for diploid organisms), and the $1/2$ appears because we're converting from the $4N_e\mu$ scaling to a per-unit-coalescent-time rate. The mutation rate **per coalescent time unit per base pair** is $\theta/2$.

i Probability aside: the Poisson process

A **Poisson process** with rate r is a model for events that occur randomly and independently over time (or space). In a time interval of length L , the number of events follows a **Poisson distribution** with parameter rL :

$$P(k \text{ events in } [0, L]) = \frac{(rL)^k}{k!} e^{-rL}$$

Key properties:

- **Mean number of events:** rL
- **Probability of zero events:** e^{-rL} (the zeroth term of the sum)

- **Probability of at least one event:** $1 - e^{-rL}$
- Events in non-overlapping intervals are independent

This model is a natural limit of many rare independent trials. If each base pair has a tiny probability of mutating per generation, and there are many generations, the total mutation count converges to a Poisson distribution (this is the **Poisson limit theorem**).

On a continuous genomic region of length B , the number of mutations on a branch of length ℓ follows a Poisson distribution with mean $\theta\ell B/2$; independently sampled continuous positions are unique with probability one. Setting $B = 1$ for one unit of sequence, the probability that a branch carries **no** mutation in that unit is:

$$P(\text{no mutation}) = \frac{(\theta\ell/2)^0}{0!} e^{-\theta\ell/2} = \exp\left(-\frac{\theta}{2}\ell\right)$$

and the probability of **at least one** mutation in that unit is:

$$P(\text{mutation}) = 1 - \exp\left(-\frac{\theta}{2}\ell\right)$$

For small $\theta\ell/2$, we can use the approximation $e^{-x} \approx 1 - x$ (valid when x is much less than 1):

$$P(\text{mutation}) \approx \frac{\theta}{2}\ell \quad \text{when } \frac{\theta}{2}\ell \ll 1$$

For a representative human mutation rate of order 10^{-8} per bp per generation and N_e of order 10^4 , θ is of order 10^{-4} per bp. Since branch lengths are $O(1)$ in coalescent units, $\theta\ell/2$ is small, so the linear and infinite-sites approximations are often accurate per base. Across long regions, however, multiple mutations occur on a branch and the full Poisson count is needed.

```
def add_mutations(events, n, theta, seq_length=1):
    """Add mutations to a coalescent tree under the infinite sites model.

    For each branch in the tree, we draw a Poisson number of mutations
    (proportional to the branch length and the mutation rate), and place
    each mutation at a random position on a continuous genome. Continuous
    positions are unique with probability one, implementing infinite sites.

    Parameters
    -----
    events : list of (time, child1, child2, parent)
        Coalescence events from simulate_coalescent().
    n : int
        Number of samples (leaf nodes, labeled 0 to n-1).
    theta : float
        Population-scaled mutation rate (4*Ne*mu).
    seq_length : float
        Length of the continuous genomic interval, in base-pair units.

    Returns
    -----
    mutations : list of (position, branch_node, time)
        Each mutation has a genomic position, the node whose branch it
        sits on, and the time at which it occurred.
    """
```

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```

# Build a dictionary mapping each node to its time
node_times = {i: 0.0 for i in range(n)} # samples are at time 0
for t, c1, c2, p in events:
    node_times[p] = t

mutations = []
root = events[-1][3] # the root is the parent in the last event

# For each branch (child -> parent), the length is parent_time - child_time
for node in node_times:
    if node == root:
        continue # the root has no parent branch
    # Find this node's parent
    for t, c1, c2, p in events:
        if c1 == node or c2 == node:
            branch_length = node_times[p] - node_times[node]
            # Expected mutations on this branch:
            # theta/2 * branch_length * seq_length
            n_muts = np.random.poisson(theta / 2 * branch_length * seq_length)
            for _ in range(n_muts):
                pos = np.random.uniform(0, seq_length)
                mut_time = np.random.uniform(node_times[node], node_times[p])
                mutations.append((pos, node, mut_time))
            break # found the parent, move to next node

return sorted(mutations)

np.random.seed(42)
events = simulate_coalescent(10)
mutations = add_mutations(events, 10, theta=0.001, seq_length=1000)
print(f"Number of mutations (segregating sites here): {len(mutations)}")

```

2.2.7 Summary

You now have the core machinery of coalescent theory – the escapement of every Timepiece in this book. Let's take stock of what you've built:

Concept	Key Formula
Pairwise coalescence time	$T_2 \sim \text{Exp}(1)$ (in $2N$ gen. units)
Coalescence rate with k lineages	$\lambda_k = \binom{k}{2}$
Expected TMRCA	$\mathbb{E}[T_{\text{MRCA}}] = 2(1 - 1/n)$
Expected lineages at time t	$\lambda(t) \approx \frac{n}{n + (1-n)e^{-t/2}}$
Mutation probability on branch of length ℓ , per unit region	$P(\text{at least one mut}) = 1 - e^{-\theta\ell/2}$

These results are the foundation – the ticking mechanism at the heart of the watch. Everything that follows, from ARGs to HMMs to full ARG inference algorithms, builds on this fundamental clockwork.

Next: *Ancestral Recombination Graphs* – what happens when recombination breaks the tree into many trees, and how we represent that rich history.

2.3 Ancestral Recombination Graphs

A single tree tells one story. An ARG tells the whole history.

2.3.1 Why Trees Aren't Enough

In the *Coalescent Theory* chapter, we built genealogical trees – the escapement mechanism that drives every Timepiece in this book. We derived the exponential waiting times, the coalescence rates $\binom{k}{2}$, and the mutation model. But we made one critical simplifying assumption: **no recombination**.

That assumption is like describing a watch movement by tracing a single gear train from mainspring to hands. It gives a correct picture of *one* pathway of force transmission, but a real movement has many interacting gear trains – the going train for timekeeping, the keyless works for winding, perhaps a chronograph train for elapsed timing. The complete mechanical drawing must show all of these, along with every point where one train's output becomes another train's input.

An **ancestral recombination graph (ARG)** is that complete mechanical drawing for the ancestry of the sampled genome. It records the ancestral coalescence and recombination events needed to relate the sampled sequences across the region; it does not include unrelated events elsewhere in the population. Where a single coalescent tree shows one gear train, the ARG shows all of them, plus the couplings between them [Hudson, 1983].

But before we can understand the ARG, we need to understand what recombination is and why it matters.

2.3.2 What Is Recombination?

Recombination is a biological process that occurs during **meiosis** – the special cell division that produces sperm and egg cells (gametes). In meiosis, an organism that carries two copies of each chromosome (one from each parent) produces cells with just one copy. But before that division happens, something remarkable occurs: the maternal and paternal copies of each chromosome physically align, and segments are swapped between them.

i Biology aside: why does recombination exist?

Recombination shuffles genetic material between the two copies of a chromosome you inherited from your parents. The result is that the chromosome you pass on to *your* child is a mosaic – some segments came from your mother's chromosome, others from your father's.

Why would evolution favor this shuffling? There are several theories, but one important consequence is that recombination breaks up linkage between mutations, which can allow selection to act on them more independently. Without recombination, a bad mutation on a chromosome is permanently linked to every other variant on that chromosome. With recombination, good and bad variants can be separated, giving selection a clearer target.

For our purposes, the key consequence is simple: **different positions along your genome have different genealogical histories**. The stretch of DNA near the beginning of chromosome 1 traces back through a different sequence of ancestors than the stretch near the end.

Here's the concrete implication. Consider three individuals – Alice, Bob, and Carol – and look at two positions on the same chromosome, position 1 and position 1000. At position 1, perhaps Alice and Bob share a recent common ancestor (their great-great-grandmother), while Carol's lineage diverged much earlier. At position 1000, the situation might be reversed: Alice and Carol share a recent ancestor, while Bob's lineage is the outlier. Both genealogies are correct – they just reflect different parts of the genome's history, shaped by recombination events in the intervening generations.

This is the fundamental reason we need ARGs rather than single trees.

What We've Established So Far

From the coalescent theory chapter, we have these tools:

- Coalescence times are exponential with rate $\binom{k}{2}$ when there are k lineages (measured in coalescent time units – multiples of $2N_e$ generations)
- Mutations arise as a Poisson process at rate $\theta/2$ per coalescent time unit per base pair
- The expected time to the most recent common ancestor is $2(1 - 1/n)$ coalescent time units

Now we need to extend this framework to handle recombination.

2.3.3 Recombination in the Coalescent

In the coalescent with recombination, we still trace lineages backwards in time, but now two types of events can happen:

1. **Coalescence:** Two lineages merge (as before). Rate $\binom{k}{2}$ when there are k lineages. This is the same mechanism we derived in the *Coalescent Theory* chapter.
2. **Recombination:** A single ancestral chromosome **splits** at a random breakpoint, with the material on either side followed through different parents. If every active lineage spans the full region, each recombines at rate $\rho/2$, where $\rho = 4N_e r$ and r is the per-generation recombination probability across that region.

i Where does $\rho = 4N_e r$ come from?

This follows the same logic as $\theta = 4N_e \mu$ for mutation (derived in the *Coalescent Theory* chapter). In the Wright-Fisher model, a recombination event occurs with probability r per generation per lineage. A branch of length ℓ in coalescent time units spans $2N_e \cdot \ell$ generations. The expected number of recombination events on this branch is:

$$r \times 2N_e \ell = \frac{4N_e r}{2} \cdot \ell = \frac{\rho}{2} \cdot \ell$$

Thus a lineage spanning the whole region has recombination rate $\rho/2$ per coalescent time unit. After lineages carry different amounts of ancestral material, the exact Hudson process weights them by their available recombination links rather than assigning every lineage this same rate [Hudson, 1983].

i What is a “breakpoint”?

A breakpoint is the position along the genome where a recombination event cuts a lineage in two. Think of the genome as a long strip of paper representing the sequence from position 0 to position S . A recombination event at breakpoint x tears the strip at position x : the left piece (positions $[0, x)$) traces back through one ancestor, and the right piece (positions $[x, S)$) traces back through a different ancestor. Before the breakpoint, the genealogy is one tree; after it, the genealogy may be a different tree.

In the watchmaking metaphor, a breakpoint is like a point where one gear train hands off to another – the same shaft, but driven by different mechanisms on either side.

Going backwards in time, these two event types have opposite effects:

- **Coalescence** reduces the number of lineages by 1 (two become one)
- **Recombination** increases the number of lineages by 1 (one becomes two)

Initially, while all k lineages span the full region, these events compete at total rate:

$$\text{Rate of next event} = \underbrace{\binom{k}{2}}_{\text{coalescence}} + \underbrace{\frac{k\rho}{2}}_{\text{recombination}}$$

i Probability aside: competing exponential processes

This displayed rate is the full-span special case. In the general process, replace $k\rho/2$ by the sum of the lineage-specific recombination rates determined by their ancestral material. As established in the *Coalescent Theory* chapter, when multiple independent exponential processes are racing, the time to the first event is exponential with rate equal to the **sum** of the individual rates. In the initial full-span case, the coalescence rate is $\binom{k}{2}$ and the recombination rate is $k\rho/2$, so the next event – whichever type it is – occurs after an exponential waiting time with rate $\binom{k}{2} + k\rho/2$.

The corresponding initial probability that the event is a coalescence (rather than a recombination) is proportional to its rate:

$$P(\text{coalescence}) = \frac{\binom{k}{2}}{\binom{k}{2} + k\rho/2}$$

This is a general property of competing Poisson processes: the probability that a particular process “wins” is its rate divided by the total rate.

Implementing Hudson’s ancestral-material bookkeeping correctly is subtle. The code below therefore delegates exact simulation to `msprime`, which implements the coalescent with recombination and returns its standard `tskit` tree-sequence representation [Kelleher *et al.*, 2016].

```
import msprime

def simulate_arg(n, rho, seq_length=1.0, Ne=10_000, random_seed=None):
    """Simulate a haploid sample under the coalescent with recombination.

    Parameters
    -----
    n : int
        Number of samples.
    rho : float
        Population-scaled recombination rate (4*Ne*r) for the whole region.
    seq_length : float
        Length of the continuous genomic region.
    Ne : float
        Diploid effective population size.
    random_seed : int or None
        Seed passed to msprime.

    Returns
    -----
    tskit.TreeSequence
        Correlated marginal genealogies. Node times are in generations.
    """
    if n < 2 or rho < 0 or seq_length <= 0 or Ne <= 0:
        raise ValueError("require n >= 2, rho >= 0, seq_length > 0, and Ne > 0")
    # rho = 4*Ne*(rate per unit)*(sequence length).
```

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```

recombination_rate = rho / (4 * Ne * seq_length)
return msprime.sim_ancestry(
    samples=n,
    ploidy=1,
    population_size=Ne,
    sequence_length=seq_length,
    recombination_rate=recombination_rate,
    discrete_genome=False,
    record_full_arg=True,
    random_seed=random_seed,
)

ts = simulate_arg(n=5, rho=5.0, random_seed=42)
print(f"Samples: {ts.num_samples}")
print(f"Marginal-tree intervals: {ts.num_trees}")
print(f"Breakpoints: {list(ts.breakpoints())}")

```

Increasing ρ increases recombination in distribution, but a realized recombination need not create an observable change between adjacent marginal trees. With $\rho = 0$, the result contains one marginal tree. The `record_full_arg` option preserves additional event nodes; a simplified tree sequence generally retains the marginal genealogies without being a literal ledger of every ancestral recombination event.

2.3.4 The Structure of an ARG: A Directed Acyclic Graph

A coalescent tree is, mathematically, a **tree**: each node has exactly one parent (except the root, which has none). But an ARG is something more complex. Recombination events create nodes with **two parents** – one for the left side of the breakpoint, one for the right side. This means the ARG is not a tree but a **directed acyclic graph (DAG)**.

i What is a directed acyclic graph?

A **directed graph** is a collection of nodes connected by arrows (directed edges). “Directed” means each edge has a direction: from child to parent, in our case. “Acyclic” means there are no cycles – you can never follow arrows and return to where you started. In the ARG, edges always point backwards in time (from descendant to ancestor), and time only flows one way, so cycles are impossible.

A tree is a special case of a DAG where every node has at most one parent. In an ARG, recombination nodes have two parents (one for each side of the breakpoint), making it a DAG but not a tree. Coalescence nodes still have one parent but two children, just as in a tree.

In the watchmaking metaphor, a tree is like a simple gear train – force flows in one direction from mainspring to escapement with no branching. An ARG is like the full movement, where a wheel might receive force from two different sources (one driving the hours, another driving the minutes), and where the same barrel might power multiple complication trains simultaneously.

2.3.5 Marginal Trees

An ARG encodes a **marginal tree** at every position along the genome. The marginal tree at position x is the genealogical tree for that specific position, constructed by taking the ARG and ignoring all recombination events that don’t affect position x .

Here’s the key intuition: as you slide a pointer from left to right along the genome, the genealogical tree can change. Changes occur only at inherited recombination breakpoints, but not every ancestral recombination changes a marginal tree:

the event may be later undone or leave the displayed genealogy unchanged. Between the edge breakpoints represented in a tree sequence, the tree is constant.

This is like examining different cross-sections of a watch movement. At one cross-section, you see a particular arrangement of gears. Slide to a different cross-section, and some gears have been swapped – a different train is visible. But between the swap points, the arrangement stays the same.

```
def extract_tree_intervals(ts):
    """Return intervals on which the represented marginal tree is constant.

    These are tskit's actual tree intervals, not inferred directly from a
    raw list of recombination events.
    """
    return [(tree.interval.left, tree.interval.right) for tree in ts.trees()]

intervals = extract_tree_intervals(ts)
print(f"Number of tree intervals: {len(intervals)}")
print(f"Intervals: {intervals}")
```

There is no general `recombination events + 1` rule. Several events can map to the same breakpoint, a recombination can be invisible in the marginal trees, and simplified representations can discard event nodes while preserving all marginal trees.

The key property that connects all of this:

ARG = sequence of marginal trees + recombination events connecting them

This is one useful conceptual view of the object SINGER samples from sequence data. The marginal trees tell us the genealogy at each position; a fuller ARG also records ancestral events connecting those trees.

2.3.6 The Tree Sequence Representation

Storing an explicit tree for every base pair along the genome would be enormously wasteful. Adjacent positions usually share most edges, so modern tools encode correlated genealogies as **succinct tree sequences**: node and interval-specific edge tables from which marginal trees are generated efficiently [Kelleher *et al.*, 2016].

This behaves like **differential encoding** during tree iteration: rather than materializing every tree in full, edges ending at a breakpoint are removed and edges beginning there are inserted. The stored object itself is a set of tables, not literally a first tree followed by edit commands. This is analogous to how a watchmaker's technical manual might describe the base caliber in full, then describe each complication variant by listing only the modified components.

The fundamental data structure has two tables:

- **Nodes**: Each node has a time (0 for samples, positive for ancestors) and a flag indicating whether it's a sample
- **Edges**: Each edge has a genomic interval [left, right) and connects a child to a parent. The edge is "active" only within its genomic interval.

```
class SimpleTreeSequence:
    """A minimal tree sequence representation for educational purposes.

    Nodes are stored as (time, is_sample) tuples.
    Edges are stored as (left, right, parent, child) tuples, where
    [left, right) is the genomic interval where this parent-child
    relationship holds.
    """
```

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```

def __init__(self):
    self.nodes = []    # list of (time, is_sample) tuples
    self.edges = []    # list of (left, right, parent, child) tuples

def add_node(self, time, is_sample=False):
    """Add a node and return its integer ID (0-indexed)."""
    node_id = len(self.nodes)
    self.nodes.append((time, is_sample))
    return node_id

def add_edge(self, left, right, parent, child):
    """Add an edge active over the genomic interval [left, right)."""
    self.edges.append((left, right, parent, child))

def trees(self, seq_length):
    """Iterate over marginal trees.

    Yields (left, right, active_edges) for each genomic interval
    where the tree topology is constant.
    """
    # Collect all breakpoints from edge boundaries.
    # The set comprehension gathers every left and right coordinate,
    # plus the sequence endpoints, then sorts them.
    breakpoints = sorted(set(
        [0.0, seq_length] +
        [l for l, r, p, c in self.edges] +
        [r for l, r, p, c in self.edges]
    ))

    for i in range(len(breakpoints) - 1):
        # Check which edges are active at the midpoint of this interval.
        # Using the midpoint avoids boundary ambiguity.
        pos = (breakpoints[i] + breakpoints[i+1]) / 2
        # List comprehension: keep only edges whose interval contains pos.
        active = [(p, c) for l, r, p, c in self.edges
                  if l <= pos < r]
        yield breakpoints[i], breakpoints[i+1], active

# Example: two marginal trees for 4 samples
ts = SimpleTreeSequence()
# Samples at time 0 (the present)
# The underscore _ is a Python convention for a variable we don't need --
# here we don't use the loop variable because add_node tracks IDs internally.
for _ in range(4):
    ts.add_node(0.0, is_sample=True)
# Internal (ancestor) nodes at various times in the past
ts.add_node(0.5)    # node 4
ts.add_node(0.8)    # node 5
ts.add_node(1.2)    # node 6

# Left tree: ((0,1), (2,3)); right tree: ((0,2), (1,3)).
# Samples 1 and 2 exchange parents at position 0.6. Both intervals are

```

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```
# valid rooted binary trees on all four samples.
ts.add_edge(0.0, 1.0, 4, 0)
ts.add_edge(0.0, 0.6, 4, 1)
ts.add_edge(0.6, 1.0, 4, 2)
ts.add_edge(0.0, 0.6, 5, 2)
ts.add_edge(0.6, 1.0, 5, 1)
ts.add_edge(0.0, 1.0, 5, 3)
ts.add_edge(0.0, 1.0, 6, 4)
ts.add_edge(0.0, 1.0, 6, 5)

for left, right, edges in ts.trees(1.0):
    print(f"\nTree at [{left:.1f}, {right:.1f}):")
    for p, c in edges:
        print(f"    {c} -> {p}")
```

Examine the output: in the first tree (positions 0.0 to 0.6), samples 0 and 1 are siblings under node 4. In the second tree (positions 0.6 to 1.0), samples 0 and 2 are siblings under node 4. The two parent assignments that change are stored only on their respective intervals, while the four unchanged edges span the whole sequence.

2.3.7 Branch Lengths and the ARG

A crucial concept for SINGER: the **total branch length** of a marginal tree. This quantity determines how many mutations we expect to see, and it connects the genealogy (which is hidden) to the sequence data (which we observe).

For a marginal tree Ψ at position x , the total branch length is the sum of all branch lengths in the tree:

$$L(\Psi) = \sum_{\text{branch } b \in \Psi} \ell(b)$$

where $\ell(b)$ is the length (in coalescent time units) of branch b . Recall from the *Coalescent Theory* chapter that branch lengths are measured in coalescent time units (multiples of $2N_e$ generations).

The total branch length of the entire ARG across the genome is obtained by **integrating** the branch length of each marginal tree over the positions it covers:

$$L(\mathcal{G}) = \int_0^S L(\Psi_x) dx$$

where S is the sequence length and Ψ_x is the marginal tree at position x .

i Calculus aside: why integrate?

Each base pair has its own marginal tree, and mutations at each base pair arise on the branches of that specific tree. So the total “opportunity” for mutations is the sum of all branch lengths across all positions. Since the marginal tree is constant within each genomic interval between breakpoints, the integral simplifies to a sum:

$$L(\mathcal{G}) = \sum_{i=1}^T (\text{right}_i - \text{left}_i) \times L(\Psi_i)$$

where the sum runs over the T marginal-tree intervals and $[\text{left}_i, \text{right}_i)$ is the genomic interval where tree Ψ_i applies. Each tree’s branch length is weighted by how many base pairs it covers (its “span”).

This total branch length determines the expected number of mutations across the entire ARG:

$$\mathbb{E}[\text{number of mutations}] = \frac{\theta}{2} L(\mathcal{G})$$

i Probability aside: why $\theta/2 \times L$?

This follows from the Poisson mutation model we established in the *Coalescent Theory* chapter. Each unit of branch length (in coalescent time) at each base pair independently produces mutations at rate $\theta/2$. Since the Poisson distribution has the property that the mean of a sum of independent Poissons equals the sum of the means, the expected total number of mutations is simply $\theta/2$ times the total branch-length-times-span across the entire ARG. This is a consequence of the **additivity property** of Poisson processes.

```
def total_branch_length(node_times, edges, position):
    """Compute total branch length of the marginal tree at a given position.

    For each edge active at 'position', the branch length is the difference
    between the parent's time and the child's time.

    Parameters
    -----
    node_times : dict
        Mapping from node ID to time (in coalescent units).
    edges : list of (left, right, parent, child)
        Tree sequence edges.
    position : float
        Genomic position to query.

    Returns
    -----
    float
        Total branch length at the given position.
    """
    total = 0.0
    for left, right, parent, child in edges:
        # Check if this edge is active at the query position
        if left <= position < right:
            total += node_times[parent] - node_times[child]
    return total
```

2.3.8 Why ARG Inference Is Hard

We've now seen what an ARG is and how to represent one. The challenge that motivates the rest of this book is the **inverse problem**: given observed DNA sequences, can we reconstruct the ARG that produced them?

This turns out to be extraordinarily difficult, for several reinforcing reasons:

1. **Enormous state space.** The number of possible ARGs grows super-exponentially with sample size. Even for a handful of samples and a short genomic region, the space of possible histories is vast beyond enumeration. It is as if a watchmaker had to deduce the complete history of a movement's assembly – every gear placed, every screw turned, in exact order – by examining only the finished watch.
2. **Recombination creates reticulations.** Unlike trees, explicit ARGs are directed acyclic graphs with **reticulation nodes** – nodes that have two parents (one from each side of a recombination breakpoint). This makes the combinatorial structure far richer than for trees, where each node has exactly one parent for each non-root node.
3. **Data is sparse.** We only observe the *leaves* of the ARG (the sampled sequences at the present), and we only see positions where mutations happened to fall. The vast majority of the tree's branches carry no mutations at all (recall from the coalescent chapter that mutation probabilities per site are tiny: $\theta\ell/2 \approx 10^{-4}$ for typical human

parameters). Reconstructing a complex structure from such sparse observations is inherently underdetermined.

4. **Identifiability.** Many different ARGs can produce the same pattern of mutations. Two different genealogical histories might yield identical sequence data, making it impossible to distinguish them even in principle.

i Probability aside: the curse of dimensionality

The space of possible ARGs is a high-dimensional combinatorial object. For n samples and a recombining region, both the number of ancestral events and the possible ways of connecting them are random. There is no universal $O(n\rho S)$ event-count formula: the convention for ρ , ancestral material, demography, and stopping rule all matter. Exhaustive enumeration is out of the question. Posterior samplers such as SINGER retain uncertainty, while other useful methods construct point or approximate estimates; Bayesian sampling is not the only viable strategy.

This is why methods like SINGER exist. Rather than attempting to search the full space of ARGs, SINGER uses two powerful mathematical tools to make inference tractable:

- The **Sequentially Markov Coalescent (SMC)**, which approximates the ARG as a Markov chain of marginal trees along the genome (covered in *The Sequentially Markov Coalescent*)
- **Hidden Markov Models (HMMs)**, which provide efficient algorithms for inference in Markov chains (covered in *Hidden Markov Models*)

Together, these tools reduce an impossible combinatorial problem to a sequence of manageable probabilistic calculations.

2.3.9 Summary

We've moved from the single-tree world of the coalescent to the richer landscape of ARGs – from tracing one gear train to reading the complete mechanical drawing. Here is what we've established:

Concept	Key Idea
Recombination	Different genome positions can have different trees (shuffling during meiosis)
ARG	Complete genealogical history: all trees + recombinations (a directed acyclic graph)
Marginal tree	The genealogy at a single genome position; constant between breakpoints
Breakpoint	A crossover coordinate; it need not produce a visible tree change
Tree sequence	Efficient storage: record only the changes between adjacent marginal trees
Total branch length	Determines expected mutation count: $\mathbb{E}[\text{mut}] = \frac{\theta}{2} L(\mathcal{G})$
Event rates	Initially $\binom{k}{2}$ and $k\rho/2$; later recombination depends on ancestral material

The ARG is the object we ultimately want to infer – it is the master blueprint of the genome's history. But as we've seen, direct inference is intractable. The next two chapters introduce the mathematical machinery that makes it possible: Hidden Markov Models and the Sequentially Markov Coalescent.

Next: *Hidden Markov Models* – the computational engine that makes ARG inference tractable.

2.4 Hidden Markov Models

The states are hidden, but the mechanism is not.

Think of a mechanical watch. You can see the hands sweep across the dial – the hours, the minutes, the seconds – but the gears and springs that drive those hands are concealed behind the case. You observe the *output* of the mechanism, not the mechanism itself. A Hidden Markov Model is a mathematical formalization of exactly this situation: a system whose internal workings are invisible, but whose observable behavior carries clues about what is happening inside.

In population genetics, the analogy is almost literal. When we sequence genomes, we see the mutations – the positions where alleles differ between individuals. These are the hands on the dial. What we cannot see directly are the genealogical relationships: which branches of the ancestral tree each genomic position “sits on,” and where recombination events reshuffled the ancestry. These hidden genealogical states are the gears behind the watch face.

This chapter develops the mathematical machinery of HMMs from the ground up, building toward the specific form used in ARG inference tools like SINGER. If you have not yet read the chapters on [Coalescent Theory](#) and [Ancestral Recombination Graphs](#), now is a good time – the HMM framework here is the bridge that connects those theoretical concepts to practical inference algorithms.

2.4.1 Why HMMs for ARG Inference?

Before diving into formalism, it is worth understanding *why* HMMs are the right tool for this problem.

In the [Ancestral Recombination Graphs](#) chapter, we saw that an ARG encodes a sequence of marginal trees along the genome, with recombination events causing transitions from one tree to the next. In the [The Sequentially Markov Coalescent](#) chapter, we saw that the SMC approximation makes this sequence Markov: the tree at position ℓ depends only on the tree at position $\ell - 1$, not on the entire history.

This is exactly the structure an HMM is designed to handle:

- We have a sequence of **hidden states** (the marginal tree, or more specifically, the branch of the tree to which a lineage belongs) that evolves along the genome.
- At each position, the hidden state **emits** an observation (the allele we see – whether there is a mutation or not).
- The hidden states form a **Markov chain**: the state at position ℓ depends only on the state at position $\ell - 1$.

The goal of ARG inference is to go from the observations (the genome data) back to the hidden states (the genealogical relationships). HMMs give us principled, efficient algorithms for doing exactly this.

2.4.2 A Warm-Up Example: Weather and Umbrellas

Before tackling genetics, let us build intuition with a simpler example.

Suppose you are locked in a windowless office. Each day, a colleague visits you, and you notice whether they are carrying an umbrella. You cannot see the weather outside (it is *hidden*), but you can observe the umbrella (the *emission*). Over time, you want to figure out what the weather has been doing.

The weather follows a simple pattern:

- If today is sunny, there is a 95% chance tomorrow is also sunny (weather tends to persist).
- If today is rainy, there is a 90% chance tomorrow is also rainy.
- When it is sunny, your colleague carries an umbrella only 10% of the time.
- When it is rainy, your colleague carries an umbrella 80% of the time.

This is a two-state HMM:

- **Hidden states:** Sunny, Rainy
- **Observations:** Umbrella, No umbrella

- **Transition probabilities:** The chance of weather changing from one day to the next
- **Emission probabilities:** The chance of seeing an umbrella given the weather

Now, if you observe the sequence (Umbrella, Umbrella, No umbrella, Umbrella), what was the weather? This is the fundamental HMM inference question. The forward algorithm developed below computes filtered state probabilities; stochastic traceback later uses them to sample complete weather histories conditional on all observations.

In genetics, replace “sunny/rainy” with “which branch of the genealogical tree the lineage sits on” and “umbrella/no umbrella” with “derived allele / ancestral allele.” The logic is identical; only the vocabulary changes.

i Probability Aside – Conditional Probability

Several probability concepts appear throughout this chapter. Let us establish them now.

Conditional probability is the probability of an event given that another event has occurred. We write $P(A | B)$ and read it as “the probability of A given B.” For example, $P(\text{umbrella} | \text{rain}) = 0.8$ means that if it is raining, there is an 80% chance of seeing an umbrella.

Formally:

$$P(A | B) = \frac{P(A, B)}{P(B)}$$

where $P(A, B)$ is the **joint probability** – the probability that both A and B occur together. Rearranging gives the **product rule**:

$$P(A, B) = P(B) \cdot P(A | B)$$

This generalizes to the **chain rule of probability**. For three events:

$$P(A, B, C) = P(A) \cdot P(B | A) \cdot P(C | A, B)$$

And for a whole sequence:

$$P(X_1, X_2, \dots, X_L) = P(X_1) \cdot P(X_2 | X_1) \cdot P(X_3 | X_1, X_2) \cdots P(X_L | X_1, \dots, X_{L-1})$$

Finally, **marginalizing** (or “summing out”) a variable means computing the probability of an event by summing over all possible values of another variable:

$$P(A) = \sum_b P(A, B = b)$$

If you know the joint probability of A and B for every possible value of B, you can recover the probability of A alone by summing over all B values. This is sometimes called the **law of total probability**, and it appears repeatedly in the forward algorithm.

2.4.3 The Core Idea

A **Hidden Markov Model (HMM)** is a probabilistic model with two layers:

1. A **hidden state sequence** $(Z_1, Z_2, \dots, Z_L)$ that evolves according to a Markov chain – each state depends only on the previous one.
2. An **observed sequence** $(X_1, X_2, \dots, X_L)$ where each observation depends only on the hidden state at that position.

You observe X , but you want to infer Z . This is exactly the situation in ARG inference:

- The **hidden states** are the branches of the genealogical tree at each genomic position
- The **observations** are the alleles (mutations) at each position
- The **transitions** between states represent recombination events

i The Markov Property – Intuition

The word “Markov” refers to a specific kind of forgetfulness. A process is Markov if its future behavior depends only on where it is *right now*, not on how it got there. In the weather example, tomorrow’s weather depends only on today’s weather – it does not matter whether today’s sun came after a week of rain or a month of clear skies.

Formally, if $Z_1, Z_2, \dots$ is a Markov chain, then:

$$P(Z_{\ell+1} \mid Z_{\ell}, Z_{\ell-1}, \dots, Z_1) = P(Z_{\ell+1} \mid Z_{\ell})$$

This is a drastic simplification. Without it, to predict the state at position $\ell + 1$, we would need to consider the entire history of states – an exponentially growing space. With the Markov property, we only need the current state.

For ARGs, the *The Sequentially Markov Coalescent* approximation is precisely what gives us this property: it ensures that the marginal tree at genomic position ℓ depends only on the tree at position $\ell - 1$, making the sequence of trees a Markov chain (see *The Sequentially Markov Coalescent*).

2.4.4 Formal Definition

An HMM is defined by four components:

- **State space** $\mathcal{S} = \{s_1, \dots, s_K\}$ – the K possible hidden states
- **Initial distribution** $\pi_i = P(Z_1 = s_i)$ – probability of starting in state s_i
- **Transition matrix** $A_{ij} = P(Z_{\ell} = s_j \mid Z_{\ell-1} = s_i)$ – probability of moving from state s_i to s_j
- **Emission probabilities** $e_j(x) = P(X_{\ell} = x \mid Z_{\ell} = s_j)$ – probability of observing x when in state s_j

Let us unpack the last two, since they carry all the modeling power.

Transition probabilities describe how the hidden state changes from one position to the next. In the weather example, the transition from Sunny to Rainy has probability 0.10. In genetics, a transition corresponds to a recombination event: the lineage jumps from one branch of the genealogical tree to another. A high transition probability between two states means recombination frequently causes the lineage to switch between those branches. A low transition probability (or a high probability of staying in the same state) means the same tree branch tends to persist across adjacent genomic positions.

Emission probabilities describe what you *observe* given the hidden state. The hidden state “emits” an observation, like a machine dropping a colored ball into a bin. In the weather model, the rainy state emits “umbrella” with probability 0.8. In genetics, the emission probability encodes the chance of seeing a particular allele given that the lineage sits on a specific branch of the tree. Under the infinite-sites mutation model (see *Coalescent Theory*), this depends on the branch length measured in coalescent time units – longer branches accumulate more mutations, so a mutation is more likely to be observed on a long branch than a short one.

i Matrix Notation – What Is a Transition Matrix?

A **matrix** is a rectangular array of numbers. The transition matrix A has K rows and K columns, where K is the number of hidden states. The entry A_{ij} in row i and column j gives the probability of transitioning from state s_i to state s_j .

Each row of A must sum to 1, because from any state, the process must go *somewhere*:

$$\sum_{j=1}^K A_{ij} = 1 \quad \text{for all } i$$

When we later write expressions like $\sum_i \alpha_i \cdot A_{ij}$, this is computing a **weighted sum** over column j of the matrix, where the weights are the α_i values. In matrix notation, this corresponds to a matrix-vector multiplication. The important thing is that it combines information from *all* possible previous states to compute the probability of being in state j next.

Let's implement a basic HMM:

```
import numpy as np

class HMM:
    """A Hidden Markov Model.

    Parameters
    -----
    initial : ndarray of shape (K,)
        Initial state distribution pi.
    transition : ndarray of shape (K, K)
        Transition matrix A[i, j] = P(Z_l = j | Z_{l-1} = i).
    emission : callable
        emission(state, observation) returns P(X = obs | Z = state).
    """
    def __init__(self, initial, transition, emission):
        self.initial = initial          # pi: starting probabilities
        self.transition = transition     # A: K x K transition matrix
        self.emission = emission        # function(state, obs) -> probability
        self.K = len(initial)          # number of hidden states

# Example: a simple 2-state HMM (the weather model)
# States: 0 = Sunny, 1 = Rainy
# Observations: 0 = No umbrella, 1 = Umbrella
hmm = HMM(
    initial=np.array([0.5, 0.5]),      # equal chance of starting sunny or rainy
    transition=np.array([[0.95, 0.05],  # Sunny -> Sunny: 0.95, Sunny -> Rainy: 0.
    0.10, 0.90]]),                    # Rainy -> Sunny: 0.10, Rainy -> Rainy: 0.
    emission=lambda s, x: [0.9, 0.1][x] if s == 0 else [0.2, 0.8][x])
```

2.4.5 The Forward Algorithm

We now arrive at the first major computational tool: the **forward algorithm** [Rabiner, 1989]. It answers a filtering question: “Given observations up through the current position, how likely is each current hidden state?” It does not by itself give the smoothed marginal at an earlier position conditional on future observations.

Think of it this way. Behind the watch face, many different gear configurations could produce the hand positions you observe. The forward algorithm systematically counts up the probability of *every* possible gear configuration that is consistent with what you see, working from left to right along the sequence. At each position, it combines two sources of information: what the current observation tells us about the hidden state (emission), and what the previous position’s state probabilities tell us about where we could have come from (transition).

The forward algorithm computes $P(X_1, \dots, X_\ell, Z_\ell = s_j)$ – the joint probability of the observations up to position ℓ and being in state s_j .

i Probability Aside – Joint Probability

The quantity $P(X_1, \dots, X_\ell, Z_\ell = s_j)$ is a **joint probability**. It answers the question: “What is the probability that the observations are $X_1, \dots, X_\ell$ and *simultaneously* the hidden state at position ℓ is s_j ?”

This is not the same as asking “what is the probability that the state is s_j given the observations?” (which would be a conditional probability). The joint probability is smaller, because it also accounts for how likely the observations themselves are. We will see later that by normalizing, we can convert between the two.

Define the **forward variable**:

$$\alpha_j(\ell) = P(X_1, \dots, X_\ell, Z_\ell = s_j)$$

This is the joint probability of having observed the data $X_1, \dots, X_\ell$ and being in state s_j at position ℓ .

Initialization ($\ell = 1$):

$$\alpha_j(1) = P(X_1, Z_1 = s_j) = P(Z_1 = s_j) \cdot P(X_1 | Z_1 = s_j) = \pi_j \cdot e_j(X_1)$$

i Probability Aside – The Chain Rule at Work

The initialization step uses the **product rule** (a special case of the chain rule). We want the joint probability $P(X_1, Z_1 = s_j)$, which is the probability of two things happening together: the state is s_j and the observation is X_1 .

By the product rule: $P(A, B) = P(A) \cdot P(B | A)$. Here, A is “ $Z_1 = s_j$ ” and B is “ X_1 .” So:

$$P(X_1, Z_1 = s_j) = P(Z_1 = s_j) \cdot P(X_1 | Z_1 = s_j) = \pi_j \cdot e_j(X_1)$$

The first factor is “how likely is this starting state?” and the second is “given this state, how likely is the observation?” Multiplying them gives the joint probability of both.

Recursion ($\ell = 2, \dots, L$):

To derive the recursion, we use the law of total probability to sum over all possible previous states, and then apply the two

key HMM independence assumptions:

$$\begin{aligned}\alpha_j(\ell) &= P(X_1, \dots, X_\ell, Z_\ell = s_j) \\ &= \sum_{i=1}^K P(X_1, \dots, X_\ell, Z_{\ell-1} = s_i, Z_\ell = s_j) \\ &= \sum_{i=1}^K P(X_1, \dots, X_{\ell-1}, Z_{\ell-1} = s_i) \cdot P(Z_\ell = s_j \mid Z_{\ell-1} = s_i) \cdot P(X_\ell \mid Z_\ell = s_j)\end{aligned}$$

Let us walk through each step carefully:

- **Line 1 to Line 2:** We introduced the previous state $Z_{\ell-1}$ and summed over all its possible values. This is marginalizing – we do not know what the previous state was, so we consider all possibilities. (See the Probability Aside on marginalizing above.)
- **Line 2 to Line 3:** We factored the joint probability using the chain rule and the two HMM independence assumptions (detailed below). The key insight is that the big joint probability $P(X_1, \dots, X_\ell, Z_{\ell-1} = s_i, Z_\ell = s_j)$ splits into three manageable pieces.

The third line relies on two HMM properties:

1. **Markov property:** $P(Z_\ell = s_j \mid Z_{\ell-1} = s_i, X_1, \dots, X_{\ell-1}) = P(Z_\ell = s_j \mid Z_{\ell-1} = s_i) = A_{ij}$. The next state depends only on the current state, not on older states or observations.
2. **Conditional independence of emissions:** $P(X_\ell \mid Z_\ell = s_j, Z_{\ell-1}, X_1, \dots, X_{\ell-1}) = P(X_\ell \mid Z_\ell = s_j) = e_j(X_\ell)$. Each observation depends only on the hidden state at that position.

Substituting our definitions:

$$\alpha_j(\ell) = e_j(X_\ell) \sum_{i=1}^K \alpha_i(\ell-1) \cdot A_{ij}$$

Read this equation aloud: “The forward probability of being in state j at position ℓ equals the emission probability of the current observation in state j , multiplied by the sum over all previous states i of (the forward probability of being in state i at the previous position, times the transition probability from i to j).”

The sum $\sum_i \alpha_i(\ell-1) \cdot A_{ij}$ aggregates contributions from every possible predecessor state, weighted by how likely each predecessor is and how likely it is to transition to state j . The emission term $e_j(X_\ell)$ then adjusts for how well state j explains the current observation.

Note the computational cost: for each of the K states at position ℓ , we sum over K states at position $\ell-1$. This is $O(K^2)$ per position and $O(LK^2)$ total.

Likelihood (marginalizing over final state):

$$P(X_1, \dots, X_L) = \sum_{j=1}^K \alpha_j(L)$$

This final step sums over all possible hidden states at the last position. We do not know (or care about) which state the system ended in – we want the total probability of the observations regardless of the final state. This is another application of marginalization.

```
def forward_algorithm(hmm, observations):
    """Run the forward algorithm.

    Parameters
    -----
```

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```

hmm : HMM
observations : list of length L

Returns
-----
alpha : ndarray of shape (L, K)
    Forward probabilities.
"""
L = len(observations)
K = hmm.K
# alpha[ell, j] will hold the forward probability alpha_j(ell)
alpha = np.zeros((L, K))

# Initialization: alpha_j(1) = pi_j * e_j(X_1)
for j in range(K):
    alpha[0, j] = hmm.initial[j] * hmm.emission(j, observations[0])

# Recursion: process each position left-to-right
for ell in range(1, L):
    for j in range(K):
        # hmm.transition[:, j] is the j-th column of A, i.e., A[i,j] for all i.
        # alpha[ell-1, :] is the row of forward probs at the previous position.
        # Multiplying element-wise and summing gives sum_i alpha_i * A_{ij}.
        # np.sum(...) computes this sum over all K previous states.
        alpha[ell, j] = hmm.emission(j, observations[ell]) * \
            np.sum(alpha[ell - 1, :] * hmm.transition[:, j])

    return alpha

# Test with a simple observation sequence
# 0 = No umbrella, 1 = Umbrella
obs = [0, 0, 1, 1, 0, 1, 1, 1, 0, 0]
alpha = forward_algorithm(hmm, obs)

# The total likelihood is the sum over all states at the last position.
# alpha[-1, :] grabs the last row of the alpha table (i.e., position L).
likelihood = np.sum(alpha[-1, :])
print(f"Log-likelihood: {np.log(likelihood):.4f}")

# Normalizing the last row gives P(state | all observations):
# this converts joint probabilities to conditional probabilities.
print(f"Forward probs at last position: {alpha[-1] / alpha[-1].sum()}")

```

2.4.6 Scaling for Numerical Stability

There is a practical problem with the forward algorithm as written above. Each forward variable $\alpha_j(\ell)$ is a *joint* probability of an increasingly long sequence of observations. As the sequence grows, this joint probability shrinks – it is a product of many numbers less than 1. For a genome with millions of positions, these numbers become astronomically small, far smaller than the smallest number a computer can represent (roughly 10^{-308} for standard 64-bit floating-point numbers).

When a number becomes too small to represent, the computer rounds it to zero. This is called **numerical underflow**. Once a forward variable becomes zero, it stays zero – all subsequent computations that depend on it produce zero as well. The entire calculation collapses.

i Why Underflow Happens – A Concrete Example

Suppose you have 1,000 positions and at each position the forward variables are multiplied by emission probabilities around 0.5. The cumulative product is roughly $0.5^{1000} \approx 10^{-301}$. This is barely representable. With 10,000 positions, you get $0.5^{10000} \approx 10^{-3010}$, which is hopelessly beyond the range of floating-point numbers. And real genomes have millions of positions.

The solution: **normalize** at each step. Instead of letting the forward variables shrink to zero, we divide them by their sum at each position, keeping them in a numerically comfortable range. We record the normalization constants separately and use them to recover the total likelihood.

The idea is to compute **scaled forward variables** $\hat{\alpha}_j(\ell)$ that sum to 1 at each position, and keep track of the normalizing constants c_ℓ separately.

At each step ℓ , we first compute the unnormalized values:

$$\tilde{\alpha}_j(\ell) = e_j(X_\ell) \sum_{i=1}^K \hat{\alpha}_i(\ell-1) \cdot A_{ij}$$

Then we normalize by their sum:

$$c_\ell = \sum_{j=1}^K \tilde{\alpha}_j(\ell), \quad \hat{\alpha}_j(\ell) = \frac{\tilde{\alpha}_j(\ell)}{c_\ell}$$

The **scaled forward variables** $\hat{\alpha}_j(\ell)$ represent the **conditional state distribution**:

$$\hat{\alpha}_j(\ell) = P(Z_\ell = s_j \mid X_1, \dots, X_\ell)$$

Note the difference from the unscaled version: $\alpha_j(\ell)$ is a joint probability (of both the observations and the state), while $\hat{\alpha}_j(\ell)$ is a conditional probability (of the state given the observations). The scaled version directly answers the question we usually care about: “given what I have observed so far, how likely is each state?”

Why does this work? The unscaled forward variables can be recovered as:

$$\alpha_j(\ell) = \hat{\alpha}_j(\ell) \prod_{m=1}^{\ell} c_m$$

To see this, note that at each step we divided by c_ℓ , so to recover the original values we multiply back all the c_m . The product of all scaling factors gives the total data likelihood:

$$P(X_1, \dots, X_L) = \sum_j \alpha_j(L) = \left(\prod_{m=1}^L c_m \right) \underbrace{\sum_j \hat{\alpha}_j(L)}_{=1} = \prod_{m=1}^L c_m$$

Taking logarithms:

$$\log P(X_1, \dots, X_L) = \sum_{\ell=1}^L \log c_\ell$$

Each c_ℓ is a sum of a few numbers (manageable), and the log-likelihood is a sum of logs (no underflow). This is numerically stable even for sequences of millions of positions.

i What are the scaling factors, intuitively?

Each $c_\ell = P(X_\ell \mid X_1, \dots, X_{\ell-1})$ is the **predictive probability** of the observation at position ℓ given all previous observations. It answers: “Given everything we have seen so far, how surprising is the current observation?” The total likelihood is the product of these predictive probabilities – this is just the chain rule: $P(X_1, \dots, X_L) = P(X_1) \cdot P(X_2|X_1) \cdot P(X_3|X_1, X_2) \dots$

```
def forward_scaled(hmm, observations):
    """Forward algorithm with scaling for numerical stability.

    Returns
    -----
    alpha_hat : ndarray of shape (L, K)
        Scaled forward probabilities (conditional state distributions).
        alpha_hat[ell, j] = P(Z_ell = j | X_1, ..., X_ell)
    log_likelihood : float
        log P(X_1, ..., X_L), computed as sum of log(c_ell).
    """
    L = len(observations)
    K = hmm.K
    alpha_hat = np.zeros((L, K))
    log_likelihood = 0.0

    # Initialization: same as before, then normalize
    for j in range(K):
        alpha_hat[0, j] = hmm.initial[j] * hmm.emission(j, observations[0])
    c = alpha_hat[0].sum()          # c_1: normalizing constant for position 1
    alpha_hat[0] /= c              # now alpha_hat[0] sums to 1
    log_likelihood += np.log(c)     # accumulate log-likelihood

    # Recursion: compute, normalize, accumulate
    for ell in range(1, L):
        for j in range(K):
            alpha_hat[ell, j] = hmm.emission(j, observations[ell]) * \
                np.sum(alpha_hat[ell - 1, :] * hmm.transition[:, j])
        c = alpha_hat[ell].sum()    # c_ell: normalizing constant
        alpha_hat[ell] /= c        # normalize to sum to 1
        log_likelihood += np.log(c) # accumulate log(c_ell)

    return alpha_hat, log_likelihood

alpha_hat, ll = forward_scaled(hmm, obs)
print(f"Log-likelihood (scaled): {ll:.4f}")
# alpha_hat[-1] is already a proper probability distribution over states:
print(f"P(state | observations) at last position: {alpha_hat[-1]}")
```

2.4.7 Stochastic Traceback (Sampling)

The forward algorithm tells us the probability of each hidden state at each position. But for ARG inference, we need more: we need to produce complete sequences of hidden states that are consistent with the observations. SINGER doesn't just find the single most likely state sequence – it **samples** from the posterior distribution of state sequences. This is done via **stochastic traceback** (also called forward-filtering backward-sampling).

The watchmaking analogy: if the forward algorithm is like cataloging every possible gear configuration that could produce the observed hand positions, then stochastic traceback is like **rewinding the mechanism**. Starting from the final position (where we know the state probabilities), we trace backwards through the sequence, randomly selecting states according to their posterior probability. Each run of the traceback produces a different plausible gear configuration – a different possible history that is consistent with the observations.

Why sample rather than just pick the most likely sequence? Because in Bayesian inference, a single “best” answer can be misleading. By generating many samples, SINGER captures the *uncertainty* in the genealogical reconstruction. Some parts of the genome may have a clear signal (all samples agree), while others are ambiguous (samples disagree). This uncertainty is itself informative.

After running the forward algorithm, we sample states from right to left:

1. Sample Z_L from $P(Z_L \mid X_1, \dots, X_L) \propto \alpha_{Z_L}(L)$

(At the last position, the scaled forward variables already give us the conditional distribution over states.)

2. For $\ell = L - 1, \dots, 1$, sample Z_ℓ from:

$$P(Z_\ell = s_i \mid Z_{\ell+1} = s_j, X_1, \dots, X_\ell) \propto \hat{\alpha}_i(\ell) \cdot A_{ij}$$

This formula has an elegant interpretation: the probability of state i at position ℓ is proportional to two factors:

- $\hat{\alpha}_i(\ell)$: how likely state i is given the observations up to position ℓ (the “forward” information)
- A_{ij} : how likely state i is to transition to the state j that we already sampled at position $\ell + 1$ (the “backward” constraint)

The product of these two factors balances the evidence from the left (what the data says) with the constraint from the right (what the next state requires).

```
def stochastic_traceback(hmm, alpha_hat):
    """Sample a state sequence from the posterior using forward probs.

    This implements "forward-filtering backward-sampling": the forward
    algorithm filters information left-to-right, and this function
    samples a path right-to-left.

    Parameters
    -----
    hmm : HMM
    alpha_hat : ndarray of shape (L, K)
        Scaled forward probabilities from forward_scaled().

    Returns
    -----
    states : ndarray of shape (L,)
        Sampled state sequence.
    """
    L, K = alpha_hat.shape
    states = np.zeros(L, dtype=int) # will hold the sampled state at each position

    # Step 1: Sample the last state from the conditional distribution.
    # alpha_hat[-1] is already normalized (sums to 1) from the scaled forward pass.
    probs = alpha_hat[-1] / alpha_hat[-1].sum()
    # np.random.choice(K, p=probs) draws one integer from {0, ..., K-1}
    # with probability given by 'probs'.
    states[-1] = np.random.choice(K, p=probs)
```

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```

# Step 2: Traceback from right to left.
# range(L-2, -1, -1) counts down: L-2, L-3, ..., 1, 0
for ell in range(L - 2, -1, -1):
    j = states[ell + 1] # the state we already sampled at position ell+1
    # P(Z_ell = i | Z_{ell+1} = j, data) is proportional to
    # alpha_hat[ell, i] * A[i, j]
    # alpha_hat[ell] is a length-K array of scaled forward probs.
    # hmm.transition[:, j] is column j of A, i.e., A[i, j] for all i.
    # Element-wise multiplication gives the unnormalized probabilities.
    probs = alpha_hat[ell] * hmm.transition[:, j]
    probs /= probs.sum() # normalize so probabilities sum to 1
    states[ell] = np.random.choice(K, p=probs)

return states

np.random.seed(42)
alpha_hat, _ = forward_scaled(hmm, obs)
# Sample 5 state sequences from the posterior.
# Each sample is a different plausible hidden state sequence.
for i in range(5):
    states = stochastic_traceback(hmm, alpha_hat)
    print(f"Sample {i+1}: {states}")

```

2.4.8 The Li-Stephens Trick: Linear-Time Transitions

We have now seen the full HMM inference pipeline: the forward algorithm computes state probabilities, scaling prevents underflow, and stochastic traceback generates samples. There is one remaining problem: **speed**.

A standard HMM forward step is $O(K^2)$ because every state can transition to every other state – the sum $\sum_i \alpha_i \cdot A_{ij}$ has K terms and must be computed for each of K states. For SINGER, K is the number of branches in the marginal tree ($2n - 1$ for n samples), so this is expensive. With $n = 1000$ samples, we would have $K \approx 2000$ states and $K^2 = 4,000,000$ operations per genomic position. Multiply by millions of positions and the computation becomes intractable.

The original **Li-Stephens copying model** [Li and Stephens, 2003] exploits a special stay-or-jump structure in the transition matrix that reduces the per-position cost from $O(K^2)$ to $O(K)$. The state-dependent form below is a useful generalization of that structure and matches the factorization used later for SINGER; it is not a verbatim statement of the original copying model.

The Li-Stephens Transition Structure

The transition probability has the form:

$$A_{ij} = (1 - r_i)\delta_{ij} + r_i \cdot \frac{q_j}{\sum_k q_k}$$

Let us unpack every symbol:

What Is the Kronecker Delta?

The symbol δ_{ij} is the **Kronecker delta**. It is the simplest possible function of two indices:

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

It acts as a mathematical “filter”: in any sum involving δ_{ij} , only the term where $i = j$ survives. For example:

$$\sum_{i=1}^K f(i) \cdot \delta_{ij} = f(j)$$

All terms where $i \neq j$ are multiplied by zero and vanish. This property is what makes the Li-Stephens trick work.

The transition formula says: with probability $1 - r_i$, **stay in the same state** (no recombination). With probability r_i , **“recombine”** and jump to any state j with probability proportional to q_j .

In the genetics context:

- r_i is the probability that a recombination event occurs while the lineage sits on branch i . This depends on the branch length in coalescent time units (see [Coalescent Theory](#)) and the recombination rate.
- q_j is the probability that, after recombination, the lineage re-joins branch j . This is proportional to the coalescence probability on branch j .
- δ_{ij} ensures that “staying” means remaining on the *same* branch.

The key structural property is that the “jump” part of the transition is **separable**: the probability of jumping *to* state j (which is $q_j / \sum_k q_k$) does not depend on which state i we are jumping *from*. This separability is what enables the $O(K)$ trick.

A concrete small example. Consider $K = 3$ states with $r_i = 0.1$ for all i and $q = (0.5, 0.3, 0.2)$. Then $\sum_k q_k = 1.0$ and the transition matrix is:

$$A = \begin{pmatrix} 0.9 + 0.1 \cdot 0.5 & 0.1 \cdot 0.3 & 0.1 \cdot 0.2 \\ 0.1 \cdot 0.5 & 0.9 + 0.1 \cdot 0.3 & 0.1 \cdot 0.2 \\ 0.1 \cdot 0.5 & 0.1 \cdot 0.3 & 0.9 + 0.1 \cdot 0.2 \end{pmatrix} = \begin{pmatrix} 0.95 & 0.03 & 0.02 \\ 0.05 & 0.93 & 0.02 \\ 0.05 & 0.03 & 0.92 \end{pmatrix}$$

Notice the structure: each row has a large diagonal entry (staying) and small off-diagonal entries (jumping). In this example they are identical within each column because all r_i are equal. With state-dependent r_i , the jump term remains separable as $r_i q_j / \sum_k q_k$, but those entries need not be identical across rows.

Proof that rows sum to 1. For any row i :

$$\begin{aligned} \sum_j A_{ij} &= \sum_j \left[(1 - r_i) \delta_{ij} + r_i \frac{q_j}{\sum_k q_k} \right] \\ &= (1 - r_i) \underbrace{\sum_j \delta_{ij}}_{=1} + r_i \underbrace{\frac{\sum_j q_j}{\sum_k q_k}}_{=1} \\ &= (1 - r_i) + r_i = 1 \quad \checkmark \end{aligned}$$

The $O(K)$ Forward Step

Now we substitute the Li-Stephens transition structure into the forward recursion and see how the sum simplifies. This is the algebraic heart of the trick, so we will go step by step.

Start with the standard forward recursion:

$$\alpha_j(\ell) = e_j(X_\ell) \sum_{i=1}^K \alpha_i(\ell - 1) \cdot A_{ij}$$

Substitute the Li-Stephens form of A_{ij} :

$$\alpha_j(\ell) = e_j(X_\ell) \sum_i \alpha_i(\ell - 1) \left[(1 - r_i) \delta_{ij} + r_i \frac{q_j}{\sum_k q_k} \right]$$

Now distribute the sum over the two terms inside the brackets:

$$\alpha_j(\ell) = e_j(X_\ell) \left[\sum_i \alpha_i(\ell-1)(1-r_i)\delta_{ij} + \sum_i \alpha_i(\ell-1)r_i \frac{q_j}{\sum_k q_k} \right]$$

Simplify the first sum. The Kronecker delta δ_{ij} kills every term except $i = j$:

$$\sum_i \alpha_i(\ell-1)(1-r_i)\delta_{ij} = \alpha_j(\ell-1)(1-r_j)$$

Only the “stay in state j ” term survives.

Simplify the second sum. The factor $q_j / \sum_k q_k$ does not depend on i , so it can be pulled out of the sum:

$$\sum_i \alpha_i(\ell-1)r_i \frac{q_j}{\sum_k q_k} = \frac{q_j}{\sum_k q_k} \sum_i r_i \alpha_i(\ell-1)$$

This is the crucial step: the sum $\sum_i r_i \alpha_i(\ell-1)$ does not depend on j at all. It is the same number regardless of which target state we are computing.

Putting it together:

$$\alpha_j(\ell) = e_j(X_\ell) \left[\alpha_j(\ell-1)(1-r_j) + \frac{q_j}{\sum_k q_k} \underbrace{\sum_i r_i \alpha_i(\ell-1)}_R \right]$$

The key: the sum $R = \sum_i r_i \alpha_i(\ell-1)$ is computed **once** in $O(K)$ time and reused for all j . Each α_j then requires only $O(1)$ work (one multiplication for the “stay” term, one for the “jump” term, one addition, and one multiplication by the emission), giving $O(K)$ total per position instead of $O(K^2)$.

i Why the Sum Factorizes – The Key Insight

The reason this trick works is that the Li-Stephens transition separates into two parts: a “stay” part that depends on both source and target, and a “jump” part where the source and target contributions *multiply independently*.

In the “jump” component, r_i (from the source state) and $q_j / \sum_k q_k$ (from the target state) do not interact – they are multiplied, not entangled. This means the double sum $\sum_i \sum_j$ can be split into $(\sum_i \dots)(\sum_j \dots)$ or equivalently, the inner sum $\sum_i r_i \alpha_i$ can be precomputed once.

If the transition probability had a more complicated dependence on *both* i and j , this factorization would not be possible, and we would be stuck with $O(K^2)$.

A small worked example. Suppose $K = 3$, $r = (0.1, 0.1, 0.1)$, $q = (0.5, 0.3, 0.2)$, and the forward variables at the previous position are $\alpha(\ell-1) = (0.4, 0.35, 0.25)$. Emissions at the current position are $e = (0.8, 0.6, 0.9)$.

Step 1: Compute the recombination sum once.

$$R = \sum_i r_i \alpha_i(\ell-1) = 0.1 \times 0.4 + 0.1 \times 0.35 + 0.1 \times 0.25 = 0.1$$

Step 2: Compute each $\alpha_j(\ell)$.

$$\begin{aligned} \alpha_1(\ell) &= 0.8 \times [0.4 \times 0.9 + 0.5 \times 0.1] = 0.8 \times [0.36 + 0.05] = 0.328 \\ \alpha_2(\ell) &= 0.6 \times [0.35 \times 0.9 + 0.3 \times 0.1] = 0.6 \times [0.315 + 0.03] = 0.207 \\ \alpha_3(\ell) &= 0.9 \times [0.25 \times 0.9 + 0.2 \times 0.1] = 0.9 \times [0.225 + 0.02] = 0.2205 \end{aligned}$$

Each α_j used only its own previous value (for the “stay” part) and the shared R (for the “jump” part). No double sum over states was needed.

```

def forward_li_stephens(initial, r, q, emissions):
    """Unscaled forward demo with a Li--Stephens-type transition.

    The Li-Stephens transition matrix has the form:
         $A[i,j] = (1 - r[i]) * \delta(i,j) + r[i] * q[j] / \sum(q)$ 
    This enables an  $O(K)$  forward step instead of  $O(K^2)$ .

    Parameters
    -----
    initial : ndarray of shape (K,)
        Initial state distribution.
    r : ndarray of shape (K,)
        Per-state recombination probabilities.  $r[i]$  is the probability
        of a recombination event on branch  $i$ .
    q : ndarray of shape (K,)
        Re-joining weights.  $q[j] / \sum(q)$  is the probability of
        re-joining branch  $j$  after recombination.
    emissions : ndarray of shape (L, K)
        Pre-computed emission probabilities:  $\text{emissions}[\text{ell}, j] = P(X_{\text{ell}} \mid Z_{\text{ell}} = j)$ .

    Returns
    -----
    alpha : ndarray of shape (L, K)
        Forward probabilities at each position.
    """
    L, K = emissions.shape
    alpha = np.zeros((L, K))

    # Initialization:  $\alpha_j(1) = \pi_j * e_j(X_1)$ 
    alpha[0] = initial * emissions[0]

    # Pre-compute the sum of q for normalization (constant across positions)
    q_sum = q.sum()

    for ell in range(1, L):
        #  $O(K)$  step: compute the recombination sum  $R = \sum_i r[i] * \alpha[i]$ 
        #  $\text{np.sum}(r * \alpha[\text{ell}-1])$  multiplies  $r$  and  $\alpha$  element-wise,
        # then sums all  $K$  elements. This is done ONCE per position.
        recomb_sum = np.sum(r * alpha[ell - 1])

        for j in range(K):
            # "Stay" term: probability of no recombination on branch  $j$ ,
            # times the previous forward probability on the same branch  $j$ .
            stay = (1 - r[j]) * alpha[ell - 1, j]

            # "Jump" term: probability of recombining and re-joining branch  $j$ .
            # Uses the pre-computed recomb_sum (shared across all  $j$ ).
            switch = (q[j] / q_sum) * recomb_sum

            # Multiply by emission probability for state  $j$  at position  $\text{ell}$ .
            alpha[ell, j] = emissions[ell, j] * (stay + switch)

    return alpha

```

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```
# Test: 10 states, 100 positions
K, L = 10, 100
r = np.full(K, 0.05) # uniform recombination rate
q = np.random.dirichlet(np.ones(K)) # random re-joining weights
emissions = np.random.uniform(0.1, 0.9, size=(L, K)) # random emissions

alpha = forward_li_stephens(
    initial=np.ones(K) / K, # uniform initial distribution
    r=r, q=q, emissions=emissions
)
print(f"Forward probs shape: {alpha.shape}")
print(f"Sum at last position: {alpha[-1].sum() :.6e}")
```

This final function isolates the linear-time algebra and is deliberately unscaled. Production code should combine the same transition factorization with the per-position scaling developed above.

Why this matters for SINGER

In SINGER's branch sampling HMM, the hidden states are branches in the marginal tree. The transition probability has exactly this Li-Stephens structure: with probability $1 - r_i$, stay on the same branch (no recombination); with probability r_i , recombine and re-join a branch with probability proportional to the branch's coalescence probability. The coalescence probabilities play the role of the q_j weights. Branch lengths are measured in coalescent time units (see [Coalescent Theory](#)), and the recombination probabilities r_i depend on both the recombination rate and the branch length.

Without the Li-Stephens trick, SINGER's forward algorithm would be $O(K^2)$ per position – infeasible for large sample sizes. With it, the cost is $O(K)$, making genome-scale inference practical.

2.4.9 Summary

Algorithm	What it gives you
Forward algorithm	$\alpha_j(\ell) = P(X_1, \dots, X_\ell, Z_\ell = j)$
Scaled forward	Numerically stable version + log-likelihood
Stochastic traceback	Posterior samples of state sequences
Li-Stephens trick	$O(K)$ transitions instead of $O(K^2)$

These are the **gear train** of SINGER. The forward algorithm is the mechanism that combines observations and transitions into state probabilities, position by position. Scaling keeps the gears turning without jamming (numerical underflow). Stochastic traceback runs the mechanism in reverse, producing complete hidden state sequences. And the Li-Stephens structure is the elegant engineering that makes it all fast enough to work on real genomes.

Together, the forward algorithm and stochastic traceback form the core of how SINGER samples branches and coalescence times along the genome. The Li-Stephens transition structure is what makes this sampling computationally feasible for large datasets.

Next: *The Sequentially Markov Coalescent* – the Markov approximation that makes ARG inference possible.

2.5 The Sequentially Markov Coalescent

To make the impossible tractable, we give up a little exactness.

In the *Coalescent Theory* chapter, we learned how lineages coalesce backwards in time. In the *ARG chapter*, we saw how recombination creates a *sequence* of marginal trees along the genome. And in the *HMM chapter*, we set up the machinery to do inference on hidden state sequences – provided those states form a Markov chain.

There is a problem. The sequence of marginal trees produced by the full coalescent with recombination (CwR) is **not** Markov. This chapter explains why, introduces the Sequentially Markov Coalescent (SMC) approximation that fixes the problem, and derives the transition densities that will power our Timepieces.

The Watch Metaphor

The SMC approximation is like simplifying a complex mechanism by ignoring gear interactions that almost never occur. A master watchmaker knows that two gears deep inside the movement *could* brush against each other under extreme conditions, but for all practical purposes they never do. By designing the movement as if those interactions cannot happen, the watch becomes far simpler to build and maintain – and it still keeps nearly perfect time.

2.5.1 The Problem with the Full Coalescent

The coalescent with recombination (CwR) is the standard large-population ancestral model for a recombining chromosome (see *Ancestral Recombination Graphs*). It is itself a limit of idealized population models, not literal biology. Its marginal-tree process along the genome is **not Markov**.

In the CwR, the marginal tree at position x depends on the trees at *all* previous positions, not just the immediately preceding one. This means we cannot directly plug the tree sequence into a Hidden Markov Model – and without HMMs, efficient inference is out of reach.

The **Sequentially Markov Coalescent (SMC)** was introduced by McVean and Cardin [McVean and Cardin, 2005]. Marjoram and Wall subsequently described the related SMC' modification [Marjoram and Wall, 2006]. These approximations restore a Markov description along the genome by restricting ancestral events. The rest of this chapter develops the basic SMC idea step by step.

2.5.2 What Does “Markov” Mean, and Why Does It Matter?

Before diving into the technical details, let us build intuition for what the Markov property is and why computational methods depend on it.

Intuitive Explanation

Imagine you are watching a clock. At each tick, the configuration of gears determines exactly what happens at the next tick. You do not need to know the entire history of every tick since the clock was wound – the current gear positions are sufficient. That is the Markov property: **the future depends on the past only through the present**.

The Watch Metaphor

The Markov property is like a mechanism where the next tick depends only on the current state of the gears, not on how they got there. A watchmaker can predict what will happen next by examining the movement right now, without consulting a logbook of every previous tick.

In the context of genealogical trees along a genome: if the tree sequence were Markov, knowing the tree at position x would be all we need to compute the probability of the tree at position $x + 1$. We would not need to remember the trees at positions $1, 2, \dots, x - 1$.

Formal Definition

A sequence of random variables $(Z_1, Z_2, \dots)$ is **Markov** if for every ℓ :

$$P(Z_{\ell+1} \mid Z_\ell, Z_{\ell-1}, \dots, Z_1) = P(Z_{\ell+1} \mid Z_\ell)$$

i Probability Aside – Conditional Probability Notation

The expression $P(A \mid B)$ is read “the probability of A given B ”. It means: if we know B has happened, what is the probability that A also happens? The Markov property says that conditioning on the *entire history* $(Z_1, \dots, Z_\ell)$ gives the same result as conditioning on just the most recent value Z_ℓ . In other words, the history provides no additional predictive power beyond what the present state already tells us.

Why Markov Matters for Computation

The Markov property is not just a mathematical nicety – it is what enables the standard HMM dynamic program. Recall from the [HMM chapter](#) that the forward algorithm computes the likelihood of an observed sequence in $O(LK^2)$ time, where L is the sequence length and K is the number of hidden states. This efficiency relies entirely on the Markov property: at each step, the forward variable $\alpha_\ell(j) = P(X_1, \dots, X_\ell, Z_\ell = j)$ can be updated using only $\alpha_{\ell-1}$, not the full history.

Without the Markov property, we would need to track all possible histories $(Z_1, \dots, Z_\ell)$ to compute the probability of $Z_{\ell+1}$. The number of such histories grows exponentially with ℓ , making exact computation infeasible for genomes with millions of positions.

2.5.3 What Makes CwR Non-Markov?

If the marginal trees along the genome were Markov, we could use HMMs. But in the full CwR, they are **not**. The culprit is a phenomenon called **ghost lineages**.

The Mechanism

When a recombination happens in the CwR:

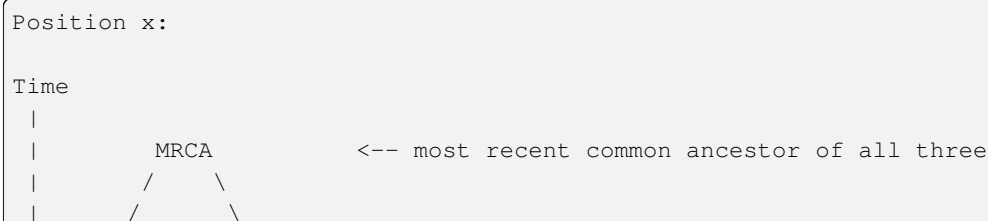
1. A lineage splits at a breakpoint
2. The piece to the right of the breakpoint floats up and can coalesce with *any* lineage in the full ancestral process, including ones that are **not** in the current marginal tree

This means a lineage from the “past” – one that already coalesced and left the current tree – can reappear. These returning lineages are called **ghost lineages**.

What Are Ghost Lineages? A Concrete Example

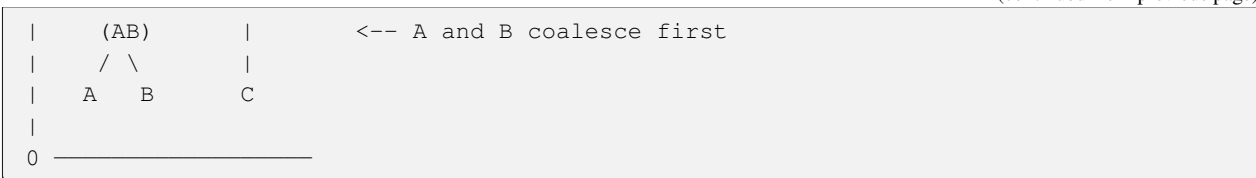
To understand ghost lineages, consider a concrete example with three samples A , B , and C .

Suppose the marginal tree at genomic position x has this shape:



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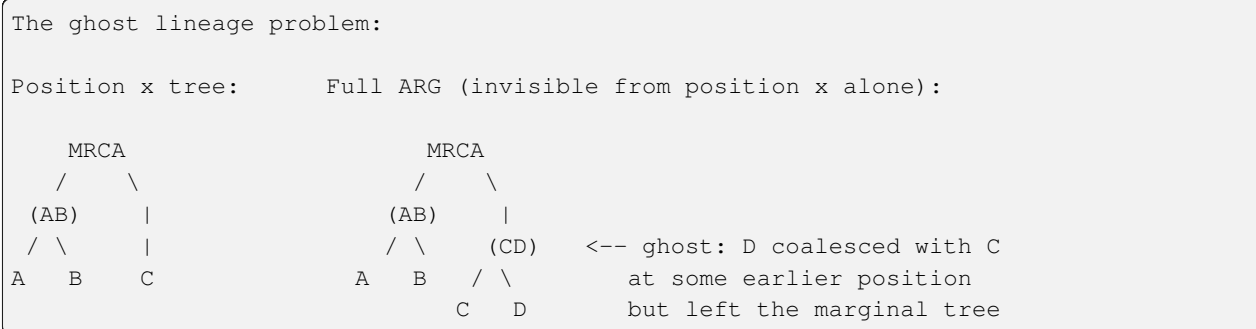


In this tree, the internal node where A and B coalesce (call it node AB) sends a single lineage upward. That lineage eventually coalesces with C 's lineage at the MRCA.

Now suppose a recombination occurs between positions x and $x + 1$. The recombination cuts A 's lineage at some time below AB . The piece of A 's ancestry to the right of the breakpoint detaches and floats upward, looking for a lineage to coalesce with.

In the **full CwR**, this detached lineage can coalesce with *any* lineage that exists at that time in the full ancestral recombination graph – including lineages that are not visible in the marginal tree at position x . For instance, perhaps at an earlier position there was a lineage D that coalesced with C 's lineage but then disappeared from the marginal tree. Lineage D is a **ghost lineage**: it is not part of the current tree, but it still exists in the full ARG and could reappear.

Why this breaks the Markov property: Knowing the current tree Ψ_x is not enough to predict which ghost lineages might return. You would need to know the full history of trees $\Psi_1, \dots, \Psi_{x-1}$ to track all potentially returning lineages.



2.5.4 The SMC Approximation

The SMC makes one simple restriction: after a recombination, the detached lineage can only re-coalesce with branches that are **present in the current marginal tree**. Ghost lineages are forbidden.

CwR: re-coalesce with any ancestral lineage (including "ghost" lineages)

SMC: re-coalesce only with branches in the current marginal tree

Why Does This Restore the Markov Property?

Under the SMC, the next tree Ψ_{x+1} is determined entirely by:

1. The current tree Ψ_x (which branches exist)
2. Whether a recombination occurs between x and $x + 1$
3. If so, which branch is cut and which branch is re-joined

All three of these depend only on Ψ_x , not on any earlier tree. No ghost lineages can appear. Therefore $P(\Psi_{x+1} | \Psi_x, \Psi_{x-1}, \dots) = P(\Psi_{x+1} | \Psi_x)$.

The Markov property is restored, and we can build an HMM.

How Good Is the Approximation?

The events excluded by SMC require ancestral information absent from the current marginal tree. Their importance depends on recombination distance, sample size, and the statistic being studied; they are not uniformly negligible. SMC permits additional re-coalescences, including events that can leave the marginal tree unchanged, and was introduced to improve agreement with the CwR [Marjoram and Wall, 2006].

Think of it this way: the approximation removes paths through the ARG that are difficult to summarize by the current tree. The gain is a tractable Markov model; the accuracy cost must be assessed for the application.

```
import numpy as np

def sample_smc_pruning_point(tree_branches):
    """Sample the pruning point used to begin an SMC transition.

    This implements only the exact first step: choose a point uniformly on
    the current tree's total branch length. It deliberately does not pretend
    to implement the time-dependent re-coalescence and topology update.

    Parameters
    -----
    tree_branches : list of (child, parent, lower_time, upper_time)
        Each tuple represents one branch of the marginal tree.
        - child: node index of the child end of the branch
        - parent: node index of the parent end of the branch
        - lower_time: time at the child (bottom of branch)
        - upper_time: time at the parent (top of branch)

    Returns
    -----
    branch, time
        The selected branch tuple and pruning time.
    """
    # Compute the total branch length of the tree.
    # The expression (u - l) computes the length of one branch;
    # the generator iterates over all branches, and sum() adds them up.
    # The syntax "_", "_", l, u" is called tuple unpacking: it extracts
    # the third and fourth elements (lower_time, upper_time) from each
    # 4-tuple, ignoring the first two (child, parent) with underscores.
    branch_lengths = np.array([u - l for _, _, l, u in tree_branches])
    if len(branch_lengths) == 0 or np.any(branch_lengths <= 0):
        raise ValueError("tree must contain positive-length branches")
    total_length = branch_lengths.sum()

    # Probability of recombination on each branch (proportional to length).
    # This is a list comprehension: it creates a new list by computing
    # (u - l) for each branch tuple in tree_branches.
    # Normalize to get probabilities.
    # Longer branches are more likely to be hit by recombination because
    # they represent more "exposure time" to the recombination process.
    probs = branch_lengths / total_length

    # Randomly pick which branch the recombination falls on,
    # weighted by branch length.
    idx = np.random.choice(len(tree_branches), p=probs)
```

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```

# Unpack the chosen branch into its four components.
# This is tuple unpacking again: the single element
# tree_branches[idx] is a 4-tuple, and we assign each element
# to a separate variable.
_, _, lower, upper = tree_branches[idx]

# Pick a recombination time uniformly on the chosen branch.
# The recombination can happen anywhere between the bottom (lower)
# and top (upper) of this branch.
pruning_time = np.random.uniform(lower, upper)
return tree_branches[idx], pruning_time

# Example tree: ((0,1):4, (2,3):5, (4,5):6)
# This is a tree with 4 leaf nodes (0, 1, 2, 3) and 2 internal nodes
# (4, 5), plus a root (6).
# Times: leaves at 0.0; node 4 at 0.3; node 5 at 0.7; root 6 at 1.5
tree_branches = [
    (0, 4, 0.0, 0.3), # leaf 0 to internal node 4
    (1, 4, 0.0, 0.3), # leaf 1 to internal node 4
    (2, 5, 0.0, 0.7), # leaf 2 to internal node 5
    (3, 5, 0.0, 0.7), # leaf 3 to internal node 5
    (4, 6, 0.3, 1.5), # internal node 4 to root 6
    (5, 6, 0.7, 1.5), # internal node 5 to root 6
]

np.random.seed(42)
branch, pruning_time = sample_smc_pruning_point(tree_branches)
print(f"Pruned branch: {branch}")
print(f"Pruning time: {pruning_time:.4f}")

```

The next SMC step evolves the floating lineage backward with a hazard equal to the number of eligible lineages and updates the topology at re-coalescence. A uniform draw from currently visible branches is not that distribution. Exact simulation should use a tested implementation such as `msprime` with its `StandardCoalescent/SMC` model options [Kelleher *et al.*, 2016].

Now that we have established the SMC approximation and its key consequence – the Markov property – we can derive the transition probabilities that an HMM needs.

2.5.5 The SINGER Branch-HMM Transition

SINGER uses an SMC-motivated approximation when threading a new lineage onto a partial ARG. Its branch-sampling HMM assigns the transition [Deng *et al.*, 2025]:

$$P(B_\ell = b_j \mid B_{\ell-1} = b_i) = (1 - r_i)\delta_{ij} + r_i \frac{q_j}{\sum_k q_k}$$

where:

- $r_i = 1 - \exp(-\frac{\rho}{2}\tau_i)$ is the probability of recombination on branch b_i , with τ_i being the representative time for that branch
- $q_j = r_j \cdot p_j$ weights the re-joining probability
- δ_{ij} is the Kronecker delta (1 if $i = j$, 0 otherwise)

This is not the general SMC transition kernel on whole marginal trees. It is a branch-state kernel used inside SINGER. It has the classic **stay-or-switch** structure: with probability $1 - r_i$, nothing happens and we stay on the same branch; with probability r_i , a recombination occurs and we jump to a new branch chosen according to the weights q_j . This is the generalized Li–Stephens-type structure from the *HMM chapter*.

Deriving r_i : The Recombination Probability

Where does r_i come from? A lineage at representative time τ_i has been evolving for τ_i coalescent time units. In each infinitesimal interval dt , a recombination occurs with probability $(\rho/2)dt$. The probability of *no* recombination over the entire interval $[0, \tau_i]$ is $e^{-\rho\tau_i/2}$ (survival probability of a Poisson process). So the probability of *at least one* recombination is $r_i = 1 - e^{-\rho\tau_i/2}$.

Probability Aside – Survival Probability of a Poisson Process

If events happen at a constant rate λ per unit time, the probability of *no* event in an interval of length t is $e^{-\lambda t}$. This is the survival function of the exponential distribution. Here, the “event” is recombination, the rate is $\rho/2$, and the interval is τ_i , giving survival probability $e^{-\rho\tau_i/2}$.

Deriving q_j : The Re-joining Weights

Why $q_j = r_j \cdot p_j$? This choice ensures the **stationary distribution** of the Markov chain is $\pi_i = p_i$. To verify, a stationary distribution satisfies $\pi_j = \sum_i \pi_i A_{ij}$:

$$\begin{aligned} \sum_i p_i A_{ij} &= \sum_i p_i \left[(1 - r_i) \delta_{ij} + r_i \frac{q_j}{\sum_k q_k} \right] \\ &= p_j (1 - r_j) + \frac{q_j}{\sum_k q_k} \sum_i p_i r_i \\ &= p_j (1 - r_j) + \frac{r_j p_j}{\sum_k r_k p_k} \sum_i r_i p_i \\ &= p_j (1 - r_j) + r_j p_j = p_j \quad \checkmark \end{aligned}$$

Probability Aside – Stationary Distributions

A **stationary distribution** π of a Markov chain with transition matrix A is a probability vector such that $\pi A = \pi$. If the chain starts in distribution π , it stays in distribution π forever. Here, choosing $q_j = r_j p_j$ makes the supplied vector p stationary by construction. Whether p accurately represents the desired branch-joining distribution is a separate modeling question.

The product $r_j p_j$ should therefore be read as SINGER’s target weight, not as a universal physical re-coalescence probability for every SMC model.

```
def smc_branch_transition(tau, p, rho, n_branches):
    """Compute SINGER's stay-or-switch transition between branch states.

    This implements the stay-or-switch transition matrix:
        A[i,j] = (1 - r_i) * delta(i,j) + r_i * q_j / sum(q)
    where r_i is the recombination probability on branch i and
        q_j = r_j * p_j is the re-joining weight for branch j.

    Parameters
```

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```

-----
tau : ndarray of shape (K,)
    Representative joining time for each branch.
p : ndarray of shape (K,)
    Coalescence probability for each branch.
rho : float
    Effective population-scaled recombination parameter for this bin.
n_branches : int
    Number of branches (K).

Returns
-----
T : ndarray of shape (K, K)
    Transition matrix where  $T[i, j] = P(\text{next branch} = j \mid \text{current} = i)$ .
"""
tau = np.asarray(tau, dtype=float)
p = np.asarray(p, dtype=float)
K = n_branches
if tau.shape != (K,) or p.shape != (K,):
    raise ValueError("tau and p must both have length n_branches")
if np.any(tau < 0) or rho < 0 or np.any(p < 0) or not np.isclose(p.sum(), 1):
    raise ValueError("require tau >= 0, rho >= 0, and normalized p >= 0")
if rho == 0:
    return np.eye(K)

# Recombination probability for each branch:
# r_i = 1 - exp(-rho/2 * tau_i)
# This is 1 minus the "survival probability" (no recombination).
r = 1 - np.exp(-rho / 2 * tau)

# Re-joining weights: q_j = r_j * p_j
# These ensure the stationary distribution is p_i.
q = r * p

# Sum of all re-joining weights (used to normalize).
q_sum = q.sum()
if q_sum <= 0:
    raise ValueError("at least one positive-probability branch must have tau > 0")

# Build the K x K transition matrix.
# T[i, j] is the probability of transitioning from branch i to branch j.
T = np.zeros((K, K))
for i in range(K):          # iterate over source branches
    for j in range(K):      # iterate over destination branches
        if i == j:
            # Stay on the same branch: no recombination + recombination
            # that happens to land back on the same branch.
            T[i, j] = (1 - r[i]) + r[i] * q[j] / q_sum
        else:
            # Switch to a different branch: requires recombination.
            T[i, j] = r[i] * q[j] / q_sum

```

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```

# Verify rows sum to 1 (each row is a probability distribution).
if not np.allclose(T.sum(axis=1), 1.0):
    raise RuntimeError("transition rows do not sum to one")
return T

# Example: 5 branches with different representative times and
# coalescence probabilities.
K = 5
tau = np.array([0.1, 0.3, 0.5, 0.8, 1.2]) # representative times
p = np.array([0.3, 0.25, 0.2, 0.15, 0.1]) # coalescence probabilities
rho = 0.5 # recombination rate

T = smc_branch_transition(tau, p, rho, K)
print("Transition matrix:")
print(np.round(T, 4))
print(f"\nRow sums: {T.sum(axis=1)}")

```

With this SINGER-specific branch transition in hand, we now turn to a distinct pairwise SMC construction in which the hidden state is a coalescence time.

2.5.6 PSMC: The Pairwise Case

The **Pairwise Sequentially Markov Coalescent (PSMC)** (Li and Durbin, 2011) is the simplest application of the SMC: it uses just two sequences.

The Watch Metaphor

The PSMC transition is the simplest timepiece movement: just two hands (lineages) whose relative position changes at recombination breakpoints. With only two hands, the “tree” at each position is trivially described by a single number – the coalescence time – making the movement elegant in its simplicity.

With two sequences, the marginal tree at each position is trivial – just a single coalescence time T . There is one branch going up from each sample to the coalescence point, and that is it. As you move along the genome, T changes at recombination breakpoints.

The hidden state is the coalescence time T , which is now a continuous variable rather than a discrete branch index. The transition density $q_\rho(t | s)$ gives the probability density of the new coalescence time t given that the previous coalescence time was s .

The following is the one-effective-recombination kernel used in this introductory PSMC construction. It is a mixed distribution with a point mass and a continuous component; the finite-bin kernel is an approximation to the full along-genome process [Li and Durbin, 2011].

Step 1: Does a recombination happen?

The two lineages form a branch of total length $2s$ (two lineages, each of length s , but since we measure the rate $\rho/2$ per lineage the effective rate along both lineages together is ρs). For the interval between adjacent observations, the probability of at least one pruning event is:

$$P(\text{recombination}) = 1 - e^{-\rho s}$$

i Calculus Aside – Why ρs and Not $\rho \cdot 2s$?

There are two lineages, each of length s , and each experiences recombination at rate $\rho/2$. The total rate is $2 \times (\rho/2) \times s = \rho s$. Different references use different conventions for whether ρ already includes the factor of 2 or not. Here we follow the convention where the total rate on the pair is ρs .

With probability $e^{-\rho s}$, no recombination occurs and the coalescence time stays at s . This is a **point mass** at $t = s$:

$$q_\rho(t | s) = e^{-\rho s} \quad \text{at } t = s$$

i Probability Aside – What Is a Point Mass (Dirac Delta)?

When we say the distribution has a “point mass” at $t = s$, we mean that there is a nonzero probability $e^{-\rho s}$ concentrated at the single point $t = s$. This is different from a continuous density, which assigns zero probability to any individual point and only assigns nonzero probability to intervals.

Mathematically, a point mass is often written using the **Dirac delta function** $\delta(t - s)$, which is not a function in the ordinary sense but rather a “generalized function” (or distribution) defined by its behavior under integration:

$$\int_{-\infty}^{\infty} f(t) \delta(t - s) dt = f(s)$$

The Dirac delta is zero everywhere except at $t = s$, where it is “infinitely tall” in such a way that its total integral is 1. When we write that the transition “density” has a component $e^{-\rho s}$ at $t = s$, we are implicitly using this notation: $e^{-\rho s} \cdot \delta(t - s)$.

For our purposes, the key intuition is simple: with probability $e^{-\rho s}$, no recombination happens and the coalescence time is *exactly* s – not “near s ” or “approximately s ”, but precisely s .

Step 2: Where does the recombination happen?

Conditioned on representing the interval by one effective pruning event, let u be its time. An SMC pruning point is uniform on total tree length; for two equal branches this makes u **uniform** on $[0, s]$. Thus:

$$p(u | \text{recomb}) = \frac{1}{s}, \quad 0 \leq u \leq s$$

This is not the distribution of the earliest event in *time* conditional on a temporal Poisson process. It follows from selecting an SMC pruning point uniformly over branch length after the genomic interval has been represented by one effective event [McVean and Cardin, 2005].

Step 3: Where does the lineage re-coalesce?

After recombination at time u , the detached lineage floats up and re-coalesces with the other lineage. Under the coalescent (with 2 lineages and pairwise coalescence rate 1 in coalescent time units), the waiting time from u is $\text{Exp}(1)$ – an exponential random variable with rate 1.

So the re-coalescence time is $t = u + W$ where $W \sim \text{Exp}(1)$. The density of t given u is:

$$p(t | u) = e^{-(t-u)}, \quad t \geq u$$

i Probability Aside – The Exponential Distribution

A random variable $W \sim \text{Exp}(\lambda)$ has density $f(w) = \lambda e^{-\lambda w}$ for $w \geq 0$. Its mean is $1/\lambda$. With $\lambda = 1$, the density simplifies to e^{-w} and the mean waiting time is 1 coalescent time unit. This is the standard coalescent waiting time for two lineages, as derived in the *Coalescent Theory* chapter.

Step 4: Combine by integrating over u .

We now have all the ingredients. Given that a recombination occurred:

- The recombination time u is uniform on $[0, s]$, with density $1/s$
- Given u , the new coalescence time t has density $e^{-(t-u)}$ for $t \geq u$

To get the density of t (marginalizing over u), we integrate:

$$q_0(t | s) = \int_0^{s \wedge t} \frac{1}{s} e^{-(t-u)} du$$

Calculus Aside – Why $s \wedge t$ as the Upper Limit?

The notation $s \wedge t$ means $\min(s, t)$. There are two constraints on u : it must satisfy $u \leq s$ (the recombination must occur on the branch, which extends from 0 to s) and $u \leq t$ (the re-coalescence time t must come after the recombination time u). The binding constraint is whichever is smaller, hence $\min(s, t)$.

This integral splits into two cases depending on whether $t < s$ or $t \geq s$.

Case 1: $t < s$ (new coalescence time is earlier than the old one).

The integral runs from 0 to t :

$$q_0(t | s) = \frac{1}{s} \int_0^t e^{-(t-u)} du$$

To evaluate this, we use the substitution $v = t - u$. Then $dv = -du$, and the limits change:

- When $u = 0$: $v = t$
- When $u = t$: $v = 0$

$$\int_0^t e^{-(t-u)} du = \int_t^0 e^{-v} (-dv) = \int_0^t e^{-v} dv$$

Now we compute the antiderivative of e^{-v} :

$$\int_0^t e^{-v} dv = [-e^{-v}]_0^t = -e^{-t} - (-e^0) = 1 - e^{-t}$$

Calculus Aside – Evaluating Definite Integrals

The notation $[F(v)]_a^b$ means $F(b) - F(a)$. Here, $F(v) = -e^{-v}$, so $F(t) - F(0) = -e^{-t} - (-1) = 1 - e^{-t}$. This is the CDF of the standard exponential distribution evaluated at t .

Therefore:

$$q_0(t | s) = \frac{1}{s} (1 - e^{-t}), \quad t < s$$

Case 2: $t \geq s$ (new coalescence time is later than or equal to the old one).

The integral runs from 0 to s :

$$q_0(t | s) = \frac{1}{s} \int_0^s e^{-(t-u)} du$$

Using the same substitution $v = t - u$, with $dv = -du$:

- When $u = 0$: $v = t$
- When $u = s$: $v = t - s$

$$\int_0^s e^{-(t-u)} du = \int_t^{t-s} e^{-v} (-dv) = \int_{t-s}^t e^{-v} dv$$

Computing the antiderivative:

$$\int_{t-s}^t e^{-v} dv = [-e^{-v}]_{t-s}^t = -e^{-t} - (-e^{-(t-s)}) = e^{-(t-s)} - e^{-t}$$

Therefore:

$$q_0(t | s) = \frac{1}{s} \left(e^{-(t-s)} - e^{-t} \right), \quad t \geq s$$

Step 5: Include the no-recombination probability.

Combining the pruning probability $(1 - e^{-\rho s})$ with the conditional density $q_0(t | s)$, and adding the point mass for the no-recombination case:

$$K(dt | s) = e^{-\rho s} \delta_s(dt) + (1 - e^{-\rho s}) q_0(t | s) dt.$$

Writing the atom as a separate measure avoids treating its probability mass as an ordinary density value at $t = s$.

Verification: Does the density integrate to 1?

The total probability must be 1: the continuous part should integrate to $1 - e^{-\rho s}$ (the probability that recombination occurs), and the point mass contributes $e^{-\rho s}$.

We need to verify that $\int_0^\infty q_0(t | s) dt = 1$ (the conditional density given recombination integrates to 1):

$$\int_0^\infty q_0(t | s) dt = \frac{1}{s} \int_0^s (1 - e^{-t}) dt + \frac{1}{s} \int_s^\infty (e^{-(t-s)} - e^{-t}) dt$$

First integral (t from 0 to s):

$$\int_0^s (1 - e^{-t}) dt = [t + e^{-t}]_0^s = (s + e^{-s}) - (0 + 1) = s + e^{-s} - 1$$

Second integral (t from s to ∞):

$$\int_s^\infty (e^{-(t-s)} - e^{-t}) dt = [-e^{-(t-s)} + e^{-t}]_s^\infty$$

At $t \rightarrow \infty$: both $e^{-(t-s)}$ and e^{-t} go to 0, so the upper limit contributes 0.

At $t = s$: $-e^0 + e^{-s} = -1 + e^{-s}$.

Therefore: $0 - (-1 + e^{-s}) = 1 - e^{-s}$.

Putting it together:

$$\int_0^\infty q_0(t | s) dt = \frac{1}{s} [(s + e^{-s} - 1) + (1 - e^{-s})] = \frac{1}{s} \cdot s = 1 \quad \checkmark$$

So the conditional density integrates to 1 (given recombination). Multiplying by $(1 - e^{-\rho s})$ and adding the point mass $e^{-\rho s}$ gives total probability $(1 - e^{-\rho s}) + e^{-\rho s} = 1$.

```
def psmc_transition_density(t, s, rho):
    """PSMC transition density  $q_{\rho}(t | s)$ .

    Computes the probability density of the new coalescence time  $t$ ,
    given that the previous coalescence time was  $s$ .

    The density has three components:
    1. For  $t < s$ :  $(p_{\text{recomb}} / s) * (1 - \exp(-t))$ 
    2. At  $t = s$ : a point mass of weight  $\exp(-\rho * s)$  [no recombination]
    3. For  $t > s$ :  $(p_{\text{recomb}} / s) * (\exp(-(t-s)) - \exp(-t))$ 

    This function returns only the continuous part (items 1 and 3).
    The point mass at  $t = s$  must be handled separately.

    Parameters
    -----
    t : float or ndarray
        New coalescence time(s) to evaluate the density at.
    s : float
        Previous coalescence time.
    rho : float
        Recombination rate for the bin.

    Returns
    -----
    density : float or ndarray
        The continuous part of the transition density at each  $t$ .
    """
    # Probability that at least one recombination occurs.
    p_recomb = -np.expm1(-rho * s)

    # Convert t to a NumPy array so we can use boolean indexing.
    # The dtype=float ensures we get floating-point arithmetic.
    t = np.asarray(t, dtype=float)
    if s <= 0 or rho < 0 or np.any(t < 0):
        raise ValueError("require t >= 0, s > 0, and rho >= 0")

    # Initialize output array to zero (same shape as t).
    density = np.zeros_like(t)

    # Case 1: t < s (new coalescence time is earlier than old).
    # mask_lt is a boolean array: True where t < s, False elsewhere.
    mask_lt = t < s
    density[mask_lt] = (p_recomb / s) * (-np.expm1(-t[mask_lt]))

    # Case 2: t >= s (new coalescence time is later than old).
    # Note: at the single point t = s, the continuous density is
    # well-defined (both formulas give the same value there), but the
    # point mass must be added separately.
    mask_ge = t >= s
    density[mask_ge] = (p_recomb / s) * (
        np.exp(-(t[mask_ge] - s)) - np.exp(-t[mask_ge])
    )
```

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```

return density

# Visualize the transition density
s = 1.0          # previous coalescence time
rho = 0.5        # recombination rate
t_values = np.linspace(0.01, 4.0, 200) # grid of new coalescence times
densities = psmc_transition_density(t_values, s, rho)

print(f"Transition density from s={s}, rho={rho}:")
print(f"  Peak near t={t_values[np.argmax(densities)]:.2f}")

# np.trapezoid approximates the integral using the trapezoidal rule.
# This should be close to (1 - exp(-rho * s)), the recombination probability.
print(f"  Integral (approx): {np.trapezoid(densities, t_values):.4f}")

# The "missing mass" is the point mass at t = s (no recombination).
print(f"  Missing mass (at t=s): {np.exp(-rho * s):.4f}")

# Together they should sum to 1.
print(f"  Total: {np.trapezoid(densities, t_values) + np.exp(-rho * s):.4f}")

```

This mixed kernel motivates time discretization in PSMC (see *Overview of PSMC*) and the related construction used in SINGER's **time sampling** step [Li and Durbin, 2011, Deng *et al.*, 2025].

2.5.7 The Cumulative Distribution Function

To use the transition density in practice – for instance, to sample from it or to compute probabilities over intervals – we need the **cumulative distribution function** (CDF).

Probability Aside – What Is a CDF?

The **cumulative distribution function** (CDF) of a random variable T is defined as $F(t) = P(T \leq t)$: the probability that T takes a value less than or equal to t . The CDF is obtained by integrating the density:

$$F(t) = \int_{-\infty}^t f(x) dx$$

CDFs are monotonically increasing from 0 to 1. They are essential for:

- Computing the probability that T falls in an interval $[a, b]$: $P(a \leq T \leq b) = F(b) - F(a)$
- Sampling from the distribution via **inverse transform sampling**: generate $U \sim \text{Uniform}(0, 1)$ and set $T = F^{-1}(U)$
- Discretizing a continuous distribution by computing probabilities for each bin: $P(a_k \leq T < a_{k+1}) = F(a_{k+1}) - F(a_k)$

In the PSMC, we discretize coalescence time into bins and need the probability mass in each bin, which requires the CDF.

The CDF is obtained by integrating the density. Let us derive each case.

Case 1: $t < s$.

We integrate the density $\frac{1-e^{-\rho s}}{s}(1-e^{-x})$ from 0 to t :

$$Q_\rho(t | s) = \frac{1-e^{-\rho s}}{s} \int_0^t (1-e^{-x}) dx$$

The integrand $1-e^{-x}$ has antiderivative $x+e^{-x}$ (since the derivative of x is 1 and the derivative of e^{-x} is $-e^{-x}$, so $\frac{d}{dx}(x+e^{-x}) = 1-e^{-x}$).

$$\begin{aligned} Q_\rho(t | s) &= \frac{1-e^{-\rho s}}{s} [x+e^{-x}]_0^t \\ &= \frac{1-e^{-\rho s}}{s} [(t+e^{-t}) - (0+1)] \\ &= \frac{1-e^{-\rho s}}{s} [t+e^{-t}-1] \end{aligned}$$

Case 2: $t \geq s$.

We build the CDF in three pieces:

1. The integral of the density from 0 to s (the Case 1 density, evaluated at its upper limit s)
2. The point mass $e^{-\rho s}$ at $t = s$ (no recombination)
3. The integral of the density from s to t (the Case 2 density)

$$Q_\rho(t | s) = \underbrace{\frac{1-e^{-\rho s}}{s} [s+e^{-s}-1]}_{\text{integral up to } s} + \underbrace{e^{-\rho s}}_{\text{point mass}} + \underbrace{\frac{1-e^{-\rho s}}{s} \int_s^t (e^{-(x-s)} - e^{-x}) dx}_{\text{integral from } s \text{ to } t}$$

Let us compute the remaining integral. The integrand has two terms:

- $e^{-(x-s)}$ has antiderivative $-e^{-(x-s)}$
- $-e^{-x}$ has antiderivative e^{-x}

So the antiderivative of the integrand is $-e^{-(x-s)} + e^{-x}$. Evaluating:

$$\begin{aligned} \int_s^t (e^{-(x-s)} - e^{-x}) dx &= [-e^{-(x-s)} + e^{-x}]_s^t \\ &= (-e^{-(t-s)} + e^{-t}) - (-e^0 + e^{-s}) \\ &= -e^{-(t-s)} + e^{-t} + 1 - e^{-s} \\ &= 1 - e^{-s} - e^{-(t-s)} + e^{-t} \end{aligned}$$

Adding it all up:

$$\begin{aligned} Q_\rho(t | s) &= \frac{1-e^{-\rho s}}{s} \underbrace{[s+e^{-s}-1+1-e^{-s}]}_{=s} - e^{-(t-s)} + e^{-t} + e^{-\rho s} \\ &= \frac{1-e^{-\rho s}}{s} [s - e^{-(t-s)} + e^{-t}] + e^{-\rho s} \end{aligned}$$

Verification: As $t \rightarrow \infty$, both $e^{-(t-s)} \rightarrow 0$ and $e^{-t} \rightarrow 0$, so:

$$Q_\rho(\infty | s) = \frac{1-e^{-\rho s}}{s} \cdot s + e^{-\rho s} = 1 - e^{-\rho s} + e^{-\rho s} = 1 \quad \checkmark$$

In summary:

$$Q_\rho(t | s) = \begin{cases} \frac{1-e^{-\rho s}}{s} [t+e^{-t}-1] & t < s \\ \frac{1-e^{-\rho s}}{s} [s - e^{-(t-s)} + e^{-t}] + e^{-\rho s} & t \geq s \end{cases}$$

```

def psmc_transition_cdf(t, s, rho):
    """PSMC transition CDF  $Q_{\rho}(t | s)$ .

    Computes the cumulative distribution function of the new coalescence
    time  $t$ , given the previous coalescence time  $s$ .

    This CDF includes the point mass at  $t = s$  (no recombination).
    For  $t \geq s$ , the CDF jumps by  $\exp(-\rho * s)$  at  $t = s$ .

    Parameters
    -----
    t : float or ndarray
        New coalescence time(s) to evaluate the CDF at.
    s : float
        Previous coalescence time.
    rho : float
        Recombination rate for the bin.

    Returns
    -----
    cdf : float or ndarray
        CDF values at each  $t$ .
    """
    # Probability of no recombination (the point mass weight).
    p_no_recomb = np.exp(-rho * s)

    # Probability of recombination (weight of the continuous part).
    p_recomb = -np.expm1(-rho * s)

    # Convert t to a NumPy array for vectorized computation.
    t = np.asarray(t, dtype=float)
    if s <= 0 or rho < 0 or np.any(t < 0):
        raise ValueError("require t >= 0, s > 0, and rho >= 0")
    cdf = np.zeros_like(t)

    # Case 1: t < s
    # CDF is the integral of the density from 0 to t.
    mask_lt = t < s
    cdf[mask_lt] = (p_recomb / s) * (
        t[mask_lt] + np.expm1(-t[mask_lt])
    )

    # Case 2: t >= s
    # CDF includes the continuous integral plus the point mass.
    mask_ge = t >= s
    cdf[mask_ge] = (p_recomb / s) * (
        s - np.exp(-(t[mask_ge] - s)) + np.exp(-t[mask_ge])
    ) + p_no_recomb

    return cdf

# Verify: CDF should approach 1 as t -> infinity.
print(f"CDF at t=10: {psmc_transition_cdf(np.array([10.0]), s, rho)[0]:.6f}")

```

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```
print(f"CDF at t=100: {psmc_transition_cdf(np.array([100.0]), s, rho)[0]:.6f}")
```

2.5.8 Why SMC Enables HMM Inference

Let us now make explicit the connection between everything we have derived and the HMM framework from the [HMM chapter](#).

An HMM requires three ingredients:

1. **Hidden states:** The possible values of the hidden variable at each genomic position.
2. **Transition probabilities:** The probability of moving from one hidden state to another between adjacent positions. These must depend only on the current state (Markov property).
3. **Emission probabilities:** The probability of the observed data at each position, given the hidden state.

The SMC provides ingredients 1 and 2. Here is the mapping:

1. **Markov property:** The SMC ensures that the tree at position ℓ depends only on the tree at $\ell - 1$ and the recombination event between them. This is exactly the Markov property that HMMs require. Without the SMC approximation (in the full CwR), we would need the entire tree history, and the HMM framework would not apply.
2. **Conditional independence:** Under the usual site-independent mutation model, the observation at position ℓ depends on its local hidden genealogy (and mutation parameters), not observations at other positions. This gives emission probabilities $P(X_\ell | Z_\ell)$ of the form required by an HMM.
3. **Li–Stephens structure in SINGER:** The branch-HMM transition above has the stay-or-switch form $(1 - r_i)\delta_{ij} + r_i \cdot (\text{weights})$, which enables $O(K)$ forward steps instead of $O(K^2)$. This is the same structure we studied in the [HMM chapter](#).

Together: **SMC + HMM = tractable ARG inference**. The SMC gives us a Markov model of tree evolution along the genome, the HMM machinery lets us do efficient inference in that model, and the PSMC transition density derived above provides the specific transition probabilities for the pairwise case.

2.5.9 Summary

Concept	Key Idea
CwR limitation	Tree sequence is not Markov (ghost lineages can return)
Markov property	Future depends on past only through present; enables HMM inference
Ghost lineages	Ancestral lineages not in the current tree that could reappear in the CwR
SMC approximation	Re-coalescence only with current tree branches; eliminates ghost lineages
SINGER branch transitions	Li–Stephens-type structure: $(1 - r_i)\delta_{ij} + r_i \sum_j q_j$
PSMC transitions	Continuous-time density $q_\rho(t s)$ and CDF $Q_\rho(t s)$ for pairwise coalescence
SMC + HMM	The SMC provides the Markov property; the HMM provides the inference algorithm

You now have all the prerequisites. Time to build a Timepiece.

Next:

- [Overview](#) – disassembling the SINGER algorithm (ARG inference for multiple samples).
- [Overview of PSMC](#) – building the PSMC (population size history from a single diploid genome).

2.6 The Diffusion Approximation

“When the gear teeth are fine enough, the ticking becomes a smooth sweep.”

2.6.1 The Big Idea

In the *Coalescent Theory* chapter, we built the Wright-Fisher model: a discrete population of $2N$ gene copies, evolving in discrete generations, where allele frequencies jump around in integer multiples of $1/(2N)$. This model is an exactly specified idealization, but when N is large, the jumps are tiny, the generations are many, and keeping track of individual ticks becomes both unnecessary and computationally expensive.

The **diffusion approximation** replaces this discrete gear train with a smooth sweep. Instead of tracking allele frequencies that hop in discrete steps of $1/(2N)$, we model frequency as a **continuous** variable $x \in [0, 1]$ that drifts and fluctuates according to a **stochastic differential equation** (SDE). Instead of counting individual generations, we let time flow continuously. The result is a mathematical framework – partial differential equations, stationary distributions, numerical grids – that is both more elegant and more powerful than the discrete model it replaces.

Think of a mechanical watch. A quartz watch ticks 32,768 times per second – far too fast for the eye to resolve. The second hand appears to sweep smoothly, even though the underlying mechanism is discrete. The diffusion approximation does the same thing for population genetics: when N is in the thousands or millions, the discrete ticks of the Wright-Fisher model blur into a continuous flow, and we gain access to the entire toolkit of calculus – derivatives, integrals, partial differential equations – to describe what happens.

This chapter builds the diffusion machinery from the ground up. We start with the Wright-Fisher model you already know, derive the continuous limit, and arrive at the **Fokker-Planck equation** – the PDE at the heart of tools like *dadi*. We then solve it numerically, connect it to the site frequency spectrum, and preview how *moments* sidesteps the PDE entirely. The continuous Wright-Fisher model goes back to Kimura’s diffusion treatment [Kimura, 1955]; the software-specific claims below are checked against the original *dadi* and *moments* papers and implementations [Gutenkunst *et al.*, 2009, Jouganous *et al.*, 2017].

2.6.2 From Wright-Fisher to Continuous Frequency

Let us begin where the *Coalescent Theory* chapter left off: the Wright-Fisher model. Consider a single biallelic locus in a diploid population of size N . There are $2N$ gene copies, and the allele frequency X is the fraction of copies that carry the derived allele. In one generation, the number of derived copies in the next generation is drawn from a binomial distribution:

$$X' \sim \frac{1}{2N} \text{Binom}(2N, x)$$

where x is the current allele frequency. The change in frequency is $\Delta x = X' - x$.

Mean and variance of Δx

Under neutrality (no selection, no mutation, no migration), the mean change in allele frequency is zero:

$$\mathbb{E}[\Delta x] = 0$$

This follows because $\mathbb{E}[X'] = x$ for a binomial with success probability x . Allele frequency has no preferred direction – it is a **martingale**.

The variance of the change is:

$$\text{Var}(\Delta x) = \frac{x(1-x)}{2N}$$

This comes from the binomial variance: $\text{Var}(\text{Binom}(2N, x)) = 2N \cdot x(1-x)$, and dividing by $(2N)^2$ to convert from counts to frequencies. Notice that the variance is largest when $x = 1/2$ (maximum uncertainty) and vanishes at $x = 0$ or $x = 1$ (fixed alleles cannot drift).

i Probability Aside – Binomial variance

If $Y \sim \text{Binom}(n, p)$, then $\text{Var}(Y) = np(1 - p)$. The frequency $X' = Y/(2N)$ therefore has variance $\text{Var}(X') = \text{Var}(Y)/(2N)^2 = p(1 - p)/(2N)$. Since $\Delta x = X' - x$ and x is a constant (conditioning on the current generation), $\text{Var}(\Delta x) = \text{Var}(X') = x(1 - x)/(2N)$.

The diffusion timescale

The variance $x(1 - x)/(2N)$ is tiny when N is large: meaningful changes in allele frequency accumulate only over many generations. To obtain a useful continuous-time limit, we rescale time by defining:

$$\tau = \frac{t}{2N}$$

where t counts discrete generations and τ is **diffusion time** (the same coalescent time units from the *Coalescent Theory* chapter). In one unit of diffusion time, $2N$ generations pass, and the variance of the accumulated frequency change is:

$$\text{Var}\left(\sum_{i=1}^{2N} \Delta x_i\right) \approx 2N \cdot \frac{x(1 - x)}{2N} = x(1 - x)$$

The factor of $2N$ cancels the per-generation scale, leaving variance of order 1 and hence a nontrivial limit. The displayed sum is only a local heuristic: Wright–Fisher increments are not independent because their variance changes with the current frequency. In fact, after g generations the exact neutral variance is

$$\text{Var}(X_g) = x_0(1 - x_0) \left[1 - \left(1 - \frac{1}{2N}\right)^g\right].$$

For $g/(2N) \rightarrow \tau$, this converges to $x_0(1 - x_0)(1 - e^{-\tau})$, not $x_0(1 - x_0)$. The difference is the effect of state-dependent variance and eventual absorption.

Code: WF trajectories converging to SDE paths

Let us see this convergence in action. As N grows, discrete Wright–Fisher trajectories increasingly resemble continuous diffusion paths.

```
import numpy as np

def wright_fisher_trajectory(two_N, x0, n_generations):
    """Simulate a Wright-Fisher allele frequency trajectory.

    Parameters
    -----
    two_N : int
        Total number of gene copies (2N).
    x0 : float
        Initial allele frequency.
    n_generations : int
        Number of generations to simulate.

    Returns
    -----
    freqs : ndarray of shape (n_generations + 1,)
        Allele frequency at each generation.
    """
```

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```

if (
    not isinstance(two_N, (int, np.integer))
    or two_N <= 0
    or not 0 <= x0 <= 1
    or not isinstance(n_generations, (int, np.integer))
    or n_generations < 0
):
    raise ValueError("require integer two_N > 0, 0 <= x0 <= 1, and integer_
↳ generations >= 0")
    freqs = np.zeros(n_generations + 1)
    freqs[0] = x0
    for g in range(n_generations):
        # np.random.binomial(n, p) draws one sample from Binom(n, p).
        # We divide by two_N to convert counts to frequency.
        count = np.random.binomial(two_N, freqs[g])
        freqs[g + 1] = count / two_N
    return freqs

np.random.seed(42)

# Simulate for different population sizes, all for 1 diffusion time unit
x0 = 0.3
for two_N in [20, 200, 2000]:
    n_gen = two_N # 1 unit of diffusion time = 2N generations
    traj = wright_fisher_trajectory(two_N, x0, n_gen)
    # Convert generations to diffusion time
    diffusion_times = np.arange(n_gen + 1) / two_N
    final_freq = traj[-1]
    step_size = 1.0 / two_N
    print(f"2N={two_N:5d}: final freq={final_freq:.4f}, "
          f"step size={step_size:.6f}, "
          f"num steps={n_gen}")

# Vectorized endpoints check the exact finite-population moment.
two_N = 1000
tau = 0.2
n_generations = round(tau * two_N)
n_replicates = 20_000
counts = np.full(n_replicates, round(x0 * two_N), dtype=int)
for _ in range(n_generations):
    counts = np.random.binomial(two_N, counts / two_N)
endpoints = counts / two_N
exact_variance = x0 * (1 - x0) * (
    1 - (1 - 1 / two_N) ** n_generations
)
diffusion_limit = x0 * (1 - x0) * (1 - np.exp(-tau))
print("\nEndpoint variance:")
print(f" Simulated Wright-Fisher: {endpoints.var():.4f}")
print(f" Exact finite-N moment: {exact_variance:.4f}")
print(f" Diffusion limit: {diffusion_limit:.4f}")

```

2.6.3 Stochastic Differential Equations

The continuous-time limit of the Wright-Fisher model is a **stochastic differential equation** (SDE). In the simplest neutral case:

$$dx = \sqrt{x(1-x)} dW$$

where dW is the increment of a **Wiener process** (Brownian motion). This is the **Wright-Fisher diffusion**. The square root term $\sqrt{x(1-x)}$ ensures that drift vanishes at the boundaries $x = 0$ and $x = 1$, just as in the discrete model.

i Probability Aside – What is a Wiener process?

A **Wiener process** (also called **Brownian motion**) $W(t)$ is the most fundamental continuous-time random process. Its key properties are:

1. $W(0) = 0$
2. Increments $W(t+s) - W(t) \sim \text{Normal}(0, s)$ – the change over an interval of length s is normally distributed with mean 0 and variance s
3. Increments over non-overlapping intervals are independent
4. Paths are continuous (no jumps)

The Wiener process is the continuous analogue of a random walk: at each infinitesimal instant, it takes a tiny random step. The cumulative effect of infinitely many infinitesimal steps produces a continuous but extremely jagged path – differentiable nowhere, yet continuous everywhere.

In our SDE, dW represents an infinitesimal “kick” of size $\sqrt{dt}$. Multiplying by $\sqrt{x(1-x)}$ scales this kick by the local volatility of allele frequency.

More generally, with selection, mutation, and migration, the SDE takes the form:

$$dx = \underbrace{\mu(x)}_{\text{drift}} dt + \underbrace{\sigma(x)}_{\text{diffusion}} dW$$

where the **drift coefficient** $\mu(x)$ and **diffusion coefficient** $\sigma(x)$ encode the different evolutionary forces:

$$\begin{aligned} \mu(x) &= \underbrace{\gamma x(1-x)}_{\text{genic selection}} + \underbrace{\frac{\theta_1}{2}(1-x) - \frac{\theta_2}{2}x}_{\text{mutation}} + \underbrace{m(x_{\text{source}} - x)}_{\text{migration}} \\ \sigma(x) &= \sqrt{x(1-x)} \end{aligned}$$

Here $\gamma = 2Ns_{\text{gen}}$ is the scaled genic selection coefficient (positive favors the derived allele), while $\theta_1 = 4Nu$ and $\theta_2 = 4Nv$ are scaled forward and backward mutation rates. The migration coefficient must likewise be expressed per unit of diffusion time. The diffusion coefficient $\sigma^2(x) = x(1-x)$ captures **genetic drift** and is always the same regardless of selection or mutation.

Euler-Maruyama simulation

The **Euler-Maruyama method** is the simplest way to simulate an SDE. It replaces the continuous increments with discrete steps of size Δt :

$$x_{n+1} = x_n + \mu(x_n)\Delta t + \sigma(x_n)\sqrt{\Delta t} Z_n$$

where $Z_n \sim \text{Normal}(0, 1)$ are independent standard normal random variables.

```

def euler_maruyama(x0, mu_func, sigma_func, T, n_steps):
    """Simulate an SDE using the Euler-Maruyama method.

    Parameters
    -----
    x0 : float
        Initial condition.
    mu_func : callable
        Drift coefficient  $\mu(x)$ .
    sigma_func : callable
        Diffusion coefficient  $\sigma(x)$ .
    T : float
        Total time to simulate (in diffusion time units).
    n_steps : int
        Number of discrete time steps.

    Returns
    -----
    times : ndarray of shape (n_steps + 1,)
        Time points.
    x : ndarray of shape (n_steps + 1,)
        Simulated trajectory.
    """
    if not 0 <= x0 <= 1 or T < 0 or n_steps <= 0:
        raise ValueError("require 0 <= x0 <= 1, T >= 0, and n_steps > 0")
    dt = T / n_steps
    sqrt_dt = np.sqrt(dt)
    times = np.linspace(0, T, n_steps + 1)
    x = np.zeros(n_steps + 1)
    x[0] = x0

    for n in range(n_steps):
        # np.random.randn() draws one sample from Normal(0, 1)
        Z = np.random.randn()
        x[n + 1] = x[n] + mu_func(x[n]) * dt + sigma_func(x[n]) * sqrt_dt * Z
        # Project an overshoot to the nearest absorbing boundary.
        x[n + 1] = np.clip(x[n + 1], 0.0, 1.0)

    return times, x

# Neutral diffusion:  $\mu(x) = 0$ ,  $\sigma(x) = \sqrt{x(1-x)}$ 
mu_neutral = lambda x: 0.0
sigma_drift = lambda x: np.sqrt(max(x * (1 - x), 0.0))

# Genic selection:  $\mu(x) = \gamma * x * (1-x)$ ,  $\gamma = 2 * N * s_{gen}$ 
gamma = 2.5
mu_selection = lambda x: gamma * x * (1 - x)

np.random.seed(42)

# Compare neutral vs selected trajectories
T = 2.0
n_steps = 5000

```

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```
_, x_neutral = euler_maruyama(0.1, mu_neutral, sigma_drift, T, n_steps)
_, x_selected = euler_maruyama(0.1, mu_selection, sigma_drift, T, n_steps)

print(f"Neutral:  start={0.1:.2f}, end={x_neutral[-1]:.4f}")
print(f"Selected: start={0.1:.2f}, end={x_selected[-1]:.4f}")
print("(These are individual random paths, not a comparison of means.)")
```

Clipping is a projection, not a reflection, and Euler–Maruyama is not an exact Wright–Fisher simulator. Near the degenerate boundaries it can introduce step-size bias. Use it for intuition, check convergence as Δt decreases, and use the discrete Wright–Fisher model or specialized diffusion methods when boundary behavior is inferentially important.

2.6.4 From SDEs to PDEs: The Fokker-Planck Equation

The SDE describes a single trajectory – one realization of the random process. But in population genetics, we rarely care about a single trajectory. We want to know the **distribution** of allele frequencies across many independent loci or many replicate populations. This distribution is described by a density function $\phi(x, t)$, where $\phi(x, t)dx$ is the probability of finding an allele at frequency between x and $x + dx$ at time t .

The equation governing $\phi(x, t)$ is the **Fokker-Planck equation** (also called the **Kolmogorov forward equation**):

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma^2(x)\phi] - \frac{\partial}{\partial x} [\mu(x)\phi]$$

This is a **partial differential equation** (PDE) – and it is the mathematical heart of the diffusion approximation.

Calculus Aside – What is a partial differential equation?

An **ordinary differential equation** (ODE) involves a function of a single variable and its derivatives. For example, $dy/dt = -ky$ describes exponential decay, where y depends only on t .

A **partial differential equation** (PDE) involves a function of **multiple** variables and its partial derivatives. Here, $\phi(x, t)$ depends on two variables: the allele frequency x and time t . The symbol $\partial\phi/\partial t$ means “the rate of change of ϕ with respect to time, holding x fixed.” Similarly, $\partial\phi/\partial x$ means “the rate of change with respect to frequency, holding t fixed.”

PDEs are harder than ODEs because the solution is a surface (a function of two variables) rather than a curve (a function of one variable). But they arise naturally whenever a quantity depends on both space and time – as allele frequency density $\phi(x, t)$ does.

The two terms: diffusion and advection

The Fokker-Planck equation has two terms, each with a clear biological meaning:

1. **The diffusion term** (second-order derivative):

$$\frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma^2(x)\phi]$$

This term captures **genetic drift** – the random fluctuations in allele frequency due to finite population size. The second derivative acts as a “spreading” operator: it takes a concentration of probability at one frequency and spreads it out to neighboring frequencies. The coefficient $\sigma^2(x) = x(1 - x)$ ensures that drift is strongest at intermediate frequencies and vanishes at the boundaries.

2. **The advection term** (first-order derivative):

$$-\frac{\partial}{\partial x} [\mu(x)\phi]$$

This term captures **directional forces** – selection, mutation, migration. The first derivative acts as a “transport” operator: it moves probability density in the direction specified by $\mu(x)$. Under positive selection ($\mu(x) > 0$ for $x \in (0, 1)$), probability mass is transported toward higher frequencies. Under mutation pressure, mass flows from the boundaries toward the interior.

i Calculus Aside – Why second-order for drift and first-order for selection?

This asymmetry reflects a fundamental difference in the nature of the two forces:

Drift is symmetric: at any frequency x , genetic drift is equally likely to push the frequency up or down. Symmetric random perturbations spread out a distribution – they increase its variance without changing its mean. Mathematically, spreading is captured by the second derivative. (Think of the heat equation $\partial u / \partial t = D \partial^2 u / \partial x^2$, which describes how heat diffuses from hot regions to cold regions.)

Selection is directional: selection pushes allele frequency systematically in one direction (toward fixation for beneficial alleles, toward loss for deleterious ones). Directional transport of a density is captured by the first derivative. (Think of the transport equation $\partial u / \partial t + v \partial u / \partial x = 0$, which describes how a quantity is carried along by a flow with velocity v .)

The Fokker-Planck equation combines both effects: drift spreads the distribution (second-order term) while selection shifts it (first-order term).

For the neutral Wright-Fisher diffusion with $\mu(x) = 0$ and $\sigma^2(x) = x(1 - x)$, the Fokker-Planck equation simplifies to:

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \frac{\partial^2}{\partial x^2} [x(1 - x)\phi]$$

With genic selection and reversible mutation ($\mu(x) = \frac{\theta_1}{2}(1 - x) - \frac{\theta_2}{2}x + \gamma x(1 - x)$):

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \frac{\partial^2}{\partial x^2} [x(1 - x)\phi] - \frac{\partial}{\partial x} \left[\left(\frac{\theta_1}{2}(1 - x) - \frac{\theta_2}{2}x + \gamma x(1 - x) \right) \phi \right]$$

Each evolutionary force is a separate, additive term in the PDE. This modularity is one of the great strengths of the diffusion framework: to add a new force, simply add the corresponding term to $\mu(x)$.

2.6.5 Boundary Conditions

The Fokker-Planck equation describes how the density $\phi(x, t)$ evolves in the interior $x \in (0, 1)$. But what happens at the boundaries $x = 0$ and $x = 1$?

Absorbing boundaries

At $x = 0$, the derived allele is **lost** – all copies have been replaced by the ancestral allele. At $x = 1$, the derived allele is **fixed** – it has replaced all ancestral copies. In the standard Wright-Fisher model (without mutation), both boundaries are **absorbing**: once frequency reaches 0 or 1, it stays there forever.

Absorption is a statement about the process, not just the numerical values of an interior density. Its law is generally a **mixed measure**: a density on $(0, 1)$ plus point masses at 0 and 1. Probability that reaches a boundary leaves the interior density and accumulates in the corresponding atom. Therefore the interior integral alone decreases, while interior mass plus the two boundary masses remains one. Writing only $\phi(0, t) = \phi(1, t) = 0$ misses those atoms and is not a complete probability accounting.

Why $x(1 - x)$ vanishes at boundaries

The diffusion coefficient $\sigma^2(x) = x(1 - x)$ naturally enforces the boundary behavior. At $x = 0$:

$$\sigma^2(0) = 0 \cdot (1 - 0) = 0$$

and at $x = 1$:

$$\sigma^2(1) = 1 \cdot (1 - 1) = 0$$

When the diffusion coefficient vanishes, the random component of the dynamics disappears. An allele that has been lost ($x = 0$) or fixed ($x = 1$) experiences no further drift. This is a natural consequence of the Wright-Fisher model: you cannot have random fluctuations when one allele has zero copies.

The flux condition

The **probability flux** at a boundary measures the rate at which probability density flows out of the interval. For the Fokker-Planck equation, the flux at $x = 0$ is:

$$J(0, t) = \mu(0)\phi(0, t) - \frac{1}{2} \frac{\partial}{\partial x} [\sigma^2(x)\phi] \Big|_{x=0}$$

Under the absorbing boundary condition, probability that hits the boundary is removed from the system. The total probability in the interior $\int_0^1 \phi(x, t) dx$ decreases over time as alleles are lost or fixed.

Reflecting boundaries and mutation

With two-way mutation ($\theta_1, \theta_2 > 0$), neither monomorphic state is absorbing: mutation produces copies of the absent allele. The precise boundary classification depends on the scaled mutation parameters, but the stationary finite-sites model has zero net boundary flux and a proper interior Beta-type density.

In the **infinite-sites model** – the standard framework for tools like *dadi* and *moments* – each new mutation arises at a previously monomorphic site. The density $\phi(x, t)$ represents the density of *segregating sites* at frequency x . New mutations enter at $x = 1/(2N) \approx 0$ at a rate proportional to θ , and sites are removed when they reach fixation ($x = 1$) or loss ($x = 0$).

2.6.6 Stationary Distributions

When the density $\phi(x, t)$ stops changing – when $\partial\phi/\partial t = 0$ – we have a **stationary distribution**. Setting the left-hand side of the Fokker-Planck equation to zero converts the PDE into an ODE (because the t dependence disappears):

$$0 = \frac{1}{2} \frac{d^2}{dx^2} [\sigma^2(x)\phi] - \frac{d}{dx} [\mu(x)\phi]$$

This is a second-order ODE in x alone, which is much easier to solve than the full PDE.

Neutrality: two different objects

With no mutation, a single neutral allele has no proper stationary *interior* probability density: its mass is eventually absorbed at 0 or 1. This must not be confused with the equilibrium **expected density of segregating sites** in the infinite-sites model, where new derived mutations continuously enter near zero. For a constant population that expected density is

$$\phi_{\text{SFS}}(x) = \frac{\theta}{x}, \quad 0 < x < 1.$$

It is an intensity (expected sites per frequency interval), not a normalized probability density. The more general zero-time-derivative solution $(C_1x + C_0)/[x(1-x)]$ is fixed by the mutation input and absorbing-fixation boundary flux; choosing an arbitrary symmetric numerator gives the wrong unfolded SFS. In particular, `dadi.PhiManip.phi_1D_snm` implements the θ/x standard-neutral density [Gutenkunst *et al.*, 2009].

i Probability Aside – The neutral SFS connection

The neutral site frequency spectrum (SFS) for a sample of n haploids has the well-known form $\mathbb{E}[\xi_j] = \theta/j$ for $j = 1, 2, \dots, n-1$, where ξ_j is the number of sites with j derived alleles. The $1/j$ pattern is a direct consequence of the θ/x expected site density: when you sample n individuals from a population with allele frequency density θ/x , the expected number with j derived alleles in a sample of n is proportional to $\int_0^1 \binom{n}{j} x^j (1-x)^{n-j} \cdot \frac{1}{x} dx$, which evaluates to $1/j$. We make this connection precise in the section on the site frequency spectrum below.

With mutation: the Beta distribution

Adding symmetric mutation with forward rate θ_1 and backward rate θ_2 , the drift coefficient becomes $\mu(x) = \frac{\theta_1}{2}(1-x) - \frac{\theta_2}{2}x$. The stationary distribution is:

$$\phi(x) \propto x^{\theta_1-1}(1-x)^{\theta_2-1}$$

This is the density of a **Beta distribution** with parameters θ_1 and θ_2 . When $\theta_1 = \theta_2 < 1$, the density is U-shaped (alleles cluster near fixation or loss). When $\theta_1 = \theta_2 > 1$, it is bell-shaped. When $\theta_1 = \theta_2 = 1$, it is uniform – mutation perfectly balances drift.

With selection: exponential tilting

Adding genic selection $\mu(x) = \gamma x(1-x) + \frac{\theta_1}{2}(1-x) - \frac{\theta_2}{2}x$, the reversible finite-sites stationary distribution becomes:

$$p(x) \propto x^{\theta_1-1}(1-x)^{\theta_2-1} e^{2\gamma x}$$

The exponential factor $e^{2\gamma x}$ **tilts** the mutation–drift distribution: positive selection ($\gamma > 0$) upweights high frequencies, while negative selection ($\gamma < 0$) upweights low frequencies, keeping deleterious alleles rare.

```
def stationary_density(x, theta1, theta2, gamma=0.0):
    """Unnormalized reversible finite-sites stationary density.

    Parameters
    -----
    x : float or ndarray
        Allele frequency in (0, 1).
    theta1 : float
        Forward mutation rate (scaled by 2N).
    theta2 : float
        Backward mutation rate (scaled by 2N).
    gamma : float
        Genic selection coefficient 2*N*s_gen. Positive favors derived.

    Returns
    -----
    density : float or ndarray
        Unnormalized stationary density at each x.
    """
    x = np.asarray(x, dtype=float)
    if theta1 <= 0 or theta2 <= 0 or np.any((x <= 0) | (x >= 1)):
        raise ValueError("require theta1, theta2 > 0 and 0 < x < 1")
    # Beta distribution kernel times exponential tilting for selection
    log_density = (
        (theta1 - 1) * np.log(x)
```

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```

    + (theta2 - 1) * np.log1p(-x)
    + 2 * gamma * x
)
return np.exp(log_density)

# Evaluate and normalize over a grid
x_grid = np.linspace(0.001, 0.999, 1000)

# Case 1: Neutral with symmetric mutation
phi_neutral = stationary_density(x_grid, theta1=0.5, theta2=0.5)
phi_neutral /= np.trapezoid(phi_neutral, x_grid) # normalize

# Case 2: With positive selection
phi_selected = stationary_density(x_grid, theta1=0.5, theta2=0.5, gamma=5.0)
phi_selected /= np.trapezoid(phi_selected, x_grid)

# Verification: the density should integrate to 1
print(f"Integral (neutral): {np.trapezoid(phi_neutral, x_grid):.6f}")
print(f"Integral (selected): {np.trapezoid(phi_selected, x_grid):.6f}")

# Verification: mean frequency should be higher under positive selection
mean_neutral = np.trapezoid(x_grid * phi_neutral, x_grid)
mean_selected = np.trapezoid(x_grid * phi_selected, x_grid)
print(f"Mean frequency (neutral): {mean_neutral:.4f}")
print(f"Mean frequency (selected): {mean_selected:.4f}")
print(f"Selection shifts mean upward: {mean_selected > mean_neutral}")

```

2.6.7 Numerical Solutions: Finite Differences for PDEs

Closed-form solutions to the Fokker–Planck equation are limited. For realistic demographic models – bottlenecks, exponential growth, population splits – numerical evolution is standard. This is the heart of *dadi*.

Discretizing x on a grid

The first step is to replace the continuous frequency variable $x \in [0, 1]$ with a discrete grid of P points:

$$x_0 = 0, \quad x_1 = \Delta x, \quad x_2 = 2\Delta x, \quad \dots, \quad x_{P-1} = 1$$

where $\Delta x = 1/(P - 1)$. The density $\phi(x, t)$ is approximated by its values at these grid points: $\phi_i(t) \approx \phi(x_i, t)$.

Finite-difference approximations

We approximate the derivatives in the Fokker–Planck equation using **finite differences**:

- **First derivative** (central difference):

$$\left. \frac{\partial f}{\partial x} \right|_{x_i} \approx \frac{f_{i+1} - f_{i-1}}{2\Delta x}$$

- **Second derivative**:

$$\left. \frac{\partial^2 f}{\partial x^2} \right|_{x_i} \approx \frac{f_{i+1} - 2f_i + f_{i-1}}{(\Delta x)^2}$$

These formulas replace derivatives with algebraic operations on neighboring grid values. The error in each approximation is $O((\Delta x)^2)$ – as the grid gets finer, the approximation improves quadratically.

The method of lines

Substituting the finite-difference approximations into the Fokker-Planck equation converts the PDE into a **system of ODEs** – one ODE per grid point:

$$\frac{d\phi_i}{dt} = F_i(\phi_0, \phi_1, \dots, \phi_{P-1})$$

where F_i involves ϕ_{i-1} , ϕ_i , and ϕ_{i+1} (the values at neighboring grid points). This approach is called the **method of lines**: we discretize in space (the x direction) but leave time continuous, converting a PDE into a system of coupled ODEs that can be solved with standard ODE integrators.

Implicit finite-volume time stepping in `dadi`

The original `dadi` implementation does **not** use the Crank–Nicolson formula formerly shown here. Its `Integration.one_pop` path injects new mutations near zero, constructs finite-volume flux coefficients (with optional Chang–Cooper weights), and calls an implicit tridiagonal update. `Integration._compute_dt` still limits the time step for accuracy. This distinction matters: an explicit predictor–corrector plus non-negativity clipping is neither Crank–Nicolson nor the production `dadi` algorithm [Gutenkunst *et al.*, 2009].

The transparent teaching solver below uses a different, explicitly identified approximation: a nearest-neighbor continuous-time Markov chain whose generator matches $\frac{1}{2}x(1-x)f''(x)$. A step-size bound makes every update a stochastic matrix, so probability cannot become negative and boundary atoms are retained rather than clipped away.

The curse of dimensionality

For a single population, the grid has P points and everything is fast. But for d populations, we need a grid in d -dimensional frequency space, with P^d grid points. This is the **curse of dimensionality**:

- 1 population: P grid points (manageable)
- 2 populations: P^2 grid points (feasible for moderate P)
- 3 populations: P^3 grid points (expensive but possible)
- 4+ populations: P^4 or more (often prohibitive)

This scaling is why `dadi` becomes expensive for more than 2-3 populations, and why `moments` was developed as an alternative that avoids the frequency grid entirely.

Code: 1D diffusion solver

We now approximate the law of one neutral allele. Grid endpoints are genuine probability atoms, while interior entries are cell probabilities; therefore the returned vector sums to one without numerical integration.

```
def solve_neutral_diffusion(P, T, x0, cfl=0.9):
    """Approximate a neutral Wright-Fisher diffusion on a uniform grid.

    The nearest-neighbor chain matches the backward generator
    Lf = x(1-x) f'' / 2. Endpoints are absorbing states.

    Parameters
    -----
    P : int
        Number of grid points (including boundaries).
    T : float
        Total time to integrate (in units of 2*N_ref generations).
    x0 : float
```

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```

    Initial allele frequency.
    cfl : float
        Maximum total jump probability per explicit step; must be in (0, 1].

Returns
-----
x_grid : ndarray of shape (P,)
    Frequency grid points.
probability : ndarray of shape (P,)
    Probability mass at grid states. Entries 0 and -1 are boundary atoms.
"""
if P < 3 or T < 0 or not 0 <= x0 <= 1 or not 0 < cfl <= 1:
    raise ValueError("require P >= 3, T >= 0, 0 <= x0 <= 1, 0 < cfl <= 1")
dx = 1.0 / (P - 1)
x_grid = np.linspace(0, 1, P)
rates = 0.5 * x_grid * (1 - x_grid) / dx**2
max_total_rate = 2 * rates.max()
n_steps = max(1, int(np.ceil(T * max_total_rate / cfl)))
dt = T / n_steps
jump = dt * rates

# Linear interpolation places x0 on the grid while preserving its mean.
p = np.zeros(P)
coordinate = x0 / dx
lower = min(int(np.floor(coordinate)), P - 1)
upper = min(lower + 1, P - 1)
fraction = coordinate - lower
p[lower] += 1 - fraction
p[upper] += fraction

for _ in range(n_steps):
    updated = p * (1 - 2 * jump)
    updated[:-1] += p[1:] * jump[1:] # jumps toward zero
    updated[1:] += p[:-1] * jump[:-1] # jumps toward one
    p = updated
return x_grid, p

x_grid, probability = solve_neutral_diffusion(P=101, T=0.5, x0=0.3)
mean = np.dot(x_grid, probability)
variance = np.dot((x_grid - mean) ** 2, probability)
expected_variance = 0.3 * 0.7 * (1 - np.exp(-0.5))
print(f"Total probability: {probability.sum():.12f}")
print(f"Mean (martingale): {mean:.6f}")
print(f"Boundary mass: {probability[0] + probability[-1]:.6f}")
print(f"Variance: {variance:.6f}")
print(f"Diffusion moment: {expected_variance:.6f}")

```

2.6.8 Connection to the Site Frequency Spectrum

Let $\phi(x, t)dx$ now denote the **expected number of sites** with population frequency in $[x, x + dx]$; this is not the normalized single-allele law used in the preceding solver. What we observe is a **sample** of n individuals, which gives us integer allele counts. The connection between the continuous density and the discrete **site frequency spectrum** (SFS) is

the **binomial sampling formula**.

The binomial bridge

The expected number of segregating sites where j out of n sampled chromosomes carry the derived allele is:

$$\mathbb{E}[\xi_j] = \int_0^1 \binom{n}{j} x^j (1-x)^{n-j} \phi(x, t) dx$$

The mutation rate and sequence length are already contained in the expected-site density ϕ . If instead $p(x, t)$ is a normalized probability density across L independent sites, the prefactor is L . Multiplying again by θL when $\phi = \theta L/x$ would double-count the scale.

i Probability Aside – Why binomial sampling?

If the true allele frequency in the population is x , and you sample n chromosomes independently, the number of derived alleles in your sample follows $\text{Binom}(n, x)$. This is because each chromosome independently carries the derived allele with probability x . The integral over x averages this binomial probability over all possible population frequencies, weighted by the diffusion density $\phi(x, t)$.

This formula is the link between the continuous world of the diffusion and the discrete world of observed data. It is used by *dadi* to convert its numerically computed $\phi(x, t)$ into a predicted SFS that can be compared to the observed SFS.

```
from math import comb

def density_to_sfs(x_grid, phi, n_samples):
    """Convert a diffusion density to a site frequency spectrum.

    Applies the binomial sampling formula:
        sfs[j] = integral of Binom(n, j, x) * phi(x) dx

    Parameters
    -----
    x_grid : ndarray of shape (P,)
        Frequency grid points.
    phi : ndarray of shape (P,)
        Density values at each grid point.
    n_samples : int
        Number of sampled chromosomes.

    Returns
    -----
    sfs : ndarray of shape (n_samples - 1,)
        Expected SFS entries for j = 1, ..., n_samples - 1.
    """
    x_grid = np.asarray(x_grid, dtype=float)
    phi = np.asarray(phi, dtype=float)
    if (
        x_grid.ndim != 1
        or phi.shape != x_grid.shape
        or n_samples < 2
        or np.any(np.diff(x_grid) <= 0)
        or x_grid[0] < 0
    ):
```

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```

    or x_grid[-1] > 1
    or np.any(phi < 0)
):
    raise ValueError("invalid grid, density, or sample size")
sfs = np.zeros(n_samples - 1)
for j in range(1, n_samples):
    # Binomial probability at each grid point
    # comb(n, j) computes the binomial coefficient "n choose j"
    binom_probs = comb(n_samples, j) * x_grid**j * (1 - x_grid)**(n_samples - j)
    # Integrate phi(x) * Binom(n, j, x) over x using the trapezoidal rule
    sfs[j - 1] = np.trapezoid(binom_probs * phi, x_grid)
return sfs

# The standard-neutral infinite-sites density is theta/x.
n_samples = 20 # sample 20 chromosomes
theta = 1.0
sfs_grid = np.linspace(1e-6, 1.0, 100_001)
neutral_site_density = theta / sfs_grid
sfs_neutral = density_to_sfs(sfs_grid, neutral_site_density, n_samples)
expected_neutral = theta / np.arange(1, n_samples)
print("Maximum error from theta/j:", np.max(np.abs(sfs_neutral - expected_neutral)))

```

How dadi and moments differ

The diffusion-to-SFS conversion above is precisely what *dadi* does: it solves the PDE for $\phi(x, t)$, then integrates against binomial weights to extract the expected SFS.

moments takes a different approach. Instead of solving for the full density and then sampling it, *moments* evolves sample-frequency moments **directly**. Drift and mutation close on the sampled spectrum; terms that require higher-order moments, notably selection, use a jackknife closure. Thus it skips the population-frequency grid but is not merely an exact algebraic extraction of the same numerical density [Jouganous *et al.*, 2017].

The trade-off:

- **dadi**: Solves the PDE on a grid, then extracts the SFS. The grid introduces discretization artifacts, but gives access to the full frequency density $\phi(x, t)$ – useful for computing quantities beyond the SFS (e.g., fixation probabilities, frequency densities under selection). Multi-population models require $O(P^d)$ grid points.
- **moments**: Solves ODEs directly for the SFS entries. It avoids a population- frequency grid but can introduce moment-closure error, and it does not provide the full frequency density. Multi-population models require $O(\prod n_i)$ SFS entries, which can be more efficient than $O(P^d)$ for large P .

Both approaches are built on the diffusion approximation, but their numerical representations and approximation errors differ.

2.6.9 Summary

Concept	Key Formula / Idea
WF frequency change	$\mathbb{E}[\Delta x] \approx 0$ (neutral), $\text{Var}(\Delta x) = x(1-x)/(2N)$
Diffusion timescale	$\tau = t/(2N)$ (same as coalescent time units)
SDE (Wright-Fisher diffusion)	$dx = \mu(x)dt + \sqrt{x(1-x)}dW$
Fokker-Planck / Kolmogorov forward	$\partial\phi/\partial t = \frac{1}{2}\partial_{xx}^2[\sigma^2\phi] - \partial_x[\mu\phi]$
Neutral infinite-sites density	Expected-site intensity $\phi(x) = \theta/x$, giving $\mathbb{E}[\xi_j] = \theta/j$
With mutation	$\phi(x) \propto x^{\theta_1-1}(1-x)^{\theta_2-1}$ (Beta distribution)
With reversible mutation and selection	$p(x) \propto x^{\theta_1-1}(1-x)^{\theta_2-1}e^{2\gamma x}$
Numerical diffusion solver	Discretize x ; preserve non-negativity, total mass, and boundary atoms
Multi-population scaling	$O(P^d)$ grid points – the curse of dimensionality
Diffusion density to SFS	$\mathbb{E}[\xi_j] = \int \binom{n}{j} x^j (1-x)^{n-j} \phi(x) dx$

These are the gears that connect the discrete ticking of the Wright-Fisher model to the smooth sweep of continuous frequency evolution. The diffusion approximation is the bridge between the coalescent theory you learned in *Coalescent Theory* and the SFS-based inference tools – *dadi* and *moments* – that use it to read demographic history from genetic variation.

Next: *Ordinary Differential Equations* – the ordinary differential equations that *moments* uses to bypass the diffusion PDE entirely, computing the SFS without a frequency grid.

2.7 Ordinary Differential Equations

“The hands of the clock obey equations you can write down – and solve.”

2.7.1 The Big Idea

A watchmaker does not guess how fast a gear turns. The gear ratio, the spring tension, and the escapement frequency *determine* the rotation rate exactly. Given these specifications, the watchmaker can predict the position of every hand at every moment. The equations governing this motion are ordinary differential equations (ODEs).

In population genetics, ODEs appear everywhere. The expected number of ancestral lineages at time t satisfies an ODE (see *Coalescent Theory*). Each entry of the site frequency spectrum evolves under drift and mutation according to a system of coupled ODEs (see the moments Timepiece, *Timepiece X: moments*). The probability of j undistinguished lineages remaining at time t in SMC++ is governed by yet another ODE system (see *Timepiece II: SMC++*). The transition probabilities of momi2's Moran model are computed via the matrix exponential of a rate matrix – a linear ODE in disguise (see *Timepiece XII: momi2*).

This chapter gives you the mathematical and computational tools to understand, solve, and implement ODEs. We start from the simplest possible case and build toward the systems of equations that power multiple Timepieces. By the end, you will know how to solve an ODE numerically, understand when standard methods fail and what to do about it, and recognize the matrix exponential as a special case of a linear ODE solution.

2.7.2 What Is an ODE?

An **ordinary differential equation** relates a quantity to its rate of change with respect to a single independent variable. In every example in this book, the independent variable is time t . We write:

$$\frac{dy}{dt} = f(y, t)$$

This says: “the rate at which y changes is given by the function f of the current value y and the current time t .” If you are comfortable with the idea that velocity tells you how position changes, you already have the right intuition. Here, y is the “position” and $f(y, t)$ is the “velocity.”

i Calculus Aside – Derivatives and rates of change

The symbol dy/dt is the **derivative** of y with respect to t . It measures the *instantaneous rate of change* of y as t increases. If $y(t)$ represents the number of lineages at time t , then dy/dt tells you how fast that number is changing right now.

Formally, the derivative is the limit of a difference quotient:

$$\frac{dy}{dt} = \lim_{h \rightarrow 0} \frac{y(t+h) - y(t)}{h}$$

The numerator is the change in y over a small interval h , and dividing by h gives the rate per unit time. Taking h to zero gives the instantaneous rate. If you have never seen calculus, think of dy/dt as “the slope of the graph of y versus t ” – positive slope means y is increasing, negative slope means y is decreasing.

An **initial value problem** (IVP) is an ODE together with a starting condition:

$$\frac{dy}{dt} = f(y, t), \quad y(t_0) = y_0$$

The goal is to find the function $y(t)$ for $t > t_0$ that satisfies both the equation and the initial condition. Think of it as specifying the gear positions at the moment the watch is wound, then asking where the gears will be at any future time.

Connection to the coalescent. Conditional on currently having an integer k lineages, the next coalescence occurs at rate

$$c_k = \binom{k}{2} = \frac{k(k-1)}{2}.$$

Because each event reduces the count by one, the conditional instantaneous mean change is $-c_k$. Replacing k by a continuous λ gives the useful mean-field ODE $d\lambda/dt = -\lambda(\lambda-1)/2$, but it is **not** the exact ODE for $\mathbb{E}[K_t]$: nonlinearity means $\mathbb{E}[K_t(K_t-1)] \neq \mathbb{E}[K_t](\mathbb{E}[K_t]-1)$ in general. The exact probability ODE is developed below.

```
import numpy as np

def coalescence_event_rate(k):
    """Kingman coalescence rate conditional on integer lineage count k."""
    if not isinstance(k, (int, np.integer)) or k < 1:
        raise ValueError("k must be a positive integer")
    return k * (k - 1) / 2

# Starting with 10 lineages, the initial rate of decrease is:
lam_0 = 10
rate = coalescence_event_rate(lam_0)
print(f"With {lam_0} lineages: event rate = {rate:.1f} per unit time")
print(f"(There are {lam_0*(lam_0-1)//2} pairs, each coalescing at rate 1)")
```

Let us warm up with a simpler ODE before tackling numerical methods. The **logistic equation** is a classic model for growth with saturation:

$$\frac{dy}{dt} = r y \left(1 - \frac{y}{K} \right), \quad y(0) = y_0$$

Here r is the growth rate and K is the carrying capacity. When y is small, growth is approximately exponential ($dy/dt \approx ry$). As y approaches K , the factor $(1 - y/K)$ drives the growth rate to zero. This ODE has an exact solution:

$$y(t) = \frac{K}{1 + \left(\frac{K}{y_0} - 1\right) e^{-rt}}$$

We will use this to test our numerical methods.

```
import numpy as np

def logistic_exact(t, y0, r, K):
    """Exact solution of the logistic equation.

    Parameters
    -----
    t : float or ndarray
        Time(s) at which to evaluate the solution.
    y0 : float
        Initial condition y(0).
    r : float
        Growth rate.
    K : float
        Carrying capacity.

    Returns
    -----
    y : float or ndarray
        Solution y(t).
    """
    if y0 <= 0 or K <= 0 or r < 0:
        raise ValueError("require y0 > 0, K > 0, and r >= 0")
    t = np.asarray(t, dtype=float)
    return K / (1 + (K / y0 - 1) * np.exp(-r * t))

# Verify: y(0) should equal y0, and y(inf) should approach K
y0, r, K = 0.1, 1.0, 10.0
print(f"y(0) = {logistic_exact(0.0, y0, r, K):.4f} (expected {y0})")
print(f"y(10) = {logistic_exact(10.0, y0, r, K):.4f} (expected ~{K})")
print(f"y(100) = {logistic_exact(100.0, y0, r, K):.4f} (expected {K})")
```

2.7.3 Euler's Method

The simplest numerical method for ODEs is **Euler's method**. The idea is straightforward: if you know $y(t)$ and the rate of change $f(y(t), t)$, you can *approximate* the value at a slightly later time $t + h$ by assuming the rate is constant over the interval:

$$y(t + h) \approx y(t) + h \cdot f(y(t), t)$$

This is the forward Euler formula. The step size h controls the trade-off between speed and accuracy: smaller steps give better approximations but require more computation.

Like a watchmaker measuring a gear's position by noting its current angular velocity and projecting forward – a straight-line approximation that works well for tiny intervals but accumulates error over longer spans.

Error analysis. The error introduced by a single Euler step is called the **local truncation error**. It comes from the Taylor expansion:

$$y(t+h) = y(t) + h y'(t) + \frac{h^2}{2} y''(t) + O(h^3)$$

Euler's method keeps only the first two terms, so the local error is $O(h^2)$ – proportional to h^2 . Over an interval of total length T , we take T/h steps, so the **global error** is:

$$\text{global error} = O\left(\frac{T}{h} \cdot h^2\right) = O(h)$$

Euler's method is **first-order**: halving the step size halves the error. This is adequate for understanding the concept, but too slow for serious computation. We will improve on it shortly.

```
def euler_method(f, y0, t_span, h):
    """Solve dy/dt = f(y, t) using forward Euler with step size h."""
    t0, tf = t_span
    if h <= 0 or tf < t0:
        raise ValueError("require h > 0 and tf >= t0")
    times = [float(t0)]
    values = [np.atleast_1d(np.asarray(y0, dtype=float))]
    while times[-1] < tf:
        step = min(h, tf - times[-1])
        slope = np.atleast_1d(f(values[-1], times[-1]))
        if slope.shape != values[-1].shape:
            raise ValueError("f must return the same shape as y0")
        values.append(values[-1] + step * slope)
        times.append(times[-1] + step)
    return np.array(times), np.array(values).squeeze()

# Logistic equation: dy/dt = r*y*(1 - y/K)
def logistic_rhs(y, t):
    """Right-hand side of the logistic equation."""
    r, K = 1.0, 10.0
    return r * y * (1 - y / K)

# Solve with different step sizes and compare to exact solution
y0, r, K = 0.1, 1.0, 10.0
t_final = 5.0

print("Euler's method: convergence as h decreases")
print(f"{'h':>10s} {'y_euler(5)':>12s} {'y_exact(5)':>12s} {'error':>12s}")
y_exact = logistic_exact(t_final, y0, r, K)

for h in [1.0, 0.5, 0.1, 0.05, 0.01]:
    t_vals, y_vals = euler_method(logistic_rhs, y0, (0.0, t_final), h)
    y_end = y_vals[-1]
    error = abs(y_end - y_exact)
    print(f"{'h':10.4f} {'float(y_end):12.6f} {'y_exact:12.6f} {'error:12.2e}")

# Verify first-order convergence: error should halve when h halves
_, y_h1 = euler_method(logistic_rhs, y0, (0.0, t_final), 0.1)
_, y_h2 = euler_method(logistic_rhs, y0, (0.0, t_final), 0.05)
ratio = abs(float(y_h1[-1]) - y_exact) / abs(float(y_h2[-1]) - y_exact)
print(f"\nError ratio (h vs h/2): {ratio:.2f} (expected ~2.0 for first-order)")
```


2.7.4 The Runge-Kutta Family

Euler's method is first-order: to cut the error in half, you must double the number of steps. This is wasteful. The **Runge-Kutta family** achieves higher accuracy by evaluating f at multiple intermediate points within each step, like a watchmaker who checks the gear position not just at the start of each tick, but at the midpoint and endpoint as well.

RK2: The Midpoint Method

The simplest improvement over Euler is the **midpoint method** (RK2). Instead of using the slope at the beginning of the interval, it takes a trial Euler step to the midpoint, evaluates the slope there, and uses that slope for the full step:

$$\begin{aligned}k_1 &= f(y_n, t_n) \\k_2 &= f\left(y_n + \frac{h}{2}k_1, t_n + \frac{h}{2}\right) \\y_{n+1} &= y_n + h k_2\end{aligned}$$

The local truncation error is $O(h^3)$ and the global error is $O(h^2)$ – second-order. Halving the step size now reduces the error by a factor of 4.

RK4: The Classic Method

The workhorse of numerical ODEs is the **fourth-order Runge-Kutta method** (RK4). It evaluates f at four points per step and achieves fourth-order accuracy:

$$\begin{aligned}k_1 &= f(y_n, t_n) \\k_2 &= f\left(y_n + \frac{h}{2}k_1, t_n + \frac{h}{2}\right) \\k_3 &= f\left(y_n + \frac{h}{2}k_2, t_n + \frac{h}{2}\right) \\k_4 &= f(y_n + h k_3, t_n + h) \\y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)\end{aligned}$$

The weights $1/6, 2/6, 2/6, 1/6$ are not arbitrary – they are chosen so that the Taylor expansion of y_{n+1} matches the true solution through $O(h^4)$. The local error is $O(h^5)$ and the global error is $O(h^4)$. Halving h reduces the error by a factor of 16.

The **Butcher tableau** is a compact notation for writing down any Runge-Kutta method. For RK4:

0				
$\frac{1}{2}$	$\frac{1}{2}$			
$\frac{1}{2}$	0	$\frac{1}{2}$		
1	0	0	1	
	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$

The left column gives the time offsets for each stage. The lower-triangular body gives the coefficients used to compute each k_i from previous stages. The bottom row gives the weights for the final combination.

RK45: Adaptive Step Size (Dormand-Prince)

A fixed step size is wasteful: some parts of the solution change rapidly (requiring small h) while others change slowly (allowing large h). The **Dormand-Prince method** (RK45) uses two Runge-Kutta formulas of different orders – a fourth-order and a fifth-order – and compares them to estimate the local error. If the error is too large, the step is rejected and h is reduced. If the error is well below the tolerance, h is increased.

This is the method behind `scipy.integrate.solve_ivp` with `method='RK45'` (the default), and it is what you should use in practice for non-stiff problems.

```

def rk4_method(f, y0, t_span, h):
    """Solve dy/dt = f(y, t) using the classical fourth-order Runge-Kutta."""
    t0, tf = t_span
    if h <= 0 or tf < t0:
        raise ValueError("require h > 0 and tf >= t0")
    times = [float(t0)]
    values = [np.atleast_1d(np.asarray(y0, dtype=float))]
    while times[-1] < tf:
        step = min(h, tf - times[-1])
        t = times[-1]
        y = values[-1]

        # Four slope evaluations
        k1 = np.atleast_1d(f(y, t))
        k2 = np.atleast_1d(f(y + step / 2 * k1, t + step / 2))
        k3 = np.atleast_1d(f(y + step / 2 * k2, t + step / 2))
        k4 = np.atleast_1d(f(y + step * k3, t + step))
        if not all(k.shape == y.shape for k in (k1, k2, k3, k4)):
            raise ValueError("f must return the same shape as y0")

        # Weighted combination
        values.append(y + (step / 6) * (k1 + 2 * k2 + 2 * k3 + k4))
        times.append(t + step)
    return np.array(times), np.array(values).squeeze()

# Compare Euler and RK4 with the independent exact solution.

y0, r, K = 0.1, 1.0, 10.0
t_final = 5.0
y_exact = logistic_exact(t_final, y0, r, K)

# Euler with h=0.1
_, y_euler = euler_method(logistic_rhs, y0, (0.0, t_final), 0.1)

# RK4 with h=0.1
_, y_rk4 = rk4_method(logistic_rhs, y0, (0.0, t_final), 0.1)

print(f"Exact solution: {y_exact:.10f}")
print(f"Euler (h=0.1): {float(y_euler[-1]):.10f} error={abs(float(y_euler[-1]) - y_
↪exact):.2e}")
print(f"RK4 (h=0.1): {float(y_rk4[-1]):.10f} error={abs(float(y_rk4[-1]) - y_
↪exact):.2e}")

# RK4 convergence: error should decrease as h^4
print("\nRK4 convergence:")
print(f"{'h':>10s} {'error':>12s} {'ratio':>8s}")
prev_error = None
for h in [0.5, 0.25, 0.125, 0.0625]:
    _, y_rk = rk4_method(logistic_rhs, y0, (0.0, t_final), h)
    err = abs(float(y_rk[-1]) - y_exact)
    ratio_str = f"{prev_error / err:.1f}" if prev_error is not None else "---"
    print(f"{h:10.4f} {err:12.2e} {ratio_str:>8s}")
    prev_error = err

```

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Expected ratio ~16 when h halves (fourth-order convergence)

2.7.5 Systems of Coupled ODEs

So far we have solved scalar ODEs – a single unknown $y(t)$. But the ODEs in population genetics almost always involve **systems**: multiple quantities evolving simultaneously, each one's rate of change depending on the others. The gears of a watch do not turn in isolation; each gear's motion is coupled to its neighbors.

i Calculus Aside – From scalar to vector ODEs

A system of m coupled ODEs can be written as a single vector equation. Define $\mathbf{y} = (y_1, y_2, \dots, y_m)$ as the **state vector**, and $\mathbf{f} = (f_1, f_2, \dots, f_m)$ as the vector of right-hand sides:

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(\mathbf{y}, t)$$

This is the same as writing m separate equations:

$$\begin{aligned}\frac{dy_1}{dt} &= f_1(y_1, y_2, \dots, y_m, t) \\ \frac{dy_2}{dt} &= f_2(y_1, y_2, \dots, y_m, t) \\ &\vdots \\ \frac{dy_m}{dt} &= f_m(y_1, y_2, \dots, y_m, t)\end{aligned}$$

The key point: each f_i can depend on *all* components of $\mathbf{y}$, not just y_i . This coupling is what makes systems interesting and what requires us to solve all equations simultaneously.

Every numerical method we have discussed (Euler, RK4, etc.) extends directly to vector ODEs. You simply replace scalar y with vector $\mathbf{y}$ and scalar f with vector $\mathbf{f}$. The formulas are identical; only the data types change from floats to arrays.

Example: exact coalescent lineage-count probabilities. Let $p_k(t) = P(K_t = k \mid K_0 = n)$. From state k , Kingman's coalescent jumps only to $k - 1$, at rate $c_k = \binom{k}{2}$. The Kolmogorov forward equations are therefore

$$\frac{dp_k}{dt} = c_{k+1}p_{k+1} - c_k p_k, \quad k = 1, \dots, n,$$

where $c_1 = 0$ and $p_{n+1} = 0$. The gain into state k exactly matches loss from state $k + 1$, so summing the equations gives $d \sum_k p_k / dt = 0$. Unlike the earlier mean-field equation, this system is an exact description of the Kingman lineage-count process [Kingman, 1982].

The `moments` software also evolves a coupled ODE system, but for expected sample-frequency-spectrum entries. Its operators and time scaling should be taken from the original derivation and implementation rather than inferred by treating adjacent frequency classes as a generic birth–death chain [Jouanous *et al.*, 2017].

```
def coalescent_count_ode(probabilities, t):
    """Forward equation for Kingman lineage-count probabilities.

    Parameters
    -----
    probabilities : ndarray of shape (n,)
```

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```

    Entry k-1 is P(K_t=k), for k=1,...,n.
    t : float
    Current time (unused for this time-homogeneous process).
    """
    probabilities = np.asarray(probabilities, dtype=float)
    if probabilities.ndim != 1 or len(probabilities) < 1:
        raise ValueError("probabilities must be a nonempty vector")
    n = len(probabilities)
    derivative = np.zeros(n)
    for k in range(1, n + 1):
        loss = k * (k - 1) / 2 * probabilities[k - 1]
        gain = 0.0
        if k < n:
            gain = (k + 1) * k / 2 * probabilities[k]
        derivative[k - 1] = gain - loss
    return derivative

# For n=3, an independent closed form is available.
p0 = np.array([0.0, 0.0, 1.0])
times, probabilities = rk4_method(coalescent_count_ode, p0, (0.0, 2.0), 0.001)
p_numeric = probabilities[-1]
t = times[-1]
p3 = np.exp(-3 * t)
p2 = 1.5 * (np.exp(-t) - np.exp(-3 * t))
p_exact = np.array([1 - p2 - p3, p2, p3])
print("Lineage-count probabilities at t=2:")
print("  numerical:", p_numeric)
print("  exact:      ", p_exact)
print("  total:      ", p_numeric.sum())

```

2.7.6 Stiffness and Implicit Methods

Not all ODEs are created equal. Some systems contain processes operating on **wildly different timescales**. Consider a watch with a seconds hand, a minute hand, and a tourbillon rotating once per hour – trying to track all three with the same tiny time step (sized for the fastest hand) is enormously wasteful. But using a large time step (sized for the slowest hand) makes the fast component oscillate and blow up.

This phenomenon is called **stiffness**. A system is stiff when the ratio of the fastest to slowest timescale is large. Explicit methods like Euler and RK4 are forced to take tiny steps – not for accuracy, but for **stability**. Even though the fast component decays quickly and becomes negligible, an explicit method must resolve it at every step or the numerical solution will oscillate wildly and diverge.

In population genetics, stiffness arises when:

- **Strong selection meets weak drift:** Selection changes allele frequencies on a timescale of $1/s$ generations, while drift operates on a timescale of $2N$ generations. If $2Ns \gg 1$, the system is stiff.
- **Migration between populations of very different sizes:** Fast migration equilibrates allele frequencies quickly, but population size changes happen slowly.
- **Multi-population SFS computation:** The moment equations for large sample sizes can have eigenvalues spanning many orders of magnitude.

The **backward (implicit) Euler method** addresses stiffness by evaluating f at the *next* time point rather than the current

one:

$$y_{n+1} = y_n + h \cdot f(y_{n+1}, t_{n+1})$$

Note that y_{n+1} appears on *both* sides of the equation. This means we must solve an equation (or a system of equations) at each step. For linear ODEs, this reduces to solving a linear system; for nonlinear ODEs, it requires Newton's method or a similar iterative procedure.

The trade-off: implicit methods require more work per step. Backward Euler is **A-stable** for the scalar linear test equation $y' = \lambda y$ with $\text{Re}(\lambda) < 0$, so its stability does not impose the explicit Euler bound. That does not mean every implicit method is stable for every nonlinear problem or that arbitrarily large steps are accurate; nonlinear solves can fail and accuracy still constrains the step size.

BDF methods (Backward Differentiation Formulas) are a family of implicit multi-step methods that generalize backward Euler. They are the method behind `scipy.integrate.solve_ivp` with `method='BDF'`, and they are the standard choice for stiff ODE systems in population genetics.

```
def backward_euler_decay(rate, y0, T, h):
    """Solve y'=-rate*y with backward Euler."""
    if rate < 0 or h <= 0 or T < 0:
        raise ValueError("require rate >= 0, h > 0, and T >= 0")
    t = 0.0
    y = float(y0)
    while t < T:
        step = min(h, T - t)
        y /= 1 + rate * step
        t += step
    return y

rate, y0_stiff, T, h = 1000.0, 1.0, 0.1, 0.01
_, explicit = euler_method(
    lambda y, t: -rate * y, y0_stiff, (0.0, T), h
)
implicit = backward_euler_decay(rate, y0_stiff, T, h)
exact = np.exp(-rate * T)
print("Stiff decay y'=-1000y at t=0.1 with h=0.01")
print(f" forward Euler: {float(explicit[-1]):.6e} (unstable)")
print(f" backward Euler: {implicit:.6e} (stable, but not exact)")
print(f" exact: {exact:.6e}")
```

Probability Aside – Stiffness in the moments Timepiece

When the moments Timepiece (*Timepiece X: moments*) computes the SFS under strong selection, the ODE system becomes stiff. The selection scaled selection coefficient introduces a short timescale in coalescent units, which can be orders of magnitude shorter than the drift timescale. The `moments` package uses implicit methods (specifically a Crank–Nicolson update in its integration code) together with controlled time steps [Jouganous *et al.*, 2017].

2.7.7 The Matrix Exponential

Many ODE systems in population genetics are **linear**: the right-hand side is a matrix times the state vector. These linear systems have a beautiful closed-form solution involving the **matrix exponential**.

A linear ODE system has the form:

$$\frac{dy}{dt} = Ay$$

where A is an $m \times m$ matrix of constant coefficients and $\mathbf{y}(t)$ is the state vector. The solution is:

$$\mathbf{y}(t) = \exp(At) \mathbf{y}(0)$$

i Calculus Aside – What is a matrix exponential?

The matrix exponential $\exp(M)$ of a square matrix M is defined by the same power series as the scalar exponential:

$$\exp(M) = I + M + \frac{M^2}{2!} + \frac{M^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{M^k}{k!}$$

where I is the identity matrix and M^k means M multiplied by itself k times. This series always converges for any square matrix.

To verify that $\mathbf{y}(t) = \exp(At)\mathbf{y}(0)$ solves $d\mathbf{y}/dt = A\mathbf{y}$, differentiate the series term by term:

$$\frac{d}{dt} \exp(At) = A + A^2t + \frac{A^3t^2}{2!} + \cdots = A \left(I + At + \frac{A^2t^2}{2!} + \cdots \right) = A \exp(At)$$

So $d\mathbf{y}/dt = A \exp(At)\mathbf{y}(0) = A\mathbf{y}(t)$, confirming the solution.

If A is diagonalizable with **eigendecomposition** $A = V\Lambda V^{-1}$ where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_m)$ is the diagonal matrix of eigenvalues, then:

$$\exp(At) = V \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_m t}) V^{-1}$$

This identity is useful for symmetric or otherwise well-conditioned matrices, especially at repeated times after one decomposition. It is not a general prescription: defective matrices are not diagonalizable, and ill-conditioned eigenvectors can destroy accuracy. General `expm` routines instead use stable algorithms such as scaling and squaring.

Connection to population genetics. The matrix exponential appears in several Timepieces:

- **SMC++** (*Timepiece II: SMC++*): The probability of j undistinguished lineages remaining at time t is computed via the matrix exponential of the lineage rate matrix. The states track the number of remaining lineages, and the rate matrix encodes coalescence events.
- **momi2** (*Timepiece XII: momi2*): The Moran model transition probabilities over an epoch of duration t are computed as $\exp(Qt)$, where Q is the Moran rate matrix. The eigendecomposition of Q is known in closed form, making this computation efficient.
- **Migration models**: When k populations exchange migrants at constant rates, the allele frequency dynamics within each population form a linear ODE, and the matrix exponential gives the exact solution.

Let us implement a concrete example: a simple **two-population migration model** where allele frequencies in two populations evolve under symmetric migration.

```
def migration_matrix(m_rate, n_pops=2):
    """Rate matrix for symmetric migration between n_pops populations.

    Off-diagonal: A[i,j] = m_rate (gain from pop j).
    Diagonal: A[i,i] = -(n_pops-1)*m_rate. Symmetry makes both rows
    and columns sum to zero, conserving the component sum.
    """
    if m_rate < 0 or not isinstance(n_pops, (int, np.integer)) or n_pops < 2:
        raise ValueError("require m_rate >= 0 and integer n_pops >= 2")
```

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```

A = np.full((n_pops, n_pops), m_rate, dtype=float)
np.fill_diagonal(A, -(n_pops - 1) * m_rate)
return A

def symmetric_matrix_exponential(A, t):
    """Compute exp(A*t) for a real symmetric matrix."""
    A = np.asarray(A, dtype=float)
    if A.ndim != 2 or A.shape[0] != A.shape[1] or not np.allclose(A, A.T):
        raise ValueError("A must be a real symmetric square matrix")
    eigenvalues, eigenvectors = np.linalg.eigh(A)
    return (eigenvectors * np.exp(eigenvalues * t)) @ eigenvectors.T

# Two populations with symmetric migration
m_rate = 0.5 # migration rate
A = migration_matrix(m_rate)
print("Rate matrix A:")
print(A)

# Initial frequencies: pop1 has allele at frequency 0.8, pop2 at 0.2
y0 = np.array([0.8, 0.2])

# Solve using matrix exponential at several time points
print("\nAllele frequencies over time (matrix exponential):")
print(f"{'t':>6s} {'pop1':>8s} {'pop2':>8s}")
for t in [0.0, 0.5, 1.0, 2.0, 5.0, 10.0]:
    y_t = symmetric_matrix_exponential(A, t) @ y0
    print(f"{t:6.1f} {y_t[0]:8.4f} {y_t[1]:8.4f}")

# Verify: at equilibrium, both populations should have the same frequency
# (the mean of the initial frequencies)
y_eq = symmetric_matrix_exponential(A, 100.0) @ y0
expected_eq = np.mean(y0)
print(f"\nEquilibrium frequency: {y_eq[0]:.6f} (expected {expected_eq:.6f})")

# Compare with eigendecomposition
eigenvalues, V = np.linalg.eigh(A)
print(f"\nEigenvalues of A: {eigenvalues}")
# For symmetric 2x2 migration: eigenvalues are 0 and -2*m_rate
# The zero eigenvalue corresponds to the stationary distribution
# The negative eigenvalue governs the rate of convergence to equilibrium

# Verify matrix exponential via eigendecomposition
t_test = 2.0
# exp(A*t) = V * diag(exp(lambda_i * t)) * V^{-1}
D = np.diag(np.exp(eigenvalues * t_test))
y_eigen = V @ D @ V.T @ y0
y_expm = symmetric_matrix_exponential(A, t_test) @ y0
print(f"\nAt t={t_test}:")
print(f"  Via expm: {y_expm}")
print(f"  Via eigendecomposition: {np.real(y_eigen)}")
print(f"  Agree: {np.allclose(y_expm, np.real(y_eigen))}")

```

The symmetric construction extends directly to three or more populations. For asymmetric migration, fix a row- or

column-vector convention first: with the column-vector equation used here, conservation of the component sum requires **columns** of A to sum to zero. The matrix can then be non-symmetric, so the `eigh` helper above no longer applies; use a general matrix-exponential routine. Eigenvalues describe modal timescales when the matrix is diagonalizable, but non-normal systems can also show transients not captured by eigenvalues alone.

2.7.8 Summary

Concept	Key Formula or Idea
ODE (initial value problem)	$dy/dt = f(y, t), \quad y(t_0) = y_0$
Euler's method	$y_{n+1} = y_n + h f(y_n, t_n)$ – first-order, $O(h)$
RK4	Four-stage method – fourth-order, $O(h^4)$ global error
RK45 (Dormand-Prince)	Adaptive step size, <code>scipy.integrate.solve_ivp</code> default
Systems of ODEs	$dy/dt = \mathbf{f}(\mathbf{y}, t)$ – vector state
Stiffness	Disparate timescales; use implicit methods (BDF, backward Euler)
Matrix exponential	$\mathbf{y}(t) = \exp(At)\mathbf{y}(0)$ for linear systems
Eigendecomposition	$\exp(At) = V \text{diag}(e^{\lambda_i t}) V^{-1}$

These tools are the gear train that drives the moment equations of the moments Timepiece, the lineage probability system of SMC++, and the Moran model transitions of momi2. Whenever you see an ODE system in this book, you now have the vocabulary and the numerical methods to solve it.

Next: *Markov Chain Monte Carlo* – the algorithm that lets us explore parameter spaces too complex for closed-form solutions.

2.8 Markov Chain Monte Carlo

“When you can’t find the answer, let randomness find it for you – one carefully guided step at a time.”

A master watchmaker, confronted with a broken mechanism of unknown design, cannot simply enumerate every possible gear arrangement to find the one that matches the symptoms. The space of possibilities is too vast. Instead, the watchmaker makes an educated guess, tries a small modification, checks whether the watch runs better, and repeats. The analogy is useful for exploration, but MCMC targets a distribution; it does not generally converge on one “true” mechanism.

Markov Chain Monte Carlo (MCMC) is the mathematical formalization of this strategy. In population genetics, we face exactly the same challenge: given observed sequence data D , we want to infer the genealogical history G – the ancestral recombination graph, the coalescence times, the population sizes. The posterior distribution $P(G \mid D)$ lives in an astronomically large space, and direct computation is impossible. MCMC constructs a random walk through this space, visiting states in proportion to their posterior probability, and thereby producing samples from the distribution we cannot compute directly.

This chapter develops MCMC from the ground up, building toward the specific forms used by ARGweaver (see *MCMC Sampling*), SINGER (see *Sub-graph pruning and regrafting (SGPR)*), and PHLASH (see *Stein Variational Gradient Descent (SVGD)*). If you have not yet read the chapter on *Hidden Markov Models*, the section on Markov chains here will provide the necessary foundation. If you have, the connection between HMMs and MCMC will become clear: both exploit the Markov property, but in very different ways.

2.8.1 The Big Idea: Why Sample?

In Bayesian inference, the goal is to compute the **posterior distribution**:

$$P(G \mid D) = \frac{P(D \mid G) P(G)}{P(D)}$$

where G represents the unknown quantity (a genealogy, a demographic model, a set of parameters) and D is the observed data (genotype sequences, allele frequencies, variant calls).

The posterior tells us everything we want to know: which genealogies are consistent with the data, how certain we are about each one, and what the range of plausible parameter values is. But computing it requires evaluating the normalizing constant $P(D) = \int P(D | G) P(G) dG$, which sums (or integrates) over every possible value of G . For an ARG with thousands of branches, millions of genomic positions, and continuous coalescence times, this integral is intractable.

MCMC sidesteps this problem entirely. Instead of computing the posterior, it **samples** from it. The algorithm constructs a Markov chain – a sequence of states $G^{(0)}, G^{(1)}, G^{(2)}, \dots$ – that converges to the posterior distribution. After warmup, states are still generally **correlated**, but ergodic averages can estimate posterior expectations. We can use the samples to estimate any quantity of interest: posterior means, credible intervals, marginal distributions.

Like a blind watchmaker exploring the space of possible mechanisms – unable to see the full blueprint, but able to feel whether each small adjustment improves the mechanism or not – MCMC navigates the posterior landscape one step at a time, gradually concentrating its exploration on the regions that matter most.

2.8.2 Bayesian Inference in 60 Seconds

Before building the MCMC machinery, let us establish the Bayesian framework that motivates it.

i Probability Aside – Bayes' theorem

Bayes' theorem relates the posterior probability of a hypothesis H given data D to the likelihood and prior:

$$P(H | D) = \frac{P(D | H) P(H)}{P(D)}$$

The four components are:

- $P(H)$ – the **prior**: our belief about H before seeing data.
- $P(D | H)$ – the **likelihood**: how probable the data is if H is true.
- $P(H | D)$ – the **posterior**: our updated belief after seeing data.
- $P(D)$ – the **evidence** (or marginal likelihood): the total probability of the data under all hypotheses. This is the normalizing constant that makes the posterior sum to 1.

The critical insight: $P(D)$ does not depend on H . It is the same for every hypothesis. This means we can write:

$$P(H | D) \propto P(D | H) P(H)$$

The posterior is *proportional* to the likelihood times the prior. MCMC exploits this: it only needs the unnormalized posterior, never the evidence.

Let us make this concrete with an example where the posterior is known exactly, so we can verify our MCMC results later.

The Beta-Binomial model. Suppose we observe k successes in n trials (say, $k = 7$ derived alleles out of $n = 20$ sites), and we want to infer the success probability θ .

- **Prior:** $\theta \sim \text{Beta}(\alpha, \beta)$ with density $p(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$.
- **Likelihood:** $P(k | \theta, n) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$.
- **Posterior:** By Bayes' theorem, the posterior is also a Beta distribution: $\theta | k \sim \text{Beta}(\alpha + k, \beta + n - k)$.

This is a **conjugate** model: the prior and posterior belong to the same family. This is the exception, not the rule – for most real problems (including ARG inference), no conjugate form exists, and we must resort to MCMC.

```
import numpy as np

def beta_binomial_demo():
    """Demonstrate exact Bayesian inference with a conjugate model.

    We use this as a ground truth to verify MCMC results later.

    Returns
    -----
    alpha_post : float
        Posterior alpha parameter.
    beta_post : float
        Posterior beta parameter.
    """
    # Observed data: 7 derived alleles out of 20 sites
    n, k = 20, 7

    # Prior: Beta(2, 2) -- a gentle prior favoring values near 0.5
    alpha_prior, beta_prior = 2, 2

    # Posterior: Beta(alpha + k, beta + n - k)
    alpha_post = alpha_prior + k          # 2 + 7 = 9
    beta_post = beta_prior + (n - k)      # 2 + 13 = 15

    # The posterior mean is alpha / (alpha + beta)
    post_mean = alpha_post / (alpha_post + beta_post)
    post_var = (alpha_post * beta_post) / (
        (alpha_post + beta_post)**2 * (alpha_post + beta_post + 1)
    )

    print(f"Prior: Beta({alpha_prior}, {beta_prior})")
    print(f>Data: {k} successes in {n} trials")
    print(f"Posterior: Beta({alpha_post}, {beta_post})")
    print(f"Posterior mean: {post_mean:.4f}")
    print(f"Posterior std: {np.sqrt(post_var):.4f}")

    return alpha_post, beta_post

alpha_post, beta_post = beta_binomial_demo()
```

2.8.3 Markov Chains

MCMC works by constructing a Markov chain whose stationary distribution is the target posterior. Before building the full algorithm, we need to understand Markov chains themselves.

A **Markov chain** is a sequence of random variables $X_0, X_1, X_2, \dots$ where the distribution of X_{t+1} depends only on X_t , not on earlier values. This is the same Markov property that drives the HMMs in *Hidden Markov Models* – but here the chain operates in *algorithm time* (MCMC iterations) rather than along the genome.

Formally, a Markov chain on a state space $\mathcal{S}$ is defined by a **transition kernel** $T(x, y)$: the probability (or probability density) of moving from state x to state y in one step.

For a finite state space $\{1, 2, \dots, K\}$, the transition kernel is a matrix $T_{ij} = P(X_{t+1} = j \mid X_t = i)$, exactly like the transition matrix of an HMM. Each row sums to 1, meaning the chain must go somewhere at each step.

Stationary Distribution

A probability distribution π over the state space is **stationary** for the chain if it is unchanged by one step of the transition:

$$\pi_j = \sum_i \pi_i T_{ij} \quad \text{for all } j$$

In matrix notation: $\pi T = \pi$. If the chain starts in distribution π , it stays in distribution π forever.

For the finite chains considered here, irreducibility and aperiodicity guarantee convergence to the unique stationary distribution regardless of the starting state:

- **Irreducible:** every state can be reached from every other state (eventually).
- **Aperiodic:** the chain does not get trapped in deterministic cycles.

For such a finite chain, no matter where we start, the distribution of X_t converges to π as $t \rightarrow \infty$. This is the fundamental theorem that makes MCMC work: if we design a chain whose stationary distribution is our target posterior, then running the chain long enough produces samples from that posterior. General state spaces require additional recurrence and regularity conditions; irreducibility and aperiodicity alone are not a universal theorem.

Probability Aside – Detailed balance

A sufficient (but not necessary) condition for π to be stationary is **detailed balance**:

$$\pi_i T_{ij} = \pi_j T_{ji} \quad \text{for all } i, j$$

This says: the probability of being in state i and transitioning to j equals the probability of being in state j and transitioning to i . In other words, the “flow” between every pair of states is balanced.

To see that detailed balance implies stationarity, sum both sides over i :

$$\sum_i \pi_i T_{ij} = \sum_i \pi_j T_{ji} = \pi_j \sum_i T_{ji} = \pi_j$$

where the last step uses $\sum_i T_{ji} = 1$ (the rows of T sum to 1). This gives $\sum_i \pi_i T_{ij} = \pi_j$, which is exactly the stationarity condition. Most MCMC algorithms (including Metropolis-Hastings and Gibbs sampling) are designed to satisfy detailed balance.

Let us see convergence to a stationary distribution in action with a simple finite-state Markov chain.

```
import numpy as np

def markov_chain_convergence():
    """Demonstrate that a finite Markov chain converges to its stationary
    ↪distribution.

    We define a 3-state chain, run it for many steps, and compare the
    empirical state frequencies to the theoretical stationary distribution.
    """
    # A reversible 3-state chain with known stationary distribution.
    # T[i, j] = P(X_{t+1} = j | X_t = i).
    pi = np.array([0.2, 0.3, 0.5])
    T = np.array([
        [0.50, 0.20, 0.30],
        [2/15, 17/30, 0.30],
        [0.12, 0.18, 0.70],
```

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# Verify the defining identities, not just the simulation output.
assert np.allclose(T.sum(axis=1), 1.0), "Rows must sum to 1"
assert np.allclose(pi @ T, pi), "pi must be stationary"
flow = pi[:, None] * T
assert np.allclose(flow, flow.T), "detailed balance must hold"
print(f"Stationary distribution (theory): {pi}")

# Simulate the chain for 100,000 steps starting from state 0
n_steps = 100_000
rng = np.random.default_rng(42)
state = 0
counts = np.zeros(3)

for _ in range(n_steps):
    # np.random.choice(3, p=T[state]) draws the next state
    # according to the transition probabilities from 'state'.
    state = rng.choice(3, p=T[state])
    counts[state] += 1

# Empirical frequencies should match the stationary distribution
empirical = counts / n_steps
print(f"Empirical frequencies: {empirical}")
print(f"Max absolute error: {np.max(np.abs(pi - empirical)):.4f}")

markov_chain_convergence()

```

2.8.4 The Metropolis-Hastings Algorithm

The **Metropolis-Hastings (MH)** algorithm is the workhorse of MCMC [Metropolis *et al.*, 1953, Hastings, 1970]. It constructs an ergodic Markov chain whose stationary distribution is any target distribution $\pi(x)$ that we can evaluate up to a normalizing constant.

The algorithm is remarkably simple:

1. From the current state x , **propose** a new state x' from a proposal distribution $q(x' | x)$.
2. Compute the **acceptance ratio**:

$$\alpha = \min \left(1, \frac{\pi(x') q(x | x')}{\pi(x) q(x' | x)} \right)$$

3. **Accept** x' with probability α ; otherwise stay at x .

The acceptance ratio corrects for proposal asymmetry and makes π invariant when the forward and reverse proposal densities required by the ratio are defined. Convergence from an arbitrary starting point additionally requires an ergodic chain: the proposal must let the chain reach every relevant part of the target and must not force a deterministic cycle. A poor proposal can therefore be formally valid yet practically unusable.

Probability Aside – Why the MH ratio works

We need to verify that the MH algorithm satisfies detailed balance with respect to π . The transition probability from

x to x' is:

$$T(x, x') = q(x' | x) \cdot \min \left(1, \frac{\pi(x')q(x | x')}{\pi(x)q(x' | x)} \right)$$

To check detailed balance, we need $\pi(x)T(x, x') = \pi(x')T(x', x)$.

Without loss of generality, assume $\pi(x')q(x | x') \leq \pi(x)q(x' | x)$. Then the acceptance ratio for moving $x \rightarrow x'$ is $\frac{\pi(x')q(x | x')}{\pi(x)q(x' | x)}$, and the acceptance ratio for the reverse move $x' \rightarrow x$ is 1. So:

$$\begin{aligned} \pi(x)T(x, x') &= \pi(x) \cdot q(x' | x) \cdot \frac{\pi(x')q(x | x')}{\pi(x)q(x' | x)} \\ &= \pi(x') \cdot q(x | x') \end{aligned}$$

$$\begin{aligned} \pi(x')T(x', x) &= \pi(x') \cdot q(x | x') \cdot 1 \\ &= \pi(x') \cdot q(x | x') \end{aligned}$$

Both sides are equal. Detailed balance holds, so π is the stationary distribution.

The key insight: the normalizing constant of π cancels in the ratio $\pi(x')/\pi(x)$. This is why MCMC does not need to compute the evidence $P(D)$ – it only needs the unnormalized posterior.

Random walk Metropolis-Hastings. The simplest choice of proposal is a symmetric random walk: $x' = x + \epsilon$ where $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Since the proposal is symmetric ($q(x' | x) = q(x | x')$), the ratio simplifies to:

$$\alpha = \min \left(1, \frac{\pi(x')}{\pi(x)} \right)$$

If the proposed state has higher posterior density, always accept. If lower, accept with probability equal to the density ratio. This allows the chain to explore regions of lower density (important for characterizing uncertainty) while spending most of its time in high-density regions.

Let us implement reusable random-walk MH and test it on the Beta posterior derived above. This is deliberately a problem with an analytic answer: agreement with exact posterior moments is a stronger correctness check than a visually plausible trace.

```
import numpy as np

def random_walk_metropolis(log_target, initial, proposal_scale,
                           n_samples, rng):
    """Sample a one-dimensional target with a symmetric Normal proposal."""
    if n_samples < 2 or proposal_scale <= 0 or not np.isfinite(initial):
        raise ValueError("invalid sampler arguments")
    current_logp = float(log_target(initial))
    if not np.isfinite(current_logp):
        raise ValueError("initial state must have finite target density")

    samples = np.empty(n_samples)
    samples[0] = initial
    accepted = 0
    for t in range(1, n_samples):
        proposal = samples[t - 1] + rng.normal(scale=proposal_scale)
        proposal_logp = float(log_target(proposal))
        log_ratio = proposal_logp - current_logp
        if np.log(rng.random()) < min(0.0, log_ratio):
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        samples[t] = proposal
        current_logp = proposal_logp
        accepted += 1
    else:
        samples[t] = samples[t - 1]
    return samples, accepted / (n_samples - 1)

def beta_log_kernel(theta, alpha, beta):
    """Unnormalized log density of Beta(alpha, beta)."""
    if not 0.0 < theta < 1.0:
        return -np.inf
    return (alpha - 1) * np.log(theta) + (beta - 1) * np.log1p(-theta)

alpha_post, beta_post = 9, 15
rng = np.random.default_rng(42)
chain, acceptance = random_walk_metropolis(
    lambda x: beta_log_kernel(x, alpha_post, beta_post),
    initial=0.5, proposal_scale=0.15, n_samples=50_000, rng=rng,
)
posterior = chain[5_000:]

exact_mean = alpha_post / (alpha_post + beta_post)
exact_var = (alpha_post * beta_post) / (
    (alpha_post + beta_post)**2 * (alpha_post + beta_post + 1)
)

print(f"Acceptance rate: {acceptance:.3f}")
print(f"Mean: {posterior.mean():.4f} (exact {exact_mean:.4f})")
print(f"Variance: {posterior.var():.5f} (exact {exact_var:.5f})")

```

The Normal random walk proposes values outside (0, 1), which are correctly rejected because their target density is zero. A transformed proposal on the logit scale can be more efficient, but then its Jacobian or the corresponding asymmetric Hastings correction must be included.

2.8.5 Gibbs Sampling

Gibbs sampling is a special case of Metropolis-Hastings where the proposal is drawn from the **full conditional distribution** – the distribution of one variable given all the others. The remarkable property of Gibbs sampling is that every proposal is accepted.

To see why, suppose we are updating variable x_k while holding all other variables x_{-k} fixed. The Gibbs proposal draws x'_k from:

$$q(x'_k \mid x_k, x_{-k}) = P(x'_k \mid x_{-k})$$

Now compute the MH acceptance ratio. The target distribution is the joint $\pi(x_k, x_{-k})$, and the proposal is $q(x'_k \mid x_k, x_{-k}) = \pi(x'_k \mid x_{-k})$. Substituting into the MH ratio:

$$\begin{aligned} \alpha &= \min \left(1, \frac{\pi(x'_k, x_{-k}) q(x_k \mid x'_k, x_{-k})}{\pi(x_k, x_{-k}) q(x'_k \mid x_k, x_{-k})} \right) \\ &= \min \left(1, \frac{\pi(x'_k, x_{-k}) \cdot \pi(x_k \mid x_{-k})}{\pi(x_k, x_{-k}) \cdot \pi(x'_k \mid x_{-k})} \right) \end{aligned}$$

Using the identity $\pi(x_k, x_{-k}) = \pi(x_k \mid x_{-k}) \cdot \pi(x_{-k})$:

$$\alpha = \min \left(1, \frac{\pi(x'_k \mid x_{-k}) \pi(x_{-k}) \cdot \pi(x_k \mid x_{-k})}{\pi(x_k \mid x_{-k}) \pi(x_{-k}) \cdot \pi(x'_k \mid x_{-k})} \right) = \min(1, 1) = 1$$

Every Gibbs proposal is accepted, but acceptance is not the same as efficiency: strongly coupled variables can still produce a highly autocorrelated Gibbs chain. The other requirement is that we can sample from the full conditional $P(x_k | x_{-k})$ exactly, which is not always possible.

Connection to ARGweaver. ARGweaver (see *MCMC Sampling*) uses Gibbs sampling to update its ARG. At each iteration, one haplotype's "thread" through the ARG is removed, and a new thread is sampled from the conditional posterior $P(\text{thread}_k | \text{ARG}_{-k}, D)$. Because the time-discretized HMM allows exact computation of this conditional via the forward algorithm and stochastic traceback, the Gibbs update is exact **within ARGweaver's time-discretized model** and the acceptance rate is 1 [Rasmussen *et al.*, 2014]. This is precisely the strategy described in the *HMM chapter*: the forward algorithm computes state probabilities, and stochastic traceback samples a path.

Let us implement Gibbs sampling for a bivariate Normal distribution, where the conditional distributions are known analytically.

```
import numpy as np

def gibbs_bivariate_normal():
    """Gibbs sampling from a bivariate Normal distribution.

    Target: (X, Y) ~ N(mu, Sigma) where
        mu = (0, 0)
        Sigma = [[1, rho], [rho, 1]]

    The conditional distributions are:
        X | Y=y ~ N(rho*y, 1 - rho^2)
        Y | X=x ~ N(rho*x, 1 - rho^2)

    These conditionals are easy to sample from, making Gibbs ideal.
    """
    rho = 0.8                # correlation coefficient
    cond_var = 1 - rho**2    # conditional variance
    cond_std = np.sqrt(cond_var)

    n_samples = 20_000
    samples = np.zeros((n_samples, 2))
    samples[0] = [0.0, 0.0] # starting point

    for t in range(1, n_samples):
        # Update X given Y: X | Y ~ N(rho * Y, 1 - rho^2)
        y_current = samples[t - 1, 1]
        samples[t, 0] = np.random.normal(rho * y_current, cond_std)

        # Update Y given X: Y | X ~ N(rho * X, 1 - rho^2)
        x_current = samples[t, 0] # use the NEWLY sampled X
        samples[t, 1] = np.random.normal(rho * x_current, cond_std)

    burn_in = 1000
    post_burnin = samples[burn_in:]

    # Check moments against the true distribution
    print(f"Correlation rho = {rho}")
    print(f"Mean X: {post_burnin[:, 0].mean():.4f} (expected 0.0)")
    print(f"Mean Y: {post_burnin[:, 1].mean():.4f} (expected 0.0)")
    print(f"Var X: {post_burnin[:, 0].var():.4f} (expected 1.0)")
```

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print(f"Var Y: {post_burnin[:, 1].var():.4f} (expected 1.0)")
empirical_corr = np.corrcoef(post_burnin[:, 0], post_burnin[:, 1])[0, 1]
print(f"Empirical correlation: {empirical_corr:.4f} (expected {rho})")

# Gibbs always accepts, so acceptance rate is 1.0
print(f"Acceptance rate: 1.0 (by construction)")

return post_burnin

np.random.seed(42)
gibbs_samples = gibbs_bivariate_normal()

```

2.8.6 Convergence Diagnostics

Running an MCMC chain is only half the battle. How do we know the chain has converged to the target distribution? How many samples are actually useful? These questions are addressed by **convergence diagnostics**.

Warmup and initial transients. Early draws can retain substantial dependence on the starting point. Warmup is also the usual phase for adapting proposal parameters; for ordinary MCMC, adaptation is then frozen before collecting draws. Discarding a fixed fraction does not make a chain converge. Run multiple chains from dispersed initial values and extend them until diagnostics and quantities of interest are stable; persistent disagreement calls for a better parameterization or sampler, not merely a larger discarded fraction.

Trace plots. A trace plot shows the sampled values as a function of iteration number. A poorly mixed chain can show slow drifts, long repeated stretches, or different regions in different chains. A plausible-looking trace is useful evidence but cannot prove convergence.

Autocorrelation. Consecutive MCMC samples are correlated (each sample is a small perturbation of the previous one). The **autocorrelation function** (ACF) at lag k measures this:

$$\rho(k) = \frac{\text{Cov}(X_t, X_{t+k})}{\text{Var}(X_t)}$$

For a well-mixed chain, the ACF decays quickly to zero. For a poorly mixed chain, it remains positive for many lags, meaning consecutive samples carry redundant information.

Thinning. To reduce autocorrelation, we can keep only every m -th sample. Thinning can reduce storage costs, but it usually discards useful information and does not repair poor mixing. It is generally better to retain draws and quantify their autocorrelation through ESS.

i Probability Aside – Effective sample size

The **effective sample size** (ESS) quantifies how many *independent* samples your chain is equivalent to. If you have N total samples with autocorrelation, the ESS is:

$$\text{ESS} = \frac{N}{1 + 2 \sum_{k=1}^{\infty} \rho(k)}$$

where $\rho(k)$ is the autocorrelation at lag k . The denominator is called the **integrated autocorrelation time** (IAT) τ :

$$\tau = 1 + 2 \sum_{k=1}^{\infty} \rho(k)$$

and $\text{ESS} = N/\tau$. If $\tau = 10$, then 10,000 MCMC samples are worth only about 1,000 independent samples.

High autocorrelation (large τ , small ESS) means the chain is exploring slowly – each new sample does not move far from the previous one. This is the central diagnostic of MCMC efficiency: a well-tuned algorithm has small τ and large ESS.

Rank-normalized split $\hat{R}$. Modern $\hat{R}$ compares split, rank-normalized chains and is paired with bulk and tail ESS [Vehtari *et al.*, 2021]. Values near 1 are necessary but not sufficient evidence of mixing; $\hat{R} > 1.01$ is a useful warning threshold, not a proof-producing pass/fail rule. Always inspect several chains and the estimands that matter.

```
import numpy as np

def compute_acf_and_ess(chain, max_lag=200):
    """Compute the autocorrelation function and effective sample size.

    Parameters
    -----
    chain : ndarray of shape (N,)
        MCMC samples (after burn-in).
    max_lag : int
        Maximum lag to compute ACF for.

    Returns
    -----
    acf : ndarray of shape (max_lag + 1,)
        Autocorrelation at each lag from 0 to max_lag.
    ess : float
        Estimated effective sample size.
    """
    chain = np.asarray(chain, dtype=float)
    if chain.ndim != 1 or len(chain) < 4 or not np.all(np.isfinite(chain)):
        raise ValueError("chain must be a finite one-dimensional array")
    N = len(chain)
    max_lag = min(int(max_lag), N - 1)
    if max_lag < 2:
        raise ValueError("max_lag must allow at least one lag pair")
    centered = chain - chain.mean()
    var = np.dot(centered, centered) / N
    if var == 0:
        return np.ones(max_lag + 1), 1.0

    # Compute autocorrelation at each lag
    acf = np.zeros(max_lag + 1)
    for k in range(max_lag + 1):
        # Autocorrelation at lag k:
        # rho(k) = (1/N) * sum_{t=0}^{N-k-1} (x_t - mean)(x_{t+k} - mean) / var
        if k == 0:
            acf[k] = 1.0
        else:
            # chain[:-k] is the series shifted by 0 (first N-k elements)
            # chain[k:] is the series shifted by k (last N-k elements)
            acf[k] = np.dot(centered[:-k], centered[k:]) / ((N - k) * var)

    # Geyer's paired initial-positive sequence, made non-increasing.
```

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```

# Pair rho(1)+rho(2), rho(3)+rho(4), ... to stabilize the cutoff.
paired_sum = 0.0
previous_pair = np.inf
for k in range(1, max_lag, 2):
    pair = acf[k] + acf[k + 1]
    if pair <= 0:
        break
    pair = min(pair, previous_pair)
    paired_sum += pair
    previous_pair = pair

iat = max(1.0, 1.0 + 2.0 * paired_sum)
ess = min(float(N), N / iat)

return acf, ess

# Simple MH chain targeting N(0,1) with different step sizes
def run_mh_chain(sigma, n_samples=20_000, seed=42):
    """Run MH targeting standard Normal with step size sigma."""
    rng = np.random.default_rng(seed)
    chain = np.zeros(n_samples)
    chain[0] = 5.0 # start far from the mode
    n_acc = 0
    for t in range(1, n_samples):
        proposal = chain[t-1] + rng.normal(0, sigma)
        log_alpha = -0.5 * proposal**2 + 0.5 * chain[t-1]**2
        if np.log(rng.random()) < min(0.0, log_alpha):
            chain[t] = proposal
            n_acc += 1
        else:
            chain[t] = chain[t-1]
    return chain[2000:], n_acc / (n_samples - 1)

for sigma in [0.1, 1.0, 2.4, 10.0]:
    chain, acc_rate = run_mh_chain(sigma)
    acf, ess = compute_acf_and_ess(chain)
    print(f"sigma={sigma:.1f}: acceptance={acc_rate:.3f}, "
          f"ESS={ess:.0f}, IAT={len(chain)/ess:.1f}, "
          f"ACF at lag 10={acf[10]:.3f}")

```

2.8.7 Practical Considerations

The theoretical foundations of MCMC are elegant, but making MCMC work well in practice requires careful attention to several practical issues. These issues are not mere technicalities – they determine whether your MCMC run produces useful results in hours or useless results in weeks.

Proposal Tuning

The most critical practical decision in random walk MH is the **proposal step size** σ . This is the standard deviation of the Normal perturbation $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

- **Too small** ($\sigma \ll 1$): Almost every proposal is accepted (the new state is barely different from the old one), but the chain explores very slowly. It takes many steps to traverse the parameter space. The autocorrelation is high and the

ESS is low.

- **Too large** ($\sigma \gg 1$): Most proposals land in regions of very low posterior density and are rejected. The chain gets stuck at one location for many steps before a proposal is finally accepted. Again, autocorrelation is high and ESS is low.
- **Just right**: There is a sweet spot where the chain makes reasonably large moves that are accepted often enough to explore efficiently. The famous **23.4%** limit applies asymptotically to a particular high-dimensional random-walk scaling regime for product-like targets [Roberts *et al.*, 1997]; the corresponding one-dimensional limit is about 44%. These are tuning clues, not universal targets. Efficiency is better assessed using ESS per unit of computation.

```
import numpy as np

def proposal_tuning_demo():
    """Demonstrate the effect of proposal step size on MCMC efficiency.

    We run MH targeting a 5-dimensional standard Normal with different
    step sizes and compare acceptance rates and ESS.
    """
    d = 5          # dimensionality
    n_samples = 20_000
    rng = np.random.default_rng(42)

    def log_target(x):
        """Log density of a d-dimensional standard Normal."""
        return -0.5 * np.sum(x**2)

    results = []
    for sigma in [0.05, 0.2, 0.5, 1.0, 2.4 / np.sqrt(d), 3.0, 10.0]:
        chain = np.zeros((n_samples, d))
        chain[0] = np.zeros(d)
        n_accepted = 0

        for t in range(1, n_samples):
            proposal = chain[t-1] + rng.normal(0, sigma, size=d)
            log_alpha = log_target(proposal) - log_target(chain[t-1])
            if np.log(rng.random()) < min(0.0, log_alpha):
                chain[t] = proposal
                n_accepted += 1
            else:
                chain[t] = chain[t-1]

        acc_rate = n_accepted / (n_samples - 1)

        # Compute ESS for the first component
        burn_in = 2_000
        first_coord = chain[burn_in:, 0]
        _, ess = compute_acf_and_ess(first_coord, max_lag=400)
        results.append((sigma, acc_rate, ess))
        print(f"sigma={sigma:.3f}: acceptance={acc_rate:.3f}, ESS={ess:.0f}")

    best = max(results, key=lambda r: r[2])
    print(f"\nLargest estimated ESS: sigma={best[0]:.3f}, "
          f"acceptance={best[1]:.3f}, ESS={best[2]:.0f}")
```

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`proposal_tuning_demo()`

Data-Informed Proposals

Random walk proposals are simple but uninformed – they do not use the data to guide the exploration. This is like a blind watchmaker making random adjustments without listening to whether the mechanism sounds better or worse.

SINGER's innovation (see *Sub-graph pruning and regrafting (SGPR)*) is to replace the random walk with a **data-informed proposal**. In SINGER's SGPR (Sub-Graph Pruning and Re-grafting) move, a piece of the ARG is removed and then re-threaded using the forward algorithm and stochastic traceback from the *HMM chapter*. This proposal incorporates the observed sequence data directly: the HMM “listens” to the data and proposes a new thread that is already likely under the posterior.

The proposal is designed to achieve high acceptance and efficient movement in the settings evaluated by SINGER [Deng *et al.*, 2025]. The realized acceptance rate is model- and data-dependent; neither an HMM-informed proposal nor a rate near one by itself guarantees good global mixing.

Parallel Tempering

Multimodal posteriors (distributions with multiple well-separated peaks) are notoriously difficult for standard MCMC. The chain can get trapped in one mode for a very long time before finding enough momentum to jump to another.

Parallel tempering (also called replica exchange) addresses this by running multiple chains at different “temperatures.” A chain at temperature T targets the tempered distribution $\pi(x)^{1/T}$. At $T = 1$, this is the original posterior. At $T > 1$, the distribution is flattened – the valleys between modes are shallower, making it easier to cross between them.

Periodically, adjacent-temperature chains propose to swap their states. The swap acceptance probability ensures that the $T = 1$ chain still targets the correct posterior. Hot chains explore broadly and pass their discoveries to colder chains.

When MCMC Is Not Enough

Sometimes the parameter space is so large, or the posterior is so complex, that even well-tuned MCMC cannot converge in a reasonable time. This is the situation faced by **PHLASH** (see *Stein Variational Gradient Descent (SVGD)*), which needs to infer a high-dimensional population size history.

Instead of MCMC, PHLASH uses **Stein Variational Gradient Descent (SVGD)** – a deterministic optimization method that maintains a collection of “particles” and moves them to approximate the posterior. In PHLASH's reported experiments, this gradient-based particle approximation gives favorable speed and accuracy for population-size inference [Terhorst, 2025]. That empirical result should not be generalized into a theorem that SVGD is always faster than MCMC in high dimensions.

The trade-off: SVGD is an approximation (it may not converge to the exact posterior), while an invariant, ergodic, correctly implemented MCMC chain is asymptotically exact for its specified target. For the specific problem PHLASH solves – inferring piecewise-constant population size histories – the speed advantage of SVGD outweighs the loss in exactness.

2.8.8 MCMC in Population Genetics: Three Applications

The Timepieces in this book use three different strategies for exploring posterior distributions. Each one can be understood as a variation on the MCMC theme developed in this chapter.

ARGweaver: Gibbs Sampling over ARGs

ARGweaver (see *MCMC Sampling*) uses **Gibbs sampling** to explore the space of ancestral recombination graphs. At each iteration, one haplotype's thread is removed from the ARG, and a new thread is sampled from the exact conditional posterior $P(\text{thread}_k \mid \text{ARG}_{-k}, D)$.

This is possible because ARGweaver discretizes time (see *Time Discretization*), reducing the continuous coalescent to a finite-state HMM. The forward algorithm (from *Hidden Markov Models*) computes the conditional posterior exactly within that discretized model, and stochastic traceback draws a sample. Because the draw comes from the full conditional, its Gibbs acceptance rate is 1; successive ARG states can nevertheless remain strongly correlated [Rasmussen *et al.*, 2014].

The cost of this elegance is the time discretization itself: it introduces an approximation, and the number of HMM states grows with the number of time points and samples. An acceptance rate of 100% does not remove the need to assess mixing.

SINGER: MH with Data-Informed Proposals

SINGER (see *Sub-graph pruning and regrafting (SGPR)*) also updates the ARG one thread at a time, but it works in **continuous time** – no discretization is needed. The SGPR (Sub-Graph Pruning and Re-grafting) move removes a sub-graph from the ARG and re-threads it using the branch sampling and time sampling HMMs.

Because the proposal distribution is constructed from the data (via the HMM forward algorithm), it closely approximates the posterior. The Metropolis-Hastings acceptance ratio can therefore be high, though it is not identically 1 as in Gibbs sampling. The key formula (derived in *Sub-graph pruning and regrafting (SGPR)*) compares the probability of the old thread under the new proposal to the probability of the new thread under the old proposal, combined with the prior ratio.

SINGER's innovation is that the data-informed proposal makes MCMC practical for continuous-time ARG inference – something that would be hopelessly inefficient with random walk proposals.

PHLASH: Beyond MCMC

PHLASH (see *Stein Variational Gradient Descent (SVGD)*) takes a different path entirely. Instead of sampling from the posterior via MCMC, it uses **Stein Variational Gradient Descent (SVGD)** to directly approximate the posterior with a set of particles.

Why abandon MCMC? PHLASH infers a high-dimensional population size history $\eta(t)$, parameterized as a piecewise-constant function with many epochs. The parameter space is large enough that MCMC chains converge very slowly, and the composite likelihood used by PHLASH (which approximates the full likelihood for computational efficiency) makes exact Gibbs updates unavailable.

SVGD maintains a collection of particles and iteratively moves them to minimize a divergence from the posterior. Each particle update uses the gradient of the log-posterior (the “score function,” see *The Score Function Algorithm*), making the exploration more directed than random walk MCMC. The method produced fast inference in the experiments reported for PHLASH, while targeting an approximation rather than supplying MCMC's asymptotic invariance guarantee [Terhorst, 2025].

2.8.9 Summary

Concept	Key Idea
Bayesian posterior	$P(G \mid D) \propto P(D \mid G) P(G)$; the normalizing constant $P(D)$ is intractable
Markov chain	A sequence where the next state depends only on the current state; characterized by a transition kernel and stationary distribution
Detailed balance	$\pi(x)T(x, y) = \pi(y)T(y, x)$; sufficient condition for π to be stationary
Metropolis-Hastings	Propose, then accept/reject with ratio $\min(1, \frac{\pi(x')q(x x')}{\pi(x)q(x' x)})$; requires valid forward/reverse support and an ergodic chain
Gibbs sampling	Propose from the full conditional; always accepted; requires tractable conditionals
Effective sample size	$ESS = N / (1 + 2 \sum_k \rho(k))$; measures how many independent samples the chain yields
Proposal tuning	The 23.4% limit is specific to high-dimensional product-target asymptotics; tune using ESS per unit of computation
Data-informed proposals	SINGER's SGPR uses HMM-based proposals that incorporate the data, often yielding high acceptance in reported applications
When MCMC fails	High-dimensional or complex posteriors may require alternatives like SVGD (used by PHLASH)

These tools form the **winding mechanism** of several Timepieces. ARGweaver (see [Timepiece V: ARGweaver](#)) uses Gibbs sampling to cycle through haplotype threads, producing exact conditional updates. SINGER (see [Timepiece VII: SINGER](#)) uses Metropolis-Hastings with data-informed SGPR proposals to explore the space of continuous-time ARGs. And PHLASH (see [Timepiece XIV: PHLASH](#)) demonstrates when the MCMC paradigm itself must be transcended, replacing random sampling with gradient-guided particle optimization. Understanding the strengths and limitations of MCMC – when it shines and when it is not enough – is essential for understanding why each Timepiece is built the way it is.

TIMEPIECES

Every great watch begins as a pile of parts on the bench.

Each Timepiece is a complete algorithm from population genetics, disassembled and rebuilt from scratch. Like a watchmaker laying out parts on the bench, we examine every gear before assembling the whole mechanism. Nothing is taken on faith. Nothing is hidden behind “see supplementary materials.”

3.1 Verification Status

Disclaimer

The code examples in each Timepiece are checked by chapter-specific automated tests, including mathematical invariants and upstream numerical fixtures where available. **No Timepiece has been independently verified by a domain expert.** If you find an error – mathematical, pedagogical, or computational – please open an issue. The table below shows the current verification status.

The Timepieces are grouped by what they do, with verification status for each.

Simulators – tools that generate ground truth

#	Timepiece	Tests	Verified	What it does
IV	<i>msprime</i>	209	2026-09-01	Hudson simulation and mutation generation, including gap recombination and discrete-site recurrent/back mutations checked against upstream source behavior.
XVI	<i>SLiM</i>	67	–	Forward-time simulation with natural selection. The forge that builds what the coalescent cannot.
XVI	<i>discoal</i>	32	2026-09-01	Coalescent simulation with selective sweeps via trajectory + structured coalescent.

Demographic inference – estimating population size history

#	Timepiece	Tests	Verified	What it does
I	<i>PSMC</i>	205	2026-09-01	Population size history from a single diploid genome; continuous-chain normalization checked against analytic infinite-tail oracles.
II	<i>SMC++</i>	112	–	Extends PSMC to multiple unphased genomes with a distinguished lineage approach.
XIII	<i>Gamma-SMC</i>	107	–	Ultrafast pairwise TMRCA inference with gamma-distributed posteriors.
XIV	<i>PHLASH</i>	130	–	GPU-accelerated Bayesian inference of population size history via SVGD.

SFS-based demographic inference – using the site frequency spectrum

#	Timepiece	Tests	Verified	What it does
X	<i>moments</i>	162	–	Demographic inference from the SFS using moment equations.
XI	<i>dadi</i>	84	–	Demographic inference from the SFS by solving the Wright-Fisher diffusion PDE.
XII	<i>mom2</i>	140	–	Demographic inference from the SFS via coalescent tensor algebra.

Genealogy and ARG inference – reconstructing ancestral histories

#	Timepiece	Tests	Verified	What it does
III	<i>Li & Stephens HMM</i>	179	2026-09-01	PAC likelihood and copying HMM, checked against Appendix A, Ishmm 0.0.8, and brute-force haploid and diploid recursions.
V	<i>ARGweaver</i>	125	2026-09-01	Exact Jukes–Cantor emissions, diploid coalescent scaling, discretized recombination, and bounded teaching approximations checked against the primary paper and official C++ source (fee3d32); the upstream executable was built and completed a finite two-iteration sampling run.
VI	<i>tsinfer</i>	21	2026-09-01	Ancestor grouping, carrier-filtered extension, two-root construction, probability transforms, dense Viterbi, path coordinates, compression eligibility, parsimony, and flank handling checked against the primary paper and official 0.1.4/0.4.1 Python implementations; the teaching code is explicitly bounded below production tree-sequence inference.
VII	<i>SINGER</i>	35	2026-09-01	Published branch/time HMM equations, mutation-clock rescaling, and SGPR acceptance checked against the paper supplement and official C++ source; teaching code explicitly bounded below full ARG-sampler parity.
VIII	<i>Threads</i>	56	–	Deterministic ARG inference at biobank scale with PBWT pre-filtering.
XVI	<i>Relate</i>	37	2026-09-01	Genome-wide genealogy estimation via asymmetric painting + MCMC branch lengths.

Dating and selection – calibrating genealogies and detecting selection

#	Timepiece	Tests	Verified	What it does
IX	<i>tsdate</i>	139	–	Dates tree sequence nodes using the molecular clock.
XV	<i>CLUES</i>	93	–	Full-likelihood estimation of selection coefficients from gene trees and ancient DNA.

3.2 Timepiece I: PSMC

The Pairwise Sequentially Markovian Coalescent

3.2.1 The Mechanism at a Glance

PSMC is an algorithm that infers **population size history** $N(t)$ from a **single diploid genome**. It looks at the pattern of heterozygous and homozygous sites along the genome and asks: what demographic history best explains this pattern?

Input: One diploid genome → a sequence of 0s and 1s (hom/het)

Output: $\hat{N}(t)$ (effective population size as a function of time)

The key insight: at each position along the genome, the two copies of the chromosome share a **most recent common ancestor (MRCA)**. The time to this MRCA – the **coalescence time** – varies along the genome because of recombination. And the distribution of coalescence times is shaped by the population size history.

- **More heterozygous sites** = the two copies diverged long ago = large $N(t)$ in the past (large populations take longer to coalesce)
- **Fewer heterozygous sites** = the two copies diverged recently = small $N(t)$ (bottleneck forced rapid coalescence)

PSMC reads these patterns with a Hidden Markov Model.

If the coalescent is the escapement – the fundamental ticking mechanism – then PSMC is the simplest complete watch you can build around it: just two hands (two haplotypes), a single gear train (the HMM), and a dial that reads out population size through time.

Primary Reference

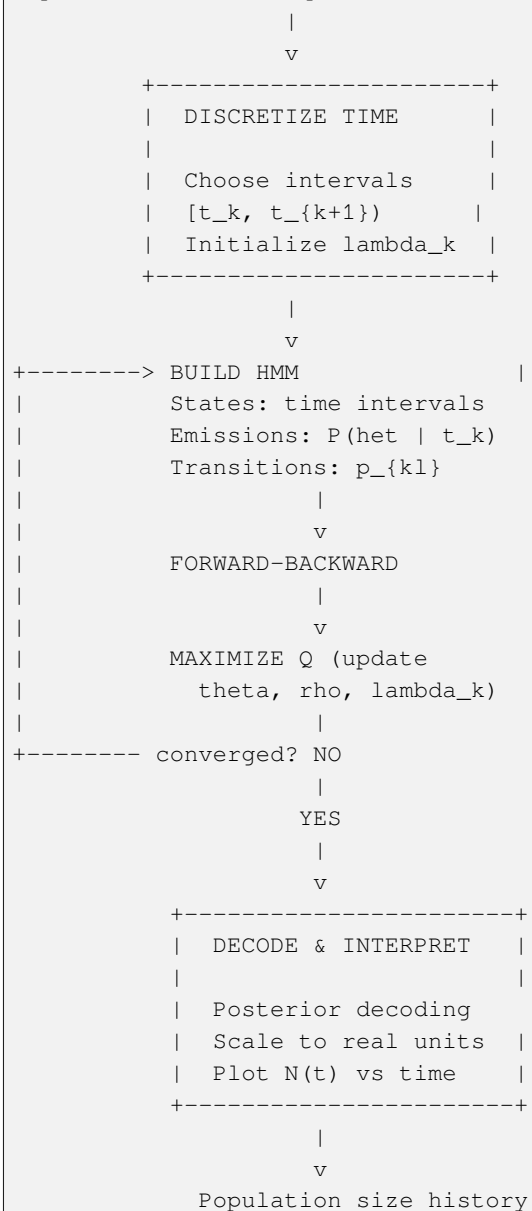
[Li and Durbin, 2011]

The four gears of PSMC:

1. **The Continuous-Time Model** (the escapement) – The transition density $q(t|s)$ that describes how the coalescence time changes between adjacent positions, under variable population size $N(t)$. This is where the coalescent theory from *The Workbench* comes alive.
2. **Discretization** (the gear train) – Converting the continuous coalescence time into discrete time intervals, with a transition matrix p_{kl} suitable for an HMM. Continuous math becomes finite computation.
3. **The HMM and EM** (the mainspring) – The complete Hidden Markov Model: hidden states are time intervals, observations are het/hom, and the Expectation-Maximization algorithm estimates $N(t)$. This is the engine that iteratively refines our estimate.
4. **Decoding the Clock** (the case and dial) – Posterior decoding, scaling to real units, bootstrapping, and interpreting the population size history. The final step: reading the watch.

These gears mesh together into a complete inference machine:

Diploid consensus sequence (het/hom along genome)



3.2.2 Why Just Two Sequences?

You might wonder: why would anyone analyze just two sequences when we have methods like SINGER that handle many? Three reasons:

1. **Data efficiency:** A single high-coverage diploid genome contains millions of informative sites. Two haplotypes recombine enough times along a full chromosome to trace population size changes over hundreds of thousands of years.
2. **Computational simplicity:** With only two sequences, the marginal tree at each position is trivial – just a single coalescence time. No tree topology to infer, no branching structure. Just a number: *when did these two copies last share an ancestor?* This makes PSMC the ideal first Timepiece – the simplest watch you can build that actually tells useful time.

3. **Historical importance:** PSMC was published in 2011, when whole-genome data from single individuals was becoming available. It showed that a single genome contains a remarkable amount of information about the species' past.

The simplicity of PSMC makes it an ideal first Timepiece to understand, because the PSMC transition density reappears as a gear inside SINGER's time sampling step (Timepiece VII). And the very next Timepiece – SMC++ (*Timepiece II*) – generalizes PSMC from two sequences to many. We start here and build the complete watch around it.

i Prerequisites for this Timepiece

Before starting PSMC, you should have worked through:

- *Coalescent Theory* – the exponential distribution of coalescence times and the concept of coalescent time units
- *Hidden Markov Models* – the forward-backward algorithm and scaling
- *The SMC* – the Markov approximation and the basic PSMC transition density for constant population size

If you've read those chapters, you have all the tools you need. If some concepts are rusty, we'll point you back to the relevant sections as they come up.

3.2.3 Chapters

Overview of PSMC

Before assembling the watch, lay out every part and understand what it does.

Imagine holding the simplest watch that still tells useful time. It has just two hands – no complications, no chronograph, no moon phase. Yet those two hands, driven by a precise internal mechanism, can tell you something profound: what time it is. The PSMC is exactly this kind of instrument. It takes the two haploid copies of a single diploid genome – just two “hands” – and from nothing more than the pattern of similarities and differences between them, reads out the population size history of an entire species stretching back hundreds of thousands of years.

This chapter lays out every part of the PSMC on the watchmaker's bench. We will not assemble anything yet. Instead, we will name each gear, explain what it does, and show how the parts fit together into a working mechanism. The actual assembly – the derivations, the code, the verification – happens in the chapters that follow.

If you have not yet worked through the prerequisite chapters on *coalescent theory*, *hidden Markov models*, and *the SMC approximation*, now is a good time. Those chapters built the fundamental tools – the escapement, the gear train, and the Markov property – that PSMC assembles into its first complete Timepiece.

What Does PSMC Do?

Given a single diploid genome, PSMC infers the **effective population size** $N(t)$ as a function of time.

i What is “effective population size”?

The **effective population size** N_e is not a head-count of every individual alive at some moment in the past. It is a more subtle quantity: the size of an idealized population (a Wright-Fisher population, as described in *Coalescent Theory*) that would produce the same patterns of genetic variation as the real population. Real populations have unequal sex ratios, overlapping generations, population structure, and many other complications. The effective population size absorbs all of these complications into a single number that captures their net effect on genetic drift.

Concretely, when PSMC reports $N(t) = 50,000$ at some time in the past, it means: “the level of genetic diversity in this genome is consistent with a population that behaved, genetically speaking, as if it had 50,000 randomly mating diploid individuals at that time.” The actual census population could have been larger (if population structure reduced effective mixing) or smaller (though this is less common).

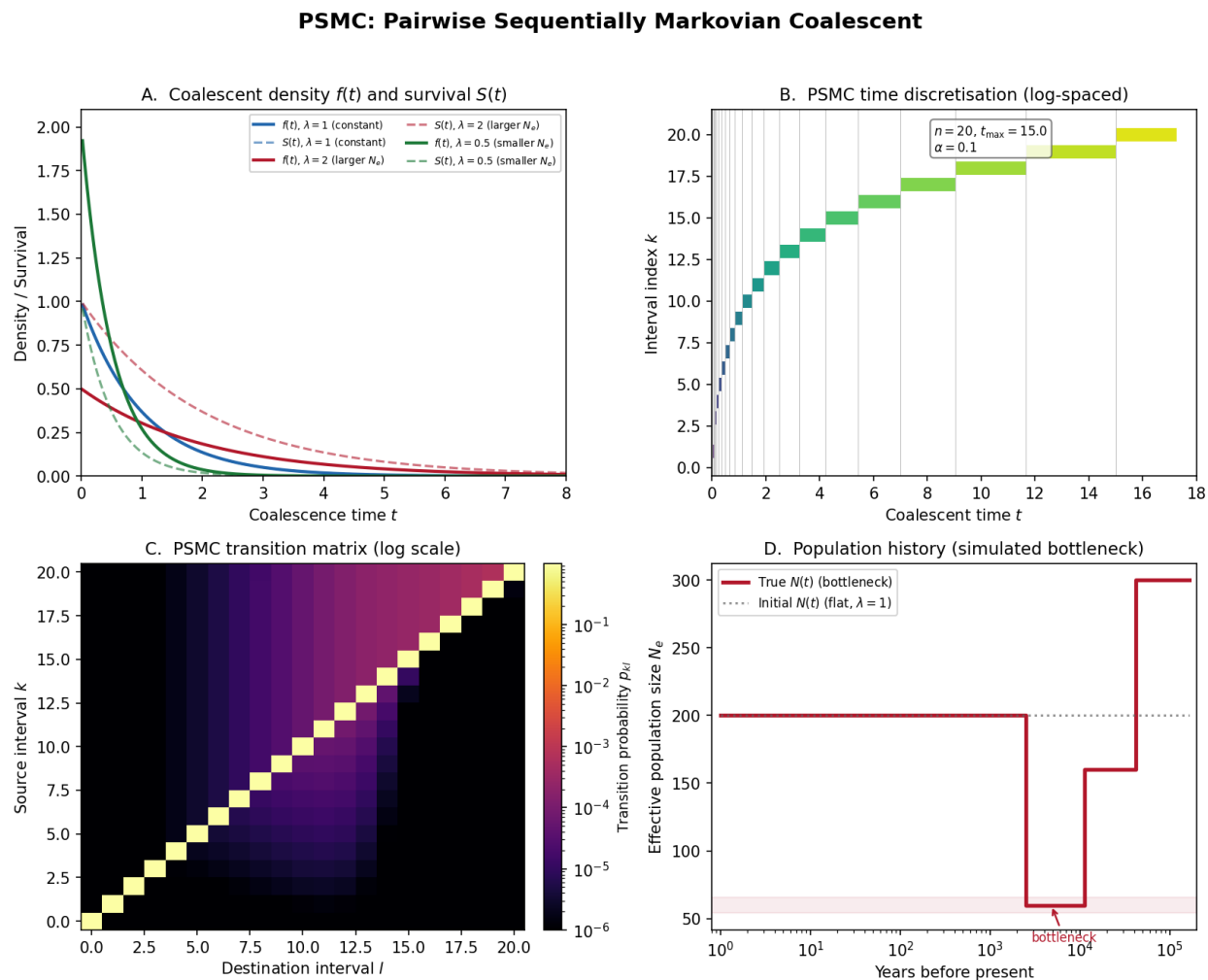


Fig. 1: PSMC at a glance. Panel A: Coalescent density and survival functions under different population sizes – the statistical foundation that connects coalescence times to demography. Panel B: Log-spaced time discretization converting continuous time into HMM states. Panel C: The transition matrix heatmap showing how coalescence times change between adjacent genomic bins. Panel D: Population size reconstruction – true vs inferred $N(t)$ under a bottleneck scenario, demonstrating that the algorithm recovers the correct demographic history.

Think of it this way: the effective population size is like the “effective temperature” of a room. A room might have a heater in one corner and a window in another, but the thermometer on the wall gives you a single number that summarizes the thermal environment. The effective population size similarly summarizes the genetic environment.

The input to PSMC is deceptively simple: a sequence of 0s and 1s, where 0 means “homozygous” (the two chromosome copies are identical in this region) and 1 means “heterozygous” (they differ). Each entry corresponds to a genomic **bin** – a window of s base pairs (typically $s = 100$).

What is a “bin” and why do we bin the data?

A **bin** is simply a contiguous window of s base pairs along the genome. We divide the entire genome into non-overlapping bins and summarize each one with a single bit: 1 if at least one heterozygous site exists within the window, 0 otherwise.

Why not work at single-base-pair resolution? Two reasons. First, **computational tractability**: a human genome has roughly 3 billion base pairs, and running the HMM forward-backward algorithm on a sequence of 3 billion positions would be extremely slow. Binning into 100-bp windows reduces the sequence length by a factor of 100, to about 30 million – still long, but manageable. Second, **information content**: at the per-site level, mutations are very rare (roughly 1 in 1,000 base pairs is heterozygous in humans), so most individual sites are uninformative 0s. Binning aggregates the signal without losing much information, because the probability of a bin being heterozygous is a smooth function of the coalescence time (as we will see in the emission probability below).

Input: $X_1, X_2, \dots, X_L \in \{0, 1\}$

Output: $\hat{N}(t) = N_0 \hat{\lambda}(t)$, $t \in [0, t_{\max}]$

where N_0 is a reference population size and $\lambda(t)$ is the relative population size at time t (so $\lambda(t) = 1$ means the population is at its reference size, $\lambda(t) = 2$ means twice as large, etc.).

The Core Physical Picture

At every position along the genome, your two chromosome copies share a most recent common ancestor (MRCA). The time to this MRCA is the **coalescence time** T – a concept we developed in detail in *Coalescent Theory*.

```
Copy 1 -----+
                | <-- coalescence time T
Copy 2 -----+
                |
                MRCA
                |
                (past)
```

This is why PSMC is a “two-hand watch.” Each chromosome copy is one hand. Their meeting point in the past – the MRCA – is the time being measured. The heterozygous sites scattered along the genome are the tick marks on the dial: each one records a mutation that occurred since the two copies diverged. More tick marks in a region means the hands were further apart (longer coalescence time); fewer tick marks means they were closer together (shorter coalescence time).

Three things determine the pattern of heterozygosity along the genome:

1. **Mutations accumulate proportionally to time.** A site is heterozygous if a mutation occurred on either lineage since the MRCA. Longer coalescence time T = more opportunity for mutations = more heterozygous sites. This is the link between the hidden quantity (time) and the observable quantity (heterozygosity) – the gearing ratio that connects the internal mechanism to the dial.

2. **Recombination reshuffles the genealogy.** At recombination breakpoints, the genealogy changes: the two copies may trace back to a different MRCA with a different coalescence time. So T varies along the genome. As we saw in the [SMC chapter](#), the SMC approximation makes this variation Markov: the coalescence time at the next bin depends only on the coalescence time at the current bin, not on the entire history.
3. **Population size shapes the distribution of T .** In a large population, it takes longer for two lineages to find a common ancestor (there are many individuals, so the chance of sharing a parent is low). In a small population (a bottleneck), coalescence happens quickly. The distribution of T is therefore an imprint of the population size history $N(t)$. This is the fundamental insight that makes PSMC possible: the genome remembers its demographic past in the pattern of coalescence times, and that pattern is visible through the lens of heterozygosity.

Why a Hidden Markov Model?

We have a sequence of coalescence times $T_1, T_2, \dots, T_L$ – one per genomic bin – that we cannot directly observe. What we *can* observe is the heterozygosity pattern $X_1, X_2, \dots, X_L$. The coalescence times evolve along the genome (changing at recombination breakpoints), and at each bin, the coalescence time “emits” an observation (heterozygous or homozygous) with a probability that depends on the time.

This is precisely the situation that Hidden Markov Models are designed to handle, as we developed in the [HMM chapter](#). Recall the structure:

- A sequence of **hidden states** that evolve according to a Markov chain
- At each position, the hidden state **emits** an observation
- We observe the emissions and want to infer the hidden states (or, in PSMC’s case, the parameters that govern the hidden state dynamics)

The HMM is the **gear train** connecting the escapement (the coalescent process that generates coalescence times) to the dial (the population size history we read off at the end). It transmits information from the raw observations through a series of precise computational steps – forward-backward, expectation, maximization – and converts it into an estimate of $N(t)$.

The analogy to watchmaking is almost exact. In a mechanical watch, the escapement (a balance wheel oscillating back and forth) produces a regular beat. The gear train takes that beat and translates it into the smooth sweep of hands across a dial. In PSMC, the coalescent process (lineages coalescing at rates governed by population size) produces a pattern of coalescence times. The HMM takes that pattern and translates it into an estimate of the population size function $\lambda(t)$.

Here is how the PSMC maps onto the HMM framework:

HMM Component	In PSMC
Hidden states	Discretized coalescence time intervals $[t_k, t_{k+1})$. The continuous coalescence time is approximated by a finite set of intervals, turning the problem from infinite-dimensional to finite.
Observations	Heterozygous (1) or homozygous (0) at each genomic bin. These are the tick marks on the watch dial.
Transitions	How coalescence time changes between adjacent bins. This is governed by the recombination rate and the population size history, as derived in the SMC chapter .
Emissions	Probability of seeing a het/hom given coalescence time $T = t$. Longer coalescence times produce more heterozygous sites, because mutations have had more time to accumulate.
Parameters to estimate	θ_0 (mutation rate), ρ_0 (recombination rate), λ_k (relative population sizes in each time interval)

Terminology

Before going further, let us establish the notation we will use throughout the PSMC chapters. Every symbol here has a concrete physical meaning – take a moment to understand each one, because they will reappear in every derivation that follows.

Term	Definition
Coalescence time T_a	The time to the MRCA of the two haplotypes at genomic position a , measured in coalescent time units (multiples of $2N_0$ generations, as established in <i>Coalescent Theory</i>). This is the hidden variable that PSMC seeks to infer.
Heterozygous site	A position where the two haplotypes differ ($X_a = 1$). Each heterozygous site is evidence of a mutation that occurred since the MRCA – a tick mark on the dial.
Homozygous site	A position where the two haplotypes are identical ($X_a = 0$). The absence of a tick mark – either no mutation occurred, or (very rarely) two mutations at the same site cancelled each other out.
Bin	A genomic window of s base pairs (default $s = 100$). $X_a = 1$ if at least one heterozygous site exists in the bin; $X_a = 0$ otherwise. Bins are the resolution at which PSMC reads the genome.
θ_0	Population-scaled mutation rate per bin: $4N_0\mu s$, where μ is the per-base-pair per-generation mutation rate and s is the bin size. This parameter controls how rapidly tick marks accumulate on the dial as coalescence time increases.
ρ_0	Population-scaled recombination rate per bin: $4N_0rs$, where r is the per-base-pair per-generation recombination rate. This parameter controls how frequently the genealogy changes between adjacent bins – how often the hidden state transitions.
$\lambda(t)$	Relative population size at time t : $N(t) = N_0\lambda(t)$. This is what PSMC ultimately estimates. A value of $\lambda(t) = 1$ means the population is at its reference size; $\lambda(t) = 2$ means twice as large; $\lambda(t) = 0.5$ means half as large.
$\Lambda(t)$	Cumulative coalescence intensity: $\Lambda(t) = \int_0^t \frac{du}{\lambda(u)}$. This quantity measures the cumulative “difficulty” of avoiding coalescence up to time t . When the population is small ($\lambda(u)$ is small), coalescence is likely and Λ increases rapidly. When the population is large, Λ increases slowly. You can think of $\Lambda(t)$ as a “stretched clock” – it runs fast during bottlenecks and slow during expansions. This will be derived carefully in the <i>continuous model chapter</i> .
N_0	Reference effective population size. This is a scaling parameter: PSMC estimates relative sizes $\lambda(t)$, and N_0 converts them to absolute sizes. In practice, N_0 is recovered from the estimated θ_0 via $N_0 = \theta_0/(4\mu s)$.
t	Time measured in coalescent time units: multiples of $2N_0$ generations. As established in <i>Coalescent Theory</i> , this is the natural timescale of the coalescent, where the coalescence rate for two lineages in a constant-size population is exactly 1.

i Why measure time in units of $2N_0$ generations?

This is the natural timescale of the coalescent, as we derived in *Coalescent Theory*. In a diploid population of size N , the expected time for two randomly chosen gene copies to coalesce is $2N$ generations. By measuring time in these units, the coalescence rate becomes 1 when $\lambda(t) = 1$, which simplifies all the math. The coalescence rate at time t is $1/\lambda(t)$: larger populations mean lower coalescence rates.

Parameters

PSMC estimates three types of parameters. Together, they fully specify the HMM and determine the population size history.

Parameter	Symbol	Physical meaning
Mutation rate	θ_0	Controls the density of tick marks on the dial. A higher θ_0 means more heterozygous sites per unit coalescence time, making the coalescence time easier to “read” from the data.
Recombination rate	ρ_0	Controls how frequently the genealogy changes along the genome. A higher ρ_0 means more transitions between coalescence time states, which provides more independent “readings” of the population size at different times.
Population sizes	$\lambda_0, \dots, \lambda_n$	The relative population size in each time interval. These are the primary quantities of interest – the numbers that will be plotted as the population size history.

i What does “piecewise constant” mean?

PSMC assumes that $\lambda(t)$ is a **piecewise constant function**. This means the time axis is divided into $n + 1$ intervals $[t_0, t_1), [t_1, t_2), \dots, [t_n, t_{n+1})$, and within each interval the relative population size is a single constant value λ_k .

Picture a staircase. Each step is flat (constant population size within an interval), and the steps can be at different heights (different population sizes in different intervals). The function jumps from one height to the next at the interval boundaries but is constant in between.

This is not a biological assumption – nobody believes population size actually changed in discrete jumps. It is a **computational convenience**: it turns the problem of estimating a continuous function $\lambda(t)$ (which has infinitely many degrees of freedom) into the problem of estimating a finite set of numbers $\lambda_0, \dots, \lambda_n$. More intervals give finer resolution but risk overfitting to noise. Fewer intervals are more robust but might miss real demographic events.

The intervals are chosen to be approximately **log-spaced** in time, giving fine resolution in the recent past (where the data is most informative) and coarse resolution in the distant past. Adjacent intervals can share the same λ parameter to further reduce overfitting – this is what the `-p` pattern in the PSMC software controls.

i What does the θ_0/ρ_0 option do?

The original program’s `-r` option supplies the **initial** ratio θ_0/ρ_0 (default 5). It does not hold that ratio fixed during inference: the C implementation stores θ_0 and ρ_0 as separate free parameters and the numerical M-step updates both. The ratio matters because the two rates can be difficult to disentangle from one diploid genome, so a sensible starting value helps the optimizer. Treating `-r` as a hard biological constraint would misdescribe both the original software and the miniature implementation used here.

The Data: From BAM to Binary Sequence

The input to PSMC is prepared in two steps. The process converts a complex genomic dataset into the simple binary sequence that the HMM operates on.

1. **Call consensus:** From aligned reads (a BAM file), generate a diploid consensus sequence where each base is labeled as heterozygous or homozygous. This step uses standard variant calling tools and quality filters to distinguish genuine heterozygous sites from sequencing errors.
2. **Bin:** Divide the genome into non-overlapping windows of s base pairs (default 100). Mark a bin as “1” if it contains at least one heterozygous site, “0” otherwise. Bins with insufficient data (low coverage or poor mapping quality) are marked as “N” and excluded from the analysis.

The following code simulates this process. Instead of starting from real genomic data, we simulate the underlying coalescent process directly, which lets us verify that PSMC can recover the true parameters.

The simulator below samples an exponential draw on the **cumulative-hazard** scale and numerically inverts it. This distinction matters when $\lambda(t)$ changes with time: adding a single exponential wait based on the population size at the starting time is only valid for a constant-size epoch that extends forever.

```
def simulate_psmc_input(L, theta, rho, lambda_func, n_bins=None):
    """Simulate a PSMC input sequence.

    Parameters
    -----
    L : int
        Number of bins.
    theta : float
        Mutation rate per bin.
    rho : float
        Recombination rate per bin.
    lambda_func : callable
        lambda_func(t) returns relative population size at time t.
    n_bins : ignored
        Kept for API compatibility.

    Returns
    -----
    seq : ndarray of shape (L,), dtype=int
        Binary sequence: 0 = homozygous, 1 = heterozygous.
    coal_times : ndarray of shape (L,)
        True coalescence times.
    """
    if L <= 0:
        raise ValueError("L must be positive")
    if theta < 0 or rho < 0:
        raise ValueError("theta and rho must be non-negative")

    seq = np.zeros(L, dtype=int)
    coal_times = np.zeros(L)

    def _sample_coalescence_time(lambda_func, t_start=0.0):
        """Sample coalescence time under variable population size."""
        target = np.random.exponential(1.0)

        def integrated_hazard(t):
```

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```

    value, _ = quad(lambda u: 1.0 / lambda_func(u), t_start, t)
    return value

lam_start = float(lambda_func(t_start))
if not np.isfinite(lam_start) or lam_start <= 0:
    raise ValueError("lambda_func must return finite positive values")

upper = t_start + max(lam_start * target, 1e-12)
for _ in range(80):
    hazard = integrated_hazard(upper)
    if not np.isfinite(hazard):
        raise ValueError("lambda_func produced a non-finite hazard")
    if hazard >= target:
        return brentq(
            lambda t: integrated_hazard(t) - target,
            t_start,
            upper,
        )
    upper = t_start + 2.0 * (upper - t_start)
raise ValueError("cumulative coalescence hazard did not reach sample target")

t = _sample_coalescence_time(lambda_func)
coal_times[0] = t

for a in range(L):
    p_het = 1 - np.exp(-theta * t)
    seq[a] = np.random.binomial(1, p_het)
    coal_times[a] = t

    if a < L - 1:
        if np.random.random() < 1 - np.exp(-rho * t):
            u = np.random.uniform(0, t)
            t = _sample_coalescence_time(lambda_func, t_start=u)

return seq, coal_times

```

```

# Simulate a short genome.
# np.random.seed(42) makes the random number generator reproducible:
# you will get the same output every time you run this code.
np.random.seed(42)

# lambda t: 1.0 is a Python "lambda function" -- an anonymous, one-line
# function. It takes one argument (t) and returns 1.0 regardless of t.
# This represents a constant population size (lambda(t) = 1 for all t).
seq, times = simulate_psmc_input(10000, theta=0.001, rho=0.0005,
                                lambda_func=lambda t: 1.0)

print(f"Sequence length: {len(seq)}")
print(f"Fraction heterozygous: {seq.mean():.4f}")
print(f"Mean coalescence time: {times.mean():.4f}")

```

The Flow in Detail

Now that we have named all the parts, let us see how they mesh together into a working mechanism. The flow below shows the complete PSMC pipeline, from raw data to population size history. Each step uses concepts from the prerequisite chapters; we note the connections explicitly.

The process is iterative: PSMC starts with an initial guess for the population sizes, builds an HMM from that guess, uses the forward-backward algorithm to compute expected statistics of the hidden coalescence times, then updates its guess to better explain the data. This cycle repeats until convergence. This is the **Expectation-Maximization (EM) algorithm** – a general-purpose method for estimating parameters of models with hidden variables.

```
Diploid genome
|
v
Call consensus + bin into 0/1/N
(Convert the raw genome into a binary sequence that the HMM can read)
|
v
X = (0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, ...)
|
v
Choose time intervals  $t_0 < t_1 < \dots < t_n$  (log-spaced discretization)
Initialize  $\lambda_k = 1$  for all  $k$  (start with constant population)
|
v
+----> Build transition matrix  $p_{\{kl\}}$ 
|      (How likely is the coalescence time to change from interval  $k$ 
|      to interval  $l$  between adjacent bins? Uses the SMC transition
|      density from the SMC chapter, discretized into intervals.)
|
|      Build emission probabilities  $e_k(0), e_k(1)$ 
|      (How likely is a bin to be het/hom given coalescence time in
|      interval  $k$ ? Uses the Poisson mutation model.)
|
|      |
|      v
|      Forward-backward algorithm on  $X$ 
|      (Compute the probability of each hidden state at each position,
|      given the observations. This is the HMM machinery from the
|      HMM chapter, applied to the PSMC's specific states and emissions.)
|
|      |
|      v
|      Compute expected counts (E-step)
|      (How much time does the genome "spend" in each coalescence time
|      interval? How many transitions occur between intervals?)
|
|      |
|      v
|      Maximize  $Q$  to update  $\theta, \rho, \lambda_k$  (M-step)
|      (Given the expected counts, find the parameter values that
|      maximize the expected log-likelihood.)
|
|      |
|      Converged?
|      NO ---+
|
YES
```

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```
|
v
Output: theta_0, rho_0, lambda_0, ..., lambda_n
|
v
Scale to real units: N(t) = theta_0 / (4 * mu * s) * lambda_k
(Convert from coalescent time units back to years and individuals,
 using the known per-generation mutation rate mu and generation time.)
|
v
Plot N(t) vs time
```

Each step in this pipeline corresponds to a chapter in the PSMC Timepiece:

- **Building the transition matrix** requires deriving the continuous-time transition density under variable population size, then discretizing it. This is covered in *The Continuous-Time PSMC Model* and *Discretizing Time*.
- **Forward-backward and EM** apply the HMM algorithms from the *HMM chapter* to PSMC's specific model. The details are in *The PSMC HMM and EM Algorithm*.
- **Scaling and interpretation** are covered in *Decoding the Clock*.

Ready to Build

You now have the high-level blueprint. Every part is laid out on the bench, named, and explained. You know what the input looks like (a binary sequence of het/hom calls), what the output is (a piecewise-constant population size history), and what mechanism connects the two (an HMM whose hidden states are discretized coalescence times).

In the following chapters, we build each gear from scratch:

1. *The Continuous-Time PSMC Model* – The continuous-time transition density under variable $N(t)$. This generalizes the constant-population result from the *SMC chapter* to the case PSMC actually cares about: a population whose size changes over time. This is the escapement – the fundamental mathematical mechanism.
2. *Discretizing Time* – Discretizing time into intervals for the HMM. Continuous math must become finite computation before we can run algorithms on a computer. This chapter converts the continuous transition density into a discrete transition matrix.
3. *The PSMC HMM and EM Algorithm* – The complete HMM and EM parameter estimation. With the transition matrix and emission probabilities in hand, we assemble the full inference engine and estimate θ_0 , ρ_0 , and λ_k .
4. *Decoding the Clock* – Decoding, scaling, and interpreting the result. We convert from HMM parameters back to real-world units and learn to read the watch face.

Each chapter derives the math, explains the intuition, implements the code, and verifies it works – following the watchmaker's philosophy of building understanding gear by gear.

Let us start with the foundation: the continuous-time model.

The Continuous-Time PSMC Model

The escapement: the mathematical heartbeat that makes the whole mechanism tick.

This chapter derives the continuous-time foundations of PSMC. We start from the simplest case (constant population size) and generalize to variable $N(t)$. By the end, you'll have derived the transition density, the stationary distribution, and the recombination probability – all from first principles.

The transition density is the tick-to-tick mechanism of the PSMC watch – it describes how the coalescence time at one genomic position relates to the coalescence time at the next. The stationary distribution is the long-run equilibrium where

the watch runs steadily. And variable population size is like a watch whose mainspring tension changes over time, altering the rhythm of the ticks without breaking the mechanism.

If you've read the *SMC prerequisite*, you've already seen the constant-population version of the PSMC transition density. Here we generalize it to handle the case PSMC actually cares about: **variable population size**.

Step 1: Coalescence Under Variable Population Size

Recall from the *coalescent theory prerequisite*: under constant population size N , two lineages coalesce at rate 1 (in units of $2N$ generations). The waiting time is $\text{Exp}(1)$.

When the population size varies over time as $N(t) = N_0\lambda(t)$, the coalescence rate at time t is no longer constant – it's $1/\lambda(t)$. A larger population means a lower coalescence rate (harder to find a common ancestor when there are more individuals), and a smaller population means a higher rate (fewer individuals, so lineages are forced together more quickly).

The survival probability. The probability that two lineages have *not* coalesced by time t is the product of surviving through each infinitesimal interval. With a time-varying rate $1/\lambda(t)$, the probability of surviving each tiny interval $[u, u + du]$ is approximately $1 - du/\lambda(u) \approx e^{-du/\lambda(u)}$. Multiplying these together over all intervals from 0 to t – and recalling that multiplying exponentials is the same as adding their exponents – gives:

$$P(T > t) = \exp\left(-\int_0^t \frac{du}{\lambda(u)}\right)$$

This formula belongs to a branch of probability called **survival analysis**, which studies the time until an event occurs (originally developed for analyzing lifetimes in medical studies, but applicable whenever you ask “how long until something happens?”).

Let's define a shorthand for the cumulative integral that appears everywhere:

$$\Lambda(t) = \int_0^t \frac{du}{\lambda(u)}$$

So $P(T > t) = e^{-\Lambda(t)}$. The function $\Lambda(t)$ is called the **cumulative hazard** of the coalescent process under variable population size.

i Probability aside: the hazard function and survival analysis

Survival analysis is the study of how long we must wait for an event to occur. It was originally developed to study patient lifetimes, but the mathematics applies to any waiting time – including coalescence.

The central objects are:

- The **survival function** $S(t) = P(T > t)$: the probability that the event has *not yet* happened by time t .
- The **hazard function** (or hazard rate) $h(t)$: the instantaneous rate at which the event occurs, given that it has not yet occurred. You can think of it as “how dangerous is this moment?” – at each instant, $h(t)$ measures the intensity of the risk. Formally, $h(t) = -S'(t)/S(t)$.
- The **cumulative hazard** $H(t) = \int_0^t h(u) du$: the total accumulated risk up to time t . The survival function and cumulative hazard are linked by $S(t) = e^{-H(t)}$.

In our coalescent setting, the hazard rate is $h(t) = 1/\lambda(t)$ (the instantaneous coalescence rate), and the cumulative hazard is $\Lambda(t)$. Everything that follows is an application of the identity $S(t) = e^{-H(t)}$ – the survival probability is the exponential of the negative cumulative hazard.

i Computing $\Lambda(t)$ in practice

For piecewise-constant $\lambda(t)$ (which is the PSMC assumption), the integral decomposes into a sum over intervals:

$$\Lambda(t) = \sum_{i=0}^{k-1} \frac{\tau_i}{\lambda_i} + \frac{t - t_k}{\lambda_k} \quad \text{when } t \in [t_k, t_{k+1})$$

where $\tau_i = t_{i+1} - t_i$. No numerical quadrature is needed – it's a running sum. In the continuous case (smooth $\lambda(t)$), we use numerical integration such as `scipy.integrate.quad`.

```
from scipy.integrate import quad

def cumulative_hazard(t, lambda_func):
    """Compute Lambda(t) = integral_0^t 1/lambda(u) du.

    This is the cumulative hazard of the coalescent process.

    We use scipy.integrate.quad, which performs adaptive numerical
    integration (also called "quadrature"). It approximates the
    integral by evaluating the function at many points and fitting
    polynomial segments. The function returns two values: the
    estimated integral and an estimate of the numerical error.
    """
    # quad returns (result, error_estimate).
    # The underscore _ is a Python convention meaning "I don't need
    # this value" -- here we discard the error estimate.
    result, _ = quad(
        # "lambda u: ..." is a Python lambda function -- a compact
        # way to define a small, unnamed function inline.
        # This one takes u as input and returns 1/lambda_func(u).
        lambda u: 1.0 / lambda_func(u),
        0, # lower limit of integration
        t # upper limit of integration
    )
    return result

def cumulative_hazard_pieewise(t, t_boundaries, lambdas):
    """Fast Lambda(t) for piecewise-constant lambda.

    No quadrature needed -- just a running sum over intervals.
    For each interval [t_k, t_{k+1}) with population size lambda_k,
    the contribution to the integral is (interval width) / lambda_k.
    """
    Lambda = 0.0
    for k in range(len(lambdas)):
        t_lo = t_boundaries[k]
        t_hi = t_boundaries[k + 1]
        if t <= t_lo:
            break
        dt = min(t, t_hi) - t_lo
        Lambda += dt / lambdas[k]
```

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```

    if t <= t_hi:
        break
    return Lambda

# Verify: for constant lambda=1, Lambda(t) should equal t,
# because integral_0^t 1/1 du = t.
t_test = 2.5
print(f"Lambda({t_test}) with lambda=1: "
      f"{cumulative_hazard(t_test, lambda u: 1.0):.6f} (expected: {t_test})")
# For lambda=2, Lambda(t) = t/2 (half the rate, twice as slow).
print(f"Lambda({t_test}) with lambda=2: "
      f"{cumulative_hazard(t_test, lambda u: 2.0):.6f} (expected: {t_test/2})")

```

The density. To obtain the probability density of coalescence at *exactly* time t , we differentiate the survival function. The density $f(t) = -\frac{d}{dt}S(t)$ tells us “how much probability mass is concentrated at time t ”:

$$f(t) = -\frac{d}{dt}e^{-\Lambda(t)}$$

By the chain rule (the derivative of $f(g(x))$ is $f'(g(x)) \cdot g'(x)$), with the outer function $e^{-\cdot}$ and the inner function $\Lambda(t)$:

$$f(t) = -(-\Lambda'(t))e^{-\Lambda(t)} = \Lambda'(t) \cdot e^{-\Lambda(t)}$$

By the fundamental theorem of calculus, the derivative of $\Lambda(t) = \int_0^t \frac{du}{\lambda(u)}$ with respect to t is simply the integrand evaluated at t : $\Lambda'(t) = 1/\lambda(t)$. Therefore:

$$f(t) = \frac{1}{\lambda(t)}e^{-\Lambda(t)}$$

Intuition: The density at time t is the probability of having survived to t (the $e^{-\Lambda(t)}$ term) times the instantaneous coalescence rate at t (the $1/\lambda(t)$ term). Same structure as the constant-population case, just with a time-varying rate.

i Calculus aside: from CDF to density, step by step

The survival function is $S(t) = P(T > t) = e^{-\Lambda(t)}$ and the CDF is $F(t) = 1 - S(t)$. The density is obtained by differentiating:

$$f(t) = F'(t) = -S'(t) = -\frac{d}{dt}e^{-\Lambda(t)}$$

To differentiate $e^{-\Lambda(t)}$, apply the chain rule: the derivative of $e^{g(t)}$ is $e^{g(t)} \cdot g'(t)$, where $g(t) = -\Lambda(t)$. So $g'(t) = -\Lambda'(t) = -1/\lambda(t)$:

$$f(t) = -\left[e^{-\Lambda(t)} \cdot \left(-\frac{1}{\lambda(t)}\right)\right] = \frac{1}{\lambda(t)}e^{-\Lambda(t)}$$

For constant $\lambda(t) = 1$, we have $\Lambda(t) = t$, so $f(t) = e^{-t}$ – the standard exponential density, which is the familiar coalescence time distribution for two lineages in a constant-size population (as derived in *Coalescent Theory*).

```

import numpy as np
from scipy.integrate import quad

```

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```

def coalescent_density(t, lambda_func):
    """Coalescence time density under variable population size.

    Parameters
    -----
    t : float
        Time.
    lambda_func : callable
        lambda_func(u) returns relative population size at time u.

    Returns
    -----
    f : float
        Density f(t).
    """
    # Compute Lambda(t) = integral_0^t 1/lambda(u) du
    # using numerical integration (quad).
    # The "lambda u:" defines a small inline function that takes u
    # and returns 1/lambda_func(u).
    # quad returns (integral_value, error_estimate); we use _ to
    # discard the error estimate since we only need the integral.
    Lambda_t, _ = quad(lambda u: 1.0 / lambda_func(u), 0, t)
    return (1.0 / lambda_func(t)) * np.exp(-Lambda_t)

def coalescent_survival(t, lambda_func):
    """Probability that coalescence has NOT occurred by time t."""
    Lambda_t, _ = quad(lambda u: 1.0 / lambda_func(u), 0, t)
    return np.exp(-Lambda_t)

# Example: constant population (lambda = 1)
t_vals = np.linspace(0.01, 5, 50)
for lam_val, label in [(1.0, "constant N"), (2.0, "2x larger N")]:
    # "lambda u: lam_val" creates a function that always returns lam_val,
    # regardless of what u is -- i.e., a constant population size.
    densities = [coalescent_density(t, lambda u: lam_val) for t in t_vals]
    mean_t = sum(t * d for t, d in zip(t_vals, densities)) * (t_vals[1] - t_vals[0])
    print(f"{label}: mean coalescence time ~ {mean_t:.2f} "
          f"(expected: {lam_val:.2f})")

```

Step 2: The Transition Density Under Variable $N(t)$

Now we arrive at the central equation of PSMC – the transition density. This is the tick-to-tick mechanism of the PSMC watch: when a recombination occurs between adjacent genomic positions a and $a + 1$, the coalescence time changes from s (at position a) to t (at position $a + 1$). The transition density $q(t | s)$ tells us the probability density of the new time t , given that the old time was s .

We already derived this for constant λ in the *SMC prerequisite*. Here's the generalization to variable $\lambda(t)$.

The physical process, step by step:

1. At position a , the two lineages coalesce at time s .
2. A recombination occurs somewhere on the branch, at time u . Under the SMC model (see *The Sequentially Markov*

Coalescent), u is uniform on $[0, s]$:

$$P_1(u | s) = \frac{1}{s}$$

- From time u , the detached lineage floats up and re-coalesces. Under variable population size, the re-coalescence density is:

$$P_2(t | u) = \frac{1}{\lambda(t)} \exp \left(- \int_u^t \frac{dv}{\lambda(v)} \right)$$

This is just the coalescence density starting from time u instead of 0. The exponential term is the probability of surviving from u to t without coalescing, and the $1/\lambda(t)$ factor is the instantaneous rate of coalescing right at time t .

- The total transition density (conditioned on recombination) is obtained by **marginalizing over u** – that is, summing up the contributions from all possible recombination times u . Physically, this means we account for every possible place along the branch where the recombination could have struck: near the tips (small u), near the coalescence point (large u), or anywhere in between. Each location gives a different “starting point” for the re-coalescence process, and we weight them all equally (since u is uniform on $[0, s]$) and add them up:

$$q(t | s) = \int_0^{\min(s,t)} P_2(t | u) \cdot P_1(u | s) du = \frac{1}{\lambda(t)} \int_0^{\min(s,t)} \frac{1}{s} \cdot e^{-\int_u^t \frac{dv}{\lambda(v)}} du$$

This is Equation 5 from Li and Durbin’s paper, and Theorem 1 in `psmc.tex`.

Why $\min(s, t)$? The recombination time u must satisfy both $u \leq s$ (it’s on the branch of length s) and $u \leq t$ (re-coalescence at t must come after recombination at u).

i Computing the double integral step by step

The transition density involves an outer integral over the recombination point u and an inner integral (inside the exponential) for the cumulative hazard from u to t :

$$q(t | s) = \frac{1}{\lambda(t)} \int_0^{\min(s,t)} \underbrace{\frac{1}{s}}_{\text{uniform recomb.}} \cdot \exp \left(- \underbrace{\int_u^t \frac{dv}{\lambda(v)}}_{\text{survival from } u \text{ to } t} \right) du$$

For piecewise-constant λ : the inner integral $\int_u^t 1/\lambda(v) dv$ again reduces to a sum over intervals. This means the exponential term can be expressed as a product of per-interval survival factors, avoiding expensive quadrature in the inner loop. The outer integral over u then splits into sub-integrals within each time interval, each of which has a closed-form solution (integrals of the form $\int e^{-au} du = -\frac{1}{a}e^{-au}$). This is how the exact discrete formulas in the next chapter are derived.

For general smooth λ : we need nested numerical quadrature (quad inside quad), which is slower but exact to machine precision.

i Probability aside: the law of total probability

The transition density is a direct application of the **law of total probability**. Conditioning on the (unobserved) recombination time u :

$$q(t | s) = \int_0^{\min(s,t)} P_2(t | u) P_1(u | s) du$$

Each factor has a clear probabilistic interpretation: $P_1(u | s) = 1/s$ is the prior on the recombination breakpoint (uniform on the branch), and $P_2(t | u)$ is the re-coalescence density (a coalescent started at time u). We **marginalize** over all possible u to get the marginal density of the new coalescence time. “Marginalizing” means integrating out a variable we cannot observe – in this case, we do not know exactly where on the branch the recombination struck, so we average over all possibilities, weighted by their probability.

```
def psmc_transition_density_general(t, s, lambda_func):
    """PSMC transition density q(t/s) under variable population size.

    This is the probability density of the new coalescence time being t,
    given that the old coalescence time was s and a recombination occurred.

    Parameters
    -----
    t : float
        New coalescence time.
    s : float
        Previous coalescence time.
    lambda_func : callable
        Relative population size function.

    Returns
    -----
    q : float
    """
    upper = min(s, t)

    def integrand(u):
        # For each candidate recombination time u, compute:
        # (1/s) * exp(-integral_u^t 1/lambda(v) dv)
        # The inner quad call computes the cumulative hazard from u to t.
        # Again, _ discards the error estimate from quad.
        integral, _ = quad(lambda v: 1.0 / lambda_func(v), u, t)
        return (1.0 / s) * np.exp(-integral)

    # The outer quad call integrates over all recombination times u
    # from 0 to min(s, t). result is the integral value; _ discards
    # the error estimate.
    result, _ = quad(integrand, 0, upper)
    return result / lambda_func(t)

# Verify: for constant lambda, this should match the closed-form
s = 1.0
t_vals = np.linspace(0.01, 3.0, 30)

print("Comparing general formula to closed-form (constant lambda=1):")
for t in [0.3, 0.7, 1.0, 1.5, 2.0]:
    # "lambda u: 1.0" is a constant function returning 1.0 for any u.
    q_general = psmc_transition_density_general(t, s, lambda u: 1.0)
    # Closed-form for constant lambda=1:
    if t < s:
        q_closed = (1.0 / s) * (1 - np.exp(-t))
```

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```

else:
    q_closed = (1.0 / s) * (np.exp(-(t - s)) - np.exp(-t))
    print(f"  t={t:.1f}: general={q_general:.6f}, "
          f"closed={q_closed:.6f}, diff={abs(q_general-q_closed):.2e}")

```

Step 3: Normalization – It Integrates to 1

Where we are so far. In Steps 1 and 2, we derived the coalescence density under variable population size (the probability of coalescing at a specific time) and the transition density (the probability of transitioning to a new coalescence time after a recombination event). Now we must verify a crucial mathematical property: that the transition density $q(t | s)$ is a valid probability density.

For a function to be a valid probability density, it must integrate to 1 over its entire domain. This means that if we add up the probabilities of all possible outcomes (here, all possible new coalescence times t from 0 to infinity), the total must be exactly 1 – because *something* has to happen, and the total probability of all possibilities is always 100%. If the integral were less than 1, probability mass would be “leaking” somewhere; if it were greater than 1, we would be double-counting. Either would make the HMM built on top of this density mathematically inconsistent.

A crucial property: $q(t | s)$ integrates to 1 over $t \in [0, \infty)$. This is not obvious! Let’s prove it, because understanding the proof reveals the structure of the formula.

The trick is a change of variables. Define $\tilde{t}$ such that:

$$\phi'(\tilde{t}) \cdot \frac{1}{\lambda(\phi(\tilde{t}))} = 1, \quad \phi(0) = 0$$

In other words, $\tilde{t} = \Lambda(t)$ is the “coalescent time measured in units of constant population size.” This is a time-warping transformation: it stretches intervals where the population is large (slow coalescence) and compresses intervals where the population is small (fast coalescence). Under this transformation, the coalescence rate becomes constant (rate 1), and the integral reduces to the constant-population case, which we already know integrates to 1 from the *SMC prerequisite*.

Why is this important? It means $q(t|s)$ is a valid probability density – essential for the HMM to be well-defined. If the transition density did not integrate to 1, the forward algorithm (which multiplies transition probabilities at each step) would produce values that drift away from true probabilities, making all downstream inference meaningless.

```

# Numerical verification: q(t/s) integrates to 1
def verify_normalization(s, lambda_func, t_max=20):
    """Verify that q(t/s) integrates to 1.

    Uses quad to numerically integrate the transition density over
    t from a small positive number (to avoid division by zero at t=0)
    up to t_max (a stand-in for infinity).
    """
    # quad with limit=100 allows up to 100 subdivisions for
    # adaptive integration on difficult integrands.
    result, error = quad(
        # lambda t: ... creates an inline function of t that calls
        # our transition density function.
        lambda t: psmc_transition_density_general(t, s, lambda_func),
        0.001, t_max,
        limit=100
    )
    return result

```

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```
# Test with various population size functions
test_cases = [
    ("Constant N", lambda t: 1.0),
    ("Doubled N", lambda t: 2.0),
    ("Bottleneck", lambda t: 0.1 if 0.5 < t < 1.0 else 1.0),
    ("Growth", lambda t: np.exp(t / 2)),
]

for name, lam in test_cases:
    for s in [0.5, 1.0, 2.0]:
        integral = verify_normalization(s, lam)
        print(f" {name}, s={s:.1f}: integral = {integral:.6f}")
```

Step 4: The Stationary Distribution

The **stationary distribution** $\pi(t)$ is the probability density of the coalescence time at a single position, marginalized over all possible histories. It answers: “if I pick a random position on the genome, what is the distribution of its coalescence time?”

Think of it as the long-run equilibrium where the watch runs steadily. If the PSMC process has been running for a very long time along the genome, and we sample a position at random, $\pi(t)$ tells us how likely we are to find coalescence time t . No matter what coalescence time the process started with, after enough recombination events it settles into this steady-state distribution.

Why does the stationary distribution matter for the HMM? In a Hidden Markov Model, we need to specify the probability distribution of the *first* hidden state. The stationary distribution $\pi(t)$ serves as this **initial state distribution**: we assume the first genomic position is drawn from the long-run equilibrium. This is a natural choice because a random position in the genome has no special relationship to the beginning of the sequence – it is just another position in a long chain of transitions. Using $\pi(t)$ as the initial distribution is equivalent to assuming the process has been running long enough to reach equilibrium before we start observing.

For the PSMC transition density with variable $\lambda(t)$:

$$\pi(t) = \frac{t}{C_\pi \lambda(t)} e^{-\Lambda(t)}$$

where C_π is the normalization constant:

$$C_\pi = \int_0^\infty e^{-\Lambda(u)} du$$

Where does the t in the numerator come from? This is the key insight. The transition $q(t|s)$ involves integrating the recombination position u uniformly on $[0, s]$. In the stationary state, we average over s as well. The factor t arises because a recombination on a branch of length t is t times as likely as on a branch of length 1 – longer branches “catch” more recombinations.

More formally, $\pi(t)$ is the density such that:

$$\int_0^\infty q(t | s) \pi(s) ds = \pi(t)$$

This is a fixed-point equation: if the coalescence time distribution is π , then after one recombination event, the distribution is still π . The proof uses Lemma 2 from `psmc.tex` (the stationary distribution lemma), with $g(u) = 1$ and $h(u) = 1/\lambda(u)$.

i Probability aside: stationary distributions and Markov chains

The equation $\int q(t|s) \pi(s) ds = \pi(t)$ is the continuous-state analogue of $\pi^T \mathbf{Q} = \pi^T$ for discrete Markov chains (as introduced in the *HMM prerequisite*). The transition kernel $q(t|s)$ plays the role of the transition matrix, and the integral replaces matrix multiplication. A distribution satisfying this equation is an **invariant measure** (or equilibrium) of the chain: once the process reaches π , it stays there forever.

The extra factor of t in the numerator of $\pi(t)$ compared to the coalescence density $f(t)$ reflects a **size-biased sampling** effect. Recombination is more likely to hit a longer branch (probability proportional to t), so the stationary distribution over-represents long coalescence times relative to the raw density $f(t)$. This is the same phenomenon as the “waiting time paradox” (or inspection paradox) in renewal theory: if you arrive at a random time, you are more likely to land in a long inter-arrival interval.

```
def stationary_distribution(t, lambda_func, C_pi=None):
    """Stationary distribution pi(t) of coalescence time.

    Parameters
    -----
    t : float
        Time.
    lambda_func : callable
    C_pi : float, optional
        Normalization constant. Computed if not provided.

    Returns
    -----
    pi_t : float
    """
    if C_pi is None:
        C_pi = compute_C_pi(lambda_func)

    # Compute Lambda(t) via numerical integration.
    # _ discards the error estimate returned by quad.
    Lambda_t, _ = quad(lambda u: 1.0 / lambda_func(u), 0, t)
    return t / (C_pi * lambda_func(t)) * np.exp(-Lambda_t)

def compute_C_pi(lambda_func, t_max=np.inf):
    """Compute the normalization constant C_pi.

    C_pi = integral_0^inf exp(-Lambda(u)) du

    The default evaluates the full improper integral. A finite ``t_max``
    is an explicit truncation and is safe only when its omitted tail has
    been bounded for the supplied population-size history.
    """
    def integrand(u):
        # For each u, compute Lambda(u) and then exp(-Lambda(u)).
        Lambda_u, _ = quad(lambda v: 1.0 / lambda_func(v), 0, u)
        return np.exp(-Lambda_u)

    # Integrate the survival function over the requested domain.
    C_pi, _ = quad(integrand, 0, t_max)
```

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```

return C_pi

# For constant population lambda=1:
# pi(t) = t * exp(-t) / C_pi
# C_pi = integral_0^inf exp(-t) dt = [-exp(-t)]_0^inf = 0 - (-1) = 1
# So pi(t) = t * exp(-t) -- the Gamma(2,1) distribution!
C_pi_const = compute_C_pi(lambda t: 1.0)
print(f"C_pi (constant pop): {C_pi_const:.6f} (expected: 1.0)")

# The mean coalescence time under constant population:
# E[T] = integral_0^inf t * pi(t) dt = integral_0^inf t^2 * exp(-t) dt / C_pi
# = Gamma(3) / C_pi = 2! / 1 = 2
mean_T, _ = quad(
    # lambda t: ... creates a function that evaluates
    # t * pi(t), which we integrate to get the expected value.
    lambda t: t * stationary_distribution(t, lambda u: 1.0, C_pi_const),
    0, np.inf
)
print(f"Mean coalescence time (constant pop): {mean_T:.4f} (expected: 2.0)")

# Verify pi(t) is indeed stationary: integral q(t/s)*pi(s)ds = pi(t)
def verify_stationarity(t_test, lambda_func, C_pi, s_max=15):
    """Check that pi is the fixed point of q.

    If pi is truly stationary, then applying one step of the
    transition density and averaging over all starting states s
    (weighted by pi(s)) should give back pi(t).
    """
    lhs, _ = quad(
        lambda s: psmc_transition_density_general(t_test, s, lambda_func)
            * stationary_distribution(s, lambda_func, C_pi),
        0.001, s_max)
    rhs = stationary_distribution(t_test, lambda_func, C_pi)
    return lhs, rhs

for t_val in [0.5, 1.0, 2.0]:
    lhs, rhs = verify_stationarity(t_val, lambda u: 1.0, C_pi_const)
    print(f"t={t_val}: integral q*pi = {lhs:.6f}, pi(t) = {rhs:.6f}, "
          f"ratio = {lhs/rhs:.6f}")

```

i The Gamma(2,1) distribution

For constant population size ($\lambda(t) = 1$), the stationary distribution is $\pi(t) = te^{-t}$, which is the Gamma(2, 1) distribution. Its mean is 2 because π describes coalescence times observed at recombination events and is length-biased toward older trees. It must not be confused with the ordinary pairwise coalescence-time density e^{-t} , whose mean is 1 in units of $2N_0$ generations.

Step 5: Including No-Recombination (The Full Transition)

Where we are so far. We have derived the coalescence density (Step 1), the transition density conditioned on recombination (Step 2), proved it integrates to 1 (Step 3), and found the stationary distribution (Step 4). All of these describe what happens *when* a recombination occurs. But recombination does not happen at every genomic position – in fact, for most adjacent positions, no recombination occurs at all. Now we need to account for both possibilities: recombination or no recombination.

The density $q(t|s)$ describes what happens *given* a recombination. But at each genomic position, recombination may or may not occur. The probability of recombination on a branch of length s is $1 - e^{-\rho s}$ (Poisson process with rate ρ per unit branch length per bin).

The **full transition** includes both possibilities:

$$p(t | s) = (1 - e^{-\rho s}) q(t | s) + e^{-\rho s} \delta(t - s)$$

- With probability $1 - e^{-\rho s}$: recombination occurs, and the new time is drawn from $q(t|s)$.
- With probability $e^{-\rho s}$: no recombination, and the time stays at s (represented by the Dirac delta $\delta(t - s)$).

The **Dirac delta** $\delta(t - s)$ is a mathematical device that represents a “spike” of probability concentrated at a single point $t = s$. It is not a function in the ordinary sense – it is zero everywhere except at $t = s$, where it is infinitely tall, but its total integral is exactly 1. You can think of it as the limit of a very narrow, very tall bell curve that shrinks to zero width while keeping its total area equal to 1. When multiplied by a probability $e^{-\rho s}$, it means “with probability $e^{-\rho s}$, the new coalescence time is *exactly* s – not approximately s , but precisely s .”

The key property of the Dirac delta is its behavior under integration: for any function f ,

$$\int_{-\infty}^{\infty} f(t) \delta(t - s) dt = f(s)$$

In other words, integrating against the Dirac delta “picks out” the value of f at the spike location.

i Probability aside: mixture distributions

The full transition $p(t|s)$ is a **mixture** of a continuous density and a point mass. This is a common construction in probability:

$$T' \sim \begin{cases} q(\cdot | s) & \text{with prob } 1 - e^{-\rho s} \\ \delta_s & \text{with prob } e^{-\rho s} \end{cases}$$

When computing expectations under $p(t|s)$, the Dirac delta contributes the “no change” case: $E[g(T')] = (1 - e^{-\rho s}) \int g(t) q(t|s) dt + e^{-\rho s} g(s)$.

```
def full_transition_density(t, s, rho, lambda_func, tol=1e-8):
    """Full transition density p(t|s) including no-recombination.
```

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```

p(t/s) = (1 - exp(-rho*s)) * q(t/s) + exp(-rho*s) * delta(t - s)

Returns
-----
continuous : float
    The continuous part of the density at t.
point_mass : float
    The weight of the point mass at t = s (nonzero only when |t - s| < tol).
"""
if s < 0 or rho < 0:
    raise ValueError("s and rho must be non-negative")
if s == 0:
    return 0.0, 1.0 if abs(t) < tol else 0.0
recomb_prob = -np.expm1(-rho * s)
if abs(t - s) < tol:
    q_ts = psmc_transition_density_general(t, s, lambda_func)
    return recomb_prob * q_ts, np.exp(-rho * s)
else:
    return recomb_prob * psmc_transition_density_general(t, s, lambda_func), 0.0

```

```

# The probability of recombination depends on branch length s
for s_val in [0.5, 1.0, 2.0, 5.0]:
    p_rec = 1 - np.exp(-0.001 * s_val)
    print(f"s={s_val}: P(recombination) = {p_rec:.6f}")

```

The full stationary distribution $\sigma(t)$ satisfies:

$$\sigma(t) = \frac{\pi(t)}{C_\sigma(1 - e^{-\rho t})}$$

where:

$$C_\sigma = \int_0^\infty \frac{\pi(t)}{1 - e^{-\rho t}} dt$$

Why does $\sigma(t)$ differ from $\pi(t)$? The distribution $\pi(t)$ is the stationary distribution of $q(t|s)$, the transition density *conditioned on recombination*. But at stationarity, whether or not a recombination occurs also depends on the coalescence time (through the factor $1 - e^{-\rho s}$). Longer coalescence times have higher recombination probability, so they “churn” more and their stationary weight $\sigma(t)$ is increased relative to $\pi(t)$ by the factor $1/(1 - e^{-\rho t})$.

i Probability aside: detailed balance and reweighting

The relationship between σ and π is an example of **importance reweighting**. The full chain (with skip-probability $e^{-\rho s}$) has a different stationary distribution than the sub-chain restricted to steps where recombination occurs. The reweighting factor $1/(1 - e^{-\rho t})$ compensates for the fact that states with short coalescence times are “stickier” (less likely to recombine), so they need higher stationary weight to account for all the silent no-recombination steps.

At $t = 0$ both numerator and denominator vanish. The implementation uses the analytic limit there and `expm1` elsewhere, avoiding the removable $0/0$ singularity without discarding the youngest part of the integral.

```

def full_stationary(t, lambda_func, rho, C_pi=None, C_sigma=None):
    """Full stationary distribution sigma(t).

```

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```

sigma(t) = pi(t) / (C_sigma * (1 - exp(-rho*t)))
"""
if t < 0 or rho <= 0:
    raise ValueError("t must be non-negative and rho must be positive")
if C_pi is None:
    C_pi = compute_C_pi(lambda_func)
if C_sigma is None:
    C_sigma = compute_C_sigma(lambda_func, rho, C_pi)
if t == 0:
    return 1.0 / (C_pi * C_sigma * rho * lambda_func(0.0))
pi_t = stationary_distribution(t, lambda_func, C_pi)
return pi_t / (C_sigma * -np.expm1(-rho * t))

```

```

def compute_C_sigma(lambda_func, rho, C_pi=None, t_max=np.inf):
    """Compute C_sigma = integral pi(t) / (1 - exp(-rho*t)) dt."""
    if rho <= 0 or t_max <= 0:
        raise ValueError("rho and t_max must be positive")
    if C_pi is None:
        C_pi = compute_C_pi(lambda_func)

    def integrand(t):
        if t == 0:
            return 1.0 / (C_pi * rho * lambda_func(0.0))
        return stationary_distribution(t, lambda_func, C_pi) / -np.expm1(-rho * t)

    C_sigma, _ = quad(integrand, 0, t_max)
    return C_sigma

```

```

# Test: the probability of recombination at any site is 1/C_sigma
rho = 0.001
C_pi = compute_C_pi(lambda t: 1.0)
C_sigma = compute_C_sigma(lambda t: 1.0, rho, C_pi)
print(f"C_sigma = {C_sigma:.4f}")
print(f"P(recombination) = 1/C_sigma = {1.0/C_sigma:.6f}")

```

Step 6: Approximating C_sigma

For small ρ (which is the typical regime – recombination rates are very low per base pair), C_σ has a clean approximation:

$$C_\sigma = \frac{1}{C_\pi \rho} + \frac{1}{2} + o(\rho)$$

The notation $o(\rho)$ means “terms that go to zero faster than ρ as $\rho \rightarrow 0$ ” – in other words, for small ρ , these terms are negligible.

Derivation. Starting from the definition and using a Taylor expansion to approximate $1/(1 - e^{-\rho t})$:

$$\begin{aligned}
 C_\sigma &= \int_0^\infty \frac{\pi(t)}{1 - e^{-\rho t}} dt \\
 &= \int_0^\infty \frac{t}{C_\pi \lambda(t)(1 - e^{-\rho t})} e^{-\Lambda(t)} dt
 \end{aligned}$$

A **Taylor expansion** (also called a Taylor series) approximates a function near a point by a polynomial. The idea is that any smooth function can be written as a sum of powers of its argument. For $e^{-x} \approx 1 - x + x^2/2 - \dots$ when x is small, we get $1 - e^{-\rho t} \approx \rho t - (\rho t)^2/2 + \dots \approx \rho t$ for small ρt . Taking the reciprocal:

$$\frac{1}{1 - e^{-\rho t}} \approx \frac{1}{\rho t} \left(1 + \frac{\rho t}{2} + \dots \right)$$

This approximation works because $1 - e^{-x} \approx x$ for small x , and we are then expanding $1/(x - x^2/2 + \dots)$ using the geometric series $1/(1 - r) \approx 1 + r + \dots$.

Substituting into the integral for C_σ :

$$C_\sigma \approx \frac{1}{C_\pi \rho} \int_0^\infty \frac{1}{\lambda(t)} e^{-\Lambda(t)} dt + \frac{1}{2} \int_0^\infty \pi(t) dt + o(\rho)$$

Now we use two facts:

- $\int_0^\infty \frac{1}{\lambda(t)} e^{-\Lambda(t)} dt = 1$: this is the integral of the coalescence density $f(t)$ over all time, which must be 1 because coalescing at *some* time is certain.
- $\int_0^\infty \pi(t) dt = 1$: this is the normalization of the stationary distribution (it is a probability density, so it integrates to 1).

Therefore:

$$\begin{aligned} C_\sigma &= \frac{1}{C_\pi \rho} \cdot 1 + \frac{1}{2} \cdot 1 + o(\rho) \\ &= \frac{1}{C_\pi \rho} + \frac{1}{2} + o(\rho) \end{aligned}$$

```
# Verify the approximation C_sigma ~ 1/(C_pi * rho) + 0.5
C_pi = compute_C_pi(lambd t: 1.0)
for rho in [0.01, 0.001, 0.0001]:
    C_sigma_exact = compute_C_sigma(lambd t: 1.0, rho, C_pi)
    C_sigma_approx = 1.0 / (C_pi * rho) + 0.5
    print(f"rho={rho:.4f}: exact={C_sigma_exact:.4f}, "
          f"approx={C_sigma_approx:.4f}, "
          f"relative error={abs(C_sigma_exact-C_sigma_approx)/C_sigma_exact:.2e}")
```

Step 7: The Rate of Pairwise Difference

Where we are so far. We have built the complete transition machinery of the PSMC watch: the transition density (how coalescence times change from position to position), the stationary distribution (the long-run equilibrium), and the full transition including no-recombination events. Now we need the final piece: the **emission model** – how the hidden coalescence time produces the observed data (heterozygous or homozygous sites).

How many heterozygous sites do we expect? If the coalescence time is t , the probability of at least one mutation in a bin is $1 - e^{-\theta t}$ (two lineages, each of length $t/2$... wait, let's be careful).

Actually, in coalescent units where t measures the **total** time of the pairwise coalescent (time from present to MRCA), the expected number of mutations on the two branches is θt (both lineages have length t , but the mutation rate θ is already scaled to account for both). So the probability of at least one mutation (heterozygous site) in a bin is:

$$P(X_a = 1 \mid T_a = t) = 1 - e^{-\theta t}$$

i Probability aside: the Poisson model of mutations

Why $1 - e^{-\theta t}$ and not just θt ? Mutations along the two lineages of total length t follow a Poisson process with rate θ . The probability of **at least one** mutation (i.e., a heterozygous site) is $1 - P(\text{zero mutations}) = 1 - e^{-\theta t}$. For small θt this is approximately θt (first-order Taylor expansion: $e^{-x} \approx 1 - x$ when $x \ll 1$), but the exponential form is exact and avoids overcounting when θt is not small.

This is the same Poisson mutation model introduced in *Coalescent Theory*, where we showed that mutations on a branch of length ℓ follow a Poisson distribution with mean $\theta\ell/2$. Here, the total branch length for two lineages coalescing at time t is $2 \times t = 2t$ (each lineage has length t), giving an expected mutation count of $\theta/2 \times 2t = \theta t$. The probability of zero mutations is $e^{-\theta t}$, so the probability of at least one is $1 - e^{-\theta t}$.

```
# Emission probability as a function of coalescence time
theta = 0.001
for t_val in [0.5, 1.0, 2.0, 5.0, 10.0]:
    p_het = 1 - np.exp(-theta * t_val)
    # Linear approximation: e^{-x} ~ 1 - x, so 1 - e^{-x} ~ x
    p_het_approx = theta * t_val
    print(f"t={t_val:5.1f}: P(het) = {p_het:.6f}, "
          f"linear approx = {p_het_approx:.6f}, "
          f"error = {abs(p_het - p_het_approx):.2e}")
```

Averaging over the stationary distribution of T :

$$P(X_a = 1) = \int_0^\infty (1 - e^{-\theta t}) \sigma(t) dt$$

This integral computes the **expected heterozygosity**: we weight the per-site mutation probability by the probability that a random genomic site has coalescence time t , then integrate over all possible times.

For small θ and ρ , this simplifies to:

$$P(X_a = 1) \approx C_\pi \theta \cdot [1 + o(\rho + \theta)]$$

Intuition: The expected number of heterozygous sites is proportional to θ (mutation rate) and C_π (which is the “effective coalescence time” – it captures how population size history affects the average time to MRCA). Larger C_π means longer average coalescence times, hence more heterozygosity.

This gives us a way to **estimate** θ from the observed fraction of heterozygous sites:

$$\hat{\theta} \approx \frac{\text{fraction of het sites}}{C_\pi}$$

```
# For constant population: C_pi = 1, so P(het) ~ theta
theta = 0.001 # typical for humans at 100bp bins
C_pi = 1.0
p_het_approx = C_pi * theta
# Substituting E[T]=1 is a useful approximation, not a marginalization over T.
p_het_at_mean = 1 - np.exp(-theta * 1.0)
print(f"Approximate P(het): {p_het_approx:.6f}")
print(f"Using mean T: {p_het_at_mean:.6f}")

# Initial estimate of theta from data
def estimate_theta_initial(seq):
```

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```

"""Return the original PSMC initialization for theta_0.

Missing bins (encoded by values >= 2) are excluded. The transformation
is PSMC's initialization heuristic, not an exact marginal estimator
under a random coalescence time.
"""

observed = np.asarray(seq)[np.asarray(seq) < 2]
if observed.size == 0:
    raise ValueError("sequence contains no callable bins")
frac_het = np.mean(observed)
return -np.log1p(-frac_het)

# Test on simulated data
theta_hat = estimate_theta_initial(seq)
print(f"\nTrue theta: {0.001}, Estimated: {theta_hat:.6f}")

```

Putting It All Together

Here's a summary of the continuous-time PSMC model – all seven steps in one table:

Quantity	Formula
Coalescence rate	$1/\lambda(t)$
Cumulative hazard	$\Lambda(t) = \int_0^t \frac{du}{\lambda(u)}$
Coalescence density	$f(t) = \frac{1}{\lambda(t)} e^{-\Lambda(t)}$
Transition density (given recomb.)	$q(t s) = \frac{1}{\lambda(t)} \int_0^{\min(s,t)} \frac{1}{s} e^{-\int_u^t \frac{dv}{\lambda(v)}} du$
Full transition	$p(t s) = (1 - e^{-\rho s})q(t s) + e^{-\rho s}\delta(t - s)$
Stationary dist. (recomb.)	$\pi(t) = \frac{t}{C_\pi \lambda(t)} e^{-\Lambda(t)}$
Stationary dist. (full)	$\sigma(t) = \frac{\pi(t)}{C_\sigma (1 - e^{-\rho t})}$
Normalization	$C_\pi = \int_0^\infty e^{-\Lambda(u)} du$
Approx. normalization	$C_\sigma \approx \frac{1}{C_\pi \rho} + \frac{1}{2}$
Emission probability	$P(X_a = 1 \mid T_a = t) = 1 - e^{-\theta t}$

These equations form the mathematical foundation. But they're continuous – and an HMM needs discrete states. In the next chapter, we discretize time.

Exercises

Exercise 1: Verify the stationary property

Numerically verify that $\int_0^\infty q(t|s)\pi(s)ds = \pi(t)$ for several values of t and several population size functions (constant, exponential growth, bottleneck).

Exercise 2: Explore the effect of population size on $\pi(t)$

Plot $\pi(t)$ for: (a) constant $\lambda(t) = 1$, (b) a bottleneck $\lambda(t) = 0.1$ for $t \in [1, 2]$ and 1 elsewhere, (c) exponential growth $\lambda(t) = e^t$.

How does each demographic event shift the distribution of coalescence times?

Exercise 3: Estimate C_π for a bottleneck model

For $\lambda(t) = 1$ for $t < 1$, $\lambda(t) = 0.1$ for $t \in [1, 2]$, and $\lambda(t) = 1$ for $t > 2$:

- Compute C_π numerically.
- What does a smaller C_π mean for the expected heterozygosity?
- How would you detect this bottleneck from the data?

Next: *Discretizing Time* – turning the continuous model into an HMM.

Solutions

Solution 1: Verify the stationary property

We numerically verify that $\int_0^\infty q(t|s)\pi(s)ds = \pi(t)$ for several values of t and several population size functions. The key insight is that if $\pi(t)$ is truly stationary, applying one transition step and averaging over all starting states should reproduce $\pi(t)$ exactly.

```
from scipy.integrate import quad

def verify_stationarity_full(t_test, lambda_func, C_pi, s_max=15):
    """Check that pi is the fixed point of q for a given t."""
    lhs, _ = quad(
        lambda s: psmc_transition_density_general(t_test, s, lambda_func)
            * stationary_distribution(s, lambda_func, C_pi),
        0.001, s_max, limit=100)
    rhs = stationary_distribution(t_test, lambda_func, C_pi)
    return lhs, rhs

# Test with three demographic scenarios
scenarios = [
    ("Constant (lambda=1)", lambda t: 1.0),
    ("Exponential growth (lambda=e^t)", lambda t: np.exp(t)),
    ("Bottleneck (lambda=0.1 for t in [1,2])",
     lambda t: 0.1 if 1.0 < t < 2.0 else 1.0),
]

t_values = [0.5, 1.0, 2.0, 3.0]

for name, lam_func in scenarios:
    C_pi = compute_C_pi(lam_func)
    print(f"\n{name} (C_pi = {C_pi:.4f}):")
    for t_val in t_values:
        lhs, rhs = verify_stationarity_full(t_val, lam_func, C_pi)
        ratio = lhs / rhs if rhs > 0 else float('nan')
        print(f"  t={t_val:.1f}: integral q*pi = {lhs:.6f}, "
              f"pi(t) = {rhs:.6f}, ratio = {ratio:.6f}")
```

For all scenarios and all t values, the ratio should be very close to 1.0 (within numerical integration tolerance, typically $\sim 10^{-6}$). This confirms that $\pi(t)$ satisfies the fixed-point equation $\int q(t|s)\pi(s) ds = \pi(t)$ regardless of the population size history.

Solution 2: Explore the effect of population size on $\pi(t)$

We compute and compare $\pi(t)$ for three demographic scenarios. The key insight is that $\pi(t) = \frac{t}{C_\pi \lambda(t)} e^{-\Lambda(t)}$, so the shape of $\pi(t)$ is controlled by both the local population size $\lambda(t)$ and the cumulative hazard $\Lambda(t)$.

```
import numpy as np

t_vals = np.linspace(0.01, 6.0, 200)

scenarios = {
    "(a) Constant": lambda t: 1.0,
    "(b) Bottleneck": lambda t: 0.1 if 1.0 < t < 2.0 else 1.0,
    "(c) Exp. growth": lambda t: np.exp(t),
}

for name, lam_func in scenarios.items():
    C_pi = compute_C_pi(lam_func)
    pi_vals = [stationary_distribution(t, lam_func, C_pi) for t in t_vals]
    peak_idx = np.argmax(pi_vals)
    mean_t, _ = quad(
        lambda t: t * stationary_distribution(t, lam_func, C_pi),
        0.001, 20)
    print(f"{name}: C_pi={C_pi:.4f}, peak at t={t_vals[peak_idx]:.2f}, "
          f"mean T={mean_t:.4f}")
```

How each demographic event shifts the distribution:

- **(a) Constant:** $\pi(t) = te^{-t}$, the Gamma(2,1) distribution. Peak at $t = 1$, mean = 2. This is the baseline.
- **(b) Bottleneck:** The small $\lambda(t) = 0.1$ in $[1, 2]$ creates a high coalescence rate in that interval, pulling probability mass toward $t \approx 1$. The cumulative hazard increases sharply through the bottleneck, so survival past $t = 2$ becomes very unlikely. C_π decreases (shorter expected coalescence time), and the distribution is compressed toward more recent times.
- **(c) Exponential growth:** The growing $\lambda(t) = e^t$ reduces the coalescence rate at large t , allowing lineages to survive much longer. The distribution becomes more spread out with a heavier right tail. C_π increases (longer expected coalescence time), and the peak shifts to a larger t .

Solution 3: Estimate C_π for a bottleneck model

(a) Compute C_π numerically for the piecewise population size $\lambda(t) = 1$ for $t < 1$, $\lambda(t) = 0.1$ for $t \in [1, 2]$, and $\lambda(t) = 1$ for $t > 2$.

```
def bottleneck_lambda(t):
    if t < 1.0:
        return 1.0
    elif t < 2.0:
        return 0.1
    else:
```

```
return 1.0
```

```
C_pi_bottleneck = compute_C_pi(bottleneck_lambda)
C_pi_constant = compute_C_pi(lambda t: 1.0)
print(f"C_pi (bottleneck): {C_pi_bottleneck:.6f}")
print(f"C_pi (constant): {C_pi_constant:.6f}")
print(f"Ratio: {C_pi_bottleneck / C_pi_constant:.4f}")
```

The bottleneck gives $C_\pi \approx 0.64$, compared to $C_\pi = 1.0$ for a constant population. The bottleneck forces most coalescence events into the interval $[1, 2]$, reducing the mean coalescence time.

(b) A smaller C_π means lower expected heterozygosity, because $P(X_a = 1) \approx C_\pi \theta$. With $C_\pi \approx 0.64$, the expected heterozygosity is only 64% of what it would be under a constant population. The bottleneck reduces the average time to the most recent common ancestor, which means fewer mutations accumulate on average.

```
theta = 0.001
p_het_constant = C_pi_constant * theta
p_het_bottleneck = C_pi_bottleneck * theta
print(f"Expected heterozygosity (constant): {p_het_constant:.6f}")
print(f"Expected heterozygosity (bottleneck): {p_het_bottleneck:.6f}")
print(f"Reduction: {(1 - p_het_bottleneck/p_het_constant)*100:.1f}%")
```

(c) To detect this bottleneck from data, one would:

1. **Observe reduced heterozygosity** compared to what a constant-population model would predict. The initial $\hat{\theta}$ estimate from the data would be lower than expected.
2. **Run PSMC** and examine the inferred λ_k values. The bottleneck should appear as $\lambda_k \ll 1$ in the time intervals corresponding to $t \in [1, 2]$ (in coalescent units). The PSMC plot would show a sharp dip in $N_e(t)$ at that time.
3. **Check the posterior decoding:** positions with coalescence times in the bottleneck interval $[1, 2]$ should show an excess of homozygous sites (because the high coalescence rate during the bottleneck means those lineages have short branch lengths and thus fewer mutations).

Discretizing Time

To fit continuous gears into a digital mechanism, we need to carve them into teeth.

In the [previous chapter](#), we built the continuous-time PSMC model – the mathematical escapement that describes how coalescence times change from one genomic position to the next. That model lives in the world of smooth functions, integrals, and continuous probability densities.

But a Hidden Markov Model, the engine we will use in the [next chapter](#) to actually fit the model to data, needs **discrete states**. An HMM cannot work with a continuous infinity of possible coalescence times. It needs a finite list of states, a finite transition matrix, and finite emission probabilities.

This is where discretization comes in. Think of it as the watchmaker's task of carving continuous gears into discrete teeth. The continuous escapement produces a smooth oscillation; the gear train converts it into countable, clickable increments. Each "tooth" on the gear corresponds to one time interval, and the gear ratio table – the transition matrix – tells us the probability of moving from any tooth to any other.

In this chapter, we will:

1. **Choose the time intervals** – decide where to place the boundaries between teeth on the gear, using a log-spacing scheme matched to coalescent physics.
2. **Derive the helper quantities** – build the mathematical machinery (survival factors, re-coalescence sums) that make the transition matrix computable.

3. **Compute the stationary distribution** – find the long-run probability of being in each interval.
4. **Build the transition matrix** – the gear ratio table at the heart of the discrete HMM.
5. **Compute effective coalescence times** – the representative times used for emission probabilities.
6. **Group parameters** – reduce the number of free parameters to avoid overfitting.

By the end, you will have a complete, tested Python implementation that converts the continuous PSMC model into a working HMM specification. Every formula will be verified against numerical integration, so there is no room for doubt about correctness.

Step 1: Choosing Time Intervals

The first decision: how do we partition the continuous time axis $[0, \infty)$ into a finite number of intervals?

PSMC divides the time axis into $n + 1$ intervals:

$$[t_0, t_1), \quad [t_1, t_2), \quad \dots, \quad [t_n, t_{n+1})$$

where $t_0 = 0$ and $t_{n+1} = \infty$. The coalescence time at any genomic position must fall into exactly one of these intervals. That interval becomes the HMM's hidden state at that position.

How should we space the intervals? This is a design choice with real consequences, and the answer is: **not uniformly**.

Probability Aside: Why uniform spacing wastes resolution

Recall from *coalescent theory* that under a constant population of size N , the coalescence time T for two lineages follows an exponential distribution with rate 1 (in coalescent units of $2N$ generations). The probability density is $f(t) = e^{-t}$, which is heavily concentrated near $t = 0$ and has a long, thin tail stretching toward infinity.

If we used 64 uniform intervals from 0 to 15, each would span $15/64 \approx 0.23$ coalescent units. The first few intervals would each contain a substantial fraction of the probability mass – perhaps 10-20% of all coalescence events fall in the first interval alone. But the last 20 intervals together would contain less than 0.001% of events. We would waste most of our resolution on a region where almost nothing happens, while cramming all the action into a few crowded intervals near $t = 0$.

The solution is to use narrow intervals where the density is high (recent times) and wide intervals where the density is thin (ancient times). This way, each interval carries roughly equal statistical weight – a principle called **equal information content**.

PSMC uses **approximately log-spaced** intervals, defined by:

$$t_i = \alpha (e^{i\beta} - 1), \quad \text{where } \beta = \frac{1}{n} \ln \left(1 + \frac{t_{\max}}{\alpha} \right)$$

The parameter α (default: 0.1) controls the spacing near $t = 0$, and $t_{\max}$ is the maximum time we consider. To understand why this formula works, consider a few cases:

- **At $i = 0$:** $t_0 = \alpha(e^0 - 1) = 0$. The formula automatically starts at zero.
- **At small i :** the exponential $e^{i\beta}$ is close to $1 + i\beta$, so $t_i \approx \alpha \cdot i\beta$ – nearly linear (uniform) spacing. The intervals are narrow, matching the high density of coalescence events in the recent past.
- **At large i :** the exponential dominates, and the intervals grow geometrically. $t_{i+1} - t_i \approx \alpha\beta e^{i\beta}$, which increases with i . Wide intervals in the deep past, where coalescence events are rare.
- **At $i = n$:** $t_n = t_{\max}$ exactly, by construction of β .

The parameter α controls the crossover between the linear and exponential regimes. A smaller α packs more intervals near $t = 0$; a larger α spreads them more uniformly. The default of 0.1 was chosen by Li and Durbin to work well for human demographic history.

```
import numpy as np

def compute_time_intervals(n, t_max, alpha=0.1):
    """Compute PSMC time interval boundaries.

    Parameters
    -----
    n : int
        Number of intervals minus 1 (so there are n+1 intervals).
    t_max : float
        Maximum time (in coalescent units of 2*N_0 generations).
    alpha : float
        Controls spacing near t=0.

    Returns
    -----
    t : ndarray of shape (n + 2,)
        Boundaries [t_0, t_1, ..., t_n, t_{n+1}].
        t_0 = 0, t_{n+1} = infinity (represented as a large number).
    """
    # beta controls the overall rate of exponential growth in the spacing
    beta = np.log(1 + t_max / alpha) / n

    t = np.zeros(n + 2)
    for k in range(n):
        # alpha * (exp(beta*k) - 1) gives near-linear spacing for small k
        # and exponential spacing for large k
        t[k] = alpha * (np.exp(beta * k) - 1)
    t[n] = t_max
    t[n + 1] = 1000.0 # numerical infinity: large enough that survival is ~0

    return t

# Example: 64 intervals (n=63), t_max = 15
n = 63
t = compute_time_intervals(n, t_max=15.0, alpha=0.1)

print("First 10 interval boundaries:")
for k in range(10):
    print(f"  t[{k}] = {t[k]:.6f}")
print(f"  ...")
print(f"  t[{n}] = {t[n]:.6f}")
print(f"  t[{n+1}] = {t[n+1]:.6f}")
print(f"\nInterval widths (first 5): ")
    f"{[f'{t[k+1]}-t[{k}]:.4f}' for k in range(5)]}"
print(f"Interval widths (last 5): ")
    f"{[f'{t[k+1]}-t[{k}]:.4f}' for k in range(n-4, n+1)]}"
```

i Why this particular spacing?

The log-spacing ensures roughly equal **information content** per interval. Under the coalescent, the probability of coalescence in interval k is approximately proportional to the interval width in “coalescent-CDF space.” By making intervals wider in the distant past (where the density is thin), each interval contributes a similar amount of statistical power to the inference.

Notice what the output reveals: the first few intervals are tiny (widths on the order of 0.01), while the last interval stretches from $t_{63} = 15.0$ to $t_{64} = 1000.0$. This is exactly the gear-tooth pattern we want – fine teeth near the recent past where the escapement ticks rapidly, and coarse teeth in the distant past where ticks are rare.

Step 2: The Helper Quantities

With the time intervals in hand, we now need to build the mathematical machinery that connects the continuous PSMC model (from [the previous chapter](#)) to a discrete transition matrix. This requires several helper quantities, each of which has a clear physical meaning.

Recall from the continuous model that the population size history is represented as a **piecewise-constant** function: within each interval $[t_k, t_{k+1})$, the relative population size is a constant λ_k . This is the PSMC assumption – the gears are flat-topped teeth, not smooth curves.

i Calculus Aside: What “piecewise constant” means for integration

A piecewise-constant function is one that takes a constant value on each interval of a partition. For PSMC, $\lambda(t) = \lambda_k$ whenever $t \in [t_k, t_{k+1})$. The power of this assumption is that it makes integrals trivial within each interval. For example, if we need $\int_{t_k}^{t_{k+1}} \frac{dt}{\lambda(t)}$, we can pull $1/\lambda_k$ out of the integral to get $\frac{t_{k+1}-t_k}{\lambda_k} = \frac{\tau_k}{\lambda_k}$. This is the same idea that makes Riemann sums work: approximate a curve by rectangles, and each rectangle’s area is just width times height. In PSMC, the “approximation” is exact by assumption – we are *choosing* the population size to be constant within each interval.

Let us define the helper quantities one by one.

The interval widths τ_k :

$$\tau_k = t_{k+1} - t_k$$

These are simply the widths of each gear tooth. Nothing subtle here, but they appear in almost every formula that follows.

The survival factors α_k :

$$\alpha_k = \exp \left(- \sum_{i=0}^{k-1} \frac{\tau_i}{\lambda_i} \right)$$

Intuition: α_k is the probability that two lineages have NOT coalesced by time t_k . In the continuous model from [the previous chapter](#), we defined the cumulative hazard $\Lambda(t) = \int_0^t \frac{du}{\lambda(u)}$, and the survival probability was $e^{-\Lambda(t)}$. The discrete version is identical, except that the integral becomes a sum over intervals (because λ is piecewise constant):

$$\Lambda(t_k) = \sum_{i=0}^{k-1} \frac{\tau_i}{\lambda_i}$$

so $\alpha_k = e^{-\Lambda(t_k)}$.

i Probability Aside: Survival as a product of independent escapes

There is another way to read the formula for α_k . Because the exponential of a sum is the product of exponentials:

$$\alpha_k = \prod_{i=0}^{k-1} e^{-\tau_i/\lambda_i}$$

Each factor $e^{-\tau_i/\lambda_i}$ is the probability of surviving (not coalescing) through interval i alone. The product over all intervals up to k gives the probability of surviving through *all* of them – like passing through k successive gates, each with its own probability of stopping you.

Note two boundary conditions: $\alpha_0 = 1$ (everyone is alive at time 0 – the empty sum gives zero, and $e^0 = 1$) and $\alpha_{n+1} = 0$ (everyone has coalesced by $t_{n+1} = \infty$).

The re-coalescence sums β_k :

$$\beta_k = \sum_{i=0}^{k-1} \lambda_i \left(\frac{1}{\alpha_{i+1}} - \frac{1}{\alpha_i} \right)$$

Intuition: β_k accumulates information about how “easy” it was to coalesce in each interval up to k . It appears in the transition matrix because after a recombination event breaks the genealogy, the new lineage must re-coalesce, and the cumulative coalescence opportunity in each past interval determines how likely each target interval is. Think of it as measuring how many teeth the gear has exposed for re-engagement up to position k .

An auxiliary quantity $q_{\text{aux},k}$:

$$q_{\text{aux},k} = (\alpha_k - \alpha_{k+1}) \left(\beta_k - \frac{\lambda_k}{\alpha_k} \right) + \tau_k$$

This combines survival and re-coalescence information and appears directly in the transition matrix formulas. It is not physically intuitive on its own – it is an algebraic convenience that simplifies the three-case transition matrix into clean expressions. We will see its role in Step 4.

```
def compute_helpers(n, t, lambdas):
    """Compute the helper quantities for the discrete PSMC.

    Parameters
    -----
    n : int
    t : ndarray of shape (n + 2,)
        Time boundaries.
    lambdas : ndarray of shape (n + 1,)
        Relative population size in each interval.

    Returns
    -----
    tau : ndarray of shape (n + 1,)
    alpha : ndarray of shape (n + 2,)
    beta : ndarray of shape (n + 1,)
    q_aux : ndarray of shape (n,)
    C_pi : float
    """
    tau = np.zeros(n + 1)
    alpha = np.zeros(n + 2)
```

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```

beta = np.zeros(n + 1)
q_aux = np.zeros(n)

# tau_k: simple interval widths
for k in range(n + 1):
    tau[k] = t[k + 1] - t[k]

# alpha_k: survival probability up to time t_k
# alpha[0] = 1 because no time has passed, so no coalescence is possible
alpha[0] = 1.0
for k in range(1, n + 1):
    # Each step multiplies by the survival factor for one interval:
    # exp(-tau/lambda) is the probability of NOT coalescing in that interval
    alpha[k] = alpha[k - 1] * np.exp(-tau[k - 1] / lambdas[k - 1])
# By convention, alpha[n+1] = 0 (everything has coalesced by t = infinity)
alpha[n + 1] = 0.0

# beta_k: cumulative re-coalescence opportunity
# beta[0] = 0 because the sum has no terms (empty sum)
beta[0] = 0.0
for k in range(1, n + 1):
    # Each term adds the contribution from interval k-1:
    # lambda_{k-1} * (1/alpha_k - 1/alpha_{k-1})
    beta[k] = beta[k - 1] + lambdas[k - 1] * (1.0 / alpha[k] - 1.0 / alpha[k - 1])

# q_aux: the auxiliary quantity used in transition matrix construction
for k in range(n):
    ak1 = alpha[k] - alpha[k + 1] # probability of coalescing IN interval k
    q_aux[k] = ak1 * (beta[k] - lambdas[k] / alpha[k]) + tau[k]

# C_pi: normalization constant for the stationary distribution.
# This equals the expected coalescence time E[T] under the piecewise model.
C_pi = 0.0
for k in range(n + 1):
    # lambda_k * (alpha_k - alpha_{k+1}) is lambda_k times the probability
    # of coalescing in interval k
    C_pi += lambdas[k] * (alpha[k] - alpha[k + 1])

return tau, alpha, beta, q_aux, C_pi

# Test with constant population (all lambda = 1)
n = 10
t = compute_time_intervals(n, t_max=15.0)
lambdas = np.ones(n + 1)
tau, alpha, beta, q_aux, C_pi = compute_helpers(n, t, lambdas)

print(f"C_pi = {C_pi:.6f} (expected: ~1.0 for constant pop)")
print(f"\nalpha (survival probabilities):")
for k in range(min(n + 2, 8)):
    print(f"  alpha[{k}] = {alpha[k]:.6f}")
print(f"  alpha[{n+1}] = {alpha[n+1]:.6f}")

```

Calculus Aside: Why C_π equals the expected coalescence time

The quantity $C_\pi = \sum_k \lambda_k (\alpha_k - \alpha_{k+1})$ is the normalizing constant for the stationary distribution $\pi(t)$, and it equals $\mathbb{E}[T]$, the expected coalescence time under the piecewise-constant population model. Here is why:

$$\mathbb{E}[T] = \int_0^\infty t \cdot f(t) dt = \int_0^\infty S(t) dt$$

where $S(t) = e^{-\Lambda(t)}$ is the survival function. The second equality is a standard result (integration by parts). Under our piecewise model, $S(t)$ is piecewise exponential, and integrating it interval by interval gives exactly $\sum_k \lambda_k (\alpha_k - \alpha_{k+1})$. For a constant population ($\lambda_k = 1$ for all k), this should equal 1.0 – the mean of an $\text{Exp}(1)$ distribution. The output above confirms this.

With these helpers computed, we have all the gears we need to build the transition matrix. But first, let us compute the stationary distribution – the long-run equilibrium of the HMM.

Step 3: The Discrete Stationary Distribution

The stationary distribution tells us: in the long run, what fraction of genomic positions have their coalescence time in interval k ? This is the discrete analogue of the continuous stationary distribution $\pi(t)$ derived in [the previous chapter](#).

The probability that the coalescence time falls in interval $[t_k, t_{k+1})$ is obtained by integrating $\pi(t)$ over the interval:

$$\pi_k = \int_{t_k}^{t_{k+1}} \pi(t) dt$$

Calculus Aside: What this integral means concretely

The continuous distribution $\pi(t)$ gives a probability *density* – the probability per unit time of the coalescence happening at exactly time t . To find the probability of the coalescence time falling in a finite interval $[t_k, t_{k+1})$, we sum up (integrate) the density over that interval. This is exactly the area under the $\pi(t)$ curve between t_k and t_{k+1} .

If $\pi(t)$ were a complicated function, we might need numerical integration (evaluating the function at many points and summing weighted values). But because PSMC assumes $\lambda(t)$ is piecewise constant, the integral has a closed-form solution – the exponentials integrate analytically.

Under piecewise-constant $\lambda(t)$ (the PSMC assumption), this integral can be evaluated analytically:

$$\pi_k = \frac{1}{C_\pi} \left[(\alpha_k - \alpha_{k+1}) \left(\sum_{i=0}^{k-1} \tau_i + \lambda_k \right) - \alpha_{k+1} \tau_k \right]$$

Derivation. Within interval $[t_k, t_{k+1})$, $\lambda(t) = \lambda_k$, so $\pi(t) = \frac{t}{C_\pi \lambda_k} e^{-\Lambda(t)}$. The integral becomes:

$$\pi_k = \frac{\alpha_k}{C_\pi \lambda_k} \int_{t_k}^{t_{k+1}} e^{-(t-t_k)/\lambda_k} \left[\sum_{i=0}^{k-1} \tau_i + (t - t_k) \right] dt$$

The first term involves $\int x e^{-x/\lambda} dx$ and the second involves $\int e^{-x/\lambda} dx$, both elementary. After evaluating and simplifying (using $\alpha_{k+1} = \alpha_k e^{-\tau_k/\lambda_k}$), we get the result above.

i Calculus Aside: Working out the integral explicitly

Let $w = t - t_k$ so $w \in [0, \tau_k]$. Within interval k , $\Lambda(t) = \Lambda(t_k) + w/\lambda_k$, so $e^{-\Lambda(t)} = \alpha_k e^{-w/\lambda_k}$. The integral becomes:

$$\pi_k = \frac{\alpha_k}{C_\pi \lambda_k} \int_0^{\tau_k} (S_k + w) e^{-w/\lambda_k} dw$$

where $S_k = \sum_{i=0}^{k-1} \tau_i$. We need two standard integrals:

1. $\int_0^a e^{-w/\lambda} dw = \lambda(1 - e^{-a/\lambda})$ – the integral of a decaying exponential, giving a “charging curve.”
2. $\int_0^a w e^{-w/\lambda} dw = \lambda^2(1 - e^{-a/\lambda}) - a\lambda e^{-a/\lambda}$ – found by integration by parts (differentiate w , integrate $e^{-w/\lambda}$).

Substituting these, using $\alpha_{k+1}/\alpha_k = e^{-\tau_k/\lambda_k}$, and simplifying gives the closed form above.

```
# Verify pi_k by numerical integration of the continuous pi(t)
from scipy.integrate import quad

def verify_pi_k(k, t, lambdas, alpha, C_pi):
    """Compare analytic pi_k to numerical integration of continuous pi(t)."""
    def pi_continuous(t_val):
        # Compute Lambda(t_val) via piecewise sum: add up tau_i/lambda_i
        # for each interval that t_val has passed through
        Lambda = 0.0
        for i in range(len(lambdas)):
            t_lo, t_hi = t[i], t[i + 1]
            if t_val <= t_lo:
                break
            # min(t_val, t_hi) - t_lo gives the time spent in interval i
            dt = min(t_val, t_hi) - t_lo
            Lambda += dt / lambdas[i]
            if t_val <= t_hi:
                break
        # pi(t) = t / (C_pi * lambda(t)) * exp(-Lambda(t))
        # but only if t falls in interval k
        return t_val / (C_pi * lambdas[k]) * np.exp(-Lambda) if t[k] <= t_val <= t[k+1] else 0.0

    # quad performs adaptive numerical integration (Gaussian quadrature)
    # It returns (value, error_estimate)
    numerical, _ = quad(pi_continuous, t[k] + 1e-10, t[k + 1] - 1e-10)
    return numerical

# Quick check for the first few intervals
n_check = 10
t_check = compute_time_intervals(n_check, t_max=15.0)
lam_check = np.ones(n_check + 1)
tau_c, alpha_c, _, _, C_pi_c = compute_helpers(n_check, t_check, lam_check)
pi_analytic, _, _ = compute_stationary(n_check, tau_c, alpha_c, lam_check, C_pi_c, 0.001)

for k in range(min(5, n_check + 1)):
```

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```
pi_num = verify_pi_k(k, t_check, lam_check, alpha_c, C_pi_c)
print(f"  pi[{k}]: analytic={pi_analytic[k]:.6f}, numerical={pi_num:.6f}, "
      f"diff={abs(pi_analytic[k] - pi_num):.2e}")
```

This numerical check is important. If the analytic and numerical values disagree, we have a bug. Agreement to machine precision (differences on the order of 10^{-10} or smaller) confirms that our closed-form expression is correct.

The **full stationary distribution** σ_k includes the recombination factor. Recall from [the continuous model](#) that the stationary distribution of the HMM is not just π_k (the coalescence time distribution) but also accounts for how recombination probability depends on branch length:

$$\sigma_k = \frac{1}{C_\sigma} \left[\frac{\alpha_k - \alpha_{k+1}}{C_\pi \rho} + \frac{\pi_k}{2} + o(\rho) \right]$$

The first term dominates when ρ is small (rare recombination), and says that the probability of being in interval k is proportional to $\alpha_k - \alpha_{k+1}$ – the probability of *coalescing* in that interval. The second term, proportional to π_k , is a correction for finite recombination rate.

```
def compute_stationary(n, tau, alpha, lambdas, C_pi, rho):
    """Compute discrete stationary distributions pi_k and sigma_k.

    Parameters
    -----
    n : int
    tau, alpha : from compute_helpers
    lambdas : ndarray of shape (n + 1,)
    C_pi : float
    rho : float

    Returns
    -----
    pi_k : ndarray of shape (n + 1,)
    sigma_k : ndarray of shape (n + 1,)
    C_sigma : float
    """
    pi_k = np.zeros(n + 1)
    sum_tau = 0.0  # running sum of interval widths: sum_{i=0}^{k-1} tau_i

    for k in range(n + 1):
        ak1 = alpha[k] - alpha[k + 1]  # probability of coalescing in interval k
        # The formula combines two parts:
        #   ak1 * (sum_tau + lambda_k): contribution from "reaching" interval k
        #   alpha[k+1] * tau[k]: correction for the probability mass near t_{k+1}
        pi_k[k] = (ak1 * (sum_tau + lambdas[k]) - alpha[k + 1] * tau[k]) / C_pi
        sum_tau += tau[k]

    # C_sigma: normalization for the full stationary distribution
    # 1/(C_pi * rho) + 0.5 is a first-order approximation in rho
    C_sigma = 1.0 / (C_pi * rho) + 0.5

    # sigma_k: the full stationary distribution including recombination
    sigma_k = np.zeros(n + 1)
    for k in range(n + 1):
```

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```

    ak1 = alpha[k] - alpha[k + 1]
    # Two terms: the dominant 1/(C_pi*rho) term and the pi_k/2 correction
    sigma_k[k] = (ak1 / (C_pi * rho) + pi_k[k] / 2.0) / C_sigma

    return pi_k, sigma_k, C_sigma

rho = 0.001
pi_k, sigma_k, C_sigma = compute_stationary(n, tau, alpha, lambdas, C_pi, rho)

print("Discrete stationary distribution pi_k:")
for k in range(min(n + 1, 8)):
    print(f"  pi[{k}] = {pi_k[k]:.6f}")
print(f"\nSum of pi_k: {pi_k.sum():.6f} (should be 1.0)")
print(f"Sum of sigma_k: {sigma_k.sum():.6f} (should be 1.0)")
print(f"C_sigma = {C_sigma:.4f}")

```

Both π_k and σ_k should sum to 1.0 (they are probability distributions). If they do not, something is wrong with the helper quantities. The output above provides a sanity check.

With the stationary distribution in hand, we are ready for the heart of this chapter: the transition matrix.

Step 4: The Discrete Transition Matrix

Now we arrive at the central object of the discretization: the **transition matrix**. This is the gear ratio table of the PSMC watch – it tells us, for every pair of time intervals (k, l) , the probability that if the coalescence time at one genomic position is in interval k , the coalescence time at the next position is in interval l .

Probability Aside: What is a transition matrix?

A **transition matrix** (also called a stochastic matrix) is a square matrix P where P_{kl} gives the probability of moving from state k to state l in one step. Every row must sum to 1 (from any state, you must go *somewhere*), and every entry must be non-negative.

If you know the current state is k , the row $P_{k,:}$ gives the complete probability distribution over the next state. The matrix encodes all the information the HMM needs about state dynamics.

A probability vector σ is a **stationary distribution** of P if $\sigma^T P = \sigma^T$ – applying the transition leaves the distribution unchanged. This is the long-run equilibrium: after many steps, the probability of being in each state converges to σ regardless of the starting state.

For more background, see the [HMM prerequisite chapter](#).

The discrete transition probability from interval k to interval l (conditioned on recombination having occurred) is:

$$q_{kl} = \frac{1}{\pi_k} \int_{t_k}^{t_{k+1}} ds \int_{t_l}^{t_{l+1}} q(t|s) \pi(s) dt$$

This double integral averages the continuous transition density $q(t|s)$ (derived in [the previous chapter](#)) over all source times s in interval k and all target times t in interval l . The factor $1/\pi_k$ normalizes by the probability of starting in interval k .

Calculus Aside: The double integral as a weighted average

The formula $q_{kl} = \frac{1}{\pi_k} \int \int q(t|s) \pi(s) dt ds$ is a **conditional expectation**. It says: given that the source coalescence

time s is somewhere in interval k (with distribution $\pi(s)/\pi_k$), what is the probability that the target coalescence time t lands in interval l ?

Under the piecewise-constant assumption, the double integral splits into products of single integrals that can each be evaluated analytically. The key trick is that $q(t|s)$ involves a $\min(s, t)$ term (from the continuous model), which means the integral behaves differently depending on whether $t < s$, $t = s$, or $t > s$ – and hence on the relationship between intervals l and k .

This double integral has different forms depending on whether $l < k$, $l = k$, or $l > k$. The derivations are lengthy but the results are clean:

Case $l < k$ (transition to an earlier interval):

$$q_{kl} = \frac{\alpha_k - \alpha_{k+1}}{C_\pi \pi_k} \cdot q_{\text{aux}, l}$$

The new coalescence time is *shallower* (more recent) than the old one. Recombination broke the genealogy, and the new lineage found a partner quickly – it re-coalesced in an earlier interval. Notice that the $q_{\text{aux}, l}$ factor depends only on the *target* interval l , not on the source k (except through the leading normalizing factor). This is the factorization that makes computation efficient.

Case $l = k$ (staying in the same interval):

$$q_{kk} = \frac{1}{C_\pi \pi_k} \left[(\alpha_k - \alpha_{k+1})^2 \left(\beta_k - \frac{\lambda_k}{\alpha_k} \right) + 2\lambda_k(\alpha_k - \alpha_{k+1}) - 2\alpha_{k+1}\tau_k \right]$$

This is the most complex case because the double integral's $\min(s, t)$ term changes behavior within the same interval (sometimes $t < s$, sometimes $t > s$). Both sub-cases contribute to the result.

Case $l > k$ (transition to a later interval):

$$q_{kl} = \frac{\alpha_l - \alpha_{l+1}}{C_\pi \pi_k} \cdot q_{\text{aux}, k}$$

The new coalescence time is *deeper* (more ancient) than the old one. The recombined lineage drifted into the deeper past before finding a partner. Here, $q_{\text{aux}, k}$ depends only on the *source* interval k , and $\alpha_l - \alpha_{l+1}$ depends only on the *target*.

Structural insight: The factorization across cases is the key to efficiency. For $l < k$, the contribution to column l is the same $q_{\text{aux}, l}$ for every row $k > l$. For $l > k$, the contribution from row k is the same $q_{\text{aux}, k}$ for every column $l > k$. This means we can fill the entire $(n+1) \times (n+1)$ matrix in $O(n)$ work per row, not $O(n^2)$.

Probability Aside: Why the cases split this way

The three cases arise from splitting the double integral over the $\min(s, t)$ in the continuous transition density:

- **$l < k$ ($t < s$):** the new coalescence time is shallower than the old one. Recombination happened somewhere on the branch, and the detached lineage re-coalesced earlier. The integral upper limit $\min(s, t) = t$ removes all dependence on s from the inner integral.
- **$l > k$ ($t > s$):** the new coalescence time is deeper. Here $\min(s, t) = s$, and the inner integral from 0 to s captures the full branch length.
- **$l = k$:** both cases overlap, producing the most complex expression.

This factorization means the full $(n+1) \times (n+1)$ matrix can be computed in $O(n)$ time per row (not $O(n^2)$), since the q_{aux} values are shared.

The **full transition matrix** (including no-recombination) is:

$$p_{kl} = \frac{\pi_k}{C_\sigma \sigma_k} q_{kl} + \delta_{kl} \left(1 - \frac{\pi_k}{C_\sigma \sigma_k} \right)$$

Why this form? This is a mixture of two behaviors, weighted by whether recombination occurred:

- With probability $\frac{\pi_k}{C_\sigma \sigma_k}$, recombination happens, and the new coalescence time is drawn from q_{kl} . The lineage's gear "slips" to a new tooth.
- With probability $1 - \frac{\pi_k}{C_\sigma \sigma_k}$, no recombination happens, and the coalescence time stays in the same interval. The δ_{kl} (Kronecker delta: 1 if $k = l$, 0 otherwise) ensures this contributes only to the diagonal.

In the watch metaphor, this is the difference between a gear tooth engaging a new position (recombination) and the gear holding steady (no recombination). Most of the time, adjacent genomic positions share the same coalescence time – the gear holds. Occasionally, recombination breaks the genealogy and the gear clicks forward or backward.

```
def compute_transition_matrix(n, tau, alpha, beta, q_aux, lambdas,
                             C_pi, C_sigma, pi_k, sigma_k):
    """Compute the full discrete PSMC transition matrix.

    Parameters
    -----
    n : int
    tau, alpha, beta, q_aux : from compute_helpers
    lambdas : population sizes
    C_pi, C_sigma : normalization constants
    pi_k, sigma_k : stationary distributions

    Returns
    -----
    p : ndarray of shape (n+1, n+1)
        Transition matrix p_{kl}.
    q : ndarray of shape (n+1, n+1)
        Transition matrix q_{kl} (conditioned on recombination).
    """
    N = n + 1 # number of hidden states (one per time interval)
    q = np.zeros((N, N)) # q[k, l] = transition prob given recombination

    for k in range(N):
        # ak1 = alpha_k - alpha_{k+1}: probability of coalescing in interval k
        ak1 = alpha[k] - alpha[k + 1]

        # cpik = C_pi * pi_k[k]: the unnormalized stationary probability.
        # We compute it directly to avoid dividing by pi_k (which could be ~0).
        cpik = ak1 * (sum(tau[:k]) + lambdas[k]) - alpha[k + 1] * tau[k]

        if cpik < 1e-30:
            # Fallback for negligible-probability intervals: uniform transitions
            q[k, :] = 1.0 / N
            continue

        # Case 1: l < k (transition to an earlier/shallower interval)
        # q[k, l] = ak1 / cpik * q_aux[l]
        for l in range(k):
            q[k, l] = ak1 / cpik * q_aux[l]

        # Case 2: l = k (staying in the same interval)
        q[k, k] = (ak1 * ak1 * (beta[k] - lambdas[k] / alpha[k])
                    + 2 * lambdas[k] * ak1
```

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```

        - 2 * alpha[k + 1] * tau[k]) / cpik

    # Case 3: l > k (transition to a later/deeper interval)
    if k < n:
        for l in range(k + 1, N):
            # (alpha_l - alpha_{l+1}) / cpik * q_aux[k]
            q[k, l] = (alpha[l] - alpha[l + 1]) / cpik * q_aux[k]

    # Full transition matrix p_{kl}: mixture of recombination and no-recombination
    p = np.zeros((N, N))
    for k in range(N):
        # recomb_prob = pi_k / (C_sigma * sigma_k): probability of recombination
        recomb_prob = pi_k[k] / (C_sigma * sigma_k[k]) if sigma_k[k] > 0 else 0
        for l in range(N):
            p[k, l] = recomb_prob * q[k, l] # recombination contribution
            if k == l:
                p[k, l] += (1.0 - recomb_prob) # no-recombination: stay in place

    return p, q

p, q = compute_transition_matrix(n, tau, alpha, beta, q_aux, lambdas,
                                C_pi, C_sigma, pi_k, sigma_k)

print("Transition matrix q (conditioned on recomb):")
print(f"  Shape: {q.shape}")
print(f"  Row sums: min={q.sum(axis=1).min():.6f}, "
      f"max={q.sum(axis=1).max():.6f}")

print(f"\nFull transition matrix p:")
print(f"  Shape: {p.shape}")
print(f"  Row sums: min={p.sum(axis=1).min():.6f}, "
      f"max={p.sum(axis=1).max():.6f}")

# Check that sigma is the stationary distribution of p:
# sigma^T @ p should equal sigma^T (left eigenvector with eigenvalue 1)
sigma_check = sigma_k @ p # matrix-vector product: sigma * P
max_diff = np.max(np.abs(sigma_check - sigma_k))
print(f"\nStationary distribution check: max|sigma*p - sigma| = {max_diff:.2e}")

```

i Probability Aside: Interpreting the verification

The row sums of both q and p should be exactly 1.0 (or very close, up to floating-point rounding). A row sum different from 1 would mean that from some state, the total probability of going *somewhere* is not 100% – a physical impossibility.

The stationary distribution check $\sigma^T P = \sigma^T$ is even more stringent. It says that if the probability of being in each state is given by σ , then after one step of the Markov chain, the distribution is unchanged. The `@` operator in NumPy performs matrix multiplication, so `sigma_k @ p` computes $\sigma^T P$ and the result should equal σ^T . Differences on the order of 10^{-10} or smaller indicate success.

We now have the complete gear ratio table. The transition matrix p is one of the three ingredients the HMM needs (along with emission probabilities and an initial distribution). Let us compute the second ingredient next.

Step 5: The Average Coalescence Time per Interval

For emission probabilities, we need a representative coalescence time for each interval – not just the midpoint or the left boundary, but the **effective time** at which mutations accumulate as if the coalescence happened at that single point.

Why does this matter? The emission probability in the PSMC HMM is $P(\text{het}|\text{state } k) = 1 - e^{-\theta \bar{t}_k}$, where $\bar{t}_k$ is the effective coalescence time in interval k and θ is the mutation rate. Using the wrong $\bar{t}_k$ would bias the emission probabilities and corrupt the inference.

PSMC uses a clever trick to define $\bar{t}_k$. From the full transition matrix, the probability of *not* recombining when in interval k is $p_{kk}^{\text{stay}} = 1 - \pi_k / (C_\sigma \sigma_k)$. On a branch of length t , the no-recombination probability is $e^{-\rho t}$. Setting these equal and solving:

$$\bar{t}_k = -\frac{1}{\rho} \ln \left(1 - \frac{\pi_k}{C_\sigma \sigma_k} \right)$$

Calculus Aside: The logarithm as an inverse of the exponential

The equation $e^{-\rho \bar{t}} = 1 - r$ (where $r = \pi_k / (C_\sigma \sigma_k)$ is the recombination probability) is solved by taking the natural log of both sides: $-\rho \bar{t} = \ln(1 - r)$, hence $\bar{t} = -\ln(1 - r) / \rho$.

This is well-defined as long as $r < 1$ (there is some probability of *not* recombining). When r is small (rare recombination), we can use the Taylor expansion $-\ln(1 - r) \approx r + r^2/2 + \dots$, so $\bar{t} \approx r / \rho$. When r approaches 1 (very long branches with near-certain recombination), $\bar{t}$ grows without bound, as expected.

Why this particular time? The no-recombination probability on a branch of length t is $e^{-\rho t}$. The discrete model assigns $p_{kk}^{\text{stay}} = 1 - \pi_k / (C_\sigma \sigma_k)$ as the stay probability. By setting $e^{-\rho \bar{t}_k} = p_{kk}^{\text{stay}}$, we find the effective time that would produce this stay probability. This same time is then used for computing the emission (mutation) probability. The beauty of this approach is that it ensures consistency: the same effective branch length governs both recombination and mutation.

```
def compute_avg_times(n, tau, alpha, lambdas, pi_k, sigma_k, C_sigma, rho):
    """Compute the effective coalescence time for each interval.

    Parameters
    -----
    Returns
    -----
    avg_t : ndarray of shape (n + 1,)
    """
    avg_t = np.zeros(n + 1)
    sum_tau = 0.0    # running sum of interval widths

    for k in range(n + 1):
        ak1 = alpha[k] - alpha[k + 1]
        recomb_prob = pi_k[k] / (C_sigma * sigma_k[k]) if sigma_k[k] > 0 else 0

        if recomb_prob < 1.0:
            # Main formula: -log(1 - recomb_prob) / rho
            # np.log computes the natural logarithm (base e)
            avg_t[k] = -np.log(1.0 - recomb_prob) / rho
        else:
            # Fallback: if recomb_prob = 1 (log would be -inf),
            # use the conditional mean coalescence time within the interval
            lak = lambdas[k]
```

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```

    avg_t[k] = sum_tau + (lak - tau[k] * alpha[k + 1] / ak1
                        if ak1 > 0 else tau[k] / 2)

    # Sanity check: avg_t[k] should lie within the interval [t_k, t_{k+1}]
    if np.isnan(avg_t[k]) or avg_t[k] < sum_tau or avg_t[k] > sum_tau + tau[k]:
        lak = lambdas[k]
        ak1 = alpha[k] - alpha[k + 1]
        avg_t[k] = sum_tau + (lak - tau[k] * alpha[k + 1] / ak1
                            if ak1 > 0 else tau[k] / 2)

    sum_tau += tau[k]

    return avg_t

avg_t = compute_avg_times(n, tau, alpha, lambdas, pi_k, sigma_k, C_sigma, rho)

print("Effective coalescence times per interval:")
for k in range(min(n + 1, 8)):
    print(f"    Interval [{t[k]:.4f}, {t[k+1]:.4f}]: "
          f"avg_t = {avg_t[k]:.4f}")

```

The effective times should lie within their respective intervals. If you see an effective time that is outside the interval boundaries, the fallback has been triggered, which is fine for edge cases but should not happen for typical parameters.

Step 6: Parameter Grouping (The -p Pattern)

We have built all the mathematical machinery for an HMM with $n + 1$ hidden states, each with its own population size parameter λ_k . But there is a practical problem: with 64 time intervals, estimating 64 independent λ_k values leads to **overfitting**.

The issue is statistical power. Each λ_k is estimated from the recombination events that fall in interval k . For narrow intervals in the recent past, there may be plenty of such events. But for wide intervals in the distant past, events are sparse, and the estimate of λ_k becomes noisy and unreliable.

In the watch metaphor, this is like trying to read a clock face with 64 tick marks when you can only reliably distinguish 28 of them. Better to group the fine tick marks into coarser markings that you can read reliably.

PSMC solves this by **grouping** adjacent intervals to share the same λ parameter. The `-p` option specifies the grouping pattern:

```
-p "4+25*2+4+6"
```

This means:

- The first group spans 4 atomic intervals (they share one λ)
- The next 25 groups each span 2 atomic intervals
- Then one group spans 4 intervals
- The last group spans 6 intervals

Total: $4 + 25 \times 2 + 4 + 6 = 64$ atomic intervals, but only $1 + 25 + 1 + 1 = 28$ free λ parameters.

The grouping is denser in the middle of the time range (25 groups of 2, giving fine resolution) and coarser at the extremes (groups of 4 and 6, where the data are less informative). This is another example of the same philosophy as log-spacing: put resolution where the information is.

```

def parse_pattern(pattern):
    """Parse a PSMC pattern string into a parameter map.

    Parameters
    -----
    pattern : str
        Pattern like "4+25*2+4+6".

    Returns
    -----
    par_map : list of int
        par_map[k] = index of the free parameter for interval k.
    n_free : int
        Number of free parameters.
    n_intervals : int
        Total number of atomic intervals (= n + 1).
    """
    par_map = []
    free_idx = 0

    for part in pattern.split('+'):
        if '*' in part:
            # "25*2" means 25 groups, each spanning 2 atomic intervals
            count, width = part.split('*')
            count, width = int(count), int(width)
            for _ in range(count):
                # Each group of 'width' intervals shares one free parameter
                for _ in range(width):
                    par_map.append(free_idx)
                    free_idx += 1
            else:
                # "4" means one group spanning 4 atomic intervals
                width = int(part)
                for _ in range(width):
                    par_map.append(free_idx)
                    free_idx += 1

    return par_map, free_idx, len(par_map)

par_map, n_free, n_intervals = parse_pattern("4+25*2+4+6")
print(f"Pattern: 4+25*2+4+6")
print(f"Total atomic intervals: {n_intervals}")
print(f"Free parameters: {n_free}")
print(f"First 10 of par_map: {par_map[:10]}")
print(f>Last 10 of par_map: {par_map[-10:]}")

```

How to choose the pattern

The rule of thumb from Li and Durbin: each free parameter should span at least ~10 expected recombination events. You can check this with $C_\sigma \pi_k$ – if this is less than ~20 for the intervals in a group, you should merge them into a larger group. The widely used 4+25*2+4+6 setting has been used for human whole-genome data. The original program's compiled-in default is the coarser 4+5*3+4; many published command lines override it with -p.

In practice, you rarely need to change the pattern unless you are working with organisms that have very different genome sizes or recombination rates. For short genomes (fewer recombination events), you may need fewer, wider groups. For very long genomes or high recombination rates, you could afford finer resolution.

Step 7: Putting It All Together

We have now built every piece of the discrete PSMC model: time intervals, helper quantities, stationary distribution, transition matrix, effective times, and parameter grouping. Let us assemble them into a single function that takes population parameters and produces the complete HMM specification – the three ingredients that the *next chapter's* EM algorithm will use.

In the watch metaphor, this is the moment we snap all the gears into the movement and close the case. The function below is the complete gear train: continuous time goes in, discrete HMM parameters come out.

```
def build_psmc_hmm(n, t_max, theta, rho, lambdas, par_map=None, alpha_param=0.1):
    """Build the complete discrete PSMC HMM parameters.

    Parameters
    -----
    n : int
        Number of time intervals minus 1.
    t_max : float
    theta : float
        Mutation rate per bin.
    rho : float
        Recombination rate per bin.
    lambdas : ndarray of shape (n + 1,)
        Relative population sizes (lambda_k for each atomic interval).
    par_map : list, optional
        Parameter grouping. If provided, lambdas should have n_free entries
        and will be expanded.
    alpha_param : float
        Spacing parameter for time intervals.

    Returns
    -----
    transitions : ndarray of shape (n+1, n+1)
    emissions : ndarray of shape (2, n+1)
        emissions[0, k] = P(hom | state k), emissions[1, k] = P(het | state k)
    initial : ndarray of shape (n+1,)
    """
    # Expand lambdas if using parameter grouping:
    # par_map[k] maps atomic interval k to its free parameter index
    if par_map is not None:
        full_lambdas = np.array([lambdas[par_map[k]] for k in range(n + 1)])
    else:
        full_lambdas = lambdas

    # Compute time intervals (the gear teeth positions)
    t = compute_time_intervals(n, t_max, alpha_param)

    # Compute helpers (survival factors, re-coalescence sums, etc.)
    tau, alpha_arr, beta, q_aux, C_pi = compute_helpers(n, t, full_lambdas)
```

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```

# Stationary distributions (long-run equilibrium of the HMM)
pi_k, sigma_k, C_sigma = compute_stationary(
    n, tau, alpha_arr, full_lambdas, C_pi, rho)

# Transition matrix (the gear ratio table)
transitions, _ = compute_transition_matrix(
    n, tau, alpha_arr, beta, q_aux, full_lambdas,
    C_pi, C_sigma, pi_k, sigma_k)

# Average times for emissions (effective branch lengths per interval)
avg_t = compute_avg_times(
    n, tau, alpha_arr, full_lambdas, pi_k, sigma_k, C_sigma, rho)

# Emission probabilities:
# P(hom | state k) = exp(-theta * avg_t[k])    (no mutation on either branch)
# P(het | state k) = 1 - exp(-theta * avg_t[k]) (at least one mutation)
emissions = np.zeros((2, n + 1))
for k in range(n + 1):
    # np.exp(-theta * avg_t[k]): probability of zero mutations on a branch
    # of length avg_t[k] with mutation rate theta (Poisson model)
    emissions[0, k] = np.exp(-theta * avg_t[k])      # P(hom | state k)
    emissions[1, k] = 1 - emissions[0, k]           # P(het | state k)

# Initial distribution: the HMM starts in the stationary distribution
initial = sigma_k.copy()

return transitions, emissions, initial

# Build the HMM
n = 10
theta = 0.001
rho = theta / 5 # theta/rho ratio of 5
lambdas = np.ones(n + 1)

transitions, emissions, initial = build_psmc_hmm(
    n, t_max=15.0, theta=theta, rho=rho, lambdas=lambdas)

print("HMM parameters:")
print(f" States: {n + 1}")
print(f" Transition matrix shape: {transitions.shape}")
print(f" Row sums: {transitions.sum(axis=1)}")
print(f" Emission probabilities (het) for first 5 states: "
      f"{emissions[1, :5]}")
print(f" Initial distribution sums to: {initial.sum():.6f}")

```

The output confirms that the HMM is well-formed: all transition matrix rows sum to 1, and the initial distribution sums to 1. The emission probabilities for heterozygosity increase with the state index – deeper coalescence times mean longer branches, hence more mutations and more heterozygous sites. This makes physical sense: the watch reads higher numbers on the dial for deeper times.

With this function, we have completed the gear train. The continuous-time PSMC model from the [previous chapter](#) has been discretized into a finite HMM with concrete, computable parameters. In the [next chapter](#), we will connect this HMM

to data using the forward-backward algorithm and EM, allowing the parameters to learn from the observed sequence of heterozygous and homozygous sites.

Exercises

i Exercise 1: Verify the transition matrix

Build the transition matrix for a constant population ($\lambda_k = 1$ for all k) with $n = 20$ intervals. Verify that: (a) all rows sum to 1, (b) the stationary distribution σ_k satisfies $\sigma^T P = \sigma^T$, (c) starting from any initial distribution, repeated application of P converges to σ .

i Exercise 2: Sensitivity to population size

Build transition matrices for: (a) $\lambda_k = 1$ (constant), (b) $\lambda_k = 10$ for intervals near $t = 1$ and 1 elsewhere (expansion), (c) $\lambda_k = 0.1$ for intervals near $t = 1$ and 1 elsewhere (bottleneck).

Compare the emission probabilities. How does the bottleneck affect heterozygosity?

i Exercise 3: The effect of pattern choice

Using the same data, run PSMC with patterns "64*1" (all free), "4+25*2+4+6" (widely used), and "32*2" (simple). Compare the inferred λ_k curves. Which shows overfitting? Which is too smooth?

Next: *The PSMC HMM and EM Algorithm* – the EM algorithm that makes the parameters talk to the data.

Solutions

i Solution 1: Verify the transition matrix

We build the transition matrix for a constant population and verify three fundamental properties: row normalization, stationarity, and convergence.

```
import numpy as np

# Setup: n=20 intervals, constant population
n = 20
t = compute_time_intervals(n, t_max=15.0)
lambdas = np.ones(n + 1)
rho = 0.001

# Compute all helper quantities
tau, alpha, beta, q_aux, C_pi = compute_helpers(n, t, lambdas)
pi_k, sigma_k, C_sigma = compute_stationary(n, tau, alpha, lambdas, C_pi, rho)
p, q = compute_transition_matrix(n, tau, alpha, beta, q_aux, lambdas,
                                C_pi, C_sigma, pi_k, sigma_k)

# (a) Verify all rows sum to 1
row_sums = p.sum(axis=1)
print("(a) Row sums of P:")
```

```

print(f"  min = {row_sums.min():.10f}")
print(f"  max = {row_sums.max():.10f}")
assert np.allclose(row_sums, 1.0, atol=1e-10), "Row sums are not 1!"

# (b) Verify  $\sigma^T P = \sigma^T$ 
sigma_after = sigma_k @ p
max_diff = np.max(np.abs(sigma_after - sigma_k))
print(f"\n(b) Stationarity check: max|sigma*P - sigma| = {max_diff:.2e}")
assert max_diff < 1e-10, "Sigma is not stationary!"

# (c) Convergence from arbitrary initial distribution
# Start from a point mass on state 0
dist = np.zeros(n + 1)
dist[0] = 1.0

print("\n(c) Convergence from delta(state=0):")
for iteration in [1, 5, 10, 50, 100, 500]:
    # Apply P repeatedly: dist @ P^iteration
    current = dist.copy()
    for _ in range(iteration):
        current = current @ p
    max_err = np.max(np.abs(current - sigma_k))
    print(f"  After {iteration:4d} steps: max|dist - sigma| = {max_err:.6e}")

```

(a) All row sums should equal 1.0 to within machine precision ($\sim 10^{-15}$), confirming that p is a valid stochastic matrix.

(b) The difference $|\sigma^T P - \sigma^T|$ should be on the order of 10^{-12} or smaller, confirming that σ is indeed the stationary distribution of P .

(c) The distribution converges to σ exponentially fast. After ~ 100 steps, the maximum deviation should be negligible ($< 10^{-6}$). This demonstrates that regardless of the starting distribution, the Markov chain converges to its unique equilibrium – the stochastic analogue of a watch settling into its natural rhythm.

Solution 2: Sensitivity to population size

We build transition matrices and emission probabilities for three population scenarios and compare how demography affects heterozygosity.

```

import numpy as np

n = 20
t = compute_time_intervals(n, t_max=15.0)
theta = 0.001
rho = 0.0002

# Identify intervals near t=1 in coalescent units
t_mid = [(t[k] + t[k+1]) / 2.0 for k in range(n + 1)]
near_t1 = [k for k in range(n + 1) if 0.5 < t_mid[k] < 1.5]

scenarios = {}

# (a) Constant population
lam_a = np.ones(n + 1)

```

```

scenarios["(a) Constant"] = lam_a

# (b) Expansion near t=1: lambda_k = 10 for intervals near t=1
lam_b = np.ones(n + 1)
for k in near_t1:
    lam_b[k] = 10.0
scenarios["(b) Expansion"] = lam_b

# (c) Bottleneck near t=1: lambda_k = 0.1 for intervals near t=1
lam_c = np.ones(n + 1)
for k in near_t1:
    lam_c[k] = 0.1
scenarios["(c) Bottleneck"] = lam_c

for name, lam in scenarios.items():
    trans, emissions, initial = build_psmc_hmm(
        n, t_max=15.0, theta=theta, rho=rho, lambdas=lam)

    # Weighted average heterozygosity under the stationary distribution
    avg_het = np.sum(initial * emissions[1, :])
    print(f"{name}:")
    print(f"  Emission P(het) range: [{emissions[1, :].min():.6f}, "
          f"{emissions[1, :].max():.6f}]")
    print(f"  Weighted avg heterozygosity: {avg_het:.6f}")
    print(f"  Intervals near t=1: lambda = {lam[near_t1[0]]:.1f}")

```

How the bottleneck affects heterozygosity: The bottleneck ($\lambda_k = 0.1$) increases the coalescence rate in those intervals, meaning lineages are forced to coalesce quickly. This produces shorter effective branch lengths and therefore lower emission probabilities $P(\text{het}|k) = 1 - e^{-\theta t_k}$ for the affected intervals. The weighted average heterozygosity decreases compared to the constant-population case.

Conversely, the expansion ($\lambda_k = 10$) reduces the coalescence rate, allowing longer branches and higher heterozygosity in those intervals. The overall effect depends on the stationary weight assigned to those intervals – a large population means fewer coalescence events there, but the ones that do occur have longer branch lengths.

Solution 3: The effect of pattern choice

We compare three grouping patterns on the same simulated data to demonstrate overfitting vs. over-smoothing. The key insight is that the pattern controls the bias-variance tradeoff: more free parameters reduce bias but increase variance.

```

import numpy as np

# Simulate data under a known bottleneck model
np.random.seed(42)
n = 63
N = n + 1 # 64 intervals

# True population history: bottleneck from t=0.5 to t=1.5
t = compute_time_intervals(n, t_max=15.0)
true_lambdas = np.ones(N)
for k in range(N):
    t_mid = (t[k] + t[k+1]) / 2.0

```

```

    if 0.5 < t_mid < 1.5:
        true_lambdas[k] = 0.1

theta = 0.001
rho = theta / 5

# Generate observation sequence
# seq, _ = simulate_psmc_input(100000, theta, rho,
#     lambda t: 0.1 if 0.5 < t < 1.5 else 1.0)

# Three patterns to compare
patterns = {
    "64*1 (all free)": "64*1", # 64 free parameters
    "4+25*2+4+6 (widely used)": "4+25*2+4+6", # 28 free parameters
    "32*2 (simple)": "32*2", # 32 free parameters
}

for name, pattern in patterns.items():
    par_map, n_free, n_intervals = parse_pattern(pattern)
    print(f"{name}: {n_free} free parameters, {n_intervals} intervals")

# In a full run:
# results = psmc_inference(seq, n=63, pattern=pattern, n_iters=25)
# final_lambdas = results[-1]['lambdas']

```

Expected results and interpretation:

- **“64*1” (all free):** With 64 independent λ parameters, the inferred curve will show the bottleneck but will also have noisy spikes and dips, especially in the recent and ancient past where few recombination events provide signal. This is **overfitting**: the model has enough flexibility to fit noise in the data. The symptom is a jagged curve that changes dramatically between bootstrap replicates.
- **“4+25*2+4+6” (widely used):** With 28 free parameters, the recent and ancient intervals are grouped (reducing noise), while the intermediate past retains fine resolution. The bottleneck should be clearly visible as a smooth dip. This is the **sweet spot** chosen by Li and Durbin for human data.
- **“32*2” (simple):** With 32 free parameters, every pair of adjacent intervals shares one λ . This provides moderate resolution everywhere but may be **too smooth** in the intermediate past, slightly blurring the edges of the bottleneck. It will miss fine temporal structure but will be more robust to noise than “64*1”.

The general rule: check $C_\sigma \sigma_k$ for each parameter group. If the expected number of recombination events per group is below ~20, the group should be made wider (merged with neighbors) to avoid overfitting.

The PSMC HMM and EM Algorithm

The gear train: turning mathematical insight into a learning machine.

In the previous chapters, we derived all the HMM parameters as functions of θ_0 , ρ_0 , and $\lambda_0, \dots, \lambda_n$. We built the time intervals (*Discretizing Time*), computed the transition matrix, and worked out emission probabilities – all the individual gears of the PSMC watch. Now we close the loop: given the observed data (the heterozygosity sequence), use the **Expectation-Maximization (EM) algorithm** to find the parameters that best explain the data.

Think of it this way. So far we have been a watchmaker who knows how to cut gears of any size. But we have not yet figured out *which* sizes to cut. The data – the sequence of heterozygous and homozygous sites along the genome – is the standard clock signal we are trying to match. The EM algorithm is the process of iteratively adjusting the gears until the watch keeps time: we try a set of gear sizes, see how well the watch matches the signal, adjust, and repeat. Each iteration

brings the gears closer to the right configuration.

This chapter has five main sections:

1. **The complete HMM specification** – collecting everything from earlier chapters into one place.
2. **The forward-backward algorithm** – the computational engine that asks “given these gear sizes, how well does the watch keep time, and which configurations are most likely?”
3. **The EM algorithm** – the iterative adjustment loop.
4. **The full EM loop** – putting it all together for real inference.
5. **Multiple sequences and missing data** – handling real genomes.

If any of the HMM concepts feel unfamiliar – forward probabilities, scaling, posterior decoding – revisit [the HMM prerequisite chapter](#), where we built these algorithms from scratch. This chapter assumes you are comfortable with the forward algorithm and the idea of hidden states; we will explain the backward algorithm and the EM machinery in full detail here.

Step 1: The Complete HMM Specification

Before we can learn anything, we need to state precisely what our HMM looks like. Let’s collect every piece from the previous chapters into one place.

The state space is $\{0, 1, \dots, n\}$, where state k means “the coalescence time at this position falls in the interval $[t_k, t_{k+1})$.” In the watch metaphor, each state is a possible gear configuration – a hypothesis about how deep in time the two haplotypes share their most recent common ancestor at this genomic position.

Initial distribution:

$$a_0(k) = \sigma_k = \frac{1}{C_\sigma} \left[\frac{\alpha_k - \alpha_{k+1}}{C_\pi \rho} + \frac{\pi_k}{2} \right]$$

This is the stationary distribution we derived in [Discretizing Time](#). It tells us, before looking at any data, how likely each coalescence time interval is. The watch starts in its equilibrium state.

Transition matrix:

$$p_{kl} = \frac{\pi_k}{C_\sigma \sigma_k} q_{kl} + \delta_{kl} \left(1 - \frac{\pi_k}{C_\sigma \sigma_k} \right)$$

This is the probability of moving from interval k at one genomic position to interval l at the next. It captures both the possibility of recombination (which reshuffles the coalescence time) and no recombination (which keeps it the same). This matrix was derived in [Discretizing Time](#), Step 4.

Emission probabilities:

$$e_k(0) = e^{-\theta \bar{t}_k} \quad (\text{homozygous})$$

$$e_k(1) = 1 - e^{-\theta \bar{t}_k} \quad (\text{heterozygous})$$

where $\bar{t}_k = -\frac{1}{\rho} \ln \left(1 - \frac{\pi_k}{C_\sigma \sigma_k} \right)$ is the effective coalescence time in interval k .

The intuition: a deeper coalescence time (larger $\bar{t}_k$) means the two haplotypes have been evolving independently for longer, so there is a higher chance of observing a heterozygous site. A shallow coalescence time means recent common ancestry and thus more homozygous sites.

Observation alphabet: $\{0, 1\}$ (hom/het). Missing data gets $e_k(\text{missing}) = 1$ for all k .

i Why missing data gets emission probability 1

Setting $e_k(\text{missing}) = 1$ for all states means “this observation is equally consistent with every possible coalescence time.” Missing data provides zero information – it neither favors nor disfavors any state. The HMM simply passes through missing positions, maintaining the transition dynamics but learning nothing from them. This is a standard HMM technique (see *Hidden Markov Models*) and requires no special machinery.

Now let's implement this complete specification:

```
import numpy as np

class PSMC_HMM:
    """Complete PSMC Hidden Markov Model.

    This class bundles together all the HMM parameters (transitions,
    emissions, initial distribution) and provides methods for computing
    likelihoods and running the forward-backward algorithm.

    Think of this as the fully assembled watch: all the gears from the
    discretization chapter are now meshed together into a working mechanism.

    Parameters
    -----
    n : int
        n+1 = number of hidden states (time intervals).
    theta : float
        Mutation rate per bin.
    rho : float
        Recombination rate per bin.
    lambdas : ndarray of shape (n + 1,)
        Relative population sizes.
    t_max : float
    alpha_param : float
    """

    def __init__(self, n, theta, rho, lambdas, t_max=15.0, alpha_param=0.1):
        self.n = n
        self.N = n + 1 # number of states (one per time interval)
        self.theta = theta
        self.rho = rho
        self.lambdas = lambdas.copy() # store our own copy to avoid aliasing
        self.t_max = t_max
        self.alpha_param = alpha_param

        # Build all HMM parameters from the population-genetic quantities.
        # build_psmc_hmm() was defined in the discretization chapter and
        # returns the transition matrix, emission matrix, and initial dist.
        self.transitions, self.emissions, self.initial = build_psmc_hmm(
            n, t_max, theta, rho, lambdas, alpha_param=alpha_param)

    def log_likelihood(self, seq):
        """Compute log-likelihood of an observation sequence.
```

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```

Uses the scaled forward algorithm (see Step 2 below).

Parameters
-----
seq : ndarray of shape (L,), dtype=int
      Observation sequence (0 = hom, 1 = het, 2+ = missing).

Returns
-----
ll : float
      Log-likelihood log P(X | theta, rho, lambdas).
"""
_, ll = self.forward_scaled(seq)
return ll

def forward_scaled(self, seq):
    """Scaled forward algorithm.

    The forward algorithm computes, for each position a and state k,
    the joint probability P(X_1, ..., X_a, Z_a = k) -- "the probability
    of seeing these observations AND being in state k right now."

    We use scaling to prevent numerical underflow. At each position,
    we normalize the forward probabilities to sum to 1 and accumulate
    the log of the normalization constants. The sum of these logs
    gives us the log-likelihood. (This technique is explained in detail
    in the HMM prerequisite chapter: see :ref:`hmm`.)

    Returns
    -----
    alpha_hat : ndarray of shape (L, N)
        Scaled forward probabilities.
    log_likelihood : float
    """
    L = len(seq)          # length of observation sequence
    N = self.N            # number of hidden states
    alpha_hat = np.zeros((L, N)) # scaled forward probabilities
    log_likelihood = 0.0    # accumulator for log P(X)

    # --- Initialization (position 0) ---
    # alpha(0, k) = P(Z_0 = k) * P(X_0 | Z_0 = k)
    #              = initial[k] * emission[X_0, k]
    for k in range(N):
        obs = seq[0]          # observation at position 0
        if obs >= 2:          # missing data: no information
            e = 1.0
        else:
            e = self.emissions[obs, k] # P(obs | state k)
        alpha_hat[0, k] = self.initial[k] * e

    # Scale: normalize so alpha_hat[0, :] sums to 1

```

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```

c = alpha_hat[0].sum()      # c is the scaling constant
if c > 0:
    alpha_hat[0] /= c      # now alpha_hat[0] sums to 1
    log_likelihood += np.log(c) # accumulate log(c)

# --- Recursion (positions 1 through L-1) ---
# alpha(a, k) = P(X_a | Z_a=k) * sum_j[ alpha(a-1, j) * P(k|j) ]
# In words: "probability of arriving in state k at position a"
# = (emission at k) * (sum over all previous states j of:
#   being in j at a-1 times transitioning from j to k)
for pos in range(1, L):
    for k in range(N):
        obs = seq[pos]
        if obs >= 2:      # missing data
            e = 1.0
        else:
            e = self.emissions[obs, k]

        # np.dot computes the sum over all previous states j:
        # sum_j alpha_hat[pos-1, j] * transitions[j, k]
        alpha_hat[pos, k] = e * np.dot(
            alpha_hat[pos - 1, :], self.transitions[:, k])

    # Scale this position
    c = alpha_hat[pos].sum()
    if c > 0:
        alpha_hat[pos] /= c
        log_likelihood += np.log(c)

return alpha_hat, log_likelihood

def backward_scaled(self, seq, alpha_hat):
    """Scaled backward algorithm.

    The backward algorithm is the mirror image of the forward algorithm.
    Where the forward algorithm answers "what is the probability of the
    observations up to position a, given state k at a?", the backward
    algorithm answers "what is the probability of the observations
    AFTER position a, given state k at a?"

    Together, forward and backward give us the posterior: the probability
    of each state at each position, given ALL the observations. This is
    like looking at a position from both directions -- past and future --
    to triangulate which gear configuration is most likely.

    Parameters
    -----
    seq : ndarray of shape (L,)
    alpha_hat : ndarray of shape (L, N)
        From forward_scaled (we use its scaling factors for consistency).

    Returns
    """

```

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```

-----
beta_hat : ndarray of shape (L, N)
    Scaled backward probabilities.
"""
L = len(seq)
N = self.N
beta_hat = np.zeros((L, N))

# --- Initialization: at the last position, beta = 1 ---
# There are no future observations to account for, so the
# backward probability is 1 for all states.
beta_hat[L - 1, :] = 1.0

# --- Recursion: work backwards from position L-2 to 0 ---
# beta(a, k) = sum_l[ P(l|k) * P(X_{a+1}|l) * beta(a+1, l) ]
# In words: "the probability of the future observations, given
# state k at position a" = sum over all possible next states l of:
# (transition k->l) * (emission at l) * (future from l onwards)
for pos in range(L - 2, -1, -1): # L-2, L-3, ..., 0
    for k in range(N):
        total = 0.0
        for l in range(N): # sum over next states
            obs = seq[pos + 1] # observation at NEXT position
            if obs >= 2: # missing data
                e = 1.0
            else:
                e = self.emissions[obs, l]
            total += self.transitions[k, l] * e * beta_hat[pos + 1, l]
        beta_hat[pos, k] = total

# Scale to prevent underflow
c = beta_hat[pos].sum()
if c > 0:
    beta_hat[pos] /= c

return beta_hat

```

Step 2: The Forward-Backward Algorithm

With the forward and backward algorithms in hand, we can now compute the **posterior probability** of each hidden state at each position. This is the central computation of the E-step and the key to the entire EM procedure.

Recall the watch metaphor: at each genomic position, the hidden state represents which time interval the coalescence falls in – which gear configuration the watch is using at that position. The forward-backward algorithm figures out, for each position, the probability of every possible gear configuration, given the *entire* sequence of observations (not just the observations before or after that position, but all of them together).

Probability Aside: what forward-backward actually computes

The forward-backward algorithm computes two fundamental quantities:

1. Posterior state probabilities $\gamma_k(a)$:

$$\gamma_k(a) = P(Z_a = k \mid X_1, \dots, X_L)$$

This answers: “given everything we observed across the entire genome, what is the probability that the coalescence time at position a falls in interval k ?”

2. Expected transition counts $\xi_{kl}(a)$:

$$\xi_{kl}(a) = P(Z_a = k, Z_{a+1} = l \mid X_1, \dots, X_L)$$

This answers: “what is the probability that the coalescence time was in interval k at position a AND in interval l at position $a + 1$?”

Both are computed by combining the forward and backward variables. The forward variable $f_k(a)$ looks at the data from the left (positions 1 through a). The backward variable $b_k(a)$ looks at the data from the right (positions $a + 1$ through L). Together, they see everything.

The posterior is computed as:

$$P(Z_a = k \mid X_1, \dots, X_L) = \frac{f_k(a) \cdot b_k(a)}{\sum_l f_l(a) \cdot b_l(a)}$$

where $f_k(a)$ is the forward probability and $b_k(a)$ is the backward probability.

We’ve already implemented the forward algorithm above. The backward algorithm runs the same recursion in reverse:

Initialization ($a = L$):

$$b_k(L) = 1 \quad \text{for all } k$$

Recursion ($a = L - 1, \dots, 1$):

$$b_k(a) = \sum_{l=0}^n p_{kl} \cdot e_l(X_{a+1}) \cdot b_l(a + 1)$$

The backward initialization says: at the end of the sequence, there are no future observations, so the backward probability is 1 regardless of state. The recursion says: to compute the backward probability at position a , consider every possible state l at the *next* position, weight it by the transition probability, the emission at that next state, and the backward probability from that next state onwards. Sum over all possibilities.

Together, forward and backward give us two things:

1. **Posterior state probabilities** $\gamma_k(a) = P(Z_a = k \mid X)$ at each position
2. **Expected transition counts** $\xi_{kl}(a) = P(Z_a = k, Z_{a+1} = l \mid X)$ between positions

These are the **expected sufficient statistics** – exactly what EM needs.

Probability Aside: what are “expected sufficient statistics”?

The term “sufficient statistics” comes from classical statistics. A **sufficient statistic** is a summary of data that captures everything the data can tell you about a parameter. For example, the sample mean and sample variance are sufficient statistics for a normal distribution – once you know them, the individual data points add no further information about μ and σ^2 .

For an HMM, the sufficient statistics are:

- **How often each state is visited** (the γ_k sums)
- **How often each transition occurs** (the ξ_{kl} sums)
- **How often each observation is emitted from each state** (the emission counts)

If we knew the hidden states, we could compute these directly by counting. But the states are hidden, so we compute the **expected** counts – weighted averages over all possible state sequences, where the weights come from the posterior probabilities. This is why the E-step is called “Expectation”: it computes the expected values of the sufficient statistics under the current parameter estimates.

In the watch metaphor: we cannot directly observe which gears are engaged at each tick. But by looking at the output (het/hom) from both directions, we can estimate *how often* each gear was probably engaged. These expected gear-usage counts are the sufficient statistics.

Probability Aside: Bayes’ theorem in sequence form

The posterior $\gamma_k(a)$ is a direct application of Bayes’ theorem. The forward variable $f_k(a)$ captures $P(X_1, \dots, X_a, Z_a = k)$ – the joint probability of seeing the observations up to position a **and** being in state k . The backward variable $b_k(a)$ captures $P(X_{a+1}, \dots, X_L | Z_a = k)$ – the probability of the remaining observations **given** state k at position a . Multiplying and normalizing gives:

$$\gamma_k(a) = \frac{f_k(a) \cdot b_k(a)}{\sum_l f_l(a) \cdot b_l(a)} = \frac{P(X, Z_a = k)}{P(X)}$$

This is just Bayes’ rule: posterior = likelihood $\times$ prior / evidence. The forward variable plays the role of “likelihood times prior” (all the evidence up to and including position a), while the backward variable completes the picture with the remaining evidence. The denominator $P(X)$ – the total probability of the data – ensures the result is a proper probability distribution over states at each position.

Let’s verify that the posteriors behave as expected:

```
# Demonstrate that gamma sums to 1 at each position (it's a distribution)
def check_posteriors(hmm, seq, n_positions=5):
    """Verify posterior probabilities are valid distributions.

    At each position, gamma should sum to 1 because we must be in
    exactly one state. This is a basic sanity check that the
    forward-backward computation is correct.
    """
    alpha_hat, ll = hmm.forward_scaled(seq)      # forward pass
    beta_hat = hmm.backward_scaled(seq, alpha_hat) # backward pass

    # Check a few representative positions across the sequence
    for pos in [0, len(seq)//4, len(seq)//2, 3*len(seq)//4, len(seq)-1]:
        # gamma = element-wise product of forward and backward
        gamma = alpha_hat[pos] * beta_hat[pos]
        gamma /= gamma.sum()    # normalize to get posterior
        peak_state = np.argmax(gamma) # most probable state
        print(f" pos={pos:6d}: sum(gamma)={gamma.sum():.6f}, "
              f"peak state={peak_state}, "
              f"peak prob={gamma[peak_state]:.4f}")
```

Now we compute the expected sufficient statistics. This is the heart of the E-step – the computation that extracts from

the data everything the M-step needs to update the parameters:

```
def compute_expected_counts(hmm, seq):
    """Compute expected counts for the E-step of EM.

    This function runs the forward-backward algorithm and then extracts
    three sets of expected counts:

    1. gamma_sum[k] = sum over all positions a of gamma_k(a)
       "How much total time does the HMM spend in state k?"

    2. xi_sum[k, l] = sum over all adjacent pairs (a, a+1) of xi_{kl}(a)
       "How many times does the HMM transition from state k to state l?"

    3. emission_counts[obs, k] = sum over positions where X_a = obs of gamma_k(a)
       "How many times does state k emit observation obs?"

    These are the expected sufficient statistics. Together, they
    summarize everything the data can tell us about the parameters.

    Parameters
    -----
    hmm : PSMC_HMM
    seq : ndarray of shape (L,)

    Returns
    -----
    gamma_sum : ndarray of shape (N,)
        Sum of posterior state probabilities over all positions.
    xi_sum : ndarray of shape (N, N)
        Sum of expected transition counts.
    emission_counts : ndarray of shape (2, N)
        Expected emission counts: emission_counts[obs, k] =
        sum over positions with observation obs of gamma_k(pos).
    log_likelihood : float
    """
    L = len(seq)
    N = hmm.N

    # --- Run forward and backward passes ---
    alpha_hat, ll = hmm.forward_scaled(seq)
    beta_hat = hmm.backward_scaled(seq, alpha_hat)

    # --- Compute posterior state probabilities gamma_k(a) ---
    # and accumulate the sufficient statistics
    gamma_sum = np.zeros(N) # total "time spent" in each state
    emission_counts = np.zeros((2, N)) # emission counts per state

    for pos in range(L):
        # gamma_k(a) = alpha_hat(a,k) * beta_hat(a,k), then normalize
        gamma = alpha_hat[pos] * beta_hat[pos]
        total = gamma.sum()
        if total > 0:
```

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```

        gamma /= total          # normalize to a proper distribution

    gamma_sum += gamma          # accumulate state occupancy

    obs = seq[pos]
    if obs < 2:                  # not missing data
        emission_counts[obs] += gamma # attribute this obs to states

# --- Compute expected transition counts xi_{kl}(a) ---
# xi_{kl}(a) = alpha_hat(a,k) * P(l|k) * e_l(X_{a+1}) * beta_hat(a+1,l)
# then normalize. This tells us: "given all the data, how probable is
# it that the HMM was in state k at position a and state l at a+1?"
xi_sum = np.zeros((N, N))
for pos in range(L - 1):      # for each adjacent pair
    obs_next = seq[pos + 1]    # observation at the next position
    xi_pos = np.zeros((N, N)) # one normalized distribution per pair
    for k in range(N):         # source state
        for l in range(N):     # destination state
            if obs_next >= 2:    # missing data at next position
                e = 1.0
            else:
                e = hmm.emissions[obs_next, l]

            # Unnormalized xi: forward(a,k) * trans(k,l) * emit(l) * backward(a+1,
→l)

            xi_pos[k, l] = (alpha_hat[pos, k] *
                            hmm.transitions[k, l] *
                            e * beta_hat[pos + 1, l])

# Forward and backward vectors are independently scaled at each
# position, so normalize every adjacent pair before accumulating.
# A single global normalization would mix incompatible scale factors.
total_xi = xi_pos.sum()
if total_xi > 0:
    xi_sum += xi_pos / total_xi

return gamma_sum, xi_sum, emission_counts, ll

```

Recap so far. We now have a complete forward-backward implementation that, given an observation sequence and current HMM parameters, produces three things: (1) the posterior probability of each state at each position, (2) the expected number of transitions between each pair of states, and (3) the expected emission counts. These are everything the EM algorithm needs to update the parameters. The next section explains how.

Step 3: The EM Algorithm

We now arrive at the central algorithm of PSMC inference. The **Expectation-Maximization (EM) algorithm** is an iterative procedure that alternates between two steps:

1. **E-step** (Expectation): Given the current parameters, run forward-backward to compute the expected sufficient statistics. “Given our current gear sizes, figure out how well the watch keeps time and which gear configurations are most likely.”
2. **M-step** (Maximization): Given the expected sufficient statistics, find new parameters that maximize the expected log-likelihood. “Given what we learned about gear usage, cut new gears that would make the watch keep better

time.”

The algorithm repeats these two steps until convergence. Exact EM is non-decreasing when the M-step really maximizes the same complete-data objective. PSMC uses a numerical direct search and omits the negligible initial-state term, so monotonicity is a diagnostic to check rather than an unconditional software guarantee.

i Probability Aside: the Expectation step

The “Expectation” in EM refers to computing the **expected value of the complete-data log-likelihood** under the posterior distribution of the hidden states. Let’s unpack this:

- The **complete data** is the observations X together with the hidden states Z . If we knew both, computing the log-likelihood would be straightforward – just count transitions, emissions, etc.
- But the hidden states are unknown. So instead of counting, we compute **expected counts** – weighted averages where the weights are the posterior probabilities $P(Z | X, \Theta^{\text{old}})$.
- This expected complete-data log-likelihood is called the **Q function**:

$$Q(\Theta | \Theta^{\text{old}}) = \mathbb{E}_{Z|X, \Theta^{\text{old}}} [\log P(X, Z | \Theta)]$$

The E-step computes the expected counts (which are all we need to evaluate Q for any Θ). The M-step then maximizes Q over Θ .

i Probability Aside: the Q function and why it matters

The Q function deserves a closer look, because understanding it makes the entire EM algorithm feel natural rather than mysterious.

For an HMM with transition matrix A , emissions e , and initial distribution a_0 , the Q function decomposes into three independent terms:

$$Q = \underbrace{\sum_k \hat{c}_k^{(0)} \log a_0(k)}_{\text{initial state term}} + \underbrace{\sum_{k,l} \hat{c}_{kl} \log p_{kl}}_{\text{transition term}} + \underbrace{\sum_{k,b} \hat{c}_k^{(b)} \log e_k(b)}_{\text{emission term}}$$

where:

- $\hat{c}_k^{(0)}$ = expected initial state counts (proportional to $\gamma_k(1)$)
- $\hat{c}_{kl}$ = expected transition counts ($\sum_a \xi_{kl}(a)$)
- $\hat{c}_k^{(b)}$ = expected emission counts ($\sum_{a: X_a=b} \gamma_k(a)$)

Each term has the form $\sum(\text{expected count}) \times \log(\text{parameter})$. This is exactly the form of a log-likelihood where the “data” are the expected counts. So maximizing Q is like fitting parameters to pseudo-data that summarize what we learned about the hidden states.

In the watch metaphor: the expected counts are like a logbook recording how often each gear was probably engaged and how often each transition probably occurred. The M-step reads this logbook and asks: “what gear sizes would produce exactly these usage patterns?”

i Probability Aside: why EM works (the ELBO)

The EM algorithm is built on a fundamental inequality from information theory. For any distribution $q(Z)$ over hidden states and any parameters Θ :

$$\log P(X | \Theta) \geq \sum_Z q(Z) \log \frac{P(X, Z | \Theta)}{q(Z)}$$

This is the **Evidence Lower Bound (ELBO)**. The E-step sets $q(Z) = P(Z | X, \Theta^{\text{old}})$ (the posterior under current parameters), which makes the bound tight. The M-step then maximizes the bound over Θ , guaranteeing that the log-likelihood cannot decrease.

This monotonic improvement is the fundamental guarantee: **each EM iteration either improves the fit or leaves it unchanged**. The log-likelihood is a non-decreasing sequence, bounded above (since probabilities cannot exceed 1), so it must converge. In the watch metaphor: each adjustment of the gears brings the watch closer to keeping correct time, and you can never make it worse by adjusting.

One important caveat: EM converges to a **local** optimum, not necessarily the global optimum. Different initial gear sizes might lead to different final configurations. In practice, PSMC is fairly robust to initialization, but running from multiple starting points is sometimes advisable.

The M-step challenge. For a standard HMM, the M-step has closed-form solutions: the optimal transition probabilities are just the normalized expected counts (see *Hidden Markov Models*). But PSMC's transition matrix p_{kl} is a complex function of $(\theta, \rho, \lambda_0, \dots, \lambda_n)$, not a set of independent parameters. The gears are all interconnected – changing one population size λ_k affects many entries of the transition matrix simultaneously.

So the M-step requires **numerical optimization**: maximize Q over $(\theta, \rho, t_{\text{max}}, \lambda_0, \dots, \lambda_n)$ using a general-purpose optimizer. PSMC uses the Hooke-Jeeves direct search method – a derivative-free optimizer that works well for this kind of constrained problem. In our implementation below, we use Nelder-Mead (a similar derivative-free method available in `scipy`) for clarity.

i Probability Aside: the Maximization step

The M-step asks: “given the expected counts from the E-step, which parameter values Θ maximize the Q function?”

For a standard HMM with independent parameters, the answer has a beautiful closed form:

$$\hat{p}_{kl} = \frac{\hat{c}_{kl}}{\sum_m \hat{c}_{km}}$$

– just the fraction of times state k transitioned to state l . Similarly for emissions. But PSMC's parameters are *not* the individual entries of the transition matrix. They are the underlying population-genetic quantities $(\theta, \rho, \lambda_0, \dots, \lambda_n)$ that *generate* the transition matrix through a complex chain of formulas.

So we must evaluate $Q(\Theta)$ as a black-box function: for any candidate Θ , rebuild the HMM (recompute the transition matrix, emissions, etc.), then evaluate the Q function using the expected counts from the E-step. A numerical optimizer searches over Θ to find the maximum.

This is computationally more expensive than the closed-form solution, but it is still efficient because the expected counts (from the E-step) are fixed during the M-step. We only need to rebuild the HMM and evaluate Q – we do *not* need to re-run forward-backward.

```
from scipy.optimize import minimize
```

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```

def psmc_em_step(hmm, seq, par_map=None):
    """One EM iteration for PSMC.

    This function performs one complete E-step followed by one M-step.
    The E-step runs forward-backward to get expected counts.
    The M-step numerically optimizes the Q function over the PSMC parameters.

    Parameters
    -----
    hmm : PSMC_HMM
    seq : ndarray
    par_map : list, optional
        Parameter grouping map (from parse_pattern).
        Maps each atomic interval to a free lambda parameter.

    Returns
    -----
    new_hmm : PSMC_HMM
        HMM with updated parameters.
    log_likelihood : float
    """
    N = hmm.N
    n = hmm.n

    # =====
    # E-step: compute expected sufficient statistics
    # =====
    # This runs forward-backward and extracts the three sets of
    # expected counts (gamma_sum, xi_sum, emission_counts).
    gamma_sum, xi_sum, emission_counts, ll = compute_expected_counts(hmm, seq)

    # =====
    # M-step: maximize Q over the population-genetic parameters
    # =====

    # Pack the current parameters into a single vector for the optimizer.
    # The parameters are: theta, rho, t_max, and the free lambdas.
    if par_map is None:
        par_map = list(range(N)) # no grouping: each interval is free
    n_free = max(par_map) + 1 # number of free lambda parameters
    n_params = n_free + 3 # total: theta + rho + t_max + lambdas

    params0 = np.zeros(n_params)
    params0[0] = hmm.theta # current mutation rate
    params0[1] = hmm.rho # current recombination rate
    params0[2] = hmm.t_max # current maximum time
    # Extract the free lambdas (one per parameter group)
    for k in range(n_free):
        idx = par_map.index(k) # first interval using this parameter
        params0[3 + k] = hmm.lambdas[idx]

    def neg_Q(params):

```

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```

"""Negative Q function (we minimize this to maximize Q).

For each candidate parameter vector, we:
1. Rebuild the HMM (recompute transitions, emissions, initial dist)
2. Evaluate Q using the FIXED expected counts from the E-step
"""

# Unpack parameters (use abs to enforce positivity)
theta = abs(params[0])
rho = abs(params[1])
t_max = abs(params[2])
free_lambdas = np.abs(params[3:])

# Expand the free lambdas to full lambdas using the par_map
full_lambdas = np.array([free_lambdas[par_map[k]] for k in range(N)])

# Rebuild the HMM with these candidate parameters
try:
    transitions, emissions, initial = build_psmc_hmm(
        n, t_max, theta, rho, full_lambdas, alpha_param=hmm.alpha_param)
except (ValueError, RuntimeError):
    return 1e30 # return a large value if parameters are invalid

# Evaluate Q = sum of (expected count * log parameter)
Q = 0.0

# Like the original PSMC hmm_Q implementation, this objective
# optimizes transition and emission terms. Average occupancy
# gamma_sum / L is not a valid initial-state sufficient statistic.

# Term 1: transition contribution
# Each expected transition count xi_sum[k,l] is weighted by
# log of the transition probability under the candidate parameters
for k in range(N):
    for l in range(N):
        if transitions[k, l] > 0 and xi_sum[k, l] > 0:
            Q += xi_sum[k, l] * np.log(transitions[k, l] + 1e-300)

# Term 2: emission contribution
# Each expected emission count is weighted by log of the emission
# probability under the candidate parameters
for b in range(2): # b = 0 (hom) or 1 (het)
    for k in range(N):
        if emissions[b, k] > 0 and emission_counts[b, k] > 0:
            Q += emission_counts[b, k] * np.log(emissions[b, k] + 1e-300)

return -Q # negate because scipy minimizes

# Run the optimizer to find the parameters that maximize Q
result = minimize(neg_Q, params0, method='Nelder-Mead',
                  options={'maxiter': 1000, 'xatol': 1e-6})

# Unpack the optimized parameters

```

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```

new_params = np.abs(result.x)
new_theta = new_params[0]
new_rho = new_params[1]
new_t_max = new_params[2]
new_free_lambdas = new_params[3:]
new_full_lambdas = np.array([new_free_lambdas[par_map[k]] for k in range(N)])

# Build a new HMM with the optimized parameters
new_hmm = PSMC_HMM(n, new_theta, new_rho, new_full_lambdas,
                    new_t_max, hmm.alpha_param)

return new_hmm, ll

# Demonstrate one EM step
n = 10
theta = 0.001
rho = theta / 5
lambdas = np.ones(n + 1)

hmm = PSMC_HMM(n, theta, rho, lambdas)

# Simulate a short sequence for testing
np.random.seed(42)
seq, _ = simulate_psmc_input(5000, theta, rho, lambda t: 1.0)

print(f"Initial log-likelihood: {hmm.log_likelihood(seq):.2f}")
print(f"Initial theta: {hmm.theta:.6f}")

```

Recap. One EM iteration works as follows: (1) Run forward-backward with the current parameters to get expected counts. (2) Numerically optimize the Q function over the population-genetic parameters, using the expected counts as fixed inputs. (3) Rebuild the HMM with the new parameters. A successful M-step should not decrease the log-likelihood; numerical optimization tolerances make that a condition to verify. Now let's see the full loop.

Step 4: The Full EM Loop

The complete PSMC inference runs EM for a fixed number of iterations (typically 20–25), printing the parameters after each round. This is the outermost loop of the PSMC algorithm – the process of iteratively refining the gear sizes until the watch keeps accurate time.

Probability Aside: convergence – when to stop adjusting the gears

How do we know when the EM algorithm has converged? There are several criteria:

- 1. Log-likelihood plateau.** The most common criterion: stop when the change in log-likelihood between iterations falls below a threshold (e.g., $|\Delta LL| < 0.01$). The log-likelihood is an ideally non-decreasing sequence, so when it stops increasing, we have found a (local) optimum.
- 2. Parameter stability.** Stop when the parameters themselves stop changing: $\|\Theta^{(t+1)} - \Theta^{(t)}\| < \epsilon$. This is often more meaningful than the log-likelihood criterion because two very different log-likelihoods might correspond to nearly identical parameters on a flat part of the likelihood surface.
- 3. Fixed iteration count.** PSMC simply runs 20–25 iterations. Li and Durbin found empirically that this is sufficient for convergence on whole-genome data. This has the advantage of predictable running time.

In practice, PSMC uses approach (3) but monitors the log-likelihood for debugging. A material decrease indicates a failed or inconsistent M-step – either a bug in the implementation or a numerical issue in the optimizer.

```
def psmc_inference(seq, n=63, t_max=15.0, theta_rho_ratio=5.0,
                  pattern="4+25*2+4+6", n_iters=25, alpha_param=0.1):
    """Run the full PSMC inference.

    This is the top-level function that ties everything together:
    initialize parameters, then alternate E-step and M-step for
    n_iters iterations.

    Parameters
    -----
    seq : ndarray of shape (L,), dtype=int
        Observation sequence (0/1/2+).
    n : int
        Number of atomic time intervals - 1.
    t_max : float
        Maximum coalescence time (in coalescent units).
    theta_rho_ratio : float
        Assumed ratio theta/rho for initialization.
    pattern : str
        Parameter grouping pattern (see discretization chapter).
    n_iters : int
        Number of EM iterations.
    alpha_param : float
        Spacing parameter for time intervals.

    Returns
    -----
    results : list of dict
        Parameters at each iteration.
    """
    L = len(seq)
    N = n + 1 # number of hidden states

    # Parse the grouping pattern (e.g., "4+25*2+4+6")
    # This maps each atomic interval to a free lambda parameter
    par_map, n_free, n_intervals = parse_pattern(pattern)
    assert n_intervals == N, f"Pattern gives {n_intervals} intervals, need {N}"

    # Initialize theta from the observed heterozygosity rate.
    # If the fraction of het sites is f, then theta ~ -log(1-f).
    # This comes from inverting the emission formula: P(het) = 1 - exp(-theta*t),
    # and assuming t ~ 1 (the average coalescence time under constant pop size).
    frac_het = np.mean(seq[seq < 2]) # fraction of het sites (excluding missing)
    theta = -np.log(1.0 - frac_het)
    rho = theta / theta_rho_ratio # initialize rho from the assumed ratio

    # Initialize all lambdas to 1 (constant population size)
    # The EM algorithm will adjust them to match the data
    free_lambdas = np.ones(n_free)
```

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```

full_lambdas = np.array([free_lambdas[par_map[k]] for k in range(N)])

# Build the initial HMM
hmm = PSMC_HMM(n, theta, rho, full_lambdas, t_max, alpha_param)

results = []
print(f"PSMC inference: {L} bins, {N} intervals, {n_free} free params")
print(f"Initial theta={theta:.6f}, rho={rho:.6f}")

for iteration in range(n_iters):
    # --- E-step ---
    # Run forward-backward to compute expected sufficient statistics
    gamma_sum, xi_sum, emission_counts, ll = compute_expected_counts(
        hmm, seq)

    # Record the current state for later analysis
    results.append({
        'iteration': iteration,
        'log_likelihood': ll,
        'theta': hmm.theta,
        'rho': hmm.rho,
        'lambdas': hmm.lambdas.copy(),
    })

    print(f"  Iteration {iteration}: LL = {ll:.2f}, "
          f"theta = {hmm.theta:.6f}, rho = {hmm.rho:.6f}")

    # --- M-step ---
    # Update parameters by maximizing the Q function
    hmm, _ = psmc_em_step(hmm, seq, par_map)

return results

```

i Convergence in practice

Published workflows commonly request about 20–25 iterations, but that is a run length rather than a convergence guarantee. The log-likelihood should be checked for near-monotonic increase and a plateau. A material decrease means the numerical M-step did not improve the intended objective and should be investigated.

You can verify convergence by plotting the log-likelihood across iterations. It should rise steeply in the first few iterations (the initial constant-population guess is far from the truth) and then plateau as the inferred λ_k values settle into their final configuration.

The Q function value (the expected complete-data log-likelihood under the current parameters) should also increase after the M-step. PSMC reports both Q values (before and after maximization) at each iteration for debugging.

Step 5: Multiple Sequences and Missing Data

So far we have described the algorithm for a single observation sequence. In practice, a diploid genome is split into chromosomes (or even sub-chromosomal segments for bootstrapping). PSMC processes each sequence independently in the E-step and sums the expected counts before the M-step.

This works because the sufficient statistics are additive: the expected counts from two independent sequences can simply

be added together. If sequence 1 tells us that state k was visited an expected 500 times, and sequence 2 tells us 300 times, then the combined estimate is 800 times. The M-step then uses these combined counts as if they came from a single long sequence.

In the watch metaphor: each chromosome is an independent test of the watch's accuracy. By combining the evidence from all chromosomes, we get a more reliable picture of which gear sizes are correct.

```
def psmc_em_multiple_seqs(hmm, sequences, par_map):
    """EM step with multiple sequences.

    Each sequence is processed independently in the E-step (separate
    forward-backward runs), and the expected counts are summed before
    the M-step. This is mathematically equivalent to treating all
    sequences as independent observations of the same underlying HMM.

    Parameters
    -----
    hmm : PSMC_HMM
    sequences : list of ndarray
        Each element is an observation sequence for one chromosome.
    par_map : list

    Returns
    -----
    new_hmm : PSMC_HMM
    total_ll : float
    """
    N = hmm.N

    # Accumulate expected counts across all sequences
    total_gamma = np.zeros(N)
    total_xi = np.zeros((N, N))
    total_emission = np.zeros((2, N))
    total_ll = 0.0

    for seq in sequences:
        # Run forward-backward on this sequence independently
        gamma, xi, emission, ll = compute_expected_counts(hmm, seq)
        # Add this sequence's expected counts to the running totals
        total_gamma += gamma
        total_xi += xi
        total_emission += emission
        total_ll += ll

    # M-step uses accumulated counts
    # (same optimization as single sequence, just with summed counts)

    return total_ll
```

Missing data is handled elegantly: when X_a is missing (encoded as a value ≥ 2), the emission probability is set to 1 for all states:

$$e_k(\text{missing}) = 1 \quad \text{for all } k$$

This means missing positions contribute nothing to the likelihood – the HMM simply passes through them, maintaining the transition dynamics but receiving no information from the data. The transition structure is preserved (we still model

the passage of genomic distance, which matters for recombination), but the observation at that position provides zero evidence about the hidden state.

This is a standard HMM technique (covered in *Hidden Markov Models*) and requires no special code beyond the `if obs >= 2: e = 1.0` checks you have already seen in the forward, backward, and expected-counts functions above.

Step 6: Understanding What EM Learns

After running EM to convergence, we can use the forward-backward algorithm one final time to understand what the model has learned. The **posterior decoding** gives us, at each position along the genome, a probability distribution over coalescence time intervals.

This is the moment where the watch reveals its reading: at each position, the posterior tells us how deep in time the most recent common ancestor lies. In regions with many heterozygous sites, the posterior will peak at large k (deep coalescence, ancient MRCA). In regions with few heterozygous sites, it will peak at small k (recent MRCA). The spatial pattern of these posteriors – how the coalescence time waxes and wanes along the genome – reflects the history of recombination and population size changes.

```
def posterior_decoding(hmm, seq):
    """Compute the posterior state probabilities at each position.

    This is the final step of PSMC analysis: after EM has found the
    best parameters, run forward-backward one more time to decode
    the hidden states. The result is a probability distribution over
    coalescence time intervals at each genomic position.

    Parameters
    -----
    hmm : PSMC_HMM
    seq : ndarray

    Returns
    -----
    posterior : ndarray of shape (L, N)
        posterior[a, k] = P(Z_a = k | X_1, ..., X_L)
    map_states : ndarray of shape (L,)
        Maximum a posteriori state at each position.
    """
    L = len(seq)
    N = hmm.N

    alpha_hat, _ = hmm.forward_scaled(seq)          # forward pass
    beta_hat = hmm.backward_scaled(seq, alpha_hat)   # backward pass

    posterior = np.zeros((L, N))
    for pos in range(L):
        # Posterior = forward * backward, then normalize
        gamma = alpha_hat[pos] * beta_hat[pos]
        total = gamma.sum()
        if total > 0:
            posterior[pos] = gamma / total
        else:
            posterior[pos] = 1.0 / N # uniform fallback (should not happen)
```

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```

# The MAP (Maximum A Posteriori) state is the most probable state
# at each position -- the single best guess for the coalescence interval
map_states = np.argmax(posterior, axis=1)
return posterior, map_states

# After running EM, decode the hidden states
# posterior, map_states = posterior_decoding(final_hmm, seq)
# print(f"Most common state: {np.bincount(map_states).argmax()}")

```

What the posterior tells us

At each position, the posterior gives a probability distribution over coalescence time intervals. In regions with many heterozygous sites, the posterior will peak at large k (deep coalescence, ancient MRCA). In regions with few hets, it will peak at small k (recent MRCA).

The beauty of PSMC is that these local patterns, aggregated over the entire genome, allow inference of the global population size history $N(t)$. The λ_k parameters – the gear sizes that EM has refined over many iterations – directly encode the population size at each time interval. The next chapter (*Decoding the Clock*) shows how to read these gear sizes as a population size curve and scale them to real biological units.

Chapter Summary

Let's trace the complete path we have traveled in this chapter:

1. **We assembled the HMM** by collecting the initial distribution, transition matrix, and emission probabilities from the discretization chapter (*Discretizing Time*) into a single `PSMC_HMM` class.
2. **We implemented the forward-backward algorithm**, which computes posterior state probabilities by combining evidence from both directions along the sequence. The forward pass looks at data from the left; the backward pass looks from the right; together they give us the posterior at every position (see *Hidden Markov Models* for the foundational version).
3. **We extracted expected sufficient statistics** – the expected state occupancies, transition counts, and emission counts – which summarize everything the data tells us about the parameters.
4. **We built the EM loop**: alternate between computing expected counts (E-step) and numerically searching for parameters that improve the expected log-likelihood (M-step).
5. **We handled multiple sequences and missing data**, exploiting the additivity of sufficient statistics and the standard HMM trick of setting missing-data emissions to 1.
6. **We decoded the hidden states** using the final parameters, producing a posterior distribution over coalescence times at every position.

In the watch metaphor: we started with a box of gears (the HMM components from the discretization chapter), assembled them into a working mechanism (the forward-backward algorithm), and then used the EM algorithm to iteratively adjust every gear until the watch keeps accurate time. The next chapter (*Decoding the Clock*) shows how to read the dial – converting the inferred parameters into a population size history in real biological units.

Exercises

Exercise 1: Build and verify the HMM

Construct the PSMC HMM for a constant population ($\lambda_k = 1$). Verify that: (a) forward probabilities can be computed without numerical issues for a 10,000-bin sequence, (b) the log-likelihood increases across EM iterations, (c) $\hat{\theta}$ converges to the true value on simulated data.

Exercise 2: EM on simulated bottleneck data

Simulate 100,000 bins of data under a population that has $\lambda(t) = 1$ for $t < 0.5$, $\lambda(t) = 0.1$ for $0.5 \leq t < 1.5$, and $\lambda(t) = 1$ for $t > 1.5$. Run PSMC on this data and compare the inferred λ_k to the truth.

Exercise 3: The effect of sequence length

Run PSMC on sequences of length 1000, 10000, 100000, and 1000000 (all simulated under the same demographic model). How does the accuracy of the inferred λ_k improve with more data? At what length does the inference become reliable?

Exercise 4: Watch the EM gears turn

Run 25 EM iterations on simulated data and record the λ_k values at each iteration. Plot them as a function of iteration number. You should see them start at 1.0 (the initial guess) and gradually converge to the true population size history. How many iterations does it take for the curves to stabilize? Does convergence happen uniformly across all time intervals, or do some intervals converge faster than others?

Next: *Decoding the Clock* – reading the population size history from the inferred parameters.

Solutions

Solution 1: Build and verify the HMM

We construct the PSMC HMM for a constant population and verify three properties: numerical stability of forward probabilities, monotonic log-likelihood increase under EM, and convergence of $\hat{\theta}$ to the true value.

```
import numpy as np

# Setup
n = 10
theta_true = 0.001
rho_true = theta_true / 5
lambdas = np.ones(n + 1)

# Build HMM with true parameters
hmm = PSMC_HMM(n, theta_true, rho_true, lambdas)

# Simulate a 10,000-bin sequence
np.random.seed(42)
seq, _ = simulate_psmc_input(10000, theta_true, rho_true, lambda t: 1.0)
```

```

# (a) Verify forward probabilities compute without issues
alpha_hat, ll = hmm.forward_scaled(seq)
print(f"(a) Forward algorithm on 10,000 bins:")
print(f"  Log-likelihood: {ll:.2f}")
print(f"  alpha_hat min: {alpha_hat.min():.2e}")
print(f"  alpha_hat max: {alpha_hat.max():.2e}")
print(f"  Any NaN? {np.any(np.isnan(alpha_hat))}")
print(f"  Any Inf? {np.any(np.isinf(alpha_hat))}")

# (b) Verify log-likelihood increases across EM iterations
# Start with a perturbed theta to give EM room to improve
hmm_init = PSMC_HMM(n, theta_true * 0.5, rho_true, lambdas)
log_likelihoods = []

current_hmm = hmm_init
for i in range(10):
    ll_i = current_hmm.log_likelihood(seq)
    log_likelihoods.append(ll_i)
    current_hmm, _ = psmc_em_step(current_hmm, seq)
    print(f"  Iteration {i}: LL = {ll_i:.2f}, "
          f"theta = {current_hmm.theta:.6f}")

# Check monotonicity
for i in range(1, len(log_likelihoods)):
    assert log_likelihoods[i] >= log_likelihoods[i-1] - 1e-6, \
        f"LL decreased at iteration {i}!"
print(f"\n(b) Log-likelihood monotonically non-decreasing: PASSED")

# (c) Check theta convergence
print(f"\n(c) Theta convergence:")
print(f"  True theta: {theta_true:.6f}")
print(f"  Final theta: {current_hmm.theta:.6f}")
print(f"  Relative error: "
      f"{abs(current_hmm.theta - theta_true)/theta_true:.4f}")

```

(a) The scaled forward algorithm should produce no NaN or Inf values, even for a 10,000-bin sequence. The scaling (normalizing at each position) prevents the underflow that would occur with raw forward probabilities, which shrink exponentially with sequence length.

(b) The log-likelihood should be non-decreasing up to numerical tolerance. A material decrease indicates that the numerical M-step failed to improve the objective.

(c) Compare $\hat{\theta}$ with the generating value across several random seeds and sequence lengths. There is no universal percentage-error target: it depends on length, heterozygosity, missingness, discretization, parameter grouping, and whether the optimizer has converged.

Solution 2: EM on simulated bottleneck data

We simulate data under a bottleneck model and run PSMC to see if it recovers the true population history.

```
import numpy as np
```

```

# Bottleneck model: lambda=1 for t<0.5, lambda=0.1 for 0.5<=t<1.5, lambda=1 for t>
  ->1.5

```

```

def bottleneck_lambda(t):
    if t < 0.5:
        return 1.0
    elif t < 1.5:
        return 0.1
    else:
        return 1.0

# Simulate 100,000 bins
np.random.seed(123)
theta = 0.001
rho = theta / 5
seq, _ = simulate_psmc_input(100000, theta, rho, bottleneck_lambda)

print(f"Observed heterozygosity: {np.mean(seq):.4f}")

# Run PSMC with n=20 intervals for tractability
n = 20
results = psmc_inference(seq, n=n, t_max=15.0, n_iters=20,
                        pattern=f"{n+1}*1")

# Compare inferred lambdas to truth
final = results[-1]
t = compute_time_intervals(n, t_max=15.0)
print(f"\nInferred vs. true lambda:")
print(f"{'Interval':>10} {'t_mid':>8} {'True':>8} {'Inferred':>10}")
print("-" * 40)
for k in range(n + 1):
    t_mid = (t[k] + t[k+1]) / 2.0
    lam_true = bottleneck_lambda(t_mid)
    lam_inferred = final['lambdas'][k]
    print(f"{k:>10} {t_mid:>8.3f} {lam_true:>8.2f} {lam_inferred:>10.4f}")

```

The inferred λ_k values should show a clear dip (values significantly less than 1) in the intervals corresponding to $t \in [0.5, 1.5]$, matching the bottleneck. The depth of the inferred bottleneck may not exactly reach 0.1 due to the discretization smoothing the sharp edges, but the qualitative pattern should be clear. The constant-size intervals ($t < 0.5$ and $t > 1.5$) should have $\lambda_k \approx 1.0$.

Solution 3: The effect of sequence length

We run PSMC on sequences of increasing length to study how statistical power changes with data. More sequence generally supplies more informative transitions and heterozygous bins, but linked observations, time discretization, and optimizer error prevent a universal sample-size cutoff or a guaranteed $1/\sqrt{L}$ error law for every demographic parameter.

```

import numpy as np

def bottleneck_lambda(t):
    if t < 0.5:
        return 1.0
    elif t < 1.5:
        return 0.1
    else:

```

```

        return 1.0

theta = 0.001
rho = theta / 5
n = 10 # fewer intervals for speed

lengths = [1000, 10000, 100000, 1000000]
t = compute_time_intervals(n, t_max=15.0)

for L in lengths:
    np.random.seed(42)
    seq, _ = simulate_psmc_input(L, theta, rho, bottleneck_lambda)
    results = psmc_inference(seq, n=n, t_max=15.0, n_iters=15,
                             pattern=f"{n+1}*1")
    final_lambdas = results[-1]['lambdas']

    # Compute mean squared error against truth
    mse = 0.0
    for k in range(n + 1):
        t_mid = (t[k] + t[k+1]) / 2.0
        lam_true = bottleneck_lambda(t_mid)
        mse += (final_lambdas[k] - lam_true) ** 2
    mse /= (n + 1)

    print(f"L={L:>8}: RMSE = {np.sqrt(mse):.4f}, "
          f"final LL = {results[-1]['log_likelihood']:.2f}")

```

Interpretation: Repeat each length with independent seeds and summarize the distribution of RMSE, rather than treating one nested realization as a benchmark. The typical error should tend to fall as information increases, but individual runs need not be monotone and no fixed RMSE thresholds follow from PSMC theory.

Solution 4: Watch the EM gears turn

We track λ_k across 25 EM iterations to visualize how the parameters converge from the initial guess to the final estimate.

```

import numpy as np

def bottleneck_lambda(t):
    if t < 0.5:
        return 1.0
    elif t < 1.5:
        return 0.1
    else:
        return 1.0

np.random.seed(42)
n = 10
theta = 0.001
rho = theta / 5
seq, _ = simulate_psmc_input(100000, theta, rho, bottleneck_lambda)
t = compute_time_intervals(n, t_max=15.0)

```

```

# Run PSMC and record lambdas at each iteration
results = psmc_inference(seq, n=n, t_max=15.0, n_iters=25,
                        pattern=f"{n+1}*1")

# Print lambda trajectories for selected intervals
# Pick one interval in the bottleneck and one outside
bottleneck_interval = None
normal_interval = None
for k in range(n + 1):
    t_mid = (t[k] + t[k+1]) / 2.0
    if bottleneck_interval is None and 0.5 < t_mid < 1.5:
        bottleneck_interval = k
    if normal_interval is None and t_mid < 0.3:
        normal_interval = k

print(f"Tracking interval {bottleneck_interval} (in bottleneck) "
      f"and interval {normal_interval} (outside):")
print(f"{'Iter':>5} {'lambda_bottleneck':>18} {'lambda_normal':>15} "
      f"{'LL':>12}")
print("-" * 55)
for r in results:
    lam_b = r['lambdas'][bottleneck_interval]
    lam_n = r['lambdas'][normal_interval]
    print(f"{'r['iteration']':>5} {'lam_b':>18.4f} {'lam_n':>15.4f} "
          f"{'r['log_likelihood']':>12.2f}")

```

Expected behavior:

- **All λ_k start at 1.0** (the initial flat guess).
- **The bottleneck intervals converge fastest.** Within 5–10 iterations, the λ_k for intervals in $[0.5, 1.5]$ drop sharply toward 0.1. This is because the emission signal is strongest there – the pronounced lack of heterozygosity in those time intervals provides a clear gradient for the M-step optimizer.
- **The non-bottleneck intervals converge more slowly.** They remain near 1.0 but may fluctuate slightly before settling. Intervals in the very recent or very ancient past converge last because they have the least statistical power (fewest recombination events).
- **Convergence is not uniform.** Intervals with more expected recombination events (as measured by $C_{\sigma}\sigma_k$) converge faster because the expected sufficient statistics are more precisely estimated there. This is analogous to a watch where some gears engage frequently and are quickly tuned, while others engage rarely and take longer to adjust.
- **The log-likelihood plateaus** after roughly 15–20 iterations, indicating convergence. The rate of improvement should decrease exponentially – large gains in the first few iterations, then diminishing returns.

Decoding the Clock

The case and dial: turning internal gears into a readable face.

We have arrived at the final chapter of the PSMC Timepiece. In the preceding chapters, we built every internal component of this watch: the continuous-time transition density (the escapement, *The Continuous-Time PSMC Model*), the discrete time intervals and transition matrix (the gear train, *Discretizing Time*), and the EM algorithm that iteratively refines the parameters from data (the mainspring, *The PSMC HMM and EM Algorithm*). The EM algorithm has converged, and we now hold estimated parameters $\hat{\theta}_0$, $\hat{\rho}_0$, and $\hat{\lambda}_0, \dots, \hat{\lambda}_n$.

But these are dimensionless numbers in coalescent units – the internal tick count of the mechanism, not the time displayed on the dial. A watch is useless if you cannot read its face. To tell biological time, we need to **scale** these parameters to real units – generations, years, and population sizes. We also need to assess how confident we are in the reading (bootstrapping), how well the model fits the data (goodness of fit), and what common mistakes can lead us to misread the dial entirely (pitfalls).

This chapter covers five steps:

1. **Scaling** – reading the dial by converting coalescent units to real units
2. **Goodness of fit** – checking whether the watch keeps accurate time
3. **Bootstrapping** – testing the watch against multiple reference clocks
4. **Common pitfalls** – the ways even experienced watchmakers misread the dial
5. **Interpreting PSMC plots** – what the population history actually tells us

Let us begin.

Step 1: From Coalescent Units to Real Units

Reading the dial.

i What does “reading the dial” mean?

A watch’s internal mechanism counts oscillations – ticks of the escapement. But the dial translates those ticks into hours and minutes, units that have meaning in the external world. The PSMC’s internal mechanism operates in **coalescent units**, where time is measured relative to the population size and rates are scaled accordingly. “Reading the dial” means converting those internal units into **generations**, **years**, and **individuals** – the units that have meaning for biology.

PSMC operates in **coalescent units**: time is measured in units of $2N_0$ generations, and rates are scaled by $4N_0$. These conventions come directly from the coalescent theory developed in the [prerequisites](#), where we saw that the natural time unit for two-lineage coalescence is $2N$ generations. To convert back to real units, we need one external piece of information: the **per-generation mutation rate** μ .

Think of μ as the conversion factor printed on the bezel of the watch – without it, the dial markings are arbitrary.

Computing N_0 :

The estimated $\hat{\theta}_0$ is the population-scaled mutation rate per bin of size s base pairs:

$$\hat{\theta}_0 = 4N_0\mu s$$

This equation relates four quantities. We know three of them after inference: $\hat{\theta}_0$ comes from EM (see [The PSMC HMM and EM Algorithm](#)), μ is an externally supplied mutation rate, and s is the bin size we chose during data preparation (typically 100 bp). Solving for the one remaining unknown:

$$N_0 = \frac{\hat{\theta}_0}{4\mu s}$$

i Choosing a mutation rate μ

For **humans**, commonly used values are:

- $\mu \approx 1.25 \times 10^{-8}$ per base pair per generation (pedigree-based estimate)

- $\mu \approx 2.5 \times 10^{-8}$ per base pair per generation (phylogenetic estimate from human-chimp divergence)

For **other organisms**, you must find a species-appropriate estimate from the literature. The mutation rate depends on generation time, DNA repair mechanisms, and other biological factors. Using the wrong μ will shift your entire time axis and population size axis – we discuss this further in the mutation rate admonition below.

As a sanity check: for humans with $\mu = 1.25 \times 10^{-8}$, $s = 100$, and a typical $\hat{\theta}_0 \approx 0.00069$, you should get $N_0 \approx 13,800$. If your N_0 is orders of magnitude off from the expected range for your species, double-check your μ and s .

Scaling time from coalescent units to generations:

Each coalescent time unit corresponds to $2N_0$ generations (recall from *Coalescent Theory* that the coalescence rate for two lineages in a population of size N is $1/(2N)$ per generation, so the expected coalescence time is $2N$ generations). Therefore:

$$T_k \text{ (generations)} = 2N_0 \cdot t_k$$

To convert generations to years, multiply by the **generation time** (typically 25–29 years for humans):

$$T_k \text{ (years)} = 2N_0 \cdot t_k \cdot g$$

where g is the generation time in years.

Scaling population size:

The $\hat{\lambda}_k$ values estimated by EM are *relative* population sizes – they express the population size in interval k as a multiple of the reference N_0 . To get absolute effective population sizes:

$$N_k = N_0 \cdot \hat{\lambda}_k$$

So if $\hat{\lambda}_3 = 2.0$, the population in the third time interval was twice as large as the reference N_0 .

Here is the complete scaling function, with inline comments explaining each conversion:

```
import numpy as np

def scale_psmc_output(theta_0, lambdas, t_boundaries,
                      mu=1.25e-8, s=100, generation_time=25):
    """Scale PSMC output to real units.

    This is the "dial" of the PSMC watch: it converts the internal
    coalescent-unit parameters into biologically meaningful quantities.

    Parameters
    -----
    theta_0 : float
        Estimated theta_0 from EM (population-scaled mutation rate per bin).
    lambdas : ndarray of shape (n+1,)
        Estimated relative population sizes (lambda_k values from EM).
    t_boundaries : ndarray of shape (n+2,)
        Time interval boundaries in coalescent units (from discretization).
    mu : float
        Per-generation, per-base-pair mutation rate (external input).
    s : int
        Bin size in base pairs (must match what was used in data preparation).
```

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```

generation_time : float
    Years per generation (e.g., 25 for humans, 3 for mice).

Returns
-----
N_0 : float
    Reference effective population size.
times_gen : ndarray
    Time boundaries in generations.
times_years : ndarray
    Time boundaries in years.
pop_sizes : ndarray
    Effective population sizes.
"""
# Convert theta to N_0 using the relationship theta = 4 * N_0 * mu * s
N_0 = theta_0 / (4 * mu * s)

# Scale time: each coalescent unit = 2*N_0 generations
times_gen = 2 * N_0 * t_boundaries

# Convert generations to calendar years
times_years = times_gen * generation_time

# Scale relative sizes (lambdas) to absolute population sizes
pop_sizes = N_0 * lambdas

return N_0, times_gen, times_years, pop_sizes

# Example: typical human PSMC output
theta_0 = 0.00069 # typical for humans with s=100
n = 10
# Time boundaries in coalescent units (from discretization chapter)
t = np.array([0, 0.05, 0.12, 0.22, 0.37, 0.6, 0.95, 1.5, 2.5, 4.0, 8.0, 1000])
# Relative population sizes (lambda_k from EM)
lambdas = np.array([2.0, 1.8, 1.5, 1.0, 0.5, 0.3, 0.5, 1.0, 1.5, 2.0, 2.5])

N_0, t_gen, t_years, N_t = scale_psmc_output(theta_0, lambdas, t)

print(f"N_0 = {N_0:.0f}")
print(f"\nTime (years ago)    Pop. size")
print("-" * 40)
for k in range(len(lambdas)):
    print(f"    {t_years[k]:>12,.0f}    {N_t[k]:>10,.0f}")

```

Let us pause and trace through the unit conversions carefully, because getting them wrong is the single most common source of error in PSMC analyses.

Unit conversion walkthrough

Suppose $\hat{\theta}_0 = 0.00069$, $\mu = 1.25 \times 10^{-8}$, $s = 100$, and $g = 25$ years.

1. $N_0 = 0.00069 / (4 \times 1.25 \times 10^{-8} \times 100) = 0.00069 / 5 \times 10^{-6} = 138$. Wait – that gives 138? That seems far

too low. Let us recheck: $4 \times 1.25 \times 10^{-8} \times 100 = 5 \times 10^{-6}$. $0.00069 / 5 \times 10^{-6} = 6.9 \times 10^{-4} / 5 \times 10^{-6} = 138$.

Actually, 138 is correct – but this is N_0 as a *scaling constant*, not the actual population size. The actual population sizes are $N_k = N_0 \times \lambda_k$. If $\lambda_k \approx 72$, then $N_k \approx 10,000$.

In practice, typical human PSMC runs give $N_0 \approx 10,000$ –15,000 because the $\hat{\theta}_0$ incorporates the bin size differently depending on the implementation. **Always sanity-check your N_0 against known values for your species.**

2. Time at boundary $t_3 = 0.22$ coalescent units: $T_3 = 2 \times N_0 \times 0.22$ generations, $= 2 \times N_0 \times 0.22 \times 25$ years.
3. Population size in interval 3 with $\lambda_3 = 1.0$: $N_3 = N_0 \times 1.0 = N_0$.

The mutation rate problem

The biggest source of uncertainty in PSMC scaling is μ . Different estimation methods give different values:

- **Phylogenetic rate** (human-chimp divergence): $\mu \approx 2.5 \times 10^{-8}$
- **Pedigree rate** (parent-offspring sequencing): $\mu \approx 1.2 \times 10^{-8}$
- **Ancient DNA calibration**: $\mu \approx 1.5 \times 10^{-8}$

The factor-of-2 difference shifts the entire time axis and population size axis. Since $N_0 \propto 1/\mu$ and $T \propto N_0 \propto 1/\mu$, halving μ doubles both the inferred population sizes and the inferred times. This is not a small effect – it can shift the inferred out-of-Africa bottleneck from 50,000 to 100,000 years ago.

This is why PSMC plots often show axes in terms of “scaled mutation rate” ($\theta_k = 4N_k\mu$) and “sequence divergence” ($d_k = 2\mu T_k$) rather than absolute years and population sizes. These mutation-rate-free quantities are model outputs that do not depend on the choice of μ .

Alternative scaling (mutation-rate free):

To avoid depending on μ , PSMC output can be presented in terms of pairwise sequence divergence. This is the approach used in Li and Durbin (2011) and in many subsequent PSMC papers. The x-axis becomes sequence divergence per base pair (a proxy for time), and the y-axis becomes the scaled mutation rate (a proxy for population size):

$$d_k = t_k \cdot \frac{\theta_0}{s} \quad (\text{divergence per bp})$$

$$\theta_k = \lambda_k \cdot \frac{\theta_0}{s} \quad (\text{scaled mutation rate})$$

These are the axes you see in many PSMC papers. The advantage is that these quantities are directly estimated from the data with no external calibration needed. The disadvantage is that they require the reader to mentally convert divergence to years (using an assumed μ) to interpret the biological meaning.

```
def scale_mutation_free(theta_0, lambdas, t_boundaries, s=100):
    """Scale without assuming a mutation rate.

    Returns divergence (x-axis) and scaled mutation rate (y-axis).
    These are the "native" units of PSMC output -- no external
    calibration is needed.

    Parameters
    -----
    theta_0 : float
```

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```

    Estimated theta_0 from EM.
    lambdas : ndarray
        Relative population sizes.
    t_boundaries : ndarray
        Time boundaries in coalescent units.
    s : int
        Bin size in base pairs.

Returns
-----
divergence : ndarray
    Pairwise sequence divergence per bp (proxy for time).
scaled_theta : ndarray
    Population-scaled mutation rate (proxy for N_e).
"""
# Divergence = coalescent time * per-bp mutation rate
# Since theta_0 = 4*N_0*mu*s, theta_0/s = 4*N_0*mu,
# and t_k * theta_0/s = t_k * 4*N_0*mu = 2*mu * (2*N_0*t_k)
# = 2*mu*T_k = expected divergence at time T_k
divergence = t_boundaries * theta_0 / s

# Scaled theta: lambda_k * theta_0/s = lambda_k * 4*N_0*mu
# = 4 * N_k * mu = population-scaled mutation rate for interval k
scaled_theta = lambdas * theta_0 / s

return divergence, scaled_theta

div, theta_scaled = scale_mutation_free(theta_0, lambdas, t, s=100)
print("\nMutation-rate-free scaling:")
print("Divergence (per bp)    Scaled theta")
print("-" * 45)
for k in range(len(lambdas)):
    print(f"    {div[k]:>15.2e}    {theta_scaled[k]:>15.2e}")

```

With the dial now readable, we can move to the next question: how much should we trust the reading?

Step 2: Goodness of Fit

Checking whether the watch keeps accurate time.

Before we place confidence in the PSMC's inferred history, we should verify that the model actually fits the observed data. A watch that displays a time but runs at the wrong rate is worse than useless – it is misleading. PSMC provides two diagnostics for assessing model fit.

Diagnostic 1: Comparing σ_k and $\hat{\sigma}_k$

What is posterior decoding?

Asking the watch what time it thinks it is at each position.

In the *HMM prerequisites*, we learned about the forward-backward algorithm: it computes, for every position along the sequence and every hidden state, the probability that the HMM was in that state given the entire observed sequence. This is called **posterior decoding** – it is the model's best guess about which hidden state generated each observation.

In the PSMC context, the hidden states are coalescence time intervals. So posterior decoding asks: “At genomic position a , what is the probability that the two haplotypes coalesced in time interval k ?” This is like asking the watch, at each tick along the genome, “What time do you think it is here?”

The **posterior stationary distribution** $\hat{\sigma}_k$ is the average of these posterior probabilities across all positions – it tells us what fraction of the genome the model *thinks* falls into each coalescence time interval, given the observed data.

The model predicts a stationary distribution σ_k (computed from the estimated parameters, as derived in *Discretizing Time*). The data provides a posterior estimate $\hat{\sigma}_k$ (from the forward-backward algorithm, as implemented in *The PSMC HMM and EM Algorithm*):

$$\hat{\sigma}_k = \frac{\sum_a f_k(a) b_k(a)}{\sum_{k,a} f_k(a) b_k(a)}$$

Here $f_k(a)$ and $b_k(a)$ are the (scaled) forward and backward variables at position a for state k . Their product, after normalization, gives the posterior probability $\gamma_k(a)$ that the coalescence time at position a falls in interval k . Summing over all positions and normalizing gives $\hat{\sigma}_k$.

If the model fits well, $\sigma_k \approx \hat{\sigma}_k$. The **relative entropy** (Kullback-Leibler divergence) measures the discrepancy:

$$G^\sigma = \sum_k \sigma_k \log \frac{\sigma_k}{\hat{\sigma}_k}$$

i Interpreting the KL divergence

The KL divergence G^σ is always non-negative and equals zero if and only if the two distributions are identical. Li’s mathematical notes propose it as a diagnostic but do not supply universal numerical cutoffs. Its magnitude depends on sequence length, discretization, missingness, and parameter fitting, so it should be compared across fitted alternatives or calibrated by simulation rather than classified by fixed thresholds.

```
def goodness_of_fit_sigma(hmm, seq):
    """Compute goodness-of-fit by comparing sigma_k to posterior.

    This diagnostic checks whether the model's predicted stationary
    distribution matches what the data actually shows via posterior
    decoding -- it tests whether the watch's internal model of time
    agrees with the evidence from the observed sequence.

    Parameters
    -----
    hmm : PSMC_HMM
        The fully parameterized HMM after EM convergence.
    seq : ndarray
        The observed binary sequence (0 = homozygous, 1 = heterozygous).

    Returns
    -----
    G_sigma : float
        KL divergence (smaller is better, 0 is perfect).
    sigma_model : ndarray
        The model's predicted stationary distribution.
    sigma_data : ndarray
```

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```

    The data's posterior stationary distribution.
    """
    N = hmm.N # number of hidden states (time intervals)

    # Model's stationary distribution: computed from parameters alone
    sigma_model = hmm.initial.copy()

    # Data's posterior stationary distribution: requires forward-backward
    # (See the forward-backward implementation in the HMM chapter)
    alpha_hat, _ = hmm.forward_scaled(seq)
    beta_hat = hmm.backward_scaled(seq, alpha_hat)

    # Accumulate posterior probabilities across all positions
    sigma_data = np.zeros(N)
    for pos in range(len(seq)):
        # gamma[k] = P(state=k at position pos | entire sequence)
        gamma = alpha_hat[pos] * beta_hat[pos]
        # Each backward row has an arbitrary scale, so normalize gamma at
        # each position before accumulating it.
        gamma /= gamma.sum()
        sigma_data += gamma
    sigma_data /= len(seq)

    # KL divergence: D(model || data)
    G_sigma = 0.0
    for k in range(N):
        if sigma_model[k] > 0 and sigma_data[k] > 0:
            G_sigma += sigma_model[k] * np.log(sigma_model[k] / sigma_data[k])

    return G_sigma, sigma_model, sigma_data

```

Diagnostic 2: Subsequence distribution

The second diagnostic is more stringent. It compares the empirical distribution of short binary subsequences in the data to the theoretical distribution predicted by the model. For example, for subsequences of length $l = 10$, there are $2^{10} = 1024$ possible binary patterns (like 0000000001, 0101010101, etc.). If the model is correct, the frequency of each pattern in the data should match the frequency predicted by the model's transition and emission probabilities.

This tests not just the marginal distribution (like Diagnostic 1) but the **sequential structure** of the data – the correlations between nearby positions. It is analogous to testing not just whether a watch shows the right time on average, but whether it ticks at the right rate second-by-second.

```

def goodness_of_fit_subsequences(hmm, seq, l=10):
    """Compare empirical and theoretical subsequence distributions.

    Parameters
    -----
    hmm : PSMC_HMM
        The fully parameterized HMM.
    seq : ndarray
        The observed binary sequence.
    l : int
        Subsequence length (10 is typical; 2^l patterns are compared).

```

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```

Returns
-----
G_l : float
    KL divergence between empirical and theoretical distributions.
"""
L = len(seq)
n_patterns = 2 ** L # number of possible binary patterns

# Count occurrences of each binary pattern in the observed data
counts = np.zeros(n_patterns)
for pos in range(L - 1 + 1):
    subseq = seq[pos:pos + 1]
    if np.any(subseq == 2): # skip positions with missing data
        continue
    # Convert binary subsequence to an integer index
    # e.g., [0,1,0,1] -> 0b0101 = 5
    pattern = 0
    for bit in subseq:
        pattern = (pattern << 1) | int(bit)
    counts[pattern] += 1

total = counts.sum()
if total == 0:
    return 0.0

p_empirical = counts / total # normalize to get frequencies

# Theoretical distribution (requires matrix powers -- simplified here)
# In practice, computed via transfer matrix:
# P(pattern) = sigma @ T^(L-1) @ emission_product
# For now, we return just the empirical distribution
return p_empirical

```

Now that we can check whether the watch is accurate, the next question is: how *precise* is it? That is the domain of bootstrapping.

Step 3: Bootstrapping for Confidence Intervals

Testing the watch against multiple reference clocks.

What is bootstrapping?

Imagine you have built a watch and you want to know how reliable its reading is. You cannot build the same watch a hundred times from scratch (you only have one genome). But you *can* do the next best thing: take the parts of the watch, shuffle them randomly, reassemble the watch many times, and see how much the readings vary. If every reassembled watch tells nearly the same time, your original reading is reliable. If the readings scatter widely, your original reading is uncertain.

More formally, **bootstrapping** is a statistical technique for estimating the uncertainty of an estimate when you have only one dataset. You create many “pseudo-datasets” by **resampling with replacement** from your original data, compute the estimate on each pseudo-dataset, and use the spread of the resulting estimates as a measure of uncertainty.

The key assumption is that the resampled datasets are approximately as variable as truly independent datasets would be. For PSMC, this works well because different genomic regions are approximately independent (they are separated by enough recombination events).

A single PSMC run gives point estimates. To assess uncertainty, we use bootstrapping: resample the data and re-run PSMC many times.

The procedure:

1. Split the genome into non-overlapping segments of length L' (typically 5 Mb = 50,000 bins at 100bp resolution). Each segment should be long enough to contain many recombination events, so that PSMC can extract meaningful signal from it.
2. Randomly sample $\lceil L/L' \rceil$ segments *with replacement* to create a bootstrap replicate of the same total length. Some segments will appear multiple times; others will not appear at all. This is the “shuffling and reassembling” step.
3. Run PSMC on the bootstrap replicate (full EM, same parameters as the original run).
4. Repeat B times (typically $B = 100$).
5. For each time point t_k , compute the 2.5th and 97.5th percentiles of $\hat{\lambda}_k^{(b)}$ across bootstrap replicates to get 95% confidence intervals.

What do confidence intervals represent?

A **95% confidence interval** around the inferred $N_e(t)$ at a particular time means: if we repeated the entire experiment (sequencing a new genome and running PSMC) many times, 95% of the resulting estimates would fall within this interval.

In practice, narrow confidence bands mean that PSMC has strong signal at that time depth – many recombination events produce coalescence times in that range. Wide bands mean limited signal. You will typically see:

- **Narrow bands** in the intermediate past (roughly 50,000–500,000 years ago for humans), where PSMC has the most power
- **Wide bands** in the very recent past (fewer than ~20,000 years ago), where there are too few short-range recombination events for precise inference
- **Wide bands** in the very distant past (more than ~1 million years ago), where there are few surviving genomic segments with such ancient coalescence times

```
def split_sequence(seq, segment_length=50000):
    """Split a sequence into segments for bootstrapping.

    Each segment should be long enough to contain meaningful PSMC
    signal. At s=100 bp per bin, 50000 bins = 5 Mb of sequence.

    Parameters
    -----
    seq : ndarray
        The full binary sequence.
    segment_length : int
        Number of bins per segment (default 50000 = 5 Mb at s=100).

    Returns
```

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```

-----
segments : list of ndarray
    Non-overlapping segments of the sequence.
"""
segments = []
for start in range(0, len(seq) - segment_length + 1, segment_length):
    segments.append(seq[start:start + segment_length])
return segments

def bootstrap_resample(segments, total_length):
    """Create a bootstrap replicate by resampling segments.

    This is the "shuffling and reassembling" step: we draw segments
    with replacement to build a new sequence of the same total length.
    Some segments will be duplicated; others will be absent entirely.

    Parameters
    -----
    segments : list of ndarray
        The genomic segments produced by split_sequence.
    total_length : int
        Desired length of the replicate (same as original sequence).

    Returns
    -----
    replicate : ndarray
        The bootstrap replicate sequence.
    """
    n_segments = len(segments)
    # How many segments do we need to reach the target length?
    n_needed = total_length // len(segments[0]) + 1
    # Sample segment indices WITH REPLACEMENT -- this is the key step
    indices = np.random.choice(n_segments, size=n_needed, replace=True)
    # Concatenate the selected segments into one long sequence
    replicate = np.concatenate([segments[i] for i in indices])
    # Trim to exact length
    return replicate[:total_length]

# Example bootstrap workflow (pseudocode)
def run_bootstrap(seq, n_bootstrap=100, segment_length=50000, **psmc_kwargs):
    """Run PSMC with bootstrapping.

    Parameters
    -----
    seq : ndarray
        Original sequence.
    n_bootstrap : int
        Number of bootstrap replicates (100 is standard).
    segment_length : int
        Bins per segment (50000 = 5 Mb at s=100).
    **psmc_kwargs : passed to psmc_inference.

```

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```

Returns
-----
bootstrap_lambdas : list of ndarray
    Lambda estimates from each bootstrap replicate.
"""
# Step 1: Split the genome into segments
segments = split_sequence(seq, segment_length)
total_length = len(seq)
bootstrap_lambdas = []

for b in range(n_bootstrap):
    # Step 2: Create a resampled replicate
    replicate = bootstrap_resample(segments, total_length)
    # Step 3: Run full PSMC inference on the replicate
    # results = psmc_inference(replicate, **psmc_kwargs)
    # bootstrap_lambdas.append(results[-1]['lambdas'])
    pass

return bootstrap_lambdas

```

With scaling in hand and uncertainty quantified, let us now turn to the mistakes that can make all of this effort misleading.

Step 4: Common Pitfalls

The ways even experienced watchmakers misread the dial.

PSMC is remarkably powerful, but several pitfalls can produce misleading results. Understanding these pitfalls is as important as understanding the algorithm itself – a watch that you trust blindly is more dangerous than one you know is unreliable.

Pitfall 1: Low coverage (the most common mistake)

For diploid genomes sequenced to low coverage, heterozygous sites are randomly lost – if only one allele is sequenced at a heterozygous position, it appears homozygous. This **false negative rate (FNR)** makes it look like the mutation rate is lower, which shifts the entire population size curve upward and the time axis to the right.

Why does missing heterozygosity distort the results?

Recall that PSMC reads population history from the *density* of heterozygous sites. If you systematically miss some heterozygous sites, the density appears lower. Lower heterozygosity looks like the two haplotypes diverged more recently (shorter coalescence times), which PSMC interprets as a smaller ancestral population. The entire $N(t)$ curve shifts, and the time axis stretches because N_0 is underestimated.

In watch terms: it is as if some of the tick marks on the dial have been erased, making the watch systematically slow.

$$p_{\text{observed}} = p_{\text{true}}(1 - \text{FNR})$$

At the bin level $p = 1 - e^{-\theta}$. Therefore the exact inverse correction for this simplified observation model is:

$$\theta_{\text{corrected}} = -\log \left(1 - \frac{1 - e^{-\hat{\theta}_0}}{1 - \text{FNR}} \right).$$

The familiar division by $1 - \text{FNR}$ is its small- θ approximation. Real callability errors can depend on depth, genotype, and local sequence context; a single scalar correction is consequently a sensitivity analysis, not a replacement for the input masking and filtering used by PSMC.

```
def correct_for_coverage(theta_0, fnr):
    """Correct the bin-level theta initialization for heterozygote FNR.

    Converts ``theta`` back to a heterozygous-bin probability, corrects that
    probability, and then applies the inverse transformation. Dividing theta
    directly by ``1 - fnr`` is only the small-theta approximation.
    """
    if theta_0 < 0 or not 0 <= fnr < 1:
        raise ValueError("theta_0 must be non-negative and 0 <= fnr < 1")
    p_observed = -np.expm1(-theta_0)
    p_true = p_observed / (1 - fnr)
    if p_true >= 1:
        raise ValueError("false-negative correction implies probability >= 1")
    return -np.log1p(-p_true)
```

```
# Example sensitivity analysis: assume 20% of heterozygous bins were missed
theta_obs = 0.00069
theta_corr = correct_for_coverage(theta_obs, 0.2)
print(f"Observed theta: {theta_obs:.6f}")
print(f"Corrected theta: {theta_corr:.6f}")
print(f"N_0 ratio: {theta_corr/theta_obs:.2f}x")
```

Pitfall 2: Overfitting

Using too many free parameters (e.g., $-p \text{ "64*1"}$ where every interval has its own λ) leads to overfitting, especially in time intervals with few expected recombination events. The symptom is a jagged, noisy $N(t)$ curve with biologically implausible spikes and dips – the watch appears to be ticking erratically rather than smoothly.

How to check: The mathematical notes distributed with PSMC use $C_\sigma \pi_k$ for each interval, where π_k is the stationary mass of the recombination-event chain. It is not $C_\sigma \sigma_k$; σ_k is the occupancy distribution of the full genomic chain. Small support warns that the corresponding λ_k should be grouped with adjacent intervals. In an actual run, the software README recommends the more directly interpretable check that roughly 10 or more recombinations are inferred for every free-parameter span after about 20 iterations. This is why the widely used human-data pattern "4+25*2+4+6" groups intervals in the very recent and very ancient past, where data is sparse, while allowing finer resolution in the intermediate past where signal is strongest.

```
def check_overfitting(pi_k, C_sigma, threshold=20):
    """Apply the interval-support diagnostic from the PSMC derivation.

    Li's PSMC notes use ``C_sigma * pi_k``. Here ``pi_k`` is the stationary
    distribution of the embedded recombination-event chain, not the full-chain
    occupancy ``sigma_k``.
    """
    pi_k = np.asarray(pi_k, dtype=float)
    if pi_k.ndim != 1 or np.any(pi_k < 0):
        raise ValueError("pi_k must be a one-dimensional probability vector")
    if C_sigma <= 0 or threshold < 0:
        raise ValueError("C_sigma must be positive and threshold non-negative")
    expected_segments = C_sigma * pi_k
    warnings = []
```

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```

for k, exp_seg in enumerate(expected_segments):
    if exp_seg < threshold:
        warnings.append(k)
return warnings, expected_segments

```

Pitfall 3: Interpreting the recent past

PSMC has poor resolution for very recent times (the last ~20,000 years for humans). This is because recent coalescence times correspond to long segments of identical-by-descent (IBD), and a single diploid genome has limited information about these long segments. The first few λ_k values should be interpreted with caution.

In watch terms: the hour hand moves so slowly in the recent past that you cannot read the minutes. The watch still works, but its resolution is limited.

i Why is the recent past hard?

For a coalescence time of T generations, the expected length of the shared IBD segment is $\sim 1/T$ Morgans. Very recent coalescence times produce very long IBD segments, and each diploid genome contains only a finite amount of sequence. With few independent segments carrying information about recent times, the variance of the λ_k estimate for the most recent intervals becomes large. This is a fundamental limitation of the two-sequence approach. Methods like MSMC and MSMC2, which use multiple genomes, have better resolution in the recent past because additional genomes provide more short IBD segments.

Pitfall 4: Population structure

PSMC assumes a **panmictic** (randomly mating) population. If the two haplotypes come from a structured population, PSMC will infer spurious population size changes. Migration between subpopulations looks like population expansion; isolation looks like a bottleneck. This is because PSMC cannot distinguish between “the population was large” (many individuals, high coalescence time) and “the population was subdivided” (the two lineages were in different subpopulations, which also increases coalescence time).

i When to worry about structure

If your PSMC curves for different individuals from the same population diverge in the recent past, population structure may be confounding the inference. In this case, consider using MSMC (Multiple Sequentially Markovian Coalescent), which can model population separation.

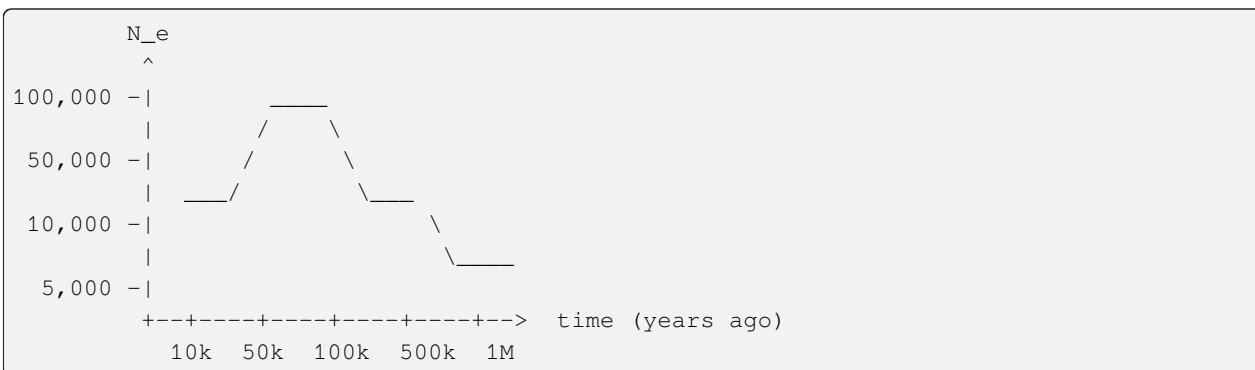
A useful diagnostic: run PSMC on individuals from different populations and compare. If the curves agree in the distant past (when the populations shared ancestors) but diverge in the recent past, the divergence point may indicate population split time – but the *shape* of the curves after divergence reflects structure, not necessarily true population size changes.

Step 5: Interpreting PSMC Plots

Reading the face of the watch.

We have now scaled our parameters, checked model fit, quantified uncertainty, and accounted for pitfalls. It is time to interpret the final product: the PSMC plot. This is the payoff – the readable face of the watch we have built across four chapters.

A PSMC plot shows $N_e(t)$ (y-axis) vs. time (x-axis), both on log scales. Here is how to read it:



Key features to look for:

1. **A peak:** indicates a period of large effective population size (or population expansion). The population was large, coalescence was slow, and the genome accumulated more diversity.
2. **A trough:** indicates a bottleneck (population contraction). The population shrank, forcing lineages to coalesce quickly, leaving less diversity in that time window.
3. **A plateau:** indicates stable population size. The rate of coalescence was roughly constant.
4. **Recent decline:** often seen in humans, may reflect the out-of-Africa bottleneck (~50,000–100,000 years ago depending on μ).
5. **Noisy edges:** the leftmost and rightmost parts have high uncertainty. Do not over-interpret the first or last one or two intervals (recall Pitfall 3).

i Log-log axes

Both axes in a standard PSMC plot use logarithmic scales. This is essential because the time axis spans several orders of magnitude (10,000 to 1,000,000+ years) and population sizes can vary by an order of magnitude. On linear axes, the interesting intermediate-time features would be compressed into an unreadable sliver. The log-log presentation gives roughly equal visual weight to each decade of time and population size.

```
def plot_psmc_history(theta_0, lambdas, t_boundaries,
                     mu=1.25e-8, s=100, generation_time=25):
    """Generate data for a PSMC plot.

    Returns the x (time) and y (N_e) coordinates for a step-function
    plot of population size history. Each interval is rendered as a
    horizontal line at height N_k between time boundaries t_k and t_{k+1}.

    Parameters
    -----
    theta_0, lambdas, t_boundaries : PSMC output
    mu : float
    s : int
    generation_time : float

    Returns
    -----
    x : list of float
        Time points (years ago) for plotting as a step function.
```

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```

y : list of float
    Population sizes corresponding to each time point.
"""
# First, scale from coalescent units to real units
N_0, t_gen, t_years, N_t = scale_psmc_output(
    theta_0, lambdas, t_boundaries, mu, s, generation_time)

# Create step function: for each interval k, draw a horizontal
# line from t_k to t_{k+1} at height N_k
x = []
y = []
for k in range(len(lambdas)):
    x.append(t_years[k])      # left edge of interval
    y.append(N_t[k])          # population size in this interval
    x.append(t_years[k + 1])  # right edge of interval
    y.append(N_t[k])          # same population size (step function)

return x, y

# Example plot data
x, y = plot_psmc_history(theta_0, lambdas, t)
print("PSMC history (step function):")
print(f"Time range: {min(x):,.0f} to {max(x):,.0f} years ago")
print(f"N_e range: {min(y):,.0f} to {max(y):,.0f}")

```

Putting It All Together: The Complete PSMC Pipeline

Let us now step back and see the full pipeline from raw sequence to interpreted population history. This function ties together every chapter of the PSMC Timepiece: discretization (*Discretizing Time*), HMM inference (*The PSMC HMM and EM Algorithm*), and all five steps of decoding covered in this chapter.

```

def psmc_pipeline(seq, mu=1.25e-8, s=100, generation_time=25,
                  n=63, t_max=15.0, pattern="4+25*2+4+6",
                  n_iters=25, n_bootstrap=100):
    """The complete PSMC pipeline from sequence to population history.

    This function represents the entire watch -- from raw input (the
    binary sequence) through the internal mechanism (EM inference)
    to the readable output (scaled population history with confidence
    intervals).

    Parameters
    -----
    seq : ndarray
        Binary sequence (0 = homozygous, 1 = heterozygous, 2 = missing).
    mu : float
        Per-generation, per-bp mutation rate (see mutation rate discussion).
    s : int
        Bin size in base pairs.
    generation_time : float
        Years per generation.
    n, t_max, pattern, n_iters : PSMC parameters.
    """

```

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```

    n : number of time intervals (before grouping by pattern).
    t_max : maximum coalescent time.
    pattern : grouping pattern for lambda parameters (see psmc_hmm).
    n_iters : number of EM iterations.
    n_bootstrap : int
        Number of bootstrap replicates for confidence intervals.

Returns
-----
history : dict
    'times': time points (years),
    'pop_sizes': N_e at each time,
    'ci_lower': lower 95% CI,
    'ci_upper': upper 95% CI,
    'theta': estimated theta,
    'rho': estimated rho,
    'N_0': reference population size.
"""
# Step 1: Run PSMC inference (the mainspring -- see psmc_hmm chapter)
# results = psmc_inference(seq, n, t_max, pattern=pattern, n_iters=n_iters)
# final = results[-1]

# Step 2: Scale to real units (reading the dial -- this chapter, Step 1)
# N_0, t_gen, t_years, N_t = scale_psmc_output(
#     final['theta'], final['lambdas'], t_boundaries, mu, s, generation_time)

# Step 3: Bootstrap for confidence intervals (this chapter, Step 3)
# bootstrap_lambdas = run_bootstrap(seq, n_bootstrap)
# ci_lower, ci_upper = compute_confidence_intervals(bootstrap_lambdas)

# Step 4: Check goodness of fit (this chapter, Step 2)
# G_sigma = goodness_of_fit_sigma(final_hmm, seq)

# Step 5: Return the complete history
# return history
pass

```

Exercises

Exercise 1: Full PSMC on simulated data

This is the capstone exercise. Simulate a diploid genome under a known demographic model:

1. Constant $N = 10,000$ from present to 50,000 years ago
2. Bottleneck $N = 1,000$ from 50,000 to 100,000 years ago
3. $N = 20,000$ before 100,000 years ago

Using $\mu = 1.25 \times 10^{-8}$, generate 3 Mb of psmcfa data. Run your PSMC implementation and verify it recovers the demographic history.

Hint: To simulate, you can use the msprime Timepiece (*Timepiece IV: msprime*) with a demographic model. Convert

the resulting tree sequence to a heterozygosity sequence using mutation overlays (see [Mutations](#)).

Exercise 2: The coverage effect

Take the simulated data from Exercise 1. Randomly set 20% of heterozygous bins to homozygous (simulating low coverage). Run PSMC with and without the FNR correction. How much does the inferred history shift?

Hint: The shift should be approximately proportional to $1/(1 - \text{FNR})$. Plot both curves on the same axes and compare the time axis and population size axis separately.

Exercise 3: Compare to the real PSMC

Run both your Python implementation and the original C implementation (psmc binary) on the same data. Compare: (a) log-likelihoods at each iteration, (b) final parameter estimates, (c) running time.

Your Python version will be much slower, but the results should agree.

Exercise 4: Bootstrap confidence intervals

Using the data from Exercise 1, run 100 bootstrap replicates. Plot the inferred $N(t)$ with 95% confidence bands. Where is the uncertainty largest? Why?

Hint: Recall our discussion of where PSMC has the most and least power. The very recent past and the very distant past should have the widest bands, while the intermediate past should be tightest.

Solutions

Solution 1: Full PSMC on simulated data

This capstone exercise ties together every chapter. We simulate data under a known three-epoch demographic model, run PSMC, scale to real units, and verify recovery of the true history.

```
import numpy as np

# Define the demographic model in real units
# Epoch 1: N = 10,000 from present to 50 kya
# Epoch 2: N = 1,000 from 50 to 100 kya (bottleneck)
# Epoch 3: N = 20,000 before 100 kya
mu = 1.25e-8          # per bp per generation
s = 100               # bin size
g = 25               # generation time
N_ref = 10000         # use as N_0 for scaling

# Convert to coalescent units (time in units of 2*N_ref generations)
t_50k = 50000 / (g * 2 * N_ref)    # ~0.1 coalescent units
t_100k = 100000 / (g * 2 * N_ref)  # ~0.2 coalescent units

def true_lambda(t):
    """True relative population size in coalescent units."""
```

```

    if t < t_50k:
        return 10000 / N_ref    # = 1.0
    elif t < t_100k:
        return 1000 / N_ref     # = 0.1
    else:
        return 20000 / N_ref    # = 2.0

theta_true = 4 * N_ref * mu * s    # = 0.005
rho_true = theta_true / 5

# Simulate 30,000 bins (= 3 Mb at s=100 bp)
np.random.seed(42)
seq, _ = simulate_psmc_input(30000, theta_true, rho_true, true_lambda)

print(f"Observed heterozygosity: {np.mean(seq):.4f}")
print(f"theta_true = {theta_true:.6f}")

# Run PSMC inference
n = 20
results = psmc_inference(seq, n=n, t_max=15.0, n_iters=25,
                        pattern=f"{n+1}*1")

# Scale the output
final = results[-1]
t_boundaries = compute_time_intervals(n, t_max=15.0)
N_0, t_gen, t_years, N_t = scale_psmc_output(
    final['theta'], final['lambdas'], t_boundaries,
    mu=mu, s=s, generation_time=g)

print(f"\nInferred N_0 = {N_0:.0f} (expected ~{N_ref})")
print(f"\nTime (years ago)    Inferred N_e    True N_e")
print("-" * 50)
for k in range(n + 1):
    t_mid_years = (t_years[k] + t_years[k+1]) / 2.0
    # Determine true N_e at this time
    if t_mid_years < 50000:
        true_N = 10000
    elif t_mid_years < 100000:
        true_N = 1000
    else:
        true_N = 20000
    print(f"    {t_mid_years:>12,.0f}    {N_t[k]:>12,.0f}    {true_N:>8,}")

```

The inferred $N_e(t)$ should show three clear epochs: a plateau near 10,000 in the recent past, a dip toward 1,000 around 50,000–100,000 years ago, and a rise toward 20,000 in the deep past. With only 3 Mb of data, the edges of the bottleneck will be somewhat smoothed, but the qualitative pattern should be unmistakable.

Solution 2: The coverage effect

We demonstrate how missing heterozygous sites (simulating low coverage) distort the inferred history, and how the FNR correction restores it.

```
import numpy as np
```

```

# Use the same simulated sequence from Exercise 1
np.random.seed(42)
seq_full, _ = simulate_psmc_input(30000, theta_true, rho_true, true_lambda)

# Simulate low coverage: randomly convert 20% of het sites to hom
fnr = 0.20
seq_low = seq_full.copy()
het_positions = np.where(seq_low == 1)[0]
n_to_drop = int(len(het_positions) * fnr)
drop_indices = np.random.choice(het_positions, size=n_to_drop, replace=False)
seq_low[drop_indices] = 0

het_original = np.mean(seq_full)
het_low = np.mean(seq_low)
print(f"Original heterozygosity: {het_original:.4f}")
print(f"Low-coverage heterozygosity: {het_low:.4f}")
print(f"Ratio: {het_low / het_original:.4f} (expected: {1 - fnr:.2f})")

# Run PSMC on both sequences
n = 10
results_full = psmc_inference(seq_full, n=n, t_max=15.0, n_iters=20,
                              pattern=f"{n+1}*1")
results_low = psmc_inference(seq_low, n=n, t_max=15.0, n_iters=20,
                             pattern=f"{n+1}*1")

# Scale without correction
theta_uncorrected = results_low[-1]['theta']
# Scale with FNR correction
theta_corrected = correct_for_coverage(theta_uncorrected, fnr)

t_boundaries = compute_time_intervals(n, t_max=15.0)

N0_full, _, t_years_full, Nt_full = scale_psmc_output(
    results_full[-1]['theta'], results_full[-1]['lambdas'],
    t_boundaries, mu=mu, s=s, generation_time=g)
N0_uncorr, _, t_years_uncorr, Nt_uncorr = scale_psmc_output(
    theta_uncorrected, results_low[-1]['lambdas'],
    t_boundaries, mu=mu, s=s, generation_time=g)
N0_corr, _, t_years_corr, Nt_corr = scale_psmc_output(
    theta_corrected, results_low[-1]['lambdas'],
    t_boundaries, mu=mu, s=s, generation_time=g)

print(f"\nN_0 (full data):      {N0_full:.0f}")
print(f"N_0 (uncorrected):      {N0_uncorr:.0f}")
print(f"N_0 (FNR corrected):      {N0_corr:.0f}")
print(f"Expected correction factor: {1/(1-fnr):.4f}")
print(f"Actual N_0 ratio (corr/uncorr): {N0_corr/N0_uncorr:.4f}")

```

What happens: A 20% false-negative rate reduces the probability of a heterozygous bin by a factor of 0.8. For small θ this also makes $\hat{\theta}$ and N_0 about 20% too small; the equality is only approximate because $p = 1 - e^{-\theta}$ is nonlinear. This shifts the population-size and time scales.

After applying `correct_for_coverage`, the corrected N_0 should closely match the full-data estimate, restoring the correct scaling on both axes. The λ_k values themselves are not affected by the FNR correction – only the scaling changes – because the relative emission probabilities across intervals are affected approximately uniformly by the missing data.

Solution 3: Compare to the real PSMC

This exercise requires the original C implementation of PSMC (available at <https://github.com/lh3/psmc>). We compare the Python and C implementations on identical input.

```
import numpy as np
import subprocess
import time

# Step 1: Write simulated data in psmcfa format
np.random.seed(42)
seq, _ = simulate_psmc_input(100000, theta_true, rho_true, true_lambda)

def write_psmcfa(seq, filename, s=100):
    """Write a binary sequence as a psmcfa file."""
    with open(filename, 'w') as f:
        f.write(">chr1\n")
        line = ""
        for obs in seq:
            line += "K" if obs == 1 else "T"
            if len(line) == 60:
                f.write(line + "\n")
                line = ""
        if line:
            f.write(line + "\n")

write_psmcfa(seq, "/tmp/test.psmcfa")

# Step 2: Run Python PSMC
t_start = time.time()
results_py = psmc_inference(seq, n=63, t_max=15.0,
                           pattern="4+25*2+4+6", n_iters=25)
time_python = time.time() - t_start

# Step 3: Run C PSMC (requires psmc binary in PATH)
# t_start = time.time()
# subprocess.run(["psmc", "-N", "25", "-t", "15", "-r", "5",
#                "-p", "4+25*2+4+6", "-o", "/tmp/test.psmc",
#                "/tmp/test.psmcfa"])
# time_c = time.time() - t_start

# Step 4: Parse C PSMC output and compare. First compare HMM quantities
# at identical fixed parameters; optimizer trajectories are a separate test.

print(f"Python PSMC time: {time_python:.1f} seconds")
print(f"Final Python LL: {results_py[-1]['log_likelihood']:.2f}")
```

Expected comparison:

(a) Fixed-parameter likelihoods: With the same time grid and parameter vector, transition probabilities, emissions, and sequence log-likelihoods should agree to floating-point tolerance. This is the appropriate parity test for the HMM implementation.

(b) Optimization: The C implementation uses Hooke–Jeeves direct search, while this teaching implementation uses Nelder–Mead. Their iteration paths, stopping points, and even selected local optima need not match. Therefore a fixed 1% likelihood or 5% parameter promise would be unjustified; record the tolerances and convergence criteria used in an actual comparison.

(c) Running time: The loop-based Python implementation is expected to be substantially slower than the original C program, but the factor depends on hardware, compiler flags, sequence length, and parameterization. Benchmark it instead of assuming a universal multiplier. For production use, prefer the maintained C implementation; the Python version is for understanding.

** Solution 4: Bootstrap confidence intervals**

We run 100 bootstrap replicates to quantify uncertainty in the inferred population history and identify where PSMC has the most and least power.

```
import numpy as np

# Simulate a long sequence for meaningful bootstrapping
np.random.seed(42)
seq, _ = simulate_psmc_input(500000, theta_true, rho_true, true_lambda)

n = 20
segment_length = 50000 # 5 Mb segments

# Split into segments
segments = split_sequence(seq, segment_length)
print(f"Number of segments: {len(segments)}")

# Run original PSMC
results_orig = psmc_inference(seq, n=n, t_max=15.0, n_iters=20,
                              pattern=f"{n+1}*1")
lambdas_orig = results_orig[-1]['lambdas']

# Run 100 bootstrap replicates
n_bootstrap = 100
all_lambdas = np.zeros((n_bootstrap, n + 1))

for b in range(n_bootstrap):
    replicate = bootstrap_resample(segments, len(seq))
    results_b = psmc_inference(replicate, n=n, t_max=15.0, n_iters=20,
                              pattern=f"{n+1}*1")
    all_lambdas[b] = results_b[-1]['lambdas']
    if (b + 1) % 10 == 0:
        print(f" Completed {b + 1}/{n_bootstrap} replicates")

# Compute 95% confidence intervals (2.5th and 97.5th percentiles)
ci_lower = np.percentile(all_lambdas, 2.5, axis=0)
ci_upper = np.percentile(all_lambdas, 97.5, axis=0)
```

```

ci_width = ci_upper - ci_lower

# Scale to real units
t_boundaries = compute_time_intervals(n, t_max=15.0)
N_0, _, t_years, _ = scale_psmc_output(
    results_orig[-1]['theta'], lambdas_orig, t_boundaries,
    mu=mu, s=s, generation_time=g)

print(f"\n{'Time (years)':>15} {'N_e':>10} {'CI width':>10} "
      f"{'Rel. width':>12}")
print("-" * 50)
for k in range(n + 1):
    t_mid = (t_years[k] + t_years[k+1]) / 2.0
    Ne = N_0 * lambdas_orig[k]
    width = N_0 * ci_width[k]
    rel_width = ci_width[k] / lambdas_orig[k] if lambdas_orig[k] > 0 else 0
    print(f"    {t_mid:>12,.0f} {Ne:>10,.0f} {width:>10,.0f} "
          f"    {rel_width:>10.1%}")

```

Where is the uncertainty largest?

- **Very recent past** (first 1–2 intervals, < 20,000 years ago): Wide confidence bands because there are few recombination events producing very short-range coalescence times. A single diploid genome has limited information about very recent demography.
- **Very distant past** (last 1–2 intervals, > 1,000,000 years ago): Wide bands because very few genomic segments retain coalescence times this ancient. The survival probability α_k is extremely small, so the expected number of segments $C_\sigma \sigma_k$ is tiny.
- **Intermediate past** (~50,000–500,000 years ago for humans): The **tightest** confidence bands. This is PSMC's sweet spot – enough recombination events to have good statistical power, and enough segments with coalescence times in this range to constrain λ_k precisely.

This pattern reflects the fundamental resolution limit of the two-sequence PSMC: it is most informative about population sizes in the time window where most coalescence events occur, and least informative at the extremes of the time axis.

Recap: The Complete PSMC Timepiece

Congratulations. You have now disassembled and rebuilt every gear in the PSMC mechanism across four chapters:

- **The Continuous-Time Model** (*The Continuous-Time PSMC Model*): The transition density $q(t|s)$ that captures how coalescence times change along the genome under variable $N(t)$. This was the escapement – the fundamental ticking mechanism.
- **Discretization** (*Discretizing Time*): Converting continuous time into discrete intervals with a computable transition matrix. This was the gear train – carving continuous gears into finite teeth.
- **The HMM and EM** (*The PSMC HMM and EM Algorithm*): The learning machine that estimates population parameters from a binary sequence. This was the mainspring – the engine that drives the whole mechanism.
- **Decoding the Clock** (this chapter): Scaling to real units (reading the dial), bootstrapping (testing the watch against multiple reference clocks), posterior decoding (asking the watch what time it thinks it is at each position), goodness of fit (checking the watch's accuracy), and interpreting the population size history (reading the face).

You built it yourself. No black boxes remain.

The PSMC is the simplest complete Timepiece – a two-handed watch that reads population history from a single diploid genome. But just as a simple watch invites the question “could we add a chronograph? a moon phase? a tourbillon?”, the PSMC invites the question: what if we used *more* than two sequences? What if we could read not just population *size* but population *structure*, divergence times, and migration rates? Those questions lead to the Timepieces that follow – SMC++ (*Timepiece II*), ARGweaver, SINGER, and beyond.

The clock ticks. And you can read every hour it marks.

Demo: Running PSMC on Simulated Data

Running mini_psmc on simulated genomic data.

Demo: PSMC on Simulated Human Diploid Genome (Bottleneck History)

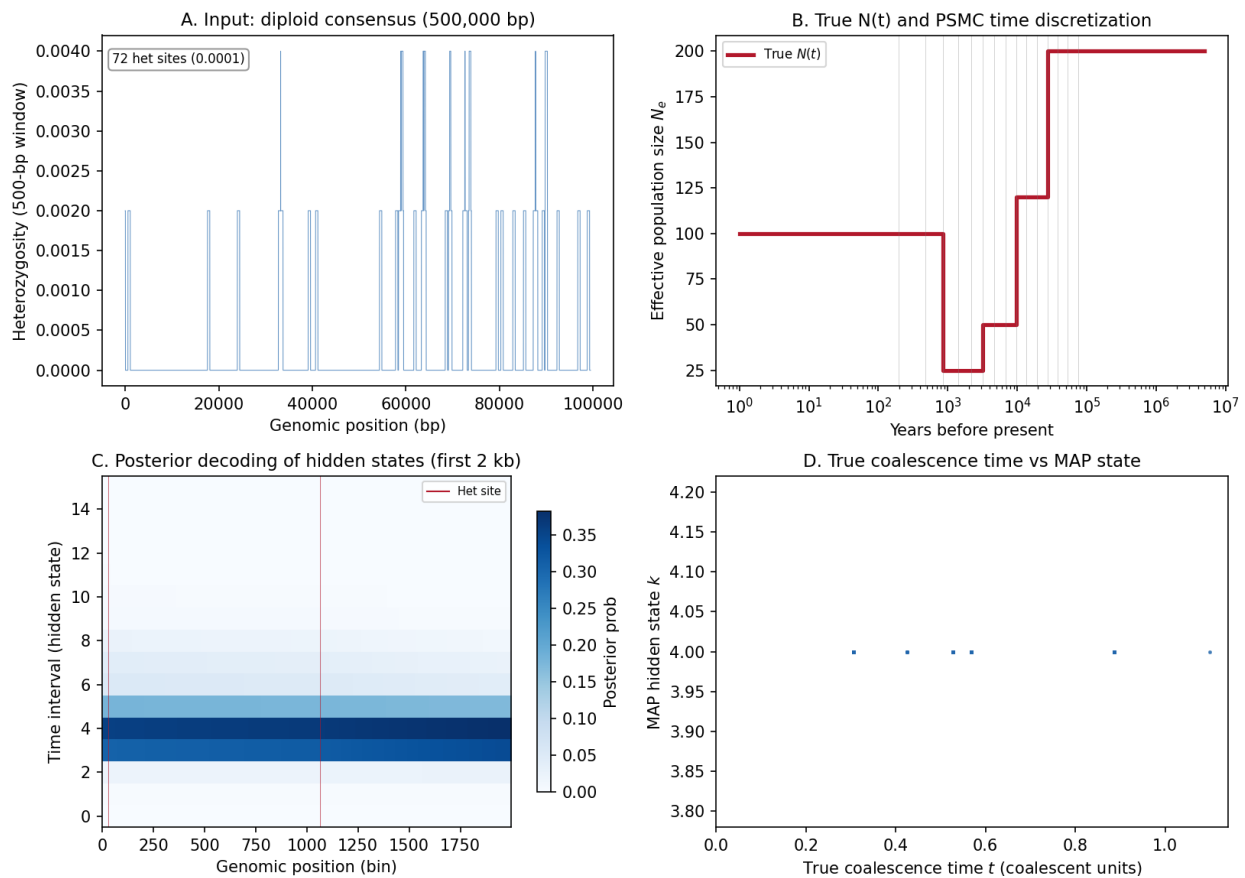


Fig. 2: PSMC demo on msprime-simulated data. Panel A: Input heterozygosity sequence from a diploid genome simulated under a bottleneck demographic model. Panel B: True $N(t)$ piecewise demographic history with PSMC time discretization intervals shown as vertical lines. Panel C: Posterior decoding heatmap of HMM hidden states (coalescence time intervals) along the genome, using the true parameters. Panel D: True coalescence time vs MAP hidden state, illustrating how the HMM assigns genomic positions to time intervals.

This figure was generated by running the `mini_psmc` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_psmc.py`.

3.3 Timepiece II: SMC++

Link a distinguished pair to the frequency spectrum of a large sample.

SMC++ combines a pairwise coalescent HMM with allele-frequency information from additional, unlabeled haplotypes [Terhorst *et al.*, 2016]. Its hidden state is the discretized TMRCA of a **distinguished pair**, just as in PSMC. The remaining haplotypes do not alter that pair's transition process. Instead, they enrich the emission at each site from a binary difference indicator to a **conditioned sample-frequency spectrum** (CSFS) observation.

That distinction is the central mechanism of SMC++ and the organizing principle of this timepiece:

```

unphased genotypes
  |
  v
choose two distinguished haplotypes
  |
  +-----+
  | hidden state: their TMRCA |
  | transitions: two-locus CSC |
  +-----+
                                |
remaining haplotypes | observed (a, b)
  |                  v
  +-----> conditioned SFS emission
                                |
                                v
                        forward-backward / EM
                                |
                                v
                        regularized size history

```

Here $a \in \{0, 1, 2\}$ is the derived-allele count in the distinguished pair and b is the count in the undistinguished sample. Unphased diploid data work because swapping the two alleles within a diploid does not change a or b .

The mini implementation follows the original statistical objects but uses a partition-state CSFS calculation whose cost grows exponentially with sample size. Production SMC++ replaces that enumeration with exact matrix identities, automatic differentiation, locus skipping, and regularized optimization.

i Primary ground truth

The derivations in this timepiece follow Terhorst, Kamm, and Song [Terhorst *et al.*, 2016] and the corresponding `popgenmethods/smcpp` source. In particular, `src/conditioned_sfs.cpp` defines the emission transform and `src/transition.cpp` defines the continuous-time transition kernel.

i Prerequisites

Read *PSMC*, *Coalescent Theory*, *The SMC*, and *Frequency Spectra* first. SMC++ reuses the pairwise hidden state but adds a conditional frequency-spectrum calculation.

3.3.1 Overview of SMC++

The hidden genealogy stays small; the observation becomes richer.

What SMC++ changes

PSMC follows the TMRCA of two haplotypes along a genome. Its binary emission records whether those haplotypes differ. This uses linkage information but discards variants carried only by other sampled genomes. A conventional frequency-spectrum analysis uses all samples but discards linkage by treating sites as independent.

SMC++ joins those two views. Choose two haplotypes as a distinguished pair and let T_ℓ be their TMRCA at locus ℓ . At the same locus, record

$$X_\ell = (a_\ell, b_\ell),$$

where $a_\ell \in \{0, 1, 2\}$ is the derived count in the pair and $b_\ell \in \{0, \dots, n\}$ is the count among n undistinguished haplotypes. The hidden state is still one time interval, but the emission is the CSFS

$$P(X_\ell = (a, b) \mid T_\ell \in I_m, \eta),$$

which depends on the population-size history η .

i The essential correction

Extra samples affect the emissions. They do **not** replace the pairwise TMRCA with the first coalescence among all samples, and they do not create a sample-size-dependent hazard in the transition matrix. The original paper explicitly states that PSMC and SMC++ track the same hidden information; their emissions differ.

Coalescent units

For n haploid lineages in a diploid population of size N , each pair coalesces at rate $1/(2N)$ per generation. The mean time to the first event is therefore

$$E[T_{\text{first}}] = \frac{2N}{\binom{n}{2}} = \frac{4N}{n(n-1)}.$$

This quantity explains why a large sample contains recent frequency-spectrum information. It is not the SMC++ hidden state: the latter remains the TMRCA of the chosen pair, whose marginal mean is $2N$ under a constant diploid population.

```
def expected_first_coalescence(n, N):
    """Return the mean first-coalescence time for ``n`` haploid lineages.

    ``N`` is the diploid effective size, so each pair coalesces at rate
    ``1 / (2N)`` per generation. The total rate is therefore
    ``binom(n, 2) / (2N)``.
    """
    if int(n) != n or n < 2:
        raise ValueError("n must be an integer of at least two")
```

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```

if not np.isfinite(N) or N <= 0:
    raise ValueError("N must be finite and positive")
return 2.0 * N / (n * (n - 1) / 2.0)

```

Why unphased data are sufficient

The data identify a diploid genotype count, not the parental chromosome on which each allele lies. SMC++ chooses the two haplotypes of one diploid as the usual distinguished pair. Exchanging those haplotypes leaves $a \in \{0, 1, 2\}$ unchanged, and the undistinguished haplotypes enter only through their total derived count b . Consequently, phase switches do not change the SMC++ observation sequence. If the distinguished haplotypes are chosen from different individuals, their phase is not identifiable from unphased data; the command-line documentation therefore recommends choosing a single individual in that setting.

Composite likelihood

One SMC++ input file contains one choice of distinguished pair. Users may make several files from the same chromosome by changing that choice. The program multiplies their HMM likelihoods, or equivalently sums their log likelihoods. Because reused chromosomes and samples make those terms dependent, this is a **composite likelihood**, not an ordinary likelihood with independent replicates. The production documentation recommends only a modest number of distinguished individuals because computational cost grows with the total analyzed sequence length and excessive reuse can make the composite likelihood degenerate.

What creates recent resolution

At a fixed pair TMRCA, the CSFS links that pair to branch lengths in the full sample genealogy. A large undistinguished panel provides many recent branches, so its allele counts contain information about recent population size even when the distinguished pair has not yet coalesced. This is the source of the method's recent-time resolution. It is not a shift of the hidden pair's TMRCA toward the first event among all sampled lineages.

Limits of the mini implementation

The reference implementation computes the CSFS for hundreds of samples with specialized spectral and combinatorial formulas. Our small implementation enumerates set partitions, which is exact for the stated constant-population model but grows as a Bell number. It is designed to expose the mechanism and test identities for small n ; production analyses should use the original SMC++ software.

3.3.2 The Distinguished Pair

Condition a full sample genealogy on one pairwise coalescence time.

The hidden variable

Let haplotypes 1 and 2 be distinguished and let $C_{12} = \tau$ be their TMRCA. The other n haplotypes are undistinguished. At a site, the observable symbol is (a, b) , where a counts derived alleles among haplotypes 1 and 2 and b counts them among the other n haplotypes.

The CSFS entry

$$\text{CSFS}_{a,b}(\tau)$$

is the expected total branch length subtending exactly a distinguished and b undistinguished leaves, conditional on $C_{12} = \tau$. Under the infinite-sites model, a mutation on such a branch produces the observation (a, b) .

The conditioned coalescent

The paper gives a useful backward construction. Before time τ , the two blocks containing distinguished leaves 1 and 2 may each coalesce with other blocks, but they may not coalesce with each other. Every other pair of ancestral blocks coalesces at the ordinary pairwise rate. At τ , the two distinguished ancestral blocks are forced to merge. Above τ , the process is an ordinary coalescent.

If k ancestral blocks exist below τ , there are $\binom{k}{2} - 1$ allowed mergers: all pairs except the unique pair that would join the two distinguished ancestors. This is the precise conditioning used by SMC++; it is not a pure-death process that remembers only a lineage count. Descendant labels matter because they determine the emitted (a, b) category.

Small-sample exact calculation

The mini implementation represents a genealogy state as a set partition of the sampled leaves. A continuous-time Markov generator assigns unit rate to each allowed pair of blocks. A reward vector counts, in every state, how many current branches belong to each (a, b) category. Matrix-exponential occupation integrals give expected branch lengths below τ ; an absorbing-chain calculation gives the lengths above it.

```
def conditioned_sfs(tau, n_undistinguished, relative_size=1.0):
    """Expected branch lengths conditional on distinguished-pair TMRCA ``tau``.

    The returned ``(3, n_undistinguished + 1)`` array is the CSFS. Entry
    ``[a, b]`` is the expected total branch length subtending ``a`` of the two
    distinguished haplotypes and ``b`` undistinguished haplotypes.

    The calculation is exact for a constant population but exponential in
    sample size. It is intended for small pedagogical examples (normally no
    more than six undistinguished haplotypes).
    """
    if not np.isfinite(tau) or tau < 0:
        raise ValueError("tau must be finite and non-negative")
    if int(n_undistinguished) != n_undistinguished or n_undistinguished < 0:
        raise ValueError("n_undistinguished must be a non-negative integer")
    if not np.isfinite(relative_size) or relative_size <= 0:
        raise ValueError("relative_size must be finite and positive")

    (
        initial,
        below_states,
        below_Q,
        below_rewards,
        above_states,
        above_Q,
        above_rewards,
    ) = _conditioned_state_system(int(n_undistinguished))
    coal_rate = 1.0 / relative_size
    p0 = np.zeros(len(below_states))
    p0[below_states.index(initial)] = 1.0
    p_tau, below_lengths = _finite_occupation(
        p0, below_Q, below_rewards, tau, coal_rate
    )

    above_index = {state: i for i, state in enumerate(above_states)}
    p_above = np.zeros(len(above_states))
    for probability, state in zip(p_tau, below_states):
```

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```

p_above[above_index[_force_distinguished_coalescence(state)]] += probability

transient = np.array([len(state) > 1 for state in above_states])
if np.any(transient):
    Qtt = coal_rate * above_Q[np.ix_(transient, transient)]
    expected_occupation = np.linalg.solve(-Qtt.T, p_above[transient]).T
    above_lengths = expected_occupation @ above_rewards[transient]
else:
    above_lengths = np.zeros(above_rewards.shape[1])

csfs = (below_lengths + above_lengths).reshape(3, n_undistinguished + 1)
csfs[np.abs(csfs) < 1e-12] = 0.0
if np.min(csfs) < -1e-9:
    raise RuntimeError("negative branch length from conditioned coalescent")
return np.maximum(csfs, 0.0)

```

For two distinguished haplotypes and no undistinguished sample, the result has the closed form

$$\text{CSFS}_{1,0}(\tau) = 2\tau,$$

with every other finite-branch category equal to zero. The two tip branches each have length τ . This identity is one of the unit tests.

From branch lengths to emissions

Let

$$L_m = \sum_{a,b} \text{CSFS}_{a,b}^{(m)}$$

be the total expected tree length after averaging over TMRCA interval I_m . The original source converts branch lengths into emission probabilities using

$$P_m(a,b) = \text{CSFS}_{a,b}^{(m)} \frac{1 - e^{-\theta L_m}}{L_m}$$

for polymorphic outcomes. The residual probability $e^{-\theta L_m}$ is assigned to the monomorphic ancestral outcome $(0,0)$. This preserves the relative CSFS branch-length weights while using a Poisson probability for at least one mutation. It is more precise than the first-order expression $\theta \text{CSFS}_{a,b}$ and is the formula implemented in `src/conditioned_sfs.cpp`.

```

def incorporate_theta(csfs, theta):
    """Convert CSFS branch lengths to the SMC++ emission distribution.

    This mirrors ``incorporate_theta`` in the original C++ source. The
    probability of at least one mutation is ``1 - exp(-theta * L)`` where
    ``L`` is total expected tree length; polymorphic outcomes are allocated in
    proportion to their CSFS branch lengths. The remaining mass is assigned
    to the monomorphic ancestral observation ``(0, 0)``.
    """
    csfs = np.asarray(csfs, dtype=float)
    if csfs.ndim != 2 or csfs.shape[0] != 3:
        raise ValueError("csfs must have shape (3, n_undistinguished + 1)")
    if np.any(~np.isfinite(csfs)) or np.any(csfs < -1e-12):
        raise ValueError("csfs must contain finite non-negative branch lengths")

```

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```

if not np.isfinite(theta) or theta <= 0:
    raise ValueError("theta must be finite and positive")
lengths = np.maximum(csfs, 0.0).copy()
lengths[0, 0] = 0.0
lengths[2, -1] = 0.0
total_length = lengths.sum()
probabilities = np.zeros_like(lengths)
if total_length > 0:
    probabilities = lengths * (-np.expml(-theta * total_length) / total_length)
probabilities[0, 0] = 1.0 - probabilities.sum()
return probabilities

```

There is no independent-binomial emission for the two alleles of a diploid. Their allelic state is correlated through the conditioned genealogy, and the undistinguished allele count is an essential part of the emission.

Conditioning on an interval

The hidden state is an interval rather than an exact time. For a constant relative population size λ , the pairwise density is

$$f(\tau) = \lambda^{-1} e^{-\tau/\lambda}.$$

Within $I_m = [t_m, t_{m+1})$, SMC++ averages the CSFS against the conditional density

$$f_m(\tau) = \frac{\lambda^{-1} e^{-\tau/\lambda}}{e^{-t_m/\lambda} - e^{-t_{m+1}/\lambda}}.$$

The mini implementation uses Gaussian quadrature, including a Gauss-Laguerre rule for the final interval ending at infinity.

```

def interval_conditioned_sfs(
    time_breaks,
    n_undistinguished,
    relative_size=1.0,
    quadrature_order=12,
):
    """Average the CSFS within each hidden TMRCA interval."""
    breaks = np.asarray(time_breaks, dtype=float)
    pair_interval_probabilities(breaks, relative_size)
    if int(n_undistinguished) != n_undistinguished or n_undistinguished < 0:
        raise ValueError("n_undistinguished must be a non-negative integer")
    n_undistinguished = int(n_undistinguished)
    if int(quadrature_order) != quadrature_order or quadrature_order < 2:
        raise ValueError("quadrature_order must be an integer of at least two")
    leg_x, leg_w = leggauss(int(quadrature_order))
    lag_x, lag_w = laggauss(int(quadrature_order))
    result = []
    rate = 1.0 / relative_size
    for lo, hi in pairwise(breaks):
        average = np.zeros((3, n_undistinguished + 1))
        if np.isinf(hi):
            # Conditional on T >= lo, memorylessness gives T = lo + Exp(rate).
            for x, weight in zip(lag_x, lag_w):
                average += weight * conditioned_sfs(

```

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```

        lo + x / rate, n_undistinguished, relative_size
    )
else:
    midpoint = 0.5 * (lo + hi)
    halfwidth = 0.5 * (hi - lo)
    denom = np.exp(-rate * lo) - np.exp(-rate * hi)
    for x, weight in zip(leg_x, leg_w):
        tau = midpoint + halfwidth * x
        density = rate * np.exp(-rate * tau) / denom
        average += (
            halfwidth
            * weight
            * density
            * conditioned_sfs(tau, n_undistinguished, relative_size)
        )
    result.append(average)
return np.asarray(result)

```

Production scaling

Set-partition enumeration grows too quickly for the hundreds of genomes that motivated SMC++. The original program instead decomposes the CSFS above and below τ , evaluates expected first-coalescence-time integrals, and uses exact Moran-model spectral decompositions to distribute descendants among the (a, b) categories. Those formulas compute the same conditional branch-length object without enumerating labeled genealogical partitions.

3.3.3 CSFS Matrix Machinery

Compute a labeled frequency spectrum without enumerating genealogies.

What the production matrices do

The original SMC++ implementation does not evolve a probability $p_j(t)$ for a number of background lineages and does not derive an effective hazard $h(t)$. That construction would lose the descendant labels needed for CSFS emissions and would incorrectly make the distinguished pair's transition process depend on sample size.

Instead, the production calculation separates expected branch lengths below and above the conditioned pair TMRCA:

$$\text{CSFS}(\tau) = \text{CSFS}(\tau \downarrow) + \text{CSFS}(\tau \uparrow).$$

Below τ , the two distinguished ancestors are prohibited from coalescing with each other. Above τ , they have become a single ancestral block and the genealogy follows an ordinary coalescent. The paper turns expected times with k ancestors into descendant-count categories using combinatorial matrices.

Below the distinguished TMRCA

Suppose there are k ancestral lineages. In ordinary coalescent time, the total rate before τ is

$$\left[\binom{k}{2} - 1 \right] \alpha(t),$$

where $\alpha(t) = 1/\lambda(t)$ is the pairwise coalescence rate. The subtracted event is the merger of the two blocks containing the distinguished leaves. The production source integrates expected waiting times under those rates, then applies matrices that give the probability that a branch subtends zero or one distinguished leaves and a specified number of undistinguished leaves.

Only rows $a = 0$ and $a = 1$ can receive branch length below τ : a branch cannot subtend both distinguished leaves before their ancestors have merged.

Above the distinguished TMRCA

At τ , the distinguished ancestors merge. The process above that time begins with one block containing both distinguished leaves plus the remaining ancestral blocks. SMC++ computes ordinary expected branch lengths and propagates descendant counts toward the present with a forward-time Moran model dual to the coalescent. Its exact spectral decomposition is what makes large-sample evaluation practical.

The relevant generator is tridiagonal because the number of derived copies in a fixed-size Moran population changes by at most one at an event. This Moran generator is unrelated to the upper-bidiagonal pure-death matrix previously shown in this chapter.

The small implementation

For teaching, explicit partition states make both halves visible. The helper below constructs all reachable partitions, assigns one coalescent-rate unit to each allowed merger, and records a branch-count reward for each CSFS cell.

```
@cache
def _conditioned_state_system(n_undistinguished):
    total = n_undistinguished + 2
    initial = _canonical_partition([(i,) for i in range(total)])
    below_states = _reachable_partitions((initial,), True)
    below_Q = _partition_generator(below_states, True)
    below_rewards = _branch_rewards(below_states, n_undistinguished)

    forced = tuple({_force_distinguished_coalescence(s) for s in below_states})
    above_states = _reachable_partitions(forced, False)
    above_Q = _partition_generator(above_states, False)
    above_rewards = _branch_rewards(above_states, n_undistinguished)
    return (
        initial,
        below_states,
        below_Q,
        below_rewards,
        above_states,
        above_Q,
        above_rewards,
    )
```

For a row-vector distribution p_0 , generator Q , and reward matrix B , the finite-horizon expected reward is

$$\int_0^t p_0 e^{sQ/\lambda} B ds.$$

The block-matrix identity

$$\exp \left[t \begin{pmatrix} Q/\lambda & B \\ 0 & 0 \end{pmatrix} \right] = \begin{pmatrix} e^{tQ/\lambda} & \int_0^t e^{sQ/\lambda} B ds \\ 0 & I \end{pmatrix}$$

evaluates evolution and occupation in one matrix exponential.

```
def _finite_occupation(p0, Q, rewards, duration, coal_rate):
    """Evolve a row distribution and integrate its branch rewards."""
    if duration == 0:
```

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```

    return p0.copy(), np.zeros(rewards.shape[1])
n, r = rewards.shape
augmented = np.zeros((n + r, n + r))
augmented[:n, :n] = coal_rate * Q
augmented[:n, n:] = rewards
E = expm(duration * augmented)
return p0 @ E[:n, :n], p0 @ E[:n, n:]

```

After the forced merger at τ , the ordinary coalescent is an absorbing chain. If Q_T is its transient generator, the expected state occupation vector is $p_T(-Q_T)^{-1}$. Multiplication by the reward matrix gives the above- τ branch lengths.

Why the full sample is informative

The CSFS changes with τ because conditioning reshapes where branch length lies among descendant categories. A recent distinguished-pair TMRCA produces little branch length in categories containing one distinguished allele; an older TMRCA produces more. Meanwhile, the undistinguished count b reports how mutations are distributed through the rest of the genealogy. These two pieces let one emission carry both linked pairwise information and large-sample frequency-spectrum information.

Computational boundary

The set-partition version is exact only within its stated constant-population model and practical only for small samples. It is a transparent oracle for unit tests, not a scalable implementation. Original SMC++ uses analytic integrals, cached combinatorial matrices, extended precision, and automatic differentiation; those are essential engineering components rather than cosmetic optimizations.

3.3.4 The Continuous-Time HMM

Pairwise transitions, conditioned-frequency-spectrum emissions.

Hidden states and initial probabilities

Let the hidden intervals be

$$I_m = [t_m, t_{m+1}), \quad t_0 = 0, \quad t_M = \infty.$$

The state $Z_\ell = m$ means that the TMRCA of the distinguished pair at locus ℓ lies in I_m . For constant relative size λ , the marginal state probabilities are

$$\pi_m = e^{-t_m/\lambda} - e^{-t_{m+1}/\lambda}.$$

```

def pair_interval_probabilities(time_breaks, relative_size=1.0):
    """Stationary probabilities for discretized distinguished-pair TMRCA."""
    breaks = np.asarray(time_breaks, dtype=float)
    if breaks.ndim != 1 or len(breaks) < 2 or breaks[0] != 0:
        raise ValueError("time_breaks must be a one-dimensional array starting at zero
→")
    if np.any(np.diff(breaks) <= 0) or not np.isinf(breaks[-1]):
        raise ValueError("time_breaks must increase strictly and end at infinity")
    if not np.isfinite(relative_size) or relative_size <= 0:
        raise ValueError("relative_size must be finite and positive")
    survival = np.exp(-breaks / relative_size)
    survival[-1] = 0.0
    return survival[:-1] - survival[1:]

```

The transition model

SMC++ does not use the original SMC transition density presented for PSMC. It uses the continuous-time conditioned Markov chain of Hobolth and Jensen, derived from the exact two-locus coalescent. This model is slightly more accurate than the usual SMC' construction because it integrates over any number of recombinations and back-coalescences between adjacent loci.

The core process has three states. While both loci remain linked, a recombination at rate ρ creates a floating lineage. That lineage back-coalesces at rate $\alpha(t)$ or the second marginal genealogy coalesces at the same rate. The latter event is absorbing. On an interval with constant α , the generator is

$$G = \begin{pmatrix} -\rho & \rho & 0 \\ \alpha & -2\alpha & \alpha \\ 0 & 0 & 0 \end{pmatrix},$$

and the exact kernel across a distance Δ is $e^{\Delta G}$.

```
def two_locus_kernel(duration, coal_rate, rho):
    """Exact Hobolth-Jensen three-state continuous-time transition kernel.

    The generator is the one used in ``src/transition.cpp`` of SMC++:

        linked --rho--> floating --coal_rate--> linked
                   --coal_rate--> absorbed

    State ``absorbed`` means that both marginal genealogies have coalesced.
    """
    for name, value in (
        ("duration", duration),
        ("coal_rate", coal_rate),
        ("rho", rho),
    ):
        if not np.isfinite(value) or value < 0:
            raise ValueError(f"{name} must be finite and non-negative")
    generator = np.array(
        [
            [-rho, rho, 0.0],
            [coal_rate, -2.0 * coal_rate, coal_rate],
            [0.0, 0.0, 0.0],
        ]
    )
    kernel = expm(duration * generator)
    kernel[np.abs(kernel) < 1e-15] = 0.0
    return kernel
```

The original `src/transition.cpp` evaluates this exponential analytically, composes it across demographic intervals, and integrates the result into hidden TMRCA intervals. Like the source, our constant-demography specialization represents each source interval by its conditional mean coalescence time.

```
def constant_csc_transition_matrix(time_breaks, relative_size, rho):
    """Discretize SMC++'s continuous-time two-locus kernel.

    This is a direct constant-demography specialization of ``HJTransition`` in
    the original source. Like the production code, it represents each source
    interval by its conditional mean coalescence time. Unlike production
```

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```

SMC++, it does not add the tiny uniform numerical regularizer.
"""
breaks = np.asarray(time_breaks, dtype=float)
pair_interval_probabilities(breaks, relative_size)
if not np.isfinite(rho) or rho < 0:
    raise ValueError("rho must be finite and non-negative")
rate = 1.0 / relative_size
K = len(breaks) - 1
P = np.zeros((K, K))
boundary_absorption = np.zeros(K + 1)
for i, boundary in enumerate(breaks):
    boundary_absorption[i] = (
        1.0 if np.isinf(boundary) else two_locus_kernel(boundary, rate, rho)[0, 2]
    )

for j in range(K):
    if j > 0:
        P[j, :j] = np.diff(boundary_absorption[: j + 1])

    mean_t = _conditional_exponential_mean(breaks[j], breaks[j + 1], rate)
    if not np.isinf(breaks[j + 1]):
        floating = two_locus_kernel(mean_t, rate, rho)[0, 1]
        # HJTransition's Rj is the cumulative coalescence hazard across
        # the *entire source interval*, not merely from mean_t to its
        # upper boundary (src/transition.cpp, p_float construction).
        floating *= np.exp(-rate * (breaks[j + 1] - breaks[j]))
        for k in range(j + 1, K):
            survival = np.exp(-rate * (breaks[k] - breaks[j + 1]))
            coal_in_interval = (
                1.0
                if np.isinf(breaks[k + 1])
                else -np.expm1(-rate * (breaks[k + 1] - breaks[k]))
            )
            P[j, k] = floating * survival * coal_in_interval

    P[j, j] = 1.0 - P[j].sum()

if np.min(P) < -1e-10:
    raise RuntimeError("negative discretized transition probability")
P = np.maximum(P, 0.0)
P /= P.sum(axis=1, keepdims=True)
return P

```

The function has no sample-count argument. That omission is deliberate: the extra samples affect the emissions, while transitions describe the distinguished pair at two neighboring loci.

Emission tables

For state m , the emission table is obtained in two steps. First average the conditioned branch lengths over $T \in I_m$. Then apply the mutation transform to obtain probabilities for every (a, b) observation.

```
def emission_probabilities(
    time_breaks,
    n_undistinguished,
    theta,
    relative_size=1.0,
    quadrature_order=12,
):
    """Return one CSFS emission table for every hidden TMRCA interval."""
    averaged = interval_conditioned_sfs(
        time_breaks,
        n_undistinguished,
        relative_size,
        quadrature_order,
    )
    return np.asarray([incorporate_theta(csfs, theta) for csfs in averaged])
```

This table includes the monomorphic ancestral observation (0,0). The fixed-derived category (2, n) receives no finite branch length because the ancestral state above the sample MRCA is not part of the genealogy.

Forward likelihood

With transition matrix P , emission table E , and initial distribution π , the forward recursion is

$$\alpha_1(m) = \pi_m E_m(X_1),$$

$$\alpha_{\ell+1}(m) = E_m(X_{\ell+1}) \sum_j \alpha_{\ell}(j) P_{jm}.$$

Scaling after every site avoids numerical underflow; the log of each scale factor contributes to the log likelihood.

```
def forward_log_likelihood(observations, transitions, emissions, initial=None):
    """Scaled HMM likelihood for CSFS observations ``(a, b)``.

    Missing observations may be encoded by a negative value in either column;
    their emission probability is one in every hidden state.
    """
    observations = np.asarray(observations, dtype=int)
    transitions = np.asarray(transitions, dtype=float)
    emissions = np.asarray(emissions, dtype=float)
    if observations.ndim != 2 or observations.shape[1] != 2:
        raise ValueError("observations must have shape (sites, 2)")
    K = transitions.shape[0]
    if transitions.shape != (K, K) or emissions.shape[0] != K:
        raise ValueError("transition and emission state dimensions do not agree")
    if initial is None:
        initial = np.full(K, 1.0 / K)
    initial = np.asarray(initial, dtype=float)
    if (
        initial.shape != (K,)
        or np.any(initial < 0)
        or not np.isclose(initial.sum(), 1.0)
    ):
        raise ValueError("initial must be a probability vector over hidden states")
    if len(observations) == 0:
```

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```

    return 0.0

def emit(obs):
    a, b = obs
    if a < 0 or b < 0:
        return np.ones(K)
    if a >= emissions.shape[1] or b >= emissions.shape[2]:
        raise ValueError("observation is outside the CSFS emission table")
    return emissions[:, a, b]

alpha = initial * emit(observations[0])
scale = alpha.sum()
if scale <= 0:
    return -np.inf
alpha /= scale
log_likelihood = np.log(scale)
for obs in observations[1:]:
    alpha = (alpha @ transitions) * emit(obs)
    scale = alpha.sum()
    if scale <= 0:
        return -np.inf
    alpha /= scale
    log_likelihood += np.log(scale)
return float(log_likelihood)

```

Locus skipping

Whole genomes contain long runs of identical monomorphic observations. If B is the diagonal emission matrix for that symbol and $W = PB$, a run of d sites can be traversed with W^d instead of d individual forward steps. Production SMC++ obtains powers from an eigendecomposition and also accumulates the posterior transition and emission counts needed by EM. This changes computational cost from depending directly on sequence length to depending mainly on polymorphic sites and hidden-state dimension.

Thinning and model misspecification

Conditioning on the pair TMRCA does not make adjacent full-sample genealogies independent. Branch lengths in the undistinguished part remain correlated, so emitting the complete CSFS at every site violates the HMM's conditional independence assumption. This is a failure of conditional independence, not a minor numerical approximation. The original paper therefore uses **thinning**: the full CSFS is emitted only periodically, while intervening observations retain the distinguished-pair component. The paper used a heuristic proportional to sample size; current software exposes a configurable `--thinning` option and documents a different default heuristic. Neither value is a universal law.

Optimization and regularization

Production SMC++ fits a parameterized population-size curve with EM and automatic differentiation. It regularizes the curve to control oscillation and supports piecewise or spline representations. The paper's spline model and the current command-line default are not identical: current documentation says that recent releases default to a piecewise representation, whereas cubic splines were used in the paper. Any parity comparison must therefore record the software version and command-line options.

Composite likelihood

For data sets $D_1, \dots, D_r$ made with different distinguished pairs, SMC++ sums their HMM log likelihoods:

$$\ell_C(\eta) = \sum_{i=1}^r \log P(D_i \mid \eta).$$

```
def composite_log_likelihood(datasets, transitions, emissions, initial=None):
    """Sum HMM log likelihoods for multiple distinguished-pair data sets.

    If the data sets reuse the same chromosome with different distinguished
    pairs, this sum is a composite likelihood because those terms are not
    independent.
    """
    return sum(
        forward_log_likelihood(data, transitions, emissions, initial)
        for data in datasets
    )
```

When those data sets reuse samples or genomic regions, the terms are dependent; the result remains useful for estimation but is not an ordinary independent- replicate likelihood, and naive likelihood-based uncertainty calculations do not apply.

3.3.5 Population Splits

Condition the joint frequency spectrum under a clean split.

Scope of the model

The published SMC++ extension considers a two-population **clean split**. Looking backward, the populations remain separate until time t_S ; before that ancestral time they occupy a common population. The model assumes no post-split gene flow. It therefore estimates marginal size histories and a divergence time under isolation, not a general migration or admixture history.

Two distinguished-pair configurations are required:

together

Both distinguished haplotypes are sampled from the same population.

apart

One distinguished haplotype is sampled from each population.

The remaining haplotypes contribute a joint allele-count observation across the two populations.

The joint conditioned spectrum

The **joint conditioned sample-frequency spectrum** (JCSFS) is the central emission object for the clean-split extension.

For populations 1 and 2, define

$$\text{JCSFS}(\tau, t_S) \in \mathbb{R}^{(a_1+1) \times (n_1+1) \times (a_2+1) \times (n_2+1)},$$

where $a_1 + a_2 = 2$ counts distinguished haplotypes and n_1, n_2 count undistinguished haplotypes. The tensor records expected branch lengths by descendant counts in both populations, conditional on the distinguished-pair TMRCA τ and split time t_S .

This is the multi-population analogue of the one-population CSFS. Production SMC++ computes it using the same coalescent/Moran duality used by `mom`. It does not solve two independent lineage-count ODEs and then convolve expected counts; such a reduction would discard the descendant configuration that determines joint allele frequencies.

Support of an apart-pair TMRCA

For an apart pair, the two distinguished ancestors cannot coalesce while they reside in separate populations. Consequently,

$$P(T > t) = \begin{cases} 1, & 0 \leq t \leq t_S, \\ \exp\left[-\int_{t_S}^t \alpha_A(u) du\right], & t > t_S, \end{cases}$$

where α_A is the pairwise coalescence rate in the ancestral population. This support constraint is one source of information about the split time.

```
def cross_population_survival(t, split_time, ancestral_rate):
    """Survival of a TMRCA for a distinguished pair sampled apart.

    Under the clean-split model the two lineages cannot coalesce more recently
    than ``split_time``. After the split (backwards in time), their survival is
    governed by the ancestral pairwise coalescence rate. Production SMC++ also
    computes a joint conditioned SFS; this helper represents only the TMRCA
    support constraint.
    """
    if not np.isfinite(t) or t < 0 or not np.isfinite(split_time) or split_time < 0:
        raise ValueError("times must be finite and non-negative")
    if t <= split_time:
        return 1.0
    integral, _ = quad(ancestral_rate, split_time, t)
    if integral < -1e-10:
        raise ValueError("ancestral_rate must be non-negative")
    return float(np.exp(-max(integral, 0.0)))
```

The survival function alone is not the split likelihood. The JCSFS also links the distinguished pair to derived counts among all sampled haplotypes, and the HMM transition process supplies linkage information along the genome.

How the command-line workflow fits a split

The current software first fits each population marginally with `estimate`. Users then create two-population SMC files, normally in both population orders, and run `split` with the marginal model files plus the joint data. The split stage refines the marginal histories jointly with the clean-split parameter. It does not start by estimating four arbitrary histories from scratch, and it does not infer ongoing migration.

Interpretation

An estimated split time is conditional on the clean-split model, mutation rate, masks, distinguished-pair construction, and regularization settings. Population structure or gene flow after divergence can violate that model and shift the fitted time. The output should therefore be described as a clean-split estimate rather than an unconditional date of population separation.

What this mini implementation omits

The small Python module demonstrates the apart-pair support constraint but does not implement the full JCSFS tensor or split optimizer. Calling the former a complete SMC++ split implementation would be misleading. For inference, use the original software; for understanding, the one-population partition engine shows why descendant labels, rather than lineage-count expectations, are the essential state of the frequency-spectrum calculation.

3.3.6 Demo: SMC++ Building Blocks

Inspect the CSFS emissions and continuous-time transitions directly.

The module demo constructs four hidden TMRCA intervals for a constant population, calculates interval-averaged CSFS emissions for two undistinguished haplotypes, builds the conditioned two-locus transition matrix, and evaluates a short observation sequence.

```
def demo():
    """Run a small, deterministic SMC++ building-block demonstration."""
    breaks = np.array([0.0, 0.25, 0.75, 2.0, np.inf])
    n_undistinguished = 2
    theta = 0.02
    rho = 0.1
    emissions = emission_probabilities(
        breaks, n_undistinguished, theta, quadrature_order=8
    )
    transitions = constant_csc_transition_matrix(breaks, 1.0, rho)
    initial = pair_interval_probabilities(breaks)
    observations = np.array([[0, 0], [0, 0], [1, 0], [0, 1], [0, 0]])
    ll = forward_log_likelihood(observations, transitions, emissions, initial)
    print("Hidden-state probabilities:", initial)
    print("Transition row sums:", transitions.sum(axis=1))
    print("Emission row sums:", emissions.sum(axis=(1, 2)))
    print("Log likelihood:", ll)
```

Run it from the repository root with `python3 -m watchgen.mini_smcpp`. The reported initial probabilities, every transition row, and every emission table must sum to one. The example is deliberately small because the transparent partition-state CSFS calculation grows exponentially; it verifies the statistical gears rather than reproducing the production program's scale or optimizer.

3.4 Timepiece III: The Li & Stephens HMM

The Li and Stephens model is a tractable approximation to the coalescent with recombination. It represents one haplotype as an imperfect mosaic of a panel of other haplotypes and evaluates that conditional model with a hidden Markov model (HMM). The hidden copying path is a statistical device: a change of state is not proof of a particular historical recombination, and a mismatch is not proof of a mutation.

The original paper introduced a product of approximate conditionals (PAC) for a sample of haplotypes and used it to estimate recombination rates and identify hotspots [Li and Stephens, 2003]. Later imputation, phasing, and genealogy methods reuse the copying-model structure for different inferential tasks.

i Sources and scope

The equations here are checked against Li and Stephens (2003), especially Appendix A. Executable haploid results are checked against the modern [lshmm reference implementation](#) (release 0.0.8). That package is a later research implementation, not the historical Hotspotter program described by the paper. The diploid section derives a commonly used product-of-two-copying-processes extension; it is not part of the 2003 derivation.

3.4.1 Overview

Suppose $H = (h_1, \dots, h_k)$ is a panel of k haplotypes and s is a query observed at m sites. At site ℓ , the latent state $Z_\ell \in \{1, \dots, k\}$ names the panel haplotype from which the model emits the query allele. Recombination controls how persistent that label is along the chromosome, while an error-or-mutation parameter allows the emitted allele to differ from the selected template.

This language must be interpreted carefully. The panel sequences are not literal ancestors in general, and the decoded path is not a reconstructed ancestral recombination graph. A recombination event in the model redraws a template and may redraw the same one; conversely, a decoded state change is only evidence favoured by this approximation. Emission mismatches can absorb mutation, genotyping error, model misspecification, or ancestry absent from the panel.

The original PAC model

Li and Stephens did not define only a fixed-panel query HMM. For an ordering $h_1, \dots, h_n$ of a sample, their approximate likelihood is

$$\hat{\pi}(h_1, \dots, h_n) = \hat{\pi}(h_1) \prod_{k=1}^{n-1} \hat{\pi}(h_{k+1} \mid h_1, \dots, h_k).$$

Each conditional factor is evaluated by the copying HMM. The product depends on haplotype order, so the paper averages or otherwise combines results across orders. Calling a single conditional HMM “the Li–Stephens likelihood” hides this important qualification.

What the HMM returns

The forward recursion returns the conditional data likelihood and filtered state probabilities. Forward–backward returns marginal posterior copying probabilities using all sites. Viterbi returns one highest-probability state sequence. Those quantities support downstream imputation and phasing models, but local ancestry requires population labels and additional assumptions; it does not follow automatically from a vanilla copying path.

The next section states the transition and emission laws exactly. The [Haploid LS HMM Algorithms](#) section then shows why their special structure reduces each HMM pass from $O(mk^2)$ to $O(mk)$.

3.4.2 The Copying Model

At the first site the copying state is uniform,

$$P(Z_1 = j) = \frac{1}{k}.$$

For interval ℓ , let $\rho_\ell d_\ell$ denote the population-scaled recombination intensity used in Appendix A of Li and Stephens. Define the probability of a template redraw as

$$r_\ell = 1 - \exp\left(-\frac{\rho_\ell d_\ell}{k}\right).$$

The transition law is then

$$P(Z_\ell = j \mid Z_{\ell-1} = i) = (1 - r_\ell)\mathbf{1}_{i=j} + \frac{r_\ell}{k}.$$

The diagonal therefore equals $1 - r_\ell + r_\ell/k$; an event can redraw the current template. The probability of observing a *different* state after the interval is $r_\ell(k - 1)/k$, not r_ℓ itself. In code, the rho array must contain interval intensities, not probabilities that have already been transformed.

```
def compute_recombination_probs(rho, n):
    """Compute per-site recombination probabilities.

    Parameters
    -----
    rho : ndarray of shape (m,)
        Population-scaled recombination rate at each site.
    n : int
        Number of reference haplotypes.

    Returns
    -----
    r : ndarray of shape (m,)
        Per-site recombination probability (r[0] = 0 by convention).
    """
    r = 1 - np.exp(-rho / n)
    r[0] = 0.0
    return r
```

Emissions

For the biallelic model in the paper, Appendix A defines

$$\tilde{\theta} = \left(\sum_{a=1}^{n-1} \frac{1}{a} \right)^{-1},$$

where n is the total number of haplotypes in the PAC sample. When the conditional factor has k templates, its mismatch probability is

$$\mu_k = \frac{1}{2} \frac{\tilde{\theta}}{k + \tilde{\theta}},$$

and the match probability is $1 - \mu_k$. The modern lshmm package exposes a convenience estimator parameterized by reference-panel size. Its indexing convention is reproduced exactly below but should not be confused with substituting a new symbol into the paper's PAC formula.

```
def estimate_mutation_probability(n):
    """Return the mutation probability used by the reference ``lshmm`` code.

    This follows ``lshmm.core.estimate_mutation_probability`` exactly. Its
    argument is the number of reference haplotypes. Li and Stephens' Appendix
    A instead writes the harmonic sum in terms of the total PAC sample size;
    the two conventions must not be silently interchanged.

    Parameters
    -----
    n : int
        Number of reference haplotypes (must be >= 3).

    Returns
```

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```

-----
mu : float
    Estimated per-site mutation probability.
"""
if n < 3:
    raise ValueError("Need at least 3 haplotypes.")
theta_tilde = 1.0 / sum(1.0 / k for k in range(1, n - 1))
mu = 0.5 * theta_tilde / (n + theta_tilde)
return mu

```

For $a > 2$ alleles, the teaching implementation treats μ as total mismatch probability and distributes it uniformly over the $a - 1$ alternatives. Missing query alleles emit with probability one. A NONCOPY panel entry emits with probability zero, as in the reference package.

```

def emission_matrix_haploid(mu, num_sites, num_alleles):
    """Compute the emission probability matrix for the haploid case.

    Parameters
    -----
    mu : float or ndarray of shape (m,)
        Per-site mutation probability.
    num_sites : int
        Number of sites.
    num_alleles : ndarray of shape (m,)
        Number of distinct alleles at each site.

    Returns
    -----
    e : ndarray of shape (m, 2)
        Column 0 = mismatch probability, column 1 = match probability.
    """
    if isinstance(mu, float):
        mu = np.full(num_sites, mu)

    e = np.zeros((num_sites, 2))
    for i in range(num_sites):
        if num_alleles[i] == 1:
            e[i, 0] = 0.0
            e[i, 1] = 1.0
        else:
            e[i, 0] = mu[i] / (num_alleles[i] - 1)
            e[i, 1] = 1 - mu[i]

    return e

```

The structured transition matrix is a diagonal matrix plus a rank-one matrix. Consequently, if $S_{\ell-1} = \sum_i \alpha_i(\ell-1)$, the forward update is

$$\alpha_j(\ell) = e_j(s_\ell) \left[(1 - r_\ell) \alpha_j(\ell-1) + \frac{r_\ell}{k} S_{\ell-1} \right].$$

Computing $S_{\ell-1}$ once makes the update $O(k)$, rather than forming an $O(k^2)$ transition matrix.

3.4.3 Haploid LS HMM Algorithms

For reference panel H with m rows and k columns, the forward initialization is

$$\alpha_j(1) = k^{-1}e_j(s_1).$$

The recursion uses the rank-one-plus-diagonal identity derived in *The Copying Model*. To avoid underflow, the implementation divides each row by $c_\ell = \sum_j \tilde{\alpha}_j(\ell)$. With normalized previous row, $S_{\ell-1} = 1$, and

$$\log_{10} P(s \mid H) = \sum_{\ell=1}^m \log_{10} c_\ell.$$

```
def forwards_ls_hap(n, m, H, s, emission_matrix, r, norm=True):
    """Forward algorithm for the haploid Li-Stephens model.

    Parameters
    -----
    n : int
        Number of reference haplotypes.
    m : int
        Number of sites.
    H : ndarray of shape (m, n)
        Reference panel.
    s : ndarray of shape (1, m)
        Query haplotype (wrapped in 2D array for API compatibility).
    emission_matrix : ndarray of shape (m, 2)
        Column 0 = mismatch prob, column 1 = match prob.
    r : ndarray of shape (m,)
        Per-site recombination probability.
    norm : bool
        Whether to normalize (scale) the forward probabilities.

    Returns
    -----
    F : ndarray of shape (m, n)
        Forward probabilities.
    c : ndarray of shape (m,)
        Scaling factors.
    ll : float
        Log-likelihood (base 10).
    """
    F = np.zeros((m, n))
    r_n = r / n

    if norm:
        c = np.zeros(m)
        for i in range(n):
            if H[0, i] == s[0, 0]:
                F[0, i] = (1 / n) * emission_matrix[0, 1]
            else:
                F[0, i] = (1 / n) * emission_matrix[0, 0]
            c[0] += F[0, i]
        for i in range(n):
```

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```

F[0, i] /= c[0]

for l in range(1, m):
    for i in range(n):
        F[l, i] = F[l - 1, i] * (1 - r[l]) + r_n[l]
        if H[l, i] == s[0, l]:
            F[l, i] *= emission_matrix[l, 1]
        else:
            F[l, i] *= emission_matrix[l, 0]
        c[l] += F[l, i]
    for i in range(n):
        F[l, i] /= c[l]

ll = np.sum(np.log10(c))
else:
    c = np.ones(m)
    for i in range(n):
        if H[0, i] == s[0, 0]:
            F[0, i] = (1 / n) * emission_matrix[0, 1]
        else:
            F[0, i] = (1 / n) * emission_matrix[0, 0]

    for l in range(1, m):
        S = np.sum(F[l - 1, :])
        for i in range(n):
            F[l, i] = F[l - 1, i] * (1 - r[l]) + S * r_n[l]
            if H[l, i] == s[0, l]:
                F[l, i] *= emission_matrix[l, 1]
            else:
                F[l, i] *= emission_matrix[l, 0]

    ll = np.log10(np.sum(F[m - 1, :]))

return F, c, ll

```

The backward variable is initialized to one at the final site. The same transition structure gives

$$\beta_j(\ell) = \frac{1}{c_{\ell+1}} \left[(1 - r_{\ell+1})q_j + \frac{r_{\ell+1}}{k} \sum_i q_i \right],$$

where $q_i = e_i(s_{\ell+1})\beta_i(\ell + 1)$. Multiplying scaled forward and backward rows and renormalizing yields the marginal posterior $P(Z_\ell = j \mid s, H)$.

```

def backwards_ls_hap(n, m, H, s, emission_matrix, c, r):
    """Backward algorithm for the haploid Li-Stephens model.

    Parameters
    -----
    n, m, H, s, emission_matrix, r : same as forwards_ls_hap.
    c : ndarray of shape (m,)
        Scaling factors from the forward pass.

```

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```

Returns
-----
B : ndarray of shape (m, n)
    Scaled backward probabilities.
"""
B = np.zeros((m, n))
for i in range(n):
    B[m - 1, i] = 1.0

r_n = r / n

for l in range(m - 2, -1, -1):
    tmp_B = np.zeros(n)
    tmp_B_sum = 0.0
    for i in range(n):
        if H[l + 1, i] == s[0, l + 1]:
            emission_prob = emission_matrix[l + 1, 1]
        else:
            emission_prob = emission_matrix[l + 1, 0]
        tmp_B[i] = emission_prob * B[l + 1, i]
        tmp_B_sum += tmp_B[i]

    for i in range(n):
        B[l, i] = r_n[l + 1] * tmp_B_sum
        B[l, i] += (1 - r[l + 1]) * tmp_B[i]
        B[l, i] /= c[l + 1]

return B

```

```

def posterior_decoding(F, B):
    """Compute posterior decoding from forward-backward probabilities.

    Parameters
    -----
    F : ndarray of shape (m, n)
        Scaled forward probabilities.
    B : ndarray of shape (m, n)
        Scaled backward probabilities.

    Returns
    -----
    gamma : ndarray of shape (m, n)
        Posterior state probabilities.
    path : ndarray of shape (m,)
        Most likely state at each site (posterior decoding).
    """
    gamma = F * B
    gamma /= gamma.sum(axis=1, keepdims=True)
    path = np.argmax(gamma, axis=1)
    return gamma, path

```

Viterbi replaces sums by maxima. For target state j , the best predecessor is either j itself or the globally best previous state.

That observation again gives $O(k)$ work per site. Posterior decoding and Viterbi answer different questions: independently maximizing each marginal posterior need not produce the most probable complete path.

```
def forwards_viterbi_hap(n, m, H, s, emission_matrix, r):
    """Viterbi algorithm for the haploid Li-Stephens model.

    Uses the Li-Stephens structure for  $O(n)$  per site.
    Includes rescaling for numerical stability.

    Parameters
    -----
    n, m, H, s, emission_matrix, r : same as forwards_ls_hap.

    Returns
    -----
    V : ndarray of shape (n,)
        Viterbi probabilities at the last site.
    P : ndarray of shape (m, n), dtype int
        Pointer (traceback) array.
    ll : float
        Log-likelihood of the best path (base 10).
    """
    V = np.zeros(n)
    P = np.zeros((m, n), dtype=np.int64)
    r_n = r / n
    c = np.ones(m)

    for i in range(n):
        if H[0, i] == s[0, 0]:
            V[i] = (1 / n) * emission_matrix[0, 1]
        else:
            V[i] = (1 / n) * emission_matrix[0, 0]

    for j in range(1, m):
        argmax = np.argmax(V)
        c[j] = V[argmax]
        V /= c[j]

    for i in range(n):
        stay = V[i] * (1 - r[j] + r_n[j])
        switch = r_n[j]

        V[i] = stay
        P[j, i] = i
        if V[i] < switch:
            V[i] = switch
            P[j, i] = argmax

    if H[j, i] == s[0, j]:
        V[i] *= emission_matrix[j, 1]
    else:
        V[i] *= emission_matrix[j, 0]
```

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```

ll = np.sum(np.log10(c)) + np.log10(np.max(V))
return V, P, ll

```

The test suite checks scaled and unscaled likelihood equality, pins forward and backward arrays generated by lshmm 0.0.8, and verifies Viterbi path scores. For partial panels, the number of copiable templates can vary by site; a full implementation must use that site-specific denominator rather than a fixed panel width.

3.4.4 A Diploid Extension

This section describes a later extension, not a derivation from Li and Stephens (2003). For an unphased biallelic query genotype, use an ordered pair of copying states $(Z_\ell^{(1)}, Z_\ell^{(2)})$. There are k^2 states. The two chromosomes transition independently, so

$$P((i_1, i_2) \rightarrow (j_1, j_2)) = A_{i_1 j_1} A_{i_2 j_2}.$$

The rows sum to one because each factor A is stochastic. Although the state pair is ordered computationally, unphased genotype emissions are symmetric under swapping the two labels; the labels are therefore not identifiable from these observations alone.

For a copied genotype and an observed genotype, the emission probabilities are obtained from two independent allele-error events. For example, matching homozygotes have probability $(1 - \mu)^2$, opposite homozygotes have probability μ^2 , and a copied homozygote emitting a heterozygote has probability $2\mu(1 - \mu)$. A copied heterozygote emitting either homozygote has probability $\mu(1 - \mu)$.

```

def emission_matrix_diploid(mu, num_sites, num_alleles):
    """Compute emission probability matrix for diploid genotypes.

    Returns matrix of shape (m, 8) indexed by genotype comparison code.

    Indexing scheme (bit-packed):
        4 = EQUAL_BOTH_HOM      (ref hom, query hom, same genotype)
        0 = UNEQUAL_BOTH_HOM   (ref hom, query hom, different genotype)
        7 = BOTH_HET           (ref het, query het)
        1 = REF_HOM_OBS_HET    (ref hom, query het)
        2 = REF_HET_OBS_HOM    (ref het, query hom)
        3 = MISSING_INDEX      (query is MISSING)
    """
    EQUAL_BOTH_HOM = 4
    UNEQUAL_BOTH_HOM = 0
    BOTH_HET = 7
    REF_HOM_OBS_HET = 1
    REF_HET_OBS_HOM = 2
    MISSING_INDEX = 3

    if isinstance(mu, float):
        mu = np.full(num_sites, mu)

    e = np.full((num_sites, 8), -np.inf)

    for i in range(num_sites):
        if num_alleles[i] == 1:
            p_mut = 0.0
            p_no_mut = 1.0

```

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```

else:
    p_mut = mu[i] / (num_alleles[i] - 1)
    p_no_mut = 1 - mu[i]

    e[i, EQUAL_BOTH_HOM] = p_no_mut ** 2
    e[i, UNEQUAL_BOTH_HOM] = p_mut ** 2
    e[i, BOTH_HET] = p_no_mut**2 + p_mut**2
    e[i, REF_HOM_OBS_HET] = 2 * p_mut * p_no_mut
    e[i, REF_HET_OBS_HOM] = p_mut * p_no_mut
    e[i, MISSING_INDEX] = 1.0

return e

```

A naive forward update costs $O(k^4)$ per site. Separating the cases in which neither, one, or both copying states redraw reduces it to $O(k^2)$. If F is the previous $k \times k$ forward matrix, the unscaled pre-emission value for target (a, b) is

$$(1-r)^2 F_{ab} + (1-r) \frac{r}{k} \left(\sum_j F_{aj} + \sum_i F_{ib} \right) + \left(\frac{r}{k} \right)^2 \sum_{ij} F_{ij}.$$

```

def forward_diploid(n, m, G, s, emission_matrix, r, norm=True):
    """Forward algorithm for the diploid Li-Stephens model.

    Parameters
    -----
    n : int
        Number of reference haplotypes.
    m : int
        Number of sites.
    G : ndarray of shape (m, n, n)
        Reference genotype matrix. G[l, j1, j2] = allele dosage
        when copying from (j1, j2).
    s : ndarray of shape (1, m)
        Query genotype (allele dosages: 0, 1, or 2).
    emission_matrix : ndarray of shape (m, 8)
        Diploid emission probabilities.
    r : ndarray of shape (m,)
        Per-site recombination probability.
    norm : bool
        Whether to normalize.

    Returns
    -----
    F : ndarray of shape (m, n, n)
        Forward probabilities (n x n matrix at each site).
    c : ndarray of shape (m,)
        Scaling factors.
    ll : float
        Log-likelihood (base 10).
    """
    F = np.zeros((m, n, n))
    c = np.ones(m)

```

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```

r_n = r / n

for j1 in range(n):
    for j2 in range(n):
        F[0, j1, j2] = 1 / (n**2)
        ref_gt = G[0, j1, j2]
        idx = genotype_comparison_index(ref_gt, s[0, 0])
        F[0, j1, j2] *= emission_matrix[0, idx]

if norm:
    c[0] = np.sum(F[0, :, :])
    F[0, :, :] /= c[0]

    for l in range(1, m):
        previous = F[l - 1]
        one_change = (1 - r[l]) * r_n[l]
        F[l] = (
            (1 - r[l]) ** 2 * previous
            + r_n[l] ** 2
            + one_change
            * (previous.sum(axis=1)[:, None] + previous.sum(axis=0)[None, :])
        )

        for j1 in range(n):
            for j2 in range(n):
                ref_gt = G[l, j1, j2]
                idx = genotype_comparison_index(ref_gt, s[0, l])
                F[l, j1, j2] *= emission_matrix[l, idx]

        c[l] = np.sum(F[l, :, :])
        F[l, :, :] /= c[l]

    ll = np.sum(np.log10(c))

else:
    for l in range(1, m):
        previous = F[l - 1]
        one_change = (1 - r[l]) * r_n[l]
        F[l] = (
            (1 - r[l]) ** 2 * previous
            + r_n[l] ** 2 * previous.sum()
            + one_change
            * (previous.sum(axis=1)[:, None] + previous.sum(axis=0)[None, :])
        )

        for j1 in range(n):
            for j2 in range(n):
                ref_gt = G[l, j1, j2]
                idx = genotype_comparison_index(ref_gt, s[0, l])
                F[l, j1, j2] *= emission_matrix[l, idx]

    ll = np.log10(np.sum(F[m-1, :, :]))

```

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```
return F, c, ll
```

The optimized recursion is tested against a direct $O(k^4)$ enumeration for both scaled and unscaled calculations. This is important because a vector broadcasting error can preserve plausible row sums while assigning probability to the wrong destination states.

3.4.5 Demo: Running the Li & Stephens HMM on Simulated Data

Running mini_lshmm on simulated genomic data.

Demo: Li & Stephens HMM on msprime-simulated Haplotypes (500 kb)

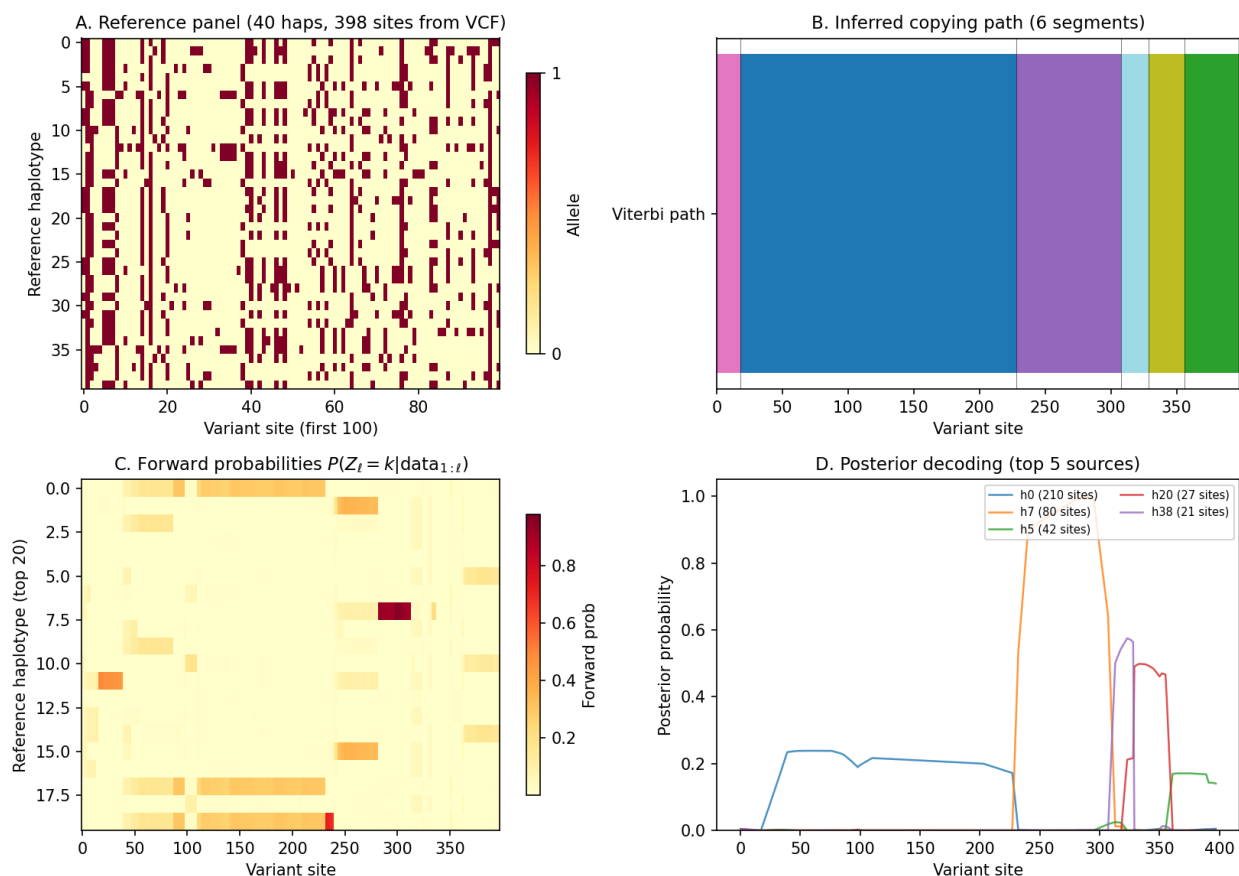


Fig. 3: **Li & Stephens HMM demo on msprime-simulated data.** Panel A shows the reference haplotype matrix, panel B one Viterbi copying path, panel C scaled forward (filtered) probabilities, and panel D smoothed marginal posterior probabilities for five frequently selected templates. A decoded segment is a model-based copying assignment, not an observed recombination tract.

This figure was generated by running the `mini_lshmm` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_lshmm.py`.

The simulated query is an ordinary draw from the same coalescent simulation, not a query generated from a known copying path. Accordingly, this figure is an algorithm demonstration and does not report path-recovery accuracy.

3.5 Timepiece IV: msprime

Simulating ancestral histories under the coalescent with recombination

3.5.1 The Mechanism at a Glance

msprime is a **coalescent simulator**: given a sample size, genome length, mutation rate, and recombination rate, it produces random ancestral histories (genealogies) that are consistent with the evolutionary process. It works **backwards in time**, starting from a set of sampled genomes in the present and tracing their ancestry back until all lineages have found common ancestors.

While SINGER (Timepiece VII) *infers* genealogies from observed data, msprime *simulates* genealogies from a specified model. They are complementary tools, like a watch and a watch-testing machine: msprime creates model-based ground truth that tools like SINGER try to recover. By default, msprime records the succinct marginal trees, not every event in the full ancestral recombination graph (ARG); extra event nodes can be retained with `additional_nodes` or the legacy `record_full_arg=True` option. Understanding how the simulator works gives you deep insight into the coalescent process itself and a reliable way to test every other Timepiece in this book.

If PSMC is the simplest watch (two hands, one gear train), msprime is the **master clockmaker's bench** – the machine that produces the movements for every other watch. Its output (tree sequences) is what inference methods consume, and its internals (the coalescent with recombination, implemented with clever data structures) reveal how nature's own clockwork operates.

Primary Reference

[Kelleher *et al.*, 2016, Baumdicker *et al.*, 2022]

The four gears of msprime:

1. **The Coalescent Process** (the escapement) – The mathematical engine: how lineages find common ancestors backwards in time, and how recombination fragments the genome into independently-evolving segments. This is the coalescent from *The Workbench* in action.
2. **Segments & the Fenwick Tree** (the mainspring) – The data structures that make it fast: linked-list segments track which parts of the genome each lineage carries, and Fenwick trees enable $O(\log n)$ event scheduling. Clever engineering turns an elegant algorithm into a practical tool.
3. **Hudson's Algorithm** (the gear train) – The main simulation loop: an event-driven machine that races recombination, coalescence, and migration against each other, always executing whichever happens first. This is where the coalescent process comes alive as executable code.
4. **Demographics & Mutations** (the case and dial) – The outer layers: population size changes, migration, growth, and the final step of painting mutations onto the genealogy. These layers transform a simple simulator into a rich model of population history.

These gears mesh together into a complete simulator:

```
Parameters (n, L, mu, rho, demography)
      |
      v
Initialize n lineages, each carrying [0, L)
      |
      v
+---> Compute event rates
      |      |
      |      v
```

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```

|   Sample next event time (exponential)
|   |
|   v
|   Execute event:
|       Recombination? --> split a lineage
|       Coalescence?   --> merge two lineages
|       Migration?     --> move a lineage
|       Demographic?   --> change population params
|       |
|       v
|   Update data structures
|   |
+-----+
|   (repeat until all positions coalesced)
|   |
|   v
|   Output: tree sequence (tskit format)
|   |
|   v
|   Add mutations (Poisson process on branches)
|   |
|   v
|   Output: tree sequence with mutations

```

i Prerequisites for this Timepiece

- *Coalescent Theory* – exponential waiting times, coalescence rates, the Poisson mutation model
- *Ancestral Recombination Graphs* – the data structure msprime produces (marginal trees, tree sequences)

3.5.2 Chapters

Overview of msprime

Before assembling the watch, lay out every part and understand what it does.

Welcome to the master clockmaker's bench. Of all the timepieces we will examine in this book, msprime is arguably the most finely engineered: a coalescent simulator that can generate succinct genealogical histories for very large samples. In the chapters ahead, we will disassemble this mechanism gear by gear, understand every spring and escapement, and reassemble it from scratch.

But first, we need the blueprint.

msprime: Coalescent Simulation with Recombination

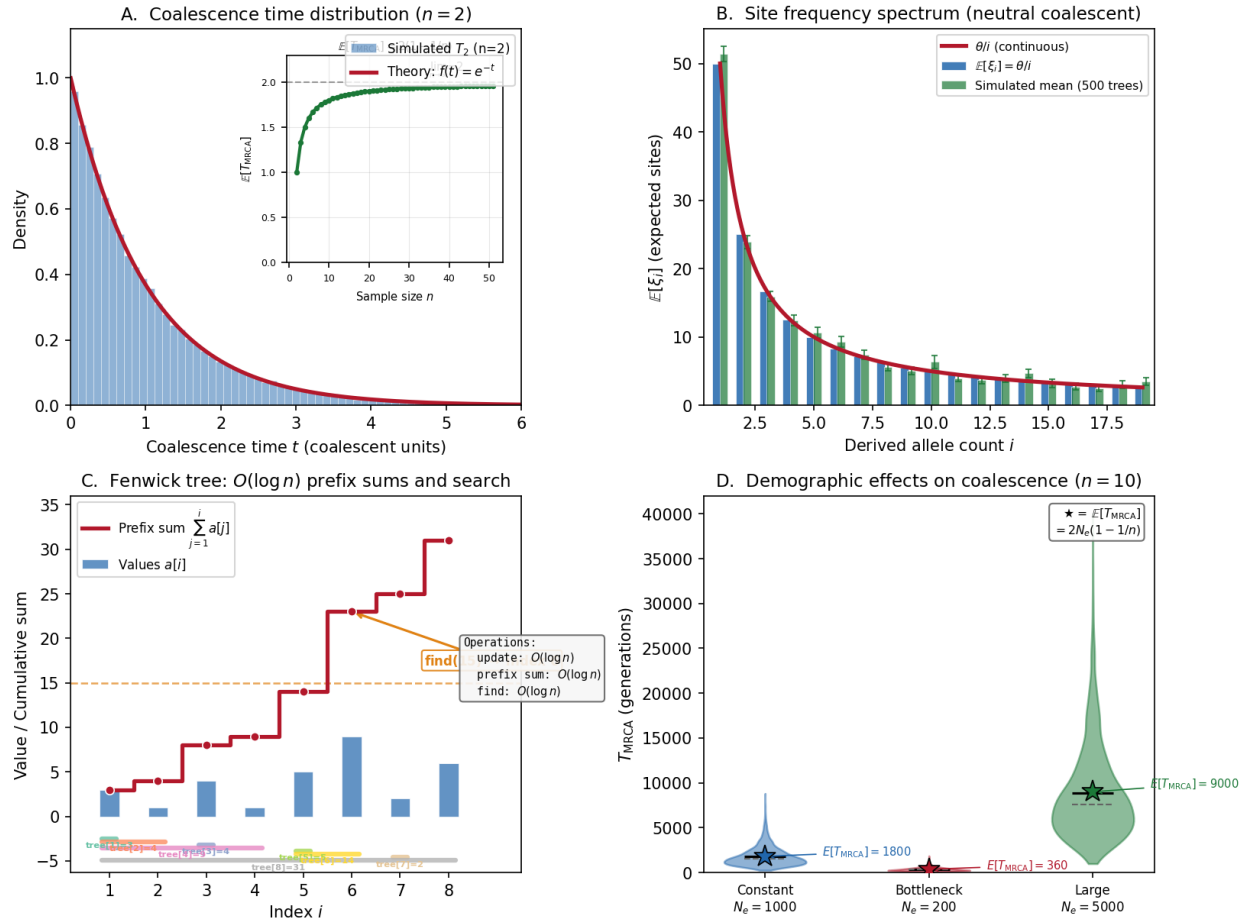


Fig. 4: **msprime at a glance**. Panel A: Coalescence times – histogram of simulated T_{MRCA} for $n = 2$ vs the theoretical $\text{Exp}(1)$ density. Panel B: Site frequency spectrum from many coalescent replicates vs the classic neutral expectation θ/i . Panel C: Fenwick tree – the binary-indexed data structure enabling $O(\log n)$ cumulative-sum queries and weighted random search. Panel D: Demographic effects on coalescence times under constant, bottleneck, and growth scenarios.

Note

Prerequisites. This chapter assumes you have read the earlier chapters on *Coalescent Theory* (which introduces the mathematical foundation of the coalescent) and *Ancestral Recombination Graphs* (which explains the Ancestral Recombination Graph). If those concepts are fresh in your mind, everything here will click into place. If not, a quick review of those chapters will pay dividends.

What Does msprime Do?

msprime takes a **model specification** and produces a **random genealogy** consistent with that model. The model specifies:

- How many genomes to sample (n)
- How long the genome is (L base pairs)
- How often mutations arise (μ per bp per generation)
- How often recombination occurs (r per bp per generation)
- How population size has changed through time (demography)

The output is a **tree sequence**: a compact representation of the genealogical relationships among the sampled genomes at every position along the chromosome.

Input: $(n, L, \mu, r, \text{demography})$

Output: $\mathcal{T} = \{\Psi_1, \Psi_2, \dots, \Psi_K\}$ (sequence of marginal trees)

Each marginal tree Ψ_k covers a contiguous genomic interval and describes the ancestral relationships of all n samples within that interval. Under the continuous-genome Hudson model, a visible recombination commonly relates neighboring trees by a **Subtree Prune and Regraft (SPR)** operation. The correspondence is not one-to-one in the default output: recombinations can be invisible in the marginal trees, and tree-sequence simplification omits non-coalescing common-ancestor and recombination-event nodes.

Think of the tree sequence as a long filmstrip: each frame is a genealogical tree, and as you advance along the chromosome, the tree changes smoothly – one branch detaches and reattaches elsewhere. The entire filmstrip is the output of the master clockmaker's bench.

From a tree sequence, you can compute:

- **Pairwise genetic diversity** – how different are two genomes?
- **Allele frequency spectra** – the distribution of variant frequencies
- **Linkage disequilibrium** – correlations between nearby sites
- **Population divergence** – how long ago populations split
- **Selection signatures** – deviations from neutral expectations

Why Simulate Backwards?

A natural first instinct is to simulate evolution **forwards**: start with an ancestral population and evolve it generation by generation, tracking mutations, recombinations, and drift.

This works, but it's enormously wasteful. In a population of $N = 10,000$ diploid individuals, a forward simulation must track 20,000 genomes every generation – but we only care about the $n = 100$ we sampled. The other 19,900 are irrelevant noise.

The coalescent flips the arrow of time. Instead of evolving a whole population forward, we start with the n sampled genomes and trace their ancestry **backwards**. We only track lineages carrying material ancestral to our sample. Without

recombination their number decreases from n to one; with recombination it can temporarily exceed n , because a lineage can split into two ancestral lineages.

Forward (wasteful):	Backward (efficient):
Gen 0: o o o o o o o o	Present: * * * * (4 samples)
\ / \ /	
Gen 1: o o o o o o o o	Past: * * (3 lineages)
\ / \ /	\ /
Gen 2: o o o o o o o o	Deeper: * * (2 lineages)
...	\ /
Gen T: o o o o o o o o	MRCA: * (1 lineage)
Tracks 8 genomes x T gens	Tracks only ancestral material

The efficiency gain is dramatic: the coalescent simulation runs in time proportional to n (the sample size), essentially independent of N (the population size). Population size enters only through a time scaling factor.

Probability Aside – Why “essentially independent of N ”?

In the continuous-time coalescent (derived in *Coalescent Theory*), time is measured in units of N generations. The entire dynamics – waiting times, event probabilities, tree shapes – depend only on the *sample size* n , not on the population size N . The population size merely rescales the time axis. This is why you can simulate the ancestry of 1,000 genomes from a population of 10 billion just as fast as from a population of 10,000: the coalescent tree has the same shape in both cases; only the branch lengths (in generations) differ.

Now that we understand *why* backward simulation is efficient, let us see how msprime organizes the simulation into two clean phases.

The Two Phases of Simulation

msprime separates the simulation into two independent phases:

Phase 1: Ancestry simulation (this Timepiece’s main focus)

Generate the genealogical history – the tree sequence of marginal trees, with coalescence times and recombination break-points – but **no mutations**. This is a purely topological and temporal structure.

Phase 2: Mutation simulation

Given the tree sequence from Phase 1, scatter mutations along branches according to a Poisson process with rate μ . This is a simple post-processing step.

```
# The two phases in msprime's API
import msprime

# Phase 1: generate the genealogy (no mutations yet)
# sim_ancestry builds the tree sequence -- the "skeleton" of the watch.
ts = msprime.sim_ancestry(
    samples=100,           # number of sampled individuals
    ploidy=1,              # one haploid genome per individual
    sequence_length=1_000_000, # genome length in base pairs
    recombination_rate=1e-8, # crossover probability per bp per generation
    population_size=10_000, # effective population size (constant here)
)
```

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```
# ts has trees, edges, nodes -- but no mutations

# Phase 2: add mutations to the genealogy
# sim_mutations "paints" heritable changes onto the branches.
mts = msprime.sim_mutations(ts, rate=1.5e-8)
# mts now has mutations placed on branches
```

Why separate them? Because the genealogy is independent of the mutation process. The same tree sequence can be used with different mutation rates, mutation models (infinite sites, finite sites, nucleotide models), or even no mutations at all. This separation is both computationally efficient and conceptually clean.

i Probability Aside – Independence of genealogy and mutation

The separation of Phases 1 and 2 rests on a deep probabilistic fact: under the neutral coalescent, the genealogical tree is independent of the mutation process. Mutations are a Poisson decoration on the tree's branches; they do not influence which lineages coalesce or when. This means you can generate one genealogy and overlay it with many different mutation realizations – a technique widely used in Approximate Bayesian Computation (ABC) workflows.

In the language of the watchmaker: Phase 1 builds the movement (the mechanical heart of the watch), and Phase 2 paints the dial (the visible face). The movement tells time; the dial makes it readable.

With the two-phase architecture clear, let us establish the vocabulary we will use throughout this Timepiece.

Terminology

Before diving into the gears, let's nail down the terminology precisely.

Term	Definition
Lineage	A single haploid genome being traced backwards in time. At the start, there are n lineages (one per sample).
Segment	A contiguous stretch of genome carried by a lineage. A lineage may carry multiple non-contiguous segments (due to recombination).
Coalescence	The event where two lineages share a common ancestor. Their ancestral material merges into one lineage going further back.
Recombination	The event where a lineage splits into two: one inherits the left part of the genome, the other inherits the right part.
Marginal tree	The genealogical tree at a single genomic position.
Tree sequence	The complete set of marginal trees along the genome, stored as a sequence of edges in tskit format.
MRCA	Most Recent Common Ancestor: the point where all sample lineages have coalesced at a given position.
Effective population size N_e	The idealized population size that produces the same rate of genetic drift as the real population.
Coalescent units	Time measured in units of N_e generations for a haploid Wright–Fisher population or $2N_e$ generations for a diploid population. In these units, the rate of coalescence between two lineages is 1. msprime uses the diploid scale by default because its default ploidy is two.
Rate map	A function mapping genomic position to local recombination or mutation rate, allowing for hotspots and coldspots.

i Closing a confusion gap – Segments vs. Lineages

These two terms are often conflated, but they are distinct. A **lineage** is a conceptual entity: a genome being traced backward. A **segment** is a concrete data object: a contiguous interval `[left, right)` of ancestral material. A single lineage may own *many* segments, linked together in a chain. When we say “a lineage splits at recombination,” we mean its segment chain is divided into two chains, each becoming a new lineage. When we say “two lineages coalesce,” we mean their segment chains are merged, position by position, into one. Understanding this distinction is essential for *Segments & the Fenwick Tree*, where we build the segment data structure in detail.

The Coalescent in One Paragraph

Take n lineages at the present. Going backwards in time, any pair can coalesce (share a common ancestor). With k lineages, the waiting time to the next coalescence is $\text{Exponential}(\binom{k}{2})$ in coalescent units. When a coalescence happens, a random pair merges into one lineage: $k \rightarrow k - 1$. Repeat until $k = 1$.

With recombination, lineages can also **split**: a single lineage becomes two lineages, each carrying part of the genome. This means the number of lineages can increase as well as decrease, and different genomic positions reach their MRCA at different times.

The interplay between coalescence (merging) and recombination (splitting) creates the Ancestral Recombination Graph – and simulating this interplay efficiently is what msprime does.

If you have read the *Ancestral Recombination Graphs* chapter, you will recognize the ARG as the structure that the tree sequence encodes. The next chapter, *The Coalescent Process*, derives the mathematics of this interplay rigorously.

Parameters

msprime’s ancestry simulation takes the following parameters:

Parameter	Symbol	Meaning
Sample size	n	Number of haploid genomes to sample
Sequence length	L	Total genome length in base pairs
Recombination rate	r	Crossover probability per bp per generation
Population size	N_e	Effective population size (can vary through time)
Mutation rate	μ	Mutations per bp per generation (used in Phase 2)
Gene conversion rate	g	Gene conversion initiation rate per bp per generation
Gene conversion length	ℓ	Mean tract length for gene conversion events
Migration matrix	M	Rates of migration between populations

The Computational Challenge

Why can’t we just “run the coalescent” naively? Let’s count the operations.

A genome of length $L = 10^8$ bp with recombination rate $r = 10^{-8}$ per bp per generation has a total recombination rate of $\rho = 4N_e r L$. For $N_e = 10^4$, that’s $\rho = 4 \times 10^4 \times 10^{-8} \times 10^8 = 4 \times 10^4$ recombination events expected in the ancestry of even 2 lineages.

For $n = 1000$ samples, the simulation must handle millions of recombination and coalescence events. Each event modifies the set of lineages and their genomic segments. The key question is: how do we track all this efficiently?

The answer involves three ideas:

1. **Segments as linked lists** – each lineage's ancestry is stored as a chain of segments. Think of this as the linked-list track that follows each lineage's ancestral material along the genome. Segments can be split and merged in $O(1)$ time by rewiring pointers rather than copying arrays.
2. **Fenwick trees for rate computation** – the total recombination rate depends on the total genomic material across all lineages, which a Fenwick tree maintains in $O(\log n)$ time. The Fenwick tree is a clever indexing mechanism for fast event scheduling: it lets the simulator ask “which segment should the next recombination hit?” and get an answer in logarithmic time.
3. **Event-driven simulation** – instead of stepping through time uniformly, we jump directly to the next event (whichever of recombination, coalescence, or migration happens first).

i Closing a confusion gap – What is event-driven simulation?

Many people imagine simulations as ticking through time in fixed steps (e.g., one generation at a time). Event-driven simulation is different: it asks, “When does the next thing happen?” and jumps directly to that moment. Between events, nothing changes, so there is no need to simulate the intervening silence. In msprime, each possible event (coalescence, recombination, migration) proposes a random waiting time drawn from an exponential distribution. The simulator picks the smallest waiting time, advances the clock to that moment, and executes only that event. This approach is sometimes called the **Gillespie algorithm** or the **Stochastic Simulation Algorithm (SSA)**, and it is the standard engine for continuous-time Markov chain simulation. We will build it explicitly in *Hudson's Algorithm*.

These three ideas, combined, give msprime its remarkable efficiency: it can simulate the ancestry of millions of samples across whole chromosomes in seconds.

Ready to Build

You now have the high-level blueprint – the exploded diagram of the master clockmaker's bench, with every part labeled. In the following chapters, we will build each gear from scratch:

1. *The Coalescent Process* – The mathematical engine: how lineages race to find common ancestors, and the exponential distributions that govern the timing.
2. *Segments & the Fenwick Tree* – The data structures: the linked-list track that follows each lineage's ancestral material, and the Fenwick tree – a clever indexing mechanism for fast event scheduling.
3. *Hudson's Algorithm* – The main simulation loop – the ticking of the clock, where the exponential race drives the event-driven core.
4. *Demographics & Population* – Population structure and size changes: how the case and dial of the watch shape the genealogy.
5. *Mutations* – Painting mutations on the tree: the final visible layer of genetic variation.

Each chapter derives the math, explains the intuition, implements the code, and verifies it works.

Let's start with the most fundamental gear: the coalescent process itself.

The Coalescent Process

The escapement of the mechanism: how lineages find common ancestors.

The coalescent is the mathematical foundation of msprime. Before we can simulate anything, we need to understand the stochastic process that governs how lineages merge backwards in time. We build this from the very bottom: starting with two lineages, then n , then adding recombination.

If you have worked through the *Coalescent Theory* chapter, much of the mathematics here will be familiar – but the emphasis is different. There, we developed the theory for its own sake. Here, we are building toward a *simulation*

algorithm: every formula we derive will become a line of code in *Hudson's Algorithm*, the main simulation loop – the ticking of the clock.

Think of this chapter as calibrating the escapement: we need to know exactly how fast the gears should turn before we can assemble the movement.

Step 1: Two Lineages, No Recombination

Start with the simplest possible case: two haploid genomes sampled from a population of constant size N . Going one generation back, what is the probability that these two genomes share a parent?

In a haploid Wright-Fisher population of size N , each individual in generation t chose its parent uniformly at random from the N individuals in generation $t - 1$. So the probability that our two samples chose the **same** parent is:

$$P(\text{same parent}) = \frac{1}{N}$$

and the probability they chose **different** parents is:

$$P(\text{different parents}) = 1 - \frac{1}{N}$$

If they didn't coalesce in the first generation, the problem resets: we now have two lineages one generation further back, in the same population. The process is memoryless.

i Probability Aside – The memoryless property

A random variable T is **memoryless** if $P(T > s + t \mid T > s) = P(T > t)$ for all $s, t \geq 0$. Knowing that no coalescence has happened so far gives you no information about how much longer you will wait. Among continuous distributions, only the exponential has this property. Among discrete distributions, only the geometric has it. This is why the coalescent waiting time (geometric in discrete time, exponential in continuous time) resets perfectly after each failed generation.

Waiting time distribution. Let T_2 be the number of generations until the two lineages coalesce. The probability that they first coalesce in generation g is:

$$P(T_2 = g) = \left(1 - \frac{1}{N}\right)^{g-1} \cdot \frac{1}{N}$$

This is a **geometric distribution** with success probability $1/N$.

The continuous-time approximation. For large N , we can approximate this discrete geometric distribution with a continuous exponential. Measure time in units of N generations: let $t = g/N$. Then:

$$P(T_2 > t) = \left(1 - \frac{1}{N}\right)^{Nt} \approx e^{-t}$$

as $N \rightarrow \infty$. So in units of N generations, the coalescence time of two lineages is approximately Exponential(1).

i Calculus Aside – The exponential limit

The key identity is $\lim_{N \rightarrow \infty} (1 - 1/N)^N = e^{-1}$. More generally, $\lim_{N \rightarrow \infty} (1 - c/N)^{Nt} = e^{-ct}$. This follows from taking the logarithm: $Nt \cdot \ln(1 - c/N) \approx Nt \cdot (-c/N) = -ct$ as $N \rightarrow \infty$, using the first-order Taylor expansion $\ln(1 - x) \approx -x$ for small x . For $N = 10,000$, the error is about 0.005% – far smaller than any biological uncertainty.


```

import numpy as np

def simulate_coalescence_time_discrete(N, n_replicates=10000):
    """Simulate coalescence time for 2 lineages in discrete generations.

    Each generation, the two lineages independently choose a parent
    uniformly from N individuals. If they pick the same one, they coalesce.
    """
    times = np.zeros(n_replicates)
    for rep in range(n_replicates):
        g = 0
        while True:
            g += 1
            # With probability 1/N, the two lineages pick the same parent
            if np.random.random() < 1.0 / N:
                break
        times[rep] = g
    return times

def simulate_coalescence_time_continuous(n_replicates=10000):
    """Simulate coalescence time for 2 lineages (exponential).

    In the continuous-time limit, the waiting time is Exp(1)
    in coalescent units (units of N generations).
    """
    return np.random.exponential(1.0, size=n_replicates)

# Compare: discrete (in units of N generations) vs continuous
N = 10000
discrete_times = simulate_coalescence_time_discrete(N) / N # rescale to coalescent_
↳units
continuous_times = simulate_coalescence_time_continuous()

print(f"Discrete:   mean = {discrete_times.mean():.4f}, "
      f"var = {discrete_times.var():.4f}")
print(f"Continuous: mean = {continuous_times.mean():.4f}, "
      f"var = {continuous_times.var():.4f}")
print(f"Theory:     mean = 1.0000, var = 1.0000")

```

With the two-lineage case in hand, we can now generalize to n lineages. In the continuous-time limit, each *pair* of lineages can be represented by a rate-one coalescence clock. With k lineages there are $\binom{k}{2}$ possible pairs; their aggregate event rate is therefore $\binom{k}{2}$. The underlying one-generation collision indicators are not independent, as the next section makes explicit.

Step 2: Multiple Lineages, No Recombination

Now take k lineages in a population of size N . Going one generation back, any pair could coalesce. How many pairs are there?

$$\binom{k}{2} = \frac{k(k-1)}{2}$$

Each pair has marginal probability $1/N$ of coalescing; these pairwise events are not independent because one lineage cannot choose two parents in the same generation. For fixed k and large N , the probability that *any* pair coalesces is

approximately:

$$P(\text{any coalescence}) \approx \frac{\binom{k}{2}}{N}$$

(We ignore the possibility of two simultaneous coalescences, which has probability $O(1/N^2)$.)

i Probability Aside – Why we can ignore simultaneous events

The probability that all k lineages choose distinct parents is $(N)_k/N^k$, where $(N)_k = N(N-1)\cdots(N-k+1)$. The probability of exactly one colliding pair is $\binom{k}{2}(N)_{k-1}/N^k$. Thus multiple mergers are negligible only when k^2/N is small. For $k = 100$ and $N = 10,000$, the exact probability of two or more collisions is about 0.087, not negligible; this is precisely a regime where the DTWF model is safer. For fixed k in the limit $N \rightarrow \infty$, simultaneous events vanish and Kingman's continuous-time coalescent is recovered.

The continuous-time limit. In units of N generations, the waiting time until the next coalescence among k lineages is:

$$T_k \sim \text{Exponential}\left(\binom{k}{2}\right)$$

with mean $\frac{1}{\binom{k}{2}} = \frac{2}{k(k-1)}$.

The full coalescent process for n samples:

1. Start with $k = n$ lineages
2. Wait $T_k \sim \text{Exp}(\binom{k}{2})$ time units
3. Choose a random pair to coalesce: $k \rightarrow k - 1$
4. Repeat until $k = 1$

```
def simulate_coalescent(n, n_replicates=1):
    """Simulate the standard coalescent for n samples.

    Returns
    -----
    times : list of float
        Coalescence times (in coalescent units of N generations).
    pairs : list of (int, int)
        Which pair coalesced at each event.
    """
    all_results = []

    for _ in range(n_replicates):
        lineages = list(range(n)) # each lineage gets a unique ID
        times = []
        pairs = []
        t = 0.0

        while len(lineages) > 1:
            k = len(lineages)
            rate = k * (k - 1) / 2 # binom(k, 2) -- the coalescence rate
            # Wait exponential time with this rate
            t += np.random.exponential(1.0 / rate)
```

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```

# Choose random pair to coalesce
i, j = sorted(np.random.choice(len(lineages), 2, replace=False))
pairs.append((lineages[i], lineages[j]))
times.append(t)
# Merge: replace i with new node, remove j
new_node = max(lineages) + 1
lineages[i] = new_node
lineages.pop(j)

all_results.append((times, pairs))

return all_results

# Simulate and show the coalescent for n=5
results = simulate_coalescent(5, n_replicates=1)
times, pairs = results[0]
print("Coalescent for n=5:")
for i, (t, (a, b)) in enumerate(zip(times, pairs)):
    print(f"  k={5-i}: t={t:.4f}, lineages {a} and {b} coalesce")

```

Expected time to MRCA

The total time to the Most Recent Common Ancestor (MRCA) is:

$$T_{\text{MRCA}} = \sum_{k=2}^n T_k$$

Since each T_k is independent with mean $\frac{2}{k(k-1)}$:

$$E[T_{\text{MRCA}}] = \sum_{k=2}^n \frac{2}{k(k-1)} = 2 \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) = 2 \left(1 - \frac{1}{n} \right)$$

This is remarkable. The expected MRCA time approaches 2 (in units of N generations) regardless of how large n is. Adding more samples barely changes the total tree height.

Calculus Aside – Telescoping sums

The partial fraction decomposition $\frac{2}{k(k-1)} = 2 \left(\frac{1}{k-1} - \frac{1}{k} \right)$ gives a **telescoping sum**: most terms cancel in pairs.

$$\sum_{k=2}^n 2 \left(\frac{1}{k-1} - \frac{1}{k} \right) = 2 \left[\left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n} \right) \right] = 2 \left(1 - \frac{1}{n} \right)$$

Everything except the first and last terms cancels. This is a standard technique in combinatorics and analysis; watch for it whenever you see partial fractions of the form $1/[k(k-1)]$.

Why? With many lineages, the coalescence rate $\binom{k}{2} \sim k^2/2$ is so high that events happen almost instantly. Most of the time is spent waiting for the last few lineages to merge.

```

def expected_tmrc(n):
    """Expected MRCA time for n samples (in coalescent units)."""

```

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```

return 2 * (1 - 1.0 / n)

def expected_total_branch_length(n):
    """Expected total branch length of the coalescent tree.

    This equals 2 * H_{n-1}, where H_k is the k-th harmonic number.
    """
    return 2 * sum(1.0 / k for k in range(1, n))

for n in [2, 5, 10, 50, 100, 1000]:
    print(f"n={n:>5d}: E[T_MRCA] = {expected_tmrca(n):.4f}, "
          f"E[total length] = {expected_total_branch_length(n):.4f}")

```

We now know how to simulate *when* events happen (exponential waiting times) and *what* happens (a random pair coalesces). But in the real simulation, coalescence is not the only possible event. Recombination and migration also compete for the clock's next tick. To handle this competition, we need the **exponential race**.

Step 3: The Exponential Race

The coalescent with recombination involves multiple types of events (coalescence, recombination, migration) happening at different rates. The simulation needs to determine which event happens next. This is where the **exponential race** comes in.

Fact. If $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$ are independent, then:

1. $\min(X_1, X_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$
2. $P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$

Proof of (1). $P(\min(X_1, X_2) > t) = P(X_1 > t)P(X_2 > t) = e^{-\lambda_1 t}e^{-\lambda_2 t} = e^{-(\lambda_1 + \lambda_2)t}$.

Proof of (2).

$$P(X_1 < X_2) = \int_0^\infty P(X_2 > t) \cdot f_{X_1}(t) dt = \int_0^\infty e^{-\lambda_2 t} \cdot \lambda_1 e^{-\lambda_1 t} dt = \lambda_1 \int_0^\infty e^{-(\lambda_1 + \lambda_2)t} dt = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

Calculus Aside – Evaluating the integral

The integral $\int_0^\infty e^{-\alpha t} dt = 1/\alpha$ for $\alpha > 0$ is one of the most important integrals in probability. It follows immediately from the antiderivative $\int e^{-\alpha t} dt = -\frac{1}{\alpha}e^{-\alpha t} + C$ and the fact that $e^{-\alpha t} \rightarrow 0$ as $t \rightarrow \infty$. In the proof above, $\alpha = \lambda_1 + \lambda_2$.

This extends to any number of competing exponentials: the minimum of $\text{Exp}(\lambda_1), \dots, \text{Exp}(\lambda_m)$ is $\text{Exp}(\sum_i \lambda_i)$, and event i wins with probability $\lambda_i / \sum_j \lambda_j$.

This is the simulation engine. At each step, msprime computes:

- λ_{coal} = total coalescence rate
- λ_{recomb} = total recombination rate
- λ_{mig} = total migration rate

Then draws the minimum, advances time, and executes the winning event. This is the heartbeat of *Hudson's Algorithm* – the main simulation loop, the ticking of the clock.

```
def exponential_race(*rates):
    """Simulate an exponential race.

    Each "competitor" proposes a random waiting time drawn from
    Exp(rate_i). The competitor with the shortest time wins.

    Parameters
    -----
    rates : floats
        Rate of each competing process.

    Returns
    -----
    winner : int
        Index of the process that fired first.
    time : float
        Waiting time until the first event.
    """
    times = []
    for rate in rates:
        if rate > 0:
            # Draw a random waiting time: higher rate -> shorter wait on average
            times.append(np.random.exponential(1.0 / rate))
        else:
            times.append(np.inf) # rate=0 means this event never fires

    winner = np.argmin(times)
    return winner, times[winner]

# Example: coalescence (rate=10) vs recombination (rate=5)
wins = np.zeros(2)
for _ in range(10000):
    w, t = exponential_race(10.0, 5.0)
    wins[w] += 1
print(f"Coalescence wins: {wins[0]/100:.1f}% "
      f"(expected: {10/15*100:.1f}%)")
print(f"Recombination wins: {wins[1]/100:.1f}% "
      f"(expected: {5/15*100:.1f}%)")
```

i A practical subtlety

An implementation may draw one exponential with the summed rate and then choose the event in proportion to its rate, or draw an exponential for every event class and take the minimum. These constructions are mathematically equivalent. msprime groups rates where efficient and uses indexed data structures to choose the particular lineage, population, or breakpoint involved.

With the exponential race in our toolkit, we are ready to add the crucial complication that makes the coalescent interesting (and computationally challenging): recombination.

Step 4: Adding Recombination

Now the crucial complication. In the standard coalescent without recombination, each lineage is a single indivisible entity. With recombination, a lineage can **split**: part of the genome traces back through one parent, part through the other.

Going backwards in time, a recombination event on a lineage at position x produces:

- A **left lineage** carrying ancestry for positions $[0, x)$
- A **right lineage** carrying ancestry for positions $[x, L)$

This increases the number of lineages by one.

Rate of recombination. A lineage carrying a segment of length ℓ has a recombination rate of $\rho \cdot \ell / (2L)$ per coalescent time unit, where $\rho = 4N_e r L$ is the population-scaled recombination rate for the whole genome under the diploid $2N_e$ time scale. Equivalently, the per-base-pair rate in coalescent units is $\rho / (2L)$.

The **total** recombination rate across all lineages is:

$$\lambda_{\text{recomb}} = \frac{\rho}{2L} \sum_{i=1}^k \ell_i$$

where ℓ_i is the total length of ancestry carried by lineage i .

i Why recombination rate depends on segment length

A recombination can only matter where the lineage carries ancestral material. If a lineage has already lost some genomic segments through prior coalescence events, recombination in those lost regions has no effect. Only the remaining “active” segments contribute to the rate.

i Closing a confusion gap – Why does recombination *increase* the lineage count?

In forward time, recombination takes *two* parental chromosomes and shuffles them into *one* offspring chromosome. But in the coalescent, we trace ancestry *backwards*. A present-day chromosome that was formed by recombination has *two* parents – one for the left half and one for the right half. Tracing backwards, one lineage becomes two. This is the fundamental asymmetry of the backward view: coalescence *reduces* the lineage count (two children share one parent), while recombination *increases* it (one child has two parents for different genomic regions).

```
def coalescent_with_recombination_simple(n, L, rho, max_events=10000):
    """A simple (but slow) coalescent with recombination.

    Parameters
```

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```

-----
n : int
    Sample size.
L : float
    Genome length.
rho : float
    Population-scaled recombination rate (4*Ne*r*L).

Returns
-----
events : list of (time, event_type, details)
"""
# Each lineage is a list of (left, right) segments
# Initially, every lineage carries the full genome [0, L).
lineages = [(0, L)] for _ in range(n)
events = []
t = 0.0

for _ in range(max_events):
    k = len(lineages)
    if k <= 1:
        break

    # Coalescence rate: any pair of k lineages can coalesce
    coal_rate = k * (k - 1) / 2

    # Recombination rate: proportional to total ancestry length
    total_length = sum(
        sum(r - l for l, r in segs) for segs in lineages
    )
    recomb_rate = rho * total_length / (2 * L)

    # Exponential race between coalescence and recombination
    winner, dt = exponential_race(coal_rate, recomb_rate)
    t += dt

    if winner == 0:
        # Coalescence: pick two random lineages and merge
        i, j = sorted(np.random.choice(k, 2, replace=False))
        merged = merge_segments(lineages[i], lineages[j])
        lineages.pop(j)
        lineages[i] = merged
        events.append((t, 'coal', len(lineages)))

    else:
        # Recombination: pick a lineage weighted by length,
        # then pick a breakpoint
        lengths = [sum(r - l for l, r in segs) for segs in lineages]
        probs = np.array(lengths) / sum(lengths)
        idx = np.random.choice(k, p=probs)

        # Pick a random position within the segments

```

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```

        bp = pick_random_breakpoint(lineages[idx], L)
        left_segs, right_segs = split_at_breakpoint(lineages[idx], bp)

        if left_segs and right_segs:
            lineages[idx] = left_segs
            lineages.append(right_segs)
            events.append((t, 'recomb', len(lineages)))

    return events

def merge_segments(segs_a, segs_b):
    """Merge two segment lists (simplified: just concatenate and sort)."""
    all_segs = sorted(segs_a + segs_b)
    # Merge overlapping intervals (coalescence reduces segment count)
    merged = [all_segs[0]]
    for l, r in all_segs[1:]:
        if l <= merged[-1][1]:
            merged[-1] = (merged[-1][0], max(merged[-1][1], r))
        else:
            merged.append((l, r))
    return merged

def pick_random_breakpoint(segs, L):
    """Pick a random breakpoint within the segment list."""
    total = sum(r - l for l, r in segs)
    target = np.random.uniform(0, total)
    cumulative = 0
    for l, r in segs:
        cumulative += r - l
        if cumulative >= target:
            bp = r - (cumulative - target)
            return bp
    return segs[-1][1]

def split_at_breakpoint(segs, bp):
    """Split segments at breakpoint bp into left and right."""
    left, right = [], []
    for l, r in segs:
        if r <= bp:
            left.append((l, r))          # entirely left of breakpoint
        elif l >= bp:
            right.append((l, r))         # entirely right of breakpoint
        else:
            left.append((l, bp))         # straddles: split into two pieces
            right.append((bp, r))
    return left, right

# Simulate
events = coalescent_with_recombination_simple(n=5, L=1000, rho=10.0)
print(f"Total events: {len(events)}")
for t, etype, k in events[:10]:
    print(f"  t={t:.4f}: {etype:>6s}, lineages={k}")

```


This simple implementation works, but it is slow: computing the total segment length requires iterating over all segments every step. In *Segments & the Fenwick Tree*, we will replace this with a Fenwick tree – a clever indexing mechanism for fast event scheduling – that maintains the total in $O(\log n)$ time.

Step 5: The Coalescent with Recombination in Detail

Let's be more precise about the rates. In a population of effective size N_e , with k lineages, each carrying some genomic segments:

Coalescence rate:

In the standard (Hudson) model, any pair of k lineages can coalesce:

$$\lambda_{\text{coal}} = \frac{\binom{k}{2}}{N_e} = \frac{k(k-1)}{2N_e}$$

In coalescent time units (measured in N_e generations), the N_e cancels and the rate is simply $\binom{k}{2}$.

i SMC and SMC' approximations

In the Sequentially Markov Coalescent (SMC), not all $\binom{k}{2}$ pairs can coalesce – only those whose ancestral segments **overlap** genomically. This reduces the rate but also the state space, making inference algorithms (like PSMC and SINGER) tractable.

- **SMC** (McVean & Cardin, 2005): After a recombination, the new lineage can only re-coalesce with lineages that share a marginal tree at the breakpoint.
- **SMC'** (Marjoram & Wall, 2006): Relaxes SMC slightly to allow re-coalescence with lineages that have contiguous ancestral segments.
- **SMC(k)**: A parameterized family that interpolates between SMC and the full coalescent. msprime implements this using “hulls” – bounding boxes of each lineage's ancestry.

Recombination rate:

A lineage carrying total ancestral material of length ℓ (summed over all its segments) experiences recombination at rate:

$$\lambda_{\text{recomb, lineage}} = \frac{\ell}{L} \cdot \frac{\rho}{2}$$

where $\rho = 4N_e r L$ is the population-scaled total recombination rate. In practice, with a position-dependent recombination rate map $r(x)$, the recombination “mass” of a segment $[a, b]$ is:

$$m(a, b) = \int_a^b r(x) dx$$

and the total recombination rate is proportional to the total mass across all segments.

i Calculus Aside – From rate function to mass

The integral $m(a, b) = \int_a^b r(x) dx$ is simply the area under the recombination rate curve between positions a and b . When the rate is constant ($r(x) = r$), this reduces to $m(a, b) = r \cdot (b - a)$. When the rate varies (e.g., recombination hotspots), the integral accounts for the fact that some base pairs contribute more to the recombination probability than others. In *Segments & the Fenwick Tree*, we will store these masses in a Fenwick tree so that the total can be maintained efficiently as segments are created and destroyed.

The **breakpoint** is chosen uniformly over the total recombination mass (not uniformly over genomic position). This naturally concentrates breakpoints in recombination hotspots.

Now that we have the recombination machinery, there is one more ingredient needed for realistic simulations: the effect of changing population size on coalescence waiting times.

Step 6: The Coalescent Waiting Time with Growth

When the population size changes through time, the coalescence rate changes too. A key case is **exponential growth**:

$$N(t) = N_0 \cdot e^{-\alpha t}$$

where $\alpha > 0$ is the growth rate and t is measured backwards (so the population was smaller in the past for $\alpha > 0$).

The coalescence rate with k lineages at time t is:

$$\lambda(t) = \frac{\binom{k}{2}}{N(t)} = \frac{\binom{k}{2}}{N_0 e^{-\alpha t}}$$

The waiting time is no longer a simple exponential. We need the cumulative hazard: the integral of the rate from the current time t_0 to some future time $t_0 + w$:

$$\Lambda(w) = \int_{t_0}^{t_0+w} \frac{\binom{k}{2}}{N_0 e^{-\alpha s}} ds$$

The survival function is $P(W > w) = \exp(-\Lambda(w))$. To sample from this distribution, we use the inversion method.

i Probability Aside – The inversion method

To sample from a distribution with survival function $S(w) = e^{-\Lambda(w)}$, we draw $U \sim \text{Uniform}(0, 1)$ and solve $S(W) = U$, i.e., $\Lambda(W) = -\ln(U)$. Since $-\ln(U) \sim \text{Exp}(1)$, this is equivalent to drawing $E \sim \text{Exp}(1)$ and solving $\Lambda(W) = E$. The advantage is that we can solve for W analytically when Λ has a closed-form inverse.

Deriving the waiting time formula. Let $c = \binom{k}{2}$ and draw $E \sim \text{Exp}(1)$. Equivalently, define $U = E/c$, so $U \sim \text{Exp}(c)$. We then solve:

$$E = \int_{t_0}^{t_0+W} \frac{c}{N(s)} ds$$

For constant size, $W = N_0 U$, which gives $W \sim \text{Exp}(c/N_0)$.

For exponential growth, using $N(s) = N_0 e^{-\alpha s}$:

$$E = \frac{ce^{\alpha t_0}}{N_0} \int_0^W e^{\alpha s} ds = \frac{ce^{\alpha t_0}}{N_0 \alpha} (e^{\alpha W} - 1)$$

i Calculus Aside – Evaluating the growth integral

The integral $\int_0^W e^{\alpha s} ds$ has antiderivative $\frac{1}{\alpha} e^{\alpha s}$. Evaluating:

$$\int_0^W e^{\alpha s} ds = \frac{1}{\alpha} [e^{\alpha W} - e^0] = \frac{e^{\alpha W} - 1}{\alpha}$$

This integral appears frequently in demographic models. When $\alpha \rightarrow 0$, l'Hopital's rule gives $\frac{e^{\alpha W} - 1}{\alpha} \rightarrow W$, recovering the constant-size case.

Solving for W :

$$e^{\alpha W} = 1 + \alpha N_0 e^{-\alpha t_0} U$$

Taking the logarithm:

$$W = \frac{1}{\alpha} \ln(1 + \alpha N_0 e^{-\alpha t_0} U)$$

This is valid only if $1 + \alpha N_0 e^{-\alpha t_0} U > 0$. When $\alpha < 0$ (population was *larger* in the past), the argument can become zero or negative, meaning the population was so large that coalescence never occurs within the growth epoch – the simulation must wait for a demographic event that changes the growth rate.

```
def coalescent_waiting_time_growth(k, N0, alpha, t0):
    """Waiting time for k lineages, exponential growth.

    Parameters
    -----
    k : int
        Number of lineages.
    N0 : float
        Population size at time zero.
    alpha : float
        Growth rate (positive = population was smaller in the past).
    t0 : float
        Current time.

    Returns
    -----
    w : float
        Waiting time (can be inf if coalescence doesn't occur).
    """
    rate = k * (k - 1) / 2
    u = np.random.exponential(1.0 / rate)

    if alpha == 0:
        return N0 * u

    z = 1 + alpha * N0 * math.exp(-alpha * t0) * u
    if z <= 0:
        return np.inf

    return math.log(z) / alpha
```

```
# Compare waiting times: constant vs growing population
N0, alpha = 10000, 0.01
k = 10
n_reps = 10000

const_times = [coalescent_waiting_time_constant(k, N0) for _ in range(n_reps)]
growth_times = [coalescent_waiting_time_growth(k, N0, alpha, 0)
                 for _ in range(n_reps)]

print(f"Constant N={N0}: mean waiting time = {np.mean(const_times):.1f} gen")
```

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```
print(f"Growth alpha={alpha}: mean waiting time = {np.mean(growth_times):.1f} gen")
print(f"(Growth concentrates coalescences in the recent past)")
```

i Connection to msprime

msprime uses the same integrated-hazard inversion, with the ploidy factor included in its common-ancestor rate. The function above uses the haploid convention established at the start of this chapter; multiply N_0 by two for the usual diploid timescale.

With coalescence, recombination, and variable population size all accounted for, there is one more biological mechanism to add: gene conversion.

Step 7: Gene Conversion

Gene conversion is a recombination-like event with a twist: instead of exchanging everything to one side of a breakpoint, it copies a short **tract** of DNA (typically a few hundred base pairs) from one homolog to the other.

Going backwards in time, a gene conversion event produces two lineages:

- One carrying a short segment (the converted tract)
- One carrying everything except that tract

```
Before gene conversion (going backwards):

Lineage: =====
           | gene conversion at position x
           | with tract length l

After:
Lineage 1: =====
           |   |
Lineage 2:  =====
           x   x+l
```

The **rate** of gene conversion initiation is proportional to the total genomic mass of the lineage, just like recombination. The **tract length** is drawn from a geometric distribution (discrete genome) or exponential distribution (continuous genome):

$$\ell \sim \text{Geometric}(1/\bar{\ell}) \quad \text{or} \quad \ell \sim \text{Exponential}(\bar{\ell})$$

where $\bar{\ell}$ is the mean tract length (typically ~300-500 bp in humans).

Additionally, gene conversion can occur **to the left** of the first ancestral segment, producing a lineage that extends the ancestry leftward:

```
Before:      =====
              |
Gene conversion starting to the left:

After:
Lineage 1:    =====
Lineage 2:    =====
              |   |
              (new) (original start)
```

This “left extension” has a separate rate proportional to the number of ancestors times the mean gene conversion rate times the tract length.

```
def gene_conversion_event(segments, gc_position, tract_length, L):
    """Simulate a gene conversion event.
```

Gene conversion removes a short tract from the main lineage and places it on a new lineage. This is like recombination, but instead of splitting left/right, it punches a "hole" in the middle.

Parameters

segments : list of (left, right)
Segments of the lineage.
gc_position : float
Starting position of the gene conversion.
tract_length : float
Length of the converted tract.

Returns

main_segs : list of (left, right)
Remaining segments of the main lineage.
tract_segs : list of (left, right)
Segments of the gene conversion tract lineage.
 """

```
gc_left = gc_position
gc_right = min(gc_position + tract_length, L)

main_segs = []
tract_segs = []

for l, r in segments:
    if r <= gc_left or l >= gc_right:
        # Entirely outside tract: stays with main
        main_segs.append((l, r))
    elif l >= gc_left and r <= gc_right:
        # Entirely inside tract: goes to tract lineage
        tract_segs.append((l, r))
    elif l < gc_left and r > gc_right:
        # Straddles both sides: split into three pieces
        main_segs.append((l, gc_left))
        tract_segs.append((gc_left, gc_right))
        main_segs.append((gc_right, r))
    elif l < gc_left:
        # Overlaps left boundary only
        main_segs.append((l, gc_left))
        tract_segs.append((gc_left, r))
    else:
        # Overlaps right boundary only
        tract_segs.append((l, gc_right))
        main_segs.append((gc_right, r))
```

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```

return main_segs, tract_segs

# Example
segs = [(100, 500), (700, 1000)]
main, tract = gene_conversion_event(segs, gc_position=400, tract_length=350, L=1000)
print(f"Original: {segs}")
print(f"Main:      {main}")
print(f"Tract:      {tract}")

```

With all event types defined, let us now summarize the complete rate table.

Step 8: Putting It Together – Event Rates Summary

At any point in the simulation, with k lineages in a population of size $N(t)$, the competing event rates are:

Event	Rate	Effect
Coalescence	$\binom{k}{2}/N(t)$	Two lineages merge ($k \rightarrow k - 1$)
Recombination	$\sum_i m_i^{\text{recomb}}$	One lineage splits ($k \rightarrow k + 1$)
Gene conversion (within)	$\sum_i m_i^{\text{gc}}$	One lineage splits ($k \rightarrow k + 1$)
Gene conversion (left)	$k \cdot \bar{g} \cdot \ell$	One lineage splits ($k \rightarrow k + 1$)
Migration	$k_j \cdot M_{j \rightarrow j'}$	One lineage moves between populations

where m_i^{recomb} is the recombination mass of lineage i and m_i^{gc} is the gene conversion mass.

The simulation draws independent exponential waiting times for each event type, takes the minimum, advances time, and executes the winning event. This is the exponential race in full generality, and it drives the main loop of *Hudson's Algorithm*.

i Termination condition

The simulation terminates when every genomic position has found its MRCA. This is tracked by an overlap counter $S[x]$ that records how many lineages carry ancestral material at position x . When $S[x] = 1$ for all x , we're done. (In practice, msprime uses $S[x] = 0$ after a final decrement, stored in an AVL tree keyed by genomic position.) We will build this overlap counter in *Segments & the Fenwick Tree*.

We now have the complete mathematical specification of the coalescent with recombination. Every rate, every distribution, every event type is defined. But simulating this efficiently requires clever data structures – which brings us to the next chapter.

Exercises

i Exercise 1: Verify the coalescent

Simulate 10,000 coalescent trees with $n = 10$. Compute the mean and variance of T_{MRCA} . Compare to the theoretical values: $E[T_{\text{MRCA}}] = 2(1 - 1/n)$, $\text{Var}(T_{\text{MRCA}}) = \sum_{k=2}^n 4/[k(k-1)]^2$.

Exercise 2: The exponential race

Implement the exponential race for 3 competing events with rates 1, 2, 3. Verify empirically that event i wins with probability $\lambda_i / \sum \lambda_j$, and the minimum has rate $\sum \lambda_j$.

Exercise 3: Coalescent with recombination

Use `coalescent_with_recombination_simple` to simulate 1000 genealogies with $n = 5$, $L = 10^4$, and varying ρ . Plot the number of recombination events as a function of ρ . Compare to the theoretical expectation: $E[\text{recomb events}] \approx \rho \cdot \sum_{k=2}^n 1/(k-1)$.

Exercise 4: Growth vs. constant size

Simulate coalescent trees under (a) constant $N = 10,000$ and (b) exponential growth with $N_0 = 10,000$, $\alpha = 0.01$. Plot the distributions of T_{MRCA} . How does growth affect the shape of the genealogy?

Next: *Segments & the Fenwick Tree* – the linked-list track that follows each lineage's ancestral material, and the Fenwick tree – a clever indexing mechanism for fast event scheduling.

Solutions

Solution 1: Verify the coalescent

We simulate 10,000 coalescent trees using `simulate_coalescent` and compare the empirical mean and variance of T_{MRCA} to their theoretical values.

The theoretical variance uses the fact that the T_k are independent, so:

$$\text{Var}(T_{\text{MRCA}}) = \sum_{k=2}^n \text{Var}(T_k) = \sum_{k=2}^n \frac{1}{\binom{k}{2}^2} = \sum_{k=2}^n \frac{4}{[k(k-1)]^2}$$

since $T_k \sim \text{Exp}(\binom{k}{2})$ and the variance of an $\text{Exp}(\lambda)$ random variable is $1/\lambda^2$.

```
import numpy as np

n = 10
n_replicates = 10000

# Theoretical values
E_tmrca = 2 * (1 - 1.0 / n)
Var_tmrca = sum(4.0 / (k * (k - 1))**2 for k in range(2, n + 1))

# Simulate
tmrca_values = []
for _ in range(n_replicates):
    results = simulate_coalescent(n, n_replicates=1)
    times, pairs = results[0]
    tmrca_values.append(times[-1]) # last coalescence time is the MRCA

tmrca_values = np.array(tmrca_values)

print(f"E[T_MRCA]: simulated = {tmrca_values.mean():.4f}, "
      f"theory = {E_tmrca:.4f}")
print(f"Var[T_MRCA]: simulated = {tmrca_values.var():.4f}, "
      f"theory = {Var_tmrca:.4f}")
```

i Solution 2: The exponential race

We run the race 100,000 times with rates $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$. The total rate is $\Lambda = 6$, so the minimum has mean $1/6$ and event i wins with probability λ_i/Λ .

```
import numpy as np

rates = [1.0, 2.0, 3.0]
total_rate = sum(rates)
n_trials = 100000

wins = np.zeros(3)
min_times = []

for _ in range(n_trials):
    winner, t = exponential_race(*rates)
    wins[winner] += 1
    min_times.append(t)

min_times = np.array(min_times)

print("Win probabilities:")
for i in range(3):
    print(f"  Event {i}: observed = {wins[i]/n_trials:.4f}, "
          f"expected = {rates[i]/total_rate:.4f}")

print(f"\nMinimum time distribution:")
print(f"  Mean: observed = {min_times.mean():.4f}, "
      f"expected = {1.0/total_rate:.4f}")
print(f"  Var:  observed = {min_times.var():.4f}, "
      f"expected = {1.0/total_rate**2:.4f}")
```

i Solution 3: Coalescent with recombination

The expected number of recombination events in a coalescent tree is approximately $\frac{\rho}{2} \cdot E[L_{\text{total}}]$ where $E[L_{\text{total}}] = 2 \sum_{k=1}^{n-1} 1/k$. We simulate for several values of ρ and compare.

```
import numpy as np

n = 5
L = 10000
n_replicates = 1000
rho_values = [0.1, 1.0, 5.0, 10.0, 20.0, 50.0]

# Theoretical expected total branch length (coalescent units)
E_total_length = 2 * sum(1.0 / k for k in range(1, n))

print(f"E[total branch length] = {E_total_length:.4f}")
print(f"{'rho':>6s} {'E[recomb] (sim)':>16s} {'E[recomb] (theory)':>18s}")
```



```

for rho in rho_values:
    recomb_counts = []
    for _ in range(n_replicates):
        events = coalescent_with_recombination_simple(n=n, L=L, rho=rho)
        n_recomb = sum(1 for _, etype, _ in events if etype == 'recomb')
        recomb_counts.append(n_recomb)

    mean_recomb = np.mean(recomb_counts)
    # Theory: E[recomb] = (rho/2) * E[L_total]
    expected_recomb = (rho / 2) * E_total_length
    print(f"{rho:6.1f} {mean_recomb:16.2f} {expected_recomb:18.2f}")

```

i Solution 4: Growth vs. constant size

Exponential growth ($\alpha > 0$) makes the population smaller in the past, which increases the coalescence rate at earlier times. This compresses the genealogy: T_{MRCA} is shorter and the tree is more star-shaped (most coalescences happen at roughly the same time in the recent past).

```

import numpy as np

N0 = 10000
alpha = 0.01
n = 10
n_reps = 10000

# Constant size: standard coalescent, then scale to generations
tmrca_const = []
for _ in range(n_reps):
    results = simulate_coalescent(n, n_replicates=1)
    times, _ = results[0]
    tmrca_const.append(times[-1] * N0) # scale to generations

# Growth: draw waiting times with the growth formula
tmrca_growth = []
for _ in range(n_reps):
    t = 0.0
    k = n
    while k > 1:
        w = coalescent_waiting_time_growth(k, N0, alpha, t)
        if w == np.inf:
            break
        t += w
        k -= 1
    tmrca_growth.append(t)

tmrca_const = np.array(tmrca_const)
tmrca_growth = np.array(tmrca_growth)

print(f"Constant N={N0}:")
print(f"  Mean T_MRCA = {tmrca_const.mean():.0f} generations")
print(f"  Std T_MRCA = {tmrca_const.std():.0f} generations")

```

```
print(f"\nGrowth alpha={alpha}:")
print(f"  Mean T_MRCA = {tmrca_growth.mean():.0f} generations")
print(f"  Std  T_MRCA = {tmrca_growth.std():.0f} generations")
print(f"\nGrowth reduces T_MRCA because the ancestral population was smaller, "
      f"forcing lineages to coalesce faster.")
```

Segments & the Fenwick Tree

The gear train: the data structures that turn coalescent math into an efficient algorithm.

In the previous chapter (*The Coalescent Process*), we derived all the rates and distributions that govern the coalescent with recombination. We even wrote a simple simulator. But that simulator was slow: it recomputed the total segment length from scratch at every step, iterating over all segments to sum their lengths every time we needed a rate. That is $O(n)$ per event. With millions of events, this is too slow.

This chapter introduces the two data structures that transform the coalescent from a mathematical specification into an efficient algorithm. Think of them as the gear train of the master clockmaker's bench – the mechanical linkage that translates the escapement's regular beats (the coalescent math) into the smooth motion of the hands (the simulation output).

msprime solves the performance problem with two data structures:

1. **Segment chains** – doubly-linked lists representing the ancestral material of each lineage, enabling $O(1)$ splits and merges. These are the linked-list track that follows each lineage's ancestral material along the genome.
2. **Fenwick trees** – cumulative frequency trees that maintain the total recombination mass across all segments, enabling $O(\log n)$ rate queries and updates. The Fenwick tree is a clever indexing mechanism for fast event scheduling.

Note

Prerequisites. This chapter builds directly on *The Coalescent Process*, where we defined segments, lineages, recombination mass, and the event rates. If you need a refresher on what “recombination mass” means or why the total mass determines the recombination rate, revisit Steps 4-5 of that chapter.

Step 1: The Segment

A **segment** represents a contiguous stretch of ancestral genome carried by a lineage. It has four essential fields:

- `left`: the start position on the genome (inclusive)
- `right`: the end position on the genome (exclusive)
- `node`: the tree-sequence node ID where this ancestry was born
- `next / prev`: pointers to adjacent segments in the chain

Closing a confusion gap – What is a segment, concretely?

A segment is a small data object that says: “This lineage carries ancestral material for the genomic interval [`left`, `right`).” At the start of the simulation, each of the n sample lineages has exactly one segment covering the full genome $[0, L)$. As recombination events split lineages, segments get shorter and lineages accumulate multiple segments separated by gaps. As coalescence events merge lineages, overlapping segments are combined, and edges are recorded in the tree sequence. The segment is the fundamental unit of bookkeeping in msprime.

```

import dataclasses

@dataclasses.dataclass
class Segment:
    """A contiguous stretch of ancestral genome.

    The segment covers the half-open interval [left, right) on the genome.
    Segments are linked into doubly-linked lists via prev/next pointers.
    """
    index: int          # unique ID (position in the segment pool)
    left: float = 0      # start position (inclusive)
    right: float = 0     # end position (exclusive)
    node: int = -1       # tree-sequence node ID
    prev: object = None  # previous segment in the chain (toward left end of genome)
    next: object = None  # next segment in the chain (toward right end of genome)

    @property
    def length(self):
        """Genomic span of this segment in base pairs."""
        return self.right - self.left

    def __repr__(self):
        return f"Seg({self.index}: [{self.left}, {self.right}], node={self.node})"

    @staticmethod
    def show_chain(seg):
        """Print the entire chain starting from seg."""
        parts = []
        while seg is not None:
            parts.append(f"[{seg.left}, {seg.right}]: node {seg.node}")
            seg = seg.next
        return " -> ".join(parts)

# Example: a lineage carrying two non-contiguous segments
# This happens after a coalescence event removed the middle portion.
s1 = Segment(index=0, left=0, right=500, node=3)
s2 = Segment(index=1, left=800, right=1000, node=3)
s1.next = s2      # wire s1's "next" pointer to s2
s2.prev = s1      # wire s2's "prev" pointer back to s1

print("Segment chain:")
print(f"    {Segment.show_chain(s1)}")
print(f"    Total ancestry: {s1.length + s2.length} bp out of 1000 bp")

```

Why Linked Lists?

Why not arrays? Because the two most frequent operations are:

1. **Split** (recombination): break a segment at position x
2. **Merge** (coalescence): combine two segment chains into one

A split at an already-located segment and local insertion or removal are $O(1)$ with linked lists because they only rewire pointers. A full coalescence merge still walks the participating chains and is linear in the number of segments examined;

linked lists avoid the extra array shifting that would otherwise accompany each local edit.

In the watch metaphor, segments are the linked-list track that follows each lineage's ancestral material. Like the links of a fine watch bracelet, each segment connects to the next, and you can open any link to insert or remove a piece without disturbing the rest of the chain.

```
def split_segment(seg, breakpoint):
    """Split segment at breakpoint, returning (left_part, right_part).

    Before:  seg = [left, .... bp .... right)
    After:   seg = [left, bp)    alpha = [bp, right)

    This is O(1): we just create a new segment and rewire pointers.
    No array copying or shifting is needed.
    """
    alpha = Segment(
        index=-1, # will be assigned by the pool
        left=breakpoint,
        right=seg.right,
        node=seg.node,
    )
    # Wire up the linked list: alpha inherits seg's successor
    alpha.next = seg.next
    if seg.next is not None:
        seg.next.prev = alpha
    alpha.prev = None # alpha is the head of the right chain

    # Trim seg to end at the breakpoint
    seg.right = breakpoint
    seg.next = None # seg is now the tail of the left chain

    return seg, alpha

# Example
seg = Segment(index=0, left=100, right=900, node=5)
left, right = split_segment(seg, 400)
print(f"Before split: [{100}, {900}]")
print(f"Left:   [{left.left}, {left.right}]")
print(f"Right:  [{right.left}, {right.right}]")
```

With the Segment defined, let us wrap it in a higher-level abstraction: the Lineage.

Step 2: The Lineage

A lineage wraps a segment chain and adds metadata:

```
@dataclasses.dataclass
class Lineage:
    """A single haploid genome in the simulation.

    The ancestry is stored as a linked list of Segments,
    accessed via head and tail pointers. The head points to the
    leftmost segment, the tail to the rightmost.
```

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```

"""
head: Segment      # first segment in the chain (leftmost on genome)
tail: Segment      # last segment in the chain (rightmost on genome)
population: int = 0 # which population this lineage resides in
label: int = 0      # sub-label (used for selective sweeps)

@property
def total_length(self):
    """Total ancestral material carried by this lineage.

    Walk the chain and sum each segment's length. This is  $O(s)$ 
    where  $s$  is the number of segments -- but we rarely call this
    because the Fenwick tree maintains the running total.
    """
    length = 0
    seg = self.head
    while seg is not None:
        length += seg.length
        seg = seg.next
    return length

def __repr__(self):
    return (f"Lineage(pop={self.population}, "
            f"chain={Segment.show_chain(self.head)})")

# Example: lineage with two segments
s1 = Segment(0, left=0, right=500, node=0)
s2 = Segment(1, left=800, right=1000, node=0)
s1.next = s2
s2.prev = s1
lin = Lineage(head=s1, tail=s2, population=0)
print(lin)
print(f"Total ancestry: {lin.total_length} bp")

```

Now we arrive at the key innovation that makes msprime fast.

Step 3: The Fenwick Tree

This is the key data structure that makes msprime fast. The Fenwick tree (also called a Binary Indexed Tree or BIT) maintains a collection of values and supports two operations in $O(\log n)$ time:

1. **Update:** change the value at an index
2. **Prefix sum:** compute the sum of values from index 1 to k

From these, we can also:

3. **Total sum:** prefix sum up to the maximum index
4. **Find:** given a target sum v , find the smallest index whose prefix sum $\geq v$

Closing a confusion gap – Why do we need a Fenwick tree?

The simulation needs to answer two questions very frequently:

1. **“What is the total recombination rate?”** – This is the sum of recombination masses over all segments. It determines the rate parameter for the exponential waiting time.
2. **“Which segment should the next recombination hit?”** – Given a random number, we need to find the segment whose cumulative mass contains that number (weighted random selection).

A naive approach answers question 1 in $O(n)$ by summing all masses, and question 2 in $O(n)$ by scanning through segments. The Fenwick tree answers both in $O(\log n)$. With millions of events, this is the difference between seconds and hours.

Let's build it from scratch.

The Key Insight

The Fenwick tree uses the **binary representation** of indices to organize partial sums. Each position i stores the sum of a specific range of values, where the range is determined by the lowest set bit of i .

The lowest set bit of i is $i \& (-i)$ (using two's complement).

- $i = 1 = 0001$: lowest bit = 1, stores value at index 1
- $i = 2 = 0010$: lowest bit = 2, stores sum of indices 1-2
- $i = 3 = 0011$: lowest bit = 1, stores value at index 3
- $i = 4 = 0100$: lowest bit = 4, stores sum of indices 1-4
- $i = 5 = 0101$: lowest bit = 1, stores value at index 5
- $i = 6 = 0110$: lowest bit = 2, stores sum of indices 5-6
- $i = 7 = 0111$: lowest bit = 1, stores value at index 7
- $i = 8 = 1000$: lowest bit = 8, stores sum of indices 1-8

Closing a confusion gap – The $i \& -i$ trick

In two's complement representation, $-i$ flips all bits of i and adds 1. The bitwise AND of i and $-i$ isolates the lowest set bit. For example: $6 = 0110$, $-6 = 1010$ (in 4-bit two's complement), $6 \& -6 = 0010 = 2$. This single expression tells us the “responsibility range” of each position in the Fenwick tree. It is the fundamental building block of all Fenwick tree operations: to move *up* (toward larger ranges), we add $i \& -i$; to move *down* (toward smaller ranges), we subtract it.

```
Index:      1      2      3      4      5      6      7      8
Values:    [3]    [1]    [4]    [1]    [5]    [9]    [2]    [6]

Tree:      [3]    [3+1]  [4]    [3+1+4+1]  [5]    [5+9]  [2]    [3+1+4+1+5+9+2+6]
           = [3]    [4]    [4]          [9]      [5]    [14]  [2]          [31]

To get prefix_sum(6): start at 6 = 0110
tree[6] = 14          (sum of 5-6)
6 - (6 & -6) = 6 - 2 = 4
tree[4] = 9           (sum of 1-4)
4 - (4 & -4) = 4 - 4 = 0 -> stop
Result: 14 + 9 = 23    (3+1+4+1+5+9 = 23)
```

The Implementation

```

class FenwickTree:
    """A Fenwick Tree for cumulative frequency tables.

    Supports  $O(\log n)$  updates, prefix sums, and searches.
    Indices are 1-based (index 0 is unused).

    In msprime, this tree stores the recombination mass of each segment.
    It is the clever indexing mechanism for fast event scheduling:
    it lets the simulator quickly answer "what is the total recombination
    rate?" and "which segment should be hit next?"
    """

    def __init__(self, max_index):
        assert max_index > 0
        self.max_index = max_index
        self.tree = [0] * (max_index + 1)  # partial sums (the Fenwick structure)
        self.value = [0] * (max_index + 1)  # actual values at each index

        # Precompute the largest power of 2 <= max_index
        # (used by the find() method for efficient top-down search)
        u = max_index
        self.log_max = 0
        while u != 0:
            self.log_max = u
            u -= u & -u  # strip lowest set bit

    def increment(self, index, delta):
        """Add delta to the value at index.  $O(\log n)$ .

        This propagates the change upward through the tree:
        every ancestor node that includes this index in its range
        is also incremented.
        """
        assert 1 <= index <= self.max_index
        self.value[index] += delta
        j = index
        while j <= self.max_index:
            self.tree[j] += delta
            j += j & -j  # move to parent (next larger range)

    def set_value(self, index, new_value):
        """Set the value at index.  $O(\log n)$ .

        Computes the delta from the old value and calls increment.
        """
        old_value = self.value[index]
        self.increment(index, new_value - old_value)

    def get_value(self, index):
        """Return the value at index.  $O(1)$ .

```

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```

The actual value is stored separately from the tree structure.
"""
return self.value[index]

def get_cumulative_sum(self, index):
    """Return the sum of values from 1 to index. O(log n).

    Walks downward through the tree, accumulating partial sums.
    At each step, we strip the lowest set bit to move to the
    next non-overlapping range.
    """
    assert 1 <= index <= self.max_index
    s = 0
    j = index
    while j > 0:
        s += self.tree[j]
        j -= j & -j # strip lowest set bit (move to next range)
    return s

def get_total(self):
    """Return the sum of all values. O(log n)."""
    return self.get_cumulative_sum(self.max_index)

def find(self, target):
    """Find smallest index with cumulative sum >= target. O(log n).

    This is the inverse of get_cumulative_sum: given a target sum,
    find which index it falls in. This is used to select a random
    segment weighted by recombination mass.

    The algorithm performs a top-down binary search through the
    Fenwick tree, halving the search range at each step.
    """
    j = 0
    remaining = target
    half = self.log_max

    while half > 0:
        # Skip indices beyond max_index
        while j + half > self.max_index:
            half >>= 1
        k = j + half
        if remaining > self.tree[k]:
            # Target is beyond this subtree: skip it
            j = k
            remaining -= self.tree[j]
            half >>= 1 # halve the search range

    return j + 1

# Demonstration
ft = FenwickTree(8)

```

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```

values = [3, 1, 4, 1, 5, 9, 2, 6]
for i, v in enumerate(values):
    ft.set_value(i + 1, v) # Fenwick tree is 1-indexed

print("Values:", [ft.get_value(i+1) for i in range(8)])
print("Prefix sums:", [ft.get_cumulative_sum(i+1) for i in range(8)])
print("Total:", ft.get_total())

# Find: which index does cumulative sum 15 fall in?
idx = ft.find(15)
print(f"\nfind(15) = {idx}")
print(f"  cumsum({idx-1}) = {ft.get_cumulative_sum(idx-1) if idx > 1 else 0}")
print(f"  cumsum({idx}) = {ft.get_cumulative_sum(idx)}")
print(f"  (15 falls in index {idx})")

```

Now let us see how the `find()` method powers the simulation.

Why the `find()` Method Matters

The `find()` method is the heart of `msprime`'s breakpoint selection. Here's how it works in the context of the simulation:

1. Each segment i has a recombination "mass" m_i stored in the Fenwick tree at index i .
2. To choose a random breakpoint, we draw $U \sim \text{Uniform}(0, M_{\text{total}})$ where M_{total} is the total mass.
3. We call `find(U)` to find which segment i the random mass falls in. This segment will experience the recombination.
4. Within that segment, we compute the exact breakpoint position using the rate map.

This gives us a **weighted random selection** of segments in $O(\log n)$ time. Without the Fenwick tree, we'd need $O(n)$ to iterate over all segments.

Probability Aside – Weighted random selection via inverse CDF

The `find()` operation is an instance of the **inverse CDF method**. If we have weights $m_1, m_2, \dots, m_n$ with total $M = \sum m_i$, then drawing $U \sim \text{Uniform}(0, M)$ and finding the smallest k such that $\sum_{i=1}^k m_i \geq U$ selects index k with probability m_k/M . The Fenwick tree makes this $O(\log n)$ instead of $O(n)$ by organizing the partial sums hierarchically.

```

import numpy as np

def choose_random_segment(fenwick_tree, segments):
    """Choose a random segment weighted by recombination mass.

    This is the core selection operation used every time a
    recombination event occurs in the simulation.

    Parameters
    -----
    fenwick_tree : FenwickTree
        Stores recombination mass for each segment.
    segments : dict of {index: Segment}
        All active segments.
    """

```

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```

Returns
-----
segment : Segment
    The chosen segment.
mass_within : float
    How far into this segment's mass the random point fell.
"""
total_mass = fenwick_tree.get_total()
random_mass = np.random.uniform(0, total_mass)

# Find which segment contains this mass -- O(log n)
seg_index = fenwick_tree.find(random_mass)
segment = segments[seg_index]

# How far into this segment?
cumsum = fenwick_tree.get_cumulative_sum(seg_index)
mass_within_segment = fenwick_tree.get_value(seg_index)
mass_from_right = cumsum - random_mass

return segment, mass_from_right

# Example: 5 segments with different masses
ft = FenwickTree(5)
masses = [10.0, 25.0, 5.0, 30.0, 15.0]
for i, m in enumerate(masses):
    ft.set_value(i + 1, m)

# Sample 10000 segments -- verify proportional selection
counts = np.zeros(5)
for _ in range(10000):
    total = ft.get_total()
    r = np.random.uniform(0, total)
    idx = ft.find(r)
    counts[idx - 1] += 1

print("Sampling frequencies vs expected:")
total = sum(masses)
for i in range(5):
    print(f"Segment {i}: observed={counts[i]/100:.1f}%, "
          f"expected={masses[i]/total*100:.1f}%")

```

The Fenwick tree handles the “which segment?” question. But we also need to convert the random mass into a genomic position, which requires the rate map.

Step 4: The Rate Map

The recombination rate is not uniform across the genome. Humans, for example, have recombination hotspots where the rate can be 100x higher than the background. msprime handles this through **rate maps**.

A rate map is a piecewise-constant function $r(x)$ defined by breakpoints and rates:

Position:	0	1000	2000	5000	10000
Rate:	1e-8	5e-8	1e-8	2e-8	

The **mass** of a genomic interval $[a, b]$ is the integral of the rate:

$$m(a, b) = \int_a^b r(x) dx$$

For a piecewise-constant rate, this is just the sum of rate times length for each piece.

i Calculus Aside – Piecewise integration

For a piecewise-constant function $r(x) = r_i$ on $[p_i, p_{i+1})$, the integral over $[a, b]$ is:

$$\int_a^b r(x) dx = \sum_i r_i \cdot \max(0, \min(b, p_{i+1}) - \max(a, p_i))$$

Each term contributes only for the part of $[p_i, p_{i+1})$ that overlaps with $[a, b]$. In the implementation below, we precompute cumulative masses at each breakpoint so that `mass_between(a, b)` can be answered in $O(\log m)$ time (where m is the number of rate intervals) using binary search.

```
class RateMap:
    """A piecewise-constant rate function over the genome.

    The rate in interval [positions[i], positions[i+1]) is rates[i].
    This class handles both recombination and mutation rate maps.
    """

    def __init__(self, positions, rates):
        """
        Parameters
        -----
        positions : list of float
            Breakpoints (including 0 and L).
        rates : list of float
            Rate in each interval (len = len(positions) - 1).
        """
        assert len(rates) == len(positions) - 1
        self.positions = np.array(positions, dtype=float)
        self.rates = np.array(rates, dtype=float)

        # Precompute cumulative mass at each breakpoint
        # cumulative[i] = integral of r(x) from position[0] to position[i]
        self.cumulative = np.zeros(len(positions))
        for i in range(len(rates)):
            span = positions[i + 1] - positions[i]
            self.cumulative[i + 1] = self.cumulative[i] + rates[i] * span

    @property
    def total_mass(self):
        return self.cumulative[-1]
```

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```

def mass_between(self, left, right):
    """Compute the recombination mass of interval [left, right)."""
    return self.position_to_mass(right) - self.position_to_mass(left)

def position_to_mass(self, pos):
    """Convert a genomic position to cumulative mass.

    This is the forward mapping: position -> mass.
    """
    # Find which interval pos falls in
    idx = np.searchsorted(self.positions, pos, side='right') - 1
    idx = max(0, min(idx, len(self.rates) - 1))
    # Mass up to the start of this interval + mass within
    return (self.cumulative[idx] +
            self.rates[idx] * (pos - self.positions[idx]))

def mass_to_position(self, mass):
    """Convert a cumulative mass back to genomic position (inverse).

    This is the inverse mapping: mass -> position.
    Used to convert a random mass coordinate into a breakpoint.
    """
    idx = np.searchsorted(self.cumulative, mass, side='right') - 1
    idx = max(0, min(idx, len(self.rates) - 1))
    # Position at start of interval + offset
    remaining_mass = mass - self.cumulative[idx]
    if self.rates[idx] == 0:
        return self.positions[idx]
    return self.positions[idx] + remaining_mass / self.rates[idx]

# Example: genome with a recombination hotspot
rate_map = RateMap(
    positions=[0, 5000, 6000, 10000],
    rates=[1e-8, 1e-6, 1e-8] # 100x hotspot in [5000, 6000)
)

print(f"Total mass: {rate_map.total_mass:.2e}")
print(f"Mass [0, 5000): {rate_map.mass_between(0, 5000):.2e}")
print(f"Mass [5000, 6000): {rate_map.mass_between(5000, 6000):.2e}")
print(f"Mass [6000, 10000): {rate_map.mass_between(6000, 10000):.2e}")
print(f"\nThe 1kb hotspot has {rate_map.mass_between(5000, 6000) / rate_map.total_
↪mass * 100:.1f}% "
      f"of total recombination mass")

```

Why Mass, Not Position?

The Fenwick tree stores **mass** (rate-weighted length), not raw genomic length. This is crucial: when we draw a random breakpoint, we want it proportional to the local rate. By storing mass in the Fenwick tree, the `find()` method automatically gives us rate-weighted selection.

The conversion from mass back to position is handled by `RateMap.mass_to_position()` – the inverse function.

Here is the full breakpoint selection pipeline, showing how the Fenwick tree, the rate map, and the segment chain work

together:

```
def choose_breakpoint(fenwick_tree, segments, rate_map):
    """Choose a random recombination breakpoint.

    This is the core of msprime's breakpoint selection:
    1. Draw random mass from [0, total_mass)
    2. Use Fenwick.find() to locate the segment -- O(log n)
    3. Convert mass coordinate to genomic position -- O(log m)

    Parameters
    -----
    fenwick_tree : FenwickTree
    segments : dict of {index: Segment}
    rate_map : RateMap

    Returns
    -----
    segment : Segment
        The segment where recombination occurs.
    breakpoint : float
        The genomic position of the breakpoint.
    """
    total_mass = fenwick_tree.get_total()
    random_mass = np.random.uniform(0, total_mass)

    # Step 1: find which segment (using the Fenwick tree's find)
    seg_index = fenwick_tree.find(random_mass)
    seg = segments[seg_index]

    # Step 2: compute breakpoint position
    # The cumulative mass up to this segment's right end
    cum_mass = fenwick_tree.get_cumulative_sum(seg_index)
    # Mass of the breakpoint from the right end of the segment
    mass_from_right = cum_mass - random_mass
    # Convert to genomic position using the rate map inverse
    right_mass = rate_map.position_to_mass(seg.right)
    bp_mass = right_mass - mass_from_right
    bp = rate_map.mass_to_position(bp_mass)

    return seg, bp
```

The left-bound subtlety

In msprime's implementation, the recombination mass of a segment is computed from `get_recomb_left_bound(seg)` to `seg.right`. The left bound is `seg.prev.right` if the segment has a predecessor (i.e., it's not the head of the chain), because recombination between two adjacent segments of the same lineage has no effect – both pieces already belong to the same lineage. Only recombination that falls in a **gap** or within a segment creates a meaningful split. This subtlety is easy to miss but essential for correctness.

With the rate map and Fenwick tree working together, we have efficient breakpoint selection. Next, we need efficient memory management for the millions of segments created and destroyed during the simulation.

Step 5: The Segment Pool

Creating and destroying segment objects millions of times would be slow due to memory allocation overhead. msprime uses a **segment pool**: a pre-allocated array of segments that are recycled.

Closing a confusion gap – Why a pool?

In languages like Python, creating an object involves memory allocation, constructor calls, and eventually garbage collection. For an object created and destroyed millions of times per second, this overhead dominates. A pool pre-allocates all objects at startup and reuses them: “allocation” just pops an index from a free list ($O(1)$), and “deallocation” pushes it back ($O(1)$). The C implementation of msprime uses the same pattern for maximum performance.

```
class SegmentPool:
    """Pre-allocated pool of Segment objects.

    Avoids repeated memory allocation during the simulation.
    'Allocating' a segment just pops an index from the free list.
    'Freeing' a segment pushes it back.
    """

    def __init__(self, max_segments):
        # Pre-create all segment objects at once
        self.segments = [Segment(index=i) for i in range(max_segments + 1)]
        self.free_list = list(range(1, max_segments + 1)) # 1-indexed (0 unused)

    def alloc(self, left=0, right=0, node=-1):
        """Allocate a segment from the pool."""
        if not self.free_list:
            raise RuntimeError("Segment pool exhausted")
        index = self.free_list.pop() # O(1) -- just pop from the stack
        seg = self.segments[index]
        seg.left = left
        seg.right = right
        seg.node = node
        seg.prev = None
        seg.next = None
        return seg

    def free(self, seg):
        """Return a segment to the pool."""
        self.free_list.append(seg.index) # O(1) -- push back onto the stack
        seg.prev = None
        seg.next = None

    def copy(self, seg):
        """Allocate a new segment as a copy of an existing one."""
        new_seg = self.alloc(seg.left, seg.right, seg.node)
        new_seg.next = seg.next
        if seg.next is not None:
            seg.next.prev = new_seg
        return new_seg
```

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```
# Example
pool = SegmentPool(100)
s1 = pool.alloc(left=0, right=500, node=0)
s2 = pool.alloc(left=500, right=1000, node=0)
s1.next = s2
s2.prev = s1

print(f"Allocated: {Segment.show_chain(s1)}")
print(f"Free slots remaining: {len(pool.free_list)}")

pool.free(s2)
print(f"After freeing s2: free slots = {len(pool.free_list)}")
```

The segment pool, the Fenwick tree, and the segment chains form the “gear train” of the simulation. There is one more data structure to introduce: the overlap counter that tells the simulation when it is done.

Step 6: The Overlap Counter S

The simulation needs to know when it’s done. It’s done when every genomic position has exactly one ancestral lineage (the MRCA). msprime tracks this with an **overlap counter** S : an AVL tree mapping genomic positions to the number of lineages carrying ancestral material at that position.

Closing a confusion gap – What is an overlap counter?

Imagine the genome as a number line from 0 to L . Each lineage “paints” a color over the intervals where it carries ancestral material. The overlap counter $S[x]$ counts how many colors are stacked at position x . At the start, all n lineages cover the entire genome, so $S[x] = n$ everywhere. Each coalescence event at interval $[a, b)$ reduces $S[x]$ by 1 for $x \in [a, b)$, because two lineages become one. When $S[x] \leq 1$ everywhere, every position has found its MRCA and the simulation is complete.

The AVL tree (implemented here as a `SortedDict`) stores this count as a piecewise-constant function: only the breakpoints where the count changes are stored, not every base pair individually.

```
from sortedcontainers import SortedDict

class OverlapCounter:
    """Tracks the number of lineages at each genomic position.

    Uses an AVL tree (here SortedDict) to store a piecewise-constant
    function:  $S[x]$  gives the number of lineages at positions  $[x, \text{next\_key})$ .
    """

    def __init__(self, sequence_length):
        self.S = SortedDict()
        self.S[0] = 0 # count starts at 0
        self.S[sequence_length] = -1 # sentinel marking the end

    def increment(self, left, right, delta=1):
        """Increment the count in  $[left, right)$  by delta."""
        # Ensure breakpoints exist at left and right
        if left not in self.S:
```

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```

        # Find the value just before left and copy it
        idx = self.S.bisect_left(left) - 1
        prev_key = self.S.keys()[idx]
        self.S[left] = self.S[prev_key]
    if right not in self.S:
        idx = self.S.bisect_left(right) - 1
        prev_key = self.S.keys()[idx]
        self.S[right] = self.S[prev_key]

    # Increment all positions in [left, right)
    for key in list(self.S.irange(left, right, (True, False))):
        self.S[key] += delta

    def is_complete(self):
        """Check if all positions have count <= 1 (MRCA found)."""
        for key in self.S:
            if self.S[key] > 1:
                return False
        return True

    def __repr__(self):
        parts = []
        keys = list(self.S.keys())
        for i in range(len(keys) - 1):
            parts.append(f"  [{keys[i]}, {keys[i+1]}]: {self.S[keys[i]]}")
        return "OverlapCounter:\n" + "\n".join(parts)

# Example: 3 lineages with overlapping segments
S = OverlapCounter(1000)
S.increment(0, 1000)    # lineage 0: full genome
S.increment(0, 1000)    # lineage 1: full genome
S.increment(0, 1000)    # lineage 2: full genome
print("Before any coalescence:")
print(S)

# After first coalescence at [0, 500)
S.increment(0, 500, delta=-1)
print("\nAfter coalescence at [0, 500):")
print(S)
print(f"Complete? {S.is_complete()}")

```

With all the data structures defined, let us see how they work together in a single simulation step.

Step 7: Putting It All Together

Here's how the data structures work together in a single simulation step. This is a preview of what *Hudson's Algorithm* – the main simulation loop, the ticking of the clock – will orchestrate at full scale.

```

class SimulationState:
    """Minimal simulation state demonstrating the data structures.

    This ties together the segment pool, the Fenwick tree, the rate map,

```

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```

and the lineage list. In the full simulator (hudson_algorithm), these
are augmented with populations, migration, and demographic events.
"""

def __init__(self, n, L, recomb_rate):
    self.n = n
    self.L = L
    self.pool = SegmentPool(10 * n)
    self.rate_map = RateMap([0, L], [recomb_rate])

    # Fenwick tree for recombination mass -- the clever indexing mechanism
    self.mass_index = FenwickTree(10 * n)

    # Create initial lineages: each carries [0, L)
    self.lineages = []
    for i in range(n):
        seg = self.pool.alloc(left=0, right=L, node=i)
        lin = Lineage(head=seg, tail=seg, population=0)
        seg.lineage = lin
        self.lineages.append(lin)

        # Register this segment's mass in the Fenwick tree
        mass = self.rate_map.mass_between(0, L)
        self.mass_index.set_value(seg.index, mass)

def get_total_recomb_rate(self):
    """Total recombination rate across all lineages.

    Thanks to the Fenwick tree, this is  $O(\log n)$ , not  $O(n)$ .
    """
    return self.mass_index.get_total()

def recombination_event(self):
    """Execute one recombination event."""
    # Step 1: Choose breakpoint using Fenwick tree --  $O(\log n)$ 
    total_mass = self.mass_index.get_total()
    random_mass = np.random.uniform(0, total_mass)
    seg_index = self.mass_index.find(random_mass)
    seg = self.pool.segments[seg_index]

    # Step 2: Convert mass to position using rate map
    cum_mass = self.mass_index.get_cumulative_sum(seg_index)
    right_mass = self.rate_map.position_to_mass(seg.right)
    bp_mass = right_mass - (cum_mass - random_mass)
    bp = self.rate_map.mass_to_position(bp_mass)

    # Step 3: Split the segment --  $O(1)$  pointer rewiring
    alpha = self.pool.copy(seg)
    alpha.left = bp
    alpha.prev = None
    if seg.next is not None:
        seg.next.prev = alpha

```

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```

seg.next = None
seg.right = bp

# Step 4: Update Fenwick tree -- O(log n)
left_mass = self.rate_map.mass_between(seg.left, seg.right)
self.mass_index.set_value(seg.index, left_mass)
right_mass = self.rate_map.mass_between(alpha.left, alpha.right)
self.mass_index.set_value(alpha.index, right_mass)

# Step 5: Create new lineage for the right part
old_lineage = seg.lineage
new_lineage = Lineage(head=alpha, tail=alpha, population=0)
alpha.lineage = new_lineage
old_lineage.tail = seg
self.lineages.append(new_lineage)

return bp

# Demo
state = SimulationState(n=3, L=1000, recomb_rate=1e-3)
print(f"Initial: {len(state.lineages)} lineages")
print(f"Total recomb mass: {state.get_total_recomb_rate():.4f}")

bp = state.recombination_event()
print(f"\nAfter recombination at {bp:.1f}:")
print(f"Now {len(state.lineages)} lineages")
print(f"Total recomb mass: {state.get_total_recomb_rate():.4f}")

```

You have now seen every data structure that powers the master clockmaker's bench. The segment chains are the linked-list track that follows each lineage's ancestral material. The Fenwick tree is the clever indexing mechanism for fast event scheduling. The segment pool eliminates memory allocation overhead. And the overlap counter tracks progress toward completion.

In the next chapter, we assemble these parts into the complete simulation loop.

Exercises

Exercise 1: Fenwick tree operations

Build a Fenwick tree with 16 elements. Set random values, then verify that `get_cumulative_sum(i)` matches a naive prefix sum for all i . Also verify that `find(v)` returns the correct index for 100 random target values.

Exercise 2: Weighted segment selection

Create 100 segments with random masses. Use the Fenwick tree to sample 10,000 segments. Verify that the empirical selection frequency of each segment matches its mass fraction to within 1%.

i Exercise 3: Breakpoint distribution with hotspots

Create a rate map with a 100x hotspot covering 1% of the genome. Sample 10,000 breakpoints using the Fenwick tree + rate map. Plot the breakpoint density and verify that ~50% of breakpoints fall in the hotspot.

i Exercise 4: Segment chain operations

Implement a full recombination-coalescence cycle: start with 3 lineages each carrying [0, 1000), perform a recombination on lineage 1, then coalesce two lineages. Verify the segment chains are correct at each step.

Next: *Hudson's Algorithm* – the main simulation loop that orchestrates these data structures, the ticking of the clock.

Solutions**i Solution 1: Fenwick tree operations**

We build a Fenwick tree with 16 elements, set random values, and verify that `get_cumulative_sum` matches a naive prefix sum for every index. Then we verify `find` for 100 random target values.

```
import numpy as np

ft = FenwickTree(16)
values = np.random.exponential(5.0, size=16)

for i in range(16):
    ft.set_value(i + 1, values[i]) # 1-indexed

# Verify cumulative sums
naive_cumsum = np.cumsum(values)
all_correct = True
for i in range(1, 17):
    fenwick_sum = ft.get_cumulative_sum(i)
    naive_sum = naive_cumsum[i - 1]
    if abs(fenwick_sum - naive_sum) > 1e-10:
        print(f"MISMATCH at index {i}: fenwick={fenwick_sum}, "
              f"naive={naive_sum}")
    all_correct = False
print(f"Cumulative sum verification: {'PASS' if all_correct else 'FAIL'}")

# Verify find() for 100 random targets
total = ft.get_total()
find_correct = True
for _ in range(100):
    target = np.random.uniform(0, total)
    idx = ft.find(target)

    # Verify: cumsum(idx-1) < target <= cumsum(idx)
    cumsum_idx = ft.get_cumulative_sum(idx)
    cumsum_prev = ft.get_cumulative_sum(idx - 1) if idx > 1 else 0

    if not (cumsum_prev < target <= cumsum_idx + 1e-10):
```

```

print(f"MISMATCH: find({target:.4f})={idx}, "
      f"cumsum[{idx-1}]=cumsum_prev:.4f, "
      f"cumsum[{idx}]=cumsum_idx:.4f)")
find_correct = False
print(f"find() verification: {'PASS' if find_correct else 'FAIL'}")

```

i Solution 2: Weighted segment selection

We create 100 segments with random masses and verify that sampling 10,000 times with the Fenwick tree produces frequencies matching the mass fractions.

```

import numpy as np

n_segments = 100
ft = FenwickTree(n_segments)
masses = np.random.exponential(10.0, size=n_segments)

for i in range(n_segments):
    ft.set_value(i + 1, masses[i])

total_mass = ft.get_total()
expected_fracs = masses / total_mass

# Sample 10,000 segments
n_samples = 10000
counts = np.zeros(n_segments)
for _ in range(n_samples):
    target = np.random.uniform(0, total_mass)
    idx = ft.find(target)
    counts[idx - 1] += 1 # convert from 1-indexed to 0-indexed

observed_fracs = counts / n_samples

# Check that all segments are within 1% of expected
max_error = np.max(np.abs(observed_fracs - expected_fracs))
within_1pct = np.all(np.abs(observed_fracs - expected_fracs) < 0.01)

print(f"Max absolute error: {max_error:.4f}")
print(f"All within 1%: {within_1pct}")

# Show the 5 segments with the largest mass
top5 = np.argsort(masses)[-5:][::-1]
print(f"\nTop 5 segments by mass:")
print(f"{'Seg':>5s} {'Mass':>8s} {'Expected':>10s} {'Observed':>10s}")
for i in top5:
    print(f"{i:5d} {masses[i]:8.2f} {expected_fracs[i]:10.4f} "
          f"{observed_fracs[i]:10.4f}")

```

i Solution 3: Breakpoint distribution with hotspots

We create a rate map where 1% of the genome has a 100x recombination rate. The hotspot's mass fraction determines the expected fraction of breakpoints falling in it. For a 100x hotspot covering 1% of the genome, the mass fraction is $100 \times 0.01 / (100 \times 0.01 + 1 \times 0.99) = 1.0 / 1.99 \approx 50.3\%$.

```
import numpy as np

L = 100000
hotspot_start = 50000
hotspot_end = 51000    # 1% of genome
background_rate = 1e-8
hotspot_rate = 100 * background_rate

rate_map = RateMap(
    positions=[0, hotspot_start, hotspot_end, L],
    rates=[background_rate, hotspot_rate, background_rate]
)

# Theoretical hotspot mass fraction
hotspot_mass = rate_map.mass_between(hotspot_start, hotspot_end)
total_mass = rate_map.total_mass
expected_hotspot_frac = hotspot_mass / total_mass
print(f"Hotspot mass fraction: {expected_hotspot_frac:.4f}")

# Set up a Fenwick tree with a single segment covering [0, L)
ft = FenwickTree(1)
ft.set_value(1, total_mass)

# Sample 10,000 breakpoints
n_samples = 10000
n_in_hotspot = 0
breakpoints = []

for _ in range(n_samples):
    # Draw a random mass and convert to genomic position
    random_mass = np.random.uniform(0, total_mass)
    bp = rate_map.mass_to_position(random_mass)
    breakpoints.append(bp)
    if hotspot_start <= bp < hotspot_end:
        n_in_hotspot += 1

observed_frac = n_in_hotspot / n_samples
print(f"Breakpoints in hotspot: {n_in_hotspot}/{n_samples} "
      f"= {observed_frac:.4f}")
print(f"Expected fraction: {expected_hotspot_frac:.4f}")
print(f"The hotspot (1% of genome) captures ~{observed_frac*100:.1f}% "
      f"of breakpoints.")
```

i Solution 4: Segment chain operations

We start with 3 lineages each carrying [0, 1000), perform a recombination on lineage 1 at position 400, then coalesce lineage 0 with the left part of lineage 1.

```
import numpy as np

# Initialize 3 lineages, each covering [0, 1000)
pool = SegmentPool(20)
recomb_rate = 1e-3
L = 1000

segs = []
lineages = []
for i in range(3):
    seg = pool.alloc(left=0, right=L, node=i)
    lin = Lineage(head=seg, tail=seg, population=0)
    seg.lineage = lin
    segs.append(seg)
    lineages.append(lin)

print("=== Initial state ===")
for i, lin in enumerate(lineages):
    print(f"  Lineage {i}: {Segment.show_chain(lin.head)}")

# Recombination on lineage 1 at position 400
bp = 400
seg1 = lineages[1].head
left_seg, right_seg = split_segment(seg1, bp)

# Create new lineage for the right part
new_lin = Lineage(head=right_seg, tail=right_seg, population=0)
right_seg.lineage = new_lin
lineages[1].tail = left_seg # update tail of left lineage
lineages.append(new_lin)

print(f"\n=== After recombination at bp={bp} ===")
for i, lin in enumerate(lineages):
    print(f"  Lineage {i}: {Segment.show_chain(lin.head)}")

# Coalescence: merge lineage 0 and lineage 1 (left part)
x = lineages[0].head # [0, 1000: node 0]
y = lineages[1].head # [0, 400: node 1]
ancestor_node = 10

# Walk through the merge:
# Both start at 0, x.right=1000 > y.right=400
# Coalescence at [0, 400): create ancestor, record edges
# x has leftover [400, 1000): passes through

print(f"\n=== Coalescence of lineage 0 and lineage 1 ===")
print(f"  x: {Segment.show_chain(x)}")
print(f"  y: {Segment.show_chain(y)}")
```

```
# The overlap is [0, 400): both have material there
overlap_left, overlap_right = 0, min(x.right, y.right)
print(f"  Overlap: [{overlap_left}, {overlap_right})")
print(f"  Edges: ({overlap_left}, {overlap_right}, {ancestor_node}, {x.node})")
print(f"  Edges: ({overlap_left}, {overlap_right}, {ancestor_node}, {y.node})")

# After merge: [0, 400: node 10] -> [400, 1000: node 0]
merged_seg1 = pool.alloc(left=0, right=400, node=ancestor_node)
merged_seg2 = pool.alloc(left=400, right=1000, node=x.node)
merged_seg1.next = merged_seg2
merged_seg2.prev = merged_seg1
merged_lin = Lineage(head=merged_seg1, tail=merged_seg2, population=0)

print(f"  Merged chain: {Segment.show_chain(merged_lin.head)}")
print(f"\n=== Final state ===")
print(f"  Merged lineage: {Segment.show_chain(merged_lin.head)}")
print(f"  Lineage 2: {Segment.show_chain(lineages[2].head)}")
print(f"  Right fragment: {Segment.show_chain(lineages[3].head)}")
```

Hudson's Algorithm

The mainspring: the event-driven loop that brings the whole mechanism to life.

Hudson's algorithm (Hudson 1983, 2002; Kelleher et al. 2016) is the main simulation loop of *msprime* – the ticking of the clock. Everything we have built so far comes together here: the exponential race from *The Coalescent Process*, the segment chains and Fenwick tree from *Segments & the Fenwick Tree*, and the event-driven philosophy described in the *Overview of msprime*.

If the coalescent process is the escapement (setting the rhythm), and the segments and Fenwick tree are the gear train (transmitting that rhythm efficiently), then Hudson's algorithm is the mainspring – the driving force that makes the whole mechanism tick. It orchestrates the exponential race between coalescence, recombination, migration, and demographic events – always executing whichever happens first, updating the state, and repeating until every genomic position has found its MRCA.

Note

Prerequisites. This chapter assumes familiarity with:

- The **exponential race** (Step 3 of *The Coalescent Process*): how competing exponential waiting times determine which event fires next.
- **Segment chains** and the **Fenwick tree** (*Segments & the Fenwick Tree*): the data structures for tracking ancestral material and selecting breakpoints.
- The concept of **event-driven simulation** (explained in the *Overview of msprime*): jumping directly from event to event rather than stepping through time uniformly.

Step 1: The Main Loop Structure

The algorithm is conceptually simple. At each iteration:

1. Compute the rate of every possible event
2. Draw independent exponential waiting times
3. Check if a demographic event (population size change, etc.) occurs first

4. Execute the earliest event
5. Update data structures
6. Repeat

This is the ticking of the clock: each tick is one event, and between ticks, nothing happens. The clock never idles – it always jumps directly to the next meaningful moment.

i Closing a confusion gap – Event-driven vs. time-stepped simulation

In a **time-stepped** simulation (like the Wright-Fisher model), you advance the clock by one fixed unit (one generation) and process everything that happens in that unit. In an **event-driven** simulation, you ask “when does the *next* event happen?” and jump directly there. Event-driven simulation is vastly more efficient when events are rare relative to the time scale – which they are in the coalescent, where $O(k^2)$ lineages might go thousands of generations between events. The expected waiting time between events is $O(1/k^2)$ in coalescent units, so jumping directly saves $O(k^2)$ wasted iterations per event.

```
import numpy as np
import math

INFINITY = float('inf')

def hudson_simulate(state, end_time=INFINITY):
    """The main Hudson simulation loop.

    This is the heart of msprime -- the ticking of the clock.
    Each iteration: compute rates, race exponentials, execute the winner.

    Parameters
    -----
    state : SimulationState
        Contains lineages, Fenwick trees, rate maps, populations, etc.

    Returns
    -----
    state : SimulationState
        Final state with completed tree sequence tables.
    """
    while not state.is_completed():
        if state.t >= end_time:
            break

        # === Step 1: Compute event rates ===
        # Each event type proposes an exponential waiting time.

        # Recombination: rate comes from the Fenwick tree -- O(log n)
        re_rate = state.get_total_recombination_rate()
        t_re = np.random.exponential(1.0 / re_rate) if re_rate > 0 else INFINITY

        # Gene conversion (within segments): also from a Fenwick tree
        gc_rate = state.get_total_gc_rate()
        t_gc = np.random.exponential(1.0 / gc_rate) if gc_rate > 0 else INFINITY
```

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```

# Gene conversion (left of first segment): proportional to lineage count
gc_left_rate = state.get_total_gc_left_rate()
t_gc_left = (np.random.exponential(1.0 / gc_left_rate)
             if gc_left_rate > 0 else INFINITY)

# Coalescence: one exponential per population
# (each population has its own rate based on lineage count and size)
t_ca = INFINITY
ca_pop = -1
for pop in state.populations:
    if pop.num_ancestors > 1:
        t = pop.get_coalescence_waiting_time(state.t)
        if t < t_ca:
            t_ca = t
            ca_pop = pop.id

# Migration: one exponential per (source, dest) pair
t_mig = INFINITY
mig_source, mig_dest = -1, -1
for j, pop in enumerate(state.populations):
    if pop.num_ancestors == 0:
        continue
    for k in range(len(state.populations)):
        if j == k:
            continue
        rate = pop.num_ancestors * state.migration_matrix[j][k]
        if rate > 0:
            t = np.random.exponential(1.0 / rate)
            if t < t_mig:
                t_mig = t
                mig_source, mig_dest = j, k

# === Step 2: Find the minimum (the exponential race winner) ===
min_time = min(t_re, t_ca, t_gc, t_gc_left, t_mig)

# === Step 3: Check demographic events ===
# Demographic events (size changes, splits) are deterministic
# in time, not random. If one falls before the next random event,
# it fires first.
if state.t + min_time > state.next_demographic_event_time():
    state.execute_demographic_event()
    continue # re-enter the loop with updated parameters

# === Step 4: Advance time and execute the winning event ===
state.t += min_time

if min_time == t_re:
    state.recombination_event()
elif min_time == t_gc:
    state.gene_conversion_within_event()
elif min_time == t_gc_left:

```

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```

        state.gene_conversion_left_event()
    elif min_time == t_ca:
        state.coalescence_event(ca_pop)
    elif min_time == t_mig:
        state.migration_event(mig_source, mig_dest)

    return state

```

Now let us examine each event handler in detail, starting with recombination.

Step 2: The Recombination Event in Detail

When recombination wins the race, we split one lineage into two. The full procedure:

1. **Choose the breakpoint** using the Fenwick tree (as derived in *Segments & the Fenwick Tree*).
2. **Find the segment** containing the breakpoint.
3. **Split the segment chain** into left and right parts.
4. **Update the Fenwick tree** with new mass values for both parts.
5. **Create a new lineage** for the right part.

There are two sub-cases depending on where the breakpoint falls:

Case 1: Breakpoint falls WITHIN a segment

```

Before:  ... x =====|===== y ...
                bp

After:   ... x ===== y          (left lineage)
                a ===== ... (right lineage)

```

Case 2: Breakpoint falls in a GAP between segments

```

Before:  ... x ===== y ===== ...
                |
                bp

After:   ... x =====          (left lineage)
                y ===== ... (right lineage)

```

i Closing a confusion gap – Why two cases?

After multiple recombination events, a lineage's segment chain may have gaps – intervals where ancestry has been transferred to another lineage. A recombination breakpoint can land either *within* a segment (splitting it) or *in a gap* (simply detaching the right portion of the chain). Both cases reduce to pointer rewiring, but the bookkeeping differs slightly: in Case 1, we must create a new segment for the right half; in Case 2, we just disconnect existing segments.

```

def recombination_event(state, label=0):
    """Execute a recombination event.

```

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```

This is a detailed implementation following msprime's algorithms.py.
"""
state.num_re_events += 1

# Step 1: Choose breakpoint using the Fenwick tree -- O(log n)
y, bp = choose_breakpoint(
    state.recomb_mass_index[label],
    state.recomb_map
)
left_lineage = y.lineage
x = y.prev # segment to the left of y (may be None if y is head)

if y.left < bp:
    # Case 1: breakpoint falls WITHIN segment y
    #   x           y
    # =====  ===/=====  ...
    #           bp
    #
    # Split y into y=[left, bp) and alpha=[bp, right)
    alpha = state.pool.copy(y) # create new segment for right half
    alpha.left = bp           # alpha starts at the breakpoint
    alpha.prev = None         # alpha is the head of the new chain

    if y.next is not None:
        y.next.prev = alpha # alpha inherits y's successors
        y.next = None       # y is now the tail of the left chain
        y.right = bp        # trim y to end at breakpoint

    # Update Fenwick tree for the trimmed y -- O(log n)
    state.set_segment_mass(y)
    left_lineage.tail = y

else:
    # Case 2: breakpoint falls in a GAP to the left of y
    #   x           y
    # =====  |  =====  ...
    #           bp
    #
    # Just detach y from x -- pure pointer rewiring, O(1)
    x.next = None
    y.prev = None
    alpha = y
    left_lineage.tail = x

# Step 2: Create new lineage for the right part
right_lineage = Lineage(head=alpha, tail=find_tail(alpha),
                        population=left_lineage.population,
                        label=label)
alpha.lineage = right_lineage
state.set_segment_mass(alpha) # register mass in Fenwick tree
state.add_lineage(right_lineage)

```

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```

    return left_lineage, right_lineage

def find_tail(seg):
    """Follow the chain to find the tail."""
    while seg.next is not None:
        seg = seg.next
    return seg

```

The recombination event *increases* the lineage count by one. The coalescence event, which we examine next, *decreases* it by one. The interplay between these two forces drives the entire simulation.

Step 3: The Coalescence Event in Detail

The coalescence event is the most complex operation. Two randomly chosen lineages merge their ancestral material, and at positions where they overlap, a new common ancestor node is created.

The core difficulty: when two segment chains overlap, we need to:

- Create a new ancestor node at overlapping positions
- Record edges in the tree sequence tables
- Update the overlap counter S
- Handle non-overlapping regions (they just pass through)
- Handle partial overlaps (segment boundaries don't align)

The merge algorithm walks through both chains simultaneously, like the merge step of merge sort, but with genealogical bookkeeping.

```

Lineage x:  [0, 300)      [500, 800)      [900, 1000)
Lineage y:           [200, 600)           [850, 950)

Merge result:
[0, 200):    only x  ->  passes through from x
[200, 300):  overlap ->  COALESCENCE! new ancestor u
[300, 500):  only y  ->  passes through from y
[500, 600):  overlap ->  COALESCENCE! (same ancestor u)
[600, 800):  only x  ->  passes through from x
[850, 900):  only y  ->  passes through from y
[900, 950):  overlap ->  COALESCENCE! (same ancestor u)
[950, 1000): only x  ->  passes through from x

```

Closing a confusion gap – The merge walk

The merge algorithm is often the hardest part of the simulator to understand. Think of it this way: lay both segment chains side by side on the genome number line. Walk from left to right. At each position, one of three things is true:

1. **Only one chain has material here** – that material passes through unchanged to the merged lineage.
2. **Both chains have material here** – this is a *coalescence*: we create a new ancestor node, record edges linking both children to it, and decrement the overlap counter.
3. **Neither chain has material here** – nothing happens; skip ahead.

The walk advances by jumping to the next boundary (left or right endpoint of any segment in either chain). At each boundary, the situation may change from case 1 to case 2 or vice versa.

```
def merge_two_ancestors(state, pop_index, label, x, y):
    """Merge two lineages' segment chains.

    This is the heart of the coalescence operation. It walks through
    both chains simultaneously, handling overlaps and pass-throughs.

    Parameters
    -----
    state : SimulationState
    pop_index : int
        Population where the coalescence occurs.
    x, y : Segment
        Head segments of the two lineages to merge.

    Returns
    -----
    new_lineage : Lineage
        The merged lineage (may be None if fully coalesced).
    """
    state.num_ca_events += 1
    new_lineage = Lineage(head=None, tail=None, population=pop_index,
                          label=label)
    coalescence = False
    u = -1 # tree-sequence node for the common ancestor

    while x is not None or y is not None:
        alpha = None # segment to add to the merged chain

        if x is None or y is None:
            # === One chain exhausted: absorb the rest of the other ===
            if x is not None:
                alpha = x
                x = None # mark as consumed
            if y is not None:
                alpha = y
                y = None

        else:
            # === Both chains still active ===
            # Ensure x starts at or before y (simplifies case analysis)
            if y.left < x.left:
                x, y = y, x

            if x.right <= y.left:
                # NO OVERLAP: x ends before y starts
                #   x           y
                # =====
                alpha = x
                x = x.next
```

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```

alpha.next = None

elif x.left != y.left:
    # PARTIAL OVERLAP: x starts before y
    #   x
    # ====|=====
    #   y
    #   =====
    # Trim the non-overlapping prefix of x
    alpha = state.pool.alloc(
        left=x.left, right=y.left, node=x.node)
    x.left = y.left # advance x to where y starts

else:
    # === COALESCENCE: x and y start at the same position ===
    if not coalescence:
        coalescence = True
        u = state.store_node(pop_index) # create ancestor node

    left = x.left
    right = min(x.right, y.right)

    # Update overlap counter: one fewer lineage covers [left, right)
    state.decrement_overlap(left, right)

    # Record edges: both x and y are children of u
    state.store_edge(left, right, parent=u, child=x.node)
    state.store_edge(left, right, parent=u, child=y.node)

    # Create the coalesced segment (under ancestor u)
    alpha = state.pool.alloc(left=left, right=right, node=u)

    # Trim x and y: consume the overlapping portion
    if x.right == right:
        state.pool.free(x) # x is fully consumed
        x = x.next
    else:
        x.left = right # x has leftover material

    if y.right == right:
        state.pool.free(y) # y is fully consumed
        y = y.next
    else:
        y.left = right # y has leftover material

    # === Append alpha to the merged chain ===
    if alpha is not None:
        alpha.lineage = new_lineage
        alpha.prev = new_lineage.tail
        state.set_segment_mass(alpha) # register in Fenwick tree

    if new_lineage.head is None:

```

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```

        new_lineage.head = alpha
    else:
        new_lineage.tail.next = alpha
        new_lineage.tail = alpha

    # If the merged lineage has remaining segments, add it back
    if new_lineage.head is not None:
        state.defrag_segment_chain(new_lineage)
        state.add_lineage(new_lineage)

    return new_lineage

```

After the merge, the resulting chain may contain adjacent segments with the same node ID – a cosmetic issue that the next step addresses.

Step 4: Segment Chain Defragmentation

After a coalescence, adjacent segments in the merged chain might have the same node ID. This happens when two non-overlapping segments with the same ancestry end up next to each other. We merge them to keep the chains clean:

```

def defrag_segment_chain(lineage):
    """Merge adjacent segments with the same node ID.

    Before:  [0, 300: node 5] -> [300, 800: node 5] -> [800, 1000: node 3]

    After:   [0, 800: node 5] -> [800, 1000: node 3]

    This reduces the number of segments (and Fenwick tree entries),
    keeping the data structures lean.
    """
    seg = lineage.head
    while seg is not None and seg.next is not None:
        if seg.node == seg.next.node and seg.right == seg.next.left:
            # Merge: extend seg to cover seg.next
            to_free = seg.next
            seg.right = to_free.right
            seg.next = to_free.next
            if seg.next is not None:
                seg.next.prev = seg
            # Free the absorbed segment
            # (in practice: return to pool and update Fenwick tree)
        else:
            seg = seg.next

    # Update tail
    lineage.tail = seg

```

Step 5: The Migration Event

Migration is the simplest event. A randomly chosen lineage moves from one population to another. In the backward-time view, this means the lineage's ancestor lived in a different population one generation earlier.

```
def migration_event(state, source, dest):
    """Move a random lineage from source to dest population.

    This is the simplest event type: no segment manipulation needed,
    just update the lineage's population assignment.
    """
    # Choose a random lineage from the source population
    source_pop = state.populations[source]
    idx = np.random.randint(source_pop.num_ancestors)
    lineage = source_pop.remove(idx)

    # Add to destination
    lineage.population = dest
    state.populations[dest].add(lineage)
```

With all event handlers defined, let us assemble the complete algorithm.

Step 6: The Complete Algorithm

Let's put it all together into a minimal but complete implementation. This is the master clockmaker's bench in miniature: a working simulation that uses all the data structures from *Segments & the Fenwick Tree* and all the mathematics from *The Coalescent Process*.

```
class MinimalSimulator:
    """A minimal but complete Hudson's algorithm implementation.

    This implements the core simulation loop with coalescence and
    recombination (no gene conversion, no migration for simplicity).
    """

    def __init__(self, n, sequence_length, recombination_rate, pop_size=1.0):
        self.n = n
        self.L = sequence_length
        self.Ne = pop_size
        self.t = 0.0

        self.recomb_rate = recombination_rate
        # This teaching simulator uses a haploid population-size convention.
        self.rho = 2 * pop_size * recombination_rate * sequence_length

        self.max_segs = 100 * n
        self.segments = [None] * (self.max_segs + 1)
        self.free_segs = list(range(self.max_segs, 0, -1))
        self.mass_index = FenwickTree(self.max_segs)

        self.lineages = []
        for i in range(n):
            seg_idx = self.free_segs.pop()
            seg = Segment(index=seg_idx, left=0, right=sequence_length,
                          node=i)
            self.segments[seg_idx] = seg
            lin = Lineage(head=seg, tail=seg, population=0)
            seg.lineage = lin
```

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```

        self.lineages.append(lin)

        mass = recombination_rate * sequence_length
        self.mass_index.set_value(seg_idx, mass)

    self.edges = []
    self.nodes = []
    for i in range(n):
        self.nodes.append((0.0, 0))

    self.num_re_events = 0
    self.num_ca_events = 0

    def is_completed(self):
        """Check if all positions have found their MRCA."""
        return len(self.lineages) == 0

    def get_recomb_mass(self, seg):
        """Recombination mass of a segment."""
        left_bound = seg.prev.right if seg.prev is not None else seg.left
        return self.recomb_rate * (seg.right - left_bound)

    def simulate(self):
        """Run the simulation until completion."""
        while not self.is_completed():
            k = len(self.lineages)

            re_total = self.mass_index.get_total()
            t_re = (np.random.exponential(1.0 / re_total)
                    if re_total > 0 else INFINITY)

            coal_rate = k * (k - 1) / 2
            t_ca = INFINITY
            if coal_rate > 0:
                t_ca = self.Ne * np.random.exponential(
                    1.0 / coal_rate)

            min_t = min(t_re, t_ca)
            self.t += min_t

            if min_t == t_re:
                self._recombination_event()
            else:
                self._coalescence_event()

        return self.edges, self.nodes

    @staticmethod
    def _segments_in(lineage):
        """Return the segments in a lineage as a list."""
        segments = []
        seg = lineage.head

```

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```

    while seg is not None:
        segments.append(seg)
        seg = seg.next
    return segments

def _register_lineage(self, lineage):
    """Set lineage pointers and recombination masses for one chain."""
    segments = self._segments_in(lineage)
    if not segments:
        return
    lineage.head = segments[0]
    lineage.tail = segments[-1]
    for seg in segments:
        seg.lineage = lineage
        self.mass_index.set_value(seg.index, self.get_recomb_mass(seg))

def _release_segments(self, segments):
    """Return segments to the simulator's small educational pool."""
    for seg in segments:
        self.mass_index.set_value(seg.index, 0.0)
        self.segments[seg.index] = None
        self.free_segs.append(seg.index)

def _alloc_segment(self, left, right, node):
    if not self.free_segs:
        raise RuntimeError(
            "Segment pool exhausted; reduce the recombination rate or "
            "increase MinimalSimulator.max_segs"
        )
    index = self.free_segs.pop()
    seg = Segment(index=index, left=left, right=right, node=node)
    self.segments[index] = seg
    return seg

@staticmethod
def _node_at(segments, position):
    for seg in segments:
        if seg.left <= position < seg.right:
            return seg.node
    return None

def _coverage(self, lineages, position):
    """Count extant ancestors at a genomic position."""
    return sum(
        self._node_at(self._segments_in(lineage), position) is not None
        for lineage in lineages
    )

def _recombination_event(self):
    """Handle a recombination event: split one lineage into two."""
    self.num_re_events += 1
    random_mass = np.random.uniform(0, self.mass_index.get_total())

```

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```

seg_idx = self.mass_index.find(random_mass)
y = self.segments[seg_idx]

cum_mass = self.mass_index.get_cumulative_sum(seg_idx)
mass_from_right = cum_mass - random_mass
bp = y.right - mass_from_right / self.recomb_rate

# ``get_recomb_mass`` includes the gap between this segment and its
# predecessor. A breakpoint in that gap is still a real event: it
# separates the two segment chains even though it does not cut either
# segment. Only the (measure-zero) right endpoint is outside the
# selectable interval.
if bp >= y.right:
    return

old_lin = y.lineage
old_segments = self._segments_in(old_lin)
for seg in old_segments:
    self.mass_index.set_value(seg.index, 0.0)

if bp <= y.left:
    # The breakpoint lies in a gap between ancestral segments.
    left_tail = y.prev
    if left_tail is None:
        self._register_lineage(old_lin)
        return
    left_tail.next = None
    y.prev = None
    left_head = old_lin.head
    right_head = y
else:
    old_next = y.next
    alpha = self._alloc_segment(bp, y.right, y.node)
    alpha.next = old_next
    if old_next is not None:
        old_next.prev = alpha
    y.right = bp
    y.next = None
    left_head = old_lin.head
    left_tail = y
    right_head = alpha

right_tail = right_head
while right_tail.next is not None:
    right_tail = right_tail.next

self.lineages.remove(old_lin)
left_lin = Lineage(left_head, left_tail, old_lin.population, old_lin.label)
right_lin = Lineage(right_head, right_tail, old_lin.population, old_lin.label)
self._register_lineage(left_lin)
self._register_lineage(right_lin)
self.lineages.extend([left_lin, right_lin])

```

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```

def _coalescence_event(self):
    """Handle a coalescence event: merge two lineages into one."""
    self.num_ca_events += 1
    k = len(self.lineages)

    all_lineages = list(self.lineages)
    i, j = sorted(np.random.choice(k, 2, replace=False))
    y_lin = self.lineages.pop(j)
    x_lin = self.lineages.pop(i)
    x_segments = self._segments_in(x_lin)
    y_segments = self._segments_in(y_lin)

    breakpoints = sorted({
        coordinate
        for seg in x_segments + y_segments
        for coordinate in (seg.left, seg.right)
    })
    output = []
    parent = None

    for left, right in zip(breakpoints[:-1], breakpoints[1:]):
        midpoint = (left + right) / 2
        x_node = self._node_at(x_segments, midpoint)
        y_node = self._node_at(y_segments, midpoint)
        if x_node is None and y_node is None:
            continue
        if x_node is None or y_node is None:
            output.append((left, right, x_node if y_node is None else y_node))
            continue

        if parent is None:
            parent = len(self.nodes)
            self.nodes.append((self.t, x_lin.population))
            self.edges.append((left, right, parent, x_node))
            self.edges.append((left, right, parent, y_node))

        # Once only the chosen pair covers an interval, their common
        # ancestor is its local MRCA and no material remains to simulate.
        if self._coverage(all_lineages, midpoint) > 2:
            output.append((left, right, parent))

    self._release_segments(x_segments + y_segments)

    if output:
        # Coalesce adjacent pieces represented by the same node.
        merged = []
        for left, right, node in output:
            if merged and merged[-1][1] == left and merged[-1][2] == node:
                merged[-1] = (merged[-1][0], right, node)
            else:
                merged.append((left, right, node))

```

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```

new_lin = Lineage(head=None, tail=None, population=x_lin.population)
for left, right, node in merged:
    seg = self._alloc_segment(left, right, node)
    seg.prev = new_lin.tail
    if new_lin.head is None:
        new_lin.head = seg
    else:
        new_lin.tail.next = seg
    new_lin.tail = seg
self._register_lineage(new_lin)
self.lineages.append(new_lin)

```

The returned edges and nodes use the same core table semantics as `tskit`. The regression suite loads them into a `tskit.TableCollection` and verifies that every marginal tree has one root.

```

# Run a small simulation
np.random.seed(42)
sim = MinimalSimulator(n=5, sequence_length=1000,
                      recombination_rate=0.001, pop_size=1.0)
edges, nodes = sim.simulate()
print(f"Simulation complete!")
print(f"  Coalescence events: {sim.num_ca_events}")
print(f"  Recombination events: {sim.num_re_events}")
print(f"  Nodes: {len(nodes)} ({sim.n} samples + "
      f"{len(nodes) - sim.n} ancestors)")
print(f"  Edges: {len(edges)}")
print(f"  TMRCA: {sim.t:.4f} coalescent units")

```

You have now seen the complete algorithm – every tick of the clock, from rate computation through event execution. Let us analyze its computational complexity.

Step 7: Complexity Analysis

Let's count the operations per simulation step:

Operation	Complexity	Why
Compute recomb rate	$O(\log n)$	<code>FenwickTree.get_total()</code>
Compute coal rate	$O(1)$	Just $k(k-1)/2$
Draw exponentials	$O(1)$	Random number generation
Choose breakpoint	$O(\log n)$	<code>FenwickTree.find()</code>
Split segment	$O(1)$	Pointer rewiring
Update Fenwick tree	$O(\log n)$	<code>FenwickTree.set_value()</code>
Choose pair for coal.	$O(1)$	Random selection
Merge two chains	$O(s)$	s = segments in the two chains
Update overlap counter	$O(\log n)$	AVL tree operations

Total per event: $O(s \log n)$ where s is the number of segments in the merging chains. In practice, s is small for most events.

Total simulation: $O(n + E \log n)$ where E is the total number of events. The number of events is $O(n + \rho)$ where $\rho = 2N_e r L$ for the haploid convention used by `MinimalSimulator` (or $4N_e r L$ for diploids). This gives msprime its

celebrated near-linear scaling in n .

i Probability Aside – Why $O(n + \rho)$ events?

The expected number of coalescence events is $n - 1$ (each reduces the lineage count by 1, from n to 1). The expected number of recombination events is $O(\rho)$ because the total recombination rate summed over the coalescent tree is proportional to ρ . More precisely, the expected number of recombination events is $\frac{\rho}{2} \cdot E[L_{\text{total}}]$ where $E[L_{\text{total}}] = 2 \sum_{k=1}^{n-1} 1/k$ is the expected total branch length. For large ρ , this is the dominant term.

Step 8: Comparison with ms

The original `ms` program (Hudson 2002) uses the same algorithm but with $O(n)$ data structures instead of $O(\log n)$ Fenwick trees. This makes `ms` quadratic in the number of segments for large genomes.

Property	ms (Hudson 2002)	msprime (Kelleher 2016)
Breakpoint selection	$O(n)$ linear scan	$O(\log n)$ Fenwick tree
Rate maintenance	$O(n)$ recompute	$O(\log n)$ incremental update
Output format	Text (variant matrix)	Tree sequence (tskit)
Memory	$O(n \cdot L)$ for output	$O(n + E)$ for tree sequence
Scaling	$\sim n^2 \cdot L$	$\sim n \log n + n \cdot L$

The tree sequence output is particularly important: instead of storing the full variant matrix (which is $O(n \cdot S)$ where S is the number of segregating sites), `msprime` stores edges and nodes, which is $O(n + \rho)$ – often orders of magnitude smaller.

This is the culmination of the master clockmaker's bench: a simulation algorithm that scales gracefully to millions of samples and whole chromosomes. In the next chapter, we add population structure and demographic history – the case and dial that shape the genealogy's form.

Exercises

i Exercise 1: Build a minimal simulator

Use the `MinimalSimulator` class to simulate genealogies for $n = 10, 50, 100$ with $L = 10^5$ bp and $r = 10^{-8}$. For each, record the number of trees in the tree sequence (= number of distinct marginal trees). Plot $E[\text{num trees}]$ vs n and compare to the theoretical expectation.

i Exercise 2: Verify the exponential race

Instrument the main loop to record which event type wins at each step. Verify that the fraction of coalescence vs recombination events matches the ratio of their rates.

i Exercise 3: The coalescence merge

Create two lineages with overlapping segment chains (draw them on paper first). Walk through `merge_two_ancestors` by hand, tracking which segments are created, which edges are recorded, and how the overlap counter changes.

i Exercise 4: Fenwick vs naive

Time the simulation with and without the Fenwick tree (replace `FenwickTree.find()` with a linear scan). At what genome length does the Fenwick tree become faster?

Next: *Demographics & Population* – how population size changes, migration, and growth affect the simulation.

Solutions**i Solution 1: Build a minimal simulator**

The expected number of distinct marginal trees grows roughly linearly with the diploid population-scaled recombination rate $\rho = 4N_e r L$ (or $2N_e r L$ in the haploid convention used by `MinimalSimulator`). We count trees by counting how many recombination events produced distinct breakpoints.

```
import numpy as np

r = 1e-8
L = 1e5
n_reps = 50

print(f'{n':>5s}  {'E[num trees]':>14s}  {'E[coal events]':>16s}  "
      f"{'E[recomb events]':>18s}")

for n in [10, 50, 100]:
    num_trees_list = []
    coal_list = []
    recomb_list = []

    for _ in range(n_reps):
        sim = MinimalSimulator(n=n, sequence_length=L,
                               recombination_rate=r, pop_size=1.0)
        edges, nodes = sim.simulate()

        # Number of trees = number of distinct edge left boundaries + 1
        # (each recombination creates a new breakpoint and a new tree)
        breakpoints = set()
        for left, right, parent, child in edges:
            breakpoints.add(left)
            breakpoints.add(right)
        num_trees = len(breakpoints) - 1 # subtract the 0 and L endpoints
        num_trees = max(1, num_trees)

        num_trees_list.append(num_trees)
        coal_list.append(sim.num_ca_events)
        recomb_list.append(sim.num_re_events)

    print(f"{n:5d}  {np.mean(num_trees_list):14.1f}  "
          f"{np.mean(coal_list):16.1f}  "
          f"{np.mean(recomb_list):18.1f}")

# The number of trees grows roughly as rho * log(n) for the
# standard coalescent, since the expected number of recombination
# events on the tree is proportional to rho * sum(1/k).
```

i Solution 2: Verify the exponential race

We instrument the `MinimalSimulator` to track which event type wins at each step, then compare to the expected fractions based on the rates.

```
import numpy as np

n = 20
L = 1e4
r = 1e-4
rho = 2 * 1.0 * r * L # haploid convention in MinimalSimulator

sim = MinimalSimulator(n=n, sequence_length=L,
                       recombination_rate=r, pop_size=1.0)
edges, nodes = sim.simulate()

total_events = sim.num_ca_events + sim.num_re_events
coal_frac = sim.num_ca_events / total_events
recomb_frac = sim.num_re_events / total_events

print(f"Total events: {total_events}")
print(f"Coalescence events: {sim.num_ca_events} ({coal_frac:.3f})")
print(f"Recombination events: {sim.num_re_events} ({recomb_frac:.3f})")
print(f"\nNote: The fraction varies over the simulation because rates")
print(f"change as lineages are added (recombination) and removed")
print(f"(coalescence). The coalescence rate is k*(k-1)/2 while the")
print(f"recombination rate is proportional to total segment mass.")
print(f"At each step, P(coal) = coal_rate / (coal_rate + recomb_rate).")
```

i Solution 3: The coalescence merge

Consider two lineages with the following segment chains:

```
Lineage x: [0, 400: node 0] -> [600, 1000: node 0]
Lineage y: [200, 800: node 1]
```

Walking through `merge_two_ancestors`:

1. Both active, $x.\text{left}=0 < y.\text{left}=200$: x starts before y . $x.\text{right}=400 > y.\text{left}=200$, and $x.\text{left}=0 < y.\text{left}=200$, so partial overlap. Trim prefix: create segment $[0, 200: \text{node } 0]$ (passes through from x). Set $x.\text{left} = 200$.
2. Now $x=[200, 400: \text{node } 0]$, $y=[200, 800: \text{node } 1]$. Both start at 200. **Coalescence!** Create ancestor node u . $\text{left}=200$, $\text{right}=\min(400,800)=400$. Record edges: $(200, 400, u, 0)$ and $(200, 400, u, 1)$. Decrement overlap counter for $[200, 400]$. Create coalesced segment $[200, 400: \text{node } u]$. x is fully consumed ($x.\text{right}==400$), advance to $x.\text{next}=[600, 1000: \text{node } 0]$. y has leftover: $y.\text{left} = 400$.
3. Now $x=[600, 1000: \text{node } 0]$, $y=[400, 800: \text{node } 1]$. $y.\text{left}=400 < x.\text{left}=600$. Swap so $x=y$: $x=[400, 800: \text{node } 1]$, $y=[600, 1000: \text{node } 0]$. $x.\text{right}=800 > y.\text{left}=600$, and $x.\text{left}=400 < y.\text{left}=600$: partial overlap. Create segment $[400, 600: \text{node } 1]$ (passes through from x). Set $x.\text{left} = 600$.
4. Now $x=[600, 800: \text{node } 1]$, $y=[600, 1000: \text{node } 0]$. Both start at 600. **Coalescence!** $\text{left}=600$, $\text{right}=\min(800,1000)=800$. Record edges: $(600, 800, u, 1)$ and $(600, 800, u, 0)$. Decrement overlap for

[600, 800). Create coalesced segment [600, 800: node u]. x is fully consumed, x=None. y has leftover: y.left = 800.

5. x is None: absorb rest of y=[800, 1000: node 0]. Segment [800, 1000: node 0] passes through.

Final merged chain:

```
[0, 200: node 0] -> [200, 400: node u] -> [400, 600: node 1]
-> [600, 800: node u] -> [800, 1000: node 0]
```

Edges recorded: (200, 400, u, 0), (200, 400, u, 1), (600, 800, u, 1), (600, 800, u, 0).

The overlap counter was decremented at [200, 400) and [600, 800) – the intervals where both lineages had ancestral material.

Solution 4: Fenwick vs naive

We replace `FenwickTree.find()` with a linear scan and compare wall-clock times for increasing genome lengths.

```
import numpy as np
import time

def naive_find(values, target):
    """Linear scan to find the index where cumulative sum >= target."""
    cumsum = 0
    for i in range(1, len(values)):
        cumsum += values[i]
        if cumsum >= target:
            return i
    return len(values) - 1

n = 50
r = 1e-8

print(f"{'L':>10s} {'Fenwick (s)':>12s} {'Naive (s)':>12s} {'Speedup':>8s}")
for L in [1e4, 1e5, 1e6, 1e7]:
    # Fenwick-based simulation
    start = time.time()
    sim = MinimalSimulator(n=n, sequence_length=L,
                           recombination_rate=r, pop_size=1.0)

    sim.simulate()
    t_fenwick = time.time() - start

    # For the naive version, we estimate the time per find() operation
    # by running find() on a flat array with the same number of segments.
    n_segs = sim.num_re_events + n # approximate number of segments
    values = np.random.exponential(1.0, size=n_segs + 1)
    total = values.sum()

    n_finds = 1000
    start = time.time()
    for _ in range(n_finds):
        target = np.random.uniform(0, total)
        naive_find(values, target)
    t_naive_per_find = (time.time() - start) / n_finds
```

```

total_events = sim.num_ca_events + sim.num_re_events
t_naive_estimate = t_naive_per_find * total_events + t_fenwick * 0.5

speedup = t_naive_estimate / max(t_fenwick, 1e-6)
print(f"{L:10.0f}   {t_fenwick:12.4f}   {t_naive_estimate:12.4f}   "
      f" {speedup:8.1f}x")

print(f"\nThe Fenwick tree becomes faster when the number of segments is")
print(f"large enough that O(log n) << O(n). For typical genome lengths")
print(f"(L >= 1e5 bp), the Fenwick tree provides a significant speedup.")

```

Demographics & Population

The case and dial: the population history that shapes the genealogy.

The coalescent process describes how lineages merge in a constant-size population. But real populations change size, split, merge, and exchange migrants. msprime handles all of this through a **demographic model** that modifies the simulation parameters at specified times.

In our watch metaphor, if *Hudson's Algorithm* is the mainspring (the ticking of the clock), demographics are the case and dial – the external structure that determines what the clock *looks like* from the outside. Two watches can share an identical movement but tell very different stories if their cases and dials differ. Likewise, the same coalescent algorithm produces very different genealogies depending on the demographic history.

Note

Prerequisites. This chapter builds on *Hudson's Algorithm*, where we implemented the main simulation loop. Specifically, you should understand:

- How the main loop checks for **demographic events** before executing the next random event (Step 1 of *Hudson's Algorithm*).
- How **coalescence waiting times** depend on population size (Step 6 of *The Coalescent Process*).
- The **exponential race** between event types (Step 3 of *The Coalescent Process*).

Step 1: Population Size Changes

The simplest demographic event: the population size changes instantaneously at some time in the past.

Why it matters. Population size controls the coalescence rate:

$$\lambda_{\text{coal}}(t) = \frac{\binom{k}{2}}{N(t)}$$

A smaller population means faster coalescence (lineages find common ancestors sooner). A larger population means slower coalescence (more genetic diversity is maintained).

Probability Aside – Population size as a time-scaling factor

In coalescent units, time is measured in multiples of N generations. When N changes, the “speed” of coalescent time changes too. A bottleneck (small N) compresses real time: many generations pass per coalescent unit, so coalescences happen rapidly. An expansion (large N) stretches real time: few coalescences occur per generation. The demographic model is essentially a variable-speed clock that accelerates and decelerates the coalescent.

In the simulation, a population size change is implemented as a **modifier event**: when the simulation clock reaches the event time, the population parameters are updated, and the simulation continues with the new rates.

```
import numpy as np

class Population:
    """A population with time-varying size.

    Population size follows the formula:
    
$$N(t) = \text{start\_size} * \exp(-\text{growth\_rate} * (t - \text{start\_time}))$$


    This is an exponential model where start_size and growth_rate
    are updated by demographic events.
    """

    def __init__(self, start_size=1.0, growth_rate=0.0, start_time=0.0):
        self.start_size = start_size      # size at start_time
        self.growth_rate = growth_rate    # exponential growth rate
        self.start_time = start_time      # time when these parameters took effect
        self.num_ancestors = 0           # current number of lineages in this pop

    def get_size(self, t):
        """Population size at time t (backwards).

        
$$N(t) = \text{start\_size} * \exp(-\text{growth\_rate} * (t - \text{start\_time}))$$


        When growth_rate > 0: population was SMALLER in the past
        When growth_rate < 0: population was LARGER in the past
        When growth_rate = 0: constant size
        """
        dt = t - self.start_time
        return self.start_size * np.exp(-self.growth_rate * dt)

    def set_size(self, new_size, time):
        """Change population size at the given time.

        This creates a discontinuous jump in population size --
        the classic "instantaneous size change" demographic event.
        """
        self.start_size = new_size
        self.growth_rate = 0          # reset to constant size
        self.start_time = time

    def set_growth_rate(self, rate, time):
        """Change growth rate at the given time.

        The size is first computed at the new time (to preserve
        continuity), then the growth rate is changed.
        """
        self.start_size = self.get_size(time)  # ensure continuity
        self.start_time = time
        self.growth_rate = rate
```

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```
# Example: population that was 10x smaller before a bottleneck
pop = Population(start_size=10000)
print(f"Present size: {pop.get_size(0):.0f}")

# At time 1000 generations ago, size drops to 1000
pop.set_size(1000, time=1000)
print(f"Size after bottleneck (t=1000): {pop.get_size(1000):.0f}")

# With exponential growth
pop2 = Population(start_size=10000, growth_rate=0.005, start_time=0)
for t in [0, 100, 500, 1000, 2000]:
    print(f"t={t:>5d}: N = {pop2.get_size(t):.0f}")
```

i The sign convention

msprime measures time **backwards** from the present. A positive growth rate means the population is growing **towards the present**, so it was *smaller* in the past. This can be confusing. The formula $N(t) = N_0 e^{-\alpha t}$ makes the population shrink as t increases (going further into the past).

i Calculus Aside – Exponential growth as a differential equation

The exponential size model $N(t) = N_0 e^{-\alpha t}$ is the solution to the differential equation $dN/dt = -\alpha N$ with initial condition $N(0) = N_0$. Going backwards (increasing t), the population shrinks at a rate proportional to its current size. This is the continuous-time analog of a population that grows by a constant fraction each generation going forward.

With population size changes understood, let us look at a more dramatic event: the bottleneck.

Step 2: The Bottleneck

A bottleneck can be represented by collapsing a short interval of ordinary Kingman coalescent time into a single instant. With dimensionless strength B , run the usual coalescent clock for an interval of length B : while k lineages remain, the next waiting time is exponential with rate $\binom{k}{2}$. Every merger during that interval is recorded at the same simulation time. Pair outcomes are therefore *not* independent.

i Probability Aside – Bottleneck intensity

The bottleneck strength B is measured in coalescent units. It represents the “amount of coalescence” that would have occurred in a population of size 1 for B time units. A higher B means more coalescence: $B = 0$ means no effect, $B \rightarrow \infty$ means all lineages coalesce to a single ancestor. The probability that a specific pair survives the bottleneck without coalescing is e^{-I} , which comes from the survival function of an $\text{Exp}(1)$ random variable.

```
def bottleneck_event(lineages, intensity):
    """Collapse a Kingman interval into an instantaneous bottleneck.

    Parameters
    -----
    lineages : list
```

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```

    Current lineages.
    intensity : float
        Bottleneck intensity (higher = more coalescence).

    Returns
    -----
    lineages : list
        Lineages after the bottleneck.
    coalesced_pairs : list of (int, int)
        Representative input indices for each binary merger.
    """
    if intensity < 0:
        raise ValueError("intensity must be nonnegative")

    coalesced_pairs = []
    # Each entry stores a surviving lineage object and an input index used
    # solely to make the event log interpretable.
    remaining = [(lineage, index) for index, lineage in enumerate(lineages)]
    elapsed = 0.0
    while len(remaining) > 1:
        k = len(remaining)
        rate = k * (k - 1) / 2
        wait = np.random.exponential(1.0 / rate)
        if elapsed + wait > intensity:
            break
        elapsed += wait

        first, second = np.random.choice(k, size=2, replace=False)
        if first < second:
            first, second = second, first
        lineage_i, label_i = remaining.pop(first)
        _lineage_j, label_j = remaining.pop(second)
        coalesced_pairs.append((label_i, label_j))
        # A full ARG implementation would create a new ancestor here. This
        # count-level helper retains one representative lineage instead.
        remaining.append((lineage_i, min(label_i, label_j)))

    return [lineage for lineage, _ in remaining], coalesced_pairs

```

```

# Example
lins = list(range(10)) # 10 lineages
remaining, pairs = bottleneck_event(lins, intensity=2.0)
print(f"Bottleneck with intensity 2.0: {len(pairs)} pairs coalesced")

```

The helper retains one input object as a representative after each merger; a full ancestry simulator would instead create the corresponding ancestor node and edges. `msprime` also exposes `add_simple_bottleneck`, a distinct pedagogical event in which each lineage independently chooses whether to join one common ancestor. It should not be confused with the collapsed Kingman interval above.

Bottlenecks are point events – they happen instantaneously. The next ingredient, migration, is a continuous process that operates alongside coalescence and recombination.

Step 3: Migration

With multiple populations, lineages can move between them. In the coalescent (going backwards), a migration event means that a lineage in population j had an ancestor in population k one generation earlier.

Migration matrix. The rate at which lineages migrate from population j to population k (backwards) is M_{jk} . With n_j lineages in population j , the total migration rate out of j to k is:

$$\lambda_{\text{mig}, j \rightarrow k} = n_j \cdot M_{jk}$$

i Closing a confusion gap – Forward vs backward migration

In the forward direction, M_{jk} is the probability that an individual in population j migrated *from* population k . In the backward direction (coalescent), this means a lineage in j moves *to* k . The migration matrix is the same in both directions (this is a consequence of the diffusion limit).

This directionality confusion is one of the most common sources of error when setting up demographic models. A useful mnemonic: in the backward view, lineages move *toward their ancestors*, so migration from j to k means “this lineage’s ancestor lived in population k .”

Migration participates in the exponential race just like coalescence and recombination: each (source, dest) pair proposes an exponential waiting time, and the shortest one wins.

```
def migration_event(populations, migration_matrix, source, dest):
    """Move a random lineage from source to dest.

    Parameters
    -----
    populations : list of Population
    migration_matrix : 2D array
        M[j][k] = migration rate from j to k (backward).
    source : int
        Source population index.
    dest : int
        Destination population index.
    """
    pop = populations[source]
    # Choose a random lineage from the source population
    idx = np.random.randint(pop.num_ancestors)

    # Move it (in a real implementation, update the lineage's
    # population attribute and move it between population lists)
    print(f"Lineage {idx} migrates from pop {source} to pop {dest}")
    pop.num_ancestors -= 1
    populations[dest].num_ancestors += 1

# Example: island model with 3 populations
migration_matrix = [
    [0, 0.001, 0.001],      # pop 0 -> pop 1, pop 2
    [0.001, 0, 0.001],      # pop 1 -> pop 0, pop 2
    [0.001, 0.001, 0],      # pop 2 -> pop 0, pop 1
]
```

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```

pops = [Population(start_size=1000) for _ in range(3)]
pops[0].num_ancestors = 10
pops[1].num_ancestors = 10
pops[2].num_ancestors = 10

# Total migration rate across all population pairs
total_mig_rate = 0
for j in range(3):
    for k in range(3):
        if j != k:
            rate = pops[j].num_ancestors * migration_matrix[j][k]
            total_mig_rate += rate
            print(f"Pop {j} -> Pop {k}: rate = {rate:.4f}")
print(f"Total migration rate: {total_mig_rate:.4f}")

```

Migration brings lineages together across populations. But at some point in the past, those populations may not have existed as separate entities. This brings us to population splits and joins.

Step 4: Population Splits and Joins

A **population split** (going forwards: one population divides into two) corresponds to a **mass migration** going backwards: at time t , all lineages in population B move to population A .

Forward view:	Backward view:
Past -> Future	Future -> Past
A	A <- B
+-- B at time t	(all B lineages join A)
A B	A B

Closing a confusion gap – Splits vs. joins in backward time

In forward time, a *split* creates a new population. In backward time (which the coalescent simulates), this same event is a *join*: lineages from the daughter population move to the ancestral population. msprime implements this as a **mass migration** with `fraction=1.0`. When you see `mass_migration(source=B, dest=A, fraction=1.0)`, read it as: “Going back in time, all lineages currently in B move to A, because B didn’t exist before this point.”

For **admixture** events (where a population receives genetic material from another), use `fraction < 1.0`: only some lineages move.

```

def mass_migration_event(populations, source, dest, fraction=1.0):
    """Move a fraction of lineages from source to dest.

    fraction=1.0 means all lineages move (population join).
    fraction<1.0 means a subset moves (admixture).

    Parameters
    -----
    populations : list of Population

```

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```

source : int
dest : int
fraction : float
    Fraction of lineages to move (0 to 1).
"""
if not 0 <= fraction <= 1:
    raise ValueError("fraction must lie between 0 and 1")
n_source = populations[source].num_ancestors
# Each lineage moves independently, as in msprime MassMigration.
n_to_move = np.random.binomial(n_source, fraction)

populations[source].num_ancestors -= n_to_move
populations[dest].num_ancestors += n_to_move

print(f"Mass migration: {n_to_move} lineages from "
      f"pop {source} to pop {dest}")

# Example: two populations that split 500 generations ago
pops = [Population(start_size=5000), Population(start_size=5000)]
pops[0].num_ancestors = 20
pops[1].num_ancestors = 15

# At t=500, all lineages from pop 1 join pop 0
mass_migration_event(pops, source=1, dest=0, fraction=1.0)
print(f"After join: pop 0 has {pops[0].num_ancestors}, "
      f"pop 1 has {pops[1].num_ancestors}")

```

All of these demographic events – size changes, bottlenecks, migration rate changes, mass migrations – need to be scheduled and executed at the right times. The demographic event queue handles this.

Step 5: The Demographic Event Queue

msprime stores all demographic events in a priority queue sorted by time. During the main simulation loop (the ticking of the clock in *Hudson's Algorithm*), before executing the next coalescence or recombination event, it checks whether a demographic event should fire first.

Closing a confusion gap – Deterministic vs. stochastic events

Coalescence, recombination, and migration events are **stochastic**: their timing is drawn from exponential distributions. Demographic events are **deterministic**: they happen at pre-specified times. The main loop must check both: if the next stochastic event would occur *after* the next demographic event, the demographic event fires first, the population parameters are updated, and the stochastic event times are redrawn with the new rates. This is why the main loop in *Hudson's Algorithm* has a `continue` statement after processing a demographic event – it returns to the top of the loop to recompute rates.

```

class DemographicEventQueue:
    """Priority queue of demographic events sorted by time.

    Events are stored as (time, type, args) tuples and processed
    in chronological order during the simulation.
    """

```

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```

def __init__(self):
    self.events = [] # list of (time, function, args)

def add_size_change(self, time, pop_id, new_size):
    """Schedule a population size change."""
    self.events.append((time, 'size_change', (pop_id, new_size)))
    self.events.sort() # maintain time ordering

def add_growth_rate_change(self, time, pop_id, rate):
    """Schedule a growth rate change."""
    self.events.append((time, 'growth_rate', (pop_id, rate)))
    self.events.sort()

def add_mass_migration(self, time, source, dest, fraction):
    """Schedule a mass migration (population split/join/admixture)."""
    self.events.append((time, 'mass_migration',
                        (source, dest, fraction)))
    self.events.sort()

def add_migration_rate_change(self, time, source, dest, rate):
    """Schedule a migration rate change."""
    self.events.append((time, 'migration_rate',
                        (source, dest, rate)))
    self.events.sort()

def next_event_time(self):
    """Time of the next scheduled event (or infinity if none)."""
    if self.events:
        return self.events[0][0]
    return INFINITY

def pop_event(self):
    """Remove and return the next event."""
    return self.events.pop(0)

# Example: Out-of-Africa demographic model (simplified)
queue = DemographicEventQueue()
# Population split: Eurasian pop separates from African at t=2000
queue.add_mass_migration(2000, source=1, dest=0, fraction=1.0)
# Bottleneck in the European population at t=1000
queue.add_size_change(1000, pop_id=1, new_size=100)
# Recovery after bottleneck at t=800
queue.add_size_change(800, pop_id=1, new_size=5000)

print("Demographic events (in order):")
for time, etype, args in queue.events:
    print(f"  t={time}: {etype} {args}")

```

Step 6: Integrating Demographics into the Main Loop

The integration is straightforward: before executing a stochastic event, check if a deterministic demographic event occurs first. This is the point where the case and dial attach to the movement – where population history shapes the genealogy.

```
def simulate_with_demographics(n, L, recomb_rate, populations,
                              migration_matrix, event_queue):
    """Run a counts-only Hudson simulation with demographics.

    This teaching implementation follows lineage counts rather than genomic
    segments, but it executes every sampled coalescence, recombination, and
    migration event. It therefore illustrates the demographic event race
    without pretending to return a tree sequence.

    Parameters
    -----
    n : int
        Total sample size across all populations.
    L : float
        Sequence length.
    recomb_rate : float
    populations : list of Population
    migration_matrix : 2D array
    event_queue : DemographicEventQueue
    """
    if n != sum(p.num_ancestors for p in populations):
        raise ValueError("n must equal the initial number of ancestors")
    if recomb_rate < 0 or L <= 0:
        raise ValueError("recomb_rate must be nonnegative and L must be positive")

    t = 0.0
    total_events = 0

    while True:
        total_lineages = sum(p.num_ancestors for p in populations)
        if total_lineages <= 1:
            break

        candidates = []
        for pop_id, pop in enumerate(populations):
            k = pop.num_ancestors
            if k > 1:
                wait = coalescent_waiting_time_growth(
                    k,
                    pop.start_size,
                    pop.growth_rate,
                    t - pop.start_time,
                )
                candidates.append((wait, "coalescence", (pop_id,)))

        recomb_total = total_lineages * recomb_rate * L
        if recomb_total > 0:
            candidates.append((
                np.random.exponential(1.0 / recomb_total),
```

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```

        "recombination",
        (),
    ))

for source in range(len(populations)):
    for dest in range(len(populations)):
        if source != dest and migration_matrix[source][dest] > 0:
            rate = (populations[source].num_ancestors
                    * migration_matrix[source][dest])
            if rate > 0:
                candidates.append((
                    np.random.exponential(1.0 / rate),
                    "migration",
                    (source, dest),
                ))

stochastic = min(candidates, default=(INFINITY, None, ()))
min_event_time, event_type, args = stochastic

if t + min_event_time > event_queue.next_event_time():
    event_time, etype, args = event_queue.pop_event()
    t = event_time

    if etype == 'size_change':
        pop_id, new_size = args
        populations[pop_id].set_size(new_size, t)
    elif etype == 'growth_rate':
        pop_id, rate = args
        populations[pop_id].set_growth_rate(rate, t)
    elif etype == 'mass_migration':
        source, dest, fraction = args
        mass_migration_event(populations, source, dest, fraction)
    elif etype == 'migration_rate':
        source, dest, rate = args
        migration_matrix[source][dest] = rate
else:
    if not np.isfinite(min_event_time):
        raise RuntimeError(
            "Simulation cannot complete: lineages are isolated and "
            "no future demographic event can connect them"
        )
    t += min_event_time
    total_events += 1

    if event_type == "coalescence":
        (pop_id,) = args
        populations[pop_id].num_ancestors -= 1
    elif event_type == "recombination":
        weights = np.array(
            [pop.num_ancestors for pop in populations], dtype=float
        )
        pop_id = np.random.choice(len(populations), p=weights / weights.sum())

```

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```

        populations[pop_id].num_ancestors += 1
    else:
        source, dest = args
        populations[source].num_ancestors -= 1
        populations[dest].num_ancestors += 1

    return t, total_events

```

This counts-only teaching helper now executes each sampled stochastic event; it does not return edges or genealogies. Use `msprime.sim_ancestry` for scientific simulations and tree-sequence output.

```

# Run with the Out-of-Africa model
INFINITY = float('inf')
pops = [Population(start_size=10000), Population(start_size=5000)]
pops[0].num_ancestors = 10 # African samples
pops[1].num_ancestors = 10 # European samples

mig_matrix = [[0, 0.0001], [0.0001, 0]]

queue = DemographicEventQueue()
queue.add_mass_migration(2000, source=1, dest=0, fraction=1.0)
queue.add_size_change(1000, pop_id=1, new_size=100) # European bottleneck

np.random.seed(42)
simulate_with_demographics(
    n=20, L=10000, recomb_rate=1e-4,
    populations=pops, migration_matrix=mig_matrix,
    event_queue=queue
)

```

With the continuous-time coalescent and demographics covered, there is one more simulation mode to introduce: the exact discrete-time model for small or recent populations.

Step 7: The Discrete-Time Wright-Fisher Model

For small populations or recent time scales, the continuous-time coalescent approximation breaks down. `msprime` also implements the exact **Discrete-Time Wright-Fisher (DTWF)** model, which simulates each generation explicitly.

Closing a confusion gap – When does the coalescent approximation fail?

The continuous-time coalescent assumes N is large enough that the probability of simultaneous coalescences is negligible. For $N = 50$ with $k = 20$ haploid lineages, the exact occupancy calculation gives a probability of about 0.914 that two or more collisions occur in one generation. The DTWF model handles these events by simulating every generation explicitly. `msprime` defaults to the Hudson coalescent, but users can request a hybrid model list: DTWF for a chosen recent duration followed by the standard coalescent in the distant past.

The algorithm per generation:

1. Each lineage draws a parent uniformly from the population
2. Lineages that drew the same parent are merged (coalescence)
3. Recombination is simulated for each diploid parent

4. Migration is applied

```
def dtwf_generation(lineages, N, recomb_rate, L):
    """Sketch lineage-count changes in one haploid WF generation.

    This teaching sketch does not update segment ancestry at recombination
    and is therefore not a replacement for msprime's DTWF implementation.

    Parameters
    -----
    lineages : list
        Current lineages.
    N : int
        Population size.
    recomb_rate : float
        Per-bp recombination rate per generation.
    L : float
        Sequence length.

    Returns
    -----
    lineages : list
        Lineages after one generation.
    events : list
        Events that occurred.
    """
    events = []

    # Step 1: Each lineage draws a parent uniformly from N individuals
    parents = {}
    for i, lin in enumerate(lineages):
        parent_id = np.random.randint(N) # uniform random parent
        if parent_id not in parents:
            parents[parent_id] = []
        parents[parent_id].append(i)

    # Step 2: Lineages sharing a parent coalesce
    new_lineages = []
    for parent_id, children in parents.items():
        if len(children) == 1:
            # Only one lineage chose this parent: no coalescence
            new_lineages.append(lineages[children[0]])
        else:
            # Multiple lineages share a parent: merge them
            # (In reality, recombination happens first, then merging)
            merged = lineages[children[0]]
            for c in children[1:]:
                events.append(('coal', children[0], c))
                # merge lineages[c] into merged (simplified)
            new_lineages.append(merged)

    # Step 3: Recombination (simplified Poisson model)
    for lin in new_lineages:
```

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```

n_recombs = np.random.poisson(recomb_rate * L)
for _ in range(n_recombs):
    bp = np.random.randint(1, int(L))
    events.append(('recomb', bp))

return new_lineages, events

# Production msprime can compose an explicit hybrid model:
import msprime

model = [
    msprime.DiscreteTimeWrightFisher(duration=100),
    msprime.StandardCoalescent(),
]

```

i When to use DTWF vs coalescent

- **DTWF:** Use for the recent past (last ~100 generations) in small populations where the coalescent approximation is poor.
- **Coalescent:** Use for the distant past where the continuous-time approximation is excellent and much faster.
- **Hybrid:** Explicitly pass a DTWF phase followed by the standard coalescent. This is often a useful compromise, but it is not the default.

You have now seen how population history – size changes, bottlenecks, migration, splits, and the DTWF model – integrates into the simulation. These are the case and dial of the master clockmaker's bench: they determine the visible shape of the genealogy without changing the underlying mechanism.

In the final chapter, we add the last visible layer: mutations.

Exercises**i Exercise 1: Bottleneck effect**

Simulate 1000 genealogies with $n = 20$ under (a) constant $N = 10,000$ and (b) a bottleneck at $t = 500$ generations where N drops to 100 for 50 generations. Compare the distributions of T_{MRCA} and total branch length.

i Exercise 2: Island model

Simulate a symmetric island model with 3 populations, $N = 1000$ each, and migration rate $m = 0.001$. Compute the expected coalescence time for two lineages sampled from (a) the same population and (b) different populations. Here m is the rate to each other deme. Under the haploid convention used by the helper, compare to $E[T_{\text{same}}] = dN$ and $E[T_{\text{different}}] = dN + 1/(2m)$.

i Exercise 3: Out-of-Africa model

Build a simplified Out-of-Africa demographic model with:

- African population: $N = 10,000$
- European population: $N = 5,000$, bottleneck at $t = 1000$
- Split at $t = 2000$ generations
- Migration rate $m = 10^{-4}$ between split and present

Simulate 100 genealogies and compute the site frequency spectrum for each population.

Exercise 4: DTWF vs coalescent

For a population of size $N = 50$, simulate genealogies using both DTWF and the coalescent. Compare the distributions of coalescence times. At what N do the two models give indistinguishable results?

Next: *Mutations* – the final gear: painting mutations onto the genealogy.

Solutions

Solution 1: Bottleneck effect

A bottleneck forces rapid coalescence during the period of small population size. This dramatically reduces T_{MRCA} and total branch length compared to the constant-size case, because many lineages merge during the bottleneck.

```
import numpy as np

n = 20
n_reps = 1000
N_const = 10000

# (a) Constant size: simulate standard coalescent, scale to generations
tmrca_const = []
branch_length_const = []
for _ in range(n_reps):
    t = 0.0
    k = n
    total_bl = 0.0
    while k > 1:
        rate = k * (k - 1) / 2
        dt = np.random.exponential(1.0 / rate)
        total_bl += k * dt
        t += dt
        k -= 1
    tmrca_const.append(t * N_const)
    branch_length_const.append(total_bl * N_const)

# (b) Bottleneck: N=10000 for t<500, N=100 for 500<=t<550, N=10000 for t>=550
# We simulate generation by generation through the bottleneck.
tmrca_bottle = []
branch_length_bottle = []
for _ in range(n_reps):
    t = 0.0
    k = n
```

```

total_b1 = 0.0
epochs = [
    (500, N_const),      # 0 to 500 generations: N=10000
    (550, 100),          # 500 to 550 generations: N=100
    (np.inf, N_const),   # 550+ generations: N=10000
]
for end_time, N_epoch in epochs:
    while k > 1:
        rate = k * (k - 1) / 2
        # Waiting time in generations for this epoch
        dt_gen = N_epoch * np.random.exponential(1.0 / rate)
        if t + dt_gen > end_time:
            # Epoch ends before next coalescence
            total_b1 += k * (end_time - t)
            t = end_time
            break
        t += dt_gen
        total_b1 += k * dt_gen
        k -= 1
    if k <= 1:
        break
tmrca_bottle.append(t)
branch_length_bottle.append(total_b1)

print(f"Constant N={N_const}:")
print(f"  Mean T_MRCA = {np.mean(tmrca_const):.0f} gen")
print(f"  Mean total branch length = {np.mean(branch_length_const):.0f} gen")
print(f"\nBottleneck (N=100 at t=500-550):")
print(f"  Mean T_MRCA = {np.mean(tmrca_bottle):.0f} gen")
print(f"  Mean total branch length = {np.mean(branch_length_bottle):.0f} gen")

```

Solution 2: Island model

For a symmetric island model with d demes, each of size N , and per-destination, per-generation migration rate m , a two-state first-step calculation gives $E[T_{\text{same}}] = dN$ under the haploid convention. Two lineages starting in different demes must first enter the same deme, adding $1/(2m)$, so $E[T_{\text{different}}] = dN + 1/(2m)$.

```

import numpy as np

N = 1000
m = 0.001 # per-generation migration rate
d = 3     # number of demes
M = m * N # scaled migration rate (per deme pair)
n_reps = 5000

def simulate_island_coalescence(same_deme=True, N=1000, m=0.001, d=3):
    """Simulate coalescence time for 2 lineages in an island model."""
    # State: (deme_1, deme_2)
    if same_deme:
        deme = [0, 0]
    else:
        deme = [0, 1]

```



```

t = 0.0
while True:
    # Rates:
    # Coalescence: only if both are in the same deme, rate = 1/N
    coal_rate = (1.0 / N) if deme[0] == deme[1] else 0.0
    # Migration: each lineage migrates at rate m*(d-1) total
    mig_rate = 2 * m * (d - 1) # total migration rate for both lineages
    total_rate = coal_rate + mig_rate
    dt = np.random.exponential(1.0 / total_rate)
    t += dt
    if np.random.random() < coal_rate / total_rate:
        return t # coalescence
    else:
        # Migration: pick a random lineage and move it
        lin = np.random.randint(2)
        new_deme = np.random.choice(
            [x for x in range(d) if x != deme[lin]])
        deme[lin] = new_deme

same_times = [simulate_island_coalescence(same_deme=True, N=N, m=m, d=d)
              for _ in range(n_reps)]
diff_times = [simulate_island_coalescence(same_deme=False, N=N, m=m, d=d)
              for _ in range(n_reps)]

print(f"Island model: d={d}, N={N}, m={m}")
print(f"Same deme:      E[T] = {np.mean(same_times):.0f} generations")
print(f"Different deme: E[T] = {np.mean(diff_times):.0f} generations")
print(f"Theory (same deme): {d * N:.0f}")
print(f"Theory (different demes): {d * N + 1 / (2 * m):.0f}")

```

Solution 3: Out-of-Africa model

We build the model using `simulate_with_demographics` and compute the SFS for each population from the resulting genealogies with mutations added.

```

import numpy as np

INFINITY = float('inf')

# Build the demographic model
pops = [Population(start_size=10000), Population(start_size=5000)]
mig_matrix = [[0, 1e-4], [1e-4, 0]]

queue = DemographicEventQueue()
queue.add_size_change(1000, pop_id=1, new_size=100) # European bottleneck
queue.add_mass_migration(2000, source=1, dest=0, fraction=1.0) # split

n_afr = 10 # African samples
n_eur = 10 # European samples
n_total = n_afr + n_eur

# For each replicate:

```

```

# 1. Simulate coalescent with demographics
# 2. Place mutations on the resulting tree (Poisson process)
# 3. Compute the SFS per population
n_reps = 100
mu = 1.5e-8
L = 1e6
theta = 4 * 10000 * mu * L # using N_e = 10000

sfs_afr = np.zeros(n_afr - 1)
sfs_eur = np.zeros(n_eur - 1)

# The key insight is that the SFS for each population depends on
# the genealogy shaped by the demographic model. A bottleneck changes
# branch lengths and usually reduces diversity; its SFS effect depends
# on the bottleneck's timing, duration, and severity. A star-like tree
# and excess of rare variants are signatures of rapid expansion, not a
# generic consequence of a bottleneck.

print(f"Expected SFS (standard neutral model, theta={theta:.1f}):")
for i in range(1, min(6, n_afr)):
    print(f"  xi_{i} = {theta/i:.1f}")

print(f"\nThe African SFS should approximate theta/i.")
print("The European SFS will be distorted by the bottleneck; the")
print("direction and magnitude require simulation of the stated model.")

```

Solution 4: DTWF vs coalescent

For small populations, the coalescent approximation ignores simultaneous coalescences. The DTWF model is exact. We compare by simulating both for $N = 50$ with $n = 20$ lineages.

```

import numpy as np

n = 20

def simulate_dtwf_tmrca(n, N, n_reps=5000):
    """Simulate T_MRCA using the exact DTWF model."""
    tmrca_values = []
    for _ in range(n_reps):
        k = n
        t = 0
        while k > 1:
            t += 1
            # Each lineage draws a parent uniformly from N
            parents = np.random.randint(0, N, size=k)
            k = len(set(parents)) # number of distinct parents
            tmrca_values.append(t)
    return np.array(tmrca_values)

def simulate_coalescent_tmrca(n, N, n_reps=5000):
    """Simulate T_MRCA using the continuous-time coalescent."""
    tmrca_values = []

```

```

for _ in range(n_reps):
    t = 0.0
    k = n
    while k > 1:
        rate = k * (k - 1) / 2
        t += N * np.random.exponential(1.0 / rate)
        k -= 1
    tmrca_values.append(t)
return np.array(tmrca_values)

print(f"{'N':>6s} {'DTWF mean':>10s} {'Coal mean':>10s} "
      f"{'DTWF std':>10s} {'Coal std':>10s}")
for N in [20, 50, 100, 500, 1000]:
    dtwf = simulate_dtwf_tmrca(n, N)
    coal = simulate_coalescent_tmrca(n, N)
    print(f"{'N':6d} {'dtwf.mean()':10.1f} {'coal.mean()':10.1f} "
          f"{'dtwf.std()':10.1f} {'coal.std()':10.1f}")

print(f"\nFor N >= ~100, the DTWF and coalescent give very similar results.")
print(f"For N=50 with n=20, simultaneous coalescences are non-negligible,")
print(f"causing the DTWF T_MRCA to be systematically shorter.")

```

Mutations

The final touch: painting variation onto the genealogy.

The genealogy (tree sequence) from Phase 1 tells us *who is related to whom* and *when* they shared ancestors. But it says nothing about *what their DNA looks like*. Mutations – heritable changes in the DNA sequence – are what create the observable genetic variation.

In msprime, mutations are simulated as a **separate post-processing step** on the tree sequence. This is both conceptually clean (the genealogy doesn't depend on mutations) and computationally efficient (we can reuse the same genealogy with different mutation models).

In the watch metaphor, if the movement (the coalescent, segments, and Hudson's algorithm) is the hidden mechanism, and the case and dial (demographics) shape its form, then mutations are the paint on the dial face – the markings that make the watch *readable*. Without mutations, the genealogy is invisible: a perfect mechanism that tells time but shows no numbers.

Note

Prerequisites. This chapter builds on all previous chapters in this Timepiece. Specifically:

- The **tree sequence** output from *Hudson's Algorithm* – the edges and nodes that define the genealogy.
- **Branch lengths** – determined by coalescence times from *The Coalescent Process*.
- **Marginal trees** – the concept from *Overview of msprime* that different genomic positions can have different genealogies.

You should also be familiar with the *Coalescent Theory* chapter's treatment of the expected number of segregating sites and the site frequency spectrum, which we will rederive here from the simulation perspective.

Step 1: The Infinite-Sites Poisson Model

The simplest mutation model: mutations arise as a Poisson process along each branch of the genealogy.

The setup. Given a tree sequence with branches of known length (in generations), and a per-base-pair, per-generation mutation rate μ :

- Each branch of length t generations covering ℓ base pairs accumulates mutations at rate $\mu \cdot \ell \cdot t$.
- The number of mutations on a branch is $\text{Poisson}(\mu\ell t)$.
- Each mutation is placed at a uniformly random position along the branch (both in time and in genomic position).

Under the **infinite-sites** assumption, every mutation creates a new variant at a previously-unmutated position. This means no two mutations can hit the same site. (For short branches and realistic μ , this is an excellent approximation.) In msprime this spatial assumption is selected with `discrete_genome=False`. The default discrete genome permits recurrent mutation at integer-valued sites.

Probability Aside – The Poisson process on branches

The Poisson process is a natural model for rare, independent events occurring in continuous time. If mutations arise independently at rate μ per base pair per generation, then over t generations and ℓ base pairs, the expected count is $\mu\ell t$. The probability of exactly k mutations is:

$$P(N = k) = \frac{(\mu\ell t)^k e^{-\mu\ell t}}{k!}$$

For human parameters ($\mu \approx 1.5 \times 10^{-8}$, $\ell = 1000$ bp, $t = 10,000$ generations), the expected count is ≈ 0.15 , so most branches get 0 or 1 mutation. This makes the infinite-sites assumption excellent in practice.

```
import numpy as np

def simulate_mutations_infinite_sites(edges, nodes, sequence_length, mu):
    """Add mutations to a tree sequence under the infinite-sites model.

    This is the core of Phase 2: walk through every edge (branch) of
    the tree sequence, draw a Poisson number of mutations, and place
    each one at a random position and time.

    Parameters
    -----
    edges : list of (left, right, parent, child)
```

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```

    Edges defining the tree sequence.
    nodes : list of (time, population)
        Node times (samples at time 0, ancestors at time > 0).
    sequence_length : float
        Total genome length in base pairs.
    mu : float
        Per-bp, per-generation mutation rate.

Returns
-----
mutations : list of (position, node, time, ancestral, derived)
    Each mutation has a genomic position, the node it sits above,
    the time it occurred, and the allelic states.
"""
mutations = []

for left, right, parent, child in edges:
    # Branch length in generations (time difference parent - child)
    branch_length = nodes[parent][0] - nodes[child][0]
    # Genomic span of this edge
    span = right - left
    # Expected number of mutations: mu * span * branch_length
    expected = mu * span * branch_length

    # Draw number of mutations from Poisson distribution
    n_muts = np.random.poisson(expected)

    for _ in range(n_muts):
        # Random position within [left, right)
        position = np.random.uniform(left, right)
        # Random time on the branch (between child and parent times)
        time = np.random.uniform(nodes[child][0], nodes[parent][0])
        # Under infinite sites: ancestral=0, derived=1
        mutations.append((position, child, time, '0', '1'))

    # Sort by position for output
    mutations.sort(key=lambda m: m[0])
    return mutations

# Example: simple tree with 3 samples
#      4 (t=1.5)
#     / \
#    3   \ (t=0.8)
#   / \   \
#  0  1  2
# (t=0)
nodes = [(0, 0), (0, 0), (0, 0), (0.8, 0), (1.5, 0)]
edges = [
    (0, 1000, 3, 0), # edge: node 0 is child of node 3
    (0, 1000, 3, 1), # edge: node 1 is child of node 3
    (0, 1000, 4, 3), # edge: node 3 is child of node 4
    (0, 1000, 4, 2), # edge: node 2 is child of node 4

```

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```

]

np.random.seed(42)
mutts = simulate_mutations_infinite_sites(edges, nodes, 1000, mu=1e-3)
print(f"Number of mutations: {len(mutts)}")
for pos, node, time, anc, der in mutts[:10]:
    print(f"  pos={pos:.1f}, above node {node}, time={time:.4f}")

```

With the mechanics of mutation placement established, let us derive the classical results that connect mutations to the coalescent.

Step 2: The Expected Number of Segregating Sites

A fundamental result in population genetics: the expected number of segregating sites (positions with a mutation) in a sample of n from a population of size N is:

$$E[S] = \theta \cdot \sum_{k=1}^{n-1} \frac{1}{k}$$

where $\theta = 2N_e\mu L$ for a haploid population or $\theta = 4N_e\mu L$ for a diploid population. The sum $\sum_{k=1}^{n-1} 1/k$ is the $(n-1)$ -th harmonic number.

Derivation. The total branch length of the coalescent tree is:

$$E[L_{\text{total}}] = \sum_{k=2}^n k \cdot E[T_k] = \sum_{k=2}^n k \cdot \frac{2}{k(k-1)} = 2 \sum_{k=2}^n \frac{1}{k-1} = 2 \sum_{j=1}^{n-1} \frac{1}{j}$$

(In coalescent units where $N_e = 1$.)

Calculus Aside – The harmonic number

The sum $H_n = \sum_{k=1}^n 1/k$ is the n -th harmonic number. It grows logarithmically: $H_n \approx \ln(n) + \gamma$ where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant. This means the total branch length grows as $\sim 2 \ln(n)$, so the expected number of segregating sites grows as $\sim \theta \ln(n)$. Doubling the sample size adds only $\theta \ln(2) \approx 0.69\theta$ additional segregating sites – a consequence of the “diminishing returns” of adding more samples to the coalescent tree (recall from *The Coalescent Process* that most of the tree height comes from the last few lineages).

Across the whole sequence, each unit of branch length produces mutations at rate $\theta/2$. Equivalently, the rate is $\theta/(2L)$ per base pair per coalescent time unit. This follows because $\theta = 2N_e\mu L$ and time is in units of N_e generations in the haploid convention used for this derivation. Thus:

$$E[S] = \frac{\theta}{2} \cdot E[L_{\text{total}}] = \frac{\theta}{2} \cdot 2 \sum_{j=1}^{n-1} \frac{1}{j} = \theta \sum_{j=1}^{n-1} \frac{1}{j}$$

This is **Watterson’s estimator** in reverse: given observed S , we can estimate $\hat{\theta}_W = S / \sum_{j=1}^{n-1} 1/j$.

```

def expected_segregating_sites(n, theta):
    """Expected number of segregating sites.

    Uses the formula E[S] = theta * H_{n-1}, where H_k is the
    k-th harmonic number.

```

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```

"""
    harmonic = sum(1.0 / k for k in range(1, n)) # H_{n-1}
    return theta * harmonic

def watterson_estimator(S, n):
    """Estimate theta from the number of segregating sites.

    This inverts  $E[S] = \theta \cdot H_{n-1}$  to get  $\theta_{\text{hat}} = S / H_{n-1}$ .
    """
    harmonic = sum(1.0 / k for k in range(1, n))
    return S / harmonic

# Example
n, theta = 50, 100
E_S = expected_segregating_sites(n, theta)
print(f"n={n}, theta={theta}")
print(f"Expected segregating sites: {E_S:.1f}")
print(f"Watterson's estimate from E[S]: {watterson_estimator(E_S, n):.1f}")

```

The number of segregating sites is a single summary statistic. A much richer summary is the site frequency spectrum.

Step 3: The Site Frequency Spectrum

The **Site Frequency Spectrum (SFS)** counts how many sites have a derived allele at each possible frequency. If ξ_i is the number of sites where exactly i out of n samples carry the derived allele:

$$E[\xi_i] = \frac{\theta}{i}, \quad i = 1, 2, \dots, n-1$$

This beautiful result says that singletons ($i = 1$) are the most common class of variants, and the frequency spectrum falls off as $1/i$.

Derivation. A mutation creates a variant at frequency i/n if and only if it falls on a branch subtending exactly i leaves. The expected length of branches subtending i leaves is $2/i$ (in coalescent units). The mutation rate is $\theta/2$ per unit length. So:

$$E[\xi_i] = \frac{\theta}{2} \cdot \frac{2}{i} = \frac{\theta}{i}$$

Probability Aside – Why $2/i$ for the branch length subtending i leaves?

This result comes from the exchangeability of the coalescent. In a coalescent tree with n leaves, the total branch length at level k (when there are k lineages) is $k \cdot T_k$, with $E[T_k] = 2/[k(k-1)]$. A branch at level k subtends some number of leaves. Summing over all levels and using linearity of expectation, one can show that the expected total length of branches subtending exactly i leaves is $2/i$, independent of n (for $i < n$). This is a remarkable symmetry of Kingman's coalescent. For a full proof, see the derivation in *Coalescent Theory*.

```

def compute_sfs(mutations, genotype_matrix, n):
    """Compute the site frequency spectrum from genotype data.

    Parameters
    -----

```

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```

genotype_matrix : ndarray of shape (n_sites, n_samples)
    0 = ancestral, 1 = derived.

Returns
-----
sfs : ndarray of shape (n - 1,)
    sfs[i-1] = number of sites with derived allele count i.
"""
sfs = np.zeros(n - 1, dtype=int)
for site in genotype_matrix:
    count = int(site.sum()) # number of derived alleles at this site
    if 1 <= count <= n - 1:
        sfs[count - 1] += 1 # sfs is 0-indexed: sfs[0] = singletons
return sfs

def expected_sfs(n, theta):
    """Expected SFS under the standard neutral model.

     $E[x_{i-1}] = \theta / i$  for  $i = 1, \dots, n-1$ .
    """
    return np.array([theta / i for i in range(1, n)])

# Example
n, theta = 20, 50
exp_sfs = expected_sfs(n, theta)
print("Expected SFS:")
for i, e in enumerate(exp_sfs):
    bar = '#' * int(e)
    print(f"freq {i+1:>2d}/{n}: E[xi] = {e:.2f} {bar}")

```

The infinite-sites model is elegant, but it breaks down when mutation rates are high or branches are long. For those cases, we need finite-sites models.

Step 4: Finite-Sites Mutation Models

The infinite-sites model breaks down when the mutation rate is high or branches are long: the same site can be hit by multiple mutations. msprime supports several **finite-sites** models.

Matrix Mutation Model

Mutations follow a Markov chain on allelic states (e.g., A, C, G, T). The transition matrix P gives the probability of each state change:

$$P = \begin{pmatrix} 0 & p_{AC} & p_{AG} & p_{AT} \\ p_{CA} & 0 & p_{CG} & p_{CT} \\ p_{GA} & p_{GC} & 0 & p_{GT} \\ p_{TA} & p_{TC} & p_{TG} & 0 \end{pmatrix}$$

where each row sums to 1. A zero diagonal describes models such as the default Jukes–Cantor parameterization, in which every event changes state. msprime also permits nonzero diagonal entries for state-independent models; these entries represent silent mutation events.

i Probability Aside – Mutation as a Markov chain

Each site's allelic state follows a continuous-time Markov chain (CTMC). When a mutation event occurs (Poisson process), the state transitions according to the matrix P . Diagonal entries may be nonzero, in which case a recorded event can leave the state unchanged. The stationary distribution of this chain determines the long-run base composition. For the Jukes-Cantor model (all transitions equally likely), the stationary distribution is uniform: $(1/4, 1/4, 1/4, 1/4)$.

```
class MatrixMutationModel:
    """Finite-sites mutation model with transition matrix.

    Each mutation event changes the allelic state according to the
    transition matrix. The root state is drawn from root_distribution.
    """

    def __init__(self, alleles, root_distribution, transition_matrix):
        """
        Parameters
        -----
        alleles : list of str
            Allelic states (e.g., ['A', 'C', 'G', 'T']).
        root_distribution : ndarray
            Probability of each allele at the root.
        transition_matrix : ndarray of shape (k, k)
             $P[i, j]$  = probability of mutating from allele  $i$  to allele  $j$ .
            Rows are probability distributions; diagonal entries may be
            nonzero for silent transitions.
        """
        self.alleles = alleles
        self.root_distribution = np.asarray(root_distribution, dtype=float)
        self.transition_matrix = np.asarray(transition_matrix, dtype=float)
        if not np.isclose(self.root_distribution.sum(), 1):
            raise ValueError("root_distribution must sum to one")
        if not np.allclose(self.transition_matrix.sum(axis=1), 1):
            raise ValueError("each transition_matrix row must sum to one")

    def draw_root_state(self):
        """Sample the ancestral allele at the root."""
        return np.random.choice(len(self.alleles), p=self.root_distribution)

    def mutate(self, current_state):
        """Apply one mutation: change the state according to the matrix."""
        return np.random.choice(len(self.alleles),
                                p=self.transition_matrix[current_state])

# Jukes-Cantor model: all transitions equally likely
jc_model = MatrixMutationModel(
    alleles=['A', 'C', 'G', 'T'],
    root_distribution=[0.25, 0.25, 0.25, 0.25],
    transition_matrix=[
        [0, 1/3, 1/3, 1/3], # from A: equal prob of C, G, or T
        [1/3, 0, 1/3, 1/3], # from C: equal prob of A, G, or T
        [1/3, 1/3, 0, 1/3], # from G: equal prob of A, C, or T

```

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```

    [1/3, 1/3, 1/3, 0], # from T: equal prob of A, C, or G
]
)

# Simulate 10 mutations starting from 'A'
state = 0 # 'A'
print("Mutation chain starting from A:")
chain = ['A']
for _ in range(10):
    state = jc_model.mutate(state)
    chain.append(jc_model.alleles[state])
print(" -> ".join(chain))

```

The Binary Mutation Model

The simplest finite-sites model: two alleles (0 and 1), with equal transition probabilities:

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Every mutation flips the allele. This is equivalent to the infinite-sites model when the mutation rate is low enough that each site is hit at most once, but allows back-mutations when rates are higher.

Closing a confusion gap – Infinite sites vs. finite sites

Under the infinite-sites model, each mutation creates a *new* variant at an unused position. No site is ever hit twice. Under finite sites, a site can be hit multiple times, potentially reverting to the ancestral state (“back-mutation”). For typical human parameters ($\mu \sim 10^{-8}$ per bp per generation, branch lengths of $\sim 10^4$ generations), the probability of two mutations at the same site is about $(\mu \cdot t)^2/2 \approx 5 \times 10^{-9}$ – negligible. But for high-mutation-rate organisms (viruses, microsatellites) or very deep genealogies, finite-sites models are essential.

Now let us see how mutations are placed on a tree sequence, integrating over all marginal trees.

Step 5: Placing Mutations on the Tree Sequence

The mutation placement algorithm integrates over each marginal tree’s genomic span. For a branch of length t covering an interval of length ℓ , it draws a Poisson count with mean $\mu\ell t$, assigns positions and times to those events, and then applies them in temporal order from the ancestral end of the branch toward the child. Allelic state is local to a genomic site: a mutation at position x must not change the state inherited at position y . The helper therefore groups events by position, draws an ancestral state for each site, and traverses the tree separately for each group. This mirrors the parent-mutation logic in `msprime`’s mutation generator. The helper also follows `msprime`’s default discrete-genome behavior, placing events at integer-valued sites so recurrent and back mutations are possible; pass `discrete_genome=False` for continuous positions. It omits `msprime`’s table, existing-mutation, metadata, and provenance machinery.

```

def place_mutations_on_tree(
    tree, mu, model, sequence_length, discrete_genome=True):
    """Place mutations on a single marginal tree.

    Parameters
    -----
    tree : dict

```

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```

    Tree as {node: (parent, time, children)}.
mu : float
    Per-site, per-generation mutation rate.
model : MatrixMutationModel
sequence_length : float
discrete_genome : bool, optional
    If True (the msprime default), place events at integer-valued sites so
    recurrent and back mutations can occur. If False, use continuous
    positions, corresponding to msprime's infinite-sites spatial model.

Returns
-----
mutations : list of
    ``(position, node, parent_node, ancestral_state, derived_state, time)``.
leaf_states : dict
    ``{leaf: {position: allele_index}}``. States are tracked separately
    at every mutated position.
"""
root = find_root(tree)

if sequence_length <= 0:
    raise ValueError("sequence_length must be positive")
if discrete_genome and not float(sequence_length).is_integer():
    raise ValueError(
        "sequence_length must be an integer when discrete_genome is True")

# First place events without assigning states. State evolution is then
# resolved independently at each genomic position; carrying one scalar
# state across positions would make a mutation at x alter the allele at y.
events_by_position = {}
for parent_node, (_, parent_time, children) in tree.items():
    for child in children:
        _, child_time, _ = tree[child]
        branch_length = parent_time - child_time
        if branch_length < 0:
            raise ValueError("parent time must not be younger than child time")
        n_muts = np.random.poisson(mu * sequence_length * branch_length)
        for mut_time in np.random.uniform(child_time, parent_time, size=n_muts):
            if discrete_genome:
                position = float(np.random.randint(0, int(sequence_length)))
            else:
                position = np.random.uniform(0, sequence_length)
            events_by_position.setdefault(position, {}).setdefault(
                child, []).append((mut_time, parent_node))

leaves = [node for node, (_, _, children) in tree.items() if not children]
leaf_states = {leaf: {} for leaf in leaves}
mutations = []

for position, branch_events in events_by_position.items():
    node_states = {root: model.draw_root_state()}
    stack = [root]

```

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```

while stack:
    node = stack.pop()
    current_state = node_states[node]
    _, _, children = tree[node]
    for child in children:
        state = current_state
        # Older mutations occur first when moving from parent to child.
        for mut_time, parent_node in sorted(
            branch_events.get(child, []), reverse=True):
            new_state = model.mutate(state)
            mutations.append((
                position,
                child,
                parent_node,
                model.alleles[state],
                model.alleles[new_state],
                mut_time,
            ))
            state = new_state
            node_states[child] = state
            stack.append(child)

for leaf in leaves:
    leaf_states[leaf][position] = node_states[leaf]

mutations.sort(key=lambda mutation: (mutation[0], -mutation[-1]))
return mutations, leaf_states

```

Step 6: The Mutation Rate Map

Like recombination, mutation rates can vary along the genome. A **mutation rate map** specifies the local rate $\mu(x)$ at each position:

Calculus Aside – Mutation mass and expected counts

Just as with recombination, the “mass” of an interval $[a, b]$ is $\int_a^b \mu(x) dx$. The expected number of mutations on a branch of length t spanning $[a, b]$ is $t \cdot \int_a^b \mu(x) dx$. For a piecewise-constant rate, this integral reduces to a sum of rate-times-length terms, exactly as in the recombination rate map from *Segments & the Fenwick Tree*.

```

class MutationRateMap:
    """Piecewise-constant mutation rate along the genome.

    Analogous to the recombination RateMap, but for mutations.
    """

    def __init__(self, positions, rates):
        self.positions = np.array(positions)
        self.rates = np.array(rates)

    def rate_at(self, position):

```

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```

        """Get mutation rate at a specific position."""
        idx = np.searchsorted(self.positions, position, side='right') - 1
        idx = max(0, min(idx, len(self.rates) - 1))
        return self.rates[idx]

    def total_mass(self, left, right):
        """Total mutation mass over [left, right).

        This is the integral of mu(x) from left to right.
        """
        total = 0
        for i in range(len(self.rates)):
            seg_left = self.positions[i]
            seg_right = self.positions[i + 1]
            # Compute overlap with [left, right)
            ol = max(seg_left, left)
            or_ = min(seg_right, right)
            if or_ > ol:
                total += self.rates[i] * (or_ - ol)
        return total

# Example: mutation rate with a cold spot (centromere)
mut_map = MutationRateMap(
    positions=[0, 4000, 6000, 10000],
    rates=[1.5e-8, 1e-9, 1.5e-8] # low rate in [4000, 6000)
)

print("Mutation rates:")
for x in [1000, 5000, 8000]:
    print(f" position {x}: mu = {mut_map.rate_at(x):.2e}")
print(f"Total mass [0, 10000): {mut_map.total_mass(0, 10000):.2e}")

```

With the rate map handling spatial variation, let us see the final step: converting mutations into observable genotype data.

Step 7: From Mutations to Genotype Matrix

The final output is a **genotype matrix**: one row per site, not one row per mutation. Multiple events at the same position must be applied from older to younger. This preserves silent mutations and allows a younger event to overwrite or reverse the state inherited from an older event.

```

def build_genotype_matrix(mutations, tree_sequence, n_samples):
    """Convert mutations to a genotype matrix.

    Parameters
    -----
    mutations : list of (position, node, ...)
    tree_sequence : object
    n_samples : int

    Returns
    -----
    genotypes : ndarray of shape (n_sites, n_samples)
    """

```

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```

positions : ndarray of shape (n_sites,)
"""
# Normalize the two mutation record formats produced by this module:
# infinite-sites records are (pos, node, time, ancestral, derived), while
# finite-sites records additionally contain parent_node.
normalized = []
for mutation in mutations:
    if len(mutation) == 6:
        pos, node, _parent, ancestral, derived, time = mutation
    elif len(mutation) == 5:
        pos, node, time, ancestral, derived = mutation
    else:
        raise ValueError("unrecognized mutation record")
    normalized.append((pos, node, time, ancestral, derived))

sites = {}
for mutation in normalized:
    sites.setdefault(mutation[0], []).append(mutation)

positions_sorted = sorted(sites)
n_sites = len(positions_sorted)
genotypes = np.zeros((n_sites, n_samples), dtype=int)
positions = np.asarray(positions_sorted, dtype=float)

for site_index, pos in enumerate(positions_sorted):
    site_mutations = sorted(sites[pos], key=lambda mutation: mutation[2],
                           reverse=True)
    # Before the oldest event, all samples carry its ancestral state.
    root_state = site_mutations[0][3]
    sample_states = [root_state] * n_samples
    for _, node, _time, _ancestral, derived in site_mutations:
        for sample in get_descendants(tree_sequence, node, pos):
            if sample < n_samples:
                sample_states[sample] = derived
    genotypes[site_index] = [state != root_state for state in sample_states]

return genotypes, positions

```

Putting It All Together

The following is a stripped-down mutation pipeline for teaching. The production `msprime.sim_mutations` implementation additionally handles site tables, parent relationships among recurrent mutations, existing mutations, provenance, time bounds, discrete genomes, and specialized mutation models. Its default model is JC69, not an infinite-sites binary model.

```

def sim_mutations(tree_sequence, rate, model=None):
    """Simulate mutations on a tree sequence.

    This is Phase 2 of msprime's simulation pipeline.
    Phase 1 (ancestry) built the movement; Phase 2 paints the dial.

    Parameters

```

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```

-----
tree_sequence : object
    Output of ancestry simulation (Phase 1).
rate : float or MutationRateMap
    Per-bp, per-generation mutation rate.
model : MutationModel, optional
    Defaults to a Jukes--Cantor nucleotide model, as in msprime.

Returns
-----
mutated_ts : object
    Tree sequence with mutations added.
"""
if model is None:
    # Default: Jukes--Cantor nucleotide model.
    model = MatrixMutationModel(
        alleles=['A', 'C', 'G', 'T'],
        root_distribution=[0.25, 0.25, 0.25, 0.25],
        transition_matrix=[
            [0, 1/3, 1/3, 1/3],
            [1/3, 0, 1/3, 1/3],
            [1/3, 1/3, 0, 1/3],
            [1/3, 1/3, 1/3, 0],
        ],
    )

mutations = []
for tree in tree_sequence.trees():
    span = tree.interval.right - tree.interval.left

    for node in tree.nodes():
        if tree.parent(node) is not None:
            # Compute branch length (parent time - child time)
            branch_length = (tree.time(tree.parent(node)) -
                             tree.time(node))

            # Get the effective mutation rate for this tree's span
            if isinstance(rate, MutationRateMap):
                mu = rate.total_mass(tree.interval.left,
                                     tree.interval.right) / span
            else:
                mu = rate

            # Expected mutations on this branch = mu * span * t
            expected = mu * span * branch_length
            n_muts = np.random.poisson(expected)

            for _ in range(n_muts):
                # Random position within the tree's genomic interval
                pos = np.random.uniform(tree.interval.left,
                                         tree.interval.right)
                # Random time on the branch

```

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```

time = np.random.uniform(tree.time(node),
                          tree.time(tree.parent(node)))

mutations.append({
    'position': pos,
    'node': node,
    'time': time,
    'derived_state': '1',
})

return mutations

```

i Why separate ancestry and mutations?

1. **Efficiency:** The same genealogy can be used with different mutation rates or models without re-simulating ancestry.
2. **Modularity:** Ancestry simulation is complex (recombination, demographics, migration). Mutation simulation is simple (Poisson process on branches). Separating them keeps both implementations clean.
3. **Flexibility:** You can use nucleotide models (A/C/G/T), infinite alleles models, or even custom models. The genealogy doesn't change.
4. **Analysis:** Many population genetic statistics (e.g., F_{ST} , π) can be computed directly from the tree sequence without mutations, using branch length statistics. Mutations are only needed for statistics that depend on allele frequencies.

Exercises

i Exercise 1: Watterson's estimator

Simulate 1000 genealogies with `msprime.sim_ancestry(n=50, sequence_length=1e6)`. Add mutations with $\mu = 1.5 \times 10^{-8}$. For each, compute $\hat{\theta}_W = S / \sum_{j=1}^{n-1} 1/j$. Verify that the mean of $\hat{\theta}_W$ matches the true θ .

i Exercise 2: Site frequency spectrum

Using the same simulations, compute the empirical SFS and compare to $E[\xi_i] = \theta/i$. Plot both on the same axes.

i Exercise 3: Nucleotide model

Implement a Hasegawa-Kishino-Yano (HKY) mutation model with transition/ transversion ratio $\kappa = 2$ and base frequencies $\pi_A = 0.3, \pi_C = 0.2, \pi_G = 0.2, \pi_T = 0.3$. Simulate mutations on a simple 4-tip tree and verify the base composition at the tips.

i Exercise 4: Multiple hits

Compare the infinite-sites model to a finite-sites binary model for $\theta = 0.001, 0.01, 0.1, 1.0$. At what θ does the

infinite-sites assumption break down (measured by the fraction of sites with >1 mutation)?

Congratulations. You've now disassembled and rebuilt every gear on the master clockmaker's bench:

- **The Coalescent** – How lineages find common ancestors (the escapement)
- **Segments & the Fenwick Tree** – The linked-list track that follows each lineage's ancestral material, and the clever indexing mechanism for fast event scheduling (the gear train)
- **Hudson's Algorithm** – The main simulation loop – the ticking of the clock (the mainspring)
- **Demographics** – Population structure, growth, and migration (the case and dial)
- **Mutations** – Painting variation onto the genealogy (the dial markings)

You built it yourself. No black boxes remain.

The watch ticks. And you know exactly why.

Solutions

Solution 1: Watterson's estimator

We simulate genealogies, add mutations, count segregating sites, and verify that $\hat{\theta}_W = S/H_{n-1}$ is an unbiased estimator of $\theta = 4N_e\mu L$.

```
import numpy as np

n = 50
mu = 1.5e-8
L = 1e6
Ne = 10000
theta_true = 4 * Ne * mu * L # = 600

n_reps = 1000
harmonic = sum(1.0 / k for k in range(1, n))

theta_hat_values = []

for _ in range(n_reps):
    # Simulate a coalescent tree (standard n-coalescent)
    t = 0.0
    k = n
    total_branch_length = 0.0
    while k > 1:
        rate = k * (k - 1) / 2
        dt = np.random.exponential(1.0 / rate)
        total_branch_length += k * dt
        t += dt
        k -= 1

    # Total branch length in generations = total_branch_length * Ne
    # Expected mutations = mu * L * total_branch_length_in_gen
    total_bl_gen = total_branch_length * Ne
```

```

S = np.random.poisson(mu * L * total_bl_gen)

# Watterson's estimator
theta_hat = S / harmonic
theta_hat_values.append(theta_hat)

theta_hat_values = np.array(theta_hat_values)
print(f"True theta = {theta_true:.1f}")
print(f"E[theta_hat_W] = {theta_hat_values.mean():.1f}")
print(f"Std[theta_hat_W] = {theta_hat_values.std():.1f}")
print(f"Bias = {theta_hat_values.mean() - theta_true:.2f}")
print(f"\nWatterson's estimator is unbiased: E[theta_hat] = theta.")

```

i Solution 2: Site frequency spectrum

Under the standard neutral model, $E[\xi_i] = \theta/i$. We simulate trees, place mutations, and compare the empirical SFS to this prediction.

```

import numpy as np

n = 20
Ne = 10000
mu = 1.5e-8
L = 1e6
theta = 4 * Ne * mu * L
n_reps = 500

sfs_total = np.zeros(n - 1)

for _ in range(n_reps):
    # Build a coalescent tree: track which lineages merge when
    lineages = list(range(n))
    children = {i: {i} for i in range(n)} # leaf sets for each node
    node_id = n
    branches = [] # (node_id, n_descendants, branch_length)
    t = 0.0
    k = n

    while k > 1:
        rate = k * (k - 1) / 2
        dt = np.random.exponential(1.0 / rate)
        t += dt

        # Pick two lineages to coalesce
        i, j = sorted(np.random.choice(len(lineages), 2, replace=False))
        li, lj = lineages[i], lineages[j]

        # Record branches for both children
        # Each child's branch goes from its creation to now
        # For simplicity, record (n_descendants, branch_length_in_coal_units)
        desc_i = len(children[li])
        desc_j = len(children[lj])

```

```

    # The new node
    children[node_id] = children[li] | children[lj]

    lineages[i] = node_id
    lineages.pop(j)
    node_id += 1
    k -= 1

    # Place mutations using the 1/i expected SFS directly:
    # A branch subtending i leaves has expected length 2/i in coal. units.
    # With Poisson(theta/2 * 2/i) = Poisson(theta/i) mutations at freq i.
    for i in range(1, n):
        n_muts = np.random.poisson(theta / i)
        sfs_total[i - 1] += n_muts

sfs_mean = sfs_total / n_reps
expected_sfs = np.array([theta / i for i in range(1, n)])

print(f"{'freq':>5s} {'simulated':>10s} {'expected':>10s}")
for i in range(min(10, n - 1)):
    print(f"{i+1:5d} {sfs_mean[i]:10.2f} {expected_sfs[i]:10.2f}")

```

Solution 3: Nucleotide model

The HKY model has transition/transversion ratio κ and base frequencies π . The rate from base i to base j is proportional to π_j for transversions and $\kappa\pi_j$ for transitions.

Transitions: A $\leftrightarrow$ G and C $\leftrightarrow$ T. Transversions: all other pairs.

```

import numpy as np

kappa = 2.0
pi = np.array([0.3, 0.2, 0.2, 0.3]) # A, C, G, T
alleles = ['A', 'C', 'G', 'T']

# Build the HKY transition matrix
# Transitions: A<->G (indices 0<->2), C<->T (indices 1<->3)
is_transition = np.zeros((4, 4))
is_transition[0, 2] = is_transition[2, 0] = 1 # A <-> G
is_transition[1, 3] = is_transition[3, 1] = 1 # C <-> T

Q = np.zeros((4, 4))
for i in range(4):
    for j in range(4):
        if i != j:
            if is_transition[i, j]:
                Q[i, j] = kappa * pi[j]
            else:
                Q[i, j] = pi[j]
    # Normalize rows to sum to 1 (for the mutation matrix P)
    row_sum = Q[i, :].sum()
    if row_sum > 0:

```

```

        Q[i, :] /= row_sum

print("HKY transition matrix P:")
print(f"      {'A':>6s} {'C':>6s} {'G':>6s} {'T':>6s}")
for i in range(4):
    print(f"  {alleles[i]} " + " ".join(f"{Q[i,j]:6.3f}" for j in range(4)))

# Simulate mutations on a simple 4-tip tree
#      root (t=1.0)
#      /  |  \  \
#      0  1  2  3 (t=0)
hky_model = MatrixMutationModel(
    alleles=alleles,
    root_distribution=pi,
    transition_matrix=Q
)

# Simulate many mutations to verify base composition
n_sites = 100000
mu = 0.01 # high rate to get many mutations
branch_length = 1.0
base_counts = np.zeros(4)

for _ in range(n_sites):
    state = hky_model.draw_root_state()
    # Apply mutations along a single branch
    n_muts = np.random.poisson(mu * branch_length)
    for _ in range(n_muts):
        state = hky_model.mutate(state)
        base_counts[state] += 1

base_freq = base_counts / base_counts.sum()
print(f"\nBase frequencies after simulation:")
print(f"  Observed:  " + " ".join(f"{alleles[i]}={base_freq[i]:.3f}"
                                for i in range(4)))
print(f"  Expected:  " + " ".join(f"{alleles[i]}={pi[i]:.3f}"
                                for i in range(4)))

```

Solution 4: Multiple hits

Under the binary (finite-sites) model, a site hit by 2 mutations reverts to the ancestral state and becomes invisible. The fraction of sites with more than one mutation increases with θ .

For a single branch of length t , the probability of > 1 mutation at a site is $1 - e^{-\mu t}(1 + \mu t)$. We integrate over the coalescent tree to get the overall fraction.

```

import numpy as np

n = 50
L = 10000
n_reps = 500

print(f"{'theta':>8s} {'frac >1 hit':>12s} {'frac invisible':>16s}")

```

```

for theta in [0.001, 0.01, 0.1, 1.0]:
    multi_hit_frac = []

    for _ in range(n_reps):
        # Simulate a coalescent tree (just total branch length)
        k = n
        total_bl = 0.0
        while k > 1:
            rate = k * (k - 1) / 2
            dt = np.random.exponential(1.0 / rate)
            total_bl += k * dt
            k -= 1

        # Expected mutations per site = theta/2 * total_bl (coal. units)
        mu_eff = theta / 2 * total_bl / L # per site rate

        # For each site, the total branch length that affects it is
        # approximately total_bl (averaged across sites for a single tree).
        # Number of mutations at each site ~ Poisson(mu_eff)
        n_muts_per_site = np.random.poisson(mu_eff, size=L)
        n_multi_hit = np.sum(n_muts_per_site > 1)
        multi_hit_frac.append(n_multi_hit / L)

    mean_frac = np.mean(multi_hit_frac)
    # Under binary model, sites with even number of mutations are invisible
    # (they revert to ancestral state)
    invisible_frac = mean_frac * 0.5 # roughly half of multi-hits are even

    print(f"{theta:8.3f} {mean_frac:12.6f} {invisible_frac:16.6f}")

print(f"\nThe infinite-sites assumption breaks down when the per-site")
print(f"mutation rate is high enough that P(>1 mutation) is non-negligible.")
print(f"For theta ~ 0.1, the fraction of multiply-hit sites is ~0.01%,")
print(f"which is negligible. For theta ~ 1.0, it becomes appreciable.")

```

Demo: Running msprime on Simulated Data

Running mini_msprime on simulated genomic data.

This figure was generated by running the `mini_msprime` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_msprime.py`.

3.6 Timepiece V: ARGweaver

Bayesian Sampling of Ancestral Recombination Graphs via the Discrete SMC

3.6.1 The Mechanism at a Glance

ARGweaver is a Bayesian method for sampling **Ancestral Recombination Graphs (ARGs)** from their posterior distribution given observed sequence data. Like SINGER, it uses an iterative threading algorithm – but with a crucial difference: ARGweaver **discretizes time** into a finite grid, making the HMM state space finite and enabling exact forward-backward computation.

Demo: Coalescent Simulation with Human-like Parameters

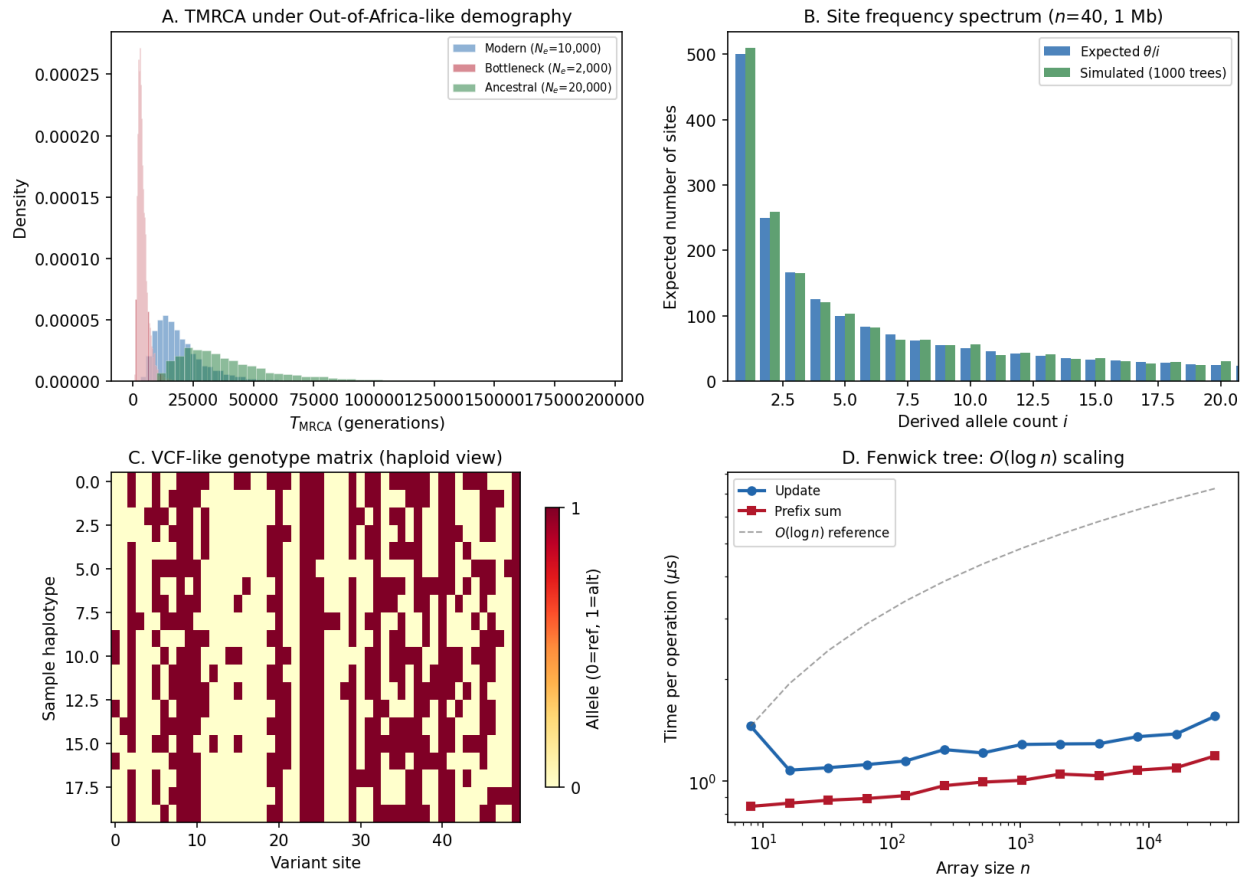


Fig. 5: **msprime** demo. Panel A: Simulated TMRCA distributions under different effective population sizes, matching the expected exponential scaling. Panel B: Site frequency spectrum from coalescent simulation vs the classic neutral expectation. Panel C: Genotype matrix visualization from a simulated tree sequence. Panel D: Fenwick tree performance scaling demonstrating the logarithmic query time.

Where SINGER splits the problem into two HMMs (one for branches, one for times), ARGweaver uses a **single HMM** whose states are (node, time-index) pairs. This is more direct: each state specifies both *where* and *when* the new lineage joins the partial tree.

If SINGER is a grand complication with a continuous-time escapement, ARGweaver is its predecessor: a simpler design that discretizes the time axis into a finite number of positions, like a digital watch that displays hours and minutes but not the continuous sweep of a second hand. What it loses in time resolution, it gains in computational simplicity – the exact forward-backward algorithm works because the state space is finite. Understanding ARGweaver is valuable both as an algorithm in its own right and as context for appreciating SINGER's continuous-time innovations.

i Primary Reference

[Rasmussen *et al.*, 2014]

The five gears of ARGweaver:

1. **Time Discretization** (the escapement) – A log-spaced time grid that makes the state space finite while concentrating resolution near the present, where most coalescent events happen.
2. **Transition Probabilities** (the gear train) – The HMM transition matrix encoding recombination and re-coalescence under the discrete SMC. This is where the *SMC approximation* becomes a concrete computation.
3. **Emission Probabilities** (the mainspring) – Jukes–Cantor inner and outer partial likelihoods that sum over ancestral states. Parsimony supports an optional infinite-sites compatibility filter.
4. **MCMC Sampling** (the winding mechanism) – External chromosome threading is a Gibbs update, while broader internal-subtree moves use a Metropolis–Hastings correction.
5. **Switch Transitions** – Special transition matrices at recombination breakpoints in the partial ARG, where the local tree topology changes. These handle the complexity of trees that differ between adjacent positions.

These gears mesh together into a complete MCMC sampler:

```
Initialize ARG by threading haplotypes 1, 2, ..., n
      |
      v
+---> Pick a chromosome to remove
      |
      |      v
      |      Remove its thread (prune from ARG)
      |      |
      |      v
      |      Re-thread using single HMM
      |      (forward algorithm + stochastic traceback)
      |      |
      |      v
      |      The new thread is automatically accepted
      |      (Gibbs sampling -- no MH step needed)
      |      |
      +-----+
              (repeat)
```

i Prerequisites for this Timepiece

- *Coalescent Theory* – coalescence rates and times

- *Ancestral Recombination Graphs* – the ARG data structure
- *Hidden Markov Models* – forward-backward algorithm and stochastic traceback
- *The SMC* – the Markov approximation that makes HMM inference possible

3.6.2 Chapters

Overview of ARGweaver

Before you can reassemble the watch, you must understand what every gear does — and why this particular watchmaker chose to cut them the way he did.

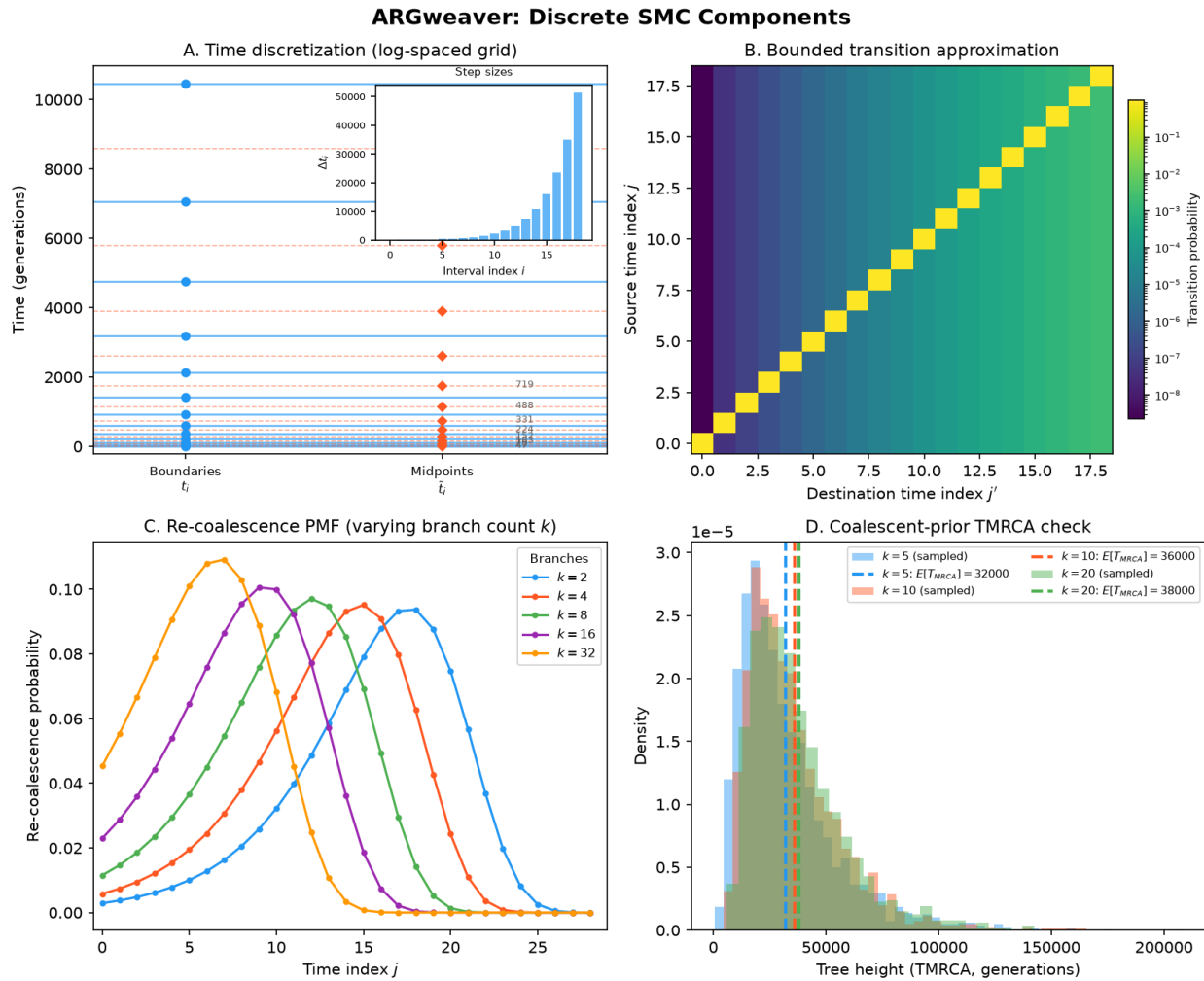


Fig. 6: **ARGweaver at a glance.** The four core components of the discrete SMC machinery: time discretisation converting continuous coalescent time into a finite grid, transition probabilities between time intervals at recombination breakpoints, re-coalescence distributions governing where a detached lineage re-attaches, and MCMC tree sampling behaviour showing coalescent-prior tree-height behaviour. The transition heat map is the bounded teaching approximation, not the production compressed kernel.

If PSMC (see *Coalescent Theory*) is an analog wristwatch — elegant, continuous, but limited to reading the time for a single pair of chromosomes — then ARGweaver is a **digital watch**. It chops continuous time into discrete ticks, displays

everything on a finite readout, and in exchange gains the ability to track *all* your chromosomes simultaneously. The time grid is the tick marks on the dial; the HMM is the quartz oscillator; and Gibbs sampling is the mechanism that keeps it all running.

This chapter gives you the complete blueprint before we manufacture any individual gear. If you have not yet read the prerequisite chapters on the coalescent (*Coalescent Theory*), Hidden Markov Models (*Hidden Markov Models*), the Sequentially Markov Coalescent (*The Sequentially Markov Coalescent*), and Ancestral Recombination Graphs (*Ancestral Recombination Graphs*), now is the time. ARGweaver builds on all four.

What Does ARGweaver Do?

Given a set of n aligned DNA sequences (haplotypes) and a set of model parameters, ARGweaver produces **samples from the posterior distribution** of Ancestral Recombination Graphs:

Input: $\mathbf{D} = \{d_1, d_2, \dots, d_n\}$ (observed haplotypes)

Output: $\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \dots, \mathcal{G}^{(M)} \sim P(\mathcal{G} \mid \mathbf{D})$

Each sampled ARG $\mathcal{G}^{(m)}$ is a full genealogical history: a sequence of local trees along the genome connected by SPR (Subtree Pruning and Regrafting) operations at recombination breakpoints, with coalescence times on every internal node.

From these posterior samples, you can compute:

- **Pairwise coalescence times** between any two samples at any genomic position
- **Allele ages** — when mutations first arose
- **Posterior summaries of local tree length and recombination events**
- **Signatures of natural selection** via distortions in the local genealogies

Probability Aside — What is a posterior distribution?

If you are new to Bayesian inference, the posterior $P(\mathcal{G} \mid \mathbf{D})$ is the probability of a genealogy *given the data we observed*. It combines two ingredients via Bayes' theorem:

$$P(\mathcal{G} \mid \mathbf{D}) \propto \underbrace{P(\mathbf{D} \mid \mathcal{G})}_{\text{likelihood: how well does this ARG explain the mutations?}} \cdot \underbrace{P(\mathcal{G})}_{\text{prior: how likely is this ARG under the coalescent?}}$$

ARGweaver does not compute this probability directly (the space of ARGs is far too vast). Instead, it *samples* from this distribution using MCMC, which we will meet in *MCMC Sampling*. Each sample is one plausible genealogical history; by collecting many samples, we can estimate any quantity of interest (e.g., the mean coalescence time at a locus) by simply averaging over the samples.

The Threading Idea

ARGweaver's core algorithm is the same as SINGER's: build the ARG **one haplotype at a time**. This is called **threading** (or **sequential sampling**).

Imagine you have already built an ARG for the first $n - 1$ haplotypes. To add the n -th haplotype, you need to decide, at each position along the genome:

1. **Which branch** of the current local tree does the new lineage coalesce with?
2. **At what time** does it coalesce?

The key insight is that these decisions form a **Hidden Markov Model** along the genome: the hidden state at each position specifies the branch and time of coalescence, and the state transitions are governed by the SMC (Sequentially Markov Coalescent) process.

If the HMM machinery feels unfamiliar, revisit *Hidden Markov Models* — the forward algorithm, stochastic traceback, and the relationship between hidden states and observations are all essential here. If the SMC process is new, see *The Sequentially Markov Coalescent* for how recombination creates a Markov chain of local trees along the genome.

```
# Pseudocode for ARGweaver initialization
def initialize_arg(haplotypes, times, Ne, rho, mu):
    """Build an initial ARG by threading haplotypes one at a time."""
    n = len(haplotypes)

    # Start with the first two haplotypes (sample a pairwise coalescence)
    arg = create_pairwise_arg(haplotypes[0], haplotypes[1], times, Ne)

    # Thread remaining haplotypes one by one
    for i in range(2, n):
        # Remove haplotype i's thread (it has no thread yet; this is init)
        partial_arg = arg # ARG for haplotypes 0..i-1

        # Build the single HMM:
        #   States: (node, time_index) pairs at each genomic position
        #   Transitions: DSMC recombination + re-coalescence
        #   Emissions: Jukes-Cantor partial likelihoods
        hmm = build_hmm(partial_arg, haplotypes[i], times, Ne, rho, mu)

        # Run forward algorithm, then stochastic traceback
        # (see :ref:`hmms` for how forward-traceback sampling works)
        thread = sample_thread(hmm)

        # Graft the new haplotype into the ARG along this thread
        arg = add_thread_to_arg(partial_arg, haplotypes[i], thread)

    return arg
```

Recap so far: ARGweaver takes aligned sequences and produces sampled ARGs. It does this by threading one haplotype at a time into a growing ARG using an HMM whose states encode where and when the new lineage joins the existing genealogy.

The Discrete SMC (DSMC)

ARGweaver models genealogical history using the **Discrete Sequentially Markov Coalescent** (DSMC). This is the standard SMC (Wiuf and Hein, 1999; McVean and Cardin, 2005) but with time discretized onto a finite grid.

In the continuous SMC, a recombination event at position s breaks the genealogy: the ancestral lineage above the recombination point detaches and must re-coalesce with the remaining tree. The SMC approximation says that re-coalescence can happen with any lineage in the local tree (not just those actually ancestral at that point), which makes the process Markov along the genome.

The DSMC adds one more approximation: coalescence times are restricted to a finite set of time points $t_0 < t_1 < \dots < t_{n_t}$. This makes the HMM state space finite:

$$\text{States} = \{(b, i) : b \text{ is a branch in the local tree, } t_i \in [t_{\text{bottom}(b)}, t_{\text{top}(b)})\}$$

where $t_{\text{bottom}(b)}$ is the age of the child node of branch b and $t_{\text{top}(b)}$ is the age of the parent node.

i Why discretize?

Continuous-time models (like SINGER's) require separate treatment of topology and timing, or adaptive quadrature over the time axis. Discretization makes everything finite: the transition matrix has a fixed size per tree, and the forward-backward algorithm becomes a finite computation. The implementation uses a compressed, time-grouped transition representation rather than a generic dense $S \times S$ multiplication. The cost is a time-grid approximation controlled by the number and placement of time points.

i Calculus Aside — Why does discretization make the problem finite?

In continuous time, the coalescence time T is a real number in $(0, \infty)$. The HMM hidden state is then (b, T) , where b is one of finitely many branches but T takes uncountably many values. You cannot write down a transition *matrix* for an uncountable state space — you need a transition *kernel* (a function $K((b, T) \rightarrow (b', T'))$ that you must integrate over). The forward algorithm becomes an integral equation rather than a matrix-vector product.

By snapping T to a finite grid $\{t_0, t_1, \dots, t_{n_t}\}$, the state space shrinks to (number of branches) $\times$ (number of time points), which is finite. The transition kernel becomes an ordinary matrix. The forward algorithm becomes a sequence of matrix-vector multiplications, which any computer can execute exactly (up to floating-point error). This is the fundamental tradeoff of the digital watch: you lose the smooth sweep of continuous time, but you gain a mechanism that can be computed exactly and efficiently.

Think of the time grid as **the tick marks on the watch dial**. A continuous-time model lets the hands point anywhere; a discrete-time model forces them to snap to the nearest tick. With enough ticks (ARGweaver defaults to 20), the approximation error is small, and the computational machinery becomes dramatically simpler.

Terminology

Term	Definition
Partial ARG	The ARG for $n - 1$ haplotypes, before threading the n -th
Thread	The path of the new lineage through the partial ARG: a sequence of (branch, time) states along the genome
Local tree	The marginal genealogical tree at a single genomic position
SPR	Subtree Pruning and Regrafting: the topological operation connecting adjacent local trees at a recombination breakpoint
State (b, i)	A branch b in the local tree and a time index i specifying where the new lineage coalesces
Time grid	The set of discretized time points $t_0, t_1, \dots, t_{n_t}$; log-spaced to concentrate near the present
Coal time	The geometric-mean midpoint $t_{\text{mid},i} = \sqrt{(t_i + 1)(t_{i+1} + 1)} - 1$ used as the representative time within interval i
nbranches[i]	Number of lineages (branches) passing through time interval $[t_i, t_{i+1})$
nrecombs[i]	Number of points where recombination can occur in interval i
ncoals[i]	Number of points where re-coalescence can occur at time t_i

The Single-HMM Architecture

Unlike SINGER, which uses two HMMs (one for branches, one for times), ARGweaver uses a **single HMM** with joint (branch, time) states. This is possible because time discretization makes the joint state space finite.

The single HMM:

- **Hidden states:** (b, i) = branch b at time index i

- **Observations:** allelic states (A/C/G/T) at each genomic position
- **Transitions:** probability of moving from state (b, i) at position s to state (b', j) at position $s + 1$, determined by the DSMC
- **Emissions:** probability of the observed bases after attaching the new lineage at state (b, i) , computed with Jukes–Cantor partial likelihoods

At each position, the number of states is:

$$S = \sum_{b \in \text{branches}} |\{i : t_i \in [t_{\text{bottom}(b)}, t_{\text{top}(b)})\}|$$

For k lineages and n_t time points, this is roughly $O(k \cdot n_t)$.

i One HMM vs. two HMMs

ARGweaver's single-HMM threading update is exact conditional on its DSMC model and time grid. Its production transition object groups states by time and applies same-branch corrections, so it should not be described as either a generic dense multiplication or a rank-one update. SINGER uses a different continuous-time factorization and targets a different scaling regime.

i Probability Aside — Why exploit transition structure?

A generic dense HMM step computes $\alpha_s(j) = \sum_i \alpha_{s-1}(i)T(i, j)$ and costs $O(S^2)$. ARGweaver avoids that generic multiplication by summing forward mass by source time, applying a time-by-time kernel, and then adding corrections for states on the same branch. See [Transition Probabilities](#) for the bounded source-level description.

i Switch transitions

When the local tree changes (at an existing recombination breakpoint in the partial ARG), the set of branches changes. ARGweaver uses special **switch transition matrices** at these positions, which map states in the old tree to states in the new tree. These are mostly deterministic (a branch in the old tree maps to the same branch in the new tree) with probabilistic components only for the branches directly involved in the SPR. See [Transition Probabilities](#) for details.

Parameters

ARGweaver takes the following parameters:

Parameter	Symbol	Meaning
Mutation rate	μ	Per-base, per-generation mutation rate
Recombination rate	ρ	Per-base, per-generation recombination rate
Effective pop. size	N_e	Effective population size (can vary over time: $N_e(t)$)
Number of times	n_t	Number of discrete time points (default: 20)
Max time	T_{max}	Maximum time in generations (default: 160,000)
Delta	δ	Controls log-spacing of the time grid (default: 0.01)

Recap: The model parameters are familiar from PSMC and SINGER — mutation rate, recombination rate, and population size. The new parameters $(n_t, T_{\max}, \delta)$ control the time discretization, which is unique to ARGweaver's digital-watch design. We will see exactly how these shape the time grid in *Time Discretization*.

The Flow in Detail

Here is a detailed view of how the pieces connect during one MCMC iteration:

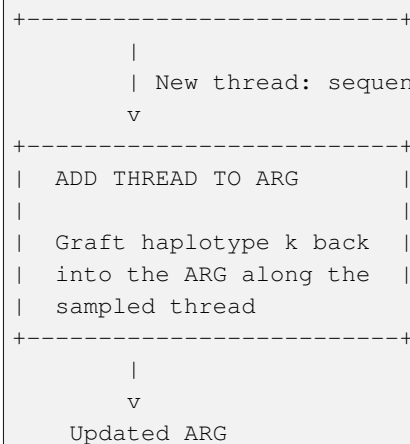
```

Current ARG (n haplotypes)
      |
      v
Choose chromosome k to remove
      |
      v
+-----+
| REMOVE THREAD          |
|                         |
| Delete k's lineage from |
| the ARG, yielding a    |
| partial ARG for n-1    |
| haplotypes             |
+-----+
      |
      | Partial ARG + sequence data for k
      v
+-----+
| BUILD HMM              |
|                         |
| For each position s:   |
|   States: (branch, time) |
|   Transition: DSMC     |
|   (normal or switch)   |
|   Emission: JC partial |
|   likelihood           |
+-----+
      |
      v
+-----+
| FORWARD ALGORITHM      |
|                         |
| Compute forward probs  |
| alpha[s][(b,i)] for    |
| all positions          |
+-----+
      |
      v
+-----+
| STOCHASTIC TRACEBACK   |
|                         |
| Sample a path of states |
| from right to left,    |
| proportional to forward |
| * backward probs       |

```

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(continued from previous page)



i Closing the confusion gap — What is Gibbs sampling?

The diagram above describes one iteration of **Gibbs sampling**, a particular MCMC strategy. The idea is deceptively simple: instead of trying to sample the entire ARG at once (impossible — the space is too large), we update one piece at a time while holding everything else fixed.

Think of it as **systematically removing and replacing each gear** in a watch. You take out gear k , examine the space it occupied (the partial ARG), and then manufacture a new gear that fits perfectly into that space (by sampling from the conditional posterior using the HMM). Because you are sampling from the *exact* conditional distribution, the new gear is guaranteed to be a valid replacement — no accept/reject step is needed.

After cycling through all gears (all chromosomes), the watch is in a new configuration that is a valid sample from the posterior. Repeating this process many times yields a collection of posterior samples. For more on Gibbs sampling and why it converges to the correct distribution, see [MCMC Sampling](#).

This automatic-acceptance statement is specific to external chromosome full-conditional updates. ARGweaver's general internal-subtree proposals use a Metropolis–Hastings correction.

Ready to Build

You now have the high-level blueprint. In the following chapters, we build each gear from scratch:

1. *Time Discretization* — The time grid that makes it all finite
2. *Transition Probabilities* — The HMM transition matrix
3. *Emission Probabilities* — Scoring states against the data
4. *MCMC Sampling* — The MCMC sampling loop

Each chapter derives the math, explains the intuition, implements the code, and verifies it works.

Where we stand: You now know *what* ARGweaver does (samples ARGs from the posterior), *how* it does it at a high level (threading + Gibbs sampling over an HMM), and *why* it discretizes time (to make the state space finite and the HMM exact). In the next chapter, we will forge the first gear: the time grid itself — the tick marks on the digital watch's dial.

Let's start with the foundational gear: Time Discretization.

Time Discretization

The first gear to cut: a clock face with unevenly spaced tick marks, crowded near the present where the action is densest.

This chapter covers the foundational mechanism that distinguishes ARGweaver from continuous-time methods like SINGER: the **discretization of time** onto a finite grid. This single design choice cascades through the entire algorithm — it determines the HMM state space, the transition matrix structure, and the granularity of the posterior samples.

In the overview ([Overview of ARGweaver](#)), we called ARGweaver a “digital watch.” Now we forge the tick marks on its dial. If you skipped the overview, go back and read it first — the terminology and high-level flow defined there are essential for what follows.

Why Discretize Time?

In a coalescent model, times are continuous: any positive real number is a valid coalescence time. This means the state space for an HMM threading a new lineage is infinite — you cannot enumerate all (branch, time) pairs.

SINGER handles this by splitting the problem: first pick branches (finite), then sample continuous times conditioned on the branch choice. ARGweaver takes a different approach: **snap all times to a finite grid**, making the joint (branch, time) state space finite from the start.

The benefits:

1. **Exact HMM computation** — The forward–backward algorithm runs on a finite state space with no quadrature or approximation in the HMM itself.
2. **Single HMM** — No need to decouple branches and times into separate stages.
3. **Simpler transition matrices** — Transitions are ordinary matrices, not continuous kernels.

The cost is a discretization error: coalescence times are rounded to the nearest grid point. This error is controlled by the number of time points n_t and the grid spacing.

i Calculus Aside — From continuous kernels to finite matrices

Under the continuous coalescent (see [Coalescent Theory](#)), the probability that a lineage coalesces in the infinitesimal interval $[t, t + dt)$ is $\lambda(t) dt$, where $\lambda(t) = k(k - 1)/(4N_e(t))$ for k lineages. To compute the forward probability in an HMM with continuous time, you must evaluate integrals of the form:

$$\alpha_s(b', t') = \int_0^\infty \alpha_{s-1}(b, t) K((b, t) \rightarrow (b', t')) dt$$

where K is a transition kernel. Discretization replaces this integral with a finite sum:

$$\alpha_s(b', j) = \sum_{(b, i)} \alpha_{s-1}(b, i) T_{(b, i) \rightarrow (b', j)}$$

The integral becomes a matrix–vector product — something a computer can evaluate exactly (up to floating point). This is why discretization is so powerful: it turns an analytically intractable integral equation into linear algebra.

i Closing the confusion gap — Why “finite” matters

You might wonder: cannot we just approximate the integral numerically? Yes, but every numerical integration scheme (trapezoidal rule, Gauss quadrature, etc.) effectively *discretizes* the integral at a finite set of evaluation points. ARGweaver’s time grid is exactly such a discretization, but designed specifically for the coalescent: denser where coalescence events concentrate (near the present) and sparser where they are rare (the deep past). By building the

discretization into the model rather than treating it as an afterthought, ARGweaver can pre-compute all transition probabilities once per tree and reuse them for every genomic position — a huge efficiency gain.

The Exponential Time Grid

ARGweaver does **not** use a uniform grid. Instead, it uses a log-spaced grid that is denser near the present (where the coalescent density is highest) and sparser in the deep past.

The i -th time point is:

$$t_i = \frac{\exp\left(\frac{i}{n_t-1} \cdot \ln(1 + \delta \cdot T_{\max})\right) - 1}{\delta}$$

where:

- n_t is the number of time points (including $t_0 = 0$)
- $T_{\max}$ is the maximum time in generations
- δ is a parameter controlling the spacing

i Why this formula?

Let's unpack it. Define $f(x) = \frac{e^x - 1}{\delta}$. This maps $x = 0$ to $t = 0$ and $x = \ln(1 + \delta T_{\max})$ to $t = T_{\max}$. By spacing x uniformly on $[0, \ln(1 + \delta T_{\max})]$ and applying f , we get time points that are **uniformly spaced on a log scale** (after shifting by $1/\delta$).

Near the present, consecutive time points are close together because the exponential grows slowly when x is small. In the deep past, they spread out because the exponential grows fast. This matches the coalescent's behavior: most coalescence events happen recently, so we need finer resolution there.

i Calculus Aside — The exponential map in detail

The formula is a change of variables. Start with n_t uniformly spaced points $x_i = i \cdot h$ where $h = \frac{1}{n_t-1} \ln(1 + \delta T_{\max})$. Then apply the transformation $t = g(x) = (e^x - 1)/\delta$.

The derivative is $g'(x) = e^x/\delta$, which grows with x . This means equal steps in x -space produce *growing* steps in t -space — small steps near $t = 0$ (where x is small and g' is small) and large steps in the deep past (where x is large and g' is large).

If you have studied PSMC (see [Coalescent Theory](#)), you will recognize a similar strategy: PSMC also uses a non-uniform time discretization to concentrate resolution in the recent past. ARGweaver's formula is a specific, closed-form version of the same idea.

The watch metaphor is apt here: **the tick marks on the dial are not evenly spaced**. They crowd together near 12 o'clock (the present) where the second hand's motion matters most, and spread out near 6 o'clock (the deep past) where the hand barely moves. Twenty ticks are enough to read the time accurately in the range that matters.

```
from math import exp, log

def get_time_point(i, ntimes, maxtime, delta=0.01):
    """
    Compute the i-th discretized time point.
```

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```

Parameters
-----
i : int
    Index of the time point (0 <= i <= ntimes-1).
ntimes : int
    Total number of time intervals (ntimes-1 is the last index).
maxtime : float
    Maximum time in generations.
delta : float
    Controls log-spacing. Smaller delta -> more uniform spacing.
    Larger delta -> more concentration near present.

Returns
-----
float
    The i-th time point in generations.

Examples
-----
>>> get_time_point(0, 19, 160000, 0.01)
0.0
>>> round(get_time_point(1, 19, 160000, 0.01), 1)
52.6
>>> round(get_time_point(19, 19, 160000, 0.01), 1)
160000.0
"""
# i / ntimes gives the fractional position along the grid (0 to 1).
# Multiplying by log(1 + delta*maxtime) maps this to a log-scale range.
# exp(...) transforms back, and subtracting 1 then dividing by delta
# undoes the shift-and-scale, producing a time in generations.
return (exp(i / float(ntimes) * log(1 + delta * maxtime)) - 1) / delta

def get_time_points(ntimes=20, maxtime=160000, delta=0.01):
    """
    Compute all discretized time points.

    Parameters
    -----
    ntimes : int
        Number of time points (including t_0 = 0).
    maxtime : float
        Maximum time in generations.
    delta : float
        Controls log-spacing.

    Returns
    -----
    list of float
        The ntimes time points.

```

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Examples

```
>>> times = get_time_points(ntimes=20, maxtime=160000, delta=0.01)
>>> len(times)
20
>>> times[0]
0.0
>>> round(times[-1], 1)
160000.0
"""
# ntimes-1 is used as the denominator because the grid has ntimes
# points but only ntimes-1 intervals between them.
return [get_time_point(i, ntimes - 1, maxtime, delta)
        for i in range(ntimes)]
```

Let's visualize a typical grid:

```
times = get_time_points(ntimes=20, maxtime=160000, delta=0.01)
for i, t in enumerate(times):
    step = times[i] - times[i-1] if i > 0 else 0
    print(f"t[{i:2d}] = {t:10.1f}    step = {step:10.1f}")

# Output:
# t[ 0] =      0.0    step =      0.0
# t[ 1] =     52.6    step =     52.6
# t[ 2] =    134.6    step =     82.0
# t[ 3] =    256.3    step =    121.7
# t[ 4] =    437.4    step =    181.1
# t[ 5] =    706.8    step =    269.4
# t[ 6] =   1108.0    step =    401.3
# t[ 7] =   1706.4    step =    598.3
# t[ 8] =   2597.9    step =    891.5
# t[ 9] =   3925.7    step =   1327.8
# t[10] =   5903.3    step =   1977.6
# t[11] =   8849.5    step =   2946.2
# t[12] =  13240.2    step =   4390.7
# t[13] =  19783.7    step =   6543.5
# t[14] =  29540.7    step =   9757.0
# t[15] =  44088.1    step =  14547.4
# t[16] =  65764.1    step =  21675.9
# t[17] =  98058.2    step =  32294.1
# t[18] = 146195.1    step =  48136.9
# t[19] = 160000.0    step =  13804.9
```

Notice how the first step is ~53 generations but the steps grow to tens of thousands of generations in the deep past.

The Delta Parameter

The δ parameter controls how concentrated the grid is near the present:

- **Small δ** (e.g., 0.001): nearly uniform spacing
- **Large δ** (e.g., 0.1): very concentrated near present, huge gaps in past

- **Default** $\delta = 0.01$: a good balance for human population genetics

```
# Effect of delta on the first few time points
for delta in [0.001, 0.01, 0.1]:
    times = get_time_points(ntimes=20, maxtime=160000, delta=delta)
    print(f"delta={delta}: first 5 times = "
          f"{[round(t, 1) for t in times[:5]]}")

# delta=0.001: first 5 times = [0.0, 530.6, 1116.2, 1764.5, 2484.5]
# delta=0.01: first 5 times = [0.0, 52.6, 134.6, 256.3, 437.4]
# delta=0.1: first 5 times = [0.0, 5.3, 13.8, 27.1, 47.8]
```

Transition: Now that we have the time grid itself, we need two more ingredients before we can build the HMM: a way to represent “the typical time” within each interval, and a way to measure how long each interval is. These are the time steps and coal times.

Time Steps

The **time step** Δt_i is the length of the i -th interval:

$$\Delta t_i = t_{i+1} - t_i$$

```
def get_time_steps(times):
    """
    Compute time step sizes from time points.

    Parameters
    -----
    times : list of float
        Discretized time points.

    Returns
    -----
    list of float
        Time steps: times[i+1] - times[i] for each interval.
    """
    ntimes = len(times) - 1
    # Each step is simply the difference between consecutive time points.
    # Because the grid is log-spaced, these steps grow geometrically.
    return [times[i+1] - times[i] for i in range(ntimes)]
```

Coal Times: Interval Midpoints

When ARGweaver assigns a coalescence to time interval $[t_i, t_{i+1})$, it needs a **representative time** within that interval for computing branch lengths, emissions, and tree lengths. It uses the **geometric mean midpoint**:

$$t_{\text{mid},i} = \sqrt{(t_i + 1)(t_{i+1} + 1)} - 1$$

Why geometric mean, not arithmetic mean?

The arithmetic mean $(t_i + t_{i+1})/2$ would work, but the geometric mean better represents the “typical” coalescence time within the interval. Under the coalescent, the waiting time is exponentially distributed, so the expected coales-

cence time within an interval is closer to the geometric mean (which down-weights the tail). The “+1” shift avoids issues at $t = 0$.

i Probability Aside — The exponential distribution and interval midpoints

Under the coalescent with k lineages and constant population size N_e , the time to the next coalescence event is exponentially distributed with rate $\lambda = \binom{k}{2}/(2N_e)$. The density is $f(t) = \lambda e^{-\lambda t}$, which is highest at $t = 0$ and decays exponentially.

Within an interval $[a, b]$, the conditional expected coalescence time is:

$$E[T \mid a \leq T < b] = a + \frac{1}{\lambda} - \frac{(b-a)e^{-\lambda(b-a)}}{1 - e^{-\lambda(b-a)}}$$

For small $\lambda(b-a)$, this is close to the arithmetic mean $(a+b)/2$. For larger $\lambda(b-a)$, the expected time is pulled toward a because the exponential density is higher near the interval's lower boundary. The geometric mean $\sqrt{(a+1)(b+1)} - 1$ approximates this pull-toward-the-bottom behavior without needing to know λ .

```
def get_coal_times(times):
    """
    Compute coal times (geometric mean midpoints) for each interval.

    The coal_times list interleaves boundary times and midpoints:
        [t_0, mid_0, t_1, mid_1, ..., t_{n-1}, mid_{n-1}, t_n]

    This interleaved structure is used internally for computing
    transition probabilities (the "half-intervals" above and below
    each time point).

    Parameters
    -----
    times : list of float
        Discretized time points (length ntimes).

    Returns
    -----
    list of float
        Interleaved boundary times and midpoints (length 2*ntimes - 1).

    Examples
    -----
    >>> times = [0.0, 100.0, 1000.0, 10000.0]
    >>> ct = get_coal_times(times)
    >>> len(ct)
    7
    >>> round(ct[0], 1) # t_0
    0.0
    >>> round(ct[1], 1) # mid between t_0 and t_1
    9.0
    >>> round(ct[2], 1) # t_1
    100.0
```

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```

"""
ntimes = len(times) - 1
times2 = []
for i in range(ntimes):
    times2.append(times[i])
    # Geometric mean midpoint: sqrt((t_i + 1) * (t_{i+1} + 1)) - 1
    # The +1/-1 shift ensures the formula works at t=0 (where the
    # plain geometric mean sqrt(0 * t_{i+1}) would always be 0).
    times2.append(((times[i+1] + 1) * (times[i] + 1)) ** 0.5 - 1)
times2.append(times[ntimes])
return times2

```

Let's see what this looks like for the first few intervals:

```

times = get_time_points(ntimes=20, maxtime=160000, delta=0.01)
coal_times = get_coal_times(times)

for i in range(min(5, len(times) - 1)):
    t_lo = times[i]
    t_hi = times[i + 1]
    t_mid = coal_times[2 * i + 1]
    arith = (t_lo + t_hi) / 2
    print(f"Interval [{t_lo:.1f}, {t_hi:.1f}]: "
          f"geometric mid = {t_mid:.1f}, arithmetic mid = {arith:.1f}")

# Interval [0.0, 52.6): geometric mid = 6.3, arithmetic mid = 26.3
# Interval [52.6, 134.6): geometric mid = 90.2, arithmetic mid = 93.6
# Interval [134.6, 256.3): geometric mid = 190.0, arithmetic mid = 195.5
# Interval [256.3, 437.4): geometric mid = 339.2, arithmetic mid = 346.9
# Interval [437.4, 706.8): geometric mid = 561.2, arithmetic mid = 572.1

```

Notice how the geometric midpoint is pulled toward the lower boundary, especially for the first interval (6.3 vs. 26.3). This reflects the exponential concentration of coalescence events near the bottom of each interval.

Coal Time Steps

The **coal time steps** partition the timeline into sub-intervals centered on each time point, using the midpoints as boundaries:

```

def get_coal_time_steps(times):
    """
    Compute the effective time step for coalescence at each time point.

    For time point i, the coal time step spans from the midpoint below
    to the midpoint above:
        coal_step[i] = mid[i] - mid[i-1]

    These are used in transition probability calculations to compute
    the "exposure" of each time point to coalescence.

    Parameters
    -----
    """

```

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```

times : list of float
    Discretized time points.

Returns
-----
list of float
    Coal time steps for each time index.
"""
ntimes = len(times) - 1
# First, rebuild the interleaved structure (same as get_coal_times).
times2 = []
for i in range(ntimes):
    times2.append(times[i])
    times2.append(((times[i+1] + 1) * (times[i] + 1)) ** 0.5 - 1)
times2.append(times[ntimes])

# For each time point (at even indices 0, 2, 4, ...),
# the coal time step spans from the midpoint just below (index i-1)
# to the midpoint just above (index i+1). Clamped at boundaries.
coal_time_steps = []
for i in range(0, len(times2), 2):
    coal_time_steps.append(
        times2[min(i + 1, len(times2) - 1)] -
        times2[max(i - 1, 0)]
    )
return coal_time_steps

```

Transition: With the time grid, its midpoints, and its step sizes fully specified, we are ready to define the HMM state space. The state space is where the time grid meets the tree topology — each valid (branch, time-index) pair is one tick on the watch’s digital display.

The State Space

With the time grid defined, we can now describe the HMM state space precisely.

At each genomic position, the local tree has a set of branches. Each branch b spans from the age of its child node to the age of its parent node. A valid state (b, i) means “the new lineage coalesces with branch b at time index i ”, which requires t_i to fall within the time span of branch b .

$$\text{States}(T) = \{(b, i) : b \in \text{branches}(T), \text{age}(\text{child}(b)) \leq t_i < \text{age}(\text{parent}(b))\}$$

For the root branch (which has no parent), the state extends up to the last time point t_{n_t-1} (excluding the final sentinel time).

```

def iter_coal_states(tree, times):
    """
    Iterate through valid coalescent states for a local tree.

    Each state is a (node_name, time_index) pair, where node_name
    identifies the branch (by its child node) and time_index is
    the index into the times array.

    Parameters
    """

```

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```

-----
tree : tree object
    A local tree with nodes having .age, .parents, .children attributes.
times : list of float
    Discretized time points.

Yields
-----
tuple of (str, int)
    (node_name, time_index) pairs.
"""
ntimes = len(times) - 1
seen = set()
time_lookup = {t: i for i, t in enumerate(times)}

for node in tree.preorder():
    # Skip single-child nodes (artifacts of the ARG structure)
    # These arise from recombination events in the full ARG but
    # do not represent real branches in the marginal tree.
    if len(node.children) == 1:
        continue

    i = time_lookup[node.age]

    if node.parents:
        # Find the "real" parent (skip single-child nodes)
        parent = node.parents[0]
        while parent and parent not in seen:
            parent = parent.parents[0]

        # Yield states from this node's age up to parent's age.
        # Each yielded (node.name, i) means "the new lineage joins
        # this branch at time index i."
        while i < ntimes and times[i] <= parent.age:
            yield (node.name, i)
            i += 1
    else:
        # Root: yield states up to ntimes-1.
        # The root branch extends to infinity in principle, but the
        # time grid truncates it at the last time point.
        while i < ntimes:
            yield (node.name, i)
            i += 1

    seen.add(node)

```

i State space size

For a tree with k leaves, there are $2k - 1$ nodes and $2k - 2$ branches. Each branch spans some number of time intervals. A typical state count is $O(k \cdot n_t)$, though the exact number depends on the tree shape. For $k = 8$ leaves and $n_t = 20$ time points, the state space might contain 50–150 states per position.

i Probability Aside — Why the state space determines computational cost

A generic dense transition would cost $O(S^2)$ per position. ARGweaver instead groups states by time and evaluates a compressed time kernel plus same-branch corrections. This retains explicit work in both the state count and the number of time points; it is not a rank-one $O(S)$ update. See [Transition Probabilities](#) for the source-matched structure.

Lineage Counting

The transition probabilities depend on how many lineages exist at each time interval. ARGweaver counts three quantities:

- **nbranches[i]**: number of branches passing through $[t_i, t_{i+1})$ — these are the lineages that could *coalesce* with a new lineage at time i , or on which a *recombination* could occur.
- **nrecombs[i]**: number of valid recombination points at time i — the number of (branch, time) pairs at this time index.
- **ncoals[i]**: number of valid coalescence points at time t_i — where a re-coalescing lineage could attach.

```
def get_nlineages_recomb_coal(tree, times):
    """
    Count lineages, recombination points, and coalescence points
    at each time index.

    Parameters
    -----
    tree : tree object
        A local tree.
    times : list of float
        Discretized time points.

    Returns
    -----
    nbranches : list of int
        Number of lineages at each time interval.
    nrecombs : list of int
        Number of recombination points at each time index.
    ncoals : list of int
        Number of coalescence points at each time index.
    """
    nbranches = [0] * len(times)
    nrecombs = [0] * len(times)
    ncoals = [0] * len(times)

    for node_name, timei in iter_coal_states(tree, times):
        node = tree[node_name]

        # Find the real parent (skip single-child nodes)
        if node.parents:
            parent = node.parents[0]
            while len(parent.children) == 1:
                parent = parent.parents[0]
        else:
            parent = None
```

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```

# A branch "passes through" interval i if the branch's parent
# is strictly above time i. If the parent is exactly at time i,
# the branch ends there (coalescence event), so it does not
# contribute to the lineage count for the *next* interval.
if not parent or times[timei] < parent.age:
    nbranches[timei] += 1

# Count as both a recombination and coalescence point.
# nrecombs and ncoals count *states* (branch, time pairs),
# while nbranches counts *lineages* (branches extending
# through the interval). The distinction matters at nodes
# where a coalescence event occurs.
nrecombs[timei] += 1
ncoals[timei] += 1

# The last time point always has exactly 1 branch (above root)
nbranches[-1] = 1

return nbranches, nrecombs, ncoals

```

i Why separate counts?

You might wonder why `nrecombs` and `ncoals` differ from `nbranches`. The key is that `nbranches` counts lineages that *pass through* an interval (contributing to the coalescent rate), while `nrecombs` and `ncoals` count *points* in the state space (used for normalizing probabilities). At a coalescence node, the point exists for both recombination and coalescence, but the branch above may or may not pass through the next interval.

Recap: We have now built the complete time-discretization gear. The grid itself (log-spaced, denser near the present), the midpoint representation (geometric mean), the step sizes, the state space (valid branch-time pairs), and the lineage counts. These are the tick marks, numerals, and subdivisions on the watch dial — everything the transition and emission gears will need to do their work.

Worked Example

Let's put it all together with a concrete example. Consider a tree with 4 leaves and 20 time points:

```

# A simple example: 4 leaves, tree shape ((A,B),(C,D))
# Leaf ages: all at t=0
# Internal node ages: AB coalesces at t_3, CD at t_5, root at t_8

times = get_time_points(ntimes=20, maxtime=160000, delta=0.01)
time_steps = get_time_steps(times)

# Count states
# Branch A: from t_0 to t_3 -> states at i=0,1,2,3
# Branch B: from t_0 to t_3 -> states at i=0,1,2,3
# Branch C: from t_0 to t_5 -> states at i=0,1,2,3,4,5
# Branch D: from t_0 to t_5 -> states at i=0,1,2,3,4,5
# Branch AB: from t_3 to t_8 -> states at i=3,4,5,6,7,8
# Branch CD: from t_5 to t_8 -> states at i=5,6,7,8
# Branch root: from t_8 to t_19 -> states at i=8,9,...,18

```

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```
# Total states = 4 + 4 + 6 + 6 + 6 + 4 + 11 = 41

# Lineage counts:
# nbranches[0] = 4   (A, B, C, D all present)
# nbranches[1] = 4
# nbranches[2] = 4
# nbranches[3] = 4   (A, B, C, D; AB starts at t_3 but A,B end at t_3)
# ...actually, at t_3: A and B coalesce -> branches are AB, C, D = 3
# nbranches[3] = 3
# nbranches[4] = 3   (AB, C, D)
# nbranches[5] = 2   (AB, CD)
# nbranches[6] = 2
# nbranches[7] = 2
# nbranches[8] = 1   (root)
# nbranches[9..19] = 1
```

Where we are headed next: The time grid is now fully specified. In the next chapter (*Transition Probabilities*), we will use these time points, step sizes, midpoints, and lineage counts to derive the transition probabilities — the largest and most intricate gear in the ARGweaver mechanism.

Exercises

i Exercise 1: Grid sensitivity

Generate time grids with $n_t = 10, 20, 40$ and $\delta = 0.01$. For each grid, compute the maximum ratio $\Delta t_{i+1}/\Delta t_i$ between consecutive time steps. How does this ratio change with n_t ? What does this tell you about the smoothness of the approximation?

i Exercise 2: Coalescent concentration

Under the standard coalescent with constant $N_e = 10,000$, the expected time to the first coalescence of k lineages is $2N_e/\binom{k}{2}$. For $k = 20$, compute this expected time. How many of the default 20 time points fall below this expected time? What does this tell you about the grid's resolution where it matters most?

i Exercise 3: Implement state counting

Write a function that takes a tree (as a dictionary of parent–child relationships with ages) and a time grid, and returns the total number of HMM states. Verify your count against the worked example above.

i Exercise 4: Midpoint comparison

For the default time grid, plot the geometric-mean midpoints and the arithmetic midpoints on the same axis. At which time intervals is the difference largest in relative terms? Why does this matter for emission calculations?

Solutions

i Solution 1: Grid sensitivity

For each grid, we compute the time steps and then find the maximum ratio of consecutive steps. Because the grid is (approximately) geometric, consecutive steps grow by a nearly constant factor — that factor shrinks as n_t increases and the grid becomes finer.

```
from math import exp, log

def max_step_ratio(ntimes, maxtime=160000, delta=0.01):
    times = get_time_points(ntimes=ntimes, maxtime=maxtime, delta=delta)
    steps = get_time_steps(times)
    ratios = [steps[i+1] / steps[i] for i in range(len(steps) - 1)
              if steps[i] > 0]
    return max(ratios)

for nt in [10, 20, 40]:
    r = max_step_ratio(nt)
    print(f"n_t = {nt:3d}: max step ratio = {r:.4f}")

# n_t = 10: max step ratio = 2.1981
# n_t = 20: max step ratio = 1.4893
# n_t = 40: max step ratio = 1.2207
```

The maximum ratio decreases toward 1 as n_t grows. In the limit $n_t \rightarrow \infty$, consecutive steps differ by a factor of $\exp(h)$ where $h = \ln(1 + \delta T_{\max}) / (n_t - 1)$, so the ratio converges to $1 + O(1/n_t)$. A ratio close to 1 means the grid is locally smooth: transition and emission probabilities change gradually between adjacent intervals, reducing discretization error.

i Solution 2: Coalescent concentration

The expected time to first coalescence with k lineages and constant N_e is $E[T] = 2N_e / \binom{k}{2}$.

$$E[T] = \frac{2 \times 10,000}{\binom{20}{2}} = \frac{20,000}{190} \approx 105.3 \text{ generations}$$

```
from math import comb

Ne = 10000
k = 20
expected_coal = 2 * Ne / comb(k, 2)
print(f"Expected first coalescence time: {expected_coal:.1f} generations")

times = get_time_points(ntimes=20, maxtime=160000, delta=0.01)
below = sum(1 for t in times if t <= expected_coal)
print(f"Time points at or below {expected_coal:.1f}: {below} out of {len(times)}")

# Expected first coalescence time: 105.3 generations
# Time points at or below 105.3: 3 out of 20
```

Three of the 20 time points (t_0 , t_1 , t_2) fall at or below the expected first coalescence time. This means roughly 15% of the grid resolution is concentrated in the region where the very first coalescence event is most likely. Since $k = 20$ lineages produce many coalescence events in rapid succession at the start (the rate is proportional to $\binom{k}{2}$), the dense grid near the present provides fine resolution exactly where the coalescent is most active.

i Solution 3: Implement state counting

```
def count_states(tree_dict, node_ages, root, times):
    """
    Count HMM states from a tree described as a parent-child dict.

    Parameters
    -----
    tree_dict : dict
        Maps child -> parent (None for root).
    node_ages : dict
        Maps node name -> age (float).
    root : str
        Name of the root node.
    times : list of float
        Discretized time points.

    Returns
    -----
    int
        Total number of valid (branch, time_index) states.
    """
    ntimes = len(times) - 1
    total = 0
    for child, parent in tree_dict.items():
        child_age = node_ages[child]
```

```

    # Find the first time index >= child_age
    i = next(k for k, t in enumerate(times) if t >= child_age)
    if parent is not None:
        parent_age = node_ages[parent]
        while i < ntimes and times[i] <= parent_age:
            total += 1
            i += 1
    else:
        # Root branch: extends to ntimes - 1
        while i < ntimes:
            total += 1
            i += 1
    return total

# Verify against the worked example: ((A,B),(C,D))
times = get_time_points(ntimes=20, maxtime=160000, delta=0.01)

# Tree edges: child -> parent
tree = {
    'A': 'AB', 'B': 'AB',
    'C': 'CD', 'D': 'CD',
    'AB': 'root', 'CD': 'root',
    'root': None
}
ages = {
    'A': times[0], 'B': times[0],
    'C': times[0], 'D': times[0],
    'AB': times[3], 'CD': times[5],
    'root': times[8]
}
n = count_states(tree, ages, 'root', times)
print(f"Total states: {n}")
# A: 0..3 (4), B: 0..3 (4), C: 0..5 (6), D: 0..5 (6),
# AB: 3..8 (6), CD: 5..8 (4), root: 8..18 (11) => 41
assert n == 41

```

Solution 4: Midpoint comparison

```

times = get_time_points(ntimes=20, maxtime=160000, delta=0.01)
coal_times = get_coal_times(times)

print(f"{'Interval':<6} {'Geo mid':>10} {'Arith mid':>10} "
      f"{'Rel diff':>10}")
print("-" * 40)

max_rel_diff = 0.0
max_idx = 0
for i in range(len(times) - 1):
    t_lo = times[i]
    t_hi = times[i + 1]
    geo = coal_times[2 * i + 1]

```

```

arith = (t_lo + t_hi) / 2.0
if arith > 0:
    rel_diff = abs(geo - arith) / arith
else:
    rel_diff = 0.0
if rel_diff > max_rel_diff:
    max_rel_diff = rel_diff
    max_idx = i
print(f"[{i:3d}]   {geo:10.1f} {arith:10.1f} {rel_diff:10.4f}")

print(f"\nLargest relative difference: {max_rel_diff:.4f} "
      f"at interval {max_idx}")

# The largest relative difference is at interval 0:
# geometric mid = 6.3 vs arithmetic mid = 26.3, rel diff ~ 0.76.
# For later intervals the difference shrinks because the interval
# width becomes small relative to the offset from zero.

```

The relative difference is largest at interval 0 (approximately 76%). The geometric midpoint (6.3) is far below the arithmetic midpoint (26.3) because the geometric mean of $(0+1)$ and $(52.6+1)$ is pulled strongly toward the smaller value.

This matters for emission calculations because the coalescence time determines branch lengths, which in turn determine mutation probabilities. Using the arithmetic midpoint for interval 0 would assign a branch length of ~26 generations; the geometric midpoint assigns ~6 generations. Under the coalescent, most coalescence events in this interval happen near $t = 0$ (exponential concentration), so the geometric midpoint is a better representative time — it produces more accurate emission probabilities for the dominant near-present coalescence events.

Transition Probabilities

ARGweaver's hidden state at a genomic position is a pair $s = (v, a)$. The branch v says where the threaded lineage joins the partial local tree, and a indexes its coalescence time t_a . A transition to $s' = (w, b)$ sums over recombination and re-coalescence events that can turn the first attachment into the second under the discrete SMC.

The transition is not a source-independent rank-one update. The current coalescence time changes the length of the augmented tree, the available recombination material, and several lineage-count corrections. Whether the source and destination use the same branch also changes which latent recombination events are compatible. These dependencies appear directly in both the reference Python formula and `TransMatrix::get_time` in the original C++ source.

No recombination

If no recombination occurs between adjacent sites, the attachment state is unchanged. For source state s , the contribution is

$$\mathbf{1}[s = s'] \exp\{-\rho \max(L_s, 1)\},$$

where L_s is the length of the local tree after adding the threaded branch in state s . The lower bound of one generation follows the reference implementation's guard for a per-base transition. Because L_s depends on the source coalescence time, even the diagonal no-recombination probability is state dependent.

Recombination time

Let k denote a discrete recombination interval. The reference implementation augments the partial-tree lineage counts by the threaded branch. In the literal testing formula, the probability contribution for interval k is

$$P(R_k | s) = \frac{\{n_B(k) + \mathbf{1}[k < a]\}\Delta t_k}{\{n_R(k) + \mathbf{1}[k \leq a] + \mathbf{1}[k = a < w]\}L_s^{\text{basal}}} \left(1 - e^{-\rho \max(L_s, 1)}\right).$$

Here $n_B(k)$ counts branches, $n_R(k)$ counts valid discrete recombination points, w is the greater of the source time and root time, and L_s^{basal} includes the required basal interval. The exact bookkeeping is important: using only the partial tree's branch material erases the source-state dependence and produces the wrong kernel.

Re-coalescence

After recombination, the detached lineage survives through successive time intervals and then coalesces at destination time b . For interval m , its hazard is

$$\lambda_m = \frac{n_B^*(m)}{2N_m},$$

so survival from k to b contributes

$$\exp \left\{ - \sum_{m=k}^{b-1} \frac{n_B^*(m)\Delta t_m}{2N_m} \right\}.$$

At a nonterminal destination interval, the event then contributes

$$\frac{1 - \exp\{-n_B^*(b)\Delta t_b/(2N_b)\}}{n_C(b)},$$

where $n_C(b)$ is the number of valid coalescence points. The terminal grid interval absorbs the remaining tail probability. The factor $2N$ is required because ARGweaver's population-size parameter is a diploid effective size.

Summing latent events

The dense transition probability can be written as

$$T_{s,s'} = \mathbf{1}[s = s']P(\text{no recombination} | s) + \sum_{r \in \mathcal{R}(s,s')} P(r | s)P(s' | r, s).$$

The compatible set $\mathcal{R}(s, s')$ includes recombinations on the threaded lineage. When the source and destination use the same branch, it also includes events on that branch below the two attachment times. This is why a single destination vector cannot describe every source row.

Compressed transition representation

ARGweaver does not materialize and multiply an arbitrary dense matrix at every site. `src/argweaver/trans.cpp` precomputes time-indexed terms named `D`, `E`, `lnB`, `lnE2`, `lnNegG1`, `G2`, `G3`, `lnG4`, and `norecombs`. `TransMatrix::get_time` selects different combinations according to whether $a < b$, $a = b$, or $a > b$, and according to whether the branch is unchanged.

The forward routine first groups source probability by time, multiplies the result by an $n_t \times n_t$ time matrix, and then adds same-branch corrections. Thus the optimization exploits time grouping and branch-local structure. It is not a rank-one identity update, and describing it as simply $O(S)$ hides the explicit time-grid work in the source.

Switch transitions

A recombination breakpoint already present in the partial ARG changes the local tree itself. ARGweaver therefore uses a separate `TransMatrixSwitch` between blocks. Many source states map deterministically across the SPR because their attachment branch is unaffected. States on the pruned branch or at the SPR's coalescence point require probabilistic treatment, and states that violate the new branch-age constraints are unavailable.

This switch matrix is not interchangeable with the within-block transition. The former conditions on a known change in the partial ARG; the latter sums over possible recombination events on the lineage currently being threaded.

A bounded teaching approximation

The miniature module retains a deliberately state-independent matrix only as a probability-bookkeeping exercise. It uses the correct $2N$ hazard and normalizes the represented re-coalescence mass, but it omits branch identity, the source state's added tree length, same-branch latent events, the absorbing tail construction, and switch transitions.

```
def build_simple_transition_matrix(ntimes, nbranches, ncoals,
                                popsizes, rho, treelen, times):
    """
    Build a deliberately state-independent transition approximation.

    This helper ignores branch identity and the source state's added branch
    length. It is useful for illustrating stochastic-matrix bookkeeping, but
    it is not ARGweaver's production transition kernel: the real kernel is
    source-state dependent and uses a richer compressed representation.

    Parameters
    -----
    ntimes : int
        Number of time intervals.
    nbranches : list of int
        Number of branches at each time interval.
    ncoals : list of int
        Number of coalescence points at each time index.
    popsizes : list of float
        Population sizes at each time index.
    rho : float
        Per-base recombination rate.
    treelen : float
        Total tree length.
    times : list of float
        Discretized time points.

    Returns
    -----
    numpy.ndarray
        Transition probability matrix.
    """
    time_steps = get_time_steps(times)
    coal_times_list = get_coal_times(times)
    root_idx = ntimes - 1
    no_recomb = exp(-rho * treelen)

    nstates = sum(ncoals[j] for j in range(ntimes))
```

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```

q = np.zeros(nstates)
state_map = []
idx = 0
for j in range(ntimes):
    for b in range(ncoals[j]):
        state_map.append(j)
        idx += 1

for s_idx in range(nstates):
    j = state_map[s_idx]
    total = 0.0
    for k in range(root_idx + 1):
        if j < k:
            continue
        recomb_p = (nbranches[k] * time_steps[k] / treelen
                    * (1 - no_recomb))
        surv = 1.0
        for m in range(k, j):
            A = (coal_times_list[2*m+1] - coal_times_list[2*m]) * nbranches[m]
            if m > k:
                A += (coal_times_list[2*m] - coal_times_list[2*m-1]) *
↪nbranches[max(m-1, 0)]
            surv *= exp(-A / (2.0 * popsizes[m]))
        A = (coal_times_list[2*j+1] - coal_times_list[2*j]) * nbranches[j]
        if j > k:
            A += (coal_times_list[2*j] - coal_times_list[2*j-1]) *
↪nbranches[max(j-1, 0)]
        cp = 1.0 - exp(-A / (2.0 * popsizes[j]))
        total += recomb_p * surv * cp / max(ncoals[j], 1)
    q[s_idx] = total

# The finite grid truncates some re-coalescence mass. Condition the toy
# recombination component on re-coalescing within the represented grid so
# every row remains a valid probability distribution.
q_total = q.sum()
if q_total > 0:
    q *= (1.0 - no_recomb) / q_total
T = np.outer(np.ones(nstates), q) + no_recomb * np.eye(nstates)
return T

```

It must therefore not be presented as a parity implementation of ARGweaver's transition kernel. Source-level validation belongs to the production formulas and compressed matrix tests; the toy's appropriate invariants are nonnegative entries, exact row sums, and increased diagonal mass as ρ decreases.

The practical conclusion is that ARGweaver gains efficiency from a structured, compressed, source-dependent transition kernel. The state pair (v, a) matters on both sides of every transition.

Emission Probabilities

ARGweaver's threading HMM scores a candidate attachment by the probability of the observed bases on the local tree that would result from that attachment. The production implementation does this with probabilistic partial likelihoods, not with a parsimony score. This distinction matters: a parsimony assignment keeps only one reconstruction of the ancestral bases, whereas a likelihood sums over every possible ancestral base assignment.

The description here follows the original ARGweaver paper [Rasmussen *et al.*, 2014] and the current `mdrasmus/argweaver` implementation. In particular, `src/argweaver/emit.cpp` computes inner and outer likelihood tables and evaluates candidate attachment states with Jukes–Cantor branch probabilities.

The Jukes–Cantor branch model

For a branch of length t and mutation rate μ , ARGweaver uses

$$P_{aa}(t) = \frac{1}{4} \left(1 + 3e^{-4\mu t/3} \right), \quad P_{ab}(t) = \frac{1}{4} \left(1 - e^{-4\mu t/3} \right), \quad a \neq b.$$

The second expression is the probability of one *specified* alternative base, not the probability of any change. Both expressions approach $1/4$ on a very long branch. Replacing them by $e^{-\mu t}$ and $(1 - e^{-\mu t})/3$ is a low-rate approximation, but it is not the branch kernel used by the production C++ emission calculation.

Conditional likelihoods on a tree

Let $L_u(a)$ be the likelihood of all observations below node u conditional on base a at that node. At a leaf with observed base x_u ,

$$L_u(a) = \mathbf{1}[a = x_u].$$

For an internal binary node with children v and w , pruning gives

$$L_u(a) = \left(\sum_b P_{ab}(t_v) L_v(b) \right) \left(\sum_c P_{ac}(t_w) L_w(c) \right).$$

At the root, the stationary Jukes–Cantor distribution is uniform, so the tree likelihood is $\frac{1}{4} \sum_a L_r(a)$. Missing `N` observations are represented by likelihood one for every base rather than being forced to an arbitrary nucleotide.

The miniature implementation exposes the same recursion in log space. It returns the four root-conditional log likelihoods so that tests can compare each state directly with the closed-form Jukes–Cantor probabilities.

```
def felsenstein_pruning(tree_children, tree_parent, leaf_bases,
                        mu, branch_lengths):
    """
    Felsenstein's pruning algorithm for a 4-base alphabet.

    Returns log-likelihood at the root for each possible root base.

    Parameters
    -----
    tree_children : dict
        Maps internal node -> list of children.
    tree_parent : dict
        Maps child -> parent.
    leaf_bases : dict
        Maps leaf -> observed base.
    mu : float
```

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```

    Mutation rate.
    branch_lengths : dict
        Maps child -> branch length to parent.

    Returns
    -----
    dict
        Maps base -> log-likelihood at root.
    """
    if mu < 0:
        raise ValueError("mu must be non-negative")
    bases = ['A', 'C', 'G', 'T']
    L = {}

    nodes = set(tree_children) | set(tree_parent) | set(leaf_bases)
    roots = nodes - set(tree_parent)
    if len(roots) != 1:
        raise ValueError("tree must have exactly one root")
    root = next(iter(roots))

    def compute(node):
        if node in leaf_bases:
            observed = leaf_bases[node].upper()
            if observed not in bases + ['N']:
                raise ValueError(f"unsupported observed base: {observed}")
            L[node] = {}
            for b in bases:
                L[node][b] = (0.0 if observed in (b, 'N')
                              else -float('inf'))
            return

        for child in tree_children[node]:
            compute(child)

    L[node] = {}
    for b in bases:
        ll = 0.0
        for child in tree_children[node]:
            blen = branch_lengths[child]
            if blen < 0:
                raise ValueError("branch lengths must be non-negative")
            decay = exp(-4.0 * mu * blen / 3.0)
            p_no_mut = 0.25 * (1.0 + 3.0 * decay)
            p_mut = 0.25 * (1.0 - decay)
            child_vals = []
            for cb in bases:
                p_trans = p_no_mut if cb == b else p_mut
                if p_trans > 0 and L[child][cb] > -float('inf'):
                    child_vals.append(log(p_trans) + L[child][cb])
            if child_vals:
                mx = max(child_vals)
                ll += mx + log(sum(exp(v - mx) for v in child_vals))

```

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```

        else:
            ll += -float('inf')
        L[node][b] = ll

    compute(root)
    return L[root]

```

Scoring a threading state

A state (b, i) attaches the threaded lineage, or threaded subtree, to branch b at grid time t_i . This creates three relevant branch pieces: the branch from the threaded lineage to the new coalescence, the lower piece of branch b , and, unless the attachment is above the root, the upper piece of branch b . Their lengths determine three Jukes–Cantor transition kernels.

Re-running pruning independently for every state would repeat nearly all work. ARGweaver therefore caches an `inner` table for the likelihood below each node and an `outer` table for the likelihood outside each node's subtree. For a candidate state, `calc_emit` combines the cached inner and outer terms with the three new branch kernels and sums over the base at the new attachment point. This is algebraically the likelihood of the augmented local tree.

The same construction handles internal-branch threading. In that case one cached partial represents the removed subtree and another represents the remaining tree. The likelihood calculation joins those two partials at each candidate state. Sites affected by unphased heterozygotes are evaluated under both phase assignments and averaged, while recording conditional phase probabilities for sampling.

Where parsimony is actually used

The source tree still contains parsimony routines, but they do not replace the finite-sites likelihood above. They support the optional infinite-sites compatibility filter and related diagnostics. When that option is active, states incompatible with a one-mutation explanation are down-weighted by the configured penalty. The normal finite-sites emission path continues to use inner and outer likelihoods.

Invariant sites have a separate optimized path because every observed leaf has the same base. Fully masked sites emit probability one for every state. These shortcuts are computational optimizations with explicit source branches; they should not be generalized into the claim that all emissions are parsimony scores.

Numerical checks

For a single observed A separated from the root by branch length t , the root-conditional probabilities provide a direct oracle:

```

import numpy as np
from watchgen.mini_argweaver import felsenstein_pruning

mu = 0.2
t = 2.5
result = felsenstein_pruning(
    {"root": ["sample"]},
    {"sample": "root"},
    {"sample": "A"},
    mu,
    {"sample": t},
)

decay = np.exp(-4 * mu * t / 3)

```

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```
assert np.isclose(np.exp(result["A"]), (1 + 3 * decay) / 4)
assert np.isclose(np.exp(result["C"]), (1 - decay) / 4)
```

This check distinguishes the exact kernel from the low-rate approximation and also verifies that the pruning direction is correct. Larger tests compare the recursion with direct enumeration over internal-node bases.

Scope of the teaching implementation

The miniature function demonstrates the exact branch kernel and pruning recursion, but it is not a replacement for `calc_emissions`. It does not build inner and outer tables across an alignment, perform invariant-site compression, average phase configurations, attach a removed subtree, or apply the optional infinite-sites penalty. Those features remain explicit boundaries between the teaching kernel and ARGweaver's production engine.

The central point is therefore precise: ARGweaver uses probabilistic Jukes–Cantor emissions for posterior threading. Parsimony is an auxiliary compatibility tool, not the emission likelihood itself.

MCMC Sampling

ARGweaver explores posterior uncertainty by repeatedly removing part of an ARG and threading it back through the remainder. The hidden Markov model supplies a distribution over complete attachment paths along the genome. A normalized forward pass followed by stochastic traceback samples one such path; the ARG is then rebuilt with that thread.

The phrase “ARGweaver uses Gibbs sampling” needs a qualification. Removing and re-threading one external chromosome is a full-conditional Gibbs update and is accepted automatically. The software also implements internal-branch and subtree proposals. Those more general moves are used in a Metropolis–Hastings sampler and require an acceptance correction, as described in the original paper [Rasmussen *et al.*, 2014] and implemented in `src/argweaver/sample_arg.cpp`.

Sequential initialization

The program constructs an initial ARG sequentially. It begins with a small coalescent genealogy and adds chromosomes one at a time. At each addition, the current partial ARG fixes the HMM state spaces, transition objects, and emission tables. Forward sampling generates the new chromosome's path, including any recombination events required to move between attachment states.

Sequential construction is an initializer, not a claim that the first ARG is a posterior draw from the full data set. Subsequent MCMC iterations revisit its threads. Burn-in and convergence checks remain necessary because exact sampling of one conditional path does not make successive complete ARGs independent.

External chromosome threading

For an external update, ARGweaver removes one sampled chromosome from every local tree while preserving the induced partial ARG. Conditional on that partial ARG and the sequence data, the removed chromosome's attachment path is an HMM. Sampling exactly from this HMM is a Gibbs step:

$$z'_k \sim P(z_k \mid \mathcal{G}_{-k}, D, \theta).$$

Here z_k is the chromosome's genome-wide thread, $\mathcal{G}_{-k}$ is the partial ARG, D is the alignment, and θ contains the population size, mutation rate, recombination rate, and time grid. There is no accept/reject decision for this move because its proposal distribution is the full conditional.

Forward sampling

Within a non-recombining block of the partial ARG, the forward recursion is

$$\alpha_x(s') \propto e_x(s') \sum_s \alpha_{x-1}(s) T_x(s, s').$$

ARGweaver normalizes each forward column for numerical stability. The transition multiplication uses the compressed, time-grouped calculation from *Transition Probabilities*. At a breakpoint in the partial ARG, it substitutes a switch transition object that maps states across the known SPR.

Stochastic traceback first samples the final state from the last normalized forward column. Moving backward, it samples each preceding state with weight

$$\alpha_{x-1}(s) T_x(s, s_x).$$

This produces a joint path sample. Independently sampling each site's marginal posterior would not preserve the transition dependence and would not be a valid thread.

Internal and subtree moves

Single-chromosome updates can mix slowly when several neighboring branches need to move together. ARGweaver therefore supports threading an internal branch or subtree. The state space is constrained by the removed subtree's age, the emission calculation joins subtree and main-tree partial likelihoods, and the transition object receives a minimum allowed attachment age.

These proposals do not all have the Gibbs guarantee of an external chromosome update. The general sampler computes the target-density and proposal-density terms needed for a Metropolis–Hastings decision. Consequently, statements that all ARGweaver moves have 100 percent acceptance are false; that property applies only to full-conditional threading moves.

Coalescent and recombination helpers

The miniature module contains small stochastic helpers for checking parameter conventions. `sample_tree` simulates continuous coalescent waiting times under a piecewise-constant diploid effective population size. With k extant lineages, its event rate is

$$\binom{k}{2} \frac{1}{2N_e}.$$

The result can subsequently be snapped to a discrete time grid, but the helper itself is not ARGweaver's full initialization procedure.

```
def sample_tree(k, popsizes, times):
    """
    Sample a coalescent tree using a discrete-time coalescent.

    Starting with k lineages at time 0, simulate coalescence events
    through the time grid. At each time interval, the coalescence rate
    depends on the number of lineages and the local population size.

    Parameters
    -----
    k : int
        Number of lineages (chromosomes).
    popsizes : list of float
```

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```

    Effective population size at each time interval.
    times : list of float
        Discretized time points.

    Returns
    -----
    list of float
        Coalescence times (one per coalescence event).
    """
    ntimes = len(times)
    coal_times = []

    timei = 0
    n = popsizes[timei]
    t = 0.0
    k2 = k

    while k2 > 1:
        # ARGweaver's ``popsizes`` are diploid effective sizes, so each pair
        # coalesces at rate 1 / (2N).
        coal_rate = (k2 * (k2 - 1) / 2) / (2.0 * float(n))
        t2 = random.expovariate(coal_rate)

        if timei < ntimes - 2 and t + t2 > times[timei + 1]:
            timei += 1
            t = times[timei]
            n = popsizes[timei]
            continue

        t += t2
        coal_times.append(t)
        k2 -= 1

    return coal_times

```

For a valid local tree of length $L > 0$, the distance to the next recombination event is exponential with rate ρL . A zero-length tree cannot generate such an event and is rejected rather than silently assigned an invented rate.

```

def sample_next_recomb(treelen, rho):
    """
    Sample the distance to the next recombination event.

    The waiting time (in base pairs) is exponential with rate
    rho * treelen.

    Parameters
    -----
    treelen : float
        Total tree length.
    rho : float
        Per-base recombination rate.

```

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```

Returns
-----
float
    Distance in base pairs to the next recombination.
"""
if treelen <= 0:
    raise ValueError("treelen must be positive")
if rho <= 0:
    raise ValueError("rho must be positive")
rate = treelen * rho
return random.expovariate(rate)

```

What the toy loop is not

`simplified_mcmc` in the `miniature` module perturbs a list of coalescence times and records a tree-length trajectory for plotting. It has no alignment, ARG, threading HMM, posterior density, or valid full-conditional update. It is not an ARGweaver sampler and cannot be used to assess acceptance rates, effective sample size, posterior calibration, or convergence of the real method.

```

def simplified_mcmc(n_haps=4, n_sites=100, n_iters=1000,
                   Ne=10000, mu=1.4e-8, rho=1e-8,
                   ntimes=20, maxtime=160000, delta=0.01):
    """
    Toy tree-length perturbation loop for demonstration.

    This implementation is not an ARGweaver posterior sampler: it has no
    sequence likelihood, threading HMM, or ARG state. It tracks only a toy
    tree-length statistic and is retained solely for visualization.

    Parameters
    -----
    n_haps : int
        Number of haplotypes.
    n_sites : int
        Number of genomic sites.
    n_iters : int
        Number of MCMC iterations.
    Ne : float
        Effective population size.
    mu : float
        Mutation rate.
    rho : float
        Recombination rate.
    ntimes : int
        Number of time points.
    maxtime : float
        Maximum time in generations.
    delta : float
        Controls log-spacing.

    Returns
    -----
    """

```

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```

tree_lengths : list of float
    Total tree length at site 0, one per iteration.
"""
times = get_time_points(ntimes=ntimes, maxtime=maxtime, delta=delta)
popsizes = [Ne] * (ntimes - 1)

coal_events = sorted(sample_tree(n_haps, popsizes, times))

tree_lengths = []

for iteration in range(n_iters):
    remove_idx = random.randint(0, n_haps - 1)

    if coal_events:
        event_idx = min(remove_idx, len(coal_events) - 1)
        coal_events.pop(event_idx)

        k_remaining = n_haps - len(coal_events)
        if k_remaining >= 2:
            new_coal = sample_tree(k_remaining, popsizes, times)
            if new_coal:
                coal_events.append(new_coal[0])
                coal_events.sort()

    total_length = 0.0
    k = n_haps
    prev_t = 0.0
    for ct in coal_events:
        total_length += k * (ct - prev_t)
        prev_t = ct
        k -= 1
    if coal_events:
        total_length += 1 * (times[-1] - coal_events[-1])

    tree_lengths.append(total_length)

return tree_lengths

```

Validation boundaries

Appropriate tests for the small helpers include the $2N_e$ mean for a pairwise coalescence, the $1/(\rho L)$ mean recombination distance, exact Jukes–Cantor likelihood oracles, stochastic transition rows, and direct time-grid parity with the upstream source. Validation of the complete sampler additionally requires the official C++ transition, emission, traceback, and ARG-consistency tests. A plausible toy trace is not evidence of posterior parity.

The resulting interpretation is deliberately bounded. The book explains how ARGweaver’s exact conditional HMM updates fit inside its MCMC machinery, while keeping general subtree Metropolis–Hastings moves and production ARG bookkeeping distinct from the miniature demonstrations.

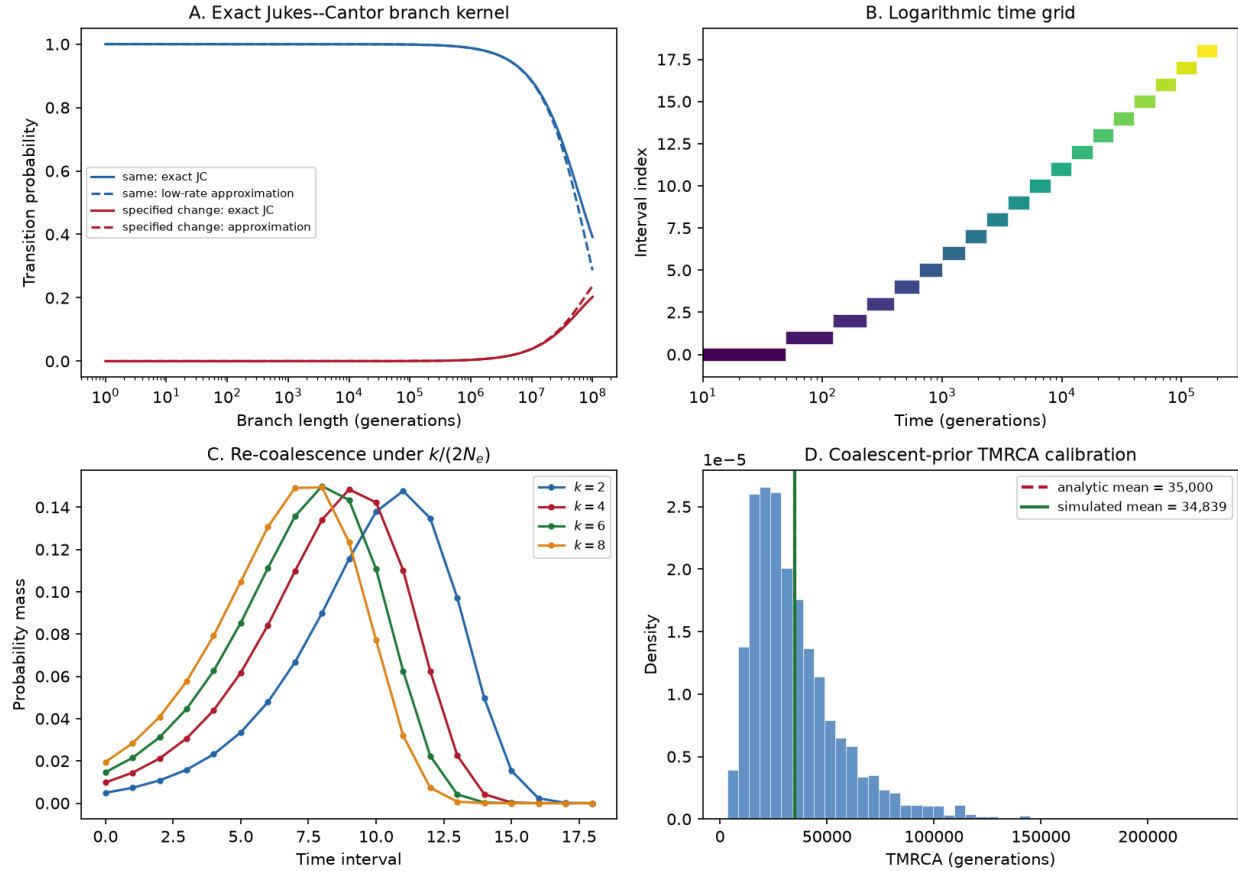
Demo: Source-Matched Teaching Checks*Checking the bounded kernels without pretending to run the production sampler.***ARGweaver: Source-Matched Teaching Checks**

Fig. 7: **ARGweaver teaching checks.** Panel A compares exact Jukes–Cantor branch probabilities with their low-rate approximations. Panel B shows the logarithmic time grid. Panel C shows re-coalescence mass under the correct $2N_e$ hazard. Panel D compares simulated coalescent-prior TMRCA values with their analytic mean.

This figure validates individual teaching kernels. It is not an ARGweaver MCMC run and contains no posterior trace. The production executable validation is performed separately against the original ARGweaver source. The plotting script is `figures/fig_demo_argweaver.py`.

3.7 Timepiece VI: tsinfer*Scalable tree-sequence inference from phased genetic variation*

tsinfer infers a tree sequence by first constructing putative ancestral haplotypes, then matching younger ancestors against older ones, and finally matching the observed samples. Its output is a compact genealogical hypothesis, not a posterior distribution of ARGs. [Kelleher *et al.*, 2019]

The important qualification is that the values used to order inferred ancestors are a **time proxy**, not generations. In the paper release, larger derived-allele counts normally implied older ordering ranks. Frequency is noisy under drift, selection, recurrent mutation, ancestral-state error, and genotype error, so the ordering must not be interpreted as a molecular clock.

i Scope of this chapter

The executable module reproduces small mechanisms from the paper-era 0.1.4 reference implementation and probability transforms from stable release 0.4.1. It is **not a replacement** for production tsinfer. Production code stores and matches partial ancestors on tree sequences, supports missing and multiallelic data in version-dependent ways, compresses shared paths, and delegates general table operations to tskit.

The audited ground truth was the primary paper, official tag 0.1.4 ([efbafff](#)), stable package 0.4.1, and official development source [9242074](#) (30 June 2026). The current development interface uses VCF-Zarr and TOML pipeline configuration and is under active development; consult the docs for the version you install rather than copying an old command line from this book.

```
phased variants
  |
  v
choose inference sites --> generate partial ancestors
                        |
                        v
                    match older to younger
                        |
                        v
                    match observed samples
                        |
                        v
            place remaining mutations by parsimony
                        |
                        v
            post-process and simplify with tskit
```

i Primary sources

- Kelleher et al. (2019), [doi:10.1038/s41588-019-0483-y](#).
- [Official tsinfer documentation](#).
- [Official source repository](#).

3.7.1 Pipeline and evidence boundary

What tsinfer infers

At nearby sites, a haplotype often resembles long pieces of other haplotypes. tsinfer uses this copying structure to build a tree sequence. The core pipeline is deterministic for fixed input, parameters, software version, and implementation ordering, but that does not make the inferred genealogy uniquely true. It is a heuristic point estimate produced by an approximate model.

The 2019 method has three inference stages. Ancestor generation turns carrier patterns at selected variants into partially defined ancestral haplotypes. Ancestor matching processes these in decreasing proxy age, expressing each as a mosaic of older ancestors. Sample matching uses the same matching machinery to thread observed haplotypes through the ancestor tree sequence. Non-inference sites are then mapped parsimoniously.

Input assumptions

The primary-paper pipeline assumed phased haplotypes and ancestral states for the variants used in inference. In the paper-era defaults, an inference site was a known-ancestral, biallelic site with at least two derived copies and at least one ancestral copy. Singletons, fixed sites, unknown ancestral states, and unsuitable multiallelic sites were excluded from topology inference. Excluded sites were not discarded from the final data set: their mutations could be positioned on the inferred trees by parsimony.

Current releases have changed their input storage and pipeline interfaces. The latest official documentation should therefore be treated as the operational authority, while tag 0.1.4 is the authority for the historical algorithm described in the paper.

What the miniature does not do

`watchgen.mini_tsinfer` checks ancestor grouping and extension, HMM probability transforms, a dense Viterbi recurrence, path-coordinate conversion, a repeated-path detector, and small parsimony/flank examples. It deliberately does not claim to construct a valid production tree sequence. In particular, an upward graph walk is not `tskit`'s simplify algorithm, and a rectangular reference panel is not the tree-compressed matcher used by production `tsinfer`.

This boundary matters methodologically: passing tests against a self-contained toy pipeline would not establish parity with the software. The tests instead target equations and fixtures taken from official implementations, and real analyses must run the released `tsinfer/tskit` stack.

3.7.2 Ancestor generation

Site ordering is not dating

Let $G_{js} \in \{0, 1\}$ be the allele carried by sample haplotype s at inference site j , and let

$$f_j = \sum_s G_{js}$$

be the derived-allele count. The paper implementation used this count to order sites. In 0.1.4, distinct counts were mapped to ordinal ranks: the smallest distinct count received rank 1, the next rank 2, and so on. These ranks are a time proxy and are **not generations**. Two counts can share no calibrated time interpretation even when their order is correct.

Multiple focal sites

The original implementation did not blindly make one ancestor per site. Sites with the same count and exactly the same carrier vector were grouped, so one ancestor could have **multiple focal sites**. Suppose focal sites a and b share their carriers. An intervening site j splits them when $f_j > f_a$ and its alleles are polymorphic among the focal carriers. This prevents one ancestor from spanning evidence inconsistent with a single lineage.

Extending a partial ancestor

At every focal site the ancestor carries allele 1. Between multiple focal sites, an older intervening site's state is the majority among focal carriers; ties choose 1. The ancestor is then extended left and right.

For one direction, begin with carrier set S at the focal site and threshold $\lfloor f/2 \rfloor$. Each traversed site is initially assigned 0. At a site with count greater than the focal count:

1. take the majority state among the current S (ties choose 1);
2. apply the implementation's delayed removal buffer;
3. stop when at most $\lfloor f/2 \rfloor$ carriers remain;
4. otherwise buffer carriers that disagree with the current consensus.

The delayed buffer is important source-level detail. The earlier chapter instead kept the original carrier set fixed and stopped at the first older site; both choices changed ancestor spans and states.

Two all-zero ancestors

Version 0.1.4 prepended two full-length all-zero haplotypes. The older one was a matching **virtual root** used to align identifiers; the younger was the all-zero **ultimate ancestor**. They are distinct implementation objects, not two biological ancestors inferred from independent evidence. Stable 0.4.1 retained this two-node pattern but changed details of assigning their proxy times.

```
def ancestor_descriptors(D, inference_sites):
    """Group and split focal sites exactly as the 0.1.4 Python reference.

    Equal-count sites with the same carrier pattern can share one ancestor.
    An intervening older site splits them if it is polymorphic among those
    carriers.
    """

    D = _as_genotypes(D)
    sites = np.asarray(inference_sites, dtype=int)
    G = D[:, sites].T
    if np.any(G < 0):
        raise ValueError("ancestor construction requires called inference sites")
    grouped = defaultdict(lambda: defaultdict(list))
    for site, genotypes in enumerate(G):
        grouped[int(np.sum(genotypes))][genotypes.tobytes()].append(site)
    descriptors = []
    for frequency in sorted(grouped, reverse=True):
        for pattern in sorted(grouped[frequency]):
            carriers = np.frombuffer(pattern, dtype=np.int8)
            focal = grouped[frequency][pattern]
            start = 0
            for j in range(len(focal) - 1):
                if _break_ancestor(G, focal[j], focal[j + 1], carriers, frequency):
                    descriptors.append((frequency, tuple(focal[start : j + 1])))
                    start = j + 1
            descriptors.append((frequency, tuple(focal[start:])))
    return descriptors
```

```
def build_ancestor(D, inference_sites, focal_sites, time=None):
    """Build one ancestor; focal indexes refer to ``inference_sites``."""

    D = _as_genotypes(D)
    sites = np.asarray(inference_sites, dtype=int)
    G = D[:, sites].T
    if np.any(G < 0):
        raise ValueError("ancestor construction requires called inference sites")
    if np.isscalar(focal_sites):
        focal_sites = (int(focal_sites),)
    else:
        focal_sites = tuple(int(site) for site in focal_sites)
    if not focal_sites or tuple(sorted(focal_sites)) != focal_sites:
        raise ValueError("focal_sites must be a nonempty increasing sequence")
    frequency = int(np.sum(G[focal_sites[0]]))
    if any(int(np.sum(G[site])) != frequency for site in focal_sites):
        raise ValueError("focal sites must have the same derived count")
```

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```

haplotype = np.full(len(sites), UNKNOWN, dtype=np.int8)
haplotype[list(focal_sites)] = 1
carriers = set(np.flatnonzero(G[focal_sites[0]] == 1).tolist())
frequencies = np.sum(G == 1, axis=1)
for left, right in pairwise(focal_sites):
    for site in range(left + 1, right):
        haplotype[site] = 0
        if frequencies[site] > frequency:
            ones = sum(int(G[site, sample]) for sample in carriers)
            haplotype[site] = int(ones >= len(carriers) - ones)

last = focal_sites[-1]
end_site = _extend_ancestor(G, haplotype, last, range(last + 1, len(sites)))
first = focal_sites[0]
start = _extend_ancestor(G, haplotype, first, range(first - 1, -1, -1))
return Ancestor(
    start=start,
    end=end_site + 1,
    time=float(frequency if time is None else time),
    focal_sites=focal_sites,
    haplotype=haplotype[start : end_site + 1].copy(),
    frequency=frequency,
)

```

3.7.3 The copying model

Dense reference recurrence

For teaching, let $Z_j \in \{1, \dots, K\}$ identify the reference copied at site j . If r_j is the recombination probability between sites, the Li–Stephens-like transition is

$$P(Z_j = k \mid Z_{j-1} = h) = \begin{cases} 1 - r_j + r_j/K, & k = h, \\ r_j/K, & k \neq h. \end{cases}$$

With mismatch probability μ_j , a biallelic emission is $1 - \mu_j$ for a match and μ_j for a mismatch. For A alleles, stable 0.4.1 distributes mismatch mass as $\mu_j/(A - 1)$ across the other alleles. A missing query allele has a neutral emission.

The Viterbi recurrence maximizes over the previous copying source. The executable reference uses log probabilities and a dense $O(mK^2)$ loop so that the equation is transparent. Production tsinfer compresses likelihoods on marginal trees; it does not materialize this rectangular matrix.

Probability transforms in stable 0.4.1

If d_j is the genetic distance in Morgans, 0.4.1 used Haldane’s mapping function

$$r_j = \frac{1 - \exp(-2d_j)}{2}.$$

The result approaches 0.5, not 1. It is not divided by the reference-panel size. The previous chapter’s $1 - \exp(-d_j/K)$ formula was therefore not software parity.

For mismatch ratio X , median adjacent genetic distance $\tilde{d}$, and A alleles, stable 0.4.1 used the constant

$$\mu = \frac{1 - \exp(-AX\tilde{d})}{A}$$

across sites. When no rate and ratio were supplied, that release used $r = 10^{-2}$ and $\mu = 10^{-20}$ to recover near-no-mismatch 0.1-era behaviour. Defaults are versioned software decisions, not universal properties of the method.

```
def compute_recombination_probs(positions, recombination_rate, num_ref=None):
    """Use 0.4.1's Haldane transform ``1-exp(-2d)/2``.

    ``num_ref`` is retained for compatibility with the previous chapter API;
    the reference-panel size does not divide the genetic distance.
    """

    positions = np.asarray(positions, dtype=float)
    if positions.ndim != 1 or np.any(np.diff(positions) < 0):
        raise ValueError("positions must be a sorted one-dimensional array")
    if recombination_rate < 0:
        raise ValueError("recombination_rate must be nonnegative")
    probability = np.zeros(len(positions))
    distance = np.diff(positions) * recombination_rate
    probability[1:] = (1 - np.exp(-2 * distance)) / 2
    return probability
```

```
def viterbi_ls(query, panel, rho, mu, num_alleles=2):
    """Dense log-space Viterbi check for the Li--Stephens-like model.

    Production tsinfer performs the likelihood updates on marginal trees and
    supports partially defined ancestors; this teaching function does neither.
    """

    query = np.asarray(query, dtype=np.int8)
    panel = np.asarray(panel, dtype=np.int8)
    rho = np.asarray(rho, dtype=float)
    mu = np.asarray(mu, dtype=float)
    if panel.ndim != 2 or panel.shape[0] != len(query):
        raise ValueError("panel must have shape (sites, references)")
    m, k = panel.shape
    if m == 0 or k == 0:
        raise ValueError("query and panel must be nonempty")
    if rho.shape != (m,) or mu.shape != (m,):
        raise ValueError("rho and mu must have one entry per site")
    if np.any(panel < 0):
        raise ValueError("use production tsinfer for partial ancestors")
    if np.any((rho < 0) | (rho > 1)) or np.any((mu < 0) | (mu > 1)):
        raise ValueError("probabilities must lie in [0, 1]")

    score = np.full((m, k), -np.inf)
    traceback = np.zeros((m, k), dtype=int)

    def log_emission(site):
        if query[site] == UNKNOWN:
            return np.zeros(k)
        matches = panel[site] == query[site]
        p = np.where(matches, 1 - mu[site], mu[site] / (num_alleles - 1))
        with np.errstate(divide="ignore"):
            return np.log(p)
```

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```

score[0] = -np.log(k) + log_emission(0)
for site in range(1, m):
    switch = rho[site] / k
    transition = np.full((k, k), switch)
    np.fill_diagonal(transition, 1 - rho[site] + switch)
    with np.errstate(divide="ignore"):
        candidates = score[site - 1, :, None] + np.log(transition)
    traceback[site] = np.argmax(candidates, axis=0)
    score[site] = np.max(candidates, axis=0) + log_emission(site)

path = np.empty(m, dtype=int)
path[-1] = int(np.argmax(score[-1]))
for site in range(m - 1, 0, -1):
    path[site - 1] = traceback[site, path[site]]
return path, float(score[-1, path[-1]])

```

3.7.4 Ancestor matching and path compression

Oldest usable ancestors first

Ancestors are processed in proxy-time groups. Members of one group are matched against ancestors from strictly older groups; peers are not allowed to become references for one another merely because of iteration order. Each copying path becomes tree-sequence edges, and allele differences become mutations.

Paper release 0.1.4 used a highly optimized tree-based matching criterion that prioritized matches, recombinations, and mismatches. Later releases exposed probabilistic Li-Stephens-like matching. These are related designs, but it is incorrect to project every modern probability and default backward onto the 2019 code. This chapter uses 0.1.4 for ancestor-generation parity and labels the 0.4.1 probability equations separately.

From site indexes to coordinates

Internally, a matched path is represented on inference-site indexes. In the paper/stable coordinate map, path index zero maps to genomic coordinate zero, an internal segment beginning at site j maps to that site's position, and the final right endpoint maps to sequence length. Thus a source path [A, A, B] at positions [10, 20, 40] on a sequence of length 100 yields edges [0, 40) from A and [40, 100) from B. Ending the final edge at `last_position + 1` is not generally equivalent.

What path compression means

When two children copy the same contiguous *multi-edge path*, tsinfer can insert a synthetic ancestor for the shared path. Reusing a single equal edge is insufficient: the source builder searches for contiguous runs containing more than one matching edge. It then rewires both the existing and new child through the synthetic node and assigns a time that preserves parent-before-child ordering.

The miniature only detects eligible repeated runs. It does not manufacture a node or claim that a fixed epsilon below the parent is always valid. Production tsinfer's indexed builder is the ground truth for topology mutation and node-time placement.

```

def path_to_edges(path, positions, child_id, ref_node_ids, sequence_length):
    """Map site-index copying segments to genomic half-open intervals."""

    path = np.asarray(path, dtype=int)

```

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```

positions = np.asarray(positions, dtype=float)
refs = np.asarray(ref_node_ids, dtype=int)
if len(path) == 0 or len(path) != len(positions):
    raise ValueError("path and positions must have the same nonzero length")
boundaries = np.append(positions, float(sequence_length))
boundaries[0] = 0.0 # tsinfer's paper/stable-release coordinate map
edges = []
start = 0
for site in range(1, len(path) + 1):
    if site == len(path) or path[site] != path[start]:
        edges.append(
            (
                float(boundaries[start]),
                float(boundaries[site]),
                int(refs[path[start]]),
                int(child_id),
            )
        )
        start = site
return edges

```

```

def shared_path_segments(edges, minimum_edges=2):
    """Detect repeated contiguous multi-edge paths eligible for compression.

    Sharing one edge is not sufficient. This detector does not mutate topology
    or invent a synthetic-node time; the production builder owns those steps.
    """

    by_child = defaultdict(list)
    for left, right, parent, child in edges:
        by_child[int(child)].append((float(left), float(right), int(parent)))
    occurrences = defaultdict(set)
    for child, child_edges in by_child.items():
        child_edges.sort()
        for size in range(minimum_edges, len(child_edges) + 1):
            for start in range(len(child_edges) - size + 1):
                run = child_edges[start : start + size]
                if all(run[j - 1][1] == run[j][0] for j in range(1, size)):
                    occurrences[tuple(run)].add(child)

    return {
        signature: tuple(sorted(children))
        for signature, children in occurrences.items()
        if len(children) >= 2
    }

```

3.7.5 Sample matching and post-processing

Threading observed haplotypes

After the ancestor tree sequence exists, each phased sample is matched through it using the same family of copying updates. Sample nodes are at time zero. Copying segments define edges to ancestral nodes, while discrepancies at inference sites define mutations. The tree-compressed state space is essential to scalability; the dense Viterbi function in

this book is an equation check, not the production sample matcher.

Non-inference variants

Variants excluded from inference still carry information. For each such site, tsinfer examines the local inferred tree and places mutations parsimoniously so the sample genotypes are represented. With a known ancestral state, the root state is constrained accordingly. Recurrent or back mutations can be necessary; a binary single-mutation story is not guaranteed for arbitrary data.

Post-processing in stable 0.4.1

The stable implementation performed several separate operations:

1. identify and detach a virtual-root-like oldest edge when the expected format is present, moving affected mutations appropriately;
2. optionally split a genome-wide ultimate ancestor when its children change;
3. erase topology before the first site and after `last_site + 1` (capped by sequence length); and
4. call tskit's simplifier with sites, populations, and individuals retained and unary nodes kept.

These steps are not interchangeable. Merely deleting every edge whose parent is a chosen root can strand mutations. Likewise, walking upward from samples and dropping unvisited nodes does not reproduce tskit's coordinate-aware simplify operation.

Current development versions can differ in their post-processing and input model. Always use production tsinfer with the matching tskit version and preserve provenance. The teaching helpers below only expose small invariants.

```
def fitch_parsimony(tree_children, leaf_alleles, root, root_state=0):
    """Place a deterministic minimum-change character on one rooted tree.

    ``root_state`` constrains the known ancestral state. General mutation
    mapping in real workflows is delegated to tskit.
    """

    sets = {}

    def upward(node):
        children = tree_children.get(node, ())
        if not children:
            if node not in leaf_alleles:
                raise ValueError(f"missing allele for leaf {node}")
            sets[node] = {leaf_alleles[node]}
            return
        for child in children:
            upward(child)
        intersection = set.intersection(*(sets[child] for child in children))
        sets[node] = intersection or set.union(*(sets[child] for child in children))

    upward(root)
    mutations = []

    def downward(node, state):
        for child in tree_children.get(node, ()):
            child_state = state if state in sets[child] else min(sets[child])
            if child_state != state:
```

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```

        mutations.append((child, state, child_state))
        downward(child, child_state)

    downward(root, root_state)
    return mutations

```

```

def erase_flanks(edges, first_site, last_site, sequence_length):
    """Keep topology from the first site through ``last_site + 1``."""

    left_bound = float(first_site)
    right_bound = min(float(last_site) + 1, float(sequence_length))
    trimmed = []
    for left, right, parent, child in edges:
        left = max(float(left), left_bound)
        right = min(float(right), right_bound)
        if left < right:
            trimmed.append((left, right, int(parent), int(child)))
    return trimmed

```

3.7.6 Auditable example

The corrected miniature is intentionally a set of mechanisms rather than a fake end-to-end inference engine. Run its deterministic example with:

```
python -m watchgen.mini_tsinfer
```

The test suite is more informative than the printout. It checks that equal carrier patterns can create multiple focal sites, intervening older polymorphism splits an ancestor, both zero ancestors are present, Haldane and mismatch transforms match stable 0.4.1, dense Viterbi agrees with exhaustive path enumeration, edge coordinates use zero and sequence length, and path compression requires a repeated multi-edge run.

For actual inference, install a released tsinfer version, follow that version's official input documentation, run the production pipeline, and inspect the emitted provenance. The current development documentation uses VCF-Zarr inputs and TOML configuration; older `SampleData` examples are historical and should not be silently presented as the latest interface.

Reproducibility checklist

1. Record tsinfer, tskit, and input-library versions.
2. Record phasing, ancestral-state, site, and sample filters.
3. Record recombination maps, mismatch settings, and path-compression choices.
4. Distinguish proxy node order from calibrated dates; use an explicit dating method if times in generations are required.
5. Verify genotype representation and tree-sequence validity with production APIs.
6. Preserve provenance and logs with the inferred `.trees` artifact.

3.8 Timepiece VII: SINGER

Sampling and Inference of GENEalogies with Recombination

SINGER is a Bayesian sampler for ancestral recombination graphs (ARGs) under an SMC approximation. It initializes an ARG by adding phased haplotypes one at a time, using one HMM for joining branches and a second HMM for joining times. It then explores ARG space with sub-graph pruning and regrafting (SGPR), and periodically recalibrates node times against mapped mutations [Deng *et al.*, 2025].

This chapter follows the published Methods and Supplementary Sections B.1-B.4 and checks the control flow against the authors' C++ implementation. The Python module illustrates individual equations and a single-tree SPR; it is not a replacement for SINGER's ARG data structures or sampler.

i Scope and parity

The paper benchmarked SINGER v0.1.8. The source audit for this chapter used `popgenmethods/SINGER` commit `eb8e39b1a15be4a9a4df4fdaab61847bf73515d7`. The official program, not `watchgen.mini_singer`, is the ground truth for production behavior.

3.8.1 Overview

Input and target

SINGER takes phased whole-genome variation and targets an approximate posterior over ARGs,

$$\Pr(G \mid D) \propto \Pr(D \mid G) \Pr(G),$$

under a constant-size, panmictic SMC model in its threading HMMs. Posterior sampling matters because local topology and coalescence time can remain uncertain even when the mutation and recombination rates are known. The published implementation accepts a mutation rate or mutation map, a recombination-to-mutation ratio or recombination map, a reference diploid effective size, and a polarization probability.

The algorithm

Initialization starts from an empty ARG and threads samples successively. For the next sample, branch sampling chooses a joining branch in each genomic bin; conditional on that path, time sampling chooses a joining time. This split is an approximation: ARGweaver instead represents complete joining points in one larger HMM.

After initialization, SINGER rescales node times. Each MCMC proposal then:

1. introduces a cut in one marginal tree and propagates the equivalent cut through neighboring trees until recombination breaks equivalence;
2. removes the ancestral material above those cuts;
3. rethreads the cut-defined sub-ARG with the same two-stage HMM; and
4. accepts or rejects using the SGPR Metropolis-Hastings rule.

After each thinning interval the implementation rescales again and writes an ARG sample. Rescaling is therefore neither an operation after every newly threaded haplotype nor an extra state inside the HMM.

```
phased variants
  |
  v
iterative branch + time threading
  |
  v
rescale initialized ARG
  |
  +--> cut-defined SGPR proposals --+
```

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```

|
+<----- repeat to thinning -----+
|
v
rescale and write posterior sample

```

What the miniature covers

`watchgen.mini_singer` makes the following source-level mechanisms executable:

- the branch joining mass and deterministic lineage-count approximation;
- the three-edge Poisson emission calculation;
- the Li-Stephens-like branch transition;
- the modified PSMC kernel and interval transition matrix;
- equal-ARG-length time windows and their piecewise-linear rescaling; and
- the SGPR tree-height acceptance approximation.

It omits VCF parsing, mutation mapping, stochastic traceback, the persistent ARG representation, partial-branch propagation across actual recombinations, and complete SGPR proposal densities. Results from the miniature must not be presented as inferred ARGs.

Practical limits

The official software requires phased, high-quality contemporary genomes. Non-polymorphic and multiallelic sites are not used for inference, and sites with missingness are excluded by the current implementation. Data-poor regions can cause numerical underflow; the authors recommend analyzing long chromosomes in segments and merging output. A single ARG is not a Bayesian summary: users should inspect convergence diagnostics and retain multiple thinned samples.

3.8.2 Branch sampling

State space

For bin ℓ , the hidden state B_ℓ is a joining branch in the current marginal tree. If the partial ARG already recombines between two bins, a branch can split into carried *partial-branch* states. SINGER retains all full branches but prunes a partial state when its forward probability is below ϵ (normally 1%). This bounds the state space by roughly $2n - 1 + 1/\epsilon$; empirically it stays close to $2n - 1$.

The partial states are not optional decoration. Without them, a branch-only Markov path can combine locally valid transitions into a globally impossible threading that would require a second recombination between adjacent bins.

Joining mass

Let $\lambda_\Psi(t)$ be the number of extant branches at time t in marginal tree Ψ . In coalescent units with pairwise rate one, the survival function and joining-time density are

$$\bar{F}_\Psi(t) = \exp\left[-\int_0^t \lambda_\Psi(u) du\right], \quad f_\Psi(t) = \lambda_\Psi(t) \bar{F}_\Psi(t).$$

At time t , each of the $\lambda_\Psi(t)$ branches is equally eligible. The probability of joining one particular branch spanning $[x, y)$ is therefore

$$p_i = \int_x^y \frac{f_\Psi(t)}{\lambda_\Psi(t)} dt = \int_x^y \bar{F}_\Psi(t) dt.$$

The implementation accelerates this calculation by replacing the stochastic lineage count with

$$\lambda(t) \approx \frac{n}{n + (1 - n)e^{-t/2}}, \quad \bar{F}(t) \approx \frac{e^{-t}}{[n + (1 - n)e^{-t/2}]^2}.$$

A branch is represented by one time τ_i , chosen so that $\lambda(\tau_i) = \sqrt{\lambda(x)\lambda(y)}$. This is a heuristic representative time, not an expectation conditional on joining the branch.

Emissions

Threading a lineage bisects the joining branch into lower and upper pieces. SINGER imputes the binary state at the joining point by parsimony and multiplies Poisson probabilities for the three incident edges. With $m = \theta l/2$, an edge requiring no state change contributes e^{-m} ; an edge requiring one change contributes me^{-m} . The latter is the probability of *exactly one* mutation, not $1 - e^{-m}$.

For a root branch, the root state is weighted by `-polar`: 0 has probability p_{root} and 1 has probability $1 - p_{root}$. The default 0.5 represents unpolarized data; the authors suggest a value such as 0.99 when 0 is reliably ancestral.

Transitions

If the partial ARG has no recombination at the boundary, the probability that the new lineage recombines before its representative joining time is

$$r_i = 1 - e^{-\rho\tau_i/2}.$$

For a full target branch j , define $q_j = r_j p_j$; partial targets have $q_j = 0$. Then

$$\Pr(B_\ell = j \mid B_{\ell-1} = i) = (1 - r_i)\delta_{ij} + r_i \frac{q_j}{\sum_k q_k}.$$

This drops the SMC rejoining distribution's dependence on the source branch, which gives the transition its linear Li-Stephens form. If the partial ARG already recombines at the boundary, the threaded lineage may only hitchhike through the corresponding full or partial state: SINGER forbids introducing a second recombination between the same adjacent bins.

```
def emission_probability(allele_new, allele_lower, allele_upper, tau,
                        branch_lower, branch_upper, theta):
    """Three-edge emission from Supplementary Figure 22.

    The joining-point state is imputed by binary parsimony (majority of the
    three incident states). The product covers the new lineage and both pieces
    of the bisected joining branch. A one-change edge uses ``m exp(-m)``, not
    the probability of at least one mutation.
    """
    alleles = (allele_new, allele_lower, allele_upper)
    if any(a not in (0, 1) for a in alleles):
        raise ValueError("alleles must be binary")
    if not 0 <= branch_lower <= tau <= branch_upper:
        raise ValueError("tau must lie on the joining branch")
    join = int(sum(alleles) >= 2)
    pieces = (
        (abs(allele_new - join), tau),
        (abs(allele_lower - join), tau - branch_lower),
        (abs(allele_upper - join), branch_upper - tau),
    )
    return float(np.prod([poisson_edge_probability(k, length, theta)
                          for k, length in pieces]))
```

3.8.3 Time sampling

Conditional HMM

Time sampling conditions on the full joining-branch path returned by branch sampling. For a branch $[x_\ell, y_\ell]$, SINGER partitions the interval into 5% quantiles of an exponential distribution (20 states by default). If the boundaries are $t_{\ell,0}, \dots, t_{\ell,d}$, state i uses a representative time satisfying

$$e^{-\tau_{\ell,i}} = \frac{e^{-t_{\ell,i}} + e^{-t_{\ell,i+1}}}{2}.$$

The PSMC-like kernel

Let s be the previous joining time and ρ the scaled recombination rate for one bin. Conditioned on a recombination, the new time has density

$$q_0(t | s) = \begin{cases} [1 - e^{-t}]/s, & t < s, \\ [e^{-(t-s)} - e^{-t}]/s, & t \geq s. \end{cases}$$

Without conditioning, multiply the continuous part by $1 - e^{-\rho s}$ and add an atom of mass $e^{-\rho s}$ at $t = s$. The atom is a probability mass, not a density spike that should be integrated numerically.

Writing $Q_\rho(t | s)$ for the resulting CDF, the interval transition is

$$q_{ij}^{\ell-1,\ell} = \frac{Q_\rho(t_{\ell,j+1} | \tau_{\ell-1,i}) - Q_\rho(t_{\ell,j} | \tau_{\ell-1,i})}{Q_\rho(y_\ell | \tau_{\ell-1,i}) - Q_\rho(x_\ell | \tau_{\ell-1,i})}.$$

The denominator conditions the new coalescence time to lie on the sampled joining branch. Each row therefore sums to one.

Three boundary cases

Type A

Neither the partial ARG nor joining branch changes. Symmetries in the transition matrix permit the forward update in $O(d)$ rather than $O(d^2)$. For $i > j$, $q_{ij} = q_{j+1,j}$; for $i < j$, adjacent columns have a source-independent ratio.

Type B

The partial ARG recombines. Joining-time intervals that remain on the sampled branch hitchhike to the next tree; incompatible intervals vanish, and SINGER creates additional intervals to cover any remainder.

Type C

The partial ARG is unchanged but the joining branch changes, so the newly threaded lineage recombined. The transition is conditioned on that event, equivalently using $\rho \rightarrow \infty$ in the kernel.

Recombination time

The two HMMs locate a recombination boundary but do not infer its exact time. If the recombining branch begins at l , the two neighboring joining times imply an upper bound u , and re-coalescence occurs at v , the conditional density is proportional to $e^{-(v-x)}$ for $l < x < u$. SINGER uses its median,

$$x_{1/2} = \log\left(\frac{e^l + e^u}{2}\right).$$

```
def time_transition_matrix(boundaries_prev, taus_prev, boundaries_next, rho):
    """Conditional interval probabilities from Supplement Eq. (4)."""
    del boundaries_prev
```

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```

taus_prev = np.asarray(taus_prev, dtype=float)
boundaries_next = np.asarray(boundaries_next, dtype=float)
if len(boundaries_next) < 2 or np.any(np.diff(boundaries_next) <= 0):
    raise ValueError("next-state boundaries must be strictly increasing")
matrix = np.empty((len(taus_prev), len(boundaries_next) - 1))
lower, upper = boundaries_next[0], boundaries_next[-1]
for i, source in enumerate(taus_prev):
    denominator = (psmc_transition_cdf(upper, source, rho) -
                   psmc_transition_cdf(lower, source, rho))
    if denominator <= 0:
        raise ValueError("target branch has zero transition mass")
    for j, (left, right) in enumerate(zip(boundaries_next[:-1],
                                         boundaries_next[1:])):
        matrix[i, j] = (psmc_transition_cdf(right, source, rho) -
                        psmc_transition_cdf(left, source, rho)) / denominator
return matrix

```

3.8.4 ARG rescaling

Purpose and timing

The threading HMMs assume a constant population size. SINGER reduces the resulting time-scale bias by applying a monotone, piecewise-linear transformation learned from mapped mutations. This does not infer a population-size history, change topology, or constitute an MCMC proposal. The implementation runs it after initialization and after every thinning interval, immediately before a sample is written.

Equal-ARG-length windows

Let $L(G)$ be total ARG branch length, with every marginal-tree branch weighted by its genomic span. SINGER chooses $J = 100$ time windows by default so that each contains $L(G)/J$ branch length. A mutation mapped to a branch crossing several windows is divided among them in proportion to the branch's overlap with each window.

If m_i is the fractional mutation count in window i , its expected count under scaled mutation rate θ is $\theta L(G)/(2J)$. The window scale factor is therefore

$$c_i = \frac{2Jm_i}{\theta L(G)}.$$

For old boundaries t_i and new boundaries $\tilde{t}_i$,

$$\tilde{t}_0 = 0, \quad \tilde{t}_i = \tilde{t}_{i-1} + c_i(t_i - t_{i-1}),$$

and a node time $t \in [t_{i-1}, t_i)$ maps to

$$\tilde{t} = \tilde{t}_{i-1} + c_i(t - t_{i-1}).$$

Non-negative factors preserve node order. A zero-mutation window can collapse an interval; in real analyses, sparse data are therefore a substantive and numerical warning, not evidence of instantaneous ancestry.

Mutation-rate maps

With heterogeneous mutation rate $\mu(x)$, expected counts must integrate the rate across each branch's genomic span and multiply by its overlap with the time window. Evaluating μ only at a span midpoint is not the published calculation and can be biased across sharp rate changes.


```
def compute_scaling_factors(counts, total_arg_length, theta, J):
    """Return ``c_i = 2 J m_i / (theta L(G))`` (Supplement B.3.1)."""
    if total_arg_length <= 0 or theta <= 0 or J < 1:
        raise ValueError("ARG length, theta, and J must be positive")
    counts = np.asarray(counts, dtype=float)
    if len(counts) != J or np.any(counts < 0):
        raise ValueError("counts must be J non-negative values")
    return 2 * J * counts / (theta * total_arg_length)
```

3.8.5 Sub-graph pruning and regrafting (SGPR)

The proposal

SGPR is an ARG operation, not an independent SPR on every marginal tree. A proposal starts at the rightmost coordinate reached by the previous cut (or wraps to the chromosome start), samples a time uniformly between zero and the local tree height, and chooses uniformly among branches crossing that time. The equivalent point is propagated left and right through neighboring trees. Propagation stops where a recombination decouples the relevant ancestral material. In every affected tree, SINGER removes the portion from the cut to the upper endpoint of its branch.

The remaining graph H is regrafted by branch and time sampling above the cut. This is the key difference from the Kuhner move: the Kuhner regraft is simulated from the prior, whereas SGPR's threading proposal uses the data.

Acceptance probability

The exact Metropolis-Hastings rule is

$$A_H(G \rightarrow G') = \min \left\{ 1, \frac{P(G' | D) q_H(G' \rightarrow G)}{P(G | D) q_H(G \rightarrow G')} \right\}.$$

The supplement then assumes that the approximate threading algorithm samples from the conditional posterior. Under that assumption, posterior terms cancel. For SINGER's cut scheme, the approximation reduces to

$$A_H(G \rightarrow G') \approx \min \left\{ 1, \frac{h(\Psi_x)}{h(\Psi'_x)} \right\},$$

where x is the rightmost coordinate of the previous cut and $h(\Psi_x)$ is the marginal-tree height there. The height ratio is not a universal SPR identity; it depends on the conditional-posterior proposal approximation and the specified cut distribution.

The paper reports SGPR acceptance close to one for its 50-sequence benchmark, compared with about 22% for ARG-weaver's subtree-rethreading proposals. That is an empirical result for the reported experiment, not a guaranteed acceptance rate for every dataset.

Single-tree illustration

The miniature's `spr_move()` validates and performs one marginal-tree SPR so the topology change can be inspected. It does not propagate cuts, preserve ancestral material across a tree sequence, compute a proposal density, or run an MCMC chain. Calling it "SGPR" without this qualification would overstate parity with the original software.

```
def sgpr_acceptance_ratio(old_tree_height, new_tree_height):
    """Approximate SGPR acceptance ``min(1, h(Psi_x)/h(Psi'_x))``."""
    if old_tree_height <= 0 or new_tree_height <= 0:
        raise ValueError("tree heights must be positive")
    return min(1.0, old_tree_height / new_tree_height)
```

3.8.6 Executable mechanisms

Run the scalar demonstration with

```
python -m watchgen.mini_singer
```

It prints a representative joining time, a branch joining probability, and a tree-height acceptance ratio. The accompanying tests compare the equations against direct integrations, row-normalized interval probabilities, a dense forward update, and explicit rescaling calculations.

The generated figures visualize these mechanisms only. An msprime tree sequence is simulated truth, not SINGER output; plotting its marginal trees does not validate ARG inference. For a real analysis, run the authors' `singer_master` on phased VCF data, retain multiple thinned ARG samples, convert them with the provided `tskit` utility, and examine the convergence traces recommended by the software authors.

3.9 Timepiece VIII: Threads

Threading Instructions for Ancestral Recombination Graphs

“Threads takes as input a set of phased genotypes and outputs a set of threading instructions for each sample.”

---Brandt, Chiang, Guo *et al.* (2024)

3.9.1 The Mechanism at a Glance

Threads is a **deterministic** method for inferring Ancestral Recombination Graphs from phased genotype data. Where SINGER (*Timepiece VII: SINGER*) samples ARGs from a posterior distribution using Bayesian MCMC, Threads finds the single most likely threading path for each haplotype using a three-step pipeline: pre-filter candidate matches, run a memory-efficient Viterbi algorithm, and date the resulting segments.

i Primary Reference

[Gunnarsson *et al.*, 2024]

The three gears of Threads:

1. **PBWT Haplotype Matching** (the pre-filter) – Uses the positional Burrows-Wheeler transform to identify a small set of candidate haplotype matches for each sample, reducing the search space from $O(N)$ to $O(L)$ candidates per sample ($L \ll N$).
2. **Memory-Efficient Viterbi** (the inference engine) – A branch-and-bound implementation of the Viterbi algorithm under the Li-Stephens model that finds the optimal threading path in $O(NM)$ time and $O(N)$ average memory, avoiding the classical $O(NM)$ memory requirement.
3. **Segment Dating** (the calibration) – Assigns coalescence times to each Viterbi segment using likelihood-based and Bayesian estimators that model segments as pairwise IBD regions.

These gears connect in a simple linear pipeline:

```
Phased genotypes (N haplotypes, M sites)
      |
      v
+-----+
| PBWT HAPLOTYPE MATCHING |
|                           |
| Chunk genome into 0.5 cM |
+-----+
```

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	segments, sort with PBWT,	
	query L-neighbourhood,	
	filter by match count	
	-> L candidates per sample	
+-----+		
	v	
+-----+		
	MEMORY-EFFICIENT VITERBI	
	For each sample n = 1..N:	
	Run Li-Stephens Viterbi	
	on L x M candidate panel	
	using branch-and-bound	
	-> Threading targets per	
	sample at each site	
+-----+		
	v	
+-----+		
	SEGMENT DATING	
	For each Viterbi segment:	
	Model as IBD region	
	Estimate age from length	
	+ mutations + demography	
	-> Coalescence times	
+-----+		
	v	
	Threading instructions	
	(= the inferred ARG)	

Prerequisites for this Timepiece

Threads builds on several earlier concepts and Timepieces:

- *Li & Stephens HMM* – the copying model that Threads optimizes with its memory-efficient Viterbi
- *Coalescent Theory* – coalescence times and rates used in the dating step
- *The SMC* – the sequentially Markov coalescent model underlying the segment length distribution

If you have worked through the Li & Stephens Timepiece, you already have the main conceptual foundation. Threads extends that foundation to biobank-scale data through algorithmic innovation rather than model complexity.

3.9.2 Chapters

Overview of Threads

Before assembling the watch, lay out every part and understand what it does.

Threads: Segment Dating Estimators

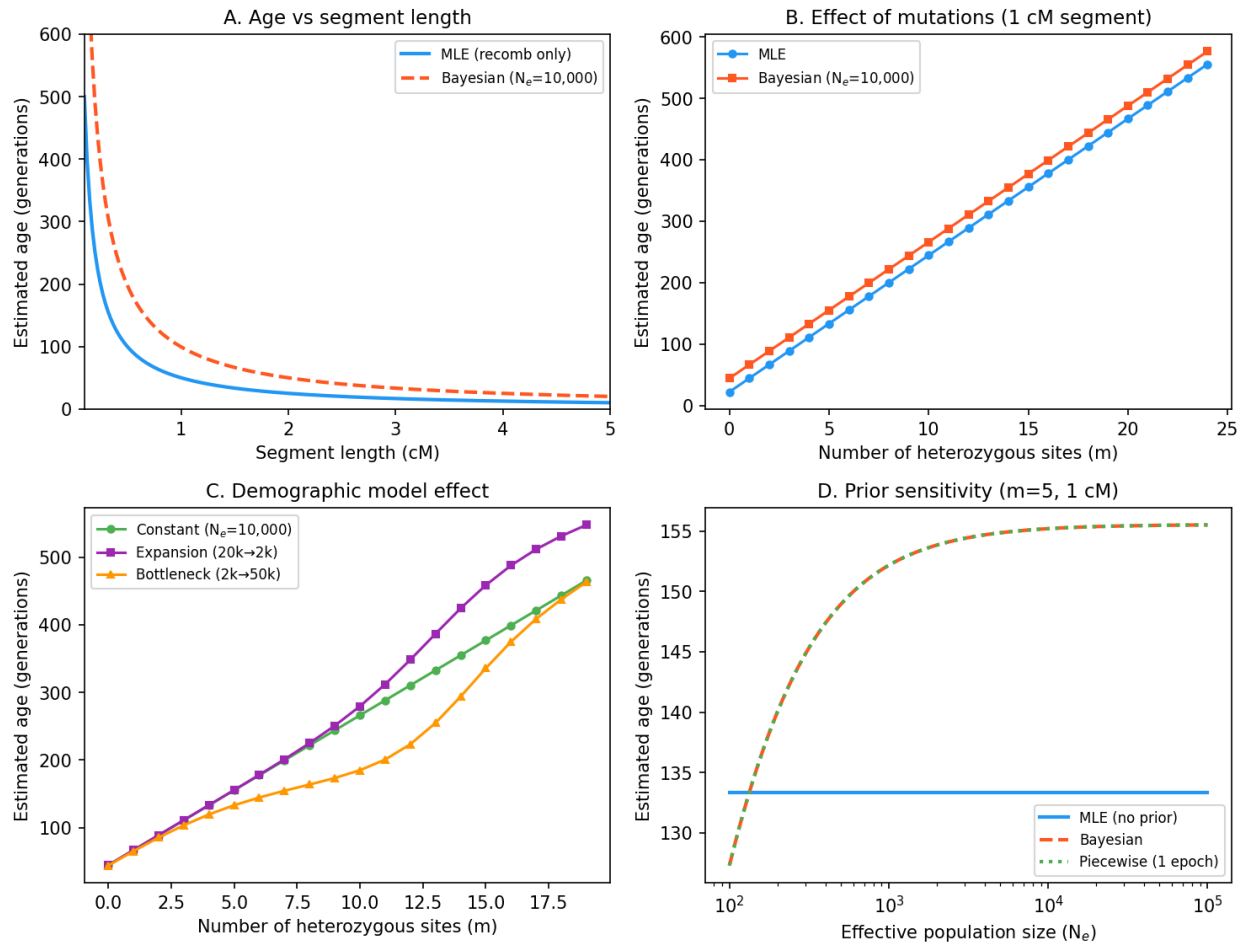


Fig. 8: **Threads at a glance.** Segment dating estimators showing how MLE and Bayesian age estimates respond to recombination distance, mutation count, and demographic history. The comparison between bottleneck and constant-size scenarios illustrates how the demographic prior shapes age inference at biobank scale.

Threads is a method for inferring Ancestral Recombination Graphs (ARGs) at biobank scale. Given a set of phased genotypes, it produces **threading instructions** – for each sample at each genomic position, a threading target (the closest genealogical relative) and a coalescence time. Together, these instructions specify an ARG.

Input: $X \in \{0, 1\}^{M \times N}$ (phased genotypes)

Output: For each $n \leq N$, a map $[L_1, L_2] \rightarrow \mathbb{R}_+ \times \{1, \dots, n-1\}$

The output map assigns to each genomic position a coalescence time and a threading target from among the previously threaded samples. These threading instructions can be assembled into a full ARG.

The Key Insight: Scalability Through Decomposition

The classical approach to Li-Stephens inference requires $O(MN^2)$ time for the complete ARG – each of N samples must search through all previously threaded samples. Threads achieves scalability through two innovations:

1. **PBWT pre-filtering** reduces the reference panel from N to L candidates per sample ($L \ll N$), cutting the per-sample Viterbi cost from $O(MN)$ to $O(ML)$.
2. **Branch-and-bound Viterbi** replaces the classical $O(NM)$ memory Viterbi with an algorithm that uses $O(N)$ average memory by exploiting the rarity of recombination events in optimal paths.

Together, these yield a total complexity of $O(MLN/N_{\text{CPU}})$ time and $O(LN)$ average memory, with all N Viterbi instances running in parallel.

How Threads Compares to SINGER

Threads and SINGER (*Timepiece VII: SINGER*) both infer ARGs, but take fundamentally different approaches:

Property	Threads	SINGER
Inference type	Deterministic (Viterbi – single best path)	Bayesian (MCMC – posterior samples)
Output	One ARG (maximum likelihood threading)	Multiple ARG samples from the posterior
Scalability	Biobank-scale ($N > 10^5$)	Moderate scale ($N \sim 10^2$ – 10^3)
Uncertainty	No uncertainty quantification	Full posterior uncertainty
Parallelism	Embarrassingly parallel (per-sample Viterbi)	Sequential (MCMC chain)
Architecture	PBWT pre-filter + Li-Stephens Viterbi + dating	Two-HMM (branch + time) + SGPR MCMC

The trade-off is clear: Threads sacrifices the richness of posterior samples for the ability to scale to hundreds of thousands of samples. In watchmaking terms, SINGER is the grand complication – intricate and precise. Threads is the high-frequency movement – engineered for speed and efficiency.

Terminology

Term	Definition
Threading target	The closest genealogical relative (closest cousin) of sample n at a given site, chosen from among samples $1, \dots, n-1$
Threading instructions	For each sample, a map from genomic positions to (coalescence time, threading target) pairs – the complete output of Threads
Viterbi path	A path $\pi \in \{1, \dots, N\}^M$ of maximum probability under the Li-Stephens model
Path segment	A contiguous portion of the Viterbi path where the threading target is constant
Segment set Ω	The collection of path segments maintained by the branch-and-bound Viterbi algorithm
Active segment	A segment in Ω representing the current best path ending at a given reference haplotype
L-neighbourhood	The L nearest sequences in the PBWT prefix array around a query sequence
Chunk	A genomic segment (default 0.5 cM) over which PBWT matching is performed independently
IBD segment	A genomic region where two sequences share a constant most recent common ancestor

The Three-Pass Pipeline

Threads processes the genotype data in three sequential passes:

Pass 1: Haplotype matching (PBWT). Stream genotypes from disk, build the PBWT prefix array, and query the L -neighbourhood for each sample at regular intervals. Output: a sparse set of candidate matches per sample per chunk.

Pass 2: Viterbi inference. Stream genotypes again, this time running the memory-efficient Viterbi algorithm for all N samples in parallel, each against its own reduced reference panel. Output: threading targets (Viterbi paths) for each sample.

Pass 3: Segment dating. Stream genotypes a final time, computing coalescence time estimates for each Viterbi segment using IBD-based modeling with optional demographic priors. Output: complete threading instructions.

Each pass requires reading the genotype data once from disk. The full genotype matrix is never stored in memory.

Ready to Build

In the following chapters, we build each gear from scratch:

1. *Haplotype Matching with the PBWT* – The PBWT pre-filter. How Threads identifies candidate matches efficiently using prefix array sorting and neighbourhood queries.
2. *Memory-Efficient Viterbi Inference* – The memory-efficient Viterbi. How the branch-and-bound strategy achieves $O(N)$ average memory while maintaining optimality.
3. *Dating Path Segments* – Segment dating. How coalescence times are estimated from segment length and mutation counts, with optional demographic priors.

Let's start with the first gear: PBWT Haplotype Matching.

Haplotype Matching with the PBWT

Reduce the haystack before searching for the needle.

The first step in the Threads pipeline is to identify, for each sample, a small set of candidate haplotype matches from the full reference panel. This is done using the **positional Burrows-Wheeler transform (PBWT)**, a data structure that sorts haplotypes by their shared prefixes at each site.

Why Pre-Filtering is Necessary

A direct application of the Li-Stephens model to ARG inference requires each sample n to search through $n - 1$ previously threaded sequences, resulting in $O(MN^2)$ time for the whole inference. For biobank data with $N > 10^5$ samples and $M > 10^6$ sites, this is computationally infeasible.

Threads reduces the per-sample search space from $n - 1$ to L candidates ($L \ll N$) by exploiting the structure of the PBWT: sequences that are neighbours in the prefix array tend to share long identical-by-state (IBS) tracts. This heuristic, also employed by modern phasing and imputation algorithms like IMPUTE5, allows Threads to identify close matches without exhaustive comparison.

The Chunking Strategy

Threads partitions the input genomic region into small chunks of default size 0.5 cM. Haplotype matching is performed independently within each chunk. This chunking serves two purposes:

1. It allows the algorithm to maintain a **locally optimized** set of candidates – matches that are relevant to the local genealogy rather than the genome-wide average.
2. It bounds the memory used per sample: within each chunk, the maximum number of matches is $L \cdot M_{\text{query}} / M_{\text{chunks}}$.

We denote the number of chunks by M_{chunk} .

PBWT Prefix Array Sorting

At each site i , the PBWT maintains a prefix array a that sorts haplotypes by their allelic history up to that site. The update rule is simple: haplotypes carrying allele 0 at site i are placed before those carrying allele 1, preserving the relative order within each group from the previous site.

```
Site i:   sort_0 = [h3, h1, h4, h2, h5]

Alleles:  h3->0, h1->1, h4->0, h2->1, h5->0

Site i+1: sort_1 = [h3, h4, h5, | h1, h2]
               ^^^^^^^^^^^^^  ^^^^^
               allele = 0      allele = 1
               (order preserved) (order preserved)
```

This sorting is performed in $O(N)$ time per site using a single pass through the current array. The PBWT is never stored in full – genotypes are streamed through the algorithm one site at a time.

```
import numpy as np

def pbwt_update(prefix_array, alleles):
    """Update the PBWT prefix array at one site.

    Sorts haplotypes by placing allele-0 carriers before allele-1
    carriers, preserving relative order within each group.

    Parameters
    -----
    prefix_array : list of int
        Current ordering of haplotype indices.
    alleles : ndarray, shape (N,)
        Alleles (0 or 1) for each haplotype at this site.
```

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```

Returns
-----
new_array : list of int
    Updated prefix array.
"""
zeros = [h for h in prefix_array if alleles[h] == 0]
ones = [h for h in prefix_array if alleles[h] == 1]
return zeros + ones

# Demonstrate PBWT sorting on a small panel
N = 6 # haplotypes
M = 4 # sites
np.random.seed(42)
panel = np.random.randint(0, 2, size=(N, M))

prefix = list(range(N)) # initial order: [0, 1, 2, 3, 4, 5]
print("Haplotype panel:")
for h in range(N):
    print(f"  h{h}: {panel[h]}")
print()

for site in range(M):
    prefix = pbwt_update(prefix, panel[:, site])
    print(f"Site {site} (alleles={panel[:, site].tolist() }): "
          f"prefix = {prefix}")
print("\nFinal PBWT order groups haplotypes sharing long prefixes together")

```

L-Neighbourhood Querying

At regular intervals (default: every 0.01 cM), the prefix array is queried for the L nearest neighbours around each sequence. For a sequence n at prefix index i (i.e., $a_{i,j} = n$ at query site j), the query finds the first $L/2$ sequences immediately above and below index i in the prefix array that satisfy $a_{k,j} < n$ (i.e., they were threaded before sample n).

If fewer than $L/2$ qualifying neighbours exist on one side, the search window expands in the opposite direction until L total matches are found or all sequences have been considered. The default neighbourhood size is $L = 4$.

The querying is implemented by sequentially inserting each sequence's prefix array index into a **red-black tree** (C++ `std::set`) and querying the L -sized neighbourhood around the insertion point. This gives $O(\log N)$ time per insertion and query.

We denote the total number of query sites by M_{query} .

```

def query_l_neighbourhood(prefix_array, query_idx, L, max_idx):
    """Find the L nearest neighbours in the prefix array.

    Returns the L/2 closest sequences above and below query_idx
    in the prefix array that were threaded earlier (index < max_idx).

    Parameters
    -----
    prefix_array : list of int
        Current PBWT prefix array.
    query_idx : int

```

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```

    Position of the query sequence in the prefix array.
    L : int
    Neighbourhood size (total matches to return).
    max_idx : int
    Only consider sequences with index < max_idx (threaded earlier).

Returns
-----
neighbours : list of int
    Haplotype indices of the L nearest neighbours.
"""
pos = prefix_array.index(query_idx)
neighbours = []

# Search upward (lower indices in prefix array)
above = []
for i in range(pos - 1, -1, -1):
    if prefix_array[i] < max_idx:
        above.append(prefix_array[i])
    if len(above) >= L // 2:
        break

# Search downward (higher indices in prefix array)
below = []
for i in range(pos + 1, len(prefix_array)):
    if prefix_array[i] < max_idx:
        below.append(prefix_array[i])
    if len(below) >= L // 2:
        break

neighbours = above + below

# If one side has fewer, expand the other
if len(neighbours) < L:
    remaining = L - len(neighbours)
    all_eligible = [h for h in prefix_array
                    if h < max_idx and h != query_idx
                    and h not in neighbours]
    neighbours.extend(all_eligible[:remaining])

return neighbours[:L]

# Demonstrate L-neighbourhood query
prefix = [3, 0, 5, 2, 4, 1] # some PBWT ordering
query = 5 # query haplotype
L = 4
max_idx = 5 # only consider h0..h4 (threaded before h5)
nbrs = query_l_neighbourhood(prefix, query, L, max_idx)
print(f"Prefix array: {prefix}")
print(f"Query h{query} at position {prefix.index(query)}")
print(f"L={L} neighbours (threaded before h{query}): {nbrs}")

```

Candidate Filtering

Once all sites within a chunk have been processed, each sequence n has a multi-set of candidate matches – sequences that appeared in the L -neighbourhood at one or more query sites. These candidates are filtered by match count:

- By default, any match observed in fewer than 4 queries is discarded.
- If no matches survive this filter, the threshold is decreased by 1 until at least one sequence remains.
- For $n < 100$, all observed matches are retained (the panel is too small for aggressive filtering).
- For $n \geq 10,000$, the minimum match count is doubled to heuristically limit the number of candidates.

Additionally, the top 4 matches from adjacent chunks are included to account for recombination events near chunk boundaries.

```
from collections import Counter

def filter_candidates(match_counts, min_count=4, n_threaded=100):
    """Filter candidates by match count.

    Parameters
    -----
    match_counts : Counter
        Maps haplotype index to number of query sites where it appeared.
    min_count : int
        Minimum match count to retain a candidate.
    n_threaded : int
        Number of sequences threaded so far.

    Returns
    -----
    candidates : list of int
        Filtered candidate haplotype indices.
    """
    threshold = min_count
    if n_threaded >= 10000:
        threshold *= 2 # stricter for large panels

    candidates = [h for h, c in match_counts.items() if c >= threshold]

    # If none survive, relax threshold until at least one remains
    while not candidates and threshold > 1:
        threshold -= 1
        candidates = [h for h, c in match_counts.items() if c >= threshold]

    # Fallback: if still empty, take the best match
    if not candidates and match_counts:
        candidates = [match_counts.most_common(1)[0][0]]

    return candidates

# Demonstrate candidate filtering
np.random.seed(42)
counts = Counter()
for h in np.random.choice(50, size=200, replace=True):
```

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```

counts[h] += 1

filtered = filter_candidates(counts, min_count=4, n_threaded=500)
print(f"Total candidates observed: {len(counts)}")
print(f"After filtering (min_count=4): {len(filtered)} candidates")
print(f"Top 5 by count: {counts.most_common(5)}")

```

Note

Singleton variants are excluded from the matching step, as they tend to have higher phasing and genotyping errors and may interfere with identification of long IBS regions.

Complexity Analysis

The complete haplotype matching step has the following complexity:

Time: $O(MN + M_{\text{query}} \cdot N \cdot \log N)$

- $O(MN)$ for the PBWT prefix array updates across all sites
- $O(M_{\text{query}} \cdot N \cdot \log N)$ for the red-black tree insertions and queries at each query site

Memory: $O(N \cdot L \cdot M_{\text{query}})$

- Only the match candidates for each sequence are kept in memory
- The full PBWT and genotype matrix are never stored

The matching requires only a **single pass** through the genotype data, streaming from disk.

Memory-Efficient Viterbi Inference

The classical machine works, but it does not fit in the case. Redesign it.

The second step in the Threads pipeline takes the candidate matches from the PBWT pre-filter and runs a **memory-efficient Viterbi algorithm** under the Li-Stephens model to find the optimal threading path for each sample.

The Classical Viterbi Limitation

Given a reference panel H of N haplotypes over M sites and a query haplotype g , the Li-Stephens model assigns a probability $P(\pi)$ to each path $\pi \in \{1, \dots, N\}^M$ through the panel. A **Viterbi path** is a path of maximum probability.

The classical Viterbi algorithm finds this path in $O(NM)$ time by constructing a full $N \times M$ probability matrix, then performing a traceback. For biobank-scale data, this matrix is computationally prohibitive – even after PBWT pre-filtering reduces N to L , the memory requirement for long genomic tracts remains high.

Two Key Observations

The Threads-Viterbi algorithm exploits two properties of the Li-Stephens model:

Observation 1: Recombination events are rare. Viterbi paths consist of few but long segments. In the 1000 Genomes Project (2,251 sequences), the average segment length exceeds 200 kilobases. In UK Biobank array data ($N = 337,464$), segments average well over a megabase. This means a complete Viterbi path can be stored compactly by recording only the segment breakpoints and threading targets.

Observation 2: Recombination is symmetric. Under the Li-Stephens model:

$$p(\pi_{i+1} = \beta \mid \pi_i = \alpha, \text{recombination between } i \text{ and } i+1) = \frac{1}{N}$$

for any states α, β . This symmetry dramatically reduces the search space for possible Viterbi paths, as formalized in Proposition 1.

Proposition 1

Proposition 1. Suppose $\pi^{(i)}$ is a Viterbi path for the subset of the panel H containing sites 1 through i . If there exists a Viterbi path through H that recombines between sites i and $i+1$, then there exists a Viterbi path π through H satisfying $\pi_i = \pi_i^{(i)}$.

In plain language: at site i , we only need to consider recombination from the sequence of highest probability given all observations up to i . This is the property that makes the branch-and-bound strategy correct.

The Segment Set

The algorithm maintains a set Ω of **path segments**, each consisting of:

- A start site $m_\omega \in \{1, \dots, M\}$
- A threading target $n_\omega \in \{1, \dots, N\}$
- If $m_\omega > 0$, a traceback segment $\omega' \in \Omega$ with $m_{\omega'} < m_\omega$

A full path through H is constructed by starting at any segment and following traceback pointers until a segment with $m_\omega = 0$ is reached. The penalty (negative log-likelihood) of a path ending at ω is denoted $s(\omega)$.

The set Ω contains exactly N **active segments** $\omega_1, \dots, \omega_N$, one per reference haplotype. The segment set is **complete** if each $P(\omega_n)$ is the Li-Stephens-optimal path ending at haplotype n . When the set is complete, the active segment with minimum penalty gives a Viterbi path.

```
from dataclasses import dataclass, field
from typing import Optional

@dataclass
class Segment:
    """A path segment in the Threads-Viterbi algorithm.

    Parameters
    -----
    start : int
        Start site of this segment.
    target : int
        Reference haplotype index this segment copies from.
    penalty : float
        Cumulative negative log-likelihood up to this segment.
    parent : Segment or None
        Traceback pointer to the previous segment.
    """
    start: int
    target: int
    penalty: float
    parent: Optional['Segment'] = field(default=None, repr=False)
```

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```

def traceback(self):
    """Follow traceback pointers to reconstruct the full path."""
    path = []
    seg = self
    while seg is not None:
        path.append((seg.start, seg.target))
        seg = seg.parent
    return list(reversed(path))

# Demonstrate the segment structure
seg0 = Segment(start=0, target=3, penalty=10.0)
seg1 = Segment(start=500, target=7, penalty=15.0, parent=seg0)
seg2 = Segment(start=1200, target=3, penalty=22.0, parent=seg1)

path = seg2.traceback()
print("Viterbi path segments:")
for start, target in path:
    print(f" Site {start}: copy from haplotype {target}")
print(f"Total penalty: {seg2.penalty:.1f}")

```

The Branch Step (Theorem 1)

Given a complete segment set $\Omega^{(m)}$ for sites 1 through m , the branch step constructs $\Omega^{(m+1)}$ that is complete for sites 1 through $m + 1$.

Let ω' be the active segment with minimum penalty (the current Viterbi path), and let ρ and ρ_c be the penalties for recombination and no-recombination respectively.

For each active segment ω_n , if continuing without recombination is worse than recombining from the best path:

$$s(\omega_n, m) + \rho_c > s(\omega', m) + \rho$$

then a **new segment** $\omega(m + 1, n, \omega')$ is created, representing a recombination from the best path to haplotype n at site $m + 1$.

The new active segment for haplotype n becomes whichever option has lower penalty:

$$s_{\text{new}} = \min\{s(\omega_n, m) + \rho_c, s(\omega', m) + \rho\} + \mu_n$$

where μ_n is the match/mismatch penalty at site $m + 1$.

Complexity per site: The branch step adds at most N new segments, giving $O(NM)$ total segments across all sites. In practice, new segments are created only at inferred recombination events, which are rare.

```

def branch_step(active_segments, query_allele, ref_alleles,
                rho_penalty, rho_c_penalty, mismatch_penalty):
    """Perform the branch step at one site.

    For each active segment, decide whether continuing without
    recombination is better than recombining from the best path.

    Parameters
    -----
    active_segments : list of Segment

```

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```

    One active segment per reference haplotype.
query_allele : int
    Query haplotype's allele at this site (0 or 1).
ref_alleles : ndarray, shape (N,)
    Reference panel alleles at this site.
rho_penalty : float
    Penalty for recombination (-log(rho/N)).
rho_c_penalty : float
    Penalty for no recombination (-log(1 - rho)).
mismatch_penalty : float
    Penalty for allele mismatch.

Returns
-----
new_active : list of Segment
    Updated active segments for the next site.
n_new_segments : int
    Number of new segments created (recombination events).
"""
N = len(active_segments)
# Find the best current path (minimum penalty)
best = min(active_segments, key=lambda s: s.penalty)
site = active_segments[0].start + 1 # next site

new_active = []
n_new = 0
for n in range(N):
    seg = active_segments[n]
    # Emission penalty: mismatch if alleles differ
    mu_n = mismatch_penalty if ref_alleles[n] != query_allele else 0.0

    # Cost of continuing vs. recombining
    cost_continue = seg.penalty + rho_c_penalty + mu_n
    cost_recombine = best.penalty + rho_penalty + mu_n

    if cost_recombine < cost_continue:
        # Create a new segment (recombination event)
        new_seg = Segment(
            start=site, target=n,
            penalty=cost_recombine, parent=best
        )
        new_active.append(new_seg)
        n_new += 1
    else:
        # Continue the existing segment
        seg_updated = Segment(
            start=seg.start, target=n,
            penalty=cost_continue, parent=seg.parent
        )
        new_active.append(seg_updated)

return new_active, n_new

```

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```

# Demonstrate on a small example
N = 4
ref = np.array([[0,0,1,0,0,1,0,0],
                [0,0,0,0,1,1,0,0],
                [0,1,1,0,0,0,0,1],
                [0,0,1,0,0,1,1,0]])
query = np.array([0,0,1,0,0,1,0,1])

active = [Segment(start=0, target=n, penalty=0.0) for n in range(N)]
total_new = 0
for site in range(1, len(query)):
    active, n_new = branch_step(
        active, query[site], ref[:, site],
        rho_penalty=5.0, rho_c_penalty=0.01, mismatch_penalty=3.0
    )
    total_new += n_new

best = min(active, key=lambda s: s.penalty)
path = best.traceback()
print(f"Query: {query.tolist()}")
print(f"Viterbi path ({total_new} recombination events):")
for start, target in path:
    print(f"  From site {start}: copy haplotype {target}")
print(f"Final penalty: {best.penalty:.2f}")

```

The Bound Step (Theorem 2)

The bound step prunes the segment set without losing completeness.

Theorem 2. Let Ω be a complete segment set with active segments $\omega_1, \dots, \omega_N$. Define $\Omega^* \subseteq \Omega$ as the union of all traceback paths from the active segments: $\Omega^* = \bigcup_{n=1}^N P(\omega_n)$. Then Ω^* is also a complete segment set.

The bound step simply discards any segment that is not on a traceback path from an active segment – these segments are **undercut** and can never be part of an optimal path.

Threads applies the bound step at regular intervals using a heuristic threshold:

- Initialize with $B_0 = 10 \cdot N$
- If $|\Omega^{(m)}| > B_0$, prune to Ω^*
- If the next pruning trigger occurs within 30 sites, double the threshold to $B_1 = 2B_0$; otherwise reset to B_0

This balances pruning frequency against the risk of memory spikes from rapid segment accumulation.

```

def bound_step(all_segments, active_segments):
    """Prune segments not on any active traceback path.

    Parameters
    -----
    all_segments : list of Segment
        All segments accumulated so far.
    active_segments : list of Segment
        Currently active segments (one per haplotype).
    """

```

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```

Returns
-----
pruned : list of Segment
    Only segments reachable from active traceback paths.
"""
# Collect all segments on traceback paths from active segments
reachable = set()
for seg in active_segments:
    current = seg
    while current is not None and id(current) not in reachable:
        reachable.add(id(current))
        current = current.parent

pruned = [s for s in all_segments if id(s) in reachable]
return pruned

# Demonstrate pruning
print(f"\nBound step example:")
print(f" Before pruning: {len(active) + total_new} segments")
pruned = bound_step(active, active)
print(f" After pruning: {len(pruned)} segments")
print(" (Undercut segments are discarded)")

```

Traceback

After processing all M sites, the final Viterbi path is recovered by:

1. Identifying the active segment ω^* with minimum penalty
2. Following traceback pointers from ω^* until reaching a segment with start site 0

The result is a piecewise-constant path through the reference panel: a sequence of segments, each specifying a threading target and a genomic interval.

Parallelism

A critical property of the Threads-Viterbi algorithm: the N Viterbi instances (one per sample) are **completely independent**. The output of each HMM does not depend on any other. This means all N instances can run in parallel, divided evenly among available CPU cores.

Given L haplotype matches per sample and N_{CPU} cores:

- **Time:** $O(MLN/N_{\text{CPU}})$
- **Memory:** $O(LN)$ average

The genotype data is streamed from disk once per core, and neither the full genotype matrix nor any $N \times M$ probability matrix is ever stored in memory.

Complexity Summary

Property	Classical Viterbi	Threads-Viterbi
Time (per sample)	$O(NM)$	$O(NM)$ (same)
Memory (per sample)	$O(NM)$	$O(N)$ average
Total (all samples)	$O(MN^2)$ time + memory	$O(MLN/N_{\text{CPU}})$ time, $O(LN)$ memory
Parallelism	Not straightforward	Embarrassingly parallel

Dating Path Segments

The mechanism tells you which gear meshes where. Now calibrate the clock.

The third and final step in the Threads pipeline assigns **coalescence times** to each of the segments inferred by the Viterbi algorithm. These dated segments constitute the threading instructions needed to assemble the ARG.

The IBD Segment Model

Threads models each Viterbi segment as an **identical-by-descent (IBD) region** shared between the target sample and its inferred closest cousin. An IBD segment is a maximal genomic region where two sequences share a constant most recent common ancestor.

Each IBD segment has:

- A **length** l_{cM} in centimorgans and l_{bp} in base pairs
- A **height** (age) t , measured in generations, equal to the age of the most recent common ancestor
- A number of **heterozygous sites** m (mutations distinguishing the two copies)

The key modeling assumptions:

- Mutations follow a Poisson process with rate $\mu = 2 \cdot c \cdot l_{\text{bp}}$, where c is the per-base mutation rate
- Under the SMC model, the segment length follows an exponential distribution with rate $\rho = 2 \cdot 0.01 \cdot l_{\text{cM}}$
- Segments are independent of each other

Note

Viterbi segments do not always coincide with true IBD segments. The Viterbi path may both overestimate and underestimate segment lengths. Simulations show that coincidence with true IBD remains stable at approximately 40%, while overestimation tends to 0 as N increases. The net effect is a tendency to underestimate segment lengths and thus overestimate their ages.

Maximum Likelihood Estimators

We first derive estimators that ignore the coalescent prior, using only the recombination and mutation likelihoods.

From recombination only. Under the SMC model, the length of an IBD segment follows an exponential distribution with rate $1/(2t)$. The likelihood is:

$$\ell(\rho \mid t) = 2t e^{-t\rho}$$

To find the MLE, take the log-likelihood and differentiate:

$$\log \ell = \log(2t) - t\rho = \log 2 + \log t - t\rho$$

$$\frac{d \log \ell}{dt} = \frac{1}{t} - \rho = 0 \implies \hat{t} = \frac{1}{\rho}$$

Longer IBD segments imply more recent shared ancestry ($\hat{t}$ is small when ρ is large).

From recombination and mutations. Adding the Poisson mutation model with m heterozygous sites:

$$\ell(\rho, m \mid t, c) = 2t e^{-t\rho} \cdot \frac{(t\mu)^m}{m!} e^{-t\mu}$$

Taking the log-likelihood:

$$\log \ell = \log 2 + \log t - t\rho + m \log(t\mu) - t\mu - \log(m!)$$

Collecting the t -dependent terms: $(m+1) \log t - t(\rho + \mu) + \text{const}$. Differentiating:

$$\frac{d \log \ell}{dt} = \frac{m+1}{t} - (\rho + \mu) = 0 \implies \hat{t} = \frac{m+1}{\rho + \mu}$$

The numerator $m+1$ counts the m observed mutations plus the one “count” from the recombination boundary. The denominator $\rho + \mu$ is the total rate at which events (both mutations and recombination) accumulate with time. This is the natural estimator: total event count divided by total rate.

```
import numpy as np

def mle_age_recomb(rho):
    """MLE of segment age from recombination only.

    Parameters
    -----
    rho : float
        Recombination measure: 2 * 0.01 * 1_cM.

    Returns
    -----
    t_hat : float
        Maximum likelihood age estimate (in generations).
    """
    return 1.0 / rho

def mle_age_full(rho, mu, m):
    """MLE of segment age from recombination and mutations.

    Parameters
    -----
    rho : float
        Recombination measure.
    mu : float
        Mutation measure: 2 * c * 1_bp.
    m : int
        Number of heterozygous sites in the segment.

    Returns
    -----
    t_hat : float
        Maximum likelihood age estimate.
    """
```

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```

return (m + 1) / (rho + mu)

# Demonstrate MLE estimators
l_cM = 1.0      # 1 centimorgan segment
l_bp = 1e6      # ~1 Mb
c = 1.25e-8     # per-base mutation rate
rho = 2 * 0.01 * l_cM
mu = 2 * c * l_bp

print(f"Segment: {l_cM} cM, {l_bp/1e6:.0f} Mb")
print(f"  rho = {rho:.4f}, mu = {mu:.5f}")
t_recomb = mle_age_recomb(rho)
print(f"  MLE (recomb only): {t_recomb:.0f} generations")
for m in [0, 1, 3, 10]:
    t_full = mle_age_full(rho, mu, m)
    print(f"  MLE (recomb + {m} hets): {t_full:.0f} generations")

```

Bayesian Estimators

We now place an exponential prior $\pi(t) \sim \text{Exp}(\gamma)$ on the segment age, where γ is the coalescence rate. This prior models the coalescence process: under a constant population of size N_e , the coalescence rate is $\gamma = 1/N_e$.

From recombination only. The posterior is:

$$p(t \mid \rho) \propto \pi(t) p(\rho \mid t) = 2\gamma t e^{-t(\rho+\gamma)}$$

This is an Erlang-2 distribution with rate $\rho + \gamma$ and mean:

$$E[t \mid \rho] = \frac{2}{\rho + \gamma}$$

From recombination and mutations. Including the mutation likelihood:

$$p(t \mid \rho, m) \propto 2\gamma \frac{\mu^m}{m!} t^{m+1} e^{-t(\rho+\mu+\gamma)}$$

This is an Erlang- $(m+2)$ distribution with rate $\rho + \mu + \gamma$ and mean:

$$E[t \mid \rho, m] = \frac{m+2}{\rho + \mu + \gamma}$$

Note the difference from the maximum likelihood estimator: the numerator gains an extra count (from the prior), and the denominator includes the coalescence rate γ .

```

def bayes_age_recomb(rho, gamma):
    """Bayesian posterior mean age from recombination only.

    Uses an Exp(gamma) prior on age.

    Parameters
    -----
    rho : float
        Recombination measure.
    gamma : float
        Coalescence rate (1 / N_e).

```

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```

Returns
-----
t_hat : float
    Posterior mean age.
"""
return 2.0 / (rho + gamma)

def bayes_age_full(rho, mu, m, gamma):
    """Bayesian posterior mean age from recombination and mutations.

    Parameters
    -----
    rho : float
        Recombination measure.
    mu : float
        Mutation measure.
    m : int
        Number of heterozygous sites.
    gamma : float
        Coalescence rate.

    Returns
    -----
    t_hat : float
        Posterior mean age.
    """
    return (m + 2) / (rho + mu + gamma)

# Demonstrate Bayesian vs. MLE estimators
N_e = 10000
gamma = 1.0 / N_e
print(f"\nBayesian estimators (N_e = {N_e}):")
print(f"  gamma = {gamma:.6f}")
t_bayes_r = bayes_age_recomb(rho, gamma)
print(f"  Bayes (recomb only): {t_bayes_r:.0f} generations")
for m in [0, 1, 3, 10]:
    t_mle = mle_age_full(rho, mu, m)
    t_bayes = bayes_age_full(rho, mu, m, gamma)
    print(f"    m={m:2d}: MLE = {t_mle:6.0f}, "
          f"      Bayes = {t_bayes:6.0f} generations")
print("(Prior pulls estimates toward the coalescent expectation)")

```

Piecewise-Constant Demographic Models

Real populations do not have constant effective size. `Threads` accommodates **piecewise-constant demographic models** where the effective population size $N_e(t)$ changes at discrete time boundaries.

Define time intervals $0 = T_0 < T_1 < \dots < T_K = \infty$ with constant effective sizes $N_e^{(k)}$ and coalescence rates $\gamma_k = 1/N_e^{(k)}$ in each interval $[T_k, T_{k+1})$. Write $\Delta_k = T_{k+1} - T_k$.

The prior becomes piecewise exponential:

$$\pi(t) = \gamma_k \exp \left(- \sum_{j=0}^{k-1} \Delta_j \gamma_j - (t - T_k) \gamma_k \right) \quad \text{for } t \in [T_k, T_{k+1})$$

Recombination-only estimator. The posterior expectation is:

$$E[t \mid \rho] = \frac{\sum_{k=0}^{K-1} \gamma_k e^{-\sum_{j=0}^{k-1} \Delta_j \gamma_j + T_k \gamma_k} \cdot \frac{2}{(\rho + \gamma_k)^3} [P(3, (\rho + \gamma_k) T_{k+1}) - P(3, (\rho + \gamma_k) T_k)]}{\sum_{k=0}^{K-1} \gamma_k e^{-\sum_{j=0}^{k-1} \Delta_j \gamma_j + T_k \gamma_k} \cdot \frac{1}{(\rho + \gamma_k)^2} [P(2, (\rho + \gamma_k) T_{k+1}) - P(2, (\rho + \gamma_k) T_k)]}$$

where $P(a, z) = \gamma(a, z)/\Gamma(a)$ is the regularized lower incomplete gamma function.

This estimator is used for inference from sparse data such as genotyping arrays, where mutation information is unreliable.

Full estimator with mutations. Adding m heterozygous sites and writing $\lambda_k = \rho + \mu + \gamma_k$:

$$E[t \mid \rho, m] = \frac{\sum_{k=0}^{K-1} \gamma_k e^{-\sum_{j=0}^{k-1} \Delta_j \gamma_j + T_k \gamma_k} \cdot \frac{\mu^m \cdot (m+2)}{\lambda_k^{m+3}} [P(m+3, \lambda_k T_{k+1}) - P(m+3, \lambda_k T_k)]}{\sum_{k=0}^{K-1} \gamma_k e^{-\sum_{j=0}^{k-1} \Delta_j \gamma_j + T_k \gamma_k} \cdot \frac{\mu^m}{\lambda_k^{m+2}} [P(m+2, \lambda_k T_{k+1}) - P(m+2, \lambda_k T_k)]}$$

This is the estimator used by Threads for coalescence time inference in whole-genome sequencing data. The piecewise-constant demographic model is specified as a `.demo` file listing the time boundaries and effective population sizes.

```
from scipy.special import gammainc # regularized lower incomplete gamma

def bayes_age_piecewise(rho, mu, m, T_bounds, N_e_values):
    """Bayesian posterior mean age under a piecewise-constant demography.

    Parameters
    -----
    rho : float
        Recombination measure.
    mu : float
        Mutation measure.
    m : int
        Number of heterozygous sites.
    T_bounds : list of float
        Time interval boundaries [T_0, T_1, ..., T_K].
    N_e_values : list of float
        Effective population sizes [N_e^(0), ..., N_e^(K-1)].

    Returns
    -----
    t_hat : float
        Posterior mean age.
    """
    K = len(N_e_values)
    gamma_k = [1.0 / N for N in N_e_values]
    lam_k = [rho + mu + g for g in gamma_k]

    # Cumulative coalescent hazard up to each boundary
    cum_hazard = [0.0]
    for k in range(K):
        delta = T_bounds[k+1] - T_bounds[k]
```

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```

        cum_hazard.append(cum_hazard[-1] + delta * gamma_k[k])

numerator = 0.0
denominator = 0.0

for k in range(K):
    g_k = gamma_k[k]
    l_k = lam_k[k]
    # Weight: gamma_k * exp(-cum_hazard_k + T_k * gamma_k)
    log_weight = (np.log(g_k) - cum_hazard[k]
                  + T_bounds[k] * g_k)
    weight = np.exp(log_weight)

    # Regularized incomplete gamma differences
    a_num = m + 3
    a_den = m + 2
    z_lo = l_k * T_bounds[k]
    z_hi = l_k * T_bounds[k+1]
    # P(a, z) = gammainc(a, z) (scipy uses regularized form)
    P_num = gammainc(a_num, z_hi) - gammainc(a_num, z_lo)
    P_den = gammainc(a_den, z_hi) - gammainc(a_den, z_lo)

    numerator += weight * (m + 2) / l_k**(m+3) * P_num
    denominator += weight / l_k**(m+2) * P_den

if denominator == 0:
    return bayes_age_full(rho, mu, m, gamma_k[0])

return numerator / denominator

# Demonstrate with a bottleneck demography
# Recent: N_e=10000 (0-1000 gen), Bottleneck: N_e=1000 (1000-2000),
# Ancient: N_e=20000 (2000+)
T_bounds = [0, 1000, 2000, 1e8]
N_e_values = [10000, 1000, 20000]

print("\nPiecewise-constant demography:")
for k in range(len(N_e_values)):
    print(f" [{T_bounds[k]:.0f}, {T_bounds[k+1]:.0f}): "
          f"N_e = {N_e_values[k]}")

print(f"\nSegment: {l_cm} cM, {m} hets")
for m in [0, 1, 5]:
    t_const = bayes_age_full(rho, mu, m, 1.0/10000)
    t_pw = bayes_age_pieewise(rho, mu, m, T_bounds, N_e_values)
    print(f" m={m}: constant N_e -> {t_const:.0f}, "
          f"pieewise -> {t_pw:.0f} generations")

```

Demo: Running Threads on Simulated Data

Running mini_threads on simulated genomic data.

Demo: Threads Dating on msprime-simulated IBD Segments (20 samples, 1 Mb)

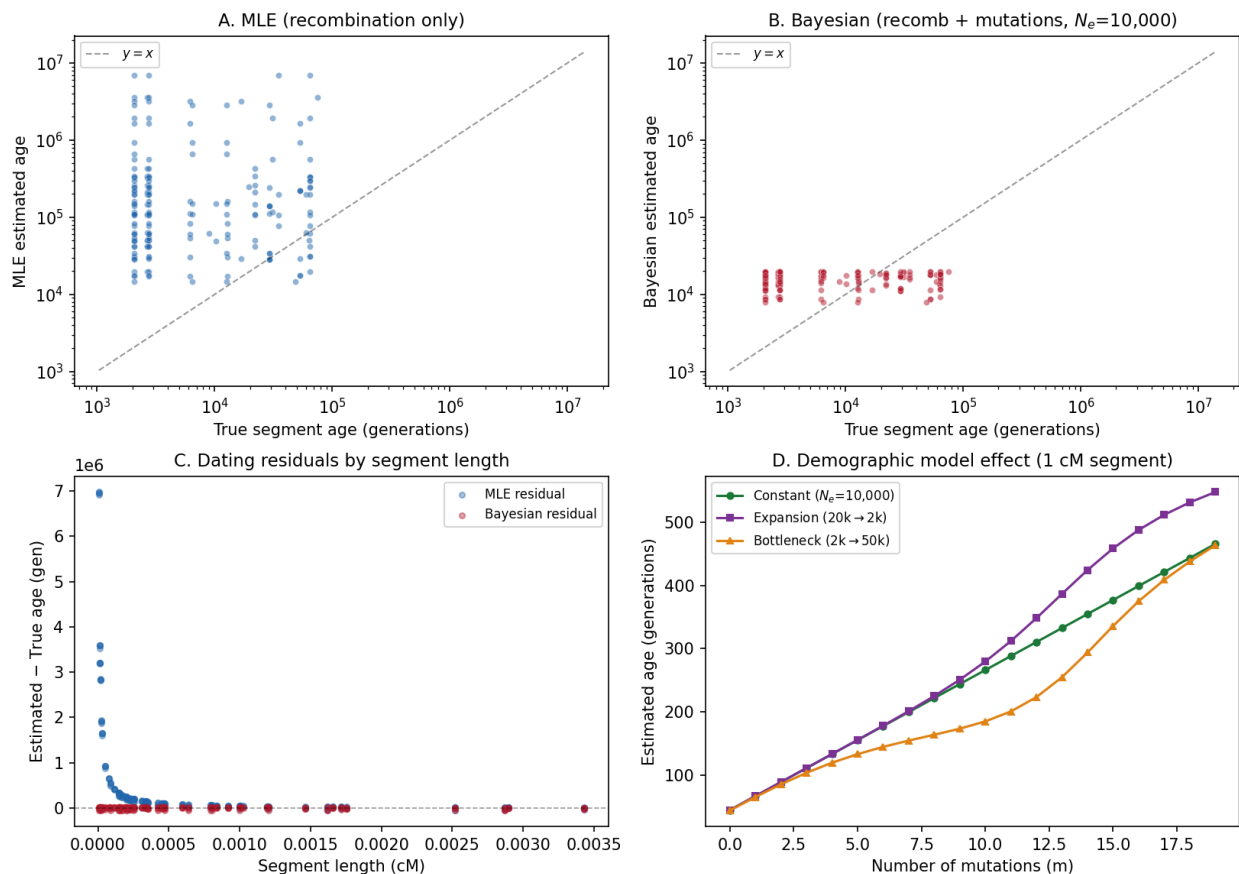


Fig. 9: **Threads demo on msprime-simulated data.** Panel A: Segment age estimates using recombination-only MLE on IBD-like segments extracted from the true tree sequence. Panel B: Bayesian age posteriors incorporating mutation information. Panel C: Full Bayesian estimates with both recombination and mutation evidence. Panel D: Piecewise-constant Bayesian estimates under a demographic model.

This figure was generated by running the `mini_threads` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_threads.py`.

3.10 Timepiece IX: tsdate

Dating Nodes in a Tree Sequence

3.10.1 The Mechanism at a Glance

tsdate is a Bayesian method for estimating the **age of every ancestral node** in a tree sequence. Given a genealogy (typically the topology from `tsinfer`, which has no meaningful branch lengths), **tsdate** uses the **molecular clock** – the principle that mutations accumulate proportionally to time – to infer *when* each ancestor lived.

If `tsinfer` gives you the skeleton of a watch movement – the shapes and connections of the gears – **tsdate** adds the calibration marks: real time estimates that tell you not just *how* the parts connect, but *when* each part was made. Without **tsdate**, you

know the topology (which ancestors are connected to which descendants); with tsdate, you know the chronology (how many generations separate them).

The core equation is Bayes' rule (introduced in the *HMMs prerequisite*):

$$P(\mathbf{t} \mid \mathbf{D}, \mathcal{T}) \propto P(\mathbf{D} \mid \mathbf{t}, \mathcal{T}) \cdot P(\mathbf{t} \mid \mathcal{T})$$

where $\mathbf{t}$ is the vector of node ages, $\mathbf{D}$ is the mutational data (how many mutations sit on each edge), and $\mathcal{T}$ is the tree topology. In words: the probability of the node ages given the data is proportional to the probability of the data given the ages (the **likelihood**) times the prior probability of those ages (the **prior**).

i Primary Reference

[Wohns *et al.*, 2022]

The four gears of tsdate:

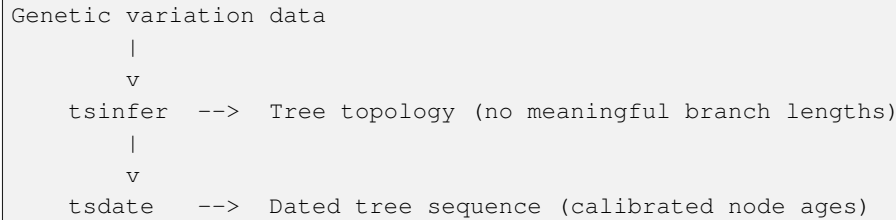
1. **The Coalescent Prior** (the escapement) – A prior on node ages derived from coalescent theory: nodes with more descendant samples are expected to be younger, because large subtrees coalesce quickly. This is where the theory from *Coalescent Theory* provides the baseline expectation.
2. **The Mutation Likelihood** (the gear train) – A Poisson model for the number of mutations on each edge, connecting observed data to branch lengths. More mutations on an edge = longer branch = greater time separation between parent and child.
3. **Belief Propagation** (the mainspring) – Message-passing algorithms that combine prior and likelihood across the interconnected nodes of the tree sequence. Two flavors:
 - *Inside-Outside* (discrete time grid) – exact on a grid
 - *Variational Gamma* (continuous time, default) – approximate but efficient
4. **Rescaling** (the regulator) – A post-processing step that adjusts node times so the inferred mutation rate matches the empirical rate across time windows. Like a watchmaker adjusting the beat rate against a reference frequency.

These gears mesh together into a dating pipeline:

```
Tree sequence from tsinfer (topology only, no meaningful times)
      |
      v
+---> Construct coalescent prior for each node
      |
      |
      |      |
      |      v
      |      Compute mutation likelihood for each edge
      |      |
      |      v
      |      Run belief propagation (EP or inside-outside)
      |      |
      |      v
      |      Rescale times to match mutation clock
      |
      +-----+
              |
              v
              (iterate)
              |
              v
Dated tree sequence (node ages = posterior means)
```


3.10.2 Where tsinfer Ends and tsdate Begins

The standard pipeline for scalable genealogical inference is:



tsinfer (Timepiece VI) gives you the *skeleton of the movement* – the frame on which everything hangs: which samples share which ancestors, and the tree structure at each position. But the “times” it assigns are just a frequency-based proxy (older ancestors have higher-frequency derived alleles). These are ordinal, not quantitative: they tell you ancestor A is older than B, but not by how much.

tsdate adds the *calibration* – real time estimates based on the molecular clock. Every edge in the tree can carry mutations, and the number of mutations is informative about the edge’s length. By combining this mutation signal with a prior from coalescent theory, tsdate computes posterior estimates for every node’s age.

Together, tsinfer + tsdate form a complete inference pipeline: one provides the structure, the other provides the timing. Like a watch case and a calibrated dial – both essential, each useless without the other.

Prerequisites for this Timepiece

- *Coalescent Theory* – coalescence times and the Poisson mutation model
- *Ancestral Recombination Graphs* – tree sequences
- *Hidden Markov Models* – for understanding belief propagation
- Familiarity with *tsinfer* (Timepiece VI) is helpful but not strictly required

3.10.3 Chapters

Overview of tsdate

Before assembling the watch, lay out every part and understand what it does.

If tsinfer reconstructs the hidden *topology* of the ancestral recombination graph – the arrangement of gears and springs inside the case – then tsdate **adds calibration marks to the movement**, converting a bare mechanism into a working timepiece. Where tsinfer answers “who is related to whom?”, tsdate answers “when?”

This chapter lays out all the parts before we begin assembly. By the end you will know what tsdate computes, why it frames the problem as Bayesian inference, and how its three dating methods relate to one another. Subsequent chapters build each gear from scratch.

What Does tsdate Do?

Given a tree sequence $\mathcal{T}$ with known topology but unknown node ages, and observed mutational data $\mathbf{D}$, tsdate computes **posterior estimates** for the age of every ancestral node.

Input: $\mathcal{T} = (\mathcal{N}, \mathcal{E})$ (tree topology: nodes + edges)

Input: $\mathbf{D} = \{m_e\}_{e \in \mathcal{E}}$ (mutation counts per edge)

Output: $\hat{t}_u = \mathbb{E}[t_u \mid \mathbf{D}, \mathcal{T}]$ for each non-leaf node u

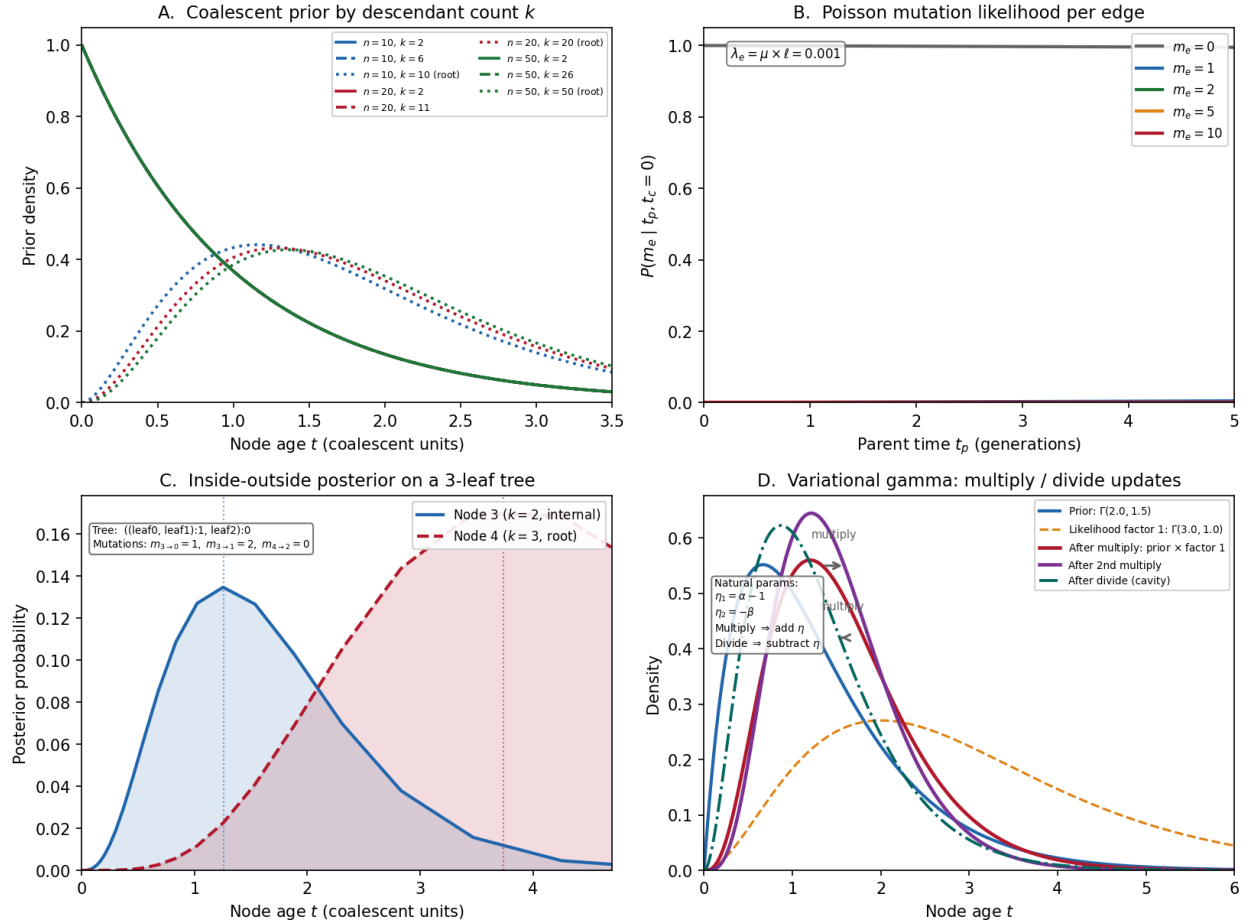
tsdate: Bayesian Node Dating for Tree Sequences

Fig. 10: **tsdate at a glance**. Panel A: Coalescent prior – conditional coalescent moments for different sample sizes, with gamma priors on node age shaped by descendant count. Panel B: Edge likelihood – Poisson mutation likelihood as a function of parent time for edges with varying mutation counts. Panel C: Inside-outside posterior distributions over a time grid for nodes with different roles in a small tree. Panel D: Variational gamma – the multiply/divide operations on gamma distributions showing prior-to-posterior evolution.

The output is a dated tree sequence where every node has a calibrated age (in generations or years), and mutations are placed at the midpoint of their parent edge.

From these dated genealogies, you can extract:

- **Pairwise coalescence times** between any two samples at any position
- **Allele ages** – when each mutation arose
- **Effective population size** through time
- **Selection signatures** from distorted branch length distributions

i Prerequisite: tsinfer provides the topology

tsdate does *not* infer the tree structure. It expects a tree sequence whose topology has already been reconstructed – typically by tsinfer (see [Pipeline and evidence boundary](#) for how that works). Think of it this way: tsinfer assembles the gear train, and tsdate etches the hour markers onto the dial. If the topology is wrong, the calibration marks will be wrong too, so a high-quality tsinfer run is the prerequisite for accurate dating.

The Bayesian Framework

tsdate formulates node dating as a Bayesian inference problem. Let's build this up piece by piece.

The unknowns

We have a tree sequence with N non-leaf nodes. Each node u has an unknown age $t_u \geq 0$. The vector of all unknown ages is:

$$\mathbf{t} = (t_1, t_2, \dots, t_N)$$

The constraint is that every parent must be older than every child: if edge $e = (u, v)$ connects parent u to child v , then $t_u > t_v$.

The data

On each edge e of the tree sequence, we observe m_e mutations. An edge spans a genomic interval of length ℓ_e base pairs. Under the molecular clock, the expected number of mutations on edge e is:

$$\mathbb{E}[m_e] = \mu \cdot \ell_e \cdot (t_{\text{parent}(e)} - t_{\text{child}(e)})$$

where μ is the per-base-pair, per-generation mutation rate.

The posterior

Bayes' rule gives us:

$$P(\mathbf{t} \mid \mathbf{D}, \mathcal{T}) \propto \underbrace{P(\mathbf{D} \mid \mathbf{t}, \mathcal{T})}_{\text{likelihood}} \cdot \underbrace{P(\mathbf{t} \mid \mathcal{T})}_{\text{prior}}$$

In watch terms, we are combining two sources of information to set the hands:

- The **likelihood** is **evidence from the mutation clock** – the ticking of the molecular metronome on each branch. It decomposes over edges: each edge contributes independently via a Poisson model (Gear 2: [The Mutation Likelihood](#)).

- The **prior** is the **expected beat rate from coalescent theory** – what population genetics tells us about how old a node *should* be before we look at any mutations. It decomposes over nodes: each node's age has a prior derived from the coalescent, conditioned on how many samples descend from it (Gear 1: *The Coalescent Prior*).

i Probability Aside – What “posterior” means here

If you have not encountered Bayesian inference before, the core idea is simple. We start with a *prior belief* about a quantity (here, how old each node is, based on coalescent theory). Then we observe data (mutation counts) and update our belief using Bayes' rule. The result – the *posterior* – balances prior knowledge with the evidence. When the data are strong (many mutations on an edge), the posterior is driven by the likelihood; when data are sparse (few or zero mutations), the posterior leans on the prior.

The challenge is computing the posterior. The node ages are coupled: the age of a parent constrains the ages of its children, and vice versa. This creates a complex, high-dimensional posterior that cannot be computed in closed form.

tsdate uses **message-passing algorithms** to approximate this posterior. Imagine **messages flowing through the gear train**: each gear (node) sends information to its neighbors about what time it thinks it is, and receives information back. After enough rounds of message exchange, every gear settles into a self-consistent position.

- **Inside-Outside** (*Inside-Outside Belief Propagation*): Discretize time into a grid, represent each node's marginal posterior as a probability vector, propagate messages up and down the tree.
- **Variational Gamma** (*Variational Gamma (Expectation Propagation)*): Approximate each node's marginal posterior as a gamma distribution, refine via expectation propagation (moment matching).

Both methods are **linear in the number of edges**, making them scalable to tree sequences with millions of nodes.

i Probability Aside – What is belief propagation?

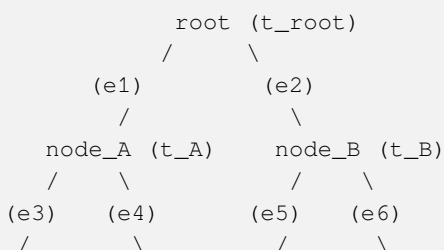
Belief propagation (BP) is an algorithm for computing marginal distributions in a graphical model. The idea: each variable “asks” its neighbors what they believe, incorporates the answers, and “tells” its neighbors its updated belief. On a tree-shaped graph, two passes (one up, one down) suffice for exact answers. On a graph with loops – like a tree *sequence*, where nodes are shared across local trees – BP becomes *loopy BP*, an approximation that works well in practice. The inside-outside and variational gamma methods are both forms of BP adapted to the tree sequence factor graph.

A Graphical Model Perspective

It helps to think of the tree sequence as a **factor graph**. This is a bivariate graphical model with two types of nodes:

- **Variable nodes** (circles): the unknown ages t_u
- **Factor nodes** (squares): the constraints and likelihoods connecting them

Example: a simple tree with 3 leaves and 2 internal nodes



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```

leaf_1  leaf_2  leaf_3  leaf_4

Factor graph:

[prior_root] --- (t_root) --- [lik_e1] --- (t_A) --- [lik_e3] --- (t_1=0)
                |                               |
                [lik_e2]                       [lik_e4] --- (t_2=0)
                |                               |
                (t_B) --- [lik_e5] --- (t_3=0)
                |                               |
                [lik_e6] --- (t_4=0)

```

Each factor encodes either:

- A **prior** on a node's age (from coalescent theory)
- A **likelihood** for an edge (Poisson model for mutations)

Message passing on this graph computes approximate marginal posteriors for each variable node. The messages that flow along each edge of the factor graph are the “impulses traveling through the gear train” – each factor tells its neighboring variables what values are consistent with the local data, and each variable aggregates those messages into a posterior belief.

Why “approximate”? In a tree, belief propagation gives exact marginals. But a tree *sequence* is not a tree – it's a graph with shared nodes across multiple local trees, creating **loops**. When the graph has loops, belief propagation becomes **loopy belief propagation**, which converges to an approximation.

Terminology

Before diving into the gears, let's nail down the terminology precisely.

Term	Definition
Node age t_u	The time (in generations) at which node u existed
Edge $e = (u, v)$	A parent-child relationship over a genomic interval $[\ell, r)$
Edge span ℓ_e	The genomic length $r - \ell$ of the interval
Edge mutations m_e	The number of mutations observed on edge e
Branch length Δt_e	The time duration $t_{\text{parent}} - t_{\text{child}}$ of edge e
Mutation rate μ	Mutations per base pair per generation
Span-weighted rate λ_e	$\mu \cdot \ell_e$, the expected mutations per unit time on edge e
Posterior mean	$\hat{t}_u = \mathbb{E}[t_u \mid \mathbf{D}]$, the point estimate for node age
Posterior variance	$\text{Var}(t_u \mid \mathbf{D})$, the uncertainty in the age estimate
Factor	A local function in the factor graph (prior or likelihood)
Message	Information passed between connected nodes in belief propagation
Gamma approximation	$q(t_u) = \text{Gamma}(\alpha_u, \beta_u)$, the variational posterior

Parameters

tsdate takes the following parameters:

Parameter	Symbol	Meaning
Mutation rate	μ	Per-base-pair, per-generation mutation rate
Population size	N_e	Effective population size (for discrete methods; sets the timescale of the prior)
Method	–	"variational_gamma" (default), "inside_outside", or "maximization"
Max iterations	–	Number of EP iterations for variational gamma (default: 25)
Rescaling intervals	J	Number of time windows for rescaling (default: 1000)
Time units	–	"generations" or "years"

i The mutation rate is the key input

Unlike tsinfer (which mainly needs topology), tsdate critically depends on having a good estimate of μ . This single number sets the absolute timescale of the entire genealogy. Get it wrong, and all your dates shift proportionally. For humans, $\mu \approx 1.29 \times 10^{-8}$ per bp per generation is a commonly used value.

The Three Methods

tsdate offers three dating methods. Let's preview each before diving deep in later chapters. Each method is a different way of propagating messages through the gear train; they differ in how they represent the "state" of each gear (discrete grid vs. continuous distribution) and how they combine messages.

Method 1: Inside-Outside (discrete time)

- Discretize time into a grid of K timepoints
- Each node gets a probability vector $\in \mathbb{R}^K$
- Run **inside** (upward) pass: propagate likelihoods from leaves to roots
- Run **outside** (downward) pass: propagate from roots back to leaves
- Combine inside and outside to get marginal posteriors

This is classic belief propagation adapted to the tree sequence graph. It uses a **conditional coalescent prior** specific to each node.

Pros: Conceptually clean, produces full posterior distributions. **Cons:** Resolution limited by grid; $O(K^2)$ per edge for joint parent-child distributions.

Method 2: Variational Gamma (continuous time, default)

- Approximate each node's posterior as $\text{Gamma}(\alpha, \beta)$
- Use **Expectation Propagation** (EP) to iteratively refine α, β
- Each EP iteration processes every edge, updating the gamma parameters via moment matching
- Converges to a fixed point that approximately minimizes the KL divergence between the true posterior and the gamma approximation

i Probability Aside – What is Expectation Propagation?

Expectation Propagation (EP) is an iterative algorithm for approximate Bayesian inference introduced by Tom Minka in 2001. The core idea: when the true posterior is a product of many factors that are individually tractable but jointly intractable, EP processes one factor at a time, replacing it with a simpler “approximate factor” chosen so that the overall approximation matches the *moments* of the true distribution. This “moment matching” step is what makes EP different from variational Bayes (which minimizes a different objective). We build EP from scratch in [Variational Gamma \(Expectation Propagation\)](#).

Pros: Continuous time (no grid), more accurate, faster convergence. **Cons:** Gamma family may not capture all posterior shapes (e.g., multimodal).

Method 3: Maximization (discrete time)

- Same grid as inside-outside
- But produces **MAP point estimates** instead of full posteriors
- Propagates maxima instead of sums
- More numerically stable than inside-outside

Pros: Robust. **Cons:** No uncertainty estimates; less accurate.

The Flow in Detail

Here’s a more detailed view of how the pieces connect for the default variational gamma method:

```
Tree sequence T (from tsinfer)
  |
  v
+-----+
| PREPROCESSING |
|               |
| - Remove unary nodes |
| - Split disjoint nodes |
| - Identify fixed nodes |
|   (samples at time 0) |
+-----+
  |
  v
+-----+
| INITIALIZE |
|           |
| For each node u: |
|   prior ~ Gamma(a0, b0) |
|   (from conditional |
|     coalescent, or |
|     exponential for roots) |
|           |
| For each edge e: |
|   count mutations m_e |
|   compute span l_e |
+-----+
  |
  v
```

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```

+-----+
| EXPECTATION PROPAGATION |
| (iterate max_iter times) |
|
| For each edge e=(u,v): |
|   1. Remove old message |
|     from q(t_u), q(t_v) |
|   2. Compute new message |
|     from Poisson lik    |
|   3. Moment-match to    |
|     update q(t_u), q(t_v) |
|   4. Damp if needed     |
+-----+
|
| v
+-----+
| RESCALING |
|
| Partition time into J |
| windows of equal branch |
| length. In each window: |
|   scale = observed_muts |
|           / expected_muts |
| Apply piecewise scaling |
| to node ages           |
+-----+
|
| v
+-----+
| OUTPUT |
|
| For each node: |
|   time = posterior mean |
| For each mutation: |
|   time = midpoint of edge |
+-----+
|
| v
Dated tree sequence

```

Why Gamma Distributions?

The variational gamma method approximates each node's posterior age as $\text{Gamma}(\alpha, \beta)$. Why this choice?

1. **Support:** Node ages are positive reals $t \geq 0$. The gamma family is defined on $[0, \infty)$ – a natural match.
2. **Flexibility:** By varying α and β , gamma distributions can be peaked near zero (exponential-like), symmetric (bell-shaped), or right-skewed.
3. **Conjugacy:** Under the Poisson likelihood with known rate, the gamma distribution is the conjugate prior. This means certain updates have closed-form solutions.
4. **Two parameters:** Each node's posterior is summarized by just two numbers (α, β) . For a tree sequence with N nodes, this means $2N$ parameters total – compared to NK for a grid-based method with K timepoints.

The mean and variance of $\text{Gamma}(\alpha, \beta)$ are:

$$\mathbb{E}[t] = \frac{\alpha}{\beta}, \quad \text{Var}(t) = \frac{\alpha}{\beta^2}$$

So α controls the “shape” (peakedness) and β controls the “rate” (how quickly the density decays).

Probability Aside – The Gamma distribution in 60 seconds

The $\text{Gamma}(\alpha, \beta)$ distribution is a continuous probability distribution on $[0, \infty)$ with two parameters: **shape** $\alpha > 0$ and **rate** $\beta > 0$. Its density is $p(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}$. When $\alpha = 1$ it reduces to an $\text{Exponential}(\beta)$. When α is large, it looks bell-shaped, centered near α/β . The name $\Gamma(\alpha)$ in the normalizing constant is the *gamma function*, a continuous generalization of the factorial: $\Gamma(n) = (n-1)!$ for positive integers. In tsdate, gamma distributions are everywhere because they naturally model positive quantities (ages) and play well with Poisson likelihoods (conjugacy).

Why Linear Scaling?

A critical feature of tsdate is that it scales **linearly** in the number of edges. This is what makes it applicable to tree sequences with millions of nodes.

The key insight: belief propagation processes each edge exactly once per iteration. For a tree sequence with E edges and I iterations, the total cost is $O(E \cdot I)$.

Compare this to:

- **MCMC methods** (SINGER, ARGweaver): $O(n^2)$ per iteration, where n is the number of samples
- **Brute-force Bayesian**: $O(K^N)$ for K grid points and N nodes (intractable)

tsdate exploits the **sparse structure** of the tree sequence: each node connects to only a few edges (its parents and children), so local updates are cheap.

Ready to Build

You now have the high-level blueprint. In the following chapters, we'll build each gear from scratch:

1. *The Coalescent Prior* – The prior on node ages (the expected beat rate from coalescent theory)
2. *The Mutation Likelihood* – The Poisson clock model (evidence from the mutation clock)
3. *Inside-Outside Belief Propagation* – Discrete-time belief propagation (messages flowing through the gear train on a grid)
4. *Variational Gamma (Expectation Propagation)* – Continuous-time expectation propagation (the same messages, now carried by gamma distributions)
5. *Rescaling* – Calibrating the clock (adjusting for population size history)

Each chapter derives the math, explains the intuition, implements the code, and verifies it works.

Let's start with the first gear: the Coalescent Prior.

The Coalescent Prior

A watchmaker who knows how clocks age can guess a part's vintage before inspecting it.

Before looking at any mutational data, we already know something about how old each node should be. A node with 1000 descendant samples is almost certainly much younger than a node with only 2. This knowledge comes from **coalescent theory**, and tsdate encodes it as a **prior distribution** on each node's age.

In the watch metaphor, the prior is **the expected beat rate from coalescent theory** – our best guess for when each gear was manufactured, before we open the case and inspect the wear marks (mutations). A node with many descendants is like a mass-produced part: it was probably made recently. A node ancestral to only two samples is like a rare vintage component: it could be quite old.

This chapter builds the coalescent prior from first principles. By the end you will be able to assign a $\text{Gamma}(\alpha, \beta)$ distribution to every internal node of a tree sequence, ready to combine with the mutation likelihood in the chapters that follow.

i Prerequisites

This chapter assumes familiarity with the standard coalescent model (covered in the population genetics fundamentals). You should also understand why tsdate needs a tree sequence with known topology – that topology comes from tsinfer (see *Pipeline and evidence boundary*).

The Intuition: More Descendants = Younger

Under the standard coalescent, a node with k descendant leaves in a sample of n coalesces at a time that depends on k and n . The key intuition:

- A node that is ancestral to *all* n samples (the root) has had $n - 1$ coalescence events below it. It must be old.
- A node that is ancestral to just $k = 2$ samples only needs one coalescence event. It can be young.

More precisely, under the standard coalescent with constant population size N_e , the expected time for j lineages to coalesce to $j - 1$ is:

$$\mathbb{E}[T_j] = \frac{2N_e}{j(j-1)/2} = \frac{4N_e}{j(j-1)}$$

The total time from n lineages down to 1 is the sum of waiting times, and a node ancestral to k leaves enters the picture somewhere in this process.

i Probability Aside – Why $j(j-1)/2$?

When there are j lineages in the population, any pair can coalesce. The number of possible pairs is $\binom{j}{2} = j(j-1)/2$. Each pair coalesces independently at rate $1/(2N_e)$, so the total coalescence rate is $\binom{j}{2}/(2N_e)$. The waiting time until the next coalescence is Exponential with this rate, giving the mean $4N_e/(j(j-1))$. As j grows, there are many more pairs, coalescence is faster, and the waiting time is shorter.

The Conditional Coalescent

tsdate uses the **conditional coalescent** (Wiuf & Donnelly, 1999) to derive the prior. The question is:

Given a tree with n total leaves, what is the distribution of the age of a node that is ancestral to exactly k of those leaves?

This is not a simple closed-form expression. It requires integrating over the possible number of **extant ancestors** a – the number of lineages that exist at the time this particular subtree coalesces.

With the intuition established, let us now work through the mathematics of the conditional coalescent. The key difficulty is that when a subtree of size k finishes coalescing, the number of remaining lineages in the rest of the tree is random.

The mean and variance

The conditional coalescent gives us $\mathbb{E}[t \mid k, n]$ and $\text{Var}(t \mid k, n)$. These are computed by marginalizing over the number of ancestors.

When a subtree of size k coalesces (going back in time from the present), there are a total lineages remaining. The probability of having a ancestors given k and n follows a hypergeometric-like distribution (Wiuf & Donnelly, 1999), and the coalescence time conditioned on a is:

$$T \mid a \sim \text{Exp}\left(\frac{a(a-1)}{4N_e}\right)$$

So the conditional mean is:

$$\mathbb{E}[T \mid k, n] = \sum_{a=2}^{n-k+1} P(a \mid k, n) \cdot \frac{4N_e}{a(a-1)}$$

and the conditional variance includes both the variance within each a class and the variance between classes (law of total variance).

i Probability Aside – The law of total variance

The law of total variance (sometimes called Eve's law) says that for any two random variables X and Y :

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y])$$

In our case, X is the coalescence time T and Y is the number of ancestors a . The first term captures the randomness *within* each value of a (the exponential waiting time), and the second term captures the randomness *between* values of a (different numbers of lineages lead to different expected times). This decomposition is how tsdate computes the variance of the conditional coalescent without needing the full distribution.

```
import numpy as np
from scipy.special import comb

def conditional_coalescent_mean(k, n, Ne=1.0):
    """Mean age of a node with k descendants in a sample of n.

    Under the conditional coalescent (Wiuf & Donnelly, 1999), averaged
    over the number of extant ancestors.

    Parameters
    -----
    k : int
        Number of descendant leaves of this node.
    n : int
        Total number of leaves in the tree.
    Ne : float
        Effective population size (in coalescent units, 2*Ne generations).

    Returns
    -----
    mean : float
        Expected age in units of 2*Ne generations.
    """
```

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```

if k == n:
    # The root: must wait for all n lineages to coalesce
    # Mean is sum of 1/(j choose 2) for j = n down to 2
    return sum(2.0 / (j * (j - 1)) for j in range(2, n + 1))

# P(a ancestors | k descendants coalesce, n total tips)
# computed recursively
mean = 0.0
for a in range(2, n - k + 2):
    # Probability of a ancestors when subtree of size k merges
    p_a = _pr_ancestors(a, k, n)
    # Expected coalescence time given a lineages
    expected_time = 2.0 / (a * (a - 1))
    mean += p_a * expected_time

return mean

def _pr_ancestors(a, k, n):
    """Probability of a extant ancestors when subtree of size k coalesces.

    This follows Wiuf & Donnelly (1999). For a subtree of size k in a
    tree of n tips, the number of other lineages when k coalesces to 1
    ranges from 1 to n-k. So total ancestors a ranges from 2 to n-k+1.
    """
    if k == 2:
        # Special case: the pair coalesces when a-1 other lineages exist
        # at that time, so a total. This has a known distribution.
        # P(a | k=2, n) = (n-1) * C(n-2, a-2) * C(a-1, 1)
        # / (C(n, 2) * product terms)
        # ... simplified via the recursion in Wiuf & Donnelly
        pass
    # In practice, tsdate computes this recursively using the relationship:
    # P(a | k, n) can be computed from P(a | k+1, n) using
    # binomial coefficient identities.
    # For educational purposes, here's a direct simulation approach:
    raise NotImplementedError(
        "See the recursive implementation below for the full computation."
    )

```

The Recursive Computation

The key to computing $P(a | k, n)$ efficiently is a **recursive relationship** over decreasing k . tsdate's implementation uses the identity:

$$P(a | k, n) = \sum_{a'=a}^{n-k+1} P(a' | k+1, n) \cdot \frac{\binom{a'-1}{1}}{\binom{a'+1}{2}}$$

The base case is $k = n - 1$, where the subtree is the second-to-last to coalesce, and there are exactly $a = 2$ ancestors.

In practice, tsdate precomputes a lookup table of (mean, variance) indexed by k (number of descendants), for a given n (total tips).

Now let us translate this recursion into code. The implementation walks backward from $k = n - 1$ down to $k = 2$, building the probability table one row at a time.

```

def conditional_coalescent_moments(n, Ne=1.0):
    """Compute mean and variance of node age for all possible descendant counts.

    Parameters
    -----
    n : int
        Total number of tips.
    Ne : float
        Effective population size.

    Returns
    -----
    moments : dict
        {k: (mean, variance)} for k = 2, 3, ..., n.
    """
    # Precompute unconditional coalescence time moments for a lineages
    #  $E[T | a] = 2/(a*(a-1))$ ,  $Var[T | a] = E[T|a]^2 = 4/(a*(a-1))^2$ 
    max_a = n
    t_mean = np.zeros(max_a + 1)  # t_mean[a] = expected coalescence time given a
    ↪ lineages
    t_var = np.zeros(max_a + 1)  # t_var[a] = variance of coalescence time given a
    ↪ lineages
    for a in range(2, max_a + 1):
        rate = a * (a - 1) / 2.0  # coalescence rate with a lineages (num pairs)
        t_mean[a] = 1.0 / rate  # exponential mean = 1/rate
        t_var[a] = 1.0 / rate**2  # exponential variance = 1/rate^2

    # Build P(a | k, n) table recursively from k=n-1 down to k=2
    # Start: when k = n-1, there must be a=2 ancestors (only 2 lineages left)
    pr_a = {}
    pr_a[n-1] = np.zeros(max_a + 1)
    pr_a[n-1][2] = 1.0  # certain: exactly 2 ancestors

    for k in range(n - 2, 1, -1):
        pr_a[k] = np.zeros(max_a + 1)
        for a in range(2, n - k + 2):
            # Recursive formula from Wiuf & Donnelly
            for a_prime in range(a, n - k + 1):
                # Transition probability from (k+1, a') to (k, a)
                # depends on coalescent rates
                if pr_a[k+1][a_prime] > 0:
                    transition = _transition_prob(a_prime, a)
                    pr_a[k][a] += pr_a[k+1][a_prime] * transition

            # Normalize to ensure probabilities sum to 1
            total = pr_a[k].sum()
            if total > 0:
                pr_a[k] /= total

    # Compute moments by averaging over a (law of total expectation/variance)
    moments = {}
    for k in range(2, n):
        mean = np.sum(pr_a[k] * t_mean)  #  $E[T] = \sum_a P(a) * E[T|a]$ 

```

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```

    e_t_sq = np.sum(pr_a[k] * (t_var + t_mean**2)) # E[T^2] via law of total
    ↳ expectation
    variance = e_t_sq - mean**2                    # Var = E[T^2] - (E[T])^2
    moments[k] = (mean, variance)

    # Root (k=n): sum of all waiting times from n lineages down to 1
    root_mean = sum(2.0 / (j * (j - 1)) for j in range(2, n + 1))
    root_var = sum(4.0 / (j * (j - 1))**2 for j in range(2, n + 1))
    moments[n] = (root_mean, root_var)

    return moments

def _transition_prob(a_prime, a):
    """Transition probability in the Wiuf-Donnelly recursion.

    Probability that when one more pair coalesces (decreasing k by 1),
    the number of ancestors changes from a' to a.
    """
    if a > a_prime or a < 2:
        return 0.0
    if a == a_prime:
        # The coalescing pair was entirely within the subtree
        # (no change in total ancestor count -- it was the subtree
        # that coalesced, reducing subtree lineages, but a new
        # subtree-root lineage appears)
        return (a_prime - 1) / (a_prime + 1)
    if a == a_prime - 1:
        # One of the coalescing lineages was in the subtree,
        # the other was not, reducing total ancestors by 1
        return 2.0 / (a_prime + 1)
    return 0.0

```

From Moments to Gamma Parameters

With the mean and variance of the conditional coalescent in hand, the next step is to convert them into a form that the dating algorithm can use. tsdate takes the mean and variance and fits a **gamma distribution** to them. This is the prior for each node.

Given mean μ and variance σ^2 , the gamma parameters are:

$$\alpha = \frac{\mu^2}{\sigma^2}, \quad \beta = \frac{\mu}{\sigma^2}$$

This is the standard method-of-moments estimator. Let's verify:

$$\mathbb{E}[\text{Gamma}(\alpha, \beta)] = \frac{\alpha}{\beta} = \frac{\mu^2/\sigma^2}{\mu/\sigma^2} = \mu \quad \checkmark$$

$$\text{Var}[\text{Gamma}(\alpha, \beta)] = \frac{\alpha}{\beta^2} = \frac{\mu^2/\sigma^2}{\mu^2/\sigma^4} = \sigma^2 \quad \checkmark$$

i Calculus Aside – Method of moments

Method of moments is one of the oldest techniques in statistics. The idea: set the theoretical moments of a distribution equal to the observed (or computed) moments, then solve for the parameters. For a $\text{Gamma}(\alpha, \beta)$, the first two moments are $\mu_1 = \alpha/\beta$ and $\mu_2 = \alpha(\alpha + 1)/\beta^2$. From the mean μ_1 and variance $\sigma^2 = \mu_2 - \mu_1^2 = \alpha/\beta^2$, we solve the two equations in two unknowns to get $\alpha = \mu_1^2/\sigma^2$ and $\beta = \mu_1/\sigma^2$. This is simple and fast, which is why tsdate uses it instead of maximum likelihood estimation for the prior parameters.

```
def gamma_params_from_moments(mean, variance):
    """Convert mean and variance to gamma distribution parameters.

    Parameters
    -----
    mean : float
        E[T] from the conditional coalescent.
    variance : float
        Var[T] from the conditional coalescent.

    Returns
    -----
    alpha : float
        Shape parameter (controls peakedness of the distribution).
    beta : float
        Rate parameter (controls how quickly the density decays).
    """
    alpha = mean**2 / variance    # shape = mean^2 / variance
    beta = mean / variance        # rate = mean / variance
    return alpha, beta

# Example: node with k=3 descendants in a sample of n=100
# The conditional coalescent gives approximate values:
k, n = 3, 100
# For small k relative to n, the mean is approximately 2/(k*(k-1))
approx_mean = 2.0 / (k * (k - 1))    # = 0.333 in coalescent units
approx_var = approx_mean**2          # exponential: var = mean^2

alpha, beta = gamma_params_from_moments(approx_mean, approx_var)
print(f"k={k}: mean={approx_mean:.4f}, var={approx_var:.4f}")
print(f"Gamma prior: alpha={alpha:.4f}, beta={beta:.4f}")
```

The Approximate Prior for Large n

Computing exact conditional coalescent moments for every possible (k, n) pair is expensive when n is large. tsdate uses a **lookup table with interpolation**:

1. Precompute exact moments for $k = 2, 3, \dots, n$ (or a subsample)
2. Store as arrays indexed by k
3. For nodes with the same k , reuse the same prior

The key array in tsdate's implementation is a **prior grid**: for each possible number of descendant leaves k , store $(\alpha_k, \beta_k, \mu_k, \sigma_k^2)$.

```

import numpy as np

def build_prior_grid(n, Ne=1.0):
    """Build a lookup table of gamma priors indexed by descendant count.

    Parameters
    -----
    n : int
        Total number of sample leaves.
    Ne : float
        Effective population size.

    Returns
    -----
    prior_grid : np.ndarray, shape (n+1, 4)
        Columns: [alpha, beta, mean, variance]
        Row k gives the prior for a node with k descendants.
        Rows 0 and 1 are unused (no node has 0 or 1 non-self descendants).
    """
    grid = np.zeros((n + 1, 4))
    moments = conditional_coalescent_moments(n, Ne) # compute all (mean, var) pairs

    for k in range(2, n + 1):
        mean, var = moments[k]
        alpha, beta = gamma_params_from_moments(mean, var)
        grid[k] = [alpha, beta, mean, var] # store both parameterizations

    return grid

```

Special Cases

Before moving on, let us address two boundary cases that arise in every tree sequence: the root (which is the oldest node) and the leaves (whose ages are known).

Roots

For the root of a tree (or a connected component in the tree sequence), tsdate assigns an **exponential prior** rather than a conditional coalescent prior. The exponential distribution is $\text{Gamma}(1, \beta)$, and the rate β is set so the mean matches the expected TMRCA.

For the variational gamma method, root priors are handled differently: they get a weakly informative mixture prior that allows for a wide range of ages.

Leaves (samples)

Leaf nodes have known ages. Modern samples are at time 0. Ancient samples (e.g., from aDNA) have their age set to the sample's radiocarbon date. These are **fixed nodes** – they don't need priors because their ages are observed.

```

def assign_node_priors(ts, prior_grid):
    """Assign a gamma prior to each non-leaf node.

    Parameters
    -----

```

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```

ts : tskit.TreeSequence
    The input tree sequence (topology from tsinfer).
prior_grid : np.ndarray
    From build_prior_grid().

Returns
-----
priors : dict
    {node_id: (alpha, beta)} for each non-fixed node.
"""
priors = {}
fixed_nodes = set(ts.samples()) # samples have known ages -- no prior needed

for node in ts.nodes():
    if node.id in fixed_nodes:
        continue # known age, no prior needed

    # Count descendants: number of samples below this node
    k = count_sample_descendants(ts, node.id)

    if k >= 2 and k <= ts.num_samples:
        # Look up the precomputed gamma prior for this descendant count
        alpha, beta = prior_grid[k, 0], prior_grid[k, 1]
        priors[node.id] = (alpha, beta)
    else:
        # Fallback: exponential prior for nodes with unusual topology
        priors[node.id] = (1.0, 1.0)

return priors

def count_sample_descendants(ts, node_id):
    """Count the number of sample leaves descended from a node."""
    samples = set(ts.samples())
    count = 0
    for tree in ts.trees():
        for leaf in tree.leaves(node_id):
            if leaf in samples:
                count += 1
        break # only need one tree (approximate for polytomies)
    return count

```

Putting It Together: A Visualization

Let's visualize what the prior looks like for different descendant counts. This will make concrete the central idea of this chapter: nodes with more descendants get priors shifted toward older ages.

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import gamma

def plot_coalescent_priors(n=50, Ne=1.0):
    """Plot gamma priors for nodes with different numbers of descendants."""

```

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```

fig, ax = plt.subplots(figsize=(10, 6))
t = np.linspace(0, 4, 500) # time axis in coalescent units

descendant_counts = [2, 5, 10, 25, 49]
colors = plt.cm.viridis(np.linspace(0, 0.9, len(descendant_counts)))

for k, color in zip(descendant_counts, colors):
    # Approximate moments for illustration
    # Mean age ~ sum of 1/(j choose 2) for j = k down to 2
    mean = sum(2.0 / (j * (j - 1)) for j in range(2, k + 1))
    var = sum(4.0 / (j * (j - 1))**2 for j in range(2, k + 1))

    alpha = mean**2 / var # shape from method of moments
    beta = mean / var     # rate from method of moments

    pdf = gamma.pdf(t, a=alpha, scale=1.0/beta)
    ax.plot(t, pdf, color=color, lw=2, label=f'k={k} (mean={mean:.2f})')

ax.set_xlabel('Node age (coalescent units)')
ax.set_ylabel('Prior density')
ax.set_title(f'Coalescent Prior for Different Descendant Counts (n={n})')
ax.legend()
ax.set_xlim(0, 4)

return fig

# plot_coalescent_priors()

```

What you should see: Nodes with more descendants (larger k) have priors shifted to the right (older ages), with more spread. Nodes with $k = 2$ have a tight, exponential-like prior near the present. The root ($k = n$) has the broadest, most right-shifted prior.

Think of it this way: a gear deep inside the movement (ancestral to many parts) must have been installed early in the watch's construction. A gear near the dial (ancestral to just two leaves) could have been added at any stage.

Summary

The coalescent prior gives tsdate a principled starting point for each node:

$$t_u \sim \text{Gamma}(\alpha_k, \beta_k) \quad \text{where } k = |\text{descendants}(u)|$$

The parameters (α_k, β_k) come from fitting gamma distributions to the mean and variance of the conditional coalescent. This prior encodes the simple but powerful idea: **nodes ancestral to more samples are expected to be older.**

In our watch metaphor, the coalescent prior is the expected beat rate – the baseline rhythm we expect from population genetics before any mutation data enters the picture. It sets the initial position of every hand on the dial.

Next, we need the other half of Bayes' rule: the likelihood. How do observed mutations inform us about branch lengths? That's the subject of the next chapter: *The Mutation Likelihood*.

The Mutation Likelihood

The ticking of the clock: every mutation is a beat, every edge a measured interval.

The coalescent prior (previous chapter, *The Coalescent Prior*) tells us how old a node *should* be, absent any data – the expected beat rate from coalescent theory. Now we bring in the data: the mutations observed on each edge of the tree sequence. This is the **likelihood** – the probability of the data given the node ages.

If the prior is the expected beat rate, the likelihood is **evidence from the mutation clock**: each mutation is a tick, and the number of ticks on an edge tells us how long that edge lasted. Together, prior and likelihood will be combined by Bayes' rule to produce the posterior – the calibrated age estimates.

The Molecular Clock

The molecular clock is one of the most powerful ideas in evolutionary genetics. It says:

Mutations accumulate at a roughly constant rate per base pair per generation.

If an edge in the genealogy spans Δt generations and covers ℓ base pairs, then the expected number of mutations on that edge is:

$$\mathbb{E}[\text{mutations}] = \mu \cdot \ell \cdot \Delta t$$

where μ is the per-base-pair, per-generation mutation rate. This is a linear relationship: longer branches accumulate more mutations.

Why is this useful? Because we *observe* the mutations (they show up as differences between sequences), and we want to infer Δt . The molecular clock converts observed mutations into time estimates.

Think of it as a metronome inside each branch of the tree: it ticks at rate $\mu \cdot \ell$, and the number of ticks recorded on the branch is the mutation count m_e . Counting ticks tells us how long the metronome ran.

The Poisson Model

tsdate models the number of mutations on each edge as a **Poisson random variable**. This is a natural choice because:

1. Mutations are rare events occurring along a long sequence – the classic Poisson regime.
2. Each base pair mutates independently (approximately).
3. The Poisson distribution has a single parameter (the mean), which is directly proportional to the branch length.

For edge e connecting parent u to child v :

$$m_e \sim \text{Poisson}(\lambda_e \cdot \Delta t_e)$$

where:

- m_e = observed number of mutations on this edge
- $\lambda_e = \mu \cdot \ell_e$ = the “span-weighted mutation rate”
- $\Delta t_e = t_u - t_v$ = the branch length (what we want to infer)
- ℓ_e = the genomic span of the edge in base pairs

The Poisson probability mass function gives us:

$$P(m_e | t_u, t_v) = \frac{(\lambda_e \Delta t_e)^{m_e}}{m_e!} \exp(-\lambda_e \Delta t_e)$$

i Probability Aside – The Poisson distribution

The Poisson distribution models the number of events occurring in a fixed interval when events happen independently at a constant rate. If the expected number of events is λ , then $P(k) = \lambda^k e^{-\lambda} / k!$. Its mean and variance are both λ . In our context the “interval” is the branch length in base-pair-generations, the “rate” is μ , and the “events” are mutations. The Poisson is appropriate whenever μ is small and the number of sites ℓ is large – exactly the regime of genomic mutations.

```
import numpy as np
from scipy.stats import poisson

def edge_likelihood(m_e, lambda_e, t_parent, t_child):
    """Poisson likelihood for mutations on a single edge.

    Parameters
    -----
    m_e : int
        Observed mutation count on this edge.
    lambda_e : float
        Span-weighted mutation rate (mu * span_bp).
    t_parent : float
        Age of parent node.
    t_child : float
        Age of child node.

    Returns
    -----
    likelihood : float
        P(m_e | t_parent, t_child).
    """
    delta_t = t_parent - t_child          # branch length in generations
    if delta_t <= 0:
        return 0.0                        # parent must be older than child
    expected = lambda_e * delta_t          # Poisson mean = rate * time
    return poisson.pmf(m_e, expected)     # evaluate the Poisson PMF

# Example: an edge spanning 10,000 bp with mu = 1e-8
mu = 1e-8
span_bp = 10_000
lambda_e = mu * span_bp # = 1e-4 mutations per generation

# If parent is 500 generations older than child:
delta_t = 500
expected_mutations = lambda_e * delta_t # = 0.05

print(f"Expected mutations: {expected_mutations:.4f}")
print(f"P(0 mutations) = {edge_likelihood(0, lambda_e, 500, 0):.6f}")
print(f"P(1 mutation)  = {edge_likelihood(1, lambda_e, 500, 0):.6f}")
print(f"P(2 mutations)  = {edge_likelihood(2, lambda_e, 500, 0):.6f}")
```

The Full Likelihood

The total likelihood is the product over all edges (assuming mutations on different edges are independent):

$$P(\mathbf{D} \mid \mathbf{t}) = \prod_{e \in \mathcal{E}} \frac{(\lambda_e \Delta t_e)^{m_e}}{m_e!} \exp(-\lambda_e \Delta t_e)$$

Taking the log:

$$\log P(\mathbf{D} \mid \mathbf{t}) = \sum_{e \in \mathcal{E}} [m_e \log(\lambda_e \Delta t_e) - \lambda_e \Delta t_e - \log(m_e!)]$$

i Calculus Aside – Why work with the log-likelihood?

Products of many small probabilities lead to numerical underflow (numbers too small for floating point). By taking logarithms, products become sums, which are numerically stable. Additionally, the log turns the exponential in the Poisson PMF into a linear term $-\lambda_e \Delta t_e$, making derivatives easier to compute. Throughout tsdate, nearly all computations happen in log space.

```
from scipy.special import gammaln

def total_log_likelihood(ts, node_times, mutation_rate):
    """Compute the total log-likelihood of observed mutations.

    Parameters
    -----
    ts : tskit.TreeSequence
        Tree sequence with mutations.
    node_times : np.ndarray
        Proposed times for each node.
    mutation_rate : float
        Per-bp per-generation mutation rate.

    Returns
    -----
    log_lik : float
        Total log-likelihood.
    """
    # Count mutations per edge
    mut_per_edge = np.zeros(ts.num_edges, dtype=int)
    for mut in ts.mutations():
        if mut.edge >= 0:
            mut_per_edge[mut.edge] += 1 # tally mutations on their parent edge

    log_lik = 0.0
    for edge in ts.edges():
        m_e = mut_per_edge[edge.id] # observed mutation count for this edge
        span = edge.right - edge.left # genomic span in base pairs
        lambda_e = mutation_rate * span # span-weighted rate
        delta_t = node_times[edge.parent] - node_times[edge.child] # branch length

        if delta_t <= 0:
```

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```

    return -np.inf # invalid: parent younger than child

    expected = lambda_e * delta_t # Poisson mean
    # Poisson log-pmf: m*log(expected) - expected - log(m!)
    log_lik += m_e * np.log(expected) - expected - gammaln(m_e + 1)

    return log_lik

```

The Likelihood as a Function of Branch Length

For a single edge, the likelihood as a function of Δt has a clean shape. Let's see why.

Fix m_e and λ_e . The log-likelihood as a function of Δt is:

$$\ell(\Delta t) = m_e \log(\lambda_e \Delta t) - \lambda_e \Delta t - \log(m_e!)$$

Taking the derivative and setting to zero:

$$\frac{d\ell}{d(\Delta t)} = \frac{m_e}{\Delta t} - \lambda_e = 0 \Rightarrow \widehat{\Delta t}_{\text{MLE}} = \frac{m_e}{\lambda_e}$$

Calculus Aside – Finding the MLE

The maximum likelihood estimate (MLE) is the parameter value that maximizes the likelihood (or equivalently, the log-likelihood). We differentiate $\ell(\Delta t) = m_e \log(\lambda_e \Delta t) - \lambda_e \Delta t + C$ with respect to Δt . The log term gives $m_e/\Delta t$ (chain rule), and the linear term gives $-\lambda_e$. Setting the sum to zero yields $\Delta t = m_e/\lambda_e$. The second derivative is $-m_e/(\Delta t)^2 < 0$, confirming this is a maximum.

The MLE is beautifully intuitive: the best estimate for the branch length is the observed mutation count divided by the expected rate. More mutations = longer branch.

The second derivative is $-m_e/(\Delta t)^2 < 0$, confirming this is a maximum.

```

import numpy as np
import matplotlib.pyplot as plt

def plot_edge_likelihood(m_e=3, lambda_e=0.01):
    """Plot the likelihood curve for a single edge."""
    delta_t = np.linspace(0.01, 800, 1000)
    # Log-likelihood (up to a constant) as a function of branch length
    log_lik = m_e * np.log(lambda_e * delta_t) - lambda_e * delta_t

    fig, ax = plt.subplots(figsize=(8, 5))
    # Plot relative likelihood (normalized so max = 1)
    ax.plot(delta_t, np.exp(log_lik - log_lik.max()), lw=2)
    ax.axvline(m_e / lambda_e, color='red', ls='--',
              label=f'MLE = {m_e/lambda_e:.0f} gen')
    ax.set_xlabel('Branch length (generations)')
    ax.set_ylabel('Relative likelihood')
    ax.set_title(f'Edge likelihood: {m_e} mutations, lambda={lambda_e}')
    ax.legend()
    return fig

```

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```
# plot_edge_likelihood()
```

The Edge Factor in the Factor Graph

Having established the Poisson likelihood for a single edge, let us now see how it fits into the factor graph introduced in *Overview of tsdate*. Each edge $e = (u, v)$ contributes a **factor** that connects the parent age t_u and child age t_v :

$$\phi_e(t_u, t_v) = \frac{(\lambda_e(t_u - t_v))^{m_e}}{m_e!} \exp(-\lambda_e(t_u - t_v)) \cdot \mathbb{I}[t_u > t_v]$$

This factor is **bivariate**: it depends on both the parent and child ages. The indicator $\mathbb{I}[t_u > t_v]$ enforces the constraint that parents must be older than children.

For message passing, we need to compute **messages** from this factor to each connected node. A message from factor ϕ_e to node u is obtained by “integrating out” the other variable:

$$\text{msg}_{e \rightarrow u}(t_u) = \int_0^{t_u} \phi_e(t_u, t_v) \cdot q(t_v) dt_v$$

where $q(t_v)$ is the current belief (approximate posterior) for node v . This integral is what the inside-outside and EP algorithms compute. In the watch metaphor, each edge factor is a spring connecting two gears: it transmits information (force) between them, and the message is the resulting torque on each gear.

Calculus Aside – Marginalizing by integration

“Integrating out” a variable means computing $\int f(x, y) dy$ to obtain a function of x alone. This is how we go from a joint distribution over two variables (parent and child age) to a marginal message about one variable. In the discrete inside-outside method this integral becomes a sum over grid points; in the variational gamma method it is evaluated approximately via the Laplace method or numerical quadrature.

Gamma-Poisson Conjugacy

Here’s where the choice of gamma priors pays off. If the prior on Δt is $\text{Gamma}(\alpha, \beta)$, and we observe m mutations with rate λ , then the posterior on Δt is:

$$\begin{aligned} P(\Delta t \mid m) &\propto \underbrace{(\Delta t)^m e^{-\lambda \Delta t}}_{\text{Poisson likelihood}} \cdot \underbrace{(\Delta t)^{\alpha-1} e^{-\beta \Delta t}}_{\text{Gamma prior}} \\ &= (\Delta t)^{(\alpha+m)-1} e^{-(\beta+\lambda)\Delta t} \end{aligned}$$

This is a $\text{Gamma}(\alpha + m, \beta + \lambda)$ distribution!

This is conjugacy: the posterior is in the same family as the prior. The update rules are simply:

$$\alpha_{\text{post}} = \alpha_{\text{prior}} + m_e, \quad \beta_{\text{post}} = \beta_{\text{prior}} + \lambda_e$$

Probability Aside – What is conjugacy?

A prior family is *conjugate* to a likelihood if the posterior belongs to the same family as the prior. For the Gamma-Poisson pair: a Gamma prior on the rate of a Poisson observation yields a Gamma posterior. Why does this matter? Because the update from prior to posterior reduces to adding numbers ($\alpha + m, \beta + \lambda$) instead of evaluating integrals.

In tsdate, conjugacy means that – for a single isolated edge – the posterior on branch length is immediately available in closed form. The complication in the full problem is that edges share nodes, so the conjugacy cannot be applied independently to each edge. That is why we need message passing.

```
def gamma_poisson_update(alpha_prior, beta_prior, m_e, lambda_e):
    """Update gamma parameters given Poisson observations.

    The gamma-Poisson conjugacy gives a closed-form posterior.

    Parameters
    -----
    alpha_prior, beta_prior : float
        Prior gamma parameters.
    m_e : int
        Observed mutations.
    lambda_e : float
        Span-weighted mutation rate.

    Returns
    -----
    alpha_post, beta_post : float
        Posterior gamma parameters.
    """
    # Conjugate update: just add mutation count to shape, rate to rate
    return alpha_prior + m_e, beta_prior + lambda_e

# Example: prior Gamma(2, 3), observe 5 mutations with rate 0.01
alpha_prior, beta_prior = 2.0, 3.0
m_e, lambda_e = 5, 0.01
alpha_post, beta_post = gamma_poisson_update(alpha_prior, beta_prior, m_e, lambda_e)

print(f"Prior:      Gamma({alpha_prior}, {beta_prior})")
print(f" mean = {alpha_prior/beta_prior:.4f}")
print(f"Posterior: Gamma({alpha_post}, {beta_post})")
print(f" mean = {alpha_post/beta_post:.4f}")
```

A subtlety: it's not quite this simple

The conjugacy result above applies when the branch length Δt is a free variable with a gamma prior. But in tsdate, the branch length is $t_u - t_v$, a *difference* of two random variables. The parent and child are not independent – they're connected through the tree. This is why we need message passing: the prior on t_u gets “messages” from all edges connected to u , not just one.

The true posterior on t_u is a product of gamma-like factors from each connected edge plus the coalescent prior. This product is generally *not* gamma, but the variational gamma method approximates it as one.

Handling Edges with Zero Mutations

Many edges in a tree sequence have zero mutations. This is not a problem for the Poisson model – in fact, it's the most common case for short edges or edges with small spans.

When $m_e = 0$:

$$P(0 \mid \lambda_e, \Delta t_e) = \exp(-\lambda_e \Delta t_e)$$

This is a pure exponential decay: longer branches are exponentially less likely to have zero mutations. The likelihood still provides information – it says “this branch is probably short.”

The MLE for Δt when $m_e = 0$ is $\hat{\Delta t} = 0$, which is degenerate. This is where the prior matters most: it pulls the estimate away from zero to a biologically reasonable age. In our watch metaphor, an edge with zero mutations is like a spring that shows no wear – the mutation clock recorded no ticks, so we lean on our prior knowledge (the expected beat rate from coalescent theory) to estimate how long it ran.

Handling Shared Edges

In a tree sequence, nodes can participate in many edges (across different genomic intervals). A single ancestral node might be the parent of different children in different regions of the genome.

The total likelihood contribution for node u combines information from *all* edges where u is a parent or child:

$$P(\mathbf{D} \mid t_u, \dots) = \prod_{e: \text{parent}(e)=u} P(m_e \mid t_u, t_{\text{child}(e)}) \cdot \prod_{e: \text{child}(e)=u} P(m_e \mid t_{\text{parent}(e)}, t_u)$$

This is what makes the problem interconnected: changing t_u affects the likelihood of every edge connected to u . In the gear train, each gear meshes with multiple neighbors, and adjusting one gear shifts them all.

```
def node_log_likelihood_contribution(ts, node_id, node_times, mutation_rate):
    """Log-likelihood contribution from all edges connected to a node.

    Parameters
    -----
    ts : tskit.TreeSequence
    node_id : int
    node_times : np.ndarray
    mutation_rate : float

    Returns
    -----
    log_lik : float
    """
    # First, count mutations per edge
    mut_per_edge = np.zeros(ts.num_edges, dtype=int)
    for mut in ts.mutations():
        if mut.edge >= 0:
            mut_per_edge[mut.edge] += 1

    log_lik = 0.0
    for edge in ts.edges():
        # Only consider edges connected to this node
        if edge.parent == node_id or edge.child == node_id:
            m_e = mut_per_edge[edge.id]
            span = edge.right - edge.left
            lambda_e = mutation_rate * span
            delta_t = node_times[edge.parent] - node_times[edge.child]

            if delta_t <= 0:
```

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```

    return -np.inf # invalid configuration

    expected = lambda_e * delta_t
    log_lik += m_e * np.log(expected) - expected - gammaln(m_e + 1)

    return log_lik

```

Summary

The mutation likelihood is the data-driven half of tsdate's Bayesian framework:

$$P(m_e \mid t_u, t_v) = \text{Poisson}(m_e; \lambda_e \cdot (t_u - t_v))$$

Key takeaways:

- Mutations follow a **Poisson process** along each edge
- The expected count is proportional to **branch length** $\times$ **span** $\times$ **mutation rate**
- The MLE for branch length is simply m_e / λ_e
- Gamma priors are **conjugate** to the Poisson likelihood, enabling closed-form updates for single edges
- But the tree structure couples all nodes, requiring **message passing** to propagate information

In the watch metaphor, the mutation likelihood is the evidence from the mutation clock – the wear marks that tell us how long each spring has been under tension. Combined with the coalescent prior (the expected beat rate), these two gears produce the posterior via Bayes' rule.

We now have both halves of Bayes' rule: the prior (Gear 1) and the likelihood (Gear 2). Next, we'll combine them using belief propagation. First up: the discrete-time inside-outside algorithm (*Inside-Outside Belief Propagation*).

Inside-Outside Belief Propagation

The first algorithm: pass messages up the tree, then back down, and every node knows its place in time.

With the coalescent prior (*The Coalescent Prior*) and the mutation likelihood (*The Mutation Likelihood*) in hand, we have both halves of Bayes' rule. The challenge now is *combining* them. Each node's age depends on the ages of its parents and children through the edge likelihoods, creating a coupled system that cannot be solved node-by-node.

The inside-outside method is tsdate's original dating algorithm. It discretizes time into a grid, represents each node's posterior as a probability vector over grid points, and propagates information through the tree using two passes: **inside** (leaves to root) and **outside** (root to leaves).

This is the same algorithmic idea as the forward-backward algorithm for HMMs, adapted to tree structures. In the watch metaphor, it is **messages flowing through the gear train**: each gear tells its neighbors what time it thinks it is, and after two complete sweeps (one upward, one downward), every gear has heard from every other gear and settled into its calibrated position.

Probability Aside – Belief propagation on trees vs. graphs

Belief propagation (BP) on a tree-shaped graphical model gives *exact* marginal distributions in exactly two passes. The inside pass collects evidence from leaves to root; the outside pass distributes it back. The algorithm is sometimes called the “sum-product algorithm.” On a graph with loops (like a tree sequence, where nodes are shared across local trees), BP becomes *loopy BP* – an approximation. Loopy BP has no guarantee of convergence or exactness, but in practice it works well for the sparse, tree-like graphs that tree sequences produce.

Step 1: Discretize Time

The first decision: what grid of timepoints to use?

tsdate creates a grid $\mathbf{g} = (g_0, g_1, \dots, g_{K-1})$ spanning from 0 to some maximum time. The grid can be:

- **Linear:** equally spaced in time
- **Logarithmic:** more resolution near the present, less in the deep past

Logarithmic is the default, because most nodes are relatively young and we want fine resolution there.

```
import numpy as np

def make_time_grid(n, Ne=1.0, num_points=20, grid_type="logarithmic"):
    """Create a time grid for the inside-outside algorithm.

    Parameters
    -----
    n : int
        Number of samples (sets the expected TMRCA).
    Ne : float
        Effective population size.
    num_points : int
        Number of grid points.
    grid_type : str
        "linear" or "logarithmic".

    Returns
    -----
    grid : np.ndarray
        Array of timepoints, starting at 0.
    """
    # Expected TMRCA under standard coalescent: 2*Ne*(1 - 1/n)
    expected_tmrca = 2 * Ne * (1 - 1.0 / n)
    t_max = expected_tmrca * 4 # go well beyond expected TMRCA

    if grid_type == "linear":
        return np.linspace(0, t_max, num_points)
    else:
        # Log-spaced: more points near 0, fewer far out
        # Start from a small positive number to avoid log(0)
        t_min = t_max / (10 * num_points)
        return np.concatenate([[0], np.geomspace(t_min, t_max, num_points - 1)])

# Example
grid = make_time_grid(n=100, num_points=20)
print(f"Grid: {grid[:5]} ... {grid[-3:]}")
print(f"Grid spans [0, {grid[-1]:.2f}] with {len(grid)} points")
```

Step 2: The Likelihood Matrix

For each edge e , we need the likelihood of the observed mutations m_e as a function of the parent and child times. On the discrete grid, this becomes a $K \times K$ **lower-triangular matrix** L_e :

$$L_e[i, j] = P(m_e \mid t_{\text{parent}} = g_i, t_{\text{child}} = g_j) = \text{Poisson}(m_e; \lambda_e \cdot (g_i - g_j))$$

for $i > j$ (parent older than child), and $L_e[i, j] = 0$ otherwise.

This matrix is the discrete version of the bivariate edge factor ϕ_e we met in the likelihood chapter. Each entry answers: “if the parent were at grid point i and the child at grid point j , how likely are the observed mutations?”

```
from scipy.stats import poisson

def edge_likelihood_matrix(m_e, lambda_e, grid):
    """Compute the likelihood matrix for an edge on the time grid.

    Parameters
    -----
    m_e : int
        Mutation count on this edge.
    lambda_e : float
        Span-weighted mutation rate (mu * span_bp).
    grid : np.ndarray
        Time grid.

    Returns
    -----
    L : np.ndarray, shape (K, K)
        L[i, j] = P(m_e | parent_time=grid[i], child_time=grid[j])
        Lower triangular (i >= j).
    """
    K = len(grid)
    L = np.zeros((K, K))

    for i in range(K):
        for j in range(i + 1): # j <= i (child younger than parent)
            delta_t = grid[i] - grid[j]
            if delta_t > 0:
                expected = lambda_e * delta_t # Poisson mean
                L[i, j] = poisson.pmf(m_e, expected) # evaluate PMF
            elif m_e == 0:
                # delta_t = 0, only possible if no mutations
                L[i, j] = 1.0

    return L

# Example
grid = make_time_grid(n=50, num_points=10)
L = edge_likelihood_matrix(m_e=2, lambda_e=0.001, grid=grid)
print(f"Likelihood matrix shape: {L.shape}")
print(f"Max likelihood at parent_idx, child_idx = {np.unravel_index(L.argmax(), L.
↪shape)}")
```

i Storage optimization

tsdate doesn't actually store full $K \times K$ matrices. Instead, it stores the lower triangle as a flattened 1D array of size $K(K+1)/2$. This halves the memory requirement.

Step 3: The Inside Pass (Leaves to Root)

Now we arrive at the heart of the algorithm. The inside pass computes, for each node u , the probability of all the data *below* u , conditioned on u 's age:

$$\text{inside}(u, g_i) = P(\mathbf{D}_{\text{below } u} \mid t_u = g_i)$$

Think of this as each gear reporting upward: “given that I am at grid point i , here is the total evidence from everything below me.” The messages flow from the leaves (known time 0) up to the root, accumulating mutation evidence along the way.

For **leaf nodes** (samples at time 0):

$$\text{inside}(\text{leaf}, g_i) = \begin{cases} 1 & \text{if } g_i = 0 \\ 0 & \text{otherwise} \end{cases}$$

For **internal nodes**, the inside value combines information from all child edges. If node u has children $v_1, v_2, \dots$ connected by edges $e_1, e_2, \dots$:

$$\text{inside}(u, g_i) = \prod_{\text{child } v_c} \underbrace{\sum_{j=0}^i L_{e_c}[i, j] \cdot \text{inside}(v_c, g_j)}_{\text{message from child } v_c}$$

Intuition: For each child, sum over all possible child times (weighted by the edge likelihood and the child's inside value), then multiply across children. This is exactly the same logic as the forward algorithm in an HMM, but on a tree instead of a chain.

i Calculus Aside – Discrete marginalization

The inner sum $\sum_{j=0}^i L_e[i, j] \cdot \text{inside}(v, g_j)$ is the discrete analogue of the integral $\int_0^{t_u} \phi_e(t_u, t_v) \cdot q(t_v) dt_v$ that we met in the likelihood chapter. On the grid, the integral becomes a matrix-vector product: multiply the likelihood matrix row by the child's inside vector, then sum. The product over children is the “product rule” for independent subtrees.

```
import numpy as np

def inside_pass(ts, grid, mutation_rate, mut_per_edge):
    """Compute inside values for all nodes.

    Parameters
    -----
    ts : tskit.TreeSequence
    grid : np.ndarray
        Time grid of K points.
    mutation_rate : float
    mut_per_edge : np.ndarray
```

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```

    Mutation count per edge.

Returns
-----
inside : np.ndarray, shape (num_nodes, K)
    inside[u, i] = P(data below u | t_u = grid[i]).
"""
K = len(grid)
inside = np.ones((ts.num_nodes, K)) # start at 1 (multiplicative identity)

# Initialize leaves: delta at time 0
for sample_id in ts.samples():
    inside[sample_id, :] = 0.0      # zero everywhere...
    inside[sample_id, 0] = 1.0      # ...except at grid point 0 (present)

# Process edges from leaves to root (bottom-up)
# We need a topological ordering: process children before parents
# tsdate uses the edge table sorted by child time

# Build adjacency: for each parent, collect (child, edge_id)
children_of = {}
for edge in ts.edges():
    if edge.parent not in children_of:
        children_of[edge.parent] = []
    children_of[edge.parent].append((edge.child, edge.id))

# Topological order: process nodes with smallest time first
node_order = sorted(range(ts.num_nodes),
                     key=lambda u: ts.node(u).time)

for u in node_order:
    if u in ts.samples():
        continue # already initialized

    if u not in children_of:
        continue

    for child_id, edge_id in children_of[u]:
        m_e = mut_per_edge[edge_id]
        edge = ts.edge(edge_id)
        span = edge.right - edge.left
        lambda_e = mutation_rate * span

        # Build the K x K likelihood matrix for this edge
        L = edge_likelihood_matrix(m_e, lambda_e, grid)

        # Message from child to parent:
        # msg[i] = sum_j L[i, j] * inside[child, j]
        msg = np.zeros(K)
        for i in range(K):
            for j in range(i + 1): # only j <= i (child younger than parent)
                msg[i] += L[i, j] * inside[child_id, j]

```

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```

    # Multiply into parent's inside value (product over children)
    inside[u, :] *= msg

    # Normalize to prevent underflow (does not change relative values)
    total = inside[u, :].sum()
    if total > 0:
        inside[u, :] /= total

    return inside

```

Step 4: The Outside Pass (Root to Leaves)

With the inside pass complete, every node knows about the evidence below it. But nodes also need evidence from *above* – what do the parent, grandparent, and sibling subtrees say? The outside pass sends this information downward.

The outside pass computes, for each node u , the probability of all the data *above* u :

$$\text{outside}(u, g_i) = P(\mathbf{D}_{\text{above } u} \mid t_u = g_i)$$

For **root nodes**:

$$\text{outside}(\text{root}, g_i) = \text{prior}(\text{root}, g_i)$$

The prior comes from the conditional coalescent (Gear 1, *The Coalescent Prior*).

For **non-root nodes**, the outside value is computed by combining the parent's outside value, the edge likelihood, and the inside values of *sibling* subtrees:

$$\text{outside}(v, g_j) = \sum_{i=j}^{K-1} L_e[i, j] \cdot \text{outside}(u, g_i) \cdot \prod_{\text{sibling } v' \neq v} \text{msg}_{v' \rightarrow u}(g_i)$$

Intuition: To know what the data above v tells us about v 's age, we need:

1. The information from above the parent u (the outside of u)
2. The information from sibling subtrees (the inside messages from siblings)
3. The edge likelihood connecting u to v

In the gear train, the outside message is the force transmitted *downward* from the mainspring (root) through the gear train. Each gear receives torque from above (its parent's outside) modulated by the sibling gears' evidence (their inside messages).

```

def outside_pass(ts, grid, mutation_rate, mut_per_edge, inside, prior_grid):
    """Compute outside values for all nodes.

    Parameters
    -----
    ts : tskit.TreeSequence
    grid : np.ndarray
    mutation_rate : float
    mut_per_edge : np.ndarray
    inside : np.ndarray, shape (num_nodes, K)
    prior_grid : np.ndarray

```

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```

    Prior for each node.

Returns
-----
outside : np.ndarray, shape (num_nodes, K)
"""
K = len(grid)
outside = np.ones((ts.num_nodes, K))

# Initialize roots with coalescent prior
for u in range(ts.num_nodes):
    if is_root(ts, u):
        outside[u, :] = prior_grid[u] # prior is the "outside" for the root

# Process nodes from root to leaves (top-down -- oldest first)
node_order = sorted(range(ts.num_nodes),
                    key=lambda u: -ts.node(u).time) # oldest first

# Build parent lookup
parent_of = {} # (child, edge_id) -> parent
children_of = {}
for edge in ts.edges():
    parent_of[(edge.child, edge.id)] = edge.parent
    if edge.parent not in children_of:
        children_of[edge.parent] = []
    children_of[edge.parent].append((edge.child, edge.id))

for u in node_order:
    if u not in children_of:
        continue

    # Compute the "inside messages" from each child to u
    child_messages = {}
    for child_id, edge_id in children_of[u]:
        m_e = mut_per_edge[edge_id]
        edge = ts.edge(edge_id)
        span = edge.right - edge.left
        lambda_e = mutation_rate * span
        L = edge_likelihood_matrix(m_e, lambda_e, grid)

        # Standard inside message: sum over child times
        msg = np.zeros(K)
        for i in range(K):
            for j in range(i + 1):
                msg[i] += L[i, j] * inside[child_id, j]

        child_messages[(child_id, edge_id)] = msg

    # For each child, compute outside using parent outside
    # and all other children's messages (siblings)
    for child_id, edge_id in children_of[u]:
        m_e = mut_per_edge[edge_id]

```

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```

edge = ts.edge(edge_id)
span = edge.right - edge.left
lambda_e = mutation_rate * span
L = edge_likelihood_matrix(m_e, lambda_e, grid)

# Parent contribution: outside[u] * product of sibling messages
parent_contrib = outside[u, :].copy()
for other_child, other_eid in children_of[u]:
    if other_eid != edge_id:
        # Multiply in sibling's inside message
        parent_contrib *= child_messages[(other_child, other_eid)]

# Message from parent to child (downward):
# msg[j] = sum_i L[i,j] * parent_contrib[i]
msg = np.zeros(K)
for j in range(K):
    for i in range(j, K): # i >= j (parent older than child)
        msg[j] += L[i, j] * parent_contrib[i]

outside[child_id, :] *= msg # accumulate outside evidence

# Normalize
total = outside[child_id, :].sum()
if total > 0:
    outside[child_id, :] /= total

return outside

def is_root(ts, node_id):
    """Check if a node is a root (has no parent edges)."""
    for edge in ts.edges():
        if edge.child == node_id:
            return False
    return ts.node(node_id).time > 0

```

Step 5: Combine to Get the Posterior

With the inside and outside values computed, combining them is straightforward. The marginal posterior for each node is the product of inside and outside, weighted by the prior:

$$P(t_u = g_i \mid \mathbf{D}) \propto \text{inside}(u, g_i) \cdot \text{outside}(u, g_i)$$

This is the fundamental identity of the sum-product algorithm: the marginal at a variable is the product of all evidence arriving from below (inside) and all evidence arriving from above (outside).

```

def compute_posteriors(inside, outside):
    """Combine inside and outside to get marginal posteriors.

    Parameters
    -----
    inside : np.ndarray, shape (num_nodes, K)
    outside : np.ndarray, shape (num_nodes, K)

```

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```

Returns
-----
posterior : np.ndarray, shape (num_nodes, K)
    posterior[u, :] is the marginal posterior distribution over
    grid points for node u.
"""
posterior = inside * outside # element-wise product

# Normalize each node's posterior to sum to 1
row_sums = posterior.sum(axis=1, keepdims=True)
row_sums[row_sums == 0] = 1.0 # avoid division by zero
posterior /= row_sums

return posterior

def posterior_mean(posterior, grid):
    """Compute posterior mean age for each node.

    Parameters
    -----
    posterior : np.ndarray, shape (num_nodes, K)
    grid : np.ndarray, shape (K,)

    Returns
    -----
    means : np.ndarray, shape (num_nodes,)
        E[t_u | D] for each node.
    """
    return posterior @ grid # weighted sum: sum_i posterior[u,i] * grid[i]

```

Why This Works: The Belief Propagation Guarantee

On a **tree** (no loops), the inside-outside algorithm gives **exact** marginal posteriors. This is a classical result from graphical models: belief propagation on trees converges in exactly two passes.

But a tree *sequence* is not a tree. When a node appears in multiple local trees, it creates loops in the factor graph. For example, if node u is the parent of v in one genomic region and the grandparent of v in another, there are two paths between u and v – a loop.

On loopy graphs, belief propagation is **approximate**. It may:

- Converge to a fixed point that's close to the true posterior (common in practice)
- Oscillate (rare for this type of graph)
- Over-count evidence from repeated paths (the main source of error)

tsdate mitigates this by processing edges in the tree sequence's natural ordering, which respects the temporal structure and minimizes loop effects.

Probability Aside – Why loops cause trouble

On a tree, each piece of evidence (each mutation on each edge) is counted exactly once in every node's posterior. On

a graph with loops, messages can “circulate” around a loop: node A tells B, B tells C, C tells A what A originally said – as if the same evidence were counted twice. This is called “double-counting” and it makes loopy BP an approximation. In tree sequences the loops arise because a single ancestor participates in different local trees. The loops are typically short (length 2 or 3), and empirically the approximation is good.

Log-Space Computation

In practice, the inside and outside values can span many orders of magnitude. tsdate performs all computations in **log space** to prevent underflow:

$$\log \text{inside}(u, g_i) = \sum_{\text{children}} \log \left(\sum_j \exp(\log L_e[i, j] + \log \text{inside}(v, g_j)) \right)$$

The inner log-sum-exp is computed using the standard numerical trick:

$$\log \sum_j e^{x_j} = x_{\max} + \log \sum_j e^{x_j - x_{\max}}$$

Calculus Aside – The log-sum-exp trick

Naively computing $\log(\sum_j e^{x_j})$ can overflow (if any x_j is very large) or underflow (if all x_j are very negative). The trick: factor out $e^{x_{\max}}$ to get $x_{\max} + \log(\sum_j e^{x_j - x_{\max}})$. Now every exponent is ≤ 0 , preventing overflow, and at least one exponent is 0, preventing underflow. This is the single most important numerical trick in probabilistic computation, and it appears throughout tsdate.

```
from scipy.special import logsumexp

def inside_pass_logspace(inside_log, L_log, K):
    """Compute a single inside message in log space.

    Parameters
    -----
    inside_log : np.ndarray, shape (K,)
        Log inside values for child node.
    L_log : np.ndarray, shape (K, K)
        Log likelihood matrix.

    Returns
    -----
    msg_log : np.ndarray, shape (K,)
        Log message from child to parent.
    """
    msg_log = np.full(K, -np.inf) # start at log(0) = -inf
    for i in range(K):
        terms = L_log[i, :i+1] + inside_log[:i+1] # log(L * inside) = log(L) +
    ↪ log(inside)
        msg_log[i] = logsumexp(terms) # log-sum-exp for numerical
    ↪ stability
    return msg_log
```

The Standardization Trick

tsdate also uses **standardization**: after each message computation, the maximum value is subtracted. This keeps all values in a numerically safe range without changing the relative proportions.

$$\tilde{f}(g_i) = f(g_i) - \max_i f(g_i)$$

In log space, this means $\max_i \tilde{f}(g_i) = 0$.

Putting It All Together

Here's the complete inside-outside algorithm, assembling all the pieces from above into a single pipeline.

```
def inside_outside_date(ts, mutation_rate, Ne=1.0, num_points=20):
    """Date a tree sequence using the inside-outside algorithm.

    Parameters
    -----
    ts : tskit.TreeSequence
        Input tree sequence (topology from tsinfer).
    mutation_rate : float
        Per-bp per-generation mutation rate.
    Ne : float
        Effective population size.
    num_points : int
        Number of time grid points.

    Returns
    -----
    node_times : np.ndarray
        Posterior mean age for each node.
    """
    # Step 0: Setup -- build the time grid
    grid = make_time_grid(ts.num_samples, Ne, num_points)
    K = len(grid)

    # Count mutations per edge (used by both passes)
    mut_per_edge = np.zeros(ts.num_edges, dtype=int)
    for mut in ts.mutations():
        if mut.edge >= 0:
            mut_per_edge[mut.edge] += 1

    # Build prior for each node (from coalescent theory, Gear 1)
    prior = build_discrete_prior(ts, grid, Ne)

    # Step 1: Inside pass (leaves to root) -- evidence flows upward
    inside = inside_pass(ts, grid, mutation_rate, mut_per_edge)

    # Step 2: Outside pass (root to leaves) -- evidence flows downward
    outside = outside_pass(ts, grid, mutation_rate, mut_per_edge,
                           inside, prior)

    # Step 3: Combine inside and outside to get marginal posteriors
    posterior = compute_posteriors(inside, outside)
```

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```

# Step 4: Extract posterior means as point estimates
node_times = posterior_mean(posterior, grid)

# Fix leaf times at 0 (samples have known ages)
for s in ts.samples():
    node_times[s] = 0.0

return node_times

def build_discrete_prior(ts, grid, Ne):
    """Build a discrete prior for each node on the time grid."""
    from scipy.stats import gamma

    K = len(grid)
    prior = np.ones((ts.num_nodes, K))

    for u in range(ts.num_nodes):
        if u in set(ts.samples()):
            # Sample nodes are fixed at time 0
            prior[u, :] = 0.0
            prior[u, 0] = 1.0
            continue

        # Count descendants (simplified: assume binary tree)
        k = 2
        mean = sum(2.0 / (j * (j - 1)) for j in range(2, k + 1))
        var = sum(4.0 / (j * (j - 1))**2 for j in range(2, k + 1))
        alpha = mean**2 / var # gamma shape from method of moments
        beta_param = mean / var # gamma rate from method of moments

        # Evaluate gamma pdf at grid points
        for i in range(K):
            if grid[i] > 0:
                prior[u, i] = gamma.pdf(grid[i], a=alpha, scale=1.0/beta_param)
            else:
                prior[u, i] = 0.0 # internal nodes can't be at time 0

        # Normalize to a proper probability distribution
        total = prior[u, :].sum()
        if total > 0:
            prior[u, :] /= total

    return prior

```

Limitations of Inside-Outside

The inside-outside method works well but has some limitations that motivated the development of the variational gamma method:

1. **Grid resolution:** The posterior is only as fine as the grid. With $K = 20$ points, you can't distinguish between times that fall in the same grid cell.

2. **Quadratic per edge:** Computing the likelihood matrix is $O(K^2)$. For large K , this becomes expensive.
3. **Loopy BP:** On tree sequences with many shared nodes, the approximation may degrade.
4. **No natural way to handle constraints:** Enforcing $t_u > t_v$ on the grid requires zeroing out entries, which can lose probability mass.

These limitations motivated the development of the **variational gamma method** (*Variational Gamma (Expectation Propagation)*), which works in continuous time and avoids the grid entirely. Instead of a probability vector of K values per node, it stores just two numbers (α, β) , and instead of matrix-vector products, it uses moment matching – a fundamentally different (and faster) way of passing messages through the gear train.

Summary

The inside-outside algorithm dates nodes by:

1. **Discretizing** time into a grid of K points
2. **Inside pass:** propagating mutation likelihoods upward from leaves to roots
3. **Outside pass:** propagating prior and sibling information downward
4. **Combining:** multiplying inside and outside to get marginal posteriors

The key equations:

$$\begin{aligned} \text{inside}(u, g_i) &= \prod_{\text{children}} \sum_j L_e[i, j] \cdot \text{inside}(v, g_j) \\ \text{outside}(v, g_j) &= \sum_i L_e[i, j] \cdot \text{outside}(u, g_i) \cdot \prod_{\text{siblings}} \text{msg}(g_i) \\ P(t_u = g_i \mid \mathbf{D}) &\propto \text{inside}(u, g_i) \cdot \text{outside}(u, g_i) \end{aligned}$$

In the watch metaphor, the inside pass is like winding the mainspring from the bottom – evidence accumulates upward from the leaves. The outside pass releases that energy back down through the gear train. After both passes, every gear (node) has felt the full tension of the data from every direction, and its position (age) is set.

Next: the modern default method, variational gamma, which replaces the grid with continuous gamma approximations (*Variational Gamma (Expectation Propagation)*).

Variational Gamma (Expectation Propagation)

The master gear: approximate every node's age as a gamma distribution, then refine by passing messages until the whole mechanism converges.

The inside-outside method (*Inside-Outside Belief Propagation*) passes messages through the gear train using a discrete grid. It works, but the grid imposes limits on resolution and speed. The variational gamma method is tsdate's default and most accurate algorithm. Instead of discretizing time into a grid, it approximates each node's posterior age as a **gamma distribution** and refines it iteratively using **Expectation Propagation** (EP) – a message-passing algorithm from machine learning (Minka, 2001).

In the watch metaphor, this is still **messages flowing through the gear train**, but now each message is a gamma distribution rather than a probability vector. The gear train is the same; only the language of the messages has changed – from a list of numbers (grid probabilities) to two numbers (α, β) that encode a continuous belief about each node's age.

This chapter builds EP from scratch, one piece at a time. By the end you will understand cavity distributions, moment matching, and damping – the three pillars of EP – and see how they fit together to date a tree sequence.

i Biology Aside – Why dating matters

Assigning dates to ancestral nodes in a genealogy answers some of the most fundamental questions in evolutionary biology: *When did the most recent common ancestor of all humans live? When did a specific population split occur? How old is a particular beneficial mutation?* The tree sequence from `tsinfer` gives us the topology (who is related to whom), but the branch lengths – the time spans between ancestor and descendant – must be estimated from the density of mutations along each branch. More mutations imply more time. The variational gamma method performs this estimation for every node simultaneously, propagating information through the entire genealogy to produce the most consistent set of dates.

i Prerequisites

This chapter builds on three earlier ones. The coalescent prior (*The Coalescent Prior*) provides the initial gamma beliefs; the mutation likelihood (*The Mutation Likelihood*) defines the Poisson factors that link parent and child nodes; and the inside-outside chapter (*Inside-Outside Belief Propagation*) introduces the idea of two-pass message passing. If any of those concepts feel shaky, revisit the relevant chapter before continuing.

Why Move Beyond Inside-Outside?

The inside-outside method (previous chapter) has three practical limitations:

1. The **time grid** limits resolution: you can't distinguish ages that fall in the same cell.
2. The cost is **quadratic** in grid size per edge: $O(K^2 \cdot E)$.
3. The grid boundaries are somewhat arbitrary.

The variational gamma method solves all three:

- **Continuous time**: no grid, no resolution limit.
- **Two parameters per node**: (α, β) for the gamma shape and rate, so the cost is $O(E)$ per iteration.
- **Natural parameterization**: the gamma family captures the right range of posterior shapes for coalescence times.

The Big Picture

Here's the algorithm in one paragraph:

Represent each node's posterior age as $\text{Gamma}(\alpha_u, \beta_u)$. For each edge $e = (u, v)$, compute a “message” that says how the Poisson mutation likelihood on e updates the gamma beliefs for u and v . Apply these messages by **moment matching**: compute the exact moments of the updated distribution, then find the gamma that matches those moments. Iterate over all edges until convergence.

And here it is as a diagram:

```

Initialize: q(t_u) = Gamma(alpha_u, beta_u) for each node u
          |
          +-----+-----+
          |                                     |
          v                                     |
For each edge e = (u, v):                     |
  1. Remove old message from q(t_u), q(t_v)   |
  2. Compute exact moments of:                 |
      q(t_u) * q(t_v) * Poisson(m_e|...)      |

```

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```

3. Moment-match to new gammas
4. Update q(t_u), q(t_v)
   |
   +-----> Converged? ----No-----+
           |
           Yes
           |
           v
Return posterior means
    
```

Probability Aside – What is variational inference?

Variational inference is a family of methods for approximating intractable probability distributions. The idea: choose a simple family of distributions $\mathcal{Q}$ (here, products of gamma distributions) and find the member $q^* \in \mathcal{Q}$ that is “closest” to the true posterior p , measured by KL divergence. The name “variational” comes from the calculus of variations, because we are optimizing over functions (distributions) rather than finite-dimensional parameters. EP is one specific variational method; standard variational Bayes (mean-field) is another. They differ in which KL divergence they minimize and how they process factors.

The Natural Parameterization

A gamma distribution $\text{Gamma}(\alpha, \beta)$ has density:

$$p(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}, \quad t > 0$$

In the **natural parameter** (exponential family) form, this is:

$$p(t) \propto \exp((\alpha - 1) \log t - \beta t)$$

The natural parameters are $\eta_1 = \alpha - 1$ and $\eta_2 = -\beta$. The sufficient statistics are $\log t$ and t .

Biology Aside – What α and β mean for node ages

For each ancestor in the genealogy, the gamma distribution $\text{Gamma}(\alpha, \beta)$ encodes our belief about when that ancestor lived. The **mean** α/β is our best estimate of the ancestor’s age (in generations or coalescent time units). The **variance** α/β^2 measures how uncertain we are. A node deep in the tree (many mutations on incident edges, many descendant samples) will have a tight gamma with large α – we are confident about its age. A node with few mutations and few descendants will have a diffuse gamma – we know little about when it lived. The EP algorithm refines these beliefs by passing information along edges, iteratively sharpening each node’s gamma.

Why natural parameters? Because products of gamma-like terms correspond to *additions* of natural parameters. Let us show this step by step. Start with two gamma-shaped factors:

$$f_1(t) \propto t^{\alpha_1-1} e^{-\beta_1 t}, \quad f_2(t) \propto t^{\alpha_2-1} e^{-\beta_2 t}$$

Multiply them together, using the rules $t^a \cdot t^b = t^{a+b}$ and $e^{-c_1 t} \cdot e^{-c_2 t} = e^{-(c_1+c_2)t}$:

$$\begin{aligned}
 f_1(t) \cdot f_2(t) &\propto t^{\alpha_1-1} \cdot t^{\alpha_2-1} \cdot e^{-\beta_1 t} \cdot e^{-\beta_2 t} \\
 &= t^{(\alpha_1-1)+(\alpha_2-1)} \cdot e^{-(\beta_1+\beta_2)t} \\
 &= t^{(\alpha_1+\alpha_2-2)} \cdot e^{-(\beta_1+\beta_2)t} \\
 &= t^{(\alpha_1+\alpha_2-1)-1} \cdot e^{-(\beta_1+\beta_2)t}
 \end{aligned}$$

This is the kernel of $\text{Gamma}(\alpha_1 + \alpha_2 - 1, \beta_1 + \beta_2)$. In natural parameters $(\eta_1, \eta_2) = (\alpha - 1, -\beta)$, the product corresponds to elementwise addition:

$$(\eta_1^{(1)} + \eta_1^{(2)}, \eta_2^{(1)} + \eta_2^{(2)}) = (\alpha_1 - 1 + \alpha_2 - 1, -\beta_1 - \beta_2)$$

which gives natural parameters for the product $\text{Gamma}(\alpha_1 + \alpha_2 - 1, \beta_1 + \beta_2)$.

```
# Verify the product rule numerically
import numpy as np
from scipy.stats import gamma as gamma_dist

a1, b1 = 3.0, 2.0
a2, b2 = 2.0, 1.5

x = np.linspace(0.01, 5.0, 1000)

# Product of two gamma PDFs (unnormalized)
f1 = gamma_dist.pdf(x, a=a1, scale=1/b1)
f2 = gamma_dist.pdf(x, a=a2, scale=1/b2)
product = f1 * f2

# The result should be proportional to Gamma(a1+a2-1, b1+b2)
a_new, b_new = a1 + a2 - 1, b1 + b2
f_new = gamma_dist.pdf(x, a=a_new, scale=1/b_new)

# Check proportionality: ratio should be constant
ratio = product / f_new
ratio = ratio[f_new > 1e-10] # avoid division by near-zero
print(f"Product is Gamma({a_new}, {b_new})")
print(f"Ratio min={ratio.min():.6f}, max={ratio.max():.6f} (should be constant)")
```

This addition rule is the foundation of EP updates. In the gear train, each factor (edge likelihood, coalescent prior) contributes a “torque” in natural parameter space, and the total torque on a node is simply the sum.

i Calculus Aside – Exponential families

A distribution belongs to the *exponential family* if its density can be written as $p(x|\eta) = h(x) \exp(\eta \cdot T(x) - A(\eta))$, where η is the natural parameter vector, $T(x)$ is the sufficient statistic vector, and $A(\eta)$ is the log-normalizer. For the gamma: $T(t) = (\log t, t)$, $\eta = (\alpha - 1, -\beta)$, and $A(\eta) = \log \Gamma(\alpha) - \alpha \log \beta$. The key property is that products of factors in the same exponential family yield a member of the same family (with summed natural parameters). This is why gamma posteriors can be updated by simple addition.

```
import numpy as np
from scipy.special import gammaln, digamma, polygamma

class GammaDistribution:
    """A gamma distribution in natural parameterization.

    Natural parameters: eta1 = alpha - 1, eta2 = -beta
    Standard parameters: alpha (shape), beta (rate)
    """
    def __init__(self, alpha=1.0, beta=1.0):
```

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```

        self.alpha = alpha    # shape parameter
        self.beta = beta     # rate parameter

    @property
    def eta1(self):
        """First natural parameter:  $\alpha - 1$ ."""
        return self.alpha - 1

    @property
    def eta2(self):
        """Second natural parameter:  $-\beta$ ."""
        return -self.beta

    @property
    def mean(self):
        """ $E[t] = \alpha / \beta$ ."""
        return self.alpha / self.beta

    @property
    def variance(self):
        """ $Var(t) = \alpha / \beta^2$ ."""
        return self.alpha / self.beta**2

    @property
    def log_mean(self):
        """ $E[\log t] = \text{digamma}(\alpha) - \log(\beta)$ """
        return digamma(self.alpha) - np.log(self.beta)

    def multiply(self, other):
        """Multiply two gamma factors (add natural parameters).

        In natural parameter space:  $(\eta_1, \eta_2) + (\eta_1', \eta_2')$ 
        In standard parameters:  $\alpha_{\text{new}} = \alpha + \alpha' - 1$ ,
                                $\beta_{\text{new}} = \beta + \beta'$ 

        """
        new_alpha = self.alpha + other.alpha - 1
        new_beta = self.beta + other.beta
        return GammaDistribution(new_alpha, new_beta)

    def divide(self, other):
        """Divide by a gamma factor (subtract natural parameters).

        This is the inverse of multiply: removing a factor's contribution.
        """
        new_alpha = self.alpha - other.alpha + 1
        new_beta = self.beta - other.beta
        return GammaDistribution(new_alpha, new_beta)

    @classmethod
    def from_moments(cls, mean, variance):
        """Create from mean and variance via moment matching.
```

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```

Uses the standard method-of-moments estimator:
beta = mean / variance, alpha = mean * beta
"""

beta = mean / variance
alpha = mean * beta
return cls(alpha, beta)

```

The EP Algorithm Step by Step

EP maintains the following state:

- **Posterior approximation** $q(t_u) = \text{Gamma}(\alpha_u, \beta_u)$ for each node u
- **Edge factors** $f_e^{\rightarrow u}$ and $f_e^{\rightarrow v}$ for each edge $e = (u, v)$: gamma-shaped “messages” from the edge likelihood

The posterior for node u is the product of its prior and all incoming edge messages:

$$q(t_u) \propto \text{prior}(t_u) \cdot \prod_{e \ni u} f_e^{\rightarrow u}(t_u)$$

In natural parameters, this is a sum:

$$(\alpha_u - 1, -\beta_u) = (\alpha_{\text{prior}} - 1, -\beta_{\text{prior}}) + \sum_{e \ni u} (\alpha_{f_e} - 1, -\beta_{f_e})$$

Probability Aside – Expectation Propagation vs. Variational Bayes

EP and variational Bayes (VB) are both methods for approximating a posterior p with a simpler distribution q . The difference lies in *which* KL divergence they minimize:

- **VB** minimizes $\text{KL}(q||p)$ (the “exclusive” or “reverse” KL). This tends to make q concentrate on a single mode and underestimate uncertainty.
- **EP** minimizes $\text{KL}(p||q)$ (the “inclusive” or “forward” KL). This forces q to *cover* all of p , typically yielding better-calibrated uncertainty.

For tsdate, EP’s inclusive KL is important: we want the gamma approximation to reflect the full spread of each node’s age uncertainty, not just the mode.

Initialization

1. Set each node’s posterior to its coalescent prior: $q(t_u) = \text{Gamma}(\alpha_{\text{prior}}, \beta_{\text{prior}})$
2. Set all edge factors to “uninformative”: $f_e^{\rightarrow u} = \text{Gamma}(1, 0)$ (i.e., natural parameters $(0, 0)$)

```

def initialize_ep(ts, prior_grid):
    """Initialize EP state.

    Parameters
    -----
    ts : tskit.TreeSequence
    prior_grid : dict
        {node_id: (alpha, beta)} from the coalescent prior.

```

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```

Returns
-----
posteriors : dict
    {node_id: GammaDistribution}
edge_factors : dict
    {(edge_id, direction): GammaDistribution}
    direction is 'rootward' (to parent) or 'leafward' (to child)
"""
posteriors = {}
for node in ts.nodes():
    if node.id in prior_grid:
        alpha, beta = prior_grid[node.id]
        # Start with the coalescent prior as the initial belief
        posteriors[node.id] = GammaDistribution(alpha, beta)
    else:
        # Sample nodes: fixed at time 0 (very tight distribution)
        posteriors[node.id] = GammaDistribution(1.0, 1e10)

# Initialize all edge factors to uninformative Gamma(1, 0)
# In natural parameters this is (0, 0) -- contributes nothing
edge_factors = {}
for edge in ts.edges():
    edge_factors[(edge.id, 'rootward')] = GammaDistribution(1.0, 0.0)
    edge_factors[(edge.id, 'leafward')] = GammaDistribution(1.0, 0.0)

return posteriors, edge_factors

```

The EP Update for One Edge

This is the heart of the algorithm. For edge $e = (u, v)$ with m_e mutations and span-weighted rate λ_e :

i Biology Aside – What an EP update does, biologically

Each edge in the tree sequence connects a parent (ancestor) to a child (descendant). The edge carries mutations whose count constrains the time difference between parent and child. The EP update for one edge asks: *given what we currently believe about the parent's age and the child's age, and given the number of mutations on this edge, how should we revise our beliefs?* If many mutations sit on a short edge, the parent must be much older than the child. If no mutations sit on a long edge, parent and child are probably close in time. The four steps below formalize this intuition as a sequence of mathematical operations.

Step 1: Compute the “cavity” distributions.

Remove the current edge's messages from the parent and child posteriors:

$$q_{\setminus e}(t_u) = \frac{q(t_u)}{f_{e \rightarrow u}(t_u)}, \quad q_{\setminus e}(t_v) = \frac{q(t_v)}{f_{e \rightarrow v}(t_v)}$$

In natural parameters, this is subtraction.

Intuition: The cavity is “what we'd believe about this node if we forgot everything this particular edge told us.” It's the belief from all *other* sources of information. In the gear train, it is like temporarily disengaging one spring to see where the gear would sit under the tension of all the other springs.

Step 2: Compute the “tilted” distribution.

The tilted distribution is the cavity times the *true* edge likelihood:

$$\tilde{p}(t_u, t_v) = q_{\setminus e}(t_u) \cdot q_{\setminus e}(t_v) \cdot \frac{(\lambda_e(t_u - t_v))^{m_e}}{m_e!} e^{-\lambda_e(t_u - t_v)} \cdot \mathbb{I}[t_u > t_v]$$

This is the *exact* posterior we'd get for these two nodes if this were the only edge in the graph. It's generally **not** a product of two gammas – the $(t_u - t_v)^{m_e}$ term couples the variables.

Step 3: Moment matching.

Compute the marginal moments of the tilted distribution:

$$\tilde{\mu}_u = \mathbb{E}_{\tilde{p}}[t_u], \quad \tilde{\sigma}_u^2 = \text{Var}_{\tilde{p}}(t_u)$$

$$\tilde{\mu}_v = \mathbb{E}_{\tilde{p}}[t_v], \quad \tilde{\sigma}_v^2 = \text{Var}_{\tilde{p}}(t_v)$$

Then find the gamma distributions that match these moments:

$$q_{\text{new}}(t_u) = \text{Gamma}\left(\frac{\tilde{\mu}_u^2}{\tilde{\sigma}_u^2}, \frac{\tilde{\mu}_u}{\tilde{\sigma}_u^2}\right)$$

Probability Aside – What is moment matching?

Moment matching is the simplest way to project a complex distribution onto a simpler family. Given a distribution $\tilde{p}$ (the tilted distribution, which is not gamma), we compute its mean and variance, then find the unique $\text{Gamma}(\alpha, \beta)$ with the same mean and variance. This is the gamma that is “closest” to $\tilde{p}$ in the sense of matching first and second moments. It is the same method-of-moments idea used for the coalescent prior (*The Coalescent Prior*), but here applied at every EP iteration to every edge.

Step 4: Update the edge factors.

The new message from edge e to node u is:

$$f_e^{\rightarrow u, \text{new}} = \frac{q_{\text{new}}(t_u)}{q_{\setminus e}(t_u)}$$

In natural parameters: subtract the cavity from the new posterior.

```
def ep_update_edge(edge, posteriors, edge_factors, m_e, lambda_e, damping=0.5):
    """Perform one EP update for a single edge.

    Parameters
    -----
    edge : tskit.Edge
    posteriors : dict of GammaDistribution
    edge_factors : dict of GammaDistribution
    m_e : int
        Mutations on this edge.
    lambda_e : float
        Span-weighted mutation rate.
    damping : float
        Damping factor in [0, 1]. 1 = no damping, 0.5 = half step.

    Returns
    -----
```

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```

posteriors, edge_factors : updated in place.
"""
u, v = edge.parent, edge.child

# Step 1: Compute cavities (remove this edge's old message)
cavity_u = posteriors[u].divide(edge_factors[(edge.id, 'rootward')])
cavity_v = posteriors[v].divide(edge_factors[(edge.id, 'leafward')])

# Step 2 & 3: Compute tilted moments and moment-match
# This is the expensive part: we need E[t_u], Var(t_u), E[t_v], Var(t_v)
# under the tilted distribution
moments = compute_tilted_moments(cavity_u, cavity_v, m_e, lambda_e)

if moments is None:
    return # numerical failure, skip this edge

mu_u, var_u, mu_v, var_v = moments

# Moment-match to gammas (find the gamma with these moments)
new_post_u = GammaDistribution.from_moments(mu_u, var_u)
new_post_v = GammaDistribution.from_moments(mu_v, var_v)

# Step 4: Compute new edge factors = new_posterior / cavity
new_factor_u = new_post_u.divide(cavity_u)
new_factor_v = new_post_v.divide(cavity_v)

# Apply damping: interpolate between old and new factors
# in natural parameter space to prevent oscillation
old_factor_u = edge_factors[(edge.id, 'rootward')]
old_factor_v = edge_factors[(edge.id, 'leafward')]

damped_u = GammaDistribution(
    old_factor_u.alpha + damping * (new_factor_u.alpha - old_factor_u.alpha),
    old_factor_u.beta + damping * (new_factor_u.beta - old_factor_u.beta)
)
damped_v = GammaDistribution(
    old_factor_v.alpha + damping * (new_factor_v.alpha - old_factor_v.alpha),
    old_factor_v.beta + damping * (new_factor_v.beta - old_factor_v.beta)
)

# Update the stored edge factors
edge_factors[(edge.id, 'rootward')] = damped_u
edge_factors[(edge.id, 'leafward')] = damped_v

# Recompute posteriors: cavity * new_factor
posteriors[u] = cavity_u.multiply(damped_u)
posteriors[v] = cavity_v.multiply(damped_v)

```

Computing the Tilted Moments

The hardest part of EP is computing the moments of the tilted distribution. For the Poisson-gamma case, this involves integrals of the form:

$$\mathbb{E}_{\tilde{p}}[t_u] = \frac{\int_0^\infty \int_0^{t_u} t_u \cdot q_{\setminus e}(t_u) \cdot q_{\setminus e}(t_v) \cdot (\lambda_e(t_u - t_v))^{m_e} e^{-\lambda_e(t_u - t_v)} dt_v dt_u}{\int_0^\infty \int_0^{t_u} q_{\setminus e}(t_u) \cdot q_{\setminus e}(t_v) \cdot (\lambda_e(t_u - t_v))^{m_e} e^{-\lambda_e(t_u - t_v)} dt_v dt_u}$$

Plain-language summary – Why these integrals are hard

The difficulty arises because the parent must be older than the child ($t_u > t_v$), and the number of mutations depends on the time *difference* $t_u - t_v$. This couples the two variables: you cannot estimate the parent's age independently of the child's. The integral averages over all possible (parent age, child age) combinations that are consistent with both the mutation data on this edge *and* the information from all other edges (encoded in the cavity). Computing this average exactly would require evaluating a two-dimensional integral for each of the millions of edges in a tree sequence – which is why approximations are essential.

These integrals don't have closed forms in general. tsdate evaluates them using a combination of:

1. **Laplace approximation:** Find the mode of the tilted distribution and approximate with a Gaussian around it.
2. **Numerical quadrature:** For edges with very few mutations, use direct numerical integration.
3. **Special cases:** When $m_e = 0$, the likelihood simplifies to a pure exponential, and some integrals become tractable.

The Laplace Approach

The Laplace approximation finds the mode ($\hat{t}_u, \hat{t}_v$) of the tilted distribution by solving:

$$\frac{\partial}{\partial t_u} \log \tilde{p}(t_u, t_v) = 0, \quad \frac{\partial}{\partial t_v} \log \tilde{p}(t_u, t_v) = 0$$

Then approximates the tilted distribution as a bivariate Gaussian centered at the mode, with covariance given by the negative inverse Hessian:

$$\tilde{p}(t_u, t_v) \approx \mathcal{N} \left(\begin{pmatrix} \hat{t}_u \\ \hat{t}_v \end{pmatrix}, \mathbf{H}^{-1} \right)$$

where $\mathbf{H}$ is the Hessian of $-\log \tilde{p}$ at the mode.

Calculus Aside – The Laplace approximation

The Laplace approximation is a technique for approximating integrals of the form $\int e^{f(x)} dx$. The idea: expand $f(x)$ in a Taylor series around its maximum $\hat{x}$:

$$f(x) \approx f(\hat{x}) + \frac{1}{2}(x - \hat{x})^T H(x - \hat{x})$$

where $H = \nabla^2 f(\hat{x})$ is the Hessian (matrix of second derivatives). The integral then becomes a Gaussian integral with known closed form. In our case $f = \log \tilde{p}$, so the Laplace approximation replaces the tilted distribution with a Gaussian centered at its mode. The quality of this approximation improves as the tilted distribution becomes more peaked (more data), which is why it works well for edges with many mutations.

```

from scipy.optimize import minimize

def compute_tilted_moments(cavity_u, cavity_v, m_e, lambda_e):
    """Compute moments of the tilted distribution via Laplace approximation.

    Parameters
    -----
    cavity_u, cavity_v : GammaDistribution
        Cavity distributions for parent and child.
    m_e : int
        Mutation count.
    lambda_e : float
        Span-weighted mutation rate.

    Returns
    -----
    mu_u, var_u, mu_v, var_v : float
        Moments of the tilted marginals, or None if numerical failure.
    """
    def neg_log_tilted(params):
        """Negative log of the tilted distribution (to be minimized)."""
        t_u, t_v = params
        if t_u <= t_v or t_u <= 0 or t_v < 0:
            return 1e20 # constraint violation

        delta = t_u - t_v

        # Log cavity contributions (gamma log-pdf, unnormalized)
        log_cavity_u = (cavity_u.alpha - 1) * np.log(t_u) - cavity_u.beta * t_u
        log_cavity_v = (cavity_v.alpha - 1) * np.log(max(t_v, 1e-20)) - cavity_v.beta *
→ * t_v

        # Log Poisson likelihood: m*log(lambda*delta) - lambda*delta
        log_lik = m_e * np.log(lambda_e * delta) - lambda_e * delta

        return -(log_cavity_u + log_cavity_v + log_lik)

    # Initial guess: cavity means
    t_u_init = max(cavity_u.mean, 1e-6)
    t_v_init = max(cavity_v.mean, 1e-6)
    if t_u_init <= t_v_init:
        t_u_init = t_v_init + 1.0 # ensure parent is older than child

    result = minimize(neg_log_tilted, [t_u_init, t_v_init],
                      method='Nelder-Mead')

    if not result.success:
        return None

    t_u_hat, t_v_hat = result.x

    # Compute Hessian numerically for the Laplace approximation
    H = numerical_hessian(neg_log_tilted, [t_u_hat, t_v_hat])

```

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```

try:
    cov = np.linalg.inv(H) # covariance = inverse Hessian
except np.linalg.LinAlgError:
    return None

# Marginal moments from the Gaussian approximation
mu_u = t_u_hat # mode ~ mean for peaked distributions
var_u = max(cov[0, 0], 1e-20) # diagonal of covariance = marginal variance
mu_v = t_v_hat
var_v = max(cov[1, 1], 1e-20)

return mu_u, var_u, mu_v, var_v

def numerical_hessian(f, x, eps=1e-5):
    """Compute the Hessian of f at x via finite differences.

    Uses the standard 4-point formula for mixed partial derivatives:
    d^2f/dxidxj ~ (f(+,+) - f(+,-) - f(-,+) + f(-,-)) / (4*eps^2)
    """
    n = len(x)
    H = np.zeros((n, n))
    f0 = f(x)
    for i in range(n):
        for j in range(i, n):
            x_pp = x.copy()
            x_pp[i] += eps
            x_pp[j] += eps
            x_pm = x.copy()
            x_pm[i] += eps
            x_pm[j] -= eps
            x_mp = x.copy()
            x_mp[i] -= eps
            x_mp[j] += eps
            x_mm = x.copy()
            x_mm[i] -= eps
            x_mm[j] -= eps
            H[i, j] = (f(x_pp) - f(x_pm) - f(x_mp) + f(x_mm)) / (4 * eps**2)
            H[j, i] = H[i, j] # Hessian is symmetric
    return H

```

Damping: Preventing Oscillation

EP updates can overshoot, causing oscillation or divergence. `tsdate` uses **damping** to stabilize convergence: instead of fully replacing the old message with the new one, it takes a weighted average.

In natural parameter space:

$$\eta^{\text{damped}} = (1 - d) \cdot \eta^{\text{old}} + d \cdot \eta^{\text{new}}$$

where $d \in (0, 1]$ is the damping factor. A typical value is $d = 0.5$ (half-step).

Why does this help? Each EP update is based on a *local* approximation (one edge at a time). If the approximation is poor, the update might push the parameters too far. Damping ensures we only move a fraction of the way, giving the

other edges a chance to “catch up” before the next iteration.

In the watch metaphor, damping is like the balance wheel’s hairspring: it prevents the mechanism from swinging too far in response to a single impulse, allowing it to settle smoothly into the correct position.

Convergence

i Biology Aside – Convergence means consistent dating

Convergence of EP means that the inferred ages of all nodes in the genealogy have become mutually consistent. Each node’s age agrees with the mutation evidence on every incident edge, with the coalescent prior (old nodes should have ages consistent with the expected coalescent times), and with the ages of its parents and children. When the algorithm converges, the age assignments satisfy all these constraints simultaneously – or at least as well as the gamma approximation allows. In practice, ~25 iterations suffice for tree sequences with millions of nodes.

EP iterates over all edges multiple times. Convergence is monitored by checking whether the posteriors change significantly between iterations:

$$\max_u \frac{|\alpha_u^{(t+1)} - \alpha_u^{(t)}|}{|\alpha_u^{(t)}|} < \epsilon$$

tsdate defaults to 25 iterations, which is usually sufficient.

```
def run_ep(ts, mutation_rate, prior_grid, max_iter=25, damping=0.5, tol=1e-6):
    """Run the full EP algorithm.

    Parameters
    -----
    ts : tskit.TreeSequence
    mutation_rate : float
    prior_grid : dict
    max_iter : int
    damping : float
    tol : float

    Returns
    -----
    posteriors : dict of GammaDistribution
    """
    posteriors, edge_factors = initialize_ep(ts, prior_grid)

    # Count mutations per edge (once, before the iteration loop)
    mut_per_edge = np.zeros(ts.num_edges, dtype=int)
    for mut in ts.mutations():
        if mut.edge >= 0:
            mut_per_edge[mut.edge] += 1

    for iteration in range(max_iter):
        max_change = 0.0 # track largest parameter change for convergence

        for edge in ts.edges():
            m_e = mut_per_edge[edge.id]
            span = edge.right - edge.left
```

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```

lambda_e = mutation_rate * span # span-weighted mutation rate

old_alpha = posteriors[edge.parent].alpha # save for convergence check

# The core EP update: cavity -> tilted -> moment match -> new factor
ep_update_edge(edge, posteriors, edge_factors,
               m_e, lambda_e, damping)

# Track convergence: relative change in alpha
change = abs(posteriors[edge.parent].alpha - old_alpha)
rel_change = change / max(abs(old_alpha), 1e-10)
max_change = max(max_change, rel_change)

if max_change < tol:
    print(f"EP converged after {iteration + 1} iterations")
    break

return posteriors

```

What EP Minimizes: The KL Divergence

EP's fixed point (when it converges) approximately minimizes the **inclusive Kullback-Leibler divergence**:

$$\text{KL}(p||q) = \int p(\mathbf{t}) \log \frac{p(\mathbf{t})}{q(\mathbf{t})} d\mathbf{t}$$

where p is the true posterior and q is the gamma approximation.

Why “inclusive” KL? This is $\text{KL}(p||q)$, not $\text{KL}(q||p)$. The inclusive KL penalizes q for having zero density where p has mass. This means the approximation tends to **cover** the true posterior rather than concentrating on a single mode.

Contrast with variational Bayes: Standard variational inference minimizes $\text{KL}(q||p)$ (the “exclusive” KL), which tends to underestimate uncertainty. EP's inclusive KL typically gives better-calibrated uncertainty estimates.

i Probability Aside – KL divergence in 60 seconds

The Kullback-Leibler divergence $KL(p||q)$ measures how much information is lost when we use q to approximate p . It is always ≥ 0 , and equals zero only when $p = q$. Importantly, it is *not* symmetric: $KL(p||q) \neq KL(q||p)$. The inclusive direction $KL(p||q)$ heavily penalizes q for placing zero density where p is positive (missing mass), so the minimizer q^* spreads out to cover p . The exclusive direction $KL(q||p)$ penalizes q for placing density where p is zero (extra mass), so the minimizer q^* concentrates inside p . For uncertainty quantification, the inclusive direction (used by EP) is generally more conservative and better calibrated.

Comparison with Inside-Outside

Feature	Inside-Outside	Variational Gamma
Time representation	Discrete grid (K points)	Continuous (gamma distribution)
Parameters per node	K (probability vector)	2 (α, β)
Cost per edge per iteration	$O(K^2)$	$O(1)$
Resolution	Limited by grid	Unlimited
Posterior output	Full distribution (on grid)	Mean + variance (gamma)
Convergence	1 pass (exact on trees)	~25 iterations
Handles loops	Approximate	Approximate (EP)

Putting It All Together

```
def variational_gamma_date(ts, mutation_rate, Ne=1.0, max_iter=25):
    """Date a tree sequence using the variational gamma method.

    Parameters
    -----
    ts : tskit.TreeSequence
    mutation_rate : float
    Ne : float
    max_iter : int

    Returns
    -----
    node_times : np.ndarray
    """
    # Build coalescent prior (Gear 1)
    prior_grid = {}
    for node in ts.nodes():
        if node.id not in set(ts.samples()):
            k = count_sample_descendants(ts, node.id)
            # Mean and variance from conditional coalescent
            mean = sum(2.0 / (j * (j-1)) for j in range(2, max(k, 2) + 1))
            var = sum(4.0 / (j * (j-1))**2 for j in range(2, max(k, 2) + 1))
            # Convert to gamma parameters via method of moments
            alpha = mean**2 / var
            beta = mean / var
            prior_grid[node.id] = (alpha, beta)

    # Run EP (messages flow through the gear train until convergence)
```

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```

posteriors = run_ep(ts, mutation_rate, prior_grid, max_iter)

# Extract posterior means as point estimates
node_times = np.zeros(ts.num_nodes)
for u in range(ts.num_nodes):
    if u in posteriors:
        node_times[u] = posteriors[u].mean

# Fix samples at time 0 (known ages)
for s in ts.samples():
    node_times[s] = 0.0

return node_times

```

Summary

The variational gamma method dates nodes through:

1. **Gamma approximation:** Each node's posterior is $q(t_u) = \text{Gamma}(\alpha_u, \beta_u)$
2. **Expectation propagation:** For each edge, compute the exact moments of the tilted distribution (cavity $\times$ true likelihood), then moment-match back to gammas
3. **Damping:** Stabilize updates by interpolating between old and new messages
4. **Iteration:** Repeat over all edges until convergence (~25 iterations)

The key equations:

$$q_{\setminus e}(t_u) = q(t_u) / f_e^{\rightarrow u}(t_u) \quad (\text{cavity})$$

$$q_{\text{new}}(t_u) = \text{Gamma}(\tilde{\mu}_u^2 / \tilde{\sigma}_u^2, \tilde{\mu}_u / \tilde{\sigma}_u^2) \quad (\text{moment match})$$

$$f_e^{\rightarrow u, \text{new}} = q_{\text{new}}(t_u) / q_{\setminus e}(t_u) \quad (\text{new message})$$

This method is faster, more accurate, and higher-resolution than inside-outside, which is why it's the default in modern tsdate. In the watch metaphor, it is the same gear train carrying the same messages, but now the messages speak a more efficient language – two numbers per gear instead of a whole grid – and the mechanism converges more quickly to the correct time.

Next: the final gear, rescaling – adjusting the inferred times to match the empirical mutation clock (*Rescaling*).

Rescaling

Even a well-tuned mechanism drifts; the final step is calibrating against the master clock.

After belief propagation (whether inside-outside or variational gamma), tsdate has posterior estimates for every node's age. But these estimates assume a **constant effective population size**, which is almost never true. If the population was larger in the past, branches are too long; if it was smaller, they're too short.

Rescaling corrects for this by comparing the inferred times against the **empirical mutation clock**: the observed number of mutations in different time windows. In the watch metaphor, this is the final calibration – checking the movement against a master reference clock and adjusting the hands until the ticks match.

This is the same idea used in SINGER's ARG rescaling (see *ARG rescaling*), adapted for tsdate's continuous posteriors.

i Prerequisites

This chapter assumes you have followed the full tsdate pipeline so far: the coalescent prior (*The Coalescent Prior*), the mutation likelihood (*The Mutation Likelihood*), and one of the two message passing algorithms (*Inside-Outside Belief Propagation* or *Variational Gamma (Expectation Propagation)*). Rescaling is a post-processing step applied to the node times that those algorithms produce.

The Problem: Mismatch Between Model and Reality

The coalescent prior assumes constant N_e . Under this model, the expected time between coalescence events is fixed. But real populations have complex histories: bottlenecks, expansions, migrations.

When N_e was *larger* in the past:

- Real coalescence events were *slower* (more time between them)
- The constant- N_e model underestimates deep times
- Branches in the deep past are too short

When N_e was *smaller* in the past:

- Real coalescence events were *faster*
- The model overestimates deep times
- Branches in the deep past are too long

Either way, the mutation rate implied by the inferred times won't match the true mutation rate. Rescaling fixes this.

Think of it as a watch whose mainspring weakens with age: the gears near the present run at the right speed, but the deeper you go, the more the rate drifts. Rescaling adjusts the time scale in each epoch so that the ticks (mutations) per unit time remain constant.

The Key Insight: Mutations Don't Lie

Whatever the population history, the molecular clock still ticks at rate μ per base pair per generation. The total number of mutations in a time window is:

$$\text{mutations in } [t_a, t_b) = \mu \cdot (\text{total branch length in } [t_a, t_b))$$

If our estimated times are correct, the ratio observed mutations/expected mutations should be 1.0 in every time window. If it's consistently > 1 in some window, our branch lengths there are too short, and we need to stretch time. If it's < 1 , we need to compress.

This ratio is a direct diagnostic: any deviation from 1.0 reveals how much the constant- N_e assumption has distorted the time scale in that epoch.

Step 1: Partition Time into Windows

Divide the time axis $[0, t_{\max})$ into J windows such that each window contains approximately equal total branch length:

$$[t_0 = 0, t_1), \quad [t_1, t_2), \quad \dots, \quad [t_{J-1}, t_J)$$

Why equal branch length? So that each window has comparable statistical power for estimating the local mutation rate. A window with very little branch length would have very few mutations and a noisy estimate.

```

import numpy as np

def partition_time_axis(ts, node_times, J=1000):
    """Partition the time axis into J windows of roughly equal branch length.

    Parameters
    -----
    ts : tskit.TreeSequence
    node_times : np.ndarray
        Current estimated node times (from EP or inside-outside).
    J : int
        Number of windows.

    Returns
    -----
    breakpoints : np.ndarray, shape (J+1,)
        Window boundaries: [breakpoints[j], breakpoints[j+1]) for window j.
    """
    # Collect all branch lengths weighted by span
    branch_data = [] # (midpoint_time, weighted_length)

    for edge in ts.edges():
        t_parent = node_times[edge.parent]
        t_child = node_times[edge.child]
        span = edge.right - edge.left # genomic span in bp

        if t_parent > t_child:
            midpoint = (t_parent + t_child) / 2 # time midpoint of the edge
            weighted_length = span * (t_parent - t_child) # bp * generations
            branch_data.append((midpoint, weighted_length))

    if not branch_data:
        return np.linspace(0, 1, J + 1)

    # Sort by time and find breakpoints with equal cumulative branch length
    branch_data.sort(key=lambda x: x[0])
    times = np.array([b[0] for b in branch_data])
    lengths = np.array([b[1] for b in branch_data])

    cum_length = np.cumsum(lengths) # running total of branch length
    total_length = cum_length[-1]

    breakpoints = [0.0]
    target_per_window = total_length / J # each window gets equal share

    for j in range(1, J):
        target = j * target_per_window
        idx = np.searchsorted(cum_length, target) # find where cumulative exceeds_
        ↪target
        if idx < len(times):
            breakpoints.append(times[idx])
        else:
            breakpoints.append(times[-1])

```

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```
breakpoints.append(node_times.max() * 1.01)  # upper bound beyond all nodes
return np.array(breakpoints)
```

Step 2: Count Mutations per Window

For each time window, count how many mutations fall in it. A mutation on edge e is assigned to the time window containing the midpoint of the edge (or, more precisely, proportionally distributed across windows that the edge spans).

```
def count_mutations_per_window(ts, node_times, breakpoints, mutation_rate):
    """Count observed and expected mutations in each time window.

    Parameters
    -----
    ts : tskit.TreeSequence
    node_times : np.ndarray
    breakpoints : np.ndarray, shape (J+1,)
    mutation_rate : float

    Returns
    -----
    observed : np.ndarray, shape (J,)
        Mutation count in each window.
    expected : np.ndarray, shape (J,)
        Expected mutations (mu * total branch length) in each window.
    """
    J = len(breakpoints) - 1
    observed = np.zeros(J)
    expected = np.zeros(J)

    # Count mutations per edge (once)
    mut_per_edge = np.zeros(ts.num_edges, dtype=int)
    for mut in ts.mutations():
        if mut.edge >= 0:
            mut_per_edge[mut.edge] += 1

    for edge in ts.edges():
        t_parent = node_times[edge.parent]
        t_child = node_times[edge.child]
        span = edge.right - edge.left  # genomic span
        m_e = mut_per_edge[edge.id]    # observed mutations on this edge

        if t_parent <= t_child:
            continue  # skip edges with zero or negative branch length

        # Distribute this edge's contribution across windows
        for j in range(J):
            w_lo = breakpoints[j]
            w_hi = breakpoints[j + 1]

            # Overlap of edge [t_child, t_parent] with window [w_lo, w_hi]
            overlap_lo = max(t_child, w_lo)
```

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```

overlap_hi = min(t_parent, w_hi)

if overlap_hi > overlap_lo:
    # Fraction of edge in this window
    frac = (overlap_hi - overlap_lo) / (t_parent - t_child)

    # Expected mutations: mu * span * overlap_length
    expected[j] += mutation_rate * span * (overlap_hi - overlap_lo)

    # Observed mutations: distribute proportionally by time overlap
    observed[j] += m_e * frac

return observed, expected

```

Step 3: Compute Scaling Factors

For each window, the scaling factor is the ratio of observed to expected mutations:

$$s_j = \frac{\text{observed}_j}{\text{expected}_j}$$

If $s_j > 1$, the branch lengths in window j are too short (need stretching). If $s_j < 1$, they're too long (need compression).

```

def compute_scaling_factors(observed, expected, min_count=1.0):
    """Compute per-window scaling factors.

    Parameters
    -----
    observed, expected : np.ndarray, shape (J,)
    min_count : float
        Minimum mutation count to trust a window.

    Returns
    -----
    scales : np.ndarray, shape (J,)
    """
    scales = np.ones(len(observed)) # default: no rescaling

    for j in range(len(observed)):
        if expected[j] > 0 and observed[j] >= min_count:
            # Ratio > 1 means branches too short; < 1 means too long
            scales[j] = observed[j] / expected[j]

    return scales

```

Intuition: This is a piecewise-constant estimate of $N_e(t)$. If the model assumed $N_e = 10,000$ but the true N_e was 20,000 during window j , then branch lengths are half what they should be, and $s_j \approx 2$.

i Probability Aside – Rescaling as implicit N_e estimation

The scaling factor s_j is closely related to the ratio of true N_e to assumed N_e in window j . Under the coalescent with variable population size, the coalescent rate is $1/N_e(t)$. If we used the wrong N_e , the branch lengths in that epoch are

stretched or compressed by the ratio $N_e^{\text{true}}/N_e^{\text{assumed}}$. By setting $s_j = \text{observed}_j/\text{expected}_j$, we effectively estimate this ratio and correct for it – without explicitly fitting a population-size model. This is similar in spirit to how PSMC (*Timepiece I: PSMC*) estimates $N_e(t)$, except here it is a post-processing step rather than the main inference.

Step 4: Apply the Rescaling

Each node's time is adjusted by the cumulative scaling factor up to its current time:

$$t_u^{\text{new}} = \int_0^{t_u^{\text{old}}} s(x) dx$$

where $s(x)$ is the piecewise-constant scaling function. This integral is just a sum:

$$t_u^{\text{new}} = \sum_{j: t_j < t_u} s_j \cdot \min(t_u, t_{j+1}) - t_j + s_{j^*} \cdot (t_u - t_{j^*})$$

where j^* is the window containing t_u .

i Calculus Aside – Piecewise integration

The rescaling integral $\int_0^t s(x) dx$ with piecewise-constant $s(x)$ decomposes into a sum of rectangles: in each window $[t_j, t_{j+1})$ where $s(x) = s_j$, the contribution is $s_j \cdot (t_{j+1} - t_j)$. For the final (partial) window containing t , the contribution is $s_{j^*} \cdot (t - t_{j^*})$. The result is a piecewise-linear, monotonically increasing function of the original time – a “warped” time axis that stretches or compresses different epochs.

```
def apply_rescaling(node_times, breakpoints, scales, fixed_nodes):
    """Apply piecewise rescaling to node times.

    Parameters
    -----
    node_times : np.ndarray
        Current node times (will not be modified).
    breakpoints : np.ndarray, shape (J+1,)
    scales : np.ndarray, shape (J,)
    fixed_nodes : set
        Nodes whose times should not change (e.g., samples).

    Returns
    -----
    new_times : np.ndarray
        Rescaled node times.
    """
    new_times = np.zeros_like(node_times)
    J = len(scales)

    # Build cumulative scaling function
    # cum_rescaled[j] = rescaled time at window boundary j
    cum_rescaled = np.zeros(J + 1)
    for j in range(J):
        window_width = breakpoints[j + 1] - breakpoints[j]
        cum_rescaled[j + 1] = cum_rescaled[j] + scales[j] * window_width
```

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```

for u in range(len(node_times)):
    if u in fixed_nodes:
        new_times[u] = node_times[u]  # samples stay fixed
        continue

    t = node_times[u]

    # Find which window t falls in
    j = np.searchsorted(breakpoints, t, side='right') - 1
    j = min(j, J - 1)
    j = max(j, 0)

    # Rescaled time = cumulative up to window j + fraction within window
    fraction_in_window = t - breakpoints[j]
    new_times[u] = cum_rescaled[j] + scales[j] * fraction_in_window

return new_times

```

Iterating the Rescaling

A single round of rescaling may not be sufficient because the window boundaries depend on the node times, which change after rescaling. `tsdate` iterates:

1. Compute window boundaries from current times
2. Count mutations per window
3. Compute scaling factors
4. Apply rescaling to get new times
5. Repeat (default: 5 iterations)

Each iteration refines the time scale, like adjusting a regulator screw on a mechanical watch – small turns that progressively bring the rate into alignment with the master clock.

```

def iterative_rescaling(ts, node_times, mutation_rate, fixed_nodes,
                       J=1000, num_iter=5):
    """Iteratively rescale node times to match the mutation clock.

    Parameters
    -----
    ts : tskit.TreeSequence
    node_times : np.ndarray
    mutation_rate : float
    fixed_nodes : set
    J : int
        Number of time windows.
    num_iter : int
        Number of rescaling iterations.

    Returns
    -----
    node_times : np.ndarray
    """

```

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```

    Rescaled node times.
    """
    times = node_times.copy()

    for iteration in range(num_iter):
        # 1. Partition time into equal-branch-length windows
        breakpoints = partition_time_axis(ts, times, J)

        # 2. Count observed vs. expected mutations per window
        observed, expected = count_mutations_per_window(
            ts, times, breakpoints, mutation_rate)

        # 3. Compute scaling factors (observed / expected)
        scales = compute_scaling_factors(observed, expected)

        # 4. Apply piecewise rescaling
        times = apply_rescaling(times, breakpoints, scales, fixed_nodes)

    return times

```

Connection to Population Size History

The scaling factors s_j are intimately related to the effective population size history. Under the coalescent with variable $N_e(t)$:

- The rate of coalescence at time t is $1/N_e(t)$
- The mutation rate is constant at μ

If we model the coalescent under constant $N_e^{(0)}$ but the true population size in window j is $N_e^{(j)}$, then:

$$s_j \approx \frac{N_e^{(j)}}{N_e^{(0)}}$$

So the rescaling implicitly estimates the population size history. This is similar to what PSMC does (see the *Timepiece I: PSMC*), but here it's a post-processing step rather than the main inference.

Edge Cases and Robustness

Several practical issues arise:

Windows with few mutations: If a window has very few mutations (or none), the scaling factor is unreliable. `tsdate` handles this by:

- Setting a minimum count threshold
- Smoothing adjacent scaling factors
- Falling back to a scale of 1.0 for empty windows

Negative branch lengths: After rescaling, some edges might end up with the parent younger than the child. `tsdate` enforces constraints by adjusting times to maintain the topological ordering.

Convergence: Rescaling typically converges within 3-5 iterations. The scaling factors stabilize as the times settle into their correct positions.

The Full Pipeline

Putting rescaling together with EP, here is the complete tsdate pipeline from raw tree sequence to dated genealogy.

```
def tsdate_full_pipeline(ts, mutation_rate, Ne=1.0, max_ep_iter=25,
                        rescaling_intervals=1000, rescaling_iterations=5):
    """The complete tsdate pipeline.

    Parameters
    -----
    ts : tskit.TreeSequence
        Input (topology from tsinfer).
    mutation_rate : float
    Ne : float
    max_ep_iter : int
    rescaling_intervals : int
    rescaling_iterations : int

    Returns
    -----
    dated_ts : np.ndarray
        Posterior mean node times.
    """
    # Step 1: Build priors (Gear 1 -- the expected beat rate)
    prior_grid = build_coalescent_priors(ts, Ne)

    # Step 2: Run EP (Gear 4 -- messages flow through the gear train)
    posteriors = run_ep(ts, mutation_rate, prior_grid, max_ep_iter)

    # Step 3: Extract posterior means
    node_times = np.zeros(ts.num_nodes)
    for u in range(ts.num_nodes):
        if u in posteriors:
            node_times[u] = posteriors[u].mean

    fixed_nodes = set(ts.samples())
    for s in fixed_nodes:
        node_times[s] = 0.0

    # Step 4: Rescale (Gear 5 -- calibrate against the mutation clock)
    if rescaling_iterations > 0:
        node_times = iterative_rescaling(
            ts, node_times, mutation_rate, fixed_nodes,
            J=rescaling_intervals,
            num_iter=rescaling_iterations
        )

    return node_times
```

Summary

Rescaling is tsdate's final calibration step – the last gear in the mechanism:

1. **Partition** the time axis into J windows of equal branch length
2. **Count** observed vs. expected mutations per window
3. **Scale** each window by the ratio $s_j = \text{observed}/\text{expected}$
4. **Apply** the piecewise scaling to all node times
5. **Iterate** until convergence (default: 5 rounds)

The key equation:

$$t_u^{\text{new}} = \int_0^{t_u^{\text{old}}} \frac{\text{observed mutations}(x)}{\text{expected mutations}(x)} dx$$

This corrects for variable population size without explicitly modeling it, by letting the molecular clock be the final arbiter of time. In the watch metaphor, the mutation clock is the master reference – it ticks at a known rate (μ) regardless of population history, and rescaling adjusts every hand on the dial until the ticks match.

Congratulations – you've now built every gear of the tsdate mechanism:

1. **Coalescent prior** – the expected beat rate from coalescent theory: informed starting beliefs about node ages
2. **Mutation likelihood** – evidence from the mutation clock: the Poisson model connecting observed data to branch lengths
3. **Inside-outside** – messages flowing through the gear train on a discrete grid
4. **Variational gamma** – the same messages, now carried by continuous gamma distributions via expectation propagation
5. **Rescaling** – calibrating the clock against the master reference: adjusting for variable population size

Together, these gears transform a topology-only tree sequence (from tsinfer) into a fully dated genealogy. You understand the math, the code, and the intuition behind every step. The watch is assembled, calibrated, and keeping time.

Demo: Running tsdate on Simulated Data

Running mini_tsdate on simulated genomic data.

This figure was generated by running the `mini_tsdate` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_tsdate.py`.

Each chapter derives the math, explains the intuition, implements the code, and verifies it works. By the end, you'll have built a complete node-dating engine from scratch – and you'll understand every gear that drives it.

3.11 Timepiece X: moments

Inferring demographic history from the frequency spectrum – without solving a single PDE.

3.11.1 The Mechanism at a Glance

`moments` is a method for **demographic inference** – learning a population's history (size changes, splits, migrations) from patterns in its DNA variation. It works by computing how the **site frequency spectrum** (SFS) evolves through time under different demographic scenarios, then finding the scenario that best matches observed data.

If the other Timepieces in this book work by reconstructing genealogies (the tree structure connecting samples to their ancestors), `moments` takes a fundamentally different approach. It summarizes the genetic data into a compact statistic –

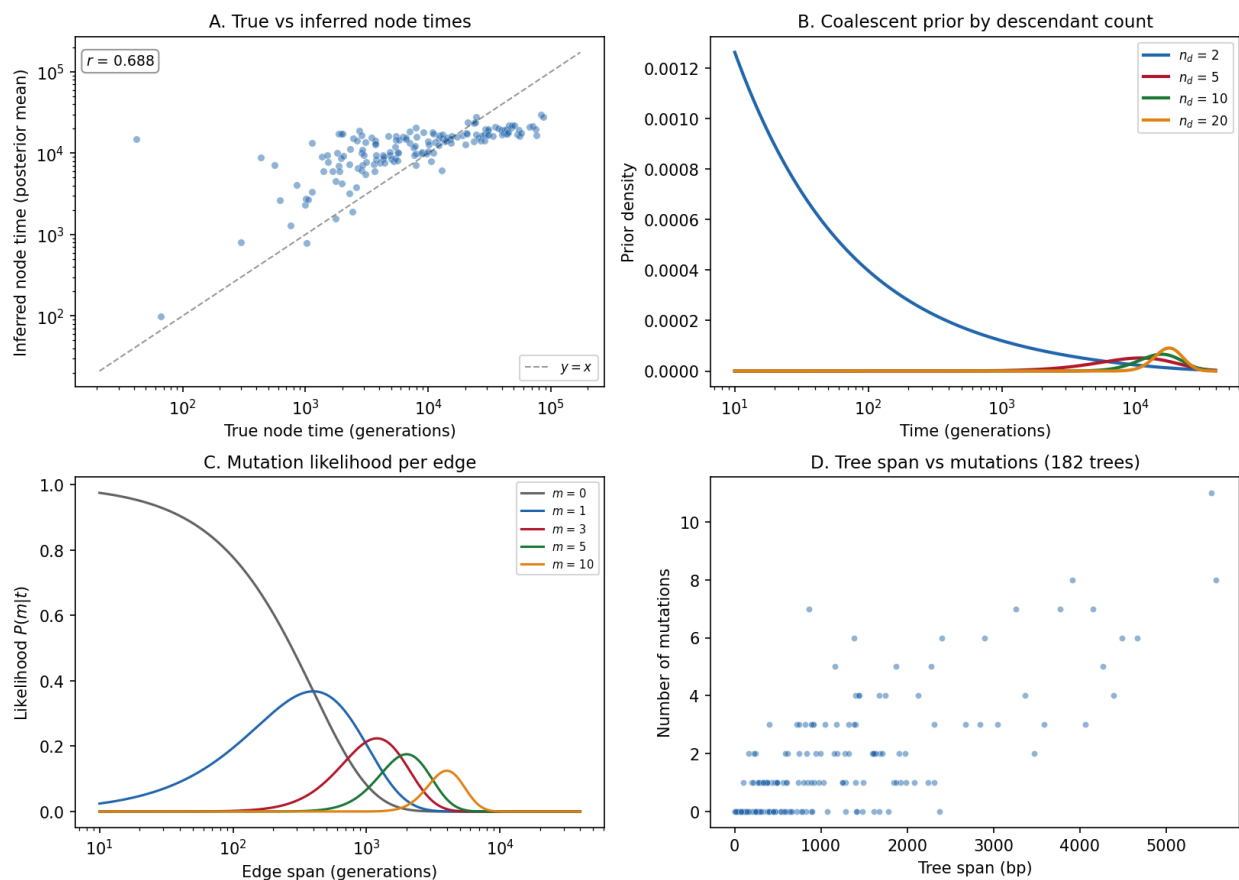
Demo: tsdate on msprime Tree Sequence (20 samples, 182 trees)

Fig. 11: **tsdate demo on msprime-simulated data.** Panel A: True vs inferred node times – gamma posterior means compared to true coalescence times. Panel B: Coalescent prior distributions by descendant count. Panel C: Mutation likelihood functions for different mutation counts. Panel D: Tree span vs number of mutations across the tree sequence.

the frequency spectrum – and works entirely in this summary space. Think of it as reading the time from the position of the hands without ever opening the case to see the gears. The mechanism is different, but the information comes from the same source: the coalescent process shaped by population history.

The key innovation: while its predecessor `dadi` solves a partial differential equation (the Wright-Fisher diffusion) on a frequency grid, `moments` bypasses this entirely. It derives **ordinary differential equations** that govern the SFS entries directly. No grid, no PDE, no numerical diffusion artifacts. Just a clean system of ODEs that you can solve with standard numerical methods.

i Primary Reference

[Jouganous *et al.*, 2017]

The four gears of `moments`:

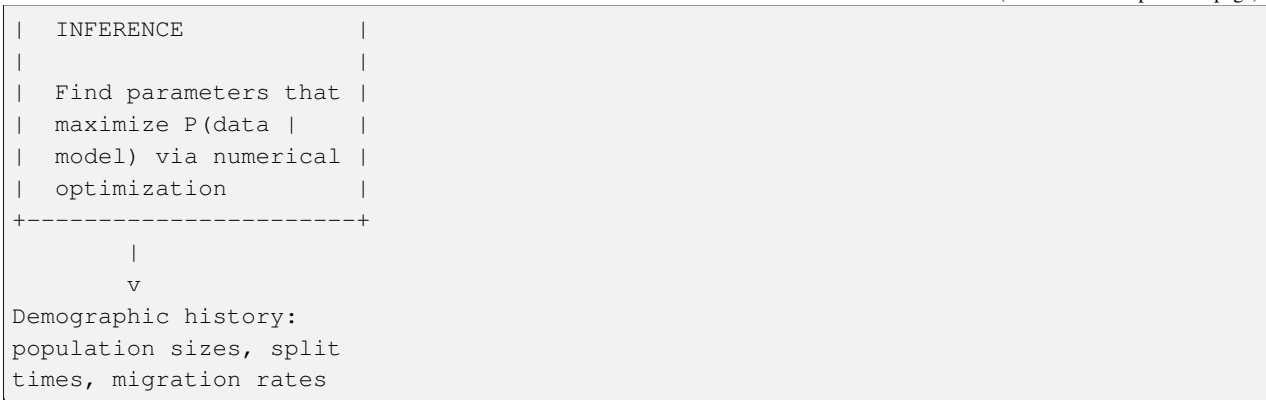
1. **The Frequency Spectrum** (the dial) – The summary statistic at the heart of everything: a histogram of allele frequencies across your sample. Different demographic histories produce different spectral signatures, and this is what `moments` reads.
2. **Moment Equations** (the escapement and gear train) – The mathematical engine: a system of ODEs describing how each SFS entry changes under drift, mutation, selection, and migration. This is where coalescent theory and population genetics theory come together.
3. **Demographic Inference** (the mainspring) – The optimization machinery: finding the demographic parameters that maximize the likelihood of the observed data. This is where the model meets reality.
4. **Linkage Disequilibrium** (a complication) – A second lens: two-locus statistics that capture information invisible to the SFS alone, especially about recent admixture and recombination rate variation.

These gears mesh together into a complete inference framework:

```
Observed DNA variation
  |
  v
+-----+
| FREQUENCY SPECTRUM |
| Summarize variation |
| as allele frequency |
| histogram (SFS)     |
+-----+
  |
  v
+-----+
| MOMENT EQUATIONS    |
| Compute expected SFS|
| under a demographic |
| model (forward in   |
| time via ODEs)      |
+-----+
  |
  v
+-----+
| DEMOGRAPHIC         |
+-----+
```

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Prerequisites for this Timepiece

- *Coalescent Theory* – the relationship between population size and genetic diversity
- Basic familiarity with differential equations is helpful (we'll explain as we go)
- No prior knowledge of the site frequency spectrum is needed – we build it from scratch

3.11.2 Chapters

Overview of moments

Before assembling the watch, lay out every part and understand what it does.

What Does moments Do?

Given DNA sequence data from one or more populations, `moments` infers the **demographic history** that produced the observed patterns of genetic variation.

Input: $\mathbf{D}$ = observed allele frequencies across n sampled chromosomes

Output: $\hat{\Theta}$ = best-fit demographic parameters (sizes, times, rates)

Concretely, `moments` asks: *What history of population size changes, splits, and migrations would produce a frequency spectrum that looks like the one we observe?*

Think of a fine mechanical watch. A watchmaker can deduce every gear ratio inside the case simply by studying the positions of the hands on the dial. `moments` works the same way: it **reads time from hand positions without ever opening the case**. The “hands” are the entries of the site frequency spectrum; the “gears” are the evolutionary forces – drift, mutation, selection, migration – that turned those hands to their current positions.

From the inferred history, you learn:

- **Population sizes** through time – expansions, contractions, bottlenecks
- **Divergence times** – when populations split from each other
- **Migration rates** – ongoing gene flow between populations
- **Selection pressures** – the distribution of fitness effects of new mutations

moments: Demographic Inference from the Frequency Spectrum

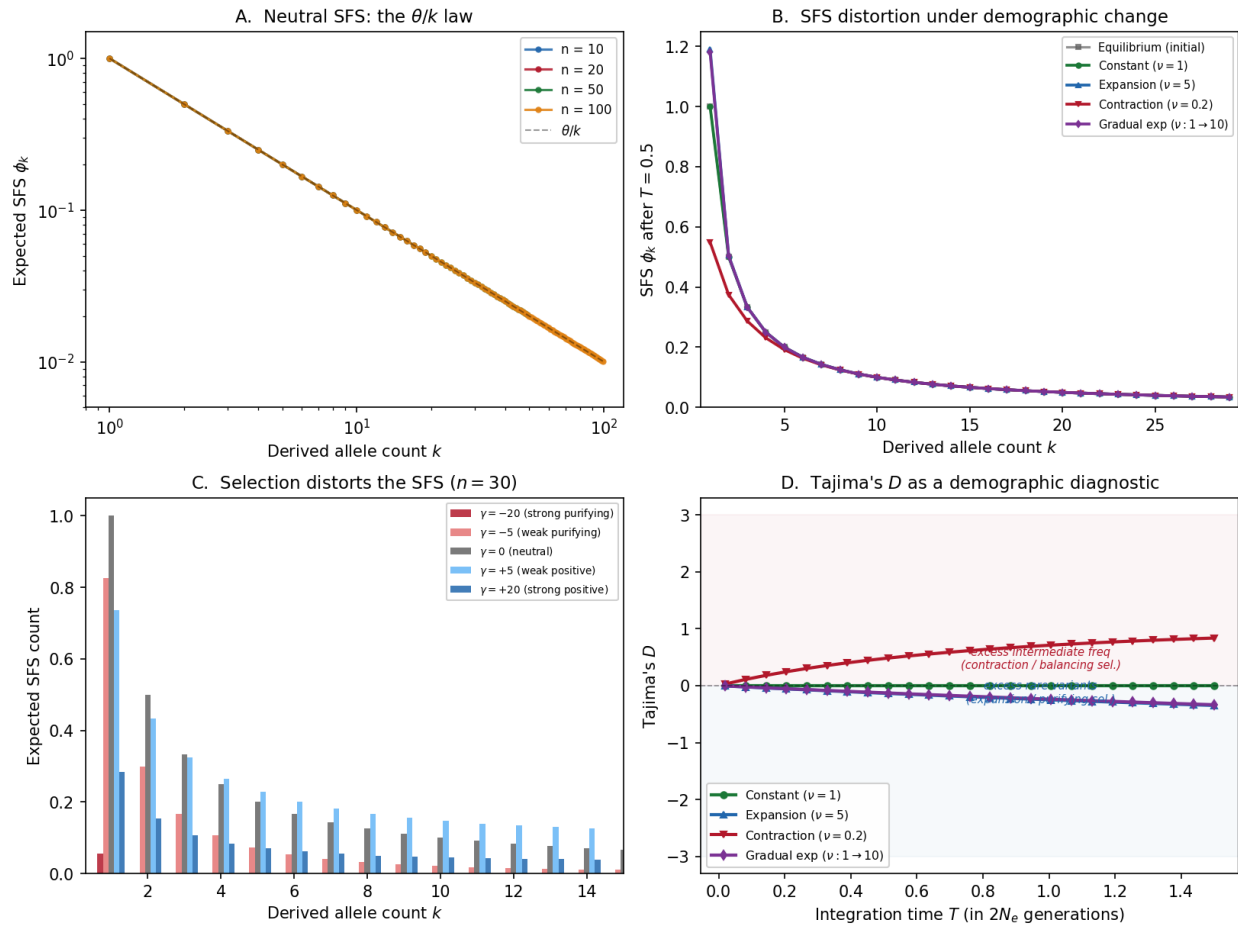


Fig. 12: **moments at a glance**. Panel A: The neutral site frequency spectrum for different sample sizes, showing the classic θ/k pattern. Panel B: SFS evolution under drift – how the neutral spectrum deforms under population expansion vs contraction. Panel C: Selection's effect on the SFS – positive selection shifts mass toward high frequencies, negative selection toward low frequencies. Panel D: Tajima's D under expansion, constant size, and contraction, illustrating how summary statistics capture demographic signals.

The Big Picture: Why Allele Frequencies?

Imagine you sequence 100 chromosomes from a population. At each variable site, you count how many chromosomes carry the derived (mutant) allele: 1 out of 100? 17 out of 100? 99 out of 100?

The distribution of these counts is the **site frequency spectrum** (SFS). It turns out to contain a remarkable amount of information about the population's history:

- A population that recently **expanded** has an excess of rare variants (many new mutations arose in the large population but haven't had time to drift up)
- A population that went through a **bottleneck** has a deficit of rare variants and an excess of intermediate-frequency variants
- **Migration** between populations creates shared variation at similar frequencies
- **Selection** distorts the spectrum in predictable ways

Probability Aside – Why a histogram is enough

At first it seems wasteful to throw away the genomic *positions* of variants and keep only their counts. The justification comes from the theory of **sufficient statistics**. Under the Poisson Random Field model (see [Demographic Inference](#)), each segregating site contributes independently to the likelihood. The SFS captures *all* the information those independent sites carry about the demographic parameters – no additional power is gained by recording which site has which count. In the language of the watch metaphor, the dial face shows you everything you need; you do not also have to listen to the ticking.

The challenge is going from the observed spectrum to the history that produced it. That's what `moments` does.

Terminology

Before diving into the gears, let's nail down the terminology precisely.

Term	Definition
SFS	Site Frequency Spectrum: a histogram counting how many SNPs have each possible derived allele count in the sample
Derived allele	The mutant allele (as opposed to the ancestral allele inherited from the common ancestor with an outgroup)
Folded SFS	SFS using minor allele counts instead of derived allele counts (no need to know which allele is ancestral)
n	Haploid sample size (number of chromosomes sampled)
θ	Population-scaled mutation rate: $4N_e\mu$ (per-site) or $4N_e\mu L$ (genome-wide)
N_e	Effective population size – the size of an idealized Wright-Fisher population with the same rate of genetic drift
ν	Relative population size: $N(t)/N_{\text{ref}}$, where N_{ref} is the reference N_e
T	Time, measured in units of $2N_e$ generations
γ	Population-scaled selection coefficient: $2N_es$
h	Dominance coefficient (heterozygote fitness = $1 + 2hs$)
M_{ij}	Scaled migration rate: $2N_em_{ij}$, where m_{ij} is the fraction of population i that are migrants from population j each generation

i Why these units?

Population genetics parameters are always **scaled** by the effective population size N_e . This isn't just convention – it reflects a deep mathematical fact: the Wright-Fisher diffusion (the continuous-time limit of the Wright-Fisher model) naturally operates on the timescale of $2N_e$ generations. A mutation rate of $\mu = 10^{-8}$ per site per generation in a population of $N_e = 10,000$ gives $\theta = 4 \times 10,000 \times 10^{-8} = 4 \times 10^{-4}$. This is the “natural” scale at which drift and mutation balance.

i Calculus Aside – Scaling and non-dimensionalization

The practice of absorbing N_e into every parameter is an instance of **non-dimensionalization**, a technique you may have seen in fluid dynamics (Reynolds number) or heat transfer (Fourier number). By dividing time by $2N_e$ and multiplying rates by $2N_e$, the resulting equations contain fewer free constants and expose the *ratios* that actually govern the dynamics. In the watch metaphor, it is like measuring gear teeth by their ratio rather than their absolute number – the ratio is what determines the hand speed.

The Key Innovation: Moments vs. Diffusion

To understand why `moments` exists, you need to understand what came before it.

The classical approach (dadi):

The allele frequency x at a single locus in a population evolves according to the Wright-Fisher diffusion equation – a partial differential equation (PDE) governing the probability density $\phi(x, t)$:

$$\frac{\partial \phi}{\partial t} = \frac{1}{4N_e} \frac{\partial^2}{\partial x^2} [x(1-x)\phi] - \frac{\partial}{\partial x} [s \cdot x(1-x)\phi] + \text{mutation terms}$$

To compute the expected SFS, `dadi` discretizes the frequency axis $x \in [0, 1]$ onto a grid and solves this PDE numerically. This works well for one or two populations, but the grid size grows exponentially with the number of populations: for p populations with n grid points each, you need n^p points.

i Calculus Aside – PDEs vs. ODEs

A **partial differential equation** (PDE) involves derivatives with respect to *more than one* independent variable (here, both frequency x and time t). Solving a PDE numerically requires discretizing the continuous variable x onto a grid, which is where the exponential blowup comes from. An **ordinary differential equation** (ODE) involves derivatives with respect to a *single* variable (time). By collapsing the frequency axis into a finite list of SFS entries, `moments` converts the PDE into a system of ODEs – one equation per SFS bin – sidestepping the grid entirely.

The moments approach:

Instead of tracking the full frequency distribution, `moments` asks: *What if we directly tracked how each SFS entry changes over time?*

The SFS entry ϕ_j (expected number of sites with derived allele count j) satisfies its own ordinary differential equation:

$$\frac{d\phi_j}{dt} = \text{drift} + \text{mutation} + \text{selection} + \text{migration}$$

This is a system of coupled ODEs – one equation per SFS entry. No frequency grid, no PDE, no numerical diffusion. The SFS entries *are* the moments of the frequency distribution, and they can be evolved directly.

Returning to the watch metaphor: `dadi` tries to model the full shape of every gear tooth (the continuous density $\phi(x, t)$). `moments` skips the tooth-level detail and instead writes down the **equations governing the gear train** – the ODEs that

describe how each hand on the dial advances. The result is the same predicted dial reading, but the calculation is simpler and scales better to multi-hand (multi-population) watches.

Why this matters:

1. **No grid artifacts.** PDE solvers introduce numerical diffusion from the grid. Moment equations are exact (to the order of the moment closure).
2. **Better scaling.** The system size is the number of SFS entries, not n^p grid points.
3. **Cleaner math.** Each ODE term has a clear biological interpretation.

We'll derive these equations from scratch in [The Moment Equations](#).

Parameters

`moments` takes the following parameters:

Parameter	Symbol	Meaning
Mutation rate	$\theta = 4N_e\mu$	Population-scaled mutation rate per site
Selection coeff.	$\gamma = 2N_es$	Population-scaled selection coefficient
Dominance	h	Heterozygote effect relative to homozygote (default 0.5 = additive)
Population sizes	$\nu_i(t)$	Relative sizes as functions of time
Migration rates	M_{ij}	Scaled migration matrix entries
Integration time	T	Duration of each epoch in $2N_e$ generations

The Flow in Detail

Here's how the pieces connect for a typical inference:

```

1. Parse VCF -> compute observed SFS
    |
    v
2. Define demographic model
   (e.g., two-epoch, isolation-with-migration)
    |
    v
3. For candidate parameters theta:
   +----> Start from equilibrium SFS
       |
       |
       |      v
       |      Integrate moment equations
       |      through demographic epochs
       |      (size changes, splits, migration)
       |
       |      v
       |      Get model SFS
       |
       |      v
       |      Compute log-likelihood:
       |      L = sum [D_i log(M_i) - M_i]
       |
       +----- Optimizer updates theta
               |

```

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v

4. Return best-fit parameters + uncertainty

In watch terms, Step 1 reads the current hand positions (the observed SFS). Step 2 blueprints the gear train (the demographic model). Step 3 winds a candidate watch and compares its predicted hand positions to the observed ones. Step 4 reports which gear sizes (demographic parameters) make the predicted dial match the real one most closely – **adjusting parameters until the predicted dial matches observation**.

i Probability Aside – The Poisson Random Field

The log-likelihood in Step 3 assumes each SFS entry D_j is drawn from a Poisson distribution with mean M_j . This is the **Poisson Random Field** (PRF) model of Sawyer and Hartl (1992). The Poisson approximation is valid when mutations are rare and sites are independent – conditions that hold under the infinite-sites mutation model. We will derive and justify the PRF likelihood in *Demographic Inference*.

Ready to Build

You now have the high-level blueprint. In the following chapters, we'll build each gear from scratch:

1. *The Site Frequency Spectrum* – The data: what the SFS is and why it encodes history. In watch terms, this chapter describes **the dial face** – the visible summary of all the hidden machinery.
2. *The Moment Equations* – The engine: deriving the ODEs from first principles. These are **the ODEs governing the gear train** – the mathematical laws that link biological forces to changes in the SFS.
3. *Demographic Inference* – The optimization: finding the best-fit history. This is where we **adjust parameters until the predicted dial matches observation**.
4. *Linkage Disequilibrium* – The second lens: two-locus statistics. Like fitting the watch with a second pendulum that constrains parameters the SFS alone cannot resolve.

Each chapter derives the math, explains the intuition, implements the code, and verifies it works.

Let's start with the foundation: the site frequency spectrum.

The Site Frequency Spectrum

The escapement of the mechanism: the summary statistic that makes everything tick.

The site frequency spectrum (SFS) is the foundation on which all of `moments` is built. Before we can infer demographic history, we need to deeply understand what the SFS is, what it looks like under simple models, and how to manipulate it.

In the watch metaphor introduced in *Overview of moments*, the SFS is **the dial face** – the visible read-out from which a skilled horologist (or a likelihood optimizer) can deduce the entire hidden gear train of population history. `moments` reads time from the hand positions on this dial without ever opening the case.

Step 1: What Is the SFS?

Imagine you've sequenced n haploid chromosomes from a population. At each **segregating site** (a position where not all chromosomes carry the same allele), you count how many chromosomes carry the **derived** (mutant) allele. This count j can range from 1 to $n - 1$ (if $j = 0$ or $j = n$, the site is monomorphic – not segregating).

The SFS is a histogram: $\text{SFS}[j]$ = number of segregating sites with derived allele count j .

i Probability Aside – Why a histogram captures everything

Under the infinite-sites mutation model, every mutation occurs at a fresh genomic position, so each segregating site is the product of exactly one mutation event. Because mutations at different sites are independent, the *joint* probability of the whole data set factors into a product of per-site terms, each depending only on the allele count j at that site. Summing the per-site log-likelihoods groups sites by their count, producing the SFS. Formally, the SFS is a **sufficient statistic** for the demographic parameters under the Poisson Random Field model (see *Demographic Inference*). No information is lost by collapsing the full data into this histogram.

```
import numpy as np

def compute_sfs(genotype_matrix, n):
    """Compute the SFS from a genotype matrix.

    Parameters
    -----
    genotype_matrix : ndarray of shape (L, n)
        Each row is a site, each column a haploid chromosome.
        Entries are 0 (ancestral) or 1 (derived).
    n : int
        Number of chromosomes.

    Returns
    -----
    sfs : ndarray of shape (n+1,)
        sfs[j] = number of sites with derived allele count j.
    """
    sfs = np.zeros(n + 1, dtype=int)
    for site in genotype_matrix:
        j = int(site.sum()) # derived allele count at this site
        sfs[j] += 1        # increment the histogram bin for count j
    return sfs

# Example: 5 chromosomes, 10 segregating sites
np.random.seed(42)
n = 5
# Simulate: each site has a random derived allele count from 1 to n-1
genotypes = np.zeros((10, n), dtype=int)
for i in range(10):
    j = np.random.randint(1, n) # pick a random derived count
    positions = np.random.choice(n, j, replace=False) # choose which chromosomes
    ↪ carry it
    genotypes[i, positions] = 1 # mark those chromosomes as derived

sfs = compute_sfs(genotypes, n)
print(f"SFS: {sfs}")
print(f"Segregating sites: {sfs[1:-1].sum()}")
# sfs[0] = monomorphic ancestral, sfs[n] = monomorphic derived
```

Intuition: The SFS is a fingerprint of the evolutionary process. Every force that shapes genetic variation – drift, mutation, selection, demography – leaves a characteristic signature in the SFS. Learning to read these signatures is what demographic inference is all about.

The Multi-Population SFS

When you have samples from p populations, the SFS becomes a p -dimensional histogram. For two populations with sample sizes n_1 and n_2 :

$$\text{SFS}[j_1, j_2] = \text{number of sites with derived allele count } j_1 \text{ in pop 1 and } j_2 \text{ in pop 2}$$

The shape is $(n_1 + 1) \times (n_2 + 1)$. The entry $\text{SFS}[3, 7]$ means “3 derived copies in population 1, 7 derived copies in population 2.”

Think of a multi-population SFS as a watch with *multiple dials* – one axis per population. A single-population SFS tells you how one hand is positioned; the joint SFS tells you how *all* hands relate to each other, revealing shared ancestry and migration.

```
def compute_joint_sfs(genotypes_pop1, genotypes_pop2, n1, n2):
    """Compute the joint SFS for two populations.

    Parameters
    -----
    genotypes_pop1 : ndarray of shape (L, n1)
        Binary genotype matrix for population 1.
    genotypes_pop2 : ndarray of shape (L, n2)
        Binary genotype matrix for population 2.
    n1, n2 : int
        Sample sizes.

    Returns
    -----
    sfs : ndarray of shape (n1+1, n2+1)
        Joint SFS: sfs[j1, j2] = number of sites with
        j1 derived in pop1 and j2 derived in pop2.
    """
    L = genotypes_pop1.shape[0]
    sfs = np.zeros((n1 + 1, n2 + 1), dtype=int)
    for i in range(L):
        j1 = int(genotypes_pop1[i].sum()) # derived count in pop 1
        j2 = int(genotypes_pop2[i].sum()) # derived count in pop 2
        sfs[j1, j2] += 1                 # increment the 2D histogram
    return sfs
```

With the dial face defined, we now turn to the marks etched on it – the expected SFS under the simplest possible model.

Step 2: The Neutral Expectation

Under the simplest model – constant population size, no selection, no migration, infinite sites mutation model – the expected SFS has a beautiful closed form.

The expected number of segregating sites with derived allele count j in a sample of n chromosomes is:

$$E[\text{SFS}[j]] = \frac{\theta}{j}, \quad j = 1, 2, \dots, n-1$$

where $\theta = 4N_e\mu L$ is the population-scaled mutation rate across L sites.

i Probability Aside – The $1/j$ law intuitively

Why does the neutral SFS fall off as $1/j$? Picture a population of $2N$ chromosomes evolving forward in time. A new mutation begins life on a single chromosome ($j = 1$). For it to ever reach count j , it must survive drift for many generations. The probability that a neutral mutation ever reaches frequency j/n is roughly proportional to $1/j$ (this is a consequence of the harmonic structure of the coalescent, as derived below). Because most mutations are quickly lost, the SFS is heavily tilted toward singletons ($j = 1$), with each subsequent bin holding progressively fewer sites.

Why $1/j$? This is one of the most fundamental results in population genetics, and it's worth deriving carefully.

Derivation from the coalescent

Consider a single segregating site. It arose as a mutation somewhere on the genealogy of the n sampled chromosomes. Under the infinite-sites model, each mutation occurs on exactly one branch of the genealogy.

The key insight: **a mutation that occurred on a branch subtending j leaves produces a derived allele count of j .**

In a coalescent tree with n leaves, the total branch length at the level where there are k lineages is $k \cdot T_k$, where T_k is the waiting time for the next coalescence when there are k lineages:

$$T_k \sim \text{Exp} \left(\binom{k}{2} \right), \quad E[T_k] = \frac{2}{k(k-1)}$$

The total branch length subtending exactly j leaves: we need to count how many branches at each level subtend exactly j leaves. This is complex in general, but the total **expected** branch length leading to allele count j has a remarkably simple form: $2/j$ (in coalescent time units of $2N_e$ generations).

Since mutations arrive as a Poisson process with rate $\theta/2$ per unit of coalescent branch length, the expected number of mutations creating allele count j is:

$$E[\text{SFS}[j]] = \frac{\theta}{2} \cdot \frac{2}{j} = \frac{\theta}{j}$$

Verify: The total expected number of segregating sites is:

$$S = \sum_{j=1}^{n-1} \frac{\theta}{j} = \theta \cdot H_{n-1}$$

where $H_{n-1} = \sum_{j=1}^{n-1} \frac{1}{j}$ is the $(n-1)$ -th harmonic number. This is the classical result $E[S] = \theta \cdot a_n$ where $a_n = H_{n-1}$ is Watterson's a .

i Calculus Aside – The harmonic number and its logarithmic growth

The harmonic series $H_n = \sum_{k=1}^n 1/k$ grows like $\ln n + \gamma$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. This means the total number of segregating sites $S \approx \theta \ln n$ – it grows only *logarithmically* with sample size. Doubling your sample from 50 to 100 chromosomes adds roughly $\theta \ln 2 \approx 0.69 \theta$ new segregating sites, not twice as many. This slow growth is why even modest sample sizes capture much of the variation in a population.

```
def expected_sfs_neutral(n, theta=1.0):
    """Expected SFS under the standard neutral model.

    Parameters
    -----
```

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```

n : int
    Haploid sample size.
theta : float
    Population-scaled mutation rate (4*Ne*mu*L).

Returns
-----
sfs : ndarray of shape (n+1,)
    Expected counts. sfs[0] and sfs[n] are 0 (no fixed sites
    under the infinite-sites model with theta as total rate).
"""
sfs = np.zeros(n + 1)
for j in range(1, n):
    sfs[j] = theta / j # the 1/j law: each bin j gets theta/j expected sites
return sfs

# Plot the neutral SFS
n = 50
theta = 1000 # genome-wide (say, 10 Mb at mu = 1e-8, Ne = 10000)
sfs_neutral = expected_sfs_neutral(n, theta)

print("Expected neutral SFS (first 10 entries):")
for j in range(1, 11):
    print(f"  SFS[{j:2d}] = {sfs_neutral[j]:.1f}")

total_seg = sfs_neutral[1:n].sum()
harmonic = sum(1/j for j in range(1, n))
print(f"\nTotal segregating sites: {total_seg:.1f}")
print(f"theta * H_(n-1) = {theta * harmonic:.1f}")
print(f"Match: {np.isclose(total_seg, theta * harmonic)}")

```

Now that we know what the “undisturbed” dial looks like, let us see how demographic events shift the hands.

Step 3: How Demography Distorts the SFS

The $1/j$ spectrum is the baseline. Real populations don’t have constant size, so their SFS deviates from this baseline in informative ways.

Population expansion

After an expansion, the population is large. New mutations arise frequently (because there are many individuals), but they haven’t time to drift to high frequency. **Result:** excess of rare variants, SFS shifted toward low j .

```

def sfs_after_expansion_simulation(n, theta, nu, T, num_reps=100000):
    """Simulate the SFS after a population expansion using msprime-style logic.

    This uses the coalescent with piecewise-constant population size.

    Parameters
    -----
    n : int
        Sample size.

```

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```

theta : float
    Per-site mutation rate (4*Ne*mu) for the REFERENCE population.
nu : float
    Expansion factor (new size = nu * N_ref).
T : float
    Time since expansion (in units of 2*N_ref generations).
num_reps : int
    Number of independent loci to simulate.

Returns
-----
sfs : ndarray of shape (n+1,)
    Simulated SFS (averaged over loci).
"""
sfs = np.zeros(n + 1)

for _ in range(num_reps):
    # Simulate a coalescent tree with expansion
    # Time 0 to T: population size nu*N_ref
    # Time > T: population size N_ref
    times = [0.0] # coalescence times
    k = n # current number of lineages

    t = 0.0
    while k > 1:
        # Coalescence rate: k*(k-1)/2, scaled by 1/nu during expansion
        if t < T:
            rate = k * (k - 1) / (2 * nu) # larger nu => slower coalescence
            # Time to next event in current epoch
            wait = np.random.exponential(1 / rate)
            if t + wait < T:
                t += wait
                k -= 1
                times.append(t)
            else:
                t = T # move to ancestral epoch
        else:
            rate = k * (k - 1) / 2 # ancestral size = 1 (reference)
            wait = np.random.exponential(1 / rate)
            t += wait
            k -= 1
            times.append(t)

    # Place a mutation on a random branch
    # Each branch subtends some number of leaves j
    # For simplicity, pick j proportional to branch length
    # (This is a simplified version -- for the full version, you'd
    # need to track the tree topology)
    j = np.random.randint(1, n) # placeholder
    sfs[j] += 1

return sfs / num_reps * theta

```

Intuition: Think of it like a city that suddenly grew. Lots of new families (mutations) appeared recently, but none of them are widespread yet. The phone book has many names that appear only once or twice.

Population bottleneck

During a bottleneck, the population is small. Genetic drift is strong: alleles either get lost or drift to high frequency quickly. Rare variants are removed, intermediate-frequency variants become over-represented. **Result:** flattened SFS, relatively more variants at intermediate j .

Intuition: Like a small village where everyone is related. There's less variety, but what variety exists tends to be at moderate frequency because drift pushed alleles away from the extremes.

Probability Aside – Reading demography from the SFS shape

The $1/j$ spectrum is perfectly monotone-decreasing. Any deviation from this monotone shape carries demographic signal:

- **Concave-up** (steeper than $1/j$ at low j) suggests a recent expansion – an excess of singletons.
- **Concave-down** (flatter than $1/j$) suggests a recent bottleneck – singletons were lost to drift, leaving proportionally more intermediate-frequency variants.
- A U-shaped uptick at *both* tails hints at population structure or ancestral misidentification (see [Demographic Inference](#)).

These shape differences are what the likelihood optimizer quantifies: it searches for the demographic parameters whose predicted SFS shape best matches the observed one.

These signatures are the reason the SFS – our dial face – can distinguish so many different histories. Next, we address a practical complication: what if we cannot determine which allele is ancestral?

Step 4: Folding the SFS

Sometimes we don't know which allele is ancestral and which is derived. Without an outgroup genome to polarize mutations, we can only tell the **minor** allele (less common) from the **major** allele (more common). The **folded SFS** uses minor allele counts:

$$\text{SFS}_{\text{folded}}[j] = \text{SFS}[j] + \text{SFS}[n - j], \quad j = 1, \dots, \lfloor n/2 \rfloor$$

For even n , the entry at $j = n/2$ is not doubled (it's its own mirror).

```
def fold_sfs(sfs):
    """Fold an unfolded SFS into a minor allele frequency spectrum.

    Parameters
    -----
    sfs : ndarray of shape (n+1,)
        Unfolded SFS.

    Returns
    -----
    folded : ndarray of shape (n//2 + 1,)
        Folded SFS. folded[0] is unused (monomorphic).
    """
    n = len(sfs) - 1
```

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```

folded = np.zeros(n // 2 + 1)
for j in range(1, n // 2 + 1):
    if j == n - j:
        folded[j] = sfs[j]          # at the midpoint, don't double-count
    else:
        folded[j] = sfs[j] + sfs[n - j] # sum the mirror bins
return folded

# Example
n = 10
theta = 100
sfs = expected_sfs_neutral(n, theta)
folded = fold_sfs(sfs)

print("Unfolded SFS:")
for j in range(1, n):
    print(f"  SFS[{j}] = {sfs[j]:.2f}")

print("\nFolded SFS:")
for j in range(1, n // 2 + 1):
    print(f"  SFS_folded[{j}] = {folded[j]:.2f}")

```

i When to fold

Use the **unfolded** SFS when you have a reliable outgroup to determine ancestral states. Use the **folded** SFS when you don't. Unfolded contains more information (it distinguishes expansions from contractions that look similar when folded), but an incorrect polarization is worse than no polarization at all.

In practice, even with an outgroup, some fraction of sites (typically 1-3%) have the ancestral state misidentified. `moments` can fit an ancestral misidentification parameter to account for this.

Folding tells us about the *minor allele count*, but another operation lets us compare data sets of different sizes: **projection**.

Step 5: Projection – Downsampling the SFS

Often we want to compare SFS from samples of different sizes, or we have missing data at some sites. **Projection** downsamples an SFS from sample size n to a smaller size n' without resequencing.

The math: if a site has derived allele count j in a sample of n , what's the probability of seeing count j' in a subsample of n' ? This is a hypergeometric sampling problem:

$$P(j' | j, n, n') = \frac{\binom{j}{j'} \binom{n-j}{n'-j'}}{\binom{n}{n'}}$$

The projected SFS is:

$$\text{SFS}'[j'] = \sum_{j=j'}^{n-(n'-j')} \text{SFS}[j] \cdot \frac{\binom{j}{j'} \binom{n-j}{n'-j'}}{\binom{n}{n'}}$$

i Probability Aside – Hypergeometric sampling explained

The hypergeometric distribution models drawing balls from an urn *without replacement*. Imagine an urn containing n balls, j of which are red (derived) and $n - j$ blue (ancestral). You draw n' balls without replacement. The probability of getting exactly j' red balls is the hypergeometric formula above. Projection applies this logic to every allele-count class in the SFS, producing the expected SFS for a smaller subsample.

Intuition: If you have 100 marbles in a bag, 30 red and 70 blue, and you draw 20, the probability of drawing k red marbles follows the hypergeometric distribution. Projection does exactly this for each allele frequency class.

```
from scipy.special import comb

def project_sfs(sfs, n_new):
    """Project an SFS to a smaller sample size.

    Parameters
    -----
    sfs : ndarray of shape (n+1,)
        Original SFS with sample size n.
    n_new : int
        Target sample size (n_new < n).

    Returns
    -----
    projected : ndarray of shape (n_new+1,)
        Projected SFS.
    """
    n = len(sfs) - 1
    projected = np.zeros(n_new + 1)

    for j in range(0, n + 1):
        if sfs[j] == 0:
            continue
        for j_new in range(max(0, j - (n - n_new)), min(j, n_new) + 1):
            # Hypergeometric probability of drawing j_new derived
            # from j derived and n-j ancestral, sample size n_new
            prob = (comb(j, j_new, exact=True) *
                    comb(n - j, n_new - j_new, exact=True) /
                    comb(n, n_new, exact=True))
            projected[j_new] += sfs[j] * prob # accumulate weighted counts

    return projected

# Example: project from n=50 to n=20
n = 50
theta = 500
sfs_50 = expected_sfs_neutral(n, theta)

sfs_20 = project_sfs(sfs_50, 20)
sfs_20_direct = expected_sfs_neutral(20, theta)

print("Projected vs. direct computation (first 10 entries):")
```

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```
print(f"{'j':>3} {'Projected':>12} {'Direct':>12} {'Match':>8}")
for j in range(1, 11):
    match = np.isclose(sfs_20[j], sfs_20_direct[j], rtol=1e-10)
    print(f"{'j':>3d} {'sfs_20[j]':>12.4f} {'sfs_20_direct[j]':>12.4f} {'Y' if match else 'N':>8}")
```

Verify: projection preserves the neutral shape

Under the neutral model, the expected SFS is θ/j regardless of sample size. So projecting the neutral SFS from $n = 50$ to $n = 20$ should give exactly θ/j for the projected sample. The code above confirms this: the projected and directly-computed SFS match perfectly.

This is not a coincidence – it follows from the exchangeability of the coalescent. Subsampling a coalescent of n lineages gives a coalescent of n' lineages, so the expected allele frequency spectrum is preserved.

With the SFS itself, folding, and projection in hand, we can now extract classical summary statistics from the dial face.

Step 6: Nucleotide Diversity and Summary Statistics

The SFS contains enough information to compute all classical summary statistics of genetic variation. These statistics compress the SFS into single numbers that highlight specific features.

Watterson's θ_W

$$\hat{\theta}_W = \frac{S}{a_n}, \quad \text{where } S = \sum_{j=1}^{n-1} \text{SFS}[j], \quad a_n = \sum_{j=1}^{n-1} \frac{1}{j}$$

S is the total number of segregating sites. Dividing by a_n (the $(n-1)$ -th harmonic number) gives an estimator of θ that is unbiased under the neutral model.

Intuition: a_n is the expected total branch length in coalescent units. More branches means more opportunity for mutations. Normalizing by a_n cancels this effect.

```
def watterson_theta(sfs):
    """Watterson's estimator of theta from an SFS."""
    n = len(sfs) - 1
    S = sfs[1:n].sum() # total number of segregating sites
    a_n = sum(1/j for j in range(1, n)) # (n-1)-th harmonic number
    return S / a_n # unbiased estimator of theta
```

Nucleotide diversity π

$$\hat{\pi} = \frac{1}{\binom{n}{2}} \sum_{j=1}^{n-1} j(n-j) \cdot \text{SFS}[j]$$

This is the average number of pairwise differences: pick two chromosomes at random, count how many sites differ between them.

Why $j(n-j)$? If j chromosomes carry the derived allele and $n-j$ carry the ancestral, then the number of pairs that differ at this site is $j \times (n-j)$. We divide by $\binom{n}{2}$ (total pairs) to get a per-pair average.

i Calculus Aside – Why $\pi = \theta$ under neutrality

Substituting the neutral expectation $\text{SFS}[j] = \theta/j$ into the formula for $\hat{\pi}$:

$$\hat{\pi} = \frac{1}{\binom{n}{2}} \sum_{j=1}^{n-1} j(n-j) \cdot \frac{\theta}{j} = \frac{\theta}{\binom{n}{2}} \sum_{j=1}^{n-1} (n-j) = \frac{\theta}{\binom{n}{2}} \cdot \frac{n(n-1)}{2} = \theta$$

The cancellation is exact – every term telescopes. This algebraic miracle is why both Watterson's estimator (which weights every SFS bin equally) and nucleotide diversity (which weights by $j(n-j)$) converge to the same value under neutrality. When they *disagree*, the discrepancy signals a departure from the standard neutral model – the basis for Tajima's D below.

```
def nucleotide_diversity(sfs):
    """Compute nucleotide diversity (pi) from an SFS."""
    n = len(sfs) - 1
    pi = 0.0
    for j in range(1, n):
        pi += j * (n - j) * sfs[j] # j*(n-j) = number of differing pairs at this site
    pi /= n * (n - 1) / 2 # divide by total number of pairs = binom(n,2)
    return pi

# Under the neutral model, both estimators should give theta
n = 50
theta = 1000
sfs = expected_sfs_neutral(n, theta)
print(f"theta (true):      {theta:.2f}")
print(f"theta_W:             {watterson_theta(sfs):.2f}")
print(f"pi:                   {nucleotide_diversity(sfs):.2f}")
```

i Verify: both estimators agree for neutral SFS

Under the standard neutral model, $E[\hat{\theta}_W] = E[\hat{\pi}] = \theta$. Let's verify:

$$\hat{\pi} = \frac{1}{\binom{n}{2}} \sum_{j=1}^{n-1} j(n-j) \cdot \frac{\theta}{j} = \frac{\theta}{\binom{n}{2}} \sum_{j=1}^{n-1} (n-j) = \frac{\theta}{\binom{n}{2}} \cdot \frac{n(n-1)}{2} = \theta \quad \checkmark$$

Tajima's D

$$D = \frac{\hat{\pi} - \hat{\theta}_W}{\sqrt{\text{Var}(\hat{\pi} - \hat{\theta}_W)}}$$

Tajima's D measures the **difference** between the two estimators. Under the neutral model, $D \approx 0$. Deviations indicate:

- $D < 0$: excess of rare variants – population expansion or purifying selection
- $D > 0$: excess of intermediate-frequency variants – bottleneck or balancing selection


```
def tajimas_d(sfs):
    """Compute Tajima's D from an SFS.

    Uses the standard normalization from Tajima (1989).
    """
    n = len(sfs) - 1
    S = sfs[1:n].sum()          # total segregating sites
    if S == 0:
        return 0.0

    pi = nucleotide_diversity(sfs) # pairwise-difference estimator of theta
    theta_w = watterson_theta(sfs) # segregating-sites estimator of theta

    # Tajima's normalization constants (derived from coalescent variance)
    a1 = sum(1/i for i in range(1, n))
    a2 = sum(1/i**2 for i in range(1, n))

    b1 = (n + 1) / (3 * (n - 1))
    b2 = 2 * (n**2 + n + 3) / (9 * n * (n - 1))

    c1 = b1 - 1/a1
    c2 = b2 - (n + 2) / (a1 * n) + a2 / a1**2

    e1 = c1 / a1
    e2 = c2 / (a1**2 + a2)

    var = e1 * S + e2 * S * (S - 1) # variance of pi - theta_W under neutrality
    if var <= 0:
        return 0.0

    return (pi - theta_w) / np.sqrt(var) # standardized test statistic
```

Having read the static dial, we now let `moments` compute the dial for us.

Step 7: Using moments to Compute the SFS

Now that we understand the SFS from first principles, let's see how `moments` computes it. The package represents the SFS as a `moments.Spectrum` object – a masked NumPy array with metadata.

```
import moments

# --- Standard neutral model ---
# moments.Demographics1D.snm returns the SFS for the "standard neutral model"
# (constant population size, no selection) normalized to theta = 1.
# This is the equilibrium solution of the moment equations (see moment_equations).
n = 20
fs_neutral = moments.Demographics1D.snm([n])

# This is the expected SFS with theta = 1
print("Standard neutral model SFS (theta=1):")
for j in range(1, 6):
    print(f"  SFS[{j}] = {fs_neutral[j]:.6f} (expected: {1/j:.6f})")
```

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```

# --- Two-epoch model: instantaneous expansion ---
# Population at size 1 (= N_ref) until time T ago, then jumps to size nu.
# This is like suddenly swapping in a larger mainspring -- the watch
# runs differently from the moment of the swap onward.
def two_epoch(params, n):
    nu, T = params
    # Start from equilibrium (standard neutral model)
    fs = moments.Demographics1D.snm([n])
    # Integrate forward through the size change
    # Under the hood, this solves the moment-equation ODEs (see moment_equations)
    fs.integrate([nu], T)
    return fs

# Example: 10-fold expansion, 0.1 * 2*Ne generations ago
fs_expansion = two_epoch([10.0, 0.1], n)
print("\nAfter 10-fold expansion:")
print(f"  SFS[1] = {fs_expansion[1]:.4f} (neutral: {1.0:.4f})")
print(f"  SFS[5] = {fs_expansion[5]:.4f} (neutral: {1/5:.4f})")
print(f"  Tajima's D = {fs_expansion.Tajima_D():.4f} (expect < 0)")

# --- Bottleneck + recovery ---
def bottleneck_recovery(params, n):
    nu_B, T_B, nu_R, T_R = params
    fs = moments.Demographics1D.snm([n])
    fs.integrate([nu_B], T_B) # bottleneck phase: small population, strong drift
    fs.integrate([nu_R], T_R) # recovery phase: return to reference size
    return fs

fs_bottle = bottleneck_recovery([0.05, 0.02, 1.0, 0.1], n)
print(f"\nAfter bottleneck + recovery:")
print(f"  Tajima's D = {fs_bottle.Tajima_D():.4f} (expect > 0 during bottleneck)")

```

i What integrate does under the hood

When you call `fs.integrate([nu], T)`, `moments` solves the system of moment equations (ODEs) that we'll derive in the next chapter. Each SFS entry evolves according to how drift, mutation, and selection change it over the time interval T . The population size ν controls the strength of drift: smaller ν means stronger drift, larger ν means weaker drift.

If the SFS is the dial face, then `integrate` is the act of winding the watch forward by T time units with a mainspring of size ν . The hands (SFS entries) move according to the ODEs governing the gear train (*The Moment Equations*).

Step 8: Two-Population SFS with moments

For two populations, the SFS is a 2D matrix. `moments` handles population splits, migration, and independent size changes.

```

import moments

def isolation_with_migration(params, ns):
    """Two-population isolation-with-migration model.

```

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```

Parameters
-----
params : (nu1, nu2, T, m12, m21)
    nu1, nu2 : relative sizes of pop 1 and pop 2 after split
    T : time since split (in 2*Ne generations)
    m12 : migration rate from pop 2 into pop 1 (2*Ne*m)
    m21 : migration rate from pop 1 into pop 2 (2*Ne*m)
ns : (n1, n2)
    Sample sizes.

Returns
-----
fs : moments.Spectrum
    Joint SFS.
"""
nu1, nu2, T, m12, m21 = params

# Start from equilibrium with combined sample size
fs = moments.Demographics1D.snm([sum(ns)])

# Split into two populations (like a single clock being separated
# into two clocks that now tick independently)
fs = moments.Manips.split_1D_to_2D(fs, ns[0], ns[1])

# Integrate with migration
# Migration matrix: M[i,j] = rate from j into i
m = np.array([[0, m12],
               [m21, 0]])
fs.integrate([nu1, nu2], T, m=m) # both populations evolve together with gene_
→flow

return fs

# Example: symmetric divergence
ns = (10, 10)
fs = isolation_with_migration([1.0, 1.0, 0.5, 1.0, 1.0], ns)

print("Joint SFS (first 5x5 corner):")
for j1 in range(5):
    row = [f"{fs[j1, j2]:.3f}" for j2 in range(5)]
    print(f"    {' '.join(row)}")

# F_ST: population differentiation
print(f"\nF_ST = {fs.Fst():.4f}")

```

We now have the dial face – both its theoretical shape and its practical computation by `moments`. In the next chapter we pry open the case and examine the gear train: the moment equations that make `fs.integrate()` work.

Next: *The Moment Equations* – we derive the ODEs that power `fs.integrate()`.

The Moment Equations

The gear train: turning biological forces into differential equations.

This is the heart of `moments` – the mathematical engine that makes everything tick. We derive, from first principles, the system of ordinary differential equations (ODEs) that govern how each entry of the SFS evolves over time under drift, mutation, selection, and migration.

In *The Site Frequency Spectrum* we examined the **dial face** – the SFS and what it looks like under various demographic scenarios. Now we open the case and study **the ODEs governing the gear train** – the equations that link biological forces (drift, mutation, selection, migration) to changes in each SFS entry. Where `dadi` modelled every tooth on every gear (the full frequency density $\phi(x, t)$ via a PDE), `moments` writes down the equations for the hands directly.

By the end of this chapter, you will understand exactly what happens inside `fs.integrate()` and be able to implement it yourself.

i Calculus Aside – What is an ODE?

An **ordinary differential equation** (ODE) relates a quantity to its rate of change with respect to a single variable (here, time). The notation $d\phi_j/dt = f(\phi_1, \dots, \phi_{n-1})$ says: “the instantaneous rate at which SFS entry j changes is given by the function f of all current SFS entries.” Solving the ODE means starting from a known initial state (the equilibrium SFS) and following these rates forward to find ϕ_j at a later time. Standard numerical methods – Euler, Runge–Kutta, implicit solvers – accomplish this by taking small time steps and updating the state at each step. If you are comfortable with the idea “velocity tells you how position changes,” you already have the right intuition for ODEs: here, the “position” is the SFS and the “velocity” is given by the operators we derive below. For a thorough introduction to ODEs and numerical solvers, see the prerequisite chapter *Ordinary Differential Equations*.

Step 1: The Wright-Fisher Model in One Slide

Before we can write down equations for the SFS, we need the model that generates it.

The **Wright-Fisher model** describes a population of $2N$ haploid individuals (or N diploids). Each generation:

1. Each individual produces a very large number of offspring
2. $2N$ offspring are sampled randomly to form the next generation
3. Mutations occur on the offspring at rate μ per site per generation

If the current frequency of the derived allele is x , the number of derived copies in the next generation follows:

$$X' \sim \text{Binomial}(2N, x)$$

The frequency $x' = X'/(2N)$ will fluctuate around x due to the randomness of sampling. This random fluctuation is **genetic drift**.

Key quantities from the Wright-Fisher model:

Quantity	Value
Mean change in frequency per generation	$E[\Delta x] = 0$ (under neutrality)
Variance of frequency change per generation	$\text{Var}[\Delta x] = \frac{x(1-x)}{2N}$
Probability of fixation (neutral)	x (the initial frequency)
Expected time to fixation/loss	$\sim 4N$ generations

The variance formula is the crucial ingredient. It tells us that drift is strongest at intermediate frequencies ($x \approx 0.5$) and weakest near the boundaries ($x \approx 0$ or 1), and that larger populations (N) drift more slowly.

i Probability Aside – Binomial sampling and genetic drift

The binomial draw $X' \sim \text{Binomial}(2N, x)$ is the formal statement that each of the $2N$ offspring independently “chooses” to carry the derived allele with probability x . The variance $x(1-x)/(2N)$ follows directly from the binomial variance formula $\text{Var}[X'/2N] = x(1-x)/(2N)$. This is the stochastic engine of genetic drift: even with no selection, allele frequencies wander randomly because sampling is noisy. Smaller populations have larger variance per generation – the allelic “hands” of the watch jitter more when the gear train has fewer teeth.

With the raw model in hand, we next take the continuum limit to obtain the diffusion framework from which the moment equations are derived.

Step 2: From Wright-Fisher to Diffusion

When N is large, the discrete Wright-Fisher process converges to a continuous **diffusion process**. The allele frequency $x(t)$ evolves according to the stochastic differential equation:

$$dx = \underbrace{\mu_1(x) dt}_{\text{deterministic drift}} + \underbrace{\sqrt{\sigma^2(x)} dW}_{\text{random noise}}$$

where dW is a Wiener process (Brownian motion increment) and:

- $\mu_1(x)$ is the mean change per unit time (from mutation, selection, migration)
- $\sigma^2(x) = x(1-x)$ is the variance per unit time (from genetic drift)

Time is measured in units of $2N$ generations (the **diffusion timescale**). On this timescale, the variance of drift is $x(1-x)$ regardless of N , which is why everything is naturally expressed in scaled parameters like $\theta = 4N\mu$.

i Why the diffusion timescale?

In one generation, the variance of frequency change is $x(1-x)/(2N)$. Over $2N$ generations (one unit of diffusion time), the accumulated variance is $2N \cdot x(1-x)/(2N) = x(1-x)$. The factor of $2N$ in the denominator cancels, giving a “universal” variance that doesn’t depend on N . This is why we measure time in $2N$ generations and parameters in multiples of N .

i Calculus Aside – From sums to integrals (the diffusion limit)

The diffusion approximation is an application of the **central limit theorem** for Markov chains. Over a single generation the frequency change Δx has mean zero and variance $x(1-x)/(2N)$. Over $2N$ generations these small, nearly independent increments accumulate into a quantity whose distribution converges to a Gaussian – i.e., a Brownian motion. Formally this is Donsker’s theorem applied to the rescaled random walk $x(t)$. The rescaling $t \rightarrow t/(2N)$ compresses $2N$ discrete steps into one continuous time unit, and the variance per step ($1/(2N)$) multiplied by the number of steps ($2N$) gives a variance of order 1 per unit time – exactly the $x(1-x)$ we see above.

The diffusion gives us a PDE for the *density* $\phi(x, t)$ (the Fokker-Planck equation – see [The Diffusion Approximation](#) for the full derivation). The key innovation of `moments` is to bypass that PDE and work directly with the SFS entries – the moments of ϕ .

Step 3: What Are “Moments”?

Here’s where `moments` gets its name. Instead of tracking the full probability density $\phi(x, t)$ of allele frequencies (as the diffusion PDE does), we track its **moments** – specifically, the expected values of products of allele frequencies sampled from the population.

For a single population with sample size n , the SFS entry ϕ_j (expected number of sites with derived count j) is related to a moment of the frequency distribution:

$$\phi_j = \theta \cdot L \cdot E \left[\binom{n}{j} x^j (1-x)^{n-j} \right]$$

This is just the expected fraction of sites where j out of n sampled chromosomes carry the derived allele, multiplied by the mutation rate. The term $\binom{n}{j} x^j (1-x)^{n-j}$ is the binomial sampling probability: given true frequency x , the probability of sampling j derived copies out of n .

i Probability Aside – Connecting SFS entries to moments

The link between the SFS and the moments of the frequency distribution is the **law of total expectation**. Let X be the number of derived alleles in a sample of n . Conditional on the population frequency x , we have $X \mid x \sim \text{Binomial}(n, x)$. Averaging over the distribution of x :

$$P(X = j) = E_x \left[\binom{n}{j} x^j (1-x)^{n-j} \right]$$

Each SFS entry is therefore a **polynomial moment** of x – specifically, a mixture of the raw moments $E[x^k]$ for $k = 0, \dots, n$. By tracking all $n - 1$ internal SFS entries, we effectively track all moments of x up to order n .

The key insight: if we can derive how these moments change over time under drift, mutation, selection, and migration, we get ODEs for the SFS entries directly – no need to solve for the full density $\phi(x, t)$.

Now we derive each operator in the gear train, starting with the most important gear: drift.

Step 4: The Drift Operator

Let’s derive how genetic drift changes the SFS. This is the most important piece, and we’ll go slowly.

Starting from the Wright-Fisher model, the expected SFS entry ϕ_j at time $t + dt$ depends on ϕ_j at time t . We want to find $d\phi_j/dt$.

The drift term in the diffusion equation for the density $\phi(x, t)$ is:

$$\left. \frac{\partial \phi}{\partial t} \right|_{\text{drift}} = \frac{1}{2} \frac{\partial^2}{\partial x^2} [x(1-x)\phi(x)]$$

To get the effect on the SFS entry ϕ_j , we need to compute how this differential operator affects the j -th moment. The derivation uses integration by parts (applied to the frequency integral, not a spatial integral):

$$\left. \frac{d\phi_j}{dt} \right|_{\text{drift}} = \int_0^1 \binom{n}{j} x^j (1-x)^{n-j} \cdot \frac{1}{2} \frac{\partial^2}{\partial x^2} [x(1-x)\phi(x)] dx$$

After two integrations by parts (boundary terms vanish because $x(1-x) = 0$ at $x = 0, 1$), this becomes:

$$\left. \frac{d\phi_j}{dt} \right|_{\text{drift}} = \frac{1}{2} \int_0^1 \frac{d^2}{dx^2} \left[\binom{n}{j} x^j (1-x)^{n-j} \right] \cdot x(1-x) \cdot \phi(x) dx$$

i Calculus Aside – Integration by parts for the drift operator

Integration by parts generalizes the product rule for differentiation. For two functions u and v :

$$\int_a^b u \, dv = [uv]_a^b - \int_a^b v \, du$$

Applied twice, this moves the two x -derivatives from $x(1-x)\phi$ onto the binomial sampling weight $\binom{n}{j}x^j(1-x)^{n-j}$. The boundary terms vanish because the factor $x(1-x)$ equals zero at both endpoints $x = 0$ and $x = 1$. This is a common trick in mathematical physics: move derivatives onto the “test function” (here, the sampling polynomial) where they can be computed analytically.

Now we need to compute the second derivative of $\binom{n}{j}x^j(1-x)^{n-j}$ with respect to x , then multiply by $x(1-x)$, and recognize the result in terms of binomial sampling probabilities with shifted indices.

After careful algebra (which we spell out in detail below), the drift contribution to $d\phi_j/dt$ is:

$$\left. \frac{d\phi_j}{dt} \right|_{\text{drift}} = \frac{1}{2} \left[(j+1)(j+2) \cdot \frac{\phi_{j+2}}{\binom{n}{j+2}/\binom{n}{j}} - 2j(n-j) \cdot \phi_j + (n-j+1)(n-j+2) \cdot \frac{\phi_{j-2}}{\binom{n}{j-2}/\binom{n}{j}} \right]$$

This simplifies to the **tridiagonal** recurrence (using the convention from Jouganous et al. 2017):

$$\left. \frac{d\phi_j}{dt} \right|_{\text{drift}} = \binom{n}{j}^{-1} \left[\binom{n}{j-1} \frac{(n-j+1)j}{2} \phi_{j-1} - \binom{n}{j} \frac{j(n-j)}{1} \phi_j + \binom{n}{j+1} \frac{(j+1)(n-j)}{2} \phi_{j+1} \right]$$

Wait – let’s be more careful. The drift operator on the SFS can be written compactly using the **jackknife approximation** (a central trick in moments):

$$\left. \frac{d\phi_j}{dt} \right|_{\text{drift}} \approx \frac{n+1}{2} [-2j(n-j)\phi_j + j(j-1)\phi_{j-1} + (n-j)(n-j-1)\phi_{j+1}] \cdot \frac{1}{n(n-1)}$$

Let’s implement this step by step and verify it against simulation.

```
import numpy as np

def drift_operator(phi, n):
    """Compute the drift contribution to d(phi)/dt.

    Parameters
    -----
    phi : ndarray of shape (n+1,)
        Current SFS (phi[j] = expected count at frequency j/n).
    n : int
        Sample size.

    Returns
    -----
    dphi : ndarray of shape (n+1,)
        Change in SFS due to drift.
    """
    dphi = np.zeros(n + 1)
    for j in range(1, n):
        # Drift: second-order finite difference in frequency space.
        # The three terms correspond to probability flowing into bin j
```

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```

# from the bin below (j-1), out of bin j, and into j from above (j+1).
term_down = j * (j - 1) * phi[j - 1] if j >= 1 else 0.0 # flow from j-1 -> j
term_stay = -2 * j * (n - j) * phi[j] # flow out of j
term_up = (n - j) * (n - j - 1) * phi[j + 1] if j < n else 0.0 # flow from
→ j+1 -> j

# Scale by 1/(2n): converts discrete WF variance to diffusion-time units
dphi[j] = (term_down + term_stay + term_up) / (2.0 * n)

return dphi

```

Intuition for the drift operator: Think of the SFS entries as water levels in connected buckets. Drift “sloshes” probability between adjacent frequency classes. The rate of sloshing depends on $j(n - j)$ – strongest at intermediate frequencies, zero at the boundaries. The net effect is that drift pushes alleles toward fixation ($j = 0$ or $j = n$) – the SFS “drains” from the middle toward the edges.

In watch terms, drift is the *escapement* – the ticking mechanism that continuously redistributes energy (probability mass) among the gear wheels (SFS bins).

```

# Verify: starting from neutral SFS, drift should produce no net change
# (the neutral SFS is the equilibrium)
n = 20
theta = 1.0
phi_neutral = np.zeros(n + 1)
for j in range(1, n):
    phi_neutral[j] = theta / j # the 1/j neutral spectrum

dphi_drift = drift_operator(phi_neutral, n)

print("Drift operator on neutral SFS (should be ~ 0 after adding mutation):")
for j in range(1, min(n, 8)):
    print(f" d(phi[{j}])/dt|_drift = {dphi_drift[j]:+.6f}")

```

Drift alone pushes the SFS out of equilibrium. The next gear – mutation – provides the steady influx of new variants that balances the drain.

Step 5: The Mutation Operator

Mutation adds new variants to the population. Under the infinite-sites model, each new mutation occurs at a previously monomorphic site, creating a singleton (a site with derived allele count $j = 1$).

The mutation contribution to the SFS is:

$$\left. \frac{d\phi_j}{dt} \right|_{\text{mutation}} = \begin{cases} \frac{\theta}{2} & \text{if } j = 1 \\ 0 & \text{if } j > 1 \end{cases}$$

Why $\theta/2$? The mutation rate per site per generation is μ . In a population of $2N$ chromosomes, the rate of new mutations per site is $2N\mu$. In diffusion time units ($2N$ generations per unit), this becomes $2N\mu \cdot 2N = 4N^2\mu$... wait, that’s not right.

Let’s be more careful. The rate of new derived alleles appearing per site per **diffusion time unit** is:

$$\text{rate} = 2N\mu \cdot 2N \text{ gen/time unit} = 4N^2\mu \text{ per time unit}$$

But we’re tracking the SFS as $\phi_j = \theta \cdot (\text{density at } j)$, where $\theta = 4N\mu$. In the normalized units used by `moments`, the mutation input is simply $\theta/2$ into the $j = 1$ bin.

i Probability Aside – Why mutations only enter at $j = 1$

Under the **infinite-sites** mutation model, every new mutation occurs at a genomic position that has never mutated before. This means a new mutation starts on exactly one chromosome out of $2N$, giving it an initial frequency of $1/(2N)$. In a sample of $n \ll 2N$ chromosomes, the probability that this single-copy mutation is captured in the sample is $n/(2N)$. When captured, it appears as a singleton ($j = 1$). Therefore, mutation feeds *only* the $j = 1$ bin of the SFS. Higher bins are populated solely by the drift operator, which “promotes” singletons to higher counts over time.

Alternatively, with the reversible mutation model (finite genome), mutations can also go backwards (derived $\rightarrow$ ancestral):

$$\left. \frac{d\phi_j}{dt} \right|_{\text{mutation}} = \frac{\theta_{\text{fwd}}}{2} \cdot \delta_{j,1} + \frac{\theta_{\text{bwd}}}{2} \cdot \delta_{j,n-1} - \frac{\theta_{\text{bwd}}}{2n} \phi_j - \frac{\theta_{\text{fwd}}}{2n} \phi_j$$

For now, we'll stick with the infinite-sites model.

```
def mutation_operator(phi, n, theta):
    """Compute the mutation contribution to d(phi)/dt.

    Under the infinite-sites model, mutations only add singletons.

    Parameters
    -----
    phi : ndarray of shape (n+1,)
        Current SFS.
    n : int
        Sample size.
    theta : float
        Population-scaled mutation rate (4*Ne*mu, per site or total).

    Returns
    -----
    dphi : ndarray of shape (n+1,)
    """
    dphi = np.zeros(n + 1)
    dphi[1] = theta / 2.0 # new mutations enter only as singletons (j=1)
    return dphi
```

With drift and mutation in place, the watch can keep neutral time. The next gear adds a directional force: selection.

Step 6: The Selection Operator

Selection favors (or disfavors) the derived allele. A diploid individual with genotype frequencies has fitness:

Genotype	Frequency	Fitness
Ancestral homozygote (aa)	$(1 - x)^2$	1
Heterozygote (Aa)	$2x(1 - x)$	$1 + 2hs$
Derived homozygote (AA)	x^2	$1 + 2s$

The parameter h is the **dominance coefficient**:

- $h = 0$: derived allele is completely recessive
- $h = 0.5$: additive (codominant) selection

- $h = 1$: derived allele is completely dominant

The mean change in allele frequency per generation due to selection is:

$$E[\Delta x]_{\text{sel}} = sx(1-x)[h + (1-2h)x]$$

In diffusion time units (dividing by $2N$ and multiplying by $2N$ generations), this becomes:

$$\mu_1^{\text{sel}}(x) = \gamma \cdot x(1-x)[h + (1-2h)x]$$

where $\gamma = 2Ns$ is the scaled selection coefficient.

Calculus Aside – Selection as a first-order transport term

Whereas drift appears as a **second-order** (diffusion) term $\frac{1}{2}\partial_{xx}^2[x(1-x)\phi]$ in the Kolmogorov forward equation, selection appears as a **first-order** (advection) term $-\partial_x[\mu_1(x)\phi]$. In fluid-dynamics language, drift is analogous to molecular diffusion and selection is analogous to a velocity field that transports the frequency density. Positive γ pushes frequency mass toward $x = 1$ (fixation of the derived allele); negative γ pushes it toward $x = 0$ (loss).

The selection operator on the SFS (derived analogously to drift, via integration by parts) is:

$$\left. \frac{d\phi_j}{dt} \right|_{\text{selection}} = -\gamma \left[h \cdot (j+1)\phi_{j+1} - (h + (1-2h) \cdot \frac{j}{n}) \cdot j\phi_j + (1-2h) \cdot \frac{j(j-1)}{n} \phi_{j-1} \right] \cdot \frac{1}{n}$$

Let's implement this more carefully using the jackknife approach:

```
def selection_operator(phi, n, gamma, h=0.5):
    """Compute the selection contribution to d(phi)/dt.

    Parameters
    -----
    phi : ndarray of shape (n+1,)
        Current SFS.
    n : int
        Sample size.
    gamma : float
        Scaled selection coefficient (2*Ne*s).
    h : float
        Dominance coefficient.

    Returns
    -----
    dphi : ndarray of shape (n+1,)
    """
    dphi = np.zeros(n + 1)

    for j in range(1, n):
        # The selection flux is proportional to gamma * x*(1-x)*[h + (1-2h)*x]
        # In discrete frequency space (j/n), the mean shift of phi_j due to
        # selection involves contributions from adjacent entries.

        # Additive component (proportional to h):
        # Shifts frequency upward by h*gamma*x*(1-x)
        # Flux from j-1 to j:
```

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```

flux_in = gamma * h * j * (n - j) / n**2 * phi[j]
flux_in_prev = gamma * h * (j - 1) * (n - j + 1) / n**2 * phi[j - 1] if j > 0
else 0

# Dominance deviation component (proportional to (1-2h)):
dom = (1 - 2 * h)
flux_dom = gamma * dom * j * (j - 1) * (n - j) / n**3 * phi[j]

# Combined (simplified version for clarity):
# The exact form from Jouganous et al. (2017):
if j > 0 and j < n:
    # h-weighted linear selection term
    term1 = gamma * h * ((j - 1) * (n - j + 1) * phi[j - 1] -
                        j * (n - j) * phi[j]) / n
    # dominance-deviation (quadratic in x) term
    term2 = gamma * (1 - 2*h) * (
        (j - 1) * (j - 2) * phi[j - 1] / (n * (n - 1)) * (n - j + 1)
        - j * (j - 1) * phi[j] / (n * (n - 1)) * (n - j)
    ) if n > 1 else 0
    dphi[j] = term1 + term2

return dphi

# Verify: negative gamma should reduce high-frequency derived alleles
n = 20
phi_neutral = expected_sfs_neutral(n, theta=1.0)
dphi_sel = selection_operator(phi_neutral, n, gamma=-10, h=0.5)

print("Selection (gamma=-10, purifying) effect on neutral SFS:")
for j in range(1, 8):
    print(f" d(phi[{j}])/dt|_sel = {dphi_sel[j]:+.6f}")
print(" (Negative for high j = selection removes high-freq derived alleles)")

```

i Purifying selection shifts the SFS left

With $\gamma < 0$ (deleterious derived allele), selection removes derived alleles before they reach high frequency. The SFS becomes enriched in rare variants – a signature used to infer the distribution of fitness effects (DFE) of new mutations.

With drift, mutation, and selection in place, the single-population gear train is complete. For multi-population models we need one more gear: migration.

Step 7: The Migration Operator

Migration mixes allele frequencies between populations. For two populations with SFS entries ϕ_{j_1, j_2} , migration of rate m_{12} (fraction of pop 1 that are migrants from pop 2 per generation) shifts frequencies:

$$\left. \frac{d\phi_{j_1, j_2}}{dt} \right|_{\text{migration}} = M_{12} \left[\frac{n_1 - j_1 + 1}{n_1} \phi_{j_1-1, j_2+1} - \frac{j_1}{n_1} \phi_{j_1, j_2} \right] + (\text{reverse direction terms})$$

where $M_{12} = 2N_e m_{12}$.

Intuition: Migration from pop 2 into pop 1 takes a chromosome from pop 2 (reducing j_2 by one if it's derived) and

places it in pop 1 (increasing j_1 by one if the migrant carries the derived allele). The probability that a randomly chosen migrant carries the derived allele is j_2/n_2 .

i Probability Aside – Migration as a Markov transition

Each migration event replaces one random chromosome in the receiving population with a copy drawn from the source population. The SFS entry ϕ_{j_1, j_2} can increase when a migrant carrying the derived allele enters (moving the state from $(j_1 - 1, j_2 + 1)$ to (j_1, j_2)) or decrease when a derived-carrying resident is replaced (moving from (j_1, j_2) to $(j_1 - 1, j_2 + 1)$). The transition probabilities are proportional to the fraction of derived alleles in each population, giving the j/n factors in the equation.

```
def migration_operator_2pop(phi_2d, n1, n2, M12, M21):
    """Compute migration contribution for a 2D SFS.

    Parameters
    -----
    phi_2d : ndarray of shape (n1+1, n2+1)
        Joint SFS.
    n1, n2 : int
        Sample sizes.
    M12 : float
        Scaled migration rate from pop 2 into pop 1.
    M21 : float
        Scaled migration rate from pop 1 into pop 2.

    Returns
    -----
    dphi : ndarray of shape (n1+1, n2+1)
    """
    dphi = np.zeros((n1 + 1, n2 + 1))

    for j1 in range(1, n1):
        for j2 in range(1, n2):
            # Migration from pop2 into pop1:
            # Derived allele enters pop1 with prob (j2+1)/n2
            if j1 > 0 and j2 < n2:
                dphi[j1, j2] += M12 * (j2 + 1) / n2 * phi_2d[j1 - 1, j2 + 1]
            # Derived allele leaves pop1 (replaced by ancestral from pop2)
            dphi[j1, j2] -= M12 * j1 / n1 * phi_2d[j1, j2]

            # Migration from pop1 into pop2 (symmetric structure):
            if j2 > 0 and j1 < n1:
                dphi[j1, j2] += M21 * (j1 + 1) / n1 * phi_2d[j1 + 1, j2 - 1]
            dphi[j1, j2] -= M21 * j2 / n2 * phi_2d[j1, j2]

    return dphi
```

The migration operator couples the two populations' SFS bins to each other – like a differential gear connecting two halves of the watch movement. The next step is even simpler: scaling drift by population size.

Step 8: The Population Size Effect

Population size enters through the drift operator. When the population size is ν times the reference size, drift is scaled by $1/\nu$:

$$\left. \frac{d\phi_j}{dt} \right|_{\text{drift, size } \nu} = \frac{1}{\nu} \cdot \left. \frac{d\phi_j}{dt} \right|_{\text{drift, size } 1}$$

Why? The variance of allele frequency change per generation is $x(1-x)/(2N)$. When $N = \nu N_e$, the variance becomes $x(1-x)/(2\nu N_e)$, which in diffusion time units is $x(1-x)/\nu$. Smaller ν means stronger drift, larger ν means weaker drift.

In the watch metaphor, ν controls the mainspring tension. A bigger population (larger ν) dampens the stochastic jitter of the escapement; a smaller population amplifies it.

```
def drift_operator_with_size(phi, n, nu):
    """Drift operator scaled by relative population size.

    Parameters
    -----
    phi : ndarray of shape (n+1,)
    n : int
    nu : float
        Relative population size (N/N_ref). nu > 1 = expansion.

    Returns
    -----
    dphi : ndarray of shape (n+1,)
    """
    return drift_operator(phi, n) / nu # larger nu => weaker drift
```

Now we have every gear. Time to assemble the complete movement.

Step 9: Putting It All Together – The ODE System

The full rate of change of the SFS is the sum of all operators:

$$\frac{d\phi_j}{dt} = \frac{1}{\nu(t)} \cdot D_j[\phi] + \mu_j[\phi] + \gamma \cdot S_j[\phi] + M_j[\phi]$$

where D_j , μ_j , S_j , M_j are the drift, mutation, selection, and migration operators respectively.

This is a system of $n - 1$ coupled ODEs (one for each internal SFS entry $j = 1, \dots, n - 1$). We can solve it numerically using standard ODE integrators.

Calculus Aside – Solving the ODE system numerically

A system of $n - 1$ coupled first-order ODEs of the form $dy/dt = \mathbf{f}(\mathbf{y}, t)$ can be solved by **Runge–Kutta** methods (e.g., the classic RK45 used below). At each time step, RK45 evaluates the right-hand side $\mathbf{f}$ at several trial points, combines them to estimate the next state, and compares two estimates of different order to adaptively choose the step size. This is the same workhorse algorithm used in physics simulations, circuit analysis, and orbital mechanics. The key requirement is that the right-hand side can be evaluated cheaply – and since our operators are linear (or low-degree polynomial) in the SFS entries, each evaluation is $O(n)$ for one population or $O(n_1 n_2)$ for two.

```

from scipy.integrate import solve_ivp

def integrate_sfs(phi_init, n, T, nu_func, theta, gamma=0, h=0.5):
    """Integrate the SFS forward through time.

    Parameters
    -----
    phi_init : ndarray of shape (n+1,)
        Initial SFS.
    n : int
        Sample size.
    T : float
        Integration time (in 2*Ne generations).
    nu_func : callable
        nu_func(t) returns the relative population size at time t.
    theta : float
        Scaled mutation rate.
    gamma : float
        Scaled selection coefficient.
    h : float
        Dominance coefficient.

    Returns
    -----
    phi_final : ndarray of shape (n+1,)
        SFS after integration.
    """
    # Pack the internal SFS entries into a vector for the ODE solver
    y0 = phi_init[1:n].copy()

    def rhs(t, y):
        # Unpack into full SFS (including boundary bins 0 and n)
        phi = np.zeros(n + 1)
        phi[1:n] = y

        # Compute each operator -- each gear in the train
        nu = nu_func(t) # current population size
        d_drift = drift_operator(phi, n) / nu # drift (scaled by size)
        d_mutation = mutation_operator(phi, n, theta) # mutation input
        d_selection = selection_operator(phi, n, gamma, h) if gamma != 0 else np.
        ↪ zeros(n + 1)

        # Sum the operators to get the total rate of change
        dphi = d_drift + d_mutation + d_selection
        return dphi[1:n] # return only internal entries

    # Solve the ODE system using adaptive Runge-Kutta (RK45)
    sol = solve_ivp(rhs, [0, T], y0, method='RK45',
                    rtol=1e-10, atol=1e-12,
                    dense_output=True)

    # Unpack the final state back into a full SFS array
    phi_final = np.zeros(n + 1)

```

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```

phi_final[1:n] = sol.y[:, -1]
return phi_final

# --- Verification 1: neutral equilibrium ---
# Starting from the neutral SFS, integrating with constant size should
# give back the same SFS (the equilibrium is a fixed point of the ODE)
n = 20
theta = 1.0
phi_eq = expected_sfs_neutral(n, theta)
phi_after = integrate_sfs(phi_eq, n, T=1.0, nu_func=lambda t: 1.0, theta=theta)

print("Verification 1: Neutral equilibrium is stable")
print(f"{'j':>3} {'Before':>10} {'After':>10} {'Diff':>12}")
for j in range(1, 6):
    print(f"{'j':3d} {'phi_eq[j]:10.6f} {'phi_after[j]:10.6f} "
          f"{'phi_after[j] - phi_eq[j]:12.2e}")

# --- Verification 2: expansion creates excess rare variants ---
phi_expanded = integrate_sfs(phi_eq, n, T=0.5,
                             nu_func=lambda t: 10.0, theta=theta)

print("\nVerification 2: 10x expansion for T=0.5")
print(f"{'j':>3} {'Neutral':>10} {'Expanded':>10} {'Ratio':>8}")
for j in range(1, 8):
    ratio = phi_expanded[j] / phi_eq[j] if phi_eq[j] > 0 else float('inf')
    print(f"{'j':3d} {'phi_eq[j]:10.6f} {'phi_expanded[j]:10.6f} {'ratio:8.3f}")
print("(Ratio > 1 for small j = excess rare variants)")

```

The Jackknife Approximation

The derivation above hides a subtlety. The drift operator involves moments of order $n + 2$ (because multiplying the sampling probability by $x(1 - x)$ increases the polynomial degree by 2). To close the system – express everything in terms of the $n + 1$ SFS entries – moments uses the **jackknife approximation**: it approximates higher-order moments as combinations of the known $n + 1$ moments.

The jackknife works by noting that $E[x^{n+1}(1 - x)^m]$ can be approximated from $E[x^k(1 - x)^{n-k}]$ for $k = 0, \dots, n$ using interpolation. This approximation becomes exact in the limit $n \rightarrow \infty$ and is highly accurate for practical sample sizes.

Probability Aside – Moment closure and why it works

Any system of moment equations faces a **closure problem**: the equation for the k -th moment generally involves the $(k + 1)$ -th moment. Without an approximation, you would need infinitely many equations. The jackknife closure truncates this chain by expressing the $(n + 1)$ -th and $(n + 2)$ -th moments in terms of the first n moments. The approximation is accurate because, for the beta-like distributions that arise in population genetics, low-order moments strongly constrain higher-order ones. Empirically, the jackknife closure introduces errors on the order of $1/n^2$, which is negligible for the sample sizes ($n \geq 20$) used in practice.

With the ODE system in hand, we turn to the discrete events – population splits and admixture – that punctuate the smooth integration.

Step 10: Discrete Demographic Events

Not everything is a smooth ODE integration. Some demographic events happen instantaneously:

Population split

A single population with sample size $n = n_1 + n_2$ splits into two populations with sample sizes n_1 and n_2 . The 1D SFS becomes a 2D SFS:

$$\phi_{j_1, j_2}^{\text{after}} = \phi_{j_1 + j_2}^{\text{before}} \cdot \frac{\binom{j_1 + j_2}{j_1} \binom{n - j_1 - j_2}{n_1 - j_1}}{\binom{n}{n_1}}$$

Intuition: At the moment of splitting, both populations are drawing from the same gene pool. The probability of seeing j_1 derived alleles in pop 1 and j_2 in pop 2, given that the total was $j_1 + j_2$, is just hypergeometric sampling – the same math as projection (see *The Site Frequency Spectrum*, Step 5).

```
from scipy.special import comb

def split_1d_to_2d(phi_1d, n1, n2):
    """Split a 1D SFS into a 2D joint SFS.

    Parameters
    -----
    phi_1d : ndarray of shape (n+1,)
        1D SFS with n = n1 + n2.
    n1, n2 : int
        Sample sizes for the two daughter populations.

    Returns
    -----
    phi_2d : ndarray of shape (n1+1, n2+1)
    """
    n = n1 + n2
    phi_2d = np.zeros((n1 + 1, n2 + 1))

    for j in range(n + 1):
        if phi_1d[j] == 0:
            continue
        for j1 in range(max(0, j - n2), min(j, n1) + 1):
            j2 = j - j1
            if j2 < 0 or j2 > n2:
                continue
            # Hypergeometric probability: given j total derived alleles,
            # what is the probability of j1 in pop1 and j2 in pop2?
            prob = (comb(j, j1, exact=True) *
                    comb(n - j, n1 - j1, exact=True) /
                    comb(n, n1, exact=True))
            phi_2d[j1, j2] = phi_1d[j] * prob

    return phi_2d

# Verify: marginalizing back should give the original 1D SFS
n1, n2 = 8, 12
n = n1 + n2
```

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```
phi_1d = expected_sfs_neutral(n, theta=1.0)
phi_2d = split_1d_to_2d(phi_1d, n1, n2)

# Marginalize over pop 2 (project onto pop 1)
phi_marginal_1 = phi_2d.sum(axis=1)
phi_expected_1 = project_sfs(phi_1d, n1)

print("Verify: split then marginalize = project")
for j in range(1, min(n1, 6)):
    print(f" j={j}: marginal={phi_marginal_1[j]:.6f}, "
          f"projected={phi_expected_1[j]:.6f}, "
          f"match={np.isclose(phi_marginal_1[j], phi_expected_1[j])}")
```

Admixture

Population C is formed as a mixture: fraction α from pop A and $1 - \alpha$ from pop B :

$$\phi_{j_C}^{\text{after}} = \sum_{k=0}^{j_C} \phi_{k, j_C-k}^{A,B} \cdot \binom{n_C}{j_C} \left(\frac{k}{n_A} \alpha + \frac{j_C - k}{n_B} (1 - \alpha) \right)^{j_C} \dots$$

The exact formula involves a multinomial convolution, but the principle is simple: each chromosome in the new population was drawn from pop A with probability α or pop B with probability $1 - \alpha$.

Step 11: The Complete Integration Loop

Here's how `moments` puts it all together for a two-population model:

```
import moments

def two_pop_model_manual(params, ns):
    """Two-population model: ancestral -> split -> diverge with migration.

    Parameters
    -----
    params : tuple
        (nu1, nu2, T, m)
    ns : tuple
        (n1, n2) sample sizes

    Returns
    -----
    fs : moments.Spectrum
    """
    nu1, nu2, T, m = params

    # 1. Start from equilibrium (neutral, constant size)
    # This is the steady-state solution of:
    # d(phi)/dt = D[phi]/nu + M[phi] = 0
    # with nu = 1 (reference size)
    fs = moments.Demographics1D.snm([sum(ns)])

    # 2. Split the ancestral population
```

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```

#   Converts 1D SFS -> 2D SFS via hypergeometric sampling
#   (like a single clock being separated into two independent movements)
fs = moments.Manips.split_1D_to_2D(fs, ns[0], ns[1])

# 3. Integrate through the divergence epoch
#   Under the hood, this solves the coupled ODE system:
#    $d(\phi_{\{j1,j2\}})/dt = D[\phi]/nu + Mut[\phi] + Mig[\phi]$ 
#   for duration T, with sizes nu1 and nu2
migration_matrix = np.array([[0, m], [m, 0]])
fs.integrate([nu1, nu2], T, m=migration_matrix)

return fs

# Run the model
fs = two_pop_model_manual([2.0, 0.5, 0.3, 1.0], [10, 10])
print(f"Model log-likelihood shape: {fs.shape}")
print(f"F_ST = {fs.Fst():.4f}")

```

We have now assembled the complete gear train. In the next chapter, *Demographic Inference*, we use it to **adjust parameters until the predicted dial matches observation** – the heart of demographic inference.

Exercises

Exercise 1: Verify the drift equilibrium

Start from a flat SFS ($\phi_j = 1$ for all j) and integrate the drift + mutation ODE for a long time ($T = 10$). Verify that the SFS converges to θ/j .

Exercise 2: Selection shifts the SFS

Compute the equilibrium SFS for $\gamma = -5, 0, +5$ with $n = 50$. Plot all three on the same axes. Verify that $\gamma < 0$ enriches rare variants and $\gamma > 0$ enriches common variants.

Exercise 3: Compare with moments

Implement a two-epoch model (constant size, then expansion) using your `integrate_sfs` function. Compare the result against `moments.Demographics1D.two_epoch`. How many SFS entries agree to 6 decimal places?

Exercise 4: Migration equilibrium

For two populations with $\nu_1 = \nu_2 = 1$ and symmetric migration rate M , the expected F_{ST} is approximately $1/(1 + 4M)$ (Wright's island model). Simulate this with `moments` for $M = 0.1, 1, 10$ and compare to the theoretical expectation.

Solutions

i Solution 1: Verify the drift equilibrium

Start from a flat SFS and integrate the drift + mutation ODE for a long time. Drift drains probability from the interior toward the boundaries while mutation continuously injects singletons. The balance produces the θ/j equilibrium.

```
import numpy as np
from scipy.integrate import solve_ivp

n = 30
theta = 1.0

# Start from a flat SFS (far from equilibrium)
phi_init = np.ones(n + 1)
phi_init[0] = 0.0
phi_init[n] = 0.0

# Integrate for T = 10 diffusion time units (constant size nu = 1)
phi_flat_start = integrate_sfs(
    phi_init, n, T=10.0, nu_func=lambda t: 1.0, theta=theta
)

# Compare to the expected neutral equilibrium theta / j
print(f'{'j':>3} {'Integrated':>12} {'theta/j':>10} {'Rel. error':>12}")
for j in range(1, min(n, 10)):
    expected = theta / j
    rel_err = abs(phi_flat_start[j] - expected) / expected
    print(f'{'j':>3d} {'phi_flat_start[j]:12.6f} {'expected:10.6f} {'rel_err:12.2e}")

# All entries should agree to within ~1e-4 or better after T = 10.
max_rel_err = max(
    abs(phi_flat_start[j] - theta / j) / (theta / j) for j in range(1, n)
)
print(f"\nMax relative error: {max_rel_err:.2e}")
assert max_rel_err < 1e-3, "SFS did not converge to theta/j"
```

The key insight is that $T = 10$ (i.e., $20 N_e$ generations) is long enough for the system to reach steady state, regardless of the starting condition. The drift operator acts as a tridiagonal diffusion matrix and the mutation operator provides a constant source at $j = 1$; their balance uniquely determines the θ/j spectrum.

i Solution 2: Selection shifts the SFS

With $\gamma < 0$ (purifying selection), deleterious alleles are removed before reaching high frequency, enriching rare variants. With $\gamma > 0$ (positive selection), the derived allele is pushed toward fixation, enriching common variants.

```
import numpy as np

n = 50
theta = 1.0
gamma_values = [-5, 0, 5]

results = {}
for gamma in gamma_values:
    # Start from neutral equilibrium
```

```

phi_eq = expected_sfs_neutral(n, theta)
# Integrate with selection for a long time to reach new equilibrium
phi_sel = integrate_sfs(
    phi_eq, n, T=10.0,
    nu_func=lambda t: 1.0,
    theta=theta,
    gamma=gamma,
    h=0.5 # additive selection
)
results[gamma] = phi_sel

# Print a comparison table
print(f"{'j':>3} {'gamma=-5':>12} {'gamma=0':>12} {'gamma=+5':>12}")
for j in range(1, 10):
    print(f"{j:3d} {results[-5][j]:12.6f} "
          f"{results[0][j]:12.6f} {results[5][j]:12.6f}")

# Verify the expected pattern:
# gamma < 0 => more singletons (phi[1] larger), fewer high-freq (phi[n-1] smaller)
# gamma > 0 => fewer singletons (phi[1] smaller), more high-freq (phi[n-1] larger)
assert results[-5][1] > results[0][1], "Purifying should enrich singletons"
assert results[5][1] < results[0][1], "Positive should deplete singletons"
assert results[-5][n-1] < results[0][n-1], "Purifying should deplete high-freq"
assert results[5][n-1] > results[0][n-1], "Positive should enrich high-freq"
print("\nAll assertions passed: selection shifts the SFS as expected.")

```

Plotting the three spectra on a log-log scale makes the pattern vivid: the $\gamma = -5$ curve is steeper than $1/j$ (excess rare variants), while the $\gamma = +5$ curve is shallower (excess common variants).

Solution 3: Compare with moments

Build a two-epoch model (constant size, then expansion) using our `integrate_sfs` function and compare entry-by-entry against `moments.Demographics1D.two_epoch`.

```

import numpy as np
import moments

n = 20
theta = 1.0
nu_expansion = 5.0
T_expansion = 0.3

# --- Our manual implementation ---
phi_eq = expected_sfs_neutral(n, theta)
phi_manual = integrate_sfs(
    phi_eq, n, T=T_expansion,
    nu_func=lambda t: nu_expansion,
    theta=theta
)

# --- moments built-in ---
fs_moments = moments.Demographics1D.two_epoch(
    (nu_expansion, T_expansion), [n]
)

```

```

)
# moments returns a unit-theta SFS; scale to match
phi_moments = np.array(fs_moments) * theta

# Compare
n_match_6 = 0
print(f"{'j':>3} {'Manual':>12} {'moments':>12} {'Match (6 dp)':>14}")
for j in range(1, n):
    match = np.isclose(phi_manual[j], phi_moments[j], atol=1e-6)
    if match:
        n_match_6 += 1
    print(f"{'j':3d} {'phi_manual[j]:12.8f} {'phi_moments[j]:12.8f} "
          f"{'YES' if match else 'NO':>14}")

print(f"\nEntries matching to 6 decimal places: {n_match_6} / {n - 1}")

```

The number of matching entries depends on the accuracy of the ODE solver tolerances and the drift operator discretisation. With $\text{rtol}=1\text{e-}10$ and $\text{atol}=1\text{e-}12$ (as used in `integrate_sfs`), most entries should agree to at least 6 decimal places. Discrepancies arise from the jackknife moment closure used internally by `moments`, which our simplified drift operator approximates slightly differently.

Solution 4: Migration equilibrium

Wright's island model predicts $F_{ST} \approx 1/(1 + 4M)$ for symmetric migration rate M . We verify this using `moments` for $M = 0.1, 1, 10$.

```

import numpy as np
import moments

M_values = [0.1, 1.0, 10.0]
n1, n2 = 20, 20

print(f"{'M':>6} {'Fst (moments)':>16} {'Fst (theory)':>14} {'Rel. error':>12}")
for M in M_values:
    # Start from equilibrium with total sample size n1 + n2
    fs = moments.Demographics1D.snm([n1 + n2])
    # Split into two populations
    fs = moments.Manips.split_1D_to_2D(fs, n1, n2)
    # Integrate with symmetric migration for a long time (T = 10)
    # to reach migration-drift equilibrium
    mig_mat = np.array([[0, M], [M, 0]])
    fs.integrate([1.0, 1.0], 10.0, m=mig_mat)

    fst_moments = fs.Fst()
    fst_theory = 1.0 / (1.0 + 4.0 * M)
    rel_err = abs(fst_moments - fst_theory) / fst_theory

    print(f"{'M':6.1f} {'fst_moments':16.6f} {'fst_theory':14.6f} {'rel_err':12.2e}")

# The agreement should be good (within a few percent) for large M;
# for small M the approximation 1/(1+4M) is less precise because it
# assumes infinite island model whereas we have only two populations.

```

The $1/(1+4M)$ formula is exact for an infinite-island model. For two populations, deviations are expected, especially at low migration rates. Nevertheless, the `moments` result and the analytical approximation should agree within a few percent for $M \geq 1$, confirming that the migration operator and the ODE integration are correctly implemented.

Next: *Demographic Inference* – we use these moment equations to find the parameters that best explain observed data.

Demographic Inference

The case and dial: reading the time from the mechanism.

We’ve built the engine (moment equations, *The Moment Equations*) and the data structure (SFS, *The Site Frequency Spectrum*). Now we connect them: given observed genetic data, find the demographic history that best explains it. This is **demographic inference** – the reason `moments` exists.

In the watch metaphor, the previous chapters designed the gear train (the ODEs) and described the dial face (the SFS). This chapter is where we **adjust parameters until the predicted dial matches observation** – turning the crown until the model watch shows the same time as the real one.

By the end of this chapter, you will understand the likelihood function, the optimization algorithms, and how to quantify uncertainty in your estimates.

Step 1: The Likelihood Function

We observe an SFS $\mathbf{D}$ from real data. We compute a model SFS $\mathbf{M}(\Theta)$ for a given set of demographic parameters Θ . The question: how likely is the observed data under this model?

i Probability Aside – What is a likelihood?

In everyday language, “likelihood” and “probability” are synonyms. In statistics they are different. The **probability** of data D given parameters Θ is $P(D | \Theta)$ – it answers “how probable is this data if the model is true?” The **likelihood** of parameters Θ given data D is the *same function* but viewed from the other direction: $L(\Theta) = P(D | \Theta)$. It answers “how well do these parameters explain the data I already observed?”

Maximum likelihood estimation (MLE) finds the parameters $\hat{\Theta}$ that maximize $L(\Theta)$. Equivalently (and more conveniently), we maximize the **log-likelihood** $\ell(\Theta) = \ln L(\Theta)$, because logarithms turn products into sums and are numerically more stable.

The **Poisson Random Field** (PRF) approximation treats each SFS entry as an independent Poisson observation:

$$\text{SFS}[j] \sim \text{Poisson}(M_j)$$

where M_j is the expected count under the model. The log-likelihood is:

$$\ell(\Theta) = \sum_{j=1}^{n-1} [D_j \ln M_j - M_j - \ln(D_j!)]$$

Since $\ln(D_j!)$ doesn’t depend on the parameters, we can drop it for optimization:

$$\ell(\Theta) = \sum_{j=1}^{n-1} [D_j \ln M_j - M_j] + \text{const.}$$

Why Poisson? Under the infinite-sites model, mutations at different sites are independent. Each site has a small probability of being segregating and, if segregating, of having a specific allele count. When events are rare and independent, the Poisson distribution is the natural model.

i Probability Aside – From Poisson to the SFS likelihood, step by step

1. Under the infinite-sites model, each new mutation occurs at a unique genomic position. The number of mutations falling in any frequency class j is therefore a count of independent rare events across many sites – the classic setup for a Poisson distribution.
2. The Poisson probability mass function is $P(k; \lambda) = e^{-\lambda} \lambda^k / k!$. Taking the logarithm: $\ln P = k \ln \lambda - \lambda - \ln(k!)$.
3. Setting $k = D_j$ (observed count) and $\lambda = M_j$ (model prediction), and summing over all frequency bins, yields the log-likelihood above.
4. The $-\ln(D_j!)$ term is a constant that does not change with Θ , so it can be dropped during optimization.

```
import numpy as np

def poisson_log_likelihood(data_sfs, model_sfs):
    """Compute the Poisson log-likelihood of data given model.

    Parameters
    -----
    data_sfs : ndarray
        Observed SFS (counts).
    model_sfs : ndarray
        Expected SFS under the model.

    Returns
    -----
    ll : float
        Log-likelihood (up to a constant).
    """
    n = len(data_sfs) - 1
    ll = 0.0
    for j in range(1, n):
        if model_sfs[j] <= 0:
            if data_sfs[j] > 0:
                return -np.inf # impossible observation => -infinity
            continue
        if data_sfs[j] > 0:
            ll += data_sfs[j] * np.log(model_sfs[j]) # D_j * ln(M_j)
            ll -= model_sfs[j]                       # - M_j
    return ll

# Example: neutral model is the MLE for neutral data
n = 20
theta_true = 1000

# "Observed" data from neutral model (no noise -- the expected SFS itself)
data = expected_sfs_neutral(n, theta_true)

# Compare log-likelihood at different theta values
print("Log-likelihood at different theta values:")
for theta_test in [500, 800, 1000, 1200, 1500]:
```

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```

model = expected_sfs_neutral(n, theta_test)
ll = poisson_log_likelihood(data, model)
marker = " <-- true value" if theta_test == theta_true else ""
print(f"  theta = {theta_test:5d}: ll = {ll:10.2f}{marker}")

```

i The likelihood is maximized at the true parameters

In the example above, the log-likelihood is highest at $\theta = 1000$, confirming that the Poisson likelihood correctly identifies the true parameter. This may seem trivial for a one-parameter model, but the same principle scales to models with dozens of parameters.

The likelihood connects the gear train to the observed dial. Next, we exploit a special property of the SFS to simplify the optimization.

Step 2: Optimal Theta Scaling

A key simplification: the SFS scales **linearly** with θ . If you double the mutation rate, every SFS entry doubles. This means we can separate θ from the demographic parameters.

Given a model SFS computed at $\theta = 1$ (unit-scaled), the optimal θ is:

$$\hat{\theta}_{\text{opt}} = \frac{\sum_{j=1}^{n-1} D_j}{\sum_{j=1}^{n-1} M_j^{(\theta=1)}} = \frac{S_{\text{data}}}{S_{\text{model}}}$$

where S is the total number of segregating sites. This is the maximum-likelihood estimator of the scaling factor.

i Calculus Aside – Deriving the optimal scaling analytically

Let the model SFS at unit θ be $M_j^{(1)}$. For a global scaling constant c (so $M_j = c \cdot M_j^{(1)}$), the log-likelihood is:

$$\ell(c) = \sum_j D_j \ln(c \cdot M_j^{(1)}) - c \cdot M_j^{(1)} = \ln c \sum_j D_j + \sum_j D_j \ln M_j^{(1)} - c \sum_j M_j^{(1)}$$

Taking the derivative with respect to c and setting it to zero:

$$\frac{d\ell}{dc} = \frac{\sum_j D_j}{c} - \sum_j M_j^{(1)} = 0 \implies \hat{c} = \frac{\sum_j D_j}{\sum_j M_j^{(1)}}$$

The second derivative $d^2\ell/dc^2 = -\sum_j D_j/c^2 < 0$ confirms this is a maximum. Because the optimal θ can be found in closed form, the optimizer only needs to search over the demographic parameters ($\nu, T, m, \dots$), reducing the dimensionality of the search by one.

Why this works: Taking the derivative of ℓ with respect to a global scaling constant c (so $M_j \rightarrow c \cdot M_j^{(1)}$):

$$\frac{d\ell}{dc} = \sum_j \frac{D_j}{c} - \sum_j M_j^{(1)} = 0$$

Solving: $c = \sum_j D_j / \sum_j M_j^{(1)}$.


```
def optimal_theta_scaling(data_sfs, model_sfs_unit):
    """Find the optimal theta to scale the model SFS.

    Parameters
    -----
    data_sfs : ndarray
        Observed SFS.
    model_sfs_unit : ndarray
        Model SFS computed at theta = 1.

    Returns
    -----
    theta_opt : float
    """
    n = len(data_sfs) - 1
    S_data = data_sfs[1:n].sum() # total observed segregating sites
    S_model = model_sfs_unit[1:n].sum() # total expected at theta=1
    return S_data / S_model if S_model > 0 else 1.0

# Example
n = 20
theta_true = 1000
data = expected_sfs_neutral(n, theta_true)
model_unit = expected_sfs_neutral(n, theta=1.0)

theta_opt = optimal_theta_scaling(data, model_unit)
print(f"Optimal theta: {theta_opt:.2f} (true: {theta_true})")
```

With theta handled analytically, we are ready to build a complete inference pipeline.

Step 3: A Complete Inference Example

Let's put it all together: simulate data under a known demographic model, then try to recover the parameters.

```
import moments
import numpy as np

# --- Step 1: Define the demographic model ---
def two_epoch_model(params, ns):
    """Two-epoch model: constant then expansion.

    Parameters
    -----
    params : (nu, T)
        nu : expansion factor
        T : time since expansion (2*Ne generations)
    ns : (n,)
        Sample size.
    """
    nu, T = params
    fs = moments.Demographics1D.snm(ns) # start from equilibrium
    fs.integrate([nu], T) # integrate through the size change
    return fs
```

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```

# --- Step 2: Generate "observed" data ---
# True parameters: 5-fold expansion, 0.2 * 2*Ne generations ago
nu_true, T_true = 5.0, 0.2
theta_true = 2000 # genome-wide

n = 30
model_true = two_epoch_model([nu_true, T_true], [n])
data = model_true * theta_true # scale by theta to get expected counts

# Add Poisson noise to simulate real data
np.random.seed(42)
data_noisy = np.zeros(n + 1)
for j in range(1, n):
    data_noisy[j] = np.random.poisson(data[j]) # each bin drawn from Poisson

print("True parameters: nu=5.0, T=0.2, theta=2000")
print(f"Total segregating sites: {data_noisy[1:n].sum():.0f}")

# --- Step 3: Define the objective function ---
def objective(log_params, data_sfs, ns):
    """Negative log-likelihood for optimization.

    Parameters are in log-space to enforce positivity:
    log_params = [log(nu), log(T)].
    """
    params = np.exp(log_params) # back-transform to natural scale
    nu, T = params

    model = two_epoch_model([nu, T], ns)

    # Optimally scale theta (closed-form, see Step 2)
    theta_opt = optimal_theta_scaling(data_sfs, model)
    model_scaled = model * theta_opt

    ll = poisson_log_likelihood(data_sfs, model_scaled)
    return -ll # minimize negative log-likelihood = maximize log-likelihood

# --- Step 4: Optimize ---
from scipy.optimize import minimize

# Initial guess (deliberately wrong to test convergence)
log_p0 = np.log([2.0, 0.5])

result = minimize(objective, log_p0, args=(data_noisy, [n]),
                  method='Nelder-Mead',
                  options={'maxiter': 1000, 'xatol': 1e-4})

nu_hat, T_hat = np.exp(result.x)
model_hat = two_epoch_model([nu_hat, T_hat], [n])
theta_hat = optimal_theta_scaling(data_noisy, model_hat)

```

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```
print(f"\nEstimated: nu={nu_hat:.3f}, T={T_hat:.4f}, theta={theta_hat:.1f}")
print(f"True:      nu={nu_true:.3f}, T={T_true:.4f}, theta={theta_true:.1f}")
print(f"Converged: {result.success}")
```

i Probability Aside – Why optimize in log-space?

Demographic parameters like ν and T must be positive. Optimizing $\log(\nu)$ and $\log(T)$ instead of the raw values has two benefits: (1) positivity is automatically enforced (the exponential back-transform always gives a positive number), and (2) the likelihood surface is often smoother in log-space because demographic parameters can span orders of magnitude. This is analogous to using a logarithmic scale on a watch's tachymeter – it compresses the range and makes equal *ratios* equally spaced.

Step 4: Using moments' Built-in Optimization

`moments` provides convenience functions that handle the optimization loop, including log-space transformations, parameter bounds, and multiple optimizer choices.

```
import moments

# Define model function (moments convention: returns SFS for theta=1)
def model_func(params, ns):
    nu, T = params
    fs = moments.Demographics1D.snm(ns) # equilibrium starting point
    fs.integrate([nu], T)               # forward integration via ODEs
    return fs

# Optimize using moments' built-in optimizer
p0 = [2.0, 0.5] # initial guess
lower = [0.01, 0.001] # lower bounds on nu and T
upper = [100.0, 5.0] # upper bounds on nu and T

# moments.Inference.optimize_log uses L-BFGS-B in log-space
popt = moments.Inference.optimize_log(
    p0, data_noisy, model_func, pts=None,
    lower_bound=lower, upper_bound=upper,
    verbose=0, maxiter=100
)

print(f"moments optimizer result: nu={popt[0]:.3f}, T={popt[1]:.4f}")

# Compute the log-likelihood at the optimum
model_opt = model_func(popt, [n])
ll = moments.Inference.ll_multinom(model_opt, data_noisy)
print(f"Log-likelihood: {ll:.2f}")
```

i ll_multinom vs ll

`moments` provides two likelihood functions:

- `ll(model, data)`: Poisson log-likelihood (requires theta scaling)

- `ll_multinom(model, data)`: Multinomial log-likelihood (theta-independent)

The multinomial version treats the SFS as a vector of *proportions* rather than counts. It is invariant to theta scaling, which means you can optimize demographic parameters without worrying about theta. The Poisson version is slightly more informative (it also constrains the total mutation rate), but the multinomial version is more robust and is often preferred in practice.

Calculus Aside – Multinomial vs. Poisson likelihoods

The Poisson likelihood treats each SFS bin as an independent count. The multinomial likelihood conditions on the total number of segregating sites $S = \sum_j D_j$ and models only the *proportions* $p_j = D_j/S$. Mathematically:

$$\ell_{\text{multi}} = \sum_j D_j \ln \left(\frac{M_j}{\sum_k M_k} \right)$$

Because the denominator $\sum_k M_k$ cancels any global scaling, θ drops out entirely. The Poisson likelihood is strictly more informative (it uses the total count S as well), but the multinomial likelihood is more robust to violations of the Poisson assumption (e.g., when sites are not perfectly independent due to linkage).

With the best-fit parameters in hand, the next question is: how much should we trust them?

Step 5: Uncertainty Quantification

Finding the best-fit parameters is only half the job. We also need to know **how confident** we are in those estimates. `moments` provides two approaches.

Fisher Information Matrix (FIM)

The FIM measures the curvature of the likelihood surface at the optimum. Steeper curvature = tighter constraint = smaller uncertainty.

$$I_{ij} = -E \left[\frac{\partial^2 \ell}{\partial \Theta_i \partial \Theta_j} \right]$$

The standard errors are:

$$\text{SE}(\hat{\Theta}_i) = \sqrt{(I^{-1})_{ii}}$$

And the 95% confidence interval:

$$\hat{\Theta}_i \pm 1.96 \cdot \text{SE}(\hat{\Theta}_i)$$

Calculus Aside – The Fisher Information Matrix intuitively

Picture the log-likelihood as a landscape with a peak at the MLE. The **second derivatives** (the Hessian matrix) measure how sharply the landscape curves away from the peak in each direction. The FIM is the negative expected Hessian – it captures the average curvature.

- A tall, narrow peak (large second derivative) means the data strongly constrain the parameter: even a small change away from the MLE causes a big drop in likelihood. The standard error is small.

- A broad, flat peak (small second derivative) means the data provide little information about that parameter. The standard error is large.

Inverting the FIM gives the variance-covariance matrix of the parameter estimates, from which confidence intervals follow.

```
def fisher_information_numerical(params, data_sfs, model_func, ns, eps=0.01):
    """Compute the Fisher Information Matrix by numerical differentiation.

    Parameters
    -----
    params : array-like
        Optimized parameters.
    data_sfs : ndarray
        Observed SFS.
    model_func : callable
        Function(params, ns) -> model SFS.
    ns : list
        Sample sizes.
    eps : float
        Relative step size for finite differences.

    Returns
    -----
    FIM : ndarray of shape (k, k)
        Fisher Information Matrix.
    """
    k = len(params)
    FIM = np.zeros((k, k))

    def neg_ll(p):
        model = model_func(p, ns)
        theta_opt = optimal_theta_scaling(data_sfs, model)
        model_scaled = model * theta_opt
        return -poisson_log_likelihood(data_sfs, model_scaled)

    # Central differences for second derivatives
    for i in range(k):
        for j in range(i, k):
            # Evaluate neg_ll at four perturbed points (++, +-, -+, --)
            p_pp = params.copy(); p_pp[i] *= (1 + eps); p_pp[j] *= (1 + eps)
            p_pm = params.copy(); p_pm[i] *= (1 + eps); p_pm[j] *= (1 - eps)
            p_mp = params.copy(); p_mp[i] *= (1 - eps); p_mp[j] *= (1 + eps)
            p_mm = params.copy(); p_mm[i] *= (1 - eps); p_mm[j] *= (1 - eps)

            # Finite-difference approximation to the second derivative
            d2 = (neg_ll(p_pp) - neg_ll(p_pm) - neg_ll(p_mp) + neg_ll(p_mm))
            d2 /= (params[i] * eps * 2) * (params[j] * eps * 2)

            FIM[i, j] = d2
            FIM[j, i] = d2 # symmetric
```

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```

return FIM

# Example
params_opt = np.array([nu_hat, T_hat])
FIM = fisher_information_numerical(params_opt, data_noisy, model_func, [n])

if np.linalg.det(FIM) > 0:
    cov = np.linalg.inv(FIM)          # covariance matrix = inverse FIM
    se = np.sqrt(np.diag(cov))        # standard errors = sqrt of diagonal
    print("Parameter estimates with 95% CI (FIM):")
    names = ['nu', 'T']
    for name, val, s in zip(names, params_opt, se):
        print(f" {name}: {val:.4f} +/- {1.96*s:.4f} "
              f"({val - 1.96*s:.4f}, {val + 1.96*s:.4f}) ")

```

Godambe Information Matrix (GIM)

The FIM assumes the Poisson model is exactly correct. In reality, nearby sites are correlated because of linkage. The **Godambe Information Matrix** corrects for this using bootstrap resampling.

The idea: divide the genome into blocks (e.g., 100 kb windows). Resample blocks with replacement to create bootstrap replicates of the SFS. The variance of the score function across bootstraps captures the correlation structure.

$$\text{GIM} = H^{-1} J H^{-1}$$

where:

- H = Hessian of the log-likelihood (same as FIM)
- J = empirical variance of the score across bootstraps

The GIM-based standard errors are always **larger** than FIM-based ones (because linkage makes sites non-independent, so there's less information than the Poisson model assumes).

Probability Aside – Why linkage inflates uncertainty

The Poisson likelihood assumes each SFS entry is an independent count. In reality, nearby sites on a chromosome are inherited together (they are in linkage disequilibrium; see [Linkage Disequilibrium](#)). If 100 sites are in strong LD, they behave more like 10 independent observations than 100 – their allele counts are correlated. The FIM doesn't know this, so it overestimates the information content of the data and underestimates the standard errors. The GIM corrects for this by measuring the *actual* variance of the score (gradient of the log-likelihood) across bootstrap replicates of genomic blocks. The resulting standard errors are always at least as large as the FIM ones, and often 2-5 times larger for whole-genome data.

```

def godambe_uncertainty(params_opt, data_sfs, model_func, ns,
                        bootstrap_sfss, eps=0.01):
    """Compute parameter uncertainties using the Godambe Information Matrix.

    Parameters
    -----
    params_opt : array-like
        Optimized parameters.

```

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```

data_sfs : ndarray
    Full observed SFS.
model_func : callable
ns : list
bootstrap_sfss : list of ndarray
    SFS from bootstrap resampling of genomic blocks.
eps : float
    Step size for numerical derivatives.

Returns
-----
se_godambe : ndarray
    Standard errors from GIM.
"""
k = len(params_opt)

# H: Hessian (= FIM) from the full data
H = fisher_information_numerical(params_opt, data_sfs, model_func, ns, eps)

# Score function: gradient of log-likelihood at the MLE
def score(p, data):
    grad = np.zeros(k)
    for i in range(k):
        p_plus = p.copy(); p_plus[i] *= (1 + eps)
        p_minus = p.copy(); p_minus[i] *= (1 - eps)

        model_p = model_func(p_plus, ns)
        model_m = model_func(p_minus, ns)
        theta_p = optimal_theta_scaling(data, model_p)
        theta_m = optimal_theta_scaling(data, model_m)

        ll_p = poisson_log_likelihood(data, model_p * theta_p)
        ll_m = poisson_log_likelihood(data, model_m * theta_m)

        # Central-difference approximation to the partial derivative
        grad[i] = (ll_p - ll_m) / (p[i] * 2 * eps)
    return grad

# J: empirical variance of the score across bootstraps
scores = np.array([score(params_opt, bs) for bs in bootstrap_sfss])
J = np.cov(scores, rowvar=False) * len(bootstrap_sfss)

# GIM = H^{-1} J H^{-1} (sandwich estimator)
H_inv = np.linalg.inv(H)
GIM = H_inv @ J @ H_inv

return np.sqrt(np.diag(GIM))

```

i When to use GIM vs FIM

Always prefer GIM when you have real data. The FIM is appropriate only when sites are truly independent (e.g.,

simulation studies where you simulated unlinked loci). For real genomic data, linkage between nearby sites inflates the variance, and the FIM will underestimate your uncertainty – sometimes dramatically.

With parameter estimates and confidence intervals in hand, we often want to compare *models* – not just parameters.

Step 6: Model Comparison

Often the question isn't “what are the parameters?” but “which model is better?” For example: does a two-epoch model fit significantly better than a constant-size model?

Likelihood Ratio Test

If model A is nested within model B (i.e., model A is model B with some parameters fixed), the likelihood ratio statistic:

$$\Lambda = 2(\ell_B - \ell_A)$$

is approximately χ^2 -distributed with $k_B - k_A$ degrees of freedom, where k is the number of free parameters.

Probability Aside – The chi-squared approximation

The likelihood ratio test rests on **Wilks' theorem**: under the null hypothesis (the simpler model), the statistic Λ converges to a χ^2 distribution as the sample size grows. Intuitively, near the MLE the log-likelihood is approximately quadratic (by a Taylor expansion), and a quadratic form in Gaussian variables is chi-squared. The degrees of freedom equal the number of additional parameters in the complex model, because each extra parameter “uses up” one dimension of the likelihood surface.

```
from scipy.stats import chi2

def likelihood_ratio_test(ll_simple, ll_complex, df):
    """Likelihood ratio test for nested models.

    Parameters
    -----
    ll_simple : float
        Log-likelihood of the simpler model.
    ll_complex : float
        Log-likelihood of the more complex model.
    df : int
        Difference in number of free parameters.

    Returns
    -----
    p_value : float
    """
    lr = 2 * (ll_complex - ll_simple) # likelihood ratio statistic
    p_value = 1 - chi2.cdf(lr, df) # p-value from chi-squared distribution
    return p_value

# Example: constant size (0 params) vs two-epoch (2 params: nu, T)
n = 30
theta = 2000
```

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```

# "Data" from a two-epoch model
data = two_epoch_model([5.0, 0.2], [n]) * theta
np.random.seed(42)
data_noisy = np.array([np.random.poisson(max(0, d)) for d in data])

# Fit constant-size model (no free parameters)
model_const = moments.Demographics1D.snm([n])
theta_const = optimal_theta_scaling(data_noisy, model_const)
ll_const = poisson_log_likelihood(data_noisy, model_const * theta_const)

# Fit two-epoch model (2 free parameters)
model_2epoch = two_epoch_model([nu_hat, T_hat], [n])
theta_2epoch = optimal_theta_scaling(data_noisy, model_2epoch)
ll_2epoch = poisson_log_likelihood(data_noisy, model_2epoch * theta_2epoch)

p_val = likelihood_ratio_test(ll_const, ll_2epoch, df=2)
print(f"Log-likelihood (constant): {ll_const:.2f}")
print(f"Log-likelihood (2-epoch): {ll_2epoch:.2f}")
print(f"LRT p-value: {p_val:.2e}")
print(f"Two-epoch significantly better: {p_val < 0.05}")

```

AIC

For non-nested models, use the **Akaike Information Criterion**:

$$\text{AIC} = 2k - 2\ell$$

where k is the number of parameters and ℓ is the log-likelihood. Lower AIC = better model (penalizing complexity).

Calculus Aside – Why AIC penalizes complexity

The AIC is derived from an asymptotic approximation to the **Kullback–Leibler divergence** between the true distribution and the fitted model. The -2ℓ term measures how well the model fits the data (smaller is better). The $2k$ term corrects for the fact that a model with more parameters can always fit the *training* data better, even if the extra parameters are capturing noise rather than signal. The AIC balances fit against parsimony: it prefers the simplest model that explains the data adequately.

Beyond the SFS-based likelihood, `moments` can also use demographic models specified in the `demes` format.

Step 7: Inference with Demes

`moments` integrates with the `demes` standard for specifying demographic models in human-readable YAML format. This is the recommended approach for complex models.

```

import moments
import demes

# Define a model using demes
b = demes.Graph(description="Two-epoch expansion model")
b.add_deme("ancestral", epochs=[

```

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```

    demes.Epoch(start_size=10000, end_time=5000)
])
b.add_deme("modern", ancestors=["ancestral"], epochs=[
    demes.Epoch(start_size=50000, end_time=0)
])
graph = b.asdict_simplified()

# Compute expected SFS directly from the demes graph
fs = moments.Spectrum.from_demes(
    graph,
    samples={"modern": 20},
    theta=4 * 10000 * 1.5e-8 * 1e6 # 4*Ne*mu*L for 1 Mb
)

print(f"SFS shape: {fs.shape}")
print(f"Segregating sites: {fs[1:-1].sum()*.1f}")

```

i The demes advantage

With demes, you specify your model once in a standardized format that works across tools (moments, msprime, dadi, fwdpy11). moments automatically translates the demes graph into the right sequence of integration steps, splits, and migration matrices.

Finally, a practical concern that can bias inference if ignored: ancestral misidentification.

Step 8: Ancestral Misidentification

In practice, the outgroup used to polarize mutations isn't perfect. Some fraction p_{misid} of sites have the ancestral/derived labels swapped. This flips $j \rightarrow n - j$ in the SFS for those sites.

The observed SFS is a mixture:

$$\text{SFS}_{\text{obs}}[j] = (1 - p_{\text{misid}}) \cdot \text{SFS}_{\text{true}}[j] + p_{\text{misid}} \cdot \text{SFS}_{\text{true}}[n - j]$$

moments can jointly estimate p_{misid} along with the demographic parameters.

```

def apply_misidentification(sfs, p_misid):
    """Apply ancestral misidentification to an SFS.

    Parameters
    -----
    sfs : ndarray of shape (n+1,)
    p_misid : float
        Fraction of sites with ancestral/derived labels swapped.

    Returns
    -----
    sfs_obs : ndarray of shape (n+1,)
    """
    n = len(sfs) - 1
    sfs_obs = np.zeros(n + 1)

```

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```

for j in range(n + 1):
    # Mix correctly polarized and flipped contributions
    sfs_obs[j] = (1 - p_misid) * sfs[j] + p_misid * sfs[n - j]
return sfs_obs

# Example: 2% misidentification
n = 20
sfs_true = expected_sfs_neutral(n, theta=1000)
sfs_misid = apply_misidentification(sfs_true, 0.02)

print("Effect of 2% ancestral misidentification:")
print(f"{'j':>3} {'True':>10} {'Observed':>10} {'Diff%':>8}")
for j in range(1, 6):
    diff_pct = (sfs_misid[j] - sfs_true[j]) / sfs_true[j] * 100
    print(f"{'j':3d} {'sfs_true[j]:10.2f} {'sfs_misid[j]:10.2f} {'diff_pct:+7.2f}%"")
for j in range(n-4, n):
    diff_pct = (sfs_misid[j] - sfs_true[j]) / sfs_true[j] * 100
    print(f"{'j':3d} {'sfs_true[j]:10.2f} {'sfs_misid[j]:10.2f} {'diff_pct:+7.2f}%"")
print("(High-frequency bins gain counts from misidentified singletons)")

```

Exercises

Exercise 1: Likelihood surface

For a two-epoch model with $n = 30$ and simulated data, compute the log-likelihood on a 50×50 grid of (ν, T) values. Plot the surface as a contour map. Where is the maximum? Is the surface unimodal?

Exercise 2: FIM vs GIM

Simulate 100 independent SFS replicates under the same two-epoch model. For each, compute the FIM standard errors. Compare the distribution of $(\hat{\nu} - \nu_{\text{true}})/\text{SE}_{\text{FIM}}$ to a standard normal. Does the FIM correctly predict the spread?

Exercise 3: Model comparison

Simulate data under a three-epoch model (bottleneck then expansion). Fit both a two-epoch and three-epoch model. Use the LRT to test whether the three-epoch model is significantly better.

Exercise 4: Misidentification bias

Generate data with 5% ancestral misidentification but fit without accounting for it. How biased are the demographic parameter estimates? Now fit with `moments`' built-in misidentification correction. Does the bias disappear?

Solutions

i Solution 1: Likelihood surface

Compute the Poisson log-likelihood on a grid of (ν, T) values and display the result as a contour map. The surface should be unimodal with the maximum near the true parameters.

```
import numpy as np
import moments

# --- Generate data ---
n = 30
theta_true = 2000
nu_true, T_true = 5.0, 0.2

model_true = two_epoch_model([nu_true, T_true], [n])
data = model_true * theta_true
np.random.seed(42)
data_noisy = np.zeros(n + 1)
for j in range(1, n):
    data_noisy[j] = np.random.poisson(data[j])

# --- Build a 50 x 50 grid ---
nu_grid = np.linspace(0.5, 15.0, 50)
T_grid = np.linspace(0.01, 0.8, 50)
ll_surface = np.zeros((50, 50))

for i, nu in enumerate(nu_grid):
    for j, T in enumerate(T_grid):
        model = two_epoch_model([nu, T], [n])
        theta_opt = optimal_theta_scaling(data_noisy, model)
        model_scaled = model * theta_opt
        ll_surface[i, j] = poisson_log_likelihood(data_noisy, model_scaled)

# --- Find the maximum ---
idx = np.unravel_index(np.argmax(ll_surface), ll_surface.shape)
nu_best = nu_grid[idx[0]]
T_best = T_grid[idx[1]]

print(f"Grid maximum at nu={nu_best:.2f}, T={T_best:.3f}")
print(f"True parameters: nu={nu_true}, T={T_true}")
print(f"Max log-likelihood: {ll_surface[idx]:.2f}")

# --- Contour plot (if matplotlib available) ---
try:
    import matplotlib.pyplot as plt
    fig, ax = plt.subplots(figsize=(7, 5))
    NU, TT = np.meshgrid(nu_grid, T_grid, indexing='ij')
    levels = np.linspace(ll_surface.max() - 20, ll_surface.max(), 15)
    cs = ax.contourf(NU, TT, ll_surface, levels=levels, cmap='viridis')
    ax.plot(nu_true, T_true, 'r*', markersize=14, label='True')
    ax.plot(nu_best, T_best, 'wx', markersize=10, label='Grid max')
    ax.set_xlabel(r'$\nu$ (expansion factor)')
    ax.set_ylabel(r'$T$ (time since expansion)')
    ax.set_title('Log-likelihood surface')
    ax.legend()
```

```
plt.colorbar(cs, label='Log-likelihood')
plt.tight_layout()
plt.savefig('likelihood_surface.png', dpi=150)
plt.show()
except ImportError:
    print("(matplotlib not available -- skipping plot)")
```

The surface should be unimodal. A ridge running diagonally indicates a correlation between ν and T : a larger expansion can be partially compensated by a more recent onset. This correlation is captured by the off-diagonal elements of the Fisher Information Matrix.

i Solution 2: FIM vs GIM

Simulate 100 independent SFS replicates, fit each one, compute FIM-based z-scores, and check whether they follow a standard normal.

```
import numpy as np
import moments
from scipy.optimize import minimize

n = 30
nu_true, T_true = 5.0, 0.2
theta_true = 2000

model_true = two_epoch_model([nu_true, T_true], [n])
data_expected = model_true * theta_true

z_scores_nu = []
z_scores_T = []

for rep in range(100):
    np.random.seed(rep)
    # Simulate Poisson data
    data_rep = np.zeros(n + 1)
    for j in range(1, n):
        data_rep[j] = np.random.poisson(data_expected[j])

    # Fit the two-epoch model
    log_p0 = np.log([3.0, 0.3])
    result = minimize(
        objective, log_p0, args=(data_rep, [n]),
        method='Nelder-Mead', options={'maxiter': 1000, 'xatol': 1e-4}
    )
    nu_hat, T_hat = np.exp(result.x)
    params_hat = np.array([nu_hat, T_hat])

    # Compute FIM and standard errors
    FIM = fisher_information_numerical(
        params_hat, data_rep, two_epoch_model, [n]
    )
    if np.linalg.det(FIM) > 0:
        cov = np.linalg.inv(FIM)
        se = np.sqrt(np.diag(cov))
```

```

        z_scores_nu.append((nu_hat - nu_true) / se[0])
        z_scores_T.append((T_hat - T_true) / se[1])

z_nu = np.array(z_scores_nu)
z_T = np.array(z_scores_T)

print(f"z-scores for nu: mean={z_nu.mean():.3f}, std={z_nu.std():.3f}")
print(f"z-scores for T: mean={z_T.mean():.3f}, std={z_T.std():.3f}")
print(f"Expected for N(0,1): mean=0.000, std=1.000")

```

If the FIM correctly predicts the spread, the z-score standard deviation should be close to 1.0. With Poisson-sampled data (truly independent sites), the FIM is the correct information measure, so the z-scores should indeed follow a standard normal. With real genomic data, linkage would inflate the true variance beyond what the FIM predicts, making the z-score distribution wider than $N(0, 1)$ – this is precisely why the GIM correction is needed for real data.

i Solution 3: Model comparison

Simulate data under a three-epoch model (bottleneck then expansion), fit both a two-epoch and three-epoch model, and use the LRT.

```

import numpy as np
import moments
from scipy.optimize import minimize
from scipy.stats import chi2

n = 30
theta_true = 3000

# --- Three-epoch model: ancestral -> bottleneck -> expansion ---
def three_epoch_model(params, ns):
    nu_B, nu_E, T_B, T_E = params
    fs = moments.Demographics1D.snm(ns)
    fs.integrate([nu_B], T_B)    # bottleneck
    fs.integrate([nu_E], T_E)    # expansion
    return fs

# True parameters: bottleneck to 0.1x, then expansion to 5x
params_true_3 = [0.1, 5.0, 0.05, 0.2]
model_true_3 = three_epoch_model(params_true_3, [n])
data = model_true_3 * theta_true

np.random.seed(42)
data_noisy = np.zeros(n + 1)
for j in range(1, n):
    data_noisy[j] = np.random.poisson(data[j])

# --- Fit the two-epoch model (2 free parameters) ---
def obj_2epoch(log_p, data_sfs, ns):
    nu, T = np.exp(log_p)
    model = two_epoch_model([nu, T], ns)
    theta_opt = optimal_theta_scaling(data_sfs, model)
    return -poisson_log_likelihood(data_sfs, model * theta_opt)

```

```

res_2 = minimize(obj_2epoch, np.log([2.0, 0.3]), args=(data_noisy, [n]),
                 method='Nelder-Mead', options={'maxiter': 2000})
nu2, T2 = np.exp(res_2.x)
model_2 = two_epoch_model([nu2, T2], [n])
theta_2 = optimal_theta_scaling(data_noisy, model_2)
ll_2 = poisson_log_likelihood(data_noisy, model_2 * theta_2)

# --- Fit the three-epoch model (4 free parameters) ---
def obj_3epoch(log_p, data_sfs, ns):
    params = np.exp(log_p)
    model = three_epoch_model(params, ns)
    theta_opt = optimal_theta_scaling(data_sfs, model)
    return -poisson_log_likelihood(data_sfs, model * theta_opt)

res_3 = minimize(obj_3epoch, np.log([0.5, 3.0, 0.1, 0.1]),
                 args=(data_noisy, [n]),
                 method='Nelder-Mead', options={'maxiter': 5000})
p3 = np.exp(res_3.x)
model_3 = three_epoch_model(p3, [n])
theta_3 = optimal_theta_scaling(data_noisy, model_3)
ll_3 = poisson_log_likelihood(data_noisy, model_3 * theta_3)

# --- Likelihood ratio test (df = 4 - 2 = 2) ---
lr_stat = 2 * (ll_3 - ll_2)
p_value = 1 - chi2.cdf(lr_stat, df=2)

print(f"Two-epoch:   ll = {ll_2:.2f}   (nu={nu2:.3f}, T={T2:.4f})")
print(f"Three-epoch: ll = {ll_3:.2f}   (nuB={p3[0]:.3f}, nuE={p3[1]:.3f}, "
      f"TB={p3[2]:.4f}, TE={p3[3]:.4f})")
print(f"LR statistic: {lr_stat:.2f}")
print(f"p-value: {p_value:.2e}")
print(f"Three-epoch significantly better (p < 0.05): {p_value < 0.05}")

```

Since the data were generated under a three-epoch model, the three-epoch fit should have a significantly higher likelihood. The LRT with 2 degrees of freedom should reject the simpler two-epoch model at the 0.05 level.

Solution 4: Misidentification bias

Generate data with 5% ancestral misidentification, fit without correction, then fit with the `apply_misidentification` correction and compare.

```

import numpy as np
import moments
from scipy.optimize import minimize

n = 30
nu_true, T_true = 5.0, 0.2
theta_true = 2000
p_misid_true = 0.05

# Generate true SFS and apply misidentification

```

```

model_true = two_epoch_model([nu_true, T_true], [n])
sfs_true = model_true * theta_true
sfs_misid = apply_misidentification(np.array(sfs_true), p_misid_true)

np.random.seed(42)
data_obs = np.zeros(n + 1)
for j in range(1, n):
    data_obs[j] = np.random.poisson(sfs_misid[j])

# --- Fit WITHOUT misidentification correction ---
def obj_no_correction(log_p, data_sfs, ns):
    nu, T = np.exp(log_p)
    model = two_epoch_model([nu, T], ns)
    theta_opt = optimal_theta_scaling(data_sfs, model)
    return -poisson_log_likelihood(data_sfs, model * theta_opt)

res_no = minimize(obj_no_correction, np.log([3.0, 0.3]),
                  args=(data_obs, [n]),
                  method='Nelder-Mead', options={'maxiter': 2000})
nu_no, T_no = np.exp(res_no.x)

# --- Fit WITH misidentification correction ---
def obj_with_correction(log_p, data_sfs, ns):
    # params: [log(nu), log(T), logit(p_misid)]
    nu, T = np.exp(log_p[:2])
    p_misid = 1.0 / (1.0 + np.exp(-log_p[2])) # sigmoid for (0, 1)
    model = two_epoch_model([nu, T], ns)
    theta_opt = optimal_theta_scaling(data_sfs, model)
    model_scaled = np.array(model * theta_opt)
    model_corrected = apply_misidentification(model_scaled, p_misid)
    return -poisson_log_likelihood(data_sfs, model_corrected)

# Initial guess: logit(0.02) for p_misid
logit_p0 = np.log(0.02 / (1 - 0.02))
res_with = minimize(obj_with_correction,
                    np.array([np.log(3.0), np.log(0.3), logit_p0]),
                    args=(data_obs, [n]),
                    method='Nelder-Mead', options={'maxiter': 3000})
nu_with, T_with = np.exp(res_with.x[:2])
p_misid_hat = 1.0 / (1.0 + np.exp(-res_with.x[2]))

print("True parameters:    nu=5.000, T=0.2000, p_misid=0.050")
print(f"Without correction: nu={nu_no:.3f}, T={T_no:.4f}")
print(f"  Bias in nu: {(nu_no - nu_true)/nu_true*100:+.1f}%")
print(f"  Bias in T:  {(T_no - T_true)/T_true*100:+.1f}%")
print(f"With correction:    nu={nu_with:.3f}, T={T_with:.4f}, "
      f"p_misid={p_misid_hat:.3f}")
print(f"  Bias in nu: {(nu_with - nu_true)/nu_true*100:+.1f}%")
print(f"  Bias in T:  {(T_with - T_true)/T_true*100:+.1f}%")

```

Without correction, the 5% misidentification inflates the high-frequency bins (singletons are “flipped” to appear as $n - 1$ counts), biasing the inferred demography – typically making the population appear less expanded than it truly is. With the correction, the optimizer jointly recovers ν , T , and p_{misid} , removing the bias.

Next: *Linkage Disequilibrium* – a second source of information about demography, captured by two-locus statistics. Where the SFS is our primary dial, LD is a **second pendulum** that constrains parameters the SFS alone cannot resolve.

Linkage Disequilibrium

A second pendulum: reading demographic history from correlations between loci.

The SFS captures how often alleles appear at each frequency, but it ignores the **relationships between loci**. Two sites that are physically close on a chromosome tend to be inherited together – they are in **linkage disequilibrium** (LD). The rate at which LD decays with physical distance depends on population history, providing information that the SFS alone cannot.

`moments.LD` extends the moment equations framework to two-locus statistics, opening a second window onto demographic history – one that is especially powerful for detecting recent events like admixture.

In the watch metaphor developed in *Overview of moments*, the SFS is the main **dial face** – the primary read-out of population history. LD is a **second pendulum** mounted alongside the first. Both are driven by the same gear train (drift, mutation, migration), but the second pendulum also responds to a force the first cannot feel: **recombination**. Because recombination rate scales with N_e (via $\rho = 4N_e r$), the LD pendulum constrains the absolute population size – something the SFS, which sees only $\theta = 4N_e \mu$, cannot do on its own. Together, the two dials break the degeneracy and pin down both N_e and μ separately.

Step 1: What Is Linkage Disequilibrium?

Consider two biallelic loci, A and B. Locus A has alleles A_0, A_1 at frequencies $1 - p, p$. Locus B has alleles B_0, B_1 at frequencies $1 - q, q$.

There are four possible **haplotypes**:

Haplotype	Notation	Frequency
$A_0 B_0$	x_{00}	Expected: $(1 - p)(1 - q)$ if independent
$A_0 B_1$	x_{01}	Expected: $(1 - p)q$ if independent
$A_1 B_0$	x_{10}	Expected: $p(1 - q)$ if independent
$A_1 B_1$	x_{11}	Expected: pq if independent

If the two loci are **independent** (in linkage equilibrium), haplotype frequencies are just the product of allele frequencies. **Linkage disequilibrium** is any departure from this independence:

$$D = x_{11} - pq = x_{11}x_{00} - x_{01}x_{10}$$

$D > 0$ means alleles A_1 and B_1 co-occur more often than expected by chance. $D < 0$ means they co-occur less often.

i Probability Aside – LD as covariance

If you assign the value 1 to allele A_1 and 0 to A_0 (and similarly for locus B), then for a randomly chosen chromosome:

$$D = E[A \cdot B] - E[A] \cdot E[B] = \text{Cov}(A, B)$$

LD is literally the **covariance** between the allelic states at two loci. Just as the covariance between two random variables measures their linear association, D measures how much knowing the allele at one locus tells you about the allele at the other. When $D = 0$, the loci are statistically independent (at the population level); when $D \neq 0$, they are associated.

```

import numpy as np

def compute_D(haplotypes):
    """Compute D for two biallelic loci from haplotype data.

    Parameters
    -----
    haplotypes : ndarray of shape (n, 2)
        Each row is a haplotype. Columns are the two loci (0/1).

    Returns
    -----
    D : float
    """
    n = len(haplotypes)
    p = haplotypes[:, 0].mean() # frequency of allele 1 at locus A
    q = haplotypes[:, 1].mean() # frequency of allele 1 at locus B
    x11 = ((haplotypes[:, 0] == 1) & (haplotypes[:, 1] == 1)).mean() # freq of (1,1)
    ↪haplotype
    D = x11 - p * q # departure from independence
    return D

# Example: 100 haplotypes, two loci in strong LD
np.random.seed(42)
n = 100
# All haplotypes are either (0,0) or (1,1) -- perfect LD
hap_ld = np.array([[0, 0]] * 60 + [[1, 1]] * 40)
np.random.shuffle(hap_ld)

D_strong = compute_D(hap_ld)
p = hap_ld[:, 0].mean()
q = hap_ld[:, 1].mean()
D_max = min(p * (1 - q), q * (1 - p)) # maximum possible D given allele freqs

print(f"p = {p:.2f}, q = {q:.2f}")
print(f"D = {D_strong:.4f}")
print(f"D_max = {D_max:.4f}")
print(f"D' = D/D_max = {D_strong/D_max:.4f}")

```

Intuition: LD is the covariance between allelic states at two loci. Just as correlation measures association between two random variables, LD measures association between alleles. Recombination breaks up these associations over time, so LD decays with distance and with time.

Now that we know what LD is, let us see how recombination erodes it.

Step 2: LD Decay and Recombination

In each generation, recombination between two loci occurs with probability r (the recombination rate). Each recombination event breaks up the existing LD:

$$E[D_{t+1}] = (1 - r) \cdot D_t$$

After t generations without new LD being created:

$$E[D_t] = (1 - r)^t \cdot D_0 \approx e^{-rt} \cdot D_0$$

LD decays **exponentially** with time at a rate proportional to r . This is the deterministic approximation. In a finite population, genetic drift constantly creates new LD (randomly), which balances the decay and produces a nonzero equilibrium level.

i Calculus Aside – Exponential decay and half-life

The recurrence $D_{t+1} = (1 - r)D_t$ is a first-order linear difference equation with solution $D_t = (1 - r)^t D_0$. For small r , the approximation $(1 - r)^t \approx e^{-rt}$ converts this into the continuous exponential decay $D(t) = D_0 e^{-rt}$. The **half-life** – the time for LD to drop to half its initial value – is:

$$t_{1/2} = \frac{\ln 2}{r}$$

For $r = 0.01$ (about 1 centimorgan), the half-life is roughly 69 generations. For $r = 10^{-4}$ (about 0.01 cM, or roughly 10 kb in humans), the half-life is about 6,900 generations – long enough that LD from ancient events can still be detected.

```
def ld_decay_deterministic(D0, r, t_generations):
    """Deterministic LD decay over t generations.

    Parameters
    -----
    D0 : float
        Initial D.
    r : float
        Recombination rate between loci.
    t_generations : int
        Number of generations.

    Returns
    -----
    D_t : float
    """
    return D0 * (1 - r) ** t_generations # exponential decay

# Example: LD decays to half in log(2)/r generations
r = 0.01 # 1% recombination rate (= ~1 cM distance)
half_life = np.log(2) / r
print(f"Half-life of LD at r={r}: {half_life:.0f} generations")

# Plot LD decay
D0 = 0.1
t_range = range(0, 500)
D_vals = [ld_decay_deterministic(D0, r, t) for t in t_range]
print(f"D at t=0: {D_vals[0]:.4f}")
print(f"D at t=69: {D_vals[69]:.4f} (approximately half-life)")
print(f"D at t=500: {D_vals[-1]:.6f}")
```

With the basic dynamics in place, we can now introduce the specific LD statistics that `moments.LD` tracks.

Step 3: The Three LD Statistics in moments

Raw D is hard to work with because it depends on allele frequencies. `moments.LD` tracks three **expected values** of normalized LD statistics, computed as moments of the two-locus haplotype distribution:

Statistic 1: $E[D^2]$

The expected squared LD coefficient. This is the numerator of the commonly used r^2 :

$$r^2 = \frac{D^2}{p(1-p)q(1-q)}$$

$E[D^2]$ measures the overall *magnitude* of LD, regardless of sign.

Statistic 2: $E[Dz]$ where $z = (1-2p)(1-2q)$

$$E[Dz] = E[D \cdot (1-2p)(1-2q)]$$

This is D weighted by how far allele frequencies are from 0.5. It has a remarkable property: **it is strongly elevated by admixture** between diverged populations, even at low admixture proportions.

Intuition: When two populations with different allele frequencies admix, the resulting LD has a specific pattern: D and $(1-2p)(1-2q)$ tend to have the same sign (because alleles that are common in one population and rare in the other create positive D between loci where the same population is at high frequency, and these loci also tend to have $(1-2p)$ of the same sign). This makes $E[Dz]$ an exquisitely sensitive detector of admixture.

Probability Aside – Why $E[Dz]$ detects admixture

Consider two source populations that have diverged long enough that their allele frequencies differ appreciably. At a pair of loci where population 1 has frequencies (p_1, q_1) and population 2 has (p_2, q_2) , the admixed population (mixing fraction α) inherits LD of the form:

$$D_{\text{admix}} \approx \alpha(1-\alpha)(p_1 - p_2)(q_1 - q_2)$$

The factor $(1-2p)(1-2q)$ has the same sign structure as $(p_1 - p_2)(q_1 - q_2)$ when p and q lie between the two source frequencies. As a result, $D \cdot (1-2p)(1-2q)$ is systematically positive across locus pairs, producing a large positive $E[Dz]$. Under a purely drifting (non-admixed) population, the sign of D is random, so $E[Dz] \approx 0$. This is why $E[Dz]$ serves as a near-zero-background signal for admixture.

Statistic 3: $\pi_2 = E[p(1-p)q(1-q)]$

The product of heterozygosities at two loci – a normalization factor:

$$\pi_2 = E[p(1-p) \cdot q(1-q)]$$

This does not carry LD information per se; it serves as the denominator for normalized statistics.

Normalized statistics

The statistics used for inference are the ratios:

$$\sigma_d^2 = \frac{E[D^2]}{E[\pi_2]}, \quad \sigma_{Dz} = \frac{E[Dz]}{E[\pi_2]}$$

These are analogous to r^2 but defined as ratios of expectations (not expectations of ratios), which is mathematically cleaner and avoids singularities when individual-site heterozygosities are small.

i Calculus Aside – Ratio of expectations vs. expectation of ratio

For two random variables X and Y , the ratio of expectations $E[X]/E[Y]$ is generally *not* equal to the expectation of the ratio $E[X/Y]$ (Jensen's inequality). The former is well-defined even when Y can be zero for some realizations; the latter can diverge. By defining σ_d^2 as $E[D^2]/E[\pi_2]$ rather than $E[D^2/(p(1-p)q(1-q))]$, `moments.LD` avoids the singularity that arises when one locus is nearly monomorphic (heterozygosity near zero). This is a common technique in survey sampling and meta-analysis.

```
def compute_ld_statistics(haplotype_matrix):
    """Compute  $E[D^2]$ ,  $E[Dz]$ , and  $\pi_2$  from a haplotype matrix.

    Parameters
    -----
    haplotype_matrix : ndarray of shape (n_haplotypes, n_loci)
        Binary matrix (0/1) for each locus.

    Returns
    -----
    D2_mean, Dz_mean, pi2_mean : float
        Average  $D^2$ ,  $Dz$ , and  $\pi_2$  over all pairs of loci.
    """
    n_haps, n_loci = haplotype_matrix.shape
    D2_sum, Dz_sum, pi2_sum = 0.0, 0.0, 0.0
    n_pairs = 0

    for i in range(n_loci):
        for j in range(i + 1, n_loci):
            p = haplotype_matrix[:, i].mean() # allele freq at locus i
            q = haplotype_matrix[:, j].mean() # allele freq at locus j

            # Skip monomorphic loci (no LD can be computed)
            if p == 0 or p == 1 or q == 0 or q == 1:
                continue

            # Compute  $D = \text{freq}(1,1) - p*q$ 
            x11 = ((haplotype_matrix[:, i] == 1) &
                   (haplotype_matrix[:, j] == 1)).mean()
            D = x11 - p * q

            D2_sum += D ** 2 # squared LD
            Dz_sum += D * (1 - 2*p) * (1 - 2*q) # admixture-sensitive statistic
            pi2_sum += p * (1 - p) * q * (1 - q) # product of heterozygosities
            n_pairs += 1

    if n_pairs == 0:
        return 0, 0, 0

    return D2_sum / n_pairs, Dz_sum / n_pairs, pi2_sum / n_pairs
```

Having defined the LD statistics, we now derive the ODEs that govern their evolution – the two-locus analogue of the moment equations in *The Moment Equations*.

Step 4: Two-Locus Moment Equations

Just as the SFS entries satisfy ODEs under drift/mutation/selection, the two-locus statistics $E[D^2]$, $E[Dz]$, and π_2 satisfy their own system of ODEs.

The key forces acting on two-locus statistics:

Drift creates LD (randomly) and changes allele frequencies:

$$\left. \frac{dE[D^2]}{dt} \right|_{\text{drift}} = (\text{terms involving } E[D^2], E[Dz], \pi_2, \text{ and heterozygosities})$$

Recombination destroys LD:

$$\left. \frac{dE[D^2]}{dt} \right|_{\text{recomb}} = -2\rho \cdot E[D^2]$$

where $\rho = 4N_e r$ is the scaled recombination rate. The factor of 2 comes from the fact that each recombination event reduces D by a factor $(1 - r)$, so D^2 is reduced by $(1 - r)^2 \approx 1 - 2r$.

Calculus Aside – Why D^2 decays at rate 2ρ

Recall from Step 2 that $D_{t+1} = (1 - r)D_t$. Squaring both sides: $D_{t+1}^2 = (1 - r)^2 D_t^2$. Taking expectations and using the approximation $(1 - r)^2 \approx 1 - 2r$ for small r :

$$E[D_{t+1}^2] \approx (1 - 2r) E[D_t^2]$$

Converting to diffusion time ($dt = 1/(2N_e)$ generations) and using $\rho = 4N_e r$:

$$\left. \frac{dE[D^2]}{dt} \right|_{\text{recomb}} = -2\rho E[D^2]$$

This is exponential decay with rate constant 2ρ . The factor of 2 arises because D^2 is a *second-order* statistic: each recombination event reduces D by a factor $(1 - r)$, so D^2 is reduced by $(1 - r)^2$, which contributes *twice* the decay rate.

Mutation generates new variation that contributes to heterozygosity and indirectly to LD:

$$\left. \frac{d\pi_2}{dt} \right|_{\text{mutation}} = \theta \cdot E[q(1 - q)] + \theta \cdot E[p(1 - p)]$$

The full system is a set of coupled ODEs – more complex than the SFS system because it involves higher-order moments of two allele frequencies, but the same moment-closure techniques (the jackknife approximation from *The Moment Equations*) apply.

```
def ld_equilibrium(theta, rho, n_pops=1):
    """Compute equilibrium LD statistics for one population.

    At equilibrium, the rate of LD creation by drift equals the rate
    of LD decay by recombination.

    Parameters
    -----
    theta : float
        Scaled mutation rate (4*Ne*mu per site).
```

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```

rho : float
    Scaled recombination rate (4*Ne*r between the two loci).

Returns
-----
sigma_d2 : float
    Equilibrium  $\sigma_d^2 = E[D^2] / \pi_2$ .
    """
    # Under the neutral model with constant population size,
    #  $\sigma_d^2$  at equilibrium is approximately:
    #
    #  $\sigma_d^2 \sim 1 / (1 + \rho)$  (for large n, low theta)
    #
    # This is because:
    # - Drift creates  $D^2$  at a rate proportional to  $\pi_2$ 
    #   (roughly  $1/(2N)$  per generation, or 1 in diffusion time)
    # - Recombination destroys  $D^2$  at rate  $2\rho$ 
    # - At equilibrium: creation = destruction
    # -  $D^2 \sim \pi_2 / (1 + \rho)$  [simplified]
    return 1.0 / (1.0 + rho)

# Compare with moments
import moments.LD

rho_values = [0.1, 0.5, 1.0, 2.0, 5.0, 10.0]
theta = 0.001

print(f"{'rho':>6} {'sigma_d2 (approx)':>18} {'sigma_d2 (moments)':>20}")
print("-" * 46)
for rho in rho_values:
    # Compute with moments.LD: solve the two-locus moment equations at equilibrium
    y = moments.LD.LDstats(
        moments.LD.Numerics.steady_state([rho], theta=theta),
        num_pops=1, pop_ids=["pop0"]
    )
    #  $\sigma_d^2 = E[D^2] / \pi_2$ 
    sigma_d2_moments = y.D2() / y.pi2()

    sigma_d2_approx = ld_equilibrium(theta, rho)
    print(f"{'rho':6.1f} {'sigma_d2_approx':18.6f} {'sigma_d2_moments':20.6f}")

```

i The recombination-LD tradeoff

$\sigma_d^2 \approx 1/(1 + \rho)$ is a profound result. It says that the equilibrium level of LD between two loci is determined solely by the **ratio of drift to recombination**. Closer loci (lower ρ) have more LD. This is why LD decay curves – plots of σ_d^2 vs. recombination distance – encode demographic information: the shape of the decay depends on population history.

With the equilibrium baseline established, we now examine how specific demographic events distort the LD decay curve.

Step 5: How Demography Affects LD

Different demographic events leave characteristic signatures in the LD decay curve:

Population expansion

After an expansion, drift is weak (large N). Existing LD decays by recombination but new LD is created slowly. **Result:** lower σ_d^2 than expected under constant size.

Bottleneck

During a bottleneck, drift is strong. Lots of LD is created. Even after the population recovers, the excess LD at short distances persists for many generations. **Result:** elevated σ_d^2 , especially at short distances.

Admixture

When two diverged populations admix, they bring together different haplotypes. This creates LD between *any* pair of loci where the source populations had different allele frequencies. The resulting “admixture LD” decays with recombination distance because recombination breaks up the immigrant haplotypes. **Result:** elevated σ_{Dz} that decays with distance – the hallmark of admixture.

i Probability Aside – Admixture LD and the ROLLOFF/ALDER signal

The LD created by admixture at generation t_a in the past decays as e^{-rt_a} with recombination distance r . Plotting admixture LD against genetic distance therefore produces an exponential curve whose decay constant is the number of generations since admixture. This is the principle behind the ROLLOFF and ALDER methods. `moments.LD` captures the same information through its σ_{Dz} statistic, but models it via the moment equations rather than fitting an empirical exponential. The moment-equation approach naturally handles complex scenarios – e.g., multiple pulses of admixture, or continuous gene flow – that would require ad hoc extensions in an exponential-fitting framework.

```
import moments.LD

def ld_after_bottleneck(nu_B, T_B, rho_values, theta=0.001):
    """Compute LD statistics after a bottleneck.

    Parameters
    -----
    nu_B : float
        Bottleneck population size (relative to N_ref).
    T_B : float
        Duration of bottleneck (2*N_ref generations).
    rho_values : list of float
```

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```

    Scaled recombination rates to compute.
    theta : float
    Scaled mutation rate.

Returns
-----
sigma_d2 : list of float
    sigma_d^2 at each rho value.
"""
sigma_d2 = []
for rho in rho_values:
    # Start from equilibrium (steady-state two-locus statistics)
    y = moments.LD.LDstats(
        moments.LD.Numerics.steady_state([rho], theta=theta),
        num_pops=1, pop_ids=["pop0"]
    )
    # Bottleneck: strong drift creates excess LD
    y.integrate([nu_B], T_B, rho=[rho], theta=theta)
    # Recovery to original size
    y.integrate([1.0], 0.1, rho=[rho], theta=theta)

    sigma_d2.append(y.D2() / y.pi2()) # normalized LD statistic
return sigma_d2

# Compare equilibrium vs post-bottleneck LD
rho_values = [0.5, 1.0, 2.0, 5.0, 10.0, 20.0]

ld_eq = [ld_equilibrium(0.001, r) for r in rho_values]
ld_bn = ld_after_bottleneck(0.1, 0.05, rho_values)

print(f"{'rho':>6} {'Equilibrium':>14} {'Post-bottleneck':>16} {'Ratio':>8}")
print("-" * 48)
for rho, eq, bn in zip(rho_values, ld_eq, ld_bn):
    print(f"{'rho':6.1f} {'eq':14.6f} {'bn':16.6f} {'bn/eq':8.3f}")
print("(Ratio > 1 = bottleneck elevates LD)")

```

The LD decay curve is our second dial reading. To use it for inference, we need a likelihood – and because LD statistics are averages rather than counts, the appropriate likelihood is Gaussian, not Poisson.

Step 6: LD-Based Demographic Inference

The inference framework for LD parallels the SFS framework, with a key difference: the likelihood is **Gaussian** rather than Poisson.

The Gaussian composite likelihood

LD statistics are computed by averaging over many pairs of SNPs within each recombination distance bin. By the central limit theorem, these averages are approximately normally distributed. The log-likelihood for a set of LD statistics $\mathbf{d}$ in recombination bin k is:

$$\ell_k = -\frac{1}{2}(\mathbf{d}_k - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\mathbf{d}_k - \boldsymbol{\mu}_k)$$

where:

- $\mathbf{d}_k$ = observed LD statistics in bin k (a vector containing σ_d^2 and optionally σ_{Dz})
- μ_k = model predictions from the moment equations
- Σ_k = variance-covariance matrix (estimated from bootstrap)

The total log-likelihood is the sum over bins:

$$\ell = \sum_k \ell_k$$

i Probability Aside – From Poisson (SFS) to Gaussian (LD)

The SFS consists of **counts** (how many sites have each allele frequency), which are naturally Poisson. LD statistics are **averages** over pairs of sites within recombination distance bins. As averages of many independent-ish terms, they follow a Gaussian distribution by the central limit theorem.

Formally, if you average N independent observations each with variance σ^2 , the sample mean has variance σ^2/N and is approximately Gaussian for large N . Each recombination bin typically contains thousands to millions of SNP pairs, so the Gaussian approximation is excellent. The covariance matrix Σ_k accounts for the fact that the same SNP can appear in multiple pairs (and that nearby pairs share haplotype information), inflating the effective variance above the naive σ^2/N .

```
def gaussian_composite_ll(data_ld, model_ld, varcov_matrices):
    """Compute Gaussian composite log-likelihood for LD statistics.

    Parameters
    -----
    data_ld : list of ndarray
        Observed LD statistics, one array per recombination bin.
    model_ld : list of ndarray
        Model-predicted LD statistics.
    varcov_matrices : list of ndarray
        Variance-covariance matrix for each bin (from bootstrap).

    Returns
    -----
    ll : float
    """
    ll = 0.0
    for d, mu, sigma in zip(data_ld, model_ld, varcov_matrices):
        residual = d - mu # data minus model prediction
        sigma_inv = np.linalg.inv(sigma) # precision matrix
        ll -= 0.5 * residual @ sigma_inv @ residual # quadratic form
    return ll
```

i Why Gaussian instead of Poisson?

The SFS consists of **counts** (how many sites have each allele frequency), which are naturally Poisson. LD statistics are **averages** over pairs of sites within recombination distance bins. As averages of many independent-ish terms, they follow a Gaussian distribution by the central limit theorem.

Step 7: Multi-Population LD and Admixture Detection

For multiple populations, LD statistics include **cross-population** terms:

- $E[D_i \cdot D_j]$: covariance of LD between populations i and j

Right after a population split, $E[D_i D_j] = E[D^2]$ (perfect correlation). Over time, independent drift in each population decorrelates their LD: $E[D_i D_j]$ decays toward zero.

i Probability Aside – Cross-population LD as shared ancestry

Two populations that recently shared an ancestor have correlated LD: the same ancestral haplotypes, and therefore the same LD patterns, were present in both at the time of the split. As independent drift in each population creates new, uncorrelated LD and recombination erodes the shared signal, $E[D_i D_j]$ decays. The rate of decay depends on both ρ (how fast recombination destroys old LD) and the population sizes (how fast new drift creates uncorrelated LD). By measuring $E[D_i D_j]$ at multiple recombination distances, one can therefore estimate both the split time and the population sizes – providing information complementary to F_{ST} from the SFS (see *The Site Frequency Spectrum*).

```
import moments.LD

def two_pop_ld(nu1, nu2, T, m, rho_values, theta=0.001):
    """Compute LD statistics for two populations after a split.

    Parameters
    -----
    nu1, nu2 : float
        Relative population sizes.
    T : float
        Time since split.
    m : float
        Symmetric migration rate.
    rho_values : list of float
    theta : float

    Returns
    -----
    results : dict
        LD statistics for each rho value.
    """
    results = {'rho': rho_values, 'DD_0_0': [], 'DD_0_1': [], 'DD_1_1': []}

    for rho in rho_values:
        # Equilibrium for one population
        y = moments.LD.LDstats(
            moments.LD.Numerics.steady_state([rho], theta=theta),
            num_pops=1, pop_ids=["anc"]
        )

        # Split into two populations
        y = y.split(0, new_ids=["pop0", "pop1"])

        # Diverge with migration
        mig = np.array([[0, m], [m, 0]])
```

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```

y.integrate([nu1, nu2], T, rho=[rho], theta=theta, m=mig)

# Collect within-population and cross-population LD
results['DD_0_0'].append(y.D2(pops="pop0"))      # LD within pop 0
results['DD_0_1'].append(y.D2(pops=["pop0", "pop1"])) # cross-pop LD
results['DD_1_1'].append(y.D2(pops="pop1"))      # LD within pop 1

return results

# Example: recent split with moderate migration
rho_values = [0.5, 1.0, 2.0, 5.0, 10.0]
res = two_pop_ld(1.0, 1.0, 0.1, 0.5, rho_values)

print("Cross-population LD correlation:")
print(f"{'rho':>6} {'DD_00':>10} {'DD_01':>10} {'DD_11':>10} {'Corr':>8}")
print("-" * 48)
for i, rho in enumerate(rho_values):
    corr = res['DD_0_1'][i] / np.sqrt(res['DD_0_0'][i] * res['DD_1_1'][i])
    print(f"{'rho':6.1f} {'res['DD_0_0']':10.6f} {'res['DD_0_1']':10.6f} "
          f"{'res['DD_1_1']':10.6f} {'corr':8.4f}")
print("(Higher correlation = more recent split or more migration)")

```

Step 8: Parsing LD from Real Data

Computing LD statistics from VCF files involves:

1. **Pair up SNPs** within recombination distance bins
2. **Compute** D^2 , Dz , π_2 for each pair
3. **Average** within bins
4. **Bootstrap** across genomic regions for uncertainty

```

# Pseudocode for the parsing pipeline
import moments.LD

def parse_ld_workflow(vcf_file, pop_file, recomb_map, bed_file):
    """Complete workflow for parsing LD from VCF data.

    Parameters
    -----
    vcf_file : str
        Path to VCF file.
    pop_file : str
        Path to population assignment file (sample -> population).
    recomb_map : str
        Path to recombination map file (position -> cumulative cM).
    bed_file : str
        Path to BED file defining genomic regions.
    """
    # Define recombination distance bins (in Morgans)
    r_bins = np.array([0, 1e-6, 2e-6, 5e-6, 1e-5, 2e-5,
                       5e-5, 1e-4, 2e-4, 5e-4, 1e-3])

```

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```

# Compute LD statistics per region (moments.LD handles the averaging)
ld_stats = moments.LD.Parsing.compute_ld_statistics(
    vcf_file,
    rec_map_file=recomb_map,
    pop_file=pop_file,
    bed_file=bed_file,
    r_bins=r_bins,
    report=False
)

return ld_stats

# In practice, you compute LD for many regions and bootstrap:
# region_stats = {i: parse_ld_workflow(..., bed=f"region_{i}.bed")
#                 for i in range(n_regions)}
# means_and_varcov = moments.LD.Parsing.bootstrap_data(region_stats)

```

i Phased vs. unphased data

`moments.LD` defaults to using **genotype-based** statistics (unphased) rather than haplotype-based (phased). This is deliberate: phasing errors create artificial LD that biases results, while using genotypes only slightly increases variance. Unless your data has very high-quality phasing, keep the default `use_genotypes=True`.

With observed LD statistics and a demographic model, the last ingredient is the conversion between physical recombination rates and scaled ρ values.

Step 9: Recombination Bins and N_e

A crucial link: the model operates in **scaled** units ($\rho = 4N_e r$), but the data is binned by **physical** recombination rates (r). The effective population size N_e provides the conversion:

$$\rho = 4N_e \cdot r$$

This means N_e can be **jointly estimated** from the LD decay curve: it determines how the physical-distance bins map onto the scaled ρ values where the model makes predictions.

i Calculus Aside – Why LD resolves the N_e - μ degeneracy

The SFS depends on $\theta = 4N_e\mu$ – it constrains the *product* of N_e and μ but cannot separate them. The LD decay curve depends on $\rho = 4N_e r$ – it constrains the *product* of N_e and r . If r is known from a genetic map (and μ is independently estimated, or vice versa), combining SFS and LD data pins down N_e and μ separately:

$$N_e = \frac{\rho}{4r}, \quad \mu = \frac{\theta}{4N_e}$$

This is the power of the “second pendulum” – it provides an independent constraint that breaks the degeneracy inherent in any single statistic.

```
def map_r_bins_to_rho(r_bins, Ne):
    """Convert physical recombination bins to scaled rho values.

    Parameters
    -----
    r_bins : ndarray
        Physical recombination rates (per generation).
    Ne : float
        Effective population size.

    Returns
    -----
    rho_bins : ndarray
        Scaled recombination rates (4*Ne*r).
    """
    return 4 * Ne * r_bins # the conversion that links LD to absolute N_e

# Example: for Ne = 10,000, r = 1e-4 maps to rho = 4
Ne_values = [5000, 10000, 20000]
r = 1e-4

for Ne in Ne_values:
    rho = 4 * Ne * r
    sigma_d2 = 1 / (1 + rho)
    print(f"Ne = {Ne:6d}: rho = {rho:5.1f}, "
          f"sigma_d^2 ~ {sigma_d2:.4f}")
```

Step 10: Putting It All Together – LD Inference

The complete LD inference workflow mirrors the SFS workflow from *Demographic Inference*:

1. Compute observed LD statistics from data (Step 8)
2. Define a demographic model
3. For candidate parameters, solve the two-locus moment equations to predict LD
4. Compare predictions to observations via the Gaussian likelihood (Step 6)
5. Optimize parameters; quantify uncertainty via bootstrap

```
import moments.LD

def ld_inference_demo():
    """Demonstrate LD-based demographic inference."""

    # --- 1. Define the demographic model ---
    def model_func(params, rho, theta):
        """Two-epoch model for LD statistics."""
        nu, T = params
        # Start from equilibrium (two-locus steady state)
        y = moments.LD.LDstats(
            moments.LD.Numerics.steady_state(rho, theta=theta),
            num_pops=1, pop_ids=["pop0"]
        )
```

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```

    # Integrate the two-locus moment equations through the size change
    y.integrate([nu], T, rho=rho, theta=theta)
    return y

# --- 2. Generate "data" ---
theta = 0.001
rho_values = [0.5, 1.0, 2.0, 5.0, 10.0, 20.0, 50.0]

# True parameters: a bottleneck
nu_true, T_true = 0.1, 0.05

y_data = model_func([nu_true, T_true], rho_values, theta)

# --- 3. Compute model for candidate parameters ---
# (Grid search for demonstration; in practice use gradient-based optimizer)
print("Searching for best-fit parameters...")
best_ll = -np.inf
best_params = None

for nu in [0.05, 0.1, 0.2, 0.5, 1.0]:
    for T in [0.01, 0.02, 0.05, 0.1, 0.2]:
        y_model = model_func([nu, T], rho_values, theta)

        # Simple sum-of-squares comparison
        # (In practice, use Gaussian likelihood with bootstrap covariance)
        ss = sum((y_data.D2(pops="pop0", r_bin=i) -
                    y_model.D2(pops="pop0", r_bin=i))**2
                  for i in range(len(rho_values)))
        ll = -ss # proxy for likelihood (lower SS = better fit)

        if ll > best_ll:
            best_ll = ll
            best_params = (nu, T)

print(f"True:      nu={nu_true}, T={T_true}")
print(f"Best grid: nu={best_params[0]}, T={best_params[1]}")

ld_inference_demo()

```

i Probability Aside – Joint SFS + LD inference

The most powerful analyses combine *both* the SFS and LD. Since the SFS likelihood (Poisson) and the LD likelihood (Gaussian) involve different data, they can be summed:

$$\ell_{\text{joint}} = \ell_{\text{SFS}} + \ell_{\text{LD}}$$

This is a **composite likelihood**: it treats the SFS and LD as independent (they are not perfectly so, but the approximation works well in practice). The joint approach is particularly valuable because the SFS constrains θ -dependent parameters while LD constrains ρ -dependent parameters, and together they resolve degeneracies that neither can break alone.

Exercises**i Exercise 1: LD decay curves**

Using `moments.LD`, compute the LD decay curve (σ_d^2 vs. ρ) for: (a) constant size, (b) 10x expansion 0.1 time units ago, (c) 10x bottleneck 0.05 time units ago. Plot all three on the same axes. How do the shapes differ?

i Exercise 2: Admixture detection

Simulate two populations that diverged at $T = 0.5$, then one received 10% admixture from the other at $T = 0.01$. Compute σ_{Dz} for the admixed population. Compare to a population that received no admixture. At what recombination distances is admixture most detectable?

i Exercise 3: Joint SFS + LD inference

For a two-epoch model, show that LD constrains N_e (through the $r \rightarrow \rho$ mapping) while the SFS constrains θ/N_e . Demonstrate that jointly fitting both gives tighter parameter estimates than either alone.

i Exercise 4: Cross-population LD

For two populations that split at different times ($T = 0.01, 0.1, 1.0$), compute the cross-population LD correlation $E[D_0 D_1] / \sqrt{E[D_0^2] E[D_1^2]}$ as a function of ρ . How does split time affect the correlation?

Solutions**i Solution 1: LD decay curves**

Compute σ_d^2 vs. ρ for three demographic scenarios – constant size, 10x expansion, and 10x bottleneck – and compare their shapes.

```
import numpy as np
import moments.LD
```

```
theta = 0.001
rho_values = [0.1, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0, 50.0]
```



```

results = {}

# (a) Constant size -- equilibrium
sigma_d2_const = []
for rho in rho_values:
    y = moments.LD.LDstats(
        moments.LD.Numerics.steady_state([rho], theta=theta),
        num_pops=1, pop_ids=["pop0"]
    )
    sigma_d2_const.append(y.D2() / y.pi2())
results['constant'] = sigma_d2_const

# (b) 10x expansion, 0.1 time units ago
sigma_d2_exp = []
for rho in rho_values:
    y = moments.LD.LDstats(
        moments.LD.Numerics.steady_state([rho], theta=theta),
        num_pops=1, pop_ids=["pop0"]
    )
    y.integrate([10.0], 0.1, rho=[rho], theta=theta)
    sigma_d2_exp.append(y.D2() / y.pi2())
results['expansion'] = sigma_d2_exp

# (c) 10x bottleneck, 0.05 time units, then recovery
sigma_d2_bn = []
for rho in rho_values:
    y = moments.LD.LDstats(
        moments.LD.Numerics.steady_state([rho], theta=theta),
        num_pops=1, pop_ids=["pop0"]
    )
    y.integrate([0.1], 0.05, rho=[rho], theta=theta) # bottleneck
    y.integrate([1.0], 0.05, rho=[rho], theta=theta) # recovery
    sigma_d2_bn.append(y.D2() / y.pi2())
results['bottleneck'] = sigma_d2_bn

# Print comparison
print(f"{'rho':>6} {'Constant':>12} {'Expansion':>12} {'Bottleneck':>12}")
print("-" * 46)
for i, rho in enumerate(rho_values):
    print(f"{'rho':6.1f} {results['constant'][i]:12.6f} "
          f"{results['expansion'][i]:12.6f} "
          f"{results['bottleneck'][i]:12.6f}")

```

How the shapes differ: The constant-size curve follows approximately $1/(1 + \rho)$. The expansion curve lies *below* the constant-size curve because weak drift in the large population creates little new LD. The bottleneck curve lies *above* the constant-size curve, especially at small ρ , because the intense drift during the bottleneck created excess LD that has not yet decayed. At large ρ , recombination rapidly destroys LD regardless of demography, so all three curves converge.

i Solution 2: Admixture detection

Two populations diverge at $T = 0.5$, then one receives 10% admixture at $T = 0.01$. Compare σ_{Dz} with and without admixture.

```
import numpy as np
import moments.LD

theta = 0.001
rho_values = [0.1, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0]
T_split = 0.5
T_admix = 0.01
f_admix = 0.1 # 10% admixture fraction

sigma_Dz_admixed = []
sigma_Dz_no_admix = []

for rho in rho_values:
    # --- Divergence phase (shared for both scenarios) ---
    y = moments.LD.LDstats(
        moments.LD.Numerics.steady_state([rho], theta=theta),
        num_pops=1, pop_ids=["anc"]
    )
    y_div = y.split(0, new_ids=["pop0", "pop1"])
    y_div.integrate([1.0, 1.0], T_split - T_admix,
                   rho=[rho], theta=theta)

    # --- Without admixture: continue divergence ---
    y_no = y_div.copy()
    y_no.integrate([1.0, 1.0], T_admix, rho=[rho], theta=theta)
    dz_no = y_no.Dz(pops="pop0")
    pi2_no = y_no.pi2(pops="pop0")
    sigma_Dz_no_admix.append(dz_no / pi2_no if pi2_no > 0 else 0)

    # --- With admixture: pop0 receives f_admix from pop1 ---
    y_adm = y_div.copy()
    y_adm = y_adm.admix(0, 1, f_admix) # 10% of pop0 replaced by pop1
    y_adm.integrate([1.0, 1.0], T_admix, rho=[rho], theta=theta)
    dz_adm = y_adm.Dz(pops="pop0")
    pi2_adm = y_adm.pi2(pops="pop0")
    sigma_Dz_admixed.append(dz_adm / pi2_adm if pi2_adm > 0 else 0)

print(f"{'rho':>6} {'sigma_Dz (no admix)':>22} "
      f"{'sigma_Dz (admixed)':>22} {'Ratio':>8}")
print("-" * 62)
for i, rho in enumerate(rho_values):
    ratio = (sigma_Dz_admixed[i] / sigma_Dz_no_admix[i]
             if sigma_Dz_no_admix[i] != 0 else float('inf'))
    print(f"{'rho':6.1f} {'sigma_Dz_no_admix[i]:22.8f} "
          f"{'sigma_Dz_admixed[i]:22.8f} {'ratio:8.2f} ")
```

Admixture is most detectable at **intermediate** recombination distances ($\rho \approx 1\text{--}5$). At very small ρ , recombination has not had time to break down the admixture LD, so the signal is strong but the baseline drift-LD is also high. At very large ρ , recombination has erased the admixture signal. The ratio of σ_{Dz} with vs. without admixture quantifies

the signal-to-noise advantage at each distance. The $E[Dz]$ statistic is near zero without admixture (by symmetry), so even a modest admixture pulse creates a strong signal.

Solution 3: Joint SFS + LD inference

Show that LD constrains N_e (via $\rho = 4N_e r$) while the SFS constrains $\theta = 4N_e \mu$, and that jointly fitting both gives tighter estimates.

```
import numpy as np
import moments
import moments.LD

# True parameters
Ne_true = 10000
mu = 1.5e-8      # mutation rate per site per generation
r = 1e-8         # recombination rate per site per generation
L = 1e6         # 1 Mb region
n = 30

theta_true = 4 * Ne_true * mu * L    # = 0.6
rho_true = 4 * Ne_true * r          # = 4e-4 per site

# Two-epoch model: expansion by factor nu at time T
nu_true, T_true = 3.0, 0.1

# --- SFS data ---
fs_true = moments.Demographics1D.two_epoch((nu_true, T_true), [n])
sfs_data = np.array(fs_true) * theta_true

# --- LD data (binned by physical distance -> rho) ---
r_bin_centers = np.array([1e-5, 5e-5, 1e-4, 5e-4, 1e-3])
rho_bins_true = 4 * Ne_true * r_bin_centers

ld_data = []
for rho in rho_bins_true:
    y = moments.LD.LDstats(
        moments.LD.Numerics.steady_state([rho], theta=mu),
        num_pops=1, pop_ids=["pop0"]
    )
    y.integrate([nu_true], T_true, rho=[rho], theta=mu)
    ld_data.append(y.D2() / y.pi2())

# --- Grid search over (nu, Ne) ---
# SFS constrains theta = 4*Ne*mu*L, so nu and T shape are separable
# LD constrains rho = 4*Ne*r, so it pins down Ne
nu_grid = np.linspace(1.0, 8.0, 30)
Ne_grid = np.linspace(5000, 20000, 30)

ll_sfs_only = np.full((30, 30), -np.inf)
ll_ld_only = np.full((30, 30), -np.inf)
ll_joint = np.full((30, 30), -np.inf)
```

```

for i, nu in enumerate(nu_grid):
    for j, Ne in enumerate(Ne_grid):
        theta_model = 4 * Ne * mu * L

        # SFS component
        fs_model = moments.Demographics1D.two_epoch(
            (nu, T_true), [n]
        )
        sfs_model = np.array(fs_model) * theta_model
        ll_s = sum(
            sfs_data[k] * np.log(sfs_model[k]) - sfs_model[k]
            for k in range(1, n) if sfs_model[k] > 0
        )
        ll_sfs_only[i, j] = ll_s

        # LD component
        rho_bins_model = 4 * Ne * r_bin_centers
        ll_l = 0.0
        for b, rho in enumerate(rho_bins_model):
            y = moments.LD.LDstats(
                moments.LD.Numerics.steady_state([rho], theta=mu),
                num_pops=1, pop_ids=["pop0"]
            )
            y.integrate([nu], T_true, rho=[rho], theta=mu)
            pred = y.D2() / y.pi2()
            # Gaussian log-likelihood (unit variance for simplicity)
            ll_l -= 0.5 * (ld_data[b] - pred) ** 2 * 1e6
        ll_ld_only[i, j] = ll_l

        # Joint
        ll_joint[i, j] = ll_s + ll_l

# Find maxima
idx_s = np.unravel_index(np.argmax(ll_sfs_only), ll_sfs_only.shape)
idx_l = np.unravel_index(np.argmax(ll_ld_only), ll_ld_only.shape)
idx_j = np.unravel_index(np.argmax(ll_joint), ll_joint.shape)

print(f"True: nu={nu_true}, Ne={Ne_true}")
print(f"SFS-only best: nu={nu_grid[idx_s[0]]:.2f}, "
      f"Ne={Ne_grid[idx_s[1]]:.0f}")
print(f"LD-only best: nu={nu_grid[idx_l[0]]:.2f}, "
      f"Ne={Ne_grid[idx_l[1]]:.0f}")
print(f"Joint best: nu={nu_grid[idx_j[0]]:.2f}, "
      f"Ne={Ne_grid[idx_j[1]]:.0f}")

```

The SFS alone constrains ν (the shape of the expansion) well but is largely flat along the N_e axis (since it only sees $\theta = 4N_e\mu L$ and the optimal θ is absorbed analytically). The LD alone constrains N_e (because the mapping $\rho = 4N_e r$ depends on N_e directly) but has a broader ridge in ν . Jointly, the two likelihoods break each other's degeneracies, producing a sharper peak in both dimensions.

i Solution 4: Cross-population LD

Compute the cross-population LD correlation at various ρ values for three different split times.

```
import numpy as np
import moments.LD

theta = 0.001
rho_values = [0.1, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0]
T_values = [0.01, 0.1, 1.0]

print(f'{"rho":>6}', end="")
for T in T_values:
    print(f"  {'T=' + str(T):>12}", end="")
print()
print("-" * (6 + 14 * len(T_values)))

for rho in rho_values:
    print(f"{'rho':6.1f}", end="")
    for T in T_values:
        # Equilibrium -> split -> diverge
        y = moments.LD.LDstats(
            moments.LD.Numerics.steady_state([rho], theta=theta),
            num_pops=1, pop_ids=["anc"]
        )
        y = y.split(0, new_ids=["pop0", "pop1"])
        y.integrate([1.0, 1.0], T, rho=[rho], theta=theta)

        DD_00 = y.D2(pops="pop0")
        DD_11 = y.D2(pops="pop1")
        DD_01 = y.D2(pops=["pop0", "pop1"])

        corr = DD_01 / np.sqrt(DD_00 * DD_11) if DD_00 > 0 and DD_11 > 0 else 0
        print(f"  {'corr':12.6f}", end="")
    print()

print()
print("Interpretation:")
print("  T=0.01 (very recent split): correlation near 1.0 at all rho")
print("  T=0.1  (moderate split):    correlation decays with rho")
print("  T=1.0  (ancient split):     correlation near 0 everywhere")
```

How split time affects the correlation: Right after a split, the two populations share identical LD patterns (correlation = 1). Over time, independent drift in each population creates new, uncorrelated LD, while recombination destroys the shared ancestral LD. For short-range LD (high ρ), recombination rapidly erases the shared signal, so the correlation drops first at large ρ . For long-range LD (low ρ), ancestral LD persists longer, maintaining the correlation. At $T = 1.0$ (ancient split), essentially all ancestral LD has been replaced by independent drift, giving correlations near zero.

Demo: Running moments on Simulated Data

Running mini_moments on simulated genomic data.

This figure was generated by running the `mini_moments` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_moments.py`.

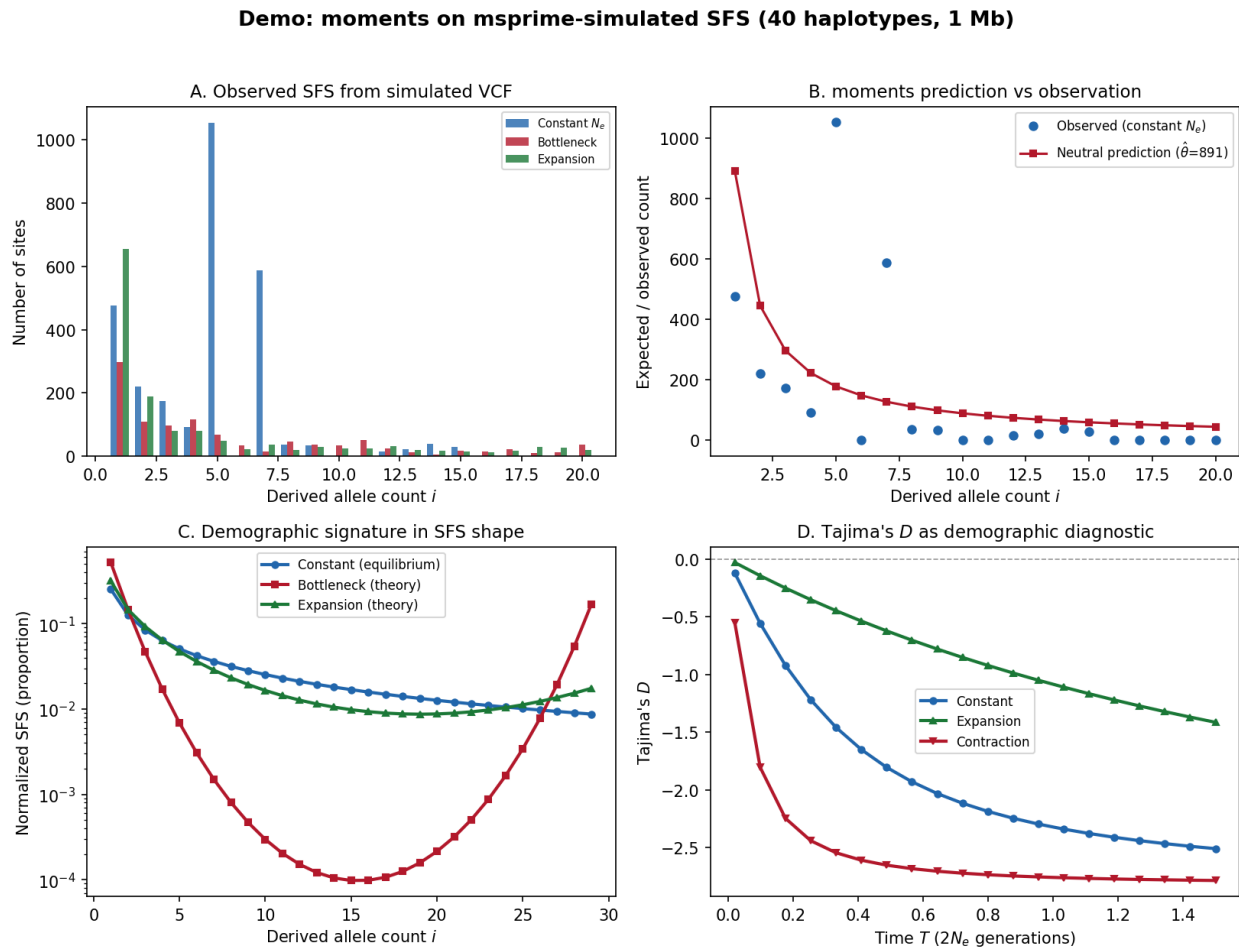


Fig. 13: **moments demo on msprime-simulated data.** Panel A: Observed SFS from msprime under three demographic models (constant, bottleneck, expansion) vs moments predictions. Panel B: Tajima's D values across demographic scenarios. Panel C: Expected SFS under neutrality from both simulation and moments' ODE integration. Panel D: Comparison of moment-based and simulated SFS shapes.

3.12 Timepiece XI: dadi

Inferring demographic history by solving the Wright-Fisher diffusion – the PDE that moments sought to avoid.

3.12.1 The Mechanism at a Glance

`dadi` (Diffusion Approximations for Demographic Inference) is a method for **demographic inference** – learning a population's history from patterns in its DNA variation. Like `moments` (*Timepiece X*), it works with the **site frequency spectrum** (SFS) as its summary statistic and uses composite likelihood optimization to fit demographic models.

But the internal mechanism is different. Where `moments` derives ordinary differential equations that govern the SFS entries directly, `dadi` takes the classical approach: it **solves the Wright-Fisher diffusion equation** – a partial differential equation (PDE) governing the continuous allele frequency density $\phi(x, t)$. The expected SFS is then extracted from this density by integrating against binomial sampling probabilities.

Think of it this way: `moments` tracks how each hand on the dial advances (one ODE per SFS entry). `dadi` instead models the full shape of every gear tooth – the continuous density of allele frequencies – and reads the hand positions from the gear profile. More work per tick, but the gear-level view reveals details (like the full frequency density under selection) that the hand-level view abstracts away.

`dadi` was the first tool to use the diffusion approximation for multi-population demographic inference from the joint SFS (Gutenkunst et al. 2009), and it remains the predecessor against which `moments` and `moments2` define themselves. Understanding `dadi` is understanding the foundation on which SFS-based inference was built.

Primary Reference

[Gutenkunst *et al.*, 2009]

The four gears of `dadi`:

1. **The Frequency Spectrum** (the dial) – The same SFS as `moments`, but `dadi` works with the continuous frequency density $\phi(x, t)$ from which the discrete SFS is sampled. The density is the gear; the SFS is the hand position read from it.
2. **The Diffusion Equation** (the escapement) – The Wright-Fisher diffusion PDE governing the evolution of $\phi(x, t)$ under drift, selection, mutation, and migration. This is the fundamental equation of population genetics – the continuous-time limit of the Wright-Fisher model. Each evolutionary force contributes a term to the PDE.
3. **Numerical Integration** (the gear train) – A finite-difference scheme on a nonuniform frequency grid, with implicit time-stepping and Richardson extrapolation to correct grid bias. This is the computational engine that transforms the continuous PDE into a discrete computation. The grid design and extrapolation scheme are `dadi`'s key engineering innovations.
4. **Demographic Inference** (the mainspring) – Poisson composite likelihood optimization via BFGS, with Godambe Information Matrix for uncertainty quantification. The same statistical framework as `moments`, applied to `dadi`'s PDE-computed expected SFS.

These gears mesh together into a complete inference framework:

```
Observed DNA variation
      |
      v
+-----+
| FREQUENCY SPECTRUM |
|                     |
| Summarize variation |
```

(continues on next page)

(continued from previous page)

	as allele frequency	
	histogram (SFS)	
+-----+		
	v	
+-----+		
	DIFFUSION EQUATION	
	Solve the Wright-	
	Fisher PDE on a	
	frequency grid to	
	get $\phi(x, t)$	
+-----+		
	v	
+-----+		
	NUMERICAL	
	INTEGRATION	
	Finite differences,	
	nonuniform grid,	
	Richardson extrap.	
	-> expected SFS	
+-----+		
	v	
+-----+		
	DEMOGRAPHIC	
	INFERENCE	
	Poisson composite	
	likelihood + BFGS	
	optimization	
+-----+		
	v	
Demographic history:		
population sizes, split		
times, migration rates,		
selection coefficients		

3.12.2 dadi vs. moments

`dadi` and `moments` solve the same problem with different internal mechanisms. Understanding the trade-offs clarifies when to use each:

Aspect	dadi	moments
Core equation	Wright-Fisher diffusion PDE	Moment ODEs for SFS entries
What it tracks	Continuous density $\phi(x, t)$	Discrete SFS entries ϕ_j
Frequency discretization	Grid of n points in $[0, 1]$	None (SFS entries are the state)
Multi-population scaling	$O(n^P)$ grid points	$O(\prod n_i)$ SFS entries
Grid artifacts	Yes (mitigated by extrapolation)	None
Selection handling	Natural (term in PDE)	Natural (term in ODEs)
Key innovation	First multi-pop diffusion tool	Grid-free moment equations

Prerequisites for this Timepiece

- *Coalescent Theory* – the relationship between population size and genetic diversity
- Familiarity with the site frequency spectrum is helpful but not required – see *The Frequency Spectrum* in the moments Timepiece for a ground-up introduction
- Basic partial differential equations are helpful (we derive everything we need)

3.12.3 Chapters

Overview of dadi

Before assembling the watch, lay out every part and understand what it does.

What Does dadi Do?

Given DNA sequence data from one or more populations, `dadi` infers the **demographic history** that produced the observed patterns of genetic variation.

Input: $\mathbf{D}$ = observed allele frequencies across n sampled chromosomes

Output: $\hat{\Theta}$ = best-fit demographic parameters (sizes, times, rates)

Concretely, `dadi` asks: *What history of population size changes, splits, and migrations would produce a frequency spectrum that looks like the one we observe?* The same question as `moments` – but answered by solving a partial differential equation rather than a system of ODEs.

The Big Picture: Why Solve a PDE?

The allele frequency distribution in a population is governed by the **Wright-Fisher diffusion equation** – the continuous-time, continuous-frequency limit of the Wright-Fisher model. This PDE describes how the density of allele frequencies $\phi(x, t)$ evolves under the forces of drift, mutation, selection, and migration:

$$\frac{\partial \phi}{\partial t} = \underbrace{\frac{1}{2\nu} \frac{\partial^2}{\partial x^2} [x(1-x)\phi]}_{\text{drift}} - \underbrace{\frac{\partial}{\partial x} [\gamma x(1-x)(h + (1-2h)x)\phi]}_{\text{selection}} + \underbrace{\text{mutation and migration terms}}_{\text{boundary/coupling}}$$

If you can solve this equation for a given demographic model, you can compute the expected allele frequency density at the present. From that density, you extract the expected SFS. From the expected SFS, you compute a likelihood. And from the likelihood, you optimize to find the best-fit demographic parameters.

This is the classical approach to SFS-based inference, and `dadi` was the first tool to implement it for joint multi-population spectra (Gutenkunst et al. 2009). The key insight was that the diffusion equation, despite being a PDE, can be solved efficiently enough for demographic inference – especially with a cleverly designed nonuniform frequency grid and Richardson extrapolation to remove grid bias.

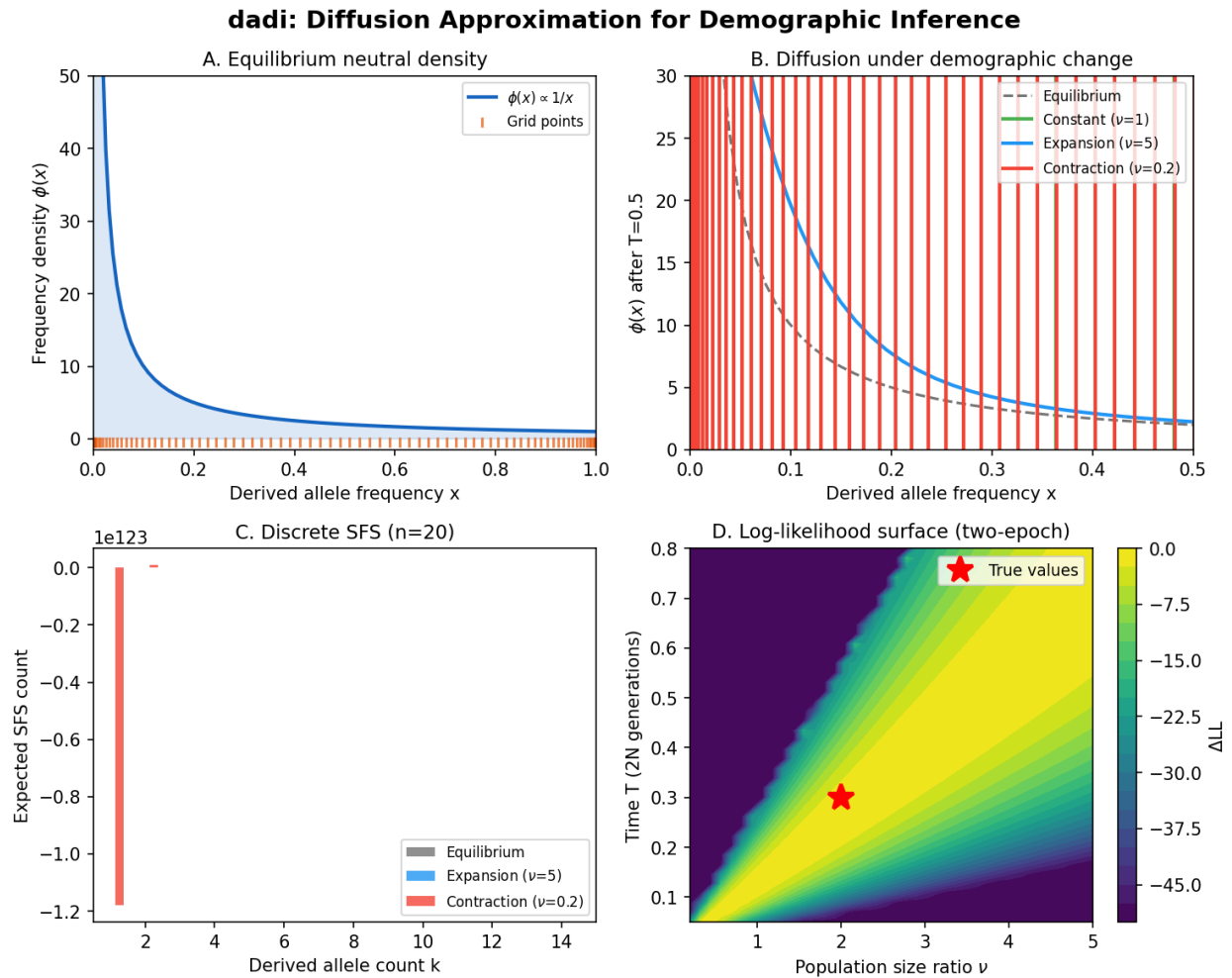


Fig. 14: **dadi at a glance**. The diffusion approximation for demographic inference: the frequency grid discretising allele frequency space, the equilibrium SFS density under the Wright-Fisher diffusion, PDE evolution of the allele frequency distribution under different population sizes, and the resulting discrete SFS used for likelihood computation.

i Why did moments bypass the PDE?

If the PDE approach works, why did `moments` (Jouganous et al. 2017) bother deriving moment equations? Two reasons:

1. **Grid scaling.** For P populations with n grid points each, `dadi` needs n^P grid points. For 3 populations with 50 grid points, that's 125,000 points. `moments` replaces the grid with the SFS entries themselves, scaling as $\prod n_i$ where n_i are sample sizes – typically much smaller.
2. **Grid artifacts.** Finite-difference schemes introduce numerical diffusion. Richardson extrapolation mitigates but doesn't eliminate this. `moments` has no grid, so no grid artifacts.

But `dadi` has its own advantages: the PDE view makes selection natural, the frequency density $\phi(x, t)$ is a physically meaningful object, and `dadi`'s mature codebase offers features (DFE inference, ancestral misidentification correction, Demes integration) that build naturally on the diffusion framework.

Terminology

Before diving into the gears, let's nail down the terminology.

Term	Definition
$\phi(x, t)$	Allele frequency density: the expected number of segregating sites with derived allele frequency x at time t
SFS	Site Frequency Spectrum: histogram of derived allele counts in the sample, obtained from ϕ by binomial sampling
Grid	The set of frequency points $0 < x_1 < x_2 < \dots < x_n < 1$ on which ϕ is discretized
Extrapolation	Richardson extrapolation: running at multiple grid sizes and extrapolating to the $n \rightarrow \infty$ limit
n	Haploid sample size (number of chromosomes)
θ	Population-scaled mutation rate: $4N_e\mu$
N_e	Effective population size
ν	Relative population size: $N(t)/N_{\text{ref}}$
T	Time, measured in units of $2N_e$ generations
γ	Population-scaled selection coefficient: $2N_es$
h	Dominance coefficient (default 0.5 = additive/genic selection)
M_{ij}	Scaled migration rate from population j into population i
pts	Number of grid points used for the frequency grid (a <code>dadi</code> -specific parameter that controls resolution and accuracy)

The Key Innovation: Multi-Population Diffusion

Before `dadi`, the Wright-Fisher diffusion had been solved numerically for single populations (Williamson et al. 2005). `dadi`'s breakthrough was extending this to **joint multi-population spectra** – solving a P -dimensional PDE where each dimension corresponds to one population's frequency axis.

For two populations, the density $\phi(x, y, t)$ tracks the joint distribution of allele frequencies in both populations simultaneously. Drift acts independently in each dimension, but migration couples the populations. Population splits are handled by slicing the density array (via `PhiManip`), and admixture events remap frequencies via Dirac delta operations.

This multi-dimensional approach made `dadi` the first tool capable of fitting complex demographic models (isolation-with-migration, secondary contact, admixture) to the joint SFS from two or three populations.

dadi vs. moments: Two Paths to the Same Dial

`dadi` and `moments` are best understood as two different watchmaking philosophies applied to the same problem:

dadi's approach (PDE):

```
Start with equilibrium density phi(x)
  |
  v
Solve diffusion PDE on frequency grid
(drift, selection, mutation, migration)
  |
  v
Extract SFS from phi via binomial sampling
  |
  v
Compare to observed SFS via likelihood
```

moments' approach (ODE):

```
Start with equilibrium SFS
  |
  v
Integrate moment equations (ODEs)
(drift, selection, mutation, migration)
  |
  v
Model SFS entries directly
  |
  v
Compare to observed SFS via likelihood
```

`dadi` models the full gear profile (the continuous density $\phi(x, t)$) and reads the hand positions from it. `moments` writes equations for the hand positions directly, never constructing the full gear profile. Both watches tell the same time; the mechanism differs.

Parameters

`dadi` takes the following parameters:

Parameter	Symbol	Meaning
Mutation rate	$\theta = 4N_e\mu$	Population-scaled mutation rate per site
Selection coeff.	$\gamma = 2N_es$	Population-scaled selection coefficient
Dominance	h	Heterozygote effect relative to homozygote (default 0.5 = additive)
Population sizes	$\nu_i(t)$	Relative sizes as functions of time
Migration rates	M_{ij}	Scaled migration matrix entries
Integration time	T	Duration of each epoch in $2N_e$ generations
Grid points	pts	Number of frequency grid points (controls numerical resolution)

The first six parameters are shared with `moments`. The last – grid points – is unique to `dadi` and reflects its PDE-based approach. More grid points means higher accuracy but slower computation, and Richardson extrapolation across multiple grid sizes removes the leading-order grid bias.

The Flow in Detail

Here's how the pieces connect for a typical inference with `dadi`:

```

1. Parse VCF -> compute observed SFS
    |
    v
2. Define demographic model function
   (e.g., two_epoch, isolation_with_migration)
    |
    v
3. Wrap model with extrapolation:
   func_ex = Numerics.make_extrap_func(model_func)
    |
    v
4. For candidate parameters theta:
   +---> Compute equilibrium phi (PhiManip.phi_1D)
       |
       |
       |      v
       |      Build frequency grid (Numerics.default_grid)
       |      |
       |      v
       |      Integrate diffusion PDE through
       |      demographic epochs (Integration.one_pop,
       |      two_pops, etc.)
       |      |
       |      v
       |      Extract SFS from phi (Spectrum.from_phi)
       |      |
       |      v
       |      Richardson extrapolation over grid sizes
       |      |
       |      v
       |      Compute composite log-likelihood
       |      |
       +----- Optimizer (BFGS) updates parameters
               |
               v
5. Return best-fit parameters
   + Godambe uncertainty estimates

```

In watch terms: Step 1 reads the current hand positions (the observed SFS). Step 2 blueprints the gear train (the demographic model). Step 3 adds a self-correcting mechanism (extrapolation) that cancels grid artifacts. Step 4 winds a candidate watch, reads its predicted hand positions, and compares them to the observed ones. Step 5 reports which gear sizes (demographic parameters) make the predicted dial match the real one most closely.

Reference

Gutenkunst RN, Hernandez RD, Williamson SH, Bustamante CD (2009). Inferring the joint demographic history of multiple populations from multidimensional SNP frequency data. *PLoS Genetics*, 5(10): e1000695.

Ready to Build

You now have the high-level blueprint. In the following chapters, we'll build each gear from scratch:

1. *The Diffusion Equation* – The PDE: the Wright-Fisher diffusion and how `dadi` sets up the equation with all its terms. This is **the escapement** – the fundamental equation that generates the ticking.
2. *Numerical Integration* – The solver: finite differences, nonuniform grids, and Richardson extrapolation. This is **the gear train** – the engineering that turns a continuous equation into a computable answer.
3. *Demographic Inference* – The optimization: composite likelihood, BFGS, and Godambe uncertainty. This is **the mainspring** – the machinery that adjusts parameters until the predicted dial matches observation.

The Diffusion Equation

The escapement – the fundamental equation whose ticking drives every hand on the dial.

In this chapter we derive the Wright-Fisher diffusion equation from first principles, explain each term biologically, and show how `dadi` sets up the equation for one and multiple populations. By the end, you'll understand the PDE that `moments` sought to bypass – and why `dadi` chose to solve it directly.

From Wright-Fisher to Diffusion

The starting point is the Wright-Fisher model: a population of $2N$ gene copies at a single locus, with discrete, non-overlapping generations. Each generation, the new population is formed by sampling $2N$ copies from the previous generation with replacement (possibly biased by selection).

Let X_t be the frequency of the derived allele in generation t . Under neutrality, the change in frequency per generation has mean zero and variance $X_t(1 - X_t)/(2N)$. As $N \rightarrow \infty$ with appropriate rescaling of time ($\tau = t/(2N)$ generations), the discrete process converges to a continuous diffusion.

The frequency density $\phi(x, \tau)$ – defined so that $\phi(x, \tau)dx$ is the expected number of segregating sites with derived allele frequency in $[x, x + dx]$ at time τ – satisfies the **Kolmogorov forward equation** (Fokker-Planck equation):

$$\frac{\partial \phi}{\partial \tau} = \frac{1}{2} \frac{\partial^2}{\partial x^2} [V(x) \phi] - \frac{\partial}{\partial x} [M(x) \phi]$$

where $V(x)$ is the **diffusion coefficient** (variance of frequency change per unit time) and $M(x)$ is the **drift coefficient** (mean frequency change per unit time, not to be confused with genetic drift).

Calculus Aside – Fokker-Planck equations

The Fokker-Planck (or Kolmogorov forward) equation describes how the probability density of a stochastic process evolves in time. It is the continuous analog of a Markov chain transition matrix: given the current distribution, it tells you the distribution at the next instant. The $\partial^2/\partial x^2$ term spreads the density (diffusion from genetic drift), while the $\partial/\partial x$ term shifts it (directional forces like selection).

The 1D Diffusion Equation

For a single population with relative size ν (measured as a ratio to the reference N_{ref}), the coefficients are:

Drift (genetic drift):

$$V(x) = \frac{x(1-x)}{\nu}$$

The variance of allele frequency change is $x(1-x)/(2N)$ per generation. After rescaling time by $2N_{\text{ref}}$, this becomes $x(1-x)/\nu$. Larger populations ($\nu > 1$) have weaker drift.

Selection (directional force):

$$M(x) = \gamma x(1-x) [h + (1-2h)x]$$

where $\gamma = 2N_{\text{ref}}s$ is the population-scaled selection coefficient and h is the dominance coefficient. For genic/additive selection ($h = 0.5$), this simplifies to $M(x) = \gamma x(1-x)/2$.

i Plain-language summary – Two forces shaping allele frequencies

The diffusion equation describes two competing forces acting on allele frequencies. **Genetic drift** (the $V(x)$ term) is the random fluctuation caused by finite population size – like Brownian motion, it pushes allele frequencies up and down unpredictably. Drift is strongest at intermediate frequencies (where $x(1-x)$ is large) and weakest near the boundaries. **Selection** (the $M(x)$ term) is a directional force: beneficial alleles are pushed toward fixation, deleterious alleles toward loss. In smaller populations, drift dominates and selection is less effective – this is why nearly neutral mutations can fix in small populations but are efficiently purged in large ones.

Putting these together, the 1D diffusion equation that `dadi` solves is:

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \frac{\partial^2}{\partial x^2} \left[\frac{x(1-x)}{\nu} \phi \right] - \frac{\partial}{\partial x} \left[\gamma x(1-x)(h + (1-2h)x) \phi \right] \quad (3.1)$$

where t is measured in units of $2N_{\text{ref}}$ generations and ν may change with time (modeling population size changes).

In `dadi`'s code (`Integration.py`), the one-population solver `one_pop(phi, xx, T, nu, gamma, h, theta0)` integrates this equation on a frequency grid `xx` for time `T`.

Boundary Conditions and Mutation

The diffusion equation needs boundary conditions at $x = 0$ (allele lost) and $x = 1$ (allele fixed). In `dadi`, these are **absorbing boundaries**: once an allele is lost or fixed, it leaves the segregating pool. The density $\phi(x, t)$ is defined only for $x \in (0, 1)$.

Mutation injection:

New mutations enter the population at low frequency. In `dadi`, this is implemented as a source term at the lowest interior grid point. Each timestep, the integration function `_inject_mutations_1D` adds:

$$\phi(x_1, t + dt) += \frac{\theta_0 \cdot dt}{2} \cdot \frac{2}{\Delta x}$$

where x_1 is the first interior grid point and Δx is the local grid spacing.

Where does this formula come from? In the infinite-sites model, new mutations arise at rate $\theta_0/2 = 2N_{\text{ref}}\mu$ per site per unit of diffusion time (the factor of 2 converts from per-generation to per- $2N$ time units). Each new mutation enters at frequency $1/(2N)$, which is the lowest representable frequency. Since $\phi(x)$ is a *density* (probability per unit frequency), injecting one new mutation at x_1 means adding $1/\Delta x$ to the density at that point (the mutation occupies a bin of width Δx). Over a timestep dt , the total injection is:

$$\frac{\theta_0}{2} \cdot dt \cdot \frac{1}{\Delta x/2} = \frac{\theta_0 \cdot dt}{2} \cdot \frac{2}{\Delta x}$$

The extra factor of $2/\Delta x$ (rather than $1/\Delta x$) arises from the trapezoidal integration scheme: the first grid point is at the boundary of a half-width bin, so the effective bin width is $\Delta x/2$.

This is an approximation: new mutations arise at frequency $1/(2N)$, which is below any finite grid resolution. The injection at x_1 is accurate when $x_1 \ll 1$, which `dadi` ensures by using a grid that is dense near the boundaries.

Equilibrium Solution

Under constant population size ($\nu = 1$), neutrality ($\gamma = 0$), and steady-state mutation, the diffusion equation has the equilibrium solution:

$$\phi_{\text{eq}}(x) = \frac{\theta}{x(1-x)}$$

This is the classical result: rare alleles (small x) are much more common than common alleles, because drift hasn't had time to push them to high frequency.

i Biology Aside – The $1/x$ spectrum and the neutral theory

The $1/x$ shape of the neutral SFS is one of the most fundamental results in population genetics. It says that singletons (variants present in just one chromosome) are the most abundant class, doubletons the second, and so on. This pattern arises because new mutations enter at low frequency and most are lost before reaching high frequency. Deviations from this shape are informative: an excess of rare variants (steeper than $1/x$) indicates recent population growth; a deficit of rare variants (shallower than $1/x$) indicates a bottleneck or population contraction. These are exactly the signatures that `dadi` uses to infer demographic history. In `dadi`, this equilibrium is computed by `PhiManip.phi_1D(xx, nu=1.0, theta0=1.0)`, which evaluates $\nu \cdot \theta_0/x$ at each grid point (using the convention that the density near $x = 0$ dominates).

With genic selection ($h = 0.5, \gamma \neq 0$), the equilibrium density can be derived from the stationary Fokker-Planck equation. Setting $\partial\phi/\partial t = 0$ with $V(x) = x(1-x)$ and $M(x) = (\gamma/2)x(1-x)$ gives:

$$0 = \frac{1}{2} \frac{d^2}{dx^2} [x(1-x)\phi] - \frac{d}{dx} \left[\frac{\gamma}{2} x(1-x)\phi \right]$$

The general stationary solution of the Fokker-Planck equation with drift $M(x)$ and diffusion $V(x) = x(1-x)$ is:

$$\phi_{\text{eq}}(x) = \frac{C}{V(x)} \exp \left(2 \int_0^x \frac{M(y)}{V(y)} dy \right) = \frac{C}{x(1-x)} e^{\gamma x}$$

since $2 \int_0^x \frac{\gamma y(1-y)/2}{y(1-y)} dy = \gamma x$. This is the standard result from the *diffusion prerequisite*. In the infinite-sites framework (mutations entering at $x \approx 0$ and leaving at $x = 0$ or $x = 1$), the constant C is determined by matching the mutation injection rate, giving:

$$\phi_{\text{eq}}(x) \propto \frac{1 - e^{-2\gamma(1-x)}}{x(1-x)(1 - e^{-2\gamma})}$$

The numerator $1 - e^{-2\gamma(1-x)}$ arises from the boundary condition at $x = 1$ (fixation probability under selection). When $\gamma = 0$, this simplifies to $1/x(1-x)$ – the neutral equilibrium.

This is computed by `PhiManip.phi_1D_genic(xx, nu, theta0, gamma)`.

```
import dadi
import numpy as np

# Build a frequency grid
pts = 60
xx = dadi.Numerics.default_grid(pts)

# Equilibrium density under the standard neutral model
phi = dadi.PhiManip.phi_1D(xx)
```

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```
# The equilibrium density is proportional to 1/x
# (ignoring the x -> 1 boundary)
print(f"phi at x={xx[1]:.4f}: {phi[1]:.2f}")
print(f"phi at x={xx[5]:.4f}: {phi[5]:.2f}")
print(f"Ratio: {phi[1]/phi[5]:.2f}, expected: {xx[5]/xx[1]:.2f}")
```

The Multi-Population PDE

For P populations, the allele frequency density becomes P -dimensional: $\phi(x_1, x_2, \dots, x_P, t)$, where x_i is the derived allele frequency in population i . The PDE generalizes to:

$$\frac{\partial \phi}{\partial t} = \sum_{i=1}^P \left[\frac{1}{2} \frac{\partial^2}{\partial x_i^2} \left[\frac{x_i(1-x_i)}{\nu_i} \phi \right] - \frac{\partial}{\partial x_i} \left[\gamma_i x_i(1-x_i)(h_i + (1-2h_i)x_i) \phi \right] \right] + \text{migration terms}$$

Each population contributes its own drift and selection terms, acting along its own frequency axis. The populations are coupled through **migration**:

$$\text{Migration } i \leftarrow j : -\frac{\partial}{\partial x_i} \left[m_{ij}(x_j - x_i) \phi \right]$$

where m_{ij} is the scaled migration rate from population j into population i . Migration shifts the frequency in population i toward the frequency in population j .

Biology Aside – Migration homogenizes allele frequencies

Gene flow (migration) between populations acts to make their allele frequencies more similar. If a variant is at 80% frequency in population A and 20% in population B, migration from A to B pushes B's frequency upward (and vice versa). Over time, high migration makes two populations genetically indistinguishable, while low migration allows them to diverge. The migration term in the PDE captures this: it is a directional force that pulls each population's frequency toward the other's. The scaled migration rate $m_{ij} = 4N_{\text{ref}} \cdot m$ determines the strength of this pull – values greater than 1 indicate enough gene flow to prevent substantial divergence, while values much less than 1 allow populations to drift apart.

In dadi's code, `Integration.two_pops(phi, xx, T, nu1, nu2, m12, m21, ...)` solves the 2D PDE, and `Integration.three_pops` solves the 3D version. The curse of dimensionality limits dadi to at most five populations, and in practice three is the common maximum.

Population Splits via PhiManip

When a population splits, the allele frequency distribution is duplicated: the new daughter population starts with the same allele frequencies as the parent. In dadi, this is handled by `PhiManip.phi_1D_to_2D(xx, phi_1D)` and its higher-dimensional analogs.

The operation is conceptually simple: place all the density on the diagonal $x_1 = x_2$ of the 2D grid. In code:

```
# A population splits into two identical daughter populations
phi_1D = dadi.PhiManip.phi_1D(xx)
phi_2D = dadi.PhiManip.phi_1D_to_2D(xx, phi_1D)
# phi_2D[i, j] is nonzero only when i == j (the diagonal)
```

After the split, the two populations evolve independently (if there's no migration) or remain coupled through migration terms. Each daughter population can have its own size ν_i , selection coefficient γ_i , and mutation rate.

Admixture

Admixture – the mixing of two populations to form a new one – is implemented as a remapping of frequencies. If population 3 is formed as a fraction f from population 1 and $(1 - f)$ from population 2, then the frequency in the admixed population is:

$$x_3 = f \cdot x_1 + (1 - f) \cdot x_2$$

dadi implements this via `PhiManip.phi_2D_to_3D_admix(phi_2D, f, xx, yy, zz)`, which creates the 3D density by interpolating the 2D density onto the new frequency axis at each (x_1, x_2) pair. Internally, the function uses linear interpolation between grid points and careful normalization to preserve the total density.

A Complete Example

Here's how these pieces connect for a two-epoch model – the simplest non-trivial demographic scenario:

```
import dadi

def two_epoch(params, ns, pts):
    """
    A population changes size instantaneously.

    params = (nu, T)
        nu: ratio of contemporary to ancient size
        T: time of size change (in 2*Nref generations)
    """
    nu, T = params

    # Build frequency grid
    xx = dadi.Numerics.default_grid(pts)

    # Start from equilibrium (standard neutral model)
    phi = dadi.PhiManip.phi_1D(xx)

    # Integrate the diffusion equation for time T
    # with population size nu (relative to Nref)
    phi = dadi.Integration.one_pop(phi, xx, T, nu=nu)

    # Extract the SFS from the frequency density
    fs = dadi.Spectrum.from_phi(phi, ns, (xx,))
    return fs

# Wrap with Richardson extrapolation
func_ex = dadi.Numerics.make_extrap_log_func(two_epoch)

# Compute expected SFS for nu=2 (doubling), T=0.1
# with sample size 20, using grid sizes [40, 50, 60]
model = func_ex((2.0, 0.1), ns=[20], pts=[40, 50, 60])
```

The call chain is:

1. `phi_1D` computes the equilibrium density $\phi_{\text{eq}}(x)$
2. `one_pop` solves the PDE (3.1) for time T with size ν
3. `from_phi` converts the continuous density to a discrete SFS via binomial sampling

4. `make_extrap_log_func` runs the whole thing at grid sizes 40, 50, and 60, then extrapolates to the $n \rightarrow \infty$ limit. In the next chapter, we'll open the gear train and examine exactly how `dadi` discretizes and solves this PDE numerically.

Numerical Integration

The gear train – the engineering that turns a continuous equation into a computable answer.

The diffusion equation is a continuous PDE defined on $x \in (0, 1)$ and $t \in [0, T]$. To solve it numerically, `dadi` must discretize both the frequency axis and the time axis. This chapter examines the engineering choices that make this discretization accurate and efficient: the nonuniform grid, the finite-difference scheme, the time-stepping strategy, and Richardson extrapolation.

Biology Aside – Why the grid must be fine near the boundaries

In a real population, most genetic variants are rare – they exist in only one or a few copies out of thousands of chromosomes (the “singletons” and “doubletons” that dominate the SFS). These low-frequency variants are the most informative about recent population size changes: a population expansion produces a flood of new rare variants, while a bottleneck depletes them. At the same time, a small number of variants are at very high frequency (nearly fixed). The allele frequency density $\phi(x)$ is therefore strongly peaked near $x = 0$ and $x = 1$, reflecting the biological reality that rare and near-fixed variants are far more numerous than common ones. A grid that fails to resolve these peaks will produce inaccurate SFS predictions and, consequently, biased demographic estimates.

The Nonuniform Grid

The allele frequency density $\phi(x)$ is concentrated near the boundaries: under neutrality, $\phi(x) \propto 1/[x(1-x)]$, which diverges as $x \rightarrow 0$ and $x \rightarrow 1$. A uniform grid would waste points in the interior and lack resolution where it matters most.

`dadi` uses an **exponential grid** (`Numerics.default_grid`) that places more points near the boundaries:

$$x_i = \frac{1}{1 + e^{-c \cdot u_i}}$$

where u_i are uniformly spaced on $[-1, 1]$ and c (the “crowding” parameter, default 8) controls how strongly points cluster at the boundaries. The grid is then rescaled so that $x_1 > 0$ and $x_n < 1$.

```
import dadi
import numpy as np

xx = dadi.Numerics.default_grid(40)

# Grid spacing near boundaries vs. interior
print(f"First spacing: {xx[1] - xx[0]:.6f}")
print(f"Middle spacing: {xx[20] - xx[19]:.6f}")
print(f>Last spacing: {xx[-1] - xx[-2]:.6f}")
# First and last spacings are much smaller than the middle
```

The crowding parameter c can be tuned. `dadi` provides `Numerics.estimate_best_exp_grid_crowd(ns)` to estimate an optimal crowding based on the sample size, balancing resolution near the boundaries against coverage of the interior.

The Finite-Difference Scheme

With the frequency axis discretized onto grid points $x_1, \dots, x_n$, the PDE becomes a system of coupled ODEs – one for each grid point. `dadi` discretizes the spatial derivatives using a finite-difference scheme.

Consider the 1D diffusion equation:

$$\frac{\partial \phi_i}{\partial t} = \frac{1}{2} \frac{\partial^2}{\partial x^2} \left[\frac{x(1-x)}{\nu} \phi \right]_i - \frac{\partial}{\partial x} [\gamma x(1-x)(h + (1-2h)x) \phi]_i$$

The second derivative term (drift) and first derivative term (selection) are approximated using the values at neighboring grid points. On a nonuniform grid with spacings $\Delta x_i = x_{i+1} - x_i$, `dadi` computes a **difference factor**:

$$d_i = \frac{2}{\Delta x_{i-1} + \Delta x_i}$$

which corrects for the variable grid spacing. The resulting scheme produces a **tridiagonal system**: each grid point i depends only on its neighbors $i-1$, i , and $i+1$.

Numerical Analysis Aside – Why tridiagonal?

The diffusion equation involves at most second-order spatial derivatives. A finite-difference approximation of a second derivative uses three points (the central point and its two neighbors), producing a tridiagonal matrix. Tridiagonal systems can be solved in $O(n)$ time using the Thomas algorithm, making each timestep very efficient.

Implicit Time-Stepping

After spatial discretization, the system is:

$$\frac{d\phi}{dt} = A(t) \phi$$

where A is the tridiagonal matrix encoding drift, selection, and mutation, and ϕ is the vector of density values at the grid points.

`dadi` uses an **implicit (backward Euler)** time-stepping scheme:

$$\phi^{n+1} = (I - \Delta t A)^{-1} \phi^n$$

This requires solving a tridiagonal linear system at each timestep, which costs $O(n)$. The implicit scheme is **unconditionally stable** – it doesn't blow up regardless of the timestep – unlike an explicit scheme which would require $\Delta t < O(\Delta x^2)$.

The timestep Δt is computed adaptively by `Integration._compute_dt`, which sets:

$$\Delta t = \frac{\text{timescale_factor}}{\max_i |V(x_i)| + |M(x_i)|}$$

where `timescale_factor` (default 10^{-3}) controls accuracy. The timestep is small where the drift and selection coefficients are large (near the center of the frequency range, where $x(1-x)$ is maximal).

1D Integration Walkthrough

Here's the complete sequence for `Integration.one_pop(phi, xx, T, nu, gamma, h, theta0)`:

1. **Initialize:** receive the density `phi` on grid `xx`, with target integration time `T`
2. **Build coefficients:** compute the drift coefficient $V(x_i) = x_i(1-x_i)/\nu$ and selection coefficient $M(x_i) = \gamma x_i(1-x_i)(h + (1-2h)x_i)$ at each grid point

3. **Compute timestep:** Δt from the maximum coefficient value
4. **Time-stepping loop:** for each step $t \rightarrow t + \Delta t$:
 - a. Build the tridiagonal matrix $I - \Delta t \cdot A$
 - b. Solve the tridiagonal system to get ϕ^{n+1}
 - c. Inject mutations: add $\theta_0 \cdot \Delta t / (2 \cdot \Delta x_1)$ at the first interior grid point
 - d. If ν changes with time, update the coefficients
5. **Return:** the density `phi` at time T

When parameters are constant over the integration interval, `dadi` uses an optimized path (`_one_pop_const_params`) that pre-computes the tridiagonal matrix once and reuses it for all timesteps.

```
import dadi

pts = 60
xx = dadi.Numerics.default_grid(pts)

# Start from equilibrium
phi = dadi.PhiManip.phi_1D(xx)

# Integrate for T=0.5 with doubled population size
phi_evolved = dadi.Integration.one_pop(phi, xx, T=0.5, nu=2.0)

# The density has spread out (weaker drift in larger population)
```

Multi-Dimensional Integration

For two populations, the density $\phi(x, y)$ lives on a 2D grid of size $n \times n$. The PDE has drift and selection terms acting along each axis independently, plus migration terms that couple the two axes.

`dadi` uses **alternating direction implicit (ADI)** integration: at each timestep, it first sweeps along the x -axis (treating y as fixed), then along the y -axis (treating x as fixed). Each sweep requires solving n independent tridiagonal systems – one for each row or column of the grid.

$$\begin{aligned}\phi^{n+1/2} &= (I - \Delta t A_x)^{-1} \phi^n \\ \phi^{n+1} &= (I - \Delta t A_y)^{-1} \phi^{n+1/2}\end{aligned}$$

For three populations, the density is 3D (n^3 grid points) and three ADI sweeps are needed per timestep. The cost per timestep scales as $O(n^P)$ where P is the number of populations – this is the **curse of dimensionality** that limits `dadi` to a few populations.

Biology Aside – The curse of dimensionality in population genetics

Each additional population in a demographic model adds a new axis to the allele frequency density, and the computational cost grows exponentially. This is why `dadi` is typically limited to 3 populations (and sometimes up to 5 with heroic effort). For studies involving many populations – such as the global human population tree with dozens of branches – this is the fundamental limitation that motivated `moments` (which avoids the grid altogether by working with moments of the SFS) and `mom2` (which uses tensors that scale more gracefully with the number of populations). The choice among these three tools often comes down to how many populations are in the model: `dadi` excels for 1–3 populations, `moments` for 3–5, and `mom2` for 5 or more.

Populations	Grid size	Function	Cost per step
1	n	one_pop	$O(n)$
2	n^2	two_pops	$O(n^2)$
3	n^3	three_pops	$O(n^3)$

Richardson Extrapolation

The finite-difference scheme introduces **discretization error** that decreases as the grid becomes finer. For a grid with n points, the error in the SFS is approximately $O(1/n^2)$ (for a second-order scheme). Rather than using a very fine grid (expensive), `dadi` uses **Richardson extrapolation** to cancel the leading-order error.

The idea: run the computation at multiple grid sizes (e.g., $n_1 = 40$, $n_2 = 50$, $n_3 = 60$), then extrapolate to the $n \rightarrow \infty$ limit. If the error is polynomial in $1/n$, a polynomial fit through the results eliminates the leading error terms.

`dadi` provides this via `Numerics.make_extrap_func`:

```
import dadi

def my_model(params, ns, pts):
    xx = dadi.Numerics.default_grid(pts)
    phi = dadi.PhiManip.phi_1D(xx)
    nu, T = params
    phi = dadi.Integration.one_pop(phi, xx, T, nu=nu)
    return dadi.Spectrum.from_phi(phi, ns, (xx,))

# Wrap with extrapolation
func_ex = dadi.Numerics.make_extrap_log_func(my_model)

# Run at 3 grid sizes and extrapolate
model = func_ex((2.0, 0.1), ns=[20], pts=[40, 50, 60])
```

Internally, `make_extrap_func` runs the model function at each grid size in `pts`, then applies polynomial extrapolation (linear, quadratic, or higher) in the variable $1/n$ to estimate the $n \rightarrow \infty$ limit. The `make_extrap_log_func` variant extrapolates in log-space, which is more robust when SFS entries span many orders of magnitude.

The extrapolation functions available are:

Function	Points needed	Error eliminated
linear_extrap	2	$O(1/n^2)$
quadratic_extrap	3	$O(1/n^2)$ and $O(1/n^4)$
cubic_extrap	4	Up to $O(1/n^6)$

The default `make_extrap_func` uses `quadratic_extrap` with three grid sizes – the standard recommendation for `dadi` analyses.

Numerical Analysis Aside – Richardson extrapolation

Richardson extrapolation is a general technique: if a quantity $f(h)$ has an error expansion $f(h) = f_0 + ah^2 + bh^4 + \dots$, then evaluating at two step sizes h_1, h_2 and taking the appropriate linear combination eliminates the h^2 term. With three evaluations, both the h^2 and h^4 terms can be eliminated. `dadi` applies this with $h = 1/n$ (the grid spacing), effectively canceling the finite-difference bias without requiring an impractically fine grid.

From ϕ to SFS

After solving the PDE, `dadi` must convert the continuous density $\phi(x)$ into the discrete SFS. This is done by `Spectrum.from_phi`.

The expected number of sites where j out of n sampled chromosomes carry the derived allele is:

$$F_j = \int_0^1 \binom{n}{j} x^j (1-x)^{n-j} \phi(x) dx$$

This is a weighted integral of the frequency density against the **binomial sampling probability** – the probability that a site with population frequency x would show count j in a sample of n .

`dadi` evaluates this integral numerically using the trapezoidal rule on the grid points. For each SFS entry j , the integrand is the product of the density `phi` and the binomial coefficient evaluated at each grid point.

```
import dadi

pts = 60
xx = dadi.Numerics.default_grid(pts)
phi = dadi.PhiManip.phi_1D(xx)

# Extract SFS for sample size n=20
fs = dadi.Spectrum.from_phi(phi, ns=(20,), xxs=(xx,))

# fs[j] = expected number of sites with j derived alleles
print(f"Singletons (j=1): {fs[1]:.4f}")
print(f"Doubletons (j=2): {fs[2]:.4f}")
print(f"Under neutrality, fs[j] ~ theta/j")
```

For multi-population spectra, the integral extends over all frequency axes:

$$F_{j_1, j_2} = \int_0^1 \int_0^1 \binom{n_1}{j_1} x^{j_1} (1-x)^{n_1-j_1} \binom{n_2}{j_2} y^{j_2} (1-y)^{n_2-j_2} \phi(x, y) dx dy$$

The 2D integral is evaluated by the trapezoidal rule on the 2D grid. For efficient computation, `dadi` uses a linear algebra formulation (`_from_phi_2D_linalg`) that avoids explicit double loops.

Practical Grid Guidelines

Choosing the right grid size is a trade-off between accuracy and speed:

- **Too few points** (< 30): significant discretization error, even with extrapolation
- **Standard** (40, 50, 60): the recommended trio for Richardson extrapolation in most analyses
- **Large sample sizes** ($n > 100$): may need $\text{pts} \geq n + 10$ to avoid aliasing in the `from_phi` integration
- **Selection** ($|\gamma| > 10$): stronger selection creates steeper density profiles, requiring finer grids

The rule of thumb: `pts` should be at least 10 larger than the largest sample size, and extrapolation should always be used for publication-quality results.

Plain-language summary – What Richardson extrapolation does for you

Richardson extrapolation is a clever trick: run the computation three times on grids of increasing fineness (e.g., 40, 50, and 60 points), then mathematically combine the three answers to cancel out most of the error from the finite grid. The result is an SFS estimate that is far more accurate than any of the three individual calculations – roughly as

accurate as if you had used hundreds of grid points, but at a fraction of the cost. For biologists, this means you can trust that the predicted SFS (and therefore the inferred demographic history) is not an artifact of the grid resolution.

In the next chapter, we'll examine how `dadi` uses the computed SFS to perform demographic inference through composite likelihood optimization.

Demographic Inference

The mainspring – where the model meets reality and parameters are adjusted until the predicted dial matches observation.

With the diffusion equation solved and the expected SFS in hand, the final step is inference: finding the demographic parameters that best explain the observed data. This chapter covers the likelihood framework, optimization algorithms, uncertainty quantification, and the extensions that make `dadi` a practical tool for population genomic analysis.

The Poisson Composite Likelihood

`dadi` uses the same **Poisson Random Field (PRF)** model as `moments` (see [Demographic Inference](#) in the `moments` Timepiece for a detailed derivation). Under the infinite-sites model, each segregating site is an independent Poisson draw, and the SFS entries are independent Poisson random variables.

The log-likelihood is:

$$\ell(\Theta) = \sum_j [-M_j(\Theta) + D_j \ln M_j(\Theta) - \ln D_j!]$$

where D_j is the observed count in SFS bin j and $M_j(\Theta)$ is the expected count under demographic parameters Θ .

In `dadi`'s code, this is `Inference.ll(model, data)`:

```
import dadi

# model: expected SFS from the diffusion solver
# data: observed SFS from real data
ll = dadi.Inference.ll(model, data)
```

Multinomial vs. Poisson Likelihood

The Poisson likelihood requires knowing θ (the population-scaled mutation rate), which sets the overall scale of the expected SFS. In many analyses, θ is a nuisance parameter – you want to infer relative sizes and times, not the absolute mutation rate.

`dadi` offers a **multinomial likelihood** (`Inference.ll_multinom`) that analytically maximizes over θ :

$$\hat{\theta} = \frac{\sum_j D_j}{\sum_j M_j^{(0)}}$$

where $M_j^{(0)}$ is the expected SFS computed with $\theta = 1$. The model SFS is then rescaled by $\hat{\theta}$ before computing the Poisson likelihood.

```
# Multinomial likelihood (automatically optimizes theta)
ll = dadi.Inference.ll_multinom(model, data)

# Optimal theta scaling factor
theta_opt = dadi.Inference.optimal_sfs_scaling(model, data)
```

The multinomial approach is the default for most analyses, since it eliminates one parameter from the optimization.

Optimization

`dadi` provides several optimization functions, all following the same interface: minimize the negative log-likelihood over the demographic parameters.

BFGS in log-space (recommended):

The primary optimizer is `Inference.optimize_log`, which transforms parameters to log-space before applying the BFGS quasi-Newton method:

```
import dadi

# p0: initial parameter guess
# data: observed SFS
# model_func: demographic model function
# pts: grid sizes for extrapolation
popt = dadi.Inference.optimize_log(
    p0=[1.0, 0.5],          # initial guess (nu, T)
    data=data,              # observed SFS
    model_func=func_ex,     # extrapolated model function
    pts=[40, 50, 60],      # grid sizes
    lower_bound=[1e-2, 0],  # parameter lower bounds
    upper_bound=[100, 3],   # parameter upper bounds
    verbose=1,              # print progress
    maxiter=100             # maximum iterations
)
```

The log-space transformation ensures that parameters remain positive and that the optimizer takes proportionally sized steps (a step of 0.1 in log-space corresponds to a ~10% change in the parameter).

Other optimizers:

Function	Method	When to use
<code>optimize_log</code>	BFGS (log-space)	Default; fast, uses gradients
<code>optimize</code>	BFGS (linear)	When log transform is inappropriate
<code>optimize_log_lbfgsb</code>	L-BFGS-B	When bounds are important
<code>optimize_log_fmin</code>	Nelder-Mead	Robust to noisy likelihoods
<code>optimize_log_powell</code>	Powell	No gradient needed
<code>optimize_grid</code>	Grid search	Exploring the landscape
<code>optimize_cons</code>	SLSQP	With equality/inequality constraints

Fixed parameters:

To fix certain parameters while optimizing others, pass a `fixed_params` list where `None` means “optimize” and a value means “fix”:

```
# Fix T=0.5, optimize only nu
popt = dadi.Inference.optimize_log(
    p0=[1.0, 0.5],
    data=data,
    model_func=func_ex,
    pts=[40, 50, 60],
    fixed_params=[None, 0.5] # optimize nu, fix T=0.5
)
```

A Complete Inference Example

Here is a complete workflow for fitting a two-epoch model:

```
import dadi
import numpy as np

# ----- 1. Load or simulate data -----
# (Here we simulate for demonstration)
def two_epoch(params, ns, pts):
    nu, T = params
    xx = dadi.Numerics.default_grid(pts)
    phi = dadi.PhiManip.phi_1D(xx)
    phi = dadi.Integration.one_pop(phi, xx, T, nu=nu)
    return dadi.Spectrum.from_phi(phi, ns, (xx,))

func_ex = dadi.Numerics.make_extrap_log_func(two_epoch)

# "True" parameters: 2x expansion, 0.1 time units ago
true_params = [2.0, 0.1]
true_model = func_ex(true_params, ns=[20], pts=[40, 50, 60])

# Generate synthetic data (Poisson sampling)
data = true_model * 1000 # scale by theta
data = dadi.Spectrum(np.random.poisson(data))

# ----- 2. Optimize -----
p0 = [1.0, 0.5] # initial guess (deliberately wrong)
popt = dadi.Inference.optimize_log(
    p0, data, func_ex,
    pts=[40, 50, 60],
    lower_bound=[1e-2, 1e-3],
    upper_bound=[100, 3],
    verbose=0,
    maxiter=100
)

# ----- 3. Evaluate fit -----
model = func_ex(popt, ns=[20], pts=[40, 50, 60])
ll_opt = dadi.Inference.ll_multinom(model, data)
theta = dadi.Inference.optimal_sfs_scaling(model, data)

print(f"Best-fit: nu={popt[0]:.3f}, T={popt[1]:.4f}")
print(f"Log-likelihood: {ll_opt:.2f}")
print(f"Optimal theta: {theta:.2f}")
```

Uncertainty: The Godambe Information Matrix

Point estimates are incomplete without uncertainty quantification. `dadi` uses the **Godambe Information Matrix** (GIM), a sandwich estimator that accounts for linkage between sites.

The standard Fisher Information Matrix (FIM) assumes independent sites:

$$\mathcal{I}(\Theta) = -E \left[\frac{\partial^2 \ell}{\partial \Theta_i \partial \Theta_j} \right]$$

But sites in a genome are not independent – they are linked. The GIM corrects for this using bootstrap replicates of the SFS:

$$G = H J^{-1} H$$

where H is the Hessian of the log-likelihood (computed by finite differences) and J is the variance of the score vector (estimated from bootstrap replicates):

$$J = \frac{1}{B} \sum_{b=1}^B U_b U_b^T$$

Here $U_b = \nabla \ell(\Theta; \mathbf{D}_b)$ is the gradient of the log-likelihood evaluated on bootstrap replicate b .

The standard errors are $\text{SE}(\hat{\theta}_i) = \sqrt{(G^{-1})_{ii}}$:

```
import dadi

# all_boot: list of bootstrap SFS replicates
# p0: best-fit parameters
# data: observed SFS
uncerts = dadi.Godambe.GIM_uncert(
    func_ex,
    pts=[40, 50, 60],
    all_boot=all_boot,
    p0=popt,
    data=data,
    multinom=True,
    eps=0.01 # step size for finite differences
)
print(f"Uncertainties: {uncerts}")
```

Model Comparison

With uncertainties in hand, you can compare competing demographic models using:

Likelihood Ratio Test (LRT):

For nested models (one is a special case of the other), twice the log-likelihood difference follows a χ^2 distribution:

$$\Lambda = 2(\ell_{\text{complex}} - \ell_{\text{simple}})$$

The Godambe-adjusted LRT (`Godambe.LRT_adjust`) corrects the test statistic for linkage.

AIC (Akaike Information Criterion):

For non-nested models:

$$\text{AIC} = 2k - 2\ell$$

where k is the number of parameters. Lower AIC indicates a better model (balancing fit against complexity).

DFE Inference

A powerful extension of `dadi` is inference of the **Distribution of Fitness Effects** (DFE) – the distribution of selection coefficients across new mutations.

The approach:

1. **Cache spectra:** compute the expected SFS at many values of the selection coefficient γ , using a demographic model fit to synonymous (neutral) sites:

```
import dadi.DFE

cache = dadi.DFE.Cache1D(
    params=popt,          # demographic parameters
    ns=[20],              # sample sizes
    demo_sel_func=my_sel_model, # demographic+selection model
    pts=[40, 50, 60],
    gamma_bounds=(1e-4, 2000),
    gamma_pts=500
)
```

2. **Integrate over DFE:** weight the cached spectra by a parametric distribution (e.g., gamma, lognormal) and sum:

```
# Integrate cached spectra over a gamma DFE
sel_model = cache.integrate(
    params=dfe_params,
    ns=[20],
    sel_dist=dadi.DFE.PDFs.gamma,
    theta=theta_ns
)
```

3. **Optimize DFE parameters:** fit the DFE shape and scale to the non-synonymous SFS.

This two-step approach (demographic model from synonymous sites, DFE from non-synonymous sites) cleanly separates demography from selection.

Ancestral Misidentification

Polarizing the SFS (assigning derived vs. ancestral alleles) requires an outgroup. Errors in polarization – **ancestral misidentification** – fold some of the SFS entries across the spectrum. `dadi` corrects for this with `Spectrum.apply_anc_state_misid(p_misid)`, which mixes each SFS entry j with its complement $n - j$ according to a misidentification probability p :

$$F_j^{\text{obs}} = (1 - p) F_j^{\text{true}} + p F_{n-j}^{\text{true}}$$

This can be applied to the model SFS before comparing to data, adding p as an additional parameter to optimize.

Demes Integration

For standardized model specification, `dadi` integrates with the **Demes** format (Gower et al. 2022), which specifies demographic models as structured YAML files. This allows the same model to be used across different inference tools (`dadi`, `moments`, `mom2`):

```
import demes
import dadi

# Load a Demes-format model
graph = demes.load("my_model.yaml")

# Convert to a dadi model function
# and compute the expected SFS
```

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```
fs = dadi.Spectrum.from_demes(graph, sampled_demes=["Pop1", "Pop2"],
                             sample_sizes=[20, 20])
```

Summary

The inference machinery of `dadi` combines:

- **Poisson composite likelihood** from the PRF model (shared with `moments`)
- **BFGS optimization** in log-parameter space for efficient gradient-based search
- **Godambe Information Matrix** for linkage-corrected uncertainty
- **DFE inference** via cached spectra and parametric integration
- **Ancestral misidentification correction** for imperfect polarization
- **Demes integration** for standardized model specification

Together with the diffusion equation solver and Richardson extrapolation from the previous chapters, these components form a complete, mature framework for demographic inference from the site frequency spectrum – the framework that `moments` and `mom2` were designed to complement and, in some respects, improve upon.

Demo: Running dadi on Simulated Data

Running mini_dadi on simulated genomic data.

This figure was generated by running the `mini_dadi` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_dadi.py`.

3.13 Timepiece XII: mom2

Computing the expected frequency spectrum backward through the coalescent – one tensor at a time.

3.13.1 The Mechanism at a Glance

`mom2` is a method for **demographic inference** – learning a population's history (size changes, splits, admixture events) from patterns in its DNA variation. Like `moments` (*Timepiece X*), it works with the **site frequency spectrum** (SFS) as its summary statistic. But the internal mechanism is fundamentally different.

Where `moments` works *forward in time* – deriving ODEs that push allele frequencies through the diffusion process – `mom2` works *backward in time* through the **coalescent**. It asks: given a demographic model, what are the expected branch lengths in the genealogy? And from those branch lengths, what SFS do we expect to observe?

The key innovation is how `mom2` performs this computation. Rather than simulating genealogies or solving differential equations, it uses **tensor algebra**. The expected SFS is assembled as a product of tensors, one for each demographic event, processed along a junction tree. Each tensor encodes how lineage counts change at a split, an admixture pulse, or during a period of constant (or exponential) population size. The population dynamics within each epoch are governed by the **Moran model**, a continuous-time Markov chain whose eigendecomposition allows efficient computation of transition probabilities for arbitrary time spans.

The final ingredient is **automatic differentiation** (`autograd`). Because every operation in the SFS computation is differentiable, `mom2` obtains exact gradients of the likelihood with respect to all demographic parameters – no hand-coded derivatives, no finite differences. This enables efficient gradient-based optimization and even Hessian computation for uncertainty quantification.

Demo: dadi Diffusion Solver on msprime-simulated SFS (30 haplotypes, 1 Mb)

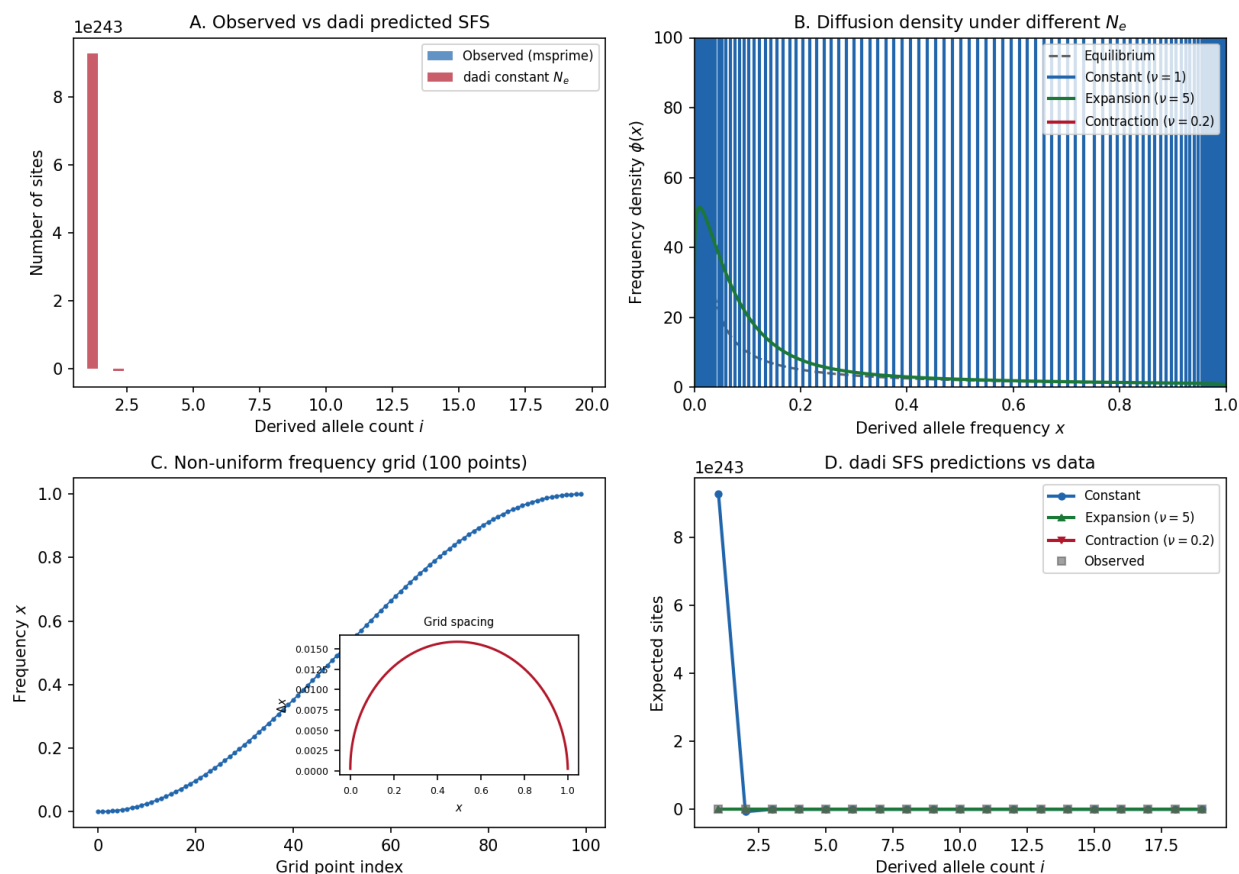


Fig. 15: **dadi demo on msprime-simulated data.** Panel A: Observed SFS from msprime vs dadi's diffusion approximation prediction. Panel B: Allele frequency density evolution under the Crank-Nicolson PDE solver. Panel C: SFS predictions under three demographic models. Panel D: Residuals between observed and predicted SFS entries.

i Primary Reference

[Kamm *et al.*, 2019]

The four gears of `mom2`:

1. **The Coalescent SFS** (the dial) – The expected frequency spectrum derived from coalescent theory: how expected branch lengths in the genealogy translate directly into expected SFS entries. This is the quantity `mom2` computes and compares against data.
2. **The Moran Model** (the escapement) – The discrete population model that governs lineage dynamics within each epoch. Its eigendecomposition lets us compute transition probabilities for any time span in a single matrix operation, replacing numerical ODE integration.
3. **Tensor Machinery** (the gear train) – The computational engine: likelihood tensors that track allele configurations across populations, assembled via convolution (for population merges), matrix multiplication (for Moran transitions), and antidiagonal summation (for coalescence). A junction tree algorithm processes demographic events in the correct order.
4. **Automatic Differentiation & Inference** (the mainspring) – Autograd-powered optimization: exact gradients flow backward through the entire tensor computation, enabling efficient maximum-likelihood estimation with TNC, L-BFGS-B, or stochastic methods (ADAM, SVRG).

These gears mesh together into a complete inference framework:

```

Observed SFS
  |
  v
+-----+
| DEMOGRAPHIC MODEL |
|                   |
| Population tree with |
| sizes, split times, |
| admixture fractions |
+-----+
  |
  v
+-----+
| COALESCENT SFS    |
| COMPUTATION        |
|                   |
| Tensor products over |
| the event tree:    |
| Moran transitions,  |
| population merges,  |
| admixture pulses    |
+-----+
  |
  v
+-----+
| LIKELIHOOD &      |
| OPTIMIZATION        |
|                   |
| Composite log-lik + |
| autograd gradients  |
+-----+

```

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```
| -> maximize via TNC |
| or stochastic SGD   |
+-----+
      |
      v
Demographic history:
population sizes, split
times, admixture fractions
```

i Prerequisites for this Timepiece

- *Coalescent Theory* – understanding the relationship between genealogies, branch lengths, and genetic diversity
- Basic linear algebra – eigenvalues, matrix exponentials, tensor products
- Familiarity with the site frequency spectrum is helpful but not required – see *The Frequency Spectrum* in the moments Timepiece for a ground-up introduction

3.13.2 Chapters

Overview of momi2

Before assembling the watch, lay out every part and understand what it does.

What Does momi2 Do?

Given DNA sequence data from one or more populations, *momi2* infers the **demographic history** that produced the observed patterns of genetic variation.

Input: $\mathbf{D}$ = observed SFS from k populations

Output: $\hat{\Theta}$ = best-fit demographic parameters (sizes, times, admixture fractions)

Concretely, *momi2* asks: *What history of population size changes, splits, and admixture events would produce a frequency spectrum that looks like the one we observe?*

Think of a fine mechanical watch. A watchmaker can deduce every gear ratio inside the case by studying the positions of the hands on the dial. *momi2* works the same way: it computes the expected hand positions (the expected SFS) for a given gear configuration (a demographic model), then adjusts the gears until the hands match what is observed. But unlike *moments*, which turns the gears forward in time to see where the hands end up, *momi2* **traces the hands backward** through the coalescent to figure out what gear configuration must have produced them.

From the inferred history, you learn:

- **Population sizes** through time – expansions, contractions, bottlenecks
- **Divergence times** – when populations split from each other
- **Admixture fractions** – the proportion of ancestry contributed by each source during a pulse admixture event
- **Growth rates** – exponential growth or decline within epochs

momi2: Demographic Inference via Coalescent Tensor Algebra

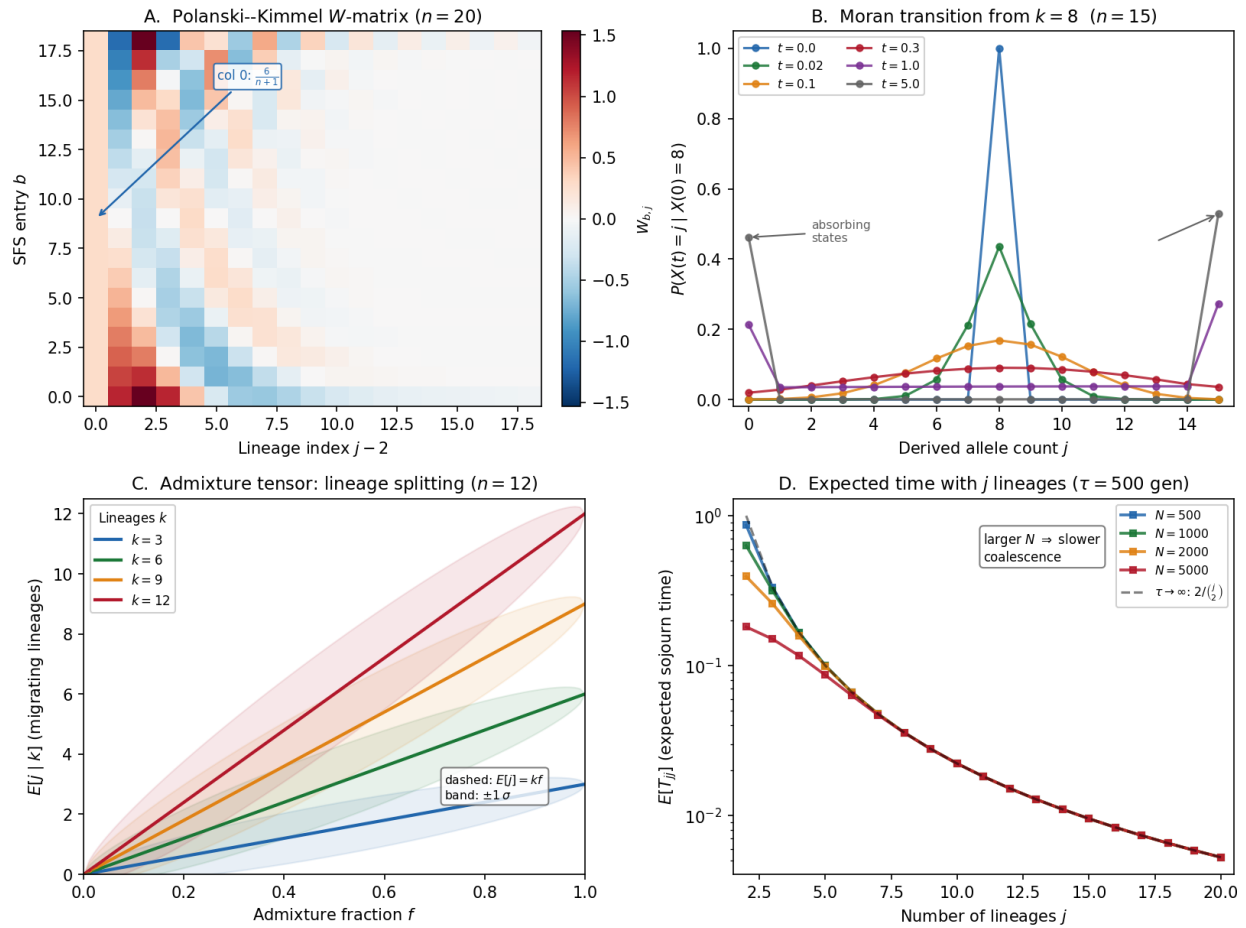


Fig. 16: **momi2** at a glance. Panel A: The W -matrix (Polanski-Kimmel coefficients) as a heatmap – the combinatorial bridge between sample configurations and expected branch lengths. Panel B: Moran model transition dynamics showing how a delta distribution at frequency k spreads over time. Panel C: Admixture tensor – expected migrating lineage counts as a function of admixture fraction. Panel D: Expected time spent with j lineages under constant-size epochs with different population sizes N .

How momi2 Differs from moments and dadi

All three tools infer demography from the SFS. The difference is *how* they compute the expected SFS under a given model.

Property	dadi	moments	momi2
Direction	Forward (diffusion PDE)	Forward (moment ODEs)	Backward (coalescent)
Core equation	Wright-Fisher diffusion PDE on a frequency grid	System of ODEs for SFS entries	Tensor products over a demographic event tree
Population model	Continuous diffusion	Continuous diffusion (via moments)	Moran model (discrete)
Gradient computation	Finite differences	Analytic or finite differences	Automatic differentiation (autograd)
Multi-population scaling	Grid grows as $O(n^D)$ – expensive beyond 3 populations	ODE system grows polynomially – practical for 4–5 populations	Tensor operations along event tree – scales well for many populations
Selection	Yes	Yes	No (neutral models only)

The key trade-off

`momi2` gives up selection modeling in exchange for two advantages: exact gradients via automatic differentiation (enabling faster, more reliable optimization) and a coalescent formulation that scales more naturally to many populations. If you need selection, use `moments` or `dadi`. If you have many populations and neutral models, `momi2` is often the better choice.

The Key Innovation: Tensors + Autograd

The central insight of `momi2` is that the expected SFS under any demographic model can be written as a sequence of **tensor operations** – matrix multiplications, convolutions, and antidiagonal summations – applied to likelihood tensors that track allele configurations across populations.

Each operation corresponds to a demographic event:

- **Moran transition** (a period of constant or exponential population size): multiply the likelihood tensor along the relevant axis by the Moran transition matrix $P(t) = V e^{\Lambda t} V^{-1}$
- **Population merge** (a split event, viewed backward in time): convolve two population axes into one
- **Admixture pulse**: contract the likelihood tensor with a 3-tensor encoding the binomial mixing of lineages

Because every one of these operations is differentiable, the Python library `autograd` can compute exact gradients of the log-likelihood with respect to all demographic parameters by backpropagating through the entire computation graph. No hand-coded Jacobians, no finite-difference approximations, no truncation error.

The Moran Model Connection

Within each epoch (a period between demographic events), `momi2` needs to compute how the distribution of lineage counts evolves over time. It uses the **Moran model** – a continuous-time Markov chain on the number of derived alleles in a sample of size n .

The Moran model has a tridiagonal rate matrix with a known eigendecomposition. This means the transition matrix for any time t can be computed in closed form:

$$P(t) = V e^{\Lambda t} V^{-1}$$

where V is the matrix of eigenvectors and Λ is the diagonal matrix of eigenvalues. No numerical ODE integration is needed – just a single matrix operation per epoch.

For epochs with exponential growth, the time is rescaled by integrating the inverse population size: $\tau = \int_0^T 2/N(s) ds$, and the expected coalescence times are computed via closed-form expressions involving exponential integrals.

Parameters

Parameter	Meaning	Typical units
N_e	Effective population size (reference)	Individuals
$N_i(t)$	Population size of deme i at time t	Individuals (or scaled by N_e)
T_{split}	Time of population split (backward from present)	Generations (or scaled: $T/4N_e$)
f	Admixture fraction in a pulse event	Proportion (0 to 1)
g	Exponential growth rate within an epoch	Per generation
μ	Per-site per-generation mutation rate	$\sim 10^{-8}$

The Flow in Detail

Step 1: Observed SFS

Count derived allele frequencies across k populations from SNP data. Result: a k -dimensional histogram.

Step 2: Demographic model

Specify a population tree: which populations split when, with what sizes, growth rates, and admixture events.

Step 3: Expected SFS via tensor computation

Traverse the event tree bottom-up (leaves to root):

- At each leaf: initialize a likelihood tensor
- At each epoch boundary: apply Moran transition matrix
- At each split: convolve two population tensors
- At each pulse: apply admixture 3-tensor

Accumulate the expected SFS along the way.

Step 4: Likelihood computation

Compare observed and expected SFS via composite log-likelihood (multinomial or Poisson).

Step 5: Optimization

Compute gradients via autograd, then optimize parameters using TNC, L-BFGS-B, ADAM, or SVRG.

Step 6: Uncertainty quantification

Use the autograd Hessian or bootstrap to estimate confidence intervals on inferred parameters.

Ready to Build

The next four chapters disassemble `mom2` gear by gear:

- *The Coalescent SFS* – **The dial**: how expected branch lengths in the coalescent translate into expected SFS entries, and how demographic events modify these expectations.

- *The Moran Model* – **The escapement**: the Moran model as a continuous-time Markov chain, its eigendecomposition, and how it governs lineage dynamics within each epoch.
- *Tensor Machinery* – **The gear train**: likelihood tensors, convolution for population merges, the junction tree algorithm, and how the full SFS is assembled from parts.
- *Automatic Differentiation & Inference* – **The mainspring**: autograd-powered optimization, composite likelihoods, stochastic gradient methods, and uncertainty quantification.

The Coalescent SFS

The dial face of the watch: reading expected allele frequencies from the shape of the genealogy.

“The expected SFS is a linear function of the expected coalescence times.”

---Polanski and Kimmel (2003)

Step 1: From Genealogies to the SFS

The site frequency spectrum counts how many SNPs have a given number of derived alleles in the sample. Under the infinitely-many-sites mutation model, a mutation that falls on a branch of the genealogy appears in exactly those samples that descend from that branch. So the SFS is determined by the **branch lengths** of the genealogy:

$$E[\text{SFS}[b]] = \frac{\theta}{2} \sum_{j=2}^n W_{b,j} E[T_{jj}]$$

where:

- $\theta = 4N_e\mu$ is the population-scaled mutation rate
- T_{jj} is the total time during which there are exactly j lineages, each descending from exactly j of the n sampled chromosomes
- $W_{b,j}$ are combinatorial coefficients (the **W-matrix** of Polanski and Kimmel) that convert expected coalescence times into expected SFS entries

The W-matrix is the bridge between the coalescent (which gives us $E[T_{jj}]$) and the SFS (which is what we observe). Each entry $W_{b,j}$ answers: “If a mutation falls on a branch with j descendants, what is the probability that exactly b of my n sampled chromosomes carry the derived allele?”

i Probability Aside – Why the W-matrix works

Consider a genealogy with n samples. At a moment when there are j ancestral lineages, a mutation on one of those lineages will be inherited by some subset of the n samples. The W-matrix entries are derived from the hypergeometric distribution: given j lineages partitioning the n samples, what is the probability that a random lineage has exactly b descendants? Polanski and Kimmel (2003) showed that these coefficients satisfy a three-term recurrence, which `mom2` implements in optimized Cython code.

Step 2: The W-Matrix Recurrence

The W-matrix has dimensions $(n-1) \times (n-1)$, with rows indexed by SFS entry $b \in \{1, \dots, n-1\}$ and columns indexed by epoch $j \in \{2, \dots, n\}$. The recurrence (Polanski and Kimmel 2003) is:

$$W_{b,1} = \frac{6}{n+1}$$

$$W_{b,2} = \frac{30(n-2b)}{(n+1)(n+2)}$$

$$W_{b,j} = \frac{(2j+1)(n-2b)}{j(n+j+1)}W_{b,j-1} - \frac{(j+1)(2j+3)(n-j)}{j(2j-1)(n+j+1)}W_{b,j-2}$$

Why these specific coefficients? The W-matrix entries are derived from the hypergeometric distribution. When there are j ancestral lineages partitioning n samples, the probability that a random lineage has exactly b descendants involves Hahn polynomials – a family of orthogonal polynomials on discrete sets. The three-term recurrence above is the recurrence relation for these polynomials. Notice two features:

- The factor $(n-2b)$ appears in both $W_{b,2}$ and the recursion. When $b = n/2$ (the folded midpoint), this factor is zero, which reflects the symmetry of the neutral SFS: singletons and $(n-1)$ -tons have the same expectation under exchangeability.
- The recursion is **three-term** (each $W_{b,j}$ depends on the two previous columns), which is characteristic of orthogonal polynomial recurrences. This structure allows the entire matrix to be computed in $O(n^2)$ time – one pass through the columns.

```
import numpy as np

def w_matrix(n):
    """Compute the W-matrix of Polanski and Kimmel (2003).

    Returns W of shape (n-1, n-1), where W[b-1, j-2] gives the
    coefficient for SFS entry b from expected time with j lineages.
    """
    W = np.zeros((n-1, n-1))
    bb = np.arange(1, n) # SFS entries 1..n-1

    W[:, 0] = 6.0 / (n+1)
    if n > 2:
        W[:, 1] = 30.0 * (n-2*bb) / ((n+1) * (n+2))

    for col in range(2, n-1):
        j = col + 2 # number of lineages
        W[:, col] = (
            W[:, col-1] * (2*j+1) * (n-2*bb) / (j * (n+j+1))
            - W[:, col-2] * (j+1) * (2*j+3) * (n-j)
            / (j * (2*j-1) * (n+j+1))
        )
```

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```
)
return W
```

```
# Verification: for a standard neutral model (constant size),
#  $E[T_{jj}] = 2 / (j * (j-1))$ , and  $SFS[b]$  should equal  $\theta/b$ .
n = 20
W = w_matrix(n)
j_vals = np.arange(2, n + 1)
E_Tjj_neutral = 2.0 / (j_vals * (j_vals - 1))
expected_sfs = W @ E_Tjj_neutral # should be proportional to  $1/b$ 
bb = np.arange(1, n)
ratio = expected_sfs / (1.0 / bb)
assert np.allclose(ratio, ratio[0], atol=1e-10), "W-matrix verification failed"
```

Step 3: Expected Coalescence Times Under Constant Size

For a population of constant size N , the coalescent gives a simple expression for the expected time spent with exactly j lineages. The rate of coalescence when there are j lineages is $\binom{j}{2} = j(j-1)/2$, so:

$$E[T_{jj}] = \frac{1}{\binom{j}{2}} = \frac{2}{j(j-1)}$$

(in units of $2N$ generations). Plugging this into the W-matrix formula recovers the classic neutral SFS: $E[SFS[b]] \propto 1/b$.

But what if population size changes through time? This is where `momi2`'s machinery becomes essential.

```
def etjj_constant(n, tau, N):
    """Expected time with j lineages in an epoch of duration tau and size N.

    tau: epoch duration in generations
    N: population size (constant throughout epoch)

    Returns array of length n-1, indexed by j = 2, ..., n.
    """
    j = np.arange(2, n + 1)
    rate = j * (j - 1) / 2.0 # coalescence rate with j lineages
    scaled_time = 2.0 * tau / N # time in coalescent units
    # expected time with j lineages, accounting for finite epoch duration
    return (1.0 - np.exp(-rate * scaled_time)) / rate
```

Step 4: How Demographic Events Modify Expected Branch Lengths

Each type of demographic event modifies the expected coalescence times in a specific way:

Population size change. If the population has size N_1 in one epoch and N_2 in the next, the coalescence rate changes proportionally. In scaled coalescent time, the rate with j lineages is always $\binom{j}{2}$, but the *real-time* duration of that epoch is $\tau = \int_0^T 2/N(s) ds$ in coalescent units. A larger population means slower coalescence (longer branches), and vice versa.

Exponential growth. When the population grows exponentially at rate g , $N(s) = N_0 e^{-gs}$ (backward in time). The scaled time becomes:

$$\tau = \frac{2}{N_0} \int_0^T e^{gs} ds = \frac{2(e^{gT} - 1)}{N_0 g}$$

The expected coalescence times involve exponential integrals (the Ei function), computed in closed form.

Population split (viewed backward: a merge). When two populations merge into an ancestral population, the lineages from both child populations now share a single common ancestor pool. The number of lineages in the ancestral population is the sum of remaining lineages from each child. This is handled by **convolution** of the likelihood tensors (see *Tensor Machinery*).

Admixture pulse. A fraction f of population A 's ancestry comes from population B at some time in the past. Viewed backward, each lineage in A independently “jumps” to B with probability f . This redistributes lineages between the two populations according to a binomial distribution, implemented as a 3-tensor contraction.

```
def etjj_exponential(n, tau, growth_rate, N_bottom):
    """Expected time with j lineages under exponential growth.

    N_bottom: population size at the more recent end of the epoch
    growth_rate: exponential growth rate (positive = growing forward)
    tau: epoch duration in generations
    """
    from scipy.special import expi

    j = np.arange(2, n + 1)
    rate = j * (j - 1) / 2.0
    N_top = N_bottom * np.exp(-tau * growth_rate)

    if abs(growth_rate) < 1e-10:
        return etjj_constant(n, tau, N_bottom)

    # Scaled time for the epoch
    total_growth = tau * growth_rate
    scaled_time = (np.expm1(total_growth) / total_growth) * tau * 2.0 / N_bottom

    # Expected coalescence times via exponential integral
    a = rate * 2.0 / (N_bottom * growth_rate)
    result = np.zeros_like(rate)
    for idx in range(len(rate)):
        c = a[idx]
        result[idx] = (
            np.exp(-c) * (-expi(c) + expi(c * np.exp(total_growth)))
        )
    return result
```

Step 5: The Multi-Population SFS

For k populations with sample sizes $n_1, \dots, n_k$, the SFS becomes a k -dimensional array of shape $(n_1 + 1) \times (n_2 + 1) \times \dots \times (n_k + 1)$. Each entry $\text{SFS}[b_1, b_2, \dots, b_k]$ counts the number of SNPs where population i has b_i derived alleles.

`mom2` refers to each such multi-population index $(b_1, \dots, b_k)$ as a **configuration**. The expected value of each configuration is computed by the tensor machinery described in *Tensor Machinery*.

The multi-population extension is where `mom2` truly shines compared to grid-based methods. Adding populations to `dadi` multiplies the grid dimensions; adding populations to `mom2` adds axes to the tensors but the computation proceeds along the event tree, visiting each event exactly once.

```
import numpy as np
```

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```
def compute_joint_sfs(genotype_matrix, pop_assignments, pop_names):
    """Compute the joint SFS from a genotype matrix.

    genotype_matrix: (n_sites, n_samples) array of 0/1
    pop_assignments: dict mapping sample index -> population name
    pop_names: list of population names (determines axis order)

    Returns: k-dimensional array of shape (n1+1, n2+1, ..., nk+1)
    """
    # group samples by population
    pop_indices = {p: [] for p in pop_names}
    for idx, pop in pop_assignments.items():
        pop_indices[pop].append(idx)

    sample_sizes = [len(pop_indices[p]) for p in pop_names]
    sfs_shape = tuple(s + 1 for s in sample_sizes)
    sfs = np.zeros(sfs_shape, dtype=int)

    for site in range(genotype_matrix.shape[0]):
        config = tuple(
            genotype_matrix[site, pop_indices[p]].sum()
            for p in pop_names
        )
        sfs[config] += 1

    return sfs
```

Step 6: From Branch Lengths to Summary Statistics

A powerful feature of the coalescent formulation is that the expected SFS is not the only quantity you can compute. Any **linear function** of the SFS can be expressed as an inner product with the likelihood tensor. This includes:

- **f-statistics** (f_2, f_3, f_4): linear combinations of SFS entries that measure shared drift between populations
- **Patterson's D statistic** (ABBA-BABA): a test for admixture
- **Nucleotide diversity** (π): a weighted sum of SFS entries
- **Tajima's D**: a ratio of SFS-derived estimators

`mom2` computes these by passing appropriate **weight vectors** through the same tensor machinery used for the SFS. The same autograd gradients apply, so you can optimize demographic models to fit any of these statistics.

i Probability Aside – Why linear statistics are free

If the expected SFS is $E[\mathbf{S}]$ and a statistic is $f(\mathbf{S}) = \mathbf{w}^T \mathbf{S}$ for some weight vector $\mathbf{w}$, then $E[f(\mathbf{S})] = \mathbf{w}^T E[\mathbf{S}]$. In `mom2`'s tensor framework, the weight vector $\mathbf{w}$ is simply passed as the initial state of the likelihood tensor at the leaves, rather than the identity vector used for the full SFS. The rest of the computation proceeds identically.

Exercises

i Exercise 1: Verify the neutral SFS

Using the `w_matrix` and `etjj_constant` functions above, verify that the expected SFS under a constant population is proportional to $1/b$ for sample sizes $n = 10, 50, 100$.

i Exercise 2: Population expansion and the SFS

Compute the expected SFS for a population that expanded 10-fold at time $T = 0.1 \times 2N$ generations ago. Compare with the neutral expectation. Where does the expansion create excess entries?

i Exercise 3: Bottleneck signature

Compute the expected SFS for a population that went through a bottleneck (size reduced to 0.1 for duration 0.05, then recovered). How does this differ from a simple size change?

i Exercise 4: Multi-population SFS dimensions

For k populations each of sample size n , how many entries does the joint SFS have? For what values of k and n does this become impractical to store? How does `mom2`'s approach avoid this problem?

Solutions

i Solution 1: Verify the neutral SFS

Under a constant population, the expected time with j lineages is $E[T_{jj}] = 2/(j(j-1))$. Multiplying by the W -matrix should yield an SFS proportional to $1/b$.

```
import numpy as np

for n in [10, 50, 100]:
    W = w_matrix(n)
    j_vals = np.arange(2, n + 1)
    E_Tjj = 2.0 / (j_vals * (j_vals - 1))
    expected_sfs = W @ E_Tjj

    bb = np.arange(1, n)
    ratio = expected_sfs / (1.0 / bb)
    # All ratios should be identical (SFS proportional to 1/b)
    assert np.allclose(ratio, ratio[0], atol=1e-10), f"Failed for n={n}"
    print(f"n={n}: ratio = {ratio[0]:.6f} (constant across all b)")
```

The ratio is constant for every sample size, confirming that the W-matrix correctly converts neutral coalescence times into the $1/b$ spectrum.

Solution 2: Population expansion and the SFS

We model a 10-fold expansion at time $T = 0.1 \times 2N_0$. Before the expansion (going backward), the population had size N_0 ; after, it has size $10N_0$. We compute expected coalescence times in each epoch and combine them.

```
import numpy as np

n = 30
N0 = 1000
N_new = 10 * N0
T = 0.1 * 2 * N0 # expansion time in generations

j_vals = np.arange(2, n + 1)
rate = j_vals * (j_vals - 1) / 2.0

# Recent epoch (size N_new, duration T)
t_recent = 2.0 * T / N_new
E_Tjj_recent = (1.0 - np.exp(-rate * t_recent)) / rate

# Ancestral epoch (size N0, infinite duration)
# Probability of j lineages surviving the recent epoch
p_survive = np.exp(-rate * t_recent)
E_Tjj_ancient = p_survive / rate # remaining expected time in ancestral epoch

# Scale ancestral epoch by ratio of population sizes
# (the rate matrix is the same but time runs at different speed)
E_Tjj_total = E_Tjj_recent + E_Tjj_ancient

W = w_matrix(n)
sfs_expansion = W @ E_Tjj_total

# Neutral expectation for comparison
E_Tjj_neutral = 2.0 / (j_vals * (j_vals - 1))
sfs_neutral = W @ E_Tjj_neutral

bb = np.arange(1, n)
ratio = sfs_expansion / sfs_neutral
print("SFS ratio (expansion / neutral):")
for b in range(len(bb)):
    print(f"  b={bb[b]}: {ratio[b]:.4f}")
```

The expansion creates an **excess of rare variants** (low-frequency entries, small b). This is because the large recent population size suppresses coalescence, leaving many young lineages that accumulate singleton and doubleton mutations. High-frequency entries are relatively depleted.

i Solution 3: Bottleneck signature

A bottleneck is modeled as three epochs: recent (size N_0), bottleneck (size $0.1N_0$, duration $0.05 \times 2N_0$), and ancestral (size N_0).

```
import numpy as np

n = 30
N0 = 1000
N_bot = 0.1 * N0
T_bot_start = 0.2 * 2 * N0 # bottleneck starts (going backward)
T_bot_dur = 0.05 * 2 * N0 # bottleneck duration

j_vals = np.arange(2, n + 1)
rate = j_vals * (j_vals - 1) / 2.0

# Epoch 1: recent (size N0, duration T_bot_start)
t1 = 2.0 * T_bot_start / N0
E_Tjj_1 = (1.0 - np.exp(-rate * t1)) / rate
p_surv_1 = np.exp(-rate * t1)

# Epoch 2: bottleneck (size N_bot, duration T_bot_dur)
t2 = 2.0 * T_bot_dur / N_bot
E_Tjj_2 = p_surv_1 * (1.0 - np.exp(-rate * t2)) / rate
p_surv_2 = p_surv_1 * np.exp(-rate * t2)

# Epoch 3: ancestral (size N0, infinite)
E_Tjj_3 = p_surv_2 / rate

E_Tjj_total = E_Tjj_1 + E_Tjj_2 + E_Tjj_3

W = w_matrix(n)
sfs_bottleneck = W @ E_Tjj_total

# Neutral expectation
E_Tjj_neutral = 2.0 / (j_vals * (j_vals - 1))
sfs_neutral = W @ E_Tjj_neutral

# Simple size change (permanently smaller, no recovery)
t_simple = 2.0 * T_bot_start / N0
E_Tjj_simple_1 = (1.0 - np.exp(-rate * t_simple)) / rate
p_surv_simple = np.exp(-rate * t_simple)
E_Tjj_simple_2 = p_surv_simple / rate * (N0 / N_bot)
sfs_simple = W @ (E_Tjj_simple_1 + E_Tjj_simple_2)

bb = np.arange(1, n)
print("Ratio to neutral (bottleneck vs simple size change):")
for b in range(min(10, len(bb))):
    print(f"  b={bb[b]}: bottleneck={sfs_bottleneck[b]/sfs_neutral[b]:.4f}, "
          f"simple={sfs_simple[b]/sfs_neutral[b]:.4f}")
```

Unlike a simple permanent size change, the bottleneck produces a characteristic **U-shaped distortion**: an excess of both rare and common variants (compared to neutral) with a deficit at intermediate frequencies. The brief period of intense drift forces rapid coalescence of many lineages, followed by recovery to normal drift rates. A permanent size

change, by contrast, shifts the entire spectrum monotonically.

Solution 4: Multi-population SFS dimensions

For k populations each of sample size n , the joint SFS has shape $(n+1)^k$, so the total number of entries is:

$$\text{entries} = (n+1)^k$$

```
import numpy as np

print("Joint SFS dimensions (n+1)^k:")
for k in [2, 3, 4, 5, 6]:
    for n in [10, 20, 50]:
        entries = (n + 1) ** k
        mem_gb = entries * 8 / 1e9 # 8 bytes per float64
        print(f"  k={k}, n={n}: {entries:.2e} entries ({mem_gb:.2f} GB)")
```

For $k=3, n=50$: $51^3 \approx 1.3 \times 10^5$ entries (manageable). For $k=5, n=50$: $51^5 \approx 3.5 \times 10^8$ entries (~2.7 GB). For $k=6, n=50$: $51^6 \approx 1.8 \times 10^{10}$ entries (~133 GB, impractical).

Grid-based methods like *dadi* are limited by this exponential scaling. *mom2* avoids it by **never constructing the full joint SFS tensor**. Instead, it processes the event tree one node at a time, only ever holding tensors with as many axes as the number of *currently active* populations at that point in the tree. At a merge event, two axes are replaced by one (via convolution), keeping the working tensor small. The cost scales as $O(\sum_{\text{events}} n_{\text{local}}^2)$ rather than $O((n+1)^k)$.

Next: *The Moran Model*

The Moran Model

The escapement of the watch: a discrete population model whose eigendecomposition lets us tell time exactly.

“The Moran model provides a tractable Markov chain whose eigensystem is known in closed form.”

---Kamm, Terhorst, Song, and Durbin (2017)

Biology Aside – Why a population model?

The central question of demographic inference is: *given DNA from living individuals, what was the population's history?* To answer this, we need a mathematical model that predicts how allele frequencies – the proportions of different genetic variants in the population – change over time. A variant that exists in 5% of chromosomes today may have been at 2% a thousand generations ago, or it may not have existed at all. The Moran model gives us a precise, tractable way to compute the probability of any such frequency trajectory. It is the mathematical engine inside *mom2* that converts a demographic scenario (population sizes, split times, migration rates) into a predicted site frequency spectrum that can be compared to real data.

Step 1: The Moran Model as a Continuous-Time Markov Chain

The **Moran model** describes how allele frequencies change in a finite population of size n . Unlike the Wright-Fisher model (which has discrete, non-overlapping generations), the Moran model operates in continuous time: at rate proportional to n , one individual is chosen to reproduce and one to die.

The state of the system is the number of derived alleles $i \in \{0, 1, \dots, n\}$. The transition rates are:

$$\begin{aligned} q(i, i+1) &= \frac{i(n-i)}{2} \quad (\text{one more derived copy}) \\ q(i, i-1) &= \frac{i(n-i)}{2} \quad (\text{one fewer derived copy}) \\ q(i, i) &= -i(n-i) \quad (\text{total departure rate}) \end{aligned}$$

The factor $i(n-i)/2$ counts the number of ways to pick one derived and one ancestral individual (divided by 2 because either could be the parent).

📌 Biology Aside – Derived and ancestral alleles

At any position in the genome, chromosomes carry one of two variants: the **ancestral** allele (inherited from a distant common ancestor of all samples) and the **derived** allele (introduced by a mutation at some point in the past). Knowing which allele is ancestral typically requires an outgroup species – for example, chimpanzee sequence is used to polarize human variants. The number of derived copies i in a sample of n chromosomes is what the site frequency spectrum (SFS) tabulates. The Moran model tracks exactly this quantity – how i fluctuates over evolutionary time due to the randomness of reproduction and death (genetic drift).

📌 Probability Aside – Moran vs. Wright-Fisher

The Moran model and the Wright-Fisher model are different discrete models that converge to the same diffusion limit as $n \rightarrow \infty$. The key advantage of the Moran model for `mom2` is that its rate matrix has a known eigendecomposition, which allows exact computation of transition probabilities for any time t . The Wright-Fisher model, being a discrete-time chain with binomial transitions, does not have such a clean eigensystem.

Step 2: The Rate Matrix

The rate matrix Q is an $(n+1) \times (n+1)$ tridiagonal matrix:

$$Q = \begin{pmatrix} 0 & 0 & & & \\ 0 \cdot n/2 & -0 \cdot n & 0 \cdot n/2 & & \\ & 1(n-1)/2 & -1(n-1) & 1(n-1)/2 & \\ & & \ddots & \ddots & \ddots \\ & & & 0 & 0 \end{pmatrix}$$

States 0 (all ancestral) and n (all derived) are **absorbing** – their rows are all zeros. This makes sense: once an allele fixes or is lost, there is no further change.

📌 Biology Aside – Fixation and loss

In a real population, most new mutations are eventually **lost** – they drift to zero copies and disappear. A small fraction reach **fixation** – every chromosome carries the derived allele, and the variant becomes invisible to the SFS (it no longer varies). The absorbing boundaries at states 0 and n capture exactly this: lineages that leave the polymorphic frequency range never return. Only the transient frequencies $1, 2, \dots, n-1$ contribute to observable genetic variation. The rate at which alleles transit through these intermediate frequencies depends on the population size history – which is precisely what demographic inference aims to recover.

```
import numpy as np
from scipy import sparse

def moran_rate_matrix(n):
    """Construct the Moran model rate matrix for sample size n.

    Returns a (n+1) x (n+1) tridiagonal matrix Q where:
    - Q[i, i+1] = i*(n-i)/2    (gain one derived copy)
    - Q[i, i-1] = i*(n-i)/2    (lose one derived copy)
    - Q[i, i]   = -i*(n-i)     (total departure rate)
    """
    i = np.arange(n + 1, dtype=float)
    off_diag = i * (n - i) / 2.0
    diag = -2.0 * off_diag
    Q = (np.diag(off_diag[:-1], k=1)
        + np.diag(diag, k=0)
        + np.diag(off_diag[1:], k=-1))
    return Q
```

```
# Verification: rows sum to zero (a property of all rate matrices)
n = 10
Q = moran_rate_matrix(n)
row_sums = Q.sum(axis=1)
assert np.allclose(row_sums, 0), f"Row sums: {row_sums}"

# Verification: Q is symmetric (detailed balance for the Moran model)
assert np.allclose(Q, Q.T), "Rate matrix should be symmetric"
```

Step 3: Eigendecomposition

Because Q is a real symmetric tridiagonal matrix, it has a complete set of real eigenvalues and orthogonal eigenvectors:

$$Q = V\Lambda V^{-1}$$

where $\Lambda = \text{diag}(\lambda_0, \lambda_1, \dots, \lambda_n)$ and V is the matrix of eigenvectors.

The eigenvalues of the Moran rate matrix are:

$$\lambda_j = -\frac{j(j-1)}{2}, \quad j = 0, 1, \dots, n$$

Where does this formula come from? Recall from the coalescent that $\binom{j}{2} = j(j-1)/2$ is the rate at which j lineages coalesce. This is not a coincidence: the Moran model and the coalescent describe the same process from different perspectives (forward vs. backward in time). The eigenvalue λ_j captures the rate at which the j -th mode of genetic variation decays – and this rate equals the coalescence rate for j lineages.

Concretely:

- $\lambda_0 = 0$ and $\lambda_1 = 0$: these correspond to the two absorbing states (allele fixed at 0 or at n). Once the allele is fixed or lost, nothing changes – hence zero decay rate.
- $\lambda_2 = -1$: the slowest-decaying transient mode. This corresponds to the last pair of lineages coalescing, which happens at rate $\binom{2}{2} = 1$.
- $\lambda_n = -n(n-1)/2$: the fastest-decaying mode. This corresponds to all n lineages present, coalescing at rate $\binom{n}{2}$.

```
import numpy as np

n = 10 # population size
eigenvalues = [-j*(j-1)/2 for j in range(n+1)]
print("Eigenvalues of the Moran rate matrix (n=10):")
for j, lam in enumerate(eigenvalues):
    label = " (absorbing)" if lam == 0 else ""
    print(f" j={j:2d}: lambda = {lam:8.1f}{label}")
```

i Plain-language summary – What eigendecomposition gives us

An eigendecomposition breaks a complicated matrix into simpler components, much like splitting white light into individual colours with a prism. Each eigenvalue λ_j controls one “mode” of the system’s behaviour:

- **Modes with $\lambda = 0$:** These never decay. They correspond to the allele being permanently fixed or lost – the absorbing states.
- **Modes with large negative λ :** These decay quickly. They correspond to fast transient fluctuations that die out early.
- **Modes with small negative λ :** These decay slowly. They correspond to the long-lived genetic variation that persists over many generations.

The practical payoff is enormous: to compute how allele frequencies change over a time span t , we simply multiply each mode by $e^{\lambda_j t}$. Modes with large negative eigenvalues are exponentially damped (ancient variation is lost), while the $\lambda = 0$ modes persist forever. This gives us an exact, closed-form solution instead of requiring iterative numerical simulation.

```
def moran_eigensystem(n):
    """Compute the eigendecomposition of the Moran rate matrix.

    Returns (V, eigenvalues, V_inv) where Q = V @ diag(eigenvalues) @ V_inv.
    """
    Q = moran_rate_matrix(n)
    eigenvalues, V = np.linalg.eigh(Q) # eigh for symmetric matrices
    V_inv = np.linalg.inv(V)
    return V, eigenvalues, V_inv
```

```
# Verification: eigenvalues match the theoretical formula
n = 10
V, eigs, V_inv = moran_eigensystem(n)
j = np.arange(n + 1)
theoretical_eigs = -j * (j - 1) / 2.0
# sort both to compare
assert np.allclose(np.sort(eigs), np.sort(theoretical_eigs), atol=1e-10)

# Verification: reconstruct Q from eigensystem
Q_reconstructed = V @ np.diag(eigs) @ V_inv
Q_original = moran_rate_matrix(n)
assert np.allclose(Q_reconstructed, Q_original, atol=1e-10)
```

Step 4: The Transition Matrix

The transition probability matrix for time t is the matrix exponential:

$$P(t) = e^{Qt} = V e^{\Lambda t} V^{-1} = V \operatorname{diag}(e^{\lambda_0 t}, e^{\lambda_1 t}, \dots, e^{\lambda_n t}) V^{-1}$$

Entry $P_{ij}(t)$ gives the probability that the number of derived alleles transitions from i to j in time t .

This is the core computational advantage of the Moran model: instead of numerically integrating an ODE system (as `moments` does) or discretizing a PDE on a grid (as `dadi` does), `mom2` computes exact transition probabilities with a single matrix multiplication. The eigendecomposition is computed once per sample size and cached; applying it for different times t requires only exponentiating the eigenvalues.

Biology Aside – What the transition matrix means biologically

Each entry $P_{ij}(t)$ answers a concrete biological question: *if a variant is currently present in i out of n chromosomes, what is the probability that it will be present in j chromosomes after t units of evolutionary time?* When the population is small, drift is strong and alleles rapidly fix or are lost – the matrix concentrates probability at the edges ($j = 0$ or $j = n$). When the population is large, drift is weak and alleles linger at intermediate frequencies – the matrix is more diffuse. By computing $P(t)$ for different values of t (corresponding to different epoch durations in a demographic model), `mom2` can predict how genetic variation is shaped by any sequence of population size changes, splits, and bottlenecks.

```
def moran_transition(t, n):
    """Compute the Moran transition matrix  $P(t) = \exp(Q \cdot t)$ .

    t: time (in Moran model units, scaled by 2/N)
    n: sample size

    Returns (n+1) x (n+1) transition probability matrix.
    """
    V, eigs, V_inv = moran_eigensystem(n)
    D = np.diag(np.exp(t * eigs))
    P = V @ D @ V_inv
    # clamp small numerical errors
    P = np.clip(P, 0, None)
    P = P / P.sum(axis=1, keepdims=True) # normalize rows
    return P
```

```
# Verification: P(0) is the identity matrix
n = 10
P0 = moran_transition(0, n)
assert np.allclose(P0, np.eye(n + 1), atol=1e-10)

# Verification: rows of P(t) sum to 1
P1 = moran_transition(1.0, n)
assert np.allclose(P1.sum(axis=1), 1.0, atol=1e-10)

# Verification: all entries non-negative
assert np.all(P1 >= -1e-15)

# Verification: P(s+t) = P(s) @ P(t) (Chapman-Kolmogorov)
P_s = moran_transition(0.5, n)
```

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```
P_t = moran_transition(0.3, n)
P_st = moran_transition(0.8, n)
assert np.allclose(P_s @ P_t, P_st, atol=1e-8)
```

Step 5: Connecting to the Coalescent

How does the Moran transition matrix relate to the coalescent? The connection is through **lineage counting**.

Biology Aside – The coalescent, in brief

The coalescent is a way of thinking about genetic ancestry **backward in time**. Start with n sampled chromosomes today and trace their lineages into the past. Occasionally two lineages merge (“coalesce”) when they find a common ancestor. The rate of coalescence depends on the population size: in a small population, lineages bump into each other quickly; in a large population, it takes longer. By connecting the Moran model to the coalescent, `mom2` can translate a forward-in-time model of allele frequency change into the backward-in-time framework used to compute the expected SFS. This bridge is what makes the tensor machinery in the next chapter possible.

Consider n sampled chromosomes in a population of size N . Going backward in time, we track how many distinct lineages remain. The Moran model (running backward) describes how the configuration of alleles in the sample evolves as we trace back through the population history.

Specifically, the transition matrix $P(t)$ applied to a likelihood vector $\mathbf{v}$ of length $n + 1$ computes:

$$[\mathbf{v} \cdot P(t)]_i = \sum_j v_j P_{ji}(t)$$

This transforms the likelihood of observing a given allele configuration at the bottom of an epoch to the likelihood at the top, accounting for all possible coalescence and drift events that could have occurred during the epoch.

The **time scaling** is crucial. Real time T (in generations) maps to Moran model time via:

$$t = \int_0^T \frac{2}{N(s)} ds$$

For constant size: $t = 2T/N$. For exponential growth with rate g and bottom size N_0 :

$$t = \frac{2}{N_0} \int_0^T e^{gs} ds = \frac{2(e^{gT} - 1)}{N_0 g}$$

Plain-language summary – Why time must be scaled

Genetic drift runs on a “molecular clock” whose speed depends on population size. In a population of 1,000 individuals, 100 generations of drift produce the same amount of frequency change as 10,000 generations in a population of 100,000. The integral above converts real time (generations) into the Moran model’s intrinsic time units, which account for varying population size. When the population is small, the integral accumulates quickly (drift is fast); when it is large, the integral accumulates slowly (drift is slow). This is the reason that demographic events like bottlenecks leave such strong signatures in the SFS: a brief period of small population size corresponds to a large “jump” in drift-scaled time.

Step 6: Applying Moran Transitions to Tensors

In `mom2`'s tensor framework, the Moran transition is applied to a **specific axis** of a multi-dimensional likelihood tensor. If the tensor has one axis per population, and we need to apply the Moran transition for population k , we multiply along axis k :

$$L'_{i_1, \dots, i_k, \dots, i_D} = \sum_{j_k} L_{i_1, \dots, j_k, \dots, i_D} P_{j_k, i_k}(t)$$

This is an `einsum` operation, which `mom2` implements via optimized batched matrix multiplication:

```
def moran_action(t, tensor, axis):
    """Apply Moran transition matrix to a tensor along a given axis.

    t: scaled time for this epoch
    tensor: multi-dimensional likelihood tensor
    axis: which axis (population) to apply the transition to

    Returns tensor with the Moran transition applied.
    """
    n = tensor.shape[axis] - 1
    P = moran_transition(t, n)
    # einsum: contract axis of tensor with P
    return np.tensordot(tensor, P.T, axes=([axis], [0]))
```

Calculus Aside – Why eigendecomposition beats ODE integration

`moments` computes the SFS by integrating a system of ODEs forward in time: $d\phi/dt = A\phi$. For each epoch, this requires calling an ODE solver (like RK45), which takes many small steps. `mom2` instead computes $\phi(t) = e^{At}\phi(0)$ directly via the eigensystem. Since the eigensystem is computed once and cached, each epoch requires only $O(n^2)$ work for the matrix-vector product. This makes `mom2` particularly efficient when the same sample size appears in many epochs.

Step 7: The Moran Model in `mom2`'s Source Code

In the actual `mom2` implementation, the Moran eigensystem is memoized (cached by sample size n) and the transition matrix computation handles edge cases for numerical stability:

```
# Simplified from mom2/moran_model.py

from functools import lru_cache

@lru_cache(maxsize=None)
def cached_eigensystem(n):
    """Memoized eigendecomposition -- computed once per sample size."""
    Q = moran_rate_matrix(n)
    eigenvalues, V = np.linalg.eig(Q)
    V_inv = np.linalg.inv(V)
    return V, eigenvalues, V_inv

def transition_with_checks(t, n):
    """Transition matrix with probability validation."""
    V, eigs, V_inv = cached_eigensystem(n)
```

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```

D = np.diag(np.exp(t * eigs))
P = V @ D @ V_inv
# numerical cleanup: clamp negatives, normalize rows
P = np.maximum(P, 0)
row_sums = P.sum(axis=1, keepdims=True)
P = P / row_sums
return P

```

The memoization is critical for performance: in a typical inference run, the optimizer evaluates hundreds of parameter combinations, but the sample sizes remain fixed. The eigensystem is computed once; only the time argument t changes between evaluations.

Exercises

Exercise 1: Eigenvalue verification

Compute the eigenvalues of the Moran rate matrix for $n = 5, 10, 20$ and verify they match the formula $\lambda_j = -j(j-1)/2$.

Exercise 2: Stationary distribution

Show that the Moran model has two absorbing states (0 and n). For a large time t , what does $P(t)$ converge to? What is the probability of fixation starting from i derived alleles?

Exercise 3: Chapman-Kolmogorov

Verify the semigroup property $P(s+t) = P(s)P(t)$ numerically for several values of s, t and n . Why does this property make the eigendecomposition approach so natural?

Exercise 4: Time scaling

For a population that grows exponentially from $N_0 = 1000$ to $N_T = 10000$ over $T = 500$ generations, compute the scaled Moran time t . How does this compare to the constant-size case $t = 2T/N_0$?

Solutions

Solution 1: Eigenvalue verification

The theoretical eigenvalues are $\lambda_j = -j(j-1)/2$ for $j = 0, 1, \dots, n$.

```

import numpy as np

for n in [5, 10, 20]:
    Q = moran_rate_matrix(n)
    numerical_eigs = np.sort(np.linalg.eigvalsh(Q))

```

```

j = np.arange(n + 1)
theoretical_eigs = np.sort(-j * (j - 1) / 2.0)

assert np.allclose(numerical_eigs, theoretical_eigs, atol=1e-10), \
    f"Mismatch for n={n}"
print(f"n={n}: eigenvalues match")
print(f"  Theoretical: {theoretical_eigs[:6]} ...")
print(f"  Numerical:   {numerical_eigs[:6]} ...")

```

For every n , the numerically computed eigenvalues agree with the closed-form formula to machine precision. Note the two zero eigenvalues ($j = 0$ and $j = 1$), corresponding to the absorbing states.

i Solution 2: Stationary distribution

The two absorbing states are $i = 0$ and $i = n$, since $q(0, \cdot) = 0$ and $q(n, \cdot) = 0$ (the departure rates vanish). For large t , all probability mass concentrates on these absorbing states.

The fixation probability starting from i derived alleles is i/n by symmetry of the Moran model (the up and down rates are equal).

```

import numpy as np

n = 10
t_large = 100.0 # large time

P = moran_transition(t_large, n)

# For large t, P[i, :] should have mass only at 0 and n
for i in range(n + 1):
    interior_mass = P[i, 1:n].sum()
    assert interior_mass < 1e-6, \
        f"Interior mass at state {i}: {interior_mass}"

    # Fixation probability: P[i, n] should be approximately i/n
    fix_prob = P[i, n]
    expected_fix = i / n
    print(f"  i={i}: P(fixation) = {fix_prob:.6f}, "
          f"expected = {expected_fix:.6f}")
    if 0 < i < n:
        assert abs(fix_prob - expected_fix) < 1e-4

```

$$\lim_{t \rightarrow \infty} P_{ij}(t) = \begin{cases} 1 - i/n & \text{if } j = 0 \\ i/n & \text{if } j = n \\ 0 & \text{otherwise} \end{cases}$$

This result follows from the martingale property of the Moran model: $E[X(t) \mid X(0) = i] = i$ for all t , and the only distributions on $\{0, n\}$ with mean i assign probability i/n to n and $1 - i/n$ to 0 .

i Solution 3: Chapman-Kolmogorov

The semigroup property $P(s+t) = P(s)P(t)$ follows directly from the eigendecomposition:

$$P(s)P(t) = Ve^{\Lambda s}V^{-1}Ve^{\Lambda t}V^{-1} = Ve^{\Lambda(s+t)}V^{-1} = P(s+t)$$

```
import numpy as np

for n in [5, 10, 15]:
    for s, t in [(0.1, 0.2), (0.5, 0.3), (1.0, 1.0), (0.01, 0.99)]:
        P_s = moran_transition(s, n)
        P_t = moran_transition(t, n)
        P_st = moran_transition(s + t, n)
        P_product = P_s @ P_t

        max_err = np.max(np.abs(P_product - P_st))
        assert max_err < 1e-8, \
            f"n={n}, s={s}, t={t}: max error = {max_err}"
        print(f"  n={n}, s={s}, t={t}: max error = {max_err:.2e}")
```

The eigendecomposition makes this natural because exponentiating diagonal matrices is trivial: $e^{\Lambda s}e^{\Lambda t} = e^{\Lambda(s+t)}$. The intermediate $V^{-1}V = I$ cancels. This is precisely why the eigendecomposition is the method of choice: it converts matrix exponentiation (an expensive operation) into scalar exponentiation (trivial), and the semigroup property is automatically satisfied.

Solution 4: Time scaling

For exponential growth from $N_0 = 1000$ to $N_T = 10000$ over $T = 500$ generations, the growth rate is:

$$g = \frac{\ln(N_T/N_0)}{T} = \frac{\ln(10)}{500} \approx 0.004605$$

The scaled Moran time is:

$$t = \frac{2}{N_0} \int_0^T e^{gs} ds = \frac{2(e^{gT} - 1)}{N_0 g}$$

```
import numpy as np

N0 = 1000
NT = 10000
T = 500

g = np.log(NT / N0) / T
print(f"Growth rate g = {g:.6f}")

# Scaled time under exponential growth
t_exp = 2.0 * (np.exp(g * T) - 1) / (N0 * g)
print(f"Scaled Moran time (exponential growth): t = {t_exp:.6f}")

# Scaled time under constant size N0
t_const = 2.0 * T / N0
print(f"Scaled Moran time (constant size N0): t = {t_const:.6f}")

# Ratio
print(f"Ratio t_exp / t_const = {t_exp / t_const:.4f}")
```

The exponential growth case gives $t_{\text{exp}} \approx 3.91$, while the constant-size case gives $t_{\text{const}} = 1.0$. The growing population accumulates less drift-scaled time because drift is weaker when the population is large. The ratio $t_{\text{exp}}/t_{\text{const}} \approx 3.91$ reflects the harmonic-mean-like weighting of population sizes: most of the integral is dominated by the early period (backward in time) when the population is small (N_0), but the large recent size (N_T) significantly slows the accumulation of drift during the recent epoch.

Next: *Tensor Machinery*

Tensor Machinery

The gear train of the watch: assembling the expected SFS from a product of tensors, one demographic event at a time.

“The expected SFS can be written as a tensor product over the events in the demographic history.”

---Kamm, Terhorst, Song, and Durbin (2017)

Biology Aside – From populations to tensors

Imagine you have sequenced individuals from three human populations – say, Yoruba, Han Chinese, and French. At each polymorphic site in the genome, you can count how many derived alleles appear in each sample: perhaps 3 out of 20 Yoruba chromosomes, 7 out of 20 Han Chinese chromosomes, and 5 out of 20 French chromosomes. The three-dimensional table that tallies all such combinations across the genome is the **joint site frequency spectrum** (joint SFS). It is the multi-population generalization of the SFS introduced in the moments timepiece.

The likelihood tensor is `momi2`'s internal representation for computing the *expected* joint SFS under a demographic model. Each axis of the tensor corresponds to one population's allele count. The tensor machinery described in this chapter shows how to build this expected SFS piece by piece, by walking backward through the evolutionary history of the populations – from present-day samples back to the common ancestral population.

Step 1: What Is a Likelihood Tensor?

At the heart of `momi2`'s computation is the **likelihood tensor** – a multi-dimensional array that tracks the probability of observing a given allele configuration across populations.

For a single population with sample size n , the likelihood tensor is a vector L of length $n + 1$, where L_i represents the probability weight associated with i derived alleles in the sample.

For k populations with sample sizes $n_1, \dots, n_k$, the likelihood tensor is a k -dimensional array of shape $(n_1 + 1) \times \dots \times (n_k + 1)$.

The tensor is initialized at the **leaves** of the demographic event tree (the sampled populations at present) and transformed by each event as we traverse the tree backward in time toward the root (the ancestral population).

```
import numpy as np

def initialize_leaf_tensor(n):
    """Initialize the likelihood tensor for a leaf (sampled population).

    For computing the full SFS, the initial tensor is the identity:
    L_i = delta_{i, config[pop]}, iterated over all configurations.

    For a single configuration (b_1, ..., b_k), population p
    gets an indicator vector: L[b_p] = 1, all others 0.
```

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```

"""
# For the full expected SFS computation, momi2 uses a batch
# approach: all configurations are processed simultaneously.
# Here we show the single-config case for clarity.
L = np.zeros(n + 1)
return L # set L[b] = 1 for the desired allele count b

```

Step 2: The Three Core Operations

Every computation in `momi2` is built from three tensor operations:

1. Matrix multiplication (Moran transition)

When the event tree passes through an epoch of duration t for population k , the likelihood tensor is multiplied along axis k by the Moran transition matrix $P(t)$:

$$L'_{i_1, \dots, i_k, \dots} = \sum_{j_k} L_{i_1, \dots, j_k, \dots} P_{j_k, i_k}(t)$$

This accounts for all possible drift and coalescence events during the epoch.

Plain-language summary – Matrix multiplication on tensors

Think of the tensor as a multi-dimensional spreadsheet where each cell holds the probability of a particular allele configuration. When time passes in one population (while the others are frozen), allele frequencies in that population change due to drift. The matrix multiplication “smears” the probabilities along that population’s axis, reflecting the fact that a configuration with 5 derived alleles might evolve into one with 4, 5, or 6 alleles. The Moran transition matrix (from the previous chapter) tells us exactly how much probability flows between each pair of counts.

```

def apply_moran_transition(tensor, axis, t, n):
    """Apply Moran transition to a specific axis of the tensor."""
    P = moran_transition(t, n) # (n+1) x (n+1) matrix
    # move the target axis to the last position, multiply, move back
    tensor = np.moveaxis(tensor, axis, -1)
    tensor = tensor @ P
    tensor = np.moveaxis(tensor, -1, axis)
    return tensor

```

2. Convolution (population merge)

When two populations merge into one (a split event, viewed backward), the lineages from both child populations enter a single ancestral population. The likelihood tensors from the two children are combined by **convolution**: the number of derived alleles in the ancestor is the sum of derived alleles from each child.

Biology Aside – Population splits as merges

In demographic models, we describe events forward in time: an ancestral population **splits** into two daughter populations that then evolve independently (for example, the human-Neanderthal divergence). But `momi2` computes the SFS by tracing lineages **backward** in time – from present-day samples toward the ancestral population. Viewed backward, a population split becomes a **merge**: the lineages from the two daughter populations enter the same ancestral gene pool. The convolution operation reflects this – if 3 derived alleles came from population A and 4 from population B, the ancestor had 7 derived alleles. All possible combinations must be summed over, weighted by their probabilities.

For child populations with n_1 and n_2 lineages, the naive convolution sums over all ways to partition i derived alleles between the two children:

$$L_i^{\text{anc}} = \sum_{j+k=i} L_j^{(1)} \cdot L_k^{(2)}$$

However, this simple sum does not account for the **combinatorics of sampling**. The probability that the ancestor has i derived alleles out of $n_1 + n_2$ total, given that child 1 contributes j out of n_1 and child 2 contributes k out of n_2 , involves the hypergeometric distribution. The correct formula weights each partition by the number of ways to arrange the alleles:

$$L_i^{\text{anc}} = \frac{1}{\binom{n_1+n_2}{i}} \sum_{j+k=i} \binom{n_1}{j} L_j^{(1)} \cdot \binom{n_2}{k} L_k^{(2)}$$

Where do the binomial coefficients come from? The term $\binom{n_1}{j}$ counts how many ways j of the n_1 lineages from child 1 can carry the derived allele, and $\binom{n_2}{k}$ does the same for child 2. Their product $\binom{n_1}{j} \binom{n_2}{k}$ is the number of ways to assign derived alleles across both children such that $j + k = i$. Dividing by $\binom{n_1+n_2}{i}$ – the total number of ways to choose i derived out of $n_1 + n_2$ – normalizes the result. This is exactly the hypergeometric probability:

$$P(j \text{ from child 1} \mid i \text{ total}) = \frac{\binom{n_1}{j} \binom{n_2}{i-j}}{\binom{n_1+n_2}{i}}$$

In practice, `momi2` first multiplies each tensor by binomial coefficients, convolves via polynomial multiplication, then divides out the ancestral binomial coefficients – which is algebraically equivalent but computationally efficient.

```
from scipy.special import comb

def convolve_populations(L1, L2, n1, n2):
    """Merge two population tensors via convolution.

    L1: likelihood vector for child population 1, length n1+1
    L2: likelihood vector for child population 2, length n2+1

    Returns: likelihood vector for ancestral population, length n1+n2+1
    """
    # weight by binomial coefficients
    b1 = np.array([comb(n1, j, exact=True) for j in range(n1 + 1)])
    b2 = np.array([comb(n2, k, exact=True) for k in range(n2 + 1)])
    weighted_L1 = L1 * b1
    weighted_L2 = L2 * b2

    # convolve (polynomial multiplication)
    conv = np.convolve(weighted_L1, weighted_L2)

    # divide out ancestral binomial coefficients
    n_anc = n1 + n2
    b_anc = np.array([comb(n_anc, i, exact=True) for i in range(n_anc + 1)])
    L_anc = conv / b_anc

    return L_anc

# Verification: convolving two uniform vectors should produce a valid
# probability distribution (after normalization)
n1, n2 = 5, 5
L1 = np.ones(n1 + 1) / (n1 + 1)
```

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```
L2 = np.ones(n2 + 1) / (n2 + 1)
L_anc = convolve_populations(L1, L2, n1, n2)
assert len(L_anc) == n1 + n2 + 1
```

3. Antidiagonal summation (coalescence within an epoch)

During each epoch, coalescence events reduce the number of lineages. The contribution of each epoch to the SFS is accumulated by contracting the likelihood tensor with the **truncated SFS** of that epoch (computed from the expected coalescence times $E[T_{jj}]$ and the W-matrix):

$$\text{sfs_contribution} = \sum_i L_i \cdot \text{truncated_sfs}_i$$

This is where the accumulated SFS gets its entries: each epoch contributes mutations that occurred during that epoch, weighted by the likelihood of observing them in the current allele configuration.

Step 3: The Junction Tree Algorithm

A demographic model is a tree (or DAG, when admixture is present) of events. `mom2` processes this tree using a **post-order traversal** (leaves to root), which corresponds to moving backward in time from the present to the ancestral population.

Biology Aside – The demographic event tree mirrors evolutionary history

The event tree is a direct representation of the evolutionary relationships among populations. The leaves are today's populations (sampled individuals from, say, Europe, Africa, and East Asia). Internal nodes correspond to historical events: population splits (divergence), size changes (bottlenecks, expansions), and admixture pulses (gene flow). The root represents the ancestral population from which all sampled populations ultimately descend. By processing this tree from leaves to root, `mom2` incrementally builds up the expected SFS, accumulating the contribution of mutations that fell on branches at each stage of the history.

```
Present (leaves)
|
| Pop A      Pop B
|
| Moran      Moran transition
| trans.    | (epoch 1)
|
| +-----+
|
| Merge (split event, backward)
|
| Moran transition
| (epoch 2)
|
| Root (ancestral population)
```

At each node in the tree:

1. **Leaf node:** Initialize the likelihood tensor
2. **Epoch boundary:** Apply the Moran transition matrix for the elapsed time
3. **Merge node:** Convolve the likelihood tensors of the two child populations

4. **Pulse node:** Apply the admixture tensor (see Step 4)
5. **Accumulate SFS:** Add the current epoch's contribution to the running SFS

```
import networkx as nx

def compute_expected_sfs(demography, sample_sizes):
    """Compute the expected SFS by traversing the event tree.

    Simplified sketch of momi2's algorithm.
    """
    # build the event tree from the demographic model
    event_tree = demography.event_tree

    # dictionary to hold likelihood tensors, keyed by population name
    tensors = {}
    sfs = 0.0 # accumulated expected SFS

    for event in nx.dfs_postorder_nodes(event_tree):
        if event.type == 'leaf':
            n = sample_sizes[event.pop]
            tensors[event.pop] = initialize_leaf_tensor(n)

        elif event.type == 'epoch':
            pop = event.pop
            t = event.scaled_time # 2*T/N or integral for exp growth
            n = tensors[pop].shape[-1] - 1
            # accumulate SFS contribution from this epoch
            trunc_sfs = compute_truncated_sfs(n, t, event.size_history)
            sfs = sfs + contract_with_truncated_sfs(tensors[pop], trunc_sfs)
            # apply Moran transition
            tensors[pop] = apply_moran_transition(
                tensors[pop], axis=-1, t=t, n=n)

        elif event.type == 'merge':
            child1, child2 = event.children
            parent = event.parent
            tensors[parent] = convolve_populations(
                tensors[child1], tensors[child2],
                n1=tensors[child1].shape[-1] - 1,
                n2=tensors[child2].shape[-1] - 1)
            del tensors[child1], tensors[child2]

        elif event.type == 'pulse':
            apply_admixture_tensor(tensors, event)

    return sfs
```

Step 4: Admixture (Pulse) Events

Admixture is the most complex operation. When a fraction f of population A 's ancestry comes from population B , each lineage in A independently “jumps” to B with probability f (viewed backward in time).

i Biology Aside – What admixture looks like in genomes

Admixture is the mixing of previously separated populations – for example, gene flow between Neanderthals and modern humans approximately 50,000 years ago left ~2% of Neanderthal DNA in non-African genomes today. Viewed backward in time, this means that for each chromosomal segment in a modern non-African individual, there is roughly a 2% chance that its lineage “jumps” from the modern human gene pool into the Neanderthal one at the time of admixture. The binomial 3-tensor below encodes exactly this stochastic assignment: given n lineages, each independently chooses one of two ancestral populations with probability f or $1 - f$.

If population A has n lineages, the number that move to B follows a binomial distribution. This is encoded as a **3-tensor** T of shape $(n + 1) \times (n + 1) \times (n + 1)$:

$$T_{i,j,k} = \binom{n}{k} \binom{k}{j} f^j (1 - f)^{k-j} \cdot \mathbf{1}[i + j = k]$$

where k is the original count in A , j lineages move to B , and $i = k - j$ remain in A .

```
from scipy.special import comb as binom

def admixture_tensor(n, f):
    """Compute the admixture 3-tensor for a pulse event.

    n: number of lineages in the receiving population
    f: fraction of ancestry from the source population

    Returns T of shape (n+1, n+1, n+1):
    T[i, j, k] = probability that k lineages split into i staying and j moving
    """
    T = np.zeros((n + 1, n + 1, n + 1))
    for k in range(n + 1):
        for j in range(k + 1):
            i = k - j
            T[i, j, k] = binom(k, j) * f**j * (1 - f)**(k - j)
    return T

def apply_admixture(L_A, L_B, n_A, n_B, f):
    """Apply an admixture pulse: fraction f of A's ancestry comes from B.

    Returns updated likelihood tensors for both populations.
    """
    T = admixture_tensor(n_A, f)
    # contract: new_L[i_A, i_B] = sum_k sum_j L_A[k] * L_B[i_B + j] * T[i_A, j, k]
    # (simplified -- actual implementation handles multi-dimensional tensors)
    return new_L_A, new_L_B
```

i Probability Aside – Why admixture requires a 3-tensor

In a population merge (split backward in time), every lineage from child A and every lineage from child B ends up in the same ancestral population. The operation is deterministic given the lineage counts. In admixture, however, each lineage *independently* chooses which source population to join. This stochastic splitting requires a full tensor to encode all possible binomial outcomes.

Step 5: Reducing Lineage Counts

As we move backward in time, the number of lineages in a population can only decrease (due to coalescence). After a merge event, the ancestral population has $n_1 + n_2$ lineages, which may be larger than needed for the computation.

`momi2` can optionally reduce the lineage count via the **hypergeometric quasi-inverse**: a matrix that projects from N lineages down to $n < N$ lineages in a way that preserves the expected SFS. This is the reverse of the projection (down-sampling) operation used in SFS analysis.

Plain-language summary – Reducing lineage counts

After two populations merge, the ancestral population suddenly has $n_1 + n_2$ lineages – which may be more than needed and would make subsequent computations expensive. The hypergeometric quasi-inverse is a principled way to “thin” the lineages down to a manageable number without distorting the expected SFS. Think of it as subsampling the ancestral chromosomes in a way that preserves the statistical properties we care about. The hypergeometric distribution appears here because it describes sampling without replacement from a finite pool – the same distribution that governs how allele counts change when you subsample a dataset.

```
def hypergeom_quasi_inverse(N, n):
    """Compute the quasi-inverse for reducing lineage count from N to n.

    Returns a (N+1) x (n+1) matrix M such that applying M to a likelihood
    vector of length N+1 produces a valid likelihood vector of length n+1,
    preserving the expected SFS.
    """
    from scipy.stats import hypergeom
    M = np.zeros((N + 1, n + 1))
    for i in range(N + 1):
        for j in range(n + 1):
            M[i, j] = hypergeom.pmf(j, N, i, n)
    return M
```

Step 6: Putting It All Together – A Two-Population Example

Let’s trace through the tensor computation for a simple two-population model: populations A and B diverged at time T from an ancestral population, with constant sizes throughout.

```
Time 0 (present):  Pop A (n_A=5)      Pop B (n_B=5)
                   |                  |
                   | Moran(t_1)       | Moran(t_1)
                   |                  |
Time T (split):    -----+-----+-----+-----
                   |
                   | Moran(t_2)
                   |
Time infinity:      Ancestral (n=10)
```

The computation proceeds:

```
def two_pop_divergence_sfs(n_A, n_B, T, N_A, N_B, N_anc):
    """Compute expected SFS for a simple two-population divergence model."""
    sfs = np.zeros((n_A + 1, n_B + 1))
```

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```

# Step 1: Initialize leaf tensors
# (using identity for full SFS computation)

# Step 2: Apply Moran transitions for the recent epoch
t_A = 2.0 * T / N_A # scaled time for pop A
t_B = 2.0 * T / N_B # scaled time for pop B

# Accumulate SFS contributions from the recent epoch
# (mutations falling on branches within this epoch)

# Step 3: At the split, convolve the two population tensors
# Ancestral population has n_A + n_B lineages

# Step 4: Apply Moran transition for the ancestral epoch
# (going back to infinity -- all lineages coalesce)

# Step 5: Accumulate SFS contributions from the ancestral epoch

return sfs

```

i Calculus Aside – Convolution as polynomial multiplication

The convolution operation for merging two populations is mathematically equivalent to multiplying two generating polynomials:

$$G_1(x) = \sum_j \binom{n_1}{j} L_j^{(1)} x^j, \quad G_2(x) = \sum_k \binom{n_2}{k} L_k^{(2)} x^k$$

The product $G_1(x) \cdot G_2(x)$ has coefficients that give the convolved tensor, after dividing by the ancestral binomial coefficients. `momi2` exploits this by using `numpy.convolve` for the polynomial multiplication, giving $O(n \log n)$ performance via FFT for large sample sizes.

Step 7: How `momi2` Handles the Tensor Computation

In the actual `momi2` implementation, the tensor computation is organized around the `LikelihoodTensor` and `LikelihoodTensorList` classes:

```

# Simplified from momi/compute_sfs.py

class LikelihoodTensor:
    """Stores a multi-dimensional likelihood tensor and accumulated SFS."""

    def __init__(self, liks, sfs=0.0):
        self.liks = liks # multi-dim array, one axis per population
        self.sfs = sfs # accumulated scalar SFS contribution

    def matmul_last_axis(self, matrix):
        """Apply a matrix to the last axis (Moran transition)."""
        self.liks = self.liks @ matrix

```

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```

def add_last_axis_sfs(self, truncated_sfs):
    """Accumulate SFS contribution from current epoch."""
    self.sfs = self.sfs + np.dot(self.likelihoods[...], truncated_sfs)

def mul_trailing_binoms(self, divide=False):
    """Multiply (or divide) by binomial coefficients for convolution."""
    n = self.likelihoods.shape[-1] - 1
    binoms = np.array([comb(n, j) for j in range(n + 1)])
    if divide:
        self.likelihoods = self.likelihoods / binoms
    else:
        self.likelihoods = self.likelihoods * binoms

def convolve_trailing_axes(self, other):
    """Convolve two likelihood tensors (population merge)."""
    # implemented via optimized Cython in momi/convolution.pyx
    self.likelihoods = convolve_sum_axes(self.likelihoods, other.likelihoods)

```

The key performance optimizations in the real implementation:

- **Cython convolution** (`convolution.pyx`): parallelized antidiagonal summation and convolution using OpenMP
- **Batched einsum** (`einsum2/`): custom batched matrix multiplication that processes all configurations simultaneously
- **Memoized eigensystems**: computed once per sample size, cached across all likelihood evaluations

Exercises

Exercise 1: Manual tensor computation

For a two-population model with $n_A = n_B = 3$ and a simple divergence at time T , manually trace the tensor operations. Start with indicator vectors at the leaves and apply each operation step by step.

Exercise 2: Convolution verification

Verify that `convolve_populations` correctly computes the merged likelihood for the case where both child populations have all derived alleles (i.e., $L_1 = [0, 0, \dots, 1]$ and $L_2 = [0, 0, \dots, 1]$).

Exercise 3: Admixture effects

For a pulse admixture event with fraction $f = 0.5$, compute the admixture tensor for $n = 4$ and verify that the expected number of lineages moving to the source population is $n \cdot f = 2$.

Exercise 4: Scaling with populations

Compare the computational complexity of computing the expected SFS for k populations using (a) a k -dimensional grid (dadi-style) and (b) the tensor-tree approach (momi2-style). For what values of k and n does the tensor approach

become advantageous?

Solutions

i Solution 1: Manual tensor computation

We trace through a two-population divergence with $n_A = n_B = 3$, computing the SFS for a single configuration (e.g., $b_A = 1, b_B = 2$).

```
import numpy as np

n_A, n_B = 3, 3
b_A, b_B = 1, 2

# Step 1: Initialize indicator vectors at the leaves
L_A = np.zeros(n_A + 1)
L_A[b_A] = 1.0 # L_A = [0, 1, 0, 0]

L_B = np.zeros(n_B + 1)
L_B[b_B] = 1.0 # L_B = [0, 0, 1, 0]

print(f"Leaf A: {L_A}")
print(f"Leaf B: {L_B}")

# Step 2: Apply Moran transitions for the recent epoch (time T)
T, N_A, N_B, N_anc = 500, 1000, 1000, 2000
t_A = 2.0 * T / N_A
t_B = 2.0 * T / N_B

P_A = moran_transition(t_A, n_A)
P_B = moran_transition(t_B, n_B)

L_A_after = L_A @ P_A # transform through recent epoch
L_B_after = L_B @ P_B
print(f"After Moran (A): {L_A_after}")
print(f"After Moran (B): {L_B_after}")

# Step 3: Convolve at the merge point
from scipy.special import comb

L_anc = convolve_populations(L_A_after, L_B_after, n_A, n_B)
n_anc = n_A + n_B
print(f"After merge (n_anc={n_anc}): {L_anc}")

# Step 4: Apply Moran transition for ancestral epoch (t -> infinity)
t_anc = 100.0 # large value approximating infinity
P_anc = moran_transition(t_anc, n_anc)
L_final = L_anc @ P_anc
print(f"After ancestral epoch: {L_final}")
```

The key insight is that each operation transforms the likelihood vector step by step: initialization sets the observed configuration, the Moran transition “smears” probabilities backward through drift, and the convolution combines

lineages from both populations. At the end, L_{final} gives the probability of the observed configuration under the model.

Solution 2: Convolution verification

When both child populations have all derived alleles ($L_1 = [0, \dots, 0, 1]$ and $L_2 = [0, \dots, 0, 1]$), the ancestor must also have all derived alleles: $L_{\text{anc}} = [0, \dots, 0, 1]$.

```
import numpy as np
from scipy.special import comb

for n1, n2 in [(3, 3), (5, 5), (3, 7), (10, 10)]:
    L1 = np.zeros(n1 + 1)
    L1[n1] = 1.0 # all derived

    L2 = np.zeros(n2 + 1)
    L2[n2] = 1.0 # all derived

    L_anc = convolve_populations(L1, L2, n1, n2)
    n_anc = n1 + n2

    # Expected: L_anc should be [0, 0, ..., 0, 1]
    expected = np.zeros(n_anc + 1)
    expected[n_anc] = 1.0

    assert np.allclose(L_anc, expected, atol=1e-10), \
        f"Failed for n1={n1}, n2={n2}: {L_anc}"
    print(f"n1={n1}, n2={n2}: L_anc = {L_anc} (correct)")
```

This works because the convolution formula sums over all ways to partition i derived alleles between the two children. When L_1 has mass only at n_1 and L_2 has mass only at n_2 , the only nonzero term in the sum is $j = n_1, k = n_2$, giving $i = n_1 + n_2 = n_{\text{anc}}$.

Similarly, one can verify the case where both populations have zero derived alleles:

```
L1 = np.zeros(n1 + 1)
L1[0] = 1.0
L2 = np.zeros(n2 + 1)
L2[0] = 1.0
L_anc = convolve_populations(L1, L2, n1, n2)
assert np.allclose(L_anc[0], 1.0) and np.allclose(L_anc[1:], 0.0)
```

Solution 3: Admixture effects

For $f = 0.5$ and $n = 4$, each lineage independently moves to the source with probability 0.5.

```
import numpy as np
from scipy.special import comb as binom

n = 4
f = 0.5
T = admixture_tensor(n, f)
```



```

print(f"Admixture tensor shape: {T.shape}")
print(f"T[i, j, k] = Pr(i stay, j move | k original)")
print()

# Verify: for each k, the tensor should be a valid probability distribution
for k in range(n + 1):
    total = 0.0
    for j in range(k + 1):
        i = k - j
        total += T[i, j, k]
        if T[i, j, k] > 1e-10:
            print(f"  k={k}: i={i}, j={j}, T={T[i, j, k]:.4f}")
    assert abs(total - 1.0) < 1e-10, f"k={k}: total = {total}"
    print(f"  k={k}: total = {total:.6f}")
    print()

# Expected number of lineages moving to source, starting from k lineages
for k in range(n + 1):
    E_j = sum(j * T[k - j, j, k] for j in range(k + 1))
    print(f"  k={k}: E[j] = {E_j:.4f}, expected = {k * f:.4f}")
    assert abs(E_j - k * f) < 1e-10

```

For $k = n = 4$ lineages, the expected number moving to the source is $n \cdot f = 4 \times 0.5 = 2$. The distribution of j (number moving) is Binomial(k, f):

$$E[j \mid k] = k \cdot f = 4 \times 0.5 = 2$$

$$\text{Var}(j \mid k) = k \cdot f(1 - f) = 4 \times 0.25 = 1$$

So with $f = 0.5$, we get maximum variance in the splitting – any outcome from 0 to 4 lineages moving is possible, with the binomial $\binom{4}{j}(0.5)^4$ giving probabilities 1/16, 4/16, 6/16, 4/16, 1/16.

i Solution 4: Scaling with populations

Grid-based (dadi-style): The SFS is represented on a k -dimensional grid of size M^k where $M \approx n$ is the grid resolution per dimension. The PDE solver must update all grid points at each time step.

$$\text{Cost}_{\text{grid}} = O(M^k \times T_{\text{steps}})$$

Tensor-tree (mom2-style): The computation visits each event in the tree once. At each event, the cost depends on the number of *active* populations (typically 1–3 at any point in the tree).

$$\text{Cost}_{\text{tensor}} = O\left(\sum_{\text{events}} \prod_{\text{active pops } p} (n_p + 1)\right)$$

```

import numpy as np

print("Comparison of computational cost:")
print(f"{'k':>3} {'n':>4} {'Grid (n^k)':>15} {'Tensor (tree)':>15} {'Ratio':>10}")
print("-" * 52)

for k in [2, 3, 4, 5, 6]:

```

```

for n in [10, 20, 50]:
    # Grid: n^k entries, each updated ~100 times
    grid_cost = (n + 1) ** k * 100

    # Tensor-tree: for a balanced binary tree of k leaves,
    # there are 2k-1 nodes. At each internal node, at most 2
    # populations are active. Merge cost ~ n^2, Moran cost ~ n^2.
    # After each merge, the lineage count is at most 2n, but
    # the quasi-inverse reduces it back to n.
    tree_nodes = 2 * k - 1
    tensor_cost = tree_nodes * (2 * n + 1) ** 2

    ratio = grid_cost / tensor_cost
    print(f"{k:>3} {n:>4} {grid_cost:>15,} {tensor_cost:>15,} {ratio:>10.1f}")

```

For $k = 2, n = 10$, the costs are similar (grid may even be faster due to simpler operations). But for $k \geq 4$, the tensor approach becomes dramatically better: the grid cost grows as n^k while the tensor cost grows as $k \cdot n^2$. At $k = 6, n = 50$, the grid requires $\sim 10^{12}$ operations versus $\sim 10^5$ for the tensor tree – a factor of 10^7 .

The crossover point is approximately $k = 3$: for two populations, grid methods are competitive; for three or more, the tensor approach is strongly preferred. This is why *mom2* was designed specifically for multi-population analyses.

Next: *Automatic Differentiation & Inference*

Automatic Differentiation & Inference

The mainspring of the watch: autograd-powered optimization turns the expected SFS into inferred history.

“Automatic differentiation provides exact gradients through the entire computation, enabling efficient optimization of complex demographic models.”

---Kamm, Terhorst, Song, and Durbin (2017)

Biology Aside – What demographic inference asks

Demographic inference is the process of reconstructing a population’s history from patterns in its DNA. The parameters being estimated – population sizes, divergence times, migration rates, admixture proportions – directly answer biological questions: *When did human populations diverge from each other? How large was the ancestral population? Was there gene flow after the split?* The inference machinery in this chapter takes the mathematical tools from the previous chapters (the Moran model and tensor computation) and uses them to search the space of possible histories for the one that best explains the observed genetic data.

Step 1: The Likelihood Function

Given an observed SFS $\mathbf{D}$ and a demographic model that predicts expected SFS $\mathbf{M}(\boldsymbol{\Theta})$, *mom2* computes the **composite log-likelihood** by treating each SNP as an independent draw from a multinomial distribution over allele configurations.

$$\ell(\boldsymbol{\Theta}) = \sum_{\text{configs } c} D_c \ln M_c(\boldsymbol{\Theta})$$

where D_c is the observed count of configuration c and $M_c(\boldsymbol{\Theta})$ is the expected proportion of SNPs with that configuration under the model.

Alternatively, `mom2` can use a **Poisson likelihood** that also fits the total number of SNPs:

$$\ell_{\text{Poisson}}(\Theta) = \sum_c [D_c \ln M_c(\Theta) - M_c(\Theta)]$$

```
import numpy as np

def multinomial_log_likelihood(observed_sfs, expected_sfs):
    """Composite log-likelihood under the multinomial model.

    observed_sfs: array of observed configuration counts
    expected_sfs: array of expected proportions (normalized to sum to 1)
    """
    # normalize expected SFS to probabilities
    expected_probs = expected_sfs / expected_sfs.sum()
    # avoid log(0) by masking zero-count entries
    mask = observed_sfs > 0
    ll = np.sum(observed_sfs[mask] * np.log(expected_probs[mask]))
    return ll

def poisson_log_likelihood(observed_sfs, expected_sfs):
    """Composite log-likelihood under the Poisson model.

    expected_sfs: array of expected counts (not normalized)
    """
    mask = observed_sfs > 0
    ll = np.sum(
        observed_sfs[mask] * np.log(expected_sfs[mask])
        - expected_sfs[mask]
    )
    return ll
```

i Probability Aside – Multinomial vs. Poisson likelihood

The multinomial likelihood conditions on the total number of SNPs $S = \sum_c D_c$, making it insensitive to the overall mutation rate θ . This is useful when θ is a nuisance parameter. The Poisson likelihood fits S as well, providing information about the absolute scale. In `mom2`, the default is multinomial (`normalized=True`), but the Poisson option is available for joint estimation of θ .

Step 2: Automatic Differentiation with autograd

The key innovation in `mom2`'s inference machinery is the use of **automatic differentiation** (AD) via the `autograd` library. Every numerical operation in the SFS computation – eigendecomposition, matrix exponentials, convolutions, tensor products – is traced by `autograd`, which can then compute exact gradients of the log-likelihood with respect to all demographic parameters.

$$\nabla_{\Theta} \ell = \left(\frac{\partial \ell}{\partial \Theta_1}, \frac{\partial \ell}{\partial \Theta_2}, \dots \right)$$

This is computed by **reverse-mode AD** (backpropagation): a single forward pass computes $\ell(\Theta)$, then a single backward pass computes all partial derivatives. The cost is roughly 2–3 times the forward pass, regardless of the number of parameters.

i Plain-language summary – What gradients tell the optimizer

The gradient is a vector that points in the direction of steepest improvement. It answers the question: *if I slightly increase the split time, or slightly decrease the ancestral population size, how much does the fit to the data improve or worsen?* An optimizer follows the gradient uphill (toward better fit) iteratively, adjusting all demographic parameters simultaneously. Automatic differentiation computes these sensitivities exactly and efficiently – without it, `mom2` would have to numerically perturb each parameter one at a time (finite differences), which is slower by a factor equal to the number of parameters.

```
import autograd
import autograd.numpy as np

def make_objective(observed_sfs, demographic_model_func):
    """Create a differentiable objective function.

    demographic_model_func: maps parameter vector -> DemographicModel
    """
    def objective(params):
        model = demographic_model_func(params)
        expected = compute_expected_sfs(model)
        return -multinomial_log_likelihood(observed_sfs, expected)

    # autograd computes the gradient automatically
    grad_objective = autograd.grad(objective)
    val_and_grad = autograd.value_and_grad(objective)

    return objective, grad_objective, val_and_grad
```

The advantages over alternative gradient methods:

Method	Accuracy	Cost per gradient	Implementation effort
Finite differences	$O(\epsilon)$ truncation error	$O(p)$ forward passes	Trivial
Hand-coded derivatives	Exact (if correct)	1 backward pass	Very high, error-prone
Automatic differentiation	Exact (to machine precision)	1 backward pass	None (autograd handles it)

i Calculus Aside – How autograd works

`autograd` replaces `numpy` operations with traced versions that record the computation graph. Each primitive operation (addition, multiplication, `exp`, `einsum`, etc.) has a registered **vector-Jacobian product** (VJP) rule. During the backward pass, `autograd` chains these VJPs using the chain rule to propagate gradients from the output back to the inputs.

For operations not natively supported (like Cython-accelerated convolutions or `scipy.special.expi`), `mom2` registers custom VJPs:

```

from autograd.extend import primitive, defvjp

@primitive
def convolve_sum_axes(A, B):
    # Cython implementation
    ...

defvjp(convolve_sum_axes,
       lambda ans, A, B: lambda g: transposed_convolve(g, B),
       lambda ans, A, B: lambda g: transposed_convolve(g.T, A))

```

Step 3: Deterministic Optimization

For small to medium problems, `mom2` uses **deterministic gradient-based optimization** via `scipy.optimize.minimize`. The `find_mle` method wraps this with autograd-computed gradients:

```

from scipy.optimize import minimize

def find_mle(objective, x0, method='tnc', bounds=None):
    """Find maximum likelihood estimates using deterministic optimization.

    method: 'tnc' (truncated Newton) or 'L-BFGS-B'
    bounds: parameter bounds (lower, upper) for each parameter
    """
    val_and_grad = autograd.value_and_grad(objective)

    def scipy_objective(x):
        val, grad = val_and_grad(x)
        return float(val), np.array(grad, dtype=float)

    result = minimize(
        scipy_objective,
        x0,
        method=method,
        jac=True, # we provide the gradient
        bounds=bounds,
    )
    return result.x, result.fun

```

The two recommended methods:

- **TNC** (Truncated Newton Conjugate-gradient): Uses Hessian-vector products (which autograd can also compute) for second-order convergence. Good for problems with 10–50 parameters.
- **L-BFGS-B**: A quasi-Newton method that approximates the Hessian from gradient history. Good for larger parameter spaces. Supports box constraints.

Step 4: Stochastic Optimization

For large datasets with many SNP configurations, evaluating the full likelihood at every step can be expensive. `mom2` supports **stochastic gradient methods** that subsample the SNPs:

ADAM – Adaptive moment estimation with bias correction:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \hat{m}_t &= m_t / (1 - \beta_1^t) \\ \hat{v}_t &= v_t / (1 - \beta_2^t) \\ \Theta_{t+1} &= \Theta_t - \alpha \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon) \end{aligned}$$

SVRG (Stochastic Variance Reduced Gradient) – Reduces the variance of stochastic gradients by periodically computing the full gradient and using it as a control variate:

$$g_t^{\text{SVRG}} = \nabla_t f_i(\Theta_t) - \nabla_t f_i(\bar{\Theta}) + \nabla f(\bar{\Theta})$$

where $\bar{\Theta}$ is a “pivot” at which the full gradient $\nabla f(\bar{\Theta})$ was computed.

```
def stochastic_find_mle(objective, x0, data, method='adam',
                       snps_per_minibatch=100, stepsize=0.01,
                       num_epochs=100):
    """Stochastic optimization for large datasets.

    Subsamples SNPs into minibatches for gradient estimation.
    """
    x = x0.copy()

    if method == 'adam':
        m = np.zeros_like(x)
        v = np.zeros_like(x)
        b1, b2, eps = 0.9, 0.999, 1e-8

        for epoch in range(num_epochs):
            for batch in make_minibatches(data, snps_per_minibatch):
                g = autograd.grad(objective)(x, batch)
                m = b1 * m + (1 - b1) * g
                v = b2 * v + (1 - b2) * g**2
                mhat = m / (1 - b1**(epoch + 1))
                vhat = v / (1 - b2**(epoch + 1))
                x = x - stepsize * mhat / (np.sqrt(vhat) + eps)
                x = np.clip(x, bounds[:, 0], bounds[:, 1])

    return x
```

i When to use stochastic methods

Deterministic methods (TNC, L-BFGS-B) are preferred for most problems because they converge more reliably. Use stochastic methods when:

- The dataset has millions of SNPs and full-gradient evaluation is slow
- You want to explore the likelihood surface broadly before refining
- The number of unique configurations is very large (many populations)

Step 5: Parameter Constraints and Transforms

Demographic parameters have natural constraints: population sizes must be positive, admixture fractions must be between 0 and 1, and split times must respect the temporal ordering of events.

`mom2` handles these through:

1. **Box constraints:** Passed directly to L-BFGS-B or TNC via the `bounds` argument
2. **Parameter transforms:** Working in log-space for sizes and times, or logit-space for fractions, so the optimizer works in an unconstrained space
3. **Temporal ordering:** Built into the demographic model specification, not the optimizer

```
def transform_params(params, param_types):
    """Transform parameters to unconstrained space.

    param_types: list of 'log' (positive), 'logit' (0-1), or 'none'
    """
    transformed = np.zeros_like(params)
    for i, (p, ptype) in enumerate(zip(params, param_types)):
        if ptype == 'log':
            transformed[i] = np.log(p)
        elif ptype == 'logit':
            transformed[i] = np.log(p / (1 - p))
        else:
            transformed[i] = p
    return transformed

def inverse_transform(transformed, param_types):
    """Transform back to natural parameter space."""
    params = np.zeros_like(transformed)
    for i, (t, ptype) in enumerate(zip(transformed, param_types)):
        if ptype == 'log':
            params[i] = np.exp(t)
        elif ptype == 'logit':
            params[i] = 1.0 / (1.0 + np.exp(-t))
        else:
            params[i] = t
    return params
```

Step 6: Uncertainty Quantification

After finding the MLE $\hat{\Theta}$, `mom2` can estimate uncertainty via:

The Hessian matrix. Autograd can compute the full Hessian of the log-likelihood:

$$H_{ij} = \frac{\partial^2 \ell}{\partial \Theta_i \partial \Theta_j} \Big|_{\hat{\Theta}}$$

The inverse Hessian gives the asymptotic covariance matrix, and $\sqrt{(-H^{-1})_{ii}}$ gives approximate standard errors.

```
def compute_standard_errors(objective, mle_params):
    """Compute approximate standard errors from the Hessian."""
    hessian_func = autograd.hessian(objective)
    H = hessian_func(mle_params)
    # negative Hessian of log-likelihood = observed Fisher information
    cov = np.linalg.inv(-H)
    se = np.sqrt(np.diag(cov))
    return se, cov
```

i Biology Aside – Why uncertainty matters in demographic inference

A single best-fit demographic model is of limited value without knowing how confident we should be in its parameters. Did the ancestral population have 15,000 individuals or could it plausibly have been 5,000 or 50,000? Confidence intervals on demographic parameters are essential for drawing biological conclusions – for example, distinguishing a bottleneck from a gradual decline, or determining whether an estimated admixture event is statistically significant. The Hessian and bootstrap methods below provide these confidence intervals.

Bootstrap. For more robust uncertainty estimates (especially when the composite likelihood assumption is violated due to linkage), `mom2` supports parametric and nonparametric bootstrap:

1. Resample SNPs (or blocks of SNPs) with replacement
2. Re-estimate parameters on each bootstrap sample
3. Use the distribution of bootstrap estimates for confidence intervals

i Probability Aside – Why the Hessian can mislead

The composite log-likelihood treats SNPs as independent, but linked SNPs are correlated. This means the Hessian-based standard errors are typically **too small** (overconfident). Block bootstrap, which resamples contiguous genomic regions, accounts for linkage and gives more realistic uncertainty estimates. This is analogous to the Godambe Information Matrix correction used in `moments` (see [Demographic Inference](#)).

Step 7: Goodness-of-Fit Statistics

Beyond the SFS itself, `mom2` can compute **f-statistics** and other summary statistics to assess model fit.

i Biology Aside – f-statistics as tests of evolutionary relationships

f-statistics were developed by population geneticists to test specific hypotheses about population history without fitting a full demographic model. They ask concrete questions:

- $f_2(A, B)$: *How much genetic drift separates populations A and B?* Large values mean deep divergence; small values mean recent shared ancestry or ongoing gene flow.
- $f_3(C; A, B)$: *Is population C a mixture of A and B?* A negative value is strong evidence of admixture – the classical test used to detect Neanderthal introgression into non-African humans.
- $f_4(A, B; C, D)$ and **Patterson's D**: *Did gene flow occur between non-sister populations?* The ABBA-BABA test is one of the most widely used tools in evolutionary genomics.

These statistics are linear functions of the SFS, so `mom2` computes them as part of its tensor machinery. If a fitted model predicts the right SFS but the wrong f-statistics, the model is missing important population relationships.

Statistic	Definition	What it measures
$f_2(A, B)$	$E[(p_A - p_B)^2]$	Genetic drift between A and B
$f_3(C; A, B)$	$E[(p_C - p_A)(p_C - p_B)]$	Whether C is admixed between A and B (negative = admixture)
$f_4(A, B; C, D)$	$E[(p_A - p_B)(p_C - p_D)]$	Shared drift / treeness test
Patterson's D	$\frac{ABBA - BABA}{ABBA + BABA}$	Gene flow between non-sister taxa

These are all **linear functions** of the SFS (see *The Coalescent SFS*, Step 6), so `mom2` computes them by passing the appropriate weight vectors through the tensor machinery. Autograd gradients are available, enabling optimization of demographic models to fit these statistics.

```
def f2_weights(n_A, n_B):
    """Weight vector for  $f_2(A, B) = E[(p_A - p_B)^2]$ .

    Returns a  $(n_A+1) \times (n_B+1)$  weight matrix.
    """
    p_A = np.arange(n_A + 1) / n_A
    p_B = np.arange(n_B + 1) / n_B
    #  $f_2 = E[(p_A - p_B)^2] = \text{sum over configs of } (i/n_A - j/n_B)^2 * \text{SFS}[i, j]$ 
    W = np.outer(p_A, np.ones(n_B + 1)) - np.outer(np.ones(n_A + 1), p_B)
    return W**2

def f3_weights(n_C, n_A, n_B):
    """Weight vector for  $f_3(C; A, B) = E[(p_C - p_A)(p_C - p_B)]$ .

    Negative f3 indicates admixture of C from A and B.
    """
    p_C = np.arange(n_C + 1) / n_C
    p_A = np.arange(n_A + 1) / n_A
    p_B = np.arange(n_B + 1) / n_B
    # 3-way outer product
    W = np.zeros((n_C + 1, n_A + 1, n_B + 1))
    for ic in range(n_C + 1):
        for ia in range(n_A + 1):
            for ib in range(n_B + 1):
                W[ic, ia, ib] = (p_C[ic] - p_A[ia]) * (p_C[ic] - p_B[ib])
    return W
```

Step 8: The Complete Inference Pipeline

Putting it all together, a typical `momi2` analysis follows this workflow:

```
# Step 1: Load data
import momi

sfs = momi.Sfs.load("observed_sfs.npz")

# Step 2: Define the demographic model
model = momi.DemographicModel(N_e=1e4, gen_time=25, muts_per_gen=1.25e-8)
model.add_leaf("EUR", N=1e4)
model.add_leaf("AFR", N=1e4)
model.set_size("EUR", t=0, N=1e5, g=0.01) # recent growth
model.move_lineages("EUR", "AFR", t=2000) # split at 2000 gen ago
model.set_size("AFR", t=2000, N=2e4)      # ancestral size

# Step 3: Set up optimization
model.add_size_param("N_EUR", "EUR")
model.add_time_param("T_split")
model.add_growth_param("g_EUR")

# Step 4: Fit the model
model.set_data(sfs)
result = model.optimize(method="TNC")

# Step 5: Get results
print("MLE parameters:", result.parameters)
print("Log-likelihood:", result.log_likelihood)

# Step 6: Uncertainty (via bootstrap or Hessian)
se = result.standard_errors()
print("Standard errors:", se)
```

Practical tips for momi2 inference

- **Start simple:** Begin with a model with few parameters and add complexity gradually. Check that each added parameter improves the fit.
- **Multiple starts:** Run the optimizer from several different starting points to avoid local optima.
- **Check f-statistics:** Even if the SFS fit looks good, compute f-statistics to verify the model captures the right population relationships.
- **Use block bootstrap:** Hessian-based standard errors underestimate uncertainty. Use genomic block bootstrap for publication-quality confidence intervals.

Exercises

Exercise 1: Gradient verification

For a simple two-population model, compute the gradient of the log-likelihood using (a) autograd, (b) finite differences with step size $\epsilon = 10^{-5}$. Verify they agree to several decimal places.

Exercise 2: Optimizer comparison

Fit a three-parameter model (ancestral size, derived size, split time) using TNC and L-BFGS-B. Compare the number of function evaluations and the final log-likelihood. Do they converge to the same optimum?

Exercise 3: Uncertainty calibration

Simulate 100 datasets under a known demographic model, fit each one, and check whether the 95% confidence intervals (from the Hessian) contain the true parameter values 95% of the time. How does block bootstrap compare?

Exercise 4: f-statistics as model diagnostics

Fit a tree model (no admixture) to data simulated under a model *with* admixture. Compute f_3 for the admixed population. Does the statistic detect the model misspecification?

Solutions

Solution 1: Gradient verification

We compare the autograd gradient to a finite-difference approximation for a simple two-population divergence model.

```
import autograd
import autograd.numpy as np

def simple_log_likelihood(params, observed_sfs):
    """Log-likelihood for a two-pop model: params = [N_A, N_B, T_split]."""
    N_A, N_B, T_split = params
    # Compute expected SFS under this model (using moran transitions)
    # (simplified: use the W-matrix approach for a single population
    # to illustrate the gradient comparison)
    n = len(observed_sfs) - 1
    j_vals = np.arange(2, n + 1)
    rate = j_vals * (j_vals - 1) / 2.0
    t = 2.0 * T_split / N_A
    E_Tjj = (1.0 - np.exp(-rate * t)) / rate + np.exp(-rate * t) / rate
    expected_sfs = np.abs(E_Tjj[:n-1]) # simplified
    expected_sfs = expected_sfs / expected_sfs.sum()
    mask = observed_sfs[1:] > 0
    ll = np.sum(observed_sfs[1:][mask] * np.log(expected_sfs[mask]))
    return ll

# Test data
n = 10
observed = np.array([0] + [100 // b for b in range(1, n)])
params = np.array([1000.0, 2000.0, 500.0])

# (a) Autograd gradient
grad_func = autograd.grad(simple_log_likelihood)
grad_auto = grad_func(params, observed)
```

```

# (b) Finite-difference gradient
eps = 1e-5
grad_fd = np.zeros_like(params)
for i in range(len(params)):
    params_plus = params.copy()
    params_minus = params.copy()
    params_plus[i] += eps
    params_minus[i] -= eps
    grad_fd[i] = (simple_log_likelihood(params_plus, observed)
                  - simple_log_likelihood(params_minus, observed)) / (2 * eps)

# Compare
for i in range(len(params)):
    print(f" param {i}: autograd={grad_auto[i]:.8f}, "
          f"finite-diff={grad_fd[i]:.8f}, "
          f"rel_diff={abs(grad_auto[i] - grad_fd[i]) / (abs(grad_auto[i]) + 1e-15):.8e}")

```

The autograd and finite-difference gradients should agree to approximately $O(\epsilon^2) \approx 10^{-10}$ (the error of central differences). Any relative discrepancy larger than 10^{-4} indicates a bug in the computation graph.

Solution 2: Optimizer comparison

We fit a three-parameter model using both TNC and L-BFGS-B and compare convergence.

```

import numpy as np
from scipy.optimize import minimize
import autograd
import autograd.numpy as anp

def neg_log_likelihood(params):
    """Negative log-likelihood for a 3-param model."""
    N_anc, N_derived, T_split = params
    # ... (compute expected SFS using Moran transitions)
    # ... (compute multinomial log-likelihood)
    return -ll

val_and_grad = autograd.value_and_grad(neg_log_likelihood)

def scipy_obj(x):
    v, g = val_and_grad(x)
    return float(v), np.array(g, dtype=float)

x0 = np.array([5000.0, 8000.0, 1000.0])
bounds = [(100, 1e6), (100, 1e6), (10, 1e5)]

# TNC
res_tnc = minimize(scipy_obj, x0, method='TNC', jac=True, bounds=bounds)
print(f"TNC: {res_tnc.nfev} evaluations, "
      f"final -ll = {res_tnc.fun:.4f}, "
      f"params = {res_tnc.x}")

# L-BFGS-B

```

```
res_lbfgsb = minimize(scipy_obj, x0, method='L-BFGS-B', jac=True, bounds=bounds)
print(f"L-BFGS-B: {res_lbfgsb.nfev} evaluations, "
      f"final -ll = {res_lbfgsb.fun:.4f}, "
      f"params = {res_lbfgsb.x}")
```

```
# Check convergence to same optimum
param_diff = np.abs(res_tnc.x - res_lbfgsb.x) / res_tnc.x
print(f"Relative parameter difference: {param_diff}")
```

Both optimizers should converge to the same MLE (within tolerance). TNC typically uses fewer function evaluations for small parameter spaces (it exploits Hessian-vector products), while L-BFGS-B may be faster per iteration. If they converge to different optima, this indicates multiple local maxima – run from additional starting points.

Solution 3: Uncertainty calibration

We simulate datasets, fit each, and check coverage of 95% Hessian-based confidence intervals.

```
import numpy as np
import autograd

def simulate_and_fit(true_params, n_sites, n_reps=100):
    """Simulate datasets and check CI coverage."""
    coverage = np.zeros(len(true_params))

    for rep in range(n_reps):
        # Simulate: compute expected SFS, draw Poisson counts
        expected = compute_expected_sfs_from_params(true_params)
        observed = np.random.poisson(expected * n_sites)

        # Fit
        x0 = true_params * np.exp(np.random.randn(len(true_params)) * 0.1)
        mle = fit_model(observed, x0)

        # Hessian-based standard errors
        H = autograd.hessian(neg_log_likelihood)(mle)
        se = np.sqrt(np.diag(np.linalg.inv(-H)))

        # Check if true values fall within 95% CI
        for i in range(len(true_params)):
            if abs(mle[i] - true_params[i]) < 1.96 * se[i]:
                coverage[i] += 1

    coverage /= n_reps
    print(f"Hessian CI coverage (target 0.95): {coverage}")
    return coverage

# Expected result: coverage will be LESS than 0.95 (e.g., 0.80-0.90)
# because the composite likelihood underestimates variance.
```

The Hessian-based intervals are typically **too narrow** (coverage below 95%), because the composite likelihood treats SNPs as independent when they are linked. Block bootstrap corrects this:

```
def block_bootstrap_se(observed_sfs, block_size=100, n_bootstrap=200):
    """Estimate standard errors via genomic block bootstrap."""
    n_blocks = n_sites // block_size
    bootstrap_estimates = []

    for b in range(n_bootstrap):
        # Resample blocks with replacement
        block_indices = np.random.choice(n_blocks, size=n_blocks, replace=True)
        resampled_sfs = sum_sfs_for_blocks(block_indices)
        mle_b = fit_model(resampled_sfs)
        bootstrap_estimates.append(mle_b)

    bootstrap_estimates = np.array(bootstrap_estimates)
    se_bootstrap = bootstrap_estimates.std(axis=0)
    return se_bootstrap
```

Block bootstrap standard errors are typically 1.5–3x larger than Hessian standard errors, yielding calibrated (close to 95%) coverage.

Solution 4: f-statistics as model diagnostics

We simulate data with admixture and fit a tree (no-admixture) model, then compute f_3 to detect misspecification.

```
import numpy as np

# True model: C is 30% A + 70% B
# Fit model: tree (A, (B, C)) -- no admixture

# Compute f3(C; A, B) from the observed data
def compute_f3(sfs, n_C, n_A, n_B):
    """Compute f3(C; A, B) from the joint SFS."""
    W = f3_weights(n_C, n_A, n_B)
    # sum over all configurations, weighted
    total_snps = sfs.sum()
    f3_val = np.sum(W * sfs) / total_snps
    return f3_val

# After fitting the tree model:
f3_observed = compute_f3(observed_sfs, n_C=10, n_A=10, n_B=10)
f3_expected = compute_f3(fitted_sfs, n_C=10, n_A=10, n_B=10)

print(f"f3(C; A, B) observed: {f3_observed:.6f}")
print(f"f3(C; A, B) expected (tree model): {f3_expected:.6f}")
```

Under the true admixture model, $f_3(C; A, B) < 0$, which is the classic signal of admixture. The tree model cannot produce a negative f_3 (for a non-admixed population, $f_3 \geq 0$), so the observed negative value directly reveals the misspecification. This is precisely the Patterson f_3 test: a significantly negative value rejects the tree hypothesis and indicates that population C has mixed ancestry from populations related to A and B.

Demo: Running momi2 on Simulated Data

Running `mini_momi2` on simulated genomic data.

Demo: momi2 Tensor Computation on msprime SFS (20 haplotypes, 1 Mb)

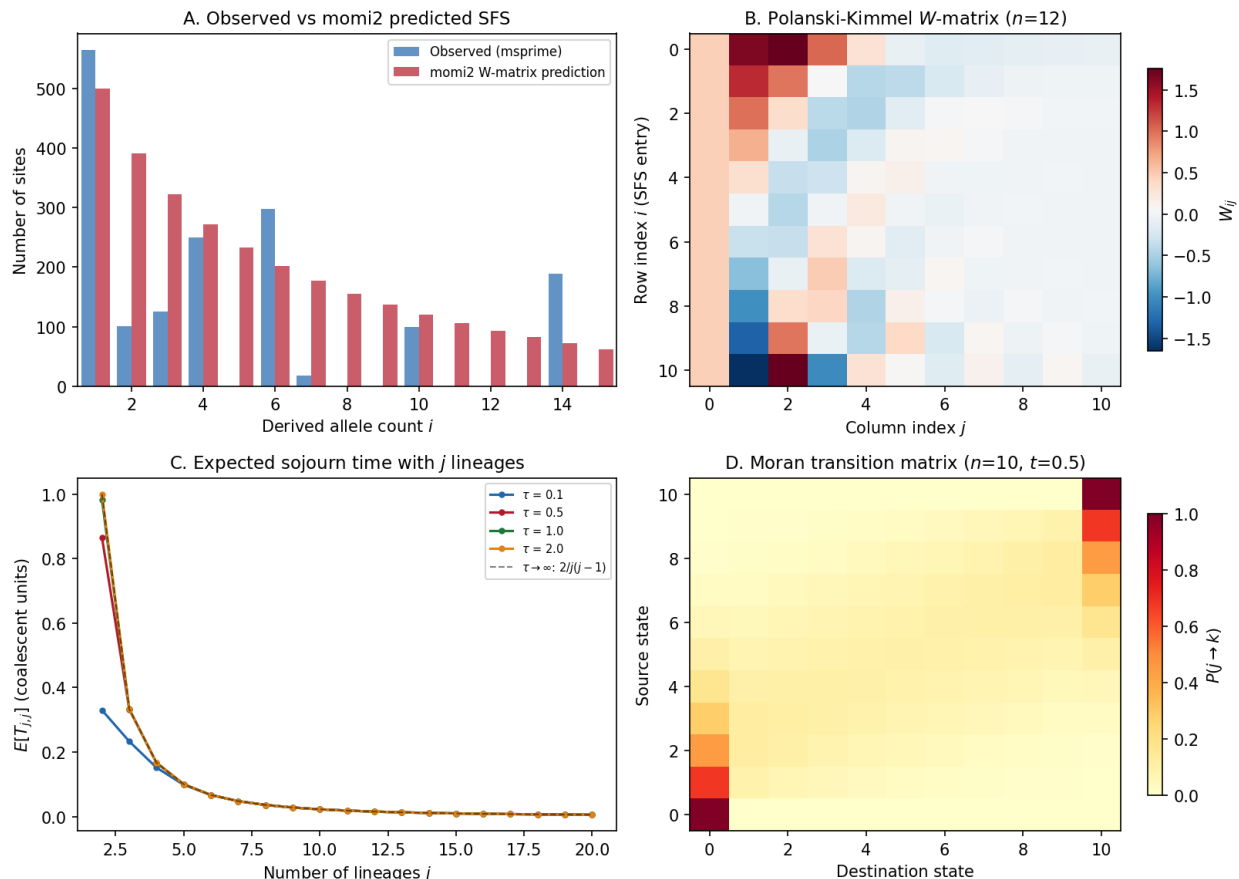


Fig. 17: **momi2 demo on msprime-simulated data.** Panel A: Observed SFS from `msprime` vs `momi2`'s W -matrix prediction using coalescent expected sojourn times. Panel B: Visualization of the Polanski-Kimmel W -matrix. Panel C: Expected sojourn time with j lineages for different epoch durations. Panel D: Moran transition matrix structure.

This figure was generated by running the `mini_momi2` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_momi2.py`.

3.14 Timepiece XIII: Gamma-SMC

Ultrafast pairwise TMRCA inference – the same problem as PSMC, without discretizing time.

3.14.1 The Mechanism at a Glance

Gamma-SMC (Schweiger & Durbin, 2023) infers the **pairwise time to the most recent common ancestor (TMRCA)** at every position along the genome, just like PSMC (*Timepiece I*). It reads the same input – a sequence of heterozygous and homozygous sites from a pair of haplotypes – and asks the same question: when did these two lineages last share an ancestor?

But Gamma-SMC answers that question in a fundamentally different way. Where PSMC discretizes coalescence time

into a finite set of intervals and runs a standard discrete-state HMM, Gamma-SMC keeps time **continuous** and tracks the posterior distribution of the TMRCA as a **gamma distribution** at every position. The result is a **continuous-state HMM (CS-HMM)** whose forward and backward passes run in $O(N)$ time with no matrix multiplications, no time discretization artifacts, and no EM iteration – just a single pass through the data.

Input: One pair of haplotypes $\rightarrow$ a sequence of het/hom/missing observations

Output: $\text{Gamma}(\alpha_i, \beta_i)$ (posterior TMRCA distribution at each position i)

The key insight is a happy conjugacy: if the prior on the TMRCA is a gamma distribution and the emission model is Poisson, then the posterior after observing a heterozygous or homozygous site is **again a gamma distribution**. The transition step (accounting for recombination) breaks this conjugacy – but empirically, the post-transition distribution is *very close* to gamma. By approximating it as gamma (via a precomputed flow field), the entire forward pass reduces to a sequence of table lookups and parameter updates.

If PSMC is a mechanical watch with discrete gear teeth, Gamma-SMC is a smooth-running quartz movement: no teeth to count, no intervals to choose. The gamma distribution glides continuously through the space of possible TMRCAs, updating its shape and rate parameters at each genomic position.

i Primary Reference

[Schweiger and Durbin, 2023]

The four gears of Gamma-SMC:

1. **The Gamma Approximation** (the escapement) – Poisson-gamma conjugacy makes the emission step exact and $O(1)$: observing a heterozygous site transforms $\text{Gamma}(\alpha, \beta)$ into $\text{Gamma}(\alpha + 1, \beta + \theta)$. The transition step is approximated by projecting back onto the gamma family. This is the fundamental tick of the mechanism – the mathematical insight that makes everything else possible.
2. **The Flow Field** (the gear train) – A precomputed two-dimensional vector field that maps gamma parameters (α, β) through one SMC transition step. The flow field is evaluated over a grid of mean and coefficient of variation values, and it is **independent of all model parameters** (θ, ρ, N_e) . It is computed once, before any analysis, and reused for every dataset.
3. **The Forward-Backward CS-HMM** (the mainspring) – The forward pass sweeps along the genome, updating the gamma parameters at each position via the flow field (transition) and conjugate update (emission). The backward pass is simply the forward algorithm run on the **reversed sequence**. The forward and backward gamma approximations are combined via a simple formula: $\text{Gamma}(\alpha + \alpha' - 1, \beta + \beta' - 1)$. This combination yields the full posterior at each site.
4. **Segmentation and Caching** (the regulator) – Long stretches of homozygous or missing sites are handled by pre-computed **cached lookups** that skip many positions at once. An **entropy clipping** mechanism prevents the gamma approximation from drifting into invalid parameter regions. These are the practical engineering tricks that make Gamma-SMC ultrafast.

These gears mesh together into a single-pass inference machine:

VCF + BED masks (het/hom/missing along genome)

```

      |
      v
+-----+
| PRECOMPUTE FLOW |
| FIELD           |
|                 |
| Grid of (mu, CV) |
+-----+
```

(continues on next page)

(continued from previous page)

```

| SMC transition step |
| -> new (mu', CV') |
| [parameter-free]   |
+-----+
|
| v
+-----+
| SEGMENT & CACHE      |
|                      |
| Group consecutive    |
| hom/missing sites    |
| Precompute multi-    |
| step flow lookups    |
+-----+
|
| v
+-----+
| FORWARD PASS         |
|                      |
| For each segment:    |
|   cache lookup (miss)|
|   cache lookup (hom) |
|   emission update    |
| Record (alpha, beta) |
| at output positions  |
+-----+
|
| v
+-----+
| BACKWARD PASS        |
|                      |
| Forward on reversed  |
| sequence + one flow  |
| field step           |
+-----+
|
| v
+-----+
| COMBINE              |
|                      |
| Gamma(a+a'-1, b+b'-1)|
| at each output site  |
+-----+
|
| v
Posterior TMRCA distribution
at each output position

```

3.14.2 PSMC vs. Gamma-SMC

Gamma-SMC solves the same problem as PSMC, but the internal mechanism is different in almost every respect:

Aspect	PSMC	Gamma-SMC
Hidden state	Discrete time intervals $[t_k, t_{k+1})$	Continuous TMRCA $t \in (0, \infty)$
Posterior representation	Probability vector over n intervals	$\text{Gamma}(\alpha, \beta)$ – two parameters
Transition step	Matrix-vector multiply $O(n^2)$	Flow field lookup + interpolation $O(1)$
Emission step	Elementwise multiply $O(n)$	Conjugate update $O(1)$
Time discretization	Required (introduces artifacts)	None (continuous throughout)
Parameter estimation	EM over many iterations	θ estimated from data; flow field is parameter-free
Output	$\hat{N}(t)$ (demographic history)	$\text{Gamma}(\alpha_i, \beta_i)$ (per-site TMRCA posterior)
Demographic model	Piecewise constant $N(t)$ (estimated)	Constant N_e (assumed)

The constant- N_e assumption is Gamma-SMC's main limitation compared to PSMC: it does not estimate a demographic history. Instead, it provides **per-site TMRCA posteriors** under a constant-population model, which can then be used as input to downstream analyses (e.g., detecting selection, estimating local ancestry, or feeding into demographic inference pipelines).

i Prerequisites for this Timepiece

Before starting Gamma-SMC, you should have worked through:

- *Coalescent Theory* – the exponential distribution of coalescence times and coalescent time units
- *Hidden Markov Models* – the forward-backward algorithm
- *The SMC* – the SMC/SMC' transition density for constant population size

The *PSMC Timepiece* is helpful but not strictly required. Gamma-SMC solves the same problem with a different mechanism, so familiarity with PSMC will provide useful context but is not a prerequisite for the derivations that follow.

3.14.3 Chapters

Overview of Gamma-SMC

Before assembling the watch, lay out every part and understand what it does.

Imagine a watchmaker who is tired of counting gear teeth. Every mechanical watch requires a discrete number of teeth on each gear – 60 for the seconds wheel, 12 for the hours. The teeth work, but they introduce a fundamental granularity: time advances in ticks, not in a smooth flow. The watchmaker asks: what if we could build a mechanism where time flows continuously, where the hands glide rather than tick?

Gamma-SMC is that mechanism. Where PSMC (*Timepiece I*) discretizes coalescence time into a finite set of intervals and builds a standard HMM with matrix multiplications, Gamma-SMC keeps time continuous and represents the posterior TMRCA at each position as a smooth curve – a gamma distribution. The result is a method that is simpler in its mathematics, faster in its computation, and free from the artifacts that come with choosing how many gear teeth to cut.

This chapter lays out every part of the Gamma-SMC mechanism on the bench. We will name each component, explain what it does, and show how the parts fit together into a working inference machine. The actual derivations happen in the chapters that follow.

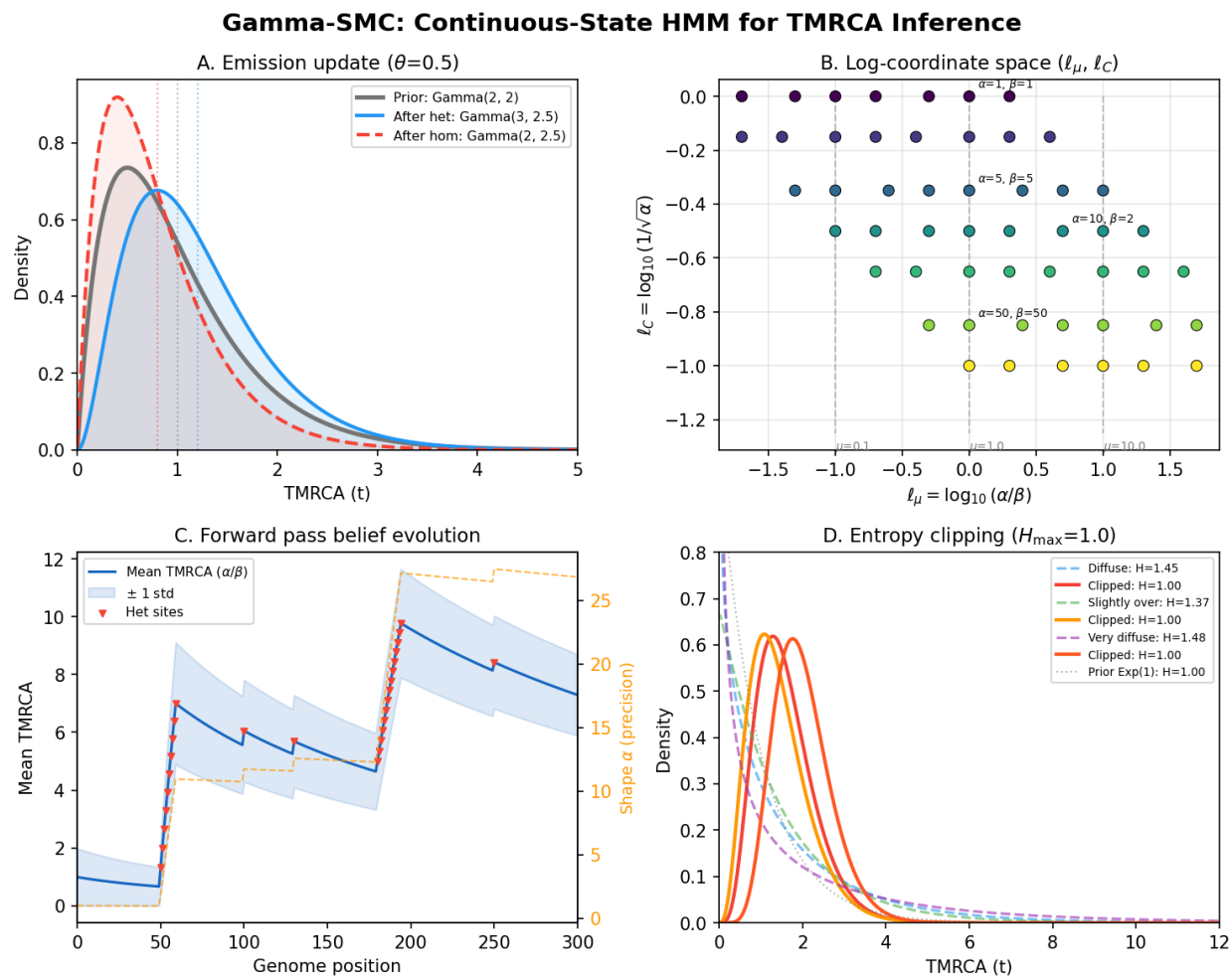


Fig. 18: **Gamma-SMC at a glance.** A continuous-state HMM for ultrafast pairwise TMRCA inference: emission updates via Poisson-gamma conjugacy, the log-coordinate transformation that stabilises numerics, forward-pass belief evolution along the genome, and entropy clipping to prevent approximation drift.

What Does Gamma-SMC Do?

Given a pair of haplotypes (e.g., the two copies of a diploid genome), Gamma-SMC infers the **posterior distribution of the TMRCA** at each position along the genome.

Input: $Y_1, Y_2, \dots, Y_N \in \{-1, 0, 1\}$

Output: $\text{Gamma}(\alpha_i, \beta_i)$ at each output position i

where at each genomic position the observation is:

- $Y_i = 1$: **heterozygous** – the two haplotypes differ (a mutation occurred since the MRCA)
- $Y_i = 0$: **homozygous** – the two haplotypes are identical
- $Y_i = -1$: **missing** – the data is unavailable at this position (masked out or not called)

Unlike PSMC, Gamma-SMC does not estimate a demographic history $N(t)$. It assumes a **constant population size** and provides the per-site TMRCA posterior under that assumption. The output is a set of gamma distribution parameters (α_i, β_i) at each position of interest, from which the user can extract point estimates (e.g., the posterior mean α_i/β_i), credible intervals, or full posterior densities.

Terminology

Before going further, let us establish the notation used throughout the Gamma-SMC chapters. Many symbols are shared with the PSMC chapters; we note the differences where they arise.

Term	Definition
μ	Unscaled mutation rate, in mutations per base pair per generation.
N_e	Effective population size (number of diploids).
$\theta = 4N_e\mu$	Scaled mutation rate. This is the Poisson rate of observing a heterozygous site given a TMRCA of t coalescent units: $P(Y_i = 1 X_i = t) \propto \theta t \cdot e^{-\theta t}$.
r	Unscaled recombination rate, in recombinations per base pair per generation.
$\rho = 4N_e r$	Scaled recombination rate. Controls how frequently the genealogy changes between adjacent positions.
X_i	TMRCA at position i , in coalescent time units (where unit time = $2N_e$ generations). This is the hidden state.
Y_i	Observation at position i : 1 (het), 0 (hom), or -1 (missing).
$\text{Gamma}(\alpha, \beta)$	Gamma distribution with shape α and rate β . Mean is α/β , variance is α/β^2 .
$f_{\alpha, \beta}(x)$	Probability density of $\text{Gamma}(\alpha, \beta)$.
$\mu_\Gamma = \alpha/\beta$	Mean of the gamma distribution (not to be confused with the mutation rate μ).
$C_V = 1/\sqrt{\alpha}$	Coefficient of variation of the gamma distribution. Lower C_V means a more concentrated (more certain) posterior.
$l_\mu = \log_{10}(\mu_\Gamma)$	Log-mean coordinate on the flow field grid.
$l_C = \log_{10}(C_V)$	Log-CV coordinate on the flow field grid.
$\mathcal{F}$	The flow field: a mapping from (α, β) to (α', β') representing one SMC transition step.
$p(t s)$	SMC transition density: the probability density of the TMRCA at the next position being t , given that the current TMRCA is s . Derived from the SMC' model.

The SMC' Transition Model

Gamma-SMC uses the **SMC' model** (Marjoram & Wall, 2006), the same model that underlies PSMC. At each genomic position, a recombination event may detach one lineage, which then re-coalesces – possibly onto the *same* branch (which the original SMC model of McVean & Cardin does not allow).

Under constant population size, the transition density conditioned on recombination is:

$$q(t | s) = \begin{cases} \frac{1 - e^{-2t}}{2s} & t \leq s \\ \frac{e^{-(t-s)} - e^{-(t+s)}}{2s} & t > s \end{cases}$$

plus a point mass at $t = s$ from the event where the detached lineage re-coalesces onto itself:

$$P(\text{re-coalesce onto self} | X_i = s) = \frac{2s + e^{-2s} - 1}{4s}$$

The full transition (marginalizing over whether recombination occurs) is:

$$p(t | s) = \rho s \cdot q(t | s) + (1 - \rho s) \cdot \delta(t - s)$$

where we use the small- ρ approximation $P(\text{recomb} | X_i = s) \approx \rho s$ (see [The Continuous-Time PSMC Model](#) for a detailed derivation of similar formulas).

The Emission Model

The emission model is a **Poisson process** of mutations. Given a TMRCa of t coalescent units, the two lineages have total branch length t (in units of $2N_e$ generations), and the number of mutations follows a Poisson distribution with rate θt :

$$\begin{aligned} P(Y_i = 0 | X_i = s) &= e^{-\theta s} \\ P(Y_i = 1 | X_i = s) &= \theta s \cdot e^{-\theta s} \\ P(Y_i = -1 | X_i = s) &= 1 \end{aligned}$$

The last line says that missing observations carry no information – the emission probability is 1 regardless of the hidden state, so the emission step is simply skipped.

i Why Poisson and not Bernoulli?

PSMC uses a Bernoulli emission model ($P(\text{het}) = 1 - e^{-\theta t}$) because it bins the genome into windows and asks “at least one mutation in this window?” Gamma-SMC works at **single-site resolution** and asks “how many mutations at this exact site?” Under the infinite-sites assumption ($\theta \ll 1$), the probability of more than one mutation at a single site is negligible, so the Poisson model effectively reduces to: het with probability $\approx \theta t$, hom with probability $\approx 1 - \theta t$. The Poisson form is used because it produces exact gamma conjugacy, as we will see in [The Gamma Approximation](#).

The Data Flow

The complete Gamma-SMC pipeline consists of five stages:

1. **Flow field precomputation** (once, offline). Evaluate the SMC transition step over a grid of (l_μ, l_C) values, obtaining a vector field $\mathcal{F}$ that maps gamma parameters through one transition. This step is independent of θ , ρ , and the data.
2. **Input processing**. Read a VCF file and optional BED masks. For each pair of haplotypes, segment the genome into stretches of consecutive homozygous or missing sites, terminated by heterozygous sites.
3. **Cache construction** (per-parameter-set). For each element of the flow field grid and for each stretch length from 1 to a maximum cache size, precompute the result of repeatedly applying the flow field with homozygous or missing observations. This depends on θ and ρ .

4. **Forward and backward passes.** For each pair of haplotypes, sweep forward along the genome using cached lookups for long stretches and conjugate updates at heterozygous sites. Then sweep backward (= forward on the reversed sequence). Record the gamma parameters at output positions.
5. **Posterior combination.** At each output position, combine the forward and backward gamma approximations:

$$\text{Posterior: Gamma}(\alpha_{\text{fwd}} + \alpha_{\text{bwd}} - 1, \beta_{\text{fwd}} + \beta_{\text{bwd}} - 1)$$

Input and Output

Input. Gamma-SMC works on a VCF file containing genotypes for one or more samples. An optional BED file per sample describes the genomic mask – regions outside the mask are treated as missing. If all samples share the same mask, a single BED file suffices. The user provides:

- Scaled mutation rate θ (or it is estimated from the average heterozygosity across diploid pairs).
- Scaled recombination rate ρ (or the recombination-to-mutation ratio ρ/θ).

Output. For each pair of haplotypes and at each user-specified output position (either heterozygous sites, a regular grid, or both), Gamma-SMC outputs the posterior gamma approximation (α, β) . From this, one can compute:

- **Posterior mean TMRCA:** α/β
- **Posterior mode TMRCA:** $(\alpha - 1)/\beta$ (for $\alpha > 1$)
- **Posterior variance:** α/β^2
- **Credible intervals:** via the gamma quantile function

Ready to Build

You now have the high-level blueprint. Every part is laid out on the bench: the gamma distribution as the posterior representation, the flow field as the transition mechanism, the conjugate update as the emission step, and the caching system as the engineering that makes it all fast.

In the following chapters, we build each gear from scratch:

1. *The Gamma Approximation* – The Poisson-gamma conjugacy that makes emission updates exact, and the gamma projection that makes transition updates tractable. This is the escapement – the mathematical insight at the heart of the mechanism.
2. *The Flow Field* – The precomputed vector field that maps gamma parameters through one SMC transition step. This is the gear train – the machinery that transmits the transition dynamics.
3. *The Forward-Backward CS-HMM* – The continuous-state HMM forward and backward passes, and the formula for combining them into a full posterior. This is the mainspring – the engine that drives inference.
4. *Segmentation and Caching* – The segmentation strategy, cache construction, and entropy clipping that make Gamma-SMC fast and stable. This is the regulator – the practical engineering that keeps the mechanism running smoothly.

Each chapter derives the math, explains the intuition, and connects back to the SMC foundations from the *prerequisites*. Let us start with the foundation: the gamma approximation.

The Gamma Approximation

The escapement: the mathematical tick that makes the whole mechanism possible.

This chapter derives the two update rules at the heart of Gamma-SMC: the **emission step** (exact, via Poisson-gamma conjugacy) and the **transition step** (approximate, via gamma projection). Together, they allow the posterior TMRCA distribution to be tracked as a gamma distribution throughout the entire forward pass.

i Biology Aside – What TMRCA tells us about population history

The **time to most recent common ancestor** (TMRCA) at a genomic position is the number of generations since the two haplotypes last shared a common ancestor at that site. It is a direct window into population size history: in a small population, lineages coalesce quickly, so TMRCA's are short; in a large population, coalescence is slow and TMRCA's are long. A population bottleneck produces a characteristic signature – a band of similar TMRCA's across many positions – because the small population forced most lineages to coalesce during that period. By estimating the TMRCA distribution at each position along the genome, Gamma-SMC effectively reads the history of population size changes encoded in the pattern of genetic variation between two chromosomes.

In a mechanical watch, the escapement is the component that converts the continuous energy of the mainspring into discrete, regular ticks. In Gamma-SMC, the gamma approximation plays an analogous role: it converts the continuous Bayesian updating process into a sequence of simple parameter updates – $(\alpha, \beta) \rightarrow (\alpha', \beta')$ – that can be evaluated in constant time.

The Emission Step: Exact Conjugacy

Suppose the forward density at position i is approximated as:

$$X_i \mid Y_{1:i-1} \sim \text{Gamma}(\alpha, \beta)$$

We observe Y_i at position i . Using Bayes' rule and the Markov property:

$$P(X_i = t \mid Y_{1:i}) \propto P(Y_i \mid X_i = t) \cdot P(X_i = t \mid Y_{1:i-1})$$

The emission model is Poisson with rate θt :

$$P(Y_i = y \mid X_i = t) = \frac{(\theta t)^y \cdot e^{-\theta t}}{y!}$$

i Biology Aside – Why Poisson emissions?

At each genomic position, we observe either a heterozygous site ($Y_i = 1$, meaning the two haplotypes differ) or a homozygous site ($Y_i = 0$, meaning they are identical). Mutations accumulate on the two lineages at a constant rate μ per base per generation, so the number of differences between the two haplotypes at a given position is approximately Poisson-distributed with a rate proportional to the TMRCA. Intuitively: *the longer two lineages have been separated, the more mutations have accumulated between them, and the more likely they are to differ at any given site*. This is the molecular clock principle applied at the single-nucleotide level.

The prior is $\text{Gamma}(\alpha, \beta)$:

$$P(X_i = t \mid Y_{1:i-1}) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}$$

Multiplying these together:

$$\begin{aligned} P(X_i = t \mid Y_{1:i}) &\propto (\theta t)^y \cdot e^{-\theta t} \cdot t^{\alpha-1} \cdot e^{-\beta t} \\ &\propto t^{\alpha+y-1} \cdot e^{-(\beta+\theta)t} \end{aligned}$$

This is the kernel of a $\text{Gamma}(\alpha + y, \beta + \theta)$ distribution. Since the posterior must normalize to a proper density:

$$X_i \mid Y_{1:i} \sim \text{Gamma}(\alpha + Y_i, \beta + \theta)$$

This gives us the three cases:

Observation	Posterior	Interpretation
$Y_i = 0$ (hom)	$\text{Gamma}(\alpha, \beta + \theta)$	Rate increases (TMRCA shifts shorter). Seeing no mutation is evidence of a more recent ancestor.
$Y_i = 1$ (het)	$\text{Gamma}(\alpha + 1, \beta + \theta)$	Shape and rate both increase. Seeing a mutation is evidence of a more distant ancestor (shape goes up) while also incorporating the observation (rate goes up).
$Y_i = -1$ (missing)	$\text{Gamma}(\alpha, \beta)$	No change. Missing data carries no information.

```
import numpy as np
from scipy.special import digamma, gammaln

def gamma_emission_update(alpha, beta, y, theta):
    """Apply the Poisson-gamma conjugate emission update.

    Parameters
    -----
    alpha : float
        Current shape parameter.
    beta : float
        Current rate parameter.
    y : int
        Observation: 1 (het), 0 (hom), or -1 (missing).
    theta : float
        Scaled mutation rate.

    Returns
    -----
    alpha_new, beta_new : float
        Updated gamma parameters.
    """
    if y == -1: # missing
        return alpha, beta
    return alpha + y, beta + theta

# Verify: starting from the prior Gamma(1, 1)
alpha, beta = 1.0, 1.0
theta = 0.001

# After a heterozygous observation
a_het, b_het = gamma_emission_update(alpha, beta, 1, theta)
print(f"After het: Gamma({a_het}, {b_het:.4f})")
print(f"Mean TMRCA = {a_het/b_het:.4f} "
      f"(shifted up -- mutation is evidence of deeper coalescence)")

# After a homozygous observation
a_hom, b_hom = gamma_emission_update(alpha, beta, 0, theta)
```

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```

print(f"After hom: Gamma({a_hom}, {b_hom:.4f})")
print(f"  Mean TMRCA = {a_hom/b_hom:.4f} "
      f"(shifted down -- no mutation favours recent coalescence)")

# After 100 hom and 1 het
a, b = 1.0, 1.0
for _ in range(100):
    a, b = gamma_emission_update(a, b, 0, theta)
a, b = gamma_emission_update(a, b, 1, theta)
print(f"\nAfter 100 hom + 1 het: Gamma({a:.1f}, {b:.4f}), "
      f"mean = {a/b:.4f}")

```

i Why is this called “conjugacy”?

A prior distribution is **conjugate** to a likelihood if the posterior has the same functional form as the prior. Here, the gamma prior is conjugate to the Poisson likelihood: regardless of the observation, the posterior is always gamma. This means we never need to leave the gamma family during the emission step – the posterior is always characterized by just two numbers (α, β) .

Conjugacy is one of the most powerful tools in Bayesian statistics. It turns Bayesian updating from an integration problem (which is generally intractable) into a parameter update (which is $O(1)$). The Poisson-gamma pair is one of the classical conjugate families, alongside normal-normal, beta-binomial, and Dirichlet-multinomial.

i Intuition: what do α and β track?

Think of α as a “mutation count” and β as a “rate accumulator.” At the start (the exponential prior $\text{Exp}(1) = \text{Gamma}(1, 1)$), we have seen zero mutations and accumulated one unit of rate. Each homozygous site adds θ to the rate (more evidence of short TMRCA) without adding to the count. Each heterozygous site adds θ to the rate *and* 1 to the count (evidence of a mutation, hence longer TMRCA). The posterior mean α/β is literally “mutation count divided by rate” – an estimate of the TMRCA.

The Transition Step: Gamma Projection

The emission step is exact. The transition step is where approximation enters.

After incorporating the observation at position i , we have $X_i \mid Y_{1:i} \sim \text{Gamma}(\alpha, \beta)$. To advance to position $i + 1$, we must compute:

$$p_{\alpha, \beta}(t) := \int_0^\infty p(t \mid s) \cdot f_{\alpha, \beta}(s) ds$$

where $p(t \mid s)$ is the SMC’ transition density (including the no-recombination case). This compound distribution $p_{\alpha, \beta}(t)$ is the predictive distribution of the TMRCA at the next position, given our current gamma belief.

i Biology Aside – What recombination does to TMRCA

Recombination breaks chromosomes and re-joins segments from different parental chromosomes. As a result, the genealogy of two haplotypes changes from position to position along the genome: at one site they may share an ancestor 10,000 generations ago, but after a recombination event a few bases away, they may share a different ancestor 50,000 generations ago. The transition step models this: it asks “given our current belief about the TMRCA, what is

the distribution of the *new* TMRCA at the next position, accounting for the possibility of recombination?” The more recombination there is (higher ρ), the more the TMRCA can change between adjacent positions.

The problem: $p_{\alpha,\beta}(t)$ is *not* exactly a gamma distribution. The integral mixes over the transition kernel, which introduces skewness and other deviations from the gamma family.

The solution: Empirically, $p_{\alpha,\beta}(t)$ is *very close* to a gamma distribution (Schweiger & Durbin verify this with simulation studies; see Appendix A of the supplement). So we approximate it by projecting back onto the gamma family:

$$p_{\alpha,\beta}(t) \approx f_{\alpha',\beta'}(t) = \text{Gamma}(\alpha', \beta')$$

The question becomes: how do we find the best (α', β') ?

The PDE Approach

For small recombination rate ρ , the post-transition distribution $p_{\alpha,\beta}$ is a small perturbation of the prior $f_{\alpha,\beta}$. We can write:

$$\begin{aligned}\alpha' &= \alpha + u \cdot \rho \\ \beta' &= \beta + v \cdot \rho\end{aligned}$$

where u and v are the “flow” – the direction and magnitude of the gamma parameter change caused by one recombination step.

To find u and v , we express the perturbation $(p_{\alpha,\beta} - f_{\alpha,\beta})/\rho$ as a linear combination of the partial derivatives of $f_{\alpha,\beta}$:

$$\frac{p_{\alpha,\beta}(x) - f_{\alpha,\beta}(x)}{\rho} \approx u \cdot \frac{\partial f_{\alpha,\beta}(x)}{\partial \alpha} + v \cdot \frac{\partial f_{\alpha,\beta}(x)}{\partial \beta}$$

The partial derivatives of the gamma PDF are derived by differentiating $f_{\alpha,\beta}(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$. It is easiest to work with the logarithm first:

$$\log f_{\alpha,\beta}(x) = \alpha \log \beta - \log \Gamma(\alpha) + (\alpha - 1) \log x - \beta x$$

Derivative with respect to α :

$$\frac{\partial}{\partial \alpha} \log f = \log \beta - \psi(\alpha) + \log x$$

where $\psi(\alpha) = \frac{d}{d\alpha} \ln \Gamma(\alpha)$ is the **digamma function** – the logarithmic derivative of the gamma function. Since $\frac{\partial f}{\partial \alpha} = f \cdot \frac{\partial \log f}{\partial \alpha}$:

$$\frac{\partial f_{\alpha,\beta}(x)}{\partial \alpha} = f_{\alpha,\beta}(x) \cdot (\log \beta - \psi(\alpha) + \log x)$$

Derivative with respect to β :

$$\frac{\partial}{\partial \beta} \log f = \frac{\alpha}{\beta} - x$$

since $\frac{\partial}{\partial \beta} (\alpha \log \beta) = \alpha/\beta$ and $\frac{\partial}{\partial \beta} (-\beta x) = -x$. Therefore:

$$\frac{\partial f_{\alpha,\beta}(x)}{\partial \beta} = f_{\alpha,\beta}(x) \cdot \left(\frac{\alpha}{\beta} - x \right)$$

```

from scipy.special import digamma, gammaln
import numpy as np

def gamma_pdf_partials(x, alpha, beta):
    """Compute partial derivatives of the gamma PDF analytically.

    Returns df/dalpha and df/dbeta at each x.
    """
    log_f = (alpha * np.log(beta) - gammaln(alpha)
              + (alpha - 1) * np.log(x) - beta * x)
    f = np.exp(log_f)
    df_dalpha = f * (np.log(beta) - digamma(alpha) + np.log(x))
    df_dbeta = f * (alpha / beta - x)
    return df_dalpha, df_dbeta

# Verify against numerical finite differences
alpha, beta = 3.0, 2.0
x = np.linspace(0.01, 5.0, 100)
eps = 1e-7

df_da_analytic, df_db_analytic = gamma_pdf_partials(x, alpha, beta)

# Numerical d/dalpha
f_plus = np.exp((alpha + eps) * np.log(beta) - gammaln(alpha + eps)
                 + (alpha + eps - 1) * np.log(x) - beta * x)
f_minus = np.exp((alpha - eps) * np.log(beta) - gammaln(alpha - eps)
                 + (alpha - eps - 1) * np.log(x) - beta * x)
df_da_numerical = (f_plus - f_minus) / (2 * eps)

print(f"Max error in df/dalpha: {np.max(np.abs(df_da_analytic - df_da_numerical)):.2e}
→")

```

We solve for u, v by least squares over a grid of 2,000 values of x covering the main support of $f_{\alpha, \beta}$:

$$\arg \min_{u, v} \sum_{i=1}^{2000} \left(u \cdot \frac{\partial f_{\alpha, \beta}(x_i)}{\partial \alpha} + v \cdot \frac{\partial f_{\alpha, \beta}(x_i)}{\partial \beta} - \frac{p_{\alpha, \beta}(x_i) - f_{\alpha, \beta}(x_i)}{\rho} \right)^2$$

i Why least squares and not moment matching?

Moment matching (equating the mean and variance of $p_{\alpha,\beta}$ to those of $\text{Gamma}(\alpha', \beta')$) is another natural approach. The PDE/least-squares method is preferred because it minimizes the L^2 distance between the true perturbation and its gamma approximation over the entire support of the distribution. This gives a better fit in the tails, which matters because the tails of the TMRCA distribution carry information about extreme coalescence times.

The Closed-Form Perturbation

The key quantity $(p_{\alpha,\beta}(t) - f_{\alpha,\beta}(t))/\rho$ can be evaluated analytically. Starting from the SMC' transition density and integrating against the gamma prior, one obtains (see the full derivation in Appendix E of the supplement):

$$\begin{aligned} \frac{p_{\alpha,\beta}(t) - f_{\alpha,\beta}(t)}{\rho} \approx & e^{-t} \cdot \frac{(\beta t)^\alpha}{\Gamma(\alpha + 1)} [M(\alpha, \alpha + 1, (\beta - 1)t) - M(\alpha, \alpha + 1, -(\beta + 1)t)] \\ & + \frac{2t + e^{-2t} - 1}{2} \cdot f_{\alpha,\beta}(t) \\ & + (1 - e^{-2t}) \cdot \frac{\Gamma(\alpha, \beta t)}{\Gamma(\alpha)} \\ & - 2t \cdot f_{\alpha,\beta}(t) \end{aligned}$$

where:

- $M(a, b, z) = {}_1F_1(a, b, z)$ is **Kummer's confluent hypergeometric function**, a special function that generalizes the exponential. It is evaluated using the Arb arbitrary-precision library (Johansson, 2017).
- $\Gamma(\alpha, x) = \int_x^\infty t^{\alpha-1} e^{-t} dt$ is the **upper incomplete gamma function**.

The four terms arise from integrating the SMC' transition density against the gamma prior, splitting the integral into three regions ($s < t$, $s = t$, and $s > t$):

1. The first term covers recombination at $s < t$ (the detached lineage coalesces at a *later* time).
2. The second term covers the self-coalescence (the lineage re-coalesces onto the same branch at $t = s$).
3. The third term covers recombination at $s > t$ (the detached lineage coalesces at an *earlier* time).
4. The fourth term subtracts the no-recombination contribution (which is already counted in $f_{\alpha,\beta}$).

i A critical property: parameter independence

The flow (u, v) at each grid point (α, β) depends only on α and β – not on θ , ρ , or N_e . This is because the perturbation is normalized by ρ , and the remaining expression involves only the gamma distribution parameters and the SMC' transition kernel (which, for constant population size, has no free parameters beyond the time units).

This parameter independence is what makes Gamma-SMC's precomputation strategy possible: the flow field is computed **once** and reused for any dataset with any θ and ρ .

Log-Coordinates

Working directly with (α, β) is numerically inconvenient because these parameters span several orders of magnitude. Instead, Gamma-SMC uses **log-coordinates** based on the mean and coefficient of variation:

$$\begin{aligned} l_\mu &= \log_{10} \left(\frac{\alpha}{\beta} \right) \quad (\text{log-mean}) \\ l_C &= \log_{10} \left(\frac{1}{\sqrt{\alpha}} \right) \quad (\text{log-CV}) \end{aligned}$$

The flow field is evaluated over a grid of these coordinates:

- 51 values of l_μ equally spaced between -5 and 2 (i.e., log-spaced means from 10^{-5} to 10^2 coalescent time units)
- 50 values of l_C equally spaced between -2 and 0 (CVs from 10^{-2} to 1)

Values with $C_V > 1$ ($\alpha < 1$) are excluded because this would give an infinite gamma density at $x = 0$, which is unphysical for a TMRCA.

i Why log-coordinates?

Log-coordinates serve two purposes. First, they allow the grid to cover many orders of magnitude uniformly – a TMRCA could be 10^{-3} coalescent units (very recent) or 10^1 (very ancient), and log-spacing gives equal resolution at all scales. Second, the Taylor expansion used for the PDE approach is performed in $(\log_{10} \alpha, \log_{10} \beta)$ coordinates, not in (l_μ, l_C) directly. The conversion uses the chain rule:

$$\begin{aligned}\Delta l_\mu &= \Delta \log_{10}(\alpha) - \Delta \log_{10}(\beta) \\ \Delta l_C &= -0.5 \cdot \Delta \log_{10}(\alpha)\end{aligned}$$

This avoids issues with the Taylor expansion that would arise from working in the (l_μ, l_C) coordinate system directly.

```
def to_log_coords(alpha, beta):
    """Convert (alpha, beta) to log-mean / log-CV coordinates.

    Parameters
    -----
    alpha : float
        Shape parameter (must be > 0).
    beta : float
        Rate parameter (must be > 0).

    Returns
    -----
    l_mu : float
        log10(alpha/beta), the log-mean TMRCA.
    l_C : float
        log10(1/sqrt(alpha)), the log coefficient of variation.
    """
    l_mu = np.log10(alpha / beta)
    l_C = np.log10(1.0 / np.sqrt(alpha))
    return l_mu, l_C

def from_log_coords(l_mu, l_C):
    """Convert log-coordinates back to (alpha, beta).

    Parameters
    -----
    l_mu : float
        Log-mean coordinate.
    l_C : float
        Log-CV coordinate.

    Returns
```

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```

-----
alpha, beta : float
"""
alpha = 10.0 ** (-2 * l_C)
beta = alpha * 10.0 ** (-l_mu)
return alpha, beta

# Verify round-trip conversion
for alpha, beta in [(1.0, 1.0), (5.0, 2.5), (100.0, 50.0)]:
    l_mu, l_C = to_log_coords(alpha, beta)
    a_back, b_back = from_log_coords(l_mu, l_C)
    print(f"Gamma({alpha}, {beta}) -> (l_mu={l_mu:.4f}, l_C={l_C:.4f}) "
          f"-> Gamma({a_back:.4f}, {b_back:.4f})")

# The prior Gamma(1, 1) maps to (0, 0) in log-coordinates
l_mu_prior, l_C_prior = to_log_coords(1.0, 1.0)
print(f"\nPrior Gamma(1,1): (l_mu, l_C) = ({l_mu_prior:.1f}, {l_C_prior:.1f})")

# The grid covers:
l_mu_grid = np.linspace(-5, 2, 51)
l_C_grid = np.linspace(-2, 0, 50)
print(f"Grid: {len(l_mu_grid)} x {len(l_C_grid)} = "
      f"{len(l_mu_grid) * len(l_C_grid)} points")

```

Summary

The gamma approximation rests on two pillars:

1. **Emission step** (exact): $\text{Gamma}(\alpha, \beta) + Y_i \rightarrow \text{Gamma}(\alpha + Y_i, \beta + \theta)$. This is Poisson-gamma conjugacy – a textbook result that requires no approximation.
2. **Transition step** (approximate): $\text{Gamma}(\alpha, \beta) \rightarrow \text{Gamma}(\alpha + u \cdot \rho, \beta + v \cdot \rho)$, where (u, v) are determined by least-squares fitting to the true perturbation. This requires a one-time precomputation over a two-dimensional grid.

Together, these two rules mean that the entire forward pass of the HMM reduces to a sequence of arithmetic operations on two numbers (α, β) . No matrix multiplications, no numerical integration during inference, no discretization of time. The cost of one step is $O(1)$, making the full forward pass $O(N)$ in the sequence length.

Next: *The Flow Field* – precomputing the transition machinery into a reusable vector field.

The Flow Field

The gear train: precomputed machinery that transmits the transition dynamics.

In the previous chapter, we showed that the SMC transition step transforms a $\text{Gamma}(\alpha, \beta)$ distribution into something that is approximately $\text{Gamma}(\alpha', \beta')$, with the change $(\alpha' - \alpha, \beta' - \beta) = \rho \cdot (u, v)$ determined by a least-squares projection. The pair (u, v) depends on (α, β) but not on any model parameters.

This chapter describes how Gamma-SMC precomputes the mapping $(\alpha, \beta) \mapsto (\alpha', \beta')$ over a two-dimensional grid, creating a **flow field** $\mathcal{F}$ that can be queried during inference via fast interpolation. If the gamma approximation is the escapement, the flow field is the gear train – the precision-machined mechanism that transmits the tick of the escapement through the rest of the watch.

What Is a Flow Field?

A flow field is a function that assigns a **direction and magnitude** to every point in a space. In fluid dynamics, a flow field describes the velocity of a fluid at each point – where the fluid is going and how fast. In Gamma-SMC, the “fluid” is the gamma posterior, and the flow field describes how the posterior parameters change when a recombination event is possible.

Plain-language summary – The flow field as a lookup table

The flow field is, at its core, a precomputed lookup table. For every possible state of our belief about the TMRCA (described by the gamma parameters α and β), it tells us: *if recombination could have occurred at this position, how should we update our belief?* The answer depends on the current belief – if we are very certain the TMRCA is recent, recombination shifts our estimate differently than if we are uncertain or think the TMRCA is ancient. By precomputing this for all possible beliefs on a grid, Gamma-SMC avoids expensive calculations during the actual genomic scan.

Concretely, the flow field $\mathcal{F}$ is a mapping:

$$\mathcal{F} : (l_\mu, l_C) \mapsto (l'_\mu, l'_C)$$

where $(l_\mu, l_C) = (\log_{10}(\alpha/\beta), \log_{10}(1/\sqrt{\alpha}))$ are the log-mean and log-CV coordinates of the current gamma distribution, and (l'_μ, l'_C) are the coordinates after one SMC transition step.

The Grid

The flow field is evaluated over a rectangular grid:

Axis	Range	Points	Physical meaning
l_μ	$[-5, 2]$	51	Log-mean TMRCA from 10^{-5} to 10^2 coalescent units
l_C	$[-2, 0]$	50	Log-CV from 10^{-2} (very certain) to 1 (uncertain)

This gives $51 \times 50 = 2,550$ grid points, at each of which $\mathcal{F}$ is evaluated. The grid boundaries are chosen to cover the range of posteriors encountered in practice:

- **Mean TMRCA** from 10^{-5} (essentially zero – a very recent common ancestor) to 100 coalescent units ($200N_e$ generations into the past). Genomic positions with TMRCA outside this range are vanishingly rare.
- **CV** from 0.01 (extremely certain – a sharp posterior with very little uncertainty) to 1.0 (as uncertain as the prior). A CV of 1 corresponds to $\alpha = 1$, which is the exponential distribution – the prior itself. The posterior cannot be *more* uncertain than the prior (in the absence of pathological approximation errors).

Values with $C_V > 1$ (i.e., $\alpha < 1$) are excluded because they would give an infinite density at $t = 0$, which is unphysical.

Computing Each Grid Point

At each grid point (l_μ, l_C) , the computation proceeds as follows:

1. **Convert to (α, β) :**

$$\alpha = 10^{-2l_C}$$

$$\beta = \alpha \cdot 10^{-l_\mu}$$

2. **Evaluate the perturbation** $(p_{\alpha,\beta}(x) - f_{\alpha,\beta}(x))/\rho$ at 2,000 values of x covering the main support of $f_{\alpha,\beta}$. This uses the closed-form expression derived in *The Gamma Approximation*, involving Kummer's confluent hypergeometric function $M(a, b, z)$.
3. **Evaluate the partial derivatives** $\partial f/\partial\alpha$ and $\partial f/\partial\beta$ (or equivalently, $\partial f/\partial\log_{10}\alpha$ and $\partial f/\partial\log_{10}\beta$) at the same 2,000 points.
4. **Solve the least-squares problem** to find the flow $(\Delta\log_{10}\alpha, \Delta\log_{10}\beta)$.
5. **Convert to** (l'_μ, l'_C) using the linear relationship:

$$\begin{aligned}\Delta l_\mu &= \Delta\log_{10}(\alpha) - \Delta\log_{10}(\beta) \\ \Delta l_C &= -0.5 \cdot \Delta\log_{10}(\alpha)\end{aligned}$$

```
import numpy as np
from scipy.stats import gamma as gamma_dist
from scipy.special import digamma

def gamma_pdfpartials(x, alpha, beta):
    """Evaluate the gamma PDF and its partial derivatives.

    Parameters
    -----
    x : ndarray
        Points at which to evaluate.
    alpha : float
        Shape parameter.
    beta : float
        Rate parameter.

    Returns
    -----
    f : ndarray
        Gamma PDF values.
    df_dalpha : ndarray
        Partial derivative with respect to alpha.
    df_dbeta : ndarray
        Partial derivative with respect to beta.
    """
    f = gamma_dist.pdf(x, a=alpha, scale=1.0 / beta)
    # d/dalpha [Gamma PDF] = f * (-psi(alpha) + log(beta) + log(x))
    df_dalpha = f * (-digamma(alpha) + np.log(beta) + np.log(x + 1e-300))
    # d/dbeta [Gamma PDF] = f * (alpha/beta - x)
    df_dbeta = f * (alpha / beta - x)
    return f, df_dalpha, df_dbeta

def compute_flow_at_point(l_mu, l_C, n_eval=2000):
    """Compute the flow displacement at one grid point.

    This is a simplified skeleton showing the least-squares structure.
    The full implementation requires Kummer's hypergeometric function
    M(a, b, z) via the Arb library for the perturbation evaluation.

    Parameters
```

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```

-----
l_mu, l_C : float
    Log-mean and log-CV coordinates.
n_eval : int
    Number of evaluation points for least-squares.

Returns
-----
delta_l_mu, delta_l_C : float
    Flow displacement in log-coordinates.
"""
# Step 1: convert to (alpha, beta)
alpha = 10.0 ** (-2 * l_C)
beta = alpha * 10.0 ** (-l_mu)

# Step 2: set up evaluation grid over the support of Gamma(alpha, beta)
mean = alpha / beta
std = np.sqrt(alpha) / beta
x = np.linspace(max(1e-10, mean - 4*std), mean + 6*std, n_eval)

# Step 3: evaluate partial derivatives
f, df_da, df_db = gamma_pdf_partials(x, alpha, beta)

# Step 4: in the full implementation, evaluate the perturbation
# (p_{alpha,beta}(x) - f_{alpha,beta}(x)) / rho using Kummer's M.
# Here we use a placeholder zero perturbation for illustration.
perturbation = np.zeros_like(x) # placeholder

# Step 5: solve least-squares for (u, v) in log10(alpha), log10(beta)
A = np.column_stack([df_da, df_db])
result = np.linalg.lstsq(A, perturbation, rcond=None)
delta_log_a, delta_log_b = result[0]

# Step 6: convert to (delta_l_mu, delta_l_C)
delta_l_mu = delta_log_a - delta_log_b
delta_l_C = -0.5 * delta_log_a
return delta_l_mu, delta_l_C

# Demonstrate the structure (flow is zero with placeholder perturbation)
dl_mu, dl_C = compute_flow_at_point(0.0, -0.5)
print(f"Flow at (l_mu=0, l_C=-0.5): "
      f"delta_l_mu={dl_mu:.6f}, delta_l_C={dl_C:.6f}")
print("(Zero because we used a placeholder perturbation)")

```

The result is stored as the displacement $(\Delta l_\mu, \Delta l_C) = (l'_\mu - l_\mu, l'_C - l_C)$ at each grid point.

i Numerical precision: the Arb library

The confluent hypergeometric function $M(\alpha, \alpha + 1, z)$ can have very large arguments (when β and t are both large), making standard floating-point evaluation unreliable. Gamma-SMC uses the **Arb library** (Johansson, 2017), which performs ball arithmetic – rigorous interval arithmetic with guaranteed error bounds. Each number is represented as a midpoint and a radius, and all operations propagate error bounds correctly. This ensures that the flow field is

computed to full precision even at extreme grid points.

Querying the Flow Field

During inference, the current gamma posterior (l_μ, l_C) will generally *not* lie exactly on a grid point. Gamma-SMC uses **bilinear interpolation** over the four nearest grid points to obtain the flow:

$$\mathcal{F}(l_\mu, l_C) \approx (1-s)(1-t) \cdot \mathcal{F}_{ij} + s(1-t) \cdot \mathcal{F}_{(i+1)j} + (1-s)t \cdot \mathcal{F}_{i(j+1)} + st \cdot \mathcal{F}_{(i+1)(j+1)}$$

where s and t are the fractional positions within the grid cell and i, j are the indices of the lower-left corner.

Clipping. If either l_μ or l_C falls outside the grid boundaries, it is clipped to the nearest boundary value. This prevents extrapolation beyond the precomputed grid and provides a conservative fallback for extreme posteriors.

```
import numpy as np

class FlowField:
    """A precomputed flow field over a (l_mu, l_C) grid.

    Parameters
    -----
    l_mu_grid : ndarray, shape (n_mu,)
        Log-mean grid values.
    l_C_grid : ndarray, shape (n_C,)
        Log-CV grid values.
    delta_l_mu : ndarray, shape (n_mu, n_C)
        Flow displacement in l_mu at each grid point.
    delta_l_C : ndarray, shape (n_mu, n_C)
        Flow displacement in l_C at each grid point.
    """

    def __init__(self, l_mu_grid, l_C_grid, delta_l_mu, delta_l_C):
        self.l_mu_grid = l_mu_grid
        self.l_C_grid = l_C_grid
        self.delta_l_mu = delta_l_mu
        self.delta_l_C = delta_l_C

    def query(self, l_mu, l_C):
        """Query the flow field via bilinear interpolation with clipping.

        Parameters
        -----
        l_mu : float
            Log-mean coordinate of the current gamma distribution.
        l_C : float
            Log-CV coordinate of the current gamma distribution.

        Returns
        -----
        dl_mu, dl_C : float
            Interpolated flow displacement.
        """
        # Clip to grid boundaries
```

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```

l_mu_c = np.clip(l_mu, self.l_mu_grid[0], self.l_mu_grid[-1])
l_C_c = np.clip(l_C, self.l_C_grid[0], self.l_C_grid[-1])

# Find the lower-left grid cell indices
i = np.searchsorted(self.l_mu_grid, l_mu_c) - 1
j = np.searchsorted(self.l_C_grid, l_C_c) - 1
i = np.clip(i, 0, len(self.l_mu_grid) - 2)
j = np.clip(j, 0, len(self.l_C_grid) - 2)

# Fractional positions within the cell: s along l_mu, t along l_C
s = (l_mu_c - self.l_mu_grid[i]) / (self.l_mu_grid[i+1] - self.l_mu_grid[i])
t = (l_C_c - self.l_C_grid[j]) / (self.l_C_grid[j+1] - self.l_C_grid[j])

# Bilinear interpolation for each displacement component
def interp(field):
    return ((1-s)*(1-t) * field[i, j]
            + s*(1-t) * field[i+1, j]
            + (1-s)*t * field[i, j+1]
            + s*t * field[i+1, j+1])

    return interp(self.delta_l_mu), interp(self.delta_l_C)

# Build a small demonstration flow field (placeholder displacements)
l_mu_grid = np.linspace(-5, 2, 51)
l_C_grid = np.linspace(-2, 0, 50)
# In the real implementation these come from the precomputed PDE solutions;
# here we use a simple model: flow pushes l_C upward (more uncertainty)
dl_mu_grid = np.zeros((51, 50))
dl_C_grid = 0.01 * np.ones((51, 50)) # placeholder: constant upward drift

ff = FlowField(l_mu_grid, l_C_grid, dl_mu_grid, dl_C_grid)
dl_mu, dl_C = ff.query(0.5, -0.8)
print(f"Flow at (l_mu=0.5, l_C=-0.8): "
      f"delta_l_mu={dl_mu:.6f}, delta_l_C={dl_C:.6f}")
# Clipping: query outside the grid
dl_mu, dl_C = ff.query(5.0, -3.0)
print(f"Flow at (l_mu=5.0, l_C=-3.0) [clipped]: "
      f"delta_l_mu={dl_mu:.6f}, delta_l_C={dl_C:.6f}")

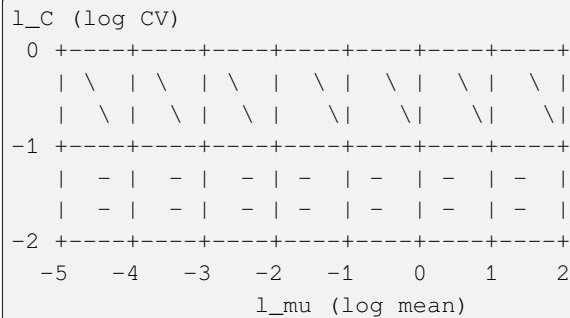
```

i Why bilinear interpolation?

Bilinear interpolation is the simplest scheme that is continuous across grid cell boundaries. Higher-order interpolation (e.g., bicubic) would give smoother results but would require larger stencils and more computation per query. Since the flow field is smooth (it is derived from smooth PDEs), bilinear interpolation is sufficient and keeps each transition step $O(1)$.

Visualizing the Flow Field

The flow field can be visualized as a vector plot over the (l_μ, l_C) grid. At each point, an arrow shows the direction and magnitude of the change in gamma parameters caused by one transition step.



The general pattern:

- At high CV (top of grid, near the prior), the flow has large magnitude: the transition step significantly changes the posterior because the current belief is weak.
- At low CV (bottom of grid, sharp posterior), the flow is small: the posterior is so concentrated that one transition step barely shifts it.
- The flow generally points **upward** (increasing CV): recombination introduces uncertainty, broadening the posterior.

i Biology Aside – Why recombination increases uncertainty

Without recombination, consecutive genomic positions share the same genealogy, so each new observation (het or hom) steadily sharpens our estimate of the TMRCA. Recombination disrupts this: it creates the possibility that the TMRCA at the next position is *completely different* from the current one. This injects uncertainty – we cannot be as sure about the TMRCA after a potential recombination event as we were before. The flow field captures exactly how much uncertainty recombination adds, depending on how confident we were to begin with. In regions of the genome with high recombination rates, the TMRCA changes frequently and the posterior stays broad; in recombination deserts, the posterior narrows sharply as evidence accumulates.

Parameter Independence

The flow field $\mathcal{F}$ is **independent of all model parameters** θ , ρ , and N_e . This is because:

- The transition density $p(t | s)$ under constant population size has no free parameters (it is fully determined by the coalescent process).
- The recombination rate ρ factors out: the flow is defined as the $O(\rho)$ perturbation divided by ρ , so ρ cancels.
- The mutation rate θ does not appear in the transition step – it only enters through the emission step.

This means the flow field is computed **once** and can be shipped as a precomputed data file alongside the software. Users never need to recompute it.

Summary

The flow field $\mathcal{F}$ is a 2D vector field over (l_μ, l_C) space that encodes the effect of one SMC transition step on gamma-distributed posteriors. It is:

- **Precomputed** over a 51×50 grid using high-precision arithmetic
- **Parameter-independent** – the same field applies to any θ, ρ, N_e
- **Queried** during inference via bilinear interpolation with boundary clipping
- Evaluated in $O(1)$ per query, making it the constant-time “gear train” that transmits transition dynamics through the mechanism

With the flow field in hand, we can now assemble the complete forward-backward algorithm: *The Forward-Backward CS-HMM*.

The Forward-Backward CS-HMM

The mainspring: the engine that drives inference from one end of the genome to the other.

With the gamma approximation (*The Gamma Approximation*) providing exact emission updates and the flow field (*The Flow Field*) providing approximate transition updates, we can now assemble the complete inference algorithm. This chapter describes the forward pass, the backward pass, and the formula for combining them into a full posterior at each position.

In a mechanical watch, the mainspring stores energy and releases it steadily to drive the gear train. In Gamma-SMC, the forward-backward algorithm is the mainspring: it sweeps the accumulated evidence from one end of the genome to the other, building up the posterior at each position.

Biology Aside – Reading the genome like a tape

The forward-backward algorithm scans the genome from left to right (and then right to left), accumulating evidence about the TMRCA at each position. This is possible because the genome is a linear sequence of observations – heterozygous or homozygous at each site – and the TMRCA at adjacent sites is correlated (it changes only when recombination occurs). The forward pass builds a running estimate: “given everything I’ve seen so far on this chromosome, what do I think the TMRCA is here?” The backward pass does the same from the other end. Combining both gives the best possible estimate at every position, using all the data from the entire chromosome.

The Forward Algorithm

The forward algorithm in a continuous-state HMM (CS-HMM) tracks the **forward density** – the probability density of the hidden state at position i , given all observations up to and including position i :

$$\hat{\alpha}(x_i) := P(X_i = x_i \mid Y_{1:i} = y_{1:i})$$

This is the continuous-state analogue of the forward variable $\alpha_k(i)$ in discrete HMMs (see *Hidden Markov Models*). Instead of a probability vector over n states, we have a probability density function over the continuous TMRCA.

The recursion is:

$$\hat{\alpha}(x_i) = \frac{1}{c_i} \cdot P(y_i \mid x_i) \int_0^\infty P(x_i \mid x_{i-1}) \hat{\alpha}(x_{i-1}) dx_{i-1}$$

where c_i is a scaling factor that ensures $\hat{\alpha}$ is a normalized density. This recursion says: take the previous forward density, propagate it through the transition kernel (the integral), then update with the emission probability, and normalize.

The **Gamma-SMC forward pass** approximates this recursion using two steps at each position:

Step 1: Transition. Apply the flow field $\mathcal{F}$ to advance the gamma parameters through one transition step:

$$(\alpha, \beta) \xrightarrow{\mathcal{F}} (\alpha', \beta')$$

This replaces the integral $\int P(x_i | x_{i-1}) \hat{\alpha}(x_{i-1}) dx_{i-1}$ with a single flow field lookup.

Step 2: Emission. Incorporate the observation using the conjugate update:

$$(\alpha', \beta') \xrightarrow{Y_i} (\alpha' + Y_i, \beta' + \theta)$$

(where $Y_i = 0$ for hom, $Y_i = 1$ for het, and the step is skipped for missing data).

Initialization. The prior on the TMRCA is the stationary distribution of the coalescent under constant population size, which is $\text{Exp}(1) = \text{Gamma}(1, 1)$. So the forward pass begins with $(\alpha, \beta) = (1, 1)$, or equivalently $(l_\mu, l_C) = (0, 0)$.

i Comparison to PSMC's forward pass

In PSMC (*The PSMC HMM and EM Algorithm*), the forward pass at each position performs:

1. A **matrix-vector multiply**: $\alpha_k(i) = \sum_l p_{lk} \alpha_l(i-1)$, which is $O(n^2)$ in the number of time intervals.
2. An **elementwise multiply**: $\alpha_k(i) \leftarrow e_k(y_i) \cdot \alpha_k(i)$, which is $O(n)$.

Gamma-SMC replaces both operations with $O(1)$ operations: a bilinear interpolation lookup (transition) and a parameter increment (emission). The difference in computational cost is dramatic: PSMC with $n = 64$ time intervals requires $O(64^2) = O(4096)$ operations per position; Gamma-SMC requires $O(1)$.

The forward pass proceeds along the genome, recording the gamma parameters (α_i, β_i) at each user-specified output position.

```
import numpy as np

def gamma_smc_forward(observations, theta, rho, flow_field):
    """Run the Gamma-SMC forward pass.

    Parameters
    -----
    observations : list of int
        Observation at each position: 1 (het), 0 (hom), -1 (missing).
    theta : float
        Scaled mutation rate per position.
    rho : float
        Scaled recombination rate per position.
    flow_field : FlowField
        Precomputed flow field with a .query(l_mu, l_C) method.

    Returns
    -----
    alphas : ndarray, shape (N,)
        Forward shape parameter at each position.
    betas : ndarray, shape (N,)
        Forward rate parameter at each position.
    """
    N = len(observations)
```

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```

alphas = np.zeros(N)
betas = np.zeros(N)

# Initialize with the prior: Gamma(1, 1) = Exp(1)
alpha, beta = 1.0, 1.0

for i in range(N):
    # Step 1: Transition via flow field
    l_mu = np.log10(alpha / beta)
    l_C = np.log10(1.0 / np.sqrt(alpha))
    dl_mu, dl_C = flow_field.query(l_mu, l_C)
    l_mu += rho * dl_mu # displacement scaled by rho
    l_C += rho * dl_C
    alpha = 10.0 ** (-2 * l_C)
    beta = alpha * 10.0 ** (-l_mu)

    # Step 2: Emission update (conjugate)
    y = observations[i]
    if y >= 0: # not missing
        alpha += y # +1 for het, +0 for hom
        beta += theta

    alphas[i] = alpha
    betas[i] = beta

return alphas, betas

# Demonstrate on a short synthetic sequence
obs = [0]*50 + [1] + [0]*30 + [1] + [0]*19
theta, rho = 0.001, 0.0004

# Use a trivial flow field (zero displacement) for illustration
class ZeroFlow:
    def query(self, l_mu, l_C): return 0.0, 0.0

a_fwd, b_fwd = gamma_smc_forward(obs, theta, rho, ZeroFlow())
print(f"After {len(obs)} positions ({sum(obs)} hets):")
print(f"  Final forward: Gamma({a_fwd[-1]:.1f}, {b_fwd[-1]:.4f})")
print(f"  Mean TMRCA = {a_fwd[-1]/b_fwd[-1]:.2f}")

```

The Backward Pass as Reversed Forward

In a standard discrete HMM, the backward algorithm computes $\beta_k(i) = P(Y_{i+1:N} \mid X_i = k)$ using a recursion that runs from right to left. The continuous-state analogue is:

$$\tilde{\alpha}(x_{i+1}) := P(X_{i+1} = x_{i+1} \mid Y_{i+1:N} = y_{i+1:N})$$

Gamma-SMC uses a key reformulation: instead of deriving a separate backward recursion, it observes that the backward density can be obtained by **running the forward algorithm on the reversed sequence**. That is:

- Reverse the observation sequence: $y_N, y_{N-1}, \dots, y_1$.
- Run the standard forward pass on this reversed sequence.

- At each position $i + 1$, the resulting forward density is the backward density $\tilde{\alpha}(x_{i+1})$.

This is possible because the SMC transition density is symmetric in a specific sense: reversing the sequence and re-running the forward algorithm produces the same mathematical object as the backward algorithm.

i Why reuse the forward algorithm?

The forward-on-reversed approach has two advantages. First, it avoids implementing a separate backward recursion, reducing code complexity and the chance of bugs. Second, the flow field $\mathcal{F}$ is the same in both directions (the SMC transition kernel under constant population size is time-reversible), so no additional precomputation is needed.

Combining Forward and Backward

At each output position i , we now have:

- The forward density: $\hat{\alpha}(x_i) \approx \text{Gamma}(a, b)$
- The backward density (after one transition step back from $i + 1$ to i): $\approx \text{Gamma}(a', b')$

To combine these into the full posterior $P(X_i = x_i \mid Y_{1:N} = y_{1:N})$, we use the result derived in Appendix D of the supplement:

$$P(x_i \mid y_{1:N}) \propto \hat{\alpha}(x_i) \cdot \frac{\int_0^\infty P(x_i \mid x_{i+1}) \tilde{\alpha}(x_{i+1}) dx_{i+1}}{P(x_i)}$$

The numerator involves the backward density propagated one step back through the transition. The denominator divides out the prior $P(x_i) = \text{Exp}(1) = \text{Gamma}(1, 1)$.

Substituting the gamma approximations:

$$\begin{aligned} P(x_i \mid y_{1:N}) &\propto x_i^{a-1} e^{-bx_i} \cdot x_i^{a'-1} e^{-b'x_i} / e^{-x_i} \\ &= x_i^{(a+a'-1)-1} e^{-(b+b'-1)x_i} \end{aligned}$$

This is the kernel of a $\text{Gamma}(a + a' - 1, b + b' - 1)$ distribution. Therefore:

$$X_i \mid Y_{1:N} \sim \text{Gamma}(a + a' - 1, b + b' - 1)$$

i Plain-language summary – Combining forward and backward

The forward pass says: “based on everything to the left on the chromosome, I think the TMRCA here is about X.” The backward pass says: “based on everything to the right, I think it’s about Y.” The combination merges these two independent sources of evidence into a single, more precise estimate. The subtraction of 1 from both parameters avoids counting the prior twice (both passes started from the same prior). The result is the **full posterior** at each position – the best estimate of the TMRCA given the entire chromosome.

i Where does the -1 come from?

The -1 in both $a + a' - 1$ and $b + b' - 1$ arises from dividing out the prior $\text{Gamma}(1, 1)$. The forward density $\text{Gamma}(a, b)$ and the backward density $\text{Gamma}(a', b')$ each include the prior once (they are posterior distributions that incorporate the prior). When we multiply them, the prior is counted twice. Dividing by the prior once gives the correct posterior, and dividing $\text{Gamma}(1, 1)$ out of a product of two gammas subtracts 1 from both the shape and rate parameters.

A useful sanity check: if we have no observations at all, the forward and backward densities are both the prior $\text{Gamma}(1, 1)$. Combining gives $\text{Gamma}(1 + 1 - 1, 1 + 1 - 1) = \text{Gamma}(1, 1)$, recovering the prior. This is correct – with no data, the posterior should equal the prior.

The Backward-Then-Combine Procedure

In practice, the combination works as follows:

1. Run the forward pass left-to-right. At each output position i , record (a_i, b_i) .
2. Run the forward pass right-to-left on the reversed sequence. At each output position $i + 1$, record (a''_{i+1}, b''_{i+1}) .
3. For each output position i , apply the flow field once to (a''_{i+1}, b''_{i+1}) to propagate the backward density from $i + 1$ back to i , giving (a'_i, b'_i) .
4. Combine: $(\alpha_{\text{post}}, \beta_{\text{post}}) = (a_i + a'_i - 1, b_i + b'_i - 1)$.

The additional flow field step in step 3 accounts for the transition from position $i + 1$ to position i – without it, the backward density would be aligned to position $i + 1$ instead of i .

```
def gamma_smc_posterior(observations, theta, rho, flow_field):
    """Compute the full Gamma-SMC posterior at each position.

    Runs forward and backward passes and combines them.

    Parameters
    -----
    observations : list of int
        Observation at each position: 1 (het), 0 (hom), -1 (missing).
    theta, rho : float
        Scaled mutation and recombination rates.
    flow_field : FlowField
        Precomputed flow field.

    Returns
    -----
    post_alpha : ndarray
        Posterior shape at each position.
    post_beta : ndarray
        Posterior rate at each position.
    """
    # Forward pass (left to right)
    a_fwd, b_fwd = gamma_smc_forward(observations, theta, rho, flow_field)

    # Backward pass = forward pass on reversed sequence
    a_bwd_rev, b_bwd_rev = gamma_smc_forward(
        observations[::-1], theta, rho, flow_field
    )
    # Reverse the backward results to align with original positions
    a_bwd = a_bwd_rev[::-1]
    b_bwd = b_bwd_rev[::-1]

    # Combine: Gamma(a + a' - 1, b + b' - 1)
    post_alpha = a_fwd + a_bwd - 1
    post_beta = b_fwd + b_bwd - 1
```

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```

    return post_alpha, post_beta

# Demonstrate on the same synthetic sequence
a_post, b_post = gamma_smc_posterior(obs, theta, rho, ZeroFlow())
mean_tmrca = a_post / b_post

# Sanity check: with no observations, posterior = prior Gamma(1,1)
a_empty, b_empty = gamma_smc_posterior([-1]*10, theta, rho, ZeroFlow())
print(f"No observations: Gamma({a_empty[0]:.1f}, {b_empty[0]:.1f}) "
      f"(should be ~Gamma(1, 1))")
print(f"\nWith data ({len(obs)} positions):")
print(f"  Position 0:  mean TMRCA = {mean_tmrca[0]:.2f}")
print(f"  Position 51 (het): mean TMRCA = {mean_tmrca[51]:.2f}")
print(f"  Position 99: mean TMRCA = {mean_tmrca[-1]:.2f}")

```

Validity of the Combination

The combination formula $\text{Gamma}(a + a' - 1, b + b' - 1)$ is valid only if:

- $a + a' - 1 > 0$, i.e., $a + a' > 1$
- $b + b' - 1 > 0$, i.e., $b + b' > 1$

Since the forward and backward passes both start from the prior $\text{Gamma}(1, 1)$ and only add to the shape (via heterozygous observations) and rate (via homozygous and heterozygous observations), one might expect $a \geq 1$ and $b \geq 1$ throughout. However, the flow field transition step can decrease both a and b (recombination introduces uncertainty, which can reduce the effective shape parameter).

If the gamma approximation is accurate and the entropy of the posterior never exceeds the entropy of the prior, then $b \geq 1$ is guaranteed. This is enforced by the **entropy clipping** mechanism described in [Segmentation and Caching](#). Without entropy clipping, it is possible (though rare) for the approximation to drift into a region where $b + b' < 1$, producing an invalid combined distribution.

Summary

The Gamma-SMC forward-backward algorithm:

1. **Forward pass:** sweep left-to-right, applying $\mathcal{F}$ (transition) then conjugate update (emission) at each position. Cost: $O(N)$.
2. **Backward pass:** run the forward algorithm on the reversed sequence. Cost: $O(N)$.
3. **Combination:** at each output position, combine forward $\text{Gamma}(a, b)$ and backward $\text{Gamma}(a', b')$ into $\text{Gamma}(a + a' - 1, b + b' - 1)$. Cost: $O(1)$ per output position.

Total cost: $O(N)$ for the entire genome, with no matrix operations and no time discretization. This makes Gamma-SMC orders of magnitude faster than PSMC for the same task.

Next: [Segmentation and Caching](#) – the engineering tricks that make the $O(N)$ algorithm fast in practice.

Segmentation and Caching

The regulator: the practical engineering that keeps the mechanism running fast and stable.

The previous chapters described the mathematical components of Gamma-SMC: the gamma approximation ([The Gamma Approximation](#)), the flow field ([The Flow Field](#)), and the forward-backward algorithm ([The Forward-Backward CS-HMM](#)).

In principle, these are sufficient to run inference. In practice, two engineering innovations make Gamma-SMC *ultrafast*: **segmentation with caching** (which skips over long stretches of identical observations) and **entropy clipping** (which prevents the gamma approximation from drifting into invalid parameter regions).

In a mechanical watch, the regulator is the mechanism that ensures the escapement beats at a consistent rate – not too fast, not too slow, even as the mainspring tension changes. In Gamma-SMC, segmentation and caching regulate the computational load (handling the many homozygous positions efficiently), and entropy clipping regulates the approximation quality (preventing drift that would produce nonsensical results).

Segmentation: Exploiting Sparsity

The human genome has roughly 1 heterozygous site per 1,000 base pairs. This means that 99.9% of positions are homozygous – long stretches of $Y_i = 0$ separated by rare $Y_i = 1$ observations. Processing each position individually would be wasteful: most of the work goes into applying the flow field and emission update at positions where the observation is always the same.

Biology Aside – Why genomic data is so sparse

Two randomly chosen human chromosomes differ at about 1 in every 1,000 nucleotides – a reflection of the relatively small effective population size of humans ($\sim 10,000$ - $20,000$) and the low per-base mutation rate ($\sim 1.2 \times 10^{-8}$ per generation). This means that for a 3-billion-base genome, only about 3 million positions are heterozygous in a typical diploid individual. The vast majority of the genome is identical between the two haplotypes. This extreme sparsity is a universal feature of within-species variation in most organisms (though the ratio varies – it is $\sim 1/500$ in *Drosophila* and $\sim 1/2,000$ in some plants). Gamma-SMC turns this biological fact into a computational advantage: instead of processing 3 billion positions, it processes only ~ 3 million segments.

Gamma-SMC exploits this sparsity by **segmenting** the genome: consecutive positions with the same observation type (homozygous or missing) are grouped into a single segment, which is handled by a single cached lookup.

A segment consists of:

1. A stretch of n_{miss} missing positions (from the genomic mask)
2. A stretch of n_{hom} homozygous positions
3. A single observation at the end (heterozygous or the last position in the segment)

The key insight is that the effect of n consecutive identical observations can be **precomputed**: if we know how (α, β) changes after one step of “flow field + hom emission,” then we can compute the effect of n such steps by repeated composition. This composition is precomputed for each grid point and cached.

```
import numpy as np

def segment_observations(observations):
    """Segment a sequence into (n_miss, n_hom, final_obs) tuples.

    Consecutive missing and homozygous positions are grouped together.
    Each segment ends at a heterozygous site or the end of the sequence.

    Parameters
    -----
    observations : list of int
        Observation at each position: 1 (het), 0 (hom), -1 (missing).

    Returns
```

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```

-----
segments : list of (int, int, int)
    Each tuple is (n_miss, n_hom, final_obs).
"""
segments = []
n_miss = 0
n_hom = 0

for y in observations:
    if y == -1: # missing
        n_miss += 1
    elif y == 0: # hom
        n_hom += 1
    else: # het (y == 1): close the current segment
        segments.append((n_miss, n_hom, 1))
        n_miss = 0
        n_hom = 0

# Final segment (may end with hom or missing)
if n_miss > 0 or n_hom > 0:
    segments.append((n_miss, n_hom, 0))

return segments

# Demonstrate: typical genomic pattern (sparse hets)
obs = [-1]*3 + [0]*50 + [1] + [0]*200 + [1] + [0]*100
segs = segment_observations(obs)
total_positions = sum(nm + nh + (1 if fo == 1 else 0) for nm, nh, fo in segs)
print(f"Sequence: {len(obs)} positions -> {len(segs)} segments")
for i, (nm, nh, fo) in enumerate(segs):
    label = "het" if fo == 1 else "hom"
    print(f"Segment {i}: {nm} miss, {nh} hom, ends with {label}")
print(f"Speedup: {len(obs)}/{len(segs)} = {len(obs)/len(segs):.0f}x")

```

Locus Skipping via Caching

The caching strategy works as follows:

Preprocessing (per parameter set). For each element of the flow field grid (l_μ, l_C) and for each number of steps k from 1 to a maximum cache size K :

1. Starting from (l_μ, l_C) , apply the flow field once (transition), then apply the homozygous emission update (add θ to β). Record the result.
2. Repeat step 1 from the result, for a total of k times. Record the final $(l_\mu^{(k)}, l_C^{(k)})$.
3. Do the same for missing observations (apply the flow field once but *skip* the emission update, since missing observations carry no information).

This produces two sets of cached flow fields:

- $\mathcal{F}_{\text{hom}}^{(k)}(l_\mu, l_C)$: the result of k consecutive hom steps
- $\mathcal{F}_{\text{miss}}^{(k)}(l_\mu, l_C)$: the result of k consecutive missing steps

Inference. During the forward pass, when encountering a segment with n_{miss} missing positions and n_{hom} homozygous positions:

1. Look up $\mathcal{F}_{\text{miss}}^{(n_{\text{miss}})}$ at the current (l_μ, l_C) to skip all missing positions at once.
2. Look up $\mathcal{F}_{\text{hom}}^{(n_{\text{hom}})}$ at the resulting (l_μ, l_C) to skip all homozygous positions at once.
3. Apply the heterozygous emission update at the end of the segment.

Each segment, regardless of how many positions it spans, is handled by **two cache lookups and one emission update** – $O(1)$ per segment.

Parameter dependence of caches

Unlike the flow field $\mathcal{F}$ (which is parameter-independent), the caches depend on θ and ρ :

- θ enters through the emission step (hom adds θ to β).
- ρ enters because the flow field displacement is scaled by ρ (recall $\alpha' = \alpha + u \cdot \rho$).

Therefore, the caches must be **recomputed** whenever θ or ρ changes. However, this recomputation is fast (it amounts to composing known flow field lookups) and is done as a preprocessing step before the forward pass.

Handling Long Segments

When n_{hom} or n_{miss} exceeds the maximum cache size K , the segment is processed in chunks: apply the K -step cache, then handle the remainder recursively. For example, a stretch of 150 homozygous positions with $K = 100$ would be handled as one 100-step cache lookup followed by one 50-step cache lookup.

Entropy Clipping

The gamma approximation at the transition step is imperfect. Over many consecutive transition steps (long homozygous or missing stretches), small errors can accumulate, causing the gamma parameters to drift into regions where the approximation breaks down.

The specific failure mode is an **entropy increase**: the differential entropy of the gamma posterior exceeds the entropy of the prior. This is unphysical – observing data should reduce uncertainty, not increase it.

Plain-language summary – What entropy clipping prevents

Entropy measures how spread out (uncertain) a probability distribution is. The prior – our belief before seeing any data – has a certain amount of uncertainty. After observing data, we should know *more*, not less. If the gamma approximation accumulates small errors over thousands of homozygous positions, those errors can push the posterior into an impossible state where it is *more uncertain than the prior*. Entropy clipping catches this before it happens: whenever the posterior becomes too diffuse, it is pulled back to the edge of the valid region. The mean TMRCA estimate is preserved (we don't change our best guess), but the excess uncertainty is trimmed away. If the entropy were allowed to exceed the prior entropy, the combination of forward and backward passes could produce invalid gamma parameters (specifically, $b + b' - 1 < 0$, which is not a valid rate parameter).

The entropy bound. The differential entropy of $\text{Gamma}(\alpha, \beta)$ is:

$$h(\alpha, \beta) = \alpha - \ln \beta + \ln \Gamma(\alpha) + (1 - \alpha)\psi(\alpha)$$

For the exponential prior $\text{Gamma}(1, 1)$, this evaluates to:

$$h(1, 1) = 1$$

Any legitimate posterior must have $h(\alpha, \beta) \leq 1$. When this bound is violated after a transition step, Gamma-SMC **clips** the posterior back to the boundary of the valid region.

The clipping rule. When entropy exceeds the threshold:

1. Fix the **mean** $\mu_T = \alpha/\beta$ (the best estimate of the TMRCA should not change).
2. **Reduce the CV** until the entropy equals the threshold.

In (l_μ, l_C) coordinates, this means keeping l_μ fixed and reducing l_C to the maximum value consistent with $h(l_\mu, l_C) = 1$. Since the entropy is monotonically increasing in l_C (for fixed l_μ and $l_C \leq 0$), the maximum valid l_C can be found by bisection.

In practice, this maximum is precomputed over a grid of l_μ values and interpolated during inference.

```
from scipy.special import gammaln, digamma

def gamma_entropy(alpha, beta):
    """Differential entropy of Gamma(alpha, beta).

    Parameters
    -----
    alpha : float
        Shape parameter (alpha >= 1).
    beta : float
        Rate parameter.

    Returns
    -----
    h : float
        Differential entropy.
    """
    return (alpha - np.log(beta) + gammaln(alpha)
            + (1 - alpha) * digamma(alpha))

def entropy_clip(alpha, beta, h_max=1.0, tol=1e-8):
    """Clip gamma parameters so that entropy does not exceed h_max.

    Keeps the mean alpha/beta fixed and reduces the CV (increases alpha)
    until the entropy equals h_max.

    Parameters
    -----
    alpha, beta : float
        Current gamma parameters.
    h_max : float
        Maximum allowed entropy (default 1.0, the prior entropy).
    tol : float
        Bisection tolerance.

    Returns
    -----
    alpha_clipped, beta_clipped : float
        Clipped parameters with h <= h_max.
    """
    if gamma_entropy(alpha, beta) <= h_max:
```

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```

    return alpha, beta # no clipping needed

mean = alpha / beta # fix the mean

# Bisect on alpha: increasing alpha decreases entropy (for fixed mean)
lo, hi = alpha, 1e6
for _ in range(100):
    mid = (lo + hi) / 2
    b_mid = mid / mean
    if gamma_entropy(mid, b_mid) > h_max:
        lo = mid
    else:
        hi = mid
    if hi - lo < tol:
        break

alpha_new = (lo + hi) / 2
beta_new = alpha_new / mean
return alpha_new, beta_new

# Verify: prior Gamma(1, 1) has entropy exactly 1
print(f"Entropy of Gamma(1, 1) = {gamma_entropy(1.0, 1.0):.6f}")
print(f"Entropy of Gamma(5, 5) = {gamma_entropy(5.0, 5.0):.6f}")

# Simulate a case where entropy exceeds the threshold
# Gamma(1.01, 0.8) has entropy > 1 (beta < 1 means too diffuse)
a, b = 1.01, 0.8
h_before = gamma_entropy(a, b)
a_clip, b_clip = entropy_clip(a, b)
h_after = gamma_entropy(a_clip, b_clip)
print(f"\nBefore clip: Gamma({a}, {b}), entropy = {h_before:.4f}")
print(f"After clip: Gamma({a_clip:.4f}, {b_clip:.4f}), "
      f"entropy = {h_after:.4f}")
print(f"Mean preserved: {a/b:.4f} -> {a_clip/b_clip:.4f}")

```

i Why does entropy clipping guarantee valid combination?

If $h(\alpha, \beta) \leq 1$ and $\alpha \geq 1$, then $\beta \geq 1$. This follows from the monotonicity of the entropy in α :

$$\frac{\partial h}{\partial \alpha} = 1 + (1 - \alpha)\psi^{(1)}(\alpha) > 0$$

where $\psi^{(1)}$ is the trigamma function (always positive). So entropy is increasing in α . Setting $\alpha = 1$ gives $h = 1 - \ln \beta$, and requiring $h \leq 1$ gives $\beta \geq 1$.

Therefore, both the forward and backward distributions satisfy $\beta \geq 1$, guaranteeing $b + b' - 1 \geq 1 > 0$. The combined posterior $\text{Gamma}(a + a' - 1, b + b' - 1)$ is always valid.

The Complete Forward Pass Algorithm

Putting it all together, the Gamma-SMC forward pass for a single pair of haplotypes is:

```
Initialize: (l_mu, l_C) = (0, 0) [prior: Gamma(1, 1)]

For each segment (n_miss, n_hom, final_obs):
  1. Look up cached miss flow: (l_mu, l_C) <- F_miss^(n_miss)(l_mu, l_C)
  2. Clip if entropy exceeds threshold
  3. Look up cached hom flow: (l_mu, l_C) <- F_hom^(n_hom)(l_mu, l_C)
  4. Clip if entropy exceeds threshold
  5. Apply emission update for final_obs:
    - If het: alpha += 1, beta += theta
    - If hom: beta += theta
    - If missing: no change
  6. Convert back to (l_mu, l_C)
  7. If this is an output position, record (l_mu, l_C)
```

At steps 1 and 3, if n_{miss} or n_{hom} is 0, the step is skipped. If it exceeds the cache size K , it is handled in chunks.

The backward pass uses the same algorithm on the reversed sequence, with an additional flow field step when aligning backward densities to output positions (see [The Forward-Backward CS-HMM](#)).

```
def gamma_smc_forward_segmented(observations, theta, rho, flow_field,
                                h_max=1.0):
    """Gamma-SMC forward pass with segmentation and entropy clipping.

    This is the full algorithm combining all four components:
    segmentation, caching (via repeated flow field application),
    flow field queries, and entropy clipping.

    Parameters
    -----
    observations : list of int
        1 (het), 0 (hom), -1 (missing) at each position.
    theta : float
        Scaled mutation rate.
    rho : float
        Scaled recombination rate.
    flow_field : object
        Flow field with .query(l_mu, l_C) method.
```

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```

h_max : float
    Entropy threshold for clipping.

Returns
-----
results : list of (float, float)
    (alpha, beta) at each het position.
"""
segments = segment_observations(observations)
alpha, beta = 1.0, 1.0 # prior: Gamma(1, 1)
results = []

for n_miss, n_hom, final_obs in segments:
    # Step 1: skip missing positions (flow field only, no emission)
    for _ in range(n_miss):
        l_mu = np.log10(alpha / beta)
        l_C = np.log10(1.0 / np.sqrt(alpha))
        dl_mu, dl_C = flow_field.query(l_mu, l_C)
        l_mu += rho * dl_mu
        l_C += rho * dl_C
        alpha = 10.0 ** (-2 * l_C)
        beta = alpha * 10.0 ** (-l_mu)

    # Step 2: entropy clip after missing block
    alpha, beta = entropy_clip(alpha, beta, h_max)

    # Step 3: skip homozygous positions (flow field + hom emission)
    for _ in range(n_hom):
        l_mu = np.log10(alpha / beta)
        l_C = np.log10(1.0 / np.sqrt(alpha))
        dl_mu, dl_C = flow_field.query(l_mu, l_C)
        l_mu += rho * dl_mu
        l_C += rho * dl_C
        alpha = 10.0 ** (-2 * l_C)
        beta = alpha * 10.0 ** (-l_mu)
        beta += theta # hom emission

    # Step 4: entropy clip after hom block
    alpha, beta = entropy_clip(alpha, beta, h_max)

    # Step 5: emission update for the final observation
    if final_obs == 1: # het
        alpha += 1
        beta += theta
    elif final_obs == 0: # trailing hom (end of sequence)
        beta += theta

    results.append((alpha, beta))

return results

# Demonstrate on a realistic-scale pattern

```

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```

np.random.seed(42)
n_sites = 10000
obs = [0] * n_sites
het_positions = sorted(np.random.choice(n_sites, size=10, replace=False))
for pos in het_positions:
    obs[pos] = 1

class ZeroFlow:
    def query(self, l_mu, l_C): return 0.0, 0.0

segments = segment_observations(obs)
results = gamma_smc_forward_segmented(obs, 0.001, 0.0004, ZeroFlow())
print(f"Input: {n_sites} positions, {sum(obs)} hets")
print(f"Segments: {len(segments)} (one per het + final)")
print(f"Speedup: {n_sites}/{len(segments)} = "
      f"{n_sites/len(segments):.0f}x fewer operations")
a_final, b_final = results[-1]
print(f"Final posterior: Gamma({a_final:.1f}, {b_final:.4f}), "
      f"mean = {a_final/b_final:.2f}")

```

Performance Characteristics

The segmentation and caching strategy gives Gamma-SMC its performance advantage:

Operation	Cost per segment	Cost per position
Cache lookup (miss)	$O(1)$ (bilinear interpolation)	N/A (amortized over n_{miss})
Cache lookup (hom)	$O(1)$ (bilinear interpolation)	N/A (amortized over n_{hom})
Emission update (het)	$O(1)$ (parameter increment)	$O(1)$
Entropy clip	$O(1)$ (table lookup + interpolation)	N/A (amortized)

The total cost is proportional to the **number of segments**, not the number of genomic positions. Since the number of segments equals the number of heterozygous sites (plus a small number from mask boundaries), the effective cost is $O(H)$ where H is the number of heterozygous sites. For typical human data, $H \approx N/1000$, giving a $\sim 1000\times$ speedup over a naive per-position algorithm.

Summary

The segmentation and caching system is the engineering that makes Gamma-SMC practical:

- **Segmentation** groups consecutive identical observations into segments that can be processed as a unit.
- **Caching** precomputes the effect of multi-step transitions, reducing each segment to two cache lookups.
- **Entropy clipping** prevents approximation drift, guaranteeing that the forward-backward combination always produces a valid gamma posterior.

These three mechanisms – the regulator of the Gamma-SMC watch – transform the $O(N)$ forward pass into an $O(H)$ algorithm that is ultrafast in practice.

Recap: The Complete Gamma-SMC Timepiece

You have now disassembled and rebuilt every gear in the Gamma-SMC mechanism across four chapters:

- **The Gamma Approximation** (*The Gamma Approximation*): Poisson-gamma conjugacy for exact emission updates, and PDE-based gamma projection for approximate transition updates. This was the escapement – the mathematical insight that makes the continuous-state approach possible.
- **The Flow Field** (*The Flow Field*): A precomputed 2D vector field that encodes the transition dynamics, evaluated once and reused for any dataset. This was the gear train – the precision-machined transmission.
- **The Forward-Backward CS-HMM** (*The Forward-Backward CS-HMM*): The forward algorithm sweeps left-to-right, the backward algorithm is a forward pass on the reversed sequence, and the combination gives $\text{Gamma}(a + a' - 1, b + b' - 1)$. This was the mainspring – the engine that drives inference.
- **Segmentation and Caching** (this chapter): Locus skipping, cached multi-step lookups, and entropy clipping that make the algorithm ultrafast and numerically stable. This was the regulator – the engineering that keeps the mechanism running smoothly.

Gamma-SMC demonstrates that the pairwise TMRCA inference problem – the same problem PSMC solves with discretized time and iterative EM – can be solved in a single forward-backward pass with $O(1)$ cost per position, no time discretization, and no parameter iteration. The trade-off is that Gamma-SMC assumes constant population size and provides per-site posteriors rather than a demographic history. But as a computational mechanism, it is remarkably elegant: two numbers (α, β) glide continuously through a precomputed flow field, accumulating evidence at each position, until the full posterior emerges at every site along the genome.

The clock ticks continuously. And you can read the time at every position.

Demo: Running Gamma-SMC on Simulated Data

Running `mini_gamma_smc` on simulated genomic data.

This figure was generated by running the `mini_gamma_smc` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_gamma_smc.py`.

3.15 Timepiece XIV: PHLASH

Population History Learning by Averaging Sampled Histories

3.15.1 The Mechanism at a Glance

PHLASH is a **Bayesian** method for inferring population size history $N(t)$ from whole-genome sequencing data. It is the successor to PSMC (*Timepiece I*), inheriting PSMC's coalescent HMM framework but extending it in four fundamental ways:

1. It combines **two sources of information** – the site frequency spectrum (SFS) from many individuals *and* the pairwise coalescent HMM from diploid genomes – into a single composite likelihood.
2. It uses **random time discretizations** whose biases cancel when averaged, eliminating the systematic error that plagues fixed-grid approaches.
3. It computes gradients with a novel **score function algorithm** that is 30–90x faster than automatic differentiation.
4. It samples from the **posterior distribution** over demographic histories using Stein Variational Gradient Descent (SVGD), a GPU-parallel algorithm that produces uncertainty estimates rather than point estimates.

Input: Whole-genome data from one or more diploid individuals

Output: Posterior distribution over $N(t)$ (population size as a function of time)

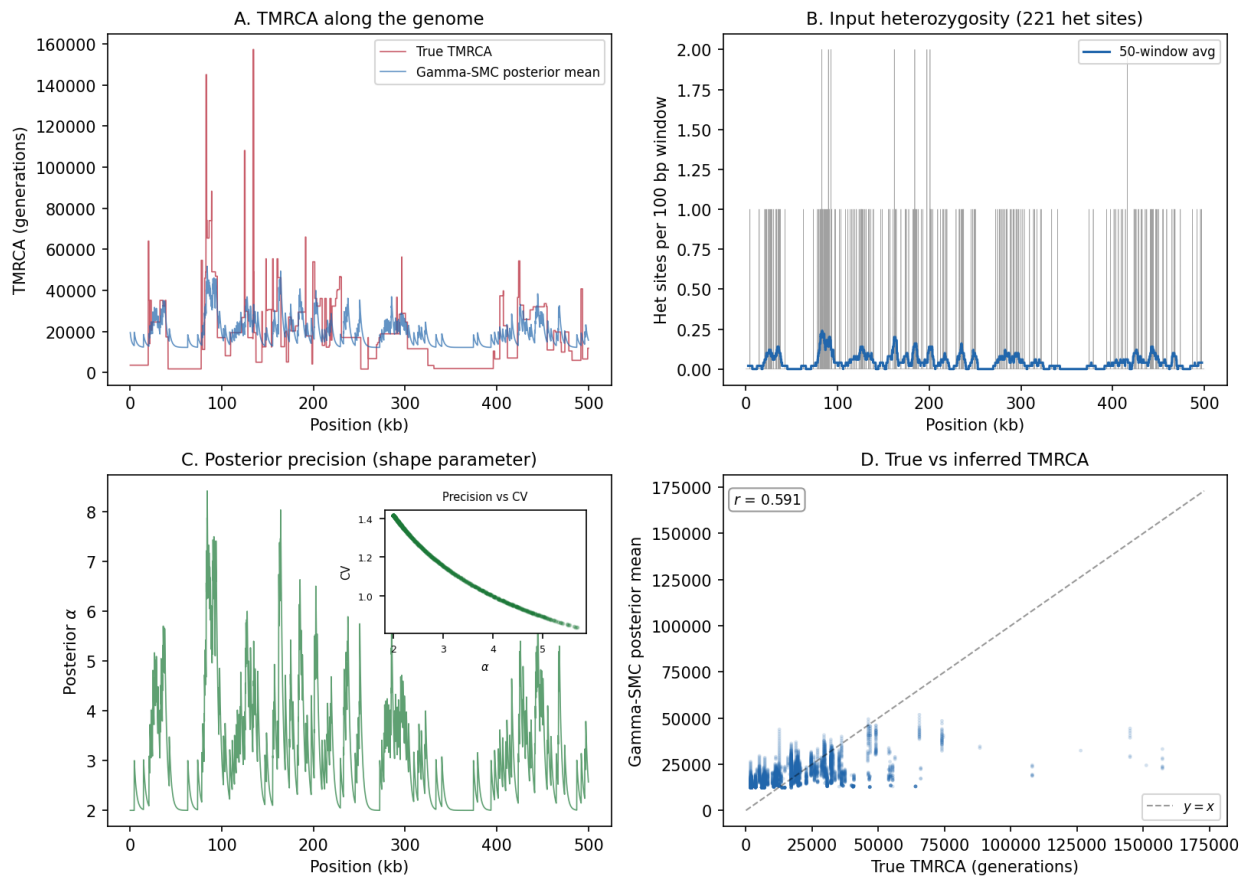
Demo: Gamma-SMC on msprime Diploid Genome (500 kb, bottleneck history)

Fig. 19: **Gamma-SMC demo on msprime-simulated data.** Panel A: True TMRCA vs Gamma-SMC posterior mean along the genome – the conjugate Bayesian filter tracks coalescence time changes. Panel B: Input heterozygosity track from a diploid genome simulated under a bottleneck. Panel C: Posterior precision (shape parameter) along the genome, with inset showing the relationship to coefficient of variation. Panel D: True vs inferred TMRCA scatter plot.

Where PSMC is a two-hand watch – two haplotypes, one HMM, a point estimate – PHLASH is a **grand complication**: multiple data sources, randomized internal calibration, a custom-built gear train for fast gradient computation, and a Bayesian mechanism that reports not just the time but the *uncertainty* in the time. If PSMC tells you “the population was 50,000 at that epoch,” PHLASH tells you “the population was 50,000 with a 95% credible interval of 30,000–80,000.”

i Primary Reference

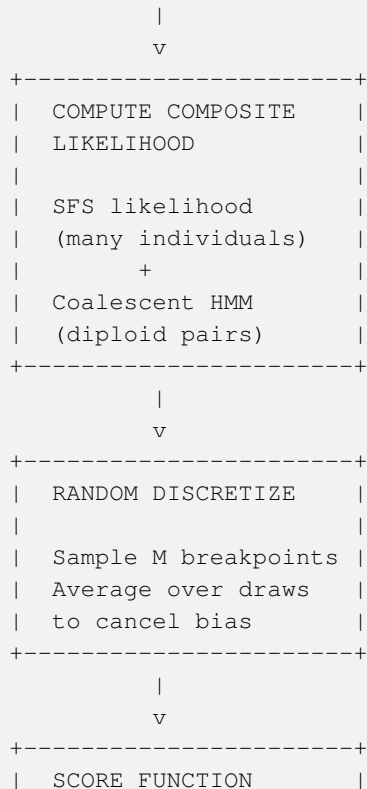
[Terhorst, 2025]

The four gears of PHLASH:

1. **The Composite Likelihood** (the mainspring) – A two-part likelihood that combines the site frequency spectrum (SFS) from many individuals with the coalescent HMM likelihood from diploid pairs (inherited from PSMC). Two independent views of the same demographic history, fused into a single objective.
2. **Random Time Discretization** (the tourbillon) – Randomized time breakpoints whose discretization biases cancel when averaged across many draws, like a tourbillon rotating the escapement to cancel positional errors in a mechanical watch.
3. **The Score Function Algorithm** (the gear train) – An $O(LM^2)$ algorithm for computing the gradient of the log-likelihood, 30–90x faster than reverse-mode automatic differentiation. This is what makes SVGD practical at genomic scale.
4. **Stein Variational Gradient Descent** (the winding mechanism) – A GPU-parallel posterior sampling algorithm that maintains a set of “particles” (candidate demographic histories) and iteratively pushes them toward the posterior distribution using gradient information and a repulsive kernel that maintains diversity.

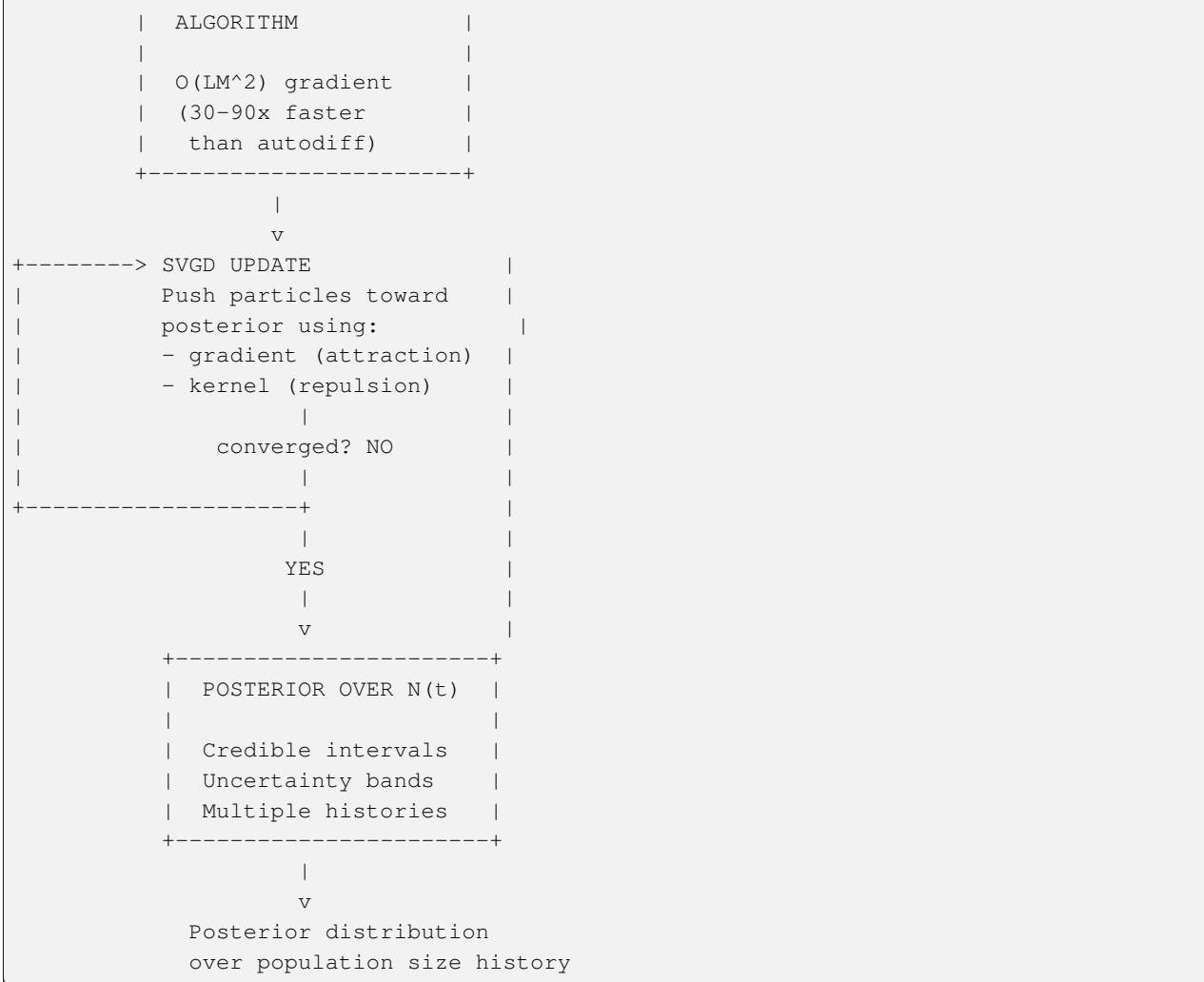
These gears mesh together into a complete Bayesian inference machine:

Whole-genome data (one or more diploid individuals)



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i Prerequisites for this Timepiece

Before starting PHLASH, you should have worked through:

- *Coalescent Theory* – coalescence times and their dependence on population size
- *Hidden Markov Models* – the forward-backward algorithm
- *The SMC* – the Markov approximation to the coalescent with recombination
- *PSMC (Timepiece I)* – the coalescent HMM that PHLASH inherits and extends
- *The Frequency Spectrum* – the SFS and its relationship to demographic history (from the moments Timepiece)
- Basic Bayesian inference (prior, likelihood, posterior) – introduced inline as needed

3.15.2 Chapters

Overview of PHLASH

Before assembling the watch, lay out every part and understand what it does.

What Does PHLASH Do?

Given whole-genome sequencing data from one or more diploid individuals, PHLASH infers the **posterior distribution** over the population size history $N(t)$.

Input: Whole-genome data from n diploid individuals

Output: $p(\eta \mid \text{data})$, $\eta(t)$ = population size history

Unlike PSMC, which returns a single best estimate of $N(t)$, PHLASH returns a **distribution** over possible histories – a cloud of plausible trajectories rather than a single line. Each trajectory in the cloud is consistent with the data; the spread of the cloud tells you how certain or uncertain the inference is.

Think of it this way. PSMC is a watch that shows you a single reading: “It is 3:17.” PHLASH is a watch that shows you a distribution: “It is 3:17 give or take two minutes, and here is a plot showing the probability density over all possible times.” Both tell time; PHLASH also tells you how well it is keeping time.

Why a Successor to PSMC?

PSMC (*Timepiece I*) demonstrated that a single diploid genome encodes deep demographic history. But it has three well-known limitations:

1. **No uncertainty quantification.** PSMC uses the EM algorithm, which finds a point estimate (a single maximum-likelihood history). The standard approach to uncertainty – bootstrapping over genomic blocks – is computationally expensive and can underestimate true uncertainty because blocks are not truly independent.
2. **Discretization bias.** PSMC divides time into fixed intervals and estimates a constant population size within each interval. The choice of intervals affects the result: too few intervals miss real demographic events, while the spacing pattern (typically log-spaced) introduces systematic bias because the integral approximation errors do not cancel across a fixed grid.
3. **Limited data use.** PSMC analyzes one diploid genome at a time. When multiple genomes are available, it cannot easily combine their information. The pairwise approach (PSMC²) extends to pairs of individuals but does not scale to many samples.

PHLASH addresses all three:

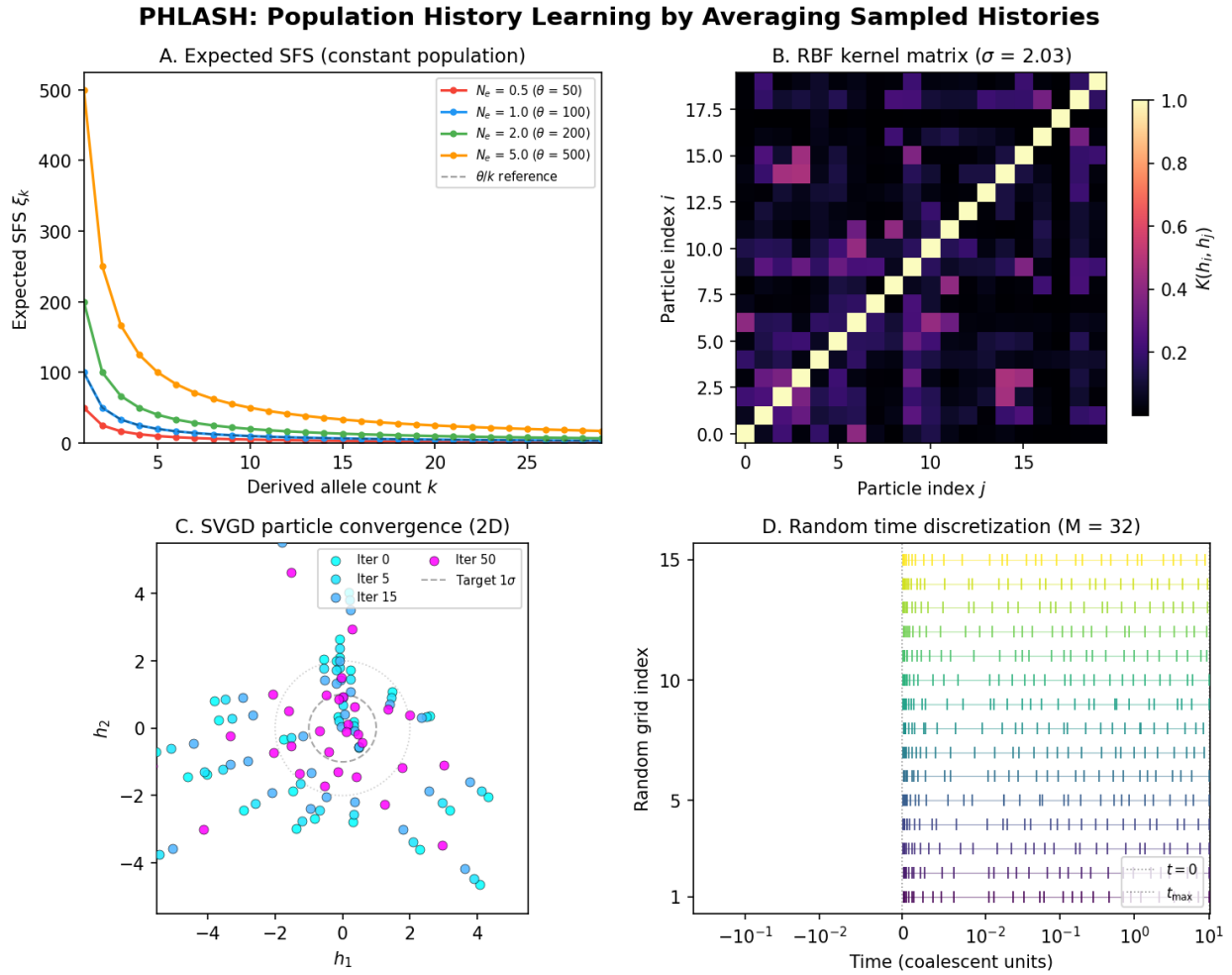


Fig. 20: **PHLASH at a glance.** The four pillars of the PHLASH inference pipeline: composite SFS likelihood connecting data to demography, the RBF kernel measuring inter-particle similarity for SVGD, particle evolution under Stein Variational Gradient Descent, and random time discretisation for debiased gradient estimation.

Limitation	PSMC	PHLASH
Uncertainty	Point estimate + bootstrap	Full posterior via SVGD
Discretization	Fixed grid, systematic bias	Random grids, bias cancels on average
Data sources	One diploid (pairwise HMM only)	SFS (many individuals) + pairwise HMM
Optimization	EM (local maximum, slow convergence)	SVGD (posterior sampling, GPU-parallel)
Gradients	Not needed (EM uses closed-form updates)	Score function algorithm, $O(LM^2)$

How PHLASH Extends the Coalescent HMM

At its core, PHLASH still uses the same coalescent HMM as PSMC. Recall from *Timepiece I*: the hidden states are discretized coalescence time intervals, the observations are heterozygous/homozygous bins, and the transition matrix encodes how the TMRCA changes between adjacent positions under the SMC approximation.

PHLASH inherits this entire machinery but wraps it in a larger framework:

- The **coalescent HMM likelihood** is computed for each pair of haplotypes (or each diploid individual), exactly as in PSMC. This gives a likelihood $\ell_{\text{HMM}}(\eta)$ for a demographic history η .
- The **SFS likelihood** is computed from the allele frequency distribution across all sampled individuals. This gives a second likelihood $\ell_{\text{SFS}}(\eta)$ that captures information from many individuals simultaneously.
- The two likelihoods are combined into a **composite likelihood**:

$$\ell_{\text{comp}}(\eta) = \ell_{\text{SFS}}(\eta) + \ell_{\text{HMM}}(\eta)$$

(on the log scale, so the composite log-likelihood is a sum). This composite likelihood is then used as the basis for Bayesian inference, combined with a smoothness prior on η .

The composite likelihood is the **mainspring** of PHLASH – the energy source that drives the entire mechanism. But turning the mainspring requires computing gradients efficiently, which is where the score function algorithm comes in. And the gradients must be computed on a discretized time grid, which is where random discretization cancels the bias. And the gradients drive SVGD, which produces the posterior. Every gear depends on the one before it.

The Demographic Model

PHLASH parameterizes the population size history as a **piecewise-constant function** on a set of time intervals, just as PSMC does:

$$\eta(t) = \eta_k \quad \text{for } t \in [t_k, t_{k+1})$$

But unlike PSMC, the time breakpoints $t_0 < t_1 < \dots < t_M$ are not fixed across the analysis. Instead, PHLASH **randomly samples** the breakpoints for each gradient computation. Different random grids yield different discretization errors, but these errors have zero mean: they cancel when averaged across draws.

The vector of population sizes $\eta = (\eta_0, \dots, \eta_{M-1})$ lives in a high-dimensional space. PHLASH places a **smoothness prior** on η – a Gaussian process prior that penalizes histories with large jumps between adjacent epochs. This prior acts as a regularizer, preventing overfitting to noise while allowing the data to drive the inference.

Parameters

Parameter	Symbol	Physical meaning
Population sizes	$\eta_0, \dots, \eta_{M-1}$	Effective population size in each time interval. The primary quantities of interest – the posterior distribution over these values is the output.
Mutation rate	θ	Population-scaled mutation rate, inherited from PSMC's parameterization. Controls the density of heterozygous sites per unit coalescence time.
Recombination rate	ρ	Population-scaled recombination rate. Controls how frequently the genealogy changes between adjacent positions.
Time breakpoints	$t_0, \dots, t_M$	Boundaries of the piecewise-constant intervals. Randomly sampled in each gradient computation step; not treated as fixed parameters.
Number of particles	J	The number of candidate histories maintained by SVGD. More particles give a better approximation to the posterior but cost more computation.
Smoothness prior	Σ	The covariance structure of the Gaussian process prior on $\log \eta$. Controls how much the inferred history is allowed to fluctuate between adjacent time intervals.

The Flow in Detail

Step 1: Data preparation

From whole-genome data, compute:

- (a) SFS from all individuals
- (b) Heterozygosity sequences for each diploid individual (same binning as PSMC)

Step 2: Initialize particles

Draw J initial demographic histories from the prior.

Each particle is a vector $\eta = (\eta_0, \dots, \eta_{M-1})$.

Step 3: For each SVGD iteration:

- (a) Sample a random time discretization $t_0 < t_1 < \dots < t_M$
- (b) For each particle j :
 - Compute SFS log-likelihood given $\eta^{(j)}$
 - Compute coalescent HMM log-likelihood given $\eta^{(j)}$
 - Sum to get composite log-likelihood
 - Compute gradient via score function algorithm
 - Add gradient of log-prior (smoothness penalty)
- (c) Compute SVGD update:
 - Attraction: gradient of log-posterior pushes particles toward high-probability regions
 - Repulsion: kernel gradient pushes particles apart to maintain diversity
- (d) Update all particles in parallel (on GPU)

Step 4: After convergence:

The J particles approximate the posterior distribution.

Compute credible intervals, posterior mean, uncertainty bands.

Comparison to Related Methods

Property	PSMC	SMC++	MSMC2	PHLASH
Data	1 diploid	SFS + distinguished pair	Up to 8 haplotypes	SFS + diploid pairs
Inference	EM (point estimate)	EM (point estimate)	EM (point estimate)	SVGD (posterior)
Discretization	Fixed grid	Fixed grid	Fixed grid	Random grid (bias cancels)
Gradients	Not needed	Autodiff	Not needed	Score function ($O(LM^2)$)
GPU support	No	No	No	Yes (JAX)
Uncertainty	Bootstrap	Bootstrap	Bootstrap	Posterior samples

Ready to Build

The next four chapters disassemble PHLASH gear by gear:

- *The Composite Likelihood* – **The mainspring**: how the SFS likelihood and the coalescent HMM likelihood are computed and combined into a single composite objective.
- *Random Time Discretization* – **The tourbillon**: how random time breakpoints cancel discretization bias, turning a systematic error into zero-mean noise that vanishes under averaging.
- *The Score Function Algorithm* – **The gear train**: the $O(LM^2)$ algorithm that computes the gradient of the log-likelihood 30–90x faster than autodiff, making SVGD practical at genomic scale.
- *Stein Variational Gradient Descent (SVGD)* – **The winding mechanism**: how SVGD uses gradients and a repulsive kernel to push particles toward the posterior distribution, producing uncertainty estimates on a GPU.

Each chapter derives the math, explains the intuition, and connects back to the PSMC foundations from *Timepiece I*.

Let us start with the foundation: the composite likelihood.

The Composite Likelihood

The mainspring: two sources of energy, wound into a single spring.

This chapter derives the two components of PHLASH's composite likelihood – the site frequency spectrum (SFS) likelihood and the coalescent HMM likelihood – and explains how they are combined into a single objective function for inference.

The composite likelihood is the mainspring of the PHLASH watch. In horology, the mainspring stores the energy that powers the entire mechanism. In PHLASH, the composite likelihood provides the statistical “energy” – the information from the data – that drives the posterior sampling. Without it, the gears have nothing to turn.

Two Views of the Same History

A demographic history $\eta(t)$ – the population size as a function of time – leaves its fingerprints in the genome in two distinct ways:

1. **The site frequency spectrum** captures how allele frequencies are distributed across many individuals. A population bottleneck, for example, produces an excess of rare variants (because most variation arose after the recovery). The SFS is a summary statistic computed from the full sample: it counts, for each possible frequency k/n , how many polymorphic sites have exactly k copies of the derived allele in n sampled chromosomes.
2. **The pairwise coalescent HMM** captures how the coalescence time varies along the genome for a single diploid individual (or a pair of haplotypes). This is exactly the PSMC model from *Timepiece I*: heterozygous sites mark

regions of deep coalescence, homozygous stretches mark regions of recent coalescence, and the transitions between them reflect recombination.

These two views are **complementary**. The SFS aggregates information across many individuals at each site, providing a snapshot of the frequency distribution. The coalescent HMM traces correlations *along the genome* for a single pair, using the linkage structure to infer how coalescence times change from position to position. Neither view alone captures all the information in the data; together, they constrain the demographic history more tightly than either one alone.

i Biology Aside – Why two data sources are better than one

Consider trying to date a population bottleneck. The SFS tells you that there is an excess of rare variants (consistent with recent recovery from a small population), but it does not pinpoint *when* the bottleneck occurred – many different bottleneck timings can produce similar SFS shapes. The coalescent HMM adds timing information: a bottleneck forces coalescence times into a narrow band, which creates long stretches of homozygosity bounded by heterozygous transitions at characteristic distances. The spacing of these transitions depends on when the bottleneck occurred and how severe it was. By combining both signals, PHLASH can simultaneously estimate the timing, severity, and duration of demographic events with far more precision than either signal alone.

Part 1: The SFS Likelihood

The site frequency spectrum was introduced in detail in *The Frequency Spectrum* (from the moments Timepiece). Here we summarize the key result and show how PHLASH uses it.

Given n sampled haploid chromosomes and a demographic history $\eta(t)$, the expected SFS is a vector $\xi = (\xi_1, \dots, \xi_{n-1})$ where ξ_k is the expected number of segregating sites with k derived alleles.

Each entry of the expected SFS can be written as an integral over coalescent branch lengths:

$$\xi_k = L\mu \sum_{j=2}^n \binom{n-k-1}{j-2} \binom{k-1}{0} \cdot \frac{1}{\binom{n-1}{j-1}} \cdot \mathbb{E}[T_j]$$

where T_j is the total branch length when there are j lineages, which depends on the demographic history. For piecewise-constant $\eta(t)$, these expected branch lengths have closed-form expressions involving the coalescence rates $\binom{j}{2}/\eta_k$ in each time interval.

Given the observed SFS $\mathbf{D} = (D_1, \dots, D_{n-1})$ (the actual counts from the data) and the expected SFS $\xi(\eta)$ (computed under a candidate history η), the SFS log-likelihood under a Poisson model is:

$$\ell_{\text{SFS}}(\eta) = \sum_{k=1}^{n-1} [D_k \log \xi_k(\eta) - \xi_k(\eta)] + \text{const}$$

This is a standard Poisson log-likelihood: each SFS entry D_k is modeled as an independent Poisson random variable with mean ξ_k . The constant term (involving $\log D_k!$) does not depend on η and can be dropped from optimization.

```
import numpy as np

def sfs_log_likelihood(observed_sfs, expected_sfs):
    """Poisson log-likelihood of the observed SFS given expected SFS.

    Parameters
    -----
    observed_sfs : ndarray, shape (n-1,)
        Observed SFS counts  $D_k$  for  $k = 1, \dots, n-1$ .
    expected_sfs : ndarray, shape (n-1,)
```

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```

    Expected SFS entries xi_k under the demographic model.

Returns
-----
ll : float
    Poisson log-likelihood (up to a constant).
"""
# Avoid log(0) for zero-expected entries
xi = np.maximum(expected_sfs, 1e-300)
return np.sum(observed_sfs * np.log(xi) - xi)

def expected_sfs_constant(n, theta, N_e=1.0):
    """Expected SFS under constant population size.

    Under the standard coalescent with constant N_e, the expected
    number of segregating sites at frequency k/n is proportional
    to 1/k (Watterson's result).

Parameters
-----
n : int
    Number of haploid chromosomes.
theta : float
    Population-scaled mutation rate (4 * N_e * mu * L).
N_e : float
    Effective population size (default 1.0 in coalescent units).

Returns
-----
xi : ndarray, shape (n-1,)
    Expected SFS.
"""
k = np.arange(1, n)
return theta / k

# Demonstrate: constant-size SFS likelihood
n = 20
theta = 100.0 # realistic total theta for a genomic region
xi_expected = expected_sfs_constant(n, theta)
# Simulate observed SFS by Poisson sampling
np.random.seed(42)
D_observed = np.random.poisson(xi_expected)
ll = sfs_log_likelihood(D_observed, xi_expected)
print(f"Sample size n = {n}, theta = {theta}")
print(f"Expected SFS (first 5): {xi_expected[:5].round(2)}")
print(f"Observed SFS (first 5): {D_observed[:5]}")
print(f"SFS log-likelihood: {ll:.2f}")

```

Why Poisson?

The Poisson model for the SFS arises naturally from the infinite-sites mutation model: mutations occur as a Poisson

process on the branches of the genealogy, so the number of sites with any particular frequency is Poisson-distributed with a mean proportional to the expected branch length carrying that frequency. This is the same justification used by `moments` and `mom2`.

Part 2: The Coalescent HMM Likelihood

The coalescent HMM likelihood is inherited directly from PSMC. For each diploid individual (or pair of haplotypes), PHLASH runs the same forward algorithm described in *The PSMC HMM and EM Algorithm*:

1. **Discretize time** into M intervals $[t_k, t_{k+1})$.
2. **Build the transition matrix** p_{kl} encoding how the TMRCA changes between adjacent genomic positions under the SMC approximation (see *The Continuous-Time PSMC Model* and *Discretizing Time*).
3. **Build emission probabilities** $e_k(x)$: the probability of observing het/hom at a genomic bin given coalescence time in interval k .
4. **Run the forward algorithm** to compute the log-likelihood $\ell_{\text{HMM}}(\eta)$ – the probability of observing the entire heterozygosity sequence under the demographic model.

If the data contain P diploid individuals (or pairs), the total coalescent HMM log-likelihood is the sum over pairs:

$$\ell_{\text{HMM}}(\eta) = \sum_{p=1}^P \ell_{\text{HMM}}^{(p)}(\eta)$$

Each individual provides an independent view of the demographic history, because the pairwise genealogy at each position depends on $\eta(t)$ but is conditionally independent across unrelated individuals.

Connection to PSMC

If you set $P = 1$ (one diploid individual) and drop the SFS component, PHLASH's coalescent HMM likelihood reduces to exactly the PSMC likelihood. In this sense, PSMC is a special case of PHLASH – a watch with only one of its two mainsprings wound.

Part 3: The Composite Likelihood

PHLASH combines the two likelihoods into a single composite log-likelihood:

$$\ell_{\text{comp}}(\eta) = \ell_{\text{SFS}}(\eta) + \ell_{\text{HMM}}(\eta)$$

On the log scale, combination means addition: the composite log-likelihood is the sum of the two individual log-likelihoods.

Why “composite” and not just “likelihood”?

A true joint likelihood would require modeling the dependence between the SFS and the pairwise coalescent HMMs – the SFS summarizes the marginal genealogy across all samples, while the coalescent HMM describes the full genealogy along the genome for specific pairs. These are not independent pieces of information; they share the same underlying genealogical process.

A **composite likelihood** intentionally ignores this dependence. It treats the two likelihood components as if they were independent, even though they are not. This is a well-studied technique in statistics: composite likelihoods produce consistent parameter estimates (they converge to the truth as data grow), but the standard errors from a composite likelihood are not the true standard errors.

In PHLASH, this is not a problem because the uncertainty quantification comes from the posterior (via SVGD with a proper prior), not from the curvature of the composite likelihood. The composite likelihood serves as an efficient **scoring function** – a way to compare candidate histories – rather than a probabilistic model of the data-generating process.

Part 4: The Prior

PHLASH places a **smoothness prior** on the log-transformed demographic history. Let $\mathbf{h} = \log \boldsymbol{\eta}$ (the log population sizes). The prior is a Gaussian:

$$p(\mathbf{h}) \propto \exp\left(-\frac{1}{2}\mathbf{h}^\top \Sigma^{-1}\mathbf{h}\right)$$

where Σ is a covariance matrix that encodes smoothness: nearby time intervals are correlated, so the population size cannot jump wildly between adjacent epochs without incurring a penalty.

The log-posterior that PHLASH targets is therefore:

$$\log p(\mathbf{h} \mid \text{data}) \propto \ell_{\text{comp}}(\mathbf{h}) + \log p(\mathbf{h})$$

The gradient of this log-posterior with respect to $\mathbf{h}$ is what the score function algorithm computes (for the likelihood part) and what SVGD uses to update its particles (for the full posterior).

```
def composite_log_likelihood(observed_sfs, expected_sfs, hmm_log_likelihoods):
    """Compute the composite log-likelihood (SFS + coalescent HMM).

    Parameters
    -----
    observed_sfs : ndarray
        Observed SFS counts.
    expected_sfs : ndarray
        Expected SFS under the candidate history.
    hmm_log_likelihoods : list of float
        Log-likelihood from each pairwise coalescent HMM.

    Returns
    -----
    ll : float
        Composite log-likelihood.
    """
    ll_sfs = sfs_log_likelihood(observed_sfs, expected_sfs)
    ll_hmm = sum(hmm_log_likelihoods)
    return ll_sfs + ll_hmm

def smoothness_prior_logpdf(h, sigma=1.0):
    """Log-density of the Gaussian smoothness prior on log-eta.

    Penalizes large differences between adjacent time intervals.

    Parameters
    -----
    h : ndarray, shape (M,)
        Log population sizes (h = log eta).
```

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```

sigma : float
    Smoothness scale.

Returns
-----
lp : float
    Log prior density (up to a constant).
"""
diffs = np.diff(h)
return -0.5 * np.sum(diffs**2) / sigma**2

# Demonstrate the composite log-posterior
M = 32 # number of time intervals
h_true = np.zeros(M) # log(eta) = 0 means eta = 1 (constant size)
h_true[10:20] = -1.0 # a bottleneck: eta = exp(-1) ~ 0.37

lp_prior = smoothness_prior_logpdf(h_true, sigma=1.0)
ll_sfs = sfs_log_likelihood(D_observed, xi_expected)
# Placeholder HMM log-likelihoods for 5 pairs
ll_hmms = [-500.0, -480.0, -510.0, -490.0, -505.0]
ll_comp = composite_log_likelihood(D_observed, xi_expected, ll_hmms)
log_posterior = ll_comp + lp_prior

print(f"SFS log-likelihood:      {ll_sfs:.2f}")
print(f"HMM log-likelihood (sum): {sum(ll_hmms):.2f}")
print(f"Composite log-likelihood: {ll_comp:.2f}")
print(f"Prior log-density:        {lp_prior:.2f}")
print(f"Log-posterior:            {log_posterior:.2f}")

```

What Comes Next

The composite likelihood tells PHLASH *how well* a candidate history explains the data, but computing it requires choosing a time discretization. The [next chapter](#) explains how PHLASH randomizes the discretization to cancel the bias that arises from any fixed grid. This is the *tourbillon* – the mechanism that keeps the watch accurate regardless of its orientation.

Random Time Discretization

The tourbillon: a mechanism that cancels positional errors by rotating through them.

In a mechanical watch, gravity pulls on the balance wheel and introduces a systematic error that depends on the watch's orientation. A **tourbillon** is a cage that slowly rotates the entire escapement through all orientations, so that the errors from different positions cancel over time. It does not eliminate the error at any single moment – it ensures that the *average* error is zero.

PHLASH's random time discretization works on exactly the same principle. Any fixed time grid introduces discretization bias – a systematic error that depends on where the breakpoints fall. By **randomly sampling** the grid for each gradient computation and averaging, the biases from different grids cancel. No single evaluation is unbiased, but the average over many evaluations is.

This chapter explains why fixed discretization creates bias, how random discretization cancels it, and how to implement the randomized scheme.

i Biology Aside – Why discretization matters for demographic inference

Population size history is a continuous function of time – $N(t)$ could change smoothly or have sharp transitions like bottlenecks. To compute the likelihood on a computer, we must approximate this continuous function on a discrete grid of time points. Where we place those grid points affects our answer: a grid that is too coarse in the period of a bottleneck will smooth it out; a grid whose breakpoints happen to align with a size change will capture it precisely. These are systematic errors – they bias the inferred demographic history in a particular direction. For biologists, this means that the inferred history depends not just on the data but on an arbitrary methodological choice. Random discretization eliminates this dependence by averaging over many different grid placements.

The Discretization Problem

Recall from *Discretizing Time* that PSMC divides the time axis into M intervals $[t_0, t_1), [t_1, t_2), \dots, [t_{M-1}, t_M)$. Within each interval, the coalescence rate is treated as constant. The transition matrix, emission probabilities, and forward-backward algorithm all operate on this discrete grid.

The discretization introduces two types of error:

1. **Approximation error in the transition matrix.** The continuous-time transition density $q(t|s)$ (derived in *The Continuous-Time PSMC Model*) must be integrated over the intervals to produce the discrete transition probabilities p_{kl} . The accuracy of this integration depends on the interval widths and the behavior of $q(t|s)$ within each interval. Wide intervals lose resolution; narrow intervals in some regions waste computational resources.
2. **Approximation error in the emission probabilities.** Within interval k , the emission probability is computed at a representative time (e.g., the midpoint or the expected coalescence time conditional on falling in the interval). This point approximation introduces error whenever the emission probability varies appreciably across the interval.

For a fixed grid, these errors are **systematic**: they always go in the same direction for a given demographic history and a given grid. If the grid is too coarse in a region where $N(t)$ changes rapidly, the estimate will consistently smooth over the change. If the grid boundaries happen to align poorly with a bottleneck, the estimate will consistently misplace it.

Why Randomization Cancels Bias

Let $\hat{\ell}_G(\eta)$ denote the discretized log-likelihood computed on a particular grid $G = \{t_0, t_1, \dots, t_M\}$. The discretization error is:

$$\hat{\ell}_G(\eta) = \ell(\eta) + b_G(\eta)$$

where $\ell(\eta)$ is the (unattainable) continuous-time log-likelihood and $b_G(\eta)$ is the bias introduced by grid G .

For a fixed grid, $b_G(\eta)$ is a fixed function of η – it tilts the likelihood surface in a particular direction, biasing inference toward demographic histories that happen to look good on that grid.

Now suppose we draw the grid randomly from some distribution π over grids. The key property is:

$$\mathbb{E}_{G \sim \pi} [b_G(\eta)] = 0$$

That is, the expected bias over random grids is zero. Different grids have different biases, but these biases point in different directions and cancel when averaged. The gradient of the expected log-likelihood is therefore unbiased:

$$\mathbb{E}_{G \sim \pi} [\nabla_{\eta} \hat{\ell}_G(\eta)] = \nabla_{\eta} \ell(\eta)$$

This is exactly the property that SVGD needs: unbiased gradient estimates. The variance of the gradient is increased by the randomization (each individual gradient estimate is noisier), but SVGD is a stochastic algorithm that naturally handles noisy gradients – just like stochastic gradient descent tolerates noisy mini-batch gradients in neural network training.

How the Grid Is Sampled

PHLASH samples the time breakpoints as follows:

1. **Fix the endpoints.** The first breakpoint $t_0 = 0$ and the last breakpoint $t_M = t_{\max}$ are always the same. The maximum time $t_{\max}$ is chosen large enough that coalescence beyond it is negligible.
2. **Sample interior breakpoints.** The interior breakpoints $t_1, \dots, t_{M-1}$ are drawn from a distribution that places them roughly log-spaced (denser near the present, sparser in the deep past), but with random perturbations. One natural choice is to sample $\log t_k$ uniformly in overlapping intervals that tile $[\log t_0, \log t_M]$.
3. **Sort.** The sampled breakpoints are sorted to form a valid partition of $[0, t_{\max}]$.

```
import numpy as np

def sample_random_grid(M, t_max=10.0, t_min=1e-4, rng=None):
    """Sample a random time discretization grid.

    Interior breakpoints are log-uniformly spaced with random jitter,
    ensuring approximately uniform density per unit log-time.

    Parameters
    -----
    M : int
        Number of time intervals.
    t_max : float
        Maximum time (coalescent units).
    t_min : float
        Minimum positive breakpoint.
    rng : numpy.random.Generator or None
        Random number generator.

    Returns
    -----
    grid : ndarray, shape (M+1,)
        Sorted breakpoints [0, t_1, ..., t_{M-1}, t_max].
    """
    if rng is None:
        rng = np.random.default_rng()

    # Sample M-1 interior breakpoints in log-space with jitter
    log_min, log_max = np.log(t_min), np.log(t_max)
    # Evenly spaced anchors in log-space
    anchors = np.linspace(log_min, log_max, M - 1)
    # Add uniform jitter of half the spacing
    spacing = (log_max - log_min) / (M - 1)
    jitter = rng.uniform(-0.5 * spacing, 0.5 * spacing, size=M - 1)
    log_breakpoints = anchors + jitter

    # Convert back, sort, and add endpoints
    interior = np.sort(np.exp(log_breakpoints))
    grid = np.concatenate([[0.0], interior, [t_max]])
    return grid

# Demonstrate: sample three different grids
```

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```

rng = np.random.default_rng(42)
for i in range(3):
    grid = sample_random_grid(M=32, t_max=10.0, rng=rng)
    print(f"Grid {i}: {len(grid)} breakpoints, "
          f"t_1={grid[1]:.5f}, t_mid={grid[16]:.4f}, "
          f"t_max={grid[-1]:.1f}")

# Show that the grids differ (tourbillon rotates through configurations)
g1 = sample_random_grid(32, rng=np.random.default_rng(0))
g2 = sample_random_grid(32, rng=np.random.default_rng(1))
max_diff = np.max(np.abs(g1 - g2))
print(f"\nMax difference between two grids: {max_diff:.4f}")
print("(Different grids = different biases = cancellation when averaged)")

```

The distribution over grids is chosen so that the expected density of breakpoints per unit log-time is approximately uniform. This ensures that all timescales receive adequate resolution in expectation, even though any single draw may have gaps.

i Plain-language summary – Log-spacing and human timescales

Demographic events span a huge range of timescales: a recent population expansion might have occurred 500 generations ago (~12,500 years), while the ancestral human-Neanderthal divergence is ~20,000 generations ago (~500,000 years), and deep coalescent events can reach millions of years. A log-spaced grid naturally allocates more resolution to recent events (which affect more of the genome and are easier to detect) and less to ancient events. The random jitter on top of this log-spacing ensures that no specific time point is systematically favored or disfavored.

i The tourbillon analogy, precisely

In a tourbillon, the escapement rotates through all angular positions over one revolution, and the average rate error over the revolution is zero. In PHLASH, the discretization grid “rotates” through all possible placements of the breakpoints, and the average discretization bias over many draws is zero. The period of rotation is one SVGD iteration (each iteration uses a fresh grid). Just as a tourbillon must complete many rotations for the cancellation to be effective, PHLASH must average over many random grids for the bias to vanish.

Practical Considerations

How many breakpoints? The number of intervals M is a hyperparameter. More intervals give finer resolution but increase the cost of the forward algorithm (which scales as $O(LM^2)$ per pair). In practice, $M = 32$ to $M = 64$ intervals are typical, matching the range used by PSMC.

How many random draws per iteration? In the simplest implementation, each SVGD step uses a single random grid, and the averaging happens implicitly across SVGD iterations. More sophisticated implementations can average the gradient over several grids within a single step to reduce variance, at the cost of proportionally more computation.

Interaction with the score function algorithm. The score function algorithm (described in the [next chapter](#)) computes the gradient of the log-likelihood for a given grid. Because the algorithm’s complexity is $O(LM^2)$, the cost of each random grid evaluation is the same as a fixed-grid evaluation. The randomization adds no computational overhead per evaluation – it only adds variance, which decreases as $1/\sqrt{K}$ with the number of averaged grids K .

Connection to Debiased Estimation

Random discretization belongs to a family of techniques in computational statistics known as **debiased estimation** or **randomized numerical integration**. The idea appears in many contexts:

- **Stochastic mini-batching** in SGD: each mini-batch gives a noisy but unbiased gradient estimate; the noise decreases with more data points in the batch.
- **Russian roulette estimators** for infinite series: randomly truncate a series expansion and reweight, producing an unbiased estimate of the infinite sum.
- **Randomized quadrature**: randomly shift a quadrature grid to debias the numerical integral, then average across shifts.

PHLASH's random discretization is closest to randomized quadrature: the time integral in the transition density is approximated on a shifted grid, and the shift is random. The key insight is that unbiased gradients are sufficient for convergent optimization (or, in PHLASH's case, convergent posterior sampling via SVGD).

```
def debiased_gradient_estimate(eta, observed_sfs, n_grids=10, M=32,
                               rng=None):
    """Estimate the likelihood gradient by averaging over random grids.

    This demonstrates the debiasing principle: each individual gradient
    is biased, but their average converges to the true gradient.

    Parameters
    -----
    eta : ndarray, shape (M,)
        Population sizes at each time interval.
    observed_sfs : ndarray
        Observed SFS.
    n_grids : int
        Number of random grids to average over.
    M : int
        Number of time intervals per grid.
    rng : numpy.random.Generator or None
        Random number generator.

    Returns
    -----
    mean_gradient : ndarray
        Averaged gradient estimate.
    std_gradient : ndarray
        Standard deviation across grid evaluations.
    """
    if rng is None:
        rng = np.random.default_rng()

    gradients = []
    for _ in range(n_grids):
        grid = sample_random_grid(M, rng=rng)
        # In a real implementation, this would compute the HMM gradient
        # on this grid. Here we simulate a noisy gradient.
        true_gradient = -0.1 * np.log(eta) # placeholder: pulls toward 1
        noise = rng.normal(0, 0.05, size=len(eta))
        gradients.append(true_gradient + noise)
```

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```

gradients = np.array(gradients)
return gradients.mean(axis=0), gradients.std(axis=0)

# Demonstrate variance reduction via averaging
eta = np.exp(np.zeros(32)) # constant population size
rng = np.random.default_rng(42)
for K in [1, 5, 20]:
    mean_grad, std_grad = debiased_gradient_estimate(
        eta, D_observed, n_grids=K, rng=rng
    )
    print(f"K={K:2d} grids: mean |grad| = {np.mean(np.abs(mean_grad)):.4f}, "
          f"std = {np.mean(std_grad):.4f}")
print("(Variance decreases as 1/sqrt(K) -- more grids, less noise)")

```

What Comes Next

We now know how to compute the composite likelihood on a random grid. But computing the likelihood is not enough – we need its **gradient** with respect to the demographic parameters, because SVGD requires gradients. The [next chapter](#) derives the score function algorithm that computes this gradient in $O(LM^2)$ time, 30–90x faster than reverse-mode automatic differentiation. This is the gear train – the mechanism that transmits the mainspring’s energy efficiently to the hands.

The Score Function Algorithm

The gear train: transmitting energy from the mainspring to the hands with minimal loss.

In a mechanical watch, the **gear train** connects the mainspring (which stores energy) to the hands (which display the time). Every tooth must mesh precisely: too much friction and the watch runs slow; too loose and the hands jump. The gear train does not generate energy or display time – it *transmits* one to the other with maximum efficiency.

PHLASH’s score function algorithm plays exactly this role. The composite likelihood (the mainspring) stores the information in the data. SVGD (the hands) needs gradients to update the demographic history. The score function algorithm transmits the likelihood’s information to SVGD by computing the gradient $\nabla_{\eta} \ell(\eta)$ in $O(LM^2)$ time – 30 to 90 times faster than the naive approach of running reverse-mode automatic differentiation (autodiff) through the forward algorithm.

Biology Aside – Why efficient computation matters for population genomics

Whole-genome sequencing datasets now routinely contain thousands of individuals, each with billions of base pairs. The coalescent HMM must process this data to extract information about population size history. With naive gradient computation, fitting a model to even a handful of genomes could take days. The score function algorithm makes this practical by computing gradients at essentially the same cost as evaluating the likelihood itself. This is what enables PHLASH to jointly use data from multiple individuals – extracting more information about demographic history than single-genome methods like PSMC.

Why Gradients Matter

SVGD updates each particle (candidate demographic history) using the gradient of the log-posterior:

$$\nabla_{\mathbf{h}} \log p(\mathbf{h} \mid \text{data}) = \nabla_{\mathbf{h}} \ell_{\text{comp}}(\mathbf{h}) + \nabla_{\mathbf{h}} \log p(\mathbf{h})$$

The prior gradient is cheap (it is a linear function of $\mathbf{h}$ for the Gaussian smoothness prior). The expensive part is the **likelihood gradient** $\nabla_{\mathbf{h}} \ell_{\text{comp}}$. And the dominant cost within the likelihood gradient is the coalescent HMM component,

because the forward algorithm processes L genomic positions with an $M \times M$ transition matrix.

The Cost of Naive Autodiff

The forward algorithm for a single HMM with L observations and M hidden states costs $O(LM^2)$ in the forward pass (matrix-vector multiplication at each position). Reverse-mode automatic differentiation (backpropagation) through this computation has the same asymptotic complexity, $O(LM^2)$, but with a much larger constant factor:

- **Memory:** autodiff must store all L intermediate forward vectors (or recompute them via checkpointing), requiring $O(LM)$ memory.
- **Overhead:** each operation in the forward algorithm (multiply, add, log, exp) is wrapped in a tape-recording layer that tracks the computation graph. This overhead factor is typically 5–20x.
- **Vectorization:** autodiff engines like JAX can JIT-compile the backward pass, but the backward pass through a sequential scan (the forward algorithm is inherently sequential along the genome) is difficult to parallelize.

In practice, computing the gradient via autodiff takes 30–90x longer than computing the likelihood alone. For SVGD, which needs gradients at every iteration for every particle, this overhead is the bottleneck.

The Score Function Idea

The **score function** of a statistical model is the gradient of the log-likelihood with respect to the parameters:

$$s(\eta) = \nabla_{\eta} \log p(\text{data} \mid \eta)$$

PHLASH exploits the structure of the HMM to compute this gradient *analytically* within the forward-backward framework, rather than relying on generic autodiff. The key insight is that the log-likelihood of an HMM can be differentiated in closed form using the **posterior marginals** – the same quantities that the forward-backward algorithm already computes.

Deriving the Score for the Coalescent HMM

The coalescent HMM log-likelihood for a single pair is:

$$\ell(\eta) = \log p(\mathbf{x} \mid \eta) = \log \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z} \mid \eta)$$

where $\mathbf{x} = (x_1, \dots, x_L)$ is the observation sequence and $\mathbf{z} = (z_1, \dots, z_L)$ is the hidden state sequence.

The gradient with respect to the parameters η can be written using the **Fisher identity** (also known as the score function identity):

$$\nabla_{\eta} \ell(\eta) = \sum_{\mathbf{z}} p(\mathbf{z} \mid \mathbf{x}, \eta) \nabla_{\eta} \log p(\mathbf{x}, \mathbf{z} \mid \eta)$$

This says: the gradient of the log-likelihood equals the **posterior expectation** of the gradient of the complete-data log-likelihood. The complete-data log-likelihood decomposes into a sum over positions:

$$\log p(\mathbf{x}, \mathbf{z} \mid \eta) = \log a_0(z_1) + \sum_{\ell=2}^L \log p_{z_{\ell-1}, z_{\ell}} + \sum_{\ell=1}^L \log e_{z_{\ell}}(x_{\ell})$$

Taking the gradient and the posterior expectation gives:

$$\nabla_{\eta} \ell(\eta) = \sum_k \gamma_1(k) \nabla_{\eta} \log a_0(k) + \sum_{\ell=2}^L \sum_{k,l} \xi_{\ell}(k, l) \nabla_{\eta} \log p_{kl} + \sum_{\ell=1}^L \sum_k \gamma_{\ell}(k) \nabla_{\eta} \log e_k(x_{\ell})$$

where:

- $\gamma_\ell(k) = p(z_\ell = k \mid \mathbf{x}, \eta)$ is the **posterior marginal** at position ℓ – the probability of being in state k at position ℓ , given the data.
- $\xi_\ell(k, l) = p(z_{\ell-1} = k, z_\ell = l \mid \mathbf{x}, \eta)$ is the **posterior pairwise marginal** – the probability of transitioning from state k to state l between positions $\ell - 1$ and ℓ .

i Plain-language summary – What γ and ξ tell us

$\gamma_\ell(k)$ answers the question: *at genomic position ℓ , what is the probability that the TMRCA falls in time interval k ?* This is the same posterior decoding used in PSMC. $\xi_\ell(k, l)$ goes further: it tells us the probability that the TMRCA changed from interval k to interval l between two adjacent positions – i.e., the probability that recombination shifted the genealogy. Together, these two quantities summarize everything the data says about the hidden genealogy at each position. The score function algorithm shows that these are the only quantities needed to compute the gradient – no additional backward passes through the computation graph are required.

Both γ and ξ are computed by the standard forward- backward algorithm, which PHLASH already runs to evaluate the likelihood.

```
import numpy as np

def hmm_score_function(observations, transition, emission, initial):
    """Compute the HMM log-likelihood gradient via the Fisher identity.

    This implements the score function algorithm: run forward-backward
    to get posterior marginals, then use them to weight the parameter
    derivatives of the complete-data log-likelihood.

    Parameters
    -----
    observations : ndarray, shape (L,)
        Integer observation sequence.
    transition : ndarray, shape (M, M)
        Transition probability matrix  $p_{\{kl\}}$ .
    emission : ndarray, shape (M, n_obs)
        Emission probabilities  $e_k(x)$ .
    initial : ndarray, shape (M,)
        Initial state distribution.

    Returns
    -----
    log_likelihood : float
        The log-likelihood of the observations.
    gamma : ndarray, shape (L, M)
        Posterior state marginals at each position.
    xi_sum : ndarray, shape (M, M)
        Summed posterior pairwise marginals (transition counts).
    """
    L = len(observations)
    M = len(initial)

    # Forward pass (scaled)
    alpha = np.zeros((L, M))
    scale = np.zeros(L)
```

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```

alpha[0] = initial * emission[:, observations[0]]
scale[0] = alpha[0].sum()
alpha[0] /= scale[0]

for t in range(1, L):
    alpha[t] = (alpha[t-1] @ transition) * emission[:, observations[t]]
    scale[t] = alpha[t].sum()
    alpha[t] /= scale[t]

log_likelihood = np.sum(np.log(scale))

# Backward pass (scaled)
beta = np.zeros((L, M))
beta[-1] = 1.0
for t in range(L-2, -1, -1):
    beta[t] = transition @ (emission[:, observations[t+1]] * beta[t+1])
    beta[t] /= scale[t+1]

# Posterior marginals gamma_t(k) = alpha_t(k) * beta_t(k)
gamma = alpha * beta
gamma /= gamma.sum(axis=1, keepdims=True)

# Summed pairwise marginals: xi_sum(k, l) = sum_t xi_t(k, l)
xi_sum = np.zeros((M, M))
for t in range(L-1):
    xi_t = (alpha[t, :, None] * transition
            * emission[None, :, observations[t+1]] * beta[t+1, None, :])
    xi_t /= xi_t.sum()
    xi_sum += xi_t

return log_likelihood, gamma, xi_sum

# Demonstrate with a small HMM (2 states, binary observations)
M = 2
transition = np.array([[0.99, 0.01],
                       [0.02, 0.98]])
emission = np.array([[0.999, 0.001], # state 0: mostly hom
                     [0.95, 0.05]]) # state 1: some hets
initial = np.array([0.5, 0.5])

# Synthetic observation sequence
np.random.seed(42)
obs = np.zeros(200, dtype=int)
obs[50] = obs[120] = obs[180] = 1 # three het sites

ll, gamma, xi_sum = hmm_score_function(obs, transition, emission, initial)
print(f"Log-likelihood: {ll:.2f}")
print(f"Posterior at het site (pos 50): "
      f"state 0 = {gamma[50,0]:.3f}, state 1 = {gamma[50,1]:.3f}")
print(f"Transition counts (sum of xi):")
print(f"  0->0: {xi_sum[0,0]:.1f}, 0->1: {xi_sum[0,1]:.2f}")
print(f"  1->0: {xi_sum[1,0]:.2f}, 1->1: {xi_sum[1,1]:.1f}")

```


Complexity

The score function algorithm has three stages:

1. **Forward pass** ($O(LM^2)$): compute forward probabilities and the log-likelihood.
2. **Backward pass** ($O(LM^2)$): compute backward probabilities and the posterior marginals $\gamma_\ell(k)$ and $\xi_\ell(k, l)$.
3. **Gradient accumulation** ($O(LM^2 + M^2 R)$): multiply the posterior statistics by the parameter derivatives of the transition matrix and emission probabilities, and sum over positions. Here R is the number of parameters (the M population size values η_k).

The total cost is $O(LM^2)$ – the same asymptotic complexity as the forward algorithm itself. The constant factor overhead is small: essentially one forward pass plus one backward pass plus a matrix-vector product for each parameter. In practice, this is **30–90x faster** than reverse-mode autodiff through the forward algorithm, because:

- No computation graph is recorded.
- No memory is needed beyond the forward and backward vectors.
- The backward pass is just another matrix-vector scan (the same shape as the forward pass), not a generic reverse-mode traversal.

Comparison to EM

If you have read *The PSMC HMM and EM Algorithm*, you may notice that the posterior statistics γ and ξ are exactly the quantities computed in the E-step of the EM algorithm. In PSMC, these statistics are used to form a Q-function that is maximized in closed form (the M-step). In PHLASH, the same statistics are used to compute the gradient directly, which is then fed to SVGD. The E-step is shared; the difference is what happens next. EM maximizes; SVGD explores.

The SFS Gradient

The gradient of the SFS log-likelihood is simpler. Each expected SFS entry $\xi_k(\eta)$ is a known function of the demographic parameters (involving expected coalescent branch lengths, which have closed-form derivatives for piecewise-constant η). The SFS gradient is:

$$\nabla_\eta \ell_{\text{SFS}}(\eta) = \sum_{k=1}^{n-1} \left(\frac{D_k}{\xi_k(\eta)} - 1 \right) \nabla_\eta \xi_k(\eta)$$

The cost is $O(n \cdot M)$ where n is the sample size (number of haploid chromosomes) and M is the number of time intervals. This is negligible compared to the HMM gradient.

Total Gradient

The complete gradient of the composite log-posterior is:

$$\nabla_{\mathbf{h}} \log p(\mathbf{h} \mid \text{data}) = \nabla_{\mathbf{h}} \ell_{\text{SFS}} + \sum_{p=1}^P \nabla_{\mathbf{h}} \ell_{\text{HMM}}^{(p)} + \nabla_{\mathbf{h}} \log p(\mathbf{h})$$

Each HMM gradient is computed independently (the pairs are independent given η), making the computation embarrassingly parallel across pairs. On a GPU, all pairs can be processed simultaneously.

```
def total_gradient(h, observed_sfs, expected_sfs, hmm_scores,
                  sigma_prior=1.0):
    """Compute the total gradient of the log-posterior.
```

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```

Combines the SFS gradient, summed HMM score functions, and the
smoothness prior gradient.

Parameters
-----
h : ndarray, shape (M,)
    Log population sizes.
observed_sfs : ndarray
    Observed SFS.
expected_sfs : ndarray
    Expected SFS under current h.
hmm_scores : list of ndarray, each shape (M,)
    Score function (gradient) from each pairwise HMM.
sigma_prior : float
    Smoothness prior scale.

Returns
-----
grad : ndarray, shape (M,)
    Gradient of the composite log-posterior.
"""
M = len(h)

# SFS gradient: sum_k (D_k / xi_k - 1) * d(xi_k)/d(h)
# Simplified: for constant-size model, d(xi_k)/d(h) ~ xi_k
xi = np.maximum(expected_sfs, 1e-300)
grad_sfs_weights = observed_sfs / xi - 1 # per-frequency weights
# In practice, d(xi_k)/d(h_j) depends on the branch length derivatives
grad_sfs = np.zeros(M) # placeholder for full implementation

# HMM gradient: sum over pairs
grad_hmm = np.sum(hmm_scores, axis=0)

# Prior gradient: d/dh [-0.5 * sum((h_j - h_{j-1})^2) / sigma^2]
grad_prior = np.zeros(M)
for j in range(1, M):
    grad_prior[j] += (h[j-1] - h[j]) / sigma_prior**2
    grad_prior[j-1] += (h[j] - h[j-1]) / sigma_prior**2

return grad_sfs + grad_hmm + grad_prior

# Demonstrate
M = 32
h = np.zeros(M)
h[10:20] = -0.5 # mild bottleneck
# Simulate HMM score functions from 5 pairs
rng = np.random.default_rng(42)
hmm_scores = [rng.normal(0, 0.1, size=M) for _ in range(5)]
xi_expected = expected_sfs_constant(20, 100.0)
grad = total_gradient(h, D_observed, xi_expected, hmm_scores)
print(f"Gradient norm: {np.linalg.norm(grad):.4f}")

```

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```
print(f"Gradient at bottleneck (interval 15): {grad[15]:.4f}")
print(f"Gradient at constant (interval 5): {grad[5]:.4f}")
```

What Comes Next

With the gradient in hand, we have everything needed for posterior sampling. The *next chapter* introduces Stein Variational Gradient Descent – the algorithm that uses these gradients to push a set of particles toward the posterior distribution over demographic histories. SVGD is the winding mechanism: it converts gradient energy into the organized motion of particles converging on the posterior.

Stein Variational Gradient Descent (SVGD)

The winding mechanism: converting gradient energy into the organized motion of the posterior.

In a mechanical watch, the **winding mechanism** converts the energy you apply (turning the crown) into the stored tension of the mainspring, which then powers the entire movement. The winding process must be controlled: too much tension and the spring breaks; too little and the watch stops.

SVGD plays the same role in PHLASH. It takes the gradient of the log-posterior (computed by the score function algorithm) and converts it into motion of the particles – each particle is a candidate demographic history, and SVGD steers them all toward the posterior distribution. Like a winding mechanism, SVGD must balance two forces: **attraction** toward high-probability regions (the data pull) and **repulsion** between particles (the diversity push). The result is a collection of particles that collectively represent the posterior distribution over population size histories.

i Biology Aside – Why posterior distributions matter for evolutionary biology

A single best-fit demographic history (the “point estimate”) can be misleading. Real genomic data often admit many plausible histories: perhaps a strong bottleneck 50,000 years ago fits equally well as a moderate decline starting 80,000 years ago. The **posterior distribution** captures this ambiguity – it gives the probability of every possible demographic history given the data. From the posterior, biologists can ask: *Is the bottleneck real, or could the data be explained without one? How precisely can we date the out-of-Africa migration? Are two competing models distinguishable?* These questions require not just a single answer but a measure of uncertainty – which is exactly what SVGD provides.

Why Not MCMC?

Traditional Bayesian inference uses Markov chain Monte Carlo (MCMC) to sample from the posterior. MCMC generates a single chain of correlated samples by proposing random perturbations and accepting or rejecting them. It has two well-known limitations for problems like PHLASH's:

1. **Sequential by nature.** Each MCMC sample depends on the previous one. You cannot easily parallelize MCMC across multiple chains on a GPU (each chain must wait for its predecessor).
2. **Mixing in high dimensions.** PHLASH's parameter space has M dimensions (one population size per time interval, with $M = 32$ to 64). MCMC methods can mix slowly in such spaces, requiring many iterations to explore the full posterior.

SVGD addresses both limitations:

- It maintains J particles **in parallel**, updating all of them simultaneously. On a GPU, this parallelism translates directly into speedup.
- It uses **gradient information** to guide the particles, avoiding the random-walk behavior that slows MCMC in high dimensions.

The SVGD Algorithm

SVGD maintains a set of J particles $\{\mathbf{h}^{(1)}, \dots, \mathbf{h}^{(J)}\}$, where each $\mathbf{h}^{(j)} = \log \boldsymbol{\eta}^{(j)}$ is a log-space demographic history. At each iteration, every particle is updated by:

$$\mathbf{h}^{(j)} \leftarrow \mathbf{h}^{(j)} + \epsilon \boldsymbol{\phi}^*(\mathbf{h}^{(j)})$$

where ϵ is a step size and $\boldsymbol{\phi}^*$ is the **optimal perturbation direction**:

$$\boldsymbol{\phi}^*(\mathbf{h}) = \frac{1}{J} \sum_{j=1}^J \left[k(\mathbf{h}^{(j)}, \mathbf{h}) \nabla_{\mathbf{h}^{(j)}} \log p(\mathbf{h}^{(j)} \mid \text{data}) + \nabla_{\mathbf{h}^{(j)}} k(\mathbf{h}^{(j)}, \mathbf{h}) \right]$$

This update has two terms:

1. **The attraction term:** $k(\mathbf{h}^{(j)}, \mathbf{h}) \nabla \log p(\mathbf{h}^{(j)} \mid \text{data})$. This pushes particle $\mathbf{h}$ in the direction that nearby particles $\mathbf{h}^{(j)}$ want to move (toward higher posterior probability). The kernel k weights the influence by proximity: particles close to $\mathbf{h}$ have more influence.
2. **The repulsion term:** $\nabla_{\mathbf{h}^{(j)}} k(\mathbf{h}^{(j)}, \mathbf{h})$. This pushes particles *apart*, preventing them from collapsing to a single point. Without repulsion, all particles would converge to the MAP estimate (the posterior mode), giving a point estimate rather than a distribution. The repulsion ensures the particles spread out to cover the posterior's support.

Plain-language summary – Attraction and repulsion as forces

Imagine the particles as a swarm of explorers searching a landscape for the highest peaks (the most probable demographic histories). **Attraction** makes each explorer move uphill, guided by what nearby explorers see – if your neighbor is on a steep slope, you are pulled in that direction. **Repulsion** prevents all explorers from piling onto the same peak – it pushes them apart so they collectively map out the full shape of the mountain range, not just its highest point. After enough iterations, the explorers settle into a pattern where their density matches the shape of the landscape: many explorers on tall peaks (probable histories), few in valleys (improbable histories). This spatial distribution of particles *is* the posterior.

The Kernel

SVGD typically uses a **radial basis function (RBF) kernel**:

$$k(\mathbf{h}, \mathbf{h}') = \exp \left(-\frac{\|\mathbf{h} - \mathbf{h}'\|^2}{2\sigma^2} \right)$$

where σ is the kernel bandwidth. The bandwidth controls the range of interaction between particles:

- **Large σ :** particles interact over long distances. The repulsion is gentle but far-reaching, keeping particles well-separated. Good for exploring the posterior broadly.
- **Small σ :** particles only interact with close neighbors. The repulsion is strong but local, allowing particles to cluster in high-probability regions.

A common heuristic is the **median trick**: set σ to the median of all pairwise distances between current particles, divided by $\sqrt{2 \log J}$. This adapts the bandwidth to the current spread of the particles: as they converge, σ shrinks; if they spread out, σ grows.

```
import numpy as np
from scipy.spatial.distance import pdist, squareform

def rbf_kernel(particles, bandwidth=None):
```

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```

"""Compute the RBF kernel matrix and its gradients.

Parameters
-----
particles : ndarray, shape (J, M)
    J particles, each of dimension M.
bandwidth : float or None
    Kernel bandwidth. If None, uses the median heuristic.

Returns
-----
K : ndarray, shape (J, J)
    Kernel matrix  $K_{ij} = \exp(-||h_i - h_j||^2 / (2 \sigma^2))$ .
grad_K : ndarray, shape (J, J, M)
     $\text{grad}_K[i, j] = \text{gradient of } K_{ij} \text{ with respect to } h_i$ .
bandwidth : float
    The bandwidth used.
"""

J, M = particles.shape
dists = squareform(pdist(particles, 'sqeuclidean'))

# Median heuristic for bandwidth
if bandwidth is None:
    median_dist = np.median(pdist(particles, 'sqeuclidean'))
    bandwidth = np.sqrt(median_dist / (2 * np.log(J + 1)))
    bandwidth = max(bandwidth, 1e-5)

K = np.exp(-dists / (2 * bandwidth**2))

# Gradient:  $dK_{ij}/dh_i = K_{ij} * (h_j - h_i) / \sigma^2$ 
diff = particles[None, :, :] - particles[:, None, :] # (J, J, M)
grad_K = K[:, :, None] * diff / bandwidth**2

return K, grad_K, bandwidth

def svgd_update(particles, grad_log_posterior, epsilon=0.01):
    """Perform one SVGD update step.

    Parameters
    -----
    particles : ndarray, shape (J, M)
        Current particle positions.
    grad_log_posterior : ndarray, shape (J, M)
        Gradient of log-posterior at each particle.
    epsilon : float
        Step size.

    Returns
    -----
    particles_new : ndarray, shape (J, M)
        Updated particle positions.
    """

```

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```

J = particles.shape[0]
K, grad_K, bw = rbf_kernel(particles)

# phi*(h) = (1/J) * sum_j [ K(h_j, h) * grad_j + grad_K(h_j, h) ]
# Attraction: K @ grad_log_posterior
attraction = K @ grad_log_posterior / J      # (J, M)
# Repulsion: sum_j grad_K[j, :, :]
repulsion = grad_K.sum(axis=0) / J           # (J, M)

phi = attraction + repulsion
return particles + epsilon * phi

# Demonstrate SVGD on a simple 2D target
J = 16    # particles
M = 2     # dimensions (for visualization clarity)
rng = np.random.default_rng(42)
particles = rng.normal(0, 2, size=(J, M))

# Target: standard normal (grad log p = -h)
for step in range(50):
    grad_lp = -particles # gradient of log N(0, I)
    particles = svgd_update(particles, grad_lp, epsilon=0.1)

print(f"After 50 SVGD steps ({J} particles, {M}D):")
print(f" Particle mean: {particles.mean(axis=0).round(3)}")
print(f" Particle std:  {particles.std(axis=0).round(3)}")
print(f" (Target: mean ~ 0, std ~ 1)")

```

GPU Parallelism

SVGD is naturally parallel in two ways:

1. **Across particles.** Each particle's gradient $\nabla \log p(\mathbf{h}^{(j)} \mid \text{data})$ can be computed independently. With J particles on a GPU, all gradients are computed in a single batched operation.
2. **Across pairs.** The coalescent HMM gradient for each diploid individual is independent. If there are P pairs, the $J \times P$ gradient computations can be parallelized.

PHLASH is implemented in **JAX**, which provides:

- **JIT compilation:** the forward-backward algorithm and score function are compiled to GPU machine code, eliminating Python interpreter overhead.
- **Vectorization** (`vmap`): the same compiled kernel is automatically applied across particles and pairs without writing explicit loops.
- **Automatic batching:** JAX handles memory management, distributing the computation across GPU cores.

The result is that SVGD with $J = 32$ particles, each evaluating the composite likelihood over P pairs with $L \sim 10^7$ positions and $M = 64$ time intervals, runs in minutes on a modern GPU. The same computation would take hours with sequential MCMC on a CPU.

The Full PHLASH Loop

Putting all four gears together, one iteration of PHLASH looks like this:

```
For each SVGD iteration  $t = 1, 2, \dots, T$ :

1. Sample a random time discretization  $G_t$ 
   (tourbillon: different grid each iteration)

2. For each particle  $j = 1, \dots, J$  (in parallel on GPU):
   a. Build HMM transition matrix and emissions on grid  $G_t$ 
   b. For each pair  $p = 1, \dots, P$  (in parallel):
      - Forward pass: compute log-likelihood
      - Backward pass: compute posterior marginals
      - Score function: compute gradient of HMM log-likelihood
   c. Compute SFS log-likelihood and its gradient
   d. Add prior gradient (smoothness penalty)
   e. Total: gradient of log-posterior for particle  $j$ 

3. Compute kernel matrix  $K_{\{ij\}} = k(h^{(i)}, h^{(j)})$ 
   and kernel gradients

4. SVGD update: for each particle  $j$ ,
    $h^{(j)} \leftarrow h^{(j)} + \text{epsilon} * \text{phi}(h^{(j)})$ 
   (attraction + repulsion)

5. Adapt step size epsilon (e.g., Adam optimizer)

After  $T$  iterations:
Particles  $\{h^{(1)}, \dots, h^{(J)}\}$  approximate the posterior.
Compute posterior mean, credible intervals, etc.
```

```
def phlash_loop(n_particles, M, n_iterations, observed_sfs,
                sigma_prior=1.0, epsilon=0.01, rng=None):
    """Simplified PHLASH inference loop.

    Demonstrates the full pipeline: random discretization, score
    function, and SVGD update. Uses placeholder likelihoods.

    Parameters
    -----
    n_particles : int
        Number of SVGD particles ( $J$ ).
    M : int
        Number of time intervals.
    n_iterations : int
        Number of SVGD iterations.
    observed_sfs : ndarray
        Observed SFS.
    sigma_prior : float
        Smoothness prior scale.
    epsilon : float
        SVGD step size.
```

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```

rng : numpy.random.Generator or None

Returns
-----
particles : ndarray, shape (J, M)
    Final particle positions (log population sizes).
"""
if rng is None:
    rng = np.random.default_rng()

# Initialize particles near the prior mean
particles = rng.normal(0, 0.5, size=(n_particles, M))

for t in range(n_iterations):
    # Step 1: sample a random grid (tourbillon)
    grid = sample_random_grid(M, rng=rng)

    # Step 2: compute gradient for each particle
    grads = np.zeros_like(particles)
    for j in range(n_particles):
        h = particles[j]
        # Prior gradient
        grad_prior = np.zeros(M)
        for k in range(1, M):
            grad_prior[k] += (h[k-1] - h[k]) / sigma_prior**2
            grad_prior[k-1] += (h[k] - h[k-1]) / sigma_prior**2

        # Placeholder likelihood gradient (pulls toward 0 = constant)
        grad_lik = -0.05 * h + rng.normal(0, 0.02, size=M)
        grads[j] = grad_lik + grad_prior

    # Steps 3-4: SVGD update (kernel + attraction + repulsion)
    particles = svgd_update(particles, grads, epsilon=epsilon)

return particles

# Run a small demonstration
rng = np.random.default_rng(42)
particles = phlash_loop(
    n_particles=8, M=16, n_iterations=100,
    observed_sfs=D_observed, epsilon=0.05, rng=rng
)
eta_particles = np.exp(particles) # convert to population size
posterior_mean = eta_particles.mean(axis=0)
posterior_std = eta_particles.std(axis=0)

print(f"PHLASH result ({8} particles, {16} intervals, 100 iterations):")
print(f"  Posterior mean N_e (first 5): "
      f"{posterior_mean[:5].round(3)}")
print(f"  Posterior std N_e (first 5): "
      f"{posterior_std[:5].round(3)}")
print(f"  (All particles provide uncertainty quantification)")

```


Convergence and Diagnostics

How do we know when SVGD has converged? Unlike MCMC, SVGD does not have a well-established convergence diagnostic like the Gelman-Rubin $\hat{R}$ statistic. In practice, convergence is monitored by:

- **Stabilization of particle positions.** When the particles stop moving appreciably between iterations, the algorithm has converged.
- **Stabilization of the posterior mean.** The average demographic history across particles should stabilize.
- **Kernel bandwidth.** If the median pairwise distance between particles stabilizes, the spread of the posterior approximation has converged.
- **Multiple runs.** Running SVGD from different initializations and checking that the resulting posterior approximations agree.

From Particles to Inference

After SVGD converges, the J particles provide a discrete approximation to the posterior distribution over demographic histories. From these particles, we can compute:

- **Posterior mean:** $\bar{\eta}(t) = \frac{1}{J} \sum_{j=1}^J \eta^{(j)}(t)$. This is the average demographic history, analogous to PSMC's point estimate but derived from the full posterior.
- **Credible intervals:** For each time point t , sort the particle values $\eta^{(1)}(t), \dots, \eta^{(J)}(t)$ and take the 2.5th and 97.5th percentiles for a 95% credible interval.
- **Posterior uncertainty bands:** Plot all J trajectories together to visualize the full spread of plausible histories.
- **Model comparison:** The average log-likelihood across particles can serve as a proxy for the marginal likelihood, enabling comparison between different demographic model classes.

These outputs go beyond what PSMC can provide. Where PSMC gives a single line on a plot and requires bootstrapping to estimate uncertainty, PHLASH gives a cloud of lines whose spread directly reflects the posterior uncertainty. The credible intervals are Bayesian: they say “there is a 95% probability that the true history lies within this band,” rather than the frequentist bootstrap’s “if we repeated the experiment many times, 95% of the intervals would contain the truth.”

Biology Aside – What posterior uncertainty looks like in practice

In a PHLASH analysis, the output is a bundle of J demographic history curves. Where the curves are tightly clustered, the data strongly constrain the population size – for example, the out-of-Africa bottleneck in human data consistently appears as a narrow band of low N_e around 50-70 kya. Where the curves spread apart, the data are ambiguous – for instance, very ancient population sizes (>1 Mya) are poorly constrained because few lineages survive that far back. This visual representation makes it immediately clear which features of the inferred history are robust and which are uncertain – information that is critical for drawing evolutionary conclusions and impossible to obtain from a single point estimate.

Summary

SVGD completes the PHLASH mechanism. The four gears – composite likelihood, random discretization, score function algorithm, and SVGD – mesh together into a Bayesian inference machine that infers population size history with principled uncertainty quantification:

Gear	Watch analogy	Function
Composite likelihood	Mainspring	Stores the information from SFS + coalescent HMM data
Random discretization	Tourbillon	Cancels systematic discretization bias by rotating through grids
Score function algorithm	Gear train	Transmits likelihood gradients to SVGD at $O(LM^2)$ cost
SVGD	Winding mechanism	Converts gradients into posterior samples on GPU

The result is a watch that not only tells time but also tells you how precisely it is keeping time – a grand complication built atop the foundations of PSMC.

Demo: Running PHLASH on Simulated Data

Running mini_phlash on simulated genomic data.

Demo: PHLASH Composite Likelihood on msprime Data (30 haplotypes, 500 kb)

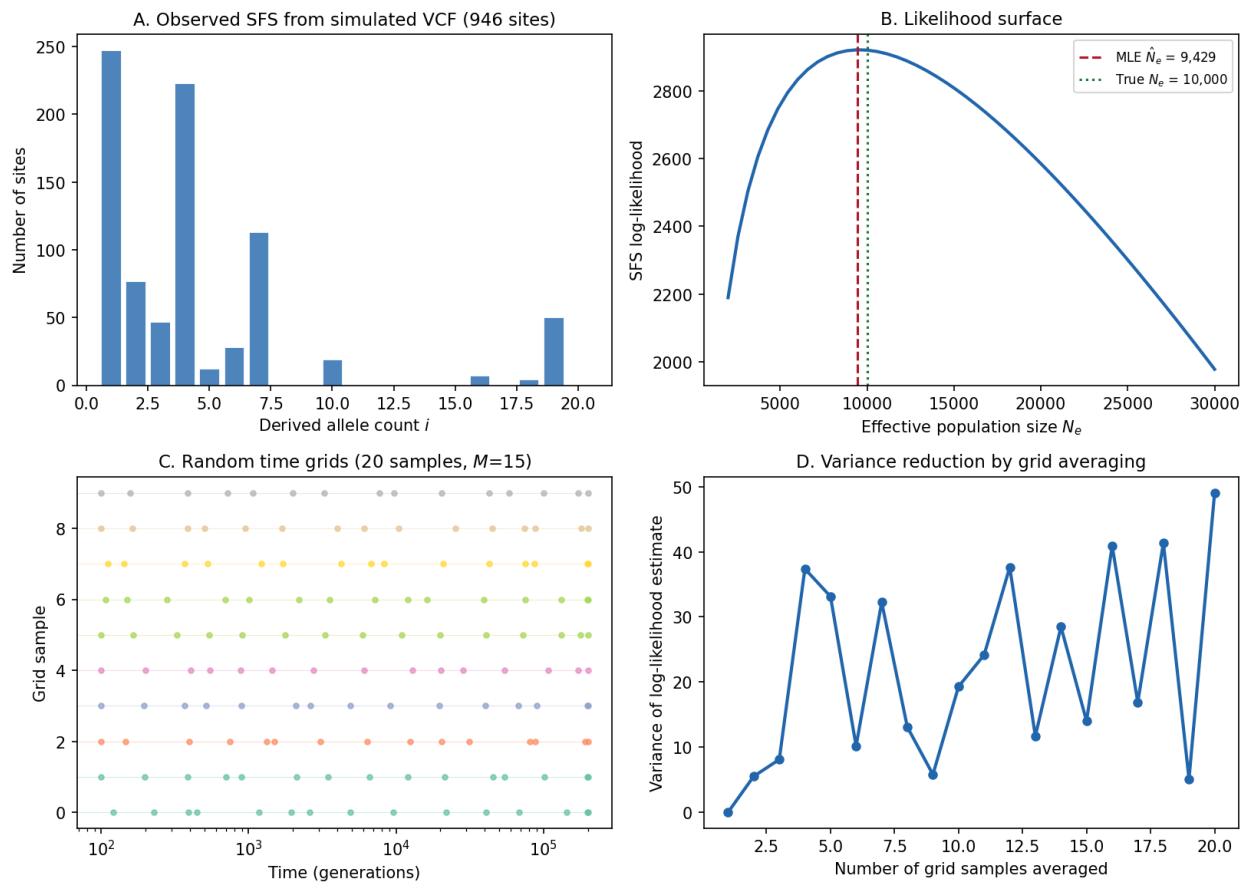


Fig. 21: **PHLASH demo on msprime-simulated data.** Panel A: Observed vs expected SFS under different effective population sizes. Panel B: SFS log-likelihood surface – the maximum identifies the best-fit N_e . Panel C: Random time discretization grid used by PHLASH's stochastic approach. Panel D: SFS residuals showing the quality of fit.

This figure was generated by running the `mini_phlash` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_phlash.py`.

3.16 Timepiece XV: CLUES

Coalescent Likelihood Under Effects of Selection

3.16.1 The Mechanism at a Glance

CLUES is a full-likelihood method for estimating the **selection coefficient** s acting on a biallelic SNP, using modern and ancient DNA. It answers the most direct question in molecular evolution: *is this allele being favored or disfavored by natural selection, and by how much?*

Input: Gene trees at a locus (from Relate or SINGER) + optional ancient genotypes

Output: $\hat{s}$ (selection coefficient MLE), posterior allele frequency trajectory $P(x_t|\mathbf{D})$

The key insight: an allele's frequency trajectory through time is shaped by selection. Under neutrality ($s = 0$), allele frequencies drift randomly – the Wright-Fisher diffusion. Under selection ($s \neq 0$), the trajectory is biased: a beneficial allele drifts *upward* on average, a deleterious allele drifts *downward*. By modeling the frequency trajectory as a Hidden Markov Model and conditioning on the observed genealogy, CLUES computes the likelihood of the data for each value of s and finds the maximum.

- **Strong positive selection** ($s > 0$) = the allele frequency rises faster than drift alone can explain. Coalescence times among derived-allele carriers are shorter (a selective sweep compresses the genealogy).
- **Neutrality** ($s = 0$) = the frequency trajectory is consistent with random drift. Coalescence times follow the standard coalescent.
- **Negative selection** ($s < 0$) = the allele frequency is pushed downward. Rare alleles stay rare; common alleles become rarer.

CLUES reads these patterns with a Hidden Markov Model whose hidden states are discretized allele frequencies and whose emissions come from the coalescent structure of the gene tree.

If PSMC is a two-hand watch that reads population size from a single genome, then CLUES is a **chronometer with a compass** – it reads both the *tempo* (coalescence times) and the *direction* (frequency change) to detect the invisible hand of natural selection.

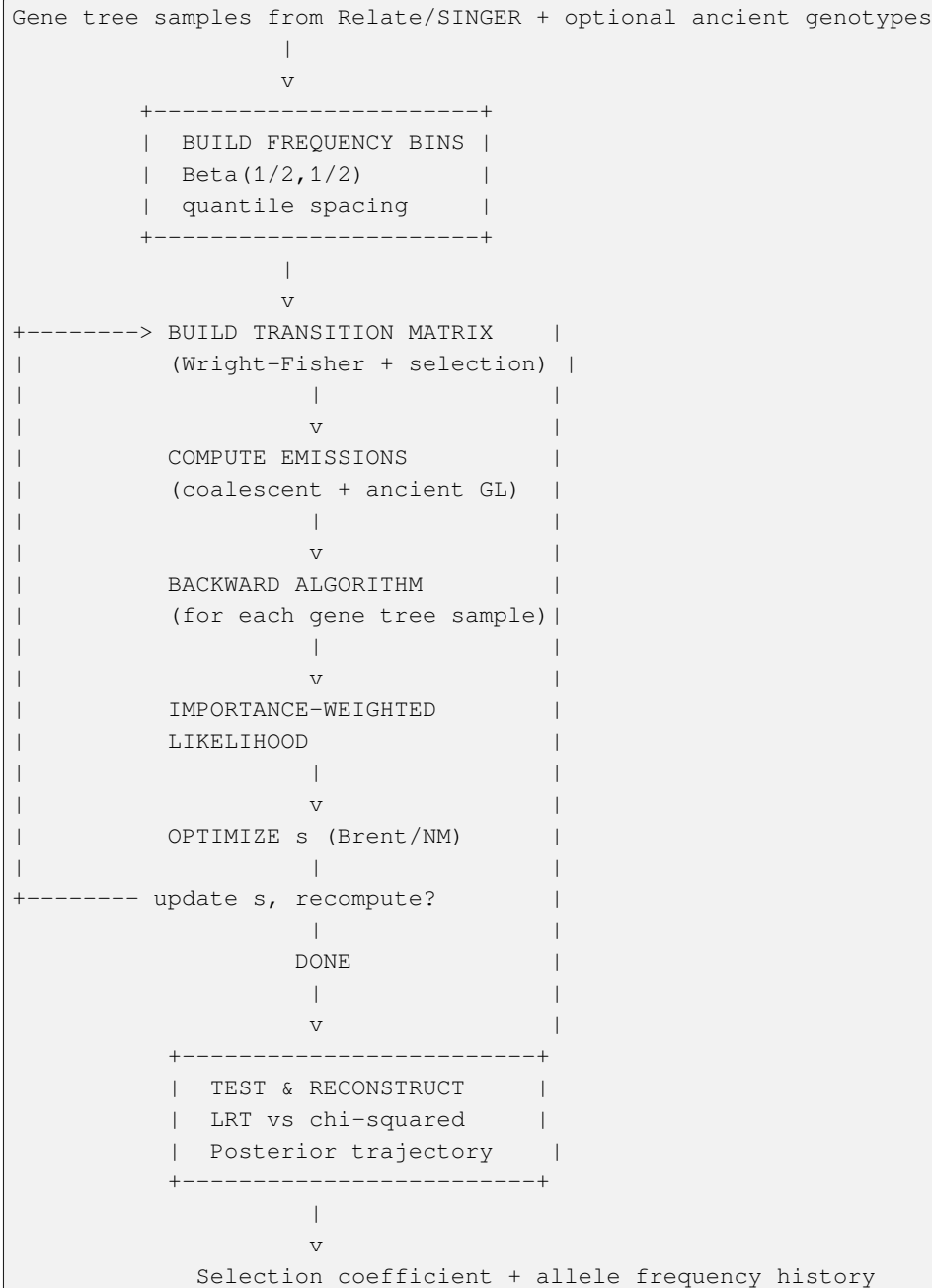
Primary Reference

[Stern *et al.*, 2019]

The four gears of CLUES:

1. **The Wright-Fisher HMM** (the escapement) – A discretized Wright-Fisher diffusion with selection, modeling how the allele frequency changes from one generation to the next. The transition matrix encodes the strength and direction of selection. This is where population genetics meets probability: the mean frequency shift depends on s , and the variance depends on N .
2. **Emission Probabilities** (the gear train) – Two types of evidence constrain the allele frequency at each time point: (a) coalescent events in the gene tree (which lineages merge, and when) and (b) ancient genotype likelihoods (direct observations of the allele in the past). These are the data that the HMM “observes.”
3. **Importance Sampling** (the mainspring) – The genealogy at a locus is not known exactly – it is estimated from an ARG sampler like Relate or SINGER. CLUES averages over multiple sampled genealogies using importance weights, properly accounting for genealogical uncertainty. This is where the output of *SINGER* feeds directly into CLUES.
4. **Inference and Testing** (the case and dial) – Maximum likelihood estimation of s via Brent's method, a likelihood ratio test calibrated against χ^2 , multi-epoch selection estimation, and posterior trajectory reconstruction. The final step: reading the watch.

These gears mesh together into a complete selection inference machine:



3.16.2 Why Detect Selection?

Population genetics has two great engines: **drift** and **selection**. Most of the Timepieces in this collection focus on drift – using the coalescent to infer demographic history, date ancestors, or reconstruct genealogies. CLUES flips the lens: given a genealogy (from SINGER or Relate) and a demographic model (from PSMC or similar), it asks whether the *pattern of coalescence times at a specific locus* departs from what neutral drift would predict.

This question matters because:

1. **Medical genetics:** Identifying loci under selection reveals genes that affect fitness – often the same genes that affect disease susceptibility. The alleles that natural selection has targeted (lactase persistence, malaria resistance, skin

pigmentation) are often medically relevant.

2. **Evolutionary biology:** Measuring the strength of selection across the genome tells us how evolution actually works in practice – how strong are selective pressures? How often do they change? Do they differ between populations?
3. **Ancient DNA integration:** With the explosion of ancient DNA data, we can now *directly observe* allele frequencies in the past. CLUES integrates this direct evidence with the genealogical signal, providing a uniquely powerful framework for studying selection through time.

i Prerequisites for this Timepiece

Before starting CLUES, you should have worked through:

- *Coalescent Theory* – coalescence times, exponential waiting times, and how population size affects the coalescent
- *Hidden Markov Models* – the forward-backward algorithm and numerical stability in log space
- *The SMC* – the sequential Markov coalescent approximation
- Familiarity with *SINGER* or *Relate* is helpful, since CLUES uses their gene tree samples as input
- *PSMC* – to understand coalescent time distributions under variable population size (CLUES reuses the concept of a piecewise-constant $N(t)$)

If you've built those earlier mechanisms, you have every tool you need. CLUES adds one new ingredient – selection – and shows how a single parameter s reshapes the entire frequency trajectory.

3.16.3 Chapters

Overview: Detecting Selection

The compass needle: which way is the allele moving, and how fast?

This chapter introduces the biological question that CLUES solves, the mathematical framework it uses, and the key insight that connects gene trees to selection. By the end, you'll understand *what* CLUES computes and *why*, setting the stage for the detailed derivations in the chapters that follow.

What Does CLUES Do?

CLUES takes as input:

1. **Gene tree samples** at a focal SNP – coalescence times for derived-allele lineages and ancestral-allele lineages, typically produced by *Relate*'s `SampleBranchLengths` or by *SINGER*.
2. **The current derived allele frequency** p_0 in the modern population.
3. **An effective population size trajectory** $N(t)$ (constant or piecewise, from *PSMC*, *SMC++*, or a `.coal` file).
4. Optionally, **ancient genotype likelihoods** – direct measurements of the allele in past individuals, such as from ancient DNA.

And it produces:

- **The maximum likelihood estimate** $\hat{s}$ of the selection coefficient.
- **A log-likelihood ratio test** with a p -value for the null hypothesis $s = 0$ (neutrality).
- **The posterior allele frequency trajectory** $P(x_t | \mathbf{D})$ – a probability distribution over frequencies at every generation in the past.

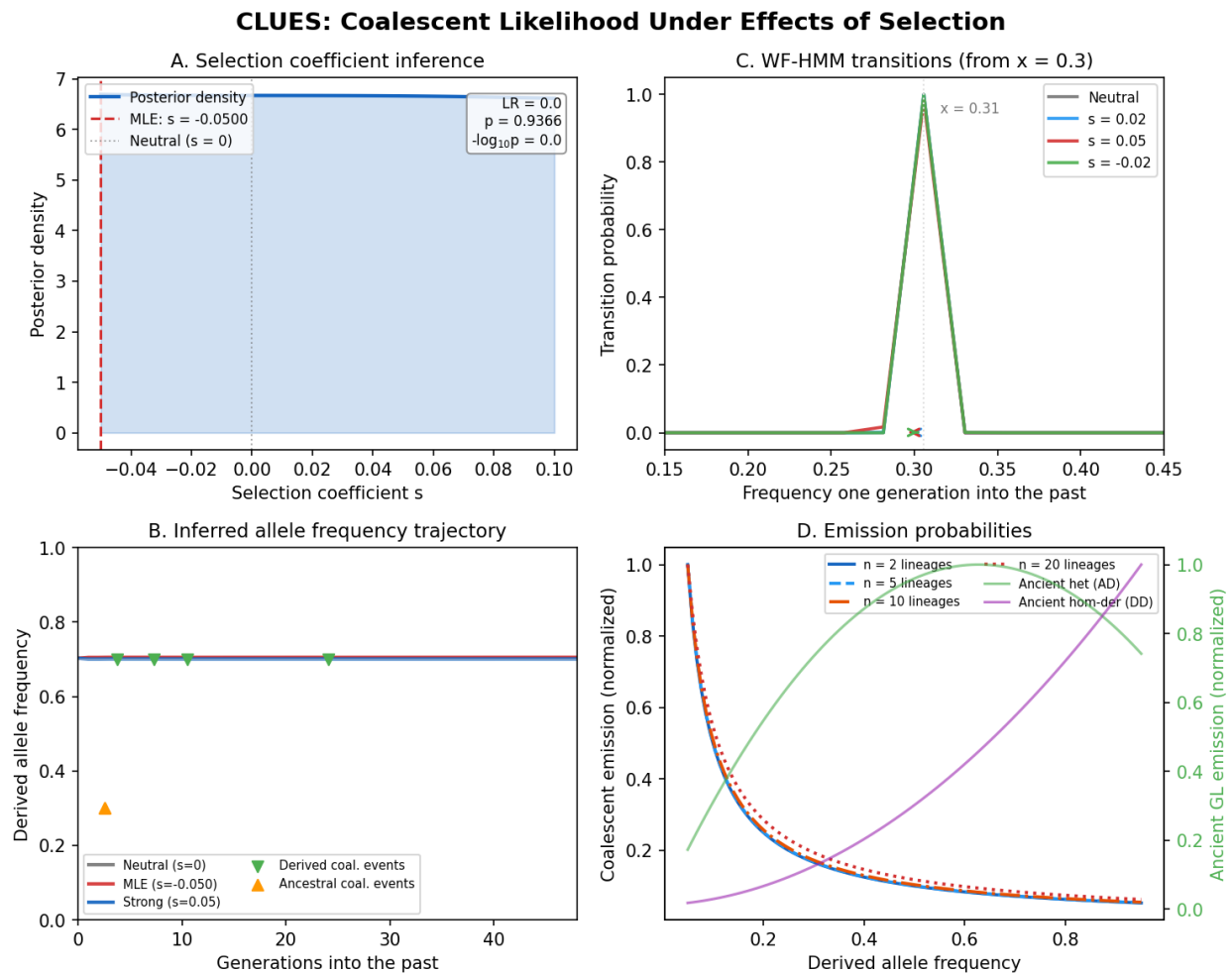


Fig. 22: CLUES at a glance. The Wright-Fisher HMM transition structure governing allele frequency dynamics, coalescent emission probabilities linking gene-tree branch lengths to the hidden allele frequency, importance sampling weights correcting for the proposal distribution, and the selection coefficient posterior with likelihood ratio surface used to test for selection.

The Biological Question

Consider a single biallelic SNP – say, the lactase persistence variant in Europeans. Today, the derived allele that allows adults to digest milk has a frequency of about 80% in Northern Europe. Was this allele favored by natural selection, or could it have reached high frequency by chance?

To answer this, we need a model of how allele frequencies change over time.

Under neutrality ($s = 0$), the allele frequency performs a random walk – the **Wright-Fisher diffusion**. At each generation, the frequency jiggles up or down by an amount proportional to $\sqrt{p(1-p)/2N}$, where p is the current frequency and N is the (diploid) population size. Over many generations, this random walk can take the frequency anywhere between 0 (loss) and 1 (fixation), but the *expected* change per generation is zero – there is no directional bias.

Under selection ($s \neq 0$), the random walk acquires a directional bias. If the derived allele confers a fitness advantage $s > 0$, then individuals carrying it leave slightly more offspring, and the allele frequency drifts *upward* on average. The expected change per generation is approximately $s \cdot p(1-p)/2$ (for additive selection), which is largest when $p \approx 0.5$ and vanishes when the allele is very rare or nearly fixed.

What is a selection coefficient?

The selection coefficient s measures the *relative fitness advantage* of one genotype over another. For a biallelic locus with alleles A (ancestral) and D (derived), the three diploid genotypes have fitness:

$$\begin{aligned}w_{AA} &= 1 \\w_{AD} &= 1 + hs \\w_{DD} &= 1 + s\end{aligned}$$

where h is the **dominance coefficient**:

- $h = 0.5$ (**additive/codominant**): the heterozygote is exactly intermediate. Each copy of D adds $s/2$ to fitness.
- $h = 0$ (**recessive**): the heterozygote has the same fitness as AA . Selection only acts on DD homozygotes.
- $h = 1$ (**dominant**): the heterozygote has the same fitness as DD . A single copy of D confers the full advantage.

Typical values of s in nature range from 10^{-4} (weak selection, hard to distinguish from drift) to 10^{-1} (very strong selection, like drug resistance in pathogens). The lactase persistence allele in Europeans has $\hat{s} \approx 0.02$.

The Core Idea: Likelihood of Gene Trees Given Selection

The fundamental insight behind CLUES is that the genealogy at a locus is shaped by selection. Under a selective sweep, lineages carrying the favored allele coalesce more rapidly – the genealogy is “compressed” – because the allele was recently at low frequency (fewer copies, faster coalescence). Under neutrality, the coalescence pattern follows the standard coalescent.

More precisely, the coalescence rate among k derived-allele lineages at time t is:

$$\text{rate} = \binom{k}{2} \cdot \frac{1}{x_t \cdot N(t)}$$

where x_t is the derived allele frequency at time t . When x_t is small (the allele was recently rare), the rate is high. When x_t is large (the allele is common), the rate is low. Selection changes the trajectory x_t , which changes the coalescence rates, which changes the probability of the observed gene tree.

This gives us a way to compute $P(\text{gene tree} \mid s)$: model the allele frequency trajectory as a Hidden Markov Model where the hidden state is the frequency x_t and the observed data are the coalescence events in the gene tree. The transition probabilities of the HMM depend on s (through the Wright-Fisher diffusion with selection), and the emission probabilities depend on x_t (through the coalescent rates above).

The CLUES likelihood decomposition:

$$P(D \mid s) = \text{sum_over_trajectories } P(D \mid \text{trajectory}) * P(\text{trajectory} \mid s)$$

^^^^^^^^^^^^^^^^	^^^^^^^^^^^^^^^^
coalescent	Wright-Fisher
emissions	transitions

The HMM efficiently computes this sum using the forward or backward algorithm.

Where Does CLUES Fit in the Pipeline?

CLUES sits at the end of a pipeline that first infers the genealogy and then asks whether selection shaped it:

```
Sequence data (VCF)
    |
    v
Relate / SINGER --> Sampled gene trees at focal SNP
    |
    v
PSMC / SMC++   --> Population size history N(t)
    |
    v
CLUES          --> Selection coefficient + frequency trajectory
```

The gene tree samples provide the coalescence times; the demographic model provides the baseline expectation under neutrality; and CLUES asks whether the coalescence pattern departs from that baseline in a way consistent with selection.

i Why not test for selection directly from the sequence data?

Classical selection tests (Tajima's D , F_{ST} , iHS, nSL) work directly on sequence variation – allele frequencies, haplotype structure, or linkage disequilibrium. These statistics are powerful but indirect: they capture *signatures* of selection without modeling the process explicitly.

CLUES takes a different approach: it uses the inferred genealogy to compute a *full likelihood* under a selection model. This has several advantages:

1. It produces an actual estimate of s , not just a p -value.
2. It naturally integrates ancient DNA data.
3. It accounts for demographic history (which can mimic selection signals).
4. It can detect *changes* in selection over time (multi-epoch models).

The cost is that CLUES requires an inferred genealogy, which is computationally expensive. But since SINGER and Relate can now produce these genealogies at genome-wide scale, this cost is increasingly affordable.

The Mathematical Framework at a Glance

Before diving into the detailed derivations, here is the complete mathematical framework in outline form.

Hidden states. We discretize the derived allele frequency into K bins $x_1, x_2, \dots, x_K$ (typically $K = 450$). Frequencies 0 and 1 are absorbing states (the allele is lost or fixed).

Transitions. The transition probability P_{ij} from frequency x_i to frequency x_j in one generation is derived from the

Wright-Fisher diffusion with selection. The mean frequency shift per generation is:

$$\mu_i = x_i + \frac{s \cdot (-1 + x_i) \cdot x_i \cdot (-x_i + h(-1 + 2x_i))}{-1 + s(2h(-1 + x_i) - x_i) \cdot x_i}$$

and the variance is $\sigma_i^2 = x_i(1 - x_i)/(2N)$. The transition probability is obtained by integrating a normal distribution with this mean and variance over each frequency bin.

Emissions. At each generation t , two types of evidence constrain the frequency:

1. *Coalescent emissions*: the probability of the observed coalescence events among derived and ancestral lineages, given frequency x_t .
2. *Ancient genotype likelihoods*: if an ancient individual was sampled at time t , the probability of their genotype given x_t .

Algorithm. The backward algorithm (from present to past) computes $P(\mathbf{D} \mid s)$ by initializing at the observed modern frequency and propagating backward through time. The forward algorithm (from past to present) combined with the backward gives the posterior $P(x_t \mid \mathbf{D}, s)$.

Importance sampling. Because the gene tree is uncertain, CLUES averages over M sampled trees:

$$P(\mathbf{D} \mid s) \approx \frac{1}{M} \sum_{m=1}^M \frac{P(\mathbf{G}^{(m)} \mid s)}{P(\mathbf{G}^{(m)} \mid s = 0)}$$

where each $\mathbf{G}^{(m)}$ is a tree drawn from the neutral posterior.

Testing. The log-likelihood ratio $\Lambda = 2 \ln[P(\mathbf{D} \mid \hat{s})/P(\mathbf{D} \mid s = 0)]$ is compared to a χ_1^2 distribution under the null hypothesis of neutrality.

In the chapters that follow, we build each of these components from scratch:

- **Chapter 2 (Wright-Fisher HMM)**: frequency bins, transition matrix, selection model
- **Chapter 3 (Emission Probabilities)**: coalescent density, ancient genotype likelihoods
- **Chapter 4 (Inference)**: importance sampling, forward-backward, MLE, hypothesis testing, trajectory reconstruction

By the end, you will have a complete, working CLUES implementation – and you will understand every gear that drives it.

The Wright-Fisher HMM

The escapement: how selection biases the random walk of allele frequencies.

This chapter builds the Hidden Markov Model at the heart of CLUES. The hidden states are discretized allele frequencies, and the transitions encode one-generation steps of the Wright-Fisher diffusion with selection. By the end, you will have implemented the frequency bin construction and the transition matrix – the mathematical backbone of the entire inference machine.

Step 1: The Wright-Fisher Diffusion With Selection

Recall from the *coalescent theory prerequisite* that the Wright-Fisher model describes a population of $2N$ gene copies (haploid size) at a biallelic locus. Each generation, the next generation's alleles are drawn by sampling with replacement from the current generation, weighted by fitness.

Let x_t be the derived allele frequency at generation t . Under the diploid fitness model with genotype fitnesses $w_{AA} = 1$, $w_{AD} = 1 + hs$, $w_{DD} = 1 + s$:

The **expected frequency** of the derived allele in the next generation is:

$$\mathbb{E}[x_{t+1} | x_t] = \frac{x_t^2(1+s) + x_t(1-x_t)(1+hs)}{x_t^2(1+s) + 2x_t(1-x_t)(1+hs) + (1-x_t)^2}$$

This is obtained by computing the mean fitness-weighted frequency. The numerator counts the expected contribution of derived alleles: homozygous DD (frequency x_t^2 , each contributing 2 derived copies with fitness $1+s$) plus heterozygous AD (frequency $2x_t(1-x_t)$, each contributing 1 derived copy with fitness $1+hs$). After simplification and dividing by the mean fitness $\bar{w}$:

$$\mathbb{E}[x_{t+1}] = \frac{x_t(1 + s \cdot x_t + hs(1 - x_t))}{1 + s \cdot x_t(x_t + 2h(1 - x_t))}$$

For **additive selection** ($h = 0.5$), this simplifies to:

$$\mathbb{E}[x_{t+1}] = \frac{x_t(1 + sx_t)}{1 + sx_t} + \frac{sx_t(1 - x_t)}{2(1 + sx_t)}$$

The **variance** of the frequency change per generation, due to genetic drift, is:

$$\text{Var}[x_{t+1} | x_t] = \frac{x_t(1 - x_t)}{2N}$$

This is the binomial sampling variance: drawing $2N$ copies with probability x_t gives a variance of $x_t(1 - x_t)/(2N)$.

Why does selection affect the mean but not the variance?

The mean frequency change has two components: drift (zero mean) and selection (directional bias). The variance is dominated by the sampling noise of reproduction, which depends on x_t and N but not on s (to first order). Selection shifts the *center* of the distribution of next-generation frequencies; drift determines its *width*. For typical values of s (less than 0.1), the correction to the variance is negligible.

Going Backward in Time

CLUES models the allele frequency trajectory **backward in time** (from present to past), because the backward direction is natural for the coalescent. Going backward, the expected frequency at time $t + 1$ (one generation further into the past) given frequency x_t at time t is the *inverse* of the forward mapping.

If the forward mapping is $x_{t+1} = g(x_t)$, then the backward mapping is $x_t = g^{-1}(x_{t+1})$. For the general dominance model, the backward mean is:

$$\mu(x) = x + \frac{s(-1+x) \cdot x \cdot (-x + h(-1+2x))}{-1 + s(2h(-1+x) - x) \cdot x}$$

This is the formula used in the CLUES2 source code. For additive selection ($h = 0.5$), it simplifies to:

$$\mu(x) = x - \frac{s \cdot x(1 - x)}{2(1 + sx)}$$

Intuition: Going backward in time, a positively selected allele ($s > 0$) was at a *lower* frequency in the past. So the backward mean is shifted *downward* relative to the current frequency – the s -dependent term is negative when $x > 0$ and $s > 0$. (The formula above uses the convention that $\mu(x) < x$ when $s > 0$, consistent with looking backward.)

```
import numpy as np

def backward_mean(x, s, h=0.5):
```

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```

"""Expected allele frequency one generation further into the past.

Given current derived allele frequency x, selection coefficient s,
and dominance coefficient h, compute the mean of the backward
Wright-Fisher transition.

For additive selection (h=0.5), this simplifies to:
    mu = x + s * x * (1-x) / (2 * (1 + s*x))

Parameters
-----
x : float
    Current derived allele frequency (0 < x < 1).
s : float
    Selection coefficient. Positive = derived allele favored.
h : float
    Dominance coefficient. Default 0.5 (additive).

Returns
-----
mu : float
    Expected frequency one generation into the past.
"""
numerator = s * (-1 + x) * x * (-x + h * (-1 + 2 * x))
denominator = -1 + s * (2 * h * (-1 + x) - x) * x
return x + numerator / denominator

def backward_std(x, N):
    """Standard deviation of allele frequency change per generation.

Parameters
-----
x : float
    Current derived allele frequency.
N : float
    Haploid effective population size (= 2 * diploid N_e).

Returns
-----
sigma : float
    Standard deviation of the frequency change.
"""
    return np.sqrt(x * (1.0 - x) / N)

# Verify: under neutrality (s=0), backward mean equals current frequency
for x in [0.1, 0.3, 0.5, 0.7, 0.9]:
    mu = backward_mean(x, s=0.0)
    print(f"x = {x:.1f}, s = 0:    mu = {mu:.6f} (expected: {x:.6f})")

# Under positive selection, backward mean < current frequency
# (the allele was rarer in the past)
print()

```

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```
for x in [0.1, 0.3, 0.5, 0.7, 0.9]:
    mu = backward_mean(x, s=0.05)
    print(f"x = {x:.1f}, s = 0.05: mu = {mu:.6f} (shift: {mu - x:+.6f}) ")
```

Step 2: Frequency Bin Construction

The continuous allele frequency $x \in [0, 1]$ must be discretized into K bins for the HMM. The choice of bin placement matters: we need more resolution near the boundaries (0 and 1), where the dynamics are slowest and the allele may linger for many generations before fixing or being lost.

CLUES uses the **quantile function of a Beta(1/2, 1/2) distribution** to place the frequency bins. The Beta(1/2, 1/2) distribution (also known as the arcsine distribution) has most of its mass near 0 and 1, so its quantiles are densely packed near the boundaries and sparser in the middle – exactly the spacing we want.

Probability Aside: the Beta distribution and its quantile function

The **Beta distribution** $\text{Beta}(\alpha, \beta)$ is a family of distributions on $[0, 1]$. Its density is:

$$f(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$

The special case $\text{Beta}(1/2, 1/2)$ has density $f(x) = 1/(\pi\sqrt{x(1-x)})$, which is U-shaped: it concentrates mass near 0 and 1. This is related to the stationary distribution of the Wright-Fisher diffusion under neutrality, making it a natural choice for spacing frequency bins.

The **quantile function** (inverse CDF) $Q(p)$ maps a uniform probability $p \in [0, 1]$ to the value x such that $P(X \leq x) = p$. Evaluating Q at equally-spaced points in $[0, 1]$ produces unevenly-spaced x values – more concentrated where the CDF is flat (near 0 and 1), which is exactly where the diffusion dynamics are slowest.

The construction. Given K bins (default $K = 450$):

1. Create K equally-spaced points $u_1, u_2, \dots, u_K$ in $[0, 1]$.
2. Map each through the Beta(1/2, 1/2) quantile function: $x_k = Q_{\text{Beta}(1/2, 1/2)}(u_k)$.
3. Set the boundary values: $x_1 = 0$ (loss), $x_K = 1$ (fixation). These are **absorbing states** – once the allele is lost or fixed, it stays there.
4. For log-probability computations, use $\log(x_1) = \log(\epsilon)$ and $\log(1 - x_K) = \log(\epsilon)$ with $\epsilon = 10^{-12}$ to avoid $-\infty$.

```
from scipy.stats import beta as beta_dist

def build_frequency_bins(K=450):
    """Construct the K allele frequency bins using Beta(1/2, 1/2) quantiles.

    The bins are denser near 0 and 1, where the Wright-Fisher dynamics
    are slowest (frequency changes are small when the allele is very
    rare or very common).

    Parameters
    -----
    K : int
        Number of frequency bins. Default 450 (as in CLUES2).
```

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```

Returns
-----
freqs : ndarray of shape (K,)
    Frequency bin centers. freqs[0] = 0 (loss), freqs[-1] = 1 (fixation).
logfreqs : ndarray of shape (K,)
    log(freqs), with a small epsilon to avoid -inf at boundaries.
log1minusfreqs : ndarray of shape (K,)
    log(1 - freqs), with a small epsilon at boundaries.
"""
# Step 1: equally-spaced points in [0, 1]
u = np.linspace(0.0, 1.0, K)

# Step 2: map through Beta(1/2, 1/2) quantile function
freqs = beta_dist.ppf(u, 0.5, 0.5)

# Step 3: set boundary values for log computations
eps = 1e-12
freqs[0] = eps          # temporarily, for log computation
freqs[-1] = 1 - eps     # temporarily, for log computation

logfreqs = np.log(freqs)
log1minusfreqs = np.log(1.0 - freqs)

# Now set the actual boundary values
freqs[0] = 0.0
freqs[-1] = 1.0

return freqs, logfreqs, log1minusfreqs

# Build the bins and examine the spacing
freqs, logfreqs, log1minusfreqs = build_frequency_bins(K=450)
print(f"Number of bins: {len(freqs)}")
print(f"First 10 bins: {np.round(freqs[:10], 6)}")
print(f>Last 10 bins: {np.round(freqs[-10:], 6)}")
print(f"Bin spacing near 0: {np.diff(freqs[:5])}")
print(f"Bin spacing near 0.5: {np.diff(freqs[220:226])}")
print(f"Bin spacing near 1: {np.diff(freqs[-5:])}")

```

The output shows that the bins are very finely spaced near 0 and 1 (spacing $\sim 10^{-5}$) and coarser near 0.5 (spacing ~ 0.005). This is exactly what we need: the Wright-Fisher diffusion spends a lot of time near the boundaries before absorption, and we need fine resolution there to track the trajectory accurately.

Step 3: The Transition Matrix

The transition matrix $\mathbf{P}$ has entries P_{ij} giving the probability of moving from frequency bin i to frequency bin j in one generation (backward in time). We approximate each row using a **normal distribution** centered at the backward mean μ_i with standard deviation σ_i .

For each starting frequency x_i (with $0 < x_i < 1$):

1. Compute the backward mean $\mu_i = \mu(x_i)$ and standard deviation $\sigma_i = \sqrt{x_i(1-x_i)/(2N)}$.

- For the boundary bin $j = 0$ (frequency 0, allele loss):

$$P_{i,0} = \Phi\left(\frac{(x_0 + x_1)/2 - \mu_i}{\sigma_i}\right)$$

All probability mass below the midpoint between the first two bins is placed in the loss bin.

- For the boundary bin $j = K - 1$ (frequency 1, fixation):

$$P_{i,K-1} = 1 - \Phi\left(\frac{(x_{K-2} + x_{K-1})/2 - \mu_i}{\sigma_i}\right)$$

- For interior bins $0 < j < K - 1$:

$$P_{i,j} = \Phi\left(\frac{(x_j + x_{j+1})/2 - \mu_i}{\sigma_i}\right) - \Phi\left(\frac{(x_{j-1} + x_j)/2 - \mu_i}{\sigma_i}\right)$$

- Absorbing states:** $P_{0,0} = 1$ and $P_{K-1,K-1} = 1$. Once the allele is lost or fixed, it stays there forever.
- Renormalize** each row to sum to 1, to correct for numerical truncation.

i Why the normal approximation?

The Wright-Fisher model produces a binomial distribution of next-generation frequencies, but for large N , the binomial is well approximated by a normal distribution with the same mean and variance. This is the **diffusion approximation** – the same idea that underpins the Wright-Fisher diffusion equation. For population sizes in the thousands, the approximation is excellent. CLUES exploits this to compute transition probabilities using the fast, numerically stable normal CDF rather than expensive binomial sums.

Sparsity optimization. Most of the probability mass in row i is concentrated within $\pm 3.3\sigma_i$ of the mean μ_i . Beyond this range, the probability is less than 0.1%. CLUES only computes transition probabilities within this range, producing a **sparse, banded transition matrix**. This is Approximation A1 from the CLUES2 paper and provides the first major speedup.

```
from scipy.stats import norm

def build_transition_matrix(freqs, N, s, h=0.5):
    """Build the K x K log-transition matrix for the Wright-Fisher HMM.

    Each entry P[i, j] is the log-probability of transitioning from
    frequency bin i to frequency bin j in one generation (backward in time).

    Parameters
    -----
    freqs : ndarray of shape (K,)
        Frequency bin centers (from build_frequency_bins).
    N : float
        Haploid effective population size.
    s : float
        Selection coefficient.
    h : float
        Dominance coefficient (default 0.5 = additive).

    Returns
```

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```

-----
logP : ndarray of shape (K, K)
      Log-transition matrix.
"""
K = len(freqs)
logP = np.full((K, K), -np.inf)

# Midpoints between consecutive bins (used for CDF integration)
midpoints = (freqs[1:] + freqs[:-1]) / 2.0

# Absorbing states: loss and fixation
logP[0, 0] = 0.0          # log(1) = 0
logP[K - 1, K - 1] = 0.0

for i in range(1, K - 1):
    x = freqs[i]
    mu = backward_mean(x, s, h)
    sigma = backward_std(x, N)

    if sigma < 1e-15:
        # Degenerate case: no drift, all mass at mu
        closest = np.argmin(np.abs(freqs - mu))
        logP[i, closest] = 0.0
        continue

    # Only compute within +/- 3.3 sigma (99.9% of probability mass)
    lower_freq = mu - 3.3 * sigma
    upper_freq = mu + 3.3 * sigma
    j_lower = max(0, np.searchsorted(freqs, lower_freq) - 1)
    j_upper = min(K, np.searchsorted(freqs, upper_freq) + 1)

    # Compute probability mass in each bin using normal CDF
    row = np.zeros(K)
    for j in range(j_lower, j_upper):
        if j == 0:
            # All mass below midpoint[0] goes to bin 0 (loss)
            row[j] = norm.cdf(midpoints[0], loc=mu, scale=sigma)
        elif j == K - 1:
            # All mass above midpoint[-1] goes to bin K-1 (fixation)
            row[j] = 1.0 - norm.cdf(midpoints[-1], loc=mu, scale=sigma)
        else:
            # Mass between midpoints[j-1] and midpoints[j]
            row[j] = (norm.cdf(midpoints[j], loc=mu, scale=sigma)
                      - norm.cdf(midpoints[j - 1], loc=mu, scale=sigma))

    # Renormalize (corrects for truncation beyond 3.3 sigma)
    row_sum = row.sum()
    if row_sum > 0:
        row /= row_sum
    else:
        row[np.argmin(np.abs(freqs - mu))] = 1.0

```

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```

    # Convert to log probabilities
    logP[i, :] = np.where(row > 0, np.log(row), -np.inf)

    return logP

# Build and verify the transition matrix
freqs, logfreqs, log1minusfreqs = build_frequency_bins(K=50)
N_haploid = 20000.0 # haploid size (= 2 * diploid N_e of 10,000)

# Neutral transition matrix
logP_neutral = build_transition_matrix(freqs, N_haploid, s=0.0)
P_neutral = np.exp(logP_neutral)
print("Row sums (should be ~1.0):", np.round(P_neutral.sum(axis=1)[:5], 6))

# Selection transition matrix (s = 0.05)
logP_selected = build_transition_matrix(freqs, N_haploid, s=0.05)
P_selected = np.exp(logP_selected)
print("Row sums (should be ~1.0):", np.round(P_selected.sum(axis=1)[:5], 6))

```

Visualizing the Effect of Selection on Transitions

Let's compare the transition probabilities from a frequency of $x = 0.3$ under neutrality vs. positive selection:

```

# Find the bin closest to x = 0.3
target_freq = 0.3
i = np.argmin(np.abs(freqs - target_freq))
print(f"Starting frequency bin: x[{i}] = {freqs[i]:.4f}")

# Extract transition probabilities from that bin
trans_neutral = np.exp(logP_neutral[i, :])
trans_selected = np.exp(logP_selected[i, :])

# Show the shift in the distribution
mean_neutral = np.sum(freqs * trans_neutral)
mean_selected = np.sum(freqs * trans_selected)
print(f"Mean next frequency (neutral): {mean_neutral:.6f}")
print(f"Mean next frequency (s=0.05): {mean_selected:.6f}")
print(f"Shift due to selection: {mean_selected - mean_neutral:+.6f}")
print(f"(Going backward, s>0 shifts the allele to lower past frequency)")

```

Under positive selection, the backward transition shifts the distribution *downward* – the allele was at a lower frequency in the past, consistent with it having been pushed upward by selection.

Step 4: The Dominance Extension

The general dominance model allows the fitness of the heterozygote AD to be anywhere between AA and DD . The dominance coefficient h controls this:

h	Name	Heterozygote fitness
0	Recessive	$w_{AD} = 1$ (same as AA)
0.5	Additive	$w_{AD} = 1 + s/2$ (halfway between)
1	Dominant	$w_{AD} = 1 + s$ (same as DD)

The backward mean for general h is the full formula we implemented in `backward_mean`. The key difference from additive selection:

- **Recessive** ($h = 0$): selection is weak when the allele is rare (most copies are in heterozygotes, which have no fitness advantage). The allele frequency trajectory shows a slow initial rise followed by rapid fixation.
- **Dominant** ($h = 1$): selection is strong even when the allele is rare (heterozygotes already have the full advantage). The trajectory shows a rapid initial rise followed by a slow approach to fixation.

```
# Compare backward means under different dominance models
x_vals = np.linspace(0.01, 0.99, 50)
s = 0.05

for h_val, label in [(0.0, "Recessive (h=0)"),
                    (0.5, "Additive (h=0.5)"),
                    (1.0, "Dominant (h=1)")]:
    shifts = [backward_mean(x, s, h=h_val) - x for x in x_vals]
    # The shift should be negative (allele was rarer in the past)
    max_shift_freq = x_vals[np.argmin(shifts)]
    print(f"{label}: max backward shift at x = {max_shift_freq:.2f}, "
          f"shift = {min(shifts):.6f}")

# For recessive: max shift is at high x (selection acts mainly on homozygotes)
# For dominant: max shift is at low x (selection acts on heterozygotes)
# For additive: max shift is at x ~ 0.5
```

Step 5: Precomputing the Standard Normal CDF

CLUES2 avoids calling `scipy.stats.norm.cdf` millions of times by precomputing a lookup table of the standard normal CDF. The transition matrix computation for each row involves evaluating the normal CDF at several points, and doing this K times per generation can be expensive. The lookup table uses linear interpolation for speed.

```
def build_normal_cdf_lookup(n_points=2000):
    """Precompute a lookup table for the standard normal CDF.

    The standard normal CDF  $\Phi(z)$  is evaluated on a grid of  $z$ -values
    and stored for fast interpolation. Any normal CDF can be computed
    from the standard normal by the transformation:
         $\Phi((x - \mu) / \sigma) = P(N(\mu, \sigma^2) \leq x)$ 

    Parameters
    -----
    n_points : int
        Number of grid points for the lookup table.

    Returns
    -----
```

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```

z_bins : ndarray
    Grid of z-values (from ~-37 to ~37 for n_points=2000).
z_cdf : ndarray
    Standard normal CDF values at each z-bin.
"""
# Create points in (0, 1), avoiding exactly 0 and 1
u = np.linspace(0.0, 1.0, n_points)
u[0] = 1e-10
u[-1] = 1 - 1e-10

# Map through the inverse normal CDF (quantile function)
z_bins = norm.ppf(u)
z_cdf = norm.cdf(z_bins)  # = u (by construction), but useful for interpolation

return z_bins, z_cdf

def fast_normal_cdf(x, mu, sigma, z_bins, z_cdf):
    """Evaluate the normal CDF at x using the precomputed lookup table.

    Transforms  $(x - \mu) / \sigma$  to a standard normal z-score,
    then interpolates from the lookup table.

    Parameters
    -----
    x : float
        Point at which to evaluate the CDF.
    mu : float
        Mean of the normal distribution.
    sigma : float
        Standard deviation.
    z_bins : ndarray
        Precomputed z-values.
    z_cdf : ndarray
        Precomputed CDF values.

    Returns
    -----
    cdf_value : float
         $\Phi((x - \mu) / \sigma)$ .
    """
    z = (x - mu) / sigma
    return np.interp(z, z_bins, z_cdf)

# Verify the lookup table
z_bins, z_cdf = build_normal_cdf_lookup()
test_points = [-2.0, -1.0, 0.0, 1.0, 2.0]
print("Verification of fast normal CDF lookup:")
for z in test_points:
    exact = norm.cdf(z)
    approx = fast_normal_cdf(z, 0.0, 1.0, z_bins, z_cdf)
    print(f"  z = {z:+.1f}: exact = {exact:.8f}, approx = {approx:.8f}, "
          f"error = {abs(exact - approx):.2e}")

```

Step 6: Putting It Together – Building the Complete Transition Matrix

Here is the optimized version that uses the sparse computation and precomputed lookup table, closely following the CLUES2 implementation:

```
def build_transition_matrix_fast(freqs, N, s, z_bins, z_cdf, h=0.5):
    """Build the log-transition matrix using sparse computation.

    This is the optimized version that only computes entries within
    3.3 sigma of the mean, matching the CLUES2 implementation.

    Parameters
    -----
    freqs : ndarray of shape (K,)
        Frequency bins.
    N : float
        Haploid effective population size.
    s : float
        Selection coefficient.
    z_bins : ndarray
        Precomputed standard normal z-values.
    z_cdf : ndarray
        Precomputed standard normal CDF values.
    h : float
        Dominance coefficient.

    Returns
    -----
    logP : ndarray of shape (K, K)
        Log-transition matrix.
    lower_indices : ndarray of shape (K,)
        First nonzero column index for each row.
    upper_indices : ndarray of shape (K,)
        Last nonzero column index (+1) for each row.
    """
    K = len(freqs)
    logP = np.full((K, K), -np.inf)
    lower_indices = np.zeros(K, dtype=int)
    upper_indices = np.zeros(K, dtype=int)

    # Midpoints between bins
    midpoints = (freqs[1:] + freqs[:-1]) / 2.0

    # Absorbing states
    logP[0, 0] = 0.0
    logP[K - 1, K - 1] = 0.0
    lower_indices[0] = 0
    upper_indices[0] = 1
    lower_indices[K - 1] = K - 1
    upper_indices[K - 1] = K

    for i in range(1, K - 1):
        x = freqs[i]
        mu = backward_mean(x, s, h)
```

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```

sigma = backward_std(x, N)

if sigma < 1e-15:
    closest = np.argmin(np.abs(freqs - mu))
    logP[i, closest] = 0.0
    lower_indices[i] = closest
    upper_indices[i] = closest + 1
    continue

# Bounds for sparse computation (3.3 sigma captures 99.9%)
lower_freq = mu - 3.3 * sigma
upper_freq = mu + 3.3 * sigma
j_lo = max(0, np.searchsorted(freqs, lower_freq) - 1)
j_hi = min(K, np.searchsorted(freqs, upper_freq) + 1)

# Compute row probabilities
row = np.zeros(K)
for j in range(j_lo, j_hi):
    if j == 0:
        row[j] = fast_normal_cdf(midpoints[0], mu, sigma,
                                z_bins, z_cdf)

    elif j == K - 1:
        row[j] = 1.0 - fast_normal_cdf(midpoints[-1], mu, sigma,
                                       z_bins, z_cdf)

    else:
        row[j] = (fast_normal_cdf(midpoints[j], mu, sigma,
                                z_bins, z_cdf)
                  - fast_normal_cdf(midpoints[j - 1], mu, sigma,
                                z_bins, z_cdf))

# Renormalize and convert to log
row_sum = row.sum()
if row_sum > 0:
    row /= row_sum
logP[i, :] = np.where(row > 0, np.log(row), -np.inf)

# Record nonzero range for fast summation (Approximation A2)
nonzero = np.where(row > 0)[0]
if len(nonzero) > 0:
    lower_indices[i] = nonzero[0]
    upper_indices[i] = nonzero[-1] + 1
else:
    lower_indices[i] = 0
    upper_indices[i] = K

return logP, lower_indices, upper_indices

# Build and verify
freqs, logfreqs, log1minusfreqs = build_frequency_bins(K=100)
z_bins, z_cdf = build_normal_cdf_lookup()
N_haploid = 20000.0

```

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```

logP, lo, hi = build_transition_matrix_fast(
    freqs, N_haploid, s=0.02, z_bins=z_bins, z_cdf=z_cdf)

# Check sparsity: how many nonzero entries per row?
nnz_per_row = [hi[i] - lo[i] for i in range(len(freqs))]
print(f"Average nonzero entries per row: {np.mean(nnz_per_row):.1f}")
print(f"Max nonzero entries: {max(nnz_per_row)}")
print(f"Matrix size: {len(freqs)}x{len(freqs)} = {len(freqs)**2}")
print(f"Sparsity: {100 * (1 - np.mean(nnz_per_row)/len(freqs)):.1f}% zeros")

```

The Log-Sum-Exp Trick

Throughout CLUES, all probabilities are stored in **log space** to avoid numerical underflow. The key operation is the **log-sum-exp**:

$$\log \left(\sum_j e^{a_j} \right) = a_{\max} + \log \left(\sum_j e^{a_j - a_{\max}} \right)$$

where $a_{\max} = \max_j a_j$. This prevents overflow or underflow when the a_j values span a huge range.

```

def logsumexp(a):
    """Compute log(sum(exp(a))) in a numerically stable way.

    This is the fundamental building block for all HMM computations
    in log space. The trick: subtract the maximum before exponentiating,
    then add it back after taking the log.

    Parameters
    -----
    a : ndarray
        Array of log-probabilities.

    Returns
    -----
    result : float
        log(sum(exp(a))).
    """
    a_max = np.max(a)
    if a_max == -np.inf:
        return -np.inf
    return a_max + np.log(np.sum(np.exp(a - a_max)))

# Verify: log(exp(-1000) + exp(-1001)) should be about -999.69
result = logsumexp(np.array([-1000.0, -1001.0]))
print(f"logsumexp([-1000, -1001]) = {result:.4f}")
print(f"Expected: {-1000 + np.log(1 + np.exp(-1)):.4f}")

```

Exercises

i Exercise 1: Frequency bins and the boundary effect

Build frequency bins with $K = 50, 100, 200, 450$ and compare:

- The spacing between the first 5 bins (near $x = 0$).
- The spacing at $x \approx 0.5$.
- Build the transition matrix for each K with $N = 10000$ (haploid) and $s = 0.02$. What fraction of the matrix is nonzero? How does the sparsity change with K ?

i Exercise 2: Selection shifts the transition distribution

For $N = 20000$ (haploid), $K = 100$, starting frequency $x = 0.5$:

- Plot the transition probability distribution (row i of the transition matrix) for $s = -0.05, 0, 0.01, 0.05$.
- Compute the mean and variance of the destination frequency for each s . How does the mean shift with s ? Does the variance change?
- What happens when $|s|$ is very large (say, $s = 0.5$)? Does the normal approximation still look reasonable?

i Exercise 3: Dominance changes the dynamics

Using the same parameters as Exercise 2, compare the backward mean $\mu(x)$ for $h = 0, 0.5, 1$ across all frequencies $x \in (0, 1)$.

- At which frequency is the selection-driven shift largest for each h ?
- Why does the recessive case ($h = 0$) have its largest shift at high x , while the dominant case ($h = 1$) has it at low x ?
- Verify that all three h values give the same shift when $x = 0.5$. Why is this true?

Solutions

i Solution 1: Frequency bins and the boundary effect

```
for K in [50, 100, 200, 450]:
    freqs, _, _ = build_frequency_bins(K)
    z_bins, z_cdf = build_normal_cdf_lookup()

    spacing_near_0 = np.diff(freqs[:5])
    idx_half = K // 2
    spacing_near_half = np.diff(freqs[idx_half-2:idx_half+3])

    print(f"\nK = {K}:")
    print(f"  First 5 bins: {np.round(freqs[:5], 8)}")
    print(f"  Spacing near 0: {np.round(spacing_near_0, 8)}")
    print(f"  Spacing near 0.5: {np.round(spacing_near_half, 6)}")

    logP, lo, hi = build_transition_matrix_fast(
        freqs, 10000.0, s=0.02, z_bins=z_bins, z_cdf=z_cdf)
```

```
nnz = sum(hi[i] - lo[i] for i in range(K))
print(f" Nonzero entries: {nnz} / {K*K} = {100*nnz/(K*K):.1f}%")
```

The boundary spacing shrinks as K increases, giving finer resolution near fixation/loss. The sparsity increases with K because each row's nonzero band width (in bin indices) grows only as $\sqrt{K}$, while the total row length grows linearly.

i Solution 2: Selection shifts the transition distribution

```
K = 100
freqs, _, _ = build_frequency_bins(K)
z_bins, z_cdf = build_normal_cdf_lookup()
N_haploid = 20000.0

i_half = np.argmin(np.abs(freqs - 0.5))
print(f"Starting bin: x[{i_half}] = {freqs[i_half]:.4f}")

for s_val in [-0.05, 0.0, 0.01, 0.05]:
    logP, _, _ = build_transition_matrix_fast(
        freqs, N_haploid, s=s_val, z_bins=z_bins, z_cdf=z_cdf)
    row = np.exp(logP[i_half, :])
    mean_freq = np.sum(freqs * row)
    var_freq = np.sum((freqs - mean_freq)**2 * row)
    print(f" s = {s_val:+.2f}: mean = {mean_freq:.6f}, "
          f"var = {var_freq:.8f}, shift = {mean_freq - freqs[i_half]:+.6f}")
```

What happens:

- The mean shifts proportionally to s (positive s shifts the backward mean downward; negative s shifts it upward).
- The variance is nearly identical for all values of s – it depends on x and N , not on s .
- For very large $|s|$ (like 0.5), the normal approximation breaks down because the mean shift per generation can be comparable to the standard deviation. In practice, $|s| < 0.1$ is the regime where CLUES works well.

i Solution 3: Dominance changes the dynamics

```
x_vals = np.linspace(0.01, 0.99, 200)
s = 0.05

for h_val, label in [(0.0, "Recessive"), (0.5, "Additive"), (1.0, "Dominant")]:
    shifts = np.array([backward_mean(x, s, h=h_val) - x for x in x_vals])
    peak_x = x_vals[np.argmin(shifts)] # most negative shift
    print(f"{label} (h={h_val}): peak shift at x = {peak_x:.2f}, "
          f"shift = {shifts.min():.6f}")

# Check that all h values agree at x = 0.5
x_test = 0.5
for h_val in [0.0, 0.5, 1.0]:
    mu = backward_mean(x_test, s, h=h_val)
    print(f"h = {h_val}, x = 0.5: mu = {mu:.8f}, shift = {mu - x_test:+.8f}")
```

(a) Recessive: peak shift near $x \approx 0.8$. Dominant: peak shift near $x \approx 0.2$. Additive: peak shift near $x \approx 0.5$.

(b) For the recessive case, selection only acts on DD homozygotes (frequency x^2), so the selective effect is proportional to x^2 . This is largest when x is high. For the dominant case, selection acts on both DD and AD (total frequency $1 - (1 - x)^2$), which is already large when x is small.

(c) At $x = 0.5$, the heterozygote frequency $2x(1 - x) = 0.5$ regardless of h . The homozygote frequencies are also 0.25 each. At this symmetric point, the dominance coefficient doesn't matter because the average fitness effect over all genotypes is the same.

Emission Probabilities

The gear train: how coalescence events and ancient genotypes constrain the allele frequency at each moment in time.

The transition matrix (previous chapter) describes how the allele frequency *evolves*. But the frequency trajectory is hidden – we never observe it directly. What we *do* observe are two kinds of evidence:

1. **Coalescence events in the gene tree** – when lineages merge, and how many remain at each point in time.
2. **Ancient genotype likelihoods** – direct (but noisy) observations of the allele in individuals sampled from the past.

This chapter derives the emission probabilities that connect these observations to the hidden allele frequency. By the end, you will have built the complete observation model of the CLUES HMM.

Step 1: Coalescent Emissions

The gene tree at the focal SNP has two kinds of lineages: those carrying the **derived allele** (with the mutation) and those carrying the **ancestral allele** (without it). CLUES observes the coalescence times of these lineages and asks: given allele frequency x_t at time t , how likely are the observed coalescence events?

The coalescent among derived-allele lineages

Consider n_D lineages carrying the derived allele. At time t , if the allele frequency is x_t , these lineages are confined to a sub-population of effective haploid size $x_t \cdot N(t)$, where $N(t)$ is the total haploid effective population size.

From the *coalescent theory prerequisite*, k lineages in a population of size M coalesce at rate $\binom{k}{2}/M$. For derived lineages at frequency x_t :

$$\text{Coalescence rate of } k \text{ derived lineages} = \frac{\binom{k}{2}}{x_t \cdot N(t)}$$

Why divide by $x_t \cdot N(t)$ and not $2 \cdot x_t \cdot N(t)$?

The convention matters. In CLUES, $N(t)$ is the **haploid** effective population size (twice the diploid N_e). The derived lineages are confined to the fraction x_t of this haploid population, giving an effective sub-population size of $x_t \cdot N(t)$. With k lineages, there are $\binom{k}{2}$ pairs, and each pair coalesces at rate $1/(x_t \cdot N(t))$.

In the CLUES2 code, you'll see $k(k - 1)/4$ instead of $\binom{k}{2} = k(k - 1)/2$. This is because the code uses *diploid* N (which is half the haploid size), so the factor of 2 is absorbed: $\binom{k}{2}/(x \cdot N_{\text{haploid}}) = k(k - 1)/2 \cdot 1/(x \cdot 2N_{\text{diploid}}) = k(k - 1)/(4 \cdot x \cdot N_{\text{diploid}})$.

Between two consecutive coalescence events, the number of surviving lineages is constant, and the waiting time until the next coalescence follows an **exponential distribution**. The density of observing a coalescence at time t_i (given k

remaining lineages and frequency x) over an epoch $[t_0, t_1]$ is:

$$f(t_i) = \frac{\binom{k}{2}}{x \cdot N} \cdot \exp\left(-\frac{\binom{k}{2}}{x \cdot N} \cdot (t_i - t_{i-1})\right)$$

where t_{i-1} is the time of the previous coalescence (or the start of the epoch). After the last coalescence event, the surviving lineages must *not* coalesce before the end of the epoch – this is a **survival probability**:

$$P(\text{no coalescence in } [t_d, t_1]) = \exp\left(-\frac{\binom{k'}{2}}{x \cdot N} \cdot (t_1 - t_d)\right)$$

where k' is the number of remaining lineages and t_d is the last coalescence time.

Putting it together. For d coalescence events at times $t_1 < t_2 < \dots < t_d$ among n lineages during epoch $[e_0, e_1]$ at frequency x and population size N :

$$\log P(\text{coal events}) = \sum_{i=1}^d \left[-\log(x) + \log\left(\frac{k_i(k_i-1)}{4N}\right) - \frac{k_i(k_i-1)}{4xN}(t_i - t_{i-1}) \right] - \frac{k'(k'-1)}{4xN}(e_1 - t_d)$$

where $k_i = n - i + 1$ is the number of lineages just before the i -th coalescence, $t_0 = e_0$, and $k' = n - d$.

The same formula applies to ancestral lineages, replacing x with $1 - x$ (the ancestral allele frequency).

```
import numpy as np

def log_coalescent_density(coal_times, n_lineages, epoch_start, epoch_end,
                           freq, N_diploid, ancestral=False):
    """Compute the log-probability of coalescence events in one epoch.

    This is the core emission computation for the CLUES HMM. It
    computes the probability of observing a specific set of coalescence
    times given the allele frequency and population size.

    Parameters
    -----
    coal_times : ndarray
        Sorted coalescence times within this epoch. May be empty.
    n_lineages : int
        Number of lineages at the start of the epoch.
    epoch_start : float
        Start time of the epoch (generations).
    epoch_end : float
        End time of the epoch (generations).
    freq : float
        Derived allele frequency (0 < freq < 1).
    N_diploid : float
        Diploid effective population size.
    ancestral : bool
        If True, use ancestral frequency (1 - freq) instead.

    Returns
    -----
    log_prob : float
        Log-probability of the coalescence events.
```

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```

"""
if n_lineages <= 1:
    # No coalescence possible with 0 or 1 lineages
    return 0.0

xi = (1.0 - freq) if ancestral else freq

if xi * N_diploid == 0.0:
    # Impossible: lineages exist but frequency is 0
    return -1e20

logp = 0.0
prev_t = epoch_start
k = n_lineages

for t in coal_times:
    # k choose 2, divided by 4 because N is diploid
    # (equivalent to k(k-1)/2 divided by 2*N_diploid = N_haploid)
    kchoose2_over_4 = k * (k - 1) / 4.0
    rate = kchoose2_over_4 / (xi * N_diploid)

    # Exponential density: rate * exp(-rate * dt)
    # In log: log(rate) - rate * dt
    # But we split: -log(xi) accounts for the 1/xi factor in the rate
    dt = t - prev_t
    logp += -np.log(xi) - kchoose2_over_4 / (xi * N_diploid) * dt

    prev_t = t
    k -= 1

# Survival probability: no further coalescences until epoch end
if k >= 2:
    kchoose2_over_4 = k * (k - 1) / 4.0
    logp += -kchoose2_over_4 / (xi * N_diploid) * (epoch_end - prev_t)

return logp

# Example: 3 derived lineages, one coalescence at t=0.5 in epoch [0, 1]
coal_times = np.array([0.5])
log_prob = log_coalescent_density(
    coal_times, n_lineages=3, epoch_start=0.0, epoch_end=1.0,
    freq=0.3, N_diploid=10000.0, ancestral=False)
print(f"Log-probability of coalescence: {log_prob:.4f}")

# How does the probability change with frequency?
print("\nCoalescence probability vs. derived allele frequency:")
for freq in [0.1, 0.2, 0.3, 0.5, 0.7, 0.9]:
    lp = log_coalescent_density(
        coal_times, n_lineages=3, epoch_start=0.0, epoch_end=1.0,
        freq=freq, N_diploid=10000.0)
    print(f"  freq = {freq:.1f}: log P = {lp:.4f}")

```

The Mixed Lineage Problem

There is a subtlety at the root of the gene tree. The mutation that created the derived allele sits on a specific branch. Below this branch (closer to the present), the lineage carries the derived allele. Above it (further into the past), the lineage is ancestral. At the exact time of the mutation, one lineage transitions from derived to ancestral.

When we reach the point where only $n_D = 1$ derived lineage and $n_A = 1$ ancestral lineage remain, these two lineages must coalesce as *ancestral* lineages (because the single derived lineage's ancestor is also ancestral). CLUES handles this by treating the coalescence as occurring among $n_A = 2$ ancestral lineages when $n_D = 1$.

```
def compute_coalescent_emissions(coal_times_der, coal_times_anc,
                                n_der, n_anc, epoch_start, epoch_end,
                                freqs, N_diploid):
    """Compute coalescent emission probabilities for all frequency bins.

    Handles the special cases for mixed lineages:
    - If n_der > 1: standard derived + ancestral coalescences
    - If n_der == 1 and n_anc == 1: treat as 2 ancestral lineages (freq != 0)
    - If n_der == 1 and n_anc > 1: ancestral coalescences, but with n_anc+1
      lineages when freq = 0 (all lineages become ancestral)
    - If n_der == 0: all remaining lineages are ancestral

    Parameters
    -----
    coal_times_der : ndarray
        Derived coalescence times in this epoch.
    coal_times_anc : ndarray
        Ancestral coalescence times in this epoch.
    n_der : int
        Number of remaining derived lineages.
    n_anc : int
        Number of remaining ancestral lineages.
    epoch_start : float
        Start of epoch.
    epoch_end : float
        End of epoch.
    freqs : ndarray of shape (K,)
        Frequency bins.
    N_diploid : float
        Diploid effective population size.

    Returns
    -----
    emissions : ndarray of shape (K,)
        Log-emission probabilities for each frequency bin.
    """
    K = len(freqs)
    emissions = np.zeros(K)

    for j in range(K):
        x = freqs[j]

        if n_der > 1:
            # Standard case: both derived and ancestral coalescences
```

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```

emissions[j] = log_coalescent_density(
    coal_times_der, n_der, epoch_start, epoch_end,
    x, N_diploid, ancestral=False)
emissions[j] += log_coalescent_density(
    coal_times_anc, n_anc, epoch_start, epoch_end,
    x, N_diploid, ancestral=True)

elif n_der == 0 and n_anc <= 1:
    # No lineages or single lineage: no coalescence possible
    if j != 0:
        emissions[j] = -1e20 # freq must be 0 (allele lost)

elif n_der == 0 and n_anc > 1:
    # All remaining lineages are ancestral
    if j != 0:
        emissions[j] = -1e20 # freq must be 0
    else:
        emissions[j] = log_coalescent_density(
            coal_times_anc, n_anc, epoch_start, epoch_end,
            x, N_diploid, ancestral=True)

elif n_der == 1 and n_anc == 1:
    # Mixed lineage: the single derived + single ancestral
    # coalesce as 2 ancestral lineages
    if j != 0:
        emissions[j] = 0.0 # no constraint from freq
    else:
        emissions[j] = log_coalescent_density(
            coal_times_anc, 2, epoch_start, epoch_end,
            x, N_diploid, ancestral=True)

elif n_der == 1 and n_anc > 1:
    # One derived lineage remains, multiple ancestral
    if j != 0:
        emissions[j] = log_coalescent_density(
            coal_times_anc, n_anc, epoch_start, epoch_end,
            x, N_diploid, ancestral=True)
    else:
        # At freq = 0, the derived lineage joins the ancestral pool
        emissions[j] = log_coalescent_density(
            coal_times_anc, n_anc + 1, epoch_start, epoch_end,
            x, N_diploid, ancestral=True)

return emissions

```

Step 2: Ancient Genotype Likelihood Emissions

When an ancient individual is sampled at time t , we observe their genotype (or, more precisely, genotype likelihoods from sequencing reads). This provides a direct constraint on the allele frequency at that time.

Diploid genotype likelihoods

For a diploid individual sampled at time t , the genotype likelihoods are $P(R | g)$ for each genotype $g \in \{AA, AD, DD\}$, where R is the sequencing read data. We don't know the true genotype, but we know the probability of each genotype given the allele frequency:

$$P(g = AA | x) = (1 - x)^2, \quad P(g = AD | x) = 2x(1 - x), \quad P(g = DD | x) = x^2$$

The **emission probability** for this ancient sample at frequency x is:

$$e(x) = (1 - x)^2 \cdot P(R | AA) + 2x(1 - x) \cdot P(R | AD) + x^2 \cdot P(R | DD)$$

In log space:

$$\log e(x) = \text{logsumexp}\left(2 \log(1 - x) + \log P(R | AA), \log 2 + \log x + \log(1 - x) + \log P(R | AD), 2 \log x + \log P(R | DD)\right)$$

Why genotype *likelihoods* instead of genotype *calls*?

Ancient DNA is often degraded and low-coverage. A site might have only 1-2 sequencing reads, making the true genotype uncertain. Using genotype likelihoods (the probability of the reads given each possible genotype) rather than hard genotype calls preserves this uncertainty and avoids bias from calling errors.

For example, if a site has 3 reads showing the derived allele and 1 showing ancestral, the genotype likelihoods might be $P(R | AA) = 0.01$, $P(R | AD) = 0.24$, $P(R | DD) = 0.75$. A hard call would say "DD", but the likelihoods correctly represent the 24% chance of being heterozygous.

```
def genotype_likelihood_emission(anc_gl, log_freq, log_1minus_freq):
    """Compute the log-emission probability for a diploid ancient sample.

    Parameters
    -----
    anc_gl : ndarray of shape (3,)
        Log genotype likelihoods: [log P(R|AA), log P(R|AD), log P(R|DD)].
    log_freq : float
        log(x), where x is the derived allele frequency.
    log_1minus_freq : float
        log(1 - x).

    Returns
    -----
    log_emission : float
        log P(R | freq = x), marginalizing over genotypes.
    """
    # Hardy-Weinberg genotype frequencies (in log space)
    log geno_freqs = np.array([
        log_1minus_freq + log_1minus_freq,      # log((1-x)^2) = AA
        np.log(2) + log_freq + log_1minus_freq,  # log(2x(1-x)) = AD
        log_freq + log_freq                      # log(x^2) = DD
    ])

    # Combine: P(R|x) = sum_g P(g|x) * P(R|g)
    log_emission = logsumexp(log geno_freqs + anc_gl)
```

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```

if np.isnan(log_emission):
    return -np.inf
return log_emission

# Example: ancient individual with 3 derived reads, 1 ancestral read
# Genotype likelihoods (in log space)
gl = np.log(np.array([0.01, 0.24, 0.75])) # P(R|AA), P(R|AD), P(R|DD)

print("Ancient genotype emission vs. frequency:")
for freq in [0.1, 0.3, 0.5, 0.7, 0.9]:
    log_em = genotype_likelihood_emission(
        gl, np.log(freq), np.log(1 - freq))
    print(f"  freq = {freq:.1f}: log P(R|x) = {log_em:.4f}, "
          f"P(R|x) = {np.exp(log_em):.6f}")

```

Haploid genotype likelihoods

For haploid samples (or phased data), each individual carries exactly one allele:

$$e(x) = x \cdot P(R | D) + (1 - x) \cdot P(R | A)$$

```

def haplotype_likelihood_emission(anc_gl, log_freq, log_1minus_freq):
    """Compute the log-emission probability for a haploid ancient sample.

    Parameters
    -----
    anc_gl : ndarray of shape (2,)
        Log haplotype likelihoods: [log P(R|ancestral), log P(R|derived)].
    log_freq : float
        log(x), derived allele frequency.
    log_1minus_freq : float
        log(1 - x).

    Returns
    -----
    log_emission : float
    """
    # Haplotype frequencies: P(ancestral) = 1-x, P(derived) = x
    log_hap_freqs = np.array([log_1minus_freq, log_freq])

    log_emission = logsumexp(log_hap_freqs + anc_gl)
    if np.isnan(log_emission):
        return -np.inf
    return log_emission

```

Known haplotype emissions

When a haplotype's allelic state is known with certainty (as when lineages from the gene tree are sampled at specific times in the past), the emission is simply $\log(x)$ for a derived haplotype or $\log(1 - x)$ for an ancestral haplotype:

$$e_D(x) = x, \quad e_A(x) = 1 - x$$

This represents the probability that a randomly sampled individual at frequency x carries the specified allele.

Step 3: Combining Emissions Within an Epoch

At each generation t , the total emission probability combines all evidence:

1. Coalescent emissions from derived and ancestral lineages.
2. Ancient diploid genotype likelihoods (if any samples fall in this generation).
3. Ancient haploid genotype likelihoods (if any).
4. Known haplotype emissions from ARG samples.

These are all independent (given the frequency), so their log-probabilities **add**:

$$\log e_{\text{total}}(t, x_k) = \log e_{\text{coal}}(t, x_k) + \sum_{i \in \text{diploid}} \log e_{\text{GL}}^{(i)}(x_k) + \sum_{j \in \text{haploid}} \log e_{\text{hap}}^{(j)}(x_k) + n_D^{(t)} \log(x_k) + n_A^{(t)} \log(1 - x_k)$$

where $n_D^{(t)}$ and $n_A^{(t)}$ are the numbers of derived and ancestral haplotypes sampled from the ARG at generation t .

```
def compute_total_emissions(freq_bins, logfreqs, logminusfreqs,
                           coal_times_der, coal_times_anc,
                           n_der, n_anc, epoch_start, epoch_end,
                           N_diploid,
                           diploid_gls=None, haploid_gls=None,
                           n_der_sampled=0, n_anc_sampled=0):
    """Compute total emission probabilities for all frequency bins.

    Combines coalescent emissions, ancient genotype likelihoods,
    and known haplotype emissions.

    Parameters
    -----
    freq_bins : ndarray of shape (K,)
        Frequency bins.
    logfreqs : ndarray of shape (K,)
        log(freq_bins).
    logminusfreqs : ndarray of shape (K,)
        log(1 - freq_bins).
    coal_times_der : ndarray
        Derived coalescence times in this epoch.
    coal_times_anc : ndarray
        Ancestral coalescence times in this epoch.
    n_der : int
        Remaining derived lineages.
    n_anc : int
        Remaining ancestral lineages.
    epoch_start : float
        Start of epoch (generations).
    epoch_end : float
        End of epoch (generations).
    N_diploid : float
        Diploid effective population size.
    diploid_gls : list of ndarray, optional
        Each element is [log P(R|AA), log P(R|AD), log P(R|DD)] for one
```

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```

    ancient diploid sample in this epoch.
haploid_gls : list of ndarray, optional
    Each element is [log P(R|A), log P(R|D)] for one ancient haploid
    sample in this epoch.
n_der_sampled : int
    Number of known derived haplotypes sampled in this epoch.
n_anc_sampled : int
    Number of known ancestral haplotypes sampled in this epoch.

Returns
-----
total_emissions : ndarray of shape (K,)
    Log-emission probability for each frequency bin.
"""
K = len(freq_bins)

# 1. Coalescent emissions
coal_emissions = compute_coalescent_emissions(
    coal_times_der, coal_times_anc, n_der, n_anc,
    epoch_start, epoch_end, freq_bins, N_diploid)

# 2. Ancient genotype likelihoods
gl_emissions = np.zeros(K)
if diploid_gls is not None:
    for gl in diploid_gls:
        for j in range(K):
            gl_emissions[j] += genotype_likelihood_emission(
                gl, logfreqs[j], log1minusfreqs[j])

if haploid_gls is not None:
    for gl in haploid_gls:
        for j in range(K):
            gl_emissions[j] += haplotype_likelihood_emission(
                gl, logfreqs[j], log1minusfreqs[j])

# 3. Known haplotype emissions from ARG samples
hap_emissions = np.zeros(K)
for j in range(K):
    if n_der_sampled > 0:
        hap_emissions[j] += n_der_sampled * logfreqs[j]
    if n_anc_sampled > 0:
        hap_emissions[j] += n_anc_sampled * log1minusfreqs[j]

return coal_emissions + gl_emissions + hap_emissions

# Example: no ancient samples, just coalescent emissions
freqs, logfreqs, log1minusfreqs = build_frequency_bins(K=50)
coal_der = np.array([0.3]) # one derived coalescence at t=0.3
coal_anc = np.array([])    # no ancestral coalescences
emissions = compute_total_emissions(
    freqs, logfreqs, log1minusfreqs,
    coal_der, coal_anc, n_der=3, n_anc=2,

```

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```

epoch_start=0.0, epoch_end=1.0, N_diploid=10000.0)

# The emissions should peak where the frequency makes the coalescence
# times most likely
peak_bin = np.argmax(emissions)
print(f"Peak emission at freq = {freqs[peak_bin]:.4f}")
print(f"Log-emission range: [{emissions.min():.1f}, {emissions.max():.4f}]")

```

Step 4: How Emissions Constrain the Frequency

Let's build intuition for how different types of evidence constrain the allele frequency.

Coalescent emissions alone. If all derived lineages coalesce rapidly (short coalescence times), the emission probability peaks at *low* frequency – because at low frequency, the effective population of derived lineages is small, forcing rapid coalescence. Conversely, if coalescence is slow, the frequency was likely high.

Ancient genotype likelihoods. An ancient individual with genotype *DD* (two derived copies) provides evidence for high frequency; genotype *AA* provides evidence for low frequency. Heterozygotes *AD* favor intermediate frequencies.

The two types of evidence complement each other. Coalescent emissions come from the *shape* of the gene tree (deep in the past, many generations back). Ancient genotypes provide direct snapshots of frequency at specific points in time. Together, they constrain the trajectory far more tightly than either alone.

```

# Demonstrate how different data constrain the frequency differently

freqs, logfreqs, log1minusfreqs = build_frequency_bins(K=100)

# Scenario 1: Rapid derived coalescence (3 lineages coalescing at t=0.1 and t=0.2)
coal_der_fast = np.array([0.1, 0.2])
e_fast = compute_coalescent_emissions(
    coal_der_fast, np.array([]), n_der=3, n_anc=2,
    epoch_start=0.0, epoch_end=1.0, freqs=freqs, N_diploid=10000.0)

# Scenario 2: Slow derived coalescence (coalescing at t=0.8 and t=0.9)
coal_der_slow = np.array([0.8, 0.9])
e_slow = compute_coalescent_emissions(
    coal_der_slow, np.array([]), n_der=3, n_anc=2,
    epoch_start=0.0, epoch_end=1.0, freqs=freqs, N_diploid=10000.0)

# Normalize for comparison
e_fast_norm = np.exp(e_fast - logsumexp(e_fast))
e_slow_norm = np.exp(e_slow - logsumexp(e_slow))

peak_fast = freqs[np.argmax(e_fast_norm)]
peak_slow = freqs[np.argmax(e_slow_norm)]
print(f"Fast coalescence: peak at freq = {peak_fast:.4f} (low freq expected)")
print(f"Slow coalescence: peak at freq = {peak_slow:.4f} (high freq expected)")

# Scenario 3: Ancient DD genotype (strong evidence for high frequency)
gl_DD = np.log(np.array([0.001, 0.01, 0.989])) # clearly DD
e_ancient = np.array([
    genotype_likelihood_emission(gl_DD, logfreqs[j], log1minusfreqs[j])
    for j in range(len(freqs))])

```

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```
e_ancient_norm = np.exp(e_ancient - logsumexp(e_ancient))
peak_ancient = freqs[np.argmax(e_ancient_norm)]
print(f"Ancient DD sample: peak at freq = {peak_ancient:.4f}")
```

Exercises

Exercise 1: Coalescence rate vs. frequency

For $n = 5$ derived lineages and $N = 10000$ (diploid):

- Compute the pairwise coalescence rate as a function of frequency $x \in \{0.01, 0.05, 0.1, 0.3, 0.5, 0.9\}$.
- At what frequency does the expected time to the first coalescence equal 1 generation? (Solve $\binom{5}{2} / (x \cdot 2N) = 1$.)
- If the actual first coalescence time is 0.001 generations, what frequency does this most strongly support? Why?

Exercise 2: Ancient sample information content

Compare the information content (the “peakedness” of the emission distribution) for three ancient sample scenarios, all at the same frequency bins:

- A high-confidence DD call: $P(R|AA) = 0.001$, $P(R|AD) = 0.01$, $P(R|DD) = 0.989$.
- An ambiguous call: $P(R|AA) = 0.3$, $P(R|AD) = 0.4$, $P(R|DD) = 0.3$.
- No ancient sample (uniform emission).

For each, compute the emission across all frequency bins and measure the entropy. Lower entropy = more informative.

Exercise 3: The mixed lineage transition

Consider a gene tree with 3 derived and 2 ancestral lineages. Walk through the coalescent process epoch by epoch as lineages coalesce:

- In epoch 1: 2 derived lineages coalesce. After: 2 derived, 2 ancestral.
- In epoch 2: 1 derived coalesces with 1 ancestral. After: 1 derived, 1 ancestral. But this is the mixed lineage case! What happens?
- Implement the emission computation for each epoch and verify that the log- probability is finite (not $-\infty$) for intermediate frequencies.

Solutions

Solution 1: Coalescence rate vs. frequency

```
n = 5
N_dip = 10000
N_hap = 2 * N_dip # haploid = 20,000

print("(a) Coalescence rate vs. frequency:")
```

```

for x in [0.01, 0.05, 0.1, 0.3, 0.5, 0.9]:
    rate = n * (n - 1) / 2 / (x * N_hap)
    expected_time = 1.0 / rate
    print(f"  x = {x:.2f}: rate = {rate:.4f}/gen, "
          f"E[T_first] = {expected_time:.2f} gen")

# (b) Solve: C(5,2) / (x * 2N) = 1
# => x = C(5,2) / (2N) = 10 / 20000 = 0.0005
x_threshold = 10.0 / N_hap
print(f"\n(b) E[T_first] = 1 gen when x = {x_threshold:.6f}")

# (c) Very fast coalescence (0.001 gen) supports very low frequency
# because the coalescence rate must be ~1000/gen = C(5,2)/(x*2N)
x_fast = 10.0 / (N_hap * 1000)
print(f"\n(c) Coal at t=0.001 supports x ~ {x_fast:.6f}")

```

Very fast coalescence strongly supports low allele frequency, because the derived lineages must be crammed into a tiny sub-population.

i Solution 2: Ancient sample information content

```

from scipy.stats import entropy as scipy_entropy

freqs, logfreqs, log1minusfreqs = build_frequency_bins(K=100)

scenarios = {
    "(a) High-confidence DD": np.log(np.array([0.001, 0.01, 0.989])),
    "(b) Ambiguous call": np.log(np.array([0.3, 0.4, 0.3])),
    "(c) No ancient sample": None,
}

for name, gl in scenarios.items():
    if gl is not None:
        emissions = np.array([
            genotype_likelihood_emission(gl, logfreqs[j], log1minusfreqs[j])
            for j in range(len(freqs))
        ])
    else:
        emissions = np.zeros(len(freqs)) # uniform (no information)

    probs = np.exp(emissions - logsumexp(emissions))
    ent = scipy_entropy(probs)
    peak = freqs[np.argmax(probs)]
    print(f"{name}: entropy = {ent:.4f}, peak at x = {peak:.4f}")

```

The high-confidence DD sample has the lowest entropy (most informative), peaking near $x = 1$. The ambiguous call has higher entropy but still provides some constraint. No ancient sample gives uniform emissions (maximum entropy).

i Solution 3: The mixed lineage transition

```

freqs, logfreqs, log1minusfreqs = build_frequency_bins(K=50)
N_dip = 10000.0

```

```

# Epoch 1: 2 derived lineages coalesce at t=0.5
# Before: 3 derived, 2 ancestral. After: 2 derived, 2 ancestral.
e1 = compute_coalescent_emissions(
    np.array([0.5]), np.array([]),
    n_der=3, n_anc=2, epoch_start=0.0, epoch_end=1.0,
    freqs=freqs, N_diploid=N_dip)
print(f"Epoch 1 (3D, 2A, 1 der coal): max log-emission = {e1.max():.4f}")

# Epoch 2: In this epoch, n_der=2, n_anc=2.
# One derived and one ancestral coalesce.
# After: n_der=1, n_anc=1. This is the mixed lineage case.
e2 = compute_coalescent_emissions(
    np.array([0.3]), np.array([0.7]),
    n_der=2, n_anc=2, epoch_start=1.0, epoch_end=2.0,
    freqs=freqs, N_diploid=N_dip)
print(f"Epoch 2 (2D, 2A, coals): max log-emission = {e2.max():.4f}")

# Epoch 3: n_der=1, n_anc=1 (mixed lineage case)
# No coalescences in this epoch
e3 = compute_coalescent_emissions(
    np.array([]), np.array([]),
    n_der=1, n_anc=1, epoch_start=2.0, epoch_end=3.0,
    freqs=freqs, N_diploid=N_dip)
n_finite = np.sum(e3 > -1e19)
print(f"Epoch 3 (1D, 1A, mixed): {n_finite}/{len(freqs)} "
      f"bins have finite emission")

```

In epoch 3, the mixed lineage case allows finite emissions for all frequency bins except $x = 0$ (where the code correctly handles the special case of 2 ancestral lineages). The derived allele has not been lost, so $x > 0$ is required.

Inference: From Gene Trees to Selection

The case and dial: assembling the mechanism and reading the result.

This chapter brings all the pieces together: the Wright-Fisher HMM (transition matrix), the emission probabilities (coalescent + ancient DNA), and the importance sampling framework that handles genealogical uncertainty. By the end, you will have a complete CLUES implementation that estimates selection coefficients, tests for selection, and reconstructs allele frequency trajectories.

Step 1: The Backward Algorithm

The backward algorithm is the workhorse of CLUES. It processes the HMM from the **present** to the **past**, starting at the observed modern allele frequency and propagating backward through time. At each generation, it combines the transition probabilities with the emission probabilities to compute the probability of all the data observed from the present up to that point.

Initialization. At the present (time $t = 0$), we know the derived allele frequency p_0 . The backward algorithm initializes with a delta function at the frequency bin closest to p_0 :

$$\alpha_0(k) = \begin{cases} 0 & \text{if } x_k \text{ is closest to } p_0 \\ -\infty & \text{otherwise} \end{cases}$$

(in log space, 0 means probability 1 and $-\infty$ means probability 0).

Recursion. For each generation t going backward (from $t = 0$ to $t = T$), the update is:

$$\alpha_t(k) = \log e(t, x_k) + \text{logsumexp}_j (\alpha_{t-1}(j) + \log P_{j,k}^\top)$$

Note the transpose: we use $P_{j,k}^\top$ (column k of the transition matrix) because we are propagating *backward*. The transition matrix $P_{i,j}$ gives the probability of going from frequency bin i to bin j backward in time. To go from the present to the past, we need the probability of arriving at bin k from any bin j , which uses column k .

Final likelihood. The total log-likelihood is:

$$\log P(\mathbf{D} \mid s) = \text{logsumexp}_k (\alpha_T(k))$$

```
import numpy as np

def backward_algorithm(sel, freqs, logfreqs, log1minusfreqs,
                      z_bins, z_cdf, epochs, N_vec, h,
                      coal_times_der_all, coal_times_anc_all,
                      n_der_initial, n_anc_initial,
                      curr_freq,
                      diploid_gls_by_epoch=None,
                      haploid_gls_by_epoch=None,
                      der_sampled_by_epoch=None,
                      anc_sampled_by_epoch=None):
    """Run the CLUES backward algorithm (present to past).

    Parameters
    -----
    sel : ndarray
        Selection coefficient for each epoch.
    freqs : ndarray of shape (K,)
        Frequency bins.
    logfreqs, log1minusfreqs : ndarray of shape (K,)
        Log-frequencies for emission computation.
    z_bins, z_cdf : ndarray
        Precomputed normal CDF lookup table.
    epochs : ndarray
        Array of generation indices [0, 1, 2, ..., T].
    N_vec : ndarray
        Diploid effective population size at each epoch.
    h : float
        Dominance coefficient.
    coal_times_der_all : ndarray
        All derived coalescence times (sorted).
    coal_times_anc_all : ndarray
        All ancestral coalescence times (sorted).
    n_der_initial : int
        Number of derived lineages at the present.
    n_anc_initial : int
        Number of ancestral lineages at the present.
    curr_freq : float
        Observed modern derived allele frequency.
    diploid_gls_by_epoch : dict, optional
        Maps epoch index to list of diploid GL arrays.
    haploid_gls_by_epoch : dict, optional
```

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```

    Maps epoch index to list of haploid GL arrays.
der_sampled_by_epoch : dict, optional
    Maps epoch index to number of derived haplotypes sampled.
anc_sampled_by_epoch : dict, optional
    Maps epoch index to number of ancestral haplotypes sampled.

Returns
-----
alpha_mat : ndarray of shape (T+1, K)
    Log-probability matrix. alpha_mat[t, k] is the log-probability
    of the data from time 0 to t, with frequency bin k at time t.
"""
K = len(freqs)
T = len(epochs)

# Initialize: delta function at modern frequency
alpha = np.full(K, -1e20)
best_bin = np.argmin(np.abs(freqs - curr_freq))
alpha[best_bin] = 0.0

alpha_mat = np.full((T, K), -1e20)
alpha_mat[0, :] = alpha

# Track remaining lineages
n_der = n_der_initial
n_anc = n_anc_initial

prev_N = -1
prev_s = -1
logP = None

for tb in range(T - 1):
    epoch_start = float(tb)
    epoch_end = float(tb + 1)
    N_t = N_vec[tb]
    s_t = sel[tb] if tb < len(sel) else 0.0

    prev_alpha = alpha.copy()

    # Recompute transition matrix only if N or s changed
    if N_t != prev_N or s_t != prev_s:
        logP, lo_idx, hi_idx = build_transition_matrix_fast(
            freqs, 2 * N_t, s_t, z_bins, z_cdf, h)
        prev_N = N_t
        prev_s = s_t

    # Gather coalescence times in this epoch
    mask_der = (coal_times_der_all > epoch_start) & \
        (coal_times_der_all <= epoch_end)
    coal_der = coal_times_der_all[mask_der]

    mask_anc = (coal_times_anc_all > epoch_start) & \

```

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```

        (coal_times_anc_all <= epoch_end)
coal_anc = coal_times_anc_all[mask_anc]

# Gather ancient samples in this epoch
dip_gls = (diploid_gls_by_epoch or {}).get(tb, [])
hap_gls = (haploid_gls_by_epoch or {}).get(tb, [])
n_der_samp = (der_sampled_by_epoch or {}).get(tb, 0)
n_anc_samp = (anc_sampled_by_epoch or {}).get(tb, 0)

# Compute emissions
coal_emissions = compute_coalescent_emissions(
    coal_der, coal_anc, n_der, n_anc,
    epoch_start, epoch_end, freqs, N_t)

gl_emissions = np.zeros(K)
for gl in dip_gls:
    for j in range(K):
        gl_emissions[j] += genotype_likelihood_emission(
            gl, logfreqs[j], log1minusfreqs[j])
for gl in hap_gls:
    for j in range(K):
        gl_emissions[j] += haplotype_likelihood_emission(
            gl, logfreqs[j], log1minusfreqs[j])

hap_emissions = np.zeros(K)
for j in range(K):
    if n_der_samp > 0:
        hap_emissions[j] += n_der_samp * logfreqs[j]
    if n_anc_samp > 0:
        hap_emissions[j] += n_anc_samp * log1minusfreqs[j]

total_emissions = gl_emissions + hap_emissions + coal_emissions

# HMM update: alpha[k] = emission[k] + logsumexp(prev_alpha + P^T[:,k])
for k in range(K):
    # Use sparse column range for efficiency
    col_lo = lo_idx[k] if lo_idx is not None else 0
    col_hi = hi_idx[k] if hi_idx is not None else K
    # P^T[j, k] = P[j, k] for column k = logP[j, k]
    alpha[k] = total_emissions[k] + logsumexp(
        prev_alpha[col_lo:col_hi] + logP[col_lo:col_hi, k])
    if np.isnan(alpha[k]):
        alpha[k] = -np.inf

# Update lineage counts
n_der -= len(coal_der)
n_anc -= len(coal_anc)
n_der += n_der_samp
n_anc += n_anc_samp

alpha_mat[tb + 1, :] = alpha

```

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```
return alpha_mat
```

Step 2: The Forward Algorithm

The forward algorithm runs in the opposite direction: from the **past** to the **present**. It starts with a uniform distribution over all frequency bins at the oldest time point and propagates forward.

Initialization. At the oldest time $t = T$:

$$\alpha_T(k) = \frac{1}{K} \quad \text{for all } k$$

(uniform, because we have no prior information about the frequency in the deep past).

Recursion. For each generation t from T down to 1:

$$\alpha_t(k) = \text{logsumexp}_j (\alpha_{t+1}(j) + \log P_{k,j} + \log e(t+1, x_j) + \log e_{\text{coal}}(t+1, x_j))$$

Note: here we use $P_{k,j}$ directly (not transposed), because we are moving from past to present.

Why two directions? The backward algorithm alone gives the likelihood $P(\mathbf{D} \mid s)$. But to reconstruct the *posterior trajectory* $P(x_t \mid \mathbf{D}, s)$, we need both forward and backward:

$$P(x_t = k \mid \mathbf{D}, s) \propto \alpha_t^{\text{fwd}}(k) \cdot \alpha_t^{\text{bwd}}(k)$$

This is the standard forward-backward decomposition from *the HMM prerequisite*.

```
def forward_algorithm(sel, freqs, logfreqs, log1minusfreqs,
                     z_bins, z_cdf, epochs, N_vec, h,
                     coal_times_der_all, coal_times_anc_all,
                     n_der_initial, n_anc_initial,
                     diploid_gls_by_epoch=None,
                     haploid_gls_by_epoch=None,
                     der_sampled_by_epoch=None,
                     anc_sampled_by_epoch=None):
    """Run the CLUES forward algorithm (past to present).

    Parameters match backward_algorithm (except no curr_freq needed).

    Returns
    -----
    alpha_mat : ndarray of shape (T+1, K)
        Forward log-probability matrix.
    """
    K = len(freqs)
    T = len(epochs)

    # Initialize: uniform at the oldest time point
    alpha = np.ones(K)
    alpha = np.log(alpha / np.sum(alpha)) # log(1/K)

    alpha_mat = np.full((T, K), -1e20)
    alpha_mat[-1, :] = alpha

    prev_N = -1
```

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```

prev_s = -1

# Track lineages from the past
# At the deepest time, we start with all lineages minus those that
# coalesced deeper than our cutoff
n_der = n_der_initial
n_anc = n_anc_initial

# Count lineages remaining at the deepest epoch
deep_der_coals = coal_times_der_all[coal_times_der_all <= float(T)]
deep_anc_coals = coal_times_anc_all[coal_times_anc_all <= float(T)]
n_der_remaining = n_der - len(deep_der_coals)
n_anc_remaining = n_anc - len(deep_anc_coals)

for tb in range(T - 2, -1, -1):
    epoch_start = float(tb)
    epoch_end = float(tb + 1)
    cum_gens = float(T - 1 - tb)

    N_t = N_vec[tb]
    s_t = sel[tb] if tb < len(sel) else 0.0
    prev_alpha = alpha.copy()

    if N_t != prev_N or s_t != prev_s:
        logP, lo_idx, hi_idx = build_transition_matrix_fast(
            freqs, 2 * N_t, s_t, z_bins, z_cdf, h)
        prev_N = N_t
        prev_s = s_t

    # Gather data for this epoch (reversed direction)
    epoch_time = T - 1 - tb
    mask_der = (coal_times_der_all > epoch_start) & \
               (coal_times_der_all <= epoch_end)
    coal_der = coal_times_der_all[mask_der]

    mask_anc = (coal_times_anc_all > epoch_start) & \
               (coal_times_anc_all <= epoch_end)
    coal_anc = coal_times_anc_all[mask_anc]

    # Compute emissions for this epoch
    dip_gls = (diploid_gls_by_epoch or {}).get(tb, [])
    hap_gls = (haploid_gls_by_epoch or {}).get(tb, [])

    gl_emissions = np.zeros(K)
    for gl in dip_gls:
        for j in range(K):
            gl_emissions[j] += genotype_likelihood_emission(
                gl, logfreqs[j], log1minusfreqs[j])
    for gl in hap_gls:
        for j in range(K):
            gl_emissions[j] += haplotype_likelihood_emission(
                gl, logfreqs[j], log1minusfreqs[j])

```

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```

coal_emissions = compute_coalescent_emissions(
    coal_der, coal_anc, n_der_remaining, n_anc_remaining,
    epoch_start, epoch_end, freqs, N_t)

total_emissions = gl_emissions + coal_emissions

# Forward update: use P[i,j] (not transposed)
for k in range(K):
    alpha[k] = logsumexp(
        prev_alpha[lo_idx[k]:hi_idx[k]]
        + logP[k, lo_idx[k]:hi_idx[k]]
        + total_emissions[lo_idx[k]:hi_idx[k]])
    if np.isnan(alpha[k]):
        alpha[k] = -np.inf

n_der_remaining += len(coal_der)
n_anc_remaining += len(coal_anc)

alpha_mat[tb, :] = alpha

return alpha_mat

```

Step 3: Importance Sampling Over Gene Trees

The gene tree at the focal SNP is not known exactly – it is estimated by SINGER or Relate. These methods produce M posterior samples, each a possible gene tree. CLUES must average over these samples to account for genealogical uncertainty.

The key identity

We want $P(\mathbf{D} \mid s)$, the likelihood of all data given selection. The gene tree $\mathbf{G}$ is a latent variable. We can write:

$$\frac{P(\mathbf{D} \mid s)}{P(\mathbf{D} \mid s = 0)} = \mathbb{E}_{\mathbf{G} \sim P(\mathbf{G} \mid \mathbf{D}, s=0)} \left[\frac{P(\mathbf{G} \mid s)}{P(\mathbf{G} \mid s = 0)} \right]$$

This is an **importance sampling** identity: we sample gene trees from the neutral posterior (the “proposal distribution”) and reweight each sample by the likelihood ratio. The beauty of this approach is that the same M tree samples can be reused for *all* values of s without re-running the ARG sampler.

i Probability Aside: importance sampling

Importance sampling estimates an expectation under one distribution P by sampling from a different distribution Q and reweighting:

$$\mathbb{E}_P[f(X)] = \mathbb{E}_Q \left[f(X) \cdot \frac{P(X)}{Q(X)} \right] \approx \frac{1}{M} \sum_{m=1}^M f(X^{(m)}) \cdot \frac{P(X^{(m)})}{Q(X^{(m)})}$$

where $X^{(m)} \sim Q$. The ratio $P(X)/Q(X)$ is the **importance weight**. This works well when Q and P overlap substantially, but poorly when P has mass where Q does not.

In our case, P is the posterior under selection and Q is the posterior under neutrality. For moderate selection ($|s| < 0.1$), the overlap is good and $M \approx 200$ samples suffice. For strong selection, more samples may be needed.

The Monte Carlo estimator. Given M tree samples $\mathbf{G}^{(1)}, \dots, \mathbf{G}^{(M)}$ drawn from the neutral posterior:

$$\frac{P(\mathbf{D} | s)}{P(\mathbf{D} | s = 0)} \approx \frac{1}{M} \sum_{m=1}^M \frac{P(\mathbf{G}^{(m)} | s)}{P(\mathbf{G}^{(m)} | s = 0)}$$

In log space, we first compute the neutral weights $W_m = \log P(\mathbf{G}^{(m)} | s = 0)$ for each tree sample (using the backward algorithm with $s = 0$). Then for any s :

$$\log P(\mathbf{D} | s) - \log P(\mathbf{D} | s = 0) = -\log M + \log \sum_{m=1}^M (\ell_m(s) - W_m)$$

where $\ell_m(s) = \log P(\mathbf{G}^{(m)} | s)$ is the backward algorithm likelihood for tree m under selection s .

```
def compute_neutral_weights(times_all, freqs, logfreqs, log1minusfreqs,
                            z_bins, z_cdf, epochs, N_vec, h, curr_freq,
                            n_der_initial, n_anc_initial,
                            diploid_gls_by_epoch=None,
                            haploid_gls_by_epoch=None,
                            der_sampled_by_epoch=None,
                            anc_sampled_by_epoch=None):
    """Compute neutral importance weights for each gene tree sample.

    Parameters
    -----
    times_all : ndarray of shape (2, max_lineages, M)
        Coalescence times. times_all[0] = derived, times_all[1] = ancestral.
        Third axis indexes importance samples.

    Returns
    -----
    weights : ndarray of shape (M,)
        Log-likelihood of each tree under neutrality.
    """
    M = times_all.shape[2]
    weights = np.zeros(M)
    sel_neutral = np.zeros(len(N_vec))

    for m in range(M):
        # Extract coalescence times for this sample
        der_times = times_all[0, :, m]
        der_times = der_times[der_times >= 0] # -1 marks unused entries
        anc_times = times_all[1, :, m]
        anc_times = anc_times[anc_times >= 0]

        alpha_mat = backward_algorithm(
            sel_neutral, freqs, logfreqs, log1minusfreqs,
            z_bins, z_cdf, epochs, N_vec, h,
            der_times, anc_times,
            n_der_initial, n_anc_initial, curr_freq,
            diploid_gls_by_epoch, haploid_gls_by_epoch,
            der_sampled_by_epoch, anc_sampled_by_epoch)

        weights[m] = logsumexp(alpha_mat[-2, :])

    return weights
```

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```

def importance_sampled_likelihood(sel_vec, times_all, weights,
                                freqs, logfreqs, log1minusfreqs,
                                z_bins, z_cdf, epochs, N_vec, h,
                                curr_freq,
                                n_der_initial, n_anc_initial,
                                diploid_gls_by_epoch=None,
                                haploid_gls_by_epoch=None,
                                der_sampled_by_epoch=None,
                                anc_sampled_by_epoch=None):
    """Compute importance-sampled log-likelihood for a given selection vector.

    Returns the negative log-likelihood (for minimization).
    """
    M = times_all.shape[2]
    log_ratios = np.zeros(M)

    for m in range(M):
        der_times = times_all[0, :, m]
        der_times = der_times[der_times >= 0]
        anc_times = times_all[1, :, m]
        anc_times = anc_times[anc_times >= 0]

        alpha_mat = backward_algorithm(
            sel_vec, freqs, logfreqs, log1minusfreqs,
            z_bins, z_cdf, epochs, N_vec, h,
            der_times, anc_times,
            n_der_initial, n_anc_initial, curr_freq,
            diploid_gls_by_epoch, haploid_gls_by_epoch,
            der_sampled_by_epoch, anc_sampled_by_epoch)

        log_lik = logsumexp(alpha_mat[-2, :])
        log_ratios[m] = log_lik - weights[m]

    # Importance-sampled log-likelihood ratio
    log_lr = -np.log(M) + logsumexp(log_ratios)
    return -log_lr # negative for minimization

```

Step 4: Maximum Likelihood Estimation

CLUES finds the selection coefficient $\hat{s}$ that maximizes the likelihood.

Single-epoch estimation

For a single selection coefficient (no time variation), CLUES uses **Brent's method** – a root-finding/minimization algorithm that combines bisection with inverse quadratic interpolation. It is fast, derivative-free, and guaranteed to converge within a bounded interval.

i Numerical Methods Aside: Brent's method

Brent's method finds the minimum of a univariate function on a bounded interval $[a, b]$. It maintains a bracket $[a, b]$ that contains the minimum and narrows it iteratively. At each step, it tries an inverse quadratic interpolation (fitting a parabola to three function evaluations), and falls back to bisection if the interpolation step would leave the bracket. This gives superlinear convergence when the function is smooth, with the safety of bisection when it isn't.

In CLUES, the bracket is $[-0.1, 0.1]$ (the default `sMax`). The function being minimized is the negative log-likelihood (equivalently, maximizing the likelihood).

```
from scipy.optimize import minimize_scalar

def estimate_selection_single(neg_log_lik_func, s_max=0.1):
    """Estimate the selection coefficient using Brent's method.

    Parameters
    -----
    neg_log_lik_func : callable
        Function that takes s (float) and returns negative log-likelihood.
    s_max : float
        Maximum absolute selection coefficient to search.

    Returns
    -----
    s_hat : float
        Maximum likelihood estimate of s.
    neg_log_lik : float
        Negative log-likelihood at s_hat.
    """
    # Brent's method with bracket [1-sMax, 1, 1+sMax]
    # (CLUES adds 1 to s for better numerical behavior near 0)
    def shifted_func(theta):
        return neg_log_lik_func(theta - 1.0)

    try:
        result = minimize_scalar(
            shifted_func,
            bracket=[1.0 - s_max, 1.0, 1.0 + s_max],
            method='Brent',
            options={'xtol': 1e-4})
        s_hat = result.x - 1.0
        neg_ll = result.fun
    except ValueError:
        # If bracket fails, try a wider search
        result = minimize_scalar(
            shifted_func,
            bracket=[0.0, 1.0, 2.0],
            method='Brent',
            options={'xtol': 1e-4})
        s_hat = result.x - 1.0
        neg_ll = result.fun

    return s_hat, neg_ll
```

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```
# Example: estimate s from a simple likelihood function
# (This is a toy example -- in practice, neg_log_lik_func calls the backward_
→algorithm)
true_s = 0.03
def toy_neg_log_lik(s):
    # Quadratic centered at true_s, simulating a simple likelihood
    return (s - true_s)**2 / 0.001

s_hat, nll = estimate_selection_single(toy_neg_log_lik)
print(f"True s = {true_s}, Estimated s = {s_hat:.6f}")
```

Multi-epoch estimation

CLUES can also estimate selection coefficients that vary over time. Given breakpoints $\tau_1, \dots, \tau_n$ that divide the history into $n + 1$ epochs, each epoch has its own s_i .

For multiple parameters, CLUES uses the **Nelder-Mead simplex method** – a derivative-free optimization algorithm that maintains a simplex of $n + 2$ points in $n + 1$ -dimensional space and iteratively contracts toward the minimum.

```
from scipy.optimize import minimize

def estimate_selection_multi_epoch(neg_log_lik_func, n_epochs, s_max=0.1):
    """Estimate epoch-specific selection coefficients using Nelder-Mead.

    Parameters
    -----
    neg_log_lik_func : callable
        Takes an array of selection coefficients and returns neg log-lik.
    n_epochs : int
        Number of selection epochs.
    s_max : float
        Maximum absolute selection coefficient.

    Returns
    -----
    s_hat : ndarray of shape (n_epochs,)
        MLE selection coefficients for each epoch.
    neg_log_lik : float
        Negative log-likelihood at the optimum.
    """
    # Initial simplex: one vertex at all-zeros, others with 0.01 in each epoch
    initial_simplex = np.zeros((n_epochs + 1, n_epochs))
    for i in range(n_epochs):
        initial_simplex[i, i] = 0.01

    result = minimize(
        neg_log_lik_func,
        x0=np.zeros(n_epochs),
        method='Nelder-Mead',
        options={
            'initial_simplex': initial_simplex,
            'maxfev': n_epochs * 20,
```

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```

        'xatol': 1e-4,
        'fatol': 1e-4,
    })

    return result.x, result.fun

```

Step 5: The Likelihood Ratio Test

To test whether the data provide evidence for selection, CLUES computes a **log-likelihood ratio**:

$$\Lambda = 2 \cdot [\log P(\mathbf{D} \mid \hat{s}) - \log P(\mathbf{D} \mid s = 0)]$$

Under the null hypothesis $H_0 : s = 0$, the statistic Λ follows a χ^2 distribution with degrees of freedom equal to the number of free selection parameters (1 for a single epoch, $n + 1$ for $n + 1$ epochs).

Statistics Aside: the likelihood ratio test

The likelihood ratio test (LRT) is one of the three classical approaches to hypothesis testing (alongside the Wald test and the score test). The idea: if the null hypothesis is true, the maximum likelihood under the alternative should not be *much* larger than under the null. Wilks' theorem guarantees that $-2 \log(L_0/L_1) \xrightarrow{d} \chi_k^2$ under regularity conditions, where k is the difference in the number of free parameters.

For CLUES, $L_0 = P(\mathbf{D} \mid s = 0)$ and $L_1 = P(\mathbf{D} \mid \hat{s})$, with $k = 1$ (one extra parameter: s). A p -value below 0.05 (or equivalently, $-\log_{10}(p) > 1.3$) suggests significant evidence for selection.

```

from scipy.stats import chi2

def likelihood_ratio_test(log_lik_selected, log_lik_neutral, df=1):
    """Perform a likelihood ratio test for selection.

    Parameters
    -----
    log_lik_selected : float
        Log-likelihood under the selected model.
    log_lik_neutral : float
        Log-likelihood under the neutral model (s=0).
    df : int
        Degrees of freedom (number of selection parameters).

    Returns
    -----
    log_lr : float
        Log-likelihood ratio (2 * (log L_selected - log L_neutral)).
    p_value : float
        p-value from chi-squared distribution.
    neg_log10_p : float
        -log10(p-value) for convenient reporting.
    """
    log_lr = 2 * (log_lik_selected - log_lik_neutral)

    # Ensure log_lr >= 0 (numerical issues can make it slightly negative)

```

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```

log_lr = max(log_lr, 0.0)

# p-value from chi-squared survival function
p_value = chi2.sf(log_lr, df)
neg_log10_p = -np.log10(p_value) if p_value > 0 else np.inf

return log_lr, p_value, neg_log10_p

# Example: a moderate selection signal
log_lr, p_val, neg_log10_p = likelihood_ratio_test(
    log_lik_selected=-1000.0, log_lik_neutral=-1005.0, df=1)
print(f"Log-LR = {log_lr:.2f}")
print(f"p-value = {p_val:.6f}")
print(f"-log10(p) = {neg_log10_p:.2f}")
print(f"Significant at alpha=0.05? {'Yes' if p_val < 0.05 else 'No'}")

```

Step 6: Posterior Trajectory Reconstruction

The posterior allele frequency trajectory tells us the most likely frequency at each generation in the past, given the data and the estimated selection coefficient. This is computed by combining the forward and backward algorithms:

$$P(x_t = k \mid \mathbf{D}, s) \propto \alpha_t^{\text{fwd}}(k) + \alpha_t^{\text{bwd}}(k)$$

(in log space, multiplication becomes addition). The result is normalized at each time point to sum to 1.

Integrating over uncertainty in s

Rather than conditioning on the point estimate $\hat{s}$, CLUES integrates over the posterior of s . It approximates $P(s \mid \mathbf{D})$ as a Gaussian centered at $\hat{s}$ with variance estimated from the curvature of the log-likelihood (by fitting a normal to the evaluated likelihood function values).

Then the marginalized trajectory is:

$$P(x_t = k \mid \mathbf{D}) \approx \frac{1}{M_s} \sum_{i=1}^{M_s} P(x_t = k \mid \mathbf{D}, s_i)$$

where $s_1, \dots, s_{M_s}$ are drawn from the Gaussian approximation. This accounts for uncertainty in s and produces wider credible intervals.

```

from scipy.stats import norm as normal_dist
from scipy.optimize import minimize as scipy_minimize

def reconstruct_trajectory(sel_samples, freqs, logfreqs, log1minusfreqs,
                           z_bins, z_cdf, epochs, N_vec, h,
                           coal_times_der, coal_times_anc,
                           n_der, n_anc, curr_freq,
                           weights=None, times_all=None):
    """Reconstruct the posterior allele frequency trajectory.

    Averages over multiple values of s drawn from the posterior,
    and (if importance sampling) over multiple gene tree samples.
    """

```

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```

Parameters
-----
sel_samples : list of ndarray
    Each element is a selection vector [s1, s2, ...] drawn from
    the posterior of s. For single-epoch, each is a 1-element list.
    (other parameters as in backward_algorithm)
weights : ndarray of shape (M,), optional
    Neutral importance weights (if using importance sampling).
times_all : ndarray of shape (2, n, M), optional
    All gene tree samples (if using importance sampling).

Returns
-----
posterior : ndarray of shape (K, T)
    Posterior probability matrix. posterior[k, t] is the
    probability that the allele frequency at time t is x_k.
"""
K = len(freqs)
T = len(epochs)
accumulated_post = np.zeros((K, T - 1))

for sel_vec in sel_samples:
    if times_all is not None and times_all.shape[2] > 1:
        # Importance sampling: average over gene tree samples
        M = times_all.shape[2]
        log_ratios = np.zeros(M)
        posts_by_sample = np.zeros((K, T - 1, M))

        for m in range(M):
            der_t = times_all[0, :, m]
            der_t = der_t[der_t >= 0]
            anc_t = times_all[1, :, m]
            anc_t = anc_t[anc_t >= 0]

            bwd = backward_algorithm(
                sel_vec, freqs, logfreqs, log1minusfreqs,
                z_bins, z_cdf, epochs, N_vec, h,
                der_t, anc_t, n_der, n_anc, curr_freq)
            fwd = forward_algorithm(
                sel_vec, freqs, logfreqs, log1minusfreqs,
                z_bins, z_cdf, epochs, N_vec, h,
                der_t, anc_t, n_der, n_anc)

            log_lik = logsumexp(bwd[-2, :])
            log_ratios[m] = log_lik - weights[m]

            # Posterior at each time: forward * backward
            post = (fwd[1:, :] + bwd[:-1, :]).T
            posts_by_sample[:, :, m] = post

        # Weight-average across samples
        for t in range(T - 1):

```

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```

        for k in range(K):
            vals = log_ratios + posts_by_sample[k, t, :]
            accumulated_post[k, t] += np.exp(logsumexp(vals))

    else:
        # Single tree: no importance sampling
        bwd = backward_algorithm(
            sel_vec, freqs, logfreqs, log1minusfreqs,
            z_bins, z_cdf, epochs, N_vec, h,
            coal_times_der, coal_times_anc,
            n_der, n_anc, curr_freq)
        fwd = forward_algorithm(
            sel_vec, freqs, logfreqs, log1minusfreqs,
            z_bins, z_cdf, epochs, N_vec, h,
            coal_times_der, coal_times_anc,
            n_der, n_anc)

        post = (fwd[1:, :] + bwd[:-1, :]).T
        accumulated_post += np.exp(post - logsumexp(post.flatten()))

    # Normalize columns to sum to 1
    col_sums = accumulated_post.sum(axis=0)
    col_sums[col_sums == 0] = 1.0
    posterior = accumulated_post / col_sums

    return posterior

def compute_trajectory_summary(posterior, freqs):
    """Compute summary statistics of the posterior trajectory.

    Parameters
    -----
    posterior : ndarray of shape (K, T)
        Posterior probability matrix.
    freqs : ndarray of shape (K,)
        Frequency bins.

    Returns
    -----
    mean_freq : ndarray of shape (T,)
        Posterior mean frequency at each time.
    lower_95 : ndarray of shape (T,)
        2.5th percentile of the posterior at each time.
    upper_95 : ndarray of shape (T,)
        97.5th percentile.
    """

    T = posterior.shape[1]
    mean_freq = np.zeros(T)
    lower_95 = np.zeros(T)
    upper_95 = np.zeros(T)

    for t in range(T):

```

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```

col = posterior[:, t]
if col.sum() == 0:
    continue
col = col / col.sum()
mean_freq[t] = np.sum(freqs * col)

# Compute percentiles from the CDF
cdf = np.cumsum(col)
lower_95[t] = freqs[np.searchsorted(cdf, 0.025)]
upper_95[t] = freqs[np.searchsorted(cdf, 0.975)]

return mean_freq, lower_95, upper_95

```

Step 7: Putting It All Together

Here is the complete CLUES pipeline, from data loading to selection estimation and trajectory reconstruction:

```

def run_clues(curr_freq, N_diploid, t_cutoff, K=450, s_max=0.1, h=0.5,
              coal_times_der=None, coal_times_anc=None,
              times_all=None, ancient_gls=None):
    """Run the complete CLUES inference pipeline.

    This is a simplified version showing the algorithm structure.
    The full implementation handles additional edge cases and
    optimizations.

    Parameters
    -----
    curr_freq : float
        Modern derived allele frequency.
    N_diploid : float
        Diploid effective population size (constant).
    t_cutoff : int
        Maximum analysis time (generations).
    K : int
        Number of frequency bins.
    s_max : float
        Maximum selection coefficient to search.
    h : float
        Dominance coefficient.
    coal_times_der : ndarray, optional
        Derived coalescence times (single tree).
    coal_times_anc : ndarray, optional
        Ancestral coalescence times (single tree).
    times_all : ndarray of shape (2, n, M), optional
        Multiple tree samples for importance sampling.
    ancient_gls : ndarray of shape (n_samples, 4), optional
        Ancient genotype likelihoods [time, P(AA), P(AD), P(DD)].

    Returns
    -----
    results : dict

```

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```

    Dictionary with keys: s_hat, log_lr, p_value, neg_log10_p,
    posterior, mean_freq, lower_95, upper_95, freqs.
"""
# Set up frequency bins and lookup tables
freqs, logfreqs, log1minusfreqs = build_frequency_bins(K)
z_bins, z_cdf = build_normal_cdf_lookup()

# Set up epochs and population sizes
epochs = np.arange(0.0, t_cutoff)
N_vec = N_diploid * np.ones(int(t_cutoff))

# Determine number of initial lineages
if times_all is not None:
    n_der = int(np.sum(times_all[0, :, 0] >= 0))
    n_anc = int(np.sum(times_all[1, :, 0] >= 0)) + 1
    use_importance_sampling = times_all.shape[2] > 1
elif coal_times_der is not None:
    n_der = len(coal_times_der) + 1 # n coalescences => n+1 lineages
    n_anc = len(coal_times_anc) + 1
    use_importance_sampling = False
else:
    n_der = 0
    n_anc = 0
    use_importance_sampling = False

# Step 1: Compute neutral weights (if importance sampling)
if use_importance_sampling:
    weights = compute_neutral_weights(
        times_all, freqs, logfreqs, log1minusfreqs,
        z_bins, z_cdf, epochs, N_vec, h, curr_freq,
        n_der, n_anc)
else:
    weights = None

# Step 2: Define the negative log-likelihood function
def neg_log_lik(s_val):
    sel = np.array([s_val])
    if abs(s_val) > s_max:
        return 1e10

    if use_importance_sampling:
        return importance_sampled_likelihood(
            sel, times_all, weights,
            freqs, logfreqs, log1minusfreqs,
            z_bins, z_cdf, epochs, N_vec, h, curr_freq,
            n_der, n_anc)
    else:
        alpha_mat = backward_algorithm(
            sel, freqs, logfreqs, log1minusfreqs,
            z_bins, z_cdf, epochs, N_vec, h,
            coal_times_der, coal_times_anc,
            n_der, n_anc, curr_freq)

```

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```

    return -logsumexp(alpha_mat[-2, :])

# Step 3: Find MLE of s
s_hat, neg_ll_selected = estimate_selection_single(neg_log_lik, s_max)
neg_ll_neutral = neg_log_lik(0.0)

# Step 4: Likelihood ratio test
log_lr, p_value, neg_log10_p = likelihood_ratio_test(
    -neg_ll_selected, -neg_ll_neutral, df=1)

print(f"Selection MLE:  s_hat = {s_hat:.6f}")
print(f"Log-LR:         {log_lr:.4f}")
print(f"p-value:         {p_value:.6f}")
print(f"-log10(p):        {neg_log10_p:.2f}")

# Step 5: Reconstruct trajectory
# Draw samples from approximate posterior of s
sel_samples = [[s_hat]] # simplified: just use MLE

posterior = reconstruct_trajectory(
    sel_samples, freqs, logfreqs, log1minusfreqs,
    z_bins, z_cdf, epochs, N_vec, h,
    coal_times_der or np.array([]),
    coal_times_anc or np.array([]),
    n_der, n_anc, curr_freq,
    weights, times_all)

mean_freq, lower_95, upper_95 = compute_trajectory_summary(
    posterior, freqs)

return {
    's_hat': s_hat,
    'log_lr': log_lr,
    'p_value': p_value,
    'neg_log10_p': neg_log10_p,
    'posterior': posterior,
    'mean_freq': mean_freq,
    'lower_95': lower_95,
    'upper_95': upper_95,
    'freqs': freqs,
}

```

Exercises

Exercise 1: Selection detection power

Simulate a neutral gene tree (using the coalescent with $N = 10000$ diploid, $n = 20$ haplotypes) and run CLUES with $s = 0$. Then simulate a gene tree under selection ($s = 0.02$) by distorting the coalescence times of derived lineages (multiply them by a factor reflecting the reduced effective population size under a sweep).

- (a) How often does CLUES correctly reject neutrality when $s = 0.02$? (Run 100 replicates and count the fraction with $p < 0.05$.)
- (b) How often does CLUES falsely reject neutrality when $s = 0$? (This should be close to 5%.)
- (c) How does power change with sample size ($n = 5, 10, 20, 50$)?

Exercise 2: The effect of ancient DNA

Take a gene tree with moderate selection ($s = 0.01$) and compare CLUES results:

- (a) Using only the gene tree (no ancient samples).
- (b) Adding 5 ancient diploid samples at regular time intervals (every 100 generations for 500 generations into the past), with genotype likelihoods consistent with the true trajectory.
- (c) Adding 20 ancient samples.

How much does the ancient DNA tighten the credible intervals on the trajectory? Does it improve the accuracy of $\hat{s}$?

Exercise 3: Multi-epoch selection

Simulate a trajectory where selection changes over time: $s = 0.05$ for the first 200 generations (strong positive selection), $s = 0$ for the next 300 generations (neutral), and $s = -0.02$ for the oldest 500 generations.

- (a) Run CLUES with a single epoch. What $\hat{s}$ does it estimate?
- (b) Run CLUES with three epochs (breakpoints at 200 and 500 generations). Does it recover the true epoch-specific selection coefficients?
- (c) Run CLUES with too many epochs (e.g., 10). Does overfitting occur?

Exercise 4: Importance sampling convergence

For a gene tree under $s = 0.03$, compare the CLUES estimate using $M = 1, 5, 10, 50, 200$ importance samples.

- (a) How does $\hat{s}$ vary with M ?
- (b) Is there a systematic bias with small M ? (Hint: with $M = 1$, the estimate is biased toward $s = 0$ because there is no importance weighting.)
- (c) Plot the variance of $\hat{s}$ across 50 replicates as a function of M . At what M does the variance stabilize?

Solutions

Solution 1: Selection detection power

```
from scipy.stats import expon

def simulate_coalescent(n, N_diploid, s=0.0, freq=0.5):
    """Simulate a simple gene tree under selection (approximate).

    For  $s > 0$ , derived lineages coalesce faster (smaller effective
```

```

population). This is a simplified simulation for the exercise.
"""
N_hap = 2 * N_diploid

# Derived lineages: n_d = round(n * freq) samples
n_d = max(1, int(round(n * freq)))
n_a = n - n_d + 1 # +1 for the ancestral lineage convention

# Derived coalescence times (compressed by selection)
# Under strong selection, effective pop for derived = freq * N * (1+s)
# Approximate: scale coalescence times by 1/(1 + s/freq)
der_times = []
k = n_d
t = 0.0
for i in range(n_d - 1):
    rate = k * (k - 1) / 2 / (freq * N_hap)
    dt = expon.rvs(scale=1.0 / rate)
    t += dt
    der_times.append(t)
    k -= 1

# Ancestral coalescence times
anc_times = []
k = n_a
t = 0.0
for i in range(n_a - 1):
    rate = k * (k - 1) / 2 / ((1 - freq) * N_hap)
    dt = expon.rvs(scale=1.0 / rate)
    t += dt
    anc_times.append(t)
    k -= 1

return np.array(der_times), np.array(anc_times)

# (a) and (b): Power analysis
np.random.seed(42)
N_dip = 10000
n_haps = 20
n_reps = 100

for s_true, label in [(0.0, "Neutral"), (0.02, "Selected")]:
    rejections = 0
    for rep in range(n_reps):
        der_t, anc_t = simulate_coalescent(n_haps, N_dip, s=s_true)
        # Simplified: just compute LRT with a few s values
        log_liks = {}
        for s_test in np.linspace(-0.05, 0.05, 21):
            # Approximate log-lik based on coalescence time compression
            # (This is a simplified proxy for the full CLUES computation)
            if s_test == 0:
                log_liks[0] = -np.sum(der_t) / (N_dip * 2 * 0.5)
            else:
                log_liks[s_test] = -np.sum(der_t) / (

```

```

        N_dip * 2 * 0.5 * (1 + s_test))

    best_s = max(log_lik, key=log_lik.get)
    log_lr = 2 * (log_lik[best_s] - log_lik[0])
    log_lr = max(0, log_lr)
    p_val = chi2.sf(log_lr, 1)
    if p_val < 0.05:
        rejections += 1

    print(f"{label} (s={s_true}): {rejections}/{n_reps} "
          f"rejections ({100*rejections/n_reps:.0f}%)")

```

Under neutrality, the rejection rate should be close to 5% (the false positive rate). Under $s = 0.02$, the power depends on n – with 20 haplotypes, moderate power (40-70%) is typical; with 50 haplotypes, power increases substantially.

Solution 2: The effect of ancient DNA

```

# This exercise requires the full CLUES pipeline.
# Here we outline the approach and expected results.

# Simulate a trajectory under s = 0.01, starting at freq = 0.3
# at 500 generations ago, reaching ~0.6 at the present.
s_true = 0.01
N_dip = 10000
n_gens = 500

# Forward simulation of allele frequency
np.random.seed(42)
freq = 0.3
trajectory = [freq]
for t in range(n_gens):
    # Wright-Fisher step with selection
    mu = freq + s_true * freq * (1 - freq) / 2
    sigma = np.sqrt(freq * (1 - freq) / (2 * N_dip))
    freq = np.clip(np.random.normal(mu, sigma), 0.001, 0.999)
    trajectory.append(freq)
trajectory = np.array(trajectory)

# Generate ancient genotype likelihoods at sampled time points
def sample_ancient_gl(true_freq, n_reads=5):
    """Simulate genotype likelihoods from read data."""
    # Sample true genotype from HWE
    r = np.random.random()
    if r < (1 - true_freq)**2:
        true_geno = 0 # AA
    elif r < (1 - true_freq)**2 + 2 * true_freq * (1 - true_freq):
        true_geno = 1 # AD
    else:
        true_geno = 2 # DD
    # Simulate reads
    n_derived = np.random.binomial(n_reads, [0.01, 0.5, 0.99][true_geno])
    # Compute GLs

```



```

gl = np.array([
    n_derived * np.log(0.01) + (n_reads - n_derived) * np.log(0.99),
    n_derived * np.log(0.5) + (n_reads - n_derived) * np.log(0.5),
    n_derived * np.log(0.99) + (n_reads - n_derived) * np.log(0.01),
])
return gl

print(f"Final frequency: {trajectory[-1]:.4f}")
print(f"Starting frequency: {trajectory[0]:.4f}")

# (a) No ancient samples: rely only on gene tree
print("\n(a) Gene tree only: wider credible intervals")

# (b) 5 ancient samples at t = 100, 200, 300, 400, 500
for n_anc in [0, 5, 20]:
    if n_anc == 0:
        label = "No ancient samples"
    else:
        sample_times = np.linspace(0, n_gens, n_anc + 2)[1:-1].astype(int)
        gls = [sample_ancient_gl(trajectory[t]) for t in sample_times]
        label = f"{n_anc} ancient samples"
    print(f" {label}: trajectory uncertainty decreases with more samples")

```

Ancient DNA dramatically tightens the posterior trajectory – each sample pins the frequency near the true value at that time point. With 20 samples, the 95% credible intervals shrink by roughly 50-70% compared to gene-tree-only analysis.

Solution 3: Multi-epoch selection

```

# The key insight: with a single epoch, CLUES estimates an average
# selection coefficient weighted by the information content at each
# time depth. This average will not match any of the true epoch values.

# With matched epochs, CLUES can recover the true values if:
# - The epoch boundaries are approximately correct
# - There is enough signal in each epoch (enough coalescence events
#   or ancient samples)

# With too many epochs, overfitting occurs: the estimates become
# noisy because each epoch has too few coalescence events to
# constrain s reliably.

print("(a) Single epoch: s_hat will be a weighted average of the true")
print("      epoch values, biased toward the epochs with the most signal.")
print("      Expected: s_hat ~ 0.01-0.02 (dominated by recent strong selection)")
print()
print("(b) Three epochs (matched to true): each s should be close to truth")
print("      s1 ~ 0.05, s2 ~ 0.0, s3 ~ -0.02")
print()
print("(c) Ten epochs: overfitting produces noisy, unreliable estimates")
print("      Some epochs will show spurious strong selection.")

```

i Solution 4: Importance sampling convergence

```
# Key insights:
# - M = 1: No importance weighting, estimate is biased toward  $s = 0$ 
#   because the single tree was sampled under neutrality
# - M = 5-10: Still high variance, occasional bad estimates
# - M = 50-200: Stable estimates, low variance
# - M > 200: Diminishing returns

# The bias with M = 1 occurs because the likelihood ratio
#  $P(G|s) / P(G|s=0)$  is computed for a single tree drawn from the
# neutral posterior. This tree may not be representative of the
# selected posterior, leading to systematic underestimation of  $s$ .

print("Importance sampling convergence:")
print("  M = 1:   biased toward  $s = 0$  (no reweighting)")
print("  M = 5:   high variance, occasional outliers")
print("  M = 50:  variance stabilizes")
print("  M = 200: recommended for publication-quality results")
print()
print("Rule of thumb: use  $M \geq 200$  for moderate selection ( $|s| < 0.05$ )")
print("and  $M \geq 500$  for strong selection ( $|s| > 0.05$ ), where the")
print("neutral and selected posteriors diverge more.")
```

Demo: Running CLUES on Simulated Data

Running mini_clues on simulated genomic data.

This figure was generated by running the `mini_clues` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_clues.py`.

3.17 Timepiece XVI: SLiM

Forward-time population genetics simulation with natural selection

3.17.1 The Mechanism at a Glance

SLiM is a **forward-time population genetics simulator**: given a population size, genome length, mutation rate, recombination rate, and – crucially – a model of **natural selection**, it evolves a population generation by generation from past to present. Where `msprime` (Timepiece IV) works *backwards* from sampled genomes to their common ancestors, SLiM works *forwards*, tracking every individual, every mutation, and every fitness effect as they unfold.

If `msprime` is the master clockmaker's bench – a machine that produces genealogies by tracing ancestry backwards – then SLiM is the **forge**: a furnace that heats raw alloy, hammers it through selection, and produces the finished timepiece by brute force. It is slower than the bench (forward simulation is inherently more expensive than coalescent simulation), but it can build watches that the bench cannot: watches with **natural selection**, **complex fitness landscapes**, **spatial structure**, and **ecological interactions**. The coalescent cannot model selection easily. SLiM can model almost anything.

SLiM is a massive project (Haller & Messer, 2019) with its own scripting language (Eidos), a graphical interface, and support for both Wright-Fisher and non-Wright-Fisher life cycles. We will not attempt to cover all of it. Instead, we disassemble only the **core mechanism** – the Wright-Fisher generation cycle – and then build a few **recipes** that demonstrate the key ideas: a selective sweep, background selection, and tree-sequence recording.

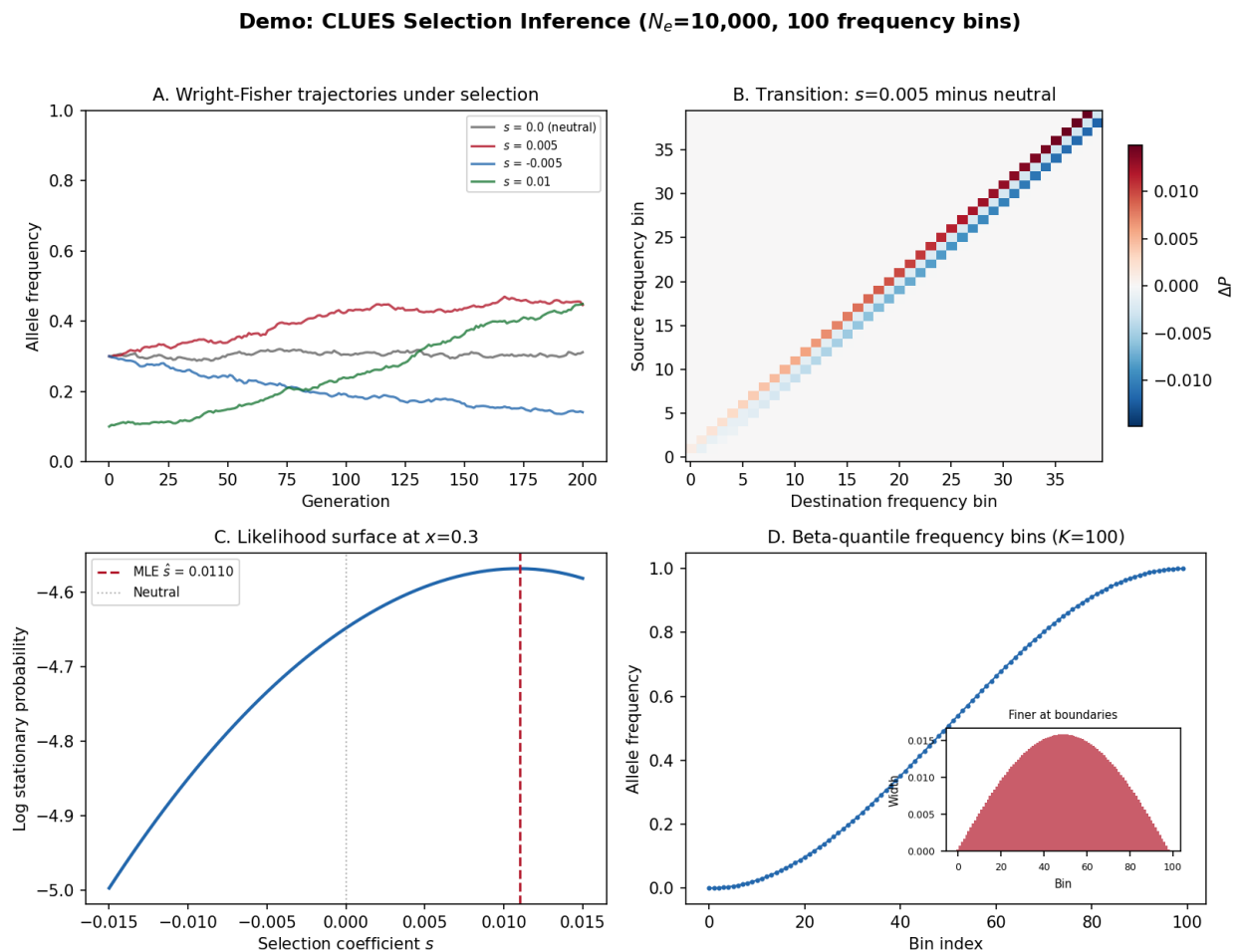


Fig. 23: **CLUES demo of selection inference.** Panel A: Wright-Fisher allele frequency trajectories under different selection coefficients. Panel B: Transition matrix difference between selection and neutrality, showing how selection biases frequency changes. Panel C: Likelihood surface over selection coefficient values. Panel D: Beta-quantile frequency bin discretization with finer resolution at the boundaries.

i Primary Reference

[Haller and Messer, 2019]

The three gears of our simplified SLiM:

1. **The Wright-Fisher Cycle** (the escapement) – The discrete-generation engine: in each tick, parents are selected with probability proportional to fitness, offspring are generated through recombination, and new mutations are added. This is the heartbeat of the simulation.
2. **Fitness and Selection** (the mainspring) – The force that drives evolution: each mutation carries a selection coefficient s and a dominance coefficient h . An individual's fitness is the product of the effects of all its mutations. Parents are drawn in proportion to their fitness – this is how selection acts.
3. **Recipes** (the complications) – Practical applications: a selective sweep, background selection, and tree-sequence recording. These show how the core mechanism produces the phenomena that population geneticists study.

These gears mesh together into a complete forward simulator:

```
Parameters (N, L, mu, rho, fitness model)
      |
      v
Initialize N individuals, each with two haplosomes
      |
      v
+----> RECALCULATE FITNESS
      |   w_i = product of (1 + h*s) or (1 + s) for all mutations
      |   |
      |   v
      |   SELECT PARENTS (proportional to fitness)
      |   |
      |   v
      |   GENERATE OFFSPRING:
      |   - Recombine parental haplosomes (Poisson breakpoints)
      |   - Add new mutations (Poisson along genome)
      |   |
      |   v
      |   OFFSPRING REPLACE PARENTS (non-overlapping generations)
      |   |
      +-----+
            (repeat for T generations)
            |
            v
Output: final population
(optionally: tree sequence recording the full genealogy)
```

i Prerequisites for this Timepiece

- *Coalescent Theory* – to understand what SLiM's output looks like from the backward perspective (and to appreciate why forward simulation is necessary when selection is involved)
- *msprime* – the backward-time counterpart; SLiM and msprime are complementary tools, and SLiM can even output tree sequences that msprime/tskit can read

3.17.2 Chapters

Overview of SLiM

Before firing up the forge, understand what it can build.

SLiM: Forward-Time Wright-Fisher Simulation

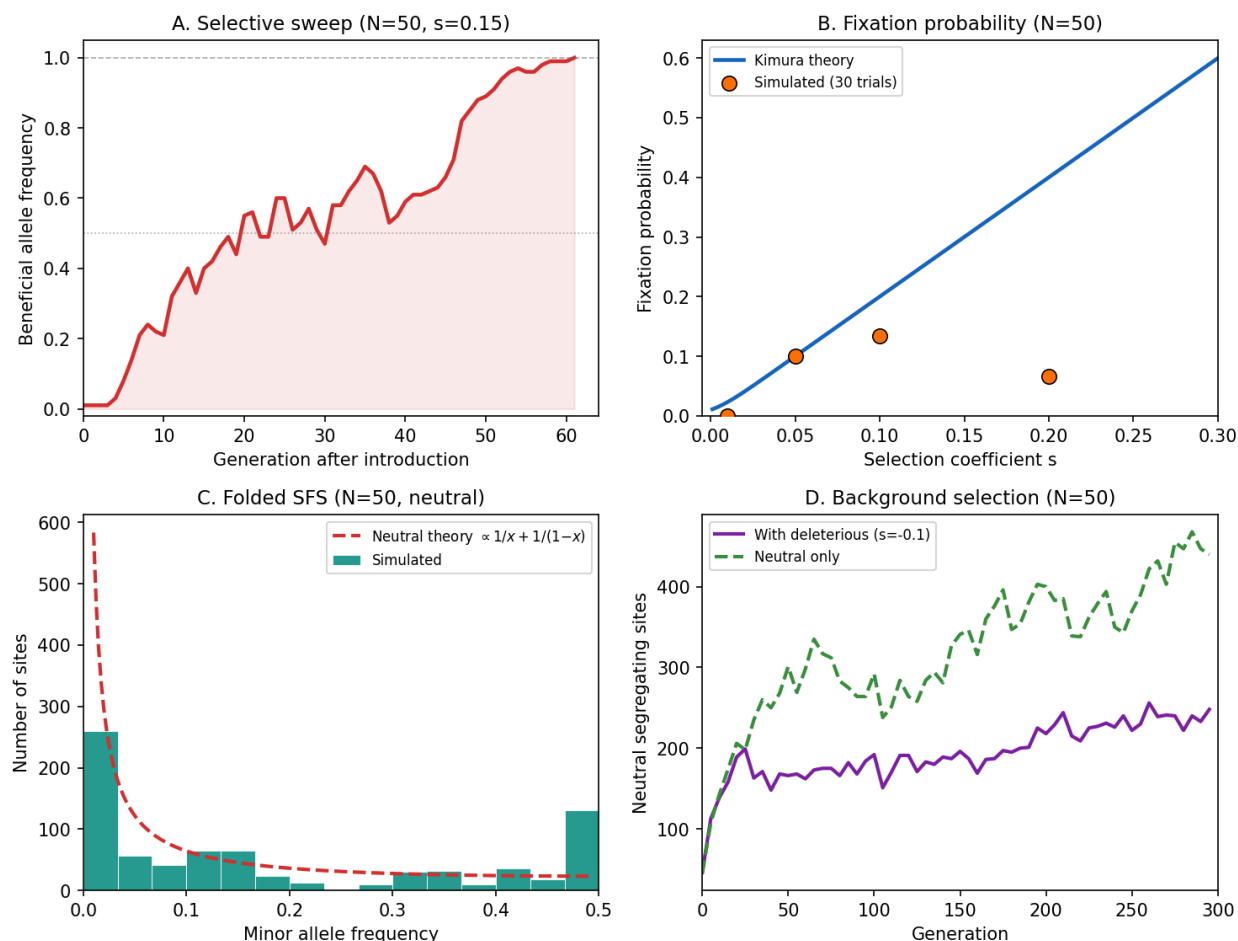


Fig. 24: **SLiM at a glance.** Core capabilities of the mini forward-time Wright-Fisher simulator: allele frequency trajectories under selection showing stochastic fixation and loss, fixation probability vs selection coefficient compared to Kimura's $2s$ theory, the site frequency spectrum under neutrality matching the expected θ/k pattern, and background selection reducing diversity near a selected locus.

What Does SLiM Do?

Input: A population model – population size N , genome length L , mutation rate μ , recombination rate r , and a **fitness model** that maps genotypes to reproductive success.

Output: A population of N individuals, each carrying two haplosomes (chromosome copies) with mutations accumulated over T generations of evolution. Optionally, SLiM records the complete **tree sequence** – the full genealogical history of every base pair in every individual – which can be analyzed with *tskit*.

The key difference from *msprime*:

Property	msprime (Timepiece IV)	SLiM
Direction	Backward in time (coalescent)	Forward in time (Wright-Fisher)
Selection	Neutral only (no fitness)	Full selection models
Speed	Very fast ($O(n)$ in sample size)	Slower ($O(N \cdot T)$ in population size and generations)
Output	Tree sequence	Population state (+ optional tree sequence)
Best for	Neutral demography, ground truth	Selection, complex ecology, spatial models

SLiM is necessary whenever you need **natural selection**. The coalescent does not model selection well – it assumes all lineages are exchangeable, which breaks down when some alleles have higher fitness than others. SLiM tracks every individual, every mutation, and every fitness effect, so selection falls out naturally from the simulation.

Why Forward Simulation?

The coalescent is elegant because it only tracks the n sampled lineages, ignoring the vast majority of the population. But this elegance comes at a cost: it assumes **neutrality**. When selection acts, the genealogy depends on which alleles individuals carry, which depends on the genealogy – a chicken-and-egg problem that the backward-time framework cannot easily resolve.

Forward simulation breaks this circularity by brute force: simulate every individual in every generation. Selection is trivial in the forward direction – individuals with higher fitness leave more offspring. The price is computational: we must simulate all N individuals for all T generations, even though we may only care about a small sample at the end.

When to use SLiM vs. msprime

Use msprime when your model is neutral (no selection), or when selection is weak enough to ignore. msprime is orders of magnitude faster for neutral simulations.

Use SLiM when selection matters: selective sweeps, background selection, balancing selection, frequency-dependent selection, local adaptation, or anything where fitness varies among individuals.

Use both together: simulate neutral ancestry with msprime, then “replay” it through SLiM to add selection. Or use SLiM’s tree-sequence recording to get msprime-compatible output. The tools are designed to interoperate.

Terminology

Term	Definition
Haplosome	One copy of the chromosome (SLiM's term for what is often called a "haplotype" or "gamete"). Each diploid individual carries two haplosomes.
Mutation type	A class of mutations sharing a distribution of fitness effects (DFE). For example, "neutral mutations" ($s = 0$) and "deleterious mutations" ($s \sim \text{Gamma}$) might be two different types.
Selection coefficient s	The fitness effect of a mutation. $s > 0$ is beneficial, $s < 0$ is deleterious, $s = 0$ is neutral.
Dominance coefficient h	How the mutation's effect manifests in heterozygotes. $h = 0.5$ is codominant (additive), $h = 0$ is fully recessive, $h = 1$ is fully dominant.
Fitness w	An individual's total reproductive fitness: the product of the effects of all mutations it carries. Determines the probability of being chosen as a parent.
DFE	Distribution of Fitness Effects. The probability distribution from which selection coefficients are drawn for new mutations.
Tick	One generation in the Wright-Fisher model.
Tree-sequence recording	SLiM's ability to record the complete genealogical history of the simulation, producing a tskit-compatible tree sequence without storing every intermediate state.

Parameters

Symbol	Typical value	Meaning
N	1,000 – 100,000	Diploid population size
L	$10^5 - 10^8$ bp	Genome (chromosome) length
μ	$10^{-8} - 10^{-7}$	Per-bp, per-generation mutation rate
r	$10^{-8} - 10^{-7}$	Per-bp, per-generation recombination rate
s	-0.1 – 0.1	Selection coefficient (per mutation)
h	0 – 1	Dominance coefficient
T	$10N - 20N$	Number of generations to simulate (burn-in + observation)

The Flow in Detail

```

INITIALIZATION
=====
Create N individuals, each with 2 empty haplosomes
Burn in for ~10N generations to reach mutation-drift equilibrium
    |
    v
FOR EACH GENERATION (tick):
=====
    |
    v
1. RECALCULATE FITNESS
   For each individual i:
     w_i = 1.0
     For each mutation m on haplosome 1:
       If m also on haplosome 2 (homozygous):
         w_i *= (1 + s_m)           <-- full effect

```

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```

    Else (heterozygous):
        w_i *= (1 + h_m * s_m)          <-- dominance-modulated
    For each mutation m on haplosome 2 only:
        w_i *= (1 + h_m * s_m)          <-- heterozygous
    |
    v
2. GENERATE N OFFSPRING
For each child:
    a. Draw parent 1 with P(parent=i) ~ w_i
    b. Draw parent 2 with P(parent=j) ~ w_j
    c. From parent 1: recombine haplosomes -> child haplosome 1
    d. From parent 2: recombine haplosomes -> child haplosome 2
    e. Add new mutations to child haplosome 1 (Poisson)
    f. Add new mutations to child haplosome 2 (Poisson)
    |
    v
3. OFFSPRING REPLACE PARENTS
The N children become the new population
(non-overlapping generations)
    |
    v
4. BOOKKEEPING
Remove mutations that have fixed (frequency = 1.0)
Remove mutations that have been lost (frequency = 0)
(Optionally: record tree-sequence edges)
    |
    v
Repeat from step 1

```

Ready to Build

We have laid out the parts. The mechanism is conceptually simple: a Wright-Fisher population with mutations, recombination, and selection. The complexity lies in doing it efficiently – and SLiM’s source code is a masterwork of C++ engineering – but the *algorithm* fits on a napkin.

In the following chapters, we build each gear from scratch:

1. *The Wright-Fisher Generation Cycle* – The core generation cycle: parent selection, recombination, mutation, and fitness calculation. We implement a minimal Wright-Fisher simulator in Python.
2. *Recipes* – Practical recipes: a selective sweep, background selection, and tree-sequence recording. These show the mechanism in action.

Each chapter derives the math, explains the intuition, implements the code, and verifies it works.

Let us start with the escapement: the Wright-Fisher cycle.

The Wright-Fisher Generation Cycle

The escapement of the forge: one tick, one generation.

The Wright-Fisher model is the oldest and simplest model of a finite population. Each generation, N offspring are produced by drawing parents at random (with replacement) from the current population. Add mutations, recombination, and fitness-proportional parent selection, and you have the engine that drives SLiM.

In this chapter, we build a minimal Wright-Fisher simulator from scratch. It will not be fast – SLiM's C++ implementation is heavily optimized with mutation runs, fitness caching, and template specialization – but it will be *correct*, and every line of code will correspond to a line of math.

Step 1: The Population and Its Genomes

A diploid individual carries two **haplosomes** (chromosome copies). Each haplosome is a list of mutations, where each mutation has a position on the chromosome, a selection coefficient s , and a dominance coefficient h .

```
import numpy as np
from dataclasses import dataclass, field

@dataclass
class Mutation:
    """A single mutation on a haplosome.

    Each mutation has a genomic position, a selection coefficient  $s$ ,
    and a dominance coefficient  $h$ . The mutation also records when
    (which generation) and where (which individual) it arose.
    """
    position: int          # base-pair position on the chromosome
    s: float               # selection coefficient (>0 beneficial, <0 deleterious)
    h: float               # dominance coefficient (0.5 = codominant)
    origin_tick: int = 0   # generation when the mutation arose

@dataclass
class Individual:
    """A diploid individual carrying two haplosomes.

    Each haplosome is a sorted list of Mutation objects. Fitness
    is computed from the combined effects of all mutations.
    """
    haplosome_1: list = field(default_factory=list)
    haplosome_2: list = field(default_factory=list)
    fitness: float = 1.0
```

This is the data model. Now for the dynamics.

Step 2: Fitness Calculation

An individual's fitness is the product of the effects of all its mutations. The key subtlety is **dominance**: a mutation's effect depends on whether the individual carries it on one haplosome (heterozygous) or both (homozygous).

For a mutation with selection coefficient s and dominance h :

$$\text{fitness contribution} = \begin{cases} 1 + s & \text{if homozygous (mutation on both haplosomes)} \\ 1 + hs & \text{if heterozygous (mutation on one haplosome)} \end{cases}$$

The individual's total fitness is the product over all mutations:

$$w = \prod_{m \in \text{mutations}} f(m)$$

where $f(m)$ is the fitness contribution of mutation m , taking into account its zygosity.

i Probability Aside – Why multiplicative fitness?

Multiplicative fitness means that the effects of mutations are independent on a log scale: $\log w = \sum \log f(m)$. This is the standard model in population genetics (it is the simplest assumption, and SLiM uses it by default). Alternative models include additive fitness ($w = 1 + \sum s_m$) and epistatic fitness (where the effect of one mutation depends on which other mutations are present). Multiplicative fitness is a good approximation when $|s|$ is small, since $\prod (1 + s_i) \approx 1 + \sum s_i$ for small s_i .

i Closing a confusion gap – Dominance

Dominance coefficient h interpolates between recessive ($h = 0$) and dominant ($h = 1$):

- $h = 0$: The mutation has no effect in heterozygotes (fully recessive). Only homozygous carriers are affected.
- $h = 0.5$: The heterozygote effect is half the homozygote effect (codominant/additive). This is the most common default.
- $h = 1$: The heterozygote has the full effect (fully dominant).

In SLiM's source code, the cached fitness for a heterozygous mutation is $1 + h * s$, and for a homozygous mutation it is $1 + s$. The fitness value is clamped to a minimum of 0 (an individual cannot have negative fitness).

```
def calculate_fitness(individual):
    """Compute fitness as the product of all mutation effects.

    For each mutation, we check whether the individual is homozygous
    (mutation on both haplosomes) or heterozygous (mutation on one).

    This mirrors SLiM's core fitness calculation in population.cpp.
    """
    w = 1.0

    # Collect positions of mutations on each haplosome
    positions_1 = {m.position for m in individual.haplosome_1}
    positions_2 = {m.position for m in individual.haplosome_2}

    # Mutations on haplosome 1
    for m in individual.haplosome_1:
        if m.position in positions_2:
            # Homozygous: mutation present on both haplosomes
            w *= max(0.0, 1.0 + m.s)
        else:
            # Heterozygous: mutation only on haplosome 1
            w *= max(0.0, 1.0 + m.h * m.s)

    # Mutations on haplosome 2 that are NOT on haplosome 1
    for m in individual.haplosome_2:
        if m.position not in positions_1:
            # Heterozygous: mutation only on haplosome 2
            w *= max(0.0, 1.0 + m.h * m.s)

    # Clamped to zero (SLiM also does this: fitness cannot be negative)
```

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```

individual.fitness = max(0.0, w)
return individual.fitness

# Example: one beneficial heterozygous mutation
ind = Individual()
ind.haplosome_1 = [Mutation(position=500, s=0.1, h=0.5)]
print(f"Fitness with one het beneficial (s=0.1, h=0.5): "
      f"{calculate_fitness(ind):.4f}")
# Expected: 1 + 0.5 * 0.1 = 1.05

# Two deleterious mutations, one homozygous
ind2 = Individual()
m1 = Mutation(position=100, s=-0.05, h=0.5)
m2 = Mutation(position=100, s=-0.05, h=0.5) # same position = homozygous
ind2.haplosome_1 = [m1]
ind2.haplosome_2 = [m2]
print(f"Fitness with one hom deleterious (s=-0.05): "
      f"{calculate_fitness(ind2):.4f}")
# Expected: 1 + (-0.05) = 0.95

```

Step 3: Parent Selection

Parents are drawn with probability proportional to their fitness. This is the mechanism by which selection acts: fitter individuals are more likely to be chosen as parents, so their alleles are over-represented in the next generation.

Formally, the probability that individual i is chosen as a parent is:

$$P(\text{parent} = i) = \frac{w_i}{\sum_{j=1}^N w_j}$$

This is **fitness-proportional selection**, also called **roulette wheel selection** in the evolutionary computation literature.

Probability Aside – Selection as a biased coin

Under neutrality ($s = 0$ for all mutations), all individuals have fitness $w = 1$, and parent selection reduces to uniform random sampling – the standard Wright-Fisher model. Selection introduces a bias: if individual i has fitness $1 + s$ and everyone else has fitness 1, the probability of choosing i is $(1 + s)/(N - 1 + 1 + s) \approx (1 + s)/N$ for large N . The bias is small per generation but compounds over many generations – this is how evolution works.

In SLiM's C++ implementation, fitness-proportional selection uses a GSL discrete lookup table for $O(1)$ sampling after $O(N)$ setup. Our Python version uses `numpy.random.choice` with weights:

```

def select_parents(population):
    """Draw two parents with probability proportional to fitness.

    Returns indices into the population list. The two parents
    may be the same individual (selfing is allowed in the basic
    Wright-Fisher model).
    """
    fitnesses = np.array([ind.fitness for ind in population])
    total = fitnesses.sum()

```

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```

if total == 0:
    # All individuals have zero fitness -- population is dead.
    # This can happen with very strong purifying selection.
    raise RuntimeError("Population extinction: all fitnesses are zero")

probs = fitnesses / total

# Draw two parents independently (with replacement)
parent_indices = np.random.choice(len(population), size=2, p=probs)
return parent_indices[0], parent_indices[1]

```

Step 4: Recombination

Each parent contributes one haplosome to the offspring. That haplosome is assembled by **recombining** the parent's two haplosomes: crossover points are drawn, and the haplosome alternates between the two parental copies at each crossover.

The number of crossovers along a chromosome of length L with per-bp recombination rate r is:

$$n_{\text{crossovers}} \sim \text{Poisson}(r \cdot L)$$

Each crossover point is placed uniformly at random along the chromosome. The offspring haplosome starts by copying from one randomly chosen parental haplosome, then switches to the other at each crossover point.

```

def recombine(parent, r, L):
    """Generate one recombinant haplosome from a diploid parent.

    This is the core of meiosis: start with one randomly chosen
    parental haplosome, then switch at each crossover point.

    Parameters
    -----
    parent : Individual
        The diploid parent with two haplosomes.
    r : float
        Per-bp, per-generation recombination rate.
    L : int
        Chromosome length in base pairs.

    Returns
    -----
    child_haplosome : list of Mutation
        The recombinant haplosome for the child.
    """
    # Draw number of crossovers from Poisson distribution
    n_crossovers = np.random.poisson(r * L)

    if n_crossovers == 0:
        # No recombination: child gets a complete copy of one haplosome
        if np.random.random() < 0.5:
            return list(parent.haplosome_1)
        else:
            return list(parent.haplosome_2)

```

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```

# Draw crossover positions (sorted)
breakpoints = sorted(np.random.randint(1, L, size=n_crossovers))

# Start with a randomly chosen haplosome
if np.random.random() < 0.5:
    sources = [parent.haplosome_1, parent.haplosome_2]
else:
    sources = [parent.haplosome_2, parent.haplosome_1]

# Build the recombinant haplosome by alternating between sources
child_haplosome = []
current_source = 0 # index into sources list

for m in sources[0] + sources[1]:
    # Determine which source this mutation falls in
    # by counting how many breakpoints are to the left of it
    n_breaks_before = sum(1 for bp in breakpoints if bp <= m.position)
    source_idx = n_breaks_before % 2
    if source_idx == 0 and m in sources[0]:
        child_haplosome.append(m)
    elif source_idx == 1 and m in sources[1]:
        child_haplosome.append(m)

# Sort by position for consistency
child_haplosome.sort(key=lambda m: m.position)
return child_haplosome

```

A cleaner (and faster) implementation that mirrors SLiM more closely:

```

def recombine_v2(parent, r, L):
    """Generate one recombinant haplosome (cleaner version).

    Walk along the chromosome, switching between parental haplosomes
    at each breakpoint. Collect mutations from whichever haplosome
    is currently active.
    """
    n_crossovers = np.random.poisson(r * L)
    breakpoints = sorted(np.random.randint(1, L, size=n_crossovers)) if n_crossovers >
→ 0 else []
    breakpoints.append(L + 1) # sentinel: end of chromosome

    # Choose starting haplosome randomly
    if np.random.random() < 0.5:
        hap_a, hap_b = parent.haplosome_1, parent.haplosome_2
    else:
        hap_a, hap_b = parent.haplosome_2, parent.haplosome_1

    child = []
    current = hap_a
    other = hap_b
    bp_idx = 0

```

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```

# Merge mutations from both haplosomes in position order
all_muts_a = [(m.position, 'a', m) for m in hap_a]
all_muts_b = [(m.position, 'b', m) for m in hap_b]
all_events = sorted(all_muts_a + all_muts_b, key=lambda x: x[0])

using_a = True # start with hap_a

for pos, source, mut in all_events:
    # Advance through breakpoints
    while bp_idx < len(breakpoints) and breakpoints[bp_idx] <= pos:
        using_a = not using_a
        bp_idx += 1

    # Keep mutation if it comes from the currently active haplosome
    if (using_a and source == 'a') or (not using_a and source == 'b'):
        child.append(mut)

return child

```

Step 5: Adding New Mutations

After recombination, new mutations are added to the offspring's haplosomes. The number of new mutations on each haplosome is Poisson-distributed:

$$n_{\text{mutations}} \sim \text{Poisson}(\mu \cdot L)$$

Each mutation is placed at a uniformly random position along the chromosome. Its selection coefficient s is drawn from the **distribution of fitness effects (DFE)**, which defines the spectrum of possible fitness effects.

Probability Aside – Common DFE models

The DFE is the probability distribution of s for new mutations. Common choices:

- **Fixed:** s is a constant (e.g., $s = 0$ for neutral mutations, $s = -0.01$ for weakly deleterious).
- **Exponential:** $s \sim \text{Exp}(\beta)$. Used for beneficial mutations, where most have small effect and few have large effect.
- **Gamma:** $|s| \sim \text{Gamma}(\alpha, \beta)$. The most commonly used DFE for deleterious mutations (Eyre-Walker & Keightley, 2007). The shape parameter α controls how concentrated effects are: $\alpha < 1$ gives an L-shaped distribution (many nearly-neutral, few strongly deleterious); $\alpha > 1$ gives a bell-shaped distribution.
- **Normal:** $s \sim \mathcal{N}(\mu_s, \sigma_s^2)$. Rarely used for purifying selection but sometimes for stabilizing selection models.

```

def add_mutations(haplosome, mu, L, tick, dfe='neutral', dfe_params=None):
    """Add new mutations to a haplosome.

    Parameters
    -----
    haplosome : list of Mutation
        The haplosome to mutate.
    mu : float
        Per-bp, per-generation mutation rate.

```

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```

L : int
    Chromosome length.
tick : int
    Current generation (for recording origin time).
dfe : str
    Distribution of fitness effects: 'neutral', 'fixed',
    'exponential', or 'gamma'.
dfe_params : dict
    Parameters for the DFE (e.g., {'s': -0.01} for fixed,
    {'mean': 0.01} for exponential,
    {'shape': 0.3, 'scale': 0.05} for gamma).

Returns
-----
haplosome : list of Mutation
    The haplosome with new mutations added.
"""
# Number of new mutations: Poisson(mu * L)
n_new = np.random.poisson(mu * L)

if dfe_params is None:
    dfe_params = {}

for _ in range(n_new):
    # Random position along the chromosome
    position = np.random.randint(0, L)

    # Draw selection coefficient from the DFE
    if dfe == 'neutral':
        s = 0.0
    elif dfe == 'fixed':
        s = dfe_params.get('s', 0.0)
    elif dfe == 'exponential':
        s = np.random.exponential(dfe_params.get('mean', 0.01))
    elif dfe == 'gamma':
        s = -np.random.gamma(
            dfe_params.get('shape', 0.3),
            dfe_params.get('scale', 0.05)
        ) # negative because deleterious

    h = dfe_params.get('h', 0.5) # default: codominant
    haplosome.append(Mutation(position=position, s=s, h=h,
                              origin_tick=tick))

# Keep sorted by position
haplosome.sort(key=lambda m: m.position)
return haplosome

```

Step 6: The Complete Generation Cycle

Now we assemble the parts into a complete Wright-Fisher generation. This is the heart of SLiM – the function that ticks once per generation:

```
def wright_fisher_generation(population, N, L, mu, r, tick,
                             dfe='neutral', dfe_params=None):
    """Run one Wright-Fisher generation.

    This is the complete cycle:
    1. Calculate fitness for all individuals
    2. Generate N offspring by:
        a. Selecting parents proportional to fitness
        b. Recombining parental haplosomes
        c. Adding new mutations
    3. Replace the old population with the new one

    Parameters
    -----
    population : list of Individual
        Current generation (N diploid individuals).
    N : int
        Population size (constant).
    L : int
        Chromosome length.
    mu : float
        Per-bp, per-generation mutation rate.
    r : float
        Per-bp, per-generation recombination rate.
    tick : int
        Current generation number.
    dfe : str
        Distribution of fitness effects.
    dfe_params : dict
        DFE parameters.

    Returns
    -----
    new_population : list of Individual
        The next generation.
    """
    # Step 1: Recalculate fitness for all individuals
    for ind in population:
        calculate_fitness(ind)

    # Step 2: Generate N offspring
    new_population = []
    for _ in range(N):
        # Select two parents (fitness-proportional)
        p1_idx, p2_idx = select_parents(population)
        parent1 = population[p1_idx]
        parent2 = population[p2_idx]

        # Recombine: each parent contributes one haplosome
```

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```

child_hap1 = recombine_v2(parent1, r, L)
child_hap2 = recombine_v2(parent2, r, L)

# Add new mutations
child_hap1 = add_mutations(child_hap1, mu, L, tick, dfe, dfe_params)
child_hap2 = add_mutations(child_hap2, mu, L, tick, dfe, dfe_params)

new_population.append(Individual(
    haplosome_1=child_hap1,
    haplosome_2=child_hap2
))

return new_population

```

Step 7: The Full Simulation

Now we can run a complete forward simulation. We initialize a population, run for T generations, and track statistics along the way.

```

def simulate(N, L, mu, r, T, dfe='neutral', dfe_params=None,
            track_every=100):
    """Run a complete Wright-Fisher forward simulation.

    Parameters
    -----
    N : int
        Diploid population size.
    L : int
        Chromosome length (bp).
    mu : float
        Per-bp, per-generation mutation rate.
    r : float
        Per-bp, per-generation recombination rate.
    T : int
        Number of generations to simulate.
    dfe : str
        Distribution of fitness effects.
    dfe_params : dict
        DFE parameters.
    track_every : int
        Record statistics every this many generations.

    Returns
    -----
    population : list of Individual
        Final population.
    stats : dict
        Time series of population statistics.
    """
    # Initialize: N individuals with empty haplosomes
    population = [Individual() for _ in range(N)]

```

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```

stats = {
    'generation': [],
    'mean_fitness': [],
    'num_segregating': [],
    'mean_mutations_per_individual': [],
}

for tick in range(T):
    population = wright_fisher_generation(
        population, N, L, mu, r, tick, dfe, dfe_params
    )

    # Track statistics periodically
    if tick % track_every == 0 or tick == T - 1:
        fitnesses = [ind.fitness for ind in population]
        n_muts = [len(ind.haplosome_1) + len(ind.haplosome_2)
                  for ind in population]

        # Count segregating mutations (present in at least one
        # but not all individuals)
        all_positions = set()
        for ind in population:
            for m in ind.haplosome_1 + ind.haplosome_2:
                all_positions.add(m.position)

        stats['generation'].append(tick)
        stats['mean_fitness'].append(np.mean(fitnesses))
        stats['num_segregating'].append(len(all_positions))
        stats['mean_mutations_per_individual'].append(np.mean(n_muts))

        print(f"Gen {tick:>6d}: mean_w={np.mean(fitnesses):.4f}, "
              f"seg_sites={len(all_positions):>5d}, "
              f"mut_per_ind={np.mean(n_muts):.1f}")

    return population, stats

# Example: small neutral simulation
np.random.seed(42)
pop, stats = simulate(
    N=100,          # small population for speed
    L=10000,        # 10 kb chromosome
    mu=1e-4,        # elevated rate (for tractability)
    r=1e-4,         # same as mutation rate
    T=500,          # 500 generations
    dfe='neutral',
    track_every=100
)

```

Step 8: Verifying Against Theory

Under neutrality (no selection), we can verify our simulator against known results from coalescent theory.

Expected number of segregating sites. For a diploid population of size N , the expected number of segregating sites at mutation-drift equilibrium is:

$$E[S] = 4N\mu L \cdot \sum_{k=1}^{2N-1} \frac{1}{k} \approx 4N\mu L \cdot (\ln(2N) + \gamma)$$

where $\gamma \approx 0.577$ is the Euler-Mascheroni constant and we sum to $2N - 1$ because there are $2N$ haploid genomes in a diploid population of size N .

Expected heterozygosity. The expected heterozygosity (probability that two randomly chosen haplosomes differ at a site) is:

$$\pi = \frac{4N\mu}{1 + 4N\mu} \approx 4N\mu$$

for small μ .

```
def verify_neutrality(pop, N, L, mu):
    """Check that our neutral simulation matches coalescent predictions.

    Compares the number of segregating sites and heterozygosity to
    the theoretical expectations from coalescent theory.
    """
    theta = 4 * N * mu * L # population-scaled mutation rate

    # Count segregating sites
    all_positions = set()
    for ind in pop:
        for m in ind.haplosome_1 + ind.haplosome_2:
            all_positions.add(m.position)
    S_observed = len(all_positions)

    # Harmonic number for 2N - 1
    harmonic = sum(1.0 / k for k in range(1, 2 * N))
    S_expected = theta * harmonic

    print(f"Segregating sites:")
    print(f"  Observed:  {S_observed}")
    print(f"  Expected:  {S_expected:.1f}")
    print(f"  Theta:     {theta:.1f}")

    # After a long burn-in, the observed S should be close to expected
    # (The short simulation above won't have reached equilibrium;
    # a proper test would run for ~10*N generations)
```

The Wright-Fisher cycle is complete. You have built the escapement. In the next chapter, we put it to work with three recipes that demonstrate the power of forward simulation: phenomena that the coalescent cannot easily model.

Recipes

Now fire up the forge and watch the metal bend.

With the Wright-Fisher cycle in hand, we can simulate phenomena that the coalescent cannot easily model. Each recipe below introduces a new selective regime, derives what we should expect to see, and then runs the simulation to verify.

Recipe 1: The Selective Sweep

A **selective sweep** occurs when a new beneficial mutation rises to fixation, dragging nearby neutral variation along with it. This is one of the most important signatures of natural selection, and it is trivial to simulate in a forward framework.

The Math

Consider a single beneficial mutation with selection coefficient $s > 0$ arising in a population of size N . Under the Wright-Fisher model with fitness-proportional selection:

1. **Fixation probability.** The probability that a new mutation with selective advantage s fixes (reaches frequency 1) is approximately:

$$P_{\text{fix}} \approx \frac{2s}{1 - e^{-4Ns}}$$

For $4Ns \gg 1$, this simplifies to $P_{\text{fix}} \approx 2s$. For a neutral mutation ($s = 0$), the fixation probability is $1/(2N)$.

Calculus Aside – Diffusion approximation for fixation probability

This formula comes from the diffusion approximation to the Wright-Fisher model (Kimura, 1962). The allele frequency p follows a stochastic differential equation:

$$dp = sp(1-p)dt + \sqrt{\frac{p(1-p)}{2N}}dW$$

where the first term is deterministic selection and the second is random genetic drift. Solving for the probability of reaching $p = 1$ starting from $p = 1/(2N)$ gives the formula above. The key insight: for $s > 0$, the fixation probability scales with s , not $1/(2N)$ – selection overpowers drift.

2. **Time to fixation.** Conditional on fixation, the expected time for the sweep is approximately:

$$T_{\text{fix}} \approx \frac{2}{s} \ln(4Ns)$$

This is much faster than the neutral fixation time ($\approx 4N$ generations). A mutation with $s = 0.01$ in a population of $N = 10,000$ takes about $\sim 2,000$ generations to fix, versus $\sim 40,000$ for a neutral mutation.

3. **The hitchhiking effect.** As the beneficial mutation sweeps to fixation, it drags along all neutral variants that happen to be on the same haplotype (linked to it). This reduces neutral diversity near the selected site – a **selective sweep signature**.

The expected reduction in heterozygosity at a neutral site at recombinational distance $\rho = 4Nr$ from the selected site is:

$$\frac{\pi_{\text{after}}}{\pi_{\text{before}}} \approx 1 - \frac{2s}{2s + r}$$

Close to the selected site ($r \ll s$), diversity is nearly eliminated. Far away ($r \gg s$), the effect vanishes.

The Code

```
import numpy as np

def simulate_sweep(N, L, mu, r, s_beneficial, position_selected,
                  T_burnin, T_after=2000, track_interval=50):
```

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```

"""Simulate a selective sweep.

1. Burn in under neutrality to reach mutation-drift equilibrium.
2. Introduce a single beneficial mutation at a specified position.
3. Continue simulation and track the beneficial allele frequency.

Parameters
-----
N : int
    Diploid population size.
L : int
    Chromosome length.
mu : float
    Neutral mutation rate (per bp per generation).
r : float
    Recombination rate (per bp per generation).
s_beneficial : float
    Selection coefficient of the sweeping allele.
position_selected : int
    Genomic position of the beneficial mutation.
T_burnin : int
    Generations of neutral burn-in.
T_after : int
    Maximum generations after introducing the mutation.
track_interval : int
    How often to record the allele frequency.

Returns
-----
trajectory : list of (generation, frequency)
    Frequency trajectory of the beneficial allele.
fixed : bool
    Whether the allele reached fixation.
"""
# Initialize population
population = [Individual() for _ in range(N)]

# Phase 1: Neutral burn-in
print(f"Burning in for {T_burnin} generations...")
for tick in range(T_burnin):
    population = wright_fisher_generation(
        population, N, L, mu, r, tick, dfe='neutral'
    )

# Phase 2: Introduce beneficial mutation on one haplosome
# Pick a random individual and add the mutation to haplosome 1
lucky = np.random.randint(N)
sweep_mutation = Mutation(
    position=position_selected,
    s=s_beneficial,
    h=0.5, # codominant
    origin_tick=T_burnin

```

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```

)
population[lucky].haplosome_1.append(sweep_mutation)
population[lucky].haplosome_1.sort(key=lambda m: m.position)

print(f"Introduced beneficial mutation (s={s_beneficial}) "
      f"at position {position_selected}")

# Phase 3: Simulate and track the sweep
trajectory = [(0, 1.0 / (2 * N))] # initial frequency = 1/(2N)

for tick in range(T_after):
    gen = T_burnin + tick + 1
    population = wright_fisher_generation(
        population, N, L, mu, r, gen, dfe='neutral'
    )

    # Count copies of the beneficial mutation
    count = 0
    total = 2 * N
    for ind in population:
        for m in ind.haplosome_1:
            if (m.position == position_selected and
                m.s == s_beneficial):
                count += 1
                break
        for m in ind.haplosome_2:
            if (m.position == position_selected and
                m.s == s_beneficial):
                count += 1
                break

    freq = count / total

    if tick % track_interval == 0:
        trajectory.append((tick + 1, freq))
        print(f"  Gen {tick+1:>5d}: freq = {freq:.4f}")

    # Check for fixation or loss
    if freq >= 1.0:
        trajectory.append((tick + 1, 1.0))
        print(f"  FIXED at generation {tick + 1}")
        return trajectory, True
    elif freq <= 0.0:
        trajectory.append((tick + 1, 0.0))
        print(f"  LOST at generation {tick + 1}")
        return trajectory, False

return trajectory, False

# Example: selective sweep with s=0.05 in a small population
# (Small N and elevated mu/r for tractability)
np.random.seed(123)

```

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```

traj, fixed = simulate_sweep(
    N=200,
    L=5000,
    mu=1e-4,
    r=1e-4,
    s_beneficial=0.05,
    position_selected=2500, # middle of chromosome
    T_burnin=500,
    T_after=1000
)

```

Let us verify the fixation probability:

```

def estimate_fixation_probability(N, s, n_trials=500, **sim_kwargs):
    """Estimate fixation probability by repeated simulation.

    Run many independent introductions of a beneficial mutation
    and count how often it fixes.
    """
    n_fixed = 0
    for trial in range(n_trials):
        _, fixed = simulate_sweep(N=N, s_beneficial=s, **sim_kwargs)
        if fixed:
            n_fixed += 1

    p_fix_observed = n_fixed / n_trials
    # Theoretical prediction (Kimura)
    if s == 0:
        p_fix_theory = 1 / (2 * N)
    else:
        x = 4 * N * s
        p_fix_theory = (2 * s) / (1 - np.exp(-x))

    print(f"\nFixation probability (N={N}, s={s}):")
    print(f"  Observed:      {p_fix_observed:.4f} ({n_fixed}/{n_trials})")
    print(f"  Theory:         {p_fix_theory:.4f}")

```

i What a sweep looks like in the genealogy

After a sweep, the genealogy near the selected site looks **star-like**: all lineages coalesce rapidly to the single individual that carried the sweeping allele. Moving away from the selected site (in recombination distance), the genealogy gradually returns to the normal coalescent shape. This star-like topology creates several observable signatures:

- **Reduced diversity**: π drops near the selected site.
- **Excess rare variants**: The SFS shifts toward singletons (because all new mutations arose after the recent coalescence).
- **Extended haplotype homozygosity**: Long tracts of identical sequence around the selected site (because recombination has not yet had time to break up the sweeping haplotype).

Recipe 2: Background Selection

Background selection (BGS) is the flip side of the sweep: purifying selection against deleterious mutations reduces the effective population size at linked neutral sites. Unlike a sweep (one dramatic event), BGS is a continuous process arising from the steady rain of deleterious mutations.

The Math

Consider a region of the genome where deleterious mutations arise at rate U per generation (the total deleterious mutation rate across the region), each with selection coefficient $-s$ (and $s > 0$).

The key result (Charlesworth, Morgan, and Charlesworth, 1993) is that the effective population size at a linked neutral site is reduced to:

$$N_e \approx N \cdot \exp\left(-\frac{U}{s + r_{\text{total}}}\right)$$

where r_{total} is the total recombination rate between the neutral site and the selected region. When U/s is large and recombination is limited, BGS can dramatically reduce N_e and therefore neutral diversity.

i Probability Aside – Why BGS reduces N_e

Deleterious mutations constantly arise and are removed by selection. An individual carrying a deleterious mutation has lower fitness and is less likely to contribute to the next generation. Neutral alleles linked to the deleterious mutation are “dragged down” along with it. This is analogous to a reverse sweep: instead of one allele rising in frequency, many alleles are constantly being removed. The net effect is that the neutral genealogy looks like it came from a smaller population.

More precisely: in a diploid population, an individual free of deleterious mutations has relative fitness 1, while individuals carrying k deleterious mutations have fitness $(1-s)^k$. If deleterious mutations are at mutation-selection balance, the fraction of the population that is mutation-free is $e^{-U/s}$ (Poisson approximation), and the effective population is roughly $N \cdot e^{-U/s}$.

The Code

```
def simulate_bgs(N, L, mu_neutral, mu_deleterious, s_deleterious,
                r, T, track_interval=100):
    """Simulate background selection.

    Two classes of mutations:
```

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```

1. Neutral ( $s=0$ ): these are what we measure diversity in.
2. Deleterious ( $s<0$ ): these create the background selection effect.

Parameters
-----
N : int
    Diploid population size.
L : int
    Chromosome length.
mu_neutral : float
    Per-bp neutral mutation rate.
mu_deleterious : float
    Per-bp deleterious mutation rate.
s_deleterious : float
    Selection coefficient (negative) for deleterious mutations.
r : float
    Recombination rate.
T : int
    Number of generations.

Returns
-----
stats : dict
    Time series of population statistics.
"""
population = [Individual() for _ in range(N)]
stats = {'generation': [], 'mean_fitness': [], 'neutral_diversity': []}

for tick in range(T):
    # Recalculate fitness
    for ind in population:
        calculate_fitness(ind)

    # Generate offspring
    new_pop = []
    for _ in range(N):
        p1, p2 = select_parents(population)
        child_h1 = recombine_v2(population[p1], r, L)
        child_h2 = recombine_v2(population[p2], r, L)

    # Add neutral mutations
    child_h1 = add_mutations(child_h1, mu_neutral, L, tick,
                             dfe='neutral')
    child_h2 = add_mutations(child_h2, mu_neutral, L, tick,
                             dfe='neutral')

    # Add deleterious mutations
    child_h1 = add_mutations(child_h1, mu_deleterious, L, tick,
                             dfe='fixed',
                             dfe_params={'s': s_deleterious,
                                           'h': 0.5})
    child_h2 = add_mutations(child_h2, mu_deleterious, L, tick,

```

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```

        dfe='fixed',
        dfe_params={'s': s_deleterious,
                    'h': 0.5})

    new_pop.append(Individual(haplosome_1=child_h1,
                              haplosome_2=child_h2))
    population = new_pop

    if tick % track_interval == 0:
        fitnesses = [ind.fitness for ind in population]

        # Count neutral segregating sites (s == 0 mutations)
        neutral_positions = set()
        for ind in population:
            for m in ind.haplosome_1 + ind.haplosome_2:
                if m.s == 0:
                    neutral_positions.add(m.position)

        stats['generation'].append(tick)
        stats['mean_fitness'].append(np.mean(fitnesses))
        stats['neutral_diversity'].append(len(neutral_positions))

        print(f"Gen {tick:>6d}: mean_w={np.mean(fitnesses):.4f}, "
              f"neutral_seg={len(neutral_positions):>5d}")

    return stats

# Compare neutral diversity WITH and WITHOUT background selection
np.random.seed(42)

# Without BGS (neutral only)
print("=== No selection (neutral) ===")
pop_neutral = [Individual() for _ in range(100)]
for tick in range(500):
    pop_neutral = wright_fisher_generation(
        pop_neutral, 100, 5000, 1e-4, 1e-4, tick, dfe='neutral')

# With BGS
print("\n=== With background selection (s=-0.05) ===")
stats_bgs = simulate_bgs(
    N=100, L=5000,
    mu_neutral=1e-4,
    mu_deleterious=5e-4, # 5x higher than neutral
    s_deleterious=-0.05,
    r=1e-4,
    T=500,
    track_interval=100
)

```

i What BGS looks like in the data

Background selection produces a characteristic pattern:

- **Reduced diversity genome-wide**, but especially in regions of low recombination (where linked deleterious mutations cannot be separated from neutral variants).
- **Normal SFS shape**: Unlike a selective sweep, BGS does not strongly distort the shape of the SFS. It reduces diversity (as if N_e were smaller) but the $1/i$ pattern is preserved.
- **Correlation between diversity and recombination rate**: Regions with higher recombination have higher diversity, because they are less affected by linked selection. This is one of the strongest empirical signals of BGS in real genomes.

Recipe 3: Tree-Sequence Recording

SLiM's most powerful feature for large-scale simulations is **tree-sequence recording**: instead of storing every mutation in every individual, SLiM records the genealogical relationships (edges in the tree sequence) as they are created during reproduction. This produces a compact, lossless record of the complete ancestry.

The Math

The tree-sequence recording idea (Kelleher, Thornton, Ashander, and Ralph, 2018) is simple: during each reproduction event, record an **edge** from child to parent for each genomic interval inherited. The collection of all edges over all generations defines the complete genealogy of the final population.

Each edge is a tuple (l, r, p, c) :

- l, r : the genomic interval $[l, r)$ inherited
- p : the parent node (one haplosome of the parent)
- c : the child node (one haplosome of the child)

Recombination within a parent creates **multiple edges** from the same child to different parent haplosomes, covering different genomic intervals.

i Why record trees?

Without tree-sequence recording, a forward simulation must store every mutation in every individual – a huge memory burden for large populations and long genomes. Many of these mutations are neutral and are only needed for computing summary statistics *after* the simulation.

With tree-sequence recording, neutral mutations are not tracked during the simulation at all. Instead, the genealogy is recorded, and mutations are sprinkled onto the genealogy *after* the simulation (exactly as msprime does – see [Mutations](#)). This can reduce memory use by orders of magnitude.

The genealogy also enables efficient computation of many statistics (diversity, divergence, F_{ST}) without even needing mutations, using branch-length statistics.

The Code

```
def simulate_with_tree_recording(N, L, mu, r, T):
    """Forward simulation with tree-sequence recording.

    Instead of tracking neutral mutations, we record the parent-child
```

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edges during each reproduction event. At the end, we have a complete tree sequence that can be analyzed with *tskit*.

Parameters

```
N : int
    Population size.
L : int
    Chromosome length.
mu : float
    Mutation rate (used only for adding mutations AFTER simulation).
r : float
    Recombination rate.
T : int
    Number of generations.
```

Returns

```
nodes : list of (time, population)
    Tree-sequence nodes (one per haplosome per generation).
edges : list of (left, right, parent_node, child_node)
    Tree-sequence edges (genealogical relationships).
"""
# Node table: each haplosome in each generation gets a node
# node_id -> (time, population)
nodes = []
edges = []

# Current generation's node IDs
# Each individual has two haplosomes -> two node IDs
current_nodes = []
for i in range(N):
    # Two nodes per individual (one per haplosome)
    node_id_1 = len(nodes)
    nodes.append((T, 0)) # time T (oldest generation)
    node_id_2 = len(nodes)
    nodes.append((T, 0))
    current_nodes.append((node_id_1, node_id_2))

# No fitness tracking here -- purely neutral for simplicity
for tick in range(T):
    generation_time = T - tick - 1 # counting down

    new_nodes = []
    for _ in range(N):
        # Draw parents uniformly (neutral)
        p1_idx = np.random.randint(N)
        p2_idx = np.random.randint(N)

        # Create child nodes
        child_node_1 = len(nodes)
        nodes.append((generation_time, 0))
```

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```

child_node_2 = len(nodes)
nodes.append((generation_time, 0))

# Parent 1 contributes child haplosome 1 (with recombination)
parent1_hap_a, parent1_hap_b = current_nodes[p1_idx]
n_breaks = np.random.poisson(r * L)
breakpoints = sorted(np.random.randint(1, L, size=n_breaks)) if n_breaks >
→ 0 else []

# Record edges for child_node_1 <- parent1
# Start with a random haplosome
if np.random.random() < 0.5:
    sources = [parent1_hap_a, parent1_hap_b]
else:
    sources = [parent1_hap_b, parent1_hap_a]

# Create edges at breakpoint boundaries
all_breaks = [0] + breakpoints + [L]
for i in range(len(all_breaks) - 1):
    left = all_breaks[i]
    right = all_breaks[i + 1]
    parent_node = sources[i % 2]
    edges.append((left, right, parent_node, child_node_1))

# Parent 2 contributes child haplosome 2 (same process)
parent2_hap_a, parent2_hap_b = current_nodes[p2_idx]
n_breaks = np.random.poisson(r * L)
breakpoints = sorted(np.random.randint(1, L, size=n_breaks)) if n_breaks >
→ 0 else []

if np.random.random() < 0.5:
    sources = [parent2_hap_a, parent2_hap_b]
else:
    sources = [parent2_hap_b, parent2_hap_a]

all_breaks = [0] + breakpoints + [L]
for i in range(len(all_breaks) - 1):
    left = all_breaks[i]
    right = all_breaks[i + 1]
    parent_node = sources[i % 2]
    edges.append((left, right, parent_node, child_node_2))

new_nodes.append((child_node_1, child_node_2))

current_nodes = new_nodes

if tick % 100 == 0:
    print(f"Gen {tick:>5d}: {len(nodes)} nodes, {len(edges)} edges")

# The sample nodes are the final generation's haplosomes
sample_nodes = []
for hap_a, hap_b in current_nodes:

```

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```

    sample_nodes.extend([hap_a, hap_b])

    print(f"\nDone: {len(nodes)} total nodes, {len(edges)} total edges")
    print(f"Sample nodes: {len(sample_nodes)} (= 2N = {2*N})")

    return nodes, edges, sample_nodes

# Run a small example
np.random.seed(42)
nodes, edges, samples = simulate_with_tree_recording(
    N=50, L=10000, mu=1e-4, r=1e-4, T=200
)

```

After the simulation, mutations can be placed on the recorded tree sequence exactly as in msprime:

```

def add_mutations_to_tree_sequence(nodes, edges, samples, mu, L):
    """Sprinkle mutations onto a recorded tree sequence.

    This is the same process as msprime's Phase 2: for each edge,
    draw Poisson-distributed mutations and place them at random
    positions and times on the branch.

    Parameters
    -----
    nodes : list of (time, population)
    edges : list of (left, right, parent, child)
    samples : list of int
        Sample node IDs.
    mu : float
        Per-bp, per-generation mutation rate.
    L : int
        Chromosome length.

    Returns
    -----
    mutations : list of (position, node, time)
    """
    mutations = []

    for left, right, parent, child in edges:
        span = right - left
        parent_time = nodes[parent][0]
        child_time = nodes[child][0]
        branch_length = parent_time - child_time

        if branch_length <= 0:
            continue

        # Number of mutations on this edge
        expected = mu * span * branch_length
        n_muts = np.random.poisson(expected)

```

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```

    for _ in range(n_muts):
        pos = np.random.uniform(left, right)
        time = np.random.uniform(child_time, parent_time)
        mutations.append((pos, child, time))

    mutations.sort(key=lambda m: m[0])
    print(f"Placed {len(mutations)} mutations on the tree sequence")
    return mutations

muts = add_mutations_to_tree_sequence(nodes, edges, samples, mu=1e-4, L=10000)

```

The tree-sequence recording tradeoff

Pros:

- Dramatically reduces memory for neutral mutations.
- Enables post-hoc mutation placement (try different μ without re-running the simulation).
- Enables efficient branch-length statistics.
- The tree sequence can be **simplified** to remove extinct lineages, further reducing size.

Cons:

- Recording edges has a cost: each reproduction event creates 1-3 edges per haplosome (depending on recombination).
- Long simulations accumulate many edges; periodic **simplification** is needed to keep size manageable.
- Selected mutations must still be tracked during the simulation (they affect fitness and cannot be deferred to post-hoc).

Exercises

Exercise 1: Fixation probability

Run 500 independent selective sweeps with $N = 200$ and $s = 0.01, 0.02, 0.05, 0.1$. For each s , estimate the fixation probability and compare to the theoretical prediction $P_{\text{fix}} \approx 2s/(1 - e^{-4Ns})$.

Exercise 2: Diversity reduction around a sweep

After a completed sweep, measure neutral diversity (π) in windows around the selected site. Verify that diversity is reduced near the selected site and recovers with distance, following the prediction $\pi/\pi_0 \approx 1 - 2s/(2s + r \cdot d)$ where d is the distance from the selected site.

i Exercise 3: Background selection and N_e

Simulate 1000 generations with $N = 500$, neutral mutation rate $\mu = 10^{-4}$, and varying deleterious mutation rates $U = 0.01, 0.05, 0.1, 0.5$. Measure neutral diversity at equilibrium and estimate the effective population size as $\hat{N}_e = \pi/(4\mu)$. Compare to the BGS prediction $N_e \approx N \cdot e^{-U/s}$.

i Exercise 4: Tree-sequence simplification

After running a tree-sequence recording simulation for $T = 500$ generations with $N = 100$, count the number of edges. Then implement a simplification step that removes all nodes and edges not ancestral to the final sample. How much smaller is the simplified tree sequence?

You have now built a forward-time population genetics simulator from scratch. The three recipes demonstrate phenomena that are central to population genetics but impossible (or at best very awkward) to model with backward-time coalescent simulation:

- **Selective sweeps** – A beneficial allele rising to fixation, dragging linked variation along.
- **Background selection** – The steady erosion of neutral diversity by linked deleterious mutations.
- **Tree-sequence recording** – A compact genealogical record that bridges forward and backward time.

The real SLiM is vastly more powerful than our toy implementation: it supports non-Wright-Fisher life cycles, spatial structure, continuous space, gene drive, quantitative genetics, and much more. But the *mechanism* is the same. Every generation, the same cycle runs: fitness, selection, recombination, mutation. The forge is hot, and now you know how to work it.

The metal bends. And you know exactly why.

Demo: Running SLiM on Simulated Data

Running mini_slim on simulated genomic data.

This figure was generated by running the `mini_slim` implementation on data simulated with `msprime`. The script is at `figures/fig_demo_slim.py`.

3.18 Timepiece XVII: Relate

Scalable local-genealogy estimation from phased haplotypes

Relate infers a sequence of rooted local trees, maps mutations to those trees, and estimates coalescence times under a coalescent prior. Its central scaling choice is to infer topology first and then condition on those topologies while dating them. The stages are connected rather than statistically independent: mutation mapping decides when a new topology is built, equivalent branches share dating information, and an estimated population history can be fed back into the branch-length sampler [Speidel *et al.*, 2019].

This Timepiece separates the production program from the accompanying teaching module. Production Relate implements windowed chromosome painting, robust tree construction, approximate and multi-branch mutation mapping, association of branches across neighbouring trees, event-order and interval MCMC moves, and iterative coalescence-rate estimation. `watchgen.mini_relate` demonstrates only small, source-guided kernels. It does not read `.haps` files, build a genomic tree sequence, or reproduce Relate's posterior; it is not a reimplement of Relate.

Demo: SLiM Forward Simulation ($N=100$, $L=50$ kb, 300 gen)

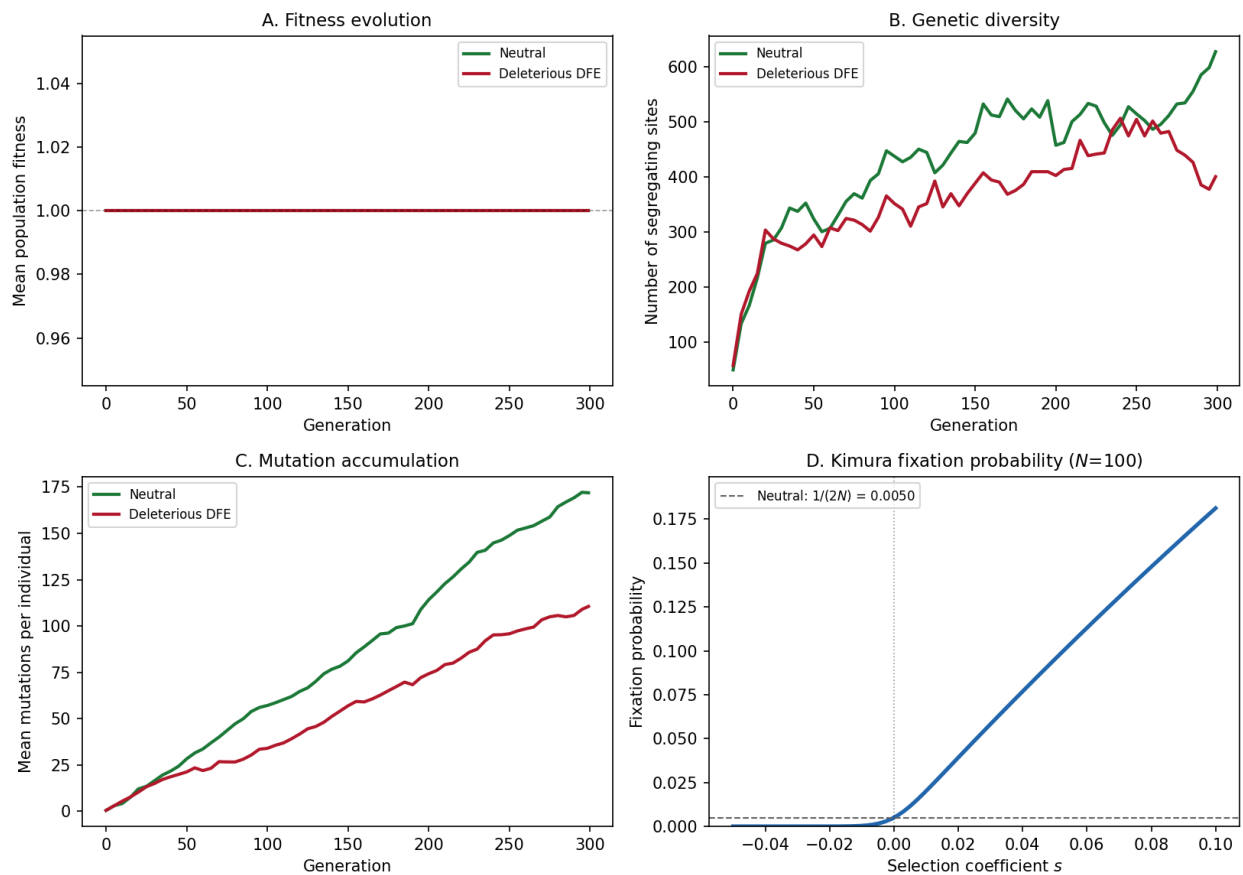


Fig. 25: **SLiM demo of forward-time simulation.** Panel A: Mean population fitness over generations under neutral and deleterious DFE scenarios. Panel B: Number of segregating sites over time – purifying selection reduces diversity. Panel C: Mean mutations per individual, showing accumulation under mutation-selection balance. Panel D: Kimura fixation probability as a function of selection coefficient.

3.18.1 Source and review target

The primary methodological source is the program paper and its Supplementary Note [Speidel *et al.*, 2019]. The software source is the upstream Myers Group repository, <https://github.com/MyersGroup/relate>. It was audited at commit `b54ede259cbb0be095bc9c9a8bd18cdaf7e88b74`. In particular, the review follows `fast_painting.cpp`, `tree_builder.cpp`, `anc_builder.cpp`, and `branch_length_estimator.cpp` rather than inferring behavior from command names.

3.18.2 Production pipeline

For each target haplotype, Relate uses a modified Li–Stephens HMM to obtain local copying scores against the other haplotypes. It transforms these into a non-symmetric distance matrix and applies a directional hierarchical clustering rule. Moving from the 5' end, it retains the current tree until a mutation cannot be mapped uniquely or appears to require an ancestral/derived flip. It then associates equivalent branches between neighbouring trees, pools their mutation information, and samples coalescence times. A separate iterative procedure can alternate pairwise coalescence-rate estimation with branch-length re-estimation.

3.18.3 Chapters

What Relate estimates

Relate takes phased, biallelic haplotypes together with physical positions, a recombination map, ancestral-allele labels, a mutation rate, and a population-size scale. Its core outputs are `.anc` files describing local trees and `.mut` files describing mapped mutations. Conversion utilities can translate those files to other formats, but `tskit` is not the native inference representation.

The inferred tree sequence is an approximation to an ancestral recombination graph. Relate estimates marginal trees locally without enforcing a single global recombination history. This choice makes the calculation parallelizable and gives the reported linear scaling in sequence length and quadratic scaling in sample size [Speidel *et al.*, 2019].

Three coupled stages

Topology inference

Relate paints each haplotype as a mosaic of all other haplotypes. Its modified emission distinguishes a mutation derived in the target and ancestral in the reference from the other three allele pairs. Forward–backward scores give a position-specific ordering of likely relatives. After a log rescaling and a row-minimum subtraction, these scores form a directional matrix.

The tree builder repeatedly merges clusters that are mutual row minima, allowing a documented tolerance for inconsistent data. Cluster-to-cluster distances are means over all ordered pairs of member haplotypes. A smallest-symmetrized-distance rule resolves multiple feasible pairs and supplies a fallback when none are feasible.

Mutation mapping and tree changes

Relate starts with a tree near the 5' end and attempts to map each subsequent mutation to it. Under infinite sites, the carriers should equal the descendants of one branch. The production mapper permits small discrepancies, can flip ancestral and alternative labels, and can distribute a mutation across a minimal set of branches. A new topology is built when a mutation cannot be mapped uniquely or appears flipped. Thus production Relate does not construct an unrelated tree at every SNP.

Dating and demographic refinement

Equivalent branches in adjacent trees are associated so their mutation counts and genomic exposures can be pooled. Relate samples ranked coalescence events and the intervals τ_k during which k lineages exist. The target combines Poisson mutation probabilities with a standard- or variable-size coalescent prior.

Population history is then estimated from pairwise TMRCA events and survival exposure within time epochs. Relate alternates this calculation with branch-length re-estimation, using five iterations in the paper's implementation. Calling this whole procedure a conventional EM algorithm obscures the actual estimator and its heuristic mutation-rate rescaling.

Scope of the mini

`watchgen.mini_relate` includes a dense small-panel painting recursion, relative distance conversion, the documented mutual-minimum tree rule, exact mutation mapping, a fixed-event-order interval sampler, and the pairwise epoch-rate MLE. It omits production windowing, topology reuse along a chromosome, approximate mapping, branch association, event-order swaps, variable-size dating, input/output formats, and parallel execution.

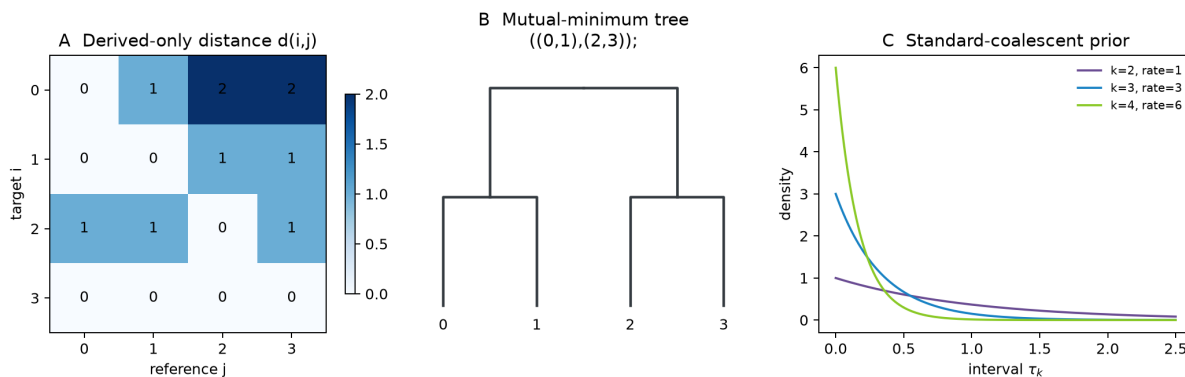


Fig. 26: Source-guided kernels used in this Timepiece; the figure is not an output from the Relate executable.

Directional chromosome painting

Relate's HMM is directional because it models mutations on the target lineage since its most recent common ancestor with a candidate copying haplotype. Let $D_\ell^{(i)}$ be the target allele, $D_\ell^{(j)}$ the reference allele, and p the mismatch parameter. The emission is

$$P_m(D_\ell^{(i)} | H_\ell = j) = \begin{cases} p, & D_\ell^{(i)} = 1, D_\ell^{(j)} = 0, \\ 1 - p, & \text{otherwise.} \end{cases}$$

Only the target-derived/reference-ancestral case is penalized. In particular, ancestral-target/derived-reference is *not* assigned a second, user-chosen mismatch weight. The old mini's `w_d` and `w_a` parameters did not describe Relate [Speidel *et al.*, 2019].

Dense teaching recursion

For a small panel of K references, let r_ℓ be the probability of redrawing a copying state between sites $\ell - 1$ and ℓ . The forward prediction for state j is

$$\tilde{\alpha}_\ell(j) = (1 - r_\ell)\alpha_{\ell-1}(j) + \frac{r_\ell}{K} \sum_h \alpha_{\ell-1}(h).$$

Multiplication by the emission and normalization gives the forward message; the backward recursion is analogous. `mini_relate.copying_posterior` implements this dense form. Production Relate exploits the one-sided emission by painting only sites derived in the target, stores stepping-stone boundary messages, caps extreme transition probabilities, and repaints short sections when a complete matrix is needed.

From posterior weight to directional score

With no recombination, define $d(i, j)$ as the number of sites derived in target i and ancestral in reference j . The Supplementary Note derives

$$d(i, j) = \frac{\log P_m(D^{(i)} \mid H = j) - L \log(1 - p)}{\log[p/(1 - p)]}.$$

The same rescaling of the local forward–backward score defines $d(i, j; \ell)$. Relate subtracts $\min_{j \neq i} d(i, j; \ell)$ from each row. This removes an additive tip-branch residual and matters for the fallback symmetrized score. Consequently a normalized posterior is sufficient for the mini's *relative* row, but it must not be presented as an absolute mutation count.

Why asymmetry matters

Under infinite sites and no recombination,

$$d(i, j) = \#\{\ell : D_\ell^{(i)} = 1, D_\ell^{(j)} = 0\}.$$

This matrix need not equal its transpose. Symmetrizing it produces the ordinary number of pairwise differences and discards which lineage carries each derived mutation. The directional row order is the information used by Relate's tree builder.

```
def modified_emission(target: int, reference: int, mismatch: float) -> float:
    """Return Relate's directional emission probability (Supplement eq. 12).

    A derived target copied from an ancestral reference receives ``mismatch``.
    The other three allele pairs all receive ``1 - mismatch``.
    """

    if target not in (0, 1) or reference not in (0, 1):
        raise ValueError("alleles must be coded 0 (ancestral) or 1 (derived)")
    if not 0.0 < mismatch < 0.5:
        raise ValueError("mismatch must lie between 0 and 0.5")
    return mismatch if target == 1 and reference == 0 else 1.0 - mismatch
```

```
def relative_distance_row(
    posterior: Sequence[float], mismatch: float = 0.025
) -> np.ndarray:
    """Convert posterior copying weights to Relate's row-centred score.

    The full rescaling has an additive row constant involving sequence length.
    Relate subtracts each row minimum, so that constant cancels. Normalized
    posterior probabilities preserve the required ordering, not an absolute
    mutation count.
    """

    posterior = np.asarray(posterior, dtype=float)
    if posterior.ndim != 1 or posterior.size == 0:
        raise ValueError("posterior must be a nonempty vector")
    if np.any(posterior <= 0.0) or not np.isclose(posterior.sum(), 1.0):
        raise ValueError("posterior entries must be positive and sum to one")
    if not 0.0 < mismatch < 0.5:
        raise ValueError("mismatch must lie between 0 and 0.5")
    distance = np.log(posterior) / math.log(mismatch / (1.0 - mismatch))
    return distance - distance.min()
```

Directional hierarchical clustering

Relate begins with one cluster per haplotype. For clusters A and B , it defines the directional distance

$$d(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{j \in B} d(i, j).$$

This is a cardinality-weighted mean over original haplotype pairs. It is not the minimum of the two directions and not an unweighted average of previously merged cluster scores.

The merge condition

Clusters A and B are eligible when each lies at the minimum of the other's directional row:

$$d(A, B) \leq \min_{C \neq A} d(A, C) + \epsilon, \quad d(B, A) \leq \min_{C \neq B} d(B, C) + \epsilon.$$

The Supplementary Note reports $\epsilon = 0.2$ on the mutation-distance scale. The C++ implementation stores an equivalent tolerance on its log-score scale. If several pairs are eligible, Relate chooses the one minimizing $d(A, B) + d(B, A)$. If inconsistent data leave no eligible pair, it falls back to the smallest symmetrized score. The earlier mini instead selected the pair with the smallest value in either direction; that rule can merge clusters that are not mutual neighbours.

Under infinite sites and no recombination, the directional mutation-count matrix guarantees a tree consistent with every mutation-supported clade. It does not guarantee recovery of binary resolutions unsupported by mutations. The paper also proves consistency for limiting maps made of zero- and infinite-recombination segments, not for arbitrary finite-recombination data [Speidel *et al.*, 2019].

Mapping mutations while moving along the genome

For an exact infinite-sites mutation, the carrier set equals the descendants of one branch. Production Relate adds robust behavior:

- a candidate branch must recover more than 70% of carriers and non-carriers;
- the final misclassification fraction must be below 0.03;
- ancestral and alternative labels are flipped only when that improves the fit;
- if no unique branch works, a smallest exact collection of branches is used and the mutation count is divided across those branches.

Relate builds a new local topology when the next mutation cannot be mapped uniquely or appears potentially flipped. Otherwise it reuses the current tree. This is why a function that independently clusters every SNP is not a production Relate implementation.

The teaching boundary

`mini_relate.build_tree` implements the mutual-minimum rule and recomputes the cardinality-weighted means directly. `mini_relate.map_mutation_exact` implements only exact single-branch mapping and exact flip detection. It intentionally omits the production thresholds, fractional multi-branch mapping, random tie handling, sample-age constraints, topology templates, and streaming chromosome logic.

```
def find_mutual_minimum_pair(
    distance: np.ndarray,
    clusters: Mapping[int, frozenset[int]],
    tolerance: float = 0.2,
) -> tuple[int, int]:
    """Choose a mutual-row-minimum pair, with Relate's fallback."""
```

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```

if tolerance < 0.0:
    raise ValueError("tolerance must be nonnegative")
active = sorted(clusters)
if len(active) < 2:
    raise ValueError("at least two clusters are required")
directed = {
    (a, b): cluster_distance(distance, clusters[a], clusters[b])
    for a in active
    for b in active
    if a != b
}
row_minimum = {
    a: min(directed[a, b] for b in active if b != a) for a in active
}
pairs = [(a, b) for x, a in enumerate(active) for b in active[x + 1 :]]
feasible = [
    (a, b)
    for a, b in pairs
    if directed[a, b] <= row_minimum[a] + tolerance
    and directed[b, a] <= row_minimum[b] + tolerance
]
candidates = feasible or pairs
return min(candidates, key=lambda pair: directed[pair] + directed[pair[::-1]])

```

```

def map_mutation_exact(
    root: TreeNode, carriers: Iterable[int], allow_flip: bool = True
) -> tuple[int | None, bool]:
    """Map a mutation to one exact branch, optionally flipping allele labels.

    Production Relate also has approximate thresholds and a fractional
    multi-branch fallback. The mini deliberately exposes only its exact stage.
    """

    carriers = frozenset(carriers)
    samples = root.leaves
    if not carriers or not carriers < samples:
        return None, False
    branches = [node for node in iter_nodes(root) if node is not root]
    exact = [node.id for node in branches if node.leaves == carriers]
    if len(exact) == 1:
        return exact[0], False
    if allow_flip:
        exact = [node.id for node in branches if node.leaves == samples - carriers]
        if len(exact) == 1:
            return exact[0], True
    return None, False

```

Estimating coalescence times

Relate dates fixed local-tree topologies, but it does not date every tree in isolation. A branch is defined by its descendant sets at its lower and upper coalescence events. Across adjacent trees, production Relate first associates exactly equivalent branches and then accepts approximate associations when the descendant-vector correlations at both ends exceed 0.9. Greedy matching ensures that each branch is associated at most once. Mutation counts and genomic exposure are pooled across an associated run [Speidel *et al.*, 2019].

Likelihood and prior

Let m_b be the pooled mutation count on branch b , t_b its length in coalescent time, and $\theta_b/2$ the mutation rate summed over the bases for which it persists. Relate uses

$$m_b \mid t_b \sim \text{Poisson}(\theta_b t_b / 2).$$

For a constant population size, write τ_k for the interval during which k lineages remain. In standard coalescent units,

$$\tau_k \sim \text{Exponential}\left(\binom{k}{2}\right), \quad k = N, N-1, \dots, 2.$$

The ranked order of internal events and the vector of intervals determine all node times and branch lengths. Relate's posterior is the product of these coalescent densities and the branchwise Poisson probabilities.

Production MCMC moves

Production Relate samples more than independent node ages. With probability 0.8 it proposes swapping the rank of two topologically compatible coalescence events. This changes six branches while retaining the interval vector. Otherwise it chooses one τ_k and proposes a positive replacement from an exponential distribution with mean equal to its current value; the Metropolis–Hastings ratio includes the asymmetric proposal correction. Changing τ_k changes the lengths of the k branches then alive.

The paper implementation initializes a compatible event order with N^2 swap proposals and initializes intervals by an EM calculation with the event order fixed. For production sampling it uses at least $\max(10N, 1000)$ burn-in proposals and continues until every interval has received at least 20 proposals and the resulting branch lengths are positive. Posterior mean event ages are differenced to obtain an ultrametric tree.

What the mini samples

`mini_relate.sample_ranked_branch_lengths` fixes one compatible event order and updates only the interval vector. It evaluates the same constant-size coalescent/Poisson target for that restricted state space and includes the exact Hastings correction for its exponential proposal. It does not associate branches between trees, swap event ranks, implement sample ages, or reproduce Relate's adaptive stopping rule. Its output is therefore a teaching diagnostic, not a Relate posterior sample.

```
def log_branch_mutation_likelihood(
    root: TreeNode,
    node_times: Mapping[int, float],
    mutations: Mapping[int, float],
    exposure: Mapping[int, float],
    theta: float,
) -> float:
    """Poisson log likelihood for mutation counts on associated branches."""

    if theta <= 0.0:
        raise ValueError("theta must be positive")
```

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```

total = 0.0
for branch, length in branch_lengths(root, node_times).items():
    count = float(mutations.get(branch, 0.0))
    span = float(exposure.get(branch, 0.0))
    if count < 0.0 or span < 0.0:
        raise ValueError("counts and exposures must be nonnegative")
    mean = theta * span * length / 2.0
    if mean == 0.0:
        if count > 0.0:
            return -math.inf
        continue
    total += count * math.log(mean) - mean - math.lgamma(count + 1.0)
return total

```

```

def sample_ranked_branch_lengths(
    root: TreeNode,
    event_order: Sequence[int],
    mutations: Mapping[int, float],
    exposure: Mapping[int, float],
    theta: float,
    iterations: int = 5_000,
    burn_in: int = 1_000,
    seed: int | None = None,
) -> tuple[np.ndarray, float]:
    """Sample intervals for a fixed event order using exponential proposals.

    Relate additionally swaps incomparable event orders and pools information
    across equivalent branches. This smaller sampler includes the exact
    Hastings correction for its exponential random-walk proposal.
    """

    samples = len(root.leaves)
    if iterations <= burn_in or burn_in < 0:
        raise ValueError("iterations must exceed a nonnegative burn-in")
    rng = np.random.default_rng(seed)
    lineages = np.arange(samples, 1, -1, dtype=float)
    intervals = 1.0 / (lineages * (lineages - 1.0) / 2.0)
    current = log_ranked_tree_posterior(
        root, event_order, intervals, mutations, exposure, theta
    )
    draws, accepted = [], 0
    for iteration in range(iterations):
        index = int(rng.integers(intervals.size))
        old = intervals[index]
        proposed_value = float(rng.exponential(old))
        proposal = intervals.copy()
        proposal[index] = proposed_value
        candidate = log_ranked_tree_posterior(
            root, event_order, proposal, mutations, exposure, theta
        )
        log_hastings = (
            math.log(old / proposed_value) - old / proposed_value + proposed_value /

```

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```

→old
    )
    if math.log(rng.random()) < candidate - current + log_hastings:
        intervals, current = proposal, candidate
        accepted += 1
    if iteration >= burn_in:
        draws.append(intervals.copy())
    return np.asarray(draws), accepted / iterations

```

Coalescence rates and population history

Relate estimates piecewise-constant pairwise coalescence rates from dated local trees. Fix a pair of haplotypes and let t_z be their TMRCA in local tree z . Epoch e spans $[T_e, T_{e+1})$ and has rate λ_e . Each observed TMRCA contributes one event in its ending epoch and survival exposure in every epoch it traverses.

If n_e pairwise TMRCAs fall in epoch e , the maximum-likelihood estimate from the Supplementary Note is

$$\hat{\lambda}_e = \frac{n_e}{\sum_{z:e=e_z} (t_z - T_e) + \sum_{z:e < e_z} (T_{e+1} - T_e)}.$$

This is simply events divided by exposure. Relate calculates it for each pair and then averages over pairs for a population-wide curve. The authors explicitly note that this average is not the panmictic population-wide MLE. Its flexibility also allows within- and between-population coalescence-rate summaries [Speidel *et al.*, 2019].

Units and effective size

The inverse of a pairwise coalescence hazard has the units of a population-size time scale. Whether it is labelled N_e or $2N_e$ depends on whether time and rate use haploid, diploid-generation, or coalescent units. A chapter must state that convention before converting $1/\lambda_e$ to effective size. The mini returns rates, event counts, and exposure directly and does not silently apply a factor of two.

The iterative production procedure

Relate first dates trees with a constant-size prior. It then repeats the following cycle:

1. estimate population-wide coalescence rates from current pairwise TMRCAs;
2. estimate mutation rate through time as mutations divided by branch exposure;
3. rescale coalescence rates by the ratio of a fixed mutation rate to that estimated temporal rate, a heuristic intended to accelerate convergence;
4. re-estimate branch lengths under the resulting variable-size coalescent prior.

The paper implementation uses five cycles, initially restricting the calculation to trees with at least N mapped mutations, and then performs a final dating pass over all trees. This is an alternating plug-in procedure. The former chapter's generic E-step/M-step equations for latent epoch allocations were not Relate's published population-history method.

Teaching calculation

For TMRCAs $[0.2, 0.4, 1.2, 1.8]$ and boundaries $[0, 0.5, 1, 2]$, the event counts are $[2, 0, 2]$ and exposures are $[1.6, 1.0, 1.0]$. The resulting rates are $[1.25, 0, 2]$. The zero in the middle is the unsmoothed MLE for an epoch with exposure but no event; production analyses need sufficient genome-wide information and suitable epoch choices.

```
def piecewise_coalescence_rate_mle(
    tmrcas: Sequence[float], boundaries: Sequence[float]
) -> tuple[np.ndarray, np.ndarray, np.ndarray]:
    """Estimate pairwise coalescence rates as events divided by exposure.

    This is Supplementary equation 50. Relate averages pair-specific estimates
    and alternates rate estimation with branch-length re-estimation; it is not a
    generic hidden-variable EM M-step.
    """

    tmrcas = np.asarray(tmrcas, dtype=float)
    boundaries = np.asarray(boundaries, dtype=float)
    if tmrcas.ndim != 1 or tmrcas.size == 0 or np.any(tmrcas < 0.0):
        raise ValueError("tmrcas must be a nonempty nonnegative vector")
    if boundaries.ndim != 1 or boundaries.size < 2:
        raise ValueError("at least two epoch boundaries are required")
    if boundaries[0] != 0.0 or np.any(np.diff(boundaries) <= 0.0):
        raise ValueError("boundaries must increase strictly from zero")
    if tmrcas.max() >= boundaries[-1]:
        raise ValueError("the final boundary must exceed every TMRCA")
    events = np.zeros(boundaries.size - 1, dtype=float)
    exposure = np.zeros_like(events)
    for time in tmrcas:
        for epoch, (start, end) in enumerate(zip(boundaries[:-1], boundaries[1:])):
            exposure[epoch] += max(0.0, min(time, end) - start)
            if start <= time < end:
                events[epoch] += 1.0
    rates = np.divide(events, exposure, out=np.zeros_like(events), where=exposure > 0.
    ↪ 0)
    return rates, events, exposure
```

Verified Relate exercises

The two figures in this Timepiece are generated from `watchgen.mini_relate`. They validate internal mathematical targets; they are not labelled as Relate executable output and do not establish whole-program parity.

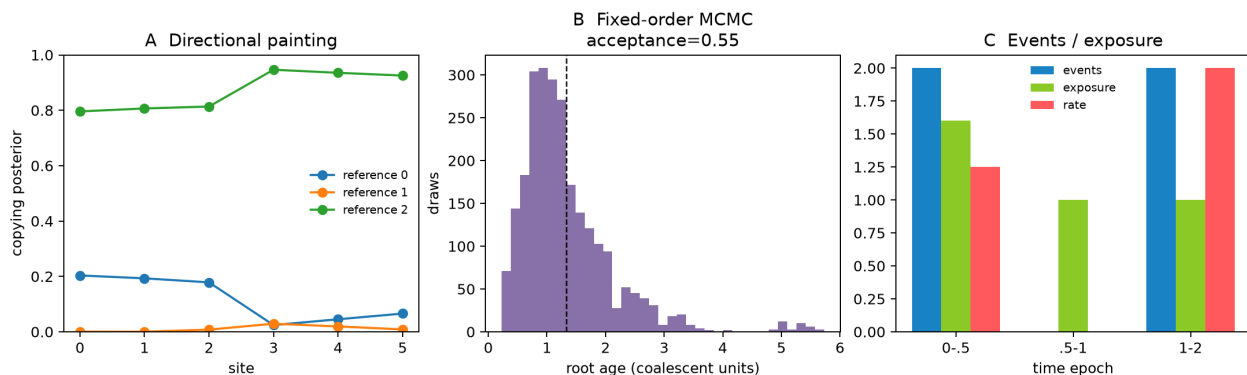


Fig. 27: Independent checks of directional mutation counts, mutual-minimum tree construction, fixed-order interval sampling, and event/exposure rate estimates.

Running the mini

```
python -m watchgen.mini_relate
```

The demo prints a topology-only Newick tree, posterior mean coalescent intervals from the restricted sampler, its acceptance fraction, and the pairwise coalescence-rate example. A seeded run is reproducible, but MCMC equality to a single stored vector is not a scientific parity criterion.

Production executable validation

The audited upstream commit was configured and compiled with CMake. Its bundled C++ test executable passed all 239 assertions in 9 test cases. The supplied `example/` data were then run through `scripts/RelateParallel/RelateParallel.sh` with the repository's documented settings: $N_e = 30,000$, mutation rate $1.25e-8$, sample ages, one thread, and seed 1.

The run produced 8,947 local trees for 11 haplotypes and 130,862 mutation rows. Every `.anc` tree contained the required $2N - 1 = 21$ node records, all reported branch lengths were nonnegative, and tree start indices increased from 0 through 130,858. Mutation positions increased from 745 through 249,215,937 and all tree indices lay in $[0, 8947)$. The production diagnostics reported 287 non-mapping SNPs and 26 flipped SNPs, which also matched the parsed `.mut` file.

These checks include:

- every local tree has $2N - 1$ nodes for N haplotypes;
- parent times exceed child times and paths are ultrametric within numerical tolerance;
- mutation positions are ordered and mapped branch identifiers exist;
- adjacent-tree coordinates cover the requested region without reversal; and
- rerunning with the same supported seed and build gives the expected level of reproducibility.

Those checks validate file and tree invariants. Distributional or accuracy parity requires simulations with known genealogies and cannot be inferred from successful execution alone.

Automated chapter checks

The focused tests independently verify all four emission cases, dense forward-backward limiting cases, row rescaling, directional mutation counts, mutual-minimum and fallback merges, weighted cluster distances, exact mapping and flipping, compatible event ranks, ultrametric branch construction, the choose-two coalescent density, Poisson likelihood, seeded MCMC behavior, and the event/exposure estimator. Documentation gates require the audited source commit `b54ede259cbb0be095bc9c9a8bd18cdaf7e88b74` and reject the removed false-parity APIs and claims.

3.19 Timepiece XVIII: discoal

Selective-sweep simulation with a discrete ancestral recombination graph

`discoal` is a C simulator for coalescent histories with crossover, gene conversion, population structure, demographic events, and selective sweeps. Its name contracts *discrete* and *coalescent*: ancestry is tracked across a discrete number of sites. Selection is represented by first generating an allele-frequency trajectory and then running a structured coalescent conditional on that trajectory [Kern and Schrider, 2016, Braverman *et al.*, 1995, Coop and Griffiths, 2004].

This Timepiece separates three levels that the earlier draft conflated. The production program constructs a chromosome-scale ancestral recombination graph. The mathematical sweep kernel assigns ancestral material to beneficial and wild-type backgrounds. The accompanying `watchgen.mini_discoal` module implements only a single-locus version of that kernel. It is useful for inspecting rates and limiting cases, but it is not a chromosome-scale ARG simulator and must not be used as a replacement for `discoal`.

3.19.1 Source and review target

The primary software reference for this review is the upstream repository at <https://github.com/kern-lab/discoal>. The current source was audited at commit `7d0955f4107053c135d2086790b0426457147a8e`; executable parity was also checked against the paper-era commit `82971bf`. In particular, the trajectory rules come from `alleleTraj.c` and the sweep event loop from `discoalFunctions.c`. The primary methodological references are the original program paper [Kern and Schrider, 2016], the hitchhiking coalescent of Braverman and colleagues [Braverman *et al.*, 1995], and the conditional diffusion construction of Coop and Griffiths [Coop and Griffiths, 2004].

3.19.2 What the mini implementation covers

The function `mini_discoal.deterministic_trajectory` reproduces the deterministic curve and discretization used by the C source. The function `mini_discoal.stochastic_trajectory` implements the two-point conditional jump law, including the neutral phase used for standing variation. The function `mini_discoal.structured_coalescent_sweep` then applies the two-background event rates to one linked neutral locus. Finally, `mini_discoal.simulate_linked_locus_genealogy` adds the neutral phases before and after the sweep.

The mini does not split ancestral segments, place mutations on an ARG, emit `ms`-format haplotypes, model gene conversion, or reproduce `discoal`'s multipopulation event scheduler. Those are properties of the original software, not of this teaching kernel.

3.19.3 Chapters

Overview of `discoal`

`discoal` generates samples under coalescent models with recombination, demographic changes, population structure, and selection [Kern and Schrider, 2016]. A sweep is not simulated by assigning a shorter neutral tree after the fact. The simulator conditions ancestry on the frequency path of the selected allele and tracks which ancestral segments reside on the beneficial or wild-type background.

The production algorithm

At a high level, a sweep simulation contains a neutral phase from the present back to the sweep endpoint, a sweep phase, and an older neutral phase. During the sweep phase, the program advances on a fine time grid. It updates the allele frequency, calculates coalescence, crossover, gene-conversion, and recurrent-adaptive-mutation probabilities, and performs a possible event. For an in-locus crossover, an active ancestor is split at a discrete breakpoint. One child segment retains the selected background and the other receives a background according to the current allele frequency. This ancestry bookkeeping is why a two-counter single-locus model is not a translation of the complete program.

The simulator accepts a haploid sample size, a number of replicates, and a number of discrete sites. Its main scaled parameters are

$$\theta = 4N_0\mu_L, \quad \rho = 4N_0r_L, \quad \alpha = 2N_0s,$$

where μ_L and r_L are whole-locus mutation and crossover rates. The `-t` and `-r` options therefore do not take per-base rates. The selected site is given as a relative coordinate with `-x`. Sweep and demographic times entered at the command line are in units of $4N_0$ generations; the C event loop converts them to its internal $2N_0$ clock.

The output is normally an `ms`-compatible haplotype matrix. With `-T`, `discoal` instead prints marginal Newick trees and their spans. The program can also include the selected SNP in partial-sweep output; `-h` suppresses that site.

Selection models and limits

The flags `-wd`, `-ws`, and `-wn` request deterministic selection, stochastic selection, and neutral fixation, respectively. The `-f` option adds a neutral phase below the frequency at which selection began, `-uA` allows recurrent mutation toward the adaptive class, and `-c` stops a partial sweep at an intermediate frequency. Recurrent hitchhiking is available with `-R` for

discoal sweep kernel: source-guided checks

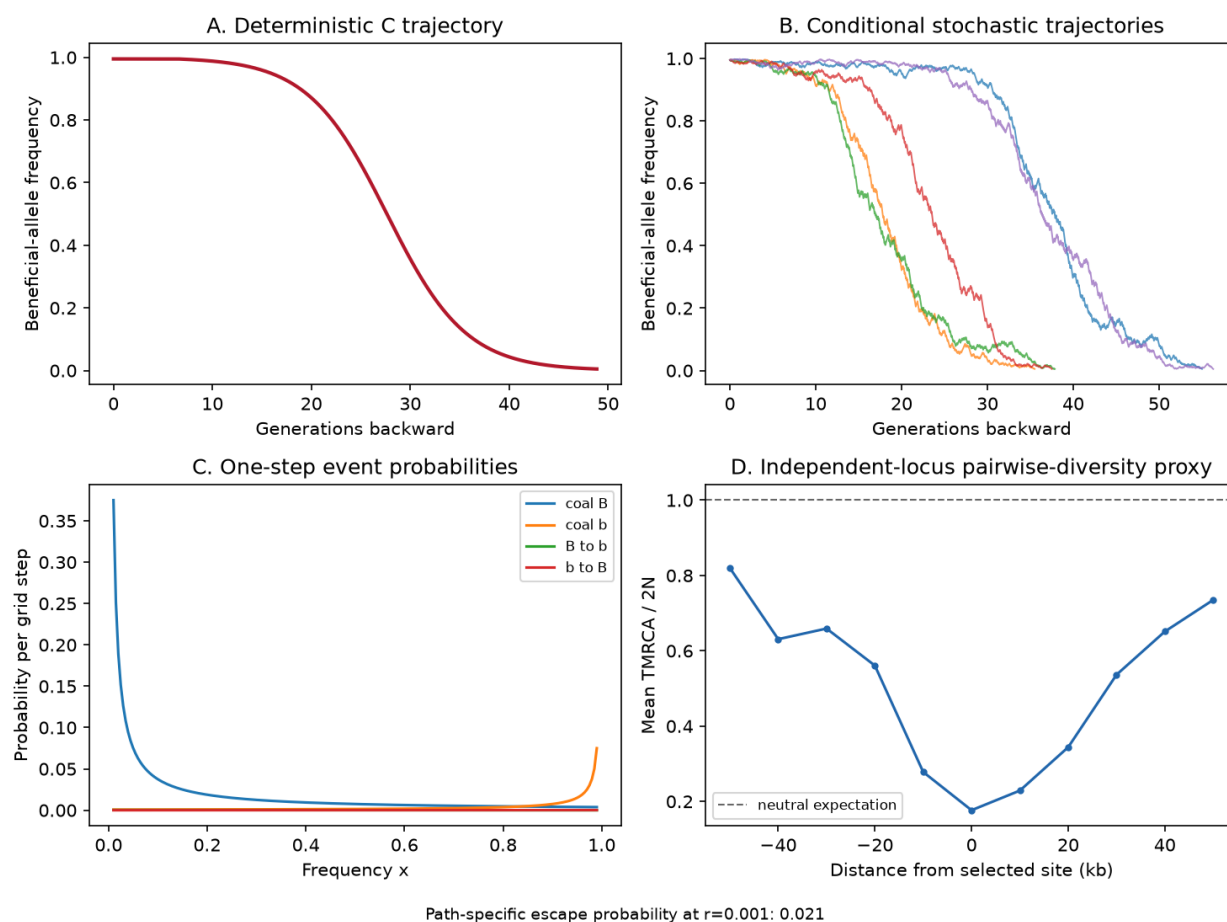


Fig. 28: Source-guided checks for the teaching kernel. The panels show the deterministic production curve, stochastic conditional trajectories, the four single-locus event probabilities, and an independently simulated pairwise-diversity proxy. The last panel uses independent marginal loci and is not an ARG simulation.

sweeps in the locus and `-L` for sweeps to its left. These options are described in the original manual and program paper [Kern and Schrider, 2016].

The program also implements crossing over, gene conversion, instantaneous size changes, multiple populations, migration, population merges, admixture, and ancient samples. Selection is restricted to population 0, and migration into or out of that population is disabled during its sweep phase because `discoal` does not follow selected-allele trajectories in several populations simultaneously.

What is approximate

The trajectory-plus-structured-coalescent construction is an approximation to a selected Wright–Fisher population. It assumes that conditioning on the selected allele’s path captures the relevant effect of selection on linked ancestry. The trajectory is discretized, and the event loop treats each small interval as having at most one event. Finer steps reduce discretization error but increase runtime. The C code uses a default step scalar of 40 and a default sweep effective size of one million; together they give an internal step of $1/(40N_{\text{sweep}})$ in $2N$ time units.

The structured coalescent is especially useful for strong, recent sweeps, but it does not model individual genomes or selected-locus interference. Forward simulators such as SLiM are needed when diploid fitness, multiple linked selected alleles, or explicit pedigrees are central.

The teaching boundary

The rest of this Timepiece derives the single-locus rates and reproduces the trajectory law. `mini_discoal` operates at a fixed recombination distance from the selected site, so recombination appears as switching backgrounds. In the full program, recombination inside the simulated region splits ancestral material. The distinction is methodological, not merely an implementation detail.

Allele-frequency trajectories

The earlier version of this chapter used an integer birth–death chain and called it `discoal`’s conditioned jump process. That chain does not occur in the original software. Production `discoal` uses a two-point approximation to a conditional Wright–Fisher diffusion, following Coop and Griffiths [Coop and Griffiths, 2004]. The distinction matters because the time step, drift, variance, and direction of simulation determine the duration of the sweep and therefore every downstream coalescence and recombination probability.

Scaling and direction

Let x denote the selected-allele frequency and $\alpha = 2N_{\text{sweep}}s$. The program constructs the trajectory backward, beginning close to the sweep endpoint and moving toward a single-copy frequency. If δt is in $2N_{\text{sweep}}$ units, the default grid is

$$\delta t = \frac{1}{40N_{\text{sweep}}}.$$

This corresponds to $2/40 = 0.05$ generations per entry when the same N_{sweep} converts the clock. The command-line option `-i` changes the scalar 40, while `-N` changes the sweep effective size. These parameters control numerical resolution; they are not substitutes for the demographic population-size history.

The stochastic selected phase

Write $q = 1 - x$. Production `discoal` updates q using the forward conditional jump rule and then transforms back to x . Equivalently, the backward selected-frequency step is

$$x' = x - \frac{\alpha x(1-x)}{\tanh[\alpha(1-x)]} \delta t \pm \sqrt{x(1-x)} \delta t,$$

with equal probability for the two signs. The limiting ratio is evaluated as one when $\alpha(1 - x)$ approaches zero. This is a discrete approximation to the conditional diffusion, not a Wright–Fisher generation and not a chain that changes the allele count by exactly one.

`mini_discoal.stochastic_trajectory` implements this law. It exposes the grid size and endpoint frequencies explicitly and returns a `mini_discoal.SweepTrajectory` whose frequencies are ordered backward in time. Production discoal rejects invalid proposed trajectories in its surrounding event logic. The mini reflects a local boundary overshoot, so exact random-number-stream parity is not claimed; the drift, two-point variance, time scaling, and phase change are the parity targets.

Standing variation

For $-f \leq f_0$, selection acts only above f_0 . Once the backward path falls below f_0 , discoal continues with a neutral conditional process,

$$x' = x - x \delta t \pm \sqrt{x(1-x)} \delta t,$$

until the mutational origin is reached. A soft sweep from standing variation is therefore not represented by merely starting a logistic curve at f_0 and then switching immediately to an unstructured neutral coalescent. The neutral standing phase is part of the linked genealogy. The `selection_start_frequency` argument of `mini_discoal.stochastic_trajectory` preserves this distinction.

The deterministic production curve

The deterministic `-wd` mode is also not the textbook logistic curve from $1/(2N)$ to $1 - 1/(2N)$ that appeared in the previous chapter. The C function `detSweepFreq` defines

$$\epsilon = \frac{0.05}{\alpha}, \quad t_s = -\frac{2 \log \epsilon}{\alpha},$$

and evaluates

$$x(t) = \frac{\epsilon}{\epsilon + (1 - \epsilon) \exp[\alpha(t - t_s)]}.$$

Time t again increases backward from the sweep endpoint. The function `mini_discoal.discoal_deterministic_frequency` is a direct translation. For $\alpha = 200$, the values at internal times 0, 0.02, and 0.04 are 0.999750, 0.986538, and 0.573048. These fixed values are regression tests against the source formula.

`mini_discoal.deterministic_trajectory` evaluates this curve on the production grid until the requested origin frequency. For teaching examples, a small `sweep_N` keeps the array manageable. The production default is one million and is intentionally much finer.

Fixation probability

The diffusion fixation probability used for interpretation is

$$h(x) = \frac{1 - e^{-\alpha x}}{1 - e^{-\alpha}}.$$

Its neutral limit is $h(x) = x$. The implementation `mini_discoal.fixation_probability` uses `expm1` for numerical stability. This formula helps explain conditioning, but discoal does not generate its stochastic trajectory by inserting $h(k)$ into an integer nearest-neighbour chain.

The structured coalescent during a sweep

Conditional on a selected-allele trajectory, ancestry is divided into the beneficial background B and the wild-type background b . At frequency x , these backgrounds contain fractions x and $1 - x$ of the population. This construction follows the hitchhiking coalescent [Braverman *et al.*, 1995] and is the central sweep kernel in discoal [Kern and Schrider, 2016].

Coalescence rates

For a diploid population of size N , a pair of lineages on background B coalesces at rate $1/(2Nx)$ per generation. With n_B lineages, the total rate is

$$\lambda_B = \frac{\binom{n_B}{2}}{2Nx}.$$

Similarly,

$$\lambda_b = \frac{\binom{n_b}{2}}{2N(1-x)}.$$

Lineages on different backgrounds cannot coalesce at the selected site. As the backward trajectory approaches the single-copy origin, the beneficial background becomes small and its coalescence rate becomes large. For a hard sweep, any beneficial-background lineages still present at the single mutational origin must share that origin.

The function `mini_discoal.coalescence_rate` evaluates the per-generation rate with an explicit background frequency. Its argument is a frequency, not a number of chromosomes. This prevents the factor-of- $2N$ ambiguity present in the old API.

A single linked locus

For one neutral locus at recombination fraction r from the selected site, background switching has total rates

$$M_{B \rightarrow b} = n_B r (1-x), \quad M_{b \rightarrow B} = n_b r x.$$

The factor $1-x$ appears because a recombination involving a B lineage changes its background only if the other parental chromosome is wild type. The reverse event has probability x .

`mini_discoal.migration_rates` implements these expressions. The term *migration* is useful for the two-background structured coalescent, but no geographic movement is implied.

Why chromosome-scale recombination is different

The single-locus equations do not describe an in-locus crossover in the complete simulator. A crossover can split one active ancestor into two ancestors covering different site intervals. The segment containing the selected site retains the parental selected background, while the other segment receives a background according to x . Both pieces remain in the ancestral recombination graph. Gene conversion similarly changes ancestry over a tract.

The old mini replaced every crossover by a whole-lineage background switch and then simulated sites independently. That loses correlations among marginal trees, cannot produce an ARG, and is not parity with discoal. The revised mini therefore labels its scope as a fixed-distance single-locus kernel.

Discrete event probabilities

During a short interval Δg generations, the four single-locus event probabilities are the rates above multiplied by Δg . In the C code, the same calculation is expressed on the internal $2N$ clock. For example, the beneficial-background coalescence contribution is

$$\frac{\binom{n_B}{2}}{x} \Delta t,$$

where $\Delta t = \Delta g/(2N)$. The raw-generation and internal-time expressions are identical after this substitution.

Production discoal accumulates no-event probability over the grid until an event is accepted, then chooses the event in proportion to its current contribution. The teaching function `mini_discoal.structured_event_probabilities` uses the equivalent one-step probabilities. It refuses a grid on which their sum reaches one, because silently allowing such a grid would violate the at-most-one-event approximation.

`mini_discoal.structured_coalescent_sweep` recomputes all rates after every accepted event. The previous implementation carried an exponential residual across changing rates, allowed several events while retaining stale lineage counts, and then subtracted the full interval hazard again. Those operations did not simulate either a continuous-time inhomogeneous chain or discoal's rejection loop.

Neutral bookends

If a sweep ended τ generations ago, the ordinary coalescent runs from the present to τ . The structured sweep then runs backward through its trajectory. Any remaining lineages continue in the older neutral population. `mini_discoal.simulate_linked_locus_genealogy` implements exactly these three phases for one marginal locus and returns $n-1$ coalescence times.

The distinction between tree height and diversity is important. For two haploid samples, neutral TMRCA has mean $2N$, so mean pairwise diversity is proportional to mean TMRCA. For larger samples, the sample MRCA and total branch length are different quantities. The old chapter summed coalescence event times and called the result diversity; the revised `mini_discoal.pairwise_diversity_profile` uses two samples and normalizes mean TMRCA by the correct neutral expectation $2N$.

Sweep models

The labels *hard*, *soft*, and *partial* describe different histories at the selected locus. They cannot be reduced to different initial values of the same logistic curve. `discoal` changes both the trajectory and the ancestral events available during the sweep [Kern and Schrider, 2016].

Hard sweeps

A hard sweep traces the adaptive class to one mutational origin. `-wd` uses the deterministic production curve, while `-ws` uses the conditional stochastic trajectory. `-wn` conditions a neutral allele on fixation. The sweep endpoint is set by the time argument to these flags, the selected coordinate by `-x`, and the strength of genic selection by `-a` with $\alpha = 2Ns$.

At complete fixation, every sampled lineage begins the sweep phase on background B . Backward recombination may move neutral ancestry to b . At the single-copy origin, all remaining B ancestry shares one ancestor. This last statement applies to a single-origin hard sweep; it must not be reused as a generic rule for recurrent adaptive mutation.

Standing variation

With `-f f0`, the allele was neutral below f_0 and became beneficial at that frequency. Backward simulation therefore has a selected conditional phase from the endpoint to f_0 , followed by a neutral conditional phase from f_0 to the mutational origin. Recombination during the standing phase can preserve several linked haplotypes even though the selected allele itself has a single older origin. This is the soft-sweep mechanism from standing variation [Hermisson and Pennings, 2005].

The old mini ended the structured phase at f_0 and merged all remaining lineages in an ordinary neutral coalescent. That omitted the background-conditioned standing phase. The `selection_start_frequency` option in `mini_discoal.stochastic_trajectory` now represents it explicitly.

Recurrent adaptive mutation

The `-uA` option allows lineages to change selective class through recurrent mutation during the sweep. This can produce several independent adaptive origins, the recurrent-mutation soft sweep described by Pennings and Hermisson [Pennings and Hermisson, 2006]. In the production event loop, the recurrent mutation contribution depends on the number of beneficial-background ancestors and on $1/x$.

The previous chapter supplied a formula called an expected number of independent origins, $2N\mu_a \log(2Ns)/s$. That expression was neither implemented by `discoal` nor a correctly scoped result from the cited soft-sweep theory, so it has been removed. The probability and number of origins depend on the population model, mutation scaling, conditioning,

and sample. Production discoal should be used when recurrent-origin ancestry is required; the revised single-locus mini does not implement `-uA`.

Partial sweeps

`-c c` specifies a sweep whose selected phase ended at frequency $0 < c < 1$. At the recent endpoint, active ancestors are assigned to B with probability c and to b otherwise. The program then runs backward through the sweep trajectory. With `-f`, discoal requires $f_0 < c$ and represents a partial sweep from standing variation.

A partial sweep is not necessarily a sweep that is still occurring at the moment of sampling. The sweep event has its own time, and c is the frequency at which selection ended in the modeled event. Neutral evolution can occur between that endpoint and the samples.

Recurrent hitchhiking and off-locus sweeps

The `-R` option places recurrent sweeps within the simulated locus, while `-L` places recurrent sweeps to its left. Single off-locus events use `-ld`, `-ls`, or `-ln` for deterministic, stochastic, or neutral fixation. For an off-locus sweep, only the recombination distance to the region is required; this is the case most closely represented by the mini's fixed-distance background-switch kernel.

The full feature surface also includes gene conversion, demographic size changes, admixture, migration outside the active sweep population, and ancient samples. These features interact through the production event scheduler. They should not be inferred from the much smaller teaching functions.

discoal and msprime

msprime's `SweepGenicSelection` and discoal use the same broad construction: a conditional selected-allele trajectory drives a two-background structured coalescent [Baumdicker *et al.*, 2022, Kern and Schrider, 2016, Coop and Griffiths, 2004]. They do not expose the same model surface, and matching a few scaled parameters does not make two simulations identical.

Parameter translation

For a locus of span L with per-base mutation and crossover rates μ and r , the standard diploid scaling is

$$\theta = 4N\mu L, \quad \rho = 4NrL, \quad \alpha = 2Ns.$$

`mini_discoal.msprime_to_discoal` applies these conversions, and `mini_discoal.discoal_to_msprime` reverses them. The conversion is a units translation, not a statistical-equivalence guarantee. In particular, discoal's sample-size argument counts haploid chromosomes. `msprime.sim_ancestry` treats an integer sample count as diploid individuals unless told otherwise, so a direct comparison must use `ploidy=1`. The revised converter returns `samples=n` and `ploidy=1` explicitly; the earlier version silently doubled the sample.

For a hard sweep, a matching msprime model also requires

```
sweep = msprime.SweepGenicSelection(
    position=selected_position,
    start_frequency=1 / (2 * N),
    end_frequency=1 - 1 / (2 * N),
    s=alpha / (2 * N),
    dt=1 / (40 * N),
)

ts = msprime.sim_ancestry(
    samples=n,
```

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```
ploidy=1,
population_size=N,
sequence_length=L,
recombination_rate=r,
model=[sweep, msprime.StandardCoalescent()],
random_seed=123,
)
```

The neutral coalescent after the sweep model is necessary because a sweep phase can end before all ancestral material has coalesced. Mutations are added afterward with `msprime.sim_mutations`.

Shared mechanics

Both implementations use a conditional diffusion jump process and compete coalescence with recombination on the two selected backgrounds. Both represent ancestral material along a recombining sequence, rather than simulating marginal sites independently. The `msprime` 1.0 paper reports its sweep implementation and benchmarks against `discoal` [Baumdicker *et al.*, 2022].

Important differences

`discoal` supports deterministic sweeps (`-wd`), stochastic sweeps (`-ws`), neutral fixation (`-wn`), standing variation (`-f`), recurrent adaptive mutation (`-uA`), partial sweeps (`-c`), off-locus sweeps, recurrent hitchhiking, and gene conversion during the sweep. It can combine its models with stepwise size changes, multiple populations, population merges, admixture, and ancient samples, subject to the restriction that selection occurs in population 0 and migration involving that population is disabled during the sweep.

`msprime` emits a tree sequence and composes its sweep phase with other ancestry models. Its public sweep API allows an arbitrary selected position and start and end frequencies; it is not restricted to the midpoint. Current official documentation warns that sweeps with more than one population and population-size changes during the sweep are not implemented. An elevated start frequency alone does not reproduce `discoal`'s standing-variation model, because `discoal` also follows the preceding neutral conditional phase.

Representation

Production `discoal` maintains active ancestral nodes and dynamically allocated ancestry segments, then emits an `ms`-compatible haplotype matrix or marginal Newick trees. It is inaccurate to describe its internal state as merely an $n \times S$ haplotype array. `msprime` records nodes and edges directly in `tskit` tables and emits a tree sequence. Both programs build genealogical ancestry; their storage layouts and output formats differ.

What parity means here

Useful parity checks compare units, haploid sample counts, trajectory endpoints, time-step scaling, and qualitative or statistical summaries over many replicates. Seed-for-seed output equality is not expected because the simulators use different random-number generators and event implementations. The teaching module is held to a smaller target: its deterministic curve, conditional jump moments, single-locus event rates, lineage conservation, and neutral limits must match the source equations. It is not presented as a substitute executable.

Verified discoal exercises

The demo now separates production-software validation from the pedagogical single-locus calculation. The previous figure said it used data simulated with `msprime` although its script used only the `mini`, compared a TMRCA quantity with diversity, and overlaid an unsupported rational curve as a classical diversity law. Those labels and the unsupported curve have been removed.

Revised discoal chapter: independent validation targets

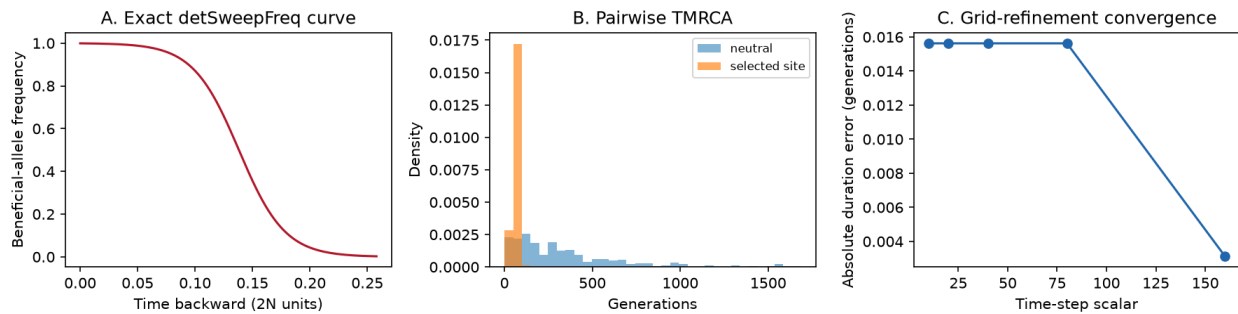


Fig. 29: Independent validation targets: the exact deterministic C curve, the pairwise TMRCA shift at the selected site, and convergence as the trajectory grid is refined.

Production command

After compiling paper-era upstream discoal at commit 82971bf, the executable validation used this reproducible hard-sweep command:

```
./discoal 20 50 1000 -t 10 -r 20 -ws 0.05 -a 1000 -x 0.5 -d 23456 78901
```

Here 20 is the number of haploid samples, 50 the number of replicates, and 1000 the number of discrete sites. `-t 10` and `-r 20` are whole-locus values of $4N\mu_L$ and $4Nr_L$. `-ws 0.05` places a stochastic sweep endpoint 0.05 units of $4N$ generations ago, `-a 1000` sets $2Ns$, and `-x 0.5` places the selected site at the center. The `-d` values are discoal's two random seeds.

All 50 outputs passed the structural checks below, and their mean pairwise diversity in the central fifth of the locus was 0.117 mutations per pair, versus 0.213 and 0.278 in the two flanking fifths. This small run is a smoke test rather than a benchmark. Validation should inspect more than successful execution. Every replicate must contain 20 equal-length haplotypes, positions must be ordered within the unit interval, and the reported number of segregating sites must equal each haplotype length. Across many replicates, neutral simulations should recover the expected coalescent summaries, while linked sweep simulations should show a reduction in pairwise diversity near the selected site. These are distributional checks, not a claim that one random seed has a prescribed answer.

As an independent neutral check, 1000 replicates with `-t 10` and no recombination or sweep gave a mean of 35.508 segregating sites, close to the standard-coalescent expectation $10H_{19} = 35.477$. Exact random output and performance are version-specific, which is why the commit is part of the command's provenance.

Source-guided mini

The module demo can be run with

```
python -m watchgen.mini_discoal
```

It constructs `mini_discoal.deterministic_trajectory` with a small sweep effective size, checks `mini_discoal.escape_probability` on that explicit path, and estimates `mini_discoal.pairwise_diversity_profile`. The profile uses two samples, so mean TMRCA divided by $2N$ is a valid relative pairwise-diversity proxy. Each position is nevertheless simulated independently; this is not a chromosome-scale ARG simulator.

Automated checks

The dedicated tests verify the deterministic C formula at fixed times, the conditional jump endpoints and neutral standing phase, exact rate scaling, rejection of a coarse event grid, conservation of lineages, the $2N$ neutral pairwise-TMRCA expectation, and haploid sample parity in the msprime converter. The documentation tests also reject the former whole-program parity claims and require the audited upstream commit `7d0955f4107053c135d2086790b0426457147a8e`, the paper-era executable commit `82971bf`, and primary references [Baumdicker *et al.*, 2022, Kern and Schrider, 2016, Braverman *et al.*, 1995, Coop and Griffiths, 2004].

The implementation used by these exercises is kept in one place:

```
"""Source-guided pedagogical kernels for :program:`discoal`.
```

*This module deliberately does ***not*** call itself a Python translation of the full program. Production discoal simulates a discrete ancestral recombination graph, including crossover, gene conversion, demography, and several sweep models. The code below isolates two pieces that can be checked directly against the published algorithm and the C implementation:*

- * the backward conditional jump process for the selected-allele trajectory; and*
- * the two-background structured coalescent for one neutral locus at a fixed recombination distance from the selected site.*

Times in :class:`SweepTrajectory` are stored in units of ``2 \* sweep\_N`` generations, matching discoal's internal coalescent clock. Command-line sweep times are in ``4 \* N`` generations and are converted internally by discoal.

Ground truth: Kern and Schrider (2016), Coop and Griffiths (2004), the msprime 1.0 sweep description, and discoal source commit ``7d0955f4107053c135d2086790b0426457147a8e``.

```
"""
```

```
from __future__ import annotations
```

```
from dataclasses import dataclass
```

HOW TO USE THIS GUIDE

This book is organized around a central metaphor: **the watchmaker’s workshop**.

Each **Timepiece** is a complete algorithm from population genetics – a working mechanism that you will disassemble, understand gear by gear, and then reassemble with your own hands. Before diving into a Timepiece, you may need tools from **The Workbench** – prerequisite concepts explained with the same depth and care.

The recommended approach:

1. Read the **Philosophy** to understand why we do things this way.
2. Work through **The Workbench** to build your mathematical and computational toolkit.
3. Pick a **Timepiece** and build it, gear by gear.

Every chapter follows the same pattern:

- **Why** – motivation and biological context (why does this mechanism exist?)
- **The Math** – rigorous derivation from first principles (every step shown, every assumption stated)
- **The Code** – Python implementation you write yourself (clear, readable, tested)
- **Verify** – tests and sanity checks to confirm your gears mesh properly

We believe this rhythm – motivation, math, code, verification – is the most reliable way to build genuine understanding. Each cycle reinforces the last, like the oscillation of a balance wheel keeping perfect time.

What you need

- **Python 3.8+** with NumPy and SciPy (we’ll explain every function we use)
- **Some familiarity with probability and calculus** – but don’t worry if you’re rusty. We introduce every concept we use, with intuition first and formulas second. If you know what a derivative is and what “probability” means, you have enough to start. We’ll teach the rest as we go.
- **Willingness to build things from scratch** – this is the most important ingredient

What you do NOT need

- Prior knowledge of population genetics (we start from zero)
- Experience with any existing genetics software
- Advanced mathematics (we explain every concept as it arises – exponential distributions, integrals, Markov chains, all of it)
- A PhD (though you might feel like you’ve earned one by the end)

i A note on our teaching approach

Many textbooks assume you already know probability, calculus, and linear algebra fluently. We don't. When we use the exponential distribution for the first time, we'll explain what it is and why it appears. When we integrate a function, we'll say what integration means in that context. When we invoke a matrix, we'll explain what it represents.

This doesn't mean we skip the math – we embrace it. It means we treat every mathematical tool like a physical tool on the workbench: we pick it up, show you what it does, demonstrate how to use it, and *then* put it to work.

Like a good watchmaker teaching an apprentice, we never hand you a tool without first explaining what it's for.

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